THE FINAL EXAM TEXTBOOK · BOOK 1 OF 9

# General Psychology

Psychology as a science, its schools and Indian psychology, sensation, perception, attention and consciousness, learning, memory, thinking and language, intelligence, motivation, emotion and stress

Written for Sihan, for the JSS AHER MA Clinical Psychology entrance, 27 October 2026. Built from every paper and mock on the site.

COLOUR EDITION · A4, DOUBLE SIDED · 21 OCTOBER 2026

## How this book works

This is Book 1 of the Final Exam Textbook: General Psychology and its theories, taught from scratch for the JSS AHER MA Clinical Psychology entrance on 27 October 2026. It is built from every past paper, question bank and practice mock on the site. Every question was sorted into the chapter that should teach it, and every option in it, wrong ones included, is explained somewhere in the book, because next year's paper can make a wrong option the right answer. Each chapter was then checked by a second reader against the source textbooks (Pinel, Eysenck and Keane, Schultz, the Psycho Didi notes and others) for names, years and facts, and against every question again for gaps.

**How to read it.** Each chapter opens with a real case or experiment, lists its sections with a learning goal for each, and then teaches them in order, the way Pinel does: the question, the idea, the study behind it and who did it, then what it means. Read the High yield sections first if time is short (the tags are explained below), and do every Scan Your Brain before moving on. Each chapter ends with the case revisited, a summary of every section, a Who's who, the key terms, real past-paper questions with the answers explained, and a one-page High-yield sheet for your last read before the exam.

Key terms are in bold colour where they are defined, and the index at the back points to those pages. People are in bold where they are introduced, with their full name and years.

### The boxes

**EXAM TRAP**

The confusable pair, the bait option, the famous misattribution. Read every one.

**KEY STUDY**

A classic experiment told as a story: the question, the method, the result, the criticism.

**CLINICAL LINK**

Where the idea matters to a clinical psychologist, and to an interview panel.

**REMEMBER**

The one idea in the section you must not leave without.

**INDIAN CONTEXT**

Indian research, Indian psychology and Indian applications.

**CASE**

A person whose story makes the concept concrete.

**SCAN YOUR BRAIN**

Fill-in-the-blank checks after every two or three sections, answers underneath. Cover the answers first.

### The colour code: how often each topic is asked

Every numbered section carries a tag, and every subsection that has been asked carries dots. Both are counted from the past papers and practice mocks on the site: each question was matched to the section that teaches it, a real past-paper question scores 2 points and a mock or question-bank item 1 point.

| HIGH YIELD · 23 QS ●●● | MEDIUM YIELD · 8 QS ●● | LOWER YIELD · 1 Q ● |
| --- | --- | --- |
| The most-asked quarter of the book's sections (at least 14 points). Read these first and know every trap. | Asked regularly (at least 6 points). Read them fully. | Asked rarely or not yet. Still in the syllabus, so still taught in full. |

"23 Qs" is the number of questions on the site that this section answers. The same tags sit next to each section in the Contents, and the page after this one shows where the marks are across the book.

### Printing

Print A4, double sided, flip on the long edge. Every chapter opens on a right-hand page, the inner margin is wider for binding, and page numbers sit on the outer corner. The colour and the black and white editions have identical pages, page numbers and index, so either can be printed.

# Where the marks are

819 questions from the past papers, question banks and practice mocks on the site are taught in this book. This is how they spread across the chapters.

| Chapter | Questions | From real papers | Share of the book | High-yield sections |
|---|---|---|---|---|
| 1 Psychology as a Science: Roots, Founders and the First Schools | 46 | 12 | 5.6% | 1 of 8 |
| 2 Schools and Systems: Behaviourism, Gestalt, Psychoanalysis, Humanism, the Cognitive Revolution and Indian Psychology | 103 | 39 | 12.6% | 4 of 9 |
| 3 Sensation and Psychophysics | 44 | 16 | 5.4% | 1 of 12 |
| 4 Perception: Organising and Interpreting the World | 37 | 25 | 4.5% | 2 of 9 |
| 5 Attention and Consciousness: Selective Attention, Sleep, Dreams, Hypnosis and Altered States | 69 | 28 | 8.4% | 3 of 11 |
| 6 Learning: Conditioning, Cognition and Observation | 144 | 78 | 17.6% | 4 of 10 |
| 7 Memory: Encoding, Storage, Retrieval and Forgetting | 111 | 72 | 13.6% | 5 of 11 |
| 8 Thinking, Problem Solving, Reasoning, Decision Making, Creativity and Language | 53 | 28 | 6.5% | 1 of 9 |
| 9 Intelligence: Theories, Measurement, Heredity and Environment | 63 | 22 | 7.7% | 1 of 8 |
| 10 Motivation: Why We Do What We Do | 76 | 36 | 9.3% | 3 of 10 |
| 11 Emotion and Stress | 73 | 48 | 8.9% | 3 of 10 |

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CHAPTER 1

# Psychology as a Science: Roots, Founders and the First Schools

IN THIS CHAPTER

1.1 ●● What Is Psychology?: define psychology as it has been defined across history, state its four goals, separate basic from applied work and name the main fields, including the one that studies animals.

1.2 ●● Philosophical Roots: A Long Past: trace nativism, rationalism, empiricism and associationism from Plato to Kant, and lay out the classic positions on the mind-body problem.

1.3 ● Physiological and Biological Roots: explain how nerve physiology, brain localisation, Darwin's evolution and Galton's study of individual differences prepared the ground for a laboratory science.

1.4 ● Measuring the Mind: Psychophysics and Mental Chronometry: show how Weber, Fechner, Donders and Ebbinghaus proved that mental events could be measured.

1.5 ●● Wilhelm Wundt and the Founding of Experimental Psychology: describe Wundt's laboratory, his voluntarism, his theory of feeling, his method and his folk psychology, and name his famous students.

1.6 ●● Structuralism and Its European Rivals: explain Titchener's structuralism, its method and its errors, and the challenges from Wurzburg, Brentano, Stumpf and G. E. Muller.

1.7 ●●● Functionalism and the American Pioneers: explain why functionalism arose, what James, Dewey, Angell, Carr and Woodworth contributed, and who built American psychology's institutions.

1.8 ● Reading the History: Zeitgeist, Great Persons and "Fathers": compare the two ways of explaining history, master the "father of" and "first" questions, and trace the arrival of psychology in India.

Weightage. 46 of the 819 questions mapped in this book (5.6%) test this chapter, 12 of them from real past papers. 1 of its 8 sections is High yield.

Leipzig, in the winter of 1879. In a small room of an old university building known as the Konvikt, a doctoral student named Max Friedrich sits in front of a clockwork timer called a Hipp chronoscope, a machine built to measure intervals in thousandths of a second. A stimulus is presented; Friedrich presses a telegraph key. Then the procedure is changed: now he must press the key not the instant he senses the stimulus, but only once he has clearly grasped what it is. The second kind of reaction takes longer, and the extra time, Friedrich and his supervisor reasoned, is the time the mind needs to *apperceive*, to bring an impression into the clear focus of attention. Trial after trial, the numbers go into a notebook.

His supervisor was a 47-year-old professor of philosophy, Wilhelm Wundt. When Wundt had arrived in Leipzig four years earlier, the university had given him this room to store the apparatus he used in lecture demonstrations. By the winter of 1879 it had become something new: a place where students carried out original experiments on the mind and published them as research. Friedrich's study of the duration of apperception became the first dissertation to come out of that room, and historians who later combed the Leipzig archives dated the start of the work to the last weeks of 1879.

Here is the puzzle that should bother you. Almost nothing in that room was new. The chronoscope had been built for physicists and astronomers. The idea of measuring the time taken by mental acts came from a Dutch physiologist. The word *apperception* came from a German philosopher who had died in 1716. And at Harvard, in 1875, the same year Wundt got his room, an American named William James had also been given space for psychological apparatus. So why does every textbook in the world, and nearly every entrance paper in India, treat Leipzig in 1879 as the moment psychology was born?

The answer to that question is the subject of this chapter. To find it you need to know where the parts in that room came from: the philosophers who argued for two thousand years about where knowledge comes from, the physiologists who timed nerves and mapped brains, the naturalists who put human beings back among the animals, and the physicists who dared to measure sensation. Then you need to see what Wundt and his students built from those parts, and what the next generation, in Germany and in America, did with it. Remember Max Friedrich pressing his key; we will come back to him at the end of the chapter.

## 1.1 What Is Psychology? MEDIUM YIELD · 5 QS

Ask ten people at a chai stall what psychology is and you will hear ten answers: reading minds, treating madness, understanding why people do what they do. Ask ten textbooks and you will get fewer answers, but you will notice something odd: the textbooks of 1890, 1920 and today do not agree with one another either. The definition of psychology has changed several times, and each change tells you which school was in charge. Entrance papers love these definitions, so we start with them.

## From soul to behaviour: a definition that kept changing ●●

The word itself is Greek: *psyche* (breath, soul) and *logos* (study or reasoned account). Literally, then, **psychology** began as "the study of the soul". The Greek philosophers did not use the word as the name of a subject; it first appears in Latin book titles of the sixteenth century (the German scholar Rudolf Goclenius used it in 1590), and the philosopher **Christian Wolff** (1679 to 1754) made it standard with two books, *Psychologia empirica* (1732) and *Psychologia rationalis* (1734). For the Greeks and the medieval scholars, the soul was the principle of life and reason; **Aristotle** (384 to 322 BCE) wrote the first systematic treatise on it, *De Anima* (*On the Soul*), which is why some older Indian textbooks call him the "father of psychology".

From the seventeenth century, philosophers spoke less of the soul and more of the *mind*: its ideas, its reasoning, its sensations. When psychology became a laboratory science in the nineteenth century, the definition narrowed again. For Wundt and his student Titchener, psychology was the science of *conscious experience*, studied by trained observers looking inward. William James, in 1890, opened his great textbook with the sentence "Psychology is the Science of Mental Life, both of its phenomena and of their conditions."

Then came the revolt. In 1905 and again in 1908 the British psychologist **William McDougall** (1871 to 1938) defined psychology as the positive science of the *conduct* or behaviour of living creatures, and in 1913 John B. Watson declared that psychology should drop consciousness altogether and study only **behaviour**, by which he meant anything an organism does that can be observed and measured from outside. (Watson and behaviourism are the subject of Chapter 2.) The new definition spread through the textbooks. **Robert Sessions Woodworth** (1869 to 1962), whom you will meet again in Section 1.7, defined psychology as the scientific study of the activities of the individual in relation to the environment. The definition you will most often find credited in Indian textbooks, "psychology is the scientific study of behaviour", is attributed to the American textbook writer **Norman L. Munn**.

Watson's definition won battles but lost the war. By the 1960s psychologists were again studying memory, imagery and thinking, and today's standard definition reunites the two halves: psychology is *the scientific study of behaviour and mental processes*. Each phrase in that definition is doing work. *Behaviour* is overt: speaking, walking, blushing, pressing a key. By contrast, **mental processes** are internal and covert: perceiving, remembering, thinking, dreaming, feeling. You cannot see a student remembering the year of Wundt's laboratory, but you can infer it from what he writes. And *scientific* means that claims are tested against systematic, public observation that someone else could repeat, not against authority or intuition.

An old classroom quip sums up the whole story: first psychology lost its soul, then it lost its mind, then it lost consciousness, and now it has behaviour, of a sort. Table 1.1 puts the sequence in order.

Table 1.1 How psychology has been defined.

| Period | Psychology is the study of | Associated with |
|---|---|---|
| Ancient Greece | the soul (psyche) | Plato, Aristotle (*De Anima*) |
| 17th to 19th century | the mind | Descartes, Locke, the associationists |
| 1879 to about 1910 | conscious experience; "mental life" | Wundt, Titchener; James (1890) |

| Period | Psychology is the study of | Associated with |
|---|---|---|
| 1905 to 1913 onward | conduct, behaviour | McDougall (1905, 1908); Watson (1913) |
| Textbook definitions | activities of the individual in relation to the environment; the scientific study of behaviour | Woodworth; N. L. Munn |
| Today | behaviour and mental processes | most modern textbooks |

EXAM TRAP

"In the definition of psychology, mental processes mean ..." The answer is *internal, covert processes*. "Outward behaviour" and "overt actions and reactions" describe the *other* half of the definition, and "only animal behaviour" describes neither. A related bank question asks who "first proposed and established psychology as the scientific study of behaviour" and keys **N. L. Munn**, because Indian textbooks quote that definition from Munn. Historically McDougall (1905, 1908) and Watson (1913) came first; if Watson appears as an option alongside Munn, read the stem closely: "established" behaviourism points to Watson, "the definition" points to Munn. Do not confuse William James (science of mental life) with any behaviour definition.

## The four goals: describe, explain, predict, control

Every science tries to do four things with the events it studies, and psychology is no exception. Take one example and push it through all four: a student who "blanks out" in the examination hall at Mysuru even though he knew the material the night before.

- **Describe.** First we must say exactly what happens, and to whom: racing heart, sweaty palms, a mind that will not retrieve answers, perhaps in the first ten minutes only. Careful description separates this from simple lack of preparation.
- **Explain.** Next we ask why. Perhaps very high arousal and catastrophic thoughts ("If I fail, everything is over") compete with retrieval for limited attention. An explanation is a cause that can be tested.
- **Predict.** A good explanation lets us forecast who will blank out and when: students with high scores on a test-anxiety scale, under time pressure, on high-stakes papers.
- **Control** (many textbooks now say *influence* or *change*). Finally we use the knowledge to change the outcome: practice under timed conditions, relaxation training, rehearsing realistic self-statements. This goal is where clinical psychology lives.

Some textbooks add a fifth goal, improving the quality of life, but the four above are the classic list. Watson put prediction and control at the very centre of his definition, a point we will return to in Chapter 2.

## Basic and applied research

Psychologists who study a problem only to understand it are doing **basic research**; those who study it to solve a practical problem are doing **applied research**. When Hermann Ebbinghaus memorised lists of meaningless syllables in the 1880s to discover how quickly memories fade, he was doing basic research; when Lightner Witmer opened a clinic in 1896 to help a boy who could not learn to spell, he was doing applied work. The line is not a wall: Witmer's clinic drew on the basic science of the laboratory, and applied problems constantly

raise new basic questions. You will see in Section 1.7 that the split between "pure" and "applied" psychology was one of the main arguments between the first two schools.

## The fields of psychology ●●●

Modern psychology is a family of specialities. You will study several of them in other books of this series, so here we need only a map (Table 1.2). Two of them deserve a sentence more, because entrance papers test the boundary between them. Put simply, **clinical psychology** is the assessment, diagnosis and treatment of psychological disorders and distress; its practitioners work with conditions ranging from anxiety to schizophrenia, in hospitals, clinics and private practice. **counselling psychology** overlaps heavily but has traditionally focused on people in the normal range who face adjustment problems: career choice, relationships, study difficulties, life transitions. In India, the Rehabilitation Council of India licenses clinical psychologists, not counsellors, to work with psychiatric disorders, which is one reason your interviewers will care whether you understand the difference.

The branch that studies the behaviour of animals is **comparative psychology**: it compares species, often in the laboratory, to understand how behaviour and learning evolved and what is common to humans and other animals. Its biological cousin, **ethology**, studies animal behaviour in natural settings and is associated with the zoologists Konrad Lorenz and Nikolaas Tinbergen. You will see in Section 1.3 that comparative psychology grew directly out of Darwin's work.

Three more names in Table 1.2 are easily confused. The first, **evolutionary psychology**, explains present-day behaviour and mental mechanisms as adaptations shaped by natural selection, that is, as part of our genetic inheritance from ancestors who faced recurring problems of survival and reproduction. The second, **behaviour genetics**, asks a narrower question: how much of the variation among people in a trait is due to differences in their genes, studied mainly through twins, families and adoption. The third, **differential psychology**, is the study of individual differences in abilities and personality; the German psychologist **William Stern** (1871 to 1938) named it in 1900, but its founder in practice was Francis Galton (Section 1.3).

Table 1.2 The main fields of psychology.

| Field | What it studies |
|---|---|
| Clinical | Assessment, diagnosis and treatment of psychological disorders |
| Counselling | Adjustment problems of people in the normal range: career, relationships, transitions |
| Health | How behaviour, stress and beliefs affect physical illness and recovery |
| Developmental | Change across the lifespan, from conception to old age |
| Social | How people think about, influence and relate to one another |
| Industrial-organisational | Behaviour at work: selection, training, motivation, leadership, organisations |
| Educational and school | Learning and teaching; children's learning problems in schools |
| Forensic | Psychology applied to law: testimony, criminal behaviour, competence |
| Comparative | Animal behaviour, compared across species |
| Neuropsychology | Brain-behaviour relations, often through people with brain damage |
| Abnormal | The nature and causes of psychological disorders (psychopathology) |

| Field | What it studies |
|---|---|
| Experimental | Basic processes (sensation, learning, memory) studied by experiment |
| Differential | Individual differences in ability and personality |
| Evolutionary | Behaviour and mind as products of natural selection |

**EXAM TRAP**

"Which branch focuses on the study of animal behaviour?" is **comparative psychology**. Abnormal psychology studies disorders; differential psychology studies individual differences among people; social psychology studies people in relation to others. "What does clinical psychology primarily focus on?" is **the assessment and treatment of mental illness**; industrial applications and organisational management belong to industrial-organisational psychology, and behavioural changes in animals to comparative psychology. The bank question asking which branch studies "behaviour in relation to genetic inheritance" keys **evolutionary psychology**; if behaviour genetics is ever offered as an option, prefer it for questions about heritability.

With a definition and a map in hand, we can ask how this discipline came to exist at all. The German memory researcher Hermann Ebbinghaus put the answer in one famous sentence in 1908: "Psychology has a long past, but only a short history." The long past is two thousand years of philosophy and a century of physiology; the short history begins in rooms like Wundt's. Figure 1.1 lays out the short history; the next three sections trace the long past that made it possible.

**Figure 1.1** The short history of psychology at a glance. Notice how the roots (left) cluster in German physiology and British biology, and how quickly the institutions (right) followed once Wundt's laboratory opened in 1879.

## 1.2 Philosophical Roots: A Long Past MEDIUM YIELD · 5 QS

Long before anyone measured a reaction time, philosophers had asked the three questions that still organise much of psychology. Where does knowledge come from? How are ideas connected to one another? And how is the mind related to the body? Psychology inherited all three, and the answers given between ancient Athens and eighteenth-century Germany are examined every year in Indian entrance papers. Let us define the main positions first, then meet the people who held them.

On the first question, there are two classic answers. The first, **nativism**, holds that some knowledge, ideas or abilities are inborn, present before any experience. The second, **empiricism**, holds that all knowledge comes from experience, delivered through the senses. A third term, **rationalism**, is often paired with nativism: it holds that reason, not the senses, is the chief source of knowledge, and that the senses can deceive. Keep the two contrasts apart. Nativism versus empiricism is a question about *origins* (inborn or learned?). Rationalism versus empiricism is a question about the *route* to knowledge (reasoning or sensing?). In practice the rationalists were usually nativists, and the empiricists were nearly always anti-nativists, which is why exam setters sometimes treat the pairs as one.

### Plato and Aristotle

**Plato** (about 427 to 347 BCE) was a nativist and a rationalist. In his dialogue *Meno*, Socrates draws a theorem of geometry out of an uneducated slave boy simply by asking him questions, and concludes that the boy was not learning but *recollecting* knowledge his soul already possessed. For Plato, the senses show us only shifting copies of the true, unchanging Forms; real knowledge comes from reason turning inward. He also divided the soul into three parts (reason, spirit and appetite), an early model of conflict within the mind that you may hear echoed when you study Freud.

Plato's student Aristotle disagreed. In *De Anima* he treated the soul not as a separate thing trapped in the body but as the form or function of a living body, the way seeing is the function of an eye. And he was an empiricist: the mind at birth, he wrote, is like a writing tablet on which nothing has yet been written, and knowledge begins with the senses. (Hold on to that tablet; John Locke will pick it up two thousand years later.)

Aristotle also gave psychology its first theory of how one idea leads to another. In his short work *On Memory and Recollection* he observed that when we try to recall something we start from a thing that is *similar* to it, *contrary* to it, or was *contiguous* with it (close to it in time or place). These are the **laws of association**: **similarity**, **contrast** and **contiguity**, to which later writers added *frequency* (ideas experienced together often are linked more strongly). Try it on yourself. Hear the name "Sachin" and "Tendulkar" follows by contiguity, because you have always met the two words together; "Virat Kohli" may follow by similarity (another great Indian batsman); and the thought of the slowest batsman you ever watched may follow by contrast.

EXAM TRAP

Aristotle is the *empiricist* of the pair and Plato the *nativist*; students often reverse them because Aristotle is called the "father of psychology" (for *De Anima*) and they assume he must have founded everything. Aristotle's three laws are **similarity, contrast, contiguity**. David Hume's three laws (below) are **resemblance, contiguity in time or place, and cause and effect**. If a question lists "contrast", it is Aristotle; if it lists "cause and effect", it is Hume.

## Rene Descartes: the body as a machine

As a young man early in the seventeenth century, **Rene Descartes** (1596 to 1650) walked in the royal gardens at Saint-Germain-en-Laye, near Paris, where hydraulic statues moved, played music and even sprayed water at visitors who stepped on hidden plates in the path. The statues seemed alive, yet they were only pipes, water and valves. Descartes drew a bold conclusion: perhaps the living body is the same kind of machine. Nerves could be tubes, and "animal spirits", a fine fluid stored in the brain's cavities, could flow through them to inflate the muscles.

Using this model, Descartes gave the first clear description of the **reflex**: an automatic, unlearned response to a stimulus that needs no mind at all. Put your hand near a fire, he said, and the heat moves particles in the skin, which tug on a thread inside the nerve, which opens a tiny valve in the brain, which releases animal spirits that flow back down to the arm muscles and pull the hand away. The details were wrong, but the idea that much behaviour is mechanical stimulus and response would run, through the Russian physiologists, all the way to behaviourism.

The body, then, is a machine, and animals, which Descartes believed had no souls, are nothing but machines (automata). But the human being has something more: a mind, a thinking thing that has no size or location and cannot be doubted. "I think, therefore I am" (in French, *je pense, donc je suis*), he wrote in his *Discourse on Method* (1637). This is **dualism**: mind and body are two different kinds of substance, the thinking substance and the extended (physical) substance. And because the two plainly affect each other (a pinprick causes pain; a decision moves an arm), Descartes was an interactionist. He placed the meeting point in the **pineal gland**, a small structure deep in the brain, chosen because, unlike most brain structures, it is single rather than paired and sits near the centre of the brain. His last book, *The Passions of the Soul* (1649), described how bodily events stir the emotions through this gland.

Descartes was also a rationalist and a nativist. He held that certain **innate ideas** are not learned from experience but are built into the mind: the ideas of God, of the self, of infinity, and the axioms of geometry. You will see why this matters when we reach Kant and, in Chapter 2, the Gestalt psychologists.

REMEMBER

Descartes left psychology four legacies: (1) **mechanism** and the reflex, which led through reflex physiology to Pavlov and behaviourism; (2) **innate ideas**, which led through Kant to the nativism of the Gestalt school; (3) **dualism**, which created the modern mind-body problem; and (4) **rationalism**, against which the British empiricists reacted. A practice question asks what Descartes' dualism "influenced" and keys *behaviourism and Gestalt psychology* on exactly this reasoning. "Empiricism and rationalism" is the wrong pairing because Descartes was himself a founder of modern rationalism, and empiricism was the reaction against him.

## The British empiricists: Hobbes, Locke, Berkeley and Hume ●●

Across the English Channel a very different tradition grew up, one that insisted the mind contains nothing that did not come in through the senses. **Thomas Hobbes** (1588 to 1679), best known for his political book *Leviathan* (1651), was its blunt forerunner. Hobbes was committed to **materialism**, the view that only matter exists; he held that sensations are motions in the body caused by motions outside it, and that thinking is a "train of thoughts" in which one motion follows another in the order in which they were first experienced. He also argued that human action is driven by the pursuit of pleasure and the avoidance of pain. Many historians call him the founder of British empiricism.

The philosopher who made empiricism a full theory of mind was **John Locke** (1632 to 1704). Locke tells us that his great book began when a group of five or six friends met in his rooms to discuss a philosophical problem and got nowhere; it struck him that before arguing about what we know, they ought to examine what the human understanding is capable of knowing. Nearly twenty years later he published *An Essay Concerning Human Understanding* (1690). Its central claim is that the mind at birth is "white paper, void of all characters", the idea usually summed up by the Latin **tabula rasa** (blank or scraped tablet). There are no innate ideas. Every idea comes from one of two sources: *sensation* (experience of the outside world) or *reflection* (the mind's experience of its own operations, such as thinking and doubting).

Locke divided ideas into **simple ideas**, which cannot be broken down further (the redness, sweetness and roundness of an apple), and **complex ideas**, which the mind builds by combining simple ones (the idea "apple" itself). This is the beginning of the "mental chemistry" that runs through the rest of this chapter: complex mental life built from simple elements.

He also distinguished two kinds of qualities in objects. The **primary qualities** (solidity, extension or size, shape, motion and number) exist in the object itself, and our ideas of them resemble the object. The **secondary qualities** (colour, sound, taste, smell and warmth) exist only as powers to produce sensations in a perceiver. Locke's own demonstration is worth doing. Put one hand in hot water and the other in cold water for a minute; then plunge both into a bowl of lukewarm water. The same water feels cold to the first hand and warm to the second. Since the water cannot be both hot and cold, warmth must be in the perceiver, not in the water. Its size and shape, by contrast, feel the same to both hands.

Finally, in the fourth edition of the *Essay* (1700) Locke added a chapter titled "Of the Association of Ideas", giving psychology the phrase. Interestingly, he used association mainly to explain irrational connections: ideas joined by chance or custom (a room where some-

thing terrible happened becomes unbearable) that produce prejudice and what he did not hesitate to call a kind of madness.

> **INDIAN CONTEXT**
>
> In 1688 the Irish scientist William Molyneux wrote to Locke with a puzzle: if a man born blind, who had learned to tell a cube from a sphere by touch, were suddenly given sight, could he tell which was which by looking alone? Locke, a good empiricist, said no: the link between seen and felt shape must be learned. For three centuries "Molyneux's problem" could not be tested. Then, in a study published in 2011, Richard Held, Pawan Sinha and colleagues working with Project Prakash in India tested children who had been blind from birth with treatable cataracts. Immediately after surgery the children could see the shapes clearly but could not pick out by sight the shape they had just felt; within days, the link appeared. Locke's answer was largely right, and the learning was fast.

**George Berkeley** (1685 to 1753), an Irish bishop, pushed Locke's argument to a startling end. If colour and warmth exist only in the perceiver, why not size and shape too? All we ever know are our own perceptions. To exist, for any object, is to be perceived: in Latin, *esse est percipi*. This position is called subjective idealism or mentalism. Berkeley also made a lasting contribution to the psychology of perception. In *An Essay Towards a New Theory of Vision* (1709) he argued that we are not born able to see distance; we learn it, by associating visual cues (such as the feeling of the eyes converging on a near object, or blur) with the experience of reaching and walking. This is the first empiricist theory of depth perception, and the nativist reply to it is one of the threads of Chapter 4.

**David Hume** (1711 to 1776), a Scot, published *A Treatise of Human Nature* in 1739 and 1740. He divided the contents of the mind into *impressions* (vivid, immediate experiences) and *ideas* (faint copies of impressions, as when you remember a taste). Ideas are joined, he said, by three laws of association: resemblance, contiguity in time or place, and cause and effect. He thought of association as a kind of gentle mental force, the mind's equivalent of Newton's gravity. Then he turned the analysis on causation itself and found that we never see one event *make* another happen; we see only that the two have always been joined (constant conjunction), and the mind forms the habit of expecting one after the other. Even the self, Hume said, is nothing but "a bundle or collection of different perceptions". Hume's scepticism woke the German philosopher Immanuel Kant, as we shall see.

### Associationism: Hartley, the two Mills and Bain

Out of British empiricism grew a doctrine with a name of its own. Its name, **associationism**, refers to the view that complex mental life, including memory, thought and emotion, is built up by linking simple elements (sensations and ideas) according to a few laws, above all contiguity and frequency. Four writers turned it into a system.

**David Hartley** (1705 to 1757), a physician, published *Observations on Man* in 1749. He was the first to give association a physiological basis: sensations are vibrations in the nerves and brain, and they leave behind miniature vibrations, which he called *vibratiuncles*, that are ideas. When two vibrations occur together often, they become linked, so that one later sets off the other. Contiguity, strengthened by repetition, was for Hartley the master law, and he distinguished simultaneous from successive associations. Historians usually credit him with founding associationism as a formal doctrine.

**James Mill** (1773 to 1836) took the doctrine to its most mechanical extreme in *Analysis of the Phenomena of the Human Mind* (1829). For him a complex idea is nothing more than the sum of the simple ideas that compose it: the idea of a house is the ideas of bricks, mortar, windows and so on, added together, and the idea of "everything" is all the other ideas at once. This view is called **mental mechanics**.

His son, **John Stuart Mill** (1806 to 1873), saw the flaw. In *A System of Logic* (1843) he argued that ideas can fuse into a whole with properties that none of the parts possessed, just as hydrogen and oxygen combine into water, which is like neither gas. His favourite example was Newton's colour disc: paint the colours of the spectrum on a wheel and spin it, and you see white, which is not experienced as "seven colours added". This is **mental chemistry**. Think of masala chai. James Mill would say your idea of its taste is tea plus milk plus ginger plus cardamom plus sugar; John Stuart Mill would say the taste of good chai is a new whole that is in none of the ingredients. The younger Mill's idea anticipates both Wundt's creative synthesis (Section 1.5) and the Gestalt slogan that the whole is different from the sum of its parts (Chapter 2).

The last of the associationists, the Scot **Alexander Bain** (1818 to 1903), brought the tradition to the edge of modern psychology. His *The Senses and the Intellect* (1855) and *The Emotions and the Will* (1859) were the standard British textbooks of psychology for half a century, and they tied every mental process to what was then known of the nervous system. Bain described how a random ("spontaneous") movement that happens to bring pleasure is retained and repeated, a trial-and-error account of learning that anticipates Thorndike. In 1876 he founded *Mind*, the first journal devoted to psychology and philosophy.

> EXAM TRAP
>
> **Mental mechanics** belongs to the father, James Mill; **mental chemistry** to the son, John Stuart Mill. Do not mix up James Mill with William James, or John Stuart Mill with John Locke. "Tabula rasa" is **Locke**, not Plato (nativist), Descartes (innate ideas) or Kant (innate categories). If asked the source of knowledge "as per empiricists", the answer is **sensory experience**; "logical reasoning" and "contemplation" are the rationalists' routes.

### The continental rationalists: Leibniz, Herbart and Kant ●●

While the British were building the mind out of sensations, philosophers in Germany insisted that the mind is active, organised and partly unconscious. Their ideas reached Wundt directly, so we need them in some detail.

**Gottfried Wilhelm Leibniz** (1646 to 1716), co-inventor of the calculus, pictured reality as made of countless **monads**: simple, indivisible, active units of force, each of which mirrors the whole universe from its own point of view. Monads do not act on one another (they are "windowless"); they are kept in step by a harmony that God arranged in advance. Applied to mind and body, this "pre-established harmony" is a form of parallelism, which we will meet below. More important for psychology was Leibniz's theory of consciousness. He argued that we receive many perceptions too weak to notice individually, which he called **petites perceptions** (small perceptions). Standing on a beach, you hear the roar of the sea, yet that roar is made of the sounds of thousands of single waves, none of which you could hear on its own. When enough small perceptions add up, they cross into clear awareness, a step Leibniz called **apperception**. Consciousness is therefore a matter of degree, not all or none, and much

mental activity goes on below awareness. Against Locke he wrote, in his *New Essays on Human Understanding*, that there is nothing in the intellect that was not first in the senses, "except the intellect itself".

**Johann Friedrich Herbart** (1776 to 1841), who succeeded to Kant's chair at Konigsberg, turned Leibniz's ideas into a kind of mental mechanics with mathematics. Ideas, he said, have strength, and they compete: incompatible ideas push each other down. An idea that is pushed down does not vanish; it sinks below a boundary he called the **threshold of consciousness** (a limen), where it remains active and strives to return. Fechner would borrow the idea of a threshold for psychophysics, and Freud's repressed ideas that push to return to awareness have a family resemblance to it.

The ideas currently in consciousness form an organised body that Herbart called the **apperceptive mass**. A new idea is understood, or apperceived, only by being taken up into this mass of existing ideas; an idea that finds nothing related to attach to is not grasped at all. Herbart was also a great educational theorist, and the classroom lesson follows directly: teach new material by linking it to what the child already knows. (His followers later turned this into the "formal steps" of lesson planning, which teacher-training colleges in India still teach.) You have experienced this yourself: the term "apperception" is far easier to learn now, attached to Leibniz's sea and Herbart's competing ideas, than it would be as an isolated word in a list.

Herbart said psychology could be empirical and mathematical, but he did not believe it could be experimental, and he kept it separate from physiology. Within forty years both claims would be overturned in Leipzig.

> EXAM TRAP
>
> An IHBAS 2025 question asks what the "apperceptive mass" constitutes and keys **pure perception and pre-existing ideas**, meaning a new perception taken up into the store of ideas already in the mind. The term is **Herbart's**; Wundt took over and reworked the idea of apperception, which is why the question names him. The distractors each belong to someone else: "a compound of elemental processes bound by association" is the associationist (and structuralist) picture of the mind; "sensation and pure perception" leaves out the existing ideas that are the whole point; "sensation plus unconscious inference" is Helmholtz's theory of perception (Section 1.3). **Apperception** as a word was coined by Leibniz.

The most influential German philosopher of all, **Immanuel Kant** (1724 to 1804), answered Hume in his *Critique of Pure Reason* (1781). Hume had shown that experience alone can never give us necessary connections such as causality. Kant replied that the mind does not wait passively for experience to write on it; it imposes its own built-in forms on whatever the senses deliver. Space and time are the forms through which we must perceive anything, and categories such as causality, unity and substance are the frames through which we must think about it. We never know things as they are in themselves, only as our minds organise them. This is nativism of a sophisticated kind, and it passed through the perceptual nativists of the nineteenth century into Gestalt psychology.

Kant also issued a challenge. Psychology, he argued, could never be a true natural science, because mental events cannot be measured, cannot be put to experiment, and are distorted by the very act of observing them. The history in the rest of this chapter can be read as a series of

replies to Kant: Herbart made psychology mathematical, Fechner made it measurable, and Wundt made it experimental. Table 1.3 gathers the philosophers of this section, with their positions and key works.

Table 1.3 The philosophers psychology inherited.

| Thinker | Position | Key ideas | Key work |
|---|---|---|---|
| Plato | Nativist, rationalist | Knowledge as recollection; three-part soul | *Meno*, *Republic* |
| Aristotle | Empiricist | Soul as function of body; laws of similarity, contrast, contiguity | *De Anima* |
| Descartes | Rationalist, nativist, dualist | Interaction via pineal gland; innate ideas; reflex | *Discourse on Method* (1637) |
| Hobbes | Empiricist, materialist | Thought as motion; pleasure and pain | *Leviathan* (1651) |
| Locke | Empiricist | Tabula rasa; simple and complex ideas; primary and secondary qualities | *Essay* (1690) |
| Berkeley | Empiricist, idealist | Esse est percipi; learned depth perception | *New Theory of Vision* (1709) |
| Hume | Empiricist, sceptic | Impressions and ideas; resemblance, contiguity, cause and effect | *Treatise* (1739 to 1740) |
| Hartley | Associationist | Vibrations and vibratiuncles; contiguity | *Observations on Man* (1749) |
| James Mill | Associationist | Mental mechanics | *Analysis* (1829) |
| J. S. Mill | Associationist | Mental chemistry | *A System of Logic* (1843) |
| Bain | Associationist | Physiology of association; trial and error; journal *Mind* | *The Senses and the Intellect* (1855) |
| Leibniz | Rationalist | Monads; petites perceptions; apperception | *Monadology* (1714) |
| Herbart | Rationalist | Threshold of consciousness; apperceptive mass | *Psychology as Science* (1824 to 1825) |
| Kant | Nativist | Innate categories; psychology cannot be a science | *Critique of Pure Reason* (1781) |

## The mind-body problem

Descartes' dualism created a puzzle that no one has fully solved: if the mind is not physical, how can it move a physical arm, and how can a physical pinprick produce a mental pain? The answers given over the next three centuries form a small set of positions that every psychology student must be able to name (Figure 1.2).

The first, **interactionism**, is Descartes' own answer: mind and body are different substances that causally affect each other. Its weakness is the one Descartes never solved, namely how something with no physical properties can push on something physical. Some of his followers, notably Nicolas Malebranche, proposed *occasionalism*: mind and body never act on each other, and God produces the bodily event on each occasion of the mental one, and the reverse.

The second, **psychophysical parallelism**, holds that mental events and bodily events run side by side, perfectly correlated, but neither causes the other. Leibniz's picture was two clocks that keep identical time because they were set together at the start, not because one drives the other. Parallelism was attractive to the first experimental psychologists because it

let them correlate mental events with physiological ones without worrying about causation; Wundt and Titchener both held versions of it.

The third, **epiphenomenalism**, holds that bodily (brain) events cause mental events, but mental events cause nothing; consciousness is a by-product. Its best-known advocate, the English biologist **Thomas Henry Huxley** (1825 to 1895), compared consciousness to the whistle of a steam engine: it accompanies the work of the engine but does nothing to drive it.

Finally, **monism** denies that there are two substances at all. Materialist monism (Hobbes) says only matter exists and mind is a process of the body. Idealist monism (Berkeley) says only minds and their perceptions exist. And **double-aspect theory**, associated with the Dutch philosopher **Baruch Spinoza** (1632 to 1677), says there is one underlying reality that shows itself in two ways: seen from the inside it is mind, seen from the outside it is body, as the same curved line is concave from one side and convex from the other. Fechner, the founder of psychophysics, held a view close to this one.

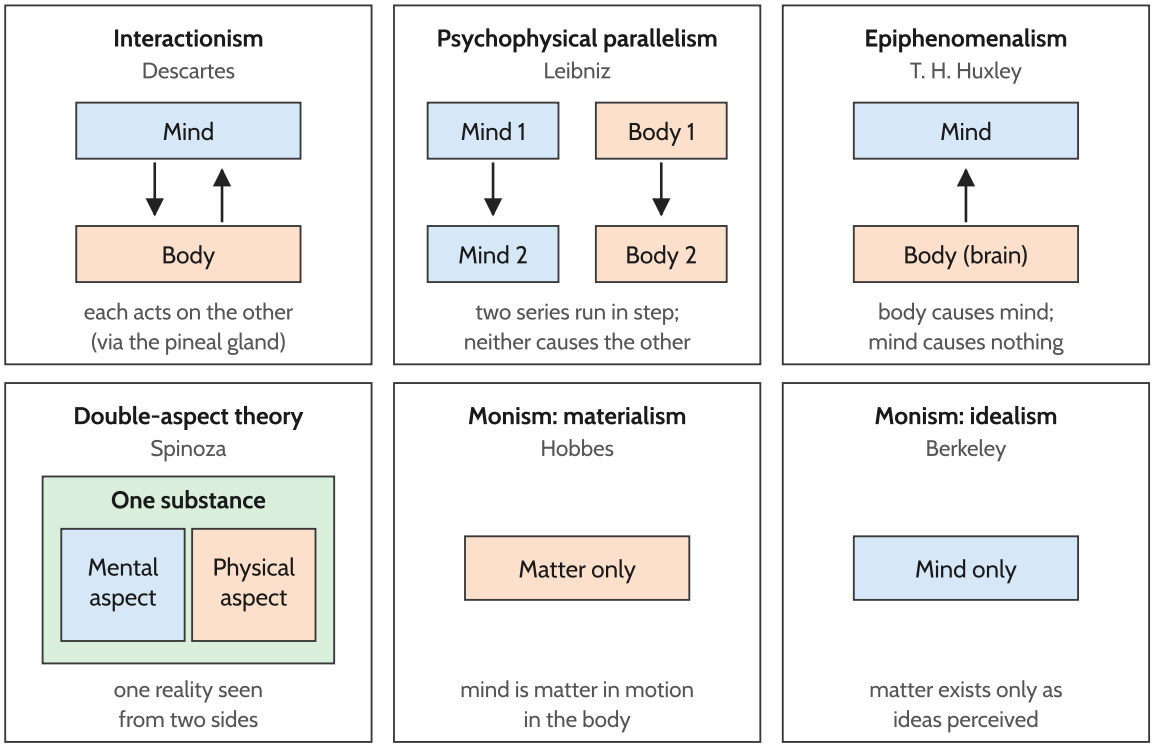


Figure 1.2 Six classic answers to the mind-body problem. Follow the arrows: two-way causation (interactionism), no causation between parallel series (parallelism), one-way causation from body to mind (epiphenomenalism), and the three monisms that deny there are two substances at all.

## A word on the philosophy of science

Questions about the mind sit inside larger philosophical questions about what exists and how we can know it, and a few terms from the philosophy of science turn up in psychology papers. The term **ontology** (from the Greek for "being") names the branch of philosophy that deals with existence: what kinds of things are real? Is the mind a thing, a process, or an illusion? By contrast, **epistemology** deals with knowledge: how do we know what we know, and what counts as good evidence? The nativist-empiricist debate above is an epistemological debate.

In the nineteenth century the French philosopher **Auguste Comte** (1798 to 1857) founded **positivism**, the view that genuine knowledge comes only from observable, verifiable facts, and that speculation about unobservable causes should be abandoned. Comte rejected introspection on the ground that the mind cannot observe itself, any more than the eye can see itself, an objection you will meet again when Wundt's method comes under fire. *Post-positivism*, associated with the twentieth-century philosopher Karl Popper, accepts that all scientific knowledge is fallible and provisional, so that theories are tested by trying to refute them rather than by piling up confirmations. *Scientific realism* holds that the unobservable entities described by our best theories (electrons, genes, perhaps memory traces) really exist. Of these terms, only ontology is defined by the question of existence.

Philosophy, then, handed psychology its questions and some brilliant guesses. What it could not supply was a way to settle the arguments. For that, psychology had to borrow the methods of a different parent: experimental physiology.

## 1.3 Physiological and Biological Roots LOWER YIELD · 1 Q

By the middle of the nineteenth century, physiologists in Germany, France and Britain were doing something philosophers never had: cutting, stimulating, timing and measuring living nervous systems. Their findings did two things for the future psychology. They showed that the machinery of mind could be studied by experiment, and they gave Wundt, who trained as a physiologist, his methods and his confidence.

### Nerves carry the message

The first great discovery was about the wiring. In 1811 the Scottish anatomist **Charles Bell** (1774 to 1842) described, in a privately printed pamphlet, evidence that the nerve roots entering the spinal cord at the front and at the back have different functions. In 1822 the French physiologist **Francois Magendie** (1783 to 1855) showed it decisively in experiments on young animals. The result is the **Bell-Magendie law**: the dorsal (back) roots of the spinal nerves carry sensory information in, and the ventral (front) roots carry motor commands out. The nervous system, in other words, has separate input and output channels, an anatomical basis for the distinction between sensation and movement, and ultimately for the stimulus-response unit. (The two men's quarrel over who deserved credit is a classic example of simultaneous discovery, which we discuss in Section 1.8.)

The most influential physiologist of the age was **Johannes Muller** (1801 to 1858) of Berlin, whose *Handbook of Human Physiology* (published in parts from 1833 to 1840) summarised the field for a generation. Muller's best-known idea is the **doctrine of specific nerve energies**: the quality of a sensation depends not on the stimulus but on which nerve is stimulated. The optic nerve, however it is excited, produces only sensations of light; the auditory nerve only sound. You can test this now: close your eyes and press gently on the outer corner of one eyelid. You will see a patch of light, though no light has entered your eye; pressure on the optic pathway is "read" as light. A blow to the head makes a boxer "see stars" for the same reason. Muller's idea meant that we never perceive the world directly, only the state of our own nerves, a physiological version of the philosophers' point that experience is not the same as the object (the modern form of the idea, "labelled lines", is covered in Chapter 3). Muller's stu-

dents included Hermann von Helmholtz and Emil du Bois-Reymond, and Wundt himself spent a semester in his Berlin laboratory.

Muller believed that the speed of the nerve impulse was so great that it might never be measured. His student proved him wrong within a decade.

KEY STUDY: HELMHOLTZ TIMES THE NERVE IMPULSE (1850)

**Question.** Is the nerve impulse effectively instantaneous, like light, or does it take measurable time? **Method. Hermann von Helmholtz** (1821 to 1894) used a frog's nerve with the muscle still attached. He stimulated the nerve first at a point far from the muscle and then at a point near it, and timed the delay before the muscle twitched. **Result.** Stimulation farther from the muscle produced a later twitch, and dividing the difference in distance by the difference in time gave a speed of only about 27 metres per second, roughly the speed of a car on a highway. **Conclusion.** Nervous activity is a physical process that takes time. If nerve conduction takes time, so may perceiving and deciding, and those times can be measured. **Caveat.** When Helmholtz tried to measure reaction times in human beings, the results varied so much from trial to trial and person to person that he gave up, an early hint of the individual differences that Galton and Cattell would later study.

Helmholtz was perhaps the greatest scientist of the century, and three more of his ideas reach psychology. First, his theory of **unconscious inference**: perception is not a simple copy of sensation but a conclusion the mind draws, quickly and without awareness, from sensory data and past experience. When you see a distant elephant as huge though its image on your retina is tiny, you are "inferring" its size from its distance. This made Helmholtz an empiricist about perception, and it is the ancestor of modern ideas of perception as hypothesis-testing (Chapter 4). Second, building on a suggestion made by the English physician **Thomas Young** (1773 to 1829) in 1802, Helmholtz developed the **trichromatic theory** of colour vision, the Young-Helmholtz theory, which holds that colour vision depends on three types of receptor, most sensitive to different parts of the spectrum. Third, he proposed a resonance (place) theory of hearing. All three are developed in Chapters 3 and 4. For this chapter, remember that Wundt worked as Helmholtz's assistant at Heidelberg for several years.

EXAM TRAP

"Helmholtz gave the trichromatic theory of colour vision" is the keyed answer to the bank question about him. The distractors are carefully chosen. The **just noticeable difference** belongs to Weber and Fechner (Section 1.4); the **law of effect** to Thorndike (Chapter 6). "Proposed the concept of the unconscious" is a trap because Helmholtz did propose *unconscious inference*, but the concept of the unconscious mind in the psychodynamic sense belongs to Freud (and its philosophical roots to Leibniz and Herbart).

## Mapping functions onto the brain

If the nerves are wires, is the brain a single organ that does everything everywhere, or a collection of specialised parts? The modern answer, **localisation of function**, emerged through a strange mixture of quackery and careful science.

The quackery came first. The Viennese physician **Franz Joseph Gall** (1758 to 1828) was a skilled brain anatomist, but he is remembered for a bolder claim: that the brain is made up of

separate organs for separate mental faculties (such as acquisitiveness, combativeness or love of offspring), that a well-developed faculty means a larger organ, and that a larger organ pushes out the skull above it. By feeling the bumps on a person's head, therefore, one could read his character. Gall's collaborator **Johann Gaspar Spurzheim** (1776 to 1832) popularised the system under the name **phrenology** and carried it across Europe and America, where "phrenological parlours" read heads for a fee. Phrenology was wrong in its method (skull shape does not reflect the brain beneath, and Gall chose his evidence selectively), but it was right in its central assumption that different parts of the brain do different things.

Phrenology's first scientific critic was the French physiologist **Pierre Flourens** (1794 to 1867). In the 1820s he used the method of **ablation** (also called extirpation): surgically removing a part of the brain of an animal, usually a pigeon or rabbit, and observing what behaviour was lost. He found real localisation at the level of large structures: damage to the cerebellum disrupted the coordination of movement, damage to the medulla stopped breathing and heartbeat, and removal of the cerebral hemispheres abolished perception and voluntary action. But within the cerebral hemispheres he found no faculties in particular places: the effects of damage seemed to depend on how much tissue was removed, and the cortex seemed to act as a whole. Flourens' ablation method became standard in physiological psychology, and his view of the cortex as acting as a whole was revived a century later by Karl Lashley (covered in the Physiological Psychology book).

Localisation within the cortex was finally demonstrated through language. In 1861 the French surgeon **Paul Broca** (1824 to 1880) examined a patient named Leborgne, known in the hospital as "Tan" because that was nearly the only syllable he could say. Tan understood much of what was said to him but could not produce speech. When he died, Broca's autopsy found damage in the left frontal lobe, in a region now called Broca's area, and by 1865 Broca had concluded that "we speak with the left hemisphere". In 1874 the 26-year-old German neurologist **Carl Wernicke** (1848 to 1905) described the opposite pattern: patients with damage to the left temporal lobe who spoke fluently but meaninglessly and could not understand speech. Broca's aphasia is effortful, non-fluent speech with relatively preserved comprehension; Wernicke's aphasia is fluent speech with poor comprehension. (The aphasias, including the transcortical and conduction forms, are covered in the Physiological Psychology book.)

Meanwhile, in 1870, the German physicians **Gustav Fritsch** (1838 to 1927) and **Eduard Hitzig** (1838 to 1907) applied weak electric currents to the exposed cortex of dogs and found a strip of the frontal lobe where stimulation produced movements of specific body parts on the opposite side of the body: the motor cortex. Electrical stimulation now joined ablation and clinical autopsy as a method of mapping the brain. Table 1.4 brings the localisation story together.

Table 1.4 The localisation pioneers.

| Who | When | Method | Finding |
|---|---|---|---|
| Gall and Spurzheim | about 1800 to 1830 | Skull bumps (phrenology) | Faculties have brain "organs"; method wrong, idea of localisation right |
| Flourens | 1820s | Ablation in pigeons and rabbits | Cerebellum: coordination; medulla: vital functions; cortex acts as a whole |
| Broca | 1861 | Clinical case plus autopsy ("Tan") | Left frontal area for speech production |

| Who | When | Method | Finding |
|---|---|---|---|
| Fritsch and Hitzig | 1870 | Electrical stimulation in dogs | Motor cortex; movements on the opposite side |
| Wernicke | 1874 | Clinical cases plus autopsy | Left temporal area for understanding speech |

CLINICAL LINK

Phrenology is the clinical psychologist's cautionary tale. It had a plausible theory, confident practitioners, satisfied customers and an apparently huge body of supporting cases, yet it was false, because its evidence was gathered selectively and never put to a fair test. Every assessment tool you will learn to use, from an intelligence test to a projective test, must earn its place by evidence of reliability and validity (the Assessment book), not by the confidence of the person using it.

### Charles Darwin: the human animal

Physiology showed that the mind had machinery. Biology showed where the machinery came from. When **Charles Darwin** (1809 to 1882) published *On the Origin of Species* in 1859, he proposed that species change over generations through **natural selection**: individuals vary, some variations help their owners survive and reproduce, and those variations become more common in later generations. In *The Descent of Man* (1871) he applied the theory to human beings, and in *The Expression of the Emotions in Man and Animals* (1872) he argued that facial expressions of emotion are inherited and universal, shared with other animals because they once served adaptive purposes (baring the teeth in anger is a remnant of preparing to bite). He even published a diary of his infant son's development (1877), one of the first "baby biographies".

Darwin gave psychology three gifts, and each grew into a branch of the discipline.

- **Continuity between species.** If humans evolved from other animals, then animal minds differ from ours in degree, not kind, and studying animals can tell us about ourselves. This is the foundation of comparative psychology and of all later animal research on learning.
- **Function.** If a trait exists because it helped survival, then the right question to ask about any mental process is what it is *for*. This question is the heart of functionalism (Section 1.7), and through functionalism, of today's evolutionary psychology.
- **Variation.** Natural selection works on differences between individuals, so those differences are worth measuring. This idea was taken up by Darwin's own half-cousin.

### Francis Galton and the measurement of individual differences

**Francis Galton** (1822 to 1911) shared a grandfather, the physician-poet Erasmus Darwin, with Charles Darwin. Independently wealthy, he never held a university post, and he could not see anything without wanting to count it: he once tried to make a "beauty map" of Britain by secretly tallying attractive and unattractive women he passed in different towns. After reading the *Origin*, he devoted himself to human variation, and in doing so he founded the study of **individual differences**.

His first major book, *Hereditary Genius* (1869), traced the families of eminent judges, statesmen, scientists and others, and found that eminence ran in families far more than chance would predict. Galton concluded that ability is inherited. The flaw is obvious to you now: eminent families also pass on wealth, education and connections, so his data could not separate heredity from environment. Galton saw the problem himself and, in 1875, proposed studying twins to separate the effects of "nature and nurture" (a phrase he popularised). He was the first to use **twin studies** for this purpose.

KEY STUDY: GALTON'S ANTHROPOMETRIC LABORATORY (1884)

**Question.** How much do people differ in basic physical and mental capacities? **Method.** At the International Health Exhibition in London in 1884, Galton set up an **anthropometric laboratory** where, for threepence, visitors were measured on height, weight, arm span, breathing capacity, strength of grip, swiftness of a blow, keenness of sight and hearing, colour discrimination, reaction time and the highest pitch they could hear (tested with a whistle he designed). More than 9,000 people passed through. **Reasoning.** Following the empiricist logic that all knowledge enters through the senses, Galton assumed that people with sharper senses and quicker reactions would be more intelligent. **Result.** He obtained the first large body of data on human individual differences, and to analyse it he had to invent new statistics. **Criticism.** The assumption linking sensory keenness to intelligence turned out to be wrong, as Cattell's followers discovered (Section 1.7); intelligence testing had to wait for Binet's very different approach (Chapter 9).

The statistics may be Galton's greatest legacy. Studying the heights of parents and children (and, earlier, the sizes of sweet-pea seeds), he noticed that very tall parents tend to have children who are tall but less extreme than themselves, a phenomenon now called **regression to the mean** (1886). From this he developed the idea of **correlation**, a measure of how closely two variables vary together (1888); his younger colleague **Karl Pearson** (1857 to 1936) later produced the formula for the correlation coefficient that you will use in the Research Methodology book. Galton also pioneered the questionnaire (asking people to describe the vividness of their mental image of that morning's breakfast table), and in 1879 he devised the **word association** test: he read a word and noted the first association it called up, and was startled to find how many of his own associations reached back to childhood and revealed thoughts he would rather not publish. Carl Jung later developed word association into a clinical technique (the Personality book).

There is a darker side. In 1883 Galton coined the word **eugenics** for the programme of improving the human stock by encouraging the "fit" to have more children and discouraging the "unfit". Eugenics was later used to justify forced sterilisation and, in Nazi Germany, mass murder. Galton's statistics are permanent tools of psychology; his eugenics is a permanent warning.

### Animal minds: from anecdote to Morgan's canon

Darwin's idea of continuity invited a new question: how intelligent are animals? Darwin's young friend **George John Romanes** (1848 to 1894) set out to answer it, and his *Animal Intelligence* (1882) is usually treated as the first book of comparative psychology. Romanes gathered stories, from naturalists, pet owners and newspaper readers, of animals doing clever things: a cat that opened a door latch, a dog that seemed to plan. This **anecdotal**

**method** had two weaknesses. Anecdotes are selected (no one writes in to report a dog doing something stupid), and they invite **anthropomorphism**, reading human thoughts and feelings into animal behaviour.

The correction came from the British psychologist **Conwy Lloyd Morgan** (1852 to 1936). Morgan's own fox terrier, Tony, learned to open the garden gate by lifting the latch with his head. A visitor who saw only the finished act would have credited Tony with understanding how latches work, but Morgan had watched the learning: it came slowly, through many chance successes while Tony poked his head through the railings. In *An Introduction to Comparative Psychology* (1894) Morgan stated the rule now called **Morgan's canon**: "In no case may we interpret an action as the outcome of the exercise of a higher psychical faculty, if it can be interpreted as the outcome of the exercise of one which stands lower in the psychological scale." It is a principle of **parsimony**: prefer the simplest explanation that fits the facts.

### KEY STUDY: CLEVER HANS (1904 TO 1907)

**The claim.** In Berlin, a retired schoolteacher, Wilhelm von Osten, exhibited a horse called Hans that seemed to add, subtract and answer questions by tapping its hoof. A commission that included the psychologist Carl Stumpf found no trickery in 1904. **Method.** Stumpf's student **Oskar Pfungst** then tested Hans systematically, varying whether the questioner knew the answer and whether Hans could see the questioner. **Result.** Hans answered correctly only when the questioner knew the answer and was visible. Without meaning to, questioners tensed as the taps began and relaxed slightly, or raised their heads, when the right number was reached, and Hans had learned to stop at that cue. **Conclusion** (published 1907). The horse was clever, but at reading people, not at arithmetic: a perfect illustration of Morgan's canon. The "Clever Hans effect" is why modern experiments use observers who do not know which condition a participant is in.

Comparative psychology soon moved from anecdote into the laboratory. Edward Thorndike's puzzle-box studies of cats (1898, Chapter 6) and Margaret Floy Washburn's textbook *The Animal Mind* (1908, Section 1.7) carried it into the twentieth century, and Watson would build behaviourism on it.

### CLINICAL LINK

Morgan's canon is good clinical sense. A child who will not read aloud in class may have a reading disability, poor eyesight, anxiety or a hearing problem before we conclude that he is "defiant" or "lazy". Test the simpler, lower-level explanations first. You will see in Section 1.7 that the first patient at the world's first psychological clinic turned out to have exactly such a simple problem.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. In the modern definition of psychology, thinking, remembering and dreaming are examples of covert _____ processes.
2. The branch of psychology that studies the behaviour of animals across species is _____ psychology.
3. Aristotle's three laws of association were similarity, contrast and _____.
4. Descartes believed that mind and body interact through the _____ gland.
5. Locke described the mind at birth as "white paper", the idea usually summed up as tabula _____.
6. For Locke, colour, taste and warmth are _____ qualities.
7. James Mill's view that a complex idea is the sum of its parts is called mental _____; his son's view is mental chemistry.
8. Herbart's term for the body of existing ideas into which a new idea is absorbed is the _____ mass.
9. The view that mental events are by-products of brain events and cause nothing is called _____.
10. The rule that we should not explain an animal's behaviour by a higher faculty if a lower one will do is Lloyd _____ canon.

**Answers:** (1) mental, (2) comparative, (3) contiguity, (4) pineal, (5) rasa, (6) secondary, (7) mechanics, (8) apperceptive, (9) epiphenomenalism, (10) Morgan's.

## 1.4 Measuring the Mind: Psychophysics and Mental Chronometry

LOWER YIELD · 4 QS

Kant had said that mental events can never be measured, and for a long time no one could prove him wrong. The proof, when it came, came from two professors at the University of Leipzig, and it took the form of a simple idea: you cannot put a ruler against a sensation, but you can measure the stimuli that produce it and find the rule that links the two. This section covers that idea and the parallel discovery that mental acts take measurable time. The full treatment of thresholds and psychophysical laws is in Chapter 3; here we need only enough to see why they made a science of mind possible.

### Ernst Weber and the just noticeable difference

**Ernst Heinrich Weber** (1795 to 1878) was a Leipzig physiologist who studied the sense of touch. Touch two points of a pair of compasses to your fingertip and you will feel two points even when they are only a few millimetres apart; touch them to the middle of your back and they must be several centimetres apart before you feel two rather than one. This "two-point threshold" was one of Weber's discoveries. The more important one came from lifting weights. Weber asked people to compare a standard weight with slightly heavier ones and found the smallest difference they could detect, the **just noticeable difference** (JND). The JND was not a fixed amount but a fixed *proportion* of the standard weight.

A worked example makes the point. Suppose that for lifted weights the proportion is one-fortieth. If you are holding 40 grams, you will just notice the change when one gram is added. If you are holding 400 grams, adding one gram is now undetectable; you need ten. In symbols, the JND (written delta I) divided by the original intensity (I) is a constant, k. Fechner later named this relation **Weber's law**. You know it from daily life: a single candle lights up a dark room, but carried into a brightly lit wedding hall it makes no visible difference.

### Gustav Fechner and the founding of psychophysics ●●

**Gustav Theodor Fechner** (1801 to 1887) was a professor of physics at Leipzig who suffered a long and severe breakdown in the late 1830s and early 1840s, made worse by damage to his eyes from gazing at the sun through coloured glass while studying afterimages. When he recovered he turned to philosophy, and in particular to the relation between mind and matter, which he believed to be two sides of one reality. On the morning of 22 October 1850, lying in bed, he had the insight that would make his name: the relation between the bodily and the mental could be expressed as a mathematical law. If equal *ratios* of stimulus intensity produce equal *steps* in sensation (as Weber's data implied), then sensation must grow with the logarithm of the stimulus. This is **Fechner's law**, usually written S = k log I, where S is the strength of the sensation and I the intensity of the stimulus.

Push a case through it. Switch on one bulb in a dark room, then a second: the room gets noticeably brighter. To get the same apparent increase again you need not three bulbs but four, and then eight, then sixteen. Sensation climbs in equal steps while the stimulus doubles at each step. That is what a logarithmic relation means.

Fechner spent ten years testing his insight and published the results as *Elements of Psychophysics* in 1860. He defined **psychophysics** as the exact science of the functional relations between body and mind, and he gave it three methods that are still taught today: the method of limits, the method of constant stimuli and the method of adjustment (average error), together with the concepts of the absolute threshold and the difference threshold (all in Chapter 3). Researchers in psychophysics still celebrate 22 October as "Fechner Day".

One point about thresholds is worth fixing now, because it matters for perception as much as for psychophysics. Whether a stimulus is perceived depends not only on how strong it is but on how long it lasts. A faint flash of light, or the contour (edge) of a briefly shown shape, is seen only if the stimulation reaches a minimum intensity *and* is present for a minimum exposure time; for very brief exposures the two trade off, so that a weaker flash must last longer to be seen (Bloch's law, 1885). This is why the early laboratories controlled exposure time so carefully with the tachistoscope. Reaction time, by contrast, measures how quickly a person responds once a stimulus has been perceived (as you will see below); it is not a condition for perceiving anything.

Why does Fechner matter so much to the history? Because he answered Kant. He showed that mental events could be measured, indirectly but precisely, and that measurement could be repeated by anyone. Wundt, Ebbinghaus and G. E. Muller all built on his book. The historian **Edwin Garrigues Boring** (1886 to 1968) even argued that 1860, not 1879, could be taken as the founding date of experimental psychology. The convention, however, is clear: Fechner founded psychophysics; Wundt founded psychology as an independent experimental discipline.

EXAM TRAP

The "founder of psychophysics" is **Fechner**, not Weber (whose work Fechner built on and named), not Wundt (who founded the laboratory) and not Ebbinghaus (who applied Fechner's spirit to memory). The JND is Weber's concept; S = k log I is Fechner's law. Keep 1850 (the insight, 22 October) separate from 1860 (the book).

## The personal equation and mental chronometry

The second line of measurement began in an observatory. In 1796 the Astronomer Royal at Greenwich, Nevil Maskelyne, dismissed his assistant David Kinnebrook because Kinnebrook's timings of the moment a star crossed a hair line in the telescope were consistently later than his own, by most of a second. Some twenty-five years later the German astronomer Friedrich Wilhelm Bessel compared the timings of several experienced astronomers and found that each had his own systematic error. No one was simply right; every observer took a characteristic amount of time to see and respond. Astronomers called the correction for each observer his **personal equation**. Here, in effect, was the first measurement of individual differences in reaction time, and the first hint that perceiving takes time.

Helmholtz had shown that nerves take time. The Dutch physiologist **Franciscus Cornelis Donders** (1818 to 1889) asked whether the time taken by a mental process itself could be measured, and in 1868 he devised a way. His **subtraction method** compares reaction times in tasks that differ by exactly one mental stage (Figure 1.3). In the *a-reaction* (simple reaction time), one light appears and the participant presses a key as fast as possible: the task needs only detection and response. In the *c-reaction* (go/no-go), one of two lights appears and the participant must respond to one of them only: now the stimulus must also be identified. In the *b-reaction* (choice reaction time), each light has its own key: the stimulus must be identified and a response must be chosen. Subtract the a-time from the c-time and you have the time needed to identify the stimulus; subtract the c-time from the b-time and you have the time needed to choose the response.

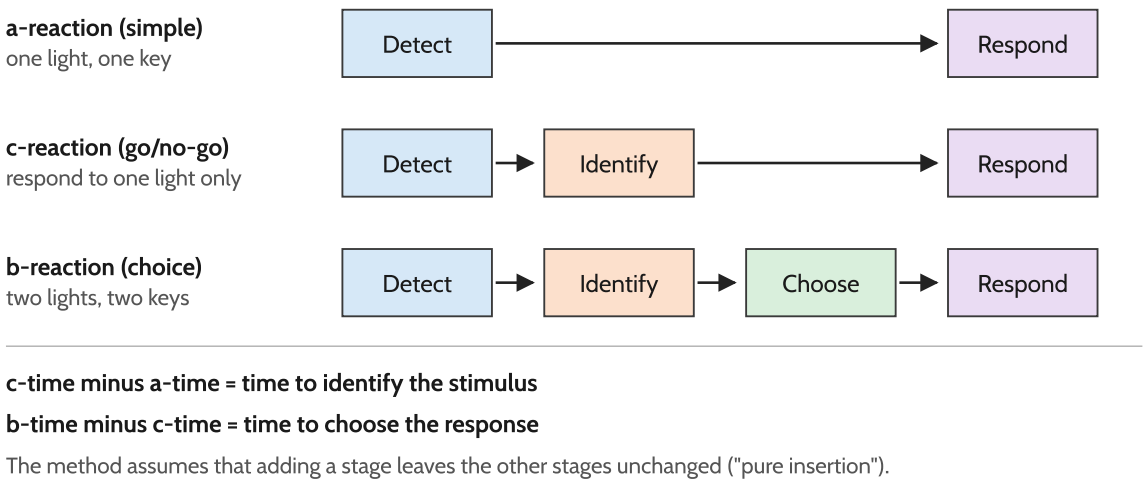


Figure 1.3 Donders' subtraction method (1868). Each task adds one mental stage to the one above it, so subtracting reaction times isolates the duration of the added stage.

Donders had founded **mental chronometry**, the measurement of the time course of mental processes through **reaction time** (the interval between the onset of a stimulus and the beginning of the response). Its weak point is marked at the foot of Figure 1.3: the method assumes that inserting a stage leaves the other stages exactly as they were, which is often untrue. A century later the cognitive psychologist Saul Sternberg replaced it with a subtler method that does not need this assumption (Chapter 8). But the idea that the mind can be studied by timing it was immediately taken up in Leipzig. When Max Friedrich pressed his key in the winter of 1879, he was running a version of Donders' experiment, trying to measure the time taken by apperception.

### Hermann Ebbinghaus: the higher processes measured ●●

Wundt, as you will see, believed that the higher mental processes such as memory and thought could not be studied by experiment. A young German scholar proved him wrong without ever working in his laboratory. **Hermann Ebbinghaus** (1850 to 1909) came across a second-hand copy of Fechner's *Elements of Psychophysics* while travelling in France and England in the late 1870s (accounts differ on whether the bookshop was in Paris or London), and resolved to do for memory what Fechner had done for sensation. To remove the effect of meaning and prior knowledge, he invented **nonsense syllables**, consonant-vowel-consonant combinations such as DAX or BOK, and memorised lists of them himself, under strictly controlled conditions, for several years. He measured memory by the **savings method**: how much less time it took to relearn a list than it had taken to learn it the first time. The result, published in *On Memory* (1885), included the first **forgetting curve**, showing rapid loss in the first hour and slower loss thereafter.

Ebbinghaus worked alone, at Berlin and later at Breslau, outside any school, and he is the author of the sentence quoted at the end of Section 1.1 ("Psychology has a long past, but only a short history", 1908). You will meet his curve and his methods properly in Chapter 7.

> **EXAM TRAP**
>
> "What do Hermann Ebbinghaus, Francis Galton and Sigmund Freud share in common?" The options are (a) each worked independently and was a pioneer, (b) each founded a formal psychology laboratory, (c) each developed novel methods to study how the mind works, and (d) (a) and (c). Option (b) is false: Freud never founded a laboratory. Both (a) and (c) are true: Ebbinghaus invented nonsense syllables and the savings method, Galton invented the questionnaire, word association and correlation, and Freud invented free association. So the best answer is **(d)**, which is how the Panwar bank keys it. The RML 2020 paper printed the same question and keyed **(a)**. Since (a) alone leaves out a true statement, (d) is the more defensible answer; if you meet this question again, choose (d), and do not be shaken if a key prefers (a).

By 1870, then, all the parts were on the bench. Philosophy had supplied the questions and the idea of mental elements and their association; physiology had supplied the experimental method and the reaction-time technique; biology had supplied the idea of function and the interest in individual differences; and psychophysics had shown that the mind could be measured. What was missing was someone to declare that these belonged together as a new, independent science, and to build an institution for it.

## 1.5 Wilhelm Wundt and the Founding of Experimental Psychology

MEDIUM YIELD · 6 QS

Psychology does have a founder in the way that few sciences do, a single person who set out deliberately to create it. Wundt is remembered as the father of experimental psychology, and also, without qualification, as the father of psychology as a science. To understand why, you need to know what he did and what he believed.

### From physiologist to psychologist ●●

**Wilhelm Maximilian Wundt** (1832 to 1920) was born on 16 August 1832 at Neckarau, near Mannheim in south-western Germany, the son of a Lutheran pastor. He was a lonely, daydreaming schoolboy who did poorly at first, but he became an intensely hard-working student of medicine at Heidelberg, with a semester in Berlin in Johannes Muller's laboratory. After qualifying in medicine he chose research over practice and spent several years at Heidelberg as an assistant to Helmholtz.

During those Heidelberg years his plan took shape. In *Contributions to the Theory of Sensory Perception* (published in parts from 1858 to 1862) he called for an experimental psychology; in 1862 he began teaching a course on psychology as a natural science; and in 1863 he published *Lectures on the Human and Animal Mind*. His decisive book was *Principles of Physiological Psychology* (*Grundzuge der physiologischen Psychologie*), published in two parts in 1873 and 1874. Its preface announced that the work was "an attempt to mark out a new domain of science". Historians rank it among the most important books in the history of modern psychology, and Wundt revised it through six editions.

After a year at Zurich, Wundt moved in 1875 to Leipzig as professor of philosophy, the post he held for the rest of his career. The room for apparatus followed, then in 1879 the research of students like Max Friedrich, then a journal, *Philosophische Studien* (*Philosophical Studies*), founded in 1881 to publish the laboratory's work and usually counted as the first journal of experimental psychology. Wundt also wrote a shorter textbook, *Outlines of Psychology* (1896), and, in his last twenty years, the ten volumes of his *Volkerpsychologie* (1900 to 1920). Boring later calculated that Wundt published more than 50,000 printed pages in his lifetime. He died on 31 August 1920, shortly after finishing his memoirs.

> EXAM TRAP
>
> Wundt's 1873 to 1874 book is *Principles of* ***Physiological*** *Psychology*; William James's 1890 book is *The Principles of Psychology*. Questions asking who wrote "Principles of Psychology" want **James**. And "physiological psychology" in Wundt's title did not mean the study of the brain. It meant psychology that borrows the *experimental methods* of physiology. Do not answer a question about Wundt's lab with "brain research".

### What Wundt studied: immediate experience and voluntarism

Every science, Wundt said, studies experience, but from different standpoints. The physicist studies *mediate* experience: experience used as a means of learning about something else, filtered through instruments and concepts. Psychology studies **immediate experience**: experience as it is directly lived, before interpretation. Look at a red hibiscus. A physicist describes the light it reflects as electromagnetic radiation of a certain wavelength; a psychologist in

Wundt's sense describes the experience itself, the quality of the redness, its intensity, how long it lasts, whether it is pleasant.

Wundt's analysis of immediate experience found two kinds of elements. *Sensations* are the elementary contents produced when a sense organ is stimulated, varying in quality and intensity. *Simple feelings* are the subjective accompaniments of sensations, such as the pleasantness of a taste or the strain of waiting. Elements combine into compounds: ideas, emotions, acts of will. Psychology's task was to analyse conscious processes into their elements, to discover how the elements are connected, and to state the laws of those connections.

So far this sounds like British associationism in a laboratory coat. But Wundt insisted that the mind is not a passive meeting place where elements stick together by contiguity. It is active, and the activity is that of attention and will. He called his system **voluntarism**, from the Latin *voluntas* (will). The central active process is apperception, the term he took from Leibniz and Herbart and made his own. Wundt distinguished the broad *field* of consciousness from its narrow *focus*. Contents that merely enter the field are *perceived*; contents brought into the focus by attention are *apperceived*. Sit in an examination hall: the whirr of the ceiling fans, a cough two rows back and the invigilator's footsteps are all perceived, at the edge of consciousness; the question in front of you is apperceived, held in the clear focus where you can work on it.

Apperception does more than select; it also combines. Wundt's principle of **creative synthesis** (also called the law of psychic resultants) states that when elements are combined by apperception, the result has properties that none of the elements possessed. A melody is not just the sum of its notes, since the same notes in a different order make a different tune; the word "SAHIL" is not just five letters. This is John Stuart Mill's mental chemistry placed at the centre of a psychology, and it is why Wundt cannot fairly be called a mere elementist.

> **REMEMBER**
>
> Wundt's own name for his system was **voluntarism**. Many Indian textbooks list Wundt as the founder of *structuralism*, because Titchener carried his work to America and gave it that name; in exam questions that pair "structuralism" with Wundt and Titchener, accept the pairing. But if a question asks for the system that stresses will, attention and apperception, the answer is voluntarism, and the person is Wundt.

### The tridimensional theory of feeling

One of Wundt's best-known ideas came from listening to a metronome. As he attended to its clicks, he noticed that some rhythmic patterns felt more pleasant than others; that while he waited for the next click he felt a slight strain, which was released when the click came; and that a fast tempo felt stirring while a slow one felt calming. From these self-observations he concluded, in 1896, that simple feelings vary along three independent dimensions: *pleasure versus displeasure*, *tension versus relaxation*, and *excitement versus depression* (sometimes translated excitement versus calm). This is the **tridimensional theory of feeling** (Figure 1.4). Any feeling can be described as a point in this three-dimensional space, and an emotion is a sequence of feelings moving through it over time. Waiting outside the result board after an examination, for example, might be unpleasant, tense and excited all at once; the moment you find your roll number, the feeling swings to pleasure and relaxation.

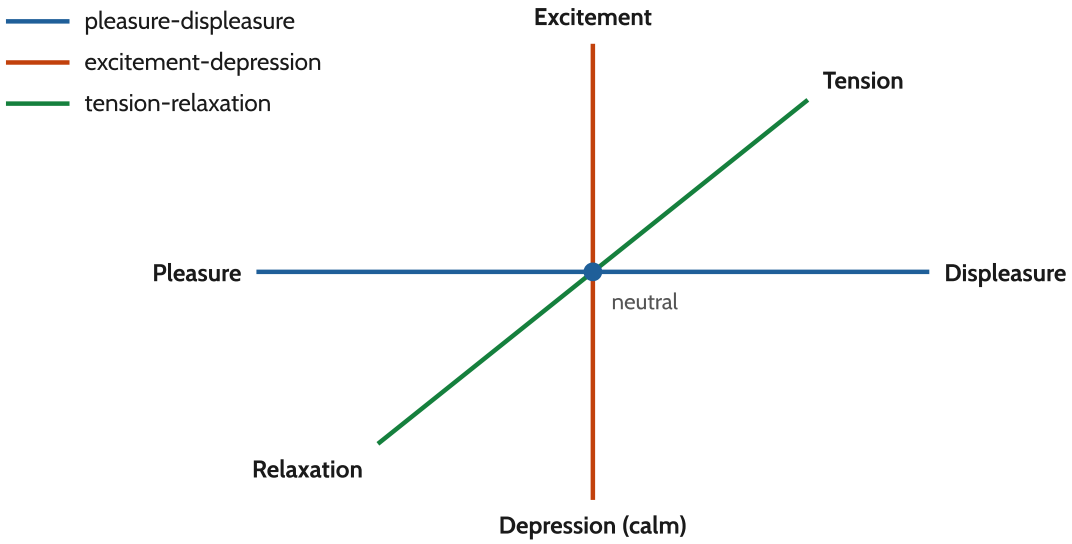


Figure 1.4 Wundt's tridimensional theory of feeling (1896). Every simple feeling is a point on three independent axes; an emotion is a path through this space. Titchener later kept only the pleasure-displeasure axis.

The theory was controversial in its day; Titchener's observers at Cornell could find only pleasantness and unpleasantness, and he rejected the other two dimensions. But notice how modern it looks. Twentieth-century research on emotion and on the meanings of words repeatedly found two or three broad dimensions, most often pleasantness (valence) and arousal, very close to Wundt's first and third axes. (Emotion theories are covered in Chapter 11.)

### The method: experimental introspection ●●

How do you study immediate experience? You have to ask the person having it. Wundt's method is called **introspection** (literally "looking within"), but it was very different from the armchair reflection of the philosophers, which Wundt distrusted as much as Comte had. His word for the armchair kind was *Selbstbeobachtung* (self-observation); the kind he accepted he called *innere Wahrnehmung* (internal perception), and it was usable only under strict experimental control. (English textbooks often call Wundt's version "experimental self-observation", which is why the terms confuse students; the point is the contrast between uncontrolled and controlled looking within.) Wundt laid down four rules for it:

1. The observer must be able to tell when the process to be observed begins (the experimenter presents the stimulus at a signalled moment).
2. The observer must be in a state of readiness, with attention fully concentrated.
3. Each observation must be repeatable many times under the same conditions.
4. The conditions must be variable, so that the stimulus can be systematically changed and the effect on experience noted.

Observers were trained, and most of their reports were short and quantitative: whether a stimulus was present or absent, louder or softer, longer or shorter, the same or different, together with reaction times measured by the chronoscope. Wundt did not ask his observers for long descriptions of their inner life; that style of introspection belongs to Titchener and, in a different form, to the Wurzburg school, as Section 1.6 will show.

EXAM TRAP

"Who is most associated with the technique of introspection?" is **Wundt**. "Which school used introspection as its method of analysis?" is **structuralism**. In the analogy "Charles Spearman is to factor analysis as Wilhelm Wundt is to ...", the answer is **introspection** (Spearman, himself a Wundt student, invented factor analysis; psychoanalysis is Freud's, functionalism is James's, humanism is Maslow's and Rogers'). Wertheimer, Watson and James are wrong answers to the first question: Wertheimer founded Gestalt psychology, Watson attacked introspection, and James used casual introspection but is not known for a technique of it.

## Volkerpsychologie: the other half of Wundt ●●

Wundt believed that the laboratory could reach only the simpler processes: sensation, feeling, reaction, attention. The higher mental processes, such as thinking, language, morality and religious belief, are shaped by culture and history, and an individual sitting in front of a chronoscope cannot reveal them. They must be studied instead through their collective products: the languages, myths, customs, arts, laws and religions of whole peoples. Wundt devoted the last twenty years of his life to this second psychology, publishing it in ten volumes between 1900 and 1920 as **Volkerpsychologie**, usually translated as "folk psychology" and better understood today as *cultural psychology*. A one-volume summary, *Elements of Folk Psychology*, appeared in 1912.

For most of the twentieth century American textbooks ignored this half of Wundt, partly because few of his American students read the later volumes, and the picture of Wundt as a narrow laboratory elementist owes much to that neglect. Today he is often praised as a forerunner of cultural psychology, which matters for a country like India where psychologists have long argued that Western laboratory findings cannot simply be transplanted.

EXAM TRAP

"Who gave the concept of folk psychology?" (RML 2024) is **Wilhelm Maximilian Wundt**. The distractors: Freud founded psychoanalysis; Pavlov discovered classical conditioning; Wilhelm Roentgen was a physicist who discovered X-rays in 1895, included only because his first name matches. Be aware that in modern philosophy of mind "folk psychology" means something else (our everyday explanations of behaviour in terms of beliefs and desires); in history questions it is Wundt's *Volkerpsychologie*.

## Leipzig as a training ground

Wundt's laboratory was above all a school. Students came from across Europe and, in great numbers, from America, took doctorates under him, and went home to found laboratories, journals and departments of their own. Wundt supervised well over a hundred doctoral students in psychology, and the first generation of professional psychologists in several countries learned the trade in his rooms. Two stories show the atmosphere. James McKeen Cattell later recalled walking into Wundt's office and announcing that Wundt needed an assistant and that he would be it; Wundt agreed, though he called Cattell's interest in individual differences in reaction time "ganz Amerikanisch" (typically American). And G. Stanley Hall, who studied at Leipzig in 1879 and 1880, liked to call himself Wundt's first American student. Table 1.5 lists the students who matter most for examinations.

Table 1.5 Wundt's students and visitors, and what they became known for.

| Student | Connection to Leipzig | Later known for |
|---|---|---|
| G. Stanley Hall | Studied there 1879 to 1880 | First US research laboratory (1883); founding president of the APA (1892) |
| James McKeen Cattell | Wundt's first assistant; PhD 1886 | Mental tests (1890); individual differences; Columbia |
| Hugo Munsterberg | PhD 1885 | Applied, forensic and industrial psychology at Harvard |
| Oswald Kulpe | PhD 1887; Wundt's assistant | Wurzburg school; imageless thought |
| Edward Bradford Titchener | PhD 1892 | Structuralism at Cornell |
| Lightner Witmer | PhD 1892 | First psychological clinic (1896); "clinical psychology" |
| Emil Kraepelin | Worked in the laboratory in the early 1880s | Classification of psychiatric disorders |
| Charles Spearman | Doctorate at Leipzig | General factor (g); factor analysis |
| Vladimir Bekhterev | Studied briefly with Wundt | Reflexology; objective psychology in Russia |

Two names in the table belong to other books. **Emil Kraepelin** (1856 to 1926) applied Wundt's experimental methods to patients and went on to build the classification of mental disorders, including the distinction between dementia praecox and manic-depressive illness, that underlies modern psychiatric diagnosis (the Psychiatric Disorders book). **Charles Spearman** (1863 to 1945) developed factor analysis and the theory of general intelligence (Chapter 9). **Vladimir Mikhailovich Bekhterev** (1857 to 1927), the Russian neurologist, gave the name "reflexology" to his objective study of behaviour, a forerunner of behaviourism (Chapter 2). The others are the founders of the two great schools and of American applied psychology, and the next two sections are about them.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. The smallest change in a stimulus that a person can detect is the just noticeable _____.
2. Fechner's insight came to him on the morning of 22 October _____; his *Elements of Psychophysics* appeared in 1860.
3. Fechner's law states that sensation grows with the _____ of stimulus intensity.
4. Donders' method of isolating the time taken by a mental stage is the _____ method.
5. Ebbinghaus measured memory by the _____ method, the reduction in time needed to relearn a list.
6. Wundt's *Principles of* _____ *Psychology* appeared in 1873 to 1874.
7. Wundt's own name for his system, stressing will and attention, was _____.
8. Bringing a content into the clear focus of consciousness is, for Wundt, _____.
9. Wundt's three dimensions of feeling are pleasure-displeasure, tension-relaxation and _____-depression.
10. Wundt's ten-volume study of language, myth and custom is his _____, or folk psychology.

**Answers:** (1) difference, (2) 1850, (3) logarithm, (4) subtraction, (5) savings, (6) Physiological, (7) voluntarism, (8) apperception, (9) excitement, (10) Volkerpsychologie.

## 1.6 Structuralism and Its European Rivals

MEDIUM YIELD · 6 QS

Wundt's psychology travelled. The version that reached America first, and that gave the first school its name, was carried by a young Englishman who had spent two years in Leipzig and who spent the rest of his life defending what he believed was Wundt's programme in its pure form.

### Edward Bradford Titchener at Cornell

**Edward Bradford Titchener** (1867 to 1927) was born in Chichester, England, studied at Oxford, and took his doctorate under Wundt in 1892. Finding no opening for an experimental psychologist in England, he accepted a post at Cornell University in New York State that same year and stayed there until his death 35 years later. He was a commanding and formal figure who lectured in his Oxford academic gown, ran his laboratory as an absolute ruler, and translated several of Wundt's books into English. His own books, especially *An Outline of Psychology* (1896), the laboratory manuals *Experimental Psychology* (1901 to 1905) and *A Textbook of Psychology* (1909 to 1910), trained a generation.

In 1904 Titchener formed his own small society, "the Experimentalists", for researchers who shared his approach; it barred women from its meetings until after his death. The irony is sharp, because his first doctoral student was a woman, Margaret Floy Washburn, whom you will meet in Section 1.7. Titchener supervised more than fifty doctoral students in all.

### The structuralist programme ●●

Titchener named his approach in an 1898 article, "The Postulates of a Structural Psychology", and the name stuck: **structuralism** is the school that studies the structure of conscious experience by analysing it into its basic elements. Titchener drew the analogy with chemistry himself. Just as the chemist analyses compounds into elements and arranges them in the periodic table, the psychologist should analyse consciousness into its elements and catalogue them. He defined his subject carefully. *Consciousness* is the sum of mental processes occurring at any given moment; *mind* is the sum of mental processes accumulated over a lifetime. And he set psychology three problems:

1. **What?** Analyse conscious processes into their simplest elements.
2. **How?** Discover how the elements combine, and the laws of their combination.
3. **Why?** Connect the elements with the physiological conditions that accompany them. (Titchener was a psychophysical parallelist: nervous processes accompany mental ones but do not cause them, so the "why" was correlation, not causation.)

Titchener found three kinds of elements, one more than Wundt. *Sensations* are the elements of perceptions; *images* are the elements of ideas (such as the image of your mother's face when she is not present); and *affections* are the elements of emotions. Each element has attributes, its basic dimensions of variation: *quality* (what distinguishes red from cold), *intensity* (how strong), *duration* (how long), *clearness* (how vivid in attention) and *extensity* (how spread out in space, possessed by sight and touch). Affections, he argued, have quality, intensity and duration but lack clearness: when you try to attend closely to a feeling of pleasantness, it fades. And he recognised only one dimension of affection, pleasantness versus unpleasantness, rejecting Wundt's tension and excitement. Working through his manuals, his laboratory catalogued more than 44,000 distinguishable qualities of sensation, the great majority of them visual and auditory.

### Introspection, the stimulus error and the meaning of meaning ●●

Titchener's method was introspection, but in a more demanding form than Wundt's. He defined it as a highly specialised form of self-observation in which the psychologist attempts to study his own consciousness scientifically. His observers were trained for months to report not short judgements but detailed descriptions of their experience in terms of elements and attributes.

The commonest fault of an untrained observer, Titchener said, was the **stimulus error**: describing the object instead of the experience. Shown an apple, the untrained person says "a red apple". But "apple" is not an experience; it is knowledge about the object. The trained observer reports what is actually in consciousness: a patch of red of a certain brightness and saturation, a bright highlight near the top, a certain extensity, perhaps a faint image of sourness and a mild pleasantness. Try this yourself with any object on your desk, and you will discover how hard it is not to name things.

If consciousness is only elements, where does meaning come from? Titchener's answer was the **context theory of meaning**. A sensation by itself has no meaning; it acquires meaning from the context of other sensations and images that the mind attaches to it. The core (say, the sound of the word "home") means something because it calls up a context of images: a kitchen, a voice, the smell of food. Often, Titchener thought, the context was kinaesthetic,

made of faint sensations of movement and posture. On the same reasoning, he defined perception as a complex of sensory and imaginal processes: sensations from the object plus images supplied by past experience. This definition is the answer to an IHBAS question that asks who defined perception in exactly these terms.

> **EXAM TRAP**
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> "Introspection is a highly specialised form of ..." is **self-observation** (Titchener), not self-perception (that is Daryl Bem's later social-psychological theory), not "self-thinking" and not attention. "Perception is a complex of sensory and imaginal processes" is **Titchener**; Jerome Bruner and Roy Schafer, two of the other names in that question, belong to the "New Look" research on perception of the 1940s and later (Chapter 4), and Gregory Kimble is known for work on learning. The aim of structuralism was to build a "periodic table" of the elements of experience; "to study behaviour and not mind" is behaviourism, so "all of the above" is wrong. "To understand the nature of consciousness" is too general to be the answer: it fits Wundt, James and the Wurzburg school just as well, whereas the periodic table of elements is structuralism's distinctive aim.

### The Wurzburg school and imageless thought

The first serious challenge to the structuralist picture came not from America but from one of Wundt's own former assistants. **Oswald Kulpe** (1862 to 1915) served as Wundt's assistant for several years, then moved to the University of Wurzburg in 1894. Unlike Wundt, he believed that thinking itself could be studied by experiment, and he extended introspection to it. His method, often called systematic experimental introspection, gave the observer a task (for example, to judge, to associate, or to answer a question) and then asked for a full, retrospective description of everything that had passed through consciousness while the task was done.

The results shocked everyone. In 1901 August Mayer and Johannes Orth reported that their observers, giving associations to words, often experienced vague states of hesitation, doubt or certainty that contained no sensations or images at all; they called these "conscious attitudes". Karl Marbe, in the same year, had observers judge which of two weights was heavier and found that they could describe the sensations of lifting perfectly well but could report nothing about how the judgement itself was made; it simply appeared. Karl Buhler later argued that thoughts themselves are elements of experience. The conclusion was **imageless thought**: thinking can proceed without any sensory or imaginal content in consciousness, which directly contradicted Titchener's claim that consciousness consists only of sensations, images and affections.

The Wurzburgers also discovered how much thinking is directed by the task. Henry J. Watt showed that the instruction or task, which he called the **Aufgabe**, determines what association a word produces. Narziss Ach (1905) called the unconscious influence that steers thinking toward the goal of the task the **determining tendency**, and the readiness it creates a **set** (in German, *Einstellung*). An example: hear the word "Mysuru" with the task "name a building" and you will think "palace"; hear the same word with the task "name its state" and you will think "Karnataka". The stimulus is identical; the set prepared before it arrived decides the answer, and you are not aware of the set doing its work.

Titchener replied that the Wurzburg observers were poorly trained and had committed the stimulus error; their "imageless" thoughts, he said, were really faint kinaesthetic sensations and images that a trained observer would detect. Wundt, for his part, dismissed the Wurzburg procedures as sham experiments, because the observer, not the experimenter, controlled what was observed. The quarrel, known as the imageless thought controversy, was never resolved, and that was the real damage: two laboratories using the same method reported different contents of consciousness, and introspection had no way to decide between them. Kulpe's influence lived on, however. The concept of set is basic to the study of problem solving, and one of Kulpe's doctoral students at Wurzburg was Max Wertheimer, the founder of Gestalt psychology (Chapter 2).

### Act psychology: Brentano, Stumpf and G. E. Muller

A different German-speaking tradition rejected the idea of conscious contents altogether. In 1874, the same year Wundt completed his *Principles*, the philosopher **Franz Brentano** (1838 to 1917) published *Psychology from an Empirical Standpoint*. Brentano argued that psychology should study mental *acts*, not mental contents. When you see a red flower, the redness is a physical phenomenon; what is mental is the act of seeing it. This is **act psychology**. Brentano's key idea was **intentionality**: every mental act is directed at, or refers to, some object beyond itself. You never just see; you see something. You never just love or fear; you love or fear someone or something. He grouped acts into three classes: ideating (sensing, imagining), judging (accepting or rejecting, perceiving, recalling) and loving-hating (feeling, wishing, resolving). His method was careful empirical observation of experience rather than laboratory experiment. Brentano taught in Vienna for two decades, where his students included Carl Stumpf, the philosopher Edmund Husserl (founder of phenomenology), Christian von Ehrenfels (whose "form qualities" anticipated Gestalt psychology) and, as an auditor of his lectures, the young Sigmund Freud.

**Carl Stumpf** (1848 to 1936), Brentano's student, studied the perception of musical tones in his *Psychology of Tone* (two volumes, 1883 and 1890) and argued with Wundt over who was the better judge of tones: Stumpf trusted trained musicians, Wundt trusted trained laboratory observers. From 1894 Stumpf directed the psychological institute at the University of Berlin, where he favoured describing experience as it naturally occurs (a phenomenological approach) rather than breaking it into elements. Three of his students, Wolfgang Kohler, Kurt Koffka and Kurt Lewin, became leaders of Gestalt psychology. You have already met Stumpf on the Clever Hans commission.

Finally, **Georg Elias Muller** (1850 to 1934) ran the second great German laboratory, at Gottingen, for forty years. Muller was a rigorous experimentalist rather than a theorist of acts, and he is grouped here because he stood outside Wundt's system. He refined Fechner's psychophysical methods, extended Ebbinghaus's memory research (with his assistant Friedrich Schumann he developed the memory drum for presenting syllables at a fixed rate), and, with Alfons Pilzecker in 1900, described retroactive interference and proposed that memories need time to consolidate (Chapter 7). His students included Edgar Rubin, whose figure-ground drawings you will meet in Chapter 4.

### The decline of structuralism, and what it left behind

Structuralism died with Titchener in 1927; no one carried it on. The reasons for its decline are examined in almost every paper, so it pays to know them as a list.

- **Introspection was unreliable.** Trained observers in different laboratories reported different things (the imageless thought controversy), and since each observer's experience is private, there was no way to settle disputes.
- **Observing changes what is observed.** As Comte had warned and William James repeated, attending to an experience alters it, and describing it afterwards depends on memory.
- **Elementism missed the whole.** The Gestalt psychologists argued that experience comes organised in wholes that analysis destroys.
- **Its subject matter was too narrow.** It excluded animals, children, the mentally ill, individual differences and the unconscious, which Freud was making impossible to ignore.
- **It refused to be useful.** Titchener regarded applied work as technology, not science, just as American society was demanding a useful psychology.

And yet structuralism mattered. It made psychology a laboratory discipline with standards of training and control, and it gave every later school something definite to argue against. Its topics have also come back. The modern approach most closely aligned with the structuralists in its interest in consciousness, mental imagery and thought processes is **cognitive psychology**, which studies the same subjects with reaction times, error patterns and brain imaging instead of trained introspection. Neuroscience studies the brain rather than the contents of experience; behaviourism banished consciousness; and functionalism, as you will now see, asked a different question altogether.

CLINICAL LINK

Introspection never left the clinic. Every self-report questionnaire, every mental status interview question ("What is going through your mind right now?") and every thought record in cognitive behaviour therapy asks a person to observe and report his own consciousness. Titchener's problems come with it: patients describe interpretations rather than experiences (a cousin of the stimulus error), and their reports are shaped by memory, expectation and the way the question is asked. Good clinicians, like good structuralists, train the observer, which is exactly what learning to separate a situation, a thought and a feeling in therapy does.

## 1.7 Functionalism and the American Pioneers

HIGH YIELD · 15 QS

Structuralism asked what the mind is made of. The question that dominated American psychology from the 1890s was different: what is the mind *for*? **functionalism** is the school that studies the mind as it functions in adapting the organism to its environment. Notice the order of words, which a bank question tests: the *mind* functions in adapting the *organism*. Functionalists studied mental processes and behaviour not as collections of elements but as activities that help a living creature survive and solve problems.

### Why functionalism grew in America ●

Three forces made America receptive. The first was Darwin, whose theory had become enormously popular in the United States, partly through the evolutionary philosophy of the British writer Herbert Spencer: if traits exist because they are useful, the obvious question about consciousness is what use it is. The second was the American philosophy of **pragmatism**, which judged ideas by their practical consequences. The third was the character of the country itself: a rapidly growing nation with expanding schools, factories and cities, and universities that had to justify their new psychology departments by doing something useful. Functionalism was never a tightly organised school with a single founder and creed. It was an attitude, shared by people at Harvard, Chicago and Columbia, and it was Titchener, in his 1898 article, who gave it its name, as the opposite of his own "structural" psychology.

### William James ●●●

**William James** (1842 to 1910), the elder brother of the novelist Henry James, is the father of American psychology and the inspiration, though not the formal founder, of functionalism. He qualified in medicine at Harvard in 1869 but never practised. Around that time he fell into a deep depression, from which he dragged himself, he said, partly by reading the French philosopher Charles Renouvier on free will and deciding that his first act of free will would be to believe in free will. He began teaching at Harvard, and in 1875 he offered what is usually called the first course in scientific psychology in America and set up a small room of apparatus for classroom demonstrations, the room mentioned at the start of this chapter. James disliked laboratory drudgery and had little patience with Fechner's psychophysics, but his teaching produced the first American doctorate in psychology (G. Stanley Hall, 1878).

In 1878 James signed a contract to write a psychology textbook in two years. It took twelve. *The Principles of Psychology* (1890), in two volumes, is the most widely read book in the history of the discipline, and its ideas are tested constantly. A shorter version followed in 1892, which students nicknamed "Jimmy" to distinguish it from the full "James". Four ideas from it matter most for you.

The first is the **stream of consciousness**. Against the structuralists' picture of consciousness as a mosaic of elements, James wrote: "Consciousness, then, does not appear to itself chopped up in bits. Such words as 'chain' or 'train' do not describe it fitly … It is nothing jointed; it flows. A 'river' or a 'stream' are the metaphors by which it is most naturally described." He listed its characteristics: thought is personal (always someone's), always changing (no state ever recurs exactly), sensibly continuous (even across gaps such as sleep), concerned with objects outside itself, and selective, choosing among the things it could attend to. The phrase passed into literature, where novelists such as James Joyce and Virginia Woolf adopted a "stream of consciousness" style. Consciousness, for James, was selective precisely because it was useful: it is a fighter for ends, which is a functionalist claim.

The second is **habit**. James argued that the "plasticity" of nervous tissue lets repeated actions wear grooves in the nervous system, so that they become automatic and free attention for other things. "Habit is thus the enormous fly-wheel of society, its most precious conservative agent." His practical advice for forming a new habit is still quoted by therapists: launch the new habit with as strong an initiative as possible, never allow an exception until it is firmly rooted, and seize the very first opportunity to act on a resolution.

The third is the **self**. James divided the self into the "Me" (the self as known) and the "I" (the self as knower). The Me has three parts: the *material self* (one's body, clothes, family, home and possessions), the *social self* (the recognition one gets from others; a man "has as many social selves as there are individuals who recognize him") and the *spiritual self* (one's inner dispositions, faculties and conscience). The I is the *pure ego*, the sense of personal identity that continues through the stream. He even offered a formula: self-esteem equals success divided by pretensions. Raise your aspirations without raising your achievements and your self-esteem falls; give up a pretension and it rises.

The fourth is his theory of emotion: in an 1884 article James argued that we do not cry because we are sorry; we are sorry because we cry. The Danish physiologist **Carl Lange** (1834 to 1900) proposed a similar theory independently in 1885, so it is known as the James-Lange theory (Chapter 11).

After the *Principles*, James turned to philosophy. He developed pragmatism, the doctrine (first stated by his friend Charles Sanders Peirce) that the meaning and truth of an idea lie in its practical consequences, in *Pragmatism* (1907); he also wrote *Talks to Teachers* (1899) and *The Varieties of Religious Experience* (1902).

> EXAM TRAP
>
> "Stream of consciousness" is **William James** (not Wundt, Jung or Skinner). "Functionalism is how people work, play and adapt to their surroundings; this viewpoint was given by ..." is **William James**: Wundt (misspelt "William Wundt" in one paper) is the voluntarist, Darwin inspired functionalism but was not a psychologist of any school, and Watson founded behaviourism. When a question asks which psychologist functionalism is "most closely associated with", the expected answer is James, not Titchener (structuralism), Watson (behaviourism) or G. Stanley Hall (who was a functionalist in spirit but is examined for institutions and adolescence). If Dewey or Angell appears as an option in a question about the *formal founding* of the functionalist *school* at Chicago, choose them.

## The Chicago school: Dewey, Angell and Carr

Functionalism became a school, with a home and a programme, at the University of Chicago. Its founding document was a paper by the philosopher **John Dewey** (1859 to 1952), "The Reflex Arc Concept in Psychology", published in the *Psychological Review* in 1896. The **reflex arc** is the physiologists' unit of behaviour: a stimulus excites a sensory (afferent) nerve, a central process connects it to a motor (efferent) nerve, and a response follows. Psychologists had adopted it as the model for all behaviour: sensation, then idea, then action. Dewey argued that this chops a single, continuous act into three disconnected pieces, "a patchwork of disjointed parts".

He made the argument with an example from James: a child sees a candle flame, reaches for it, is burnt, and pulls back. On the reflex-arc view, two separate arcs have occurred: light-reach, then burn-withdraw. Dewey pointed out that the seeing was never a bare stimulus. It was already part of a coordinated act of looking-in-order-to-reach, and the burn does not end the story; it changes the meaning of the seeing, so that next time the child sees a flame as "a light that burns" (Figure 1.5). Stimulus and response are not separate events but phases of one continuous coordination, in which each response alters the next stimulus. Behaviour, Dewey concluded, must be studied as a whole adaptive act in its context.

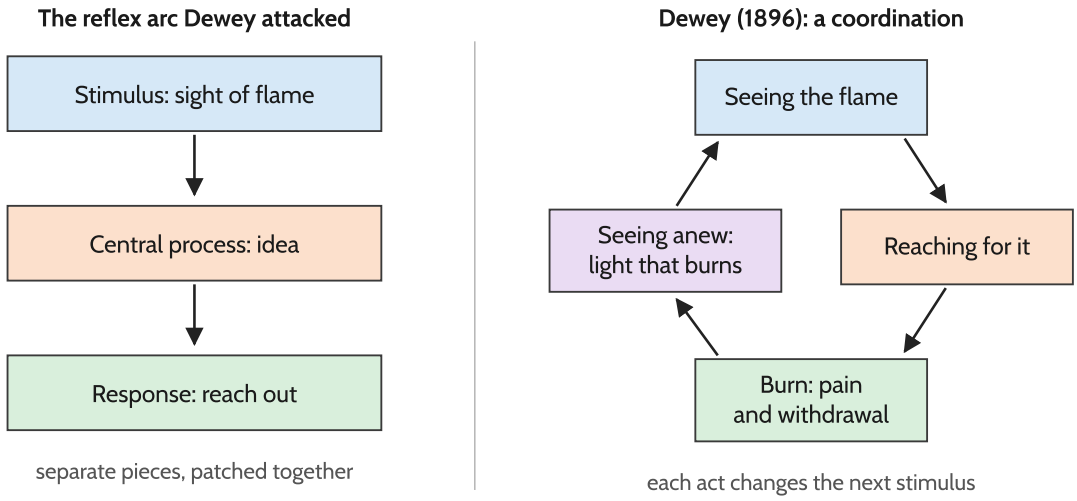


Figure 1.5 Dewey's critique of the reflex arc. On the left, the traditional view treats stimulus, idea and response as separate links in a chain; on the right, Dewey's child-and-candle example shows a continuous circuit in which the burn transforms the meaning of the next sight of a flame.

Dewey went on to become the most influential American philosopher of education. At Chicago he founded a Laboratory School where children learned by doing, and he served as president of the American Psychological Association in 1899. He left Chicago for Columbia in 1904.

> **EXAM TRAP**
>
> "According to John Dewey, ____ is the connection between sensory stimuli and motor responses." The answer is the **reflex arc**, the concept his 1896 paper examined and criticised. "Efferent" describes only the outgoing, motor half of the arc (afferent nerves carry signals in, efferent nerves carry them out), so an "efferent arc" cannot connect stimulus to response; "association arc" and "dynamic arc" are invented distractors, the second borrowing a word from Woodworth's *Dynamic Psychology*. Remember that Dewey *attacked* the reflex arc as the unit of behaviour; he did not invent it.

The man who turned Dewey's attitude into a working department was **James Rowland Angell** (1869 to 1949). In his presidential address to the American Psychological Association in 1906, "The Province of Functional Psychology" (published in 1907), Angell gave the school its clearest statement. Functional psychology, he said, is (1) the psychology of mental *operations*, as opposed to mental elements; (2) the study of the fundamental *utilities* of consciousness, the mind being what mediates between the environment and the needs of the organism; and (3) *psychophysical* psychology, which treats mind and body as a single functioning organism rather than two parallel series. Angell's Chicago department trained many students, among them a young man named John B. Watson, who took his doctorate there in 1903. Angell later became president of Yale University.

Under Angell's successor, **Harvey A. Carr** (1873 to 1954), Chicago functionalism reached its mature form. In *Psychology: A Study of Mental Activity* (1925), Carr defined the subject as mental activity: the acquisition, fixation, retention, organisation and evaluation of experiences, and their use in guiding conduct. His unit of analysis was the *adaptive act*, made up of a mo-

tive (a stimulus that persists until it is dealt with, such as hunger), a sensory situation, and a response that changes the situation so as to satisfy the motive. By Carr's time functionalism was less a rebel school than the ordinary way American psychologists worked.

### The Columbia school: Cattell, Woodworth and Thorndike

At Columbia University in New York, functionalism took a still looser form, focused on measurement and individual differences. Its senior figure was **James McKeen Cattell** (1860 to 1944), Wundt's self-appointed assistant. After Leipzig, Cattell worked with Galton in England and absorbed his interest in measuring individual differences. In 1888 he became professor of psychology at the University of Pennsylvania, often described as the first such title anywhere, and in 1891 he moved to Columbia, where he stayed until 1917, when he was dismissed for opposing the conscription of soldiers in the First World War.

In an 1890 article in *Mind*, Cattell coined the term **mental test**. His tests followed Galton's logic: strength of grip, speed of movement, two-point threshold, sensitivity to pain, discrimination of weights, reaction time to sound, time to name colours, accuracy in bisecting a line, judging a ten-second interval, and the number of letters remembered after one hearing. He hoped such tests would reveal mental ability. In 1901 his student Clark Wissler showed that scores on them bore almost no relation to college grades, and the sensory approach to intelligence collapsed. Cattell's lasting contributions lay elsewhere: in reaction time and reading research (he showed, for example, that a familiar word can be recognised about as quickly as a single letter), in the ranking method of measurement, and above all as an organiser and editor who did more than anyone to make American psychology a profession.

Robert Sessions Woodworth, whom you met in Section 1.1, spent his career at Columbia and became the great systematiser of functionalism. In *Dynamic Psychology* (1918) he insisted that psychology must study not only the mechanisms of behaviour (how it happens) but also its motivation (why), and he introduced the term "drive" to psychology (Chapter 10). He also proposed adding the organism to the stimulus-response formula. The **S-O-R** formula (stimulus, organism, response) states that the same stimulus produces different responses depending on the state of the organism: a plate of biryani means one thing to a hungry person and another to someone who has just eaten. Woodworth's *Experimental Psychology* (1938) was so widely used that students called it "the Columbia Bible", and his *Contemporary Schools of Psychology* (1931) taught a generation how to compare the systems you are studying now. (His Personal Data Sheet, an early personality questionnaire, belongs to the Assessment book.)

Columbia's third functionalist was Edward Lee Thorndike, who began studying the learning of chicks in William James's basement at Harvard and completed his doctorate under Cattell at Columbia in 1898 with the famous puzzle-box experiments on cats. His law of effect and his work on educational psychology are taught in Chapter 6; for now, notice that he came out of the functionalist tradition, as Watson did.

### The builders: Hall, Witmer and Munsterberg ●●

Functionalism's practical spirit produced the institutions of American psychology and its first applied fields. Three Wundt-trained men did most of the building.

**Granville Stanley Hall** (1844 to 1924) was a founder of almost everything. He received the first American doctorate in psychology (Harvard, 1878), studied in Leipzig in 1879 and 1880, and in 1883 opened at Johns Hopkins University the first American psychological laboratory

devoted to research (James's 1875 room had been for demonstration). In 1887 he founded the *American Journal of Psychology*, the first American psychology journal, and in 1889 he became the first president of the new Clark University in Worcester, Massachusetts. In July 1892 a small group of psychologists meeting at Clark founded the **American Psychological Association (APA)**, with Hall as its first president; its charter members numbered only a few dozen. Hall was also a pioneer of child and developmental psychology: he launched the child-study movement, using questionnaires to learn what children knew and thought, and in *Adolescence* (1904) he described the teenage years as a period of "storm and stress" (from the German *Sturm und Drang*). He explained development through **recapitulation theory**, the idea that each child, in growing up, passes through the stages of the evolution of the human race (a theory now abandoned; the Developmental Psychology book covers adolescence). In 1909 he invited Sigmund Freud and Carl Jung to lecture at Clark, Freud's only visit to America, which did much to spread psychoanalysis there.

**Lightner Witmer** (1867 to 1956) studied under Cattell at Pennsylvania, took his doctorate under Wundt in 1892, and returned to succeed Cattell at Pennsylvania. In March 1896 he opened there the world's first **psychological clinic**. At the APA meeting in December of that year he proposed a new field, which he called clinical psychology; in 1907 he founded the journal *The Psychological Clinic* and, in an article titled "Clinical Psychology" in its first issue, became the first person to use the term in print. His clinic dealt mainly with children who had learning and behaviour problems, and he insisted that clinical work rest on the methods and findings of experimental psychology.

KEY STUDY: THE FIRST CASE AT THE FIRST CLINIC (1896)

**The problem.** A schoolteacher brought Witmer a fourteen-year-old boy she described as a "chronic bad speller". He was not lazy and not dull, yet he could not learn to spell. **Method.** Witmer examined the boy as a scientist would: testing his vision, his memory, his reading and his performance under different conditions, instead of simply labelling him. **Finding.** Part of the problem turned out to be a defect of vision that made printed words hard to see clearly; with corrected vision and patient, individual teaching, the boy made progress. **Conclusion.** Witmer took two lessons from the case, and they define the profession you are entering: begin with careful assessment before treatment, and look for the simplest, most concrete cause first (Morgan's canon, applied to a child).

**Hugo Munsterberg** (1863 to 1916), another Wundt graduate, was invited by William James in 1892 to run the Harvard laboratory. Munsterberg became the great popular champion of *applied* psychology. In *On the Witness Stand* (1908) he argued that eyewitness testimony and memory are unreliable and that psychology could help the courts, making him a founder of forensic psychology. In *Psychology and Industrial Efficiency* (1913) he applied psychological methods to selecting workers (telephone operators, tram drivers) and to advertising, making him a founder of industrial psychology. He also wrote on psychotherapy and on the psychology of film. He died in 1916 while lecturing at Radcliffe College.

CLINICAL LINK

Witmer is the father of clinical psychology, and the model of the clinical psychologist that Indian training programmes follow, the scientist-practitioner who assesses carefully, treats on the basis of evidence and evaluates the results, is recognisably his. Notice too what Witmer's clinic did not do: it did not begin with psychoanalysis or with psychiatric diagnosis. Clinical psychology was born out of experimental psychology and education, not out of medicine, and that history explains why assessment remains its distinctive skill.

EXAM TRAP

The first psychological clinic and the term "clinical psychology" are **Witmer's** (clinic 1896; term proposed 1896, first in print 1907). Freud founded psychoanalysis, Wundt the laboratory, G. Stanley Hall the APA and the first US research laboratory, Cattell the mental test, and Erik Erikson the psychosocial stages. The APA was founded in **1892**; 1896 is the year of Witmer's clinic and Dewey's paper, so do not let it pull you. Hall was the APA's **first** president; Calkins its first **woman** president.

## The women pioneers

Women were excluded from most universities and all of Titchener's meetings, yet several of the founders of American psychology were women, and their stories are examined often.

**Mary Whiton Calkins** (1863 to 1930) studied psychology at Harvard with William James and Hugo Munsterberg in the early 1890s, attending as a guest because Harvard did not admit women. By 1895 she had completed every requirement for the doctorate, and her examiners recommended the degree, but Harvard refused to award a PhD to a woman; she later declined the offer of a degree from Radcliffe, Harvard's college for women. She taught at Wellesley College, where she founded a psychology laboratory in 1891, invented the *paired-associate* technique for studying memory (learning pairs of items so that one cues the other), and developed a "self psychology" that treated the self as the centre of all psychology. In 1905 she became the **first woman president of the American Psychological Association**, and later the first woman president of the American Philosophical Association as well.

**Margaret Floy Washburn** (1871 to 1939) was the first woman to be awarded a PhD in psychology, at Cornell in 1894, as Titchener's first doctoral student. She taught for most of her career at Vassar College, and her book *The Animal Mind* (1908) was the standard text of comparative psychology for a quarter of a century, gathering the experimental evidence on the sensory and mental capacities of animals. She also proposed a motor theory of consciousness, in which thought depends on small, incipient movements. In 1921 she became the second woman president of the APA.

**Leta Stetter Hollingworth** (1886 to 1939), who trained under Thorndike at Columbia, used experiments to demolish two beliefs used to keep women out of education and work. In *Functional Periodicity* (1914) she tested women on mental and motor tasks across the menstrual cycle and found no impairment at any phase. She also attacked the "variability hypothesis", the claim that men vary more than women and therefore produce more geniuses, showing that the evidence did not support it and that the difference in eminence was explained by social opportunity. She later became a leading authority on gifted children, and worked as one of the early clinical psychologists in New York.

EXAM TRAP

Calkins was the **first woman APA president** (1905); Washburn the **first woman PhD in psychology** (1894) and the second woman APA president (1921). Calkins completed every requirement for a Harvard doctorate in 1895 but was refused the degree; Washburn had received hers from Cornell in 1894, so she holds the "first PhD" title. Hollingworth is examined for the psychology of women and the gifted. Karen Horney, sometimes offered as a distractor, was a psychoanalyst who challenged Freud's view of women (the Personality book) and was never APA president.

## The legacy of functionalism ●●

Functionalism did not fade like structuralism; it dissolved into the mainstream. Its interests (learning, motivation, individual differences, development, abnormal behaviour, animals) became the ordinary subject matter of psychology, and its willingness to be useful created educational, industrial, forensic and clinical psychology. It also produced its own successors. Watson, Angell's student, took the functionalist interest in adaptive behaviour and animals to its logical end by dropping consciousness altogether; the aim of behaviourism, as Watson put it, was that given the stimulus, psychology should be able to predict the response, and given the response, to specify the stimulus that caused it (Chapter 2). And the functionalist question, what is a mental process for, inherited from Darwin, is the founding question of today's evolutionary psychology. Table 1.6 sets the two first schools side by side.

Table 1.6 Structuralism and functionalism compared.

| Issue | Structuralism | Functionalism |
|---|---|---|
| Central question | What is consciousness made of? | What is the mind for? |
| Leaders | Titchener (Wundt in many Indian texts) | James; Dewey, Angell, Carr (Chicago); Cattell, Woodworth (Columbia) |
| Main centres | Leipzig, Cornell | Harvard, Chicago, Columbia |
| Model science | Chemistry: a periodic table of mental elements | Biology: Darwin's evolution and adaptation |
| Method | Trained introspection only | Introspection plus observation, tests, animal and child studies |
| Who was studied | Normal adult human observers | Animals, children, the abnormal, individual differences |
| Applied work | Rejected as "technology" | Welcomed: education, industry, clinic, law |
| Mind and body | Parallelism | One functioning organism |
| What it led to | Laboratory standards; topics revived by cognitive psychology | Behaviourism, applied psychology, evolutionary psychology |

EXAM TRAP

Functionalism was most strongly influenced by **Darwin's theory of evolution** (not Freud's psychosexual stages, Pavlov's conditioning or the Gestalt laws). The historical perspective that "gave rise to the evolutionary perspective" is **functionalism**. The aim "given the stimulus, to predict the response, and given the response, to predict the antecedent stimulus" is **behaviourism**, and John B. Watson is the founder of behaviourism (not functionalism, rationalism or structuralism). Rationalism, remember, is the philosophical view of Descartes and Leibniz, not a psychological school.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Titchener's three elements of consciousness were sensations, images and _____.
2. Describing the object ("an apple") instead of the conscious experience is Titchener's _____ error.
3. Titchener explained meaning by his _____ theory: a core sensation acquires meaning from accompanying images.
4. The Wurzburg finding that thinking can occur without sensory or imaginal content is called _____ thought.
5. Brentano held that every mental act is directed at an object, a property he called _____.
6. James's 1890 book described consciousness as a continuous _____, not a chain of elements.
7. Dewey's 1896 paper attacked the _____ concept in psychology.
8. Woodworth inserted the organism into the stimulus-response formula, giving the _____ formula.
9. The first psychological clinic was opened in 1896 by _____ at the University of Pennsylvania.
10. The first woman president of the APA (1905) was Mary Whiton _____.

**Answers:** (1) affections, (2) stimulus, (3) context, (4) imageless, (5) intentionality, (6) stream, (7) reflex arc, (8) S-O-R, (9) Lightner Witmer, (10) Calkins.

## 1.8 Reading the History: Zeitgeist, Great Persons and "Fathers"

LOWER YIELD · 4 QS

You now know where every part in Max Friedrich's room came from. The chronoscope and the habit of timing observers came from physics and astronomy; the idea that a mental act takes measurable time came from Helmholtz and Donders; the word *apperception* came from Leibniz and Herbart; the conviction that the mind can be measured came from Fechner. So the puzzle from the start of the chapter can now be put sharply. If nearly everything in that room was borrowed, in what sense did Wundt *found* anything, and why do we give the credit to one man, one city and one year? Historians answer questions of this kind in two broad ways. The answers matter for more than history, because they decide which names end up in textbooks as "fathers", and so which names end up in your question paper.

### Two ways of explaining history

The first way puts individuals at the centre. In a series of lectures given in London in 1840 and published the next year as *On Heroes, Hero-Worship, and the Heroic in History*, the Scottish writer **Thomas Carlyle** (1795 to 1881) declared that "the history of the world is but the biography of great men". This is the **great-person theory** of history, still often called by Carlyle's own phrase, the great-man theory: progress happens because exceptional individuals, through genius and force of will, change the course of events, and without them it would not have happened, or would have happened very differently. Applied to our story, psychology began in Leipzig because Wundt was Wundt. It took a man who would publish more than 50,000 printed pages, supervise well over a hundred doctorates and work at his desk for more than sixty years to turn a scattered set of techniques into a discipline. Historians call this the *personalistic* view of history.

The second way puts the times at the centre. **Zeitgeist** is German for "the spirit of the times" (*Zeit*, time; *Geist*, spirit or mind): the whole climate of ideas, knowledge, techniques, values and social conditions of a period. On the Zeitgeist view, a discovery is made when the time is ripe for it, and if one person had not made it, someone else soon would have. Individuals still matter, but as the agents through whom their age does its work; the times choose the person more than the person chooses the times. Historians call this the *naturalistic* view. Its great champion in psychology was Edwin G. Boring, whose *A History of Experimental Psychology* (1929; second edition 1950) shaped how generations of students, and of examiners, learned the story you have just read. For Boring the founding of psychology in the 1870s was no accident. You saw at the end of Section 1.4 that by 1870 all the parts were already on the bench; someone was bound to assemble them.

What evidence could decide between the two views? The strongest argument for the Zeitgeist is the frequency of **multiple discovery**: the same discovery made independently by two or more people at about the same time. You have already met several in this chapter. Charles Bell (1811) and Francois Magendie (1822) both separated the sensory and motor roots of the spinal nerves, and quarrelled for years over the credit. William James (1884) and the Danish physiologist Carl Lange (1885) proposed the same theory of emotion without knowing of each other's work. The most famous case of all is evolution itself. In 1858, while Darwin was still writing his great book, he received from the naturalist **Alfred Russel Wallace** (1823 to 1913), then collecting specimens in the Malay Archipelago, an essay setting out the same theory of natural selection, and papers by both men were read together at a meeting of the Linnean Society in London that July. Outside the life sciences, Newton and Leibniz each invented the calculus. In 1922 two American sociologists, William Ogburn and Dorothy Thomas, listed 148 such cases, and the sociologist of science Robert K. Merton later argued that "multiples" are the normal pattern of science, not the exception. If great discoveries depended entirely on unique geniuses, such coincidences should be rare. That they are common suggests that the state of knowledge makes certain discoveries almost inevitable.

The other side of the same coin is the discovery made too early. In 1866 the monk Gregor Mendel published the laws of heredity he had worked out by breeding pea plants in his monastery garden at Brno. Almost nobody noticed. In 1900 three botanists, working separately, arrived at the same laws and only then found Mendel's paper: a multiple discovery that rescued a premature one. The Zeitgeist also works against ideas. When two Cambridge scholars proposed a laboratory for psychophysics in 1877, the university turned them down;

one objector is said to have complained that it would "insult religion by putting the human soul in a pair of scales". At that very moment the same science was flourishing in Leipzig. The ideas were identical; the climate was not.

The spirit of the times includes forces that have little to do with ideas, which historians call *contextual forces*, and you have met three of them already. The first is *economic opportunity*: American universities, schools and factories wanted a useful psychology, and functionalism, mental testing and the first clinic grew to meet the demand (Section 1.7). The second is *war*: Cattell lost his Columbia chair in 1917 for opposing conscription, while the same war gave American psychologists their first mass testing programme, the Army Alpha and Army Beta tests of 1917 (Chapter 9). The third is *prejudice*: Mary Whiton Calkins was refused the doctorate she had earned because she was a woman, Titchener's Experimentalists barred women from their meetings, and Leta Hollingworth spent the first part of her career disproving claims that were used to keep women out of education and work. Who becomes a "great person" depends partly on who is allowed into the room.

Most historians now conclude that neither view is enough on its own, and the clearest way to see why is to push one event through both. Take Lightner Witmer's clinic of 1896. The great-person version is easy to tell: a schoolteacher brought a boy who could not spell, and Witmer, unlike his colleagues, decided that a psychologist ought to try to help him. Now tell it the Zeitgeist way. American states were making schooling compulsory, so classrooms were filling with children who could not keep up; G. Stanley Hall's child-study movement had made children a respectable subject of research; Cattell had just introduced mental tests; and functionalism had made usefulness a scientific virtue. Someone was likely to open a clinic for schoolchildren around the turn of the century. Both versions are true. The Zeitgeist explains why a clinic appeared in America in the 1890s; the person explains why it appeared in Philadelphia in 1896, and why it took the shape it did, built on careful assessment rather than on medicine or hypnosis. A useful summary: the Zeitgeist decides what can happen; persons decide when, where and how it does.

> EXAM TRAP
>
> **Zeitgeist** means "spirit of the times". It is not "spirit of the place", not "worldview" (that is *Weltanschauung*) and not "form" or "organised whole" (that is *Gestalt*, Chapter 2). The historian who championed the Zeitgeist view of psychology's history is **Boring**; the "great man" view goes back to **Carlyle** (1841). If a question asks which view best explains simultaneous discoveries, such as Darwin and Wallace (1858), Bell and Magendie, or James and Lange, the answer is the **Zeitgeist** or naturalistic view. The great-person view is the **personalistic** one.

## Why Germany, why Wundt, why 1879?

The Zeitgeist explains a good deal about *where* psychology was born. Why Germany, and not Britain, France or America, all of which had brilliant scientists? Three features of German academic life made the difference. First, the German universities, reformed on the model of the University of Berlin (founded in 1810), were built around research as well as teaching. Professors were expected to discover things and enjoyed *Lehrfreiheit*, the freedom to teach whatever their research led them to; students enjoyed *Lernfreiheit*, the freedom to move between universities in search of the best teachers. Because the German states had many universities competing for prestige, a new field could find a chair, a room and a budget some-

where. Second, the German word for science, *Wissenschaft*, covered every kind of systematic knowledge, history and philology included, so a "science of mind" sounded natural rather than contradictory. Third, as Section 1.3 showed, experimental physiology was stronger in Germany than anywhere else; Muller, Helmholtz, Weber and Fechner were all German, and their methods were exactly the ones psychology needed.

Britain, by contrast, had brilliant individuals but no institutional home for them. Darwin and Galton, the two British giants of this chapter, were men of independent means who never held university posts, and the ancient universities, as the Cambridge refusal shows, were slow to accept new sciences. France had a powerful clinical tradition, and French psychology grew up in the hospital and the school rather than the laboratory: first in the study of hysteria and hypnosis, and then in the work of **Alfred Binet** (1857 to 1911), who with Theodore Simon produced the first practical intelligence test in 1905 (Chapter 9). America had energy and money, but in 1879 it had no research laboratory, and its universities were still learning the German model; G. Stanley Hall, like so many others, crossed the Atlantic to study in Leipzig.

The Zeitgeist cannot, however, explain why it was Wundt rather than one of the many Germans who had the same knowledge. Here historians draw a useful distinction between a *founder* and a *precursor* (or pioneer). Weber, Fechner, Helmholtz and Donders all did experiments that we now count as psychology, but none of them set out to create a new discipline. Weber was a physiologist of the senses; Helmholtz thought of himself as a physicist and physiologist; Donders was an eye specialist; Fechner was trying to prove a philosophy of mind and matter. Wundt alone did all four of the things that turn a set of methods into a science. He *declared* it, in print, in the preface to his *Principles* (1873 to 1874). He gave it a *place*, the Leipzig laboratory. He gave it a *voice*, the journal *Philosophische Studien* (1881). And he gave it *people*: the stream of students who carried it to Wurzburg, Cornell, Columbia, Philadelphia, Harvard and, as you will see shortly, Calcutta. A founder is not the first person to have an idea. He is the person who turns the idea into an institution that outlives him.

That leaves the year. Why 1879, when James had a room of apparatus at Harvard in 1875 and Fechner's book had appeared in 1860? Because 1875 was a room for showing known effects to students, and 1860 was a book. The winter of 1879 is the first time that students did original, publishable research in a room set aside for psychology. The date is a convention, fixed by later historians (the university did not formally recognise the laboratory as one of its institutes until some years afterwards), but it is a sensible convention, and it is the one every question paper expects.

EXAM TRAP

"Who established the first laboratory for experimental psychology?" is **Wundt** (Leipzig, 1879). Each distractor is tied to a different "first" or to no laboratory at all. **Galton** opened the first *anthropometric* laboratory (London, 1884), a centre for measuring individual differences, not a research school. **Weber** was a physiologist whose experiments on touch and lifted weights came before any psychology laboratory existed. **Jung** turned Galton's word association test into a clinical tool in Zurich (the Personality book). **James** had a demonstration room at Harvard in 1875. If the question asks for the city, it is **Leipzig**: not Heidelberg (where Wundt assisted Helmholtz), not Zurich (where he spent one year), not Vienna (Brentano and Freud), not Paris (Charcot's clinic and Binet's tests), and not Cambridge, which is both the town in Massachusetts that is home to Harvard and the English university that refused a psychophysics laboratory in 1877. The first *American* research laboratory is **Hall's** (Johns Hopkins, 1883).

## Fathers, founders and firsts

Indian entrance papers have a favourite question form: "Who is the father of ...?" The difficulty is that "father" is used loosely, to mean at least four different things. It can mean the *founder of an institution*: Wundt founded the laboratory, Witmer the clinic. It can mean the person who *named* a field or school: Titchener named both structuralism and functionalism, though he founded only the first; Witmer named clinical psychology; William Stern named differential psychology. It can mean the *most influential early figure*, as James is for American psychology. And it can mean the first person to do something *in one country*, as Sengupta is for India. Two habits will protect you. Read the qualifying word in the stem ("experimental", "American", "modern", "formal", "in India"), because it often decides between two good candidates. And when two candidates are both defensible, choose the one that standard textbooks name, which is the one given in Table 1.7.

Names attached to discoveries can mislead in a second way. Hyphenated names are often a fossil record of multiple discovery: the Bell-Magendie law, the James-Lange theory, the Young-Helmholtz theory (Thomas Young's suggestion of 1802, developed by Helmholtz half a century later) and, in Chapter 11, the Cannon-Bard theory. And some names were given by other people: "Weber's law" was named by Fechner, and Morgan's canon restates, for animal psychology, the much older rule of parsimony usually credited to the fourteenth-century philosopher William of Ockham ("Ockham's razor").

If psychology has fathers, it also has a family tree, and seeing the tree whole is the quickest way to remember who trained whom (Figure 1.6). Notice how much of it grows from one root. Wundt's students founded the Wurzburg school, structuralism, the first American research laboratory, the Columbia school, the first psychological clinic and applied psychology at Harvard. Their students, in turn, include the founder of Gestalt psychology, the first woman to earn a doctorate in psychology, the first woman to lead the APA and the founder of modern psychology in India. A Zeitgeist historian would add that a network like this is exactly how the spirit of an age travels: not through the air, but through people who learned in the same rooms.

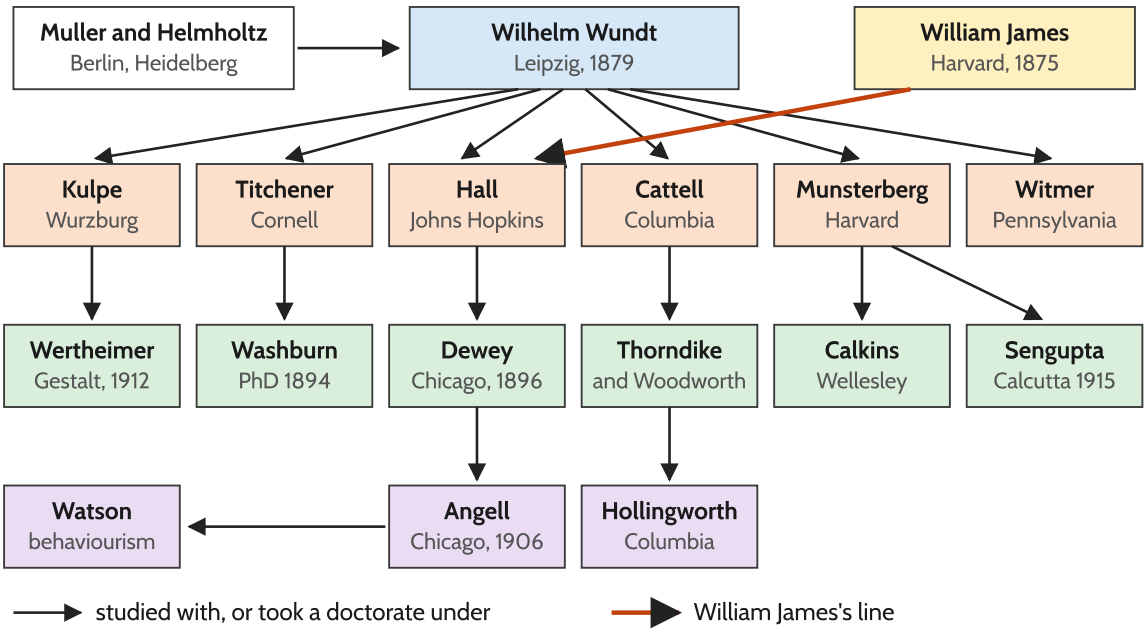


**Figure 1.6** Psychology's family tree. Arrows run from teacher to student. Nearly every founder in this chapter is one or two steps from Wundt, and the line through Munsterberg runs all the way to Calcutta. (Calkins and Thorndike also studied with William James.)

Table 1.7 gathers the "father" questions you are most likely to meet, together with the bait that usually sits beside the right answer. A few of the people in it belong to later chapters; they are here because the question form is the same, and you will meet them properly where their subject is taught. Table 1.8 does the same for the "firsts", which are examined just as often and confused just as easily.

**Table 1.7** Who is the "father" of what: the usual answers and the usual bait.

| Father or founder of | Usual answer | Remember by | Common bait |
|---|---|---|---|
| Psychology as a science; modern psychology | Wilhelm Wundt | Leipzig laboratory, 1879 | Aristotle (father of psychology in the ancient sense); James |
| Experimental psychology | Wilhelm Wundt | *Principles of Physiological Psychology* (1873 to 1874) | Fechner (Boring's case for 1860) |
| Voluntarism; folk (cultural) psychology | Wilhelm Wundt | Apperception; *Volkerpsychologie* | Titchener; Freud |
| Psychophysics | Gustav Fechner | *Elements of Psychophysics* (1860) | Weber (the JND); Wundt |
| Mental chronometry | F. C. Donders | Subtraction method (1868) | Helmholtz (speed of the nerve impulse) |
| Experimental study of memory | Hermann Ebbinghaus | *On Memory* (1885) | G. E. Muller (the memory drum) |
| Associationism as a doctrine | David Hartley | *Observations on Man* (1749) | Aristotle (the first laws) |
| Phrenology | Franz Joseph Gall | Skull "organs" of faculties | Spurzheim (spread it under that name) |

| Father or founder of | Usual answer | Remember by | Common bait |
|---|---|---|---|
| Structuralism | E. B. Titchener | Named it in 1898; Cornell | Wundt (accepted in many Indian texts) |
| Act psychology | Franz Brentano | Intentionality (1874) | Kulpe (Wurzburg school) |
| American psychology | William James | *Principles of Psychology* (1890) | G. Stanley Hall (the institution builder) |
| Functionalism | William James (its inspiration) | Formal school: Dewey (1896) and Angell (1906) | Titchener (named it, opposed it) |
| Behaviourism | John B. Watson | The 1913 manifesto | Pavlov, Thorndike, Skinner |
| Reflexology | Vladimir Bekhterev | Objective psychology in Russia | Pavlov (conditioned reflexes); Sechenov (*Reflexes of the Brain*, 1863) |
| Operant conditioning; radical behaviourism | B. F. Skinner | Chapters 2 and 6 | Watson; Thorndike |
| Psychoanalysis | Sigmund Freud | *The Interpretation of Dreams* (1900) | Breuer; Charcot |
| Gestalt psychology | Max Wertheimer | Phi phenomenon paper (1912) | Kohler and Koffka (co-founders) |
| Clinical psychology (and school psychology) | Lightner Witmer | Clinic 1896; term in print 1907 | Freud; Kraepelin; Hall |
| Differential psychology and psychometrics | Francis Galton | Anthropometric laboratory (1884); correlation | Stern (named the field, 1900) |
| The mental test (the term) | James McKeen Cattell | *Mind*, 1890 | Raymond Cattell (personality) |
| Intelligence testing | Alfred Binet | Binet-Simon scale (1905) | Galton; Cattell; Stern (IQ, 1912); Terman (1916) |
| Factor analysis | Charles Spearman | General factor g (1904) | Thurstone (Chapter 9) |
| Comparative psychology | George Romanes | *Animal Intelligence* (1882) | Lloyd Morgan (made it scientific); Darwin |
| Child psychology | Wilhelm Preyer | *The Mind of the Child* (1882) | G. Stanley Hall (child study; adolescence) |
| Educational psychology | Edward Lee Thorndike | *Educational Psychology* (1903) | Herbart (scientific pedagogy) |
| Industrial and forensic psychology | Hugo Munsterberg | Books of 1908 and 1913 | Cattell; Witmer |
| Modern social psychology | Kurt Lewin | Field theory; group dynamics | Triplett (first experiment, 1898); McDougall (textbook, 1908); Comte (sociology) |
| Humanistic psychology | Abraham Maslow, with Carl Rogers | The "third force" | Freud; Skinner |
| Cognitive psychology | Ulric Neisser | *Cognitive Psychology* (1967) | Chomsky; Miller |
| Positive psychology | Martin Seligman | APA presidency (1998) | Maslow |
| Modern psychology in India | N. N. Sengupta | Calcutta laboratory (1915) and department (1916) | G. Bose; M. V. Gopalaswamy |
| Psychoanalysis in India | Girindrasekhar Bose | Indian Psychoanalytic Society (1922) | Sengupta; Freud |

Table 1.8 The famous "firsts" of the new science.

| The first … | Who | Where | When |
|---|---|---|---|
| Psychology course and demonstration laboratory in America | William James | Harvard | 1875 |
| Journal of psychology and philosophy (*Mind*) | Alexander Bain | Britain | 1876 |
| American doctorate in psychology | G. Stanley Hall | Harvard | 1878 |
| Laboratory for psychological research | Wilhelm Wundt | Leipzig | 1879 |
| Journal of experimental psychology (*Philosophische Studien*) | Wilhelm Wundt | Leipzig | 1881 |
| American research laboratory | G. Stanley Hall | Johns Hopkins | 1883 |
| Anthropometric (testing) laboratory | Francis Galton | London | 1884 |
| American psychology journal (*American Journal of Psychology*) | G. Stanley Hall | Johns Hopkins | 1887 |
| Professor of psychology (the title) | James McKeen Cattell | Pennsylvania | 1888 |
| Use of the term "mental test" | James McKeen Cattell | In *Mind* | 1890 |
| President of the APA | G. Stanley Hall | Founded at Clark | 1892 |
| Woman awarded a PhD in psychology | Margaret Floy Washburn | Cornell | 1894 |
| Psychological clinic | Lightner Witmer | Pennsylvania | 1896 |
| Experiment in social psychology | Norman Triplett | Indiana | 1898 |
| Woman president of the APA | Mary Whiton Calkins | Wellesley | 1905 |
| Practical intelligence test | Alfred Binet and Theodore Simon | Paris | 1905 |
| "Clinical psychology" in print | Lightner Witmer | *The Psychological Clinic* | 1907 |
| Textbooks titled *Social Psychology* | William McDougall; Edward A. Ross | Britain; America | 1908 |

EXAM TRAP

**Father of psychology**: Wundt if the stem says scientific, modern or experimental; Aristotle only if it plainly means the ancient study of the soul (*De Anima*). **Father of American psychology**: James, not Hall, who built the institutions (the first American research laboratory, the first American journal, the first APA presidency). **Father of functionalism**: James, unless the stem asks who founded the functionalist *school* at Chicago, which is Dewey, with Angell as its organiser. **Father of behaviourism**: Watson. Pavlov discovered classical conditioning, Thorndike the law of effect, and Skinner developed operant conditioning and radical behaviourism; "reflexology" was **Bekhterev's** name for his system, although Pavlov's conditioned-reflex work is often grouped with it as Russian reflexology. **Father of intelligence testing**: Binet. Galton measured individual differences, Cattell coined "mental test", Stern coined the intelligence quotient (1912), and Terman's Stanford-Binet (1916) made the IQ famous.

EXAM TRAP

**Same name, different person.** James McKeen Cattell (1860 to 1944) coined "mental test"; Raymond Bernard Cattell (1905 to 1998), no relation, built the 16PF questionnaire and divided traits into **source and surface** traits (the Personality book), so a question about "Cattell's" two categories of traits means Raymond. Johannes Muller gave us specific nerve energies; G. E. Muller ran the Gottingen laboratory and studied memory. James Mill (mental mechanics) was the father of John Stuart Mill (mental chemistry), and neither is William James. Ernst Weber the physiologist is not Max Weber the sociologist. Erasmus Darwin was the grandfather of both Charles Darwin and Francis Galton. Chapter 2 adds the two Watsons.

## Psychology comes to India ●●

Reflection on the mind in India is far older than anything in this chapter. The Upanishads, the Samkhya and Yoga systems, Buddhist analyses of consciousness and the medical tradition of Ayurveda all contain detailed accounts of perception, attention, emotion and the self, and they are the subject of Sections 2.8 and 2.9. What reached India in the early twentieth century was something narrower and newer: the laboratory psychology of Leipzig and its American descendants. It arrived, as so much of modern Indian science did, through the University of Calcutta (now Kolkata).

The first step was a syllabus. In 1905, by the usual account, the philosopher **Brajendranath Seal** (1864 to 1938) drew up a course in experimental psychology for the University of Calcutta, taught within its philosophy curriculum. The person who turned the syllabus into a science was **Narendra Nath Sengupta** (1889 to 1944; also spelt Sen Gupta). After his studies in Calcutta, Sengupta went to Harvard, where he trained in experimental psychology under Hugo Munsterberg, himself a Leipzig graduate, so that the line in Figure 1.6 runs unbroken from Wundt's room to Bengal. Back in Calcutta, he set up in 1915 the first laboratory of experimental psychology in India, and in 1916 the University of Calcutta opened the first independent department of psychology in the country, with Sengupta as its moving spirit. (Some sources date the laboratory to 1916 as well; the department is 1916, and the place is always Calcutta.) His early research followed the Leipzig agenda of attention and perception, but he also turned to society: with the sociologist Radhakamal Mukerjee he wrote *Introduction to Social Psychology: Mind in Society* (1928), one of the first Indian books in the field. He took the lead in founding the Indian Psychological Association in 1925, and later moved to Lucknow University, where he built another department. He is the person Indian papers call the founder, or father, of modern psychology in India.

The second founder came to psychology from medicine. **Girindrasekhar Bose** (1887 to 1953; also written Girindra Shekhar Bose) qualified as a doctor, then took a master's degree in the new Calcutta department, and in 1921 earned what is usually counted as the first doctorate in psychology awarded by an Indian university, for a thesis published as *The Concept of Repression* (sources differ by a year on the date of the degree). He had been treating patients by psychoanalytic methods, and in 1921 he wrote to Freud, beginning a correspondence that lasted many years. In 1922 Bose founded the **Indian Psychoanalytic Society** in Calcutta, one of the earliest psychoanalytic societies outside Europe and North America, and it was soon affiliated to the International Psychoanalytical Association. Bose did not simply copy Freud. In his *theory of opposite wishes* he argued that every wish is accompanied by its opposite (the

wish to beat by the wish to be beaten, for example), and he drew on Indian philosophy to explain how such conflicts resolve; Freud read his ideas with interest but was not persuaded. Bose later headed the Calcutta department, and in 1947 he founded the journal *Samiksa*. He is the father of psychoanalysis in India. (His theory is taken further in Section 2.8 and in the Personality book.)

Other centres followed. In 1923 psychology was given its own section, together with educational science, at the annual Indian Science Congress. In 1924 **M. V. Gopalaswamy**, who had trained in London, started the department of psychology at the University of Mysore. In 1925 the Indian Psychological Association was founded, and in 1926 it launched the *Indian Journal of Psychology*. Over the following decades departments opened at universities across the country. Table 1.9 sets the founding events in order; notice how closely they follow the American sequence of laboratory, department, society and journal, a generation later.

**Table 1.9** Psychology arrives in India.

| Year | Event | Who or where |
|---|---|---|
| 1905 | Syllabus in experimental psychology, University of Calcutta | Brajendranath Seal |
| 1915 | First laboratory of experimental psychology in India | N. N. Sengupta, Calcutta |
| 1916 | First independent department of psychology in India | University of Calcutta |
| 1921 | First Indian doctorate in psychology (*The Concept of Repression*) | Girindrasekhar Bose |
| 1922 | Indian Psychoanalytic Society founded | Girindrasekhar Bose, Calcutta |
| 1923 | Psychology section at the Indian Science Congress | Indian Science Congress |
| 1924 | Department of psychology, University of Mysore | M. V. Gopalaswamy |
| 1925 | Indian Psychological Association founded | N. N. Sengupta and colleagues |
| 1926 | *Indian Journal of Psychology* begins | Indian Psychological Association |
| 1928 | *Introduction to Social Psychology* | Mukerjee and Sengupta |
| 1947 | Journal *Samiksa* founded | Girindrasekhar Bose |
| 1954 | All India Institute of Mental Health (NIMHANS from 1974) | Bangalore |
| 1968 | Indian Association of Clinical Psychologists formed | Professional body |
| 1992 | Rehabilitation Council of India Act | Parliament of India |

**INDIAN CONTEXT**

If you sit your examination in Mysuru, you sit it in one of the oldest homes of Indian psychology. M. V. Gopalaswamy's energy did not stop at the department he began at the University of Mysore in 1924. In 1935 he set up a small radio transmitter at his house in Mysore and began broadcasting, and the station came to be known as *Akashvani*, "voice from the sky", the name that All India Radio later adopted for itself. The voice of the nation's radio began in a psychologist's house.

CLINICAL LINK

Clinical psychology in India has founding events of its own, and interviewers for clinical programmes expect you to know them. The All India Institute of Mental Health was set up in Bangalore (now Bengaluru) in 1954 and began formal training in clinical psychology the following year; in 1974 it became the National Institute of Mental Health and Neuro Sciences (NIMHANS). The Indian Association of Clinical Psychologists was formed in 1968. Under the Rehabilitation Council of India Act of 1992, clinical psychologists must hold a recognised qualification and be registered with the Rehabilitation Council of India (RCI) to practise. The line from Witmer's clinic in 1896 to the course you are applying for is a direct one: assess first, then treat on the basis of evidence. (The profession itself is covered in the Clinical Psychology book.)

EXAM TRAP

**Founder (father) of modern psychology in India**: **N. N. Sengupta**. Bose is the father of Indian *psychoanalysis*; Gopalaswamy began the Mysore department (1924); "Amitosh Mukherjee", a distractor in one bank question, is not a founder of Indian psychology (the name may be a garbled form of Sir Ashutosh Mookerjee, the Calcutta vice-chancellor who backed the new laboratory and department, but he was a mathematician and administrator, not a psychologist). **First department of psychology in India**: Calcutta, **1916** (not 1922, the Psychoanalytic Society, and not 1924, Mysore). A bank question asks who "played a key role in establishing the first psychology laboratory in India", offers only Anand C. Paranjpe, Uma Chaudhuri, Sudhir Kakar and G. Misra, and keys **Uma Chaudhuri**. Historically the answer is Sengupta, who is not among the options. Uma Chaudhuri (also spelt Chowdhury) of Calcutta is known for the Indian modification of the Thematic Apperception Test (1960); a companion bank question keys "Kamala Chaudhuri" for that adaptation. The other three are later leaders of indigenous Indian psychology (Section 2.8): Paranjpe for the self in Indian thought, Sudhir Kakar for psychoanalysis and Indian culture, and Girishwar Misra for cultural psychology; Durganand Sinha, another option in the TAT question, is known for the indigenisation of psychology in India. So: if Sengupta is offered, choose him; if he is not, the key expects Uma Chaudhuri (and an IHBAS item without Sengupta keys Girindrasekhar Bose, the leading Calcutta pioneer, for the same "first laboratory"; both keys are wrong on the facts); and for the TAT adaptation choose Uma if she is offered and Kamala only if she is not. "Father of *Indian psychology*", in the sense of the indigenous, consciousness-centred tradition, is a different question, taken up in Section 2.8.

## Reading old history with care

One last lesson concerns the reader rather than the past. It is tempting to judge old ideas by what we know now: to laugh at phrenology, to dismiss introspection as naive, to treat Wundt as a dull cataloguer of sensations. In 1965 the historian George Stocking made the standard name for this habit familiar to historians of the behavioural sciences: **presentism**, interpreting and judging the past by the standards and concerns of the present. Its opposite, *historicism*, tries to understand past thinkers on their own terms: what questions they were asking, what evidence they had, what their contemporaries believed. Presentism is not always wrong (we do want to know which old ideas turned out to be true), but a purely presentist history becomes a story of heroes and fools, and it gets the facts wrong. Phrenology, judged historically, was a serious theory that got localisation right and method wrong; introspection was abandoned because it could not settle disputes, not because its users were foolish.

The fate of Wundt's reputation is the classic case. For much of the twentieth century American students met Wundt mainly through Titchener, who translated him, presented his own structuralism as the faithful continuation of Wundt's programme, and was far more of an elementist than Wundt ever was. Few Americans read the ten volumes of the *Volkerpsychologie*, and the most influential history of the period was written by Boring, who had taken his doctorate under Titchener at Cornell. The result was a Wundt who was narrow, mechanical and introspective in Titchener's style. Only in the 1970s did historians who went back to Wundt's own German texts, among them Arthur Blumenthal in "A Reappraisal of Wilhelm Wundt" (1975), recover the Wundt you met in Section 1.5: a voluntarist who stressed attention, will and creative synthesis, who limited introspection to short, controlled judgements, and who wrote a whole second psychology of culture. Translation was part of the problem, which is why the German terms in Table 1.10 are worth knowing exactly. Examiners like to put them side by side as options.

**Table 1.10** German terms from the early history of psychology.

| Term | Meaning | Associated with |
|---|---|---|
| Zeitgeist | Spirit of the times | Boring's view of history |
| Wissenschaft | Systematic knowledge; science in the broad sense | German universities |
| Lehrfreiheit, Lernfreiheit | Freedom to teach; freedom to learn | German universities |
| Grundzuge der physiologischen Psychologie | Principles of Physiological Psychology | Wundt (1873 to 1874) |
| Selbstbeobachtung | Self-observation, uncontrolled (the armchair kind) | Rejected by Wundt |
| innere Wahrnehmung | Internal perception under experimental control | Wundt's method |
| Volkerpsychologie | Folk (cultural) psychology | Wundt (1900 to 1920) |
| ganz Amerikanisch | Typically American | Wundt on Cattell's interest in individual differences |
| Bewusstseinslagen | Conscious attitudes | Mayer and Orth, Wurzburg (1901) |
| Aufgabe | Task or instruction | Watt, Wurzburg |
| Einstellung | Set | Ach, Wurzburg (1905) |
| Sturm und Drang | Storm and stress | Hall on adolescence (1904) |
| Gestalt | Form; organised whole | Wertheimer, Kohler, Koffka (Chapter 2) |

REMEMBER

Three questions will take you through almost any history item. **What is the qualifier?** "Experimental", "American", "formal", "first woman", "in India" usually decides between two strong names. **Founder or precursor?** The founder built the institution (laboratory, school, clinic, society); the precursor had the idea or the method. **Discovered, named or popularised?** Gall founded phrenology but Spurzheim spread it under that name; Titchener named functionalism but James inspired it; Weber found the law but Fechner named it.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. The view that history is made by exceptional individuals, stated by Thomas Carlyle in 1841, is the great-_____ theory.
2. The German word for "the spirit of the times" is _____.
3. The historian of experimental psychology who championed the Zeitgeist view was Edwin G. _____.
4. When two people independently make the same discovery, as Darwin and Wallace did in 1858, historians speak of a _____ discovery.
5. Weber, Fechner and Helmholtz contributed to psychology without setting out to found it, so historians call them _____ rather than founders.
6. The usual answer to "father of intelligence testing" is Alfred _____.
7. The first laboratory of experimental psychology in India was set up at Calcutta in 1915 by N. N. _____.
8. Girindrasekhar Bose founded the Indian Psychoanalytic Society in the year _____.
9. The department of psychology at the University of Mysore was started in 1924 by M. V. _____.
10. Judging past thinkers by the standards of the present is called _____.

**Answers:** (1) person (or man), (2) Zeitgeist, (3) Boring, (4) multiple, (5) precursors (or pioneers), (6) Binet, (7) Sengupta, (8) 1922, (9) Gopalaswamy, (10) presentism.

## The case of Max Friedrich's key, revisited

Go back to the winter of 1879 and look again at the doctoral student pressing his telegraph key. You can now name the ancestry of almost everything in that room. The clockwork chronoscope belonged to the physicists, and the very idea that observers differ in how long they take to respond had been forced on the astronomers by the personal equation. The logic of the experiment, comparing a simple reaction with one that requires an extra mental act and subtracting the times, was Donders' subtraction method of 1868. The mental act being timed, apperception, had been named by Leibniz and given its psychological meaning by Herbart. The belief that a mental process could be measured at all was Fechner's answer to Kant, and the assumption that each mental event has a bodily partner running alongside it was the psychophysical parallelism that Wundt inherited from Leibniz. Even the plan of asking a trained observer to report under controlled, repeatable conditions was laid down in Wundt's four rules of experimental introspection.

So what was new? Not the parts but the purpose. For the first time, a student was doing original research on the mind, in a room set aside for it, under a professor who had announced a new science and would soon give it a journal, a textbook tradition and a stream of students. That is the whole answer to the puzzle the chapter began with, and it needs both theories of history from Section 1.8: the Zeitgeist had put every part on the bench, and one determined person assembled them into an institution. Friedrich's own measurements are long forgotten, and the hope of timing apperception by simple subtraction did not survive the problem of pure insertion described in Section 1.4. But the room outlived its first experiment. Within

a generation its graduates had carried it to Wurzburg, Cornell, Columbia, Pennsylvania and Harvard. In Philadelphia in 1896 one of them, Lightner Witmer, brought the same patient, measuring attitude to a boy who could not spell, and in 1915 a student of another, Munsterberg, carried the laboratory to Calcutta. The clinical psychology you are preparing to study is a direct descendant of that winter in Leipzig.

## Chapter summary

**1.1 What Is Psychology?** The word comes from the Greek *psyche* (soul) and *logos* (study); Christian Wolff made it standard in the 1730s, and Aristotle's *De Anima* was the first treatise on the soul. The definition has changed with each school: the soul, then the mind, then conscious experience (Wundt and Titchener; James's "science of mental life", 1890), then behaviour (McDougall, 1905 and 1908; Watson, 1913; Woodworth; the definition Indian textbooks credit to N. L. Munn). Today psychology is the scientific study of behaviour, which is overt, and mental processes, which are internal and covert. Its four goals are to describe, explain, predict and control (or influence). Basic research seeks understanding for its own sake; applied research solves practical problems. Clinical psychology assesses and treats psychological disorders, while counselling psychology serves people in the normal range with adjustment problems. Comparative psychology studies animal behaviour across species, ethology studies it in natural settings, evolutionary psychology explains behaviour as the product of natural selection, behaviour genetics asks how much variation is due to genes, and differential psychology studies individual differences.

**1.2 Philosophical Roots: A Long Past.** Nativism holds that some knowledge is inborn and empiricism that all knowledge comes from experience; rationalism makes reason the chief route to knowledge. Plato was a nativist and rationalist who saw learning as recollection; Aristotle was an empiricist who gave the first laws of association (similarity, contrast, contiguity). Descartes treated the body as a machine, described the reflex, held innate ideas and proposed dualism with interaction through the pineal gland. The British empiricists followed: Hobbes the materialist; Locke with the tabula rasa, sensation and reflection, simple and complex ideas, and primary and secondary qualities; Berkeley with *esse est percipi* and a learned theory of depth; Hume with impressions, ideas and the laws of resemblance, contiguity and cause and effect. The associationists Hartley, James Mill (mental mechanics), John Stuart Mill (mental chemistry) and Bain built the mind out of linked elements. On the continent, Leibniz proposed monads, petites perceptions and apperception, Herbart the threshold of consciousness and the apperceptive mass, and Kant innate categories and the claim that psychology could never be a science. The mind-body positions are interactionism, psychophysical parallelism, epiphenomenalism (Huxley), materialist and idealist monism, and Spinoza's double-aspect theory. Ontology concerns existence and epistemology knowledge; Comte's positivism rejected introspection.

**1.3 Physiological and Biological Roots.** The Bell-Magendie law (1811, 1822) separated sensory dorsal roots from motor ventral roots. Johannes Muller's doctrine of specific nerve energies held that the quality of a sensation depends on the nerve stimulated. Helmholtz measured the speed of the nerve impulse in 1850 (about 27 metres per second) and proposed unconscious inference and the trichromatic theory. Localisation of function came through Gall and Spurzheim's phrenology (right idea, wrong method), Flourens' ablation studies, Broca's patient Tan (1861), Wernicke's sensory aphasia (1874) and Fritsch and Hitzig's electrical mapping of the motor cor-

tex (1870). Darwin's natural selection (1859) and his book on emotional expression (1872) gave psychology continuity between species, the question of function and an interest in variation. His half-cousin Galton founded the study of individual differences with *Hereditary Genius* (1869), twin studies, the anthropometric laboratory (1884), regression, correlation, word association and the questionnaire, and coined eugenics (1883). Romanes' anecdotal method invited anthropomorphism, which Lloyd Morgan's canon of parsimony (1894) corrected, as Pfungst's study of Clever Hans (1907) showed.

**1.4 Measuring the Mind: Psychophysics and Mental Chronometry.** Weber found that the just noticeable difference is a constant proportion of the standard stimulus (Weber's law). Fechner, after his insight of 22 October 1850, proposed that sensation grows with the logarithm of stimulus intensity ($S = k \log I$), published *Elements of Psychophysics* in 1860, gave the three classical psychophysical methods and is the founder of psychophysics; Boring argued that 1860 could even be taken as psychology's founding date. The astronomers' personal equation revealed individual differences in reaction time. Donders' subtraction method (1868) compared simple, go/no-go and choice reactions to time the stages of identification and choice, founding mental chronometry, though it assumes "pure insertion" of stages. Ebbinghaus extended measurement to memory with nonsense syllables, the savings method and the forgetting curve (*On Memory*, 1885).

**1.5 Wilhelm Wundt and the Founding of Experimental Psychology.** Wundt (1832 to 1920), trained in medicine and physiology and once Helmholtz's assistant, published *Principles of Physiological Psychology* (1873 to 1874), moved to Leipzig in 1875, began laboratory research in 1879 and founded *Philosophische Studien* in 1881. Psychology for Wundt studied immediate experience, analysed into sensations and simple feelings. His system was voluntarism: the mind actively organises experience through attention and will, apperception brings contents into the focus of consciousness, and creative synthesis produces wholes with new properties. His tridimensional theory of feeling (1896) described feelings as pleasure-displeasure, tension-relaxation and excitement-depression. His method was controlled internal perception (*innere Wahrnehmung*), not armchair self-observation (*Selbstbeobachtung*), governed by four rules. Higher mental processes he studied through language, myth and custom in the ten-volume *Volkerpsychologie* (1900 to 1920). His students included Hall, Cattell, Munsterberg, Kulpe, Titchener, Witmer, Kraepelin, Spearman and, briefly, Bekhterev.

**1.6 Structuralism and Its European Rivals.** Titchener carried a stricter version of Wundt's programme to Cornell and named it structuralism (1898). He sought the elements of consciousness (sensations, images and affections) and their attributes (quality, intensity, duration, clearness, extensity), kept only the pleasant-unpleasant dimension of feeling, and trained observers to avoid the stimulus error. Meaning came from context, and perception was a complex of sensory and imaginal processes. Kulpe's Wurzburg school found imageless thought and showed how the Aufgabe, the determining tendency and set (Einstellung) direct thinking; the resulting controversy exposed the unreliability of introspection. Brentano's act psychology (1874) studied mental acts and their intentionality, Stumpf studied tone and trained the Gestalt founders, and G. E. Muller extended psychophysics and memory research at Gottingen. Structuralism declined because introspection could not settle disputes, observation altered experience, elementism missed wholes, its subject matter was narrow and it refused application; cognitive psychology later revived its topics.

**1.7 Functionalism and the American Pioneers.** Functionalism studied the mind as it functions in adapting the organism to its environment, drawing on Darwin, pragmatism and American

practicality. James's *Principles of Psychology* (1890) described the stream of consciousness, habit, the material, social and spiritual selves and the pure ego, and the James-Lange theory of emotion. Dewey's reflex arc paper (1896) founded the Chicago school, Angell's "Province of Functional Psychology" (1906) defined it and Carr's adaptive act matured it. At Columbia, Cattell coined "mental test" (1890), Woodworth gave the S-O-R formula and dynamic psychology, and Thorndike began the study of animal learning. Hall took the first American doctorate (1878), opened the first American research laboratory (1883), founded the *American Journal of Psychology* (1887) and the APA (1892), and described adolescence as storm and stress. Witmer opened the first psychological clinic (1896) and named clinical psychology; Munsterberg founded forensic and industrial psychology. Calkins became the first woman APA president (1905), Washburn the first woman PhD in psychology (1894), and Hollingworth disproved claims about women's inferiority. Functionalism dissolved into the mainstream and led to behaviourism, applied psychology and evolutionary psychology.

**1.8 Reading the History: Zeitgeist, Great Persons and "Fathers".** The great-person (personalistic) theory, stated by Carlyle in 1841, explains history through exceptional individuals; the Zeitgeist (naturalistic) theory, championed by Boring, explains it through the spirit of the times, citing multiple discoveries (Bell and Magendie, Darwin and Wallace, James and Lange), premature discoveries such as Mendel's, and contextual forces such as economic opportunity, war and prejudice. Psychology began in Germany because of its research universities, its broad idea of *Wissenschaft* and its experimental physiology; it began with Wundt because he, unlike the precursors, declared the science, gave it a laboratory, a journal and students; and 1879 marks the start of original student research. "Father" can mean founder, namer, chief influence or first in a country, so the qualifier in the question decides. In India, Seal drew up a syllabus in 1905, Sengupta set up the first laboratory (1915) and the first department (1916) at Calcutta and is the founder of modern psychology in India, Bose earned the first Indian doctorate (1921) and founded the Indian Psychoanalytic Society (1922), Mysore's department began in 1924 under Gopalaswamy, and the Indian Psychological Association followed in 1925. Presentism judges the past by present standards and distorted Wundt's reputation until historians returned to his own texts.

## Who's who

**Table 1.11** The people in this chapter.

| Name | When | Examined for |
|---|---|---|
| Max Friedrich | 1879 | First Leipzig dissertation: a reaction-time study of the duration of apperception |
| Christian Wolff | 1679 to 1754 | Made "psychology" a standard term: *Psychologia empirica* (1732), *Psychologia rationalis* (1734) |
| Aristotle | 384 to 322 BCE | *De Anima*; empiricist; laws of similarity, contrast and contiguity; "father of psychology" in the ancient sense |
| William McDougall | 1871 to 1938 | Psychology as the science of conduct or behaviour (1905, 1908); early textbook of social psychology (1908) |
| John Broadus Watson | 1878 to 1958 | Founder of behaviourism (1913); Angell's student at Chicago |
| Robert Sessions Woodworth | 1869 to 1962 | S-O-R formula; *Dynamic Psychology* (1918); drive; Columbia |

| Name | When | Examined for |
|---|---|---|
| Norman L. Munn | Twentieth century | "The scientific study of behaviour", the definition credited to him in Indian textbooks |
| William Stern | 1871 to 1938 | Named differential psychology (1900); coined the intelligence quotient (1912) |
| Konrad Lorenz and Nikolaas Tinbergen | Twentieth century | Ethology: animal behaviour in natural settings |
| Plato | About 427 to 347 BCE | Nativist and rationalist; knowledge as recollection (*Meno*); three-part soul |
| Sigmund Freud | 1856 to 1939 | Founder of psychoanalysis; free association; Clark lectures (1909) |
| Rene Descartes | 1596 to 1650 | Dualism; interaction through the pineal gland; innate ideas; the reflex |
| Thomas Hobbes | 1588 to 1679 | Materialism; forerunner of British empiricism; *Leviathan* (1651) |
| John Locke | 1632 to 1704 | Tabula rasa; sensation and reflection; simple and complex ideas; primary and secondary qualities (1690) |
| George Berkeley | 1685 to 1753 | *Esse est percipi*; learned depth perception (1709) |
| David Hume | 1711 to 1776 | Impressions and ideas; resemblance, contiguity, cause and effect |
| David Hartley | 1705 to 1757 | Founder of associationism; vibrations and vibratiuncles (1749) |
| James Mill | 1773 to 1836 | Mental mechanics (1829) |
| John Stuart Mill | 1806 to 1873 | Mental chemistry (1843) |
| Alexander Bain | 1818 to 1903 | Physiological associationism; trial-and-error learning; founded *Mind* (1876) |
| Edward Lee Thorndike | 1874 to 1949 | Puzzle boxes (1898); law of effect; father of educational psychology |
| Gottfried Wilhelm Leibniz | 1646 to 1716 | Monads; petites perceptions; coined apperception; pre-established harmony |
| Johann Friedrich Herbart | 1776 to 1841 | Threshold of consciousness; apperceptive mass; scientific pedagogy |
| Immanuel Kant | 1724 to 1804 | Innate forms and categories; held that psychology could not be a science (1781) |
| Thomas Henry Huxley | 1825 to 1895 | Epiphenomenalism (consciousness as the engine's whistle) |
| Baruch Spinoza | 1632 to 1677 | Double-aspect theory |
| Nicolas Malebranche | 1638 to 1715 | Occasionalism: God links mind and body on each occasion |
| Auguste Comte | 1798 to 1857 | Positivism; rejected introspection; founder of sociology |
| Charles Bell | 1774 to 1842 | Bell-Magendie law (1811) |
| Francois Magendie | 1783 to 1855 | Bell-Magendie law (1822) |
| Johannes Muller | 1801 to 1858 | Doctrine of specific nerve energies; *Handbook of Human Physiology* |
| Hermann von Helmholtz | 1821 to 1894 | Speed of the nerve impulse (1850); unconscious inference; trichromatic theory |
| Thomas Young | 1773 to 1829 | Suggested three colour receptors (1802): the Young-Helmholtz theory |
| Franz Joseph Gall | 1758 to 1828 | Founder of phrenology |
| Johann Gaspar Spurzheim | 1776 to 1832 | Popularised phrenology under that name |
| Pierre Flourens | 1794 to 1867 | Ablation; cerebellum and medulla; cortex acts as a whole |

| Name | When | Examined for |
|---|---|---|
| Paul Broca | 1824 to 1880 | Patient "Tan" (1861); left frontal area for speech production |
| Carl Wernicke | 1848 to 1905 | Left temporal area for understanding speech (1874) |
| Gustav Fritsch | 1838 to 1927 | Motor cortex by electrical stimulation, with Hitzig (1870) |
| Eduard Hitzig | 1838 to 1907 | Motor cortex by electrical stimulation, with Fritsch (1870) |
| Charles Darwin | 1809 to 1882 | Natural selection (1859); *The Expression of the Emotions* (1872); continuity, function, variation |
| Francis Galton | 1822 to 1911 | *Hereditary Genius* (1869); anthropometric laboratory (1884); correlation; word association; twin studies; eugenics |
| Karl Pearson | 1857 to 1936 | Formula for the correlation coefficient, developing Galton's idea |
| George John Romanes | 1848 to 1894 | *Animal Intelligence* (1882); anecdotal method; founder of comparative psychology |
| Conwy Lloyd Morgan | 1852 to 1936 | Morgan's canon of parsimony (1894) |
| Oskar Pfungst | 1907 | Exposed Clever Hans as a reader of unintended cues |
| Ernst Heinrich Weber | 1795 to 1878 | Just noticeable difference; Weber's law; two-point threshold |
| Gustav Theodor Fechner | 1801 to 1887 | Founder of psychophysics; insight of 22 October 1850; *Elements of Psychophysics* (1860); S = k log I |
| Edwin Garrigues Boring | 1886 to 1968 | *A History of Experimental Psychology* (1929); champion of the Zeitgeist view |
| Nevil Maskelyne and Friedrich Wilhelm Bessel | 1796; 1820s | The personal equation in astronomers' timings |
| Franciscus Cornelis Donders | 1818 to 1889 | Subtraction method (1868); mental chronometry |
| Hermann Ebbinghaus | 1850 to 1909 | Nonsense syllables; savings method; forgetting curve (1885); "a long past, but only a short history" |
| Wilhelm Maximilian Wundt | 1832 to 1920 | Leipzig laboratory (1879); voluntarism; tridimensional theory of feeling; *Volkerpsychologie*; father of experimental psychology |
| Emil Kraepelin | 1856 to 1926 | Worked in Wundt's laboratory; classification of psychiatric disorders |
| Charles Spearman | 1863 to 1945 | Factor analysis; the general factor g |
| Vladimir Mikhailovich Bekhterev | 1857 to 1927 | Reflexology; objective psychology |
| Edward Bradford Titchener | 1867 to 1927 | Structuralism (1898) at Cornell; stimulus error; context theory of meaning |
| Oswald Kulpe | 1862 to 1915 | Wurzburg school; imageless thought; systematic experimental introspection |
| August Mayer and Johannes Orth | 1901 | Wurzburg: conscious attitudes (Bewusstseinslagen) without images |
| Karl Marbe | 1869 to 1953 | Wurzburg: judgements of weight with no reportable content (1901) |
| Henry J. Watt | 1879 to 1925 | Wurzburg: the Aufgabe (task) determines associations |
| Narziss Ach | 1871 to 1946 | Wurzburg: determining tendency; set (1905) |
| Karl Buhler | 1879 to 1963 | Wurzburg: thoughts as elements of experience |
| Max Wertheimer | 1880 to 1943 | Kulpe's doctoral student; founder of Gestalt psychology (1912) |
| Franz Brentano | 1838 to 1917 | Act psychology; intentionality (1874) |

| Name | When | Examined for |
|---|---|---|
| Carl Stumpf | 1848 to 1936 | *Psychology of Tone*; Berlin institute; Clever Hans commission; teacher of the Gestalt founders |
| Georg Elias Muller | 1850 to 1934 | Gottingen laboratory; memory drum; consolidation and retroactive interference (1900) |
| William James | 1842 to 1910 | Father of American psychology; *Principles of Psychology* (1890); stream of consciousness; habit; the self |
| Carl Lange | 1834 to 1900 | Theory of emotion (1885), independently of James: the James-Lange theory |
| Charles Sanders Peirce | 1839 to 1914 | First stated pragmatism |
| John Dewey | 1859 to 1952 | "The Reflex Arc Concept in Psychology" (1896); Chicago functionalism |
| James Rowland Angell | 1869 to 1949 | "The Province of Functional Psychology" (1906) |
| Harvey A. Carr | 1873 to 1954 | The adaptive act; mature Chicago functionalism (1925) |
| James McKeen Cattell | 1860 to 1944 | Coined "mental test" (1890); individual differences; Columbia |
| Clark Wissler | 1870 to 1947 | Showed Cattell's mental tests did not predict college grades (1901) |
| Granville Stanley Hall | 1844 to 1924 | First American PhD (1878); first American research laboratory (1883); first APA president (1892); *Adolescence* (1904) |
| Lightner Witmer | 1867 to 1956 | First psychological clinic (1896); named clinical psychology; father of clinical psychology |
| Hugo Munsterberg | 1863 to 1916 | Forensic (1908) and industrial (1913) psychology; Sengupta's teacher |
| Mary Whiton Calkins | 1863 to 1930 | First woman APA president (1905); paired associates; self psychology |
| Margaret Floy Washburn | 1871 to 1939 | First woman PhD in psychology (1894); *The Animal Mind* (1908) |
| Leta Stetter Hollingworth | 1886 to 1939 | *Functional Periodicity* (1914); against the variability hypothesis; gifted children |
| Thomas Carlyle | 1795 to 1881 | Great-man (great-person) theory of history (1841) |
| Alfred Russel Wallace | 1823 to 1913 | Natural selection, independently of Darwin (1858): a multiple discovery |
| William Ogburn and Dorothy Thomas | 1922 | Listed 148 multiple discoveries |
| Robert K. Merton | 1910 to 2003 | Multiple discoveries as the normal pattern of science |
| Alfred Binet | 1857 to 1911 | First practical intelligence test, with Theodore Simon (1905); father of intelligence testing |
| Brajendranath Seal | 1864 to 1938 | Syllabus in experimental psychology at Calcutta (1905) |
| Narendra Nath Sengupta | 1889 to 1944 | First Indian laboratory (1915) and department (1916) at Calcutta; Indian Psychological Association (1925); founder of modern psychology in India |
| Radhakamal Mukerjee | 1889 to 1968 | Co-author with Sengupta of *Introduction to Social Psychology* (1928) |
| Girindrasekhar Bose | 1887 to 1953 | First Indian doctorate in psychology (1921); Indian Psychoanalytic Society (1922); theory of opposite wishes |
| M. V. Gopalaswamy | 1924 | Department of psychology, University of Mysore; the Akashvani radio station (1935) |
| Uma Chowdhury (Chaudhuri) | 1960 | Indian modification of the Thematic Apperception Test |
| George W. Stocking | 1965 | Presentism versus historicism in the history of the behavioural sciences |

| Name | When | Examined for |
|---|---|---|
| Arthur Blumenthal | 1975 | "A Reappraisal of Wilhelm Wundt": Wundt the voluntarist |

## Key terms

psychology, behaviour, mental processes, basic research, applied research, clinical psychology, counselling psychology, comparative psychology, ethology, evolutionary psychology, behaviour genetics, differential psychology, nativism, empiricism, rationalism, laws of association, similarity, contrast, contiguity, reflex, dualism, pineal gland, innate ideas, materialism, tabula rasa, simple ideas, complex ideas, primary qualities, secondary qualities, associationism, mental mechanics, mental chemistry, monads, petites perceptions, apperception, threshold of consciousness, apperceptive mass, interactionism, psychophysical parallelism, epiphenomenalism, monism, double-aspect theory, ontology, epistemology, positivism, Bell-Magendie law, doctrine of specific nerve energies, unconscious inference, trichromatic theory, localisation of function, phrenology, ablation, natural selection, individual differences, twin studies, anthropometric laboratory, regression to the mean, correlation, word association, eugenics, anecdotal method, anthropomorphism, Morgan's canon, parsimony, just noticeable difference, Weber's law, Fechner's law, psychophysics, personal equation, subtraction method, mental chronometry, reaction time, nonsense syllables, savings method, forgetting curve, immediate experience, voluntarism, creative synthesis, tridimensional theory of feeling, introspection, Volkerpsychologie, structuralism, stimulus error, context theory of meaning, imageless thought, Aufgabe, determining tendency, set, act psychology, intentionality, functionalism, pragmatism, stream of consciousness, habit, self, reflex arc, mental test, S-O-R, recapitulation theory, psychological clinic, great-person theory, Zeitgeist, multiple discovery, presentism

## From the past papers

1. Who was the first to propose and establish Psychology as the scientific study of behaviour? (Panwar Systems & Theories bank)

   (a) R.S. Woodworth (b) J.B. William (c) Wilhelm Wundt (d) N.L. Munn

   **Answer: (d) N. L. Munn.** "Psychology is the scientific study of behaviour" is the definition Indian textbooks quote from Munn, and that is what the key rewards. Woodworth defined psychology as the study of the activities of the individual in relation to the environment, and Wundt as the science of immediate conscious experience; "J. B. William" is not a recognised name (perhaps a misprint for J. B. Watson). Historically McDougall (1905, 1908) and Watson (1913) made behaviour the subject first, so read such stems closely.

2. Which branch of psychology focuses on the study of animal behavior? (RML 2025)

   (a) Social psychology (b) Abnormal psychology (c) Comparative psychology (d) Differential psychology

   **Answer: (c) Comparative psychology.** It compares behaviour and learning across species, a branch that grew out of Darwin's idea of continuity and Romanes' *Animal Intelligence* (1882). Social psychology studies people in relation to one another, abnormal psychology studies disorders, and differential psychology studies individual differences among people.

3. What does clinical psychology primarily focus on? (IHBAS 2024)

   (a) Industrial applications (b) Assessment and treatment of mental illness (c) Behavioural changes in animals (d) Organizational management

   **Answer: (b) Assessment and treatment of mental illness.** Clinical psychology, named by Lightner Witmer, assesses, diagnoses and treats psychological disorders. Industrial applications and organisational management belong to industrial-organisational psychology, and behavioural change in animals to comparative psychology.

4. In the nature versus nurture debate, which early philosopher argued that the mind was a "tabula rasa" or blank slate at birth? (IHBAS 2024)

   (a) Plato (b) Rene Descartes (c) John Locke (d) Immanuel Kant

   **Answer: (c) John Locke.** In his *Essay Concerning Human Understanding* (1690) Locke described the mind at birth as "white paper", with every idea coming from sensation or reflection. Plato thought learning was recollection of inborn knowledge, Descartes believed in innate ideas, and Kant held that the mind imposes innate forms and categories on experience; all three are on the nativist side.

5. ____ is a source of knowledge as per Empiricists. (Panwar Systems & Theories bank)

   (a) Logical reasoning (b) Theories (c) Sensory experiences (d) Contemplations

   **Answer: (c) Sensory experiences.** Empiricism holds that all knowledge comes from experience delivered through the senses. Logical reasoning and contemplation are the rationalists' routes to knowledge (Plato, Descartes, Leibniz), and theories are products of knowledge rather than its source.

6. According to Wundt 'apperceptive mass' constitutes _____ . (IHBAS 2025)

   (a) Compound of elemental processes bound by association (b) Sensation and pure perception (c) Pure perception and preexisting ideas (d) Sensation plus unconscious inference

   **Answer: (c) Pure perception and preexisting ideas.** The apperceptive mass is the organised body of ideas already in the mind, into which a new perception is taken up and understood; the term is Herbart's, and Wundt reworked the idea of apperception. Option (a) is the associationist and structuralist picture of the mind, option (b) leaves out the existing ideas that are the whole point, and option (d) is Helmholtz's theory of perception.

7. Herman von Helmholtz (1821-94): (Panwar Systems & Theories bank)

   (a) He was a proponent of the just noticeable difference (b) He gave the trichromatic theory of colour vision (c) He proposed the concept of the unconscious (d) He gave the law of effect

   **Answer: (b) He gave the trichromatic theory of colour vision.** Building on Thomas Young's suggestion of 1802, Helmholtz developed the Young-Helmholtz theory of three receptor types. The just noticeable difference is Weber's and Fechner's, the law of effect is Thorndike's, and although Helmholtz proposed *unconscious inference* in perception, the concept of the unconscious mind belongs to Freud and, before him, to Leibniz and Herbart.

8. What do Hermann Ebbinghaus, Francis Galton and Sigmund Freud share in common? (RML 2020)

   (a) Each of them worked independently and were pioneers in the field of psychology (b) Each of them founded a formal psychology laboratory (c) Each of them developed novel observational methods to study how the mind works (d) (a) and (c)

   **Answer: (d) (a) and (c).** All three worked largely alone and were pioneers, and all three invented methods: Ebbinghaus nonsense syllables and the savings method, Galton the questionnaire, word association and correlation, Freud free association. The RML 2020 key printed (a), but since (c) is also true, (d) is the more defensible answer and is the one the Panwar bank keys. Option (b) is false because Freud never founded a laboratory.

9. Who established the first laboratory for experimental study of psychology? (Panwar Systems & Theories bank)

   (a) Galton (b) Weber (c) Wundt (d) Jung

   **Answer: (c) Wundt.** Wundt's Leipzig laboratory began research in 1879, the conventional founding date of psychology as an independent science. Galton's laboratory (London, 1884) was anthropometric, a centre for measuring individual differences; Weber was a physiologist who founded no psychology laboratory; Jung developed word association as a clinical tool in Zurich.

10. Who gave the concept of 'Folk psychology'? (RML 2024)

    (a) Wilhelm Maximilian Wundt (b) Sigmund Freud (c) Ivan Pavlov (d) Wilhelm Roentgen

    **Answer: (a) Wilhelm Maximilian Wundt.** His ten-volume *Volkerpsychologie* (1900 to 1920) studied the higher mental processes through language, myth and custom. Freud founded psychoanalysis and Pavlov discovered classical conditioning; Roentgen was the physicist who discovered X-rays, there only because he shares Wundt's first name.

11. Charles Spearman is to Factor Analysis as Wilhelm Wundt is to: (Real PYQ 2024)

    (a) Psychoanalysis (b) Introspection (c) Functionalism (d) Humanism

    **Answer: (b) Introspection.** The analogy pairs each pioneer with his signature method: Spearman (himself a Leipzig doctorate) invented factor analysis, and Wundt made controlled introspection the method of the first laboratory. Psychoanalysis is Freud's, functionalism is associated with James, and humanism with Maslow and Rogers.

12. Who among the following has defined perception as a complex of sensory and imaginal events? (IHBAS 2024)

    (a) Kimbel (b) Titchener (c) Bruner (d) Schafer

    **Answer: (b) Titchener.** For the structuralist Titchener, a perception is built from sensations supplied by the object plus images supplied by past experience, which is also the heart of his context theory of meaning. Bruner and Schafer belong to the "New Look" research of the 1940s, which showed that needs and values shape perception (Chapter 4), and Kimble is known for work on learning; none of them held the structuralist definition.

13. The current approach to psychology most closely aligned with the structuralists in its interest in topics like consciousness, mental imagery and thought process is: (Panwar Systems & Theories bank)

    (a) Neuroscience (b) Cognitive psychology (c) Behaviourism (d) Functionalism

    **Answer: (b) Cognitive psychology.** It studies the structuralists' topics, consciousness, imagery and thought, with reaction times, error patterns and brain imaging instead of trained introspection. Neuroscience studies the brain rather than the contents of experience, behaviourism banished consciousness altogether, and functionalism asked what the mind is for rather than what it is made of.

14. Functionalism, an early school of thought in psychology, was most closely associated with which psychologist? (IHBAS 2024)

    (a) Edward Titchener (b) John Watson (c) William James (d) G. Stanley Hall

    **Answer: (c) William James.** James's *Principles of Psychology* (1890) inspired the functionalist attitude, and Dewey and Angell turned it into a school at Chicago. Titchener founded the rival structuralism, Watson founded behaviourism, and Hall, though functionalist in spirit, is examined for institutions (the APA, the first American research laboratory) and for adolescence.

15. Functionalism is how people work, play, and adapt to their surroundings. This viewpoint was given by: (Amity 2023)

    (a) William James (b) William Wundt (c) Charles Darwin (d) John B. Watson

    **Answer: (a) William James.** The functionalist question of how the mind helps us adapt is James's. "William Wundt" is Wilhelm Wundt misnamed, the voluntarist of Leipzig; Darwin inspired functionalism but belonged to no psychological school; Watson founded behaviourism.

16. The aim of Functionalism was to study _____ as it functions in adapting the _____ to its environment. (Panwar Systems & Theories bank)

    (a) Organism, Mind (b) Mind, Organism (c) Thinking, Organism (d) Attention, Individual

    **Answer: (b) Mind, Organism.** The functionalists studied the mind as an instrument that adapts the whole living organism to its environment, the Darwinian question of what consciousness is for. Option (a) reverses the order, and options (c) and (d) narrow the subject to single processes, which is not how the school defined itself.

17. The term "stream of consciousness", referring to the flow of thoughts in the human mind, was coined by which psychologist? (IHBAS 2024)

    (a) Carl Jung (b) Wilhelm Wundt (c) William James (d) B.F. Skinner

    **Answer: (c) William James.** In the *Principles* (1890) James argued that consciousness "is nothing jointed; it flows", against the structuralists' picture of separate elements. Jung is known for analytical psychology and the collective unconscious, Wundt analysed consciousness into elements, and Skinner, a radical behaviourist, did not study consciousness at all.

18. The first person to use the term clinical psychology in print was (RML 2020)

    (a) Erik Erikson (b) Sigmund Freud (c) Lightner Witmer (d) Cattell

    **Answer: (c) Lightner Witmer.** Witmer opened the first psychological clinic in 1896, proposed the new field that year, and used the term in print in the first issue of his journal *The Psychological Clinic* (1907). Erikson is known for the psychosocial stages, Freud for psychoanalysis, and Cattell for coining "mental test".

19. American Psychological Association (APA) was founded in: (Panwar Systems & Theories bank)
(a) 1892 (b) 1893 (c) 1896 (d) 1897

**Answer: (a) 1892.** The APA was founded at Clark University in July 1892, with G. Stanley Hall as its first president. 1896 is the tempting distractor because it is the year of Witmer's clinic and Dewey's reflex arc paper.

20. Who among the following was the first female president of the American Psychological Association (APA)? (Panwar Systems & Theories bank)
(a) Margaret Floy Washburn (b) Leta Stetter Hollingworth (c) Karen Horney (d) Mary Calkins

**Answer: (d) Mary Calkins.** Mary Whiton Calkins became APA president in 1905. Washburn was the first woman to earn a PhD in psychology (1894) and the second woman president (1921); Hollingworth is known for the psychology of women and of gifted children; Karen Horney was a psychoanalyst who was never APA president.

21. Who is often referred to as the "Founder of Modern Psychology in India"? (Panwar Systems & Theories bank)
(a) Girindra Shekhar Bose (b) Narendra Nath Sen Gupta (c) Amitosh Mukherjee (d) M.V. Gopalswamy

**Answer: (b) Narendra Nath Sen Gupta.** Trained under Munsterberg at Harvard, Sengupta set up India's first laboratory of experimental psychology (1915) and its first department of psychology (1916) at Calcutta. Bose is the father of psychoanalysis in India (1922), Gopalaswamy began the Mysore department (1924), and "Amitosh Mukherjee" is not a founder of Indian psychology (perhaps a garbling of Sir Ashutosh Mookerjee, the vice-chancellor who backed the Calcutta department, an administrator rather than a psychologist).

22. Which of the following Indian psychologists played a key role in establishing the first psychology laboratory in India? (Panwar Systems & Theories bank)
(a) Anand C. Paranjpe (b) Uma Chaudhuri (c) Sudhir Kakar (d) G. Misra

**Answer: (b) Uma Chaudhuri, as keyed; historically, none of the four.** The first laboratory was N. N. Sengupta's, at Calcutta in 1915, and he is not among the options. Uma Chaudhuri of Calcutta is known for the Indian modification of the Thematic Apperception Test (1960), and she is the name the key expects; Paranjpe, Kakar and Misra are later leaders of indigenous Indian psychology (Section 2.8). If Sengupta is ever offered, choose him.

## High-yield sheet

*The facts this chapter's questions ask most, most-asked first. Last read before the exam.*

1 **Functionalism** is most closely associated with **William James**, its inspiration; Dewey (1896) and Angell (1906) made it a formal school at Chicago. ASKED 3 TIMES

2 **Wilhelm Wundt** opened the first psychology laboratory at **Leipzig in 1879** (not Vienna, Paris or Cambridge); he is the father of experimental psychology. ASKED 4 TIMES

3 **Introspection** is the method most associated with **Wundt**, and **structuralism** is the school built on it; Spearman is to factor analysis as Wundt is to introspection. ASKED 3 TIMES

4 **Structuralism**, named by **Titchener** (1898, Cornell), aimed at a "periodic table" of the elements of consciousness (sensations, images, affections); for Titchener introspection is a highly specialised form of **self-observation**. ASKED 3 TIMES

5 Functionalism studies the **mind** as it functions in adapting the **organism** to its environment; it was inspired by **Darwin's evolution** and gave rise to the evolutionary perspective. ASKED 3 TIMES

6 **Lightner Witmer** opened the first psychological clinic (University of Pennsylvania, 1896), named clinical psychology, and first used the term in print in 1907 (*The Psychological Clinic*). ASKED 2 TIMES

7 **William James** wrote *The Principles of Psychology* (1890) and coined "**stream of consciousness**"; Wundt's book is *Principles of Physiological Psychology* (1873 to 1874). ASKED 2 TIMES

8 For **empiricists** knowledge comes from **sensory experience**; **John Locke** called the mind at birth a **tabula rasa** ("white paper") in his *Essay Concerning Human Understanding* (1690). ASKED 2 TIMES

9 **Ebbinghaus, Galton and Freud** each worked independently as pioneers and each devised new methods; "each founded a formal laboratory" is false (Freud never did). Bank key (d) "(a) and (c)"; RML 2020 key (a). ASKED 2 TIMES

10 **Folk psychology** is Wundt's **Volkerpsychologie**, ten volumes (1900 to 1920) on language, myth and custom. ASKED ONCE

11 The **apperceptive mass** (Herbart's term, taken up by Wundt) is the store of pre-existing ideas that absorbs a new perception: keyed as "pure perception and pre-existing ideas". ASKED ONCE

12 **Titchener** defined perception as a complex of **sensory and imaginal** processes (his context theory of meaning). ASKED ONCE

13 **Comparative psychology** is the branch that studies animal behaviour across species (Romanes, 1882; Morgan's canon, 1894). ASKED ONCE

14 **Clinical psychology** focuses on the **assessment and treatment** of psychological disorders; industry and organisations belong to industrial-organisational psychology. ASKED ONCE

15 **John B. Watson** founded **behaviourism** (1913); its aim was, given the stimulus, to predict the response, and given the response, to predict the stimulus. ASKED 2 TIMES

16 **N. N. Sengupta**, trained under Munsterberg at Harvard, is the founder of modern psychology in India: first laboratory (1915) and first department (1916) at **Calcutta**. ASKED 2 TIMES

17 **Uma Chowdhury** (Chaudhuri) made the Indian modification of the TAT (1960); the bank keys "Kamala Chaudhuri" for the TAT and Uma for the first Indian laboratory (really Sengupta's). ASKED 2 TIMES

18 The **APA** was founded in July **1892** at Clark University, with **G. Stanley Hall** as first president. ASKED ONCE

19 **Mary Whiton Calkins** was the first woman APA president (1905); Washburn was the first woman PhD in psychology (1894). ASKED ONCE

20 **Gustav Fechner** founded psychophysics: *Elements of Psychophysics* (1860), insight of 22 October 1850, S = k log I; the JND is Weber's. ASKED ONCE

21 **Hermann von Helmholtz** (1821 to 1894) gave the **trichromatic** (Young-Helmholtz) theory of colour vision, timed the nerve impulse (1850) and proposed unconscious inference. ASKED ONCE

22 **John Dewey**'s "The Reflex Arc Concept in Psychology" (1896) took the **reflex arc**, the stimulus-to-response connection, as its target. ASKED ONCE

23 In "the scientific study of behaviour and mental processes", **mental processes** are internal, **covert** processes; behaviour is overt. ASKED ONCE

24 "Psychology is the scientific study of behaviour": the bank credits **N. L. Munn** (McDougall, 1905 and 1908, and Watson, 1913, came earlier). ASKED ONCE

25 **Evolutionary psychology** explains behaviour in relation to genetic inheritance and natural selection. ASKED ONCE

26 **Cognitive psychology** is the modern approach closest to the structuralists' topics: consciousness, imagery and thought. ASKED ONCE

27 **Ontology** is the branch of philosophy that deals with existence; epistemology deals with knowledge. ASKED ONCE

28 **Descartes**: dualism with interaction through the pineal gland, innate ideas and the reflex; a practice key says his dualism influenced behaviourism and Gestalt psychology. ASKED ONCE

29 Perceiving a **contour** requires a minimum stimulus intensity and a minimum exposure time (not a minimum reaction time). ASKED ONCE

CHAPTER 2

# Schools and Systems: Behaviourism, Gestalt, Psychoanalysis, Humanism, the Cognitive Revolution and Indian Psychology

IN THIS CHAPTER


2.1 ● Schools, Systems and Paradigms: Explain what makes a school of thought, how Kuhn's idea of a paradigm applies to psychology, and how the schools are numbered as "forces".

2.2 ●● Behaviourism: From Watson's Manifesto to Skinner: Trace behaviourism from its precursors through Watson, the neobehaviourists (Tolman, Hull, Guthrie) and Skinner, and explain why it declined and what it left behind.

2.3 ●●● Gestalt Psychology and Kurt Lewin: Describe how Wertheimer, Kohler and Koffka argued that wholes come first, and how Lewin carried field theory into motivation and social life.

2.4 ●● Psychoanalysis as a School: The First Force: Outline the history of psychoanalysis from Charcot and Breuer to Freud's two models of the mind and the neo-Freudians.

2.5 ●●● The Third Force and Beyond: Explain humanistic, existential, transpersonal and positive psychology, who founded each and how they differ.

2.6 ●● The Cognitive Revolution: Explain why the mind returned to psychology after 1956, the information-processing metaphor, cognitive neuroscience and connectionism.

2.7 ●● Psychology Today: Perspectives, Eclecticism and the Enduring Issues: Compare the eight contemporary perspectives and the great questions on which the schools have always divided.

2.8 ●●● Indian Psychology I: The Movement, Its Pioneers and Samkhya's Map of the Mind: Distinguish psychology in India from Indian psychology, name its pioneers, and explain purusha, prakriti, the three gunas and the antahkarana.

2.9 ●●● Indian Psychology II: Yoga, the Self and the Paths of Transformation: Explain Patanjali's yoga, the four states of consciousness, the five koshas, Buddhist and Sufi psychology, Sri Aurobindo, the doshas, karma and dharma.


**Weightage.** 103 of the 819 questions mapped in this book (12.6%) test this chapter, 39 of them from real past papers. 4 of its 9 sections are High yield.

In 1913 a twenty-six-year-old German psychologist named Wolfgang Kohler sailed to Tenerife, in the Canary Islands, to take charge of a small research station that the Prussian Academy of Sciences kept there for the study of chimpanzees. He expected a short posting. Then the First World War broke out, the sea lanes closed, and he stayed until 1920. For six years he had little to do but watch apes think.

His most gifted subject was a chimpanzee called Sultan. One day Kohler placed a banana outside the bars of Sultan's cage, beyond reach. Inside the cage lay two hollow bamboo sticks, one thin enough to slide into the other, but each too short to reach the fruit on its own. Sultan tried one stick, then the other. He pushed one stick out with the second, so that the first touched the banana, but he could not draw it in. After more than an hour Kohler gave up and left a keeper to watch. Sultan sat on a box, idly playing with the two sticks. By chance he found himself holding them in a line, and he pushed the thin one into the end of the thick one. According to Kohler's record, he jumped up at once, ran to the bars and raked in the banana with his new long stick. When the sticks came apart, he joined them again without hesitation.

What had happened inside Sultan? Kohler was sure he knew: the ape had suddenly seen the situation in a new way, the two sticks reorganised in his perception into one tool. But in the very year Kohler reached Tenerife, a young American at Johns Hopkins University, John B. Watson, was telling psychologists that "inside" questions of this kind had no place in a science at all; all that could be studied was what the animal did. In Vienna, Sigmund Freud would have asked a different question again: what hidden wish does the behaviour serve? Forty years later, a cognitive psychologist would describe Sultan as an information-processing system searching a "problem space", and today a researcher trained in the Indian tradition might ask about the state of attention in which the solution arrived.

Same ape, same sticks, five different stories. That is what this chapter is about. Between about 1900 and 1960 psychology was divided into rival *schools*, each with its own idea of what psychology should study, how it should study it, and what a human being is. You will meet each school in turn, see why it arose, what it discovered, and why it declined or was absorbed. Remember Sultan; you will meet him again at the end of the chapter.

## 2.1 Schools, Systems and Paradigms

LOWER YIELD · 3 QS

Chapter 1 ended with the first two schools of scientific psychology, structuralism and functionalism. This chapter picks up the story where they left it. Before we meet the newcomers, though, it helps to be clear about what a "school" actually is, because the word is used loosely, and the vocabulary of the field matters as much as its content.

### What makes a school of thought

A **school of thought** is a group of psychologists who share a common set of assumptions about what psychology should study (its subject matter), how it should study it (its method), and what kind of explanation counts. A school usually has a recognisable founder or leader, a

statement of its programme (often called its manifesto), a university or laboratory that serves as its home, a journal in which its members publish, a stream of students who carry its ideas into other institutions, and, just as important, a rival school against which it defines itself. Behaviourism, for example, had Watson as its founder, his 1913 paper as its manifesto, and structuralism's introspection as its enemy.

Two classic books gave the age of schools its shape. **Robert Sessions Woodworth** (1869 to 1962) of Columbia University surveyed the rivals in *Contemporary Schools of Psychology* (1931), and **Edna Heidbreder** (1890 to 1985) described *Seven Psychologies* (1933). Notice their dates. The schools flourished roughly between the 1890s and the 1950s; after that, as you will see in Section 2.7, psychology gradually traded rival schools for overlapping "perspectives".

Two related words are worth separating. A **system** of psychology is an organised and internally consistent body of assumptions, methods and theory that tries to cover the whole field; a *theory* is narrower, an explanation of one set of facts (a theory of forgetting, a theory of colour vision). Courses called "Systems and Theories" study both: the great systems, and the theories they produced.

### Kuhn's paradigms and a preparadigmatic science

Why did psychology split into warring schools when physics and chemistry did not? A helpful answer comes from a physicist turned historian, **Thomas Samuel Kuhn** (1922 to 1996). In *The Structure of Scientific Revolutions* (1962), Kuhn argued that a mature science works within a **paradigm**: a shared framework of assumptions, problems, methods and model solutions that all workers in the field accept. Under a paradigm, scientists do what Kuhn called **normal science**: puzzle-solving within the accepted framework. Over time, results accumulate that the paradigm cannot explain (anomalies). When they pile up, the field enters a crisis; a new framework appears; and if it wins, the result is a scientific revolution, or *paradigm shift*. Newton's physics giving way to Einstein's is Kuhn's model case.

A field that has no single accepted paradigm, but several competing schools, each with its own problems and methods, is in Kuhn's term a **preparadigmatic science**. Kuhn himself noticed this in the social sciences. In the preface to his book he recalled a year spent among social scientists and how struck he was by the number and extent of their open disagreements about what the legitimate problems and methods even were. The age of schools in psychology fits that description well: a structuralist and a behaviourist did not merely disagree about results, they disagreed about what counted as data. Whether psychology has since acquired a paradigm (or whether the cognitive revolution was a true Kuhnian revolution) is still debated, and we return to the question at the end of Section 2.7.

### The four forces ●●

You will often read that psychology has had three or four "forces". The numbering comes from the humanistic psychologists of the 1960s, above all Abraham Maslow, who called their own movement the **third force** to set it against the two forces that, in their view, dominated mid-century psychology. Table 2.1 sets out the sequence. Notice that the numbers describe a sequence of influence in the eyes of the humanists, not strict dates of birth: psychoanalysis (from the 1890s) is the first force and behaviourism (1913) the second.

Table 2.1 The forces in psychology.

| Force | School | Key figures | Picture of the person |
|---|---|---|---|
| First | Psychoanalysis | Freud and his successors | Driven by unconscious conflict and early childhood |
| Second | Behaviourism | Watson, Skinner | Shaped by the environment through conditioning |
| Third | Humanistic psychology | Maslow, Rogers | Free, growth-seeking, capable of self-actualisation |
| Fourth | Transpersonal psychology | Maslow, Sutich, Grof | Capable of experiences that go beyond the individual self |

Figure 2.1 lays the schools out in time, so that you can see which ones overlapped and which ones replaced each other. Keep it in mind as you read; each bar is the subject of one part of this chapter.

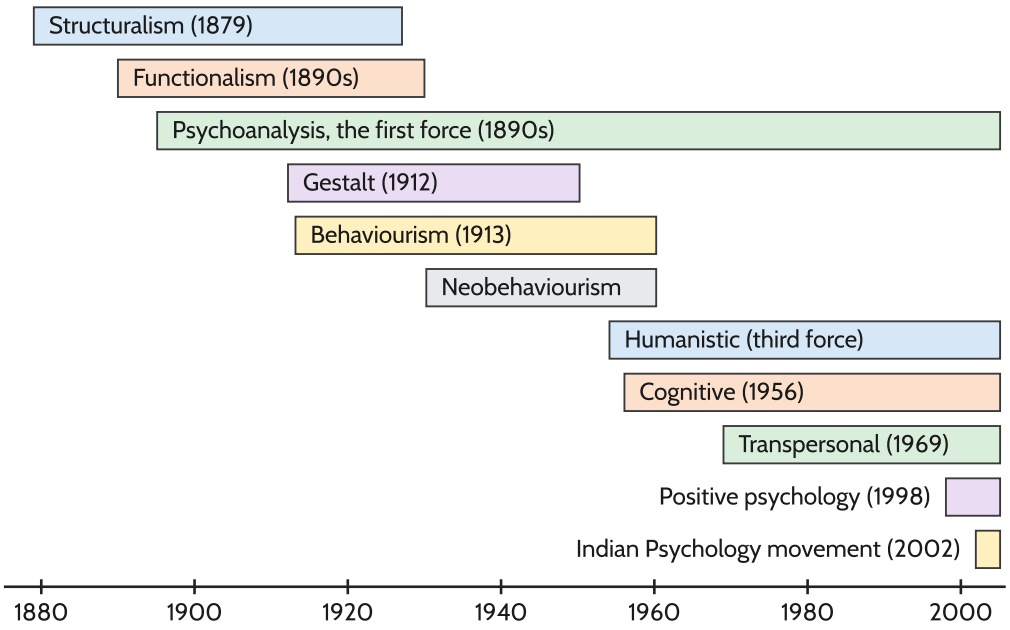


**Figure 2.1** The schools of psychology in time. Each bar runs from a school's founding event to the point where it stopped being a dominant, separate school (bars reaching the right edge are still active traditions). Notice how Gestalt and behaviourism were born a year apart, and how the cognitive and humanistic movements grew up side by side in the 1950s.

> **EXAM TRAP**
>
> "Which is the fourth force?" The answer is **transpersonal psychology**. In counselling textbooks, however, Paul Pedersen (1991) called *multiculturalism* the "fourth force" in counselling. If both appear in the options, read the stem: "in psychology" points to transpersonal; "in counselling" may point to multiculturalism. "Experiential" psychology is not one of the numbered forces.

## 2.2 Behaviourism: From Watson's Manifesto to Skinner

MEDIUM YIELD · 9 QS

In 1913, structuralism (Wundt and Titchener) and functionalism (James, Dewey and Angell) were still arguing about consciousness: what its elements were, or what it was for. Both relied

on introspection, the trained observation of one's own experience. A young professor at Johns Hopkins decided that the whole argument was a mistake. Psychology, he said, should stop studying consciousness altogether. To understand how anyone could say such a thing in 1913, and be applauded for it, you need to see the ground that had been prepared.

## The ground prepared: animal psychology and Russian objective psychology

The first preparation came from the study of animals. After **Charles Darwin** (1809 to 1882) argued for continuity between humans and other species, animal minds became a serious topic. **George John Romanes** (1848 to 1894) collected stories of clever animals (his anecdotal method), and **Conwy Lloyd Morgan** (1852 to 1936) answered with his canon of 1894: never interpret an action as the result of a higher mental faculty if it can be explained by a lower one (both are covered in Chapter 1). **Jacques Loeb** (1859 to 1924), a physiologist who taught Watson at Chicago, went further and explained animal movement as *tropisms*, forced, mechanical orientations toward or away from light, gravity and chemicals, with no mind required. And **Edward Lee Thorndike** (1874 to 1949), in his 1898 dissertation *Animal Intelligence*, put hungry cats in puzzle boxes and showed that they escaped not by reasoning but by gradual trial and error, stamped in by satisfying consequences: his law of effect (Chapter 6). If animal behaviour could be studied so precisely without asking what the animal was conscious of, why not human behaviour too?

The second preparation came from Russia, where a tradition of *objective psychology* had grown out of physiology. **Ivan Mikhailovich Sechenov** (1829 to 1905), in *Reflexes of the Brain* (1863), argued that all acts of conscious and unconscious life are reflexes in their origin. **Ivan Petrovich Pavlov** (1849 to 1936), who won the 1904 Nobel Prize for his work on digestion, discovered around the turn of the century that his dogs began to salivate to signals that preceded food; he called these conditioned reflexes and spent the rest of his life studying them (Chapter 6). **Vladimir Mikhailovich Bekhterev** (1857 to 1927) studied what he called association reflexes, conditioning motor responses such as withdrawing a limb from an electric shock, and named his whole approach **reflexology**; his *Objective Psychology* appeared between 1907 and 1912. Bekhterev's motor conditioning, more than Pavlov's salivation, gave Watson a method he could use on people.

The third preparation was home-grown. American functionalism had already pulled psychology toward practical questions (education, testing, industry) and toward the study of what organisms *do* to adapt. Behaviourism, in a sense, was functionalism with the mind removed.

## Watson's manifesto (1913) ●●

**John Broadus Watson** (1878 to 1958) grew up poor in South Carolina, earned his doctorate at the University of Chicago in 1903 with a study of learning in rats, and moved to Johns Hopkins in 1908. He had never been comfortable with introspection; he later admitted that he hated serving as a subject in introspective experiments and preferred animals, which could not be asked what they were experiencing. Early in 1913 he gave a lecture at Columbia University, published that March in *Psychological Review* as "Psychology as the Behaviorist Views It". It is the founding document of **behaviourism**, the school that defines psychology as the science of observable behaviour.

**KEY STUDY: "PSYCHOLOGY AS THE BEHAVIORIST VIEWS IT" (1913)**

Watson's paper opens with a sentence every student of the history of psychology should know almost by heart: psychology as the behaviourist views it "is a purely objective experimental branch of natural science. Its theoretical goal is the prediction and control of behavior. Introspection forms no essential part of its methods." He added that the behaviourist "recognizes no dividing line between man and brute".

The paper made four claims. (1) *Subject matter*: behaviour, not consciousness. (2) *Method*: objective observation, never introspection. (3) *Goal*: prediction and control, which made psychology useful to education, business and medicine. (4) *Continuity*: the same principles apply to humans and animals. In later books Watson phrased the goal precisely: given the stimulus, to predict the response, and given the response, to identify the stimulus that produced it. Watson was elected President of the American Psychological Association in 1915, only two years later; the manifesto had found an audience ready for it.

What methods did Watson allow? Four: observation, with or without instruments; testing (he liked tests as samples of behaviour, not as measures of mental capacities); the verbal report method, in which speech is treated as behaviour, so that what a person *says* counts as data but is not taken as a window on consciousness; and, from 1915, the conditioned reflex method borrowed from Pavlov and Bekhterev.

### Watson's programme: thinking, emotion and the "twelve infants" ●●

A science of behaviour has to say something about the things introspectionists cared about, such as thought and feeling. Watson had an answer for each, and his answers show you the spirit of the school.

**Thinking as subvocal speech.** Watson argued that thinking is nothing but talking to oneself so quietly that the movements cannot be seen: thought is **subvocal speech**, small movements of the tongue and larynx (he called them laryngeal habits). This view is called **peripheralism**, because it locates mental life in the peripheral muscles rather than in central brain processes. It led Watson to divide behaviour into *explicit* (overt) responses that anyone can see, and *implicit* (covert) responses such as subvocal speech, which cannot be seen by an observer but can, in principle, be measured with instruments. This is exactly the sense in which modern textbooks say covert behaviour is measurable but not directly observable, and why the "mental processes" in today's definition of psychology (the scientific study of behaviour and mental processes) are internal, covert processes.

Was Watson right about thought? A dramatic test came in 1947, when Scott Smith and his colleagues at the University of Utah gave Smith himself a dose of curare large enough to paralyse every skeletal muscle, keeping him alive on a respirator. If thinking were movement of the speech muscles, Smith should have been unable to think. He reported afterwards that he had remained fully aware, had understood what was said to him and could remember it. Peripheralism did not survive.

**Emotion.** Watson held that infants show just three unlearned emotional reactions: *fear* (to loud noises and sudden loss of support), *rage* (to hampering of movement) and *love* (to stroking and gentle handling). Every other emotion, he argued, is built from these three by conditioning.

**Environmentalism.** Watson's most quoted words appeared in his popular book *Behaviorism* (1925). "Give me a dozen healthy infants, well-formed, and my own specified world to bring them up in and I'll guarantee to take any one at random and train him to become any type of specialist I might select: doctor, lawyer, artist, merchant-chief and, yes, even beggar-man and thief, regardless of his talents, penchants, tendencies, abilities, vocations, and race of his ancestors." This is **environmentalism** in its purest form: the belief that behaviour is shaped almost entirely by experience rather than heredity. What is usually left out is the next sentence, in which Watson conceded that he was "going beyond my facts", adding that so had the advocates of the contrary view, and they had been doing it for thousands of years. Consistent with this, Watson first accepted, and then by the mid-1920s rejected, the idea of human instincts.

**Little Albert.** To show that emotions are conditioned, Watson and his graduate student **Rosalie Rayner** (1898 to 1935) conditioned fear in an eleven-month-old infant, "Albert B.", in 1920, pairing a white rat with a loud noise made by striking a steel bar behind his head. Albert came to fear the rat, and the fear generalised to a rabbit, a dog, a fur coat and a Santa Claus mask. The study, its method and its ethical problems are told in full in Chapter 6. The same year, Watson's affair with Rayner became public; he was forced to resign from Johns Hopkins, married Rayner, and spent the rest of his working life in advertising at the J. Walter Thompson agency. His academic career had lasted barely twenty years, but the school he named outlived him by decades.

EXAM TRAP

**Who defined psychology as the science of behaviour?** Watson (1913) made behaviour the subject matter of psychology, so for "founder of behaviourism" and "Watson defined psychology as the science of ..." the answers are behaviourism and behaviour (not soul, mind or consciousness: the soul was the subject matter of the ancient philosophers, the mind that of later philosophers, and consciousness that of Wundt and Titchener). One Panwar item, however, keys **N. L. Munn** as the first to establish psychology as "the scientific study of behaviour": Norman Leslie Munn's widely used textbooks popularised that wording. If Watson is an option, prefer Watson; if the options are Munn, Woodworth and Wundt, the key is Munn. Woodworth is remembered for the S-O-R formula and a quip often credited to him: psychology first lost its soul, then its mind, then consciousness.

**Rationalism** is a philosophical position (Descartes), not a school Watson founded. **Functionalism** belongs to James, Dewey and Angell; **structuralism** to Titchener.

## Three meanings of "behaviourism"

As the school matured, it became clear that "behaviourism" could mean three different things, and the three are easily confused. The difference lies in what each says about the mind.

Table 2.2 Three kinds of behaviourism.

| Kind | What it says about mental events | Typical figures |
|---|---|---|
| Metaphysical (eliminative) | Minds and conscious states do not exist as anything other than behaviour | Watson in his most extreme statements |

| Kind | What it says about mental events | Typical figures |
|---|---|---|
| Methodological | Mental events may exist, but they are private and cannot be studied scientifically; only public behaviour counts as data | Watson in practice; most neobehaviourists |
| Radical | Private events (thoughts, feelings) are real and are themselves behaviour, obeying the same laws; but they are not the causes of overt behaviour | Skinner, who adopted the term in 1945 (Mary Whiton Calkins had first used it, for Watson's view, in 1921) |

The word "radical" in radical behaviourism does not mean "extreme". Skinner meant it in its root sense: thoroughgoing, going to the roots. Ironically, radical behaviourism is the only one of the three that lets thoughts and feelings into the science; methodological behaviourism simply sets them aside.

## Neobehaviourism: logical positivism and operationism ●●●

Watson's crusading behaviourism dominated American psychology from about 1913 to 1930. Historians of psychology usually describe what followed as a second stage, **neobehaviourism** (about 1930 to 1960), and a third, sociobehaviourism or social learning theory (about 1960 onward). The neobehaviourists, Tolman, Hull, Guthrie and Skinner, kept Watson's demand for objectivity but wanted to make behaviourism more rigorous and more theoretical. They drew on two ideas from the philosophy of science.

The first was **logical positivism**, the philosophy of the Vienna Circle of the 1920s, which held that a statement is meaningful only if it can be verified by observation (or is true by definition). Talk of unobservable "minds" was therefore not false but meaningless unless it could be tied to observables. The second was **operationism**, proposed by the American physicist **Percy Williams Bridgman** (1882 to 1961) in *The Logic of Modern Physics* (1927): a scientific concept means nothing more than the set of operations used to measure it. The psychologist S. S. Stevens brought the idea into psychology in 1935. The legacy is the **operational definition** that every research methods course still teaches: "hunger" in a rat experiment is defined as "twenty-three hours without food", and "intelligence" as "the score on this test".

**Edward Chace Tolman's purposive behaviourism. Edward Chace Tolman** (1886 to 1959) of the University of California at Berkeley was the behaviourist who let cognition in by the back door. He called his system **purposive behaviourism** and set it out in *Purposive Behavior in Animals and Men* (1932), a book he dedicated, with typical humour, to the white rat. Tolman insisted on studying **molar behaviour**, whole goal-directed acts such as a rat running a maze or a person going home, rather than Watson's **molecular behaviour**, the individual muscle twitches and glandular secretions that make up an act. Molar behaviour, Tolman observed, "reeks of purpose": a rat that finds its usual route blocked takes another, so what it has learned is not a chain of movements but where the food is.

To explain this without returning to introspection, Tolman introduced the **intervening variable** in the mid-1930s: an unobserved process inferred between the observable stimulus and the observable response and defined operationally by the conditions that produce it. Hunger, expectancy and demand are intervening variables. On one side sit the independent variables (environmental stimuli, hours of deprivation, heredity, previous training, age); on the other, behaviour; in between, the processes that link them. In effect, Watson's S-R for-

mula became S-O-R, the formula Woodworth had used to insert the organism between stimulus and response.

Tolman's most famous intervening variable was the **cognitive map**, a mental representation of the layout of the environment, which he described in "Cognitive Maps in Rats and Men" (1948). The best evidence for it was **latent learning**, learning that occurs without reinforcement and stays hidden until there is a reason to show it. In the classic experiment of Tolman and Charles Honzik (1930), rats that explored a maze for ten days without food made many errors; once food appeared in the goal box on day 11, their errors fell by the very next day to the level of rats rewarded from the beginning. They had been learning the maze all along (the full study is in Chapter 6). For Tolman, rats learn *what leads to what* (expectancies), not stimulus-response habits.

**Clark Leonard Hull's hypothetico-deductive system.** If Tolman was the playful neobehaviourist, **Clark Leonard Hull** (1884 to 1952) of Yale University was the system-builder. Hull admired Newton's *Principia* and wanted a psychology built the same way, by the **hypothetico-deductive method**: state a small set of postulates, deduce precise testable predictions (theorems) from them, test the predictions experimentally, and revise the postulates when they fail. His *Principles of Behavior* (1943) made him for two decades the most cited psychologist in the world.

At the centre of Hull's theory is **drive reduction**. A biological need (for food, water, relief from pain) produces a drive (D), an aroused state that energises behaviour; any response that reduces the drive is reinforced. Reinforced repetitions build **habit strength**, written ${}_{S}H_{R}$, the strength of the learned bond between a stimulus and a response, which grows with the number of reinforced trials. Habit alone does not produce behaviour; it must be energised by drive. Hull's 1943 formula for **reaction potential**, the tendency to make the response, was therefore multiplicative:

$${}_{S}E_{R} = {}_{S}H_{R} \times D$$

The multiplication matters: if either habit or drive is zero, there is no behaviour, however strong the other. A well-trained but completely satiated rat does not run the maze. Later versions added incentive motivation (K, the size of the reward) and stimulus intensity (V), and subtracted inhibition (fatigue-like reactive inhibition and conditioned inhibition). Hull's students, especially Kenneth Spence, **Neal Elgar Miller** and O. Hobart Mowrer, carried his ideas into many fields. One famous offshoot was **John Dollard** and Neal Miller's *Personality and Psychotherapy* (1950), which translated Freud's ideas into Hull's vocabulary of drive, cue, response and reinforcement; it is usually called the early social learning theory, as distinct from the later social learning theory of Albert Bandura and Richard Walters (1963).

**Edwin Ray Guthrie's contiguity theory. Edwin Ray Guthrie** (1886 to 1959) of the University of Washington offered the simplest theory of all, in *The Psychology of Learning* (1935). His one law was **contiguity**: a combination of stimuli that has accompanied a movement will, on its recurrence, tend to be followed by that movement. Reinforcement is not needed; what matters is only that stimulus and movement occur together. Learning is **one-trial learning**: a stimulus-movement association reaches full strength on its first pairing. Why, then, does practice help? Because a skill is made of many movements, each attached to many stimuli, and practice attaches the act to more and more of the situations in which it must occur.

Guthrie also gave practical advice on breaking habits (the threshold, fatigue and incompatible-response methods, Chapter 6).

**Burrhus Frederic Skinner's radical behaviourism. Burrhus Frederic Skinner** (1904 to 1990) of Harvard became the most famous psychologist of the twentieth century; a 2002 survey ranking the most eminent psychologists placed him first, ahead of Piaget and Freud. Skinner distinguished respondent behaviour, elicited by a preceding stimulus as in Pavlov's experiments, from **operant** behaviour, emitted by the organism and controlled by its consequences. His tools (the operant chamber or "Skinner box", the cumulative recorder) and findings (reinforcement schedules, shaping) are covered in Chapter 6. Here what matters is his system.

Skinner was a radical behaviourist (Table 2.2) and also an anti-theorist. In a 1950 paper he asked whether theories of learning were necessary at all and answered no: psychology should describe functional relations between environmental events and behaviour, without inventing intervening variables, drives or cognitive maps. This is why Skinner's approach is sometimes called descriptive behaviourism, or the "empty organism" approach (a description Skinner disliked, since he never denied that the organism has insides; he denied that inner events explain anything). His major books are worth knowing by title and year: *The Behavior of Organisms* (1938); *Walden Two* (1948), a novel about a utopian community run on reinforcement principles; *Science and Human Behavior* (1953); *Verbal Behavior* (1957), which treated language as operant behaviour shaped by a verbal community; and *Beyond Freedom and Dignity* (1971), which argued that the idea of an autonomous, free person is a pre-scientific fiction we can no longer afford.

**Zing-Yang Kuo, the behaviourist who abolished instinct.** The most extreme environmentalist of all was a Chinese psychologist trained at Berkeley, **Zing-Yang Kuo** (1898 to 1970). In "Giving Up Instincts in Psychology" (1921) he argued that the concept of instinct explains nothing. He backed the claim with an elegant experiment (1930): kittens reared alone, kittens reared watching their mothers kill rats, and kittens reared together with rats. Most of the kittens that watched their mothers became rat-killers, while only a small minority of those raised alongside rats ever killed one, and some seemed to treat rats as companions. "Rat-killing instinct," Kuo concluded, depends on rearing.

Table 2.3 sets the main behaviourists side by side.

Table 2.3 The behaviourists compared.

| Theorist | Label | Signature ideas | Key work |
|---|---|---|---|
| Watson | Classical (methodological) behaviourism | Prediction and control; peripheralism; fear, rage, love; environmentalism | "Psychology as the Behaviorist Views It" (1913); *Behaviorism* (1925) |
| Tolman | Purposive behaviourism | Molar behaviour; intervening variables; cognitive maps; latent learning; expectancy | *Purposive Behavior in Animals and Men* (1932) |
| Hull | Hypothetico-deductive neobehaviourism | Drive reduction; habit strength; ${}_{S}E_{R} = {}_{S}H_{R} \times D$ | *Principles of Behavior* (1943) |
| Guthrie | Contiguity theory | Contiguity without reinforcement; one-trial learning | *The Psychology of Learning* (1935) |
| Skinner | Radical (descriptive) behaviourism | Operant conditioning; no inner causes; anti-theory | *The Behavior of Organisms* (1938); *Verbal Behavior* (1957) |

EXAM TRAP

**Cognitive map and latent learning** are Tolman's, not Kohler's (insight), not Harlow's (learning sets), not Skinner's and not Bandura's (observational learning). **Gould**, the fourth option in one Panwar item, did not propose the concept either: the ethologist James L. Gould tested Tolman's idea in insects, asking in 1986 whether honeybees have cognitive maps, and the evolutionary biologist Stephen Jay Gould is known for *The Mismeasure of Man* (1981), a critique of intelligence testing. **Guthrie's one-trial learning** (any stimulus-movement pairing) is not the same thing as the one-trial learning in Garcia's taste-aversion studies (Chapter 6). **Dollard and Miller** = early social learning theory (Hull plus Freud, 1950); **Bandura and Walters** = later social learning theory (1963). *Walden Two* is a novel, not a research report. **Thorndike's "connectionism"** (stimulus-response bonds) is not the connectionism of modern neural-network models (Section 2.6).

### Criticism, decline and legacy

By the 1950s behaviourism had won American psychology, and it was beginning to lose. The criticisms came from several directions at once. It could not explain language: Noam Chomsky's 1959 review of Skinner's *Verbal Behavior*, discussed in Section 2.6, argued that children produce sentences they have never heard and never been reinforced for. It ignored biology: two of Skinner's own former students, Keller and Marian Breland, reported in 1961 that the animals they trained for commercial shows kept "misbehaving", drifting back toward instinctive food-getting actions (instinctive drift), and John Garcia showed that rats learn some associations far more easily than others (Chapter 6). Its central concepts strained against its own evidence: Tolman's rats seemed to carry maps in their heads, and Kohler's apes seemed to reason. And it seemed to many to leave out exactly what makes us human: purpose, meaning and freedom. The humanists (Section 2.5) attacked from that side, the cognitive psychologists (Section 2.6) from the other.

Yet behaviourism did not so much die as dissolve into the rest of psychology. Its demand for objective, operationally defined, replicable measurement became simply the way psychology is done. Its laws of learning remain the core of Chapter 6. And it gave clinical psychology one of its most effective toolkits.

CLINICAL LINK

**Behaviour therapy** applies learning principles to psychological problems. **Mary Cover Jones** (1896 to 1987) removed a small boy's fear of rabbits in 1924 by gradually bringing a rabbit closer while he ate. **Joseph Wolpe** (1915 to 1997), in *Psychotherapy by Reciprocal Inhibition* (1958), developed **systematic desensitisation**: relaxation, which is incompatible with anxiety, is paired with an imagined hierarchy of feared situations. (The method is Wolpe's, not Skinner's, Watson's or Pavlov's.) Skinner's operant principles produced token economies and applied behaviour analysis, now widely used with children with autism. The therapies themselves are taught in the Clinical book.

## 2.3 Gestalt Psychology and Kurt Lewin HIGH YIELD · 16 QS

Behaviourism rebelled against structuralism by throwing out consciousness. A second rebellion, in Germany, began a year earlier and took the opposite route: it kept consciousness but

threw out the idea that consciousness is built from elements. Where Titchener had analysed experience into sensations, images and feelings, and Watson analysed behaviour into reflexes, the Gestalt psychologists argued that both had made the same mistake. They had taken the whole apart and assumed that the parts were more real than the whole.

### Wertheimer and the phi phenomenon (1912)

**Max Wertheimer** (1880 to 1943) was born in Prague and trained under Oswald Kulpe at Wurzburg. As the story is usually told, in the summer of 1910 he was travelling by train to a holiday in the Rhineland when an idea about seen motion struck him so forcefully that he got off at Frankfurt, bought a toy stroboscope (a device that creates the impression of movement from a series of still pictures), and began experimenting in his hotel room. Historians suspect the story has been polished over the years, but what followed is well documented. At the Psychological Institute in Frankfurt, Wertheimer used a tachistoscope (an instrument that flashes visual displays for precisely controlled intervals) to study apparent motion. His two main observers were younger psychologists, Wolfgang Kohler and Kurt Koffka. The three of them became the founders of Gestalt psychology, and Wertheimer's 1912 paper on the experimental study of seen motion is the school's founding document.

The experiment was simple. Two lines were flashed one after the other in slightly different positions. When the interval between them was very short (around 30 milliseconds), observers saw two lines at once; when it was long (around 200 milliseconds), they saw one line and then the other. But at intervals in between (around 60 milliseconds), they saw a single line move from one position to the other. Wertheimer called this apparent movement the **phi phenomenon**. (He distinguished "pure phi", a sense of motion without any moving object, from the smooth movement of the object itself, later called beta movement; the details belong to Chapter 4.)

Why did such a small effect launch a school? Because the motion was not in either stimulus. Each line, analysed on its own, was stationary. An elementist, adding up two static sensations, could never arrive at movement. The perceived motion was a property of the whole pattern of stimulation in time, and it was as real in experience as anything else. If something as basic as seen motion could not be built from elements, perhaps nothing in perception could.

The idea was not entirely new. In 1890 the Austrian philosopher **Christian von Ehrenfels** (1859 to 1932), whose lectures Wertheimer had attended in Prague, had described *form qualities* (Gestaltqualitaten): a melody played in a different key consists of entirely different notes, yet we hear the same melody. The melody, then, is something more than its notes. Ehrenfels, however, treated the form quality as an extra element added to the sensations. Wertheimer's radical step was to reverse the order: the whole comes first, and the parts get their character from their place in it. The German word **Gestalt** means form, shape or configuration, an organised whole.

### Kohler: insight, transposition and isomorphism

**Wolfgang Kohler** (1887 to 1967), born in Reval (now Tallinn in Estonia), took his doctorate in Berlin under Carl Stumpf and was the school's scientist and chief spokesman. You have already met him on Tenerife (1913 to 1920). His observations there, published in German in 1917 and in English as *The Mentality of Apes* (1925), challenged Thorndike's trial-and-error account

of animal problem-solving. Thorndike's cats, Kohler argued, had been put in puzzle boxes whose mechanisms they could not see; of course they fumbled. Give an animal a problem whose elements are all in view, and it may solve it by **insight**: a sudden reorganisation of the perceptual field so that the relations needed for a solution are seen. Sultan's two sticks, and the boxes his companions stacked to reach fruit hung from the roof, were Kohler's showpieces. Insightful solutions, he noted, appear suddenly, are carried out smoothly once found, are retained, and transfer readily to similar problems. Insight learning as a type of learning is treated in Chapter 6.

### KEY STUDY: KOHLER'S CHICKENS AND TRANSPOSITION (1918)

**The question.** When an animal learns to choose one of two stimuli, does it learn the absolute properties of the stimulus (this particular shade of grey) or a relation between stimuli (the darker of the two)?

**The method.** Kohler trained hens to peck grain from the darker of two grey papers; grain on the lighter one was out of bounds. Then he gave them a test pair: the old "correct" darker grey, now beside an even darker new grey.

**The result.** Most of the time the hens pecked at the new, darker grey, the one they had never been rewarded on, and ignored the shade that had been reinforced. Kohler found the same with chimpanzees.

**The conclusion.** The animals had learned a relation ("the darker one"), not an absolute stimulus. Kohler called this **transposition**, by analogy with a melody transposed to a new key. It was a direct challenge to the stimulus-response view that a specific stimulus becomes bonded to a response. **The criticism:** Kenneth Spence (1937) later showed that a stimulus-response account combining generalisation gradients of excitation and inhibition could predict transposition too, and that transposition breaks down when the test pair is far from the training pair.

Kohler's most ambitious idea was physiological. In *Die physischen Gestalten* ("Physical Gestalten", 1920) he proposed **isomorphism**: the organisation of experience corresponds in structure (not in literal copy) to the organisation of electrical field processes in the brain. A perceived circle does not mean a circle drawn on the cortex, but a pattern of brain activity with the same relational order. The theory was tested in the 1950s. Karl Lashley and his colleagues laid strips of gold foil across the visual cortex of monkeys (1951), and Roger Sperry inserted mica plates and tantalum wires into the cortex of cats (1955), to short-circuit or block any such electrical fields. Perception was essentially unaffected. The field version of isomorphism did not survive, although the broader idea that experience and brain activity share a structure lives on in cognitive neuroscience.

Kohler became director of the Psychological Institute at the University of Berlin in 1922, the most prestigious post in German psychology. When the Nazis began dismissing Jewish professors in 1933, he published a newspaper article in their defence, one of the last openly critical pieces to appear in Germany, and resisted the regime's interference with his institute until he resigned and emigrated in 1935. He spent the rest of his career at Swarthmore College in Pennsylvania and was President of the American Psychological Association in 1959.

### Koffka: development and the behavioural environment

**Kurt Koffka** (1886 to 1941) was the school's great communicator. His *Growth of the Mind* (German 1921, English 1924) applied Gestalt ideas to child development, arguing that infants perceive organised wholes from the beginning rather than assembling them from sensations. In 1922 he introduced Gestalt psychology to American readers in an article in the *Psychological Bulletin* titled "Perception: An introduction to the Gestalt-theorie". The article was so influential that for years Americans assumed Gestalt psychology was only about perception, which the founders found frustrating. Koffka moved to Smith College in Massachusetts in 1927, and his *Principles of Gestalt Psychology* (1935) is the school's most systematic textbook.

Koffka's most memorable lesson is a German legend he retold in the *Principles*. On a winter evening, in a driving snowstorm, a man on horseback rode across a wide, wind-swept plain and arrived at last at an inn. The innkeeper, astonished, asked where he had come from. The rider pointed back across the plain. "Do you know," said the innkeeper, "that you have ridden across Lake Constance?" The rider dropped dead at his feet. In what environment, Koffka asks, did the rider's behaviour take place? In the **geographical environment**, the physical world, he crossed thin ice over deep water. But he *behaved* as if he were crossing solid ground, because his **behavioural environment**, the world as perceived, was a frozen plain. Behaviour, Koffka argued, is governed by the behavioural environment. It is one of the cleanest arguments in psychology for studying the world as the person experiences it.

The three founders, together with the neurologist Kurt Goldstein and the psychiatrist Hans Gruhle, launched the school's journal, *Psychologische Forschung* ("Psychological Research"), in 1921.

### The central ideas of Gestalt psychology ●●●

**The whole is different from the sum of its parts.** This is the school's slogan, and it is usually quoted as "the whole is greater than the sum of its parts". Koffka himself thought "greater" was the wrong word: in the *Principles* he wrote that it is more correct to say the whole is something *else* than the sum of its parts, because summing is a meaningless procedure, whereas the whole-part relationship is meaningful. A square drawn with four lines is not "more" than four lines; it is a different kind of thing, with properties (squareness, closedness, an inside) that none of the lines possesses. Textbooks use both versions for the Gestalt principle.

**Field theory.** The Gestalt psychologists borrowed the concept of a field from nineteenth-century physics, where Faraday and Maxwell had shown that electrical and magnetic events are properties of a whole field of forces rather than of isolated particles. **Field theory** in psychology holds that perception, thought and behaviour are determined by the whole field of interacting forces present at the moment, not by isolated stimuli. Iron filings around a magnet do not each "decide" where to go; the field organises them.

**Organisation and Pragnanz.** Because experience is organised into wholes, the Gestalt psychologists set out the laws by which the organisation happens. Wertheimer's 1923 paper on the laws of perceptual organisation described proximity, similarity, closure, good continuation and common fate, all governed by an overarching law of *Pragnanz*: perceptual organisation will always be as "good" (simple, regular, symmetrical, stable) as the conditions allow. Figure-ground organisation, first analysed by the Danish psychologist **Edgar Rubin** (1886 to 1951) in 1915, was adopted as a cornerstone: the *figure* is seen as a thing, nearer, with a clear

contour and shape, usually smaller and more memorable, whereas the *ground* seems shapeless, diffuse, farther away and to continue behind the figure. The laws themselves, with demonstrations, are taught in Chapter 4.

**Productive thinking.** Wertheimer's last book, *Productive Thinking* (published in 1945, two years after his death), brought Gestalt ideas to problem-solving and education. He distinguished *reproductive thinking*, the blind application of a memorised method, from **productive thinking**, in which the thinker grasps the structure of the problem and sees why a solution works. His favourite example came from a school classroom. Children taught the formula for the area of a parallelogram (base times height) by rote could apply it to the textbook figure but were helpless when the figure was turned on its side. A child who saw that the triangle "sticking out" at one end could be moved to fill the gap at the other, turning the parallelogram into a rectangle, understood the formula and could never be fooled by a rotation. Wertheimer also interviewed Albert Einstein at length about how the theory of relativity had been discovered. Karl Duncker's work on functional fixedness (1945) continued this line (Chapter 8).

Table 2.4 Gestalt psychology against its two rivals.

| Issue | Structuralism | Behaviourism | Gestalt psychology |
|---|---|---|---|
| Subject matter | Conscious experience | Observable behaviour | Conscious experience and behaviour, as organised wholes |
| Method | Trained introspection, analysing experience into elements | Objective observation; conditioning | Phenomenological description of naive experience; experiments |
| Units | Elements (sensations, images, feelings) | Stimulus-response bonds | Wholes, fields and relations |
| Learning and thought | Association of elements | Trial and error, conditioning | Insight, restructuring, productive thinking |
| Nature-nurture leaning | Empiricist | Strongly environmentalist | Nativist: organising tendencies are built in |

Table 2.4 sets Gestalt psychology against its two rivals. Notice the method row. The Gestalt psychologists did not use Titchener's trained introspection, which taught observers to report sensations and to avoid the "stimulus error" of naming the object. They wanted exactly what Titchener forbade: a description of experience as it naturally presents itself to an untrained observer, a book as a book and not as patches of colour. This descriptive approach is called phenomenological, and it links Gestalt psychology to the phenomenology of Husserl (Section 2.5).

EXAM TRAP

**Founders.** The Gestalt trio is Wertheimer (founder), Kohler and Koffka. Watson, William James and Pavlov are favourite distractors. Lewin is Gestalt-influenced but is not one of the founders. **Most important school for perception?** Gestalt psychology.

**"The whole is greater than the sum of its parts and the parts are related to each other and to the whole"** in a management or organisational stem refers to **systems theory**, associated with Ludwig von Bertalanffy's general systems theory, not to Gestalt psychology, motivation theory, contingency theory or Fayol's administrative theory. Read the options.

**The Bender Visual Motor Gestalt Test** (Lauretta Bender, 1938) borrowed its designs from Wertheimer's 1923 figures, but it is a test of visual-motor integration (Assessment book), not a Gestalt theory.

## Kurt Lewin: field theory applied to the person ●●

The Gestalt founders worked mainly on perception and thinking. It was a younger colleague at the Berlin Institute, **Kurt Lewin** (1890 to 1947), who carried field theory into motivation, personality and, finally, social life. Lewin's starting point was that behaviour at any moment is a function of the whole psychological situation. He expressed it in a formula that appears in his *Principles of Topological Psychology* (1936):

B = f(P, E)

Behaviour (B) is a function (f) of the person (P) and the environment (E). The person and the environment are not independent: they form one interdependent field, which Lewin called the **life space** (LSp), the totality of facts that determine a person's behaviour at a given moment. So the formula is often written B = f(LSp). The life space contains the person and the *psychological* environment, the environment as it exists for that person (Koffka's behavioural environment again), not the physical world as a geographer would map it. A rule a student has never heard of is not in her life space; an exam she fears is, even though it is a week away.

To describe the life space, Lewin borrowed from a branch of mathematics called topology, which deals with regions, boundaries and connections rather than distances and sizes; hence **topological psychology**. The life space is divided into regions (activities, goals, groups) separated by boundaries, some of which act as barriers. Movement from one region to another, in thought or action, is called locomotion. Because topology alone cannot express direction and strength, Lewin added two dynamic concepts. A **valence** is the attracting (positive) or repelling (negative) quality a region has for the person; a **vector** is a force acting on the person, with a direction and a strength, toward or away from a region. Lewin later added **hodological space** (from the Greek *hodos*, path), a geometry of paths that represents the routes, including detours, by which a person can reach a goal. Figure 2.2 puts these terms together.

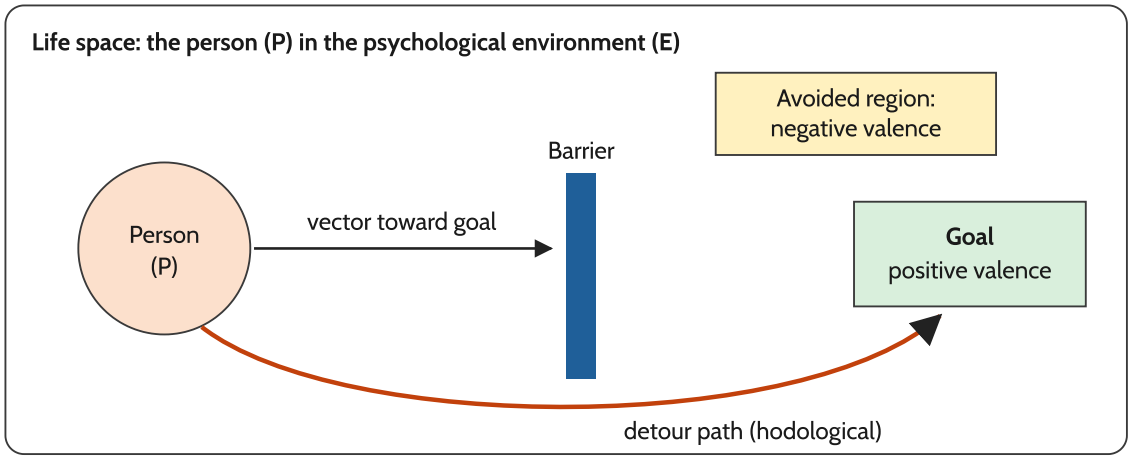


Figure 2.2 Lewin's life space. The person and the psychological environment together make up the life space. The goal has positive valence and exerts a force (vector) on the person; a barrier blocks the direct route, so the person must find a detour path; a region of negative valence pushes the person away. Notice that everything in the diagram is the situation as the person sees it.

**Tension systems and the Zeigarnik effect.** Lewin proposed that an intention to reach a goal sets up a *tension system* within the person, which is discharged only when the goal is reached. One prediction follows directly: unfinished tasks should leave tension behind them and so stay active in the mind. According to a story told by his students, Lewin noticed this one evening at a Berlin restaurant, where a waiter remembered every detail of the unpaid orders but could not recall what had been ordered at a table that had already settled its bill. His student **Bluma Zeigarnik** (1900 to 1988) tested the idea in 1927. Participants were given a series of simple tasks, half of which were interrupted before completion. Afterwards they recalled the interrupted tasks nearly twice as often as the completed ones. Better memory for interrupted than for completed tasks is the **Zeigarnik effect**. A second student, **Maria Ovsiankina** (1898 to 1993), showed in 1928 that people given the chance spontaneously go back to interrupted tasks and finish them, the **Ovsiankina effect**. Remember the pair this way: Zeigarnik, people *remember* unfinished tasks; Ovsiankina, people *resume* them.

**Contemporaneity.** A further principle of Lewin's field theory is that only the present field can influence present behaviour. The past matters only to the extent that it has left traces in the life space now; the future, only as present expectations. This principle of contemporaneity is the most direct "here-and-now" idea in Gestalt-derived psychology. Lewin also analysed conflict as a balance of valences (approach-approach, avoidance-avoidance and approach-avoidance, Chapter 10).

**Group dynamics and action research.** Lewin, who was Jewish, left Germany in 1933, worked briefly at Cornell University and then at the Iowa Child Welfare Research Station, and in 1945 founded the Research Center for Group Dynamics at the Massachusetts Institute of Technology. There he turned field theory to groups, which he saw as dynamic wholes whose properties are not the sum of their members'. He coined the term **group dynamics**. With Ronald Lippitt and Ralph White (1939) he compared boys' clubs run under autocratic, democratic and laissez-faire leadership (Social book). He also founded **action research**, research carried out in partnership with the people affected by a social problem, in a repeating

cycle of planning, acting, observing the results and reflecting, then planning again. A summer workshop he led in 1946 gave rise to the T-group (training group) and the National Training Laboratories. His three-stage model of change (unfreezing, moving, refreezing) is still taught in management courses. Lewin is widely called the father of modern social psychology, and his motto summarises his career: "There is nothing so practical as a good theory."

> EXAM TRAP
>
> **Life space** = the person plus the *psychological* environment (not society and group, not person and group, not environment and group). **Lewin founded** action research, group dynamics and modern experimental social psychology; he did not found psychoanalysis (Freud), psychophysics (Fechner) or functionalism (James, Dewey, Angell). **Zeigarnik** = recall of interrupted tasks; **Ovsiankina** = resumption of interrupted tasks. **Topological** = regions and boundaries; **hodological** = paths.

### Emigration, legacy, and a different "Gestalt" ●●●

With the rise of the Nazis, all four leading figures, Wertheimer (1933, to the New School for Social Research in New York), Koffka (already at Smith from 1927), Kohler (1935) and Lewin (1933), ended up in the United States. There they met a psychology dominated by behaviourism, taught at small colleges rather than major research universities, and wrote in a second language. Gestalt psychology never became a dominant American school. Its influence, however, was enormous and lasting. Its account of perceptual organisation is still taught almost unchanged. Its insistence on insight and understanding prepared the ground for cognitive psychology (Section 2.6). And through Lewin, his students (Leon Festinger among them) and Gestalt-minded social psychologists such as Fritz Heider and Solomon Asch, it shaped modern social psychology.

One thing Gestalt psychology did *not* produce is Gestalt therapy. **Gestalt therapy** was developed in the 1940s and 1950s by **Frederick Salomon Perls** (Fritz Perls, 1893 to 1970), a German-born psychoanalyst, with his wife Laura Perls and the writer Paul Goodman; the book *Gestalt Therapy* (1951) was written by Perls, Ralph Hefferline and Goodman. It is an experiential, humanistic therapy that emphasises awareness in the here and now, personal responsibility, and the completion of "unfinished business" (a phrase that echoes Zeigarnik's interrupted tasks). Its best-known technique is the *empty chair*, in which the client speaks to an imagined person, or a disowned part of the self, placed in an empty chair, and then changes seats to answer. Perls had worked with the holistic neurologist Kurt Goldstein, and he borrowed the word Gestalt and a general holism, but little else. The Gestalt-trained psychologist Mary Henle published a careful comparison in 1978 and concluded that the connection between the two is largely one of name. The therapy itself is taught in the Clinical book.

EXAM TRAP

**Gestalt psychology (Wertheimer, Kohler, Koffka) is not Gestalt therapy (Perls).** The *empty chair technique* belongs to Gestalt therapy, not to psychoanalysis, CBT or person-centred therapy. Questions sometimes blur the two. An assertion-reason item states "Fritz Perls was a proponent of the Gestalt perspective" with the reason "Gestalt focuses on the wholeness of individuals' experiences in the present moment"; the key treats both as true and the reason as the correct explanation, because "Gestalt perspective" is used loosely there to include the therapy. Similarly, one Panwar item keys "here-and-now" to **Gestalt psychology**: the idea is most at home in Gestalt therapy (and in existential-humanistic therapy), and within Gestalt psychology proper it corresponds to Lewin's principle of contemporaneity. If the options separate Gestalt psychology from Gestalt therapy, choose the therapy.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. A field with competing schools and no shared framework is, in Thomas Kuhn's term, a _____ science.
2. Watson's view that thinking is subvocal speech located in the muscles of the larynx is called _____.
3. The three unlearned emotions Watson described in infants were fear, rage and _____.
4. Skinner's position that private events are behaviour but not causes is called _____ behaviourism.
5. The physicist Percy Bridgman argued that a concept means the set of operations used to measure it, a doctrine called _____.
6. Tolman demonstrated learning without reinforcement, which he called _____ learning, and explained it with the concept of a cognitive _____.
7. In Hull's 1943 formula, reaction potential equals habit strength multiplied by _____.
8. The apparent movement that Wertheimer studied in 1912 is called the _____ phenomenon.
9. Kohler's hens chose the darker of two greys even when the shade had changed, showing _____.
10. Lewin's life space consists of the person and the _____ environment.

**Answers:** (1) preparadigmatic, (2) peripheralism, (3) love, (4) radical, (5) operationism, (6) latent, map, (7) drive, (8) phi, (9) transposition, (10) psychological.

## 2.4 Psychoanalysis as a School: The First Force

MEDIUM YIELD · 6 QS

Behaviourism and Gestalt psychology were both born in university laboratories, as quarrels with structuralism about how to study normal minds. Psychoanalysis came from somewhere else entirely: the consulting room of a Viennese doctor treating patients whose symptoms made no medical sense. It never really belonged to academic psychology, yet it changed the way the whole twentieth century thought about the mind. This section tells its history as a

school; Freud's theory of personality (the stages, defences and dynamics) is taught in full in the Personality book.

## From neurology to hysteria: Charcot, Bernheim and Breuer ●●

**Sigmund Freud** (1856 to 1939) did not set out to be a psychologist. He qualified in medicine at the University of Vienna in 1881 after years of research in the physiological laboratory of Ernst Brucke, a leader of the mechanistic school of German physiology, where he studied the nerve cells of fish and other animals under the microscope. Unable to support a family on a researcher's pay, he went into private practice in 1886 as a *neurologist*, a specialist in diseases of the nerves. Many of the patients who came to him suffered from **hysteria**, a condition (today classed among the conversion and dissociative disorders) in which people developed paralyses, blindness, loss of feeling or fits with no detectable damage to the nervous system.

Three encounters shaped what Freud did with these patients. From October 1885 to February 1886 he studied in Paris with **Jean-Martin Charcot** (1825 to 1893), the most famous neurologist in Europe, who showed at the Salpetriere hospital that hysterical symptoms could be produced and removed under hypnosis. That meant the symptoms had a psychological, not an anatomical, cause. Charcot's pupil **Pierre Janet** (1859 to 1947) went on to explain hysteria as *dissociation*, a splitting off of some mental functions from the rest of consciousness into what he called the subconscious; his ideas ran parallel to Freud's, and the two later disputed priority. In 1889 Freud visited **Hippolyte Bernheim** (1840 to 1919) at Nancy, the rival French school, which taught that hypnosis was simply heightened suggestibility; Freud was struck by evidence that patients could later recall what had happened under hypnosis when pressed, which suggested that forgotten memories were not lost but kept out of awareness.

The third and decisive influence was a senior Viennese physician and friend, **Josef Breuer** (1842 to 1925). Between 1880 and 1882 Breuer had treated a gifted young woman, known in the literature as **Anna O.** (her real name was Bertha Pappenheim), who suffered paralysis, disturbances of vision and speech, and periods of altered consciousness. Breuer found that when, under light hypnosis, she talked about the circumstances in which a symptom had first appeared, and expressed the feelings she had held back at the time, the symptom eased. Anna O. called the treatment "the talking cure" and "chimney sweeping". Breuer called it the **cathartic method**, from the Greek *catharsis*, purging: emotion that had been "strangulated" was released. Breuer and Freud published *Studies on Hysteria* in 1895, and that book is usually taken as the beginning of psychoanalysis. (Anna O. was Breuer's patient, not Freud's; she later became a pioneering social worker and campaigner for women's rights.)

Freud soon gave up hypnosis, partly because not all patients could be hypnotised. He replaced it first with a "pressure technique" (pressing on the patient's forehead and insisting that a memory would come) and then with **free association**, in which the patient lies on a couch and reports everything that comes to mind, however trivial, embarrassing or illogical. The places where associations stall, he argued, mark resistance, the mind's defence against painful material. By 1896 he was calling his method psychoanalysis.

## The Interpretation of Dreams and the growth of a movement

In the late 1890s, after the death of his father, Freud undertook a systematic self-analysis, much of it through his own dreams. The result was *The Interpretation of Dreams*, which appeared in November 1899 but was dated 1900 by its publisher, a fitting debut for the new cen-

tury. In it Freud argued that every dream is the disguised fulfilment of a wish, and he called the interpretation of dreams "the royal road to a knowledge of the unconscious" (dream theory is in Chapter 5). It sold only a few hundred copies in its first years. Yet a circle gathered. From 1902 a small group met at Freud's apartment on Wednesday evenings; in 1908 it became the Vienna Psychoanalytic Society, and in 1910 the International Psychoanalytical Association was founded, with Carl Jung as its first president.

International recognition came in 1909, when G. Stanley Hall invited Freud to lecture at **Clark University** in Worcester, Massachusetts, as part of the university's twentieth-anniversary celebrations. Freud travelled with Jung and Sandor Ferenczi, gave five lectures in German, and received an honorary doctorate, his first academic honour. William James, by then old and ill, came to hear him. The Clark lectures are the usual marker for the arrival of psychoanalysis in America.

## Two models of the mind ●●

**The topographic model.** In *The Interpretation of Dreams* (1900) Freud described the mind as having three regions, a **topographic model** (topography means the mapping of places). The *conscious* contains what we are aware of at this moment. The *preconscious* contains material that is not in awareness now but can easily be recalled, such as your phone number or what you ate this morning. The *unconscious* contains wishes, memories and impulses that are actively kept out of awareness because they are unacceptable, and that can reach consciousness only in disguised form, in dreams, slips of the tongue and symptoms. The popular image of the mind as an iceberg, mostly below the waterline, illustrates this model, though the image is not Freud's own.

**The structural model.** In *The Ego and the Id* (1923) Freud added a second map, the **structural model**, which describes three agencies rather than three regions. The *id* is the oldest and most primitive part of the personality, present at birth and entirely unconscious; it is the source of instinctual energy and operates on the pleasure principle, demanding immediate gratification and caring nothing for logic, time or morality (impulsive, irrational, pleasure-seeking: these words always point to the id). The *ego* develops to deal with reality; it operates on the reality principle, delays gratification, and uses realistic thought (secondary process) to find safe ways of meeting the id's demands. The *superego* is the internalised voice of parents and society, made up of the conscience and the ego-ideal. The ego, caught between the id, the superego and the outside world, protects itself against the anxiety these conflicts produce by means of defence mechanisms such as repression. The two models are not rivals: the ego and superego are partly conscious and partly unconscious, while the id is wholly unconscious.

> EXAM TRAP
>
> **Freud began his career as a neurologist**, not as a psychiatrist, philosopher or (obviously) a psychoanalyst. **Father of psychoanalysis**: Freud, not Bernheim (Nancy school of hypnosis), not Adler (individual psychology), not Jung (analytical psychology). **Psychosocial** theory is Erikson's; **psychoanalytic** theory is Freud's. **Gratification, irrational, impulsive** = id. Freud's three anxieties are realistic, neurotic (fear that id impulses will break through and be punished) and moral; "innate anxiety" is not a technical term in any of these theories; Horney's **basic anxiety** and the existentialists' **existential anxiety** are different concepts (see Table 2.5 and Section 2.5).

## Dissenters and neo-Freudians

Freud demanded loyalty to the core of his theory, above all to the sexual basis of neurosis, and some of his ablest followers broke away. Alfred Adler left in 1911 and Carl Jung in 1913 to 1914; each founded his own school. A later generation, often called the **neo-Freudians**, kept the idea of unconscious dynamics but shifted the emphasis from biological instincts to social relationships, culture and the ego. Strictly, Adler and Jung are "dissenters" and Horney, Fromm and Sullivan the neo-Freudians proper, while Anna Freud and Erikson are ego psychologists and Klein the founder of the object relations school; introductory textbooks usually group them all together. Table 2.5 gives one line each. Their theories are taught in the Personality book.

Table 2.5 The dissenters and neo-Freudians in one line each.

| Theorist | Name of the approach | Signature ideas |
|---|---|---|
| **Alfred Adler** (1870 to 1937) | Individual psychology | Inferiority feelings, striving for superiority, style of life, social interest, birth order |
| **Carl Gustav Jung** (1875 to 1961) | Analytical psychology | Collective unconscious, archetypes, introversion and extraversion, individuation |
| **Karen Horney** (1885 to 1952) | Social-cultural psychoanalysis | Basic anxiety (pervasive loneliness and helplessness in a hostile world), moving toward, against and away from people; critique of penis envy |
| **Erich Fromm** (1900 to 1980) | Humanistic psychoanalysis | *Escape from Freedom* (1941); character orientations (receptive, exploitative, hoarding, marketing, productive) |
| **Harry Stack Sullivan** (1892 to 1949) | Interpersonal psychiatry | Personality exists only in relationships; self-system; personifications |
| **Erik Homburger Erikson** (1902 to 1994) | Psychosocial theory (ego psychology) | Eight psychosocial stages across the lifespan; identity crisis |
| **Anna Freud** (1895 to 1982) | Ego psychology; child analysis | *The Ego and the Mechanisms of Defence* (1936) |
| **Melanie Klein** (1882 to 1960) | Object relations | Play technique; paranoid-schizoid and depressive positions |

## Psychoanalysis as the first force

As a school, psychoanalysis was unlike any other in this chapter. Its home was the clinic, not the university; its data were case histories, dreams and free associations, not experiments; and its aim was to relieve suffering as well as to understand it. Its central claims, that much of mental life is unconscious, that childhood experience shapes the adult, and that symptoms have meaning, entered everyday thought so completely that we hardly notice them. It is the **first force** in psychology (Table 2.1), and the broader **psychodynamic perspective** descended from it still explains behaviour, and many mental disorders, in terms of unconscious urges and the hidden struggle over their expression.

Its critics were just as important to the history of the field. The philosopher **Karl Popper** (1902 to 1994) argued that psychoanalysis was unscientific because it was unfalsifiable: any behaviour at all could be explained after the event, so no observation could ever count against it. Behaviourists dismissed its concepts as untestable, and later the humanists complained that it built a whole picture of human nature from the study of sick people. Modern psycho-

dynamic research has answered some of these charges with controlled studies, but the debate shaped how all the later schools defined themselves.

CLINICAL LINK

Short-term psychodynamic psychotherapy, which focuses on recurring relationship patterns, the transference and avoided feelings, has a growing body of controlled evidence for depression and some anxiety and personality problems. When an interviewer asks you which psychotherapy "school" you find most convincing, remember that modern practice borrows across schools; see eclecticism in Section 2.7.

## 2.5 The Third Force and Beyond: Humanistic, Existential, Transpersonal and Positive Psychology HIGH YIELD · 16 QS

By the 1950s, a student of psychology in America could choose between two great pictures of human nature. Psychoanalysis showed a creature driven by blind instincts and scarred by childhood. Behaviourism showed a creature shaped, like a laboratory rat, by its schedule of rewards. Both pictures were deterministic, and to a growing number of psychologists both seemed to leave out what mattered most: the capacity to choose, to grow, to create, to love, to look for meaning. The movement that grew out of this dissatisfaction called itself the third force.

### The birth of humanistic psychology ●●●

**Abraham Harold Maslow** (1908 to 1970) began his career as an enthusiastic behaviourist. He took his doctorate at the University of Wisconsin in 1934, studying dominance and sexual behaviour in monkeys, and as a young man he believed Watson's behaviourism could solve the world's problems. Two experiences changed his mind: the Second World War, and the birth of his first child. He later said that holding his baby made him feel small and not really in control, and that anyone who had a baby could not be a behaviourist. Maslow decided to study not sick people, as Freud had done, nor rats, but the healthiest, most fulfilled people he could find. His hierarchy of needs (1943) and his concept of **self-actualisation**, the fulfilment of one's unique potential, are taught in Chapter 10. His books *Motivation and Personality* (1954) and *Toward a Psychology of Being* (1962) became the movement's texts, and it was Maslow who named it the third force.

**Carl Ransom Rogers** (1902 to 1987) came to the same place by a different road, through counselling. In *Counseling and Psychotherapy* (1942) and *Client-Centered Therapy* (1951) he argued that every person has an *actualising tendency*, an innate drive to grow and develop, which flourishes when others offer unconditional positive regard, empathy and genuineness, and is blocked when love is given only on "conditions of worth". His theory is phenomenological: what matters is the person's own perceived world, the phenomenal field. (Rogers's theory belongs to the Personality book and his therapy to the Clinical book.)

Two other founders deserve a place. **Charlotte Buhler** (1893 to 1974), a Viennese developmental psychologist who studied the course of whole human lives through biographies, brought a European lifespan perspective and was an early president of the movement's association.

**Rollo May** (1909 to 1994) brought existential philosophy into American psychology, as you will see below.

The movement acquired the trappings of a school quickly. In 1961 Maslow and **Anthony Sutich** (1907 to 1976) launched the *Journal of Humanistic Psychology*, and the Association for Humanistic Psychology was formed in the early 1960s (sources differ on the exact year, usually giving 1961 or 1962). In November 1964 a small invitational conference at Old Saybrook, Connecticut, brought together Maslow, Rogers, May, Buhler, Gordon Allport, Henry Murray, George Kelly and others; the **Old Saybrook conference** is usually treated as the formal birth of humanistic psychology as a movement. In 1971 the American Psychological Association created a Division of Humanistic Psychology.

What did the humanists believe? **James Bugental** (1915 to 2008), the Association's first president, summarised it in five postulates in 1964:

- Human beings, as human, supersede the sum of their parts; they cannot be reduced to components.
- Human beings have their existence in a uniquely human context, which is relational: we live among other people.
- Human beings are aware, and are aware of being aware; awareness is part of what it is to be human.
- Human beings have choice, and with choice comes responsibility.
- Human beings are intentional: they seek meaning, value and creativity, and aim at goals.

Put simply, **humanistic psychology** emphasises free will, conscious experience, the whole person, the innate tendency to grow, and the reaching of maximum potential. Its method is phenomenological: understand people from inside their own experience. Its view of psychological disorder follows. Disturbance arises when growth is blocked, when people deny their real feelings to win approval and their self-concept becomes incongruent with their experience. Healing comes from relationships that offer acceptance, respect for the client and belief in the person's potential, not from reinforcement contingencies (a behavioural idea) or the uncovering of unconscious conflict (a psychoanalytic one).

### Phenomenology

Several schools in this chapter call their method "phenomenological", so the word needs a clear definition. **Phenomenology** is the philosophical method founded by **Edmund Husserl** (1859 to 1938) for describing experience exactly as it presents itself, before any theory about its causes is applied. Husserl's call was "back to the things themselves". His central technique was **bracketing** (Greek *epoche*): deliberately suspending one's assumptions about the external reality and causes of an experience, in order to describe the experience itself. He took from his teacher Franz Brentano the idea of *intentionality*, that consciousness is always consciousness *of* something. Phenomenology fed three streams in psychology: the Gestalt psychologists' description of naive experience (through Carl Stumpf, who also taught Husserl), the existential psychologists, and the humanists' focus on the subjective world of the client. Today interpretative phenomenological analysis is a respected qualitative research method (Research Methodology book).

## Existential psychology ●●●

Humanistic psychology was optimistic: people naturally grow toward fulfilment. **Existential psychology**, its close relative, is more sober. It begins from the "givens" of human existence that no one escapes, death, freedom, isolation and the absence of any ready-made meaning, and asks how people live authentically in the face of them. Its roots are in European philosophy.

**Soren Kierkegaard** (1813 to 1855), the Danish thinker usually called the father of existentialism, analysed anxiety in *The Concept of Anxiety* (1844) as the "dizziness of freedom" that comes with having to choose. **Martin Heidegger** (1889 to 1976), in *Being and Time* (1927), described human existence as **Dasein**, literally "being there": the human being is not a mind looking at a separate world but is always already *being-in-the-world*, thrown into a situation not of its choosing and living toward its own death. **Jean-Paul Sartre** (1905 to 1980) declared that for human beings "existence precedes essence": we are not born with a fixed nature but must continually create ourselves through our choices, so that we are "condemned to be free", and to deny that freedom is bad faith. The statement that "each person must continually recreate himself; the meaning of life is never fixed but constantly changes" is pure existentialism.

Two Swiss psychiatrists turned Heidegger's philosophy into a clinical method. **Ludwig Binswanger** (1881 to 1966) founded **Daseinsanalysis** (existential analysis), which tries to understand a patient's "world-design". He described three simultaneous modes of being-in-the-world: the *Umwelt* (the biological and physical world around us), the *Mitwelt* (the world with others) and the *Eigenwelt* (one's own world, the relationship with oneself). The existential therapist Emmy van Deurzen later added a fourth, the *Uberwelt*, the spiritual world of values and beliefs. **Medard Boss** (1903 to 1990), who worked closely with Heidegger himself, developed Daseinsanalysis further.

The most widely read existential psychiatrist is **Viktor Emil Frankl** (1905 to 1997) of Vienna. Frankl survived Theresienstadt, Auschwitz and Dachau's subcamps; his wife, parents and brother died in the camps. In *Man's Search for Meaning* (German 1946, English 1959) he described how prisoners who held on to a reason to live, a task to finish or a person to return to, were better able to endure. He founded **logotherapy** (from the Greek *logos*, meaning), often called the "third Viennese school of psychotherapy" after Freud's psychoanalysis and Adler's individual psychology. Where Freud saw a will to pleasure and Adler a will to power, Frankl saw the **will to meaning** as the primary human motive. When it is frustrated, people fall into an **existential vacuum**, a sense of emptiness and boredom; when the frustration produces symptoms, Frankl called the result **noogenic neurosis** (from the Greek *noos*, mind or spirit), a neurosis arising not from psychological conflict or biology but from the spiritual dimension, from problems of meaning. Meaning, Frankl taught, can be found in three ways: by creating a work or doing a deed (creative values), by experiencing something or loving someone (experiential values), and by the attitude we take toward unavoidable suffering (attitudinal values). His techniques include paradoxical intention and dereflection (Clinical book).

**Rollo May** brought European existentialism to American readers in *Existence* (1958), edited with Ernest Angel and Henri Ellenberger. In *The Meaning of Anxiety* (1950) he distinguished *normal anxiety*, which is proportionate to the threat, does not require repression and can be used constructively, from *neurotic anxiety*, which is disproportionate and involves repression. Existential anxiety, the anxiety of facing death, freedom and meaninglessness, is normal in this sense. Finally, the American psychiatrist **Irvin David Yalom** (born 1931), in

*Existential Psychotherapy* (1980), organised the field around four "ultimate concerns": death, freedom (with its responsibility), existential isolation, and meaninglessness. Table 2.6 gathers the existential thinkers and their signature terms.

Table 2.6 The existential thinkers and their signature terms.

| Thinker | Signature term or work | Meaning |
|---|---|---|
| Kierkegaard | Anxiety as the dizziness of freedom (1844) | Anxiety comes with the freedom to choose |
| Heidegger | Dasein; being-in-the-world (*Being and Time*, 1927) | Human existence is always situated in a world |
| Sartre | Existence precedes essence; bad faith | We create our own nature through choices |
| Binswanger | Daseinsanalysis; Umwelt, Mitwelt, Eigenwelt | Three modes of being-in-the-world |
| Boss | Daseinsanalysis | Heideggerian therapy |
| Frankl | Logotherapy; will to meaning; noogenic neurosis; existential vacuum | Meaning is the primary human motive |
| May | *Existence* (1958); normal versus neurotic anxiety | Brought existentialism to America |
| Yalom | Four ultimate concerns (1980) | Death, freedom, isolation, meaninglessness |

EXAM TRAP

**Humanistic or existential?** Both are third-force, phenomenological and pro-freedom. Humanism stresses growth and self-actualisation (Maslow, Rogers); existentialism stresses the givens of death, freedom, isolation and meaninglessness, and the need to create meaning (Frankl, May, Yalom). "Uncovering our unconscious fears and desires" is a *psychoanalytic* aim, not an existential construct. **Dasein** and **noogenic anxiety** belong to the existential school. **Will to meaning** = logotherapy (Frankl), not REBT (Ellis), not ACT (Hayes), and more specific than "existential-humanistic therapy". The fourth world, *Uberwelt*, is van Deurzen's addition, not Heidegger's.

## Transpersonal psychology: the fourth force

Toward the end of his life Maslow came to believe that humanistic psychology itself was a transitional stage. In the preface to the second edition of *Toward a Psychology of Being* (1968) he wrote of a still "higher" fourth psychology, transpersonal, centred in the cosmos rather than in human needs and interests. His studies of self-actualising people had shown him that many reported *peak experiences*, moments of awe, unity and self-transcendence. In 1969 Maslow and Sutich founded the *Journal of Transpersonal Psychology*. **Transpersonal psychology**, the **fourth force**, studies experiences in which the sense of identity extends beyond (trans) the individual person: mystical and peak experiences, meditation and altered states of consciousness, spiritual development and self-transcendence. Its leading figures include the psychiatrist **Stanislav Grof** (born 1931), who studied non-ordinary states of consciousness, **Charles Tart** (born 1937), who wrote on states of consciousness, **Ken Wilber** (born 1949), whose spectrum of consciousness and integral theory tried to unite Western psychology with Eastern traditions, and, earlier, the Italian psychiatrist **Roberto Assagioli** (1888 to 1974), founder of psychosynthesis. Frankl, with his stress on self-transcendence, is often

claimed as a forerunner. Because so much transpersonal psychology draws on yoga, Vedanta and Buddhism, it forms a natural bridge to Indian psychology in Sections 2.8 and 2.9.

### Positive psychology ●●●

The last movement in this family is the youngest. The phrase "positive psychology" first appeared as the title of the final chapter of Maslow's *Motivation and Personality* (1954), but it remained a passing label. The field as it exists today was launched by **Martin Elias Peter Seligman** (born 1942), the psychologist who had made his name with research on learned helplessness and depression (Chapter 6). Elected President of the American Psychological Association for 1998, he used his presidency to argue that psychology had become a science of repairing damage and had forgotten its other two missions: making the lives of ordinary people more fulfilling, and nurturing talent.

Seligman later told how the idea crystallised. He was weeding his garden with his five-year-old daughter Nikki, who was throwing weeds into the air and dancing. He shouted at her. She walked away, came back, and told him that from her third birthday to her fifth she had been a whiner, that on turning five she had decided to stop, that it was the hardest thing she had ever done, and that if she could stop whining, he could stop being such a grouch. Raising Nikki, Seligman realised, was not about correcting her whining but about recognising and nurturing the strength she had just shown. Psychology, he concluded, should do the same.

In January 2000 Seligman and **Mihaly Csikszentmihalyi** (1934 to 2021) edited a special issue of the *American Psychologist* that announced **positive psychology**: the scientific study of positive experiences, positive individual traits and the institutions that enable them (its "three pillars"). Csikszentmihalyi was already known for his research on **flow**, the absorbing state of complete engagement in an activity whose challenge matches one's skill. Other landmarks followed. *Character Strengths and Virtues* (2004), by Christopher Peterson and Seligman, classified 24 character strengths under six virtues as a kind of "manual of the sanities" to set beside the diagnostic manuals. In *Flourish* (2011) Seligman proposed the **PERMA** model of well-being: Positive emotion, Engagement, Relationships, Meaning and Accomplishment (detail in the Motivation and Emotion chapters).

Several research programmes are now considered part of positive psychology. C. R. Snyder's *hope theory* describes hope as the combination of agency (the will to reach a goal) and pathways (the ways to reach it); promoting hope is a typical positive psychology intervention. Fred Bryant and Joseph Veroff (2007) studied **savouring**, the capacity to attend to, appreciate and deliberately enhance the enjoyment of positive experiences, past, present or anticipated; savouring is to good events what coping is to bad ones. Barbara Fredrickson's broaden-and-build theory holds that positive emotions broaden attention and thinking and build lasting personal resources, and Ed Diener's research on subjective well-being measures happiness as high life satisfaction, frequent positive feeling and infrequent negative feeling. Positive psychology interventions, such as writing down three good things each day, using one's signature strengths in a new way, or delivering a gratitude letter, have been tested in randomised trials.

How does positive psychology relate to humanism? Seligman was at pains to distance his field from what he saw as the unscientific, self-help side of humanistic psychology; positive psychology is meant to be empirical. Humanistic psychologists replied, with some justice,

that positive psychology had rediscovered their themes and given them new names. Most textbooks now treat humanistic psychology as its forerunner.

EXAM TRAP

**Founder of positive psychology**: Seligman (with Csikszentmihalyi). Maslow coined the phrase in 1954 but did not found the field. Howard Gardner is multiple intelligences; Steven Pinker is language and evolutionary psychology; Bandura is self-efficacy and social cognitive theory. **Savouring** = enhancing the enjoyment of positive experiences (not avoiding negative ones, not relaxation, not reflecting on trauma). **The primary focus of positive psychology interventions** is building strengths such as hope, not behaviour therapy, psychoanalysis or DBT.

## Criticisms of the third force

Humanistic psychology has been criticised on four main grounds. First, its concepts (self-actualisation, growth, authenticity) are vague and hard to test; Maslow's description of self-actualisers rested on a small, hand-picked sample of people he admired. Second, its optimism can seem naive in the face of real human destructiveness, a criticism made from inside the movement by Rollo May. Third, it is individualistic: by concentrating so heavily on the individual's inner world, the humanistic and phenomenological approaches miss the wider social, cultural and economic contexts that shape experience and identity, and they reflect a Western, individualist ideal of the self. Fourth, some critics argued that its celebration of self-fulfilment encouraged self-absorption. Positive psychology tried to answer the first criticism with data; cross-cultural and Indian psychologists, as you will see, have pressed the third.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Freud began his professional career as a _____.
2. Anna O., the patient who called her treatment "the talking cure", was treated by _____.
3. In 1909 Freud lectured in America at _____ University.
4. Freud's 1923 model of id, ego and superego is called the _____ model.
5. Horney's term for a pervasive feeling of loneliness and helplessness is _____ anxiety.
6. The 1964 conference usually taken as the birth of humanistic psychology was held at _____.
7. Husserl's technique of suspending assumptions about an experience is called _____.
8. Heidegger's term for human existence as being-in-the-world is _____.
9. Frankl called neurosis arising from frustrated meaning _____ neurosis.
10. The fourth force in psychology is _____ psychology.

**Answers:** (1) neurologist, (2) Josef Breuer, (3) Clark, (4) structural, (5) basic, (6) Old Saybrook, (7) bracketing (epoche), (8) Dasein, (9) noogenic, (10) transpersonal.

## 2.6 The Cognitive Revolution MEDIUM YIELD · 5 QS

The humanists attacked behaviourism for leaving out the person. A second group of critics attacked it for leaving out the mind, and this second attack proved decisive in the laboratories. Between the late 1940s and the late 1960s, the study of memory, attention, language, thinking and perception, all of which behaviourism had pushed to the edges, returned to the centre of experimental psychology. The change was so rapid and so complete that it is usually called the **cognitive revolution**. Why did it happen when it did?

### Why behaviourism waned

Several pressures converged, and it helps to keep them separate.

**Anomalies inside behaviourism.** You have already met two: Tolman's rats that learned maps without reward, and Kohler's apes that solved problems by insight. A third came from **Karl Spencer Lashley** (1890 to 1958), once Watson's own student. At the Hixon Symposium on "Cerebral Mechanisms in Behavior" in 1948 (published in 1951), Lashley gave a paper on "The Problem of Serial Order in Behavior". A pianist playing a fast passage, or a person speaking a sentence, produces movements far too quickly for each one to be triggered by sensory feedback from the one before, as a chain of stimulus-response links would require. The sequence must be planned in advance, centrally, and organised hierarchically. In other words, behaviour requires internal plans.

**The war and its machines.** The Second World War posed problems that behaviourism could not answer: how radar operators could stay alert through long watches, how pilots could cope with too many dials and messages at once, how to design controls that people would not misread. Psychologists working on these problems, such as Donald Broadbent in Britain, found themselves talking about attention, capacity and the flow of information through a person.

**Information, feedback and computers.** In 1948 the engineer **Claude Elwood Shannon** (1916 to 2001) published a mathematical theory of communication that measured information in units called bits, and the mathematician **Norbert Wiener** (1894 to 1964) published *Cybernetics*, the science of control and feedback in animals and machines. A machine with a goal and a feedback loop (a thermostat, a guided missile) behaves purposefully, and nobody thinks it is mystical to say so. Then came the stored-program digital computer, and in 1950 **Alan Mathison Turing** (1912 to 1954) asked whether machines could think. Computers made internal states respectable. A computer's behaviour cannot be understood without reference to its program and its memory, which nobody can see. If a machine could be described in terms of stored symbols and rules for manipulating them, why not a person?

**Language.** The final blow came from linguistics, as you will see shortly. Seen through Kuhn's eyes (Section 2.1), these were anomalies piling up until the old framework entered a crisis.

### 1956 and after ●●

If the revolution has a birthday, it is in 1956. From 10 to 12 September of that year, a Symposium on Information Theory was held at the Massachusetts Institute of Technology. **George Armitage Miller** (1920 to 2012), one of the participants, later dated the birth of cognitive science to its second day, 11 September 1956. On that day **Allen Newell** (1927 to 1992) and **Herbert Alexander Simon** (1916 to 2001) described the *Logic Theorist*, a computer program

(written with Cliff Shaw) that proved theorems from Whitehead and Russell's *Principia Mathematica*, the first program to model human problem-solving. **Noam Chomsky** (born 1928), a young linguist, presented a paper on formal models of language. And Miller himself spoke about the limits of short-term memory, the subject of his famous 1956 paper "The Magical Number Seven, Plus or Minus Two", which showed that people can hold about seven items (between five and nine) in immediate memory, and that the limit can be stretched by grouping items into larger meaningful units, or *chunks* (Chapter 7).

The same year saw the Dartmouth summer research project at which John McCarthy, Marvin Minsky, Claude Shannon and others launched the field of *artificial intelligence* (McCarthy had coined the term in the 1955 proposal for the meeting), and the publication of *A Study of Thinking* by **Jerome Seymour Bruner** (1915 to 2016), Jacqueline Goodnow and George Austin, which analysed the strategies people use to discover concepts (Chapter 8). In 1957 Chomsky published *Syntactic Structures*, and in 1958 **Donald Eric Broadbent** (1926 to 1993) published *Perception and Communication*, which presented the filter model of attention (Chapter 5). Broadbent's book is regarded as the first major theory cast explicitly in information-processing terms: information flows through a series of stages of limited capacity, drawn as a box-and-arrow flow chart.

### KEY STUDY: CHOMSKY'S REVIEW OF *VERBAL BEHAVIOR* (1959)

**The background.** In *Verbal Behavior* (1957), Skinner had argued that language is operant behaviour like any other: children learn to speak because the verbal community reinforces some sounds and not others, and adult speech is controlled by stimuli and reinforcement histories.

**The argument.** In a long review in the journal *Language* (1959), Chomsky argued that this could not work. Children acquire the grammar of their language rapidly, at about the same age everywhere, from speech that is fragmentary and full of errors (later called the poverty of the stimulus). They constantly produce and understand sentences they have never heard, so these sentences cannot have been reinforced. Their errors, such as "I goed" or "two mouses", are not imitations of anything adults say; they are over-applications of rules. And when Skinner's terms (stimulus, response, reinforcement) are stretched to cover language, Chomsky argued, they lose their precise laboratory meaning and become a vague paraphrase of ordinary talk.

**The conclusion.** Language is rule-governed, creative and generative, and human beings must come equipped with an innate capacity for acquiring it (Chapter 8).

**The criticism.** Behaviour analysts, notably Kenneth MacCorquodale (1970), replied that Chomsky had misunderstood Skinner's position. Skinner never answered publicly. Whatever its fairness, the review was read as a manifesto for the new cognitive psychology, and it made Chomsky, a linguist rather than a psychologist, one of the movement's heroes.

The pieces came together quickly. In 1960 Miller, Eugene Galanter and Karl Pribram published *Plans and the Structure of Behavior*, which replaced the reflex arc with the TOTE unit (Test, Operate, Test, Exit), a feedback loop in which an organism compares its current state with a goal, acts, and tests again. The same year, Bruner and Miller founded the Center for Cognitive Studies at Harvard. Then, in 1967, **Ulric Neisser** (1928 to 2012) published *Cognitive Psychology*, the first textbook of the new field, which gave it its name. Neisser defined **cognition** as "all the processes by which the sensory input is transformed, reduced, elaborated,

stored, recovered, and used". He is called the father of **cognitive psychology**, the study of internal mental processes such as perception, attention, memory, language, problem-solving and reasoning. (Characteristically, Neisser later criticised his own field: in *Cognition and Reality*, 1976, he argued that laboratory cognitive psychology had lost touch with how people use their minds in real life, its ecological validity.) Table 2.7 lists the milestones of the revolution in order.

Table 2.7 Milestones of the cognitive revolution.

| Year | Landmark | Who |
|---|---|---|
| 1932 | *Remembering*: schemas and reconstructive memory | Bartlett |
| 1948 | Information theory; *Cybernetics*; Hixon Symposium (serial order) | Shannon; Wiener; Lashley |
| 1950 | "Computing Machinery and Intelligence" | Turing |
| 1956 | MIT symposium; Logic Theorist; "The Magical Number Seven"; Dartmouth AI project; *A Study of Thinking* | Newell and Simon; Miller; McCarthy and Minsky; Bruner, Goodnow and Austin |
| 1957 | *Syntactic Structures* | Chomsky |
| 1958 | *Perception and Communication*: first major information-processing theory | Broadbent |
| 1959 | Review of Skinner's *Verbal Behavior* | Chomsky |
| 1960 | *Plans and the Structure of Behavior* (TOTE); Harvard Center for Cognitive Studies | Miller, Galanter and Pribram; Bruner and Miller |
| 1967 | *Cognitive Psychology*: the field is named | Neisser |
| 1986 | *Parallel Distributed Processing* | Rumelhart, McClelland and the PDP Research Group |

## The information-processing metaphor

The core idea of the new psychology is the **information-processing approach**: the mind is treated as a system that takes in information from the senses, transforms it, stores it, retrieves it and uses it to guide action, much as a computer takes input, runs it through a program and produces output. The approach rests on four assumptions. Mental processes can be broken into a sequence of stages. Each stage has a limited capacity. Processing takes time, so the time a task takes (the reaction time) reveals what stages it involves, an idea that revived Donders's mental chronometry from the nineteenth century (Chapter 1). And mental processes can be described as operations on internal representations (symbols and rules), at the level of the "software", without first knowing the "hardware" of the brain.

A worked example shows how the approach thinks. A fast bowler releases a cricket ball at 140 kilometres an hour, and the batsman has about half a second before it reaches him. In information-processing terms: light from the ball is registered by the eyes (sensory input); attention selects the ball from the crowd, the bowler's arm and the sightscreen (selection); the line and length of the delivery are recognised by matching them against patterns stored from thousands of earlier deliveries (pattern recognition, drawing on long-term memory); a shot is selected and its motor program retrieved (decision and response selection); and the bat moves (output). Each stage takes time, and an expert's advantage lies partly in doing some stages faster and partly in anticipating the ball before it is bowled, from cues in the bowler's

run-up, which is top-down processing (Chapter 4). Figure 2.3(a) draws the classic serial version of this picture.

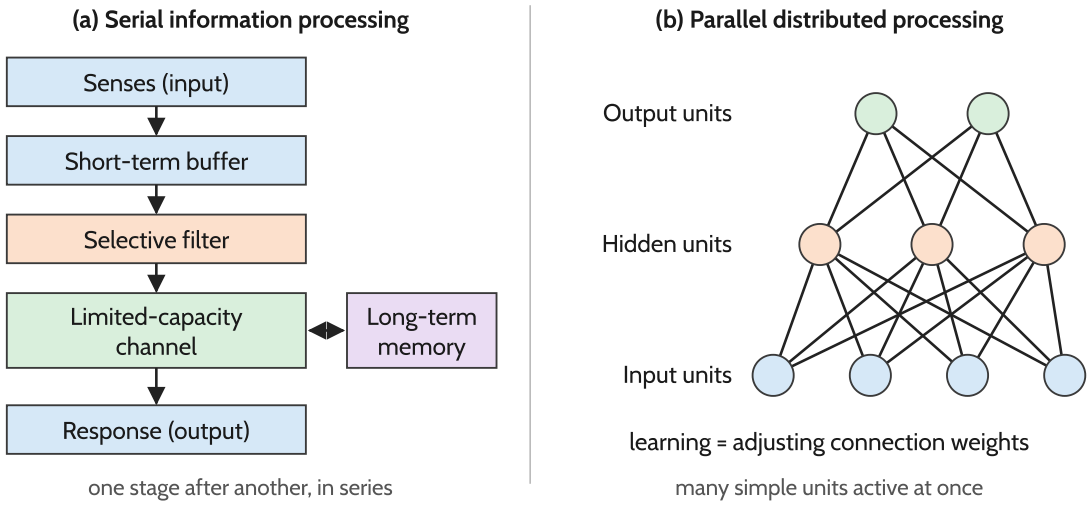


Figure 2.3 Two ways of modelling the mind. (a) A serial information-processing model in the style of Broadbent (1958): information passes through a sequence of stages, one of which has limited capacity. (b) A connectionist network: knowledge is stored in the strengths (weights) of the connections between many simple units working in parallel, not in any single box.

## Forerunners: Bartlett and Piaget ●●

The cognitive revolution was a revolution mainly in America. In Europe, the study of the mind had never stopped, and two European psychologists in particular are treated as forerunners. **Frederic Charles Bartlett** (1886 to 1969) of Cambridge, in *Remembering* (1932), asked English students to read and later recall a Native American folk tale, "The War of the Ghosts". Their recall grew shorter, lost its unfamiliar supernatural details and was rationalised into the shape of an English story. Bartlett concluded that remembering is reconstruction guided by a **schema**, an organised mental structure of knowledge built from past experience (Chapter 7).

**Jean Piaget** (1896 to 1980) of Geneva spent half a century, from the 1920s on, studying how children's thinking develops. He called his field genetic epistemology, the study of the origins and growth of knowledge, and described how children build schemas and adapt them through assimilation (fitting new experience into existing schemas) and accommodation (changing schemas to fit new experience). Piaget studied thinking directly at a time when American psychology refused to, which is why he is often named a pioneer of cognitive psychology; in the 2002 survey of eminent psychologists he ranked second only to Skinner. His stage theory is taught in the Developmental book. Other forerunners include the Gestalt psychologists (insight and productive thinking), Tolman (cognitive maps), the Wurzburg school (thought without images, Chapter 1) and the Russian psychologist **Lev Semyonovich Vygotsky** (1896 to 1934), who studied the relation between thought and language.

EXAM TRAP

**The first major information-processing theory** was Broadbent's (1958), not Treisman's (her attenuation work began around 1960), not Chomsky's (1957 was *Syntactic Structures*, a work of linguistics), and certainly not Watson's (1920 was Little Albert). **Father of cognitive psychology**: Neisser (1967), though Miller and Bruner were central figures. **Pioneer contributor to cognitive psychology** in an Amity item with options Piaget, Kohler, Chomsky and Kohlberg: the key is **Piaget**, who studied thinking from the 1920s; Chomsky was a linguist whose 1959 review fuelled the revolution; Kohler belongs to Gestalt; Kohlberg studied moral development. **Chomsky's review** was of *Verbal Behavior* (1957) and appeared in 1959.

### Cognitive science and cognitive neuroscience

Cognitive psychology soon joined a larger enterprise. **Cognitive science** is the interdisciplinary study of mind and intelligence, bringing together psychology, artificial intelligence and computer science, linguistics, philosophy, neuroscience and anthropology; it acquired its own journal and, in 1979, a Cognitive Science Society.

For its first twenty years cognitive psychology mostly ignored the brain, in keeping with the "software" assumption. That changed in the late 1970s, when, as **Michael Gazzaniga** (born 1939) tells it, he and George Miller coined the term **cognitive neuroscience** in the back of a New York taxi. Cognitive neuroscience studies how mental processes are carried out by the brain, combining the experimental tasks of cognitive psychology with methods for measuring brain activity: the study of patients with brain damage, electrical recording (EEG and event-related potentials), positron emission tomography (PET) from the 1980s, functional magnetic resonance imaging (fMRI) from the 1990s, and transcranial magnetic stimulation. Its methods are taught in the Physiological book.

### Connectionism and parallel distributed processing

The serial, box-and-arrow model had a problem. Brains are made of billions of slow neurons, each firing at most a few hundred times a second, yet people recognise a face in a fraction of a second. A strictly serial process with such slow parts could not be that fast. The answer, many researchers argued, was that the brain does many things at once.

**Connectionism** models the mind as a network of many simple, neuron-like processing units joined by connections of varying strength (weights). Each unit adds up the activation it receives and passes activation on; many units are active at the same time (parallel processing); and a piece of knowledge is stored not in one place but as a pattern of weights spread across the whole network (distributed representation). The landmark was the two-volume *Parallel Distributed Processing* (1986) by **David Everett Rumelhart** (1942 to 2011), **James Lloyd McClelland** (born 1948) and the PDP Research Group, which gave the approach its other name, **parallel distributed processing** (PDP). Figure 2.3(b) shows a simple network.

Networks learn by adjusting their weights. One idea goes back to Donald Hebb (1949), who proposed that a connection between two neurons strengthens when they are active together; another, the back-propagation rule popularised in 1986 by Rumelhart, Geoffrey Hinton and Ronald Williams, adjusts weights in proportion to the network's errors. Connectionist models have several brain-like properties. They degrade gracefully: damage a few units and performance worsens a little rather than failing completely. They are content-addressable: a par-

tial cue can retrieve a whole memory. And they generalise: Rumelhart and McClelland's 1986 model of the English past tense learned regular and irregular verbs from examples, with no rules written in, and even went through a stage of over-regularising ("goed") as children do. Connectionism has deep roots, in Warren McCulloch and Walter Pitts's model neurons (1943) and Frank Rosenblatt's perceptron (1958), and today's deep-learning systems are its descendants.

### Cognition enters behaviour therapy •

The cognitive revolution did not stop at the laboratory door. **Albert Bandura** (1925 to 2021), whose Bobo doll studies you will meet in Chapter 6, moved social learning theory steadily toward cognition. In 1977 he introduced the concept of **self-efficacy**, a person's belief in his or her capacity to carry out the actions needed to deal with a particular situation, as a unifying explanation of how people change. Expectations, not reinforcements alone, drive behaviour. His mature system is social cognitive theory (1986). At the same time, **Albert Ellis** (1913 to 2007) developed rational therapy, later renamed rational emotive behaviour therapy, from 1955, and **Aaron Temkin Beck** (1921 to 2021) developed cognitive therapy for depression in the 1960s, both based on the idea that maladaptive thoughts produce distress. The merger of these ideas with behaviour therapy produced cognitive-behavioural therapy (CBT), the therapy that focuses on changing maladaptive thought patterns.

CLINICAL LINK

Psychotherapy has gone through three "waves" that mirror this chapter's history: behaviour therapy (the legacy of behaviourism), cognitive therapy and CBT (the legacy of the cognitive revolution), and a third wave of acceptance and mindfulness-based therapies, including acceptance and commitment therapy (ACT), dialectical behaviour therapy (DBT) and mindfulness-based cognitive therapy (MBCT), which draw explicitly on Buddhist ideas (Section 2.9). Psychoanalysis is not one of the waves of CBT. Each therapy is taught in the Clinical book.

## 2.7 Psychology Today: Perspectives, Eclecticism and the Enduring Issues

MEDIUM YIELD · 6 QS

By the 1970s the age of schools was over. Few psychologists now call themselves behaviourists or Gestaltists; instead they speak of *perspectives*, ways of approaching a question that coexist and complement one another. The difference is real. A school claimed to be the right way to do psychology; a perspective claims only to be a useful angle on a problem that other perspectives also illuminate.

### The contemporary perspectives •••

Table 2.8 sets out the eight perspectives most textbooks recognise, with one running example: Ravi, a final-year student in Mysuru, becomes intensely anxious before every examination.

Table 2.8 The contemporary perspectives, applied to one case of examination anxiety.

| Perspective | Key figures | Explains behaviour through | Ravi's anxiety is due to |
|---|---|---|---|

| Perspective | Key figures | Explains behaviour through | Ravi's anxiety is due to |
|---|---|---|---|
| Biological (neurobiological) | Hebb, Sperry; modern neuroscience | Brain, nervous system, genes, hormones | An over-reactive threat system and stress hormones; perhaps a family history of anxiety |
| Evolutionary | Darwin; Buss; Cosmides and Tooby | Adaptations shaped by natural selection | An ancient alarm system, useful against predators, firing in a modern exam hall |
| Psychodynamic | Freud and the neo-Freudians | Unconscious conflict and early experience | An unconscious fear of a critical parent's disapproval |
| Behavioural | Watson, Skinner | Learning: conditioning and reinforcement | Past failures paired with humiliation; avoidance rewarded by relief |
| Cognitive | Neisser, Beck | Mental processes: attention, memory, thinking | Catastrophic thoughts ("if I fail, my life is over") |
| Humanistic | Maslow, Rogers | Free will, growth, the self-concept | Self-worth made conditional on marks (conditions of worth) |
| Sociocultural | Vygotsky; cross-cultural psychologists | Culture, society and social context | Family and community pressure in a fiercely competitive entrance system |
| Biopsychosocial | George Engel (1977) | Interaction of biological, psychological and social factors | All of these, interacting |

A few of these need a word more. The **biological perspective**, also called the neurobiological perspective, holds that behaviour is ultimately governed by the brain and nervous system. The **evolutionary perspective** asks what adaptive problem a behaviour helped our ancestors solve; it was developed into evolutionary psychology by researchers such as David Buss, Leda Cosmides and John Tooby. The **sociocultural perspective**, inspired in part by Vygotsky, studies how culture and social context shape thought and behaviour. The **biopsychosocial model** was proposed by the American psychiatrist **George Libman Engel** (1913 to 1999) in a 1977 paper in *Science*, as an alternative to the purely biomedical model of illness: any illness, physical or mental, must be understood as the product of biological, psychological and social factors acting together. It is now the standard framework of clinical psychology and psychiatry. You will also see "developmental" listed as a perspective in some textbooks, meaning the study of how people change as they mature across the lifespan; strictly, developmental psychology is a field of study that uses all the other perspectives.

Within the sociocultural approach, three terms are easily confused. **Cross-cultural psychology** compares psychological processes across different cultures, usually with the same measures, to find out which are universal and which vary (the work of Harry Triandis, John Berry and Geert Hofstede is typical). **Cultural psychology** studies, often from within a single culture, how culture and mind "make each other up", treating culture as part of the mind rather than as a variable acting on it. **Indigenous psychology** builds psychological concepts and theories from within a culture's own traditions instead of importing them; Durganand Sinha's call to indigenise psychology in India, which you will meet in Section 2.8, is an example.

EXAM TRAP

"Who studies how psychological processes may differ among people of different cultures?" The textbook answer is **cross-cultural psychologists**. One Panwar item keys *social psychologists*, presumably because culture is part of the social context; if both appear, cross-cultural psychology is the more precise answer. "Behaviour is governed by the brain and nervous system" = **neurobiological** (not psychophysical, which is Fechner's study of stimulus-sensation relations, and not neo-behaviouristic). "Unconscious processes and early childhood" = **psychodynamic**. "Stimulus-response associations through conditioning" = **behavioural**. "Internal mental processes" = **cognitive**. "Free will and personal growth" = **humanistic**.

## Eclecticism

If every perspective captures part of the truth, why choose one? **Eclecticism** is the practice of selecting concepts, methods or techniques from different schools according to what best fits the problem in hand. Frederick Thorne argued for an eclectic approach to counselling around 1950, and Arnold Lazarus later distinguished *technical eclecticism*, using techniques from any school without necessarily accepting its theory (his multimodal therapy is the best-known example), from *theoretical integration*, which tries to combine the theories themselves. A third route looks for the *common factors*, such as the therapeutic relationship and the instillation of hope, that make different therapies work. Surveys repeatedly find that eclectic or integrative is the most common orientation among practising clinicians. The danger, critics point out, is a shapeless mixture with no principle for choosing.

## The enduring issues

Schools come and go, but the questions that divided them do not. The historian of psychology **Robert Irving Watson** (1909 to 1980) argued in a 1967 paper, "Psychology: A Prescriptive Science", that since psychology lacks a single paradigm, its history is better organised by **prescriptions**: contrasting attitudes, held more or less strongly by different psychologists at different times, about what psychology should be and do. He listed eighteen pairs, among them conscious versus unconscious mentalism, determinism versus indeterminism, empiricism versus rationalism, mechanism versus vitalism, molecularism versus molarism, monism versus dualism, nomotheticism versus idiographicism, and peripheralism versus centralism. Table 2.9 shows the issues you will meet most often and where the schools you have met stand on each.

Table 2.9 The enduring issues and where the schools stand.

| Issue | The two poles | Where the schools stand |
|---|---|---|
| Mind-body | Dualism (mind and body are distinct) versus monism (one kind of reality) | Descartes dualist; behaviourism and the biological perspective materialist monist |
| Nature-nurture | Heredity versus environment | Gestalt and Chomsky nativist; Watson and Kuo extreme nurture; most modern views interactionist |
| Free will-determinism | Behaviour freely chosen versus caused | Freud (psychic determinism) and Skinner (environmental determinism) versus the humanists and existentialists |
| Mechanism-vitalism | Organisms are machines versus a special life force | Hull and the reflexologists mechanistic; purposive and humanistic views resist mechanism |

| Issue | The two poles | Where the schools stand |
|---|---|---|
| Molar–molecular | Large meaningful units versus small elements | Gestalt and Tolman molar; structuralism and Watson molecular |
| Nomothetic–idiographic | General laws versus the unique individual | Experimental psychology nomothetic; Allport, case studies and humanists idiographic |
| Objectivism–subjectivism | Public observation versus private experience | Behaviourism objectivist; structuralism, phenomenology and humanism subjectivist |
| Empiricism–rationalism | Knowledge from experience versus from reason or innate ideas | Associationists and behaviourists empiricist; Descartes, Kant and Chomsky rationalist |

Two of these pairs have a named origin worth knowing. The terms **nomothetic** (seeking general laws, from the Greek *nomos*, law) and **idiographic** (studying the unique individual case, from *idios*, one's own) were coined by the German philosopher Wilhelm Windelband in 1894 and brought into psychology by Gordon Allport in 1937. And notice how one philosopher can feed opposite schools. Rene Descartes's mechanical account of the body's reflexes runs through Sechenov and Pavlov to behaviourism, while his doctrine of innate ideas runs through nativism to Gestalt psychology and Chomsky.

EXAM TRAP

**Two Watsons.** *John B. Watson* (1878 to 1958) founded behaviourism in 1913. *Robert I. Watson* (1909 to 1980) was a historian of psychology who proposed the eighteen **prescriptions** in 1967. A question about "Watson's prescriptions" means Robert. **Kuhn** (1962) gave us paradigm, normal science, paradigm shift and preparadigmatic science; **Popper** gave us falsifiability.

Has psychology, then, found its paradigm? Some historians describe the cognitive revolution as a Kuhnian paradigm shift, behaviourism overthrown by the new science of mind. Others, notably Thomas Leahey in a 1992 paper on "the mythical revolutions of American psychology", argue that nothing so sudden happened: behaviourism had already been absorbing cognitive ideas for decades, and psychology is better described as a field with several research traditions that coexist, borrow from one another and occasionally merge. Either way, the debate itself shows how far psychology has come since the days when a structuralist and a behaviourist could not agree on what counted as a fact.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Lashley's 1948 paper argued that rapid skilled sequences require central plans; its title referred to the problem of _____ in behaviour.
2. George Miller dated the birth of cognitive science to 11 September _____.
3. Miller's 1956 paper put the span of immediate memory at seven, plus or minus _____.
4. The first major theory framed in information-processing terms was published by _____ in 1958.
5. Chomsky's famous 1959 review attacked Skinner's book _____.
6. The 1967 textbook that named cognitive psychology was written by _____.
7. Bartlett's studies of "The War of the Ghosts" led him to describe memory as reconstruction guided by a _____.
8. Models in which knowledge is stored as weights across many simple units working at once are called connectionist or _____ models.
9. The biopsychosocial model was proposed by _____ in 1977.
10. The terms nomothetic and idiographic were brought into psychology by Gordon _____.

**Answers:** (1) serial order, (2) 1956, (3) two, (4) Donald Broadbent, (5) *Verbal Behavior*, (6) Ulric Neisser, (7) schema, (8) parallel distributed processing (PDP), (9) George Engel, (10) Allport.

## 2.8 Indian Psychology I: The Movement, Its Pioneers and Samkhya's Map of the Mind HIGH YIELD · 31 QS

Every school you have met so far was born in Europe or North America, and every one of them took for granted a particular picture of the person: an individual mind, studied from the outside, whose deepest questions could in principle be settled in a laboratory. At the end of Section 2.5 you saw one consequence. Critics charged that even humanistic psychology, for all its talk of the whole person, reflected a Western, individualist ideal of the self. Yet the questions psychology asks (what the mind is, why we suffer, how attention can be trained, what the self really is) were being asked in India more than two thousand years before Wundt opened his laboratory in Leipzig.

The *Bhagavad Gita* is the most famous example. It opens with a warrior, Arjuna, standing in his chariot between two armies. Facing his own teachers and cousins across the field, he tells his charioteer Krishna that his limbs are failing, his mouth is dry, his body trembles and his hair stands on end, that his bow is slipping from his hand and his mind is reeling. He sits down and refuses to fight, because every course open to him seems wrong. The rest of the poem is a dialogue in which Krishna helps him understand his own mind and find his way back to action. A modern reader can hardly miss that it has the shape of a counselling session with a man paralysed by anxiety and moral conflict.

This section and the next tell two connected stories. The first is how modern, Western-style psychology came to India in the early twentieth century, and how Indian psychologists gradually moved from copying it to questioning it. The second is what the classical Indian sys-

tems actually say about the mind. We begin by separating two phrases that sound almost the same.

## Psychology in India and Indian psychology ●●●

*Psychology in India* means the discipline as it is taught and practised in Indian universities, hospitals and clinics. For most of its first century it was Western psychology carried on in India: laboratories on the Leipzig model, theories from Europe and America, and tests translated or adapted for Indian use. Most of this book is psychology in that sense.

**Indian psychology**, in the sense used by the movement that took shape around the year 2000, is something else: an approach to psychology rooted in the ideas and practices of the classical Indian traditions (the Upanishads, Samkhya, Yoga, Vedanta, Buddhism and Jainism, the *Bhagavad Gita* and Ayurveda), joined in later centuries by Indian Sufism and, in the twentieth century, by Sri Aurobindo. It is not the psychology *of Indians* either; a study of how Indian students cope with examination stress belongs to cross-cultural or cultural psychology (Section 2.7). Its advocates stress that, like the psychology that came out of Germany, it is meant to be universal. Its claims about attention, the self and suffering are claims about human beings anywhere, which happen to have been worked out in India.

What makes the approach distinctive? Its leading scholars usually point to five features.

- **It is consciousness-centred.** Consciousness is treated as fundamental rather than as a by-product of the brain, and a sharp line is drawn between consciousness itself, the silent witness, and the mind's changing contents and instruments (thoughts, feelings, memories), which are treated as part of nature.
- **It is holistic.** Body, mind and spirit are treated as one system, and so are the person, the family and the community; health is a state of balance among them.
- **It is transformative.** The aim is not only to describe the mind but to change it, toward freedom from suffering and the realisation of one's true nature (called *moksha*, *kaivalya* or *nirvana* in different systems). Disciplines such as yoga and meditation are its laboratory techniques.
- **It values disciplined first-person methods.** Where Western psychology after Watson distrusted introspection, the Indian traditions developed elaborate methods for training and refining it, and they treat the reports of a trained observer as evidence.
- **It is relational and value-laden.** The self is understood in relation to others and to *dharma*, the order of duties that sustains social and moral life, so ethics is part of psychology rather than outside it.

Table 2.10 sets these features against mainstream Western psychology. Remember that the contrast is one of emphasis, not a wall. Behaviourists also wanted to change behaviour, humanists also cared about values, and over the last forty years Western psychology has borrowed meditation and mindfulness from India on a large scale (Section 2.9).

Table 2.10 Mainstream Western psychology and Indian psychology: differences of emphasis.

| Question | Mainstream Western psychology | Indian psychology |
|---|---|---|
| Starting point | Behaviour and mental processes | Consciousness and the self |
| Main method | Third-person observation, experiment and measurement | Disciplined first-person inquiry alongside observation |

| Question | Mainstream Western psychology | Indian psychology |
|---|---|---|
| What the mind is | A product of brain processes | Subtle "matter", part of nature, distinct from consciousness |
| Picture of the person | A separate individual | A person embedded in family, community and cosmos |
| Aim | To describe, explain, predict and control | To understand and to transform, toward self-realisation |
| Mental health | Diagnose and treat disorder | Restore balance through practice, values and therapy together |

How, then, does Indian psychology deal with mental illness? Not by rejecting modern treatment. Its practitioners combine traditional practices (yoga postures and breathing, meditation, a regulated daily routine and diet, reflection on the values of texts such as the *Gita*) with modern psychotherapy and, where needed, medication, and they aim at the well-being of the whole person rather than the removal of a symptom alone. The Ayurvedic and yogic ideas behind this approach are taught in Section 2.9.

EXAM TRAP

**The primary focus of Indian psychology** is the integration of Indian philosophical and contemplative traditions with psychological concepts and methods. It is *not* the exclusive application of Western theories (that is psychology in India at its most imitative), *not* the study of folklore and mythology, and *not* meditation without a psychological context. **How it differs from Western psychology**: it emphasises holistic and spiritual approaches, including family and community, rather than the isolated individual; it does not rely only on ancient texts without modern relevance, and it is Indian (not Western) psychology that stresses spirituality. **Atman** is the true self, eternal consciousness; not the physical body, not Jung's collective unconscious and not a passing state of mind.

## The pioneers: how modern psychology came to India ●●●

The story of psychology in India begins in Calcutta, then the capital of British India and home of its leading university. In 1905, on the initiative of the philosopher **Brajendranath Seal** (1864 to 1938), the University of Calcutta introduced experimental psychology as a subject within its philosophy curriculum. Seal, a scholar of both Western science and Indian philosophy, wanted Indian students to learn the new laboratory psychology. The university's vice-chancellor, the mathematician and lawyer **Asutosh Mookerjee** (1864 to 1924), backed him. What was missing was someone trained to run a laboratory.

That person was **Narendra Nath Sengupta** (1889 to 1944; the name is also written Sen Gupta). Sengupta had studied at Harvard under Hugo Munsterberg, who had himself taken his doctorate with Wundt at Leipzig, so there is a direct line of teachers running from Wundt's laboratory to Calcutta. In 1915 Sengupta set up the first psychology laboratory in India at the University of Calcutta, modelled on the German laboratories, and in 1916 the university created a separate Department of Psychology (often called the Department of Experimental Psychology), the first in the country, with Sengupta in charge. With the sociologist Radhakamal Mukerjee he wrote *Introduction to Social Psychology: Mind in Society* (1928), one of the first social psychology textbooks written by Indian authors, and he later moved to

Lucknow University. For his laboratory, his department and the students he trained, Sengupta is called the father, or founder, of modern psychology in India.

Among the first generation of Calcutta psychologists was a physician who would take the subject in a completely different direction. **Girindrasekhar Bose** (1887 to 1953; also written Girindra Shekhar Bose) had qualified in medicine and become fascinated by the mind through his own medical practice. His thesis on the concept of repression, published in 1921 as *Concept of Repression*, earned him what is usually described as the first doctorate in psychology awarded by an Indian university (some Indian study notes give 1916 or 1922 for this degree; 1921, the year of the thesis, is the date most histories give). He sent a copy to Freud, and the two corresponded for the next sixteen years or so. In 1922 Bose founded the *Indian Psychoanalytical Society* in Calcutta, one of the earliest psychoanalytic societies outside Europe and North America, and in 1947 he launched its journal, *Samiksha*. When Sengupta left Calcutta, Bose succeeded him as head of the psychology department. He is the father of psychoanalysis in India.

Bose did not simply import Freud. His best-known idea, the **theory of opposite wishes**, holds that every wish comes paired with its opposite: the wish to beat goes with the wish to be beaten, the wish to be male with the wish to be female, the wish to see with the wish to be seen. One member of each pair is expressed in consciousness and behaviour while the other is repressed, and the two can change places. For Bose it was this play of opposites, rather than a fixed Oedipus complex, that lay at the centre of mental life, and he reported that the Oedipal conflict did not hold the same central place in his Indian patients as in Freud's Viennese ones. Freud read his work with interest but was not persuaded. Bose's ideas show an Indian thinker reworking psychoanalysis through his own culture's preoccupation with the reconciliation of opposites, an early example of what would later be called indigenisation.

Institutions followed the pioneers. In 1923 the Indian Science Congress opened a section for psychology and educational science. In 1925 Sengupta founded the *Indian Psychological Association*, and in 1926 the association's *Indian Journal of Psychology*, the first Indian psychology journal, began publication. Universities across the country then opened departments of their own; Table 2.11 gives the main dates.

Table 2.11 Milestones of psychology in India.

| Year | Event | Person or place |
|---|---|---|
| 1905 | Experimental psychology enters the philosophy curriculum | University of Calcutta (Brajendranath Seal) |
| 1915 | First psychology laboratory in India | Calcutta (N. N. Sengupta) |
| 1916 | First Department of Psychology in India | Calcutta (N. N. Sengupta) |
| 1921 | *Concept of Repression*; first Indian doctorate in psychology | Girindrasekhar Bose |
| 1922 | Indian Psychoanalytical Society | Calcutta (G. Bose) |
| 1923 | Psychology section of the Indian Science Congress | Psychology with educational science |
| 1924 | Psychology department and laboratory at Mysore | M. V. Gopalaswamy |
| 1925 | Indian Psychological Association | Founded by N. N. Sengupta |

| Year | Event | Person or place |
|---|---|---|
| 1926 | *Indian Journal of Psychology* begins | Indian Psychological Association |
| 1943 | Department of Psychology | University of Madras |
| 1946 | Department of Psychology | Patna (H. P. Maiti) |
| 1947 | *Samiksha*, journal of the Indian Psychoanalytical Society | G. Bose |
| 1950 | Department of Psychology | University of Pune (V. K. Kothurkar) |
| 1954 to 1955 | Psychology department and first formal clinical psychology training | All India Institute of Mental Health, Bangalore (now NIMHANS) |

### INDIAN CONTEXT: MYSURU'S PLACE IN THE STORY

The second major centre of Indian psychology was Mysore (now Mysuru). In 1924 **M. V. Gopalaswamy**, a professor at the University of Mysore, established a psychology department with an experimental laboratory there, the second in the country after Calcutta. Gopalaswamy is remembered for something else too. In 1935 he set up a small radio transmitter at his home in Mysore and named his station *Akashvani*, "voice from the sky", the name that All India Radio later adopted for itself. The history of clinical psychology in India, which grew up a few hours away in Bangalore and at Ranchi, is told in the Clinical book.

### EXAM TRAP

**First laboratory and first department.** The first psychology laboratory in India was set up by **N. N. Sengupta at Calcutta in 1915**, and the first department followed there in **1916**. 1905 is the year psychology entered the Calcutta syllabus; 1922 is Bose's Psychoanalytical Society; 1924 is Mysore. An IHBAS item keys **Girindrasekhar Bose** as the founder of the first Indian psychological laboratory, and a Panwar item keys **Uma Chaudhuri**: both keys are wrong on the facts. If Sengupta is not among the options, Bose (the leading Calcutta pioneer) is the examiners' intended answer. S. K. Mangal and A. K. Singh are authors of widely used Indian textbooks, and neither they nor G. S. Rathi, the remaining option in that item, founded a laboratory.

**Father titles.** Father of modern psychology in India: **Sengupta** (not Bose, not Gopalaswamy). Father of psychoanalysis in India: **Bose** (not Freud, not Sengupta). Bose's own theory: **opposite wishes** (not repression, which is Freud's concept even though it was the subject of Bose's thesis; not *dharma-sankata*, which is the moral dilemma of Section 2.9; and "guilt theory" is not a named theory of Bose's at all). "Amitosh Mukherjee" in one item is a garbled form of Asutosh Mookerjee, the vice-chancellor who supported the Calcutta department, not a founder of psychology.

**Sequence.** On the commonly cited dates the order is Calcutta department (1916), Mysore (1924), *Indian Journal of Psychology* (1926), Madras (1943), Pune (1950). One Panwar item keys the order Calcutta, Journal, Mysore, Madras, Pune, placing the journal before Mysore; on the dates given here that key is wrong, so choose the option that follows the dates.

## From imitation to indigenisation

For its first fifty years, psychology in India was largely imitative. Indian psychologists, many trained in Britain or America, repeated Western experiments on Indian participants and used Western tests in translation. Did the results mean what they meant in London or Chicago? Increasingly, the answer seemed to be no.

The psychologist who did most to name the problem and chart a way out was **Durganand Sinha** (1922 to 1998) of the University of Allahabad. In *Psychology in a Third World Country: The Indian Experience* (1986) he described psychology in India as passing through four broad phases: a period before independence in which the Western academic and experimental tradition was transplanted largely unchanged; a period of rapid expansion after independence, still mostly imitative; a problem-oriented phase from about the 1970s, in which psychologists turned to India's own pressing problems (poverty, deprivation, rural development, social change); and finally a phase of **indigenisation**, the effort to make psychology's concepts, theories and methods fit the culture in which they are used. Indigenous psychologists distinguish two routes, a pair of terms usually credited to the Filipino psychologist Virgilio Enriquez and used by Sinha in his later writing. *Indigenisation from without* takes a Western concept or tool and modifies it to suit local conditions. *Indigenisation from within* builds concepts, theories and methods from the culture itself, from its traditions, texts, languages and everyday life.

Test adaptation is the clearest example of indigenisation from without. The pictures of the Thematic Apperception Test (TAT), drawn in the United States, showed American faces, clothes and settings, and Indian respondents told thin, guarded stories about them. **Uma Chowdhury** (the name is also spelled Chaudhuri) of Calcutta redrew the cards with Indian figures and scenes, producing an Indian modification of the TAT around 1960 (the tests themselves are covered in the Assessment book). Two contributions from organisational psychology show indigenisation moving further, toward concepts grown from Indian conditions.

**Jai B. P. Sinha**, studying Indian organisations in the 1970s, found that neither the authoritarian leadership common in Indian offices nor the participative style recommended by American research worked best. In *The Nurturant Task Leader* (1980) he proposed the **nurturant-task leader**: a leader who combines a strong task orientation (clear goals, discipline, insistence on hard work) with nurturance (warmth, personal care and guidance, interest in the subordinate's growth), and who makes that nurturance depend on the subordinate's work. The style fits a culture that values hierarchy and personal relationships, and Sinha saw it as a stage through which a work group could move toward fuller participation as subordinates matured (leadership styles in general are covered in the Social book). **Udai Pareek** (1925 to 2010) of the Indian Institute of Management, Ahmedabad, was a founder of human resource development in Indian organisations; he developed the concepts and scales of *role stress* (he identified ten organisational role stressors, among them role overload, role ambiguity and role stagnation) and *role efficacy*, and, with G. Trivedi, a widely used socio-economic status scale for rural India (1964).

Psychoanalysis, too, was reworked from within. **Sudhir Kakar** (born 1938), a psychoanalyst influenced by Erik Erikson, wrote *The Inner World: A Psycho-analytic Study of Childhood and Society in India* (1978), which traced how the Indian family, with its long and intense early bond between mother and son and its many caretakers in the joint family, shapes an identity more relational than the Western ideal of the autonomous individual. In *Shamans, Mystics*

*and Doctors* (1982) he studied India's traditional healers as psychotherapists in their own right. Kakar is the leading figure of a cultural psychoanalysis of India, the field Bose began.

EXAM TRAP

**Two Sinhas.** *Durganand Sinha* (1922 to 1998) = indigenisation and the four phases of psychology in India (1986). *Jai B. P. Sinha* = the nurturant-task leader (1980). **Kakar** = psychoanalysis applied to Indian childhood and society; when a Panwar item asks which Indian psychologist's work "aligns with Karen Horney's conception of neurosis", the key is Kakar because he is the only psychoanalyst among the options (Paranjpe, Kamala Chaudhuri, Durganand Sinha), not because he developed Horney's concept. **Indian TAT**: Uma Chowdhury (the "Kamala Chaudhuri" of one bank item is probably a garbled form of her name); the original TAT is Murray and Morgan's.

## The Indian Psychology movement ●●

Indigenisation asked psychology to fit Indian conditions. By the 1990s a growing number of psychologists wanted something bolder: to take the Indian traditions themselves seriously as sources of psychological theory and method, on an equal footing with the Western schools. The turning point came in 2002, when a national conference on yoga and Indian approaches to psychology, held in Pondicherry (now Puducherry), closed with a statement signed by the psychologists present, the **Pondicherry Manifesto of Indian Psychology**. The manifesto argued that Indian psychologists had too long confined themselves to replicating Western research, that the Indian traditions contain a rich and systematic understanding of consciousness and the self, together with methods for exploring and transforming them, and that this knowledge should be developed into a modern psychology useful anywhere in the world. Notice that Figure 2.1 marks 2002 as the start of the Indian Psychology movement.

Several people carried the movement forward. **Matthijs Cornelissen**, a Dutch-born physician and psychologist long resident at the Sri Aurobindo Ashram, was one of its prime movers and founded the *Indian Psychology Institute* in Pondicherry, which runs courses and workshops grounded largely in Sri Aurobindo's integral psychology. **Koneru Ramakrishna Rao** (born 1932), a philosopher and parapsychologist and a former vice-chancellor of Andhra University, became its most prolific theorist; with **Anand C. Paranjpe** of Simon Fraser University in Canada, author of *Self and Identity in Modern Psychology and Indian Thought* (1998), and **Ajit K. Dalal** of the University of Allahabad, he edited the *Handbook of Indian Psychology* (2008), the movement's first comprehensive reference work, and with Paranjpe he later wrote *Psychology in the Indian Tradition* (2016). Cornelissen, **Girishwar Misra** and **Suneet Varma**, both of the University of Delhi, edited the two-volume *Foundations of Indian Psychology* (2011), one volume on theories and concepts and one on practical applications. Misra is also one of the most influential scholars of cultural and indigenous psychology in India.

The movement has its critics. Some worry that it romanticises the past, that it blurs the line between religion and science, or that claims about pure consciousness cannot be tested. Its defenders answer that the first-person methods of yoga and Buddhism can be made as disciplined as any laboratory technique, that many traditional claims (about meditation, breathing and attention, for example) are now being tested experimentally, and that Western psy-

chology's own assumptions are no less culture-bound for being familiar. The debate is a live one, and you will find both sides in Indian journals today.

EXAM TRAP

**Who led the early-2000s Indian Psychology movement and founded the Indian Psychology Institute?** Matthijs Cornelissen (not Kakar, Durganand Sinha or Paranjpe). **Handbook of Indian Psychology**: 2008 (Rao, Paranjpe and Dalal); 2002 is the Pondicherry Manifesto; 2011 is *Foundations of Indian Psychology*. "**Father of Indian psychology**" is ambiguous. If the question means modern scientific psychology in India, the answer is Sengupta. An RML 2025 item instead keys "G. R. S. Rao", a name that matches no well-known figure; it is most plausibly a garbled reference to K. Ramakrishna Rao, the leading theorist of the Indian Psychology movement. If a Rao appears beside Sengupta in a question about Indian Psychology as a movement, the Rao is the intended answer; if the question is about laboratories and departments, choose Sengupta.

## The darshanas: a map of Indian thought ●●

Before we open the first of the classical systems, you need a map. Indian philosophy is traditionally divided into schools called **darshanas**, from a root meaning "to see": each is a way of seeing reality. The schools are sorted into two groups by one criterion, their attitude to the authority of the Vedas, the oldest Indian scriptures. The **astika** (orthodox) schools accept that authority; the **nastika** (heterodox) schools reject it. Notice what the criterion is not: it is not belief in God. Classical Samkhya has no creator god, yet it is astika, while Buddhism and Jainism are nastika because they reject the Vedas. Table 2.12 lists the schools; the six astika darshanas are traditionally grouped into three pairs.

Table 2.12 The schools of classical Indian philosophy.

| Group | School | Traditional founder or text | Central concern |
|---|---|---|---|
| Astika | Nyaya | Gautama (Akshapada) | Logic and the means of valid knowledge |
| Astika | Vaisheshika | Kanada | Atoms and the categories of reality |
| Astika | Samkhya | Kapila | Dualism of consciousness (purusha) and matter (prakriti) |
| Astika | Yoga | Patanjali | Discipline of the mind, leading to liberation |
| Astika | Mimamsa (Purva Mimamsa) | Jaimini | Vedic ritual and duty (dharma) |
| Astika | Vedanta (Uttara Mimamsa) | Badarayana's *Brahma Sutras*; the Upanishads | Brahman, the atman and liberation |
| Nastika | Buddhism | Siddhartha Gautama, the Buddha | Suffering, its cause and its end; no permanent self |
| Nastika | Jainism | Mahavira, the 24th tirthankara | Non-violence, karma and the liberation of the soul |
| Nastika | Charvaka (Lokayata) | Brihaspati (traditional) | Materialism: only perception is valid; no soul beyond the body |

Most of these schools also argued carefully about how we know anything at all. The valid means of knowledge are called *pramanas*. Samkhya and Yoga accept three: *pratyaksha* (per-

ception), *anumana* (inference) and *shabda* or *agama* (reliable testimony). Nyaya adds *upamana* (comparison), and the Charvakas, the materialists of ancient India, accepted perception alone. Indian psychology thus has its own epistemology, its own theory of how knowledge is acquired, built into its foundations. Of all these schools, the one that matters most for psychology is the one whose map of the mind most of the others borrowed: Samkhya.

> EXAM TRAP
>
> **Which is an astika system?** Advaita Vedanta (and Samkhya, Yoga, Nyaya, Vaisheshika, Mimamsa). Buddhism, Jainism and Charvaka are nastika. Sufism is the mystical tradition of Islam and lies outside this classification altogether. **Where do the three gunas come from?** Mainly **Samkhya** (with the *Bhagavad Gita*), not Nyaya (logic), Mimamsa (ritual) or Charvaka (materialism).

## Samkhya: purusha and prakriti ●●●

**Samkhya** is the oldest of the systematic Indian philosophies of mind. Its name means "enumeration", because it explains the world by counting out its basic principles, twenty-five of them. Tradition names its founder as the sage **Kapila**, of whom nothing certain is known. Its oldest surviving text is the *Samkhya Karika* of **Ishvarakrishna**, a short work of about seventy verses usually dated to around the fourth or fifth century CE. Its ideas, however, are much older; they run through the Upanishads, the *Mahabharata* and the *Gita*, and the Yoga of Patanjali takes them over almost entire.

The *Samkhya Karika* begins, as any clinician might, with suffering. Human beings, it says, are tormented by three kinds of suffering: suffering from within ourselves, in body and mind (*adhyatmika*); suffering caused by other beings, people and animals (*adhibhautika*); and suffering from natural and supernatural forces such as floods and storms (*adhidaivika*). Ordinary remedies, medicine or ritual, relieve suffering only for a while. The aim of Samkhya is a lasting cure, and the cure is knowledge of what we really are.

What are we? Samkhya's answer is a strict dualism of two ultimate, eternal and independent realities.

The first, **purusha**, is pure consciousness: the seer, the witness (*sakshi*). It does not act, change, think, feel or choose; it simply is aware. It has no qualities, it is neither produced nor productive, and it is untouched by anything that happens. Samkhya holds that there are many purushas, one for each living being, because beings are born, suffer and die at different times. The second, **prakriti**, is primordial matter or nature: unconscious, active, ever-changing and one. It is the uncaused cause from which everything that is not consciousness evolves, and it is made of three strands or qualities, the gunas, which in its unmanifest state are held in perfect balance. Samkhya explains how the world arises from prakriti by its doctrine that an effect already exists, unmanifest, in its cause (*satkaryavada*), as curd exists potentially in milk or oil in sesame seeds. Evolution is the manifestation of what was latent.

When prakriti comes into the presence of purusha, its balance is disturbed and it begins to evolve, unfolding in a fixed order into twenty-three products, or evolutes. The first is *mahat*, "the great one", which in the individual is **buddhi**, the intellect. From buddhi comes **ahamkara**, the "I-maker", the sense of being a separate self. Ahamkara then evolves in two directions. On its sattvic (luminous) side it produces **manas**, the sensory mind, together with

the five organs of perception (*jnanendriyas*: hearing, touch, sight, taste and smell) and the five organs of action (*karmendriyas*: speech, grasping, movement, excretion and reproduction). On its tamasic (inert) side it produces the five subtle elements (*tanmatras*: sound, touch, form, taste and smell as pure potentials), and from them the five gross elements (*mahabhutas*: space, air, fire, water and earth), of which the physical world, including the body, is made. Rajas, the active strand, supplies the energy for both lines of evolution. Purusha, prakriti and the twenty-three evolutes together make the twenty-five principles (*tattvas*) of Samkhya. Figure 2.4 lays out the whole scheme.

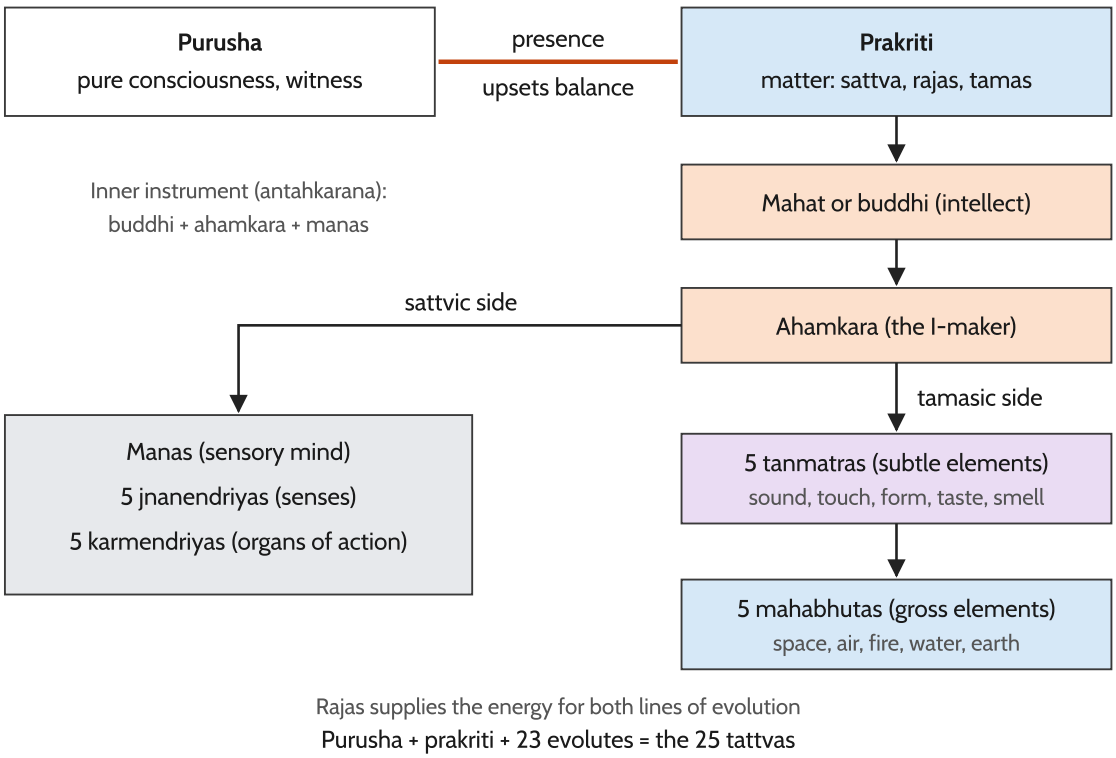


**Figure 2.4** Samkhya's map of mind and matter. In the presence of purusha, prakriti evolves into intellect, the I-maker, the sensory mind, the ten organs and the elements. Notice that everything below the top row, including thinking, feeling and the sense of "I", is a product of prakriti; only purusha is conscious.

Look at Figure 2.4 again and notice something that surprises most Western readers. Intellect, ego and mind, everything a Western psychologist would call mental, are all products of prakriti. They are part of nature, as material as the body, only subtler. Descartes drew the great dividing line between mind and body; Samkhya draws it between consciousness and everything else, with the mind on the side of matter. Why, then, does the mind seem conscious? Because, Samkhya and Yoga answer, it reflects the light of purusha, as a clear crystal placed beside a red flower appears red. Thinking, deciding and feeling are processes in prakriti; the awareness *of* them belongs to purusha alone. In modern terms, Samkhya separates the contents and operations of the mind, which cognitive psychology studies, from awareness itself, the question at the heart of what philosophers now call the hard problem of consciousness (Chapter 5).

Two images from the *Samkhya Karika* show how the system pictures their relation. Purusha and prakriti are like a lame man who can see and a blind man who can walk: the lame man rides on the blind man's shoulders, and together they find their way. And prakriti is like a

dancer who performs for an audience and stops once she has been fully seen. Why do we suffer? Because of *aviveka*, non-discrimination: consciousness mistakes the activities of prakriti for itself, so that the witness says "I am angry", "I am tired", "I am a failure", when anger, tiredness and failure belong to the body and mind. The cure is *viveka*, discriminative knowledge, the direct realisation that the seer is not the seen. The result is **kaivalya**, "aloneness" or isolation: purusha rests in its own nature, free, and prakriti ceases to act for it. The same word names the goal of Patanjali's yoga (Section 2.9).

> EXAM TRAP
>
> **Purusha is passive, unchanging and conscious; prakriti is active, ever-changing and unconscious.** So "Purusha is described as dynamic and ever-changing" is the false statement in a list about Samkhya; "prakriti is the cosmic material from which all beings emerge", "purusha is the essence of pure consciousness" and "prakriti evolves into the five great elements (pancha mahabhutas)" are all true. Prakriti is the primordial matter from which all forms of life emerge; it is *not* pure consciousness, *not* unaffected by change and *not* the intelligent self. **The gunas belong to prakriti**, not to purusha, atman or Brahman, and not to samadhi or samskara. **Ahamkara** is an evolute of prakriti, not a second principle beside it. Samkhya is **dualist** (many purushas and one prakriti); Advaita Vedanta is **monist** (Section 2.9).

## The three gunas ●●●

If everything in nature, including the mind, is made of prakriti, and prakriti is made of three strands, then every thought, feeling and act is some mixture of the three. This is the doctrine of the three **gunas**, the best-known idea in Indian psychology. The word *guna* means a strand, as in the strands twisted together to make a rope, and also a quality. The three strands are always present together and never found alone, in different proportions in different things and at different times. Like the oil, the wick and the flame of a lamp, the *Samkhya Karika* says, they are opposed in nature yet work together to produce one light.

The first, **sattva**, is the strand of lightness, clarity and illumination. In the mind it shows itself as calm, balance, harmony, knowledge, contentment and goodness. The second, **rajas**, is the strand of energy, motion and stimulation. In the mind it shows itself as activity, passion, desire, ambition and restlessness, and, when it runs to excess, as anger, greed, anxiety and attachment. The third, **tamas**, is the strand of heaviness, inertia and obstruction, which covers and darkens. In the mind it shows itself as dullness, lethargy, sleepiness, confusion, ignorance and delusion. The *Samkhya Karika* links the three to pleasure, pain and dejection respectively, and to three functions, illumination, activity and restraint. A traditional colour code, found as early as the *Shvetashvatara Upanishad*, makes sattva white, rajas red and tamas black.

It is the *Bhagavad Gita* that turned the gunas into a full psychology of personality. The *Gita* is a poem of seven hundred verses in eighteen chapters, set within the epic *Mahabharata*; its fourteenth chapter is devoted to the gunas, and chapters 17 and 18 classify faith, food, worship, charity, knowledge, action, the doer, the intellect, resolve and happiness into sattvic, rajasic and tamasic kinds. The gunas, Krishna tells Arjuna, are born of prakriti and bind the embodied self: sattva binds by attachment to happiness and knowledge, rajas by attachment to action, born of craving, and tamas by heedlessness, laziness and sleep. From sattva comes

knowledge, from rajas greed, and from tamas negligence, delusion and ignorance. Table 2.13 gathers their signatures.

**Table 2.13** The three gunas and their behavioural signatures.

| Feature | Sattva | Rajas | Tamas |
|---|---|---|---|
| Quality in nature | Light, clear, illuminating | Active, mobile, stimulating | Heavy, inert, covering |
| Feeling | Calm, contentment, joy | Passion, desire, restlessness | Dullness, dejection, apathy |
| Thinking | Clarity, knowledge, discernment | Driven, scattered, ambitious | Confusion, ignorance, delusion |
| Behaviour | Balanced, self-controlled, helpful | Busy, competitive; under stress angry or greedy | Lazy, negligent, oversleeping, withdrawn |
| Binds by (*Gita* 14) | Attachment to happiness and knowledge | Attachment to action, from craving | Heedlessness, laziness and sleep |
| Food (*Gita* 17) | Fresh, wholesome, agreeable | Very bitter, sour, salty, hot, pungent | Stale, tasteless, spoiled, leftover |
| Happiness (*Gita* 18) | Like poison at first, nectar in the end | Like nectar at first, poison in the end | Delusion from beginning to end |
| Traditional colour | White | Red | Black |

A worked example shows how the scheme is used. Three friends in Mysuru face the same entrance examination next week. Ananya has a revision plan, keeps to it, eats simply, sleeps on time and feels a quiet confidence: sattva predominates. Rohan has bought six guidebooks, switches between them every hour, drinks tea after tea, argues with his roommate and lies awake with his heart racing: rajas predominates. Karthik keeps meaning to start, sleeps until noon, scrolls his phone in bed and tells himself there is no point: tamas predominates. Now apply the Gita's three kinds of happiness. Ananya's hours of revision are hard at first and sweet at the end, when the material falls into place; Rohan's late-night binge of videos is sweet at first and bitter the next morning; Karthik's oversleeping is dull and self-deceiving from start to finish. Notice two further points. First, nobody is purely one guna; the gunas describe proportions, and the same person can be sattvic in the morning, rajasic at noon and tamasic after a heavy lunch. Second, the scheme is not only descriptive but practical, since diet, routine, company and practice can shift the balance toward sattva. The Gita's ideal is the person who has gone beyond all three, the *gunatita*, who stays even-minded in pleasure and pain, praise and blame, honour and dishonour.

### CLINICAL LINK

The gunas map surprisingly well onto clinical pictures. The person who has stopped eating properly, withdrawn from others and cannot get herself to do anything is in a predominantly tamasic state, which resembles the psychomotor slowing and anergia of severe depression; restless agitation, racing thoughts, irritability and craving are rajasic, as in anxiety and mania. Researchers have built questionnaires to measure the three gunas (David Wolf's Vedic Personality Inventory, from the late 1990s, is one), and in such studies higher sattva scores tend to go with better well-being, and higher rajas and tamas scores with more distress. The disorders themselves are taught in the Disorders book.

EXAM TRAP

**Sattva** = harmony, balance, clarity, purity (not passion and desire, which is rajas; not inertia, ignorance, darkness or confusion, which are tamas). **Rajas** = restlessness, passion, desire, over-activity. **Tamas** = lethargy, inertia, darkness, dullness, ignorance. The stem "Bavisha lost her job, stopped eating properly and cannot get herself to do anything" points to **tamas alone**; *yama* is the first limb of yoga (an ethical discipline, not a mental state), *pratyahara* is sense withdrawal and *jnana* is knowledge (Section 2.9). The three gunas are **sattva, rajas and tamas**. Distractors that are *not* gunas: **vata** (an Ayurvedic dosha), **ahimsa** (non-violence, a yama), **prakriti** (the matter the gunas compose), **ahamkara** (an evolute, the I-maker), and the triads "atman, Brahman, maya", "karma, dharma, moksha" and "prakriti, purusha, kshetra".

## The antahkarana: the inner instrument ●●

Samkhya gives a name to the part of prakriti that does the work of the mind. The **antahkarana**, literally the "inner instrument", is contrasted with the ten "outer instruments", the five senses and the five organs of action. The *Samkhya Karika* says the inner instrument is threefold (intellect, I-maker and mind) and the outer tenfold. The *Bhagavad Gita* uses the same three: Krishna lists earth, water, fire, air, space, mind, intellect and ego as the eightfold division of his lower nature, prakriti. Later Vedanta added a fourth function, *chitta*, giving the four-part antahkarana of most modern accounts. Each function deserves a careful definition, because the four are easily confused.

The first function, **manas**, is the sensory mind. It receives and coordinates the impressions brought by the senses, directs attention to them, and controls the organs of action. It is also the seat of desire and of doubt: manas weighs alternatives ("is it this or that?") without being able to settle them. The second, **ahamkara**, is the I-maker. It appropriates experience as "mine": "I see", "I want", "this is *my* pen". It is the source of individuality, and in excess of egoism and pride. The third, **buddhi**, is the intellect. It ascertains, judges, decides and discriminates; it is the highest and most sattvic of the evolutes, the one closest to purusha and the one whose clarity makes liberation possible. The fourth, **chitta**, in the fourfold scheme, is the storehouse of memory, holding the latent impressions (*samskaras*) left by every experience. (In Patanjali's Yoga, as you will see, *chitta* has a wider sense: the whole mind-stuff, all three inner instruments together.) Table 2.14 summarises the four.

Table 2.14 The four functions of the antahkarana.

| Function | Usual translation | What it does |
|---|---|---|
| Manas | Sensory mind | Receives and coordinates sense impressions, attends, desires, doubts and deliberates |
| Ahamkara | Ego, the I-maker | Refers experience to a separate self: "I", "me", "mine" |
| Buddhi | Intellect | Decides, judges, discriminates, knows with certainty |
| Chitta | Memory store | Holds the impressions (samskaras) of past experience |

Push one moment through the scheme. You are writing an examination in a silent hall when a phone rings somewhere behind you. Your ears register the sound and *manas* takes it up: a sound, a ring, a phone, should I attend or ignore it? *Chitta* supplies the memory: that tune is familiar. *Ahamkara* claims it: "that is *my* ringtone!" *Buddhi* settles the matter: "my phone is

switched off in my bag outside the hall, so it cannot be mine; keep writing". And *manas* sends your hand back to the answer sheet. If you compare this with the information-processing account of the batsman in Section 2.6 (sensory input, recognition from long-term memory, decision, response), the parallel is striking. The difference lies at the very end. For the Indian systems, none of these four functions is the knower. The awareness that witnessed the whole episode belongs to none of them; it belongs to purusha, or, in Vedanta, to the atman.

EXAM TRAP

**Which function is the ego, the I-maker?** Ahamkara (not manas, the sensory mind; not buddhi, the intellect; not chitta, memory). **The Gita's three elements of mind** are manas, buddhi and ahamkara, not "prakriti, purusha, kshetra" (*kshetra*, the "field", is the Gita's word for the body and nature that the self knows), not the purusharthas "dharma, artha, kama", and not the gunas "sattva, rajas, tamas". Do not equate ahamkara with Freud's ego: the Freudian ego is an executive that tests reality, a job closer to buddhi; ahamkara is simply the sense of "I". Some popular study notes, following modern teachers, describe manas as memory and chitta as consciousness; the classical Vedanta assignment is the one in Table 2.14, and ahamkara is the ego in every version.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. An approach to psychology rooted in classical Indian thought and practice is called Indian psychology, as distinct from psychology _____ India.
2. The first psychology laboratory in India was set up at Calcutta in 1915 by _____.
3. Girindrasekhar Bose founded the Indian Psychoanalytical Society in _____ and proposed the theory of _____ wishes.
4. Durganand Sinha's effort to make psychology fit Indian culture is called _____.
5. Jai B. P. Sinha's leader who combines task orientation with contingent care is the _____ leader.
6. The 2002 statement that launched the Indian Psychology movement is the _____ Manifesto.
7. Schools of Indian philosophy that accept the authority of the Vedas are called _____.
8. In Samkhya, pure consciousness is _____ and primordial matter is _____.
9. The guna of inertia, dullness and ignorance is _____.
10. The function of the antahkarana that decides and discriminates is _____.

**Answers:** (1) in, (2) Narendra Nath Sengupta, (3) 1922, opposite, (4) indigenisation, (5) nurturant-task, (6) Pondicherry, (7) astika, (8) purusha, prakriti, (9) tamas, (10) buddhi.

## 2.9 Indian Psychology II: Yoga, the Self and the Paths of Transformation

HIGH YIELD · 11 QS

Samkhya told you what the mind is made of. The question a clinician asks next is practical: can the mind be changed, and if so, how? The Indian traditions answered with methods, and

the most systematic of them is yoga. In September 1893 a young monk from Calcutta, Swami Vivekananda, addressed the World's Parliament of Religions in Chicago and became famous overnight; three years later he published *Raja Yoga* (1896), which presented Patanjali's yoga to Western readers not as a religion or a set of exercises but as a science of the mind. This section follows his lead. It begins with yoga, moves to the Indian theories of the self and its states and sheaths, and then surveys the other great paths of transformation: Buddhist, Sufi and integral psychology, Ayurveda, and the ideas of karma and dharma.

### Patanjali's Yoga Sutras: stilling the mind ●●

The *Yoga Sutras* of **Patanjali** are a collection of 195 or 196 aphorisms (the count varies slightly between editions), arranged in four chapters: on absorption (*samadhi*), on practice (*sadhana*), on the extraordinary powers that practice brings (*vibhuti*), and on liberation (*kaivalya*). A *sutra* is literally a thread, a compressed statement meant to be memorised and unpacked by a teacher, and the oldest surviving commentary, the *Yoga Bhashya* attributed to Vyasa, has guided readers ever since. Nothing reliable is known about Patanjali the man, and scholars date the compilation anywhere between about the second century BCE and the fourth century CE. Yoga adopts Samkhya's map of purusha and prakriti almost unchanged, adding one thing: *Ishvara*, a special purusha forever untouched by affliction and karma, who can serve as an object of devotion. For this reason Yoga is sometimes called "Samkhya with God".

The second sutra defines the whole discipline in four words: *yogas chitta vritti nirodhah*. Take them one at a time. *Yoga* is the subject. *Chitta* is the mind-stuff, the whole inner instrument of Section 2.8. *Vritti* means a whirl or fluctuation, any modification of the mind: a perception, a thought, a memory, a mood. *Nirodha* means restraint, stilling or cessation. So **chitta-vritti-nirodha**, yoga itself, is the stilling of the fluctuations of the mind. The next two sutras give the reason. When the fluctuations are stilled, "the seer abides in its own nature"; at other times the seer identifies with the fluctuations. The classic image is a lake. When wind ripples its surface, you cannot see the bottom, and the reflection of the moon breaks into a thousand pieces; when the water is still, the bottom is visible and the reflection whole. The goal of the stilling is not a blank mind for its own sake but liberation, *kaivalya*: consciousness resting in itself, no longer confused with the mind's activity. Notice how far this is from the modern image of yoga as physical fitness. Of the 195 sutras, only three deal with posture.

What are the fluctuations? Patanjali lists five kinds of **vritti**, each of which can be *klishta* (afflicted, leading to suffering) or *aklishta* (unafflicted).

1. *Pramana*, valid cognition, which comes from perception, inference and reliable testimony (the three pramanas of Section 2.8).
2. *Viparyaya*, error: false knowledge that does not correspond to the real form of its object, like taking a coiled rope in the dusk for a snake.
3. *Vikalpa*, verbal conceptualisation: an idea that follows words but has no real object, such as "the horn of a hare" or the abstractions we think with but never meet.
4. *Nidra*, sleep: a fluctuation whose content is the absence of other contents.
5. *Smriti*, memory: the retention of an object once experienced.

Read the list again with a modern eye: perception and reasoning, illusion, imagination and abstraction, sleep, memory. It is very nearly the table of contents of a general psychology textbook, drawn up some two thousand years ago.

Patanjali's commentator Vyasa added a description of five levels (*bhumis*) of the mind, which reads like a scale of attentional control. The *kshipta* mind is restless and scattered, driven by rajas; the *mudha* mind is dull and stupefied, sunk in tamas; the *vikshipta* mind is distracted but occasionally steady, the state of most of us most of the time; the *ekagra* mind is one-pointed, held on a single object; and the *niruddha* mind is fully restrained and still. Yoga proper begins at the one-pointed level and culminates in the restrained one.

How is the stilling achieved? Through two means that work together (sutra 1.12). The first, **abhyasa**, is practice: repeated effort to rest in stillness, which becomes firmly grounded only when it is kept up for a long time, without interruption and with devotion. The second, **vairagya**, is non-attachment or dispassion: freedom from craving for objects, whether seen or merely heard of. Practice supplies direction; non-attachment removes the pull of distraction. A modern psychologist would call the first sustained training of attention and the second a kind of acceptance, the willingness to let urges arise and pass without acting on them.

Why does the mind keep fluctuating? Because of five afflictions, the **kleshas**, which Patanjali lists in the second chapter.

1. *Avidya*, ignorance: taking the impermanent for permanent, the impure for pure, the painful for pleasant and the not-self for the self. It is the field in which all the others grow.
2. *Asmita*, "I-am-ness" or egoism: identifying the seer with the instruments of seeing, so that one says "I am this body, these thoughts".
3. *Raga*, attachment: the craving that follows pleasure.
4. *Dvesha*, aversion: the repulsion that follows pain.
5. *Abhinivesha*, clinging to life: the fear of death, which, Patanjali notes, is present even in the wise.

The afflictions may be dormant, weakened, interrupted or fully active. They generate actions, actions leave latent impressions (*samskaras*) in the chitta, and the impressions give rise to new fluctuations, a self-perpetuating cycle. The cycle can be weakened by *kriya yoga*, the "yoga of action", which consists of discipline (*tapas*), self-study (*svadhyaya*) and surrender to Ishvara (*ishvara pranidhana*), and the subtler impressions can be dissolved by meditation.

CLINICAL LINK

Two sutras anticipate modern therapy. Sutra 2.33 advises that when the mind is disturbed by harmful thoughts, one should cultivate their opposites (*pratipaksha bhavana*), which is close in spirit to the cognitive therapist's practice of identifying an unhelpful thought and building a balanced alternative (Section 2.6). Sutra 1.33 recommends friendliness toward the happy, compassion toward the suffering, gladness toward the virtuous and equanimity toward the wrongdoer as ways to make the mind clear, attitudes that compassion-focused therapies now teach deliberately. The therapies are taught in the Clinical book.

EXAM TRAP

**The primary goal of yoga according to Patanjali** is control of the mind (chitta-vritti-nirodha) leading to liberation (kaivalya): not physical fitness, not cognitive enhancement and not "emotional regulation only", although yoga may bring all three. **Do not mix the two lists of five.** The five *vrittis* (pramana, viparyaya, vikalpa, nidra, smriti) are kinds of mental activity; the five *kleshas* (avidya, asmita, raga, dvesha, abhinivesha) are afflictions that drive them. **Abhyasa** = practice; **vairagya** = non-attachment.

## The eight limbs of yoga ●●●

The method itself is set out in sutra 2.29 as **ashtanga yoga**, the yoga of eight limbs (*ashta*, eight; *anga*, limb). The order matters, because each limb prepares for the next, moving from conduct toward others, through the body and breath, to the innermost work of attention. Table 2.15 gives the eight with Patanjali's own definitions.

Table 2.15 The eight limbs of Patanjali's yoga, in order.

| Limb | Translation | What Patanjali means |
|---|---|---|
| 1. Yama | Restraints | Five ethical restraints toward others: non-violence, truthfulness, non-stealing, sexual restraint, non-possessiveness |
| 2. Niyama | Observances | Five personal disciplines: purity, contentment, austerity, self-study, surrender to Ishvara |
| 3. Asana | Posture | A seated posture that is steady and comfortable |
| 4. Pranayama | Breath regulation | Regulating the flow of inhalation and exhalation |
| 5. Pratyahara | Withdrawal of the senses | The senses withdraw from their objects and follow the mind inward |
| 6. Dharana | Concentration | Binding the mind to one place or object |
| 7. Dhyana | Meditation | An unbroken flow of awareness toward that object |
| 8. Samadhi | Absorption | The object alone shines forth, the mind as if empty of its own form |

The first two limbs set the ethical foundation. The **yamas** are *ahimsa* (non-violence), *satya* (truthfulness), *asteya* (non-stealing), *brahmacharya* (sexual restraint) and *aparigraha* (non-possessiveness); when they are kept universally, regardless of time, place and circumstance, Patanjali calls them the "great vow". The **niyamas** are *shaucha* (purity), *santosha* (contentment), *tapas* (austerity or disciplined effort), *svadhyaya* (self-study, including study of the scriptures) and *ishvara pranidhana* (surrender to Ishvara). Why begin with ethics? Because a mind full of conflict with others, guilt and greed cannot be stilled by any technique; the yamas and niyamas remove the largest sources of disturbance first. Asana and pranayama then steady the body and the breath.

The fifth limb, **pratyahara**, is the hinge of the system. The word means drawing back (*prati*, against, and *ahara*, food or what is taken in): the senses stop "feeding" on their objects and are drawn inward. The *Gita*'s image is a tortoise drawing its limbs into its shell. With pratyahara, attention turns from the outer world to the inner. The last three limbs are the stages of meditation proper. The first, **dharana**, is concentration: fixing the mind on one point, an object, the breath, a mantra, and bringing it back each time it wanders. The second, **dhyana**, is med-

itation: the concentration becomes an unbroken flow, so that it no longer needs to be dragged back. The third, **samadhi**, is absorption: the sense of a separate meditator falls away and only the object remains in awareness. Patanjali calls the three together, practised on one object, *samyama*.

The distinctions between the last four limbs are easy to blur, so push one practice session through them. A student sits to meditate on a candle flame in a hostel room in Mysuru. She stops her attention being pulled by the traffic outside, the smell of the mess and her phone, and turns her senses inward: that is pratyahara. She fixes her gaze and attention on the flame and, each time her mind drifts to tomorrow's class, gently brings it back: dharana. After some minutes the effort drops away and attention flows steadily to the flame without breaks: dhyana. For a moment even the sense "I am meditating" disappears and there is only the flame: samadhi.

Patanjali divides the limbs into two groups. The first five are *bahiranga*, the outer limbs, and the last three, dharana, dhyana and samadhi, are *antaranga*, the inner limbs; pratyahara is the bridge between them. Samadhi itself has grades. In *samprajnata* samadhi the mind is absorbed in an object and still has cognitive content; in *asamprajnata* samadhi there is no object at all, only the latent impressions remain, and these too are finally dissolved. The end is *kaivalya* in its yogic sense: the gunas, their purpose fulfilled, return to their source, and purusha is established in its own nature. A simple sentence will help you remember the order of the eight: "You Need A Peaceful Place; Do Daily Sitting" (yama, niyama, asana, pranayama, pratyahara, dharana, dhyana, samadhi).

CLINICAL LINK

Yoga has become one of India's most studied contributions to mental health, and since 2015 the International Day of Yoga has been observed every 21 June. At NIMHANS in Bengaluru, controlled trials have tested yoga as an add-on to medication for schizophrenia and depression, with encouraging effects on symptoms and functioning, and breathing practices drawn from pranayama are widely used for anxiety. The research question is always the same as for any treatment: does it work better than a credible control? Yoga is used alongside, not instead of, standard care.

EXAM TRAP

**Order of the last four limbs**: pratyahara, dharana, dhyana, samadhi (not dharana before pratyahara, and not dhyana before dharana). **Concentration on a single point** = dharana; **unbroken flow of meditation** = dhyana; **turning the senses inward, away from sounds, smells and sights** = pratyahara; **breath regulation** = pranayama; **personal observances** = niyama. **Karma is not one of the eight limbs** (karma yoga is a path in the Gita), and neither is **jnana** (knowledge). Bahiranga = the first five limbs; antaranga = the last three.

## The paths of the Gita: action, knowledge and devotion

Patanjali's way is the way of meditation. The *Bhagavad Gita* recognises that people differ, and offers several paths suited to different temperaments. The first, **karma yoga**, is the path of action: doing one's work and duty in the right spirit. *Jnana yoga* is the path of knowledge: discriminating the self from the not-self through study and reflection. *Bhakti yoga* is the path of devotion: love and surrender to God. The sixth chapter of the Gita also describes *dhyana yoga*,

the path of meditation, close to Patanjali's. Vivekananda later presented these as four yogas, karma, jnana, bhakti and raja (his name for Patanjali's yoga). *Hatha yoga*, the yoga of postures, breathing and cleansing practices systematised in medieval texts such as the *Hatha Yoga Pradipika* of Svatmarama (fifteenth century), is the ancestor of most yoga classes today.

At the heart of the *Gita*'s psychology of action is one famous verse (2.47): you have a right to your action alone, never to its fruits; do not let the fruits of action be your motive, and do not be attached to inaction either. This is the principle of **nishkama karma**, desireless or disinterested action: acting fully and well, as one's duty, without attachment to the outcome. The next verses define yoga as evenness of mind in success and failure and as "skill in action". Psychologically, nishkama karma is a cure for the anxiety that comes from fixing one's mind on results one cannot control. Sport psychologists who tell an athlete to focus on the process rather than the scoreboard, and therapists who help clients act on their values whatever the outcome, are teaching something very similar.

The *Gita* also offers one of the earliest analyses of how a passing thought can grow into self-destruction (2.62 to 2.63). Brooding on objects of the senses produces attachment; from attachment arises desire; from frustrated desire, anger; from anger, delusion; from delusion, the loss of memory (of what one knows to be right); from the loss of memory, the destruction of discrimination (buddhi); and with the destruction of buddhi, a person is lost. Set it beside a cognitive therapist's account of how an angry episode escalates and the resemblance is plain. Its ideal is the *sthitaprajna*, the person of steady wisdom described at the end of the second chapter, who is unshaken by sorrow, free from craving, fear and anger, and who can withdraw the senses from their objects at will. Elsewhere (12.12) the *Gita* ranks the stages of the path: knowledge is better than mere practice, meditation better than knowledge, and renunciation of the fruits of action better than meditation, because peace follows immediately on renunciation. Indian textbooks often summarise this as a progression from practice (*abhyasa*) through knowledge (*jnana*), meditation (*dhyana*) and renunciation (*tyaga*) to peace (*shanti*).

> EXAM TRAP
>
> **The principle of disinterestedness on which the Gita is based** is **nishkama karma**, action without attachment to its fruits. Karma yoga is the broader path of action that applies the principle; jnana yoga (knowledge) and bhakti yoga (devotion) are different paths. If both "karma yoga" and "nishkama karma" appear, the stem's word "principle" points to nishkama karma.

## The self in Vedanta and the four states of consciousness ●●

Who, or what, is the self that all these practices are meant to free? The Upanishads, the philosophical texts at the end of the Vedas (hence *Vedanta*, "the end of the Veda"), answer with a story. In the *Chandogya Upanishad*, the gods and the demons hear that the creator Prajapati has spoken of a self that is free from evil, old age, death, grief, hunger and thirst. Each side sends a pupil: Indra for the gods and Virochana for the demons. After thirty-two years of discipline, Prajapati tells them to look at their reflections in a pan of water, dressed in their finest clothes: that, he says, is the self. Virochana goes home satisfied and teaches the demons that the body is the self. Indra, on the way home, has a doubt. If the self is the body, it is blind when the body is blind and perishes when the body perishes. He returns, studies

another thirty-two years, and is told that the self is the one who moves about happily in dreams. Again he doubts: in dreams one is chased, beaten and weeps. After thirty-two more years he is told that the self is the one in deep, dreamless sleep. But in deep sleep, Indra objects, one knows neither oneself nor anything else; it is like being annihilated. After five more years, one hundred and one in all, Prajapati teaches him the self that is beyond body, dream and sleep, which only uses the body as its vehicle.

The story is a lesson in method: the self is found by examining, one after another, what it is *not*. The Upanishads call the true self the **atman**: pure, eternal, unchanging consciousness, the witness of body and mind, distinct from the physical body and from the fleeting contents of thought and feeling. They call the ultimate reality underlying the whole universe **Brahman**. And their boldest teaching is that the two are one. In the *Chandogya*, the sage Uddalaka repeats to his son Shvetaketu, "*tat tvam asi*", "that thou art"; in the *Brihadaranyaka*, the realisation is phrased "*aham brahmasmi*", "I am Brahman". When asked to describe the self, the Upanishadic teacher Yajnavalkya can only say "*neti, neti*", "not this, not this".

The most influential interpretation of these texts is **Advaita Vedanta**, the non-dualism systematised by **Adi Shankaracharya** (Shankara), who lived in the eighth century CE. For Advaita, Brahman alone is real; the world of many separate things is **maya**, appearance, neither simply real nor simply unreal; and the individual self (*jiva*) is in truth none other than atman, which is Brahman. Bondage is caused by ignorance (*avidya*), which superimposes the limited body-mind on the self, just as a coiled rope seen in dim light is mistaken for a snake. Liberation (*moksha*) comes not from action but from knowledge (*jnana*) that removes the error; when the light comes on, the snake is not killed, it is simply seen never to have been there. Later Vedantins disagreed: Ramanuja's *Vishishtadvaita* (qualified non-dualism) held that individual souls and the world are real parts of Brahman, and Madhva's *Dvaita* (dualism) that God and souls are eternally distinct. Notice the contrast with Samkhya too: Samkhya has many purushas and a real prakriti, a dualism, while Advaita has one consciousness and a world of appearance, a monism.

Indra's four stages, waking body, dreamer, deep sleeper and the self beyond, have a systematic counterpart in the *Mandukya Upanishad*, the shortest of the principal Upanishads (just twelve verses), which analyses the self through the four states of consciousness. In *jagrat*, the waking state, consciousness is turned outward and experiences gross physical objects; the self in this state is called Vaishvanara. In *svapna*, the dream state, consciousness is turned inward and experiences subtle objects made by the mind itself; the self here is Taijasa, "the luminous". In *sushupti*, deep dreamless sleep, there is neither desire nor dream; consciousness is an undivided mass, resting in bliss; the self here is called Prajna. The fourth, **turiya** (literally "the fourth"), is not a further state alongside the others. It is pure consciousness itself, neither inward- nor outward-turned, peaceful and non-dual, the witness present through all three states, and it is identified with the atman. The *Mandukya* maps the four onto the sacred syllable *Om*, pronounced A-U-M: A is waking, U is dreaming, M is deep sleep, and the silence that follows is turiya. Table 2.16 sets out the scheme; the modern science of sleep and dreaming is taught in Chapter 5.

Table 2.16 The four states of consciousness in the *Mandukya Upanishad*.

| State | Meaning | Name of the self | What is experienced | Part of Om |
|---|---|---|---|---|

| State | Meaning | Name of the self | What is experienced | Part of Om |
|---|---|---|---|---|
| Jagrat | Waking | Vaishvanara | Gross outer objects | A |
| Svapna | Dreaming | Taijasa | Subtle inner images | U |
| Sushupti | Deep dreamless sleep | Prajna | No objects, no desire; undivided bliss | M |
| Turiya | "The fourth" | Atman itself | Pure, non-dual consciousness, witness of the other three | The silence after |

Why should a psychologist care? Because the scheme makes a claim that can be argued about. When you wake from deep sleep you say "I slept well; I knew nothing". Who knew that nothing was known? Vedanta argues that some witnessing awareness must have persisted through the gap, and that this continuity across waking, dreaming and dreamless sleep is the best evidence that the self is not identical with any of its states. Western psychology describes the states; Vedanta asks what stays the same across them.

> EXAM TRAP
>
> **Turiya** = "the fourth", pure consciousness, the witness of all three states, not a deeper stage of sleep. **Sushupti** = deep dreamless sleep; its self is called *Prajna*, so lists sometimes give the states as jagrat, svapna, prajna and turiya. **Svapna** = dream, not deep sleep. Texts that speak of *avastha-traya* ("the three states") count waking, dream and deep sleep, with turiya as what lies beyond them. **Atman versus Brahman**: atman is the self, Brahman ultimate reality; Advaita says they are one. **Advaita Vedanta** (Shankara) = non-dualism; **Vishishtadvaita** (Ramanuja) = qualified non-dualism; **Dvaita** (Madhva) = dualism.

## The five sheaths and the three bodies ●●

If the self is pure consciousness, why do we experience ourselves as bodies with breath, moods and opinions? The *Taittiriya Upanishad* answers with a second famous story and a model that is now one of the most widely used frameworks in Indian health psychology. The young Bhrigu asks his father Varuna to teach him Brahman, and Varuna tells him only to reflect deeply. Bhrigu concludes first that food is Brahman, since beings are born from food, live on food and return to it; dissatisfied, he reflects again and decides that it is breath, then mind, then understanding, and at last bliss. Each answer goes deeper than the one before.

The **panchakosha** ("five sheaths") model turns Bhrigu's progression into a map of the person as five concentric layers covering the self, from the grossest and outermost to the subtlest and innermost.

1. *Annamaya kosha*, the food sheath: the physical body, built and sustained by food.
2. *Pranamaya kosha*, the vital sheath: the life-force (*prana*) that animates the body through breath, circulation and the other vital functions, together with the organs of action.
3. *Manomaya kosha*, the mental sheath: manas with the five senses, the layer of perception, emotion, desire and doubt, where sense impressions are received and processed.
4. *Vijnanamaya kosha*, the sheath of understanding: buddhi with the senses, the layer of discrimination, judgement, decision and the sense of being the doer.

5. *Anandamaya kosha*, the bliss sheath: the subtle joy experienced in deep sleep and deep meditation, the layer closest to the self.

The atman itself is not a sheath. It is what the sheaths cover, as a scabbard covers a sword or five lampshades of increasing thickness dim a single lamp. A person who identifies with the outermost sheath says "I am fat" or "I am ill"; with the mental sheath, "I am anxious"; with the intellectual sheath, "I am the one who decides". Vedanta teaches discrimination of the self from each sheath in turn, the method Indra used.

Vedanta groups the five sheaths into three bodies (*sharira-traya*), and links the bodies to the states of consciousness you have just met. The *sthula sharira*, the gross body, is the annamaya kosha, and it is the body we experience in waking. The **linga sharira** (also called *sukshma sharira*), the subtle body, consists of the three middle sheaths, pranamaya, manomaya and vijnanamaya, and it is the body we experience in dreams; tradition holds that it is the subtle body that carries the individual from one life to the next. The *karana sharira*, the causal body, is the anandamaya kosha, experienced in deep sleep. Turiya is the atman, beyond all three. Figure 2.5 puts the whole scheme in one picture. (Samkhya, which does not use the kosha language, describes the subtle body by its components instead: buddhi, ahamkara, manas, the ten organs and the five subtle elements.)

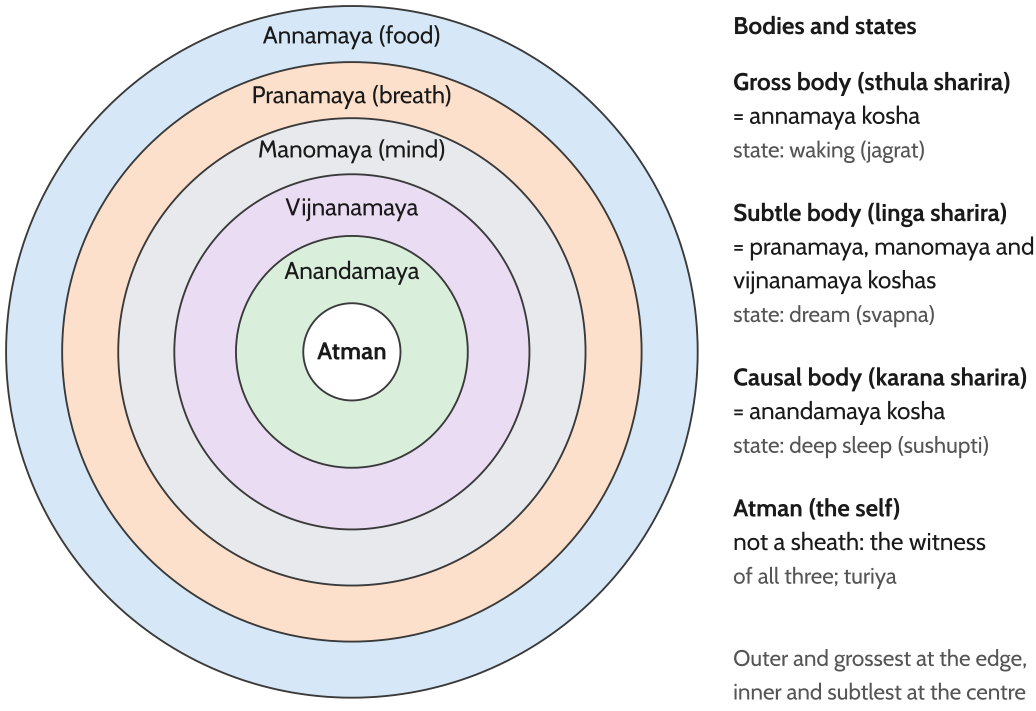


**Figure 2.5** The five sheaths (panchakosha) of the *Taittiriya Upanishad*, with Vedanta's three bodies and the *Mandukya*'s states of consciousness. Notice that the subtle body is made of three sheaths, and that the atman at the centre is not itself a sheath but the consciousness all of them cover.

The model is not only of historical interest. Yoga therapists in Bengaluru built an "integrated approach of yoga therapy" on it, prescribing practices for each sheath (postures and diet for the food sheath, breathing for the vital sheath, meditation and devotional practice for the mental sheath, study and counselling for the sheath of understanding, and practices aimed at contentment for the bliss sheath), and the panchakosha model is often used in India as a

framework for holistic, multi-level assessment of well-being, much as the biopsychosocial model (Section 2.7) is used in the West.

> **EXAM TRAP**
>
> **Outermost sheath**: annamaya (the food body); then pranamaya, manomaya, vijnanamaya, and **innermost**, anandamaya. **The sheath involved in perceptual processing** is manomaya (manas with the senses), not vijnanamaya (intellect and discrimination), pranamaya (vital energy) or anandamaya (bliss). **The subtle body (linga sharira)** corresponds to the three middle sheaths, pranamaya, manomaya and vijnanamaya. A practice item keys "pranamaya, manomaya, vijnanamaya and atman": the three sheaths are right, but the atman is the witness, never a component of any body. Ahamkara, manas, buddhi and chitta is the antahkarana, not the subtle body as the kosha scheme defines it.

## Buddhist psychology

Of all the Indian traditions, the one that has entered Western clinical psychology most deeply is Buddhism. Its founder, **Siddhartha Gautama**, the Buddha ("the awakened one"), was born a prince of the Shakya clan at Lumbini, in what is now Nepal; the traditional dates are 563 to 483 BCE, though many historians now place him somewhat later. According to the traditional story, the sight of an old man, a sick man, a corpse and a wandering ascetic led him to leave his palace in search of a way beyond suffering. Six years of extreme self-denial brought him close to death but not to freedom, and he turned instead to a "middle way" between indulgence and asceticism. Meditating under a tree at Bodh Gaya, he attained awakening, and in his first sermon, at the Deer Park in Sarnath near Varanasi, he set out the teaching on which all Buddhist psychology rests.

The **Four Noble Truths** have the structure of a physician's consultation: diagnosis, cause, prognosis and treatment.

1. *Dukkha*: life as ordinarily lived is marked by suffering or unsatisfactoriness, in birth, ageing, illness and death, in separation from what we love, contact with what we dislike and not getting what we want.
2. *Samudaya*, the origin of suffering: craving (*tanha*, literally "thirst"), whether for sensual pleasure, for existence or for non-existence.
3. *Nirodha*, the cessation of suffering: when craving ends, suffering ends. This state is *nirvana* (Pali *nibbana*), the "blowing out" of the fires of greed, hatred and delusion.
4. *Magga*, the path to the cessation of suffering: the Noble Eightfold Path.

The **Noble Eightfold Path** is not a sequence of steps but eight factors cultivated together, traditionally grouped into three trainings, as Table 2.17 shows.

Table 2.17 The Noble Eightfold Path and its three trainings.

| Training | Path factors | What they cultivate |
|---|---|---|
| Wisdom (*prajna*, Pali *panna*) | Right view (understanding); right intention (thought) | Seeing things as they are; wholesome motives |
| Ethical conduct (*sila*) | Right speech; right action; right livelihood | Harmless, honest living |
| Mental discipline (*samadhi*) | Right effort; right mindfulness; right concentration | Training of attention and the mind |

What, according to the Buddha, is the person who suffers? Here Buddhism parts company with the Upanishads. All conditioned things, he taught, bear three marks: impermanence (*anicca*), unsatisfactoriness (*dukkha*) and non-self (**anatta**, Sanskrit *anatman*). There is no permanent, unchanging self or soul behind experience. What we call a person is a constantly changing process made of five aggregates, the **skandhas** (Pali *khandhas*): form or the body (*rupa*), feeling, pleasant, unpleasant or neutral (*vedana*), perception (*samjna*), mental formations such as volitions and habits (*samskara*), and consciousness (*vijnana*). In a famous dialogue from the *Milindapanha*, the monk Nagasena asks the Indo-Greek King Milinda what a chariot is. Is it the axle? The wheels? The pole? None of them, the king admits, yet "chariot" is not something apart from them either; it is a convenient name for parts in a certain arrangement. So too, says Nagasena, is "Nagasena". Belief in a fixed self is, for Buddhism, a root of craving and therefore of suffering. Modern cognitive science, which also describes the self as a construction assembled moment by moment by many processes, has found the Buddhist view surprisingly congenial.

Buddhism also produced one of the world's first systematic psychologies. The Pali canon is divided into three "baskets" (*Tipitaka*): the *Vinaya* (rules of monastic discipline), the *Sutta* (the Buddha's discourses) and the **Abhidhamma** ("higher teaching"). The *Abhidhamma* analyses experience into momentary states of consciousness (*cittas*) and the mental factors that accompany them (*cetasikas*, fifty-two in the Theravada lists), each classed as wholesome, unwholesome or neutral. The three unwholesome roots of action are greed (*lobha*), hatred (*dosa*) and delusion (*moha*), the "three poisons"; their wholesome opposites are generosity, loving-kindness and wisdom. In its ambition to catalogue the elements of mind it resembles Titchener's structuralism, but its purpose is therapeutic: to know which states to cultivate and which to abandon. A later Mahayana school, the Yogacara ("practice of yoga") school of Asanga and Vasubandhu (around the fourth century CE), described a "storehouse consciousness" (*alaya-vijnana*) holding the seeds of past actions below awareness, an idea often compared with the unconscious.

Buddhist practice divides meditation into two kinds. *Samatha* ("calm abiding") develops concentration on a single object, much like Patanjali's dharana and dhyana. The second, **vipassana** ("insight", literally "seeing clearly"), uses that stability to observe experience as it arises and passes, until its impermanence, unsatisfactoriness and lack of self are seen directly. Its central skill is **sati**, mindfulness: the Pali word originally meant remembering or keeping in mind, and came to mean a clear, non-reactive, present-moment awareness of whatever is happening. The *Satipatthana Sutta* describes four foundations of mindfulness: the body, feelings, the mind and mental objects. Alongside these, Buddhism cultivates four boundless attitudes, the *brahmaviharas*: loving-kindness (*metta*), compassion (*karuna*), sympathetic joy (*mudita*) and equanimity (*upekkha*), the same four Patanjali recommends in sutra 1.33. In modern India, vipassana was revived on a large scale by **S. N. Goenka** (1924 to 2013), a Burmese-born teacher of Indian descent who began teaching in India in 1969, founded the centre at Igatpuri near Nashik in 1976, and made the ten-day silent residential course famous, including courses held inside Tihar Jail in Delhi in the 1990s.

CLINICAL LINK

Buddhist mindfulness is the root of the third wave of cognitive-behavioural therapy (Section 2.6). **Jon Kabat-Zinn** (born 1944) built *mindfulness-based stress reduction* (MBSR) in 1979 and defined mindfulness as paying attention in a particular way: on purpose, in the present moment and non-judgementally. *Mindfulness-based cognitive therapy* (MBCT; Segal, Williams and Teasdale, 2002) uses it to prevent relapse in recurrent depression. **Marsha Linehan** (born 1943) developed *dialectical behaviour therapy* (DBT) for chronically suicidal clients and borderline personality disorder; it blends cognitive-behavioural change strategies with acceptance and mindfulness practices drawn from Zen Buddhism and with dialectical philosophy, and teaches four skills modules: core mindfulness, distress tolerance, emotion regulation and interpersonal effectiveness. Steven Hayes's *acceptance and commitment therapy* (ACT) also builds on acceptance and mindful awareness. The therapies are taught in the Clinical book.

A word on Buddhism's sister tradition is useful here, because the two are easily confused. Jainism was shaped by **Mahavira**, an older contemporary of the Buddha whom Jains honour as the twenty-fourth and last *tirthankara* ("ford-maker"). Unlike Buddhism, it affirms an eternal, conscious soul (*jiva*), one for each living being, and treats karma as a subtle matter that clings to the soul through passions such as anger, pride, deceit and greed. Liberation, a state of perfect knowledge called *kevala jnana*, comes through the "three jewels" of right faith, right knowledge and right conduct, with *ahimsa* (non-violence) at the centre of conduct. The Jain doctrine of *anekantavada*, "many-sidedness", holds that reality has many aspects and every single viewpoint is partial, an ancient lesson in intellectual humility.

EXAM TRAP

**Anatta (Buddhism) is the opposite of atman (Upanishads)**: Buddhism denies a permanent self; Vedanta and Jainism affirm one (and Samkhya affirms many purushas). **The Abhidhamma** is the third basket of the Pali canon, the systematic psychology of mind-states; the Vinaya is monastic discipline and the Sutta the discourses. **Mindfulness** comes from Buddhist *sati* and vipassana; samatha is concentration. **DBT's core mindfulness skills** come from **Zen Buddhism** (not Stoicism, which influenced Ellis's REBT; not psychoanalysis or existentialism), and DBT blends CBT with mindfulness and acceptance practices and dialectical philosophy (not with free association, drugs or Bowen family systems theory). Its four modules are mindfulness, distress tolerance, emotion regulation and interpersonal effectiveness; cognitive restructuring and relaxation are not modules. Buddhism and Jainism are nastika.

## Sufi psychology

Sufism (*tasawwuf*) is the mystical, inward dimension of Islam, and from about the eleventh and twelfth centuries Sufi teachers made it one of the great spiritual traditions of India. The Chishti order, brought to India by Khwaja Moinuddin Chishti, whose shrine at Ajmer still draws millions, and carried on by Nizamuddin Auliya in Delhi, became famous for its tolerance and for its use of music (*sama*, the ancestor of *qawwali*) as a spiritual practice. Like yoga and Buddhism, Sufism contains a detailed psychology of the self and its transformation. The Persian theologian al-Ghazali (1058 to 1111) analysed the "diseases of the heart" (pride, envy, greed, anger) and their cures in terms any counsellor would recognise.

Sufi psychology describes the person in terms of the **nafs** (the self or ego, especially the lower self of appetites and egoism), the *qalb* (the heart, the organ of spiritual perception), the *ruh* (the spirit) and the *aql* (the intellect). Its central teaching is that the nafs can be refined, and it describes the refinement in seven stages, most of whose names come from the Quran.

1. *Nafs al-ammara*, the commanding or tyrannical self, which incites to evil and is ruled by appetite and ego.
2. *Nafs al-lawwama*, the self-reproaching or blaming self, which has begun to feel remorse and to struggle against its impulses.
3. *Nafs al-mulhama*, the inspired self, which begins to receive inspiration and to act from goodness.
4. *Nafs al-mutma'inna*, the tranquil or serene self, at peace and secure in faith.
5. *Nafs al-radiya*, the contented self, pleased with whatever comes.
6. *Nafs al-mardiyya*, the self that is pleasing to God.
7. *Nafs al-kamila* (also called *safiya*), the complete or pure self.

The journey is often described as passing through four levels, from the outer to the inner: *shariat* (the religious law, the outward practice), *tariqat* (the path of inner discipline), *haqiqat* (inner truth) and *marifat* (direct knowledge of God). Along the way the seeker passes through *maqamat*, "stations" reached by effort and stable once attained, such as repentance (*tawba*), scrupulousness (*wara*), renunciation (*zuhd*), poverty (*faqr*), patience (*sabr*), trust in God (*tawakkul*) and contentment (*rida*), and experiences *ahwal*, "states" such as love and longing, which come as gifts and pass. The principal practice is *zikr* (*dhikr*), remembrance through the rhythmic repetition of divine names, together with contemplation (*muraqaba*), under the guidance of a teacher (*murshid* or *pir*) to whom the disciple (*murid*) is closely bound, a relationship some writers compare with that of therapist and client. The goal is **fana**, the "annihilation" or passing away of the ego-self in God, followed by *baqa*, "subsistence", a return to life in the world with the self transformed.

> EXAM TRAP
>
> **Lowest stage of the nafs**: *ammara*, the commanding self; next *lawwama*, the blaming self; the serene self is *mutma'inna*. **Fana** = annihilation of the ego in God; **baqa** = subsistence after it. **Maqamat** (stations) are earned and lasting; **ahwal** (states) are given and passing. Sufism is an Islamic tradition: it is neither astika nor nastika.

### Sri Aurobindo's integral psychology

The last of the great Indian psychologies is modern. **Sri Aurobindo** (Aurobindo Ghose, 1872 to 1950) was born in Calcutta, educated in England from the age of seven and at Cambridge, and returned to India to become one of the leaders of the movement for independence. Arrested in 1908 in the Alipore bomb case, he spent a year in jail awaiting trial, and there, by his own account, had the spiritual experiences that changed the direction of his life. Acquitted in 1909, he withdrew in 1910 to Pondicherry, then a French territory, where he spent the rest of his life writing and practising yoga. His collaborator **Mirra Alfassa** (1878 to 1973), known as the Mother, organised the Sri Aurobindo Ashram and later founded the international township of Auroville (1968). His major works include *The Life Divine*, *The Synthesis of Yoga* and the epic poem *Savitri*.

Aurobindo's *integral yoga* differs from the classical yogas in one decisive respect. Their aim was release from the world; his was the transformation of life in the world. "All life is yoga", he wrote. He saw evolution as the progressive emergence of consciousness from matter, to life, to mind, and beyond mind to what he called the supermind, and he aimed to integrate the paths of knowledge, works and devotion into one discipline. The psychology built on his writings, which his disciple, the psychologist **Indra Sen**, is usually credited with naming **integral psychology**, describes the person along two dimensions.

The first is a concentric system, from the surface inward. On the surface is the *outer being*, made of three parts: the physical, the vital (Aurobindo's term for the life-energy of desires, emotions, sensations and drives) and the mental. Behind it lies a wider *inner* or subliminal being, with inner physical, vital and mental parts of far greater range than their surface counterparts. And at the centre is the **psychic being**, the innermost being: the evolving soul, a "divine spark" that grows from life to life and can, once it comes forward, guide the whole personality. The second is a vertical system. Below the waking mind lie the subconscient and the inconscient; above it rises a series of higher planes of consciousness, the higher mind, the illumined mind, intuition, the overmind and, highest, the supermind. Aurobindo described the work of yoga as a triple transformation: a *psychic* transformation, in which the psychic being comes to the front; a *spiritual* transformation, in which higher consciousness descends into the person; and a *supramental* transformation, the final change of the whole nature. Cornelissen's Indian Psychology Institute (Section 2.8) works largely within this framework, and transpersonal theorists such as Ken Wilber (Section 2.5) drew partly on it for their own use of the word "integral".

> EXAM TRAP
>
> **The innermost being in the concentric system of integral yoga** is the **psychic being**, not the mind, the vital or the body, which make up the outer being. "The vital" means life-energy, desire and emotion, not the physical body. The **supermind** belongs to the vertical system, at its top.

### Ayurveda: the doshas, the gunas and mental health

India's classical medicine, *Ayurveda* ("the knowledge of life"), is set out above all in two great compendia, the *Charaka Samhita*, on internal medicine, associated with the physician **Charaka**, and the *Sushruta Samhita*, on surgery, associated with **Sushruta.** Ayurveda's definition of health is strikingly psychological. Sushruta calls a person healthy (*svastha*) whose doshas, digestive fire, tissues and excretions are in balance, and whose self, senses and mind are *prasanna*, clear and content. Health is not merely the absence of disease but balance and contentment in body and mind together.

The body, in Ayurveda, is governed by three **doshas**, functional principles each built from two of the five elements of Samkhya. Every person is born with a characteristic balance of the three, called his or her *prakriti* or constitution (note that this is a different use of the word from Samkhya's cosmic prakriti), and illness is a departure from that balance. Each constitution has a recognisable psychological signature, set out in Table 2.18.

Table 2.18 The three doshas of Ayurveda.

| Dosha | Elements | Bodily function | Typical temperament (and when out of balance) |
|---|---|---|---|
| Vata | Space and air | Movement: breath, circulation, nerve impulses, elimination | Quick, lively, creative, changeable (worry, fear, insomnia) |
| Pitta | Fire and water | Transformation: digestion, metabolism, body heat | Sharp, focused, ambitious (irritability, anger) |
| Kapha | Earth and water | Structure and cohesion: stability, lubrication, strength | Calm, steady, patient, affectionate (sluggishness, possessiveness) |

Where do the gunas fit? Ayurveda applies them to the mind. The mind (*manas*) is made of the three gunas; sattva is the guna of mental health and clarity, while rajas and tamas are called the two doshas of the mind (*manasa doshas*), the sources of mental disturbance, just as vata, pitta and kapha are the doshas of the body. Charaka described sixteen kinds of mental constitution, seven sattvic, six rajasic and three tamasic, an early typology of personality. Ayurveda also had a branch for mental illness. *Bhuta vidya*, one of its eight branches, dealt with disorders attributed to possession and mental causes, and the *Charaka Samhita* describes *unmada* (insanity) and *apasmara* (epilepsy), distinguishing disorders arising from deranged doshas within the body from those with external causes.

Most remarkable for a clinical psychologist is Charaka's threefold classification of treatment. *Daivavyapashraya chikitsa* is spiritual or faith-based therapy: prayer, mantras, rituals and offerings. *Yuktivyapashraya chikitsa* is rational therapy: diet, medicines and physical procedures, reasoned from the cause of the illness. And *sattvavajaya chikitsa*, "conquest of the mind", is psychotherapy: restraining the mind from harmful objects and strengthening it through knowledge, understanding, courage, memory and concentration. Long before the word existed, Ayurveda recognised a treatment that works through the mind itself.

> EXAM TRAP
>
> **Doshas are not gunas.** Vata, pitta and kapha are the three doshas of the body (Ayurveda); sattva, rajas and tamas are the three gunas (Samkhya and the Gita). So **vata** is the wrong option in any question about the gunas. In Ayurveda, rajas and tamas are the doshas of the *mind*. **Vata** = space and air; **pitta** = fire and water; **kapha** = earth and water. Ayurveda's *prakriti* (a person's constitution) is not Samkhya's *prakriti* (primordial matter). **Sattvavajaya** is Charaka's psychotherapy.

## Karma and dharma as psychological ideas

Two words from Indian thought have entered every language in the world, usually in a garbled form. Both are psychological ideas, and both shape how millions of people in India explain their lives.

The word **karma** (from the root *kri*, to do) means action, and by extension the law that every action has consequences for the one who acts. Each deliberate act leaves a latent impression (*samskara*) in the mind; impressions accumulate into tendencies (*vasanas*) that incline us to act the same way again; and actions bear fruit (*phala*), in this life or, the tradition holds, a later one. Vedanta distinguishes three kinds. *Sanchita karma* is the accumulated store from the past. *Prarabdha karma* is the portion of that store which has begun to bear fruit and shapes the circumstances of the present life. *Agami* (or *kriyamana*) karma is the new karma being

made now, by present choices. Read psychologically, the doctrine of samskaras is a theory of habit: every act strengthens a disposition to repeat it, much as Hull's habit strength (Section 2.2) grows with each reinforced repetition. It is also a theory of moral responsibility and a way of finding meaning in misfortune.

That last use cuts both ways. A fatalistic reading ("it is my karma; nothing can be done") can encourage passivity and the acceptance of injustice. An agentic reading ("my present actions shape my future") encourages responsibility and hope, and the distinction between prarabdha and agami karma leaves plenty of room for choice. Indian psychologists, Ajit Dalal among them, have studied how patients explain their illnesses, and found that beliefs about karma and God often stand beside medical explanations rather than replacing them, sometimes helping people to accept what cannot be changed.

The word **dharma** (from the root *dhr*, to hold or sustain) means that which upholds: the cosmic and social order, and the duties, ethics and right conduct that sustain it. Dharma is sensitive to context. What is right depends on one's stage of life, role and situation; hence *svadharma*, one's own duty, of which the *Gita* says it is better to perform one's own dharma imperfectly than another's well. Dharma is the first of the four aims of life, the **purusharthas**: *dharma* (virtue and duty), *artha* (wealth and security), *kama* (pleasure and love) and *moksha* (liberation). Unlike Maslow's hierarchy (Section 2.5), the scheme does not climb from lower to higher needs; it places dharma as the regulator of the pursuit of wealth and pleasure, with liberation as the final aim. The tradition also describes four stages of life, the *ashramas*: the student (*brahmacharya*), the householder (*grihastha*), the retired forest-dweller (*vanaprastha*) and the renunciant (*sannyasa*), a lifespan model that Sudhir Kakar and others have compared with Erikson's psychosocial stages.

When duties collide, the result is a **dharma-sankata**, a dilemma of dharma: a situation in which every available action violates some obligation. Arjuna on the battlefield, bound by his duty as a warrior to fight and by his duty to his family not to kill his kinsmen, is the classic case, and his despair (the first chapter of the *Gita* is called "the yoga of Arjuna's despair") is the starting point of the whole poem. Krishna's answer combines several ideas you have met in this section: understand that the self is not the body (Vedanta), act according to your svadharma (dharma), and act without attachment to the fruits (nishkama karma). Indian psychologists use the concept of dharma-sankata in the study of moral reasoning and decision-making in Indian families and organisations.

CLINICAL LINK

For a clinical psychologist in India these ideas are not abstractions. Clients explain their distress in terms of karma, fate, dharma and duty to family, and a formulation that ignores those explanations will often fail. Some Indian counsellors draw explicitly on the *Gita*, treating the Krishna-Arjuna dialogue as a model of a helping relationship that moves from listening to the client's despair, through a change of perspective, to committed action. Whatever framework you use, the rule of the biopsychosocial model (Section 2.7) applies: understand the client's own cultural meanings first.

Step back and notice what these Indian systems share, for all their disagreements about the self. Each begins with suffering; each traces it to a mistake about who we are (avidya, aviveka, craving, the commanding nafs); each prescribes a disciplined training of ethics and atten-

tion; and each aims at a freedom that is also a transformation of experience. These are exactly the themes that the transpersonal psychologists of Section 2.5 and the third-wave therapists of Section 2.6 later went looking for in India.

EXAM TRAP

**The four purusharthas** are dharma, artha, kama and moksha (not the gunas, not the koshas). **The four ashramas** are brahmacharya, grihastha, vanaprastha and sannyasa. **Prarabdha** = karma already bearing fruit in this life; **sanchita** = the accumulated store; **agami** = karma being made now. **Karma** as a word means action; it is not one of Patanjali's eight limbs. **Dharma-sankata** = a moral dilemma between duties, not Bose's theory of opposite wishes.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Patanjali defines yoga as *chitta vritti* _____, the stilling of the fluctuations of the mind.
2. The five afflictions that drive the mind's fluctuations are called _____; the first of them, ignorance, is _____.
3. The fifth limb of yoga, withdrawal of the senses from their objects, is _____; the sixth, concentration on a single point, is _____.
4. The Gita's principle of action without attachment to its fruits is _____ karma.
5. In the *Mandukya Upanishad*, the fourth state, pure consciousness beyond waking, dreaming and deep sleep, is _____.
6. The outermost of the five sheaths is the _____ kosha, and the sheath of perception is the _____ kosha.
7. The Buddhist doctrine that there is no permanent self is called _____.
8. The third basket of the Pali canon, which analyses mind-states and mental factors, is the _____.
9. In Sufi psychology the lowest, commanding stage of the self is *nafs al-*_____.
10. In Sri Aurobindo's concentric system, the innermost being is the _____ being.

**Answers:** (1) nirodha (nirodhah), (2) kleshas, avidya, (3) pratyahara, dharana, (4) nishkama, (5) turiya, (6) annamaya, manomaya, (7) anatta, (8) Abhidhamma, (9) ammara, (10) psychic.

## The case of Sultan's sticks, revisited

Go back to Tenerife, to the chimpanzee sitting on his box and idly playing with two bamboo sticks until, suddenly, he fitted one into the other and raked in the banana. You can now tell all five of the stories the opening promised. A strict behaviourist would refuse to say anything about what went on inside Sultan; she would describe the solution as a response that was emitted and then reinforced, and would want to know his history of reinforcement with sticks. Kohler, the Gestalt psychologist, saw insight: a sudden restructuring of the perceptual field, two separate sticks reorganised into one tool, a solution that appeared all at once, was carried out smoothly and was repeated without hesitation. A psychoanalyst would look past the problem to the motive it served. A cognitive psychologist would model Sultan as an infor-

mation-processing system searching a problem space, holding the goal in working memory and testing operators against it. And a psychologist trained in the Indian tradition would notice what Kohler himself recorded but did not dwell on: the answer came not while Sultan was straining at the bars but after he had given up and sat quietly, when the restless, rajasic effort had subsided, much as many human solvers report that a solution arrives when they stop forcing it (incubation, Chapter 8).

The schools did not merely tell different stories; they tested one another. In 1945 Herbert Birch found that young chimpanzees with no previous experience of playing with sticks rarely solved a similar problem, but that after a few days of free play with sticks they did. Insight, it turned out, draws on a learning history, which is exactly what a behaviourist would have predicted, while the sudden reorganisation of that learning is exactly what a Gestalt psychologist would have predicted. That is the lesson of this chapter. Each school saw something real, overreached, and was corrected by its rivals; and psychology today, with its overlapping perspectives, its information-processing models, its third-wave therapies drawn from Buddhist mindfulness and its renewed interest in India's own traditions, is the sum of those corrections. Sultan's two sticks, joined end to end, are not a bad image of it.

## Chapter summary

2.1 Schools, Systems and Paradigms. A school of thought is a group sharing assumptions about psychology's subject matter, method and explanation, usually with a founder, a manifesto, a home, a journal, students and a rival; Woodworth's *Contemporary Schools of Psychology* (1931) and Heidbreder's *Seven Psychologies* (1933) surveyed them. A system is an organised body of assumptions covering the whole field; a theory explains one set of facts. Kuhn's *The Structure of Scientific Revolutions* (1962) described paradigms, normal science, anomalies, crises and paradigm shifts; a field with competing schools and no shared paradigm is a preparadigmatic science, which fits psychology's age of schools. The humanists numbered the forces: psychoanalysis first, behaviourism second, humanistic psychology third and transpersonal psychology fourth (multiculturalism is Pedersen's "fourth force" in counselling).

2.2 Behaviourism: From Watson's Manifesto to Skinner. Behaviourism was prepared by animal psychology (Darwin, Romanes, Morgan's canon, Loeb's tropisms, Thorndike's puzzle boxes of 1898), by Russian objective psychology (Sechenov, Pavlov and Bekhterev's reflexology) and by functionalism. Watson's 1913 manifesto made psychology "a purely objective experimental branch of natural science" aiming at prediction and control without introspection. He treated thought as subvocal speech (peripheralism), proposed three innate emotions (fear, rage, love), conditioned Little Albert with Rayner (1920) and made the extreme environmentalist "twelve infants" claim (1925). Behaviourism may be metaphysical, methodological or radical. The neobehaviourists drew on logical positivism and Bridgman's operationism (1927): Tolman's purposive behaviourism (molar behaviour, intervening variables, cognitive maps, latent learning), Hull's hypothetico-deductive system (drive reduction, habit strength, reaction potential = habit strength × drive), Guthrie's contiguity and one-trial learning, and Skinner's radical, anti-theoretical behaviourism (*Walden Two*, *Verbal Behavior* 1957, *Beyond Freedom and Dignity* 1971). Kuo abolished instinct. Dollard and Miller (1950) gave early social learning theory, Bandura and Walters (1963) the later one. Behaviourism declined under attack from Chomsky, the Brelands and Garcia but left operational definition and behaviour therapy (Jones 1924, Wolpe 1958).

**2.3 Gestalt Psychology and Kurt Lewin.** Wertheimer's 1912 paper on the phi phenomenon (apparent movement, with Kohler and Koffka as observers) founded Gestalt psychology, building on Ehrenfels's form qualities. Kohler's apes on Tenerife (1913 to 1920; *The Mentality of Apes*, 1917 and 1925) showed insight; his hens showed transposition; his theory of isomorphism was tested and rejected in its field form. Koffka's *Growth of the Mind*, his 1922 article and his *Principles of Gestalt Psychology* (1935) spread the school, and his Lake Constance story separated the geographical from the behavioural environment. The whole is different from (popularly, greater than) the sum of its parts; field theory, Pragnanz, figure-ground (Rubin, 1915) and productive thinking (Wertheimer, 1945) followed. Lewin's B = f(P, E) located behaviour in the life space (the person and the psychological environment), described with topological and hodological concepts, valences and vectors; his students found the Zeigarnik (recall) and Ovsiankina (resumption) effects; he founded group dynamics and action research. Gestalt therapy (Perls, empty chair) is not Gestalt psychology.

**2.4 Psychoanalysis as a School: The First Force.** Freud began as a neurologist. Charcot's demonstrations of hysteria under hypnosis (Paris, 1885 to 1886), Janet's dissociation, Bernheim's suggestion (Nancy, 1889) and above all Breuer's treatment of Anna O. (the "talking cure", the cathartic method; *Studies on Hysteria*, 1895) led Freud to free association and psychoanalysis. *The Interpretation of Dreams* (1900) called dreams the royal road to the unconscious, and the Clark University lectures (1909) brought psychoanalysis to America. The topographic model divides the mind into conscious, preconscious and unconscious; the structural model (1923) into id, ego and superego. Adler and Jung dissented; Horney, Fromm and Sullivan (the neo-Freudians), Anna Freud and Erikson (ego psychology) and Klein (object relations) followed. Popper criticised psychoanalysis as unfalsifiable.

**2.5 The Third Force and Beyond.** Maslow (self-actualisation) and Rogers (the actualising tendency, conditions of worth), with Buhler and May, founded humanistic psychology, the third force, which emphasises free will, growth, the whole person and phenomenological understanding; landmarks were the *Journal of Humanistic Psychology* (1961), the Association for Humanistic Psychology (1961 or 1962), Bugental's five postulates and the Old Saybrook conference (1964). Phenomenology (Husserl) brackets assumptions to describe experience. Existential psychology (Kierkegaard, Heidegger's Dasein, Sartre, Binswanger's Umwelt, Mitwelt and Eigenwelt, Boss, Frankl's logotherapy, will to meaning and noogenic neurosis, May, Yalom's four ultimate concerns) confronts death, freedom, isolation and meaninglessness. Transpersonal psychology (Maslow and Sutich, 1969; Grof, Tart, Wilber, Assagioli) is the fourth force. Positive psychology was launched by Seligman (APA president for 1998) with Csikszentmihalyi (flow) in 2000, and includes PERMA, character strengths, hope and savouring. Humanistic and phenomenological approaches are criticised as vague and as missing social and cultural context.

**2.6 The Cognitive Revolution.** Behaviourism waned under anomalies (Tolman, Kohler, Lashley's 1948 paper on serial order), wartime problems of attention, information theory (Shannon), cybernetics (Wiener), the computer (Turing, 1950) and linguistics. In 1956 came the MIT symposium (Newell and Simon's Logic Theorist, Chomsky, Miller's magical number seven plus or minus two), the Dartmouth project on artificial intelligence and Bruner's *A Study of Thinking*; Broadbent's *Perception and Communication* (1958) was the first major information-processing theory; Chomsky's 1959 review of *Verbal Behavior* became a manifesto; and Neisser's *Cognitive Psychology* (1967) named the field. The information-processing approach treats the mind as a series of limited-capacity stages operating on representations. Bartlett (schema) and Piaget were forerunners. Cognitive science and cognitive neuroscience followed, and connectionism

(Rumelhart and McClelland's *Parallel Distributed Processing*, 1986) modelled knowledge as weights in parallel networks. Bandura's self-efficacy (1977), Ellis and Beck brought cognition into therapy, producing CBT and later its mindfulness-based third wave.

**2.7 Psychology Today: Perspectives, Eclecticism and the Enduring Issues.** Schools gave way to overlapping perspectives: biological (neurobiological), evolutionary, psychodynamic, behavioural, cognitive, humanistic, sociocultural and Engel's biopsychosocial model (1977); developmental psychology is a field that uses them all. Cross-cultural psychology compares cultures, cultural psychology studies how culture and mind make each other up, and indigenous psychology builds theory from within a culture. Eclecticism selects from several schools (Lazarus's technical eclecticism versus theoretical integration, and common factors). The enduring issues, which Robert I. Watson (1967) called prescriptions, include mind-body, nature-nurture, free will-determinism, mechanism-vitalism, molar-molecular, nomothetic-idiographic (Windelband, Allport), objectivism-subjectivism and empiricism-rationalism; whether psychology has a Kuhnian paradigm is still debated.

**2.8 Indian Psychology I: The Movement, Its Pioneers and Samkhya's Map of the Mind.** Psychology in India is the discipline as practised in India; Indian psychology is an approach rooted in classical Indian thought, consciousness-centred, holistic, transformative, relational and open to disciplined first-person methods. Experimental psychology entered the Calcutta curriculum in 1905 (Seal); Sengupta founded the first laboratory (1915) and department (1916) and the Indian Psychological Association (1925), whose *Indian Journal of Psychology* began in 1926; Bose, father of Indian psychoanalysis, founded the Indian Psychoanalytical Society (1922) and proposed the theory of opposite wishes; Gopalaswamy began psychology at Mysore (1924). Durganand Sinha led indigenisation (from without and from within); Uma Chowdhury adapted the TAT; J. B. P. Sinha proposed the nurturant-task leader (1980); Pareek studied role stress; Kakar applied psychoanalysis to Indian childhood. The Pondicherry Manifesto (2002), Cornelissen's Indian Psychology Institute, the *Handbook of Indian Psychology* (Rao, Paranjpe and Dalal, 2008) and *Foundations of Indian Psychology* (Cornelissen, Misra and Varma, 2011) launched the Indian Psychology movement. The darshanas are astika or nastika by their acceptance of the Vedas. Samkhya (Kapila; Ishvarakrishna's *Samkhya Karika*) is a dualism of passive, conscious purusha and active, unconscious prakriti, which evolves into buddhi, ahamkara, manas, the ten organs, the subtle and the gross elements (25 tattvas); liberation (kaivalya) comes from discriminating the seer from the seen. Prakriti is made of three gunas: sattva (clarity, harmony), rajas (activity, passion, restlessness) and tamas (inertia, dullness, ignorance), elaborated in the *Gita*. The antahkarana comprises manas (sensory mind), ahamkara (I-maker), buddhi (intellect) and, in Vedanta, chitta (memory).

**2.9 Indian Psychology II: Yoga, the Self and the Paths of Transformation.** Patanjali's *Yoga Sutras* define yoga as chitta-vritti-nirodha, aiming at kaivalya; they list five vrittis (pramana, viparyaya, vikalpa, nidra, smriti), two means (abhyasa and vairagya), five kleshas (avidya, asmita, raga, dvesha, abhinivesha) and the eight limbs: yama, niyama, asana, pranayama, pratyahara, dharana, dhyana and samadhi, the first five outer and the last three inner. The *Gita* offers the paths of action, knowledge, devotion and meditation, with nishkama karma, action without attachment to its fruits. The Upanishads identify the atman with Brahman; Shankara's Advaita Vedanta treats the world as maya. The *Mandukya Upanishad* describes waking (jagrat), dreaming (svapna), deep sleep (sushupti) and turiya, the fourth; the *Taittiriya* describes five sheaths (annamaya, pranamaya, manomaya, vijnanamaya, anandamaya), grouped into gross, subtle (linga) and causal bodies. Buddhism teaches the Four Noble Truths, the Noble Eightfold Path,

anatta and the five skandhas, systematised in the Abhidhamma, with samatha, vipassana and mindfulness (sati) as practices that now underlie MBSR, MBCT, DBT and ACT. Sufi psychology describes seven stages of the nafs and the goal of fana. Sri Aurobindo's integral psychology places the psychic being at the centre of the person. Ayurveda describes three doshas (vata, pitta, kapha), treats rajas and tamas as doshas of the mind and includes Charaka's sattvavajaya psychotherapy. Karma (with samskaras and three kinds of karma) and dharma (svadharma, the purusharthas, the ashramas and dharma-sankata) shape how Indians explain their lives.

## Who's who

Table 2.19 The people in this chapter.

| Name | When | Examined for |
|---|---|---|
| Alfred Adler | 1870 to 1937 | Individual psychology; inferiority, striving for superiority, birth order |
| Mirra Alfassa (the Mother) | 1878 to 1973 | Sri Aurobindo's collaborator; the Ashram; Auroville (1968) |
| Roberto Assagioli | 1888 to 1974 | Psychosynthesis; a forerunner of transpersonal psychology |
| Sri Aurobindo | 1872 to 1950 | Integral yoga; psychic being; triple transformation |
| Albert Bandura | 1925 to 2021 | Self-efficacy (1977); later social learning theory with Walters (1963) |
| Frederic Charles Bartlett | 1886 to 1969 | *Remembering* (1932); schema; forerunner of cognitive psychology |
| Aaron Temkin Beck | 1921 to 2021 | Cognitive therapy for depression |
| Vladimir Mikhailovich Bekhterev | 1857 to 1927 | Reflexology; motor (association) reflexes |
| Hippolyte Bernheim | 1840 to 1919 | Nancy school of hypnosis as suggestion |
| Ludwig Binswanger | 1881 to 1966 | Daseinsanalysis; Umwelt, Mitwelt, Eigenwelt |
| Girindrasekhar Bose | 1887 to 1953 | Father of psychoanalysis in India; Indian Psychoanalytical Society (1922); opposite wishes |
| Medard Boss | 1903 to 1990 | Daseinsanalysis, with Heidegger |
| Josef Breuer | 1842 to 1925 | Anna O.; cathartic method; *Studies on Hysteria* (1895) |
| Percy Williams Bridgman | 1882 to 1961 | Operationism (1927) |
| Donald Eric Broadbent | 1926 to 1993 | First major information-processing theory (1958); filter model |
| Jerome Seymour Bruner | 1915 to 2016 | *A Study of Thinking* (1956); Harvard Center for Cognitive Studies |
| James Bugental | 1915 to 2008 | Five postulates of humanistic psychology (1964) |
| Charlotte Buhler | 1893 to 1974 | Lifespan humanistic psychology; early leader of the AHP |
| Mary Whiton Calkins | 1863 to 1930 | First used the term "radical behaviourism" (1921), for Watson's view |
| Charaka | Date uncertain | *Charaka Samhita*; three kinds of therapy, including sattvavajaya |
| Jean-Martin Charcot | 1825 to 1893 | Hysteria and hypnosis at the Salpetriere; taught Freud |

| Name | When | Examined for |
|---|---|---|
| Noam Chomsky | Born 1928 | Review of Skinner's *Verbal Behavior* (1959); *Syntactic Structures* (1957) |
| Uma Chowdhury | About 1960 | Indian modification of the TAT |
| Matthijs Cornelissen | 2000s | Indian Psychology movement; Indian Psychology Institute, Pondicherry |
| Mihaly Csikszentmihalyi | 1934 to 2021 | Flow; co-founder of positive psychology |
| Ajit K. Dalal | 2008 | Co-editor of the *Handbook of Indian Psychology* |
| Charles Darwin | 1809 to 1882 | Continuity of species; spur to animal psychology |
| John Dollard | 1950 | Early social learning theory, with Neal Miller |
| Christian von Ehrenfels | 1859 to 1932 | Form qualities (1890), forerunner of Gestalt |
| Albert Ellis | 1913 to 2007 | Rational emotive behaviour therapy |
| George Libman Engel | 1913 to 1999 | Biopsychosocial model (1977) |
| Erik Homburger Erikson | 1902 to 1994 | Psychosocial stages; identity crisis |
| Viktor Emil Frankl | 1905 to 1997 | Logotherapy; will to meaning; noogenic neurosis |
| Anna Freud | 1895 to 1982 | Ego psychology; defence mechanisms (1936) |
| Sigmund Freud | 1856 to 1939 | Founder of psychoanalysis; *The Interpretation of Dreams* (1900); Clark lectures (1909) |
| Erich Fromm | 1900 to 1980 | *Escape from Freedom* (1941); character orientations |
| Siddhartha Gautama, the Buddha | About 5th century BCE | Four Noble Truths; Eightfold Path; anatta |
| Michael Gazzaniga | Born 1939 | Coined "cognitive neuroscience" with George Miller |
| S. N. Goenka | 1924 to 2013 | Revival of vipassana in India; ten-day courses |
| M. V. Gopalaswamy | 1924 | Psychology department and laboratory at Mysore; Akashvani (1935) |
| Stanislav Grof | Born 1931 | Transpersonal psychology; non-ordinary states |
| Edwin Ray Guthrie | 1886 to 1959 | Contiguity; one-trial learning |
| Edna Heidbreder | 1890 to 1985 | *Seven Psychologies* (1933) |
| Martin Heidegger | 1889 to 1976 | Dasein; being-in-the-world (1927) |
| Karen Horney | 1885 to 1952 | Basic anxiety |
| Clark Leonard Hull | 1884 to 1952 | Hypothetico-deductive system; drive reduction; habit strength (1943) |
| Edmund Husserl | 1859 to 1938 | Phenomenology; bracketing |
| Ishvarakrishna | About 4th to 5th century CE | *Samkhya Karika* |
| Pierre Janet | 1859 to 1947 | Dissociation; the subconscious |
| Mary Cover Jones | 1896 to 1987 | Removing a child's fear of rabbits (1924) |
| Carl Gustav Jung | 1875 to 1961 | Analytical psychology; collective unconscious |
| Jon Kabat-Zinn | Born 1944 | Mindfulness-based stress reduction (1979) |
| Sudhir Kakar | Born 1938 | *The Inner World* (1978); psychoanalysis of Indian childhood |

| Name | When | Examined for |
|---|---|---|
| Kapila | Traditional | Legendary founder of Samkhya |
| Soren Kierkegaard | 1813 to 1855 | Father of existentialism; anxiety as the dizziness of freedom |
| Melanie Klein | 1882 to 1960 | Object relations; play technique |
| Kurt Koffka | 1886 to 1941 | Gestalt founder; behavioural environment; *Principles of Gestalt Psychology* (1935) |
| Wolfgang Kohler | 1887 to 1967 | Gestalt founder; insight in apes; transposition; isomorphism |
| Thomas Samuel Kuhn | 1922 to 1996 | Paradigms; preparadigmatic science (1962) |
| Zing-Yang Kuo | 1898 to 1970 | "Giving Up Instincts" (1921); kittens and rats (1930) |
| Karl Spencer Lashley | 1890 to 1958 | The problem of serial order in behaviour (1948) |
| Kurt Lewin | 1890 to 1947 | Field theory; life space; group dynamics; action research |
| Marsha Linehan | Born 1943 | Dialectical behaviour therapy |
| Jacques Loeb | 1859 to 1924 | Tropisms; taught Watson |
| Mahavira | About 6th to 5th century BCE | Jainism; ahimsa; the three jewels |
| Abraham Harold Maslow | 1908 to 1970 | Third force; self-actualisation; named the fourth force |
| Rollo May | 1909 to 1994 | *Existence* (1958); normal versus neurotic anxiety |
| James Lloyd McClelland | Born 1948 | Parallel distributed processing (1986) |
| George Armitage Miller | 1920 to 2012 | The magical number seven (1956); TOTE (1960) |
| Neal Elgar Miller | 1909 to 2002 | Early social learning theory, with Dollard (1950) |
| Girishwar Misra | 2011 | *Foundations of Indian Psychology*; indigenous psychology |
| Asutosh Mookerjee | 1864 to 1924 | Calcutta vice-chancellor who backed the first department |
| Conwy Lloyd Morgan | 1852 to 1936 | Morgan's canon (1894) |
| Norman Leslie Munn | Textbook author | Popularised "the scientific study of behaviour" as the definition of psychology |
| Ulric Neisser | 1928 to 2012 | *Cognitive Psychology* (1967); father of cognitive psychology |
| Allen Newell | 1927 to 1992 | Logic Theorist (1956), with Simon |
| Maria Ovsiankina | 1898 to 1993 | Resumption of interrupted tasks (1928) |
| Anand C. Paranjpe | 1998, 2008 | *Self and Identity*; co-editor of the *Handbook* |
| Udai Pareek | 1925 to 2010 | HRD; role stress; role efficacy |
| Patanjali | Uncertain (about 2nd century BCE to 4th century CE) | *Yoga Sutras*; chitta-vritti-nirodha; eight limbs |
| Ivan Petrovich Pavlov | 1849 to 1936 | Conditioned reflexes; Nobel Prize (1904) |
| Frederick Salomon (Fritz) Perls | 1893 to 1970 | Gestalt therapy; empty chair |
| Jean Piaget | 1896 to 1980 | Genetic epistemology; pioneer of cognitive psychology |
| Karl Popper | 1902 to 1994 | Falsifiability; critique of psychoanalysis |

| Name | When | Examined for |
|---|---|---|
| Koneru Ramakrishna Rao | Born 1932 | Theorist of Indian psychology; *Handbook* (2008) |
| Rosalie Rayner | 1898 to 1935 | Little Albert, with Watson (1920) |
| Carl Ransom Rogers | 1902 to 1987 | Client-centred therapy; actualising tendency |
| George John Romanes | 1848 to 1894 | Anecdotal method |
| Edgar Rubin | 1886 to 1951 | Figure-ground (1915) |
| David Everett Rumelhart | 1942 to 2011 | Parallel distributed processing (1986) |
| Jean-Paul Sartre | 1905 to 1980 | Existence precedes essence; bad faith |
| Brajendranath Seal | 1864 to 1938 | Experimental psychology in the Calcutta curriculum (1905) |
| Ivan Mikhailovich Sechenov | 1829 to 1905 | *Reflexes of the Brain* (1863) |
| Martin Elias Peter Seligman | Born 1942 | Founder of positive psychology (1998); PERMA |
| Indra Sen | 20th century | Usually credited with naming integral psychology |
| Narendra Nath Sengupta | 1889 to 1944 | First laboratory (1915) and department (1916) in India; founder of modern psychology in India |
| Adi Shankaracharya | 8th century CE | Advaita Vedanta; maya |
| Claude Elwood Shannon | 1916 to 2001 | Information theory (1948) |
| Herbert Alexander Simon | 1916 to 2001 | Logic Theorist (1956), with Newell |
| Durganand Sinha | 1922 to 1998 | Indigenisation; four phases of psychology in India (1986) |
| Jai B. P. Sinha | 1980 | Nurturant-task leader |
| Burrhus Frederic Skinner | 1904 to 1990 | Radical behaviourism; operant; *Verbal Behavior* (1957) |
| Harry Stack Sullivan | 1892 to 1949 | Interpersonal psychiatry |
| Sushruta | Date uncertain | *Sushruta Samhita*; definition of health |
| Anthony Sutich | 1907 to 1976 | Journals of humanistic (1961) and transpersonal (1969) psychology |
| Charles Tart | Born 1937 | States of consciousness; transpersonal psychology |
| Edward Lee Thorndike | 1874 to 1949 | *Animal Intelligence* (1898); puzzle boxes |
| Edward Chace Tolman | 1886 to 1959 | Purposive behaviourism; cognitive maps; latent learning |
| Alan Mathison Turing | 1912 to 1954 | "Computing Machinery and Intelligence" (1950) |
| Suneet Varma | 2011 | Co-editor of *Foundations of Indian Psychology* |
| Lev Semyonovich Vygotsky | 1896 to 1934 | Thought and language; sociocultural perspective |
| John Broadus Watson | 1878 to 1958 | Founder of behaviourism (1913); Little Albert (1920) |
| Robert Irving Watson | 1909 to 1980 | Prescriptions (1967) |
| Max Wertheimer | 1880 to 1943 | Founder of Gestalt psychology; phi phenomenon (1912) |
| Norbert Wiener | 1894 to 1964 | *Cybernetics* (1948) |
| Ken Wilber | Born 1949 | Spectrum of consciousness; integral theory |

| Name | When | Examined for |
|---|---|---|
| Joseph Wolpe | 1915 to 1997 | Systematic desensitisation (1958) |
| Robert Sessions Woodworth | 1869 to 1962 | *Contemporary Schools of Psychology* (1931); S-O-R |
| Irvin David Yalom | Born 1931 | Four ultimate concerns (1980) |
| Bluma Zeigarnik | 1900 to 1988 | Recall of interrupted tasks (1927) |

## Key terms

school of thought, system, paradigm, normal science, preparadigmatic science, third force, reflexology, behaviourism, subvocal speech, peripheralism, environmentalism, neobehaviourism, logical positivism, operationism, operational definition, purposive behaviourism, molar behaviour, molecular behaviour, intervening variable, cognitive map, latent learning, hypothetico-deductive method, drive reduction, habit strength, reaction potential, contiguity, one-trial learning, operant, phi phenomenon, Gestalt, insight, transposition, isomorphism, geographical environment, behavioural environment, field theory, productive thinking, life space, topological psychology, valence, vector, hodological space, Zeigarnik effect, Ovsiankina effect, group dynamics, action research, Gestalt therapy, hysteria, Anna O., cathartic method, free association, Clark University, topographic model, structural model, neo-Freudians, first force, psychodynamic perspective, self-actualisation, Old Saybrook conference, humanistic psychology, phenomenology, bracketing, existential psychology, Dasein, Daseinsanalysis, logotherapy, will to meaning, existential vacuum, noogenic neurosis, transpersonal psychology, fourth force, positive psychology, flow, PERMA, savouring, cognitive revolution, cognition, cognitive psychology, information-processing approach, schema, cognitive science, cognitive neuroscience, connectionism, parallel distributed processing, self-efficacy, biological perspective, evolutionary perspective, sociocultural perspective, biopsychosocial model, cross-cultural psychology, cultural psychology, indigenous psychology, eclecticism, prescriptions, nomothetic, idiographic, Indian psychology, theory of opposite wishes, indigenisation, nurturant-task leader, Pondicherry Manifesto of Indian Psychology, darshanas, astika, nastika, Samkhya, purusha, prakriti, kaivalya, gunas, sattva, rajas, tamas, antahkarana, manas, ahamkara, buddhi, chitta, chitta-vritti-nirodha, vritti, abhyasa, vairagya, kleshas, ashtanga yoga, yamas, niyamas, pratyahara, dharana, dhyana, samadhi, karma yoga, nishkama karma, atman, Brahman, Advaita Vedanta, maya, turiya, panchakosha, linga sharira, Four Noble Truths, Noble Eightfold Path, anatta, skandhas, Abhidhamma, vipassana, sati, nafs, fana, integral psychology, psychic being, doshas, karma, dharma, purusharthas, dharma-sankata

## From the past papers

1. Match List I (psychologist) with List II (theory): A. Dollard and Miller, B. Skinner, C. Bandura and Walters, D. Maslow; 1. Radical behaviorism, 2. Self-actualization, 3. Early social learning, 4. Later social learning. (IHBAS 2024)

   (a) A-1, B-2, C-3, D-4 (b) A-3, B-1, C-4, D-2 (c) A-2, B-1, C-3, D-4 (d) A-3, B-2, C-4, D-1

   **Answer: (b) A-3, B-1, C-4, D-2.** Dollard and Miller translated Freud into Hull's terms in 1950 (early social learning); Skinner's system is radical behaviourism; Bandura and Walters gave the later social learning theory (1963); Maslow is self-actualisation. Option (d) is tempting only if you forget that Skinner, not Maslow, is the radical behaviourist.

2. Psychology studies both overt and covert behaviour. Covert behaviour is: (IHBAS 2020)

   (a) Observable and measurable (b) Measurable but not observable (c) Only measurable (d) Only observable

   **Answer: (b) Measurable but not observable.** Covert (implicit) behaviour, such as Watson's subvocal speech, cannot be seen by an observer but can be measured with instruments. Overt behaviour is both observable and measurable, so (a) describes the wrong kind; (c) says much the same as (b) but less precisely, and (d) is the reverse of the truth.

3. Latent learning was demonstrated by: (IHBAS 2025)

   (a) Tolman (b) Bandura (c) Skinner (d) Kohler

   **Answer: (a) Tolman.** Tolman and Honzik (1930) showed that unrewarded rats learned a maze and revealed it at once when food appeared, evidence for a cognitive map. Bandura is observational learning, Skinner operant conditioning and Kohler insight.

4. Gestalt psychology emphasizes: (IHBAS 2025)

   (a) Reinforcement and punishment (b) Neurotransmitter balance (c) Cognitive distortions (d) The whole is greater than the sum of its parts

   **Answer: (d) The whole is greater than the sum of its parts.** This is the school's slogan (Koffka preferred "different from"). Reinforcement and punishment are behaviourist, neurotransmitters biological, and cognitive distortions belong to Beck's cognitive therapy.

5. In Lewin's topological system, emphasis was placed on the study of 'life space,' which includes: (Amity 2023)

   (a) Person and psychological environment (b) Society and group (c) Person and group (d) Environment and group

   **Answer: (a) Person and psychological environment.** The life space is the person plus the environment as it exists for that person, hence B = f(P, E). Groups were a later application of field theory (group dynamics), not part of the definition of the life space.

6. Kurt Lewin is widely titled the father of modern social psychology. Which OTHER field is he also credited with founding? (Amity recall)

   (a) Psychoanalysis (b) Action research (c) Psychophysics (d) Functionalism

   **Answer: (b) Action research.** Lewin founded action research, the cycle of planning, acting, observing and reflecting with the people affected. Psychoanalysis is Freud's, psychophysics Fechner's and functionalism belongs to James, Dewey and Angell.

7. Which therapy focuses on "empty chair technique"? (IHBAS 2025)

   (a) CBT (b) Gestalt therapy (c) Psychoanalysis (d) DBT

   **Answer: (b) Gestalt therapy.** The empty chair is Fritz Perls's technique for dialogue with an absent person or a disowned part of the self. Note that Gestalt therapy is Perls's, not the Gestalt psychology of Wertheimer, Kohler and Koffka; CBT, psychoanalysis and DBT do not use the technique as a signature.

8. Which perspective in psychology emphasizes unconscious processes and early childhood experiences? (RML 2025)

   (a) Behavioral (b) Psychodynamic (c) Humanistic (d) Cognitive

   **Answer: (b) Psychodynamic.** The psychodynamic perspective, descended from Freud, explains behaviour by unconscious conflict and early experience. Behavioural stresses conditioning, humanistic free will and growth, and cognitive internal mental processes.

9. Noogenic anxiety is described under: (IHBAS 2020)

   (a) Humanistic school (b) Gestalt school (c) Existential school (d) Psychodynamic school

   **Answer: (c) Existential school.** Frankl's noogenic neurosis (and anxiety) arises from frustrated meaning, the spiritual dimension, which places it in existential psychology. The humanistic school is a close neighbour, but the concept is Frankl's, not Maslow's or Rogers's.

10. The concept of "Dasein," which refers to human existence as "being-in-the-world," is primarily associated with which school of psychology? (Real PYQ 2025)

    (a) Behavioral psychology (b) Existential psychology (c) Cognitive psychology (d) Psychoanalytic psychology

    **Answer: (b) Existential psychology.** Dasein is Heidegger's term (*Being and Time*, 1927), made clinical by Binswanger and Boss in Daseinsanalysis. None of the other schools uses it.

11. Viktor Frankl's concept of the "will to meaning," the fundamental motivational force in humans to find purpose in life, is central to which therapeutic school? (Real PYQ 2024)

    (a) Rational Emotive Behavior Therapy (REBT) (b) Logotherapy (c) Existential-Humanistic Therapy (d) Acceptance and Commitment Therapy (ACT)

    **Answer: (b) Logotherapy.** Frankl's own school, the "third Viennese school", rests on the will to meaning. Existential-humanistic therapy is the broad family it belongs to, so (c) is the tempting but less specific answer; REBT is Ellis's and ACT is Hayes's.

12. Savoring is a concept in Positive Psychology that involves: (IHBAS 2024)

    (a) Avoiding negative experiences (b) Enhancing the enjoyment of positive experiences (c) Minimizing stress through relaxation (d) Reflecting on past traumas

    **Answer: (b) Enhancing the enjoyment of positive experiences.** Bryant and Veroff defined savouring as attending to, appreciating and intensifying positive experiences, past, present or anticipated; it is to good events what coping is to bad ones. The other options describe avoidance, relaxation and trauma work.

13. Who is a pioneer contributor to Cognitive Psychology? (Amity 2023)

    (a) Jean Piaget (b) Kohler (c) Chomsky (d) Kohlberg

    **Answer: (a) Jean Piaget.** Piaget studied the growth of thinking from the 1920s, long before the cognitive revolution, and is treated as a forerunner of cognitive psychology. Chomsky was a linguist whose 1959 review fuelled the revolution, Kohler a Gestalt psychologist and Kohlberg a theorist of moral development.

14. "Third-wave CBT" includes all EXCEPT: (IHBAS 2025)

    (a) Acceptance and Commitment Therapy (b) Dialectical Behavior Therapy (c) Mindfulness-Based Cognitive Therapy (d) Psychoanalysis

    **Answer: (d) Psychoanalysis.** The third wave comprises acceptance and mindfulness-based therapies such as ACT, DBT and MBCT, many drawing on Buddhist practice. Psychoanalysis is the first force and is not a wave of CBT at all.

15. In Dialectical Behavior Therapy (DBT), the "core mindfulness" skills are primarily derived from which tradition? (Real PYQ 2025)

    (a) Stoicism (b) Zen Buddhism (c) Psychoanalysis (d) Existentialism

    **Answer: (b) Zen Buddhism.** Marsha Linehan drew DBT's mindfulness skills from Zen practice and balanced them against cognitive-behavioural change strategies. Stoicism influenced Ellis's REBT; psychoanalysis and existentialism are unrelated to DBT's mindfulness module.

16. What is the primary focus of Indian psychology? (RML 2025)

    (a) The application of Western psychological theories exclusively (b) The promotion of meditation without psychological context (c) The study of Indian folklore and mythology (d) The integration of Indian philosophical traditions with psychological concepts

    **Answer: (d) The integration of Indian philosophical traditions with psychological concepts.** Indian psychology draws its concepts and methods from Samkhya, Yoga, Vedanta, Buddhism and related traditions. Option (a) describes imitative psychology in India, and Indian psychology is neither mythology nor meditation stripped of theory.

17. Who is regarded as the pioneer of psychology in India, establishing the first Indian psychological laboratory? (IHBAS 2024)

    (a) G. S. Rathi (b) S. K. Mangal (c) Girindrashekhar Bose (d) A. K. Singh

    **Answer as keyed: (c) Girindrashekhar Bose.** Strictly, the first laboratory was set up by N. N. Sengupta at Calcutta in 1915, and Sengupta is not among the options. Bose, Sengupta's Calcutta colleague and the father of Indian psychoanalysis, is the only pioneer listed; Mangal and A. K. Singh are textbook authors.

18. Who is considered the father of Indian psychology? (RML 2025)

    (a) Narendra Nath Sen Gupta (b) Sudhir Kakar (c) Giriswar Misra (d) G. R. S. Rao

    **Answer as keyed: (d) G. R. S. Rao.** The key appears to intend a leader of the Indian Psychology movement, most plausibly K. Ramakrishna Rao, co-editor of the *Handbook of Indian Psychology* (2008). If the question meant modern scientific psychology in India, the answer would be (a) Sengupta; Kakar is a psychoanalyst and Misra a contemporary scholar of indigenous psychology.

19. The concept of the three gunas comes mainly from which school of Indian philosophy, alongside the Bhagavad Gita? (NIMHANS 2026 recall)

    (a) Nyaya (b) Charvaka (c) Samkhya (d) Mimamsa

    **Answer: (c) Samkhya.** Samkhya's prakriti is composed of sattva, rajas and tamas, and the *Gita* builds its psychology of personality on them. Nyaya is the school of logic, Charvaka materialism, and Mimamsa the school of Vedic ritual and duty.

20. In Samkhya dualism, the principle of pure consciousness, the passive witness, is Purusha. The three gunas belong to its counterpart, which is: (Amity recall)

    (a) Atman (b) Prakriti (c) Brahman (d) Ahamkara

    **Answer: (b) Prakriti.** Prakriti, primordial matter, is made of the three gunas. Atman and Brahman are Vedantic terms for the self and ultimate reality, and ahamkara is one of prakriti's products, not its counterpart.

21. Bavisha lost her job. Since then she is stressed, has stopped eating properly and cannot get herself to do anything. According to yogic philosophy, the main issue she is dealing with is: (NIMHANS 2026 recall)

    (a) Yama alone (b) Tamas alone (c) Both yama and tamas (d) Rajas alone

    **Answer: (b) Tamas alone.** Withdrawal, loss of appetite and inability to act are the signature of tamas, the guna of inertia and dullness. Yama is the first limb of yoga (ethical restraints), not a mental state, so (a) and (c) fail; rajas would show as restless over-activity.

22. In Indian psychology, the 'inner instrument' (antahkarana) classically has four functions. Which one is the EGO or 'I-maker'? (Amity recall)

    (a) Manas (b) Buddhi (c) Ahamkara (d) Chitta

    **Answer: (c) Ahamkara.** Ahamkara refers experience to a separate "I". Manas is the sensory mind that doubts and deliberates, buddhi the intellect that decides, and chitta the store of memory impressions.

23. According to Patanjali's Yoga Sutras, the primary goal of yoga is: (JEMAS 2025)

    (a) Physical fitness (b) Control of the mind (citta vritti nirodha) and liberation (kaivalya) (c) Cognitive enhancement (d) Emotional regulation only

    **Answer: (b) Control of the mind and liberation.** Sutra 1.2 defines yoga as the stilling of the mind's fluctuations, and its aim is kaivalya, consciousness resting in itself. Fitness, sharper cognition and calmer emotions may be by-products, but none is the goal.

24. Which is the correct order of the last four of Patanjali's eight limbs? (NIMHANS 2026 recall)

    (a) Dharana, pratyahara, dhyana, samadhi (b) Pratyahara, dharana, dhyana, samadhi (c) Pratyahara, dhyana, dharana, samadhi (d) Dhyana, dharana, pratyahara, samadhi

    **Answer: (b) Pratyahara, dharana, dhyana, samadhi.** The senses must first be withdrawn (pratyahara) before attention can be fixed (dharana), the fixed attention becomes an unbroken flow (dhyana) and the flow becomes absorption (samadhi). The other options swap concentration and meditation or put withdrawal after concentration.

25. In the Pancha Kosha model, the five sheaths of the self run from gross to subtle. The OUTERMOST sheath, the physical body sustained by food, is: (Amity recall)

    (a) Annamaya kosha (b) Pranamaya kosha (c) Manomaya kosha (d) Anandamaya kosha

    **Answer: (a) Annamaya kosha.** *Anna* means food, and the food sheath is the gross body. Pranamaya is the vital sheath of breath, manomaya the mental sheath of perception and emotion, and anandamaya the innermost bliss sheath.

## High-yield sheet

*The facts this chapter's questions ask most, most-asked first. Last read before the exam.*

1 The **empty chair technique** belongs to **Gestalt therapy**, developed by Fritz Perls with Laura Perls and Paul Goodman (*Gestalt Therapy*, 1951), which stresses here-and-now awareness; it is not the Gestalt psychology of Wertheimer, Kohler and Koffka. ASKED 7 TIMES

2 **Patanjali's eight limbs** (sutra 2.29), in order: yama, niyama, asana, pranayama, pratyahara (withdrawal of the senses), dharana (concentration on one point), dhyana (unbroken flow of meditation), samadhi (absorption). Karma and jnana are not limbs. ASKED 4 TIMES

3 **Tamas** is the guna of inertia, lethargy, darkness, dullness and ignorance; a person who has stopped eating properly, withdrawn and cannot get herself to act shows tamas alone. ASKED 4 TIMES

4 **Narendra Nath Sengupta** set up India's first psychology laboratory (1915) and first Department of Psychology (1916) at the University of Calcutta, and is the father of modern psychology in India. Keys naming Bose or Uma Chaudhuri for the first laboratory are wrong. ASKED 5 TIMES

5 **Sigmund Freud** founded psychoanalysis, the first force; the psychodynamic perspective explains behaviour, and many disorders, by unconscious urges and conflict and by early childhood experience. ASKED 4 TIMES

6 **Indian psychology** integrates Indian philosophical and contemplative traditions with psychological concepts; it is holistic and spiritual, includes family and community, and treats mental health by combining traditional practices with modern therapy. ASKED 3 TIMES

7 **Humanistic psychology** (Maslow, Rogers) is the **third force**, after psychoanalysis (first) and behaviourism (second); **transpersonal psychology** is the fourth force. ASKED 5 TIMES

8 Humanistic psychology emphasises **free will, personal growth**, an innate tendency to grow and reaching one's full potential; disorder comes from blocked growth, not from reinforcement or unconscious conflict. ASKED 4 TIMES

9 The **three gunas** are sattva, rajas and tamas, the strands of prakriti (not of purusha); the doctrine comes mainly from **Samkhya**, with the *Bhagavad Gita*. Ahimsa, vata and prakriti are not gunas. ASKED 4 TIMES

10 **John B. Watson** founded behaviourism with "Psychology as the Behaviorist Views It" (1913): psychology is the science of observable behaviour, its goal prediction and control (given the stimulus, predict the response), learned through stimulus-response conditioning. ASKED 4 TIMES

11 **Viktor Frankl** founded logotherapy on the **will to meaning**; frustrated meaning produces an existential vacuum and **noogenic** neurosis, ideas of the existential school. ASKED 3 TIMES

12 **Positive psychology** was founded by Martin Seligman (APA president for 1998; *American Psychologist* special issue with Mihaly Csikszentmihalyi, 2000); its interventions build strengths such as hope, and **savouring** means enhancing the enjoyment of positive experiences. ASKED 3 TIMES

13 **Gestalt psychology**: the whole is different from (popularly, greater than) the sum of its parts. ASKED 3 TIMES

14 In **Samkhya** (Kapila), **purusha** is pure, passive, unchanging consciousness, and **prakriti** is active, ever-changing primordial matter, made of the three gunas and evolving into the five great elements. ASKED 3 TIMES

15 The **five koshas** (*Taittiriya Upanishad*), outer to inner: annamaya (food body, outermost), pranamaya (vital), manomaya (mind and senses: perception), vijnanamaya (intellect), anandamaya (bliss). The subtle body (linga sharira) is the three middle sheaths. ASKED 3 TIMES

16 **Existential psychology** confronts death, freedom, isolation and meaninglessness, holds that meaning is never fixed and must be continually created, and uses Heidegger's **Dasein** (being-in-the-world, 1927); uncovering unconscious desires is a psychoanalytic aim. ASKED 3 TIMES

17 **Kurt Lewin**: B = f(P, E); the **life space** is the person plus the psychological environment; he founded **action research** and group dynamics and is the father of modern social psychology. ASKED 2 TIMES

18 **Edward Chace Tolman** proposed the **cognitive map** ("Cognitive Maps in Rats and Men", 1948) and demonstrated **latent learning** (Tolman and Honzik, 1930). ASKED 2 TIMES

19 **Ahamkara** is the I-maker (ego) of the antahkarana; manas is the sensory mind, buddhi the intellect, chitta the memory store. The Gita's three elements of mind are manas, buddhi and ahamkara. ASKED 2 TIMES

20 **Max Wertheimer** founded Gestalt psychology with his 1912 paper on the phi phenomenon, with Kohler and Koffka; Gestalt is the school that contributed most to the study of perception. ASKED 2 TIMES

21 **Cognitive psychology** studies internal mental processes (attention, memory, problem solving) with the mind-as-information-processor metaphor; Ulric Neisser's *Cognitive Psychology* (1967) named it. ASKED 2 TIMES

22 **Covert behaviour** (thoughts, subvocal speech) is measurable but not directly observable; the "mental processes" in the definition of psychology are internal, covert processes. ASKED 2 TIMES

23 **Sattva** = harmony, balance, clarity; **rajas** = restlessness, passion, desire and over-activity. ASKED 2 TIMES

24 **Girindrasekhar Bose** founded the Indian Psychoanalytical Society (Calcutta, 1922), is the father of psychoanalysis in India, and proposed the theory of opposite wishes. ASKED 2 TIMES

25 The **Indian Psychology movement**: Matthijs Cornelissen led it and founded the Indian Psychology Institute (Pondicherry); the *Handbook of Indian Psychology* (Rao, Paranjpe and Dalal) appeared in 2008; the Pondicherry Manifesto dates from 2002. ASKED 2 TIMES

26 Patanjali's **goal of yoga** is chitta-vritti-nirodha, the stilling of the mind's fluctuations (sutra 1.2), leading to liberation (kaivalya). ASKED ONCE

27 **Atman** is the true self, eternal pure consciousness, not the body, a passing state of mind or a collective unconscious. ASKED ONCE

28 **Jean Piaget**, who studied the growth of thinking from the 1920s (genetic epistemology), is a pioneer contributor to cognitive psychology. ASKED ONCE

29 Early social learning theory is **Dollard and Miller** (1950); later social learning theory is **Bandura and Walters** (1963); radical behaviourism is Skinner's; self-actualisation is Maslow's. ASKED ONCE

30 The **developmental perspective** studies how people change as they mature across the lifespan. ASKED ONCE

CHAPTER 3

# Sensation and Psychophysics

IN THIS CHAPTER

3.1 ●● Sensation and perception: from energy to experience: distinguish sensation from perception and explain transduction, sensory coding and sensory adaptation.

3.2 ● Thresholds: how weak a stimulus, how small a difference: define the absolute threshold and the difference threshold (JND) and explain why both are defined as 50 per-cent points.

3.3 ● The psychophysical laws: Weber, Fechner and Stevens: state, apply and criticise Weber's law, Fechner's law and Stevens' power law.

3.4 ● The classical psychophysical methods: describe the methods of limits, constant stimuli and adjustment, their errors and controls, and modern staircase methods.

3.5 ●● Signal detection theory: separate sensitivity (d′) from the response criterion, read a 2 by 2 outcome table and an ROC curve, and apply the theory to real decisions.

3.6 ●●● Vision I: light, the eye and the two receptor systems: trace light through the eye and contrast rods with cones, dark adaptation, the Purkinje shift, acuity and lateral inhibition.

3.7 ● Vision II: colour: explain colour mixing, the trichromatic and opponent-process theories, their dual-process resolution and the forms of colour blindness.

3.8 ● Hearing: relate the dimensions of sound to pitch, loudness and timbre, follow sound through the ear, and compare the theories of pitch, sound localisation and the types of hearing loss.

3.9 ● The chemical senses: taste and smell: describe the five basic tastes, papillae and taste buds, the olfactory system, pheromones and flavour.

3.10 ● The skin senses and pain: describe touch, pressure and temperature, and explain pain through gate control theory, endorphins and phantom limbs.

3.11 ● The body senses: kinesthesis and balance: distinguish kinesthesis from the vestibu-lar sense and the roles of the semicircular canals and otolith organs.

3.12 ● Senses in concert and senses deprived: explain sensory interaction, synaesthesia and the effects of sensory deprivation.

Weightage. 44 of the 819 questions mapped in this book (5.4%) test this chapter, 16 of them from real past pa-pers. 1 of its 12 sections is High yield.

On the morning of 22 October 1850, a forty-nine-year-old former professor of physics lay in bed in Leipzig and worked out how to measure the mind. His name was Gustav Theodor Fechner, and for most of the previous decade he had been too ill to work at all. In the late 1830s Fechner had been studying afterimages, the ghostly colours that linger in your vision after you stare at a bright light, and he had used his own eyes as the instrument, gazing at the sun through coloured glass. The experiments damaged his eyes. A long breakdown followed: he could not bear light, could hardly eat, and could not read. He gave up his chair of physics and spent about three years withdrawn from the world, much of the time in darkened rooms.

When he recovered, Fechner turned to an old philosophical puzzle: how are mind and body related? The great philosopher Immanuel Kant had argued that psychology could never become a true natural science, because mental events cannot be measured, and without measurement there is no mathematics and no science. Fechner's insight that October morning was a way around Kant. You cannot hold a ruler against a sensation. But you can measure the physical stimulus that produces it, and you can find out how much that stimulus must change before a person notices any difference at all. If every just-noticeable step in sensation is treated as one unit, sensation can be counted, and the relation between the physical world and the mental world can be written as an equation.

Fechner spent the next decade testing the idea, lifting weights, comparing lights and judging distances, thousands of judgements at a time. He found that his Leipzig colleague Ernst Heinrich Weber had already discovered the key regularity, and he generously named it Weber's law. In 1860 Fechner published *Elemente der Psychophysik* (*Elements of Psychophysics*), and with it founded a new science. Nineteen years before Wilhelm Wundt opened his laboratory, Fechner had shown that the mind could be measured. Psychophysicists still celebrate 22 October as Fechner Day.

This chapter explores the territory Fechner opened. It asks how the sense organs turn energy into experience, how weak a stimulus can be before you miss it, how small a change you can notice, and why a "threshold" turns out to be a decision as much as a sensation. It then walks through each sense in turn: vision, hearing, taste, smell, touch, temperature, pain, body position and balance. Keep Fechner's damaged eyes in mind. At the end of the chapter you will see that the afterimages that injured him are now well understood, and that his famous law has been both honoured and revised.

## 3.1 Sensation and perception: from energy to experience

MEDIUM YIELD · 10 QS

Look up from this page for a moment. You see a room full of objects, each with a colour, a shape and a place. You probably hear something too: a fan, traffic, a voice in the next room. It all seems effortless and immediate, as if the world simply pours into your head. It does not. What actually reaches you is a shower of physical energy (photons striking the back of your eyes, pressure waves in the air, molecules drifting into your nose), and everything you experience is built by your nervous system out of that raw material. This section introduces the first steps of that construction.

## Sensation versus perception ●●●

Psychologists traditionally divide the process into two stages. **sensation** is the process by which sense organs detect physical energy from the environment and convert it into neural signals. **perception** is the process by which the brain organises and interprets those signals so that they become meaningful objects and events. Think of a batsman at the crease. Light reflected from the ball forms an image on his retinas, and receptors there fire: that is sensation. Recognising, in a fraction of a second, that the ball is a googly spinning the other way: that is perception.

The boundary between the two is not sharp, and many modern researchers treat them as one continuous process. But the distinction is useful, and examiners like it, because it tells you *where* an error happens. Consider an optical illusion such as the Muller-Lyer figure, in which two lines of identical length look unequal because one carries outward-pointing fins and the other inward-pointing fins. Your receptors register the two lines accurately; the image of each line on your retina really is the same length. The error arises later, when the brain interprets the lines in the light of the fins. Optical illusions are therefore distortions of perception, not of sensation (Chapter 4 explains why the brain makes such errors).

The idea that experience can be broken into sensory building blocks has a long history. The structuralists, led in America by **Edward Bradford Titchener** (1867 to 1927), treated sensations as the elements of consciousness and hoped to catalogue them the way chemists catalogue elements, a "periodic table" of the elements of sensation; Titchener counted more than 44,000 distinguishable sensation qualities, most of them visual or auditory. For Titchener, a perception was a complex of sensory and imaginal elements: sensations, plus the images that past experience attaches to them, with meaning supplied by this context (Chapter 1 tells the full story of structuralism).

A modern descendant of this idea treats perception as inference. **Hermann von Helmholtz** (1821 to 1894), whose work runs through this chapter, argued in the 1860s that perception involves unconscious inference: the brain draws conclusions about the world from incomplete sensory evidence, using past experience. Today this view is formalised in the **Bayesian approach to perception**, named after the eighteenth-century mathematician Thomas Bayes. According to it, every perceptual experience combines two ingredients: *prior beliefs* (what you already expect to be out there, learned from experience) and *sensory evidence* (what the receptors are reporting now). When the evidence is clear, it dominates; when it is weak or ambiguous, the prior carries more weight. Walking down a dimly lit lane at night, you may "see" a dog lying by a gate when it is really a sack of cement, because dogs by gates are common (a strong prior) and the light is poor (weak evidence). Genes, culture and emotional states can all shape a person's priors, but the Bayesian formula itself has just those two inputs: prior beliefs and sensory evidence.

> **EXAM TRAP**
>
> "Optical illusions result from distortion in ..." The answer is *perception*, not sensation: the receptors register the stimulus correctly, and the error is in interpretation. "Translation" and "adaptation" are bait. Likewise, "according to the Bayesian approach, perception is influenced by ..." has the answer *prior beliefs and sensory evidence*; options naming only genes, culture or emotions describe things that can shape priors, not the approach itself.

## Receptors and transduction •

Every sense begins with a **receptor**, a specialised cell (or the specialised ending of a neuron) that responds to one particular form of energy. The receptor's job is **transduction**: the conversion of physical energy into the electrical and chemical language of the nervous system. Light, sound and molecules cannot travel along nerves; only neural signals can. So in the eye, light changes the shape of a pigment molecule, which changes the receptor's electrical state; in the ear, the bending of tiny hairs opens ion channels; in the nose, an odour molecule binds to a receptor protein. Each is a different trick for the same task. Table 3.1 groups the receptors by the energy they transduce.

Table 3.1 Receptors classified by the energy they transduce.

| Receptor class | Energy detected | Examples | Sense |
|---|---|---|---|
| Photoreceptors | Light (electromagnetic) | Rods and cones of the retina | Vision |
| Mechanoreceptors | Mechanical bending, pressure, stretch | Hair cells of the cochlea and vestibular organs; skin corpuscles; muscle spindles | Hearing, balance, touch, kinesthesis |
| Chemoreceptors | Dissolved or airborne molecules | Taste receptor cells; olfactory receptor neurons | Taste, smell |
| Thermoreceptors | Temperature change | Warm and cold free nerve endings | Temperature |
| Nociceptors | Actual or threatened tissue damage | Free nerve endings in skin, muscle and organs | Pain |

Receptors can also be classified by where their information comes from. The British physiologist **Charles Scott Sherrington** (1857 to 1952), in *The Integrative Action of the Nervous System* (1906), distinguished **exteroceptors**, which report on the outside world (eyes, ears, skin), **interoceptors**, which report on the internal organs (a full bladder, a churning stomach), and **proprioceptors**, which report on the position and movement of the body itself (Section 3.11).

Once a receptor has transduced its stimulus, the signal travels toward the central nervous system along **afferent** (sensory, incoming) nerve fibres. Commands travel outward to muscles and glands along **efferent** (motor, outgoing) fibres. The simplest complete loop, from receptor through afferent fibre to the spinal cord and back out along an efferent fibre to a muscle, is the reflex arc: the knee jerk is the textbook example. (The philosopher-psychologist John Dewey famously criticised the reflex arc as a model of behaviour in 1896; Chapter 1 explains why.)

## Sensory coding: specific nerve energies and labelled lines •••

Here is a puzzle that troubled nineteenth-century physiologists. Every sensory nerve carries the same kind of signal: brief electrical impulses called action potentials. An impulse from the eye looks exactly like an impulse from the ear. So how does the brain know whether a message means "red" or "middle C" or "sour"?

The German physiologist **Johannes Peter Muller** (1801 to 1858) gave the classic answer in his *Handbook of Human Physiology* (1833 to 1840). His **doctrine of specific nerve energies** states that the quality of a sensation depends on which nerve is stimulated, not on what stimulates it. Each sensory nerve, however it is excited, produces its own characteristic sensation. You can test this yourself. Close your eyes and press very gently on the outer corner of one eyelid: you will see a spot of light (a phosphene), although no light has entered the eye. The pressure

excited the optic nerve's receptors, and the optic nerve can only "speak" light. A hard blow to the back of the head makes people "see stars" for the same reason. Muller drew a startling conclusion: we are never directly aware of objects in the world, only of the activity of our own nerves. Helmholtz, Muller's student, later extended the idea within a single sense, proposing that different fibres in the eye and ear carry different colours and pitches (an idea you will meet again in his theories of colour and hearing).

Modern neuroscience keeps the core of Muller's insight under the name **labelled-line coding**: the quality of a sensation is signalled by which neurons are active, because each pathway is wired to a particular brain region, as if each line carried a label. Activity in the visual pathway to the occipital lobe is experienced as sight; activity in the auditory pathway to the temporal lobe is experienced as sound. Within a sense, however, quality is often coded by the *pattern* of activity across many receptor types rather than by single labelled lines. This **across-fibre pattern coding** is especially important in taste and smell, where each receptor type responds to many chemicals and a particular flavour or odour corresponds to a particular profile of activity across the whole population.

What about intensity? A dim light and a bright light, a whisper and a shout, must also be told apart. Here a basic fact of neurophysiology matters. An action potential obeys the all-or-none law: once a neuron's threshold is reached, the impulse fires at full, constant size and travels along the axon without shrinking, and a stronger stimulus does not produce a bigger impulse. A stimulus too weak to reach threshold produces no action potential at all, not a "miniature" one (only small graded changes in voltage that fade away). Intensity therefore cannot be coded by the size of impulses. It is coded instead by **rate coding** (stronger stimuli make each neuron fire more often) and by recruitment (stronger stimuli activate more neurons, including less sensitive ones). Note one curiosity: signals can also be carried by a *decrease* in activity. The rods and cones of your retina are most active in darkness, and light actually reduces their activity (Pinel and Barnes's textbook makes this point with some relish).

A final piece of wiring is worth knowing for general psychology. Before reaching the cerebral cortex, almost every sensory pathway relays through the thalamus, the brain's sensory switching station: vision through its lateral geniculate nucleus, hearing through its medial geniculate nucleus, and touch, pain and taste through its ventral posterior nuclei. Smell is the exception: olfactory signals run from the olfactory bulb straight to the olfactory cortex and the amygdala without a thalamic relay on the way (Section 3.9). The details belong to the Physiological Psychology book. Explaining experience and behaviour through the workings of the brain and nervous system in this way is the *neurobiological* (biological) approach to psychology, whose main premise is that behaviour is governed by the brain and nervous system. It differs from the *psychophysical* approach that fills most of this chapter, which studies the lawful relation between physical stimuli and the sensations they produce without needing to know what the neurons are doing (Chapter 2 compares the approaches).

EXAM TRAP

The doctrine of specific nerve energies belongs to *Johannes Muller*, not Helmholtz (Helmholtz extended it) and not Fechner. "The principal sensory relay to the cortex for every sense except smell" is the *thalamus*; the hypothalamus regulates drives such as hunger and temperature, the medulla controls breathing and heart rate, the corpus callosum connects the two hemispheres, the basal ganglia help to control movement, and the hippocampus serves memory. The lateral geniculate nucleus is visual; the medial geniculate nucleus is auditory ("M for music").

### Sensory adaptation

When you put on a wristwatch in the morning you feel it on your skin. Ten minutes later you do not, although it is still pressing on exactly the same spot. When you walk into a home where a garland of Mysuru jasmine hangs by the door, the scent is striking; the family who have been sitting there all evening hardly notice it. This is **sensory adaptation**: a decline in sensitivity (and in receptor response) to a stimulus that remains constant. Adaptation is built into the receptors and the early stages of each sensory pathway, and it is automatic.

The English philosopher **John Locke** (1632 to 1704) described a neat demonstration in 1690. Put one hand in a bowl of hot water and the other in a bowl of cold water for a minute, then plunge both into a bowl of lukewarm water. The same lukewarm water feels cool to the first hand and warm to the second. Each hand has adapted to its starting temperature, so each judges the new water relative to that adapted level.

Why should the senses be designed to fade out unchanging stimulation? Because change is what usually matters. A constant stimulus (the pressure of your clothes, the hum of a refrigerator) rarely carries new information, whereas a change (a touch on the shoulder, a sudden silence) may signal something important. Adaptation frees the senses to respond to change. Vision is the dramatic case: if an image is held perfectly still on the retina, it fades from view within seconds, and your eyes avoid this only by moving constantly (Section 3.6).

EXAM TRAP

Do not confuse *sensory adaptation* with *habituation*. Adaptation is a reduced response of the sensory system itself to a constant stimulus. Habituation is a learned decrease in behavioural response to a stimulus that is repeated and proves unimportant (you stop jumping at the ticking clock), and it can be reversed instantly if the stimulus changes (dishabituation). Habituation is the simplest form of learning and is covered in Chapter 6. Adaptation to light and dark is called *visual adaptation*, not "visual accommodation" (accommodation is the focusing of the lens, Section 3.6).

## 3.2 Thresholds: how weak a stimulus, how small a difference

LOWER YIELD · 2 QS

Fechner's new science, **psychophysics**, is the study of the quantitative relationship between the physical properties of stimuli and the sensations they produce. It asks two basic kinds of question. The first is about detection: what is the weakest stimulus a person can detect? The

second is about discrimination: what is the smallest change in a stimulus a person can notice? The answers are called thresholds.

## The absolute threshold ●●

The **absolute threshold** is the minimum intensity of a stimulus that a person can detect. The older term for a threshold is **limen** (Latin for "threshold"), which survives in two words you should know. A stimulus below the absolute threshold is *subliminal* (below the limen); one above it is *supraliminal*.

At first sight the absolute threshold sounds like a simple switch: below a certain intensity you detect nothing, above it you always detect the stimulus. Experiments show that no such switch exists. If you present a faint tone many times at each of several intensities, there is no single intensity at which "never heard" suddenly becomes "always heard". Instead, the percentage of "yes, I heard it" responses climbs gradually, tracing an S-shaped curve called the **psychometric function** (Figure 3.1). Why the gradual climb? Because the nervous system is never silent. Neurons fire spontaneously, attention wanders, receptors vary slightly from moment to moment, and even the stimulus itself fluctuates (a very dim light delivers a variable number of light particles from one flash to the next). So the same faint tone is sometimes detected and sometimes missed.

Psychophysicists therefore define the absolute threshold statistically: it is the stimulus intensity that is detected on 50 percent of the trials on which it is presented. Not 100 percent, and not "the first time it is noticed", but half the time.

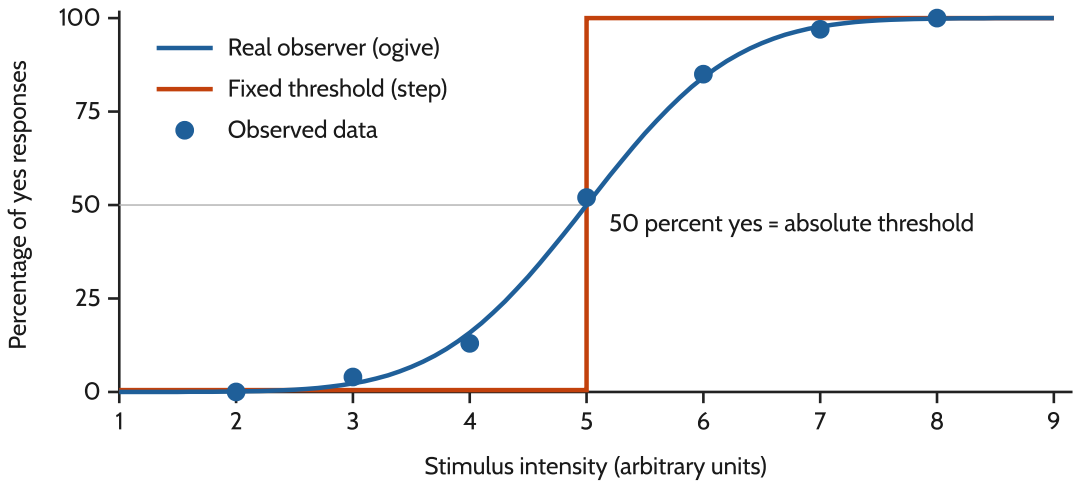


Figure 3.1 The psychometric function for detection. If thresholds were fixed switches, detection would jump from 0 to 100 percent at one intensity (dashed step). Real observers produce a gradual S-shaped curve (an ogive), so the absolute threshold is defined as the intensity detected on 50 percent of trials. The dots show data of the kind collected by the method of constant stimuli (Section 3.4).

How sensitive are human senses at their absolute thresholds? **Eugene Galanter** (1962) gave a set of everyday approximations that appear in almost every introductory text (Table 3.2). They are rough, but they make the point that our senses, at their best, are astonishingly sensitive.

Table 3.2 Approximate absolute thresholds for five senses (after Galanter, 1962).

| Sense | Approximate absolute threshold |
|---|---|
| Vision | A candle flame seen from about 48 km (30 miles) on a dark, clear night |
| Hearing | The tick of a watch from about 6 metres (20 feet) under very quiet conditions |
| Taste | One teaspoon of sugar dissolved in about 7.5 litres (two gallons) of water |
| Smell | One drop of perfume diffused through a six-room apartment |
| Touch | The wing of a bee falling on your cheek from a height of 1 centimetre |

**KEY STUDY: HOW FEW PARTICLES OF LIGHT CAN YOU SEE? (HECHT, SHLAER AND PIRENNE, 1942)**

**Question.** Light comes in tiny packets called quanta (photons). What is the smallest number of quanta a fully dark-adapted human can detect? **Method. Selig Hecht**, Simon Shlaer and Maurice Pirenne, working in New York, flashed a very brief spot of blue-green light (wavelength 510 nanometres, close to the peak sensitivity of the rods) onto a rod-rich region of the retina about 20 degrees from the centre of gaze. Observers, including the authors themselves, sat in darkness and reported whether they had seen each flash. **Result.** At threshold, the flash delivered only about 54 to 148 quanta to the front of the eye. After allowing for light lost in the eye's media and for light that passed through the retina unabsorbed, they calculated that only about 5 to 14 quanta were actually absorbed by rods. Spread across hundreds of rods, this meant that no single rod could have absorbed more than one quantum. **Conclusion.** A single rod responds to a single quantum of light, and the coincidence of a handful of such events produces the experience of seeing. The human eye at threshold works close to the limit set by physics. **Why it matters here.** Because the number of quanta in a dim flash fluctuates randomly from flash to flash, some of the variability in "seeing" a threshold flash lies in the light itself. The threshold is probabilistic by its very nature.

## The difference threshold and the JND

Detection is only half of psychophysics. Much of the time we need to tell whether something has changed: whether the dal needs more salt, whether the music has been turned up, whether one mango is heavier than another. The **difference threshold** (or difference limen) is the smallest difference between two stimuli that a person can detect 50 percent of the time. The amount of change at that threshold is called the **just noticeable difference** (JND). The terms are effectively interchangeable in examinations: the difference threshold is the JND.

Difference thresholds are measured by comparing a fixed *standard stimulus* with a variable *comparison stimulus*. If you hold a 100-gram standard weight in one hand and a series of comparison weights in the other, you might find that you can tell 102 grams from 100 grams half the time; the JND at that standard is then 2 grams. Along the way you will find a comparison value that feels exactly equal to the standard; this is the **point of subjective equality** (PSE), and it need not coincide with the true physical equality point. (A systematic gap between the PSE and the standard is called a constant error.)

Ernst Heinrich Weber measured one especially elegant kind of difference threshold on the skin. Touch the skin with the two points of a pair of compasses. If the points are far apart, you feel two touches; bring them closer and, at some separation, the two touches merge into one. The smallest separation at which two points are felt as two is the **two-point threshold**.

Weber found that it varies enormously across the body, from about 2 millimetres on the fingertips and lips to several centimetres on the back and thigh (Section 3.10 explains why). It was while measuring such thresholds, and the JNDs for lifted weights, that Weber noticed the regularity that bears his name.

> EXAM TRAP
>
> "The minimum level of a stimulus that can be detected" is the *absolute threshold*; "the smallest change that can be detected" is the *difference threshold* (JND). "Sensory adaptation point" and "perceptual boundary" are invented distractors, and so is "signal detection of response". A question defining "the point at which a stimulus is strong enough to trigger a response only 50 percent of the time" wants *threshold*; *response bias* is a signal detection term for a person's readiness to say "yes" (Section 3.5), and the *phi phenomenon* is the illusion of movement produced by lights flashing in sequence, described by Max Wertheimer in 1912 (Chapter 4).

## 3.3 The psychophysical laws: Weber, Fechner and Stevens

LOWER YIELD · 3 QS

Thresholds tell us where sensation begins and how finely it can be graded. The next question is more ambitious: is there a general law relating the size of a stimulus to the size of the sensation it produces? Three answers have been proposed, by Weber, Fechner and Stevens. Each was a real advance, each captures something true, and each fails somewhere. Learn them as a sequence, because each one grew out of the one before.

### Weber's law: the JND is a constant fraction

**Ernst Heinrich Weber** (1795 to 1878) was a professor of anatomy and physiology at Leipzig. In studies of touch and the muscle sense published in 1834 (*De Tactu*) and 1846, he asked people to compare lifted weights. He noticed, among other things, that discrimination was finer when people actively lifted the weights than when the weights simply rested on a passive hand. His enduring discovery, however, was this: the JND is not a fixed amount. With a light standard weight, a small extra amount is noticed; with a heavy standard, a much larger amount is needed. What stays constant is the *ratio* of the JND to the standard.

In symbols, **Weber's law** is

$$\Delta I / I = k$$

where $I$ is the intensity of the standard stimulus, $\Delta I$ (read "delta I") is the increase needed to produce a just noticeable difference, and $k$ is a constant for that sensory dimension, called the **Weber fraction** (or Weber's constant). A small Weber fraction means fine discrimination; a large one means coarse discrimination.

A worked example shows how the law works. Suppose the Weber fraction for lifted weights is about 0.02 (one-fiftieth). Then with a 500-gram standard, the JND is 0.02 × 500 = 10 grams: you will just notice the difference between 500 and 510 grams. With a 5-kilogram standard, the JND is 0.02 × 5,000 = 100 grams. Ten grams added to the heavier load would go completely unnoticed. Or take your morning chai. If a cup sweetened with two teaspoons of sugar needs roughly an extra half teaspoon before anyone notices it is sweeter, then a cup with four tea-

spoons needs roughly an extra whole teaspoon. The absolute amount needed doubles; the proportion stays the same.

The everyday consequences are everywhere. A single candle transforms a dark room but is invisible in sunlight. A shopkeeper knows that a fifty-rupee discount feels generous on a two-hundred-rupee item and trivial on a fifty-thousand-rupee phone. In each case what we notice is proportional change, not absolute change. Table 3.3 gives approximate Weber fractions for several dimensions.

**Table 3.3** Approximate Weber fractions in the middle range of intensities. Values differ considerably between sources and methods; learn the pattern rather than the decimals. Frequency (pitch) is discriminated most finely; the chemical senses are usually among the coarsest.

| Dimension | Approximate Weber fraction | Change needed on a standard of 100 units |
|---|---|---|
| Pitch (frequency of a middle tone) | About 0.003 (about 1/333) | About 0.3 |
| Electric shock | About 0.01 | About 1 |
| Heaviness of lifted weights | About 0.02 (1/50) | About 2 |
| Brightness of light | About 0.02 to 0.08 | About 2 to 8 |
| Loudness (sound intensity) | About 0.05 to 0.1 | About 5 to 10 |
| Pressure on the skin | About 0.14 (1/7) | About 14 |
| Taste (salt) and smell | About 0.08 to 0.3 | About 8 to 30 |

Where does Weber's law fail? It holds well across the middle range of intensities for most senses, but it breaks down at the extremes. Near the absolute threshold, the Weber fraction rises sharply: when a stimulus is barely detectable, a proportionally much larger change is needed before any difference is noticed. It can also drift at very high intensities, and for pure tones the fraction for loudness declines slowly as intensity increases, a small but systematic departure known as the "near miss" to Weber's law. Weber's law is best regarded as an excellent approximation over the range in which we usually live.

## Fechner's law: sensation grows with the logarithm of the stimulus ●●

**Gustav Theodor Fechner** (1801 to 1887) took Weber's finding and built it into a law of sensation itself. Weber's law tells you how big a JND is; it says nothing about how strong a sensation is. Fechner's bridge was a single bold assumption: *every JND represents the same increase in sensation, no matter where on the scale it occurs*. The step from 100 to 102 grams and the step from 1,000 to 1,020 grams are both one JND, so, Fechner assumed, both add one equal unit to the felt heaviness. With that assumption, the strength of any sensation can be measured simply by counting how many JNDs it lies above the absolute threshold.

Now combine this with Weber's law. Start at the threshold and climb one JND at a time. Because each JND is a constant fraction of the current intensity, the physical intensities at successive JNDs form a *geometric* series (each one multiplied by the same factor), while the sensation steps form an *arithmetic* series (1, 2, 3, 4 units). A quantity that rises arithmetically while another rises geometrically is a logarithm. The result is **Fechner's law**:

$S = k \log I$

where S is the magnitude of the sensation, I is the physical intensity (strictly, measured in multiples of the threshold intensity) and k is a constant that depends on the sense. Because it rests on Weber's law, it is often called the Weber-Fechner law.

A worked example makes the logarithm concrete. Picture a dark room on Diwali night. Light one diya and the room is transformed. Light a second and the room gets noticeably brighter. To get the same sized step again, Fechner's law says you will need not three diyas but four; then eight; then sixteen; then thirty-two. Each *doubling* of the physical light adds roughly one equal step of brightness. By the time sixty-four diyas are burning, adding one more makes no visible difference at all. Scales of this kind are familiar from science and engineering: the astronomers' magnitude scale for the brightness of stars (much older than Fechner) and the decibel scale for sound (introduced by telephone engineers in the 1920s; Section 3.8) are both logarithmic, and both fit how our senses actually grade intensity.

Fechner's achievement went beyond the equation. In the *Elemente* of 1860 he coined the word psychophysics, set out the experimental methods for measuring thresholds (Section 3.4), and distinguished "outer psychophysics" (the relation between the stimulus and sensation, which he could study) from "inner psychophysics" (the relation between brain activity and sensation, which he could not). For founding the field he is called the father of psychophysics, and his work showed the young science of psychology that mental events could be measured. One reader who took that lesson to heart was **Hermann Ebbinghaus** (1850 to 1909), who came across a second-hand copy of the *Elemente* and set out to measure memory with the same rigour (Chapter 7). Fechner's method of scaling sensation is called **indirect scaling**, because it measures sensation indirectly, through discrimination (counting JNDs), rather than by asking people how strong a sensation feels.

Where does Fechner's law fail? Its weak point is its founding assumption. There is no direct evidence that all JNDs are subjectively equal, and when people are asked to judge sensory magnitudes directly, their judgements often do not follow a logarithmic curve. For some dimensions, such as electric shock, sensation grows *faster* than intensity, which no logarithmic law can describe. That is where Stevens enters the story.

### Stevens' power law: ask people directly

**Stanley Smith Stevens** (1906 to 1973) of Harvard University argued that there was no need to count JNDs: people can judge the magnitude of their sensations directly, and their judgements are lawful. His main method, **magnitude estimation**, works like this. The experimenter presents a standard stimulus (called the modulus) and assigns it a number, say 10. The observer is then given other stimuli and asked to assign each a number proportional to how intense it seems: a light that looks twice as bright as the standard gets 20; one that looks half as bright gets 5. Across hundreds of studies, Stevens found that the results for every continuum he tested were described by a power function. In 1957 he set out what is now called **Stevens' power law**:

$$S = k\, I^{n}$$

where S is the judged magnitude of the sensation, I is the physical intensity, k is a scaling constant and the exponent $n$ is characteristic of each sensory continuum. Taking logarithms of both sides gives $\log S = \log k + n \log I$: plotted on log-log axes, every power function becomes a straight line whose slope is the exponent. That is how exponents are measured in practice.

The exponent tells you how sensation grows. When $n$ is less than 1, sensation grows more slowly than intensity, a pattern called **response compression**: brightness has an exponent of about 0.33, so to double the apparent brightness of a light you must increase its physical intensity about eightfold (because 2 raised to the power 1/0.33 is 8). When $n$ equals 1, sensation grows in proportion to intensity, as it does for the apparent length of a line. When $n$ is greater than 1, sensation grows faster than intensity, a pattern called **response expansion**: electric shock has an exponent of about 3.5, so doubling the current makes the shock feel roughly eleven times as intense (2 raised to the power 3.5 is about 11). Figure 3.2 shows the three cases and Table 3.4 lists classic exponents.

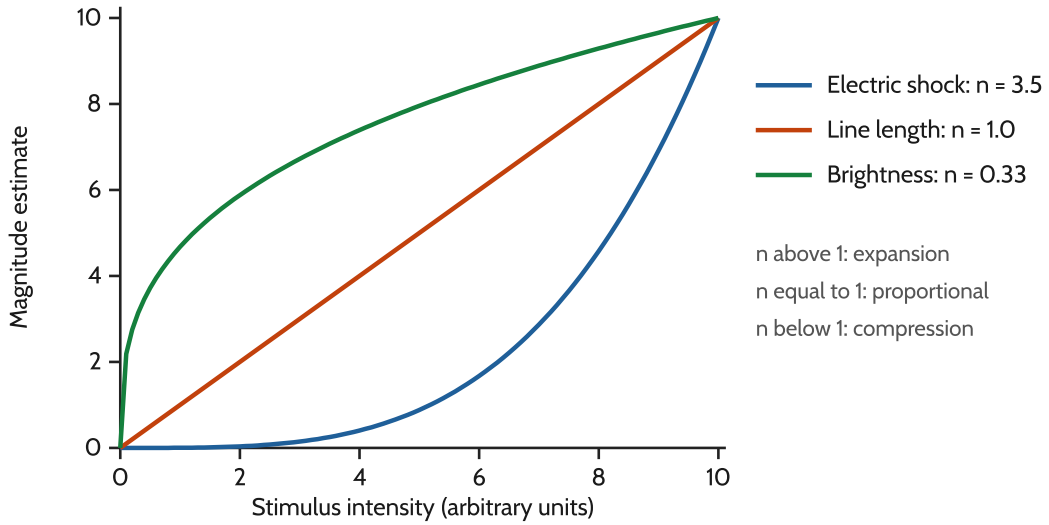


**Figure 3.2** Stevens' power functions for three continua. Brightness (exponent 0.33) shows response compression, line length (1.0) is proportional, and electric shock (3.5) shows response expansion. On logarithmic axes each curve would become a straight line whose slope is its exponent.

**Table 3.4** Representative exponents of Stevens' power law (approximate values reported by Stevens).

| Continuum | Exponent (n) | Type of growth |
|---|---|---|
| Brightness (large target, dark-adapted eye) | 0.33 | Compression |
| Smell (coffee odour) | 0.55 | Compression |
| Loudness (3,000 Hz tone, measured as sound pressure) | 0.67 | Compression |
| Apparent length of a line | 1.0 | Proportional |
| Taste (sugar) | 1.3 | Expansion |
| Heaviness of lifted weights | 1.45 | Expansion |
| Electric shock (through the fingers) | 3.5 | Expansion |

Why should the senses differ like this? A plausible functional answer is that compression lets the eye and ear handle an enormous range of physical intensities (the brightest light we can tolerate is billions of times more intense than the dimmest we can see) within a manageable range of experience, whereas expansion makes potentially harmful stimuli, such as electric current, feel alarming quickly.

Stevens strengthened his case with **cross-modality matching**. Instead of using numbers, observers match the intensity of one sense to another: for example, they squeeze a hand grip until the effort feels as strong as a sound is loud. If both senses follow power laws with known exponents, the power law predicts the exponent of the matching function (it should equal the ratio of the two exponents), and the matches came out close to the predictions. This showed that the power law was not merely an artefact of how people use numbers. Stevens was a productive figure in other ways too: in 1946 he proposed the four scales of measurement (nominal, ordinal, interval and ratio) taught in the Research Methodology book, and in 1936 he defined the sone, a unit of loudness.

Where does Stevens' law fail? Magnitude estimates are affected by the range of stimuli presented, by the choice of modulus, by the order of presentation and by individual habits with numbers, so exponents vary from study to study. Some critics argue that magnitude estimation measures how people map sensations onto numbers rather than sensation itself. Even so, the power law describes direct magnitude judgements far better than Fechner's logarithm, and in 1961 Stevens titled a famous paper "To honor Fechner and repeal his law". The two laws are best seen as answers to different questions: Fechner's law describes discriminability (how many just noticeable steps lie between two stimuli), while Stevens' law describes apparent magnitude (how big a sensation seems). Table 3.5 sets the three laws side by side.

Table 3.5 The three psychophysical laws compared.

| Law | Formula | What it says | Method behind it | Main limitation |
|---|---|---|---|---|
| Weber's law (Weber, 1834) | $\Delta I / I = k$ | The JND is a constant fraction of the standard | Discrimination of weights and touches | Fails near threshold and at extremes |
| Fechner's law (Fechner, 1860) | $S = k \log I$ | Sensation grows with the logarithm of intensity | Indirect scaling: counting JNDs | Assumes all JNDs are equal in sensation; cannot describe expansion |
| Stevens' power law (Stevens, 1957) | $S = k I^n$ | Sensation is a power function of intensity; exponent varies by sense | Direct scaling: magnitude estimation, cross-modality matching | Judgements are sensitive to range, context and number use |

EXAM TRAP

Weber gave the law of the *JND as a constant fraction*; Fechner gave the *logarithmic* law and the *assumption that every JND adds an equal unit of sensation*; Stevens gave the *power* law and *magnitude estimation*. "Who assumed that every just noticeable difference is a constant increment in sensation?" is Fechner, not Weber, even though it is Weber's JND. The founder (father) of psychophysics is Fechner, author of *Elements of Psychophysics* (1860), not Weber, not Wundt (father of experimental psychology) and not Ebbinghaus (pioneer of memory research, inspired by Fechner). Tolman, sometimes offered as a distractor, was a learning theorist (Chapter 6).

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. The conversion of physical energy into neural signals by a receptor is called _____.
2. The doctrine that the quality of a sensation depends on which nerve is stimulated was proposed by Johannes _____.
3. Because of the all-or-none law, stimulus intensity is coded largely by the _____ of neural firing.
4. The decline in sensitivity to an unchanging stimulus is called sensory _____.
5. The absolute threshold is the intensity detected on _____ percent of trials.
6. The smallest detectable change in a stimulus is the difference threshold, also called the _____.
7. Weber's law states that $\Delta I/I$ equals a _____.
8. Fechner's law states that sensation is proportional to the _____ of stimulus intensity.
9. Stevens measured sensation directly by the method of magnitude _____.
10. An exponent greater than 1 in Stevens' law, as for electric shock, produces response _____.

**Answers:** (1) transduction, (2) Muller, (3) rate (frequency), (4) adaptation, (5) 50, (6) just noticeable difference (JND), (7) constant (the Weber fraction), (8) logarithm, (9) estimation, (10) expansion.

## 3.4 The classical psychophysical methods

LOWER YIELD · NOT ASKED YET

A threshold is defined as a 50 percent point, but how do you actually find it? Fechner's *Elemente* described three procedures, and with small changes they are still in use in every vision clinic, hearing clinic and perception laboratory in the world. Each involves a trade-off between speed and accuracy, and each is vulnerable to particular errors. Understanding those errors, and how they are controlled, is as important for examinations as knowing the procedures.

A few terms first. In threshold work the experimenter presents stimuli in a *series*, which may be *ascending* (starting well below threshold and increasing step by step) or *descending* (starting well above threshold and decreasing). The point in a series at which the observer's response changes (from "no" to "yes", or the reverse) is the *transition point*. When two stimuli are compared, the observer may also show a **time-order error**: a systematic tendency to judge the second of two equal stimuli as different from the first, simply because it comes second. A similar *space error* arises when a stimulus is judged differently because it appears on the left rather than the right. Errors of this kind are controlled by **counterbalancing**: arranging the order (or position) of conditions so that each occurs equally often in each place, so that the errors cancel out.

### The method of limits

In the **method of limits** (Fechner called it the method of just noticeable differences; it is also called the method of minimal changes), the experimenter changes the stimulus in small steps, in alternating ascending and descending series, and records where the observer's re-

sponse changes. To find the absolute threshold for a faint tone, for example, a descending series starts with a clearly audible tone and lowers it step by step until the observer first says "no"; an ascending series starts with an inaudible tone and raises it until the observer first says "yes". The threshold for each series is taken as the midpoint between the last two responses, and the overall threshold is the average across all series. Table 3.6 shows a worked example.

Table 3.6 A worked example of the method of limits for the absolute threshold of a tone (intensity in arbitrary steps). The overall threshold is the mean transition point, 4.0. Notice that ascending transitions are higher than descending ones, the signature of the error of habituation.

| Series | Direction and start | Responses | Transition point |
|---|---|---|---|
| 1 | Descending from 9 | "Yes" from 9 down to 4, "no" at 3 | 3.5 |
| 2 | Ascending from 1 | "No" from 1 up to 4, "yes" at 5 | 4.5 |
| 3 | Descending from 8 | "Yes" from 8 down to 4, "no" at 3 | 3.5 |
| 4 | Ascending from 2 | "No" from 2 up to 4, "yes" at 5 | 4.5 |

The method of limits is quick and intuitive, but it suffers from two opposite errors, both caused by the predictable order of the stimuli. In the **error of habituation** (also called the error of perseveration), the observer falls into the habit of giving the same response and keeps giving it after the true threshold has been crossed: continuing to say "no" too long in an ascending series (so the ascending threshold comes out too high) and continuing to say "yes" too long in a descending series (so the descending threshold comes out too low). In the **error of anticipation** (or expectation), the observer, knowing that a change is coming, switches response too early: saying "yes" prematurely in an ascending series and "no" prematurely in a descending series. The two errors push the transition points in opposite directions.

The cure is counterbalancing. Ascending and descending series are alternated in equal numbers, so that whatever error inflates the ascending thresholds is balanced by the corresponding error in the descending ones, and the starting point of each series is varied so that the observer cannot simply count steps to the expected change. The same logic applies to other order effects in psychophysical experiments, such as primacy and recency effects: they are controlled by counterbalancing the order of presentation.

The method of limits can also measure difference thresholds. The observer compares each comparison stimulus with the standard and responds "greater", "equal" or "less". The range of comparison values judged "equal" is called the interval of uncertainty; its midpoint gives the point of subjective equality, and half its width gives the difference threshold.

### The method of constant stimuli

The **method of constant stimuli** (Fechner's method of right and wrong cases, also called the frequency method) removes the predictable order altogether. The experimenter first chooses a small set of fixed ("constant") stimulus values, usually five to nine, spaced so that the weakest is almost never detected and the strongest almost always is. Each value is then presented many times, often fifty or a hundred times, in a random order. The proportion of "yes" responses is calculated for each value and plotted against stimulus intensity to give the psychometric function (Figure 3.1). The absolute threshold is the intensity that corresponds to 50 percent detection, read off (interpolated from) the curve.

For difference thresholds, each comparison is paired with the standard and the observer says whether it is "heavier" or "lighter" (or "louder", "brighter" and so on). The comparison value judged "heavier" 50 percent of the time is the point of subjective equality. The difference threshold is conventionally taken as half the distance between the 75 percent and 25 percent points.

Because the stimuli arrive in random order, the observer cannot anticipate a change or settle into a habit, so the errors of habituation and anticipation disappear. The method of constant stimuli is therefore regarded as the most accurate and reliable of the classical methods. Its cost is time: many trials are needed, and many are spent on stimuli far from threshold, which tell the experimenter little. It also requires pilot work to choose a sensible set of values.

### The method of adjustment

In the **method of adjustment** (Fechner's method of average error, also called the method of reproduction), the observer controls the stimulus. To measure an absolute threshold, the observer turns a knob until a tone is just audible; to measure a point of subjective equality, the observer adjusts a comparison line until it looks equal in length to a standard. The procedure is repeated many times. The average setting gives the threshold or the point of subjective equality, and the variability of the settings (their average deviation or standard deviation, the "average error" that gives the method its name) is an index of the difference threshold: the more finely a person can discriminate, the more tightly the settings cluster.

The method of adjustment is fast and engaging, and it is the natural way to measure the size of an illusion: observers adjust a plain line until it matches the apparent length of the Muller-Lyer line, and the difference between the setting and the true length (the constant error) measures how strong the illusion is. Its weakness is precision. The stimulus changes continuously under the observer's own control, so motor skill, impatience and the observer's beliefs can all creep into the settings, and the experimenter loses control over exactly which stimuli are presented. Table 3.7 sets the three classical methods side by side.

Table 3.7 The three classical psychophysical methods.

| Method | Other names | Procedure | Strengths | Weaknesses |
|---|---|---|---|---|
| Limits | Method of just noticeable differences; minimal changes | Stimulus changed in small steps in alternating ascending and descending series | Quick, simple | Errors of habituation and anticipation; needs counterbalancing |
| Constant stimuli | Method of right and wrong cases; frequency method | Fixed set of values, each presented many times in random order | Most accurate; no order errors; gives full psychometric function | Slow; many wasted trials |
| Adjustment | Method of average error; reproduction | Observer adjusts the stimulus until it is just detectable or matches the standard | Fastest; ideal for measuring PSE and illusions | Least precise; observer controls the stimulus |

### Staircases and forced choice: the modern refinements

Modern laboratories and clinics mostly use adaptive procedures that combine the speed of the method of limits with the efficiency of concentrating trials near threshold. In a **staircase method**, the stimulus is made weaker after every "yes" (or correct) response and stronger after every "no" (or incorrect) response. The intensity therefore climbs and descends around the threshold like a person walking up and down a staircase, and the threshold is estimated

as the average of the reversal points, the intensities at which the direction changed. The method was popularised in psychophysics by Tom Cornsweet in 1962. Refinements change the rule to target a particular percentage: in the widely used "two-down, one-up" staircase, the stimulus is made harder only after two correct responses in a row, which makes the staircase settle at about 71 percent correct. To stop observers from guessing the rule, experimenters often interleave two staircases, one starting high and one low, at random.

A close cousin is the tracking method that **Georg von Bekesy** (1899 to 1972) built into an audiometer in 1947. The listener holds down a button whenever the tone is audible; while the button is pressed the intensity falls, and when it is released the intensity rises. The result is a zigzag trace that hugs the listener's threshold as the frequency sweeps from low to high, an audiogram drawn by the listener's own responses.

Many modern procedures also replace "yes or no" with a **forced-choice procedure**. In two-alternative forced choice, the stimulus appears in one of two intervals (or locations) and the observer must say which, even when guessing. Chance performance is 50 percent, so the threshold is usually set at the 75 percent correct point, halfway between chance and perfection. The great advantage of forced choice is that it removes the observer's willingness to say "yes" from the measurement. A cautious person and a bold person can no longer differ simply because one says "yes" more readily. That problem, the contamination of thresholds by the observer's decision style, is exactly what signal detection theory was invented to solve.

CLINICAL LINK

Every routine sensory assessment is a psychophysical experiment. Pure-tone audiometry finds the absolute threshold for tones of different frequencies, usually with an ascending method of limits or a staircase. An eye chart measures a visual threshold for detail. Automated perimetry, used to detect glaucoma, maps detection thresholds across the visual field. When a clinical psychologist tests a patient's two-point threshold or pain threshold as part of a neuropsychological examination, the same principles (50 percent points, counterbalanced series, attention to response bias) decide whether the result can be trusted.

EXAM TRAP

Match the names: method of limits = method of just noticeable differences = minimal changes; method of constant stimuli = method of right and wrong cases = frequency method; method of adjustment = method of average error = reproduction. In the method of limits, the *error of habituation* means continuing the previous response too long, and the *error of anticipation* means changing the response too early. Both, along with primacy, recency and time-order effects, are controlled by *counterbalancing* (alternating ascending and descending series), not by "randomisation", "pairing" or a "double-blind technique". Randomisation is the cure in a different method: constant stimuli.

## 3.5 Signal detection theory MEDIUM YIELD · 4 QS

During the Second World War, radar operators sat for hours in front of screens watching for the faint blip that meant an enemy aircraft. The screen was never clean. It sparkled with "noise": random flecks produced by the equipment and the atmosphere. A real aircraft at long range produced a blip barely stronger than the noise. Every few seconds the operator had to

make a decision under uncertainty: is that an aircraft or just noise? Say "aircraft" too readily and fighters are scrambled for nothing; say it too reluctantly and the bombers arrive unannounced. Two operators with equally good eyes could perform very differently because they set different standards for raising the alarm.

The classical idea of a threshold had no room for this. It assumed that a stimulus is either above the threshold (and detected) or below it (and missed), with the observer passively reporting the result. Yet experimenters had long noticed awkward facts. If they slipped "catch trials" into a detection experiment (trials on which no stimulus was presented), some observers said "yes" anyway. Observers who were paid for each detection reported more stimuli, and their measured thresholds fell, although their eyes and ears had not changed. A threshold, it seemed, measured not only what an observer could sense but also how willing the observer was to say so.

## The origins of the theory

**signal detection theory** (SDT) began in engineering. In the early 1950s, W. Wesley Peterson, Theodore G. Birdsall and W. C. Fox at the University of Michigan developed a mathematical "theory of signal detectability" for radar and radio receivers, published in 1954. In the same year **Wilson P. Tanner** Jr. and **John A. Swets** applied it to human observers in a paper titled "A decision-making theory of visual detection". The definitive statement for psychology came in the 1966 book *Signal Detection Theory and Psychophysics* by **David M. Green** and John A. Swets. The central claim is simple and powerful: performance in any detection task depends on two independent things, the observer's *sensitivity* and the observer's *decision criterion*, and the two can be measured separately.

## Noise, signals and the four outcomes

SDT starts by admitting that the observer's internal state is never quiet. **noise**, in SDT, is all the random background activity that is present whether or not a signal occurs: static on a screen, hum in a room, and above all the spontaneous firing of the observer's own neurons. On every trial the observer experiences some amount of internal "evidence" for a signal. On trials with no signal, this evidence comes from noise alone, and it varies from trial to trial, forming the *noise distribution*. On trials with a signal, the signal adds to the noise, so the evidence tends to be larger, forming the *signal-plus-noise distribution*. The two distributions overlap (Figure 3.3a). Some noise-alone trials produce more evidence than some signal trials, and the observer, who experiences only the evidence, cannot tell which kind of trial produced it.

So the observer must adopt a rule: if the evidence on this trial exceeds a certain level, say "yes, a signal was present"; otherwise say "no". That cut-off level is the **criterion**. Every trial then ends in one of four outcomes, set out in Table 3.8. A **hit** is saying "yes" when a signal is present. A **miss** is saying "no" when a signal is present. A **false alarm** is saying "yes" when only noise is present. A **correct rejection** is saying "no" when only noise is present.

Table 3.8 The four outcomes of a detection trial. The hit rate and the false alarm rate together contain all the information, because misses = 1 minus the hit rate and correct rejections = 1 minus the false alarm rate.

| What was presented | Observer says "yes" | Observer says "no" |
|---|---|---|

| What was presented | Observer says "yes" | Observer says "no" |
|---|---|---|
| Signal present (signal plus noise) | Hit | Miss |
| Signal absent (noise alone) | False alarm | Correct rejection |

Two numbers summarise performance. The *hit rate* is the proportion of signal trials on which the observer said "yes"; the *false alarm rate* is the proportion of noise trials on which the observer said "yes". Notice what a hit rate alone cannot tell you. A smoke alarm that rings constantly has a perfect hit rate: it never misses a fire. It is also useless, because its false alarm rate is perfect too. Any honest measure of detection must look at hits and false alarms together.

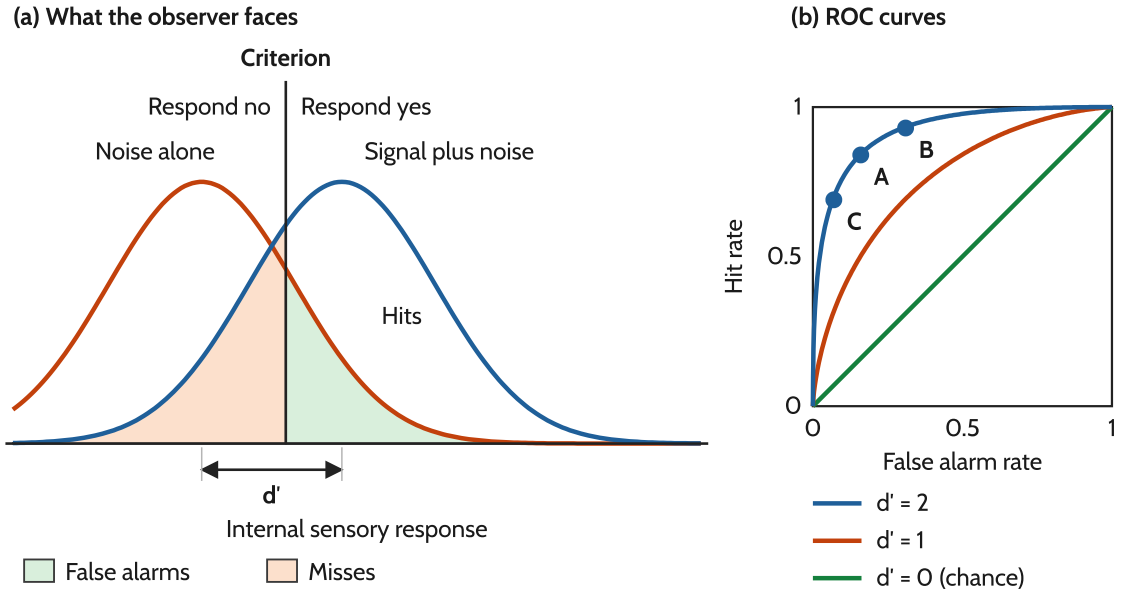


Figure 3.3 Signal detection theory. (a) On noise trials the internal response follows the noise distribution; on signal trials it follows the shifted signal-plus-noise distribution. The observer says "yes" whenever the response exceeds the criterion. The shaded areas are false alarms (noise beyond the criterion) and misses (signal short of it); the remaining areas are hits and correct rejections. The distance between the two peaks is d′. (b) ROC curves: moving the criterion traces out a curve of hit rate against false alarm rate. Doctors A, B and C from the worked example have the same sensitivity (d′ about 2) but different criteria.

## Sensitivity: d′ ●●

The first quantity SDT measures is **sensitivity**, the ability to distinguish a signal from noise. It is indexed by **d prime** (written d′), the distance between the means of the noise and signal-plus-noise distributions, expressed in standard deviation units. When the two distributions lie far apart, signal trials almost always produce more evidence than noise trials, and detection is easy (large d′). When they overlap completely, the observer cannot do better than guess (d′ = 0). Sensitivity depends on the strength of the signal, on the amount of noise and on the acuity of the observer's sensory system. Crucially, it does not depend on how willing the observer is to say "yes".

In practice d′ is calculated from the hit rate (H) and the false alarm rate (F) by converting each into a z score from the standard normal table: d′ = z(H) minus z(F). You do not need to compute it in the examination, but you must understand what makes it large or small. If the hit

rate equals the false alarm rate, the observer is saying "yes" just as often when nothing is there as when something is, and d′ is zero. The further the hit rate rises above the false alarm rate, the larger d′ becomes, so sensitivity is highest when hits clearly outnumber false alarms. (Some Indian exam notes write this loosely as "d = hits minus false alarms"; the direction is right, but the real formula subtracts the z scores of the two rates, not the raw counts.) If false alarms exceed hits, the observer is doing worse than chance, which signals confusion or a misunderstanding of the task.

### The criterion: liberal and conservative observers

The second quantity is the observer's **response bias**, the location of the criterion. It is expressed either as $\beta$ (beta), the ratio of the heights of the signal and noise distributions at the criterion, or as $c$, the distance of the criterion from the point midway between the two distributions. An observer whose criterion is low, who says "yes" on rather weak evidence, has a **liberal criterion** (also called lax or risky): such an observer earns many hits but also many false alarms ($\beta$ below 1; $c$ negative). An observer whose criterion is high, who says "yes" only when the evidence is strong, has a **conservative criterion** (strict or cautious): few false alarms, but more misses ($\beta$ above 1; $c$ positive). A neutral observer places the criterion where the two distributions cross ($\beta = 1$; $c = 0$).

What sets the criterion? Two kinds of factor, neither of them sensory. The first is expectation: if signals are common, it pays to say "yes" more readily; if they are rare, it pays to be cautious. The second is the payoff matrix, the rewards for hits and correct rejections and the costs of misses and false alarms. A smoke detector is deliberately set to a very liberal criterion because a miss (an undetected fire) is catastrophic while a false alarm (burnt toast) is merely annoying. A court of law, by contrast, demands proof "beyond reasonable doubt", a very conservative criterion, because society judges convicting an innocent person (a false alarm) to be worse than acquitting a guilty one (a miss). SDT shows that an ideal observer should shift the criterion in exactly these ways, and real observers do.

> **WORKED EXAMPLE: THREE RADIOLOGISTS, ONE SENSITIVITY**
>
> Each of four radiologists reads the same 200 mammograms, half showing a tumour.
>
> Dr A detects 84 percent of the tumours (H = 0.84) and wrongly flags 16 percent of the healthy scans (F = 0.16): d′ is about 2.0, and the criterion is neutral ($c = 0$).
> Dr B, recently criticised for missing a tumour, now flags anything suspicious: H = 0.93, F = 0.31. Her d′ is still about 2.0; only her criterion has moved, to a liberal position ($c$ about −0.5).
> Dr C, worried about unnecessary biopsies, flags only clear cases: H = 0.69, F = 0.07. Again d′ is about 2.0, with a conservative criterion ($c$ about +0.5).
> Dr D has a hit rate of 0.84, just like Dr A, but a false alarm rate of 0.50. His d′ is only about 1.0. He is genuinely less able to tell tumours from healthy tissue.
>
> Comparing hit rates alone would rank Dr B best and Dr C worst, and would rate Dr D as good as Dr A. SDT shows that A, B and C are equally skilled and differ only in caution, while D is the weakest reader. (Figure 3.3b plots A, B and C on the same ROC curve.)

### The ROC curve

If you change an observer's criterion (by changing the payoffs or the proportion of signal trials, or by asking for confidence ratings) while keeping the signal the same, the hit rate and false alarm rate move together. Plotting hit rate against false alarm rate for each criterion produces a **receiver operating characteristic** (ROC) curve, named after its origin in radar engineering (Figure 3.3b). Each curve is an isosensitivity curve: every point on it has the same d′ and differs only in bias. A conservative observer sits near the lower left; a liberal observer sits near the upper right. If the observer has no sensitivity at all, the curve is the diagonal, where hits equal false alarms. The greater the sensitivity, the more the curve bows towards the upper left corner, where hits are high and false alarms low. The area under the ROC curve is therefore a convenient bias-free index of accuracy, ranging from 0.5 (chance) to 1.0 (perfect discrimination).

ROC curves also settled the theoretical argument. A strict "high threshold" theory, in which a stimulus is either detected or not and the observer simply guesses on undetected trials, predicts ROC curves that are straight lines. Real data almost always produce smoothly bowed curves of the kind SDT predicts. This is the main reason signal detection theory replaced the idea of a fixed sensory threshold: detection is a continuous process of weighing evidence, and the 50 percent "threshold" of classical psychophysics turns out to mix two quantities, sensitivity and bias, that SDT can separate.

#### KEY STUDY: ARE FINGERPRINT EXPERTS REALLY BETTER? (THOMPSON AND COLLEAGUES, 2014)

**Question.** When a fingerprint examiner declares a "match", is that because experts discriminate better than novices, or simply because they are more willing to say "match"? **Method.** Matthew Thompson and colleagues gave experienced examiners and untrained novices genuine pairs of crime-scene prints, some matching and some only similar. **Result.** Both groups correctly said "match" on roughly 70 percent of truly matching pairs, so their hit rates were nearly identical. The difference lay in false alarms: novices said "match" to similar but non-matching pairs on 57 percent of trials, experts on fewer than 2 percent. **Conclusion.** Converted to SDT terms, the experts' d′ was far higher (roughly 2.7 against roughly 0.4 for novices), and the experts were also more conservative. A hit rate alone would have shown no difference; the false alarm rate revealed everything. **Why it matters.** The same analysis applies to eyewitness line-ups, lie detection and medical screening: never judge a detector by its hits without its false alarms.

### Signal detection in the real world ●●

Once you have the idea, you will see signal detection problems everywhere. An airport baggage screener looks for a rare weapon among thousands of harmless bags; because real threats are so rare, screeners drift towards a conservative criterion, which is why security agencies insert fake threat images to keep them alert (vigilance and its decline over long watches are discussed in Chapter 5). A polygraph examiner who calls almost everyone "deceptive" will catch nearly every liar but will brand many innocent people as liars, and the polygraph's modest sensitivity is the core of the scientific case against relying on it. In recognition memory, deciding whether a face or word is "old" or "new" has exactly the same structure, and memory researchers measure d′ and bias in the same way (Chapter 7). And whenever a claim is made that people respond to "subliminal" stimuli, the first SDT question is

whether the stimuli were truly undetectable (d′ of zero) or merely below a cautious observer's criterion (Chapter 4 covers subliminal perception).

**CLINICAL LINK**

Every screening test you will use as a clinical psychologist, from a depression questionnaire to a dementia screen, is a signal detector with a criterion: its cut-off score. Lowering the cut-off catches more true cases (higher hit rate) at the price of more false positives; raising it does the reverse. In diagnostic language the hit rate is called the test's *sensitivity*, and one minus the false alarm rate is its *specificity*, and test developers choose cut-offs by drawing exactly the ROC curves described here. The choice is not purely statistical: it depends on the payoffs, such as the cost of missing a suicidal patient versus the cost of an unnecessary referral.

**EXAM TRAP**

The ability to distinguish a signal from background noise is *sensitivity* (*d′*); the willingness to say "yes" is the *criterion* (*β* or *c*), also called response bias; the ROC curve is the plot that displays both; the JND belongs to classical psychophysics, not SDT. The parameter that SDT added to classical threshold theory is the *response criterion*. Beware a second meaning of "sensitivity": in diagnostic testing it means the hit rate alone, which is not the same as d′. SDT is attributed to Green and Swets (1966, the classic book), Tanner and Swets (1954, first psychological application) and, in Indian notes, to the engineers Peterson and Birdsall; any of these can appear as the keyed answer.

**EXAM TRAP: "RECEIVER OPERANT CHARACTERISTICS"**

IHBAS 2025 asked what "receiver-operant characteristics" refer to "in communication studies" and keyed the answer as *factors in the audience influencing persuasion*. In the communication (persuasion) framework, the "receiver" is the audience, and receiver characteristics (intelligence, self-esteem, involvement, initial attitude) are audience factors that moderate persuasion (Social Psychology book). Do not confuse this with the *receiver operating characteristic* (ROC) curve of signal detection theory, which plots hit rate against false alarm rate as the criterion changes. The option "physiological thresholds of auditory perception" is bait for students who half-remember the SDT term; it describes neither concept.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Fechner's method of right and wrong cases is better known as the method of _____ stimuli.
2. Continuing to say "no" too long in an ascending series is the error of _____.
3. Errors of habituation and anticipation are controlled by alternating ascending and descending series, a form of _____.
4. The method of average error is also called the method of _____.
5. A procedure that lowers the stimulus after each "yes" and raises it after each "no" is a _____ method.
6. In signal detection theory, saying "yes" when no signal was presented is a _____.
7. The index of sensitivity in signal detection theory is _____.
8. An observer who says "yes" on weak evidence has a _____ criterion.
9. A plot of hit rate against false alarm rate is called a _____ curve.

**Answers:** (1) constant, (2) habituation (perseveration), (3) counterbalancing, (4) adjustment, (5) staircase, (6) false alarm, (7) d′ (d prime), (8) liberal (lax), (9) receiver operating characteristic (ROC).

## 3.6 Vision I: light, the eye and the two receptor systems HIGH YIELD · 11 QS

Vision is the sense psychologists know best, and it is the sense examiners ask about most. We begin with the stimulus itself, follow light through the eye to the receptors, and then examine the remarkable fact that each human eye contains two quite different visual systems, one for daylight and one for the dark.

### Light ●●

Light is electromagnetic radiation, the same kind of energy as radio waves, X-rays and microwaves. Like all such radiation it travels as waves (and, as physicists insist, as particles called photons) at about 300,000 kilometres per second. What distinguishes light from other electromagnetic radiation is only its wavelength. The human eye responds to wavelengths between about 380 and 760 nanometres (a nanometre is a billionth of a metre); this narrow band is the *visible spectrum*. Some textbooks round it to 400 to 700 nanometres, but 380 to 760 is the standard examination answer. There is nothing physically special about this band: it is simply the range our photopigments absorb. Just below it lies ultraviolet radiation, which bees can see; just above it lies infrared, which rattlesnakes and other pit vipers can sense, letting them detect warm-blooded prey in what is darkness to us.

For any wave travelling at a fixed speed, wavelength and frequency are inversely related: speed equals frequency multiplied by wavelength, so the shorter the wavelength, the higher the frequency. Violet light (around 400 nanometres) therefore has a shorter wavelength and a higher frequency than red light (around 700 nanometres). The same inverse relation holds for sound.

Three physical properties of light map onto three psychological dimensions of visual experience. Wavelength is experienced mainly as **hue**, the quality we name with colour words:

short wavelengths look violet and blue, middle wavelengths green and yellow, long wavelengths orange and red. Intensity (the amount of light energy, the height or amplitude of the wave) is experienced mainly as **brightness**. Purity (the extent to which light consists of a single wavelength rather than a mixture) is experienced as **saturation**, the vividness of a colour: a pure red is highly saturated, while red diluted with white light looks pink, a less saturated version of the same hue. Table 3.9 summarises these mappings alongside the corresponding ones for sound.

Table 3.9 Physical properties of light and sound and the psychological dimensions they mainly determine.

| Sense | Physical property | Usual unit | Psychological dimension |
|---|---|---|---|
| Vision | Wavelength | Nanometres (nm) | Hue |
| Vision | Intensity (amplitude) | Light energy | Brightness |
| Vision | Purity (spectral composition) | Proportion of one wavelength | Saturation |
| Hearing | Frequency | Hertz (Hz) | Pitch |
| Hearing | Amplitude (intensity) | Decibels (dB) | Loudness |
| Hearing | Complexity (mixture of frequencies) | Harmonic content | Timbre |

## The eye ●●

Follow a ray of light into the eye. It first passes through the **cornea**, the transparent, curved front surface of the eye, which does most of the focusing (roughly two-thirds of the eye's bending power). It then crosses a watery fluid (the aqueous humour) and passes through the **pupil**, the opening in the centre of the coloured **iris**. The iris is a ring of muscle that changes the size of the pupil: it constricts the pupil in bright light and dilates it in dim light. This involves a trade-off. A small pupil gives a sharper image with a greater depth of focus; a large pupil lets in more light but sacrifices sharpness. The eye favours acuity when light is plentiful and sensitivity when it is scarce.

Behind the pupil lies the **lens**, which fine-tunes the focus. Its shape is adjusted by the ciliary muscles: when you look at something near, the lens becomes thicker and rounder, bending light more strongly; when you look at something distant, it is pulled flatter. This process of changing the shape of the lens to bring images into focus is called **accommodation**. With age the lens stiffens and loses the ability to round up, which is why most people need reading glasses in their forties (presbyopia). Other common focusing errors arise from the shape of the eyeball: in short-sightedness (myopia) the eyeball is too long and distant objects focus in front of the retina; in long-sightedness (hyperopia) it is too short and near objects focus behind it. A clouded lens is a cataract, which remains the leading cause of blindness in India and is corrected by replacing the lens surgically. Light then crosses the jelly-like vitreous humour and arrives at the **retina**, the thin sheet of neural tissue lining the back of the eye, where transduction takes place.

The retina is built, oddly, inside out. The receptors (rods and cones) lie at the very back, against the pigment layer, so light must pass through the other retinal layers before reaching them. The receptors connect to *bipolar cells*, which connect to *retinal ganglion cells*, whose axons run across the inner surface of the retina and gather together to leave the eye as the optic nerve (cranial nerve II). Two further cell types, *horizontal cells* and *amacrine cells*, carry sig-

nals sideways across the retina, allowing neighbouring regions to influence each other; you will see why that matters when we reach lateral inhibition.

Two regions of the retina have special names. Near the centre of the retina, in a small yellowish area called the macula, is a tiny pit, the **fovea**, where the overlying layers are pushed aside so that light falls directly on the receptors. The fovea contains only cones, packed tightly together, and it gives us our sharpest, most detailed, colour vision; when you look straight at something, you are turning your eyes so that its image falls on the fovea. A few millimetres towards the nose lies the **optic disc**, where the ganglion cell axons leave the eye and blood vessels enter. Because the optic disc has no receptors at all, it creates a **blind spot** in each eye's field of vision. (The farthest forward edge of the retina, where it ends near the ciliary body, is the ora serrata, named for its serrated outline; it has no special role in detailed vision.)

You can find your blind spot. Draw a small cross and, about 8 centimetres to its right, a dot the size of a pea. Close your left eye, stare steadily at the cross with your right eye and move the page slowly towards and away from your face. At about 25 to 30 centimetres the dot vanishes, because its image has fallen on the optic disc. Yet you never notice a hole in your vision, even with one eye closed. The brain fills the gap with whatever surrounds it, a process called completion or filling in: draw a line passing through the dot's position, and when the dot disappears the line appears unbroken.

### Rods and cones: two visual systems in one eye ●●

The human retina contains two kinds of photoreceptor, named for their shapes. The **rods** number roughly 120 million per eye; the **cones** number only about 6 to 7 million (estimates vary). Their distribution is strikingly different: the fovea contains cones and no rods at all, while the density of rods peaks about 20 degrees away from the fovea and rods dominate the periphery. This is why, on a dark night, you can often see a faint star more clearly by looking slightly to one side of it than by looking straight at it.

In the nineteenth century, researchers noticed that animals active only by day tend to have cone-rich retinas while nocturnal animals have rod-rich ones. From this observation came the **duplexity theory** of vision: rods and cones mediate two different kinds of vision. **photopic vision** is cone-mediated vision in good light; it provides high acuity and colour. **scotopic vision** is rod-mediated vision in dim light; it is extremely sensitive but lacks both detail and colour. (Twilight vision, with both systems contributing, is called mesopic.) At night, everything looks grey because only the rods are working, and rods cannot distinguish wavelengths.

Why should one system be sensitive and the other sharp? Part of the answer is chemistry and part is wiring. The rods contain a single photopigment, **rhodopsin** (sometimes called visual purple), which is so sensitive that a rod can respond to a single quantum of light, as Hecht and his colleagues showed. The cones contain three different photopigments, each most sensitive to a different range of wavelengths, which is the basis of colour vision (Section 3.7). The wiring difference is convergence. In rod pathways, the outputs of many rods (often hundreds) converge on a single ganglion cell, so the weak signals from many rods add together and a dim light becomes detectable; but the brain cannot tell which of those many rods was stimulated, so detail is lost. In the fovea, by contrast, each cone may have something close to a private line to the brain, so detail is preserved, but each cone needs more light to drive its path-

way. Sensitivity is bought at the price of acuity, and acuity at the price of sensitivity. Table 3.10 summarises the contrast.

Table 3.10 Rods and cones compared.

| Feature | Rods | Cones |
|---|---|---|
| Number per eye | About 120 million | About 6 to 7 million |
| Location | Absent from the fovea; dominate the periphery | Concentrated in the fovea; sparse in the periphery |
| Light level | Dim light (scotopic vision) | Moderate to bright light (photopic vision) |
| Sensitivity | Very high; respond to single quanta | Lower |
| Acuity | Low (high convergence) | High (low convergence) |
| Colour | None (one pigment, rhodopsin): achromatic | Colour vision (three pigments) |
| Peak spectral sensitivity | About 500 nm (blue-green) | About 555 to 560 nm overall (yellow-green) |
| Dark adaptation | Slow, about 30 minutes, but reaches far greater sensitivity | Fast, about 7 to 10 minutes, but limited |

EXAM TRAP

Scotopic (night) vision is mediated by *rods*; photopic vision by *cones*. The cones are most densely packed in the *fovea centralis*; the optic disc has *no* receptors (it is the blind spot); the peripheral retina is rod territory; the macula surrounds the fovea, so "macula but not the fovea" is wrong; the ora serrata is the retina's front edge. The false statement in "which statement about cones is incorrect?" is that cones provide only monochromatic information: it is the rods that give colourless vision. The retina and the iris are structures, not receptor types.

## Visual adaptation: from sunlight to darkness ●●

Walk from a sunny street into a dark cinema and for a minute you can hardly see the seats. Gradually the room "brightens" until you can see the faces of people around you. Nothing in the cinema has changed; your eyes have. **dark adaptation** is the gradual increase in the eye's sensitivity to light of low intensity that occurs in darkness. The reverse process, **light adaptation**, is the rapid decrease in sensitivity that occurs when you move into bright light; it takes from a few seconds to about a minute, which is why sunlight dazzles you only briefly. Together these are called visual adaptation, the process of adjusting to different intensities of light.

If you measure a person's absolute threshold for light at intervals after entering complete darkness, you obtain the classic dark adaptation curve (Figure 3.4). It has two distinct branches. For the first 7 to 10 minutes, the threshold falls quickly and then levels off: this is the cone branch, as the cones recover their sensitivity. Then there is a kink, the **rod-cone break**, after which the threshold falls again, slowly, for another 20 minutes or so: this is the rod branch, as the far more sensitive rods take over. After about 30 minutes in the dark the eye is many thousands of times more sensitive than it was in daylight. The main mechanism is chemical. Light "bleaches" rhodopsin, splitting it into its components, and in darkness the

pigment slowly regenerates; until it does, the rods cannot respond. The slower regeneration of rhodopsin explains why rods take so much longer to adapt than cones.

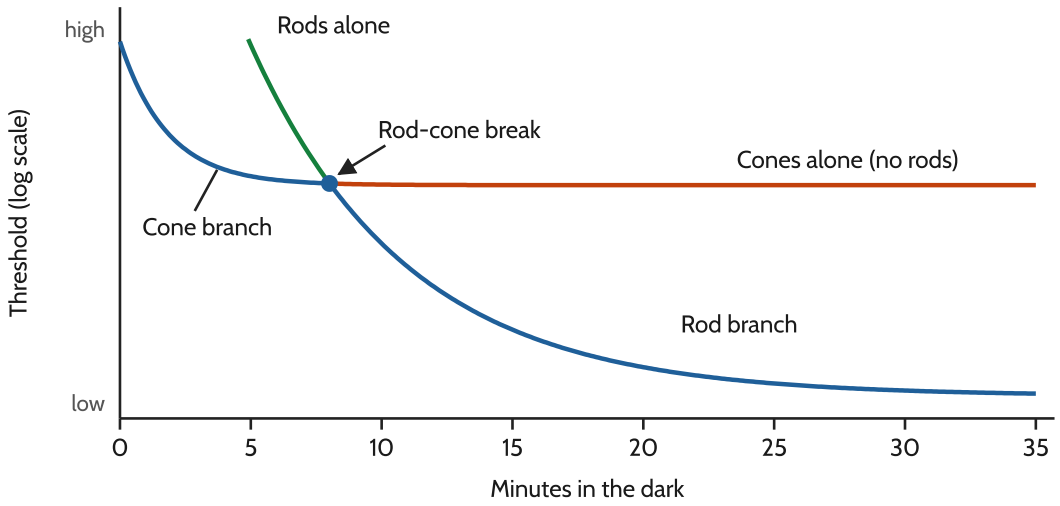


**Figure 3.4** The dark adaptation curve. The threshold for detecting light falls rapidly as the cones adapt and levels off after a few minutes (cone branch). At the rod-cone break, about 7 to 10 minutes into darkness, the more sensitive rods take over and the threshold continues to fall for 20 minutes or more (rod branch). A person with no functioning rods would follow the dashed line; one with no functioning cones would show only a rod curve.

Rods are almost insensitive to long-wavelength (red) light. This has a neat practical use. Wearing red goggles, or working under dim red light, lets the cones see well enough to read and work while the rods, which barely respond to red, dark-adapt in the meantime. Military pilots waiting to fly night missions, astronomers and darkroom workers have all used this trick, and so did radiologists in the days when X-ray images were viewed on dim fluoroscope screens.

> **INDIAN CONTEXT**
>
> Rhodopsin is built from a protein (opsin) and retinal, a molecule made from vitamin A. When vitamin A is scarce, rhodopsin cannot be regenerated fast enough and dark adaptation fails: the person develops night blindness (nyctalopia), often the first sign of deficiency. Vitamin A deficiency was long a major cause of preventable childhood blindness in India, which is why young Indian children receive periodic high-dose vitamin A supplementation through the public health system.

## The Purkinje shift

Rods and cones are not equally sensitive to all wavelengths. Under photopic (daylight) conditions the visual system is most sensitive to light of about 555 to 560 nanometres, in the yellow-green; under scotopic (night) conditions its peak sensitivity shifts to about 500 nanometres, in the blue-green, the wavelength rhodopsin absorbs best. This shift has a visible consequence that was first described in 1825 by the Czech physiologist **Jan Evangelista Purkyne** (1787 to 1869), usually spelled Purkinje in English. Walking in his garden at dusk, Purkinje noticed that his red and yellow flowers, which had looked brighter than his blue ones in day-

light, became darker than the blue ones as night fell. The **Purkinje shift** (or Purkinje effect) is this change in relative brightness as illumination falls: long-wavelength colours (reds) lose brightness sooner, and short-wavelength colours (blues and greens) look relatively brighter, as vision passes from cones to rods. It is one reason a red object, a red cricket ball included, becomes hard to pick out as daylight fades.

### Visual acuity and eye movements •

**visual acuity** is the ability to see fine detail. It is usually measured with the letter chart devised by the Dutch ophthalmologist **Herman Snellen** (1834 to 1908) in 1862. Normal acuity is recorded as 6/6 (or 20/20 in feet): at 6 metres you can read what a person with standard vision reads at 6 metres. A score of 6/12 means you need to be at 6 metres to read what a standard eye reads from 12 metres away. Acuity is highest at the fovea, falls steeply in the periphery (because of the distribution of cones and the convergence of rods) and drops in dim light. The visual system also has a temporal limit: a light flickering faster than a certain rate (the critical flicker fusion frequency, typically around 50 to 60 flashes per second in bright light) is seen as steady, which is why a film made of separate frames looks continuous.

Your eyes are never still. When you read or scan a scene, they jump from point to point in rapid movements called **saccades** (several per second), separated by brief pauses called fixations, during which information is taken in. They also follow moving objects smoothly (smooth pursuit) and turn inwards or outwards for near and far targets (vergence). Even during a fixation, when you try to hold your gaze perfectly still, the eyes make tiny involuntary movements: a fine tremor, slow drifts and small rapid flicks called microsaccades.

These tiny movements are not a defect. In the early 1950s, researchers devised ways of producing a **stabilised retinal image**, an image that moves with the eye and therefore stays fixed on the same receptors (one method mounted a tiny projector on a contact lens). Within a few seconds the image faded and disappeared, sometimes returning in fragments. Visual neurons respond mainly to change, so an unchanging image adapts out of existence. A milder version, Troxler fading (described in 1804), can be seen by fixating steadily on a point while a faint blurry patch sits in your peripheral vision: the patch fades. The constant small movements of the eyes keep the image shifting across the receptors and so prevent sensory adaptation from erasing the visual world. Saccades and microsaccades, in other words, ensure constant perception.

> **EXAM TRAP**
>
> "Tiny rapid movements of the eyes ensure …" Answer: *saccades* (strictly, microsaccades) ensure *constant perception*. A fixation is the pause between saccades, not a movement. Eye movements *prevent* sensory adaptation; they do not ensure it. Do not confuse *accommodation* (the lens changing shape to focus, Section 3.6), *adaptation* (sensitivity changing with light level) and Piaget's *assimilation and accommodation* (cognitive development, Developmental book). Getting adjusted to different intensities of light is *visual adaptation*.

### Lateral inhibition and Mach bands ••

Look at a set of grey stripes arranged in steps from light to dark, each stripe a perfectly uniform grey. Most people see each stripe as slightly lighter along the edge next to its darker neighbour and slightly darker along the edge next to its lighter neighbour, so the stripes seem

gently scalloped. These illusory light and dark bands at edges are called **Mach bands**, after the Austrian physicist and philosopher **Ernst Mach** (1838 to 1916), who described them in 1865.

The explanation is **lateral inhibition**: the tendency of active neurons in a sensory network to suppress the activity of their neighbours. In the retina, horizontal and amacrine cells carry inhibitory signals sideways, so each receptor's output is reduced in proportion to the activity of the receptors around it. Now consider a border between a light region and a dark one. A cell just on the light side of the border has some dark neighbours, so it receives less inhibition than cells deeper in the light region, and it signals "brighter": a light band. A cell just on the dark side has some light neighbours, so it receives more inhibition than cells deeper in the dark region, and it signals "darker": a dark band. The net effect is to exaggerate contrast at edges. Lateral inhibition was demonstrated directly by **Haldan Keffer Hartline** (1903 to 1983) and Floyd Ratliff in the eye of the horseshoe crab (*Limulus*), whose large receptors could be recorded one at a time; Hartline shared the 1967 Nobel Prize in Physiology or Medicine with Ragnar Granit and George Wald for work on the physiology and chemistry of vision.

Edge enhancement is useful. The boundaries of objects are where the information is, and a system that sharpens edges helps us see where one object ends and another begins. The same mechanism contributes to simultaneous contrast, in which a grey patch looks lighter on a black background than on a white one. (The Hermann grid, in which grey smudges appear at the intersections of white lines between black squares, was long explained the same way, although recent research suggests the full story is more complicated.)

> EXAM TRAP
>
> Mach bands are explained by *lateral inhibition* in the retina, which *enhances* edge perception (it does not dull edges, lower overall brightness or enhance colour). The opponent-process theory explains colour phenomena such as afterimages, not brightness bands; the rod-cone dichotomy explains differences between day and night vision; temporal summation is the adding together of weak signals over time.

## 3.7 Vision II: colour

LOWER YIELD · 1 Q

In 1666 a young Isaac Newton darkened his room, let a thin beam of sunlight through a hole in the shutter, and passed it through a glass prism. The white light spread into a band of colours from red to violet. Passed through a second prism, the colours could be recombined into white. White light, Newton concluded, is a mixture of all the colours. But he added a remark that sounds as if it were written by a psychologist: "For the Rays to speak properly are not coloured." Light has wavelengths; colour is something the nervous system creates. This section asks how.

### Mixing colours: additive and subtractive

Two very different kinds of colour mixing are easily confused. **additive colour mixing** is the mixing of lights. When two lights shine on the same spot, the wavelengths they contain are added together, and the result reaches the eye as a combination. Red light plus green light gives yellow; green plus blue gives cyan; red plus blue gives magenta; red, green and blue together give white. Your phone screen and a television work this way: each pixel contains tiny

red, green and blue elements, and every colour on the screen is an additive mixture of the three.

**subtractive colour mixing** is the mixing of paints, inks and dyes. A pigment looks coloured because it absorbs (subtracts) some wavelengths and reflects the rest. Mixing two pigments means that each absorbs its share, so the mixture reflects only the wavelengths that *both* pigments reflect. Blue paint reflects mostly blue and some green; yellow paint reflects mostly yellow and some green; mixed together they absorb everything except green, which is what you see. Mix enough pigments and almost everything is absorbed, producing a muddy dark brown or black, as anyone who has seen the colours at the end of a Holi celebration will know. Printers use cyan, magenta and yellow inks (plus black) for this reason. The rule to remember: lights add towards white; pigments subtract towards black.

Some pairs of colours are special. **complementary colours** are pairs of hues that, when mixed as lights in the right proportions, produce white or grey: blue and yellow, red and green (strictly, red and a bluish green). On a colour circle, complementary colours sit opposite each other. They will turn out to be the key to one of the two great theories of colour vision.

### The trichromatic theory

In 1802 the English physician and physicist **Thomas Young** (1773 to 1829) proposed that the eye does not need a separate receptor for every hue. Three kinds of receptor, each responding to a different part of the spectrum, would be enough, with each colour signalled by the ratio of activity across the three. Hermann von Helmholtz developed and refined the idea in 1852, and it has been known ever since as the Young-Helmholtz **trichromatic theory** (also called the component theory).

The main evidence came from colour-matching experiments. An observer sees a test light of a particular colour and adjusts the intensities of three "primary" lights until their mixture looks identical to it. With three suitably chosen primaries, any colour can be matched; with only two, many cannot. Three is the magic number because there are three kinds of receptor. One striking consequence is the existence of **metamers**: lights with physically different mixtures of wavelengths that look exactly the same colour, because they produce the same ratio of activity across the three receptor types. Your phone's yellow (a mixture of red and green light) and the yellow of a sodium street lamp (a single wavelength) are metamers.

The physiological proof came more than 160 years after Young. In 1964 two research groups, one of them led by **George Wald** (1906 to 1997), measured the light absorbed by the pigment inside single cones and found exactly three types. Today they are called short-wavelength (S) cones, peaking at about 420 nanometres (violet-blue); medium-wavelength (M) cones, peaking at about 530 nanometres (green); and long-wavelength (L) cones, peaking at about 560 nanometres (yellow-green, but they respond further into the red than the others). Each type responds to a broad range of wavelengths; what signals colour is the ratio between them. Humans share this trichromatic system with the apes and Old World monkeys; most other mammals, including dogs, have only two cone types.

### The opponent-process theory

The trichromatic theory explains colour matching beautifully, but several everyday facts fit it poorly. Stare at a red patch for half a minute, then look at a white wall: you see a green patch.

Why green, of all colours? Nobody ever sees a reddish green or a bluish yellow, although reddish blue (purple) and greenish yellow are easy to imagine. Yellow looks like a pure, unmixed colour, not like a blend of red and green. And people with colour blindness tend to confuse colours in pairs: red with green, or blue with yellow.

The German physiologist **Ewald Hering** (1834 to 1918) built these observations into the **opponent-process theory**, published in 1878. Hering proposed three opponent systems, each signalling one of a pair of opposed sensations by changing its activity in one direction or the other: a red versus green system, a blue versus yellow system and a black versus white (brightness) system. Because the two members of a pair are signalled by opposite activity in the same system, they cannot be experienced together at the same place; hence there is no reddish green.

The theory gives an elegant account of the **negative afterimage**, the image in complementary colours that appears after prolonged staring at a coloured stimulus. Staring at red fatigues the red side of the red versus green system. When you then look at a white surface, which stimulates both sides equally, the rested green side dominates, and you see green. Staring at blue produces a yellow afterimage; staring at white produces a black one. You can see this with a flag: stare for thirty seconds at a version of the Indian tricolour painted in blue (top), black (middle) and magenta (bottom), the complements of saffron, white and green, then look at a white wall, and the familiar saffron, white and green appear. (A brief, intense flash, such as a camera flash, can instead leave a positive afterimage, a short-lived image in the original colours, caused by receptors continuing to fire for a moment after the light is gone.)

Opponent cells were eventually found. In the 1950s Gunnar Svaetichin recorded opponent responses from cells in the fish retina, and in the 1960s **Russell De Valois** and his colleagues found cells in the lateral geniculate nucleus of monkeys that were excited by red and inhibited by green (or the reverse), and others that were excited by blue and inhibited by yellow. In 1957 Leo Hurvich and Dorothea Jameson had already shown that Hering's ideas could be made quantitative and could predict colour-matching data.

### The dual-process resolution

For decades, supporters of the two theories argued as though only one could be right. Both were right, about different stages of the visual system. According to the **dual-process theory** (sometimes called the zone theory), colour is coded in two stages. At the first stage, in the receptors, the three cone types work exactly as Young and Helmholtz proposed. At the second stage, in the retinal ganglion cells and the lateral geniculate nucleus, the cone signals are combined into opponent channels exactly as Hering proposed: the difference between L and M cone signals forms a red versus green channel; the difference between S cone signals and the sum of L and M forms a blue versus yellow channel; and the sum of L and M signals forms a brightness channel (Figure 3.5).

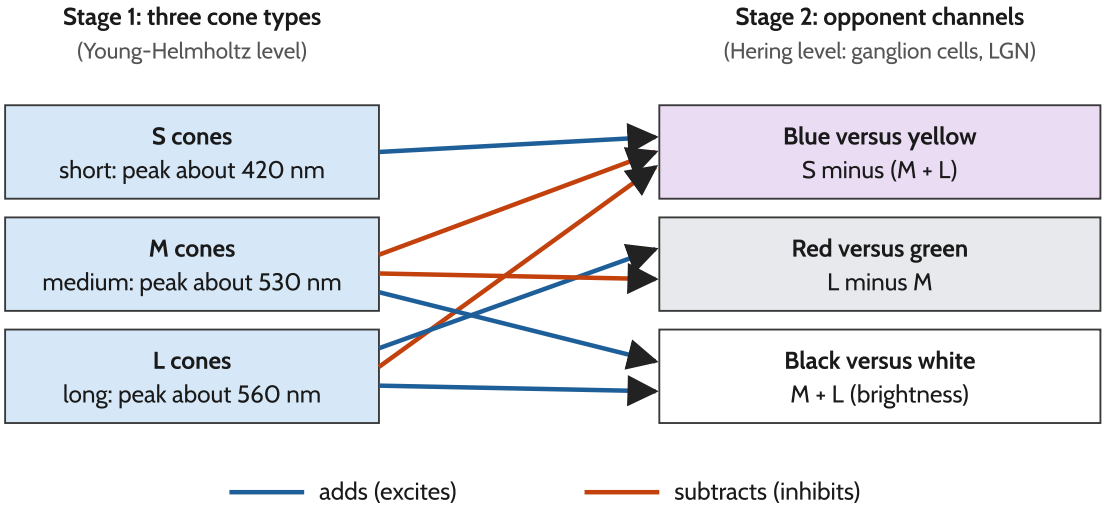


Figure 3.5 The dual-process theory of colour vision. Stage 1 is trichromatic: three cone types with different peak sensitivities. Stage 2 is opponent: ganglion cells and the lateral geniculate nucleus combine cone signals, adding some and subtracting others, into red versus green, blue versus yellow and black versus white channels. Colour matching reflects Stage 1; afterimages and the pairing of colour confusions reflect Stage 2.

Two further theories are worth knowing by name. The American logician and psychologist **Christine Ladd-Franklin** (1847 to 1930) proposed an evolutionary theory of colour vision in 1892: vision began as achromatic (black and white), later divided into blue and yellow, and later still yellow divided into red and green. Her theory neatly predicted that the most recently evolved distinction, red versus green, would be the most fragile, which is exactly the commonest form of colour blindness. And the inventor **Edwin Land** (1909 to 1991), founder of the Polaroid company, proposed the retinex theory to explain colour constancy, our tendency to see an object's colour as stable under very different lighting (Chapter 4 discusses the constancies). Table 3.11 compares the theories.

Table 3.11 Theories of colour vision.

| Theory | Proposed by | Core idea | Explains best |
|---|---|---|---|
| Trichromatic (component) | Young (1802); Helmholtz (1852) | Three receptor types; colour coded by their ratio | Colour matching, metamers, missing-cone colour blindness |
| Opponent-process | Hering (1878) | Three opponent pairs: red versus green, blue versus yellow, black versus white | Negative afterimages, no reddish green, paired colour confusions, unique yellow |
| Evolutionary | Ladd-Franklin (1892) | Achromatic, then blue versus yellow, then red versus green evolved in sequence | Why red-green deficiency is commonest |
| Dual-process (zone) | Modern synthesis (Hurvich and Jameson; De Valois) | Trichromatic at the cones, opponent at ganglion cells and LGN | Reconciles all of the above |
| Retinex | Land | Colour computed by comparing light across the whole scene | Colour constancy |

## Colour blindness

In 1794 the English chemist **John Dalton** (1766 to 1844), famous for atomic theory, gave the Manchester Literary and Philosophical Society a careful description of his own odd colour vision: a pink geranium flower looked to him almost sky blue by daylight but red by candlelight, and colours that other people told apart easily, such as reds and greens, looked much alike to him. Colour blindness was long called Daltonism in his honour. Dalton asked that his eyes be examined after his death; one was preserved, and in 1995 DNA analysis of the tissue showed that he had lacked the M (green-sensitive) cone pigment.

"Colour blindness" is a misleading name, because very few colour-blind people see no colour at all. The accurate term is **colour vision deficiency**, and it comes in three degrees (Table 3.12). In **anomalous trichromacy**, the mildest and most common form, all three cone types are present but one has a shifted sensitivity, so certain colours are matched differently from normal. In **dichromacy**, one cone type is missing altogether and the person matches all colours with two primaries. The three forms are named with Greek numbers in the order long, medium, short: *protanopia* ("first", missing L cones: reds look dim and are confused with greens), *deuteranopia* ("second", missing M cones: reds and greens confused, Dalton's condition) and *tritanopia* ("third", missing S cones: blues and yellows confused). In **monochromacy**, the rarest form, the person has at most one functioning cone type. Rod monochromats (achromatopsia) have no working cones at all; they see only shades of grey, have poor acuity and are dazzled by daylight, exactly as you would predict for someone who must rely on rods alone.

Table 3.12 Forms of colour vision deficiency.

| Form | What is wrong | Colours confused | How common |
|---|---|---|---|
| Anomalous trichromacy (protanomaly, deuteranomaly, tritanomaly) | Three cone types, one with shifted sensitivity | Mild versions of the corresponding dichromacy | Most common; deuteranomaly about 5 percent of men |
| Protanopia | L (long, "red") cones missing | Red with green; reds look dark | About 1 percent of men |
| Deuteranopia | M (medium, "green") cones missing | Red with green | About 1 percent of men |
| Tritanopia | S (short, "blue") cones missing | Blue with yellow | Very rare; equal in men and women |
| Monochromacy (rod monochromacy, achromatopsia) | No functioning cones (or only one type) | All hues; sees in greys | Very rare |

The genes for the L and M pigments lie on the X chromosome, so red-green deficiencies are inherited in an X-linked recessive pattern. A man has only one X chromosome, so a single faulty gene is enough; a woman needs faulty genes on both. In populations of northern European ancestry, about 8 percent of men and about 0.5 percent of women have some red-green deficiency. The gene for the S pigment lies on a non-sex chromosome, which is why tritan defects affect men and women equally.

The most widely used screening test is the set of plates designed by the Japanese ophthalmologist **Shinobu Ishihara** (1879 to 1963) and published in 1917. Each **Ishihara plate** shows a circle of coloured dots in which a number is formed by dots of one hue against a background of another; a person with normal colour vision reads the number easily, while a person with a

red-green deficiency sees a different number or none. The standard Ishihara plates screen for red-green defects and do not detect tritan defects. Finer diagnosis uses arrangement tests such as the Farnsworth-Munsell 100 Hue test and the anomaloscope, in which the viewer adjusts a mixture of red and green light to match a yellow light; how much red or green the person needs reveals the type and degree of deficiency. In India, colour perception standards are part of the medical examination for jobs such as railway driver, pilot and many posts in the armed forces.

> **EXAM TRAP**
>
> *Hermann von Helmholtz* (1821 to 1894) is examined for the trichromatic theory (with Young), the place (resonance) theory of hearing, the speed of the nerve impulse (1850) and unconscious inference. He did *not* propose the JND (Weber and Fechner), the unconscious mind (that is Freud's concept; Helmholtz's unconscious *inference* concerns perception) or the law of effect (Thorndike). Remember too: trichromatic = receptors, three cone types; opponent-process = afterimages, Hering; "protan, deutan, tritan" = first (L, red), second (M, green), third (S, blue); and the commonest deficiency is deuteranomaly, not total colour blindness.

**SCAN YOUR BRAIN**

Fill in the blanks. The answers are below; cover them first.

1. The visible spectrum runs from about 380 to _____ nanometres.
2. Wavelength is experienced mainly as hue, intensity as brightness and purity as _____.
3. The changing of the lens's shape to focus near and far objects is called _____.
4. The area of the retina containing only cones and giving the sharpest vision is the _____.
5. The kink in the dark adaptation curve, at about 7 to 10 minutes, is the _____ break.
6. At dusk, reds darken before blues; this is the _____ shift.
7. Mach bands are explained by lateral _____.
8. The theory that explains negative afterimages was proposed by Ewald _____ in 1878.
9. A person missing the M cones has _____.
10. Mixing blue and yellow paint gives green because paint mixing is _____.

**Answers:** (1) 760, (2) saturation, (3) accommodation, (4) fovea, (5) rod-cone, (6) Purkinje, (7) inhibition, (8) Hering, (9) deuteranopia, (10) subtractive.

## 3.8 Hearing

LOWER YIELD · 4 QS

Close your eyes at a busy chai stall and you can still tell a great deal about the world: the hiss of milk boiling over, a scooter pulling up behind you, two friends arguing about cricket at the next table, and which of them is winning the argument. Hearing works in the dark, around corners and through walls, and it carries spoken language, which is why losing it is so isolating. As with vision, we begin with the physical stimulus, follow it through the ear to the re-

ceptors, and then ask how the brain extracts pitch, loudness and direction from what the receptors report.

## Sound: the stimulus

Sound is a wave of pressure travelling through a medium such as air or water. When a tabla player strikes the drumhead, the skin moves rapidly out and in. Each outward movement pushes the neighbouring air molecules together, creating a region of slightly higher pressure (a compression); each inward movement leaves a region of slightly lower pressure (a rarefaction). These alternating compressions and rarefactions travel outward at about 340 metres per second in air, and more than four times faster in water, although each air molecule only jostles back and forth in place. Because sound needs molecules to carry it, there is no sound in a vacuum: the roaring explosions of a space film are a cinematic fiction.

Sound waves have three physical properties, and they map onto three psychological dimensions of hearing (Table 3.9 set them beside their visual counterparts; Figure 3.6 draws them). The *frequency* of a sound is the number of complete cycles of compression and rarefaction passing a point each second. It is measured in hertz (Hz), after the German physicist Heinrich Hertz: one hertz is one cycle per second. Frequency is experienced mainly as **pitch**, how high or low a sound seems: the deep thud of a dhol has a low frequency, the shriek of a whistle a high one. As with light, frequency and wavelength are inversely related, so a 100 Hz hum has a wavelength of about 3.4 metres and a 10,000 Hz hiss a wavelength of about 3.4 centimetres, a difference that will matter when we come to locating sounds. The *amplitude* (intensity) of a sound is the size of its pressure changes, the height of the wave, and it is experienced mainly as **loudness**. The *complexity* of a sound is its mixture of frequencies, and it is experienced as **timbre** (pronounced "tam-ber"), the tone quality that lets you tell one sound source from another.

Timbre needs a word of explanation. A tuning fork produces a *pure tone*, a sound of a single frequency whose pressure changes trace a smooth sine wave. Almost every other sound is complex. When a veena player plucks a string, the string vibrates as a whole at its *fundamental frequency*, which largely sets the pitch, and at the same time in halves, thirds, quarters and so on, producing *harmonics* (overtones) at two, three, four and more times the fundamental. Early in the nineteenth century the French mathematician Joseph Fourier showed that any complex wave can be described as a sum of simple sine waves, and the ear, as you will see, performs something close to this analysis mechanically. A veena, a flute and a human voice sounding the same note share a fundamental but differ in their mixtures of harmonics, which is why you can tell them apart with your eyes closed. Timbre also depends on how a sound begins and fades: play a recording of a piano backwards and it no longer sounds like a piano.

Pitch is not simply frequency in disguise. A complex tone made of 200, 300 and 400 Hz components is heard with the pitch of 100 Hz, although it contains no energy at 100 Hz at all, because 100 Hz is the fundamental the three components share. This *missing fundamental* effect is why a man's voice still sounds like a man's voice over a telephone line that cuts out almost everything below about 300 Hz. Pitch also shifts a little with loudness, and loudness depends on frequency: a 50 Hz tone must be far more intense than a 3,000 Hz tone to sound equally loud. Pitch and loudness are psychological dimensions, measured with the psychophysical

methods of Section 3.4; frequency and amplitude are physical ones, measured with instruments. Stevens built ratio scales for both: the *mel* for pitch and the *sone* for loudness.

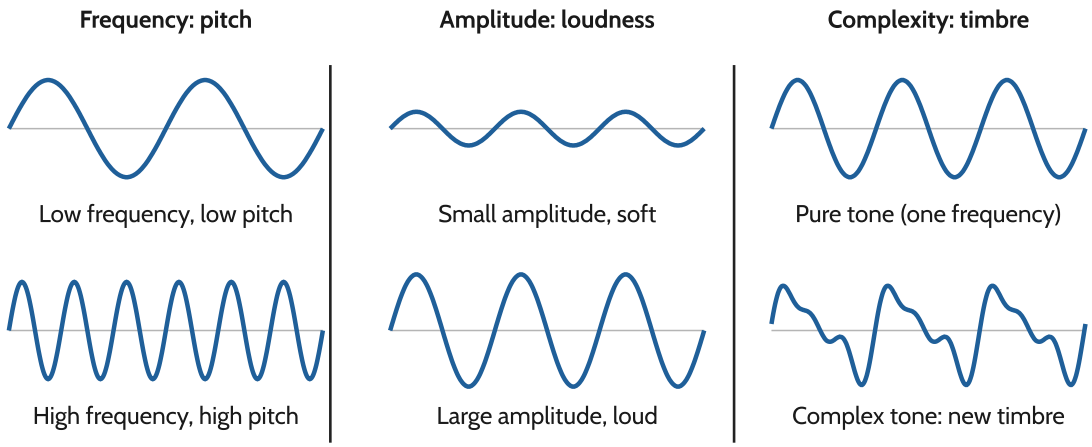


**Figure 3.6** The three physical dimensions of sound and what they are heard as. Each trace shows air pressure rising and falling over time. More cycles per second (higher frequency) is heard as higher pitch; taller waves (greater amplitude) as louder sound; adding harmonics to the same fundamental changes the timbre but not the pitch.

The healthy young human ear responds to frequencies from about 20 Hz to about 20,000 Hz. It is most sensitive between about 2,000 and 5,000 Hz, the range that carries many of the consonants of speech, and least sensitive at the extremes. Frequencies below 20 Hz (infrasound) are felt as vibration rather than heard; elephants use them to call to one another over several kilometres. Frequencies above 20,000 Hz (ultrasound) are silent to us but audible to dogs (up to about 45,000 Hz) and to bats (well above 100,000 Hz). **Francis Galton** (1822 to 1911), the Victorian polymath who founded the study of individual differences, invented a high-pitched whistle in 1876 to test the upper limit of hearing in people and animals; it is the ancestor of the "silent" dog whistle. The upper limit falls steadily with age, and by middle age many people hear little above about 12,000 to 14,000 Hz.

## Loudness and the decibel

The range of intensities the ear can handle is staggering. The most intense sound you can tolerate carries roughly a million million ($10^{12}$) times the energy of the faintest sound you can detect. A scale running from 1 to a million million is useless in daily practice, so acousticians use a logarithmic scale, the same kind of scale that Fechner's law says our senses themselves use (Section 3.3). The unit is the **decibel** (dB), one-tenth of a bel, a unit named after Alexander Graham Bell, the inventor of the telephone. Zero decibels is set at a reference pressure close to a young listener's absolute threshold for a 1,000 Hz tone. It does not mean "no sound": very sensitive ears can detect sounds slightly below 0 dB.

On the decibel scale, every increase of 10 dB is a tenfold increase in sound intensity (energy). A 20 dB sound is therefore 100 times as intense as a threshold sound, a 60 dB conversation a million times as intense, and a 120 dB rock concert a million million times as intense. (Strictly, the familiar scale of sound pressure level is defined by pressure, which grows tenfold for

every 20 dB; the conclusions are the same.) Perceived loudness grows far more slowly than intensity. As a rough rule, every 10 dB increase makes a sound seem about twice as loud, which is Stevens' power law at work: tenfold more energy raised to an exponent of about 0.3 (equivalent to the exponent of 0.67 for sound pressure in Table 3.4) is roughly a doubling. Table 3.13 gives typical levels.

**Table 3.13** Approximate levels of everyday sounds. Real values vary with distance from the source; learn the landmarks (0, 60, 85, 120 to 130).

| Sound | Approximate level | What it means for the listener |
|---|---|---|
| Faintest tone a young adult can hear (1,000 Hz) | 0 dB | Absolute threshold (by definition) |
| Rustling leaves; quiet breathing | 10 to 20 dB | Barely audible |
| Whisper at a metre or two | About 30 dB | Faint |
| Quiet library or bedroom at night | About 40 dB | Quiet |
| Normal conversation at 1 metre | About 60 dB | Comfortable |
| Heavy city traffic; noisy factory floor | About 80 to 90 dB | Hours of daily exposure above 85 dB risk lasting damage |
| Motorcycle close by; car horn at 1 metre | About 100 to 110 dB | Damage after minutes of exposure |
| Front rows of a loud concert; thunderclap overhead | About 120 dB | Discomfort; ringing in the ears afterwards |
| Firecracker burst near the ear; jet engine at 30 metres | 130 to 150 dB | Pain; a single exposure can cause permanent damage |

Two numbers are worth remembering for clinical work. Sustained exposure above about 85 dB, a working day in a noisy factory, risks permanent hearing loss, and the safe exposure time halves with every further 3 dB: about eight hours at 85 dB, two hours at 91 dB and only fifteen minutes at 100 dB. And sounds of about 120 to 130 dB reach the threshold of pain, the point at which hearing turns into hurting.

### INDIAN CONTEXT

India's Noise Pollution (Regulation and Control) Rules, 2000 set daytime limits of 55 dB in residential areas and 50 dB in "silence zones" around hospitals, schools and courts, and the environment protection rules ban firecrackers that exceed 125 dB measured four metres from the point of bursting. Festival processions, wedding bands, construction sites and traffic horns commonly far exceed these limits. Because the decibel scale is logarithmic, a firecracker at 125 dB is not "twice as loud" as a 60 dB conversation: it carries about three million times as much energy.

EXAM TRAP

Hertz (Hz) measures *frequency*, which is heard as *pitch*; decibels (dB) measure *intensity* (amplitude), which is heard as *loudness*; *complexity* (the mix of harmonics) is heard as *timbre*. The human range is 20 to 20,000 Hz. The decibel scale is logarithmic: 80 dB is 100 times as intense as 60 dB, not one-third more, and sounds about four times as loud. Decibels measure the physical stimulus; the psychological unit of loudness is the sone and of pitch the mel. Frequency and wavelength are inversely related for sound as for light.

## The ear: outer, middle and inner

The ear is a chain of energy converters (Figure 3.7). The *outer ear* collects airborne sound; the *middle ear* turns it into the vibration of three tiny bones; the *inner ear* turns that into waves in fluid and, finally, into neural signals.

*The outer ear.* The visible flap of skin and cartilage is the pinna (or auricle). Its folds funnel sound into the auditory canal and subtly alter the high frequencies of a sound according to the direction it comes from, which helps you tell whether a sound is above or below you, in front or behind. The auditory canal, about 2.5 centimetres long, resonates like an organ pipe and boosts frequencies around 2,000 to 5,000 Hz, one reason our sensitivity peaks there. At its inner end is the **tympanic membrane** (eardrum), a thin, cone-shaped membrane that vibrates in step with the pressure changes that strike it.

*The middle ear.* Behind the eardrum lies a small air-filled cavity containing the three **ossicles**, the smallest bones in the body: the malleus (hammer), attached to the eardrum; the incus (anvil); and the stapes (stirrup), whose footplate fits into the *oval window*, a membrane-covered opening into the inner ear. Why bother with bones? Because the inner ear is filled with fluid, and fluid is far harder to set in motion than air. If airborne sound struck the fluid directly, almost all of its energy (about 99.9 percent) would bounce off, which is why people underwater can barely hear someone shouting from the poolside. The middle ear solves this problem in two ways. The eardrum is about 17 to 20 times larger in area than the oval window, so the force collected over the large drum is concentrated onto the small window, and the ossicles act as a lever. Together these raise the pressure about twentyfold. Two tiny muscles in the middle ear (the tensor tympani and the stapedius) also contract reflexively to loud sounds, stiffening the chain and damping the vibration; this acoustic reflex takes a few tens of milliseconds, too slow to protect the ear from a sudden firecracker blast. Finally, the Eustachian (auditory) tube connects the middle ear to the throat and keeps the air pressure equal on both sides of the eardrum. Your ears "pop" when it opens during a flight or a fast drive down a ghat road. In young children the tube is short and nearly horizontal, which is one reason middle-ear infections are so common in childhood.

*The inner ear.* The **cochlea** (from the Greek *kokhlos*, land snail) is a coiled, fluid-filled tube about the size of a pea, making about two and a half to two and three-quarter turns and roughly 35 millimetres long if uncoiled. It is divided along its length into three chambers. The partition that forms the floor of the middle chamber is the **basilar membrane**, a flexible strip that is narrow and stiff at the *base* of the cochlea (near the oval window) and several times wider and floppier at the *apex* (the tip of the coil). Resting on the basilar membrane is the **organ of Corti**, the receptor organ of hearing, named after the Italian anatomist Alfonso Corti, who described it in 1851. It contains the **hair cells**, the auditory receptors, each topped by a tuft of tiny stiff hairs (stereocilia), with a second membrane, the tectorial membrane, ly-

ing over them. When the stapes pushes the oval window inward, the fluid of the cochlea is displaced and a wave runs along the basilar membrane; the pressure is finally released at a second, flexible window lower down, the round window, which bulges outward.

The human cochlea has about 3,500 *inner hair cells*, in a single row, and about 12,000 *outer hair cells*, in three rows. The inner hair cells are the true sensory receptors: about 95 percent of the roughly 30,000 fibres of the auditory nerve carry their signals. The outer hair cells act as an amplifier. They lengthen and shorten in time with the sound, pumping energy into the basilar membrane's vibration and sharpening its response to particular frequencies. In doing so they produce faint sounds of their own that travel back out through the middle ear. These otoacoustic emissions can be picked up by a tiny microphone placed in the ear canal, and recording them is now a quick, painless way to screen the hearing of newborn babies, who cannot answer "yes, I heard it".

From the cochlea, signals travel along the auditory nerve, the cochlear branch of the vestibulocochlear nerve (cranial nerve VIII), to the cochlear nuclei of the brainstem. From there the pathway runs to the superior olives on both sides, the first place where signals from the two ears meet, then through the inferior colliculi of the midbrain to the medial geniculate nucleus of the thalamus, and finally to the primary auditory cortex in the temporal lobe. Every stage is laid out in order of frequency, like the keys of a piano, an arrangement called *tonotopic organisation*. Because each ear sends signals to both hemispheres, damage to the auditory cortex on one side does not make the opposite ear deaf. (The Physiological Psychology book traces this pathway in detail.)

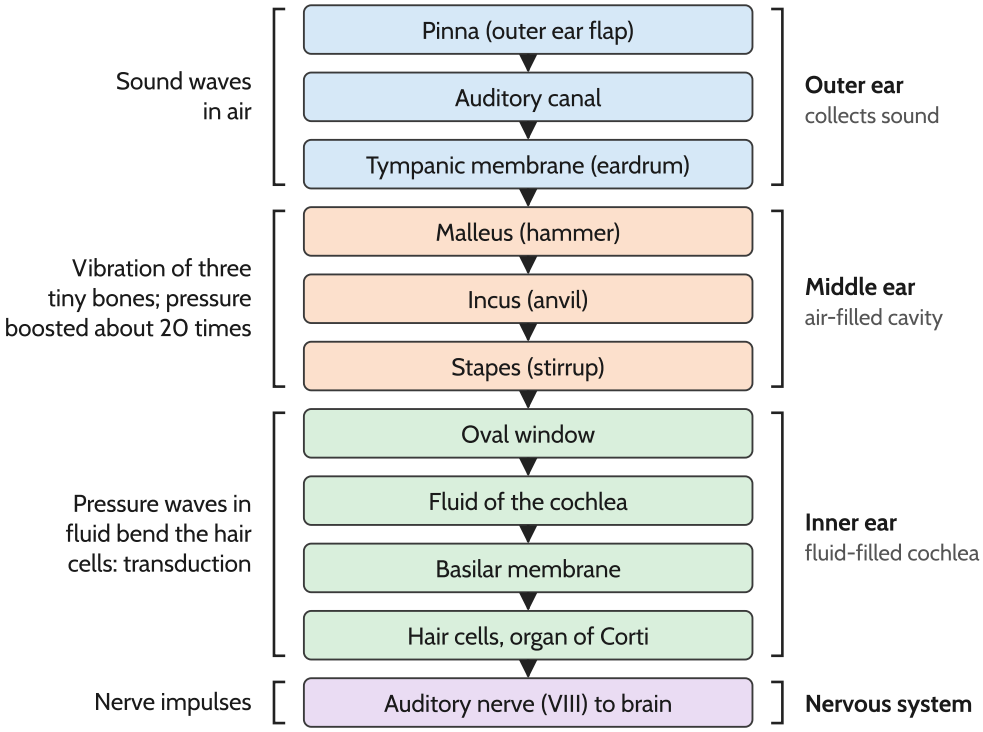


Figure 3.7 The route of sound to the auditory receptors. Air pressure waves (outer ear) become vibrations of the eardrum and the three ossicles (middle ear), then pressure waves in the fluid of the cochlea that bend the hair cells of the organ of Corti (inner ear). Only at the hair cells is sound transduced into neural signals.

### From vibration to nerve impulse: the hair cells ●

Transduction in the ear is purely mechanical, and it is among the fastest in any sense. As the basilar membrane moves up and down, the hair cells riding on it are pushed against the overlying tectorial membrane, and their stereocilia bend. The stereocilia on each hair cell stand in rows of increasing height, and the tip of each is joined to its taller neighbour by a fine filament called a tip link. Bending the bundle towards its tallest stereocilia stretches the tip links, which pull open mechanically gated ion channels. Positively charged ions, mainly potassium from the unusually potassium-rich fluid (endolymph) bathing the top of the hair cell, flow in, and the cell depolarises. Depolarisation makes the hair cell release the neurotransmitter glutamate onto the auditory nerve fibres beneath it, and they fire. Bending the bundle the other way closes the channels and the cell hyperpolarises. At low frequencies, therefore, the hair cell's voltage rises and falls in step with the sound wave itself. Because no chemical cascade intervenes between movement and channel opening, the whole process takes only microseconds.

Notice how different this is from the other senses you have met. In the rods and cones, light triggers a chemical cascade that *closes* sodium channels, so photoreceptors hyperpolarise to light (Section 3.1 noted that they are most active in the dark). In taste and smell, transduction begins when a molecule binds to a receptor (Section 3.9). Hair cells are also fragile. Intense noise can snap tip links and tear stereocilia, and in mammals, unlike birds, lost hair cells do not grow back, which is why noise-induced hearing loss is permanent.

> **EXAM TRAP**
>
> Cochlear hair cells transduce sound by the *direct mechanical bending of their stereocilia*, which opens cation channels; potassium flows in and the cell depolarises. "Closure of sodium channels" is a garbled version of what happens in photoreceptors, not hair cells; "depolarisation by GABA release" is wrong twice over (GABA is the brain's main inhibitory transmitter, and hair cells release glutamate); "chemical signalling" describes taste and smell. The correct order of structures from the outside in is pinna, auditory canal, tympanic membrane, ossicles (malleus, incus, stapes), oval window, cochlea, organ of Corti (hair cells), auditory nerve. Any option that puts the oval window before the ossicles, or the cochlea before the eardrum, is wrong.

### How we hear pitch: place, frequency and volleys ●●

The ear must code two things about every sound: how loud it is and how high it is. Loudness is the easier problem. Louder sounds make the basilar membrane vibrate more vigorously, the hair cells bend further, auditory nerve fibres fire faster (rate coding, Section 3.1), and more fibres are recruited, including less sensitive ones. Pitch is harder, and the attempt to explain it produced one of the longest-running debates in sensory psychology. There are two basic ideas: that pitch is signalled by *where* on the basilar membrane the hair cells are most stimulated, and that it is signalled by *when*, the timing of the nerve impulses.

*Place theory.* Helmholtz, whom you met as a champion of the trichromatic theory, proposed the first version in 1863 in *On the Sensations of Tone*. He thought of the basilar membrane as a set of transverse fibres of graded length, like the strings of a piano or a harp, each resonating to one particular frequency: short fibres near the base for high tones and long fibres near the apex for low tones. A tone would set its own fibre vibrating, the hair cells at that place would

fire, and pitch would be signalled by place, an application of Muller's specific nerve energies within a single sense. This **place theory** (in Helmholtz's form, the resonance or "harp" theory) had a problem: the basilar membrane is not a row of separate, tensed strings but a continuous strip, and its fibres cannot vibrate independently.

The decisive evidence came from Georg von Bekesy, the Hungarian physicist whose tracking audiometer you met in Section 3.4. Working first in the laboratory of the Hungarian telephone service in Budapest and later at Harvard, he showed that the basilar membrane responds to a tone not by resonating at one spot but with a *travelling wave*, and that the place where the wave peaks depends on frequency. For this work he received the 1961 Nobel Prize in Physiology or Medicine.

### KEY STUDY: BEKESY WATCHES THE TRAVELLING WAVE (1920S TO 1950S; NOBEL PRIZE 1961)

**Question.** What does the basilar membrane actually do when a tone enters the ear? **Method.** Bekesy opened the cochleas of human cadavers and of animals, made the delicate membrane visible under a microscope, and illuminated it with a flickering (stroboscopic) light that "froze" its rapid motion. He also built enlarged mechanical models of the cochlea. **Result.** A tone sets up a wave that travels along the membrane from the base towards the apex, like a ripple running along a shaken rope. The wave grows to a peak at one place and then dies away rapidly. High frequencies peak near the narrow, stiff base; low frequencies travel further and peak near the wide, floppy apex (Figure 3.8). **Conclusion.** Frequency is mapped onto place along the membrane: Helmholtz's conclusion was right, although his mechanism was wrong. **Criticism.** Bekesy's peaks were broad, too broad to explain how finely we can discriminate pitch, partly because he worked with dead tissue and very loud sounds. Measurements in living cochleas, beginning in the 1970s, revealed much sharper peaks, produced by the outer hair cells acting as the cochlear amplifier.

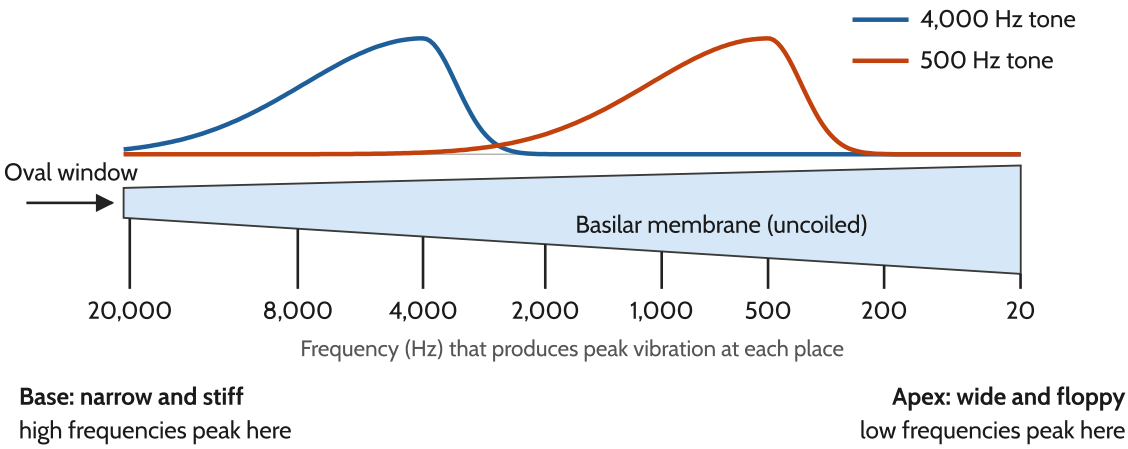


Figure 3.8 Place coding along the basilar membrane, shown uncoiled. A travelling wave rises gradually from the base, peaks at a place determined by its frequency and then dies away sharply. A 4,000 Hz tone (solid line) peaks near the narrow, stiff base; a 500 Hz tone (dashed line) peaks much nearer the wide, floppy apex. The scale beneath shows the frequency that peaks at each point.

Place coding is now firmly established. The cochlea is tonotopically organised, and so is every later stage of the auditory pathway. The most striking practical proof is the cochlear implant (described at the end of this section), which works by placing electrodes at different points

along the cochlea: stimulating near the base produces a high-pitched sensation and stimulating further towards the apex a lower one. Place theory explains *high* frequencies especially well. Its weaknesses lie at the bottom of the range. Low-frequency tones, below a few hundred hertz, set much of the membrane near the apex moving with a broad, poorly defined peak, so place alone cannot explain how precisely we tell one low tone from another. And place cannot explain the missing fundamental: a tone made of 200, 300 and 400 Hz has no energy at the 100 Hz place, yet we hear a 100 Hz pitch.

*Frequency theory.* In 1886 the British physiologist **William Rutherford** (1839 to 1899) proposed a radically different idea. He compared the ear to the newly invented telephone, whose diaphragm vibrates as a whole and sends an electrical signal down the wire at the same frequency as the sound. In the ear, he suggested, the whole basilar membrane vibrates in step with the sound, and the auditory nerve fires at the same frequency: a 200 Hz tone produces 200 impulses a second, and the brain reads pitch from the rate of firing. This **frequency theory** (also called the telephone theory, or the temporal theory) explains low pitches well, and the missing fundamental poses no problem for it, because a complex tone with components at 200, 300 and 400 Hz still repeats its overall pattern 100 times a second. Its fatal weakness is the refractory period. After firing, a neuron needs about a millisecond before it can fire again, so no single neuron can fire much more than about 1,000 times a second, yet we hear pitches up to 20,000 Hz. Frequency theory alone cannot explain pitch above about 1,000 Hz.

*The volley principle.* In 1930 two Princeton psychologists, **Ernest Glen Wever** (1902 to 1991) and **Charles William Bray** (born 1904), recorded the electrical activity of the auditory nerve of a cat while sounds were played to it. When they amplified the signal through a loudspeaker in another room, they could understand speech spoken into the cat's ear. (Much of this "Wever-Bray effect" was later traced to the cochlear microphonic, the electrical response of the hair cells themselves.) To explain how the nerve could follow frequencies beyond the limit of any single fibre, Wever and Bray proposed the **volley principle**, developed in Wever's book *Theory of Hearing* (1949). Imagine a squad of soldiers ordered to keep up a fire of 3,000 shots a minute when each soldier can fire only 1,000 times a minute: they fire in rotation. In the same way, groups of auditory nerve fibres fire in staggered volleys. Each fibre fires only on some cycles of the sound wave, but always at the same phase of the wave (a property now called phase locking), and because different fibres take different cycles, the group as a whole fires on every cycle. Volleying lets timing information follow frequencies up to about 4,000 to 5,000 Hz, above which phase locking breaks down.

*The modern resolution.* As with colour vision, both camps were partly right. Today's view, sometimes called the duplex or place-frequency theory, holds that the auditory system uses both codes. For low frequencies, below about 1,000 Hz, the timing of impulses follows the sound wave and carries pitch (frequency theory); from about 1,000 to about 4,000 or 5,000 Hz, volleys of timed firing and place work together; above about 5,000 Hz, where timing fails, pitch is signalled by place alone. The exact boundaries differ from textbook to textbook, but the pattern does not: frequency (temporal) coding for low pitches, place coding for high pitches. Table 3.14 summarises the theories.

Table 3.14 Theories of pitch perception. The boundaries in hertz are approximate and vary between textbooks.

| Theory | Proposed by | Core idea | Explains best | Main problem |
|---|---|---|---|---|

| Theory | Proposed by | Core idea | Explains best | Main problem |
|---|---|---|---|---|
| Place (resonance) theory | Helmholtz (1863) | Each part of the basilar membrane resonates to one frequency, like a piano string | That different places code different pitches | The membrane is not a set of separate tuned strings |
| Place (travelling wave) theory | Bekesy (Nobel Prize 1961) | A travelling wave peaks at a place set by frequency: high at the base, low at the apex | High pitches | Broad peaks for low tones; cannot explain the missing fundamental |
| Frequency (telephone) theory | Rutherford (1886) | The membrane vibrates as a whole; the nerve fires at the frequency of the sound | Low pitches, below about 1,000 Hz | Neurons cannot fire much faster than 1,000 times a second |
| Volley principle | Wever and Bray (1930) | Groups of fibres fire in staggered turns, together following the frequency | Pitches up to about 4,000 to 5,000 Hz | Timing breaks down above about 5,000 Hz |
| Duplex (place-frequency) theory | Modern synthesis | Timing for low pitches, both codes in the middle, place for high pitches | The whole audible range | Exact boundaries still debated |

EXAM TRAP

"The tendency of different frequencies to make different parts of the basilar membrane flex" is *place coding*. *Frequency coding* (temporal coding) is firing in time with the sound wave; *rate coding* is the coding of intensity (loudness) by how fast fibres fire; "pitch coding" is not a technical term at all, because pitch is the experience being coded. Place theory (Helmholtz; Bekesy) best explains *high* frequencies, through the location of maximum displacement; low frequencies are explained by firing rate (frequency theory); sound localisation depends on comparing the two ears (see Locating sounds, below), not on place along the membrane; and loudness is coded by firing rate and the number of fibres active. The frequency theory belongs to the physiologist *William* Rutherford (1886), not to Ernest Rutherford, the physicist of the atomic nucleus, although some answer keys and notes say "Ernest"; Indian notes sometimes pair Rutherford with Rinne or date the theory to 1880. Bekesy won the Nobel Prize in 1961 for the travelling wave, a *place* theory, and the volley principle (Wever and Bray) is a rescue of the *frequency* theory.

## Locating sounds

Hearing tells you not only what made a sound but where it is. Yet the cochlea, unlike the retina, contains no map of space: every point on the basilar membrane codes a frequency, not a direction. Direction must be computed by comparing the two ears. The English physicist **John William Strutt, Lord Rayleigh** (1842 to 1919), winner of the 1904 Nobel Prize in Physics, set out the two main binaural cues in his duplex theory of sound localisation (1907).

The first cue is the **interaural time difference**, the difference in the time at which a sound reaches the two ears. A sound directly to your right reaches your right ear about 0.6 to 0.7 milliseconds before your left ear, because it must travel around your head; a sound straight ahead reaches both ears at once. Astonishingly, the auditory system can detect time differences as small as about 10 microseconds (hundred-thousandths of a second), corresponding to a shift of only a degree or two in direction. Timing cues work best for low-frequency sounds. The second cue is the **interaural intensity difference** (or level difference), the difference in a sound's intensity at the two ears. The head casts a "sound shadow" for the ear farther from the source, but only for high frequencies, whose short wavelengths are blocked by

the head; low-frequency waves, metres long, simply bend around it. Intensity cues therefore work best for high-frequency sounds, above about 1,500 to 2,000 Hz. The two cues are first compared in the superior olives of the brainstem, the first meeting point of the two ears.

Binaural cues leave one ambiguity. A sound directly in front of you and one directly behind you produce identical time and intensity differences (both zero), and so do other pairs of positions, which is why front-back confusions are common. Two further cues resolve it. Turning the head changes the cues in different ways for front and back sources, which is why you instinctively turn towards an uncertain sound. And the folds of the pinna filter high frequencies differently for sounds from in front, behind, above and below. In 1998 Paul Hofman and his colleagues in the Netherlands fitted volunteers with plastic moulds that changed the shape of their pinnae: their ability to judge the elevation of sounds collapsed at once, and then gradually recovered over several weeks as their brains learned the new "ears". When the moulds were removed, their old skill returned immediately. Your sense of where sounds come from is learned from the shape of your own ears.

> **EXAM TRAP**
>
> "Duplex theory" has two meanings in hearing. Rayleigh's duplex theory of *localisation* (1907) says that time differences locate low-frequency sounds and intensity differences locate high-frequency sounds. The duplex (place-frequency) theory of *pitch* says that timing codes low pitches and place codes high pitches. Do not confuse either with the duplexity theory of *vision* (rods versus cones, Section 3.6). Note the neat inversion: in localisation, timing serves low frequencies and intensity serves high ones; in pitch, timing serves low frequencies and place serves high ones.

## Hearing loss

The World Health Organization estimated in 2021 that more than 1.5 billion people have some degree of hearing loss, about 430 million of them to a disabling degree. Total deafness is rare; most impairment is partial, and its type depends on where in the chain of Figure 3.7 the damage lies. Table 3.15 contrasts the two main types.

**conductive hearing loss** results from a problem in the outer or middle ear that stops sound from being conducted to the cochlea, while the cochlea and auditory nerve remain healthy. Common causes are impacted earwax or a foreign body in the canal, a perforated eardrum, fluid behind the eardrum after middle-ear infections (otitis media, very common in children), and otosclerosis, in which abnormal bone growth fixes the stapes in the oval window so that it can no longer vibrate. Sounds are heard more faintly but not distorted, and because the inner ear is intact, sound conducted directly through the bones of the skull is heard normally. Conductive loss rarely exceeds about 60 dB, is often treatable with medicine or surgery (for example, replacing the fixed stapes with a tiny prosthesis), and responds well to hearing aids, which simply make sounds louder.

**sensorineural hearing loss** (once called nerve deafness) results from damage to the cochlea, most often to the hair cells, or to the auditory nerve. Its causes include prolonged or intense noise; ageing; certain drugs that are toxic to hair cells (ototoxic drugs such as aminoglycoside antibiotics like streptomycin and gentamicin); infections such as meningitis, mumps, and rubella or cytomegalovirus caught before birth; genetic conditions; and tumours of the auditory nerve. Age-related hearing loss, *presbycusis*, typically begins with the highest frequen-

cies and creeps downward. Because many consonants (s, f, t, th) carry their information at high frequencies, older people with presbycusis often complain that others "mumble": they can hear that someone is speaking but cannot make out the words. Sensorineural loss is usually permanent, because hair cells do not regenerate, and it distorts sound as well as weakening it, so hearing aids help less. It is often accompanied by *tinnitus*, a ringing, hissing or buzzing heard in the absence of any external sound. Pinel and Barnes note that cutting the auditory nerve from a ringing ear does not stop the ringing, which suggests that tinnitus is generated by the brain's own reorganisation after loss of input, a phantom sound comparable to the phantom limbs of Section 3.10.

**Table 3.15** Conductive versus sensorineural hearing loss.

| Feature | Conductive | Sensorineural |
|---|---|---|
| Site of damage | Outer or middle ear (canal, eardrum, ossicles) | Inner ear (hair cells) or auditory nerve |
| Typical causes | Wax, perforated eardrum, otitis media, otosclerosis | Noise, ageing (presbycusis), ototoxic drugs, infections, genes, tumours |
| Bone conduction | Normal | Reduced |
| Quality of sound | Fainter but clear | Fainter and distorted; speech hard to understand |
| Treatment | Often medical or surgical; hearing aids work well | Usually permanent; hearing aids or cochlear implant |

Two simple tuning-fork tests, still taught to every medical student, separate the types at the bedside. In the *Rinne test*, a vibrating fork is held against the bone behind the ear and then beside the ear canal; a normal ear, or one with sensorineural loss, hears it longer through the air, whereas an ear with conductive loss hears it longer through bone. In the *Weber test*, the fork is placed on the top of the head; with conductive loss in one ear the sound is heard louder in the *affected* ear (it is no longer masked by room noise), whereas with sensorineural loss it is heard louder in the *better* ear. Formal assessment uses pure-tone audiometry, the clinical application of the psychophysical methods of Section 3.4.

For profound sensorineural deafness, a **cochlear implant** can restore useful hearing. A microphone and processor behind the ear split incoming sound into frequency bands, and a thin array of electrodes threaded into the cochlea stimulates the auditory nerve directly, each electrode at a different place along the membrane, bypassing the missing hair cells. The device is place theory turned into engineering. Implants do not restore normal hearing, and they need a working auditory nerve. They help most when given early: children implanted in the first two or three years of life, while the auditory brain is still highly plastic, usually develop far better spoken language than children implanted later, and adults who lost their hearing after learning to speak benefit more than adults deaf from birth.

CLINICAL LINK

Hearing loss is easily mistaken for something else. An older adult who answers questions wrongly may be judged confused, and a child who ignores instructions may be labelled inattentive or defiant, when neither can hear the high-frequency consonants that carry the words. A hearing check belongs early in any assessment of cognition, attention or behaviour. The stakes go beyond misdiagnosis: the *Lancet* Commission on dementia prevention (2020) identified midlife hearing loss as the largest single potentially modifiable risk factor for dementia, and untreated loss is linked to loneliness and depression. In a classic experiment, Philip Zimbardo and his colleagues (1981) used hypnosis to make students partially deaf without being aware of the cause; the students became more suspicious and hostile, a hint of how unrecognised deafness might feed paranoid thinking in later life.

EXAM TRAP

Damage to the ossicles or eardrum causes *conductive* loss; damage to the hair cells or auditory nerve causes *sensorineural* (nerve) loss. Otosclerosis is conductive; presbycusis and noise-induced loss are sensorineural and begin with high frequencies. A cochlear implant replaces the *hair cells*, not the auditory nerve, so it cannot help if the nerve itself is destroyed. And yes, the Weber tuning-fork test is named after the same Ernst Heinrich Weber who gave us Weber's law: do not let the shared name tempt you to link the two ideas.

## 3.9 The chemical senses: taste and smell

LOWER YIELD · 2 QS

Vision and hearing respond to energy that arrives from a distance. Taste and smell respond to matter itself: molecules that dissolve in saliva or drift into the nose and bind to receptor proteins. For this reason they are called the chemical senses. They are the oldest senses in evolutionary terms (even bacteria swim towards nutrients and away from toxins), and they still do their ancient job of standing guard at the entrance to the body, deciding what may be eaten and what must be spat out. They are also the senses most tightly bound up with emotion and memory, as anyone who has been carried back to childhood by the smell of a grandmother's kitchen will know.

### Taste ●●

The sense of taste is called **gustation**. For most of the twentieth century, Western textbooks listed four basic tastes: sweet, sour, salty and bitter. In 1908 the Japanese chemist **Kikunae Ikeda** (1864 to 1936) asked why the broth his wife made from kombu seaweed had a delicious savoury quality that fitted none of the four. He extracted the responsible substance, glutamate, and named its taste **umami** (from the Japanese for "delicious"); he also patented monosodium glutamate as a seasoning. Umami was widely accepted as a fifth basic taste only around 2000, when receptors for glutamate were found in taste cells. Today the standard list is five, and each one appears to signal something the body needs to know (Table 3.16). Researchers continue to debate further candidates, such as a taste for fat.

Table 3.16 The five basic tastes.

| Taste | Typical stimulus | What it probably signals | How it is transduced |
|---|---|---|---|
| Sweet | Sugars (jaggery, ripe mango) | Energy-rich food | G-protein-coupled receptor |

| Taste | Typical stimulus | What it probably signals | How it is transduced |
|---|---|---|---|
| Salty | Sodium chloride (table salt) | Sodium and minerals needed by the body | Ion channels |
| Sour | Acids (lemon, tamarind, curd) | Unripe or spoiled food | Ion channels (hydrogen ions) |
| Bitter | Alkaloids such as quinine and caffeine (karela) | Possible poison | About 25 different G-protein-coupled receptors |
| Umami | Glutamate (tomatoes, mushrooms, cheese, meat broths) | Protein | G-protein-coupled receptor |

Look at your tongue in a mirror and you will see that its surface is covered with small bumps. These are **papillae** (singular papilla). Many people assume the bumps are the taste buds; in fact the **taste buds** are microscopic structures buried in the walls of most papillae, each containing about 50 to 100 taste receptor cells clustered like the segments of an orange around a tiny opening, the taste pore. An adult has several thousand taste buds (estimates range from about 2,000 to 10,000), mainly on the tongue but also on the soft palate and in the throat. Tips of the receptor cells project into the taste pore, where dissolved chemicals reach them. Salty and sour tastes are transduced when ions (sodium for salt, hydrogen for sour) pass through channels into the cell; sweet, bitter and umami substances bind to receptor proteins coupled to G proteins, which trigger a chemical cascade inside the cell. Taste receptor cells survive only a few weeks at most before being replaced, which is why a scalded tongue soon recovers its sense of taste. There are four kinds of papillae, set out in Table 3.17; only three of them contain taste buds.

Table 3.17 The four kinds of papillae on the tongue.

| Papilla | Shape and location | Taste buds? |
|---|---|---|
| Fungiform | Mushroom-shaped; scattered over the front two-thirds, especially the tip and edges | Yes, a few each |
| Foliate | Leaf-like folds on the sides at the back of the tongue | Yes, many |
| Circumvallate (vallate) | About 8 to 12 large, round papillae in a V across the back of the tongue | Yes, hundreds each |
| Filiform | Tiny cone-shaped threads over most of the tongue; the most numerous | No; they sense texture (touch) |

A famous diagram, reproduced in textbooks for decades, showed a "tongue map" with sweet at the tip, salty and sour along the sides and bitter at the back. It is a myth. It began with a 1901 German study by David Hanig, who found only small differences in sensitivity around the edge of the tongue; when the Harvard psychologist and historian Edwin Boring replotted Hanig's data in 1942, the small differences were widely misread as separate zones. In reality every region of the tongue that has taste buds can detect all five tastes, and every taste bud contains receptor cells for several of them.

Taste signals leave the mouth along three cranial nerves: the facial nerve (VII) from the front two-thirds of the tongue, the glossopharyngeal nerve (IX) from the back third, and the vagus nerve (X) from the throat. They travel to the solitary nucleus of the medulla, then to the thalamus, and on to the primary gustatory cortex in the insula, deep within the lateral fissure; a

secondary gustatory area lies in the orbitofrontal cortex. Because three separate nerves carry taste, complete loss of taste (*ageusia*) is rare.

Some taste preferences are present at birth. In the 1970s the Israeli researcher Jacob Steiner photographed the faces of newborns, some only hours old and not yet fed, as drops of different solutions were placed on their tongues: sweet produced a relaxed, almost smiling face and eager licking, sour produced pursed lips, and bitter produced a grimace, gaping and spitting. A preference for sweet and an aversion to bitter are built in, sensible defaults for an animal that needs energy and must avoid poisons. People also differ in how intensely they taste. In 1931 the chemist Arthur Fox was pouring the compound phenylthiocarbamide (PTC) when some of the dust blew into the air; a colleague complained of its intense bitterness, while Fox tasted nothing at all. Sensitivity to PTC, and to the related compound PROP, turned out to be largely inherited. Building on this, **Linda Bartoshuk** identified *supertasters*, people (roughly a quarter of those tested in American samples, more often women) who have unusually many fungiform papillae and experience bitter tastes, and the burn of chilli, with special intensity. Supertasters often dislike bitter vegetables such as karela and strong black coffee.

> **INDIAN CONTEXT: SIX RASAS, ONE BURNING CHILLI**
>
> Ayurveda recognises six *rasas* (tastes): madhura (sweet), amla (sour), lavana (salty), katu (pungent), tikta (bitter) and kashaya (astringent). Four of these match modern basic tastes. The other two are not tastes in the modern sense. Pungency, the "heat" of chilli, is a pain and temperature sensation: capsaicin, the active compound in chilli, binds to a heat receptor (the TRPV1 channel) on pain fibres of the trigeminal nerve, the same receptor that responds to temperatures above about 43 °C, which is why chilli feels hot. The *bhut jolokia* of north-east India was certified in 2007 as the hottest chilli in the world. Astringency, the dry, puckering feel of strong tea, unripe banana or amla, is a touch sensation produced by tannins. And a herb from Indian forests offers a striking demonstration of receptor specificity: chewing leaves of *Gymnema sylvestre*, known in Hindi as *gurmar* ("sugar destroyer"), temporarily blocks the sweet receptors, so that sugar tastes like sand while salt, sour and bitter are unaffected.

## Smell

The sense of smell is called **olfaction**. Its receptors lie in the **olfactory epithelium** (or olfactory mucosa), a patch of mucus-covered tissue about the size of a postage stamp high in each nasal cavity, above the main stream of inhaled air. (Sniffing works by swirling air up to it.) The epithelium contains roughly 10 million olfactory receptor neurons. Each has hair-like cilia bathed in mucus, in which odour molecules dissolve and bind to receptor proteins. Unusually for neurons, olfactory receptor neurons die and are replaced every few weeks throughout life. Their axons pass upward through tiny holes in a thin plate of bone at the base of the skull, the cribriform plate, to the olfactory bulb just beneath the frontal lobes. (Together, these axons form the olfactory nerve, cranial nerve I.)

How can a few kinds of receptor distinguish thousands of odours? The answer came in 1991, when **Linda Buck** (born 1947) and **Richard Axel** (born 1946) discovered a huge family of genes for olfactory receptor proteins, about 1,000 in rats and mice. Humans have a few hundred working types (roughly 350). Each receptor neuron carries only one type of receptor protein; each type responds, more or less strongly, to many different odorants; and each odorant activates several types. An odour is therefore coded not by a single labelled line but by the *combi-*

*nation* of receptor types it activates, the across-fibre pattern coding you met in Section 3.1, and with a few hundred receptor types the number of possible combinations is astronomical. Buck and Axel received the Nobel Prize in Physiology or Medicine in 2004. In the olfactory bulb, the axons of all the receptor neurons carrying the same receptor type, several thousand of them, converge on one or two small spherical clusters called glomeruli, so each glomerulus collects the signal from one receptor type and the bulb displays the odour as a pattern of active glomeruli.

From the olfactory bulb, signals travel along the olfactory tract to the piriform cortex (the primary olfactory cortex, in the medial temporal lobe) and directly to the amygdala and the entorhinal cortex, structures central to emotion and memory. As Section 3.1 noted, smell is the only sense whose main pathway reaches the cortex without first relaying in the thalamus; a secondary route does pass through the thalamus to the orbitofrontal cortex, where odours appear to be consciously identified. These direct links to the limbic system help explain why odours evoke emotions and memories so powerfully. In the opening volume of *In Search of Lost Time* (1913), Marcel Proust describes how the taste and smell of a madeleine dipped in tea suddenly brought back his whole childhood in the village of Combray. Research has given the "Proust phenomenon" some support: memories cued by odours tend to be more emotional, and older, often from the first decade of life, than memories cued by words or pictures (Johan Willander and Maria Larsson, 2006).

Human smell is better than its reputation. We can detect some substances at concentrations of less than one part per billion. The gas in an Indian LPG cylinder has no smell of its own; gas companies add a tiny amount of a pungent sulphur compound (ethyl mercaptan) precisely so that a leak can be smelled at once. Dogs, with many times as many receptor neurons, are far more sensitive still, which is why they are used to detect explosives and drugs. People can discriminate thousands of odours (one 2014 study claimed more than a trillion odour mixtures, an estimate other researchers have challenged), yet we are poor at naming them. Presented with familiar smells such as coffee or cloves out of context, people can name only about half, and they often experience a "tip-of-the-nose" state, sure they know the smell but unable to find its name. Smell adapts quickly, too: you soon stop noticing the jasmine garland of Section 3.1, and you cannot smell your own home the way a visitor does. Smell is also important early in life. Aidan Macfarlane (1975) showed that breast-fed babies only six days old turned their heads towards a breast pad worn by their own mother more often than towards one worn by another nursing mother.

For a century, psychologists searched for a small set of "primary" odours from which all others could be built, as colours can be built from three primaries. The German psychologist **Hans Henning** in 1916 arranged six primary odours, flowery (fragrant), fruity (ethereal), spicy, resinous, burnt and putrid (foul), at the six corners of a hollow triangular prism, the **smell prism**; every other odour was supposed to lie on its surface, between the primaries it resembled. In the same year he arranged the four classical primary tastes, sweet, sour, salty and bitter, at the corners of a **taste tetrahedron** (a pyramid with four triangular faces), with other tastes lying along its edges and faces as blends. In the 1950s and 1960s the British chemist **John Amoore** proposed a stereochemical (lock-and-key) theory: odour molecules fit receptor sites according to their shape, and there are seven primary odours (camphoraceous, musky, floral, pepperminty, ethereal, pungent and putrid). Buck and Axel's discovery of hundreds of receptor types settled the matter: there is no small set of primary odours. Smell is

closer to the principle Amoore glimpsed (molecules fitting receptors) than to Henning's geometry, but on a vastly larger scale. Table 3.18 compares the attempts.

Table 3.18 Attempts to find the primary tastes and smells.

| Proposal | Who and when | Primaries | Verdict today |
|---|---|---|---|
| Taste tetrahedron | Henning (1916) | Four: sweet, sour, salty, bitter | Umami added as a fifth basic taste |
| Smell prism | Henning (1916) | Six: flowery, fruity, spicy, resinous, burnt, putrid | Historical; no small set of primaries exists |
| Stereochemical theory | Amoore (1950s and 1960s) | Seven: camphoraceous, musky, floral, peppermint, ethereal, pungent, putrid | Right that shape matters; wrong about seven |
| Receptor gene family | Buck and Axel (1991; Nobel Prize 2004) | No primaries: a few hundred receptor types, combinatorial code | Current view |

CLINICAL LINK

Loss of smell, **anosmia**, is more common and more disabling than most people realise. Its most common neurological cause is a blow to the head that shifts the brain inside the skull and shears the delicate olfactory axons where they pass through the cribriform plate. Smell also declines with age and is often among the earliest signs of Parkinson's disease and Alzheimer's disease, sometimes years before other symptoms; sudden anosmia became a hallmark of COVID-19 in 2020. People with anosmia cannot smell smoke, gas leaks or spoiled food, lose much of the pleasure of eating, and are prone to depression. Smell is assessed with scratch-and-sniff tests such as the University of Pennsylvania Smell Identification Test (Richard Doty, 1984). Smell can also be falsely *present*: a brief, often unpleasant olfactory hallucination (burning rubber, rotting flesh) can be the aura that announces a seizure starting in the medial temporal lobe, the so-called uncinate fit, a reminder of how close the olfactory cortex lies to the temporal lobe structures involved in epilepsy.

## Pheromones

In many species, smell carries social messages. **pheromones** are chemicals released by one animal that affect the behaviour or physiology of other members of the same species; the term was coined in 1959. The first to be chemically identified, in the same year, was bombykol, a sex attractant released by the female silkworm moth that males can detect from kilometres away. Mammals also use pheromones. Pinel and Barnes describe a deliciously devious experiment by Michael Murphy and Gerald Schneider (1970): male hamsters normally attack an unfamiliar male placed in their colony, but when the intruder was first swabbed with the vaginal secretions of a sexually receptive female, the residents tried to mate with it instead. Many mammals detect pheromones with a separate organ in the nasal septum, the vomeronasal organ (Jacobson's organ), which sends signals to an accessory olfactory bulb. In adult humans this organ is vestigial, and most of the genes for its receptors no longer work.

Do humans have pheromones? The evidence is weak. In 1971 Martha McClintock reported that women living together in a college dormitory gradually synchronised their menstrual cycles, an effect attributed to chemical signals, but many later studies have failed to find reliable synchrony. Studies in which women rated the smell of T-shirts worn by men suggested a preference for the odour of men whose immune-system genes differ from their own, but the

findings are inconsistent. Pinel and Barnes conclude that there is little evidence that humans release or respond to sexual pheromones, whatever the perfume advertisements claim.

### Flavour: taste plus smell

If you have ever had a heavy cold, you will know that food seems to lose its taste. In fact it loses its smell. What we call the taste of food is really **flavour**, a combination of taste from the tongue, smell from the nose, texture and temperature from the touch receptors of the mouth, and the "chemical feel" of irritants such as chilli, mustard and mint. During chewing, odour molecules travel from the mouth up the back of the throat into the nasal cavity, a route called retronasal smell, and the brain refers the resulting smell to the mouth, so that it is experienced as taste. You can demonstrate this with a handful of flavoured sweets: pinch your nose, close your eyes, and you will be able to tell sweet from sour but not lemon from orange; release your nose and the flavour floods in. Blindfolded people with blocked noses are notoriously poor at telling a slice of apple from a slice of raw potato or onion, because the textures are similar and the tastes differ little: the difference is almost all smell.

The "chemical feel" deserves its own name, chemesthesis. It is carried not by the taste or olfactory nerves but by the trigeminal nerve (cranial nerve V), which also carries touch, temperature and pain from the face. Capsaicin in chilli activates heat-sensing receptors, and menthol in mint activates cold-sensing receptors (the TRPM8 channel), which is why mint feels cool though its temperature is unchanged. David Julius and Ardem Patapoutian shared the 2021 Nobel Prize in Physiology or Medicine for discovering receptors for temperature and touch; Julius found the capsaicin receptor by asking how chilli produces its burn.

> **EXAM TRAP**
>
> Papillae belong to *taste* (gustation), not smell, touch or vision; the taste buds sit inside them (except in the filiform papillae, which have none). Smell receptors lie in the *olfactory epithelium*. The five basic tastes are sweet, salty, sour, bitter and *umami* (Ikeda, 1908); "pungent" (chilli) is pain and temperature carried by the trigeminal nerve, and the tongue map is a myth. Loss of smell is *anosmia*; loss of taste is *ageusia*; inability to recognise faces is *prosopagnosia* (Chapter 4). At least one answer key (IHBAS 2024) pairs prosopagnosia with "distinguishing tastes", which is wrong. Henning gave both the taste tetrahedron (four primaries) and the smell prism (six); Amoore gave the stereochemical theory (seven primary odours); Buck and Axel found the receptor genes. The olfactory nerve is cranial nerve I, the first of the twelve, and smell is the one sense whose main route bypasses the thalamus; the optic (II), trigeminal (V) and vestibulocochlear (VIII) nerves all feed pathways that relay in the thalamus, and taste travels in nerves VII, IX and X and does relay in the thalamus too.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. The frequency of a sound is experienced mainly as _____, and its complexity as timbre.
2. Sound intensity is measured on a logarithmic scale in units called _____.
3. The malleus, incus and stapes are the three _____ of the middle ear.
4. The receptor organ of hearing, resting on the basilar membrane, is the organ of _____.
5. Bekesy showed that high-frequency sounds produce their peak vibration near the _____ of the basilar membrane.
6. Rutherford's frequency theory fails above about 1,000 Hz because of the neuron's _____ period.
7. Wever and Bray proposed that groups of fibres fire in staggered turns, the _____ principle.
8. Loss of hearing caused by damage to the hair cells is called _____ hearing loss.
9. Ikeda named the savoury taste of glutamate _____.
10. The axons of olfactory receptor neurons pass through the cribriform plate to the olfactory _____.

**Answers:** (1) pitch, (2) decibels, (3) ossicles, (4) Corti, (5) base, (6) refractory, (7) volley, (8) sensorineural, (9) umami, (10) bulb.

## 3.10 The skin senses and pain

LOWER YIELD · 2 QS

Aristotle counted touch as one sense. It is really several. The **skin senses** (cutaneous senses) include touch and pressure, vibration, warmth, cold and pain, and in daily life they blend with one another and with the body senses of Section 3.11 into what physiologists call somatosensation, the senses of the body. The skin that houses them is the body's largest organ, about two square metres in an adult. This section looks first at touch and temperature and then at pain, the sense that turns out to be the most psychological of all.

### Touch: receptors, maps and active exploration

In the 1880s and 1890s, Magnus Blix in Sweden, Alfred Goldscheider in Germany and, most systematically, **Max von Frey** (1852 to 1932) in Germany mapped the skin point by point. Touching small areas with fine probes, they found that sensitivity is not continuous but spotty: some points yield a sensation of pressure, others of cold, others of warmth, others of pain, and many points in between yield little at all. Von Frey measured touch thresholds with a graded set of stiff hairs (von Frey hairs), each bending at a known force; their modern descendant, a nylon monofilament, is pressed against the soles of people with diabetes to screen for loss of protective sensation in the feet. Von Frey proposed that each kind of spot corresponded to a specific kind of receptor, an extension of Muller's doctrine of specific nerve energies to the skin.

The skin does contain several distinct receptors (Table 3.19). Four kinds of encapsulated or specialised endings respond to mechanical stimulation, and they differ in two important ways: in the size of the skin area each one monitors (its receptive field) and in how fast it

adapts. As Pinel and Barnes explain, when constant pressure is applied to the skin, all the receptors fire at first, but within a few hundred milliseconds only the slowly adapting ones remain active. That is why you feel your shirt as you put it on and then forget it. The largest and deepest of the mechanoreceptors are the onion-like **Pacinian corpuscles**, which adapt very rapidly and respond to vibration and sudden pressure but not to steady pressure. Alongside the mechanoreceptors are *free nerve endings*, bare branches of sensory neurons with no special structure, which respond to temperature and to potentially damaging stimuli.

Table 3.19 The main receptors of the skin.

| Receptor | Location | Responds best to | Adaptation |
|---|---|---|---|
| Meissner's corpuscles | Just under the surface of hairless skin (fingertips, lips, palms) | Light touch and flutter; slipping of objects in the grip | Fast |
| Merkel's discs | Base of the outer skin layer | Steady pressure, fine detail, edges and texture (reading Braille) | Slow |
| Ruffini endings | Deeper skin | Stretching of the skin; finger and hand position | Slow |
| Pacinian corpuscles | Deepest layers; largest receptors | Vibration and sudden pressure | Very fast |
| Hair follicle receptors | Wrapped around the roots of hairs | Movement of hairs (an ant walking on your arm) | Fast |
| Free nerve endings | Throughout the skin and body | Temperature, pain (nociceptors) and itch | Mostly slow |

Section 3.2 promised an explanation of why Weber's two-point threshold is about 2 millimetres on the fingertips and lips but several centimetres on the back. There are two reasons. First, fingertips and lips are packed with receptors that have small receptive fields, whereas the back has few receptors with large fields; two points on the back may fall within one receptor's field and be signalled as one. Second, the brain devotes far more tissue to sensitive regions. Touch signals reach the primary somatosensory cortex, a strip of the parietal lobe just behind the central fissure, which contains an orderly map of the body. From the 1930s the Canadian neurosurgeon **Wilder Penfield** (1891 to 1976), stimulating this strip in conscious patients during epilepsy surgery, found that the areas for the lips, tongue and fingers are enormous compared with those for the back or the legs. Drawn in proportion to its cortical territory, the body becomes the grotesque little figure called the sensory homunculus, with huge lips and hands. (The Physiological Psychology book covers the somatosensory cortex.)

Touch signals and pain signals also travel to the brain by different routes. Fine touch, vibration and body position ascend the spinal cord in the *dorsal column-medial lemniscus system*, while pain and temperature ascend in the *anterolateral system* (the spinothalamic tract); both relay in the thalamus on their way to the somatosensory cortex. A patient whose anterolateral system is damaged on one side may lose pain and temperature sensation while keeping fine touch.

Two further facts show how much touch depends on the brain's active use of the receptors. In 1824 a fifteen-year-old French schoolboy, Louis Braille, who had been blinded in an accident in early childhood, devised a script of raised dots arranged in cells of six. The dots are spaced about 2.5 millimetres apart, close to the fingertip's two-point threshold, and skilled

readers can read more than 100 words a minute. Remarkably, when people blind from early life read Braille, their *visual* cortex becomes active, a sign that the brain has reassigned unused visual territory to touch (Sadato and colleagues, 1996). Second, touch is far better when it is active. In 1962 James J. Gibson asked people to identify cookie cutters of various shapes. When they explored the cutters with their fingers, they identified about 95 percent correctly; when the cutters were simply pressed into their passive palms, they managed only about half. Gibson called active exploration *haptic perception*. The ability to recognise objects by touch is called stereognosis, and its loss after damage to the parietal lobe is astereognosis.

> **EXAM TRAP**
>
> Pain is detected by *free nerve endings* (nociceptors), not by Merkel's discs, Meissner's corpuscles or Ruffini endings, which are mechanoreceptors for touch, pressure and stretch. Vibration is the speciality of the *Pacinian corpuscles*. Older textbooks assign warmth to Ruffini endings and cold to Krause end bulbs, following von Frey; modern research finds that temperature is sensed by free nerve endings. The *anterolateral* (spinothalamic) system carries *nociceptive* (pain) and temperature information; the dorsal column-medial lemniscus system carries fine touch, vibration and proprioception.

### Temperature

Spot mapping showed separate cold spots and warm spots, with cold spots several times more numerous. Temperature receptors are free nerve endings that respond not to temperature as such but mainly to *changes* in skin temperature. Normal skin temperature, around 32 to 33 °C, feels neither warm nor cold; this neutral point is called *physiological zero*, and it shifts with adaptation, which is the explanation of John Locke's three bowls of water in Section 3.1. The receptors that make temperature sensing possible belong to a family of ion channels (the TRP channels), each opening over a particular range: one opens above about 43 °C, where warmth turns into burning pain (the capsaicin receptor of Section 3.9), and another opens below about 25 °C and is also opened by menthol.

Two curiosities show that temperature sensation is constructed from the pattern of activity in warm and cold fibres. In **paradoxical cold**, first described by von Frey in the 1890s, touching a cold spot with a warm probe (about 45 °C) produces a sensation of cold, because cold fibres also fire at high temperatures and the brain reads any activity in a cold fibre as "cold". And in the thermal grill illusion, a hand laid on a grill of alternating warm and cool bars, neither of them hot enough to hurt, feels a startling burning heat, even pain. Pinel and Barnes report that when people experience this illusory pain in a brain scanner, the brain area most clearly activated is the anterior cingulate cortex, a region associated with the unpleasantness of pain.

### Pain: more than a sensation

The International Association for the Study of Pain defines pain as "an unpleasant sensory and emotional experience associated with, or resembling that associated with, actual or potential tissue damage" (revised in 2020). Notice the words *emotional* and *experience*. Pain is not a simple readout of damage. Yet it is indispensable. Pinel and Barnes tell the story of Miss C., a highly intelligent Canadian university student who had never felt pain. As a child she had bitten off the tip of her tongue while chewing and burned herself badly kneeling on a hot

radiator. Because she never felt the discomfort that makes the rest of us shift our weight, turn over in our sleep or rest an injured joint, her knees, hips and spine became badly damaged, and she died at the age of 29 of massive infections. Her condition is now called **congenital insensitivity to pain**. In 2006 Geoffrey Woods, James Cox and their colleagues traced one genetic form of it in families in northern Pakistan. The first case to come to their attention was a boy who performed street theatre, pushing knives through his arms and walking on burning coals without feeling pain; he died before his fourteenth birthday after jumping from a roof. The affected families carried a mutation in a gene (SCN9A) for a sodium channel found in pain fibres.

The receptors for pain are **nociceptors**, free nerve endings that respond to stimuli that damage or threaten to damage tissue: intense pressure, extreme heat or cold, and chemicals released by injured cells. Their signals travel along two kinds of fibre. A-delta fibres are thin and lightly myelinated, conduct fairly fast, and carry sharp, well-localised "first pain". C fibres are the thinnest of all, unmyelinated and slow, and carry dull, aching, burning, poorly localised "second pain". Stub your toe and you feel both: an immediate sharp jab, then, a second or so later, a spreading throb. In the dorsal horn of the spinal cord these fibres release the neurotransmitters glutamate and substance P onto neurons that relay the message to the brain. Pain from internal organs is often felt in the skin instead, because visceral and skin fibres converge on the same spinal neurons; the pain of a heart attack is commonly felt in the left arm, shoulder or jaw. This is called *referred pain*.

The evidence that pain depends on psychological factors is overwhelming. During the Second World War the American anaesthetist **Henry Knowles Beecher** (1904 to 1976) treated soldiers badly wounded on the beachhead at Anzio in Italy and was struck by how few asked for morphine; in later comparisons, only about a quarter to a third of severely wounded soldiers wanted a painkiller, whereas more than four out of five civilians with comparable surgical wounds did. For the soldier, Beecher concluded, the wound meant survival and a ticket away from the battlefield; for the civilian, it meant disaster. Expectation matters too: a dummy pill (a placebo) relieves pain in a substantial proportion of patients. And Ronald Melzack and Kenneth Casey (1968) argued that pain has three dimensions: a *sensory-discriminative* dimension (where it is, how intense, what kind), an *affective-motivational* dimension (how unpleasant it is, and the urge to escape it) and a *cognitive-evaluative* dimension (what it means). In 1997 Pierre Rainville and his colleagues used hypnotic suggestion to make the same hot water feel more or less unpleasant without changing its felt intensity; activity changed in the anterior cingulate cortex but not in the primary somatosensory cortex, as if the two dimensions were handled by different parts of the brain.

INDIAN CONTEXT

Ronald Melzack drew attention to a hook-swinging ritual once practised in parts of India, in which a man chosen to bless the children and crops of a series of villages had two steel hooks, tied by ropes to a special cart, thrust under the skin and muscles of his back. The cart was drawn from village to village, and at the climax of the ceremony in each village he swung free, hanging only from the hooks, apparently in a state of exaltation rather than agony. Pinel and Barnes cite the same kind of ceremony as evidence that pain can be suppressed by cognitive and emotional factors. Devotees who carry the *kavadi* at Thaipusam with skewers through their cheeks, and fire-walkers at village festivals in South India, show the same phenomenon. Meaning, belief, attention and the social setting change pain: a cut finger hurts more when you are alone and worried than in the middle of a cricket match.

## Theories of pain and the gate control theory

The oldest scientific account of pain is a *specificity theory*. In a book published in 1664, Rene Descartes pictured pain as a direct line: fire touching the foot tugs a thread running up to the brain, rather as pulling a bell rope rings a bell in a church tower. Von Frey's discovery of pain spots seemed to complete the picture: specific pain receptors, a specific pain pathway and a pain centre in the brain, with the intensity of pain proportional to the amount of damage. A rival *pattern theory* (proposed by Goldscheider in 1894 and developed by John Nafe in 1929) held instead that pain arises from the intensity and pattern of firing in receptors that are not specific to pain. Neither theory could explain the facts you have just met: soldiers who feel little pain from terrible wounds, pain relieved by placebos and distraction, pain that persists after healing, and pain felt in limbs that no longer exist.

In 1965 the Canadian psychologist **Ronald Melzack** (1929 to 2019) and the British physiologist **Patrick David Wall** (1925 to 2001), who had begun working together at the Massachusetts Institute of Technology, published "Pain mechanisms: a new theory" in the journal *Science*. Their **gate control theory** proposed that a neural "gate" in the dorsal horn of the spinal cord controls how much of the pain message from the body is passed up to the brain (Figure 3.9). The theory has four main parts.

- The gate lies in the **substantia gelatinosa** (SG), a layer of small inhibitory interneurons in the dorsal horn. When these cells are active, they inhibit the *transmission cells* (T cells) that carry signals up to the brain, and the gate is closed; when they are quiet, the gate is open.
- Small-diameter fibres (the A-delta and C fibres that carry pain) excite the T cells and inhibit the SG cells, so their activity *opens* the gate.
- Large-diameter fibres (the A-beta fibres that carry touch, pressure and vibration) also excite the T cells but excite the SG cells too, so their activity tends to *close* the gate. Pain therefore depends on the balance of activity in large and small fibres.
- Descending signals from the brain, reflecting attention, emotion, expectation and past experience, can also open or close the gate. Melzack and Wall called this central control. When the output of the T cells exceeds a critical level, it activates the brain systems responsible for the experience of pain and the response to it.

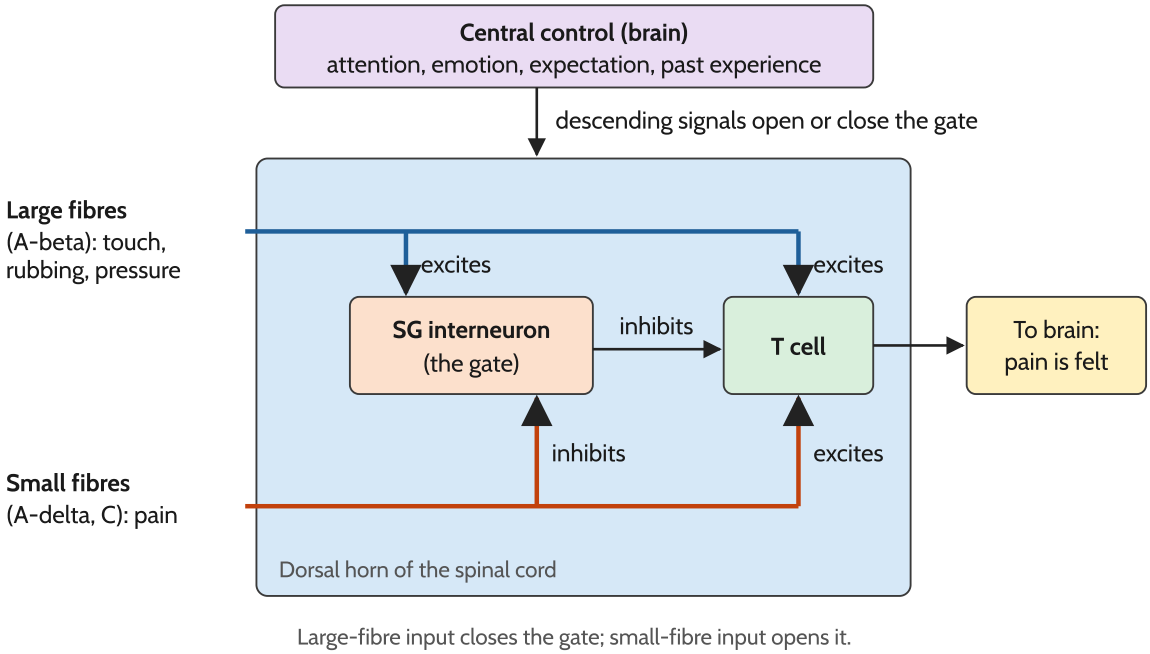


Figure 3.9 The gate control theory of pain (Melzack and Wall, 1965), simplified. Small pain fibres excite the transmission (T) cell and inhibit the substantia gelatinosa (SG) interneuron, opening the gate; large touch fibres excite both, and the SG cell inhibits the T cell, closing the gate. Descending signals from the brain can open or close the gate from above.

The theory explained a host of everyday observations at once. Why does rubbing a bumped elbow help? Rubbing activates large touch fibres, which close the gate. Why do soldiers in battle, athletes in a match and people absorbed in an emergency often feel little pain? Descending signals from a brain fully occupied elsewhere close the gate. Why do anxiety and dwelling on pain make it worse? Descending signals open the gate. The theory also led directly to treatments. In 1967 Wall and the neurosurgeon William Sweet showed that electrically stimulating large fibres through the skin relieved chronic pain, the origin of transcutaneous electrical nerve stimulation (TENS), still used in physiotherapy and childbirth. Its emphasis on central control gave psychological treatments of pain, from relaxation and distraction to cognitive behaviour therapy, a scientific basis.

Critics soon pointed out that the specific wiring Melzack and Wall proposed, especially the way the SG cells were supposed to act on the incoming fibres, was not confirmed in detail. But the core idea, that pain signals are modulated in the spinal cord and controlled from the brain, has been amply confirmed, and the gate control theory is generally regarded as the most influential theory of pain ever proposed. Melzack later extended it. In his **neuromatrix theory** (developed from about 1990), pain is produced by a widely distributed network of brain regions, the neuromatrix, which generates a characteristic pattern of activity, a "neurosignature", for the body and its sensations. Sensory input can trigger the neurosignature, but the neuromatrix can also produce pain with no input at all, an idea that becomes essential when we turn to phantom limbs. Table 3.20 compares the theories.

Table 3.20 Theories of pain.

| Theory | Proposed by | Core idea | Main weakness or strength |
|---|---|---|---|

| Theory | Proposed by | Core idea | Main weakness or strength |
|---|---|---|---|
| Specificity theory | Descartes (1664); von Frey (1890s) | Specific pain receptors and a direct line to a pain centre; pain proportional to injury | Cannot explain psychological modulation, placebo or phantom pain |
| Pattern theory | Goldscheider (1894); Nafe (1929) | Pain arises from the intensity and pattern of firing in non-specific receptors | Ignores the existence of specialised nociceptors |
| Gate control theory | Melzack and Wall (1965) | A spinal gate, opened by small fibres, closed by large fibres and by descending control | Wiring details unconfirmed, but its central idea transformed pain research and treatment |
| Neuromatrix theory | Melzack (from about 1990) | A brain-wide network generates pain as a pattern (neurosignature), with or without input | Explains phantom limbs and chronic pain; hard to test precisely |

REMEMBER: WHAT OPENS AND CLOSES THE PAIN GATE

**Opening the gate (more pain):** extensive injury and intense small-fibre input; anxiety, worry and depression; attending to and dwelling on the pain; boredom and inactivity.

**Closing the gate (less pain):** large-fibre counter-stimulation such as rubbing, massage, warmth or TENS; relaxation, positive emotions and rest; distraction and absorbing activity; analgesic drugs and the body's own endorphins.

## The body's own painkillers: endorphins

The gate control theory's "central control" was given a chemical basis within a decade. Three discoveries came together. First, in 1969 David Reynolds showed that electrically stimulating the periaqueductal grey (PAG), a region of the midbrain, produced such profound analgesia in rats that he could perform abdominal surgery on them with no anaesthetic. Second, in 1973 Candace Pert and Solomon Snyder (and, independently, other groups) discovered receptors in the brain to which morphine and other opiate drugs bind, concentrated in the PAG and in the dorsal horn. Why should the brain have receptors for a drug made from poppies? Because, third, it makes its own. In 1975 John Hughes and Hans Kosterlitz in Aberdeen isolated the first of the brain's natural opiates, the enkephalins, and others soon followed. The whole family is called the **endorphins**, a contraction of "endogenous morphine" (morphine made within the body).

Endorphins act at many points in the pain system, including a descending circuit that Allan Basbaum and Howard Fields proposed in 1978: the PAG excites serotonin-releasing neurons in the raphe nuclei of the medulla, which project down the spinal cord to interneurons that block incoming pain signals in the dorsal horn. Endorphins are released during stress and severe injury (stress-induced analgesia, which helps explain Beecher's soldiers), during prolonged vigorous exercise and during childbirth. Their role can be tested with naloxone, a drug that blocks opiate receptors (and is used in emergency rooms to reverse heroin overdoses). In 1978 Jon Levine, Newton Gordon and Howard Fields gave dental patients a placebo after the removal of wisdom teeth; the placebo relieved pain in many of them, but naloxone reduced that relief, evidence that placebo analgesia works partly by releasing endorphins. Naloxone also reduced the analgesia produced by acupuncture in a 1977 study by David Mayer and his colleagues. (The popular "runner's high" was long credited to endorphins; current research suggests that the body's cannabis-like chemicals, the endocannabinoids, play a part too.)

EXAM TRAP

"The neurotransmitters called feel-good hormones" and "the neurotransmitter associated with control of the pain response" are both keyed as *endorphins*. Dopamine is linked to reward and wanting; serotonin does take part in the descending pain-control circuit, but the classic answer for pain control is endorphins; acetylcholine and GABA are distractors. Pain *signals* are transmitted in the spinal cord by glutamate and substance P. The gate control theory is *Melzack and Wall (1965)*; the gate is in the *substantia gelatinosa of the dorsal horn* of the spinal cord, not in the thalamus or cortex; large fibres *close* it and small fibres *open* it. A phobia of pain is algophobia (Disorders book).

## Phantom limbs

After a limb is amputated, most patients continue to feel it. The missing arm may itch, tingle, feel cramped or clenched, seem to gesture during conversation, or reach out to answer a telephone. The American neurologist **Silas Weir Mitchell** (1829 to 1914), who treated amputees from the American Civil War, coined the term **phantom limb** in 1871 for this vivid sense that an amputated limb is still present. Most amputees experience a phantom at some time, and many suffer phantom limb pain, often severe and long-lasting. Phantoms are not confined to limbs: people report phantom breasts after mastectomy and phantom teeth after extraction.

What causes phantoms? The obvious idea is irritation of the cut nerve endings in the stump, and stump problems can certainly add to the pain. But as Pinel and Barnes point out, surgical attempts to cure phantom limb pain by cutting the stump again, or by cutting the nerve pathways between the stump and the brain, have generally failed. The phantom is not in the stump; it is in the brain.

KEY STUDY: RAMACHANDRAN FINDS A HAND ON THE FACE (1990S)

**Background.** In 1991 Tim Pons and his colleagues reported that in monkeys whose sensory nerves from one arm had been cut many years earlier, the region of somatosensory cortex that had once served the arm now responded to touches on the face. On the cortical body map, the face area lies right next to the hand area. **Question. Vilayanur Subramanian Ramachandran** (born 1951), an Indian-born neuroscientist trained in medicine in Chennai and working at the University of California, San Diego, wondered whether the same reorganisation happens in human amputees, and whether it might explain phantom sensations. **Method.** With a patient's eyes closed, he touched different parts of the body with a cotton bud and asked what was felt. **Result.** When he touched the face of a patient whose arm had been amputated, the patient felt touches both on his face and on specific fingers of his phantom hand; a second, orderly map of the hand was found on the upper arm near the stump. **Conclusion.** After amputation, neighbouring regions of the somatosensory cortex invade the territory of the lost limb, so that input from the face or upper arm is experienced as coming from the phantom. Phantom limb sensation is best explained by *cortical reorganisation (remapping)*.

Ramachandran then devised a treatment of startling simplicity. Some patients felt their phantom hand frozen in a painful, clenched position that they could not move. He built a *mirror box*: a box divided by a vertical mirror, into which the patient placed the intact arm on one side and the stump on the other. Looking into the mirror, the patient saw the reflection of the intact hand where the phantom should be. When the patient moved both hands to-

gether, the phantom appeared to move, and for some patients the cramp and pain eased; Pinel and Barnes describe a patient whose pain subsided after three weeks of home practice. Melzack's neuromatrix theory adds a further insight. Some people born without a limb report a phantom of the limb they never had, which suggests that the brain contains a built-in representation of the whole body that does not depend entirely on experience.

> **EXAM TRAP**
>
> Phantom limb sensation is best explained by *cortical reorganisation (remapping) in the somatosensory cortex*, associated with V. S. Ramachandran (and, for pain, by Melzack's neuromatrix). It is not caused by the severed nerve growing back into the limb (there is no limb to grow into, and cutting the nerves usually fails to stop phantoms), not by psychological denial (amputees know perfectly well the limb is gone, and phantoms occur even in people born without limbs), and not by damage to the cerebellum. The term "phantom limb" was coined by Silas Weir Mitchell (1871); the mirror box is Ramachandran's treatment.

> **CLINICAL LINK**
>
> Chronic pain, pain lasting longer than about three months, is among the commonest reasons people seek health care, and clinical psychologists are central to its treatment. Because pain has sensory, emotional and cognitive dimensions, it is assessed with tools such as the McGill Pain Questionnaire (Melzack, 1975), which asks patients to choose words describing the sensory quality of their pain ("throbbing", "burning"), its emotional quality ("exhausting", "terrifying") and its overall intensity. Treatments informed by the gate control and neuromatrix theories, including cognitive behaviour therapy, relaxation, graded activity and acceptance-based approaches, reduce disability even when they cannot remove the pain (the Clinical book describes them). Thoughts such as "this pain will never end" (catastrophising) reliably make pain worse, exactly as descending control predicts.

## 3.11 The body senses: kinesthesis and balance LOWER YIELD · 2 QS

Close your eyes and touch your nose with your fingertip. You can do it easily, although you cannot see your hand, because two senses that rarely reach awareness are constantly telling your brain where your body parts are and which way is up. They are so reliable that most people have never heard their names. They are kinesthesis and the vestibular sense, and you notice them mainly when they fail: when your leg "goes to sleep" and you stumble, or when you step off a merry-go-round and the world keeps spinning.

### Kinesthesis

**kinesthesis** (or kinaesthesis, from the Greek for "movement" and "sensation"; the term was introduced by the British neurologist Henry Charlton Bastian in 1880) is the sense of the position and movement of the parts of the body. Its information comes from the proprioceptors that Sherrington named in 1906 (Section 3.1). *Muscle spindles*, embedded in the muscles, signal changes in muscle length (stretch); they are the receptors of the knee-jerk reflex. *Golgi tendon organs*, embedded in the tendons that attach muscles to bones, signal muscle tension, the force of the muscle's pull; as Pinel and Barnes note, tendon organs respond to tension but not to length, and spindles to length but not to tension. Receptors in the joints and stretch re-

ceptors in the skin (the Ruffini endings of Table 3.19) add information about joint angle. The brain combines all of these with copies of its own motor commands to compute where each limb is.

What would life be like without kinesthesis? In 1971 a nineteen-year-old British man named Ian Waterman developed what seemed to be a bad flu. An immune reaction, probably triggered by the infection, destroyed the large sensory nerve fibres from his body below the neck, leaving him without touch and without any sense of the position of his limbs, although his muscles and his sense of pain and temperature were intact. At first he could not sit up, stand or feed himself; his limbs moved, but he did not know where they were. Over many months he taught himself to move again by watching his limbs constantly and planning every movement consciously, as if steering a puppet. He could walk, but if the lights went out, he collapsed. His physician, Jonathan Cole, told his story in *Pride and a Daily Marathon* (1991), and Oliver Sacks described a similar patient, "the disembodied lady", in *The Man Who Mistook His Wife for a Hat* (1985). Clinicians test kinesthesis with simple bedside tasks: in the Romberg test, a patient stands with the feet together and then closes the eyes; a person who has lost proprioception from the legs can stand while looking but sways and falls when vision is removed.

### The vestibular sense ●●

The **vestibular sense** is the sense of balance: it registers the position and movement of the head in relation to gravity. Its receptors lie in the vestibular apparatus of the inner ear, next to the cochlea, and like the cochlea they are hair cells whose stereocilia are bent by movement. The vestibular apparatus has two parts.

The three **semicircular canals** are fluid-filled loops set at roughly right angles to one another, so that each lies in a different plane of space (one for nodding, one for shaking the head "no", one for tilting the head towards the shoulder). At the base of each canal is a swelling, the ampulla, containing a gelatinous flap (the cupula) in which hair cells are embedded. When the head begins to rotate, the fluid in the canal (endolymph) lags behind because of its inertia, like the water in a glass that you suddenly turn; it pushes the cupula, bends the hair cells, and signals rotation. The canals therefore detect *rotational (angular) acceleration* of the head, in any direction, because the three planes between them cover every axis of rotation.

This explains a familiar experience. If you spin round and round, the fluid soon catches up with the canals and moves along with them, the cupula straightens, and the sensation of turning fades. When you stop suddenly, the fluid keeps moving for a few seconds and bends the cupula the other way, so you feel as if you are spinning in the opposite direction, and your eyes jerk rhythmically (post-rotational nystagmus). That is why a child who has been spinning in a playground staggers about, and why the room seems to whirl after a ride on a giant wheel at a Dasara fair. Dancers who must perform many rapid spins, such as the *chakkars* of Kathak, train themselves to resist the dizziness; ballet dancers use a trick called spotting, keeping their eyes fixed on one point for as long as possible during each turn and then whipping the head round, which limits the confusing signals.

The two **otolith organs**, the utricle and the saccule, lie in the vestibule between the canals and the cochlea. Their hair cells are embedded in a gelatinous layer topped with tiny crystals of calcium carbonate, the otoliths or otoconia ("ear stones"). Because the crystals are heavier than the surrounding fluid, gravity pulls on them whenever the head tilts, and they lag be-

hind whenever the body starts or stops moving in a straight line. The otolith organs therefore detect *static head position* (tilt relative to gravity) and *linear acceleration*: the utricle mainly in the horizontal plane (a car accelerating or braking), the saccule mainly in the vertical plane (a lift starting or stopping). Table 3.21 summarises the body senses.

Table 3.21 The body senses and their receptors.

| Sense | Receptors | Location | What they detect |
|---|---|---|---|
| Kinesthesis | Muscle spindles | Within muscles | Muscle length (stretch) |
| Kinesthesis | Golgi tendon organs | Tendons | Muscle tension (force) |
| Kinesthesis | Joint and skin stretch receptors | Joints and skin | Joint angle and movement |
| Vestibular sense | Hair cells in the three semicircular canals | Inner ear | Rotational (angular) acceleration of the head in three planes |
| Vestibular sense | Hair cells of the otolith organs (utricle, saccule) under calcium carbonate crystals | Inner ear (vestibule) | Head tilt (static position) and linear acceleration |

Vestibular signals travel along the vestibular branch of cranial nerve VIII to the vestibular nuclei of the brainstem and on to the cerebellum, the spinal cord (for posture) and the nuclei that control the eye muscles. That last connection drives the *vestibulo-ocular reflex*, which moves the eyes in the direction opposite to any head movement so that the visual world stays steady. Try it: shake your head while reading this sentence and the words stay legible; shake the book instead and they blur, because the reflex compensates only for movements of your own head. The Austrian physician **Robert Barany** (1876 to 1936) received the 1914 Nobel Prize in Physiology or Medicine for his work on the vestibular apparatus; his rotating chair and his caloric test (warm or cold water in the ear canal sets up currents in the canal fluid and produces nystagmus) are still used to test vestibular function.

Disorders of the vestibular system produce *vertigo*, a false sense of spinning or movement. In benign paroxysmal positional vertigo (BPPV), the commonest cause of vertigo, otolith crystals break loose and drift into a semicircular canal, so that certain head movements (rolling over in bed, looking up at a shelf) trigger brief, violent spinning; it is often cured by a sequence of head movements (the Epley manoeuvre) that floats the crystals back out of the canal. In Meniere's disease, excess fluid in the inner ear produces attacks of vertigo with ringing in the ear and fluctuating hearing loss, because the cochlea and vestibular organs share the same fluid system.

The vestibular sense also explains **motion sickness**, the nausea of the bus on a ghat road, the boat and the virtual-reality headset. According to the sensory conflict theory (James Reason and Joseph Brand, 1975), motion sickness arises when the senses disagree: reading in a moving bus, your vestibular organs signal movement while your eyes, fixed on the page, signal stillness; in a virtual-reality game, the eyes signal movement while the vestibular organs signal stillness. Why should such a conflict cause vomiting? Michel Treisman (1977) proposed an evolutionary answer: in nature, a mismatch between the eyes and the vestibular system is a typical early effect of swallowing a poison, so the brain responds to the conflict by emptying the stomach, just in case. Looking out at the horizon, sitting where motion is least (near the middle of a bus or boat) and not reading all reduce the conflict.

EXAM TRAP

Spinning and then feeling that your head is still spinning is due to the *semicircular canals*, which detect *rotational* (*angular*) *acceleration* in three planes. The *otolith organs* (utricle and saccule) detect head tilt (static position relative to gravity) and *linear* acceleration; the *otolith crystals* (otoconia) are the calcium carbonate particles inside them, not a separate organ; and *proprioceptors* in muscles, tendons and joints report the position of the limbs, not the spinning of the head. So "static head position only" and "linear acceleration only" are wrong for the canals, and "auditory pitch" belongs to the cochlea. Kinesthesis is limb position and movement; the vestibular sense is head position and movement (balance); proprioception, in Sherrington's broad sense, covers both.

One more body sense deserves a sentence. *Interoception*, served by Sherrington's interoceptors, is the sense of the internal state of the body: heartbeat, breathing, hunger, a full bladder, the churning stomach of anxiety. People differ in how accurately they perceive their own heartbeat, and the misreading of harmless bodily sensations as signs of catastrophe (a racing heart taken for a heart attack) lies at the core of cognitive models of panic disorder (Disorders and Clinical books).

## 3.12 Senses in concert and senses deprived

LOWER YIELD · 3 QS

This chapter has treated the senses one at a time, as textbooks must. The brain does not. At every moment it combines information from all the senses into a single experience of a single world, and it depends on a steady flow of that information to work normally. This final section looks at what happens when the senses work together, when they cross over into one another, and when they are taken away.

### Sensory interaction ●●

**sensory interaction** is the principle that one sense can influence another. You have already met the most familiar example: flavour, in which smell, taste, touch and temperature fuse so completely that we experience smell as taste (Section 3.9). The most dramatic demonstration involves hearing and vision. In 1976 Harry McGurk and John MacDonald dubbed a recording of a voice saying "ba-ba" onto a video of a face saying "ga-ga". Viewers heard neither: they heard "da-da", a compromise between what they heard and what they saw. Close your eyes and the sound changes back to "ba". This **McGurk effect** shows that the visible movements of the lips are part of what we "hear" (Chapter 8 discusses speech perception), and it is one reason why a noisy room is easier to follow when you can see the speaker's face.

When the senses disagree about location, vision usually wins. In the ventriloquism effect, the voice seems to come from the puppet's moving mouth, and in a cinema the dialogue seems to come from the actors' lips rather than from the loudspeakers at the side. But vision does not always dominate. In the sound-induced flash illusion (Ladan Shams and colleagues, 2000), a single flash of light accompanied by two quick beeps is seen as two flashes, because hearing is more precise than vision about timing. Even taste and appearance interact: when wine-tasting students at the University of Bordeaux were given a white wine coloured red with a tasteless dye, they described it with the vocabulary normally reserved for red wines (Gil Morrot and colleagues, 2001). And in the rubber hand illusion (Matthew Botvinick and

Jonathan Cohen, 1998), a person watches a rubber hand being stroked with a brush while their own hidden hand is stroked in exact time; within a minute or two many people feel that the rubber hand is their own.

Why does the brain sometimes trust one sense and sometimes another? Marc Ernst and Martin Banks (2002) showed that when people judge the size of an object by both sight and touch, the brain weights each sense according to how reliable it is at that moment, exactly as a statistically optimal observer would. When the visual information was made blurry, touch dominated; when it was clear, vision dominated. This is the Bayesian logic of Section 3.1 applied across the senses. At the level of single neurons, cells in the superior colliculus of the midbrain respond to sight, sound and touch, and they fire far more strongly when signals from two senses arrive together from the same place at the same time than the sum of their responses to each sense alone, especially when each signal by itself is weak. Combining the senses makes detection faster and more accurate, which is also the rationale for multisensory teaching of children with learning difficulties, in which a letter is seen, said aloud, traced with a finger and written at once (the Developmental book discusses these methods).

## Synaesthesia

For most of us, sensory interaction is subtle and automatic. For some people the senses mingle so vividly that one sense, or even an idea, produces a second, quite separate experience. **synaesthesia** (from the Greek for "sensing together") is a condition in which stimulation in one sense or of one kind automatically and consistently triggers an additional experience of another kind. In the commonest form, grapheme-colour synaesthesia, letters and digits are experienced as having colours: for one person the letter A is always crimson and 5 is always a particular green. In coloured hearing (chromesthesia), musical notes or voices trigger colours; in lexical-gustatory synaesthesia, words produce tastes in the mouth; in sequence-space synaesthesia, numbers, days or months are experienced as laid out in a particular arrangement in space. Galton described such "number forms" and coloured letters in 1880 and in his *Inquiries into Human Faculty* (1883), noting that they ran in families. The great Russian mnemonist Solomon Shereshevsky, studied for thirty years by Alexander Luria, had synaesthesia so intense that every word evoked colours, tastes and textures, which helped his extraordinary memory and also made ordinary conversation exhausting (Chapter 7).

For a long time synaesthesia was dismissed as metaphor or imagination. Two kinds of evidence established it as a genuine perceptual phenomenon. First, synaesthetic associations are remarkably consistent: asked without warning, months or even years later, to give the colour of each letter, synaesthetes give almost exactly the same answers, whereas non-synaesthetes who are asked to invent associations and remember them cannot match this consistency even after a week or two. Second, the colours affect perception. In 2001 Ramachandran and Edward Hubbard showed grapheme-colour synaesthetes a display of black 5s among which a few black 2s formed a triangle. Most people must search the display item by item; many synaesthetes saw the triangle almost at once, because for them the 2s "popped out" in a different colour. Large surveys suggest that some form of synaesthesia occurs in about 4 percent of people, and grapheme-colour synaesthesia in about 1 percent. A popular explanation is cross-activation: the region of the fusiform gyrus that processes the shapes of letters and numbers lies right next to a colour area, and in synaesthetes the connections between them may not have been pruned away during development as they usually are.

Synaesthesia may be an exaggeration of something we all share. In 1929 the Gestalt psychologist Wolfgang Kohler showed people two shapes, one rounded and one spiky, and asked which was called "baluma" and which "takete"; almost everyone matched the rounded shape with "baluma" (in the 1947 edition of his book Kohler changed the word to "maluma", the version most textbooks quote). When Ramachandran and Hubbard repeated the demonstration with the words "bouba" and "kiki", about 95 percent of people called the rounded shape "bouba" and the spiky one "kiki", across languages. Such cross-modal correspondences, linking the sound of a word to the shape of an object, are probably one of the roots of metaphor ("a sharp cheese", "a loud shirt").

> EXAM TRAP
>
> *Synaesthesia* is an automatic, consistent, involuntary experience in a second sense (letters seen in colour), present in a minority of people. *Sensory interaction* is the influence of one sense on another that occurs in everyone (flavour, the McGurk effect). Neither is a *hallucination*, which is a perception without any external stimulus, nor an *illusion*, a misperception of a real stimulus (Chapter 4). Do not confuse synaesthesia with *kinesthesis* (Section 3.11), whose name looks similar.

## Senses deprived ●●

If the senses are designed to detect change (Section 3.1), what happens when there is almost nothing to detect? In the early 1950s, at McGill University in Montreal, the Canadian psychologist **Donald Olding Hebb** (1904 to 1985) and his students William Bexton, Woodburn Heron and Thomas Scott paid male students to lie for as long as they could on a bed in a lit cubicle, wearing translucent goggles that let in only diffuse light, cotton gloves and cardboard cuffs that limited touch, with the steady hum of an air conditioner masking other sounds. Their report (Bexton, Heron and Scott, 1954) became the classic study of **sensory deprivation**, a drastic reduction in the amount and variety of sensory stimulation. Most volunteers quit within two or three days. They found it hard to think or concentrate, performed worse on simple tests, became irritable, were more easily swayed by recorded propaganda about the supernatural, and many experienced hallucinations, beginning with dots and patterns and developing into whole scenes. After release, some saw the room as curved or moving for a while. (Chapter 5 tells the full story of these studies and of the later flotation-tank research that found brief, voluntary restriction of stimulation relaxing rather than distressing.)

Hebb's conclusion was that the waking brain needs a continuous flow of varied stimulation to keep functioning normally, an idea that fed his theory of optimal arousal (Chapter 10). The hallucinations are the most revealing finding for this chapter. When normal input is removed, the sensory brain does not fall silent; it generates experience of its own. The same principle connects several phenomena you have met: the phantom limbs of amputees (Section 3.10), the phantom sounds of tinnitus after hearing loss (Section 3.8) and, as the Clinical link below explains, the phantom visions of people losing their sight.

### INDIAN CONTEXT

Indian contemplative traditions deliberately reduce sensory input rather than fear it. In Patanjali's eight-limbed yoga, the fifth limb, *pratyahara*, is the withdrawal of the senses from their objects, so that attention is no longer pulled by sounds, smells and sights and can turn inward in preparation for concentration (*dharana*). The difference between pratyahara and the McGill cubicle lies in control and meaning: chosen, gradual, purposeful withdrawal is calming, whereas imposed, open-ended deprivation is distressing (Chapters 2 and 5 discuss yoga and meditation).

### CLINICAL LINK: CHARLES BONNET SYNDROME

In 1760 the Swiss naturalist **Charles Bonnet** (1720 to 1793) described his elderly grandfather, Charles Lullin, who was almost blind from cataracts yet saw vivid, silent visions: men, women, birds, carriages and buildings, appearing and vanishing, while he remained perfectly aware that they were not real. **Charles Bonnet syndrome** is the occurrence of complex visual hallucinations in people with significant visual loss (from macular degeneration, glaucoma or cataract) who are cognitively intact and keep their insight. It is common among older people with poor sight, but many never mention it for fear of being thought mad. The hallucinations are a "release" phenomenon: deprived of input, the visual cortex generates images, just as the McGill students' brains did. Reassurance that the visions are a harmless consequence of sight loss, and not a sign of mental illness, is often the most useful treatment. Distinguish it carefully from other hallucinations a clinician will meet: visual hallucinations *with* fluctuating cognition and parkinsonism suggest dementia with Lewy bodies; auditory hallucinations (voices) are typical of schizophrenia; tactile hallucinations of insects crawling on the skin (formication) occur in alcohol withdrawal and cocaine use; and a brief olfactory hallucination may be the aura that precedes a temporal lobe seizure (the Disorders book covers these conditions). People with hearing loss can develop an auditory equivalent, hearing music that is not playing.

### EXAM TRAP

Charles Bonnet syndrome = *visual* hallucinations in people with significant *visual* loss, who are *cognitively intact* and keep *insight*. "Auditory hallucinations in dementia", "tactile hallucinations in alcohol withdrawal", "multimodal hallucinations in delirium" and "olfactory hallucinations preceding seizures" are real phenomena belonging to other conditions, and they are the distractors. The McGill sensory deprivation studies were carried out in Donald Hebb's laboratory (Bexton, Heron and Scott, 1954); the flotation tank was John Lilly's.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. The largest skin receptors, which respond to vibration and adapt very rapidly, are the _____ corpuscles.
2. Receptors that respond to actual or threatened tissue damage are called _____.
3. The gate control theory of pain was proposed in 1965 by Melzack and _____.
4. According to the gate control theory, activity in large touch fibres tends to _____ the gate.
5. The body's own opiates are called _____.
6. Ramachandran explained phantom limb sensations by cortical _____ in the somatosensory cortex.
7. Rotational acceleration of the head is detected by the _____ canals.
8. Head tilt and linear acceleration are detected by the _____ organs, the utricle and saccule.
9. Hearing "da" when the voice says "ba" and the lips say "ga" is the _____ effect.
10. Complex visual hallucinations with preserved insight in people with severe sight loss are called _____ syndrome.

**Answers:** (1) Pacinian, (2) nociceptors, (3) Wall, (4) close, (5) endorphins, (6) reorganisation (remapping), (7) semicircular, (8) otolith, (9) McGurk, (10) Charles Bonnet.

## The case of Gustav Fechner, revisited

Go back to the Leipzig professor gazing at the sun through coloured glass. You can now explain what he was seeing. When he looked away, the disc of colour that hung before him was a negative afterimage: the cones and opponent channels driven hard by the colour had adapted, so that a white surface, which stimulates every channel equally, was seen in the complementary hue (Section 3.7). The intense light also bleached his photopigments, which is why the world went dark and only slowly returned, as in the cone branch of the dark adaptation curve (Section 3.6). And because the lens focuses sunlight onto the fovea, prolonged sun-gazing can burn the retina itself, an injury now called solar retinopathy. Whether such damage explains Fechner's long collapse nobody knows; his intolerance of light came with exhaustion and deep low mood, and historians still debate what his illness was.

His law has fared rather like his eyes: damaged but far from destroyed. Weber's law, which Fechner named and made famous, holds well across the middle range of every sense. Fechner's founding assumption, that every just noticeable difference is an equal step of sensation, did not survive Stevens' decision simply to ask people how strong their sensations were, and in 1961 Stevens proposed to "honor Fechner and repeal his law". Signal detection theory then showed that Fechner's threshold mixes two things, what an observer can sense and what the observer is willing to report. Yet almost everything in this chapter descends from that October morning in 1850. The decibel scale on a sound meter is a Fechnerian logarithm; every audiogram, eye chart and pain threshold is measured with his methods or their descendants; and the idea that experience can be measured, which Kant had declared impossible, is now so ordinary that we no longer notice it. The International Society for Psychophysics still holds its annual meeting under the name Fechner Day.

## Chapter summary

**3.1 Sensation and perception: from energy to experience.** Sensation is the detection of physical energy by receptors and its conversion into neural signals (transduction); perception is the brain's organisation and interpretation of those signals, so optical illusions are distortions of perception, not sensation. Titchener treated sensations as the elements of consciousness and perception as a complex of sensory and imaginal elements; Helmholtz's unconscious inference survives in the Bayesian approach, in which perception combines prior beliefs with sensory evidence. Receptors are classified by energy (photo-, mechano-, chemo-, thermo- and nociceptors) and, after Sherrington (1906), as exteroceptors, interoceptors and proprioceptors; signals travel inward along afferent fibres. Muller's doctrine of specific nerve energies (labelled lines) holds that the quality of a sensation depends on which nerve is active; within taste and smell, across-fibre patterns code quality. Because of the all-or-none law, intensity is coded by firing rate and recruitment. Every sense except smell relays through the thalamus. Sensory adaptation, the fading of response to a constant stimulus (Locke's three bowls), lets the senses track change; it is not the same as habituation.

**3.2 Thresholds: how weak a stimulus, how small a difference.** Psychophysics, founded by Fechner, studies the quantitative relation between stimuli and sensations. The absolute threshold (limen) is the minimum intensity detected on 50 percent of trials, because detection rises gradually along an S-shaped psychometric function rather than switching on; stimuli below it are subliminal. Galanter's examples (a candle flame seen at about 48 kilometres on a dark night, a watch tick at 6 metres) show how sensitive the senses are, and Hecht, Shlaer and Pirenne (1942) found that absorbing only 5 to 14 quanta can produce a visible flash. The difference threshold, or just noticeable difference (JND), is the smallest difference detected 50 percent of the time; the point of subjective equality is the comparison value that seems equal to the standard. Weber's two-point threshold varies from about 2 millimetres on the fingertips to several centimetres on the back.

**3.3 The psychophysical laws: Weber, Fechner and Stevens.** Weber's law (1834) states that the JND is a constant fraction of the standard ($\Delta I/I = k$); Weber fractions are smallest for pitch and largest for the chemical senses, and the law fails near threshold. Fechner (1860) assumed that every JND adds an equal unit of sensation and derived $S = k \log I$ by indirect scaling; his *Elements of Psychophysics* made him the founder of the field. Stevens (1957) asked people to judge magnitudes directly (magnitude estimation) and found a power law, $S = kI^n$, with exponents below 1 producing response compression (brightness, 0.33) and above 1 producing response expansion (electric shock, 3.5); cross-modality matching confirmed it. Fechner's law describes discriminability; Stevens' law describes apparent magnitude.

**3.4 The classical psychophysical methods.** The method of limits (just noticeable differences, minimal changes) alternates ascending and descending series and suffers from errors of habituation (keeping the same response too long) and anticipation (changing too early), controlled by counterbalancing. The method of constant stimuli (right and wrong cases) presents fixed values many times in random order; it is the most accurate and yields the full psychometric function. The method of adjustment (average error) lets the observer set the stimulus and suits the point of subjective equality and illusions. Time-order and space errors are also controlled by counterbalancing. Modern staircase methods (Cornsweet, 1962), Bekesy's tracking audiometer (1947) and forced-choice procedures concentrate trials near threshold and reduce response bias.

**3.5 Signal detection theory.** Signal detection theory, applied to people by Tanner and Swets (1954) and set out by Green and Swets (1966), holds that detection depends on two independent quantities. Noise and signal-plus-noise produce overlapping distributions of internal evidence, and the observer says "yes" when evidence exceeds a criterion, producing hits, misses, false alarms and correct rejections. Sensitivity (d′), the separation of the distributions, rises as hits exceed false alarms; the criterion (β or c), or response bias, is liberal or conservative depending on expectations and payoffs. The ROC curve plots hit rate against false alarm rate as the criterion moves; bowed curves showed that the fixed threshold idea was wrong. SDT applies to radiologists, radar operators, fingerprint examiners, polygraphs and every clinical screening test.

**3.6 Vision I: light, the eye and the two receptor systems.** Visible light runs from about 380 to 760 nanometres; wavelength is seen as hue, intensity as brightness and purity as saturation, and frequency and wavelength are inversely related. Light passes through the cornea, pupil (set by the iris) and lens, which focuses by accommodation, to the retina. The fovea contains only cones and gives sharp colour vision; the optic disc has no receptors and forms the blind spot. About 120 million rods (rhodopsin, scotopic, sensitive but colourless and coarse) and 6 to 7 million cones (photopic, colour, acuity) form two systems (duplexity theory). Dark adaptation takes about 30 minutes, with a rod-cone break at 7 to 10 minutes; light adaptation is fast. The Purkinje shift (1825) darkens reds at dusk. Snellen charts measure acuity, saccades and microsaccades prevent stabilised images from fading, and lateral inhibition produces Mach bands.

**3.7 Vision II: colour.** Lights mix additively towards white; pigments mix subtractively towards black; complementary colours mix to grey. The Young-Helmholtz trichromatic theory (1802, 1852) proposes three receptor types, confirmed as S, M and L cones, and explains colour matching and metamers. Hering's opponent-process theory (1878) proposes red versus green, blue versus yellow and black versus white systems and explains negative afterimages and the absence of reddish green. The dual-process theory places trichromacy in the cones and opponency in the ganglion cells and lateral geniculate nucleus. Ladd-Franklin offered an evolutionary theory and Land the retinex theory of colour constancy. Colour vision deficiency includes anomalous trichromacy (commonest), dichromacy (protanopia, deuteranopia, tritanopia) and monochromacy; red-green defects are X-linked and are screened with Ishihara plates (1917).

**3.8 Hearing.** Sound is a pressure wave whose frequency (hertz) is heard as pitch, amplitude (decibels, a logarithmic scale on which each 10 dB is a tenfold rise in intensity) as loudness and complexity as timbre; humans hear about 20 to 20,000 Hz, and the missing fundamental shows that pitch is not simply frequency. Sound passes from the pinna and auditory canal to the tympanic membrane, the ossicles (malleus, incus, stapes), the oval window and the fluid of the cochlea, where the basilar membrane bends the hair cells of the organ of Corti, which transduce mechanically and signal along cranial nerve VIII. Place theory (Helmholtz, 1863; Bekesy's travelling wave, Nobel Prize 1961) explains high pitches; Rutherford's frequency theory (1886) explains low pitches but fails above about 1,000 Hz; Wever and Bray's volley principle (1930) extends timing codes to about 4,000 to 5,000 Hz. Sounds are located by interaural time differences (low frequencies) and intensity differences (high frequencies), as in Rayleigh's duplex theory (1907), with pinna cues for elevation. Conductive loss arises in the outer or middle ear; sensorineural loss in the hair cells or auditory nerve, where cochlear implants can help.

**3.9 The chemical senses: taste and smell.** Gustation has five basic tastes: sweet, salty, sour, bitter and umami (Ikeda, 1908). Taste buds of 50 to 100 receptor cells lie in fungiform, foliate and circumvallate papillae (filiform papillae have none); every part of the tongue detects every taste,

so the tongue map is a myth. Taste travels in cranial nerves VII, IX and X to the insula. Supertasters (Bartoshuk) taste bitterness intensely; chilli "heat" is pain carried by the trigeminal nerve. Olfaction begins in the olfactory epithelium, whose receptor neurons carry one of a few hundred receptor types (Buck and Axel, 1991; Nobel Prize 2004) and code odours combinatorially; signals reach the olfactory bulb and go directly to the piriform cortex and amygdala, bypassing the thalamus. Henning's smell prism and taste tetrahedron (1916) and Amoore's stereochemical theory sought primaries. Pheromones are well established in animals but doubtful in humans. Flavour combines taste, retronasal smell, texture and chemesthesis. Anosmia often follows head injury and can herald Parkinson's or Alzheimer's disease.

**3.10 The skin senses and pain.** Von Frey's spot mapping found separate pressure, warm, cold and pain spots. Meissner's corpuscles, Merkel's discs, Ruffini endings and Pacinian corpuscles (vibration) differ in receptive field and adaptation; free nerve endings sense temperature and pain. Two-point thresholds reflect receptor density and the size of each region on Penfield's somatosensory map. Touch travels in the dorsal columns, pain and temperature in the anterolateral system. Paradoxical cold and the thermal grill show that temperature is a pattern code. Pain, an unpleasant sensory and emotional experience, is vital, as congenital insensitivity to pain shows; nociceptors signal along A-delta (first, sharp) and C (second, dull) fibres, and pain depends on meaning (Beecher's soldiers) and has sensory, affective and evaluative dimensions. Melzack and Wall's gate control theory (1965) places a gate in the substantia gelatinosa, opened by small fibres, closed by large fibres and controlled from the brain; Melzack's neuromatrix theory followed. Endorphins, acting through a descending circuit from the periaqueductal grey, mediate stress, placebo and acupuncture analgesia. Phantom limbs (Weir Mitchell, 1871) reflect cortical reorganisation (Ramachandran) and are treated with the mirror box.

**3.11 The body senses: kinesthesis and balance.** Kinesthesis is the sense of limb position and movement, served by muscle spindles (length), Golgi tendon organs (tension) and joint receptors; Ian Waterman's loss of it showed how much movement depends on it. The vestibular sense registers head position and movement. The three semicircular canals detect rotational acceleration in three planes, which is why the world seems to spin after you stop turning; the otolith organs (utricle and saccule), with their calcium carbonate crystals, detect head tilt and linear acceleration. The vestibulo-ocular reflex steadies the eyes; Barany won the 1914 Nobel Prize for work on the vestibular apparatus; loose otoliths cause positional vertigo; and motion sickness arises from conflict between vestibular and visual signals.

**3.12 Senses in concert and senses deprived.** Sensory interaction means that one sense influences another: flavour, the McGurk effect (1976), ventriloquism, the sound-induced flash illusion and the rubber hand illusion. The brain weights each sense by its reliability (Ernst and Banks, 2002), and multisensory neurons respond most strongly to combined signals. Synaesthesia, in about 4 percent of people, is an automatic, consistent extra experience (coloured letters, coloured music), verified by consistency and pop-out tests and probably due to cross-activation between neighbouring brain areas. The McGill sensory deprivation studies in Hebb's laboratory (Bexton, Heron and Scott, 1954) produced poor concentration, suggestibility and hallucinations, showing that the brain needs varied input. Charles Bonnet syndrome, complex visual hallucinations with insight in people losing their sight, belongs with phantom limbs and tinnitus as experience generated by a brain deprived of input.

## Who's who

Table 3.22 The people in this chapter.

| Name | When | Examined for |
|---|---|---|
| Rene Descartes | 1596 to 1650 | Specificity theory of pain: the bell-rope picture (1664) |
| John Locke | 1632 to 1704 | Three-bowls demonstration of sensory (temperature) adaptation (1690) |
| Isaac Newton | 1666 | Prism experiments: white light is a mixture of colours |
| Charles Bonnet | 1720 to 1793 | Described visual hallucinations in his nearly blind grandfather (1760): Charles Bonnet syndrome |
| Immanuel Kant | 1724 to 1804 | Argued that mental events cannot be measured, the challenge Fechner answered |
| John Dalton | 1766 to 1844 | Described his own red-green colour blindness (1794), once called Daltonism |
| Joseph Fourier | 1768 to 1830 | Any complex wave is a sum of simple sine waves |
| Thomas Young | 1773 to 1829 | Trichromatic theory of colour vision (1802) |
| Jan Evangelista Purkyne (Purkinje) | 1787 to 1869 | Purkinje shift: reds darken before blues at dusk (1825) |
| Ernst Heinrich Weber | 1795 to 1878 | Weber's law, $\Delta I/I = k$ (1834); two-point threshold; Weber tuning-fork test |
| Johannes Peter Muller | 1801 to 1858 | Doctrine of specific nerve energies |
| Gustav Theodor Fechner | 1801 to 1887 | Founder of psychophysics; *Elements of Psychophysics* (1860); $S = k \log I$; the classical methods |
| Louis Braille | 1809 to 1852 | Raised-dot reading script for the blind (1824) |
| Hermann von Helmholtz | 1821 to 1894 | Trichromatic theory (with Young, 1852); place (resonance) theory of hearing (1863); unconscious inference |
| Francis Galton | 1822 to 1911 | Galton whistle (1876); early descriptions of synaesthesia and number forms (1880, 1883) |
| Alfonso Corti | 1822 to 1876 | Described the organ of Corti, the receptor organ of hearing (1851) |
| Silas Weir Mitchell | 1829 to 1914 | Coined the term "phantom limb" (1871) |
| Herman Snellen | 1834 to 1908 | Eye chart for visual acuity (1862) |
| Ewald Hering | 1834 to 1918 | Opponent-process theory of colour vision (1878) |
| Henry Charlton Bastian | 1837 to 1915 | Coined the term kinaesthesis (1880) |
| Ernst Mach | 1838 to 1916 | Mach bands (1865) |

| Name | When | Examined for |
|---|---|---|
| William Rutherford | 1839 to 1899 | Frequency (telephone) theory of pitch (1886) |
| John William Strutt, Lord Rayleigh | 1842 to 1919 | Duplex theory of sound localisation: time and intensity differences (1907) |
| Christine Ladd-Franklin | 1847 to 1930 | Evolutionary theory of colour vision (1892) |
| Alexander Graham Bell | 1847 to 1922 | Inventor of the telephone; the bel and the decibel are named after him |
| Hermann Ebbinghaus | 1850 to 1909 | Inspired by Fechner's *Elemente* to measure memory |
| Max von Frey | 1852 to 1932 | Mapping of pressure, warm, cold and pain spots; von Frey hairs; specificity theory; paradoxical cold |
| Charles Scott Sherrington | 1857 to 1952 | Exteroceptors, interoceptors and proprioceptors (1906) |
| Heinrich Hertz | 1857 to 1894 | The hertz (Hz), the unit of frequency, is named after him |
| Alfred Goldscheider | 1858 to 1935 | Mapping of skin spots; pattern theory of pain (1894) |
| John Dewey | 1859 to 1952 | Criticised the reflex arc concept (1896) |
| Kikunae Ikeda | 1864 to 1936 | Discovered umami, the taste of glutamate (1908) |
| Edward Bradford Titchener | 1867 to 1927 | Structuralism; sensations as elements; perception as a complex of sensory and imaginal elements |
| Robert Barany | 1876 to 1936 | Nobel Prize (1914) for work on the vestibular apparatus; caloric test |
| Shinobu Ishihara | 1879 to 1963 | Colour-blindness plates (1917) |
| Max Wertheimer | 1880 to 1943 | Phi phenomenon, apparent movement (1912) |
| Wolfgang Kohler | 1887 to 1967 | Rounded and spiky shapes matched to "baluma" (later "maluma") and "takete" (1929) |
| Wilder Penfield | 1891 to 1976 | Mapped the somatosensory cortex; the sensory homunculus |
| Selig Hecht | 1892 to 1947 | With Shlaer and Pirenne (1942): 5 to 14 quanta absorbed at the visual threshold |
| Georg von Bekesy | 1899 to 1972 | Travelling wave (place) theory of hearing; Nobel Prize 1961; tracking audiometer (1947) |
| Ernest Glen Wever | 1902 to 1991 | Volley principle with Bray (1930); *Theory of Hearing* (1949) |
| Haldan Keffer Hartline | 1903 to 1983 | Lateral inhibition in the *Limulus* eye; Nobel Prize 1967 |
| Charles William Bray | Born 1904 | Volley principle with Wever (1930) |
| Henry Knowles Beecher | 1904 to 1976 | Wounded soldiers at Anzio needed far less morphine than civilians: the meaning of pain |

| Name | When | Examined for |
|---|---|---|
| James Jerome Gibson | 1904 to 1979 | Active touch and the cookie-cutter study (1962); haptic perception |
| Donald Olding Hebb | 1904 to 1985 | McGill sensory deprivation studies (Bexton, Heron and Scott, 1954); optimal arousal |
| Stanley Smith Stevens | 1906 to 1973 | Power law $S = kI^n$ (1957); magnitude estimation; cross-modality matching; sone and mel |
| George Wald | 1906 to 1997 | Visual pigments and the three cone types; Nobel Prize 1967 |
| Edwin Land | 1909 to 1991 | Retinex theory of colour constancy |
| Patrick David Wall | 1925 to 2001 | Gate control theory with Melzack (1965); electrical stimulation for pain (TENS) |
| Ronald Melzack | 1929 to 2019 | Gate control theory with Wall (1965); neuromatrix theory; McGill Pain Questionnaire (1975) |
| Arthur Fox | 1931 | Discovered that some people cannot taste PTC |
| Gunnar Svaetichin | 1950s | Opponent responses recorded in the fish retina |
| W. Wesley Peterson and Theodore G. Birdsall | 1954 | Engineering theory of signal detectability (with W. C. Fox); Indian notes credit them with SDT |
| William Bexton, Woodburn Heron and Thomas Scott | 1954 | McGill sensory deprivation study in Hebb's laboratory |
| John Lilly | 1950s | Flotation (isolation) tank |
| Leo Hurvich and Dorothea Jameson | 1957 | Quantitative version of Hering's opponent-process theory |
| Tom Cornsweet | 1962 | Staircase method in psychophysics |
| Kenneth Casey | 1968 | Three dimensions of pain (with Melzack) |
| Ian Waterman | 1971 | Lost touch and limb-position sense below the neck; moves by watching his limbs |
| Martha McClintock | 1971 | Reported menstrual synchrony in women living together (later doubted) |
| Candace Pert and Solomon Snyder | 1973 | Discovered opiate receptors in the brain |
| Jacob Steiner | 1970s | Newborns' inborn facial reactions to sweet, sour and bitter |
| John Hughes and Hans Kosterlitz | 1975 | Isolated the enkephalins, the first of the endorphins |
| James Reason and Joseph Brand | 1975 | Sensory conflict theory of motion sickness |
| Harry McGurk and John MacDonald | 1976 | McGurk effect: seen lips change heard speech |
| Michel Treisman | 1977 | Poison hypothesis of motion sickness |
| Tim Pons | 1991 | Cortical reorganisation in monkeys after loss of arm input |
| Matthew Thompson | 2014 | Fingerprint experts: same hit rate as novices, far fewer false alarms |
| Wilson P. Tanner | 1954 | With Swets, first applied signal detection theory to human observers |
| John A. Swets | 1954; 1966 | Signal detection theory (with Tanner, 1954; with Green, 1966) |
| David M. Green | 1966 | *Signal Detection Theory and Psychophysics* (with Swets) |
| Eugene Galanter | 1962 | Everyday examples of absolute thresholds (candle flame at 48 kilometres) |

| Name | When | Examined for |
|---|---|---|
| Russell De Valois | 1960s | Opponent cells in the monkey lateral geniculate nucleus |
| Hans Henning | 1916 | Smell prism (six primary odours) and taste tetrahedron (four primary tastes) |
| John Amoore | 1950s to 1960s | Stereochemical (lock-and-key) theory of smell; seven primary odours |
| Linda Bartoshuk | 1990s | Supertasters |
| Richard Axel | Born 1946 | Olfactory receptor gene family with Buck (1991); Nobel Prize 2004 |
| Linda Buck | Born 1947 | Olfactory receptor gene family with Axel (1991); Nobel Prize 2004 |
| Vilayanur Subramanian Ramachandran | Born 1951 | Phantom limbs explained by cortical remapping; mirror box therapy; synaesthesia pop-out test; bouba and kiki |

## Key terms

sensation, perception, Bayesian approach to perception, receptor, transduction, exteroceptors, interoceptors, proprioceptors, afferent, efferent, doctrine of specific nerve energies, labelled-line coding, across-fibre pattern coding, rate coding, sensory adaptation, psychophysics, absolute threshold, limen, psychometric function, difference threshold, just noticeable difference, point of subjective equality, two-point threshold, Weber's law, Weber fraction, Fechner's law, indirect scaling, magnitude estimation, Stevens' power law, response compression, response expansion, cross-modality matching, time-order error, counterbalancing, method of limits, error of habituation, error of anticipation, method of constant stimuli, method of adjustment, staircase method, forced-choice procedure, signal detection theory, noise, criterion, hit, miss, false alarm, correct rejection, sensitivity, d prime, response bias, liberal criterion, conservative criterion, receiver operating characteristic, hue, brightness, saturation, cornea, pupil, iris, lens, accommodation, retina, fovea, optic disc, blind spot, rods, cones, duplexity theory, photopic vision, scotopic vision, rhodopsin, dark adaptation, light adaptation, rod-cone break, Purkinje shift, visual acuity, saccades, stabilised retinal image, Mach bands, lateral inhibition, additive colour mixing, subtractive colour mixing, complementary colours, trichromatic theory, metamers, opponent-process theory, negative afterimage, dual-process theory, colour vision deficiency, anomalous trichromacy, dichromacy, monochromacy, Ishihara plate, pitch, loudness, timbre, decibel, tympanic membrane, ossicles, cochlea, basilar membrane, organ of Corti, hair cells, place theory, frequency theory, volley principle, interaural time difference, interaural intensity difference, conductive hearing loss, sensorineural hearing loss, cochlear implant, gustation, umami, papillae, taste buds, olfaction, olfactory epithelium, smell prism, taste tetrahedron, anosmia, pheromones, flavour, skin senses, Pacinian corpuscles, paradoxical cold, congenital insensitivity to pain, nociceptors, gate control theory, substantia gelatinosa, neuromatrix theory, endorphins, phantom limb, kinesthesis, vestibular sense, semicircular canals, otolith organs, motion sickness, sensory interaction, McGurk effect, synaesthesia, sensory deprivation, Charles Bonnet syndrome.

## From the past papers

1. Optical illusions result from distortion in: (Amity 2023)
   (a) Translation (b) Sensation (c) Perception (d) Adaptation
   **Answer: (c) Perception.** In an illusion such as the Muller-Lyer figure, the receptors register the lines correctly; the error arises when the brain interprets them. Sensation (detection and transduction) is accurate; "translation" is not a stage of sensory processing, and adaptation is the fading of response to a constant stimulus.

2. According to the Bayesian approach to perception, our perceptual experiences are influenced by: (RML 2023)
   (a) Genetic predispositions (b) Cultural background (c) Prior beliefs and sensory evidence (d) Emotional states
   **Answer: (c) Prior beliefs and sensory evidence.** The Bayesian approach combines the prior (what we expect) with the likelihood (what the senses report), weighting each by its reliability. Genes, culture and emotion may shape a person's priors, but none of them alone defines the approach.

3. Who among the following has defined perception as a complex of sensory and imaginal events? (IHBAS 2024)
   (a) Kimbel (b) Titchener (c) Bruner (d) Schafer
   **Answer: (b) Titchener.** For the structuralist Titchener, a perception was sensations plus the images that past experience attaches to them. Bruner is known for the "New Look" emphasis on needs and expectations in perception (Chapter 4); the other names are distractors.

4. The minimum level of a stimulus that can be detected is called the: (RML 2025)
   (a) Difference threshold (b) Absolute threshold (c) Sensory adaptation point (d) Perceptual boundary
   **Answer: (b) Absolute threshold.** The absolute threshold is the weakest stimulus detected on 50 percent of trials. The difference threshold (JND) is the smallest detectable *change*; "sensory adaptation point" and "perceptual boundary" are invented terms.

5. The point at which a stimulus becomes strong enough to trigger a response only 50% of the time is: (Panwar Systems & Theories bank)
   (a) Threshold of response (b) Response bias (c) Signal detection of response (d) Phi phenomenon
   **Answer: (a) Threshold of response.** Thresholds are defined as 50 percent points because detection rises gradually along the psychometric function. Response bias is the signal detection term for an observer's readiness to say "yes"; the phi phenomenon is apparent movement between flashing lights (Wertheimer, 1912).

6. Every Just Noticeable Difference in stimulation constitutes a constant increment in sensation magnitude. Who among the following had given this assumption? (Panwar Systems & Theories bank)
   (a) Ebbinghaus (b) Weber (c) Tolman (d) Fechner
   **Answer: (d) Fechner.** Weber discovered that the JND is a constant fraction of the standard; Fechner added the assumption that every JND is an equal step of sensation and so derived his logarithmic law. Ebbinghaus applied Fechner's methods to memory, and Tolman was a learning theorist.

7. In signal detection theory, the ability to distinguish a stimulus from background noise is: (IHBAS 2025)

   (a) Sensitivity (d′) (b) Criterion (β) (c) ROC curve (d) JND

   **Answer: (a) Sensitivity (d′).** d′ is the separation between the noise and signal-plus-noise distributions and does not depend on willingness to say "yes". The criterion (β) is that willingness (response bias); the ROC curve is the plot of hit rate against false alarm rate; the JND belongs to classical psychophysics.

8. "Receiver-operant characteristics" in communication studies refer to: (IHBAS 2025)

   (a) Physiological thresholds of auditory perception (b) Reinforcement controlling message recall (c) Cognitive biases in message interpretation (d) Factors in the audience influencing persuasion

   **Answer: (d) Factors in the audience influencing persuasion.** In communication research the receiver is the audience, and receiver characteristics such as self-esteem, involvement and initial attitude moderate persuasion (Social Psychology book). The wording echoes the receiver *operating* characteristic (ROC) curve of signal detection theory, which is a different idea; option (a) is bait for that confusion.

9. The visible spectrum wavelength ranges from: (IHBAS 2020)

   (a) 380 to 760 nanometres (b) 200 to 700 nanometres (c) 400 to 1000 nanometres (d) 100 to 1000 nanometres

   **Answer: (a) 380 to 760 nanometres.** Shorter wavelengths are ultraviolet and longer ones infrared, both invisible to humans. Some books round the range to 400 to 700 nanometres, but no option offers that; the others include ultraviolet or infrared.

10. What is the relationship between frequency and wavelength? (RML 2024)

    (a) Direct (b) Collateral (c) Bilateral (d) Inverse

    **Answer: (d) Inverse.** Speed equals frequency multiplied by wavelength, so for light or sound travelling at a fixed speed, a higher frequency means a shorter wavelength. "Collateral" and "bilateral" are not mathematical relationships.

11. Which of the following act as receptor(s) for scotopic vision? (RML 2024)

    (a) Retina (b) Cones (c) Iris (d) Rods

    **Answer: (d) Rods.** Scotopic (dim-light) vision depends on the rods and their pigment rhodopsin. Cones serve photopic (daylight, colour) vision; the retina is the tissue that contains both receptor types, and the iris controls the pupil.

12. Which statement is INCORRECT? (Amity 2023)

    (a) Cones are most prevalent in the central retina (b) Cones are sensitive to moderate to high levels of light (c) Cones provide only monochromatic information (d) Cones provide excellent acuity

    **Answer: (c).** Cones provide colour vision through three pigments; it is the rods, with one pigment, that give colourless (monochromatic) vision. The other three statements are true of cones.

13. Dark adaptation is the process in which the eyes get more sensitive to: (RML 2024)

    (a) Green light (b) Coloured lights (c) Light with bright illumination (d) Light with low illumination

    **Answer: (d) Light with low illumination.** In darkness, regeneration of rhodopsin makes the rods thousands of times more sensitive over about 30 minutes. Becoming adjusted to bright light is light adaptation; dark-adapted rods are colour-blind, so "coloured lights" is wrong, and although rods are most sensitive to blue-green, "green light" misses the point.

14. The process of getting adjusted to different intensities of light is called: (RML 2024)

    (a) Visual adaptation (b) Visual assimilation (c) Visual accommodation (d) Visual sensation

    **Answer: (a) Visual adaptation.** Visual adaptation (dark and light adaptation) adjusts sensitivity to the light level. Accommodation is the lens changing shape to focus; assimilation belongs to Piaget's theory of cognitive development.

15. The Mach bands phenomenon is best explained by: (IHBAS 2025)

    (a) Rod-cone dichotomy in the fovea (b) Opponent-process theory of color vision (c) Lateral inhibition in the retina (d) Temporal summation in photoreceptors

    **Answer: (c) Lateral inhibition in the retina.** Neighbouring retinal cells inhibit each other, exaggerating contrast at edges and producing illusory light and dark bands. The opponent-process theory explains colour afterimages, the fovea contains no rods at all, and temporal summation is the adding together of weak signals over time.

16. Herman von Helmholtz (1821-94): (Panwar Systems & Theories bank)

    (a) He was a proponent of the just noticeable difference (b) He gave the trichromatic theory of colour vision (c) He proposed the concept of the unconscious (d) He gave the law of effect

    **Answer: (b) He gave the trichromatic theory of colour vision.** Helmholtz developed Young's three-receptor idea (the Young-Helmholtz theory) and also proposed the place theory of hearing and unconscious inference. The JND belongs to Weber and Fechner, the dynamic unconscious to Freud, and the law of effect to Thorndike.

17. Papillae is responsible for which sensory function? (RML 2024)

    (a) Smell (b) Touch (c) Taste (d) Vision

    **Answer: (c) Taste.** Papillae are the bumps on the tongue that house the taste buds (fungiform, foliate and circumvallate papillae; filiform papillae lack taste buds and serve touch). Smell receptors lie in the olfactory epithelium of the nose.

18. Narendra turns around and around in a circle. When he stops, he feels like his head is spinning. What is responsible for this sensation? (RML 2024)

    (a) Semi circular canals (b) Proprioceptors (c) Otolith organs (d) Otolith crystals

    **Answer: (a) Semicircular canals.** When he stops, the fluid in the canals keeps moving and bends the hair cells, signalling rotation in the opposite direction. The otolith organs (and the crystals inside them) detect tilt and linear acceleration, and proprioceptors report the position of the limbs.

19. What is a key feature of effective learning strategies for individuals with learning difficulties? (RML 2025)

    (a) Relying solely on written instructions (b) Providing multisensory inputs and regular repetition (c) Avoiding any accommodations or modifications (d) Focusing only on rote memorization techniques

    **Answer: (b) Providing multisensory inputs and regular repetition.** Combining sight, sound, movement and touch strengthens learning, as the research on multisensory integration predicts, and repetition consolidates it. Relying on a single channel, refusing accommodations or using rote memorisation alone all ignore the learner's difficulty.

## High-yield sheet

*The facts this chapter's questions ask most, most-asked first. Last read before the exam.*

1 The **thalamus** relays every sense except smell to the cortex: vision through the lateral geniculate nucleus, hearing through the medial geniculate nucleus, touch and taste through the ventral posterior nuclei. Smell (olfactory nerve, cranial nerve I) runs from the olfactory bulb straight to the piriform cortex and amygdala. ASKED 4 TIMES

2 **Rods** (about 120 million, one pigment, rhodopsin) give scotopic (dim-light) vision and no colour; **cones** (about 6 to 7 million, three pigments) give photopic vision, colour and fine acuity. "Cones give only monochromatic information" is false. ASKED TWICE

3 **Dark adaptation** makes the eye more sensitive to light of low illumination over about 30 minutes (cones first, a rod-cone break at 7 to 10 minutes, then rods as rhodopsin regenerates). Adjusting to different light intensities is **visual adaptation**; accommodation is the lens focusing. ASKED TWICE

4 Cones are most concentrated in the **fovea centralis**, which has no rods and gives the sharpest colour vision; the optic disc has no receptors at all (the blind spot); the peripheral retina is rod territory. ASKED 3 TIMES

5 The **absolute threshold** is the minimum stimulus intensity detected on 50 percent of trials; the difference threshold (just noticeable difference, JND) is the smallest change detected 50 percent of the time. ASKED TWICE

6 The three **semicircular canals** detect rotational (angular) acceleration of the head in three planes, so the head seems to spin after you stop turning; the otolith organs (utricle and saccule) detect head tilt and linear acceleration. ASKED TWICE

7 In signal detection theory (Green and Swets, 1966), **sensitivity (d′)** is the ability to tell a signal from background noise; it is highest when hits clearly exceed false alarms (d′ = z of the hit rate minus z of the false alarm rate). ASKED TWICE

8 **Edward Bradford Titchener** (structuralism) sought a "periodic table" of the elements of sensation and defined perception as a complex of sensory and imaginal elements. ASKED TWICE

9 **Gustav Theodor Fechner** founded psychophysics with *Elements of Psychophysics* (1860); he assumed that every JND adds an equal unit of sensation and derived S = k log I. ASKED TWICE

10 **Place theory** (Helmholtz, 1863; Georg von Bekesy's travelling wave, Nobel Prize 1961): each frequency peaks at its own place on the basilar membrane, high at the base and low at the apex, and it best explains high pitches. The frequency theory of low pitches is William (not Ernest) Rutherford's (1886). ASKED TWICE

11 **Charles Bonnet syndrome** is complex visual hallucinations in cognitively intact people with significant visual loss who keep their insight (described by Charles Bonnet in 1760). ASKED TWICE

12 The visible spectrum runs from about **380 to 760 nanometres**. ASKED ONCE

13 Frequency and wavelength are **inversely** related (speed = frequency × wavelength), for light and for sound. ASKED ONCE

14 **Papillae** on the tongue serve taste: fungiform, foliate and circumvallate papillae house the taste buds; filiform papillae have none. ASKED ONCE

15 Optical illusions are distortions of **perception**, not of sensation: the receptors register the stimulus correctly. ASKED ONCE

16 The **Bayesian approach** to perception: every percept combines prior beliefs with sensory evidence. ASKED ONCE

17 **Mach bands** (Ernst Mach, 1865) are explained by lateral inhibition in the retina, which exaggerates contrast at edges. ASKED ONCE

18 "Receiver characteristics" in communication studies are audience factors that influence persuasion; the **receiver operating characteristic (ROC)** of signal detection theory plots hit rate against false alarm rate. ASKED ONCE

19 Effective teaching for learning difficulties uses **multisensory input** (see, say, hear, trace) with regular repetition. ASKED ONCE

20 The decision parameter that signal detection theory added to classical threshold theory is the **response criterion** (β or c): liberal observers say "yes" readily, conservative observers rarely. ASKED ONCE

21 **Weber's law** (Ernst Heinrich Weber, 1834): the JND is a constant fraction of the standard stimulus, $\Delta I/I = k$, not a fixed amount. ASKED ONCE

22 **Hermann von Helmholtz** (1821 to 1894) gave the trichromatic theory (with Thomas Young, 1802; Helmholtz 1852), the place theory of hearing and unconscious inference; not the JND, the unconscious or the law of effect. ASKED ONCE

23 **Saccades** and microsaccades, tiny rapid eye movements, keep the image shifting on the retina and so ensure constant perception; a stabilised image fades. ASKED ONCE

24 The route of sound: pinna, auditory canal, **tympanic membrane**, ossicles (malleus, incus, stapes), oval window, cochlea, organ of Corti. ASKED ONCE

25 Cochlear **hair cells** transduce by direct mechanical bending of their stereocilia, which opens cation channels; potassium flows in and the cell depolarises and releases glutamate. ASKED ONCE

26 The **anterolateral (spinothalamic) system** carries pain (nociceptive) and temperature information; the dorsal column-medial lemniscus system carries fine touch, vibration and proprioception. ASKED ONCE

27 **Phantom limb** sensation (term coined by Silas Weir Mitchell, 1871) is best explained by cortical reorganisation in the somatosensory cortex (V. S. Ramachandran). ASKED ONCE

28 The **all-or-none law**: once threshold is reached, an action potential fires at full, constant amplitude; stimulus intensity is coded by firing rate. ASKED ONCE

29 **John Dewey** criticised the reflex arc, the connection between sensory stimulus and motor response, in 1896. ASKED ONCE

30 The **neurobiological approach** holds that behaviour is governed by the brain and nervous system. ASKED ONCE

CHAPTER 4

# Perception: Organising and Interpreting the World

IN THIS CHAPTER

4.1 ●● What Perception Is: distinguish bottom-up from top-down processing and compare the constructivist, Bayesian and direct (ecological) theories, and Neisser's perceptual cycle.

4.2 ●● Perceptual Organisation: The Gestalt Laws: explain figure-ground organisation and every Gestalt grouping law, with Prägnanz as the master law.

4.3 ●●● Depth and Distance Perception: classify every binocular, monocular, motion and oculomotor depth cue and evaluate the visual cliff studies.

4.4 ● Perceptual Constancy: explain size, shape, brightness, colour and location constancy, size-distance invariance and Emmert's law.

4.5 ●●● Illusions: describe the classic illusions and their explanations, and separate illusions from hallucinations and delusions.

4.6 ● Motion Perception: distinguish real, apparent, induced and autokinetic motion and the motion aftereffect.

4.7 ● Pattern and Object Recognition: compare template, feature, prototype, recognition-by-components and computational theories, and explain the word superiority effect and prosopagnosia.

4.8 ● Perceptual Set, Learning and Culture: show how expectation, need, emotion, experience and culture change what we perceive.

4.9 ● Perception Without Awareness, and Extrasensory Perception: evaluate subliminal perception and the evidence for extrasensory perception.

Weightage. 37 of the 819 questions mapped in this book (4.5%) test this chapter, 25 of them from real past papers. 2 of its 9 sections are High yield.

In 1958 a fifty-two-year-old Englishman, known in the scientific literature only by his initials, S.B., had an operation that gave him back his sight. He had been effectively blind since he was about ten months old. He had learned to get about with confidence and to use tools, and he lived a full, independent adult life by touch and hearing. Then a surgeon replaced the clouded cornea of one eye with a graft, the bandages came off, and light from the world reached a working retina for the first time in half a century.

What did he see? Not, as almost everyone expected, the world suddenly laid out before him. He heard the surgeon's voice in front of him, turned towards it, and saw a blur. Only because the voice came from it did he decide that the blur must be a face. The psychologist Richard Gregory and his colleague Jean Wallace, who studied him over the following months, recorded a string of puzzles. S.B. could read the time from a large clock almost at once, because as a blind man he had told the time by feeling the hands of a watch with no glass. He could read capital letters, which he had learned by touch at school, but not lower-case ones, which he had never felt. Looking down from his hospital window, which was several storeys up, he thought he could touch the ground with his feet if he hung from the sill by his hands.

The most famous episode came at the Science Museum in London. Gregory and Wallace showed S.B. a lathe in a glass case, a machine he had always wanted to understand. He could make nothing of it. He was then allowed to run his hands over it with his eyes shut. When he opened them again he said, "Now that I've felt it I can see."

Think about what that sentence means. The light reaching S.B.'s eyes did not change when he touched the lathe. What changed was what his brain could do with that light. This chapter is about that difference: the difference between light arriving at the eye and a person seeing a lathe, a face or a cliff edge. Keep S.B. in mind; you will meet him again at the end of the chapter.

## 4.1 What Perception Is: From Sensation to Interpretation

MEDIUM YIELD · 5 QS

Open your eyes and the world is simply there: a cup on the table, a friend across the room, the far wall at its proper distance. It feels as if nothing could be more direct. This section shows you why that feeling is misleading, and introduces the great debate about how perception actually works.

### Sensation, perception and the inverse problem

Chapter 3 followed physical energy as far as the sense organs: light focused on the retina, sound waves moving the basilar membrane, and receptors turning that energy into neural signals. That is **sensation**, the detection of physical energy by the sense organs and its conversion (transduction) into neural activity. **Perception** is what the brain then does with those signals: it selects, organises and interprets them so that they become meaningful objects and events. Sensation gives you a pattern of light; perception gives you your mother's face.

Why should perception be hard? Because the information at the eye is poorer than the world that produced it. The retina is a curved, nearly flat sheet, so the image on it has only two dimensions, yet you see a three-dimensional world. A small coin held close and a large plate held far away can throw exactly the same-sized image on the retina. A white page in shadow can send less light to your eye than a black page in sunshine. In each case many different states of the world could have produced the same retinal image. Working backwards from the image to the world that caused it is called the **inverse problem**, and every theory of perception is, at heart, a proposal about how the brain solves it.

### Bottom-up and top-down processing

Two kinds of processing help. **Bottom-up processing** (also called data-driven processing) starts with the stimulus: receptors register features such as edges, colours and movements, and these are built up step by step into more complex descriptions. Information flows in one direction, from the senses upwards. **Top-down processing** (conceptually driven processing) runs the other way: knowledge, expectations and context, stored in the brain, shape how the incoming data are interpreted.

You can feel both at work when you read a doctor's handwriting on a prescription. Some letters are clear enough to be read from their shapes alone (bottom-up). Others are scrawls that you can read only because you know what word is likely to come next, for example that "Paracet..." is probably followed by "amol" (top-down). Almost all real perception mixes the two. The theories that follow differ in how much weight they give to each.

### Perception as inference: Helmholtz, Gregory and the Bayesian brain ●●●

The oldest answer to the inverse problem came from **Hermann von Helmholtz** (1821 to 1894), the German physicist and physiologist who measured the speed of the nerve impulse and wrote the great *Handbook of Physiological Optics* (completed in 1867). Helmholtz argued that perception is the end point of **unconscious inference**: the brain draws rapid, automatic conclusions about the world from the sensory evidence plus past experience, much as a scientist draws conclusions from data, except that we are not aware of doing it. His **likelihood principle** states that we perceive the object that is most likely to have produced the pattern of stimulation we receive. A small retinal image of a person is perceived as a normal-sized person far away, not as a tiny person nearby, because that is by far the likelier cause.

A century later the British psychologist **Richard Langton Gregory** (1923 to 2010) turned this idea into the modern **constructivist approach** (also called the indirect or top-down approach) in books such as *Eye and Brain* (1966) and *The Intelligent Eye* (1970). For Gregory, a perception is a **perceptual hypothesis**: a best guess about what is out there, constructed from fragmentary sensory data and stored knowledge, and tested against further input. The senses supply evidence; the brain supplies the theory.

What is the evidence for this view? Gregory's favourite was the **hollow-face illusion**. Look at the inside of a hollow mask from a metre or so away and you see a normal face bulging towards you, not a hollow one. Your knowledge that faces stick out is so strong that it overrides the depth information from your two eyes. Illusions (Section 4.5), ambiguous figures that flip between two interpretations, and the effects of expectation (Section 4.8) all fit the idea that perception is an active construction that can be wrong.

Critics raised two problems. First, if perception is guesswork from poor data, why is it usually so accurate? Second, most illusions persist even when you know the truth; if perception were a hypothesis open to correction by knowledge, knowing that the lines of the Müller-Lyer figure are equal ought to make them look equal, and it does not.

The modern descendant of Helmholtz's idea is the **Bayesian approach** to perception, named after the English clergyman and mathematician **Thomas Bayes** (about 1701 to 1761), whose theorem shows how to update a belief in the light of new evidence. On this view the brain combines two things: **prior** beliefs (how likely each possible state of the world is, learned from experience or built in by evolution) and the **likelihood** (how well each possibility fits the current sensory evidence). The result, the posterior, is the percept. When the sensory evidence is clear it dominates; when it is weak or ambiguous the priors take over. The Indian-born neuroscientist **Vilayanur S. Ramachandran** (born 1951) gave a neat demonstration in 1988. Shaded discs that are lighter at the top look like bumps and discs that are lighter at the bottom look like dents, and turning the page upside down swaps them. The brain is applying a prior that light comes from above, because for the whole of our evolutionary history it has. The hollow face is the same story: the prior that faces are convex outweighs weak evidence that this one is concave.

> EXAM TRAP
>
> The approach that "emphasises unconscious inference and probabilistic reasoning" is the **constructivist** approach (Helmholtz, Gregory); its Bayesian form adds that the percept combines **prior beliefs and sensory evidence**. Do not pick "genetic predispositions", "cultural background" or "emotional states" for the Bayesian question: they can shape priors, but the definition is priors plus evidence. And do not confuse Helmholtz's unconscious inference with Wundt's apperception: the "apperceptive mass" (a term from Johann Friedrich Herbart that Wilhelm Wundt took over) is the store of existing ideas into which a new perception is absorbed (Chapter 1).

### Direct perception: Gibson's ecological approach ●●

The fiercest critic of the inference view was the American psychologist **James Jerome Gibson** (1904 to 1979). During the Second World War Gibson worked on how to select and train pilots, and he noticed that the cues listed in textbooks, cues taken from flat pictures viewed by a motionless observer, were of little use in explaining how a pilot lands an aircraft. A pilot is moving, and movement creates information. Over three books, *The Perception of the Visual World* (1950), *The Senses Considered as Perceptual Systems* (1966) and *The Ecological Approach to Visual Perception* (1979), he built the theory of **direct perception**, also called the **ecological approach** because it studies perception in the animal's real environment rather than in the laboratory.

Gibson's central claim is that the light reaching a moving observer is not poor at all. The **optic array**, the structured pattern of light arriving at the eye from every direction, is rich in information that specifies the layout of the world without any need for inference. Three of his ideas are examined often.

- **Invariants** are properties of the optic array that stay constant as the observer moves and so specify the world directly. The texture gradient (Section 4.3) is one: the elements of a surface get smaller and denser with distance in a lawful way. The horizon ratio is another:

for objects standing on the same ground, the horizon cuts them at the same proportion of their height whatever their distance, so objects of equal height are seen as equal.

- **Optic flow** is the pattern of movement in the optic array as you move through the world. When you walk or drive forward, the scene streams outwards from a single point, the **focus of expansion**, which marks exactly where you are heading. A pilot landing an aircraft keeps the focus of expansion on the start of the runway. No inference is needed; the information is in the light.
- **Affordances**, a word Gibson coined in the 1960s and developed in his 1979 book, are the possibilities for action that an object or surface offers to a particular animal. A chair affords sitting, a handle affords grasping, a ledge at knee height affords climbing for an adult but not for a toddler. Gibson claimed that we perceive affordances directly, so that meaning is picked up from the environment rather than added by the brain.

Gibson described the perceiver as "resonating" to this information, as a radio resonates to a broadcast. Because perception is driven by the information in the stimulus and not by stored knowledge, Gibson's theory is the classic example of a bottom-up theory.

His approach was a healthy corrective: it put the moving, acting animal back at the centre of perception and inspired work on how we steer, catch and avoid collisions. But it has weaknesses. It has little to say about illusions, which by definition are cases where the information misleads us. It explains poorly how we recognise objects whose meaning depends on culture and learning (a letter box affords posting a letter only to someone who knows what a letter box is). And "resonance" names the process without explaining it.

> **INDIAN CONTEXT**
>
> Watch a fielder at long-on under a skied hit at a cricket match in Bengaluru. He does not calculate the ball's trajectory. Studies of fielders running to catch high balls suggest that they run so as to keep the image of the ball rising smoothly in their field of view; if it starts to accelerate upwards they move back, if it starts to drop they move in. This is Gibsonian perception at its best: action guided directly by a property of the changing optic array.

### Bringing the two together: Neisser's perceptual cycle and two visual systems

Must we choose between Gregory and Gibson? **Ulric Neisser** (1928 to 2012), whose *Cognitive Psychology* (1967) gave the new field its name, argued in his later book *Cognition and Reality* (1976) that we need not. His **perceptual cycle** joins both directions of processing into a loop (Figure 4.1). A **schema**, an organised body of knowledge about what to expect, *directs* exploration: where you look, what you listen for, what you reach out to touch. Exploration *samples* the information actually available in the object or scene. The information sampled then *modifies* the schema, which directs the next round of exploration. Perception is not a single event but a continuous cycle in which top-down expectation and bottom-up information keep correcting each other.

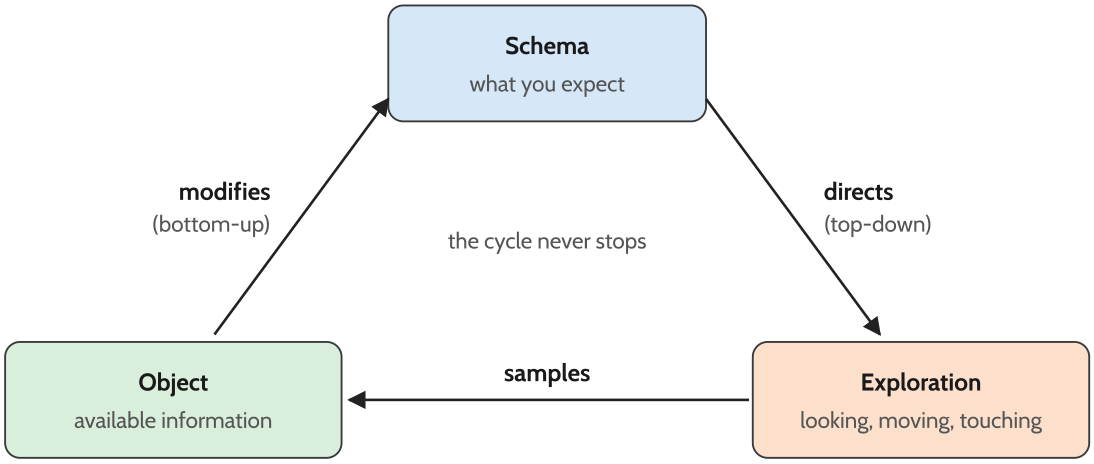


Figure 4.1 Neisser's perceptual cycle (1976). Notice that top-down and bottom-up processing are not rivals here: each arrow feeds the next, so expectation and information correct each other continuously.

A second reconciliation came from the brain. In 1995 **David Milner** and **Melvyn Goodale** proposed that the visual system has two streams with different jobs. A ventral stream, running into the temporal lobe, serves **vision-for-perception**: recognising what objects are, which needs stored knowledge, just as Gregory said. A dorsal stream, running into the parietal lobe, serves **vision-for-action**: guiding the hand and body in real time, fast and largely unconscious, very much as Gibson described. Their patient D.F., whose ventral stream had been damaged by carbon monoxide poisoning, could not say whether a slot in a disc was horizontal or vertical, yet could turn her hand correctly and post a card through it. The two classic theories may simply be describing two different visual systems (the anatomy is covered in the Physiological book).

Two further labels appear as options in examination questions. The **information-processing approach** treats perception as a series of stages, each transforming the output of the one before, much as a computer program does. The **computational approach**, associated above all with David Marr, asks what problem each stage of vision has to solve and what calculations would solve it; you will meet it in Section 4.7. Table 4.1 sets all these approaches side by side.

Table 4.1 The main approaches to perception.

| Approach | Key names | Direction | Core claim | Signature ideas |
|---|---|---|---|---|
| Constructivist (indirect) | Helmholtz (1867), Gregory (1966, 1970) | Top-down | Perception is inference from poor data plus knowledge | Unconscious inference, likelihood principle, perceptual hypotheses, hollow face |
| Bayesian | Bayes; Ramachandran (1988) | Top-down and bottom-up | Percept = prior beliefs combined with sensory evidence | Priors, likelihood, light-from-above prior |
| Direct (ecological) | J. J. Gibson (1950, 1966, 1979) | Bottom-up | The optic array specifies the world; no inference needed | Optic array, invariants, optic flow, affordances |

| Approach | Key names | Direction | Core claim | Signature ideas |
|---|---|---|---|---|
| Perceptual cycle | Neisser (1976) | Both, in a loop | Schemas direct exploration; information modifies schemas | Schema, exploration, object |
| Computational | Marr (1982) | Mainly bottom-up | Vision computes a series of representations | Primal sketch, 2.5-D sketch, 3-D model |

EXAM TRAP

"Who proposed the bottom-up theory of perception?" The answer is **James J. Gibson**; Gregory and Helmholtz are the top-down (constructivist) theorists, and Wertheimer is a Gestalt psychologist. "Affordances" is always Gibson. Do not confuse **James J. Gibson** (direct perception, affordances) with his wife **Eleanor J. Gibson** (visual cliff, perceptual learning): they are two different, equally examinable people. Neisser wrote *Cognitive Psychology* in 1967 but proposed the perceptual cycle in *Cognition and Reality* in 1976.

CLINICAL LINK

Several studies have found that people with schizophrenia are much less fooled by the hollow-face illusion than other people: they are more likely to see the mask as hollow, as it really is. One interpretation is that the balance between top-down expectation and bottom-up evidence is altered in the disorder. It is a reminder that "seeing correctly" in a laboratory task is not always a sign of a healthy brain.

## 4.2 Perceptual Organisation: The Gestalt Laws MEDIUM YIELD · 8 QS

Before you can recognise an object you must first decide which bits of the visual field belong together and which belong to something else. That sounds trivial until you try to explain how it is done. The school that took this problem most seriously, and that has contributed more to the study of perception than any other, was Gestalt psychology.

### The Gestalt starting point ●●

Gestalt psychology was founded in Germany in 1912, when **Max Wertheimer** (1880 to 1943) published his experiments on apparent movement, with **Kurt Koffka** (1886 to 1941) and **Wolfgang Köhler** (1887 to 1967) as his first subjects and lifelong collaborators (the history of the school is told in Chapter 2). The German word *Gestalt* means form, shape or configuration. The school's founding claim, aimed squarely at Wundt's and Titchener's attempt to break experience into elementary sensations, was that the whole is different from the sum of its parts. A melody is still the same melody when every note is changed by transposing it into another key; the "melodiness" lies in the relations between the notes, not in the notes themselves. In perception, said the Gestaltists, the brain spontaneously organises the visual field into wholes according to a small set of innate laws, and in 1923 Wertheimer set these out in a famous paper.

## Figure and ground •

The most basic act of organisation is to split the field into a **figure**, the object that stands out and is attended to, and a **ground**, the background against which it appears. The Danish psychologist **Edgar Rubin** (1886 to 1951) studied this **figure-ground** relationship in his 1915 doctoral work and drew the best-known demonstration of it, **Rubin's vase**: a white vase on a black background which, as you look, becomes two black faces in profile on a white background. Rubin described how figure and ground differ in experience:

- The figure has a thing-like quality (it looks like an object) and a definite shape; the ground is shapeless and looks like substance or empty space.
- The contour between them seems to belong to the figure, not the ground.
- The figure appears nearer and in front; the ground appears farther away, diffuse, and seems to continue behind the figure.
- The figure is more vivid, more easily named and better remembered.

Certain regions are more likely to become figure: smaller regions, enclosed regions, symmetrical and convex regions, and regions in the lower part of the field. Figure-ground organisation is not only visual. At a noisy wedding reception, the voice of the person talking to you is an auditory figure and the din of the hall is ground.

The vase and faces is a **reversible figure**: either region can be seen as figure, but never both at once, and the percept flips back and forth. Notice what this means. The stimulus never changes, yet the perception does. Organisation is something the perceiver does, not something in the picture.

> EXAM TRAP
>
> "Which is **not** a characteristic of a figure?" The answer is "appears diffused and spatially distant", because that describes the **ground**. Smaller size, clear outlines and greater object character (thing-like quality) are all properties of the figure. Camouflage works by breaking figure-ground segregation, so that the animal's contour is lost in the ground.

## The laws of grouping •••

Once figures are separated from ground, elements must be grouped into larger units. The Gestalt **laws of perceptual organisation** describe how. Look at Figure 4.2 as you read; each panel lets you watch one law at work in your own perception.

By the **law of proximity**, elements that are close together in space or in time are seen as belonging together. In the first panel the dots are equally spaced vertically but not horizontally, so you see three pairs of columns rather than four rows. Proximity in time works the same way: the taps of a drum heard close together form a rhythm group. By the **law of similarity**, elements that look alike (in colour, shape, size or orientation) are grouped together; in the second panel equal spacing would favour columns, but similarity makes you see rows of filled and open dots. By the **law of closure**, we tend to complete figures that have gaps, so that the broken outline in the third panel is seen as a circle, not as four separate arcs. The **law of good continuation** (also called the law of continuity) holds that elements that follow a smooth line or curve are grouped together, so that the fourth panel shows a straight line crossing a curve rather than two odd angular shapes meeting at a point. By the **law of common fate**, elements

moving together in the same direction are grouped together: a flock of birds or a column of marching soldiers is seen as one unit, and in the sixth panel the arrows split the row into two sets.

The Gestaltists also described a **law of symmetry** (symmetrical regions are grouped together and tend to be seen as figure), and Wertheimer added a law of past experience or familiarity (elements that together form a familiar object tend to be grouped). Modern researchers have added two laws. The **law of common region**, proposed by **Stephen Palmer** in 1992, states that elements inside a common enclosed region are grouped together, even when proximity says otherwise (fifth panel, top row). The law of **connectedness** (element connectedness), described by Palmer and **Irvin Rock** in 1994, states that elements physically joined by lines are seen as a unit (fifth panel, bottom row). In the same 1994 paper Palmer and Rock proposed uniform connectedness: any connected region of uniform colour, texture or lightness is organised as a single perceptual unit. Both kinds of connectedness, and common region, are powerful enough to override proximity and similarity. Table 4.2 brings all the laws together.

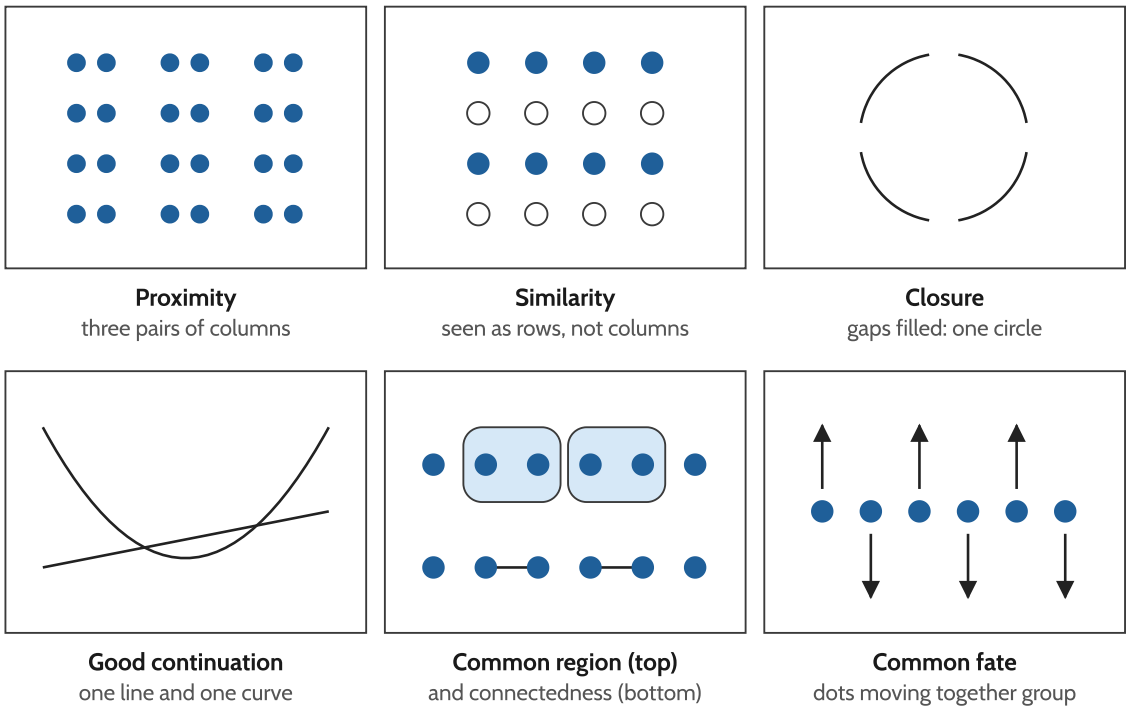


Figure 4.2 Six grouping laws you can watch in your own perception. In each panel the elements are physically identical or evenly spaced in some respect, yet you see definite groups; the grouping is supplied by your visual system.

Table 4.2 The Gestalt laws of perceptual organisation.

| Law | What it says | Everyday example |
|---|---|---|
| Figure-ground (Rubin, 1915) | The field splits into a figure in front and a ground behind | Words on this page; a friend's voice at a noisy party |
| Proximity | Elements close in space or time group together | Pairs of dots; seat rows in a cinema hall |
| Similarity | Similar elements group together | Team jerseys on a cricket field |

| Law | What it says | Everyday example |
|---|---|---|
| Closure | Gaps are filled to make complete figures | A circle drawn with breaks; logos with missing strokes |
| Good continuation (continuity) | Elements on a smooth path group together | Crossing wires seen as two smooth lines |
| Common fate | Elements moving together group together | A flock of birds; a marching contingent |
| Symmetry | Symmetrical regions group and become figure | A kolam pattern seen as one design |
| Common region (Palmer, 1992) | Elements in one enclosed region group | Items in one box on a form |
| Connectedness (Palmer and Rock, 1994) | Physically linked elements group | Stations joined on a metro map |
| Prägnanz (the master law) | We see the simplest, most stable organisation possible | Overlapping rings seen as circles, not odd shapes |

## Prägnanz, illusory contours and how good the laws are ●●

Above all the others stands the **law of Prägnanz**, from the German for conciseness or pithiness, also called the law of good figure, the law of good form or the law of simplicity. It states that every stimulus pattern is seen in such a way that the resulting structure is as simple, regular, stable and complete as the conditions allow. Two overlapping rings could in principle be seen as three oddly shaped pieces, but you see two circles because that is the simplest description. The other laws can be seen as special cases of this one, which is why Prägnanz is called the master law of Gestalt organisation and why a question about a figure "perceived not in parts but organised in the best possible whole form" is answered by Prägnanz.

The power of these laws shows most clearly when they create something that is not there. In 1955 the Italian psychologist **Gaetano Kanizsa** (1913 to 1993) drew three black discs, each with a wedge cut out, arranged so that the wedges face each other. Observers see a bright white triangle lying on top of the discs, complete with sharp edges across empty paper. These **illusory contours** (subjective contours) show closure and Prägnanz at work: the simplest interpretation of three cut discs is one triangle covering three whole discs. A related process, **amodal completion**, makes us perceive an object partly hidden behind another as continuing, complete, behind it.

How good are the Gestalt laws? They are excellent descriptions; every textbook of design, typography and advertising still uses them. But they are descriptions, not explanations. They do not say why the brain groups this way, and they are vague about what happens when two laws conflict. The Gestaltists' own physiological explanation, that perceptual organisation reflects electrical fields in the brain with the same shape as the percept (Köhler's isomorphism), was not supported by later experiments in which the cortex of animals was filled with metal strips or wires to disrupt any such fields without destroying perception. And research since then has shown that grouping is partly learned: what counts as "similar" or "familiar" depends on experience, as you will see in Section 4.8.

EXAM TRAP

The **tendency to complete a figure** is **closure**, not continuity or continuation (continuity is about smooth paths, not gaps). Items close together **in space and time** group by **proximity**. The figure "organised in the best possible whole form" is **Prägnanz**. The school that contributed most to perception is **Gestalt psychology**, not structuralism, behaviourism or psychoanalysis. And **Gestalt therapy** (Fritz Perls) has nothing to do with Gestalt psychology except the name.

EXAM TRAP

Do not confuse **interposition** (a depth cue: a nearer object blocks a farther one, Section 4.3) with **interpolation** (filling in between known points, as the visual system does when it completes a contour hidden behind an occluder) or **extrapolation** (extending a trend beyond the known points). One past paper also used **structuring** for working out a rule or organising principle from incomplete evidence and then picking out the items that satisfy it; that keyed answer fits the Gestalt idea that perception imposes structure on incomplete input.

CLINICAL LINK

In 1938 the American child psychiatrist Lauretta Bender built a test from nine of Wertheimer's grouping designs. The Bender Visual-Motor Gestalt Test asks the person to copy the designs; distortions, rotations and failures of integration can point to visuoconstructive or organic difficulties. The test itself is covered in the Psychological Assessment book.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Working backwards from a two-dimensional retinal image to the three-dimensional world that caused it is called the _____ problem.
2. Processing driven by knowledge, expectation and context is called _____ processing.
3. Helmholtz described perception as the product of _____ inference.
4. Gregory called a perception a perceptual _____.
5. In the Bayesian approach, the percept combines _____ beliefs with sensory evidence.
6. Gibson's term for the action possibilities that an object offers an animal is _____.
7. When you move forward, the scene streams outwards from the focus of _____.
8. In Neisser's perceptual cycle, the _____ directs exploration.
9. Rubin's vase is a _____ figure in which figure and ground swap.
10. The master law of Gestalt organisation, the tendency to see the simplest possible form, is the law of _____.

**Answers:** (1) inverse, (2) top-down, (3) unconscious, (4) hypothesis, (5) prior, (6) affordances, (7) expansion, (8) schema, (9) reversible, (10) Prägnanz.

## 4.3 Depth and Distance Perception HIGH YIELD · 9 QS

Reach out and pick up the cup in front of you. Your hand opens to the right width and stops at exactly the right distance, without a moment's thought. Yet the image of the cup on each retina has no depth at all. How does a flat image give you a world in three dimensions?

### The problem of the flat retina

The problem was posed clearly in 1709 by the Irish philosopher **George Berkeley** (1685 to 1753) in *An Essay Towards a New Theory of Vision*. Berkeley pointed out that a point at any distance along a line of sight falls on the same point of the retina, so distance cannot be "seen" directly. He concluded that we learn to judge distance by associating visual sensations, such as the strain of turning the eyes inwards, with the experience of reaching and walking. Modern research has replaced Berkeley's short list with a long list of **depth cues**, sources of information that correlate with distance and that the brain combines to give **depth perception**, the ability to see the world in three dimensions and to judge how far away things are. The cues fall into families according to what they need: two eyes, one eye, movement, or feedback from the eye's own muscles.

### Binocular cues: disparity and convergence ●●●

**Binocular cues** are depth cues that require both eyes working together. The most important is **retinal disparity** (also called binocular disparity). Your eyes are about 6 centimetres apart, so each receives a slightly different view of the world. Hold a finger about 20 centimetres in front of your nose and close each eye in turn: the finger jumps from side to side against the background. The difference between the positions of an object's images on the two retinas is its disparity, and the nearer the object, the larger the disparity. Objects at the fixation distance fall on corresponding points of the two retinas and have zero disparity; the imaginary curved surface through all such points is called the **horopter**. Objects nearer than the horopter have crossed disparity and objects beyond it have uncrossed disparity, so the sign of the disparity tells the brain whether an object is in front of or behind the point being looked at. The vivid impression of solid depth produced by fusing the two images is called **stereopsis**.

The English physicist **Charles Wheatstone** (1802 to 1875) proved the point in 1838 with his **stereoscope**, which presented a slightly different drawing to each eye; viewers saw a single solid object. Every 3-D film works the same way, using special glasses to send a different image to each eye. In 1960 the Hungarian-American engineer **Béla Julesz** (1928 to 2003) went further with random-dot stereograms: two fields of random dots, identical except that a central square of dots is shifted slightly in one of them. Each image alone looks like television snow. Viewed in a stereoscope, a square floats out in depth. Because there is no square to recognise in either image alone, Julesz showed that stereopsis can work before any object is recognised, a purely bottom-up process.

The second binocular cue is **convergence** (vergence): the inward turning of the two eyes to fixate a near object. Look at your fingertip as you bring it towards your nose and you can feel your eyes cross. The brain registers how much the eyes are turned in, from the commands sent to the eye muscles and feedback from them, and the greater the convergence, the nearer the object. Convergence is a binocular cue, because it needs two eyes, and also an oculomotor

cue, because it comes from the eye muscles; it is useful only for objects within a couple of metres.

Binocular cues can work only where the fields of the two eyes overlap. Each human eye sees roughly 160 degrees horizontally (about 60 degrees towards the nose and about 100 degrees towards the temple); the two together cover roughly 200 degrees, and the central overlap where both eyes see, the binocular field, is about 120 degrees wide. Predators such as cats and owls, and primates such as ourselves, have forward-facing eyes with a large overlap and good stereopsis; prey animals such as rabbits and deer have eyes on the sides of the head, a nearly all-round view, and little binocular overlap.

> EXAM TRAP
>
> A past paper asked for the "approximate visual field covered by each human eye" and keyed **180 degrees**. The better textbook figure for **one** eye is about **160 degrees** horizontally; 180 to 200 degrees is the field of **both** eyes together, and about 120 degrees is the binocular overlap. If you meet that exact item, be aware that the paper's key was 180; if the question clearly says "both eyes", 180 to 200 is right.

### Monocular cues: what a painter knows ●●●

**Monocular cues** are depth cues available to one eye alone. People who have lost an eye still judge depth well, and a flat painting or photograph can look deep, because of them. Most are **pictorial cues**, so called because artists since the Renaissance have used them to put depth on a flat canvas. Figure 4.3 shows the main ones in a single scene.

- **Relative size**: if two objects are known or assumed to be the same size, the one casting the smaller retinal image is seen as farther away. A row of identical lamp posts seems to shrink into the distance.
- **Familiar size**: when we know an object's real size, its retinal size tells us its distance. A tiny-looking autorickshaw is seen as a normal autorickshaw far away.
- **Interposition** (also called overlap or occlusion): when one object partly blocks the view of another, the blocked object is seen as farther away. Interposition gives only order (nearer or farther), not distance.
- **Linear perspective**: parallel lines, such as railway tracks or the edges of a road, appear to converge as they recede, meeting at a vanishing point on the horizon.
- **Texture gradient**: a surface with an even texture (pebbles, grass, the sleepers of a railway track, bricks in a courtyard) shows its elements as larger, coarser and more widely spaced nearby, and smaller, finer and more densely packed farther away. This was Gibson's favourite cue, because it specifies distance and the slant of a surface directly.
- **Aerial perspective** (atmospheric perspective): distant objects look hazier, less sharp, lower in contrast and often bluer, because light from them passes through more air, dust and moisture. On a clear day after the monsoon, the Nilgiri hills seem nearer than they do in hazy summer.
- **Height in the visual field** (relative height or elevation): for objects on the ground below the horizon, those higher in the visual field, that is, closer to the horizon, are seen as farther away; for objects above the horizon, such as clouds, the reverse holds.

- **Shading** (light and shadow): the pattern of light and shadow on an object reveals its three-dimensional shape, and shadows cast on the ground show where an object stands. As Section 4.1 showed, the brain assumes the light comes from above.
- Relative brightness: of two identical objects, the brighter is usually seen as nearer.

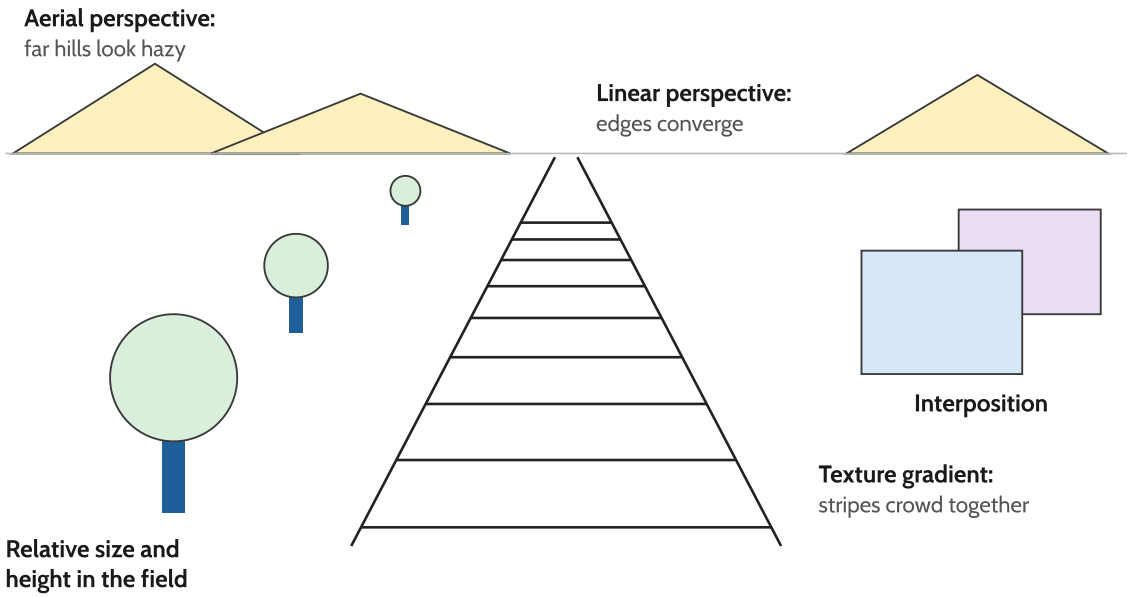


Figure 4.3 Pictorial depth cues in one scene. The road's edges converge (linear perspective), the cross stripes crowd together with distance (texture gradient), the three identical trees shrink and rise towards the horizon (relative size and height in the field), the near house hides part of the far one (interposition), and the distant hills are pale (aerial perspective).

Road engineers use these cues deliberately. Stripes painted across the approach to a toll plaza or an exit ramp, with the gaps between them getting smaller, create an exaggerated texture gradient and a heightened sense of speed, so that drivers slow down without being told.

## Cues from movement and from the eye's own muscles

Gibson's complaint that textbook cues were taken from frozen pictures applies with special force to the cues that need movement. The most powerful is **motion parallax**: when you move, objects at different distances move across your retina at different speeds. Look out of the side window of a moving train. The electric poles beside the track flash past in the direction opposite to your travel, the trees beyond move more slowly, the distant hills crawl, and the moon seems to travel along with you. The relative speed of the images tells you their relative distance. Motion parallax is a monocular cue (one eye is enough) but not a pictorial one, because a still picture cannot show it. A related cue is deletion and accretion: as you move, a nearer surface progressively covers (deletes) and uncovers (accretes) the texture of a farther one, which is interposition in motion.

**Accommodation** is the change in the shape of the lens, controlled by the ciliary muscles, that focuses the image of near or far objects on the retina (Chapter 3). The brain can use the state of these muscles as a rough distance cue for objects within about two metres. Accommodation, like convergence, is an **oculomotor cue**, but unlike convergence it works

with one eye. Table 4.3 classifies every cue in this section by the number of eyes it needs and the family it belongs to.

Table 4.3 Depth cues classified.

| Cue | Eyes needed | Family | How it signals depth |
|---|---|---|---|
| Retinal (binocular) disparity | Two | Binocular | Different images in the two eyes; more disparity = nearer |
| Convergence | Two | Binocular, oculomotor | More inward turning of the eyes = nearer |
| Accommodation | One | Monocular, oculomotor | Lens thickens for near objects |
| Motion parallax | One | Monocular, motion-based | Near objects move faster across the retina |
| Interposition (overlap) | One | Monocular, pictorial | Blocked object is farther |
| Relative size and familiar size | One | Monocular, pictorial | Smaller image = farther |
| Linear perspective | One | Monocular, pictorial | Parallel lines converge with distance |
| Texture gradient | One | Monocular, pictorial | Texture becomes finer and denser with distance |
| Aerial perspective | One | Monocular, pictorial | Distant objects are hazy and bluish |
| Height in the visual field | One | Monocular, pictorial | Closer to the horizon = farther |
| Shading | One | Monocular, pictorial | Light and shadow reveal shape |

EXAM TRAP

The binocular cues are only **retinal disparity** and **convergence**. Everything else, including accommodation and motion parallax, is monocular. So "a monocular cue" from convergence, retinal disparity, assimilation and interposition is **interposition**; "requires both eyes" from motion parallax, aerial perspective, convergence and accommodation is **convergence**; and retinal disparity is a **binocular** cue, not a pictorial, static monocular or motion parallax cue. **Assimilation** is not a depth cue at all: in Piaget's theory it means fitting new experience into an existing schema (Developmental book), and in perception it names the opposite of contrast, when a colour or grey looks more like its surroundings. Increasing density of stones, pebbles or painted stripes with distance is **texture gradient**, not aerial perspective (haze) or interposition (overlap).

## Is depth perception innate? The visual cliff ●●

Berkeley thought depth perception had to be learned. Is he right? In the 1950s **Eleanor Jack Gibson** (1910 to 2002) and **Richard Walk** (1920 to 1999), at Cornell University, built an apparatus to find out.

KEY STUDY: THE VISUAL CLIFF (GIBSON AND WALK, 1960)

**Question.** Can infants and young animals perceive depth before they have had a chance to learn it through falls and bumps? **Method.** The **visual cliff** is a large table topped by a sheet of strong glass. Under one half a checkerboard pattern is fixed directly beneath the glass (the shallow side); under the other half the same pattern lies far below the glass (the deep side), so that the table appears to end in a sheer drop, although the glass makes it perfectly safe. A narrow centre board separates the two sides. Thirty-six infants aged 6 to 14 months were placed on the centre board and their mothers called them from the shallow side and from the deep side in turn. **Results.** Twenty-seven infants moved off the board. All of them crawled onto the shallow side at least once, but only three crept, hesitantly, onto the deep side. Many crawled away from their mothers when called from the deep side, or cried because they could not reach them, and some patted the glass yet still refused to cross. Newly hatched chicks, young goats and lambs placed on the apparatus within a day of birth also avoided the deep side. **Conclusion.** By the time human infants can crawl, they can perceive depth, and in species that walk from birth, depth perception is present from birth. **Criticism.** Human infants cannot be tested until they crawl, at around six months, by which time they have had months of visual experience, so the study cannot show that depth perception in humans is innate.

Later work divided the question. When **Joseph Campos** and his colleagues in 1970 lowered infants of about two to three and a half months directly onto the deep side, their heart rate slowed, the sign of interest and attention, not the speeding up that signals fear. Young infants therefore detect the difference in depth long before they fear it. Fear of the drop appears with experience of moving about on one's own: infants of the same age who have begun crawling, or who have been given experience in baby walkers, show fear on the deep side sooner than those who have not. The most balanced conclusion is that the ability to detect depth appears very early and may well be largely built in, while the meaning of a drop, the knowledge that it is dangerous, is learned through self-produced movement.

EXAM TRAP

The visual cliff was devised by **Eleanor Gibson and Richard Walk** (1960) to study **depth perception** and whether it is innate. It did **not** study object permanence (Piaget), colour perception, imprinting (Lorenz) or language. The infants were **6 to 14 months** old (crawling age), not newborns and not "3 to 6 months", as some revision notes say. The best summary of the human results is "depth perception is present by the time infants can crawl", not "fully developed at birth" and not "only after age two".

CLINICAL LINK

Children whose eyes are misaligned (strabismus) or who have a lazy eye (amblyopia) in early childhood often grow up stereoblind, because the brain never learns to combine the two eyes' images. The neurobiologist **Susan Barry**, cross-eyed from infancy, gained stereoscopic vision in her late forties after intensive vision therapy, a case made famous by the neurologist **Oliver Sacks** (1933 to 2015). Her experience, of suddenly seeing snowflakes floating in depth, shows both how much the brain depends on early experience and how much plasticity remains in adulthood.

## 4.4 Perceptual Constancy

LOWER YIELD · 3 QS

As a friend walks away from you down a corridor, the image of her on your retina shrinks steadily; at twice the distance it is half the height. Yet she does not seem to shrink. When a door swings open, the image it casts changes from a rectangle to a narrower and narrower trapezoid; yet you see a rectangular door turning. This stability is called **perceptual constancy**: the tendency to perceive objects as having stable, unchanging properties (size, shape, lightness, colour, position) despite large changes in the sensory input they produce. Without constancy the world would be a chaos of swelling, shrinking, stretching, darkening shapes.

### Size constancy and size-distance invariance ●●

**Size constancy** is the perception of an object as remaining the same size even though the size of its retinal image changes with distance. Try this. Hold one hand at arm's length and the other halfway to your face. The retinal image of the far hand is only about half the size of the near one, yet both look like normal hands of the same size.

How is it done? The brain takes account of distance. The **size-distance invariance hypothesis** states that perceived size is proportional to retinal image size multiplied by perceived distance. If the retinal image halves while the perceived distance doubles, perceived size stays the same. It follows that size constancy should depend on depth cues, and should break down when they are removed.

KEY STUDY: HOLWAY AND BORING (1941)

**Question.** Does size constancy depend on information about distance? **Method.** **Alfred Holway** and **Edwin Garrigues Boring** (1886 to 1968) seated observers at the junction of two long corridors. Down one corridor, at distances from 10 to 120 feet, they placed test discs whose physical size was increased with distance so that every disc cast exactly the same retinal image (one degree of visual angle). In the other corridor was a comparison disc whose size the observer adjusted until it matched the test disc. Observers did this under four conditions: with both eyes, with one eye, with one eye through a small peephole, and finally through a peephole and a long black tunnel that hid almost every cue to distance. **Results.** With both eyes and full cues, observers matched the discs' physical sizes almost perfectly: size constancy was excellent. As cues were removed, their settings moved steadily towards the retinal size; with the peephole and tunnel, all the discs looked nearly the same size. **Conclusion.** Size constancy depends on information about distance, just as the size-distance invariance hypothesis predicts.

### Emmert's law

You can test size-distance invariance on yourself with an afterimage. Stare for about thirty seconds at a bright lamp or a coloured square, then look at a sheet of paper in your hand, and then at a distant wall. The afterimage, which is a fixed patch of fatigued retina and therefore has a fixed retinal size, looks small on the paper and large on the wall. This is **Emmert's law**, described by the Swiss ophthalmologist **Emil Emmert** (1844 to 1911) in 1881: the perceived size of an afterimage is proportional to the perceived distance of the surface on which it is projected (Figure 4.4). Emmert's law is the size-distance relationship laid bare, because the retinal size is held perfectly constant and only perceived distance varies.

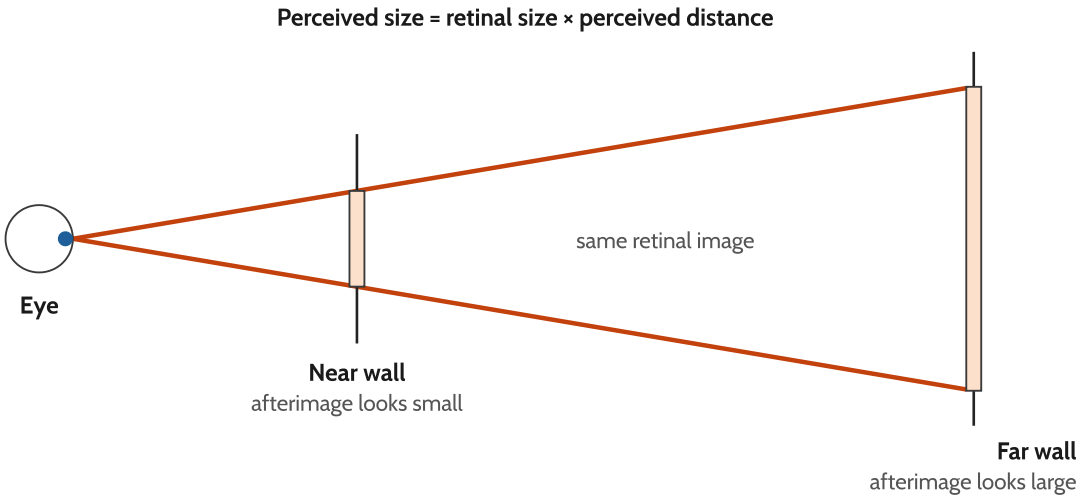


Figure 4.4 Emmert's law. The dashed lines mark one fixed visual angle, the size of the afterimage on the retina. Projected onto a near wall the afterimage looks small; onto a far wall it looks large, because the brain scales retinal size by perceived distance.

Size constancy is not perfect, and it depends on experience of distance. From the top of a tall building, people on the street look like ants, because we have little practice in judging vertical distances. In Section 4.8 you will meet a forest dweller who had never seen across open country and who took distant buffalo for insects. And when distance is misjudged, size is misjudged too; this is the key to several famous illusions (Section 4.5).

### Shape constancy

**Shape constancy** is the perception of an object as keeping its shape even when its orientation to us changes, so that the retinal image changes shape. A round dinner plate on a table casts an elliptical image unless you look straight down on it, and an opening door casts a trapezoid, yet the plate looks round and the door rectangular. The British psychologist **Robert Thouless** (1894 to 1984) showed in 1931 that constancy is usually a compromise: when observers drew the shape of a tilted circle as it appeared to them, they drew an ellipse that was rounder than the retinal image but not fully round. He called this tendency for perception to lie between the retinal image and the real object **phenomenal regression to the real object**.

### Brightness and colour constancy

A lump of coal in bright sunlight sends more light to your eye than a sheet of white paper in the shade, yet the coal still looks black and the paper white. **Brightness constancy** (more precisely, lightness constancy) is the perception of an object's lightness (its whiteness, greyness or blackness) as constant despite changes in the amount of light falling on it. The key, as **Hans Wallach** (1904 to 1998) showed in 1948, is the **ratio principle**: perceived lightness depends on the ratio of light reflected from an object to the light reflected from its surroundings. When the illumination changes, object and surroundings change together and the ratio stays the same. White paper reflects about 80 to 90 percent of the light falling on it and coal only a few percent, whatever the lighting.

The ratio principle also explains **simultaneous contrast**: a grey square on a black background looks lighter than an identical grey square on a white background, because the ratio to the surround differs. Part of the neural basis lies in **lateral inhibition**, the tendency of active neurons in the retina to suppress the activity of their neighbours. Lateral inhibition exaggerates differences at borders and so *enhances* edge perception. It explains **Mach bands**, the illusory light and dark stripes seen at the boundaries of a graded strip, described by the Austrian physicist **Ernst Mach** (1838 to 1916) in 1865, and the grey smudges seen at the intersections of the white lines of the **Hermann grid**, described by **Ludimar Hermann** (1838 to 1914) in 1870 (the retinal circuitry is in Chapter 3 and the Physiological book).

**Colour constancy** is the perception of an object's colour as constant under different coloured lights. A ripe banana looks yellow under the bluish light of a tube light, the orange light of a sodium street lamp and the reddish light of sunset, although the light it reflects changes greatly. The brain compares the light from the banana with the light from the rest of the scene and discounts the colour of the illuminant; adaptation of the cones to the prevailing light also helps. **Edwin Land** (1909 to 1991), the inventor of the Polaroid instant camera, built his retinex theory on demonstrations of this kind. Colour constancy can fail when the brain cannot tell what the illumination is. In February 2015 a photograph of a striped dress divided the internet: some people saw it as blue and black, others as white and gold. The two groups seemed to differ in whether their brains assumed the dress was lit by bluish daylight or by yellowish artificial light, and so discounted different illuminants.

### Location constancy

Every time you move your eyes, the image of the world sweeps across your retina, yet the world stays still. **Location constancy** (position constancy) is the perception of stationary objects as remaining in the same place despite movements of the eyes, head and body. You can break it yourself with a demonstration that Helmholtz described. Close one eye and press very gently on the outer corner of the other eyelid with a fingertip: the world jumps. When you move your eyes in the normal way, the brain sends a copy of the motor command to the visual system, called a **corollary discharge** by **Roger Sperry** (1913 to 1994) and an efference copy by **Erich von Holst**, both in 1950, and uses it to cancel the expected movement of the retinal image. When your finger moves the eye, there is no command and nothing to cancel, so the movement of the image is attributed to the world. The same mechanism underlies the perception of real motion (Section 4.6).

How often does the brain have to make this correction? Several times every second. When you read or scan a scene your eyes do not glide smoothly; they jump from point to point in rapid movements called saccades, separated by brief pauses called fixations, during which information is taken in. Even during a fixation the eyes are never perfectly still: they make a fine tremor, slow drifts and tiny rapid flicks called microsaccades. These small movements are not a flaw. In the 1950s researchers devised ways of keeping an image fixed on the same receptors however the eye moved (a stabilised retinal image), and the image faded from view within seconds, because receptors stimulated without change adapt (sensory adaptation, Chapter 3). By constantly refreshing the image on the retina, saccades and microsaccades ensure constant perception, while the corollary discharge keeps the world from seeming to jump with every one of them. Table 4.4 summarises the five constancies.

Table 4.4 The perceptual constancies.

| Constancy | What changes at the eye | What we perceive | Main explanation |
|---|---|---|---|
| Size | Retinal image size with distance | Constant size | Size-distance invariance; depth cues (Holway and Boring, 1941) |
| Shape | Retinal shape with orientation | Constant shape | Perceived slant; phenomenal regression (Thouless, 1931) |
| Brightness (lightness) | Amount of light reflected | Constant whiteness or blackness | Ratio principle (Wallach, 1948) |
| Colour | Wavelengths reflected under different lights | Constant colour | Discounting the illuminant; adaptation; retinex (Land) |
| Location (position) | Image movement when the eyes move | A stable world | Corollary discharge (Sperry; von Holst, 1950) |

EXAM TRAP

Your outstretched hand looks the same size as your near hand because of **size constancy**, not shape constancy, motion parallax or "depth perception" in general (depth perception is what size constancy uses, not the name of the effect). **Emmert's law** concerns **afterimages** and their apparent size. Lateral inhibition **enhances** edges; it does not dull them, lower overall brightness or improve colour vision. Tiny rapid eye movements that keep the retinal image from fading are **saccades** (strictly, microsaccades), and they ensure **constant perception**; a **fixation** is the pause between saccades, not a movement, and sensory adaptation is the fading they prevent. Constancy is a property of perception, not of sensation: the sensory input changes; the percept does not.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. The difference between the images of an object on the two retinas is called retinal _____.
2. The vivid impression of depth produced by fusing the two eyes' images is called _____.
3. The inward turning of the eyes to look at a near object is _____, a binocular cue.
4. Stones that look smaller and more densely packed with distance illustrate the cue of texture _____.
5. Near objects flashing past a train window faster than distant ones illustrate motion _____.
6. The visual cliff was devised by Eleanor Gibson and Richard _____ in 1960.
7. Perceived size equals retinal size multiplied by perceived _____.
8. An afterimage looks larger on a distant wall than on a near page, as stated by _____ law.
9. Thouless called the compromise between retinal shape and real shape phenomenal _____ to the real object.
10. Wallach explained lightness constancy by the _____ principle.

**Answers:** (1) disparity, (2) stereopsis, (3) convergence, (4) gradient, (5) parallax, (6) Walk, (7) distance, (8) Emmert's, (9) regression, (10) ratio.

## 4.5 Illusions: What Perception's Mistakes Reveal HIGH YIELD · 7 QS

Everything in this chapter so far has been a success story: a flat retina yields a deep world, a shrinking image yields a friend of constant size, a scatter of dots yields tidy groups. But a system that works by inference, as Helmholtz and Gregory argued, must sometimes infer wrongly. Those failures are far more than curiosities for puzzle books. A slip of the tongue tells you something about how speech is planned; in the same way, a visual illusion tells you which rules the visual system is following, because it catches the system applying a sensible rule in a place where the rule does not fit.

### What an illusion is ●●

An **illusion** is a perception that does not correspond to the physical properties of the stimulus: a misinterpretation of a stimulus that really exists in the outside world. Three features mark an illusion in the psychologist's sense. First, there is a real external stimulus; the lines of the Müller-Lyer figure are really there on the page. Second, the error is systematic. Nearly everyone who looks makes the same mistake, in the same direction and by roughly the same amount, so illusions can be measured with the psychophysical methods of Chapter 3. Third, most illusions are stubborn. You can measure the lines with a ruler, satisfy yourself that they are equal, look again, and still see them as unequal. This was the second objection to Gregory's hypothesis theory that you met in Section 4.1, and it tells you that much of the machinery behind illusions runs automatically, below the level that knowledge can reach.

Where exactly does the error happen? Not in the eye. Photograph the Müller-Lyer figure and the two lines in the photograph are equal; the image on your retina is just as faithful. The receptors register the light correctly, so sensation is not distorted. What goes wrong is the interpretation of that light, the organising and inferring that turn a retinal image into a seen world. Optical illusions are therefore distortions of *perception*, not of sensation, which is why textbooks reach for them whenever they want to make the difference between sensation and perception concrete.

Illusions are not only visual. Cross your index and middle fingers, close your eyes and roll a small marble between the two fingertips: you feel two marbles (Aristotle's illusion, described more than two thousand years ago). Lift a small box and a large box of exactly the same weight and the small one feels clearly heavier; this size-weight illusion was measured by the French physician Augustin Charpentier in 1891. Listen to a sequence of Shepard tones and the pitch seems to climb for ever without ever getting higher. In each case a real stimulus is correctly registered by the receptors and then misread.

Many Indian textbooks of general psychology sort illusions into two classes, and examiners still use the terms. **Personal illusions** (individual illusions) are private and variable. They depend on the observer's state, expectations and surroundings, so they differ from person to person and from one occasion to the next. The standard example is the coil of rope seen as a snake at dusk: one person sees a snake, the next sees a rope, and the same person may not be fooled the following evening. **Universal illusions**, also called permanent or general illusions, are experienced by practically everyone, every time, because they arise from the normal working of the perceptual system: railway tracks that seem to converge in the distance, the moon that looks larger at the horizon, the unequal-looking lines of the Müller-Lyer fig-

ure. Most of the illusions studied in perception laboratories are universal illusions, and that is exactly what makes them useful. A universal error must come from a universal rule.

INDIAN CONTEXT

The rope mistaken for a snake (*rajju-sarpa*) is one of the oldest illusions in Indian thought. In the Advaita Vedanta of Adi Shankaracharya (eighth century CE) it illustrates *adhyasa*, superimposition: the mind projects a snake onto the rope, just as, on that view, ignorance projects the world of appearances onto Brahman. Psychologically it is a model personal illusion. It needs dim light (weak bottom-up evidence), a setting in which snakes are expected, and a little fear (top-down influences, Section 4.8); and it vanishes the moment someone shines a torch on the rope.

EXAM TRAP

"Rail tracks appearing to converge" was keyed as a **permanent illusion**, with "personal illusion", "visual illusion" and "Müller-Lyer illusion" as the other options. The item is testing the personal versus permanent classification: everyone sees the rails converge, every time, so the illusion is permanent (universal). It is of course also a visual illusion in the broad sense, but when "permanent" is offered alongside "personal", choose permanent. It is not the Müller-Lyer illusion. Strictly, what you are seeing is the depth cue of linear perspective (Section 4.3): the retinal images of the rails really do converge, and the illusion lies only in the impression that the rails themselves draw together. Finally, illusions result from a distortion in **perception**, not in sensation, adaptation or "translation" (a loose word for transduction, the receptor's conversion of energy into nerve signals, which works perfectly well when you look at an illusion).

### The Müller-Lyer illusion and the carpentered world ●●

The most studied illusion of all was published in 1889 by the German psychiatrist and sociologist **Franz Carl Müller-Lyer** (1857 to 1916). In the **Müller-Lyer illusion**, two lines (the shafts) are of equal length. One ends in arrowheads that point outwards, away from the shaft; the other ends in fins that flare outwards like the tail of an arrow. The shaft with the fins looks clearly longer than the shaft with the arrowheads (Figure 4.5, first panel). Look at the figure, then measure it.

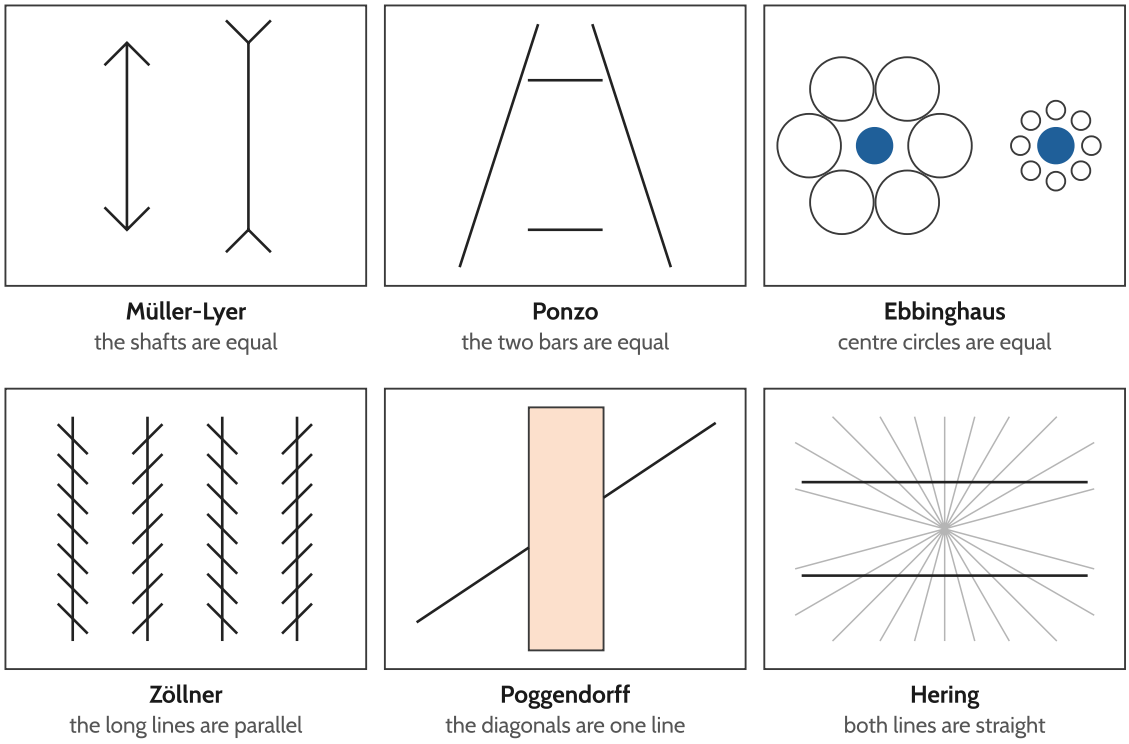


Figure 4.5 Six classic geometric illusions. In every panel the caption underneath is physically true, yet you see something else: the finned shaft looks longer, the upper bar looks longer, the circle among small circles looks larger, the long lines look tilted, the diagonal looks broken and the straight lines look bowed. Check them with a ruler.

Why should a few short strokes at the ends of a line change its apparent length? The best-known answer came from Richard Gregory in 1963, and he called it **inappropriate constancy scaling** (it is also called the misapplied size constancy theory). The shafts in Figure 4.5 are drawn vertically; take the arrowhead version first. It resembles the outside corner of a building seen from the street: the vertical edge is nearest to you and the walls slope away on either side. The finned version resembles the inside corner of a room: the vertical edge where two walls meet is the farthest point from you, and the edges of the ceiling and floor come towards you. Gregory argued that the visual system reads these depth cues automatically, even in a flat drawing, and then applies size constancy to them (Section 4.4). The shaft that is coded as farther away (the inside corner) is scaled up and looks longer; the shaft coded as nearer (the outside corner) is scaled down and looks shorter. Two lines that cast the same retinal image, one of which is judged to be farther away, must be of different sizes; that is the size-distance invariance hypothesis doing its normal job in a place where it does not belong, a flat page on which both lines are really at the same distance.

Gregory reported a striking piece of support: when Müller-Lyer figures are made luminous and shown in a dark room, where the flat surface of the paper cannot be seen, observers see them as three-dimensional corners, receding or approaching in just the way the theory predicts. But the theory has met real criticism. First, the illusion survives when the arrowheads and fins are replaced by small circles or squares at the ends of the lines, shapes that suggest no corner and no depth at all. Second, the figure does not look deep on paper, yet the theory needs depth to be registered. Third, a rival account, defended by the Australian psychologist Ross Day among others, holds that the visual system cannot completely separate the length

of the shaft from the length of the whole figure, fins included, and the finned figure as a whole is longer. A fourth finding links back to Section 4.1: in a meta-analysis of 33 studies, Nicola Bruno and colleagues (2008) found that fast pointing movements to the ends of the shafts are much less affected by the illusion than verbal judgements are, as Milner and Goodale's separation of vision-for-action from vision-for-perception would predict.

If Gregory is even partly right, the illusion should depend on experience with corners, and therefore on culture. In the late 1950s and early 1960s the psychologists **Marshall Segall** and **Donald Thomas Campbell** (1916 to 1996) and the anthropologist **Melville Jean Herskovits** (1895 to 1963) tested this idea with nearly two thousand children and adults from more than a dozen societies in Africa and the Philippines, and from the city of Evanston in the United States. Their reports (1963, and the book *The Influence of Culture on Visual Perception*, 1966) set out the **carpentered-world hypothesis**: people who grow up in a "carpentered" environment, full of rectangular rooms, buildings, windows and furniture made with saws and set squares, learn to read acute and obtuse angles in a drawing as right angles extending in depth, and so are more susceptible to the Müller-Lyer illusion. People raised in a non-carpentered environment, with round huts and few straight edges, should be less susceptible.

That is broadly what they found. The urban American sample was among the most susceptible to the Müller-Lyer illusion, and some African samples, such as the San foragers of the Kalahari, were among the least. The rural Zulu, who traditionally lived in round huts with round doorways, became the textbook example of a non-carpentered culture. The same team also found the reverse pattern for the horizontal-vertical illusion (a vertical line looks longer than an equal horizontal line): it was strongest in people who live on open plains, where long stretches of ground recede horizontally into the distance, and weakest in people who live in dense forest, which is just what an experience-based account predicts. The hypothesis is not settled. A rival explanation proposed that people with more densely pigmented retinas (pigmentation tends to go with darker skin) detect contours slightly less sharply and are therefore less affected by such figures; schooling and experience with pictures differ between groups too; and later studies gave mixed results. But the work established the main point that matters for this chapter: susceptibility to illusions is partly learned, so illusions are not purely the product of innate machinery.

> EXAM TRAP
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> The Müller-Lyer illusion is strongest in cultures with **buildings with corners** (carpentered environments), not in cultures with "a lot of trees", not in those with "structures without many corners and angles" (that is the low-susceptibility group), and "more men than women" is pure decoration. The carpentered-world hypothesis is **Segall, Campbell and Herskovits** (1963, 1966). The illusion is "best explained by" **misapplied size constancy** (Gregory's inappropriate constancy scaling). Binocular disparity cannot be the cause, because the illusion works with one eye shut and on a flat page; lateral inhibition explains Mach bands and the Hermann grid (Section 4.4), not line length; figure-ground reversal is Rubin's vase (Section 4.2).

### Ponzo, the Ames room and the moon

The same size-distance logic explains the second panel of Figure 4.5. The Italian psychologist **Mario Ponzo** (1882 to 1960) described it in 1911. In the **Ponzo illusion**, two identical horizon-

tal bars are drawn between two lines that converge towards the top, like railway tracks receding into the distance, and the upper bar looks longer. Linear perspective (Section 4.3) tells the visual system that the upper bar is farther away; it casts the same retinal image as the lower bar; so, by size-distance invariance, it must be bigger. The illusion is strong in photographs of real roads and railway lines, where the depth cues are richer, which is what a depth-scaling account predicts.

The most dramatic size illusion is a whole room. The **Ames room** was first built in 1946 (the commonly cited date; some textbooks cite Ames's account of 1952) by the American scientist **Adelbert Ames Jr.** (1880 to 1955), a lawyer turned painter turned researcher in physiological optics, developing an idea that Helmholtz had described. Seen through a peephole with one eye, it looks like an ordinary rectangular room. In fact, as Figure 4.6 shows, its back wall slopes away from the viewer, so that one back corner is about twice as far from the peephole as the other, and the walls, floor, ceiling and windows are distorted in exactly the way needed to make the room cast the same retinal image as a rectangular one. Two people of the same height standing in the two back corners now look absurdly different: the one in the near corner looks like a giant and the one in the far corner like a child. A person walking from one corner to the other seems to grow or shrink as they walk.

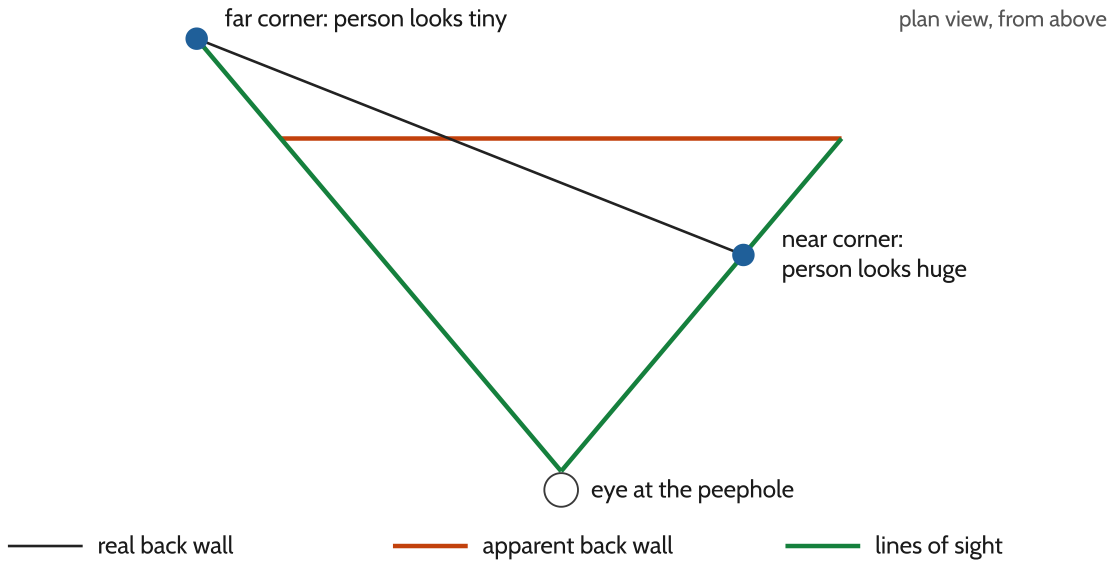


Figure 4.6 The Ames room seen from above. The real back wall (solid) slopes away, so the left corner is twice as far from the eye as the right. Each corner lies on the same line of sight as a corner of an ordinary rectangular room (dashed), so the eye assumes that both people stand at the same distance and judges their sizes from their retinal images alone.

Why does the room win? Two assumptions are in conflict: that rooms are rectangular, and that people stay the same size. The brain has met far more rectangular rooms than people who change size, yet it keeps the room and sacrifices the person. Ames, whose view of perception became known as transactionalism, argued that every percept is a kind of bet based on our past dealings with the world, and that in a peephole view the bet on rectangular rooms is the safest one available. Familiarity can shift the bet. In 1952 Warren Wittreich reported the "Honi phenomenon", named after a woman nicknamed Honi who saw strangers change size in the room but saw her own husband almost undistorted; in a story often retold

in textbooks, a researcher's wife arrived at the laboratory, saw her husband inside the room and said at once that the room was distorted. Ames also built a trapezoidal window that, when rotated steadily, seems to swing back and forth instead. **Gordon Willard Allport** (1897 to 1967) and Thomas Pettigrew found in 1957 that rural Zulu children, who had grown up with few rectangular windows, were less likely than urban children to see this illusion when the viewing conditions were poor: another sign that the "bet" is learned.

The oldest illusion of all hangs in the sky. A full moon rising over the horizon looks enormous; the same moon high overhead a few hours later looks much smaller. This is the **moon illusion**, and it is purely perceptual. The moon's retinal image is the same size in both positions (about half a degree of visual angle, roughly the size of a pea held at arm's length); photographs show identical discs; and the horizon moon is in fact very slightly *farther* from you than the overhead moon, by about the radius of the Earth. Contrary to a popular belief, the atmosphere does not magnify it.

The leading explanation, the **apparent distance hypothesis**, goes back about a thousand years to the Arab scholar **Ibn al-Haytham** (about 965 to about 1040), known in Europe as Alhazen. The sky is not perceived as a hemisphere but as a flattened dome, whose horizon seems much farther away than its zenith, because at the horizon we see the moon across a great stretch of terrain filled with depth cues (fields, trees, buildings) while overhead there is only empty sky. A moon of constant retinal size that is registered as farther away must be larger; size-distance invariance again. **Lloyd Kaufman** and Irvin Rock supported this in 1962 by showing observers the moon through optical arrangements that removed or added the terrain: the horizon moon lost much of its enlargement when the terrain was hidden, and an elevated moon looked larger when it was seen across terrain. There is a puzzle, though. Asked which moon looks nearer, most people say the horizon moon, the opposite of what the theory seems to need. One resolution is that the registered distance that drives size scaling is unconscious, while the conscious judgement of distance is made afterwards from the size that has been seen ("it looks big, so it must be close"). A second explanation, angular size contrast, adds that the horizon moon is seen next to small-looking objects such as trees and roofs, while the overhead moon is dwarfed by the vast expanse of sky, a contrast effect like the Ebbinghaus illusion that follows.

> EXAM TRAP
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> The Ponzo, Müller-Lyer, Ames room and moon illusions are all explained by **size-distance scaling going wrong**: something registered as farther away is seen as bigger. The Ames room depends on the assumption that rooms are **rectangular** and must be viewed with **one eye through a peephole**, which removes binocular and motion cues. The moon illusion is not caused by the moon being nearer at the horizon (it is slightly farther), nor by atmospheric magnification or refraction; the retinal image is the same. "Apparent distance hypothesis" is the answer for the moon; Kaufman and Rock (1962) are its classic experimenters.

### Size contrast and bent lines

Not every illusion is about depth. In the **Ebbinghaus illusion** (third panel of Figure 4.5), a circle surrounded by large circles looks smaller than an identical circle surrounded by small ones. It is named after **Hermann Ebbinghaus** (1850 to 1909), the pioneer of memory research

(Chapter 7), and was made widely known by **Edward Bradford Titchener** (1867 to 1927) in his laboratory manual of 1901, which is why it is also called the Titchener circles. The usual explanation is size contrast: the visual system judges size relative to the neighbours, just as it judges lightness relative to the surround in simultaneous contrast (Section 4.4). In 1995 Salvatore Aglioti, Joseph DeSouza and Melvyn Goodale used poker-chip versions of the figure and found that when people reached out to pick up the central disc, the opening of their finger and thumb fitted the disc's real size even though they saw it as larger or smaller; they summed this up as illusions that "deceive the eye but not the hand". Later studies have argued over how complete this immunity is, but the finding remains a classic piece of evidence for a separate vision-for-action system.

A family of nineteenth-century illusions bends or breaks straight lines. In the **Zöllner illusion**, described in 1860 by the German astrophysicist **Johann Karl Friedrich Zöllner** (1834 to 1882), long parallel lines crossed by short oblique strokes, slanting in opposite directions on alternate lines, no longer look parallel; they seem to tilt towards and away from each other. When Zöllner sent his figure to the physicist **Johann Christian Poggendorff** (1796 to 1877), editor of the journal in which it appeared, Poggendorff noticed another distortion in it. In the **Poggendorff illusion** (1860), a straight diagonal line that passes behind a vertical strip seems to come out at the wrong place, so the two visible parts look misaligned. In the **Hering illusion**, described in 1861 by the German physiologist **Ewald Hering** (1834 to 1918), the same man who proposed the opponent-process theory of colour vision (Chapter 3), two straight parallel lines drawn across a pattern of lines radiating from a central point appear to bow outwards; Wilhelm Wundt drew a variant in which the background lines converge the other way and the parallels bow inwards.

No single explanation covers this family. One idea is that the visual system tends to overestimate acute angles, perhaps because neurons tuned to neighbouring orientations inhibit one another (an orientation version of the lateral inhibition you met in Section 4.4), so each crossing line pushes the perceived orientation of the long line slightly away from itself. Another is that the figures are again read as perspective drawings: the radiating background of the Hering figure looks like a tunnel or a road stretching away. What these illusions share is the lesson that the visual system does not measure lines one at a time; it codes every element in relation to its context.

### Ambiguous and impossible figures

A different kind of trick produces not a wrong percept but two percepts that take turns. An **ambiguous figure** (reversible figure) is a single unchanging stimulus that can be seen in two or more different ways, with the percept flipping from one to the other. You have already met Rubin's vase, in which figure and ground swap (Section 4.2). In 1832 the Swiss crystallographer **Louis Albert Necker** (1786 to 1861) noticed that his drawings of crystals seemed to jump in depth, and the **Necker cube**, a line drawing of a transparent wire cube, now carries his name: look at it for a few seconds and the face that was in front suddenly lies behind, then jumps forward again. Other famous examples are the duck-rabbit, used by the American psychologist Joseph Jastrow in 1899, and "My wife and my mother-in-law", a cartoon by W. E. Hill published in 1915 and brought into psychology by Edwin Boring in 1930, in which a young woman looking away and an old woman in profile share the same lines.

Ambiguous figures are the strongest single argument for Gregory's view of perception as hypothesis testing. The stimulus does not change, so the flipping must come from the perceiver: the brain entertains one hypothesis, then the other, because the evidence fits both equally well. Why does it not settle on one? A popular explanation is that the neural representation of the current interpretation adapts or tires, letting the rival take over; another is that shifts of attention and fixation tip the balance (fixating a particular corner of the Necker cube tends to bring that corner forward). Knowledge matters too. Irvin Rock and his colleagues found in 1994 that young children rarely reverse ambiguous figures spontaneously unless they have been told that there are two interpretations to look for.

An **impossible figure** is a drawing that the visual system reads as a solid object which could not exist in three dimensions. The best known is the Penrose triangle (tribar), whose three beams each look perfectly sensible but join at right angles to make a closed triangle. It was published, together with the never-ending Penrose staircase, in a short paper of 1958 by the British psychiatrist and geneticist **Lionel Penrose** (1898 to 1972) and his son, the mathematician **Roger Penrose** (born 1931), later a Nobel laureate in physics; the Swedish artist Oscar Reutersvärd had drawn a version in 1934, and the Dutch artist M. C. Escher built his lithographs *Ascending and Descending* (1960) and *Waterfall* (1961) around the Penroses' figures. Impossible figures work because each part is interpreted locally: every corner, taken by itself, is a legitimate piece of a three-dimensional object, and the visual system does not check that the local interpretations add up to a consistent whole. Notice the contrast with ambiguous figures. An ambiguous figure gives one stimulus two possible objects; an impossible figure gives one stimulus a single object that cannot be. Table 4.5 brings together the illusions and figures of this section.

Table 4.5 The classic illusions and ambiguous figures.

| Illusion | Who and when | What you see | Leading explanation |
|---|---|---|---|
| Müller-Lyer | Müller-Lyer, 1889 | Finned shaft looks longer than arrowhead shaft | Inappropriate constancy scaling (Gregory, 1963); carpentered world (Segall and colleagues, 1966) |
| Ponzo | Ponzo, 1911 | Upper bar between converging lines looks longer | Linear perspective plus size scaling |
| Ames room | Ames, 1946 | People in the two back corners look giant and tiny | Assumption that rooms are rectangular; size from retinal image |
| Moon illusion | Described since antiquity | Horizon moon looks larger than overhead moon | Apparent distance (Ibn al-Haytham; Kaufman and Rock, 1962); size contrast |
| Ebbinghaus (Titchener circles) | Ebbinghaus; Titchener, 1901 | Circle among large circles looks smaller | Size contrast |
| Zöllner | Zöllner, 1860 | Parallel lines with oblique hatching look tilted | Orientation contrast; acute angles overestimated |
| Poggendorff | Poggendorff, 1860 | Interrupted diagonal looks misaligned | Angle and orientation misjudgement |
| Hering | Hering, 1861 | Straight lines on radiating background look bowed | Orientation contrast; perspective reading |
| Necker cube | Necker, 1832 | Wire cube flips in depth | Two equally good perceptual hypotheses |
| Penrose triangle | L. and R. Penrose, 1958 | A solid object that cannot exist | Local parts interpreted without a global check |

EXAM TRAP

Keep three confusable names apart: the **Hering illusion** (Ewald Hering, 1861, straight lines that look bowed), the **Hermann grid** (Ludimar Hermann, 1870, grey spots at grid intersections, lateral inhibition) and **Helmholtz** (unconscious inference). The **Ebbinghaus illusion** and the **Titchener circles** are the same figure. The **Necker cube** is an ambiguous (reversible) figure in depth; **Rubin's vase** is a figure-ground reversal; the **Penrose triangle** is an impossible figure, not a reversible one.

## Illusions, hallucinations and delusions ●●●

For a clinical psychologist the most important thing about illusions is what they are not. Three terms are confused constantly, in examinations and in casual speech, and the distinctions between them are part of every mental status examination.

An illusion, as you have seen, is a misinterpretation of a real external stimulus. A **hallucination** is a perception without an adequate external stimulus: the person hears a voice when no one is speaking or sees a figure in an empty room, and the experience has the full vividness of a real perception, is located in outer space and is not under voluntary control. The French psychiatrist Jean-Étienne Esquirol, in the early nineteenth century, was the first to separate the two clearly: in an illusion something is there and is misperceived; in a hallucination nothing is there at all. A **delusion** is not a perception at all. It is a false belief, held with unshakeable conviction, not open to correction by argument or evidence and out of keeping with the person's cultural and educational background: for example, the conviction that one's neighbours are broadcasting one's thoughts. Illusions and hallucinations are disorders of perception; delusions are disorders of the content of thought.

The British psychiatrist Andrew Sims, in his textbook of descriptive psychopathology *Symptoms in the Mind*, adds a finer distinction that one past paper used as its options. **Sensory distortions** are changes in the perception of a real object that is perceived as it is but altered in intensity, quality or size: sounds that seem unbearably loud, colours that seem changed, objects that look too small (micropsia) or too large (macropsia). **Sensory deceptions** are new perceptions that are false, and they include both illusions and hallucinations. So "a real perceptual object perceived in a distorted way" describes a sensory distortion; "misinterpretation of stimuli arising from an external object" is the standard psychopathology definition of an illusion; "perception without an adequate external stimulus" is a hallucination; and "vivid mental images" describes imagery, which is experienced in the mind's eye rather than in outer space and which a person can usually call up or dismiss at will.

Psychopathology recognises three kinds of illusion. A **completion illusion** depends on inattention: you read a misprinted word as the word you expected, and the error disappears as soon as you pay attention. An **affect illusion** is driven by emotion: a frightened child alone at night sees the coat hanging on the door as a man, and a grieving widow briefly "sees" her husband in a crowd. **Pareidolia** is the effortless perception of vivid, meaningful images in vague stimuli, such as faces in clouds, animals in the flames of a fire or a face on the front of a car; unlike the other two, it persists, and even grows richer, when you attend to it, and it coexists with knowing perfectly well what the stimulus really is.

A last group of "illusions" belongs to memory rather than perception. In **déjà vu** ("already seen") a new situation feels as if it has been experienced before; in **déjà entendu** ("already

heard") something heard for the first time feels as if it has been heard before; in déjà vécu ("already lived") a whole experience feels relived; and in **jamais vu** ("never seen") a familiar situation suddenly feels strange, as if it had never been seen. These are illusions of familiarity, occasional in healthy people and frequent in some forms of temporal lobe epilepsy. Do not confuse them with hypermnesia, an unusually vivid and detailed memory for past events, or with disorders of speech such as palilalia, the involuntary repetition of one's own words (Disorders book). Table 4.6 sets the main terms side by side.

Table 4.6 Illusion, hallucination and their neighbours.

| Term | Is there a real stimulus? | What happens | Example |
|---|---|---|---|
| Illusion | Yes | The real stimulus is misinterpreted | Rope seen as a snake; Müller-Lyer lines |
| Sensory distortion | Yes | The object is perceived correctly but altered in size, intensity or quality | Micropsia; sounds unbearably loud |
| Hallucination | No | A perception with no adequate external stimulus | Hearing a voice in an empty room |
| Imagery | No | A picture in the mind's eye, recognised as one's own | Imagining your home while away |
| Delusion | Not a perception | A false, fixed belief unshaken by evidence | Belief that neighbours read one's thoughts |

EXAM TRAP

"Illusions are": choose **misinterpretation of stimuli arising from an external object**, or **distortions of existing stimuli**. "Perception without an adequate external stimulus" and "not based on external reality" describe **hallucinations**; "the same as hallucinations" is wrong by definition; "vivid mental images" is imagery; "a real perceptual object perceived in a distorted way" is Sims's **sensory distortion**. "The result of innate mechanisms" is tempting but wrong as a definition: illusions arise from normal perceptual processes, and the carpentered-world studies show that learning shapes them. **Déjà entendu** is the illusion of having **heard** something before; the illusion of having **seen** it before is **déjà vu**. A **delusional perception**, one of Kurt Schneider's first-rank symptoms of schizophrenia, is a normal perception given a sudden delusional meaning; it is not an illusion (Disorders book).

CLINICAL LINK

Illusions and visual hallucinations are common in delirium, the acute confusional state seen in high fever, alcohol withdrawal and severe medical illness, and they typically get worse in the dim light of evening, when bottom-up information is weakest. Elderly people with failing eyesight may develop vivid, silent visual hallucinations (often of small people or patterns) while knowing perfectly well that they are not real, a condition named after Charles Bonnet, the Swiss naturalist who in 1760 described it in his grandfather. Recognising which phenomenon a patient is describing, and whether insight is preserved, is a basic diagnostic skill (Disorders book).

## 4.6 Motion Perception LOWER YIELD · NOT ASKED YET

For most animals, movement matters more than anything else in the visual field: it means food, danger or a mate. A hungry frog will ignore dead, motionless flies yet snap at a moving speck, and a rabbit freezes because a predator's eye is drawn to motion. Human beings are no different. A flicker at the edge of your vision pulls your eyes towards it before you know what it is. Yet the perception of motion turns out to be as constructed as the perception of depth, and the illusions of motion are as revealing as the illusions of size.

### Real motion: two systems and a motion area

**Real motion** is the perception of an object that is physically moving. You might think that seeing it is simply a matter of an image moving across the retina, and sometimes it is. When you hold your eyes still and a car crosses your field of view, its image sweeps across the retina, stimulating one receptor after another. Richard Gregory called this the **image-retina system**. But now follow a flying kite with your eyes. Its image stays more or less still on your fovea, while the image of the background streams across your retina in the opposite direction; yet you see the kite moving and the background standing still. Here the information comes from what Gregory called the **eye-head system**: the brain knows the eyes are turning because it issued the commands to turn them (the corollary discharge of Section 4.4), and it attributes movement to whatever the eyes are tracking. This is why the world stays still when you move your eyes normally, and jumps when you push your eyeball with a finger.

The image-retina system relies on neurons that respond to movement in a particular direction and not in the opposite one. Such direction-selective cells were found in the retinas of frogs and rabbits and in the visual cortex of cats and monkeys (Section 4.7 describes the work of Hubel and Wiesel). In primates, an area near the junction of the occipital and temporal lobes, the middle temporal area (MT, also called V5), is specialised for motion: nearly all its neurons are direction-selective, and damaging it in monkeys impairs the perception of motion while leaving other aspects of vision intact.

> CLINICAL LINK
>
> In 1983 Josef Zihl and his colleagues in Munich described L.M., a woman who had suffered damage to this region on both sides of the brain. She could see shapes, colours and positions but not movement, a rare condition called **akinetopsia** (motion blindness). Pouring tea was difficult because the stream of liquid looked frozen, like a glacier, and she could not see the level rising in the cup, so it overflowed. Crossing the road was frightening: a car that seemed far away a moment ago was suddenly close, with no movement in between. Conversation was strange because she could not see the speaker's lips move. Her case shows that motion is not inferred from a series of positions; it is computed by its own neural machinery, which can be lost while position perception survives.

Motion is also a powerful source of information about what things are. In 1973 the Swedish psychologist **Gunnar Johansson** (1911 to 1998) attached small lights to the shoulders, elbows, wrists, hips, knees and ankles of a person dressed in black and filmed them walking in the dark. Any single frame of the film looks like a meaningless scatter of a dozen dots. The moment the film runs, every observer sees a person walking, and later studies showed that people can tell from such point-light displays whether the walker is a man or a woman, happy or

sad, and even, sometimes, which of their friends it is. This perception of **biological motion** is an extreme case of the Gestalt law of common fate (Section 4.2): dots that move together in the pattern of a body are grouped into a body.

Movement also tells an animal when something will hit it. As an object approaches at a steady speed its retinal image expands, and the Gibsonian psychologist David Lee showed in 1976 that the ratio of the image's size to its rate of expansion, which he called tau, specifies the time remaining before contact, without any need to know the object's distance or speed. Diving seabirds appear to fold their wings at a constant value of tau, and drivers use the expanding image of the car ahead when they brake. Gibson's idea that the optic array carries information directly (Section 4.1) is at its most persuasive here.

### Apparent motion: phi, beta and the cinema

Can you see movement when nothing moves? You do every time you watch a film. **Apparent motion** is the perception of movement where there is no physical movement at all, produced by stimuli that appear in different places at different times. Its scientific study founded a school of psychology. In 1912 Max Wertheimer, working in Frankfurt with Koffka and Köhler as his observers, flashed two lines one after the other in slightly different places (the history is told in Chapter 2). When the interval between them was very short, about 30 milliseconds, observers saw two lines at once; when it was long, about 200 milliseconds, they saw one line and then the other; at intervals in between, around 60 milliseconds, they saw a single line moving from the first position to the second. The impression of movement was not in either stimulus; it arose from the relation between them, and Wertheimer took this as proof that a perceptual whole can contain something its parts do not. He called the phenomenon **phi phenomenon**.

A subtlety lies behind the name. Wertheimer distinguished two experiences. At some intervals observers saw a smooth, object-like movement of the line from one place to the other; later writers named this **beta movement**, or optimal movement. At other intervals they reported a pure sensation of movement in the gap between the lines without seeing any object travel across it, which Wertheimer called pure phi. Most introductory textbooks, and most examination papers, now use "phi phenomenon" loosely for any apparent movement of flashing lights. Strictly, the film-like movement of an object is beta movement, and phi is the objectless motion; in 2000 Robert Steinman and his colleagues wrote a paper whose title says it all, "Phi is not beta". In 1915 Adolf Korte showed that apparent motion is lawful: for example, the farther apart the two lights, the longer the interval needed between them for the best movement (Korte's laws).

Apparent motion is everywhere once you look for it. The strings of lights that seem to chase each other round a balcony at Diwali, the arrows that seem to slide along a shop sign and the running headlines of a news ticker are all rows of stationary bulbs switching on and off in sequence. **Stroboscopic motion** is the apparent motion produced when a series of still pictures, each slightly different from the last, is presented in rapid succession. A cinema film shows 24 still frames each second (the projector flashes each frame two or three times, so that the light flickers faster than the eye can follow, Chapter 3), a flip book does the same in your hands, and an animated cartoon is drawn frame by frame. The visual system's motion detectors respond to such sequences much as they respond to real motion, and brain imaging shows that apparent motion activates area MT just as real motion does. Early researchers

gave Greek letters to other varieties of apparent movement as well; the one you are most likely to meet is gamma movement, the apparent expansion of a figure when it is suddenly lit and its contraction when the light goes off.

> EXAM TRAP
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> The **phi phenomenon** was described by **Max Wertheimer in 1912**, and that paper is usually taken as the founding of Gestalt psychology. In loose usage phi means any apparent movement from lights flashed in sequence; strictly, smooth movement of an object is **beta movement** and objectless "pure" motion is **phi**. Motion pictures produce **stroboscopic motion**. None of these involves anything physically moving, which is what separates apparent motion from **real motion**. **Motion parallax** (Section 4.3) is a depth cue produced by real movement of the observer; it is not an illusion of motion.

### Induced motion and the autokinetic effect

On a windy night, look at the moon through scattered clouds. The moon seems to race across the sky while the clouds stand still. In reality the clouds are moving and the moon, for the purposes of your perception, is not. This is **induced motion**: a stationary object appears to move because the frame that surrounds it moves. The German Gestalt psychologist **Karl Duncker** (1903 to 1940), whom you will meet again in Chapter 8 for his work on problem solving, studied it in 1929. In a dark room he showed observers a luminous dot inside a luminous rectangle and moved the rectangle slowly to one side. Observers saw the rectangle as still and the dot as moving the opposite way. The visual system treats the larger, surrounding frame as the stable background against which smaller things move, which is usually a safe assumption, since rooms, landscapes and skies rarely move.

The same rule applies to your own body. Sit in a stationary train at a busy station such as Mumbai Central, and when the train on the next platform pulls out, you feel your own train begin to move, perhaps even sway, until you glance at the platform. The illusion of self-motion produced by a large moving visual field is called **vection**. It is induced motion with the observer as the "dot" and the neighbouring train as the "frame", and it is the principle behind flight simulators and some forms of motion sickness.

Now remove the frame altogether. Sit in a completely dark room and look steadily at a single small, stationary point of light. Within a few seconds the light begins to drift, wander or swing, sometimes by what seems a large distance, even though it is perfectly still. This is the **autokinetic effect** (autokinesis), long noticed by astronomers watching single stars and by night pilots who have mistaken a solitary star or distant lamp for another aircraft. Its cause is the absence of any frame of reference. In darkness the eyes drift slightly, and the small commands needed to hold fixation steady are not fully accounted for, so the brain cannot tell whether the image or the eye has moved; with no visible background against which to check, it attributes some of the uncertainty to the light. The amount and direction of movement seen differ from person to person and from moment to moment, which made the effect perfect for one of the most famous experiments in social psychology.

KEY STUDY: SHERIF'S AUTOKINETIC NORMS (1935)

**Question.** How do social norms form when a situation is ambiguous and there is no objectively correct answer? **Method.** The Turkish-American social psychologist **Muzafer Sherif** (1906 to 1988) seated participants in a completely dark room and showed them a stationary pinpoint of light, asking them on each of many trials to say how far it had moved. Some first made their judgements alone and then in groups of two or three, calling out their estimates aloud; others started in groups and were later tested alone. **Results.** Alone, each person soon settled on a personal range and a typical estimate, and these differed widely from one person to another. In groups, the estimates converged over trials on a common group norm. When people who had formed a norm in a group were later tested alone, they kept to the group's norm. **Conclusion.** When reality is ambiguous, people use one another as a frame of reference, and the norms they build together become their own private standards (Sherif reported the work in 1935 and in his book *The Psychology of Social Norms*, 1936). In 1961 Robert Jacobs and Donald Campbell showed that an arbitrary, exaggerated norm planted by a confederate survived for several "generations" of a laboratory group after the confederate had left and members had been replaced one by one. **Criticism.** Because the movement is wholly illusory, there is no right answer to resist, so the study shows informational influence (using others as a source of information), not the yielding to group pressure on an obvious question that Solomon Asch tested in 1951 with judgements of line length (Social book).

EXAM TRAP

**Sherif** used the **autokinetic effect** to study norm formation; **Asch** used **line-length** judgements to study conformity. The autokinetic effect is a **stationary light seen moving in darkness**, with no frame of reference; **induced motion** is a stationary object seen moving because its frame moves (the moon "racing" behind clouds, Duncker 1929); **vection** is illusory self-motion (the train beside yours pulls out).

## The motion aftereffect

In 1834 a visitor to the Scottish Highlands, Robert Addams, stared for some time at the waterfall at the Falls of Foyers, near Loch Ness, and then turned his eyes to the rocks beside it. The rocks seemed to drift slowly upwards, in the direction opposite to the falling water. This **motion aftereffect**, often called the waterfall illusion, is the illusory movement of a stationary scene in the direction opposite to a movement that has been watched for some time. You can produce it after staring out of the window of a moving train: when the train stops, the platform seems to creep the other way.

The explanation is adaptation in direction-selective neurons, and it parallels the colour afterimages of Chapter 3. A stationary scene normally produces roughly equal, low activity in cells tuned to downward and to upward motion, and the brain reads the balance as "no movement". Prolonged watching of downward motion fatigues the downward-tuned cells. When you then look at the still rocks, the upward-tuned cells fire at their normal resting rate while the tired downward cells fire below theirs, the balance tips towards "up", and you see upward drift. Two further observations make the aftereffect especially informative. First, the rocks seem to move without getting anywhere: they drift upwards yet stay in the same place. That paradox shows that the brain codes motion and position separately, just as L.M.'s akinetopsia suggests. Second, if you adapt one eye and then test with the other, a weaker aftereffect is still seen, so at least part of the adaptation must take place in the visual cortex, where cells

first receive input from both eyes. A rotating spiral gives a vivid version: after watching a spiral that seems to expand, a friend's face looks as if it is shrinking. Table 4.7 summarises the varieties of motion perception met in this section.

Table 4.7 The varieties of motion perception.

| Kind | What is physically there | What is seen | Example or source |
|---|---|---|---|
| Real motion | An object really moving | Movement, through image-retina or eye-head systems | A car crossing your view; tracking a kite |
| Phi (pure phi) | Two lights flashed in sequence | Objectless movement between them | Wertheimer, 1912 |
| Beta movement | Two lights flashed in sequence | One object moving smoothly between positions | Chasing Diwali lights; neon signs |
| Stroboscopic motion | Rapid series of still pictures | Continuous movement | Cinema at 24 frames a second; flip books |
| Induced motion | Stationary object in a moving frame | The object moves, the frame stays still | Moon behind clouds; Duncker, 1929 |
| Vection | Large moving visual field | One's own body moving | The neighbouring train pulls out |
| Autokinetic effect | Stationary light in darkness | The light wanders | Sherif's norm studies, 1935 |
| Motion aftereffect | Stationary scene after watching motion | Drift in the opposite direction | Waterfall illusion, Addams, 1834 |
| Biological motion | A dozen moving dots | A person walking, running or dancing | Johansson, 1973 |

EXAM TRAP

The **waterfall illusion** is the **motion aftereffect** (Addams, 1834): it comes from adaptation of direction-selective neurons, and the illusory movement is in the **opposite** direction to the movement watched. It is not stroboscopic motion, not induced motion and not autokinesis. Damage to area **MT (V5)** causes **akinetopsia**, loss of motion perception with other vision spared.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. A misinterpretation of a stimulus that really exists in the outside world is an _____.
2. Illusions that practically everyone experiences every time, such as converging rails, are called universal or _____ illusions.
3. Gregory explained the Müller-Lyer illusion as inappropriate constancy _____.
4. Segall, Campbell and Herskovits proposed the _____-world hypothesis.
5. In the _____ room, viewed through a peephole, a person in one back corner looks like a giant.
6. The Ebbinghaus illusion is also called the _____ circles.
7. A perception without an adequate external stimulus is a _____.
8. The illusion of having heard something before is déjà _____.
9. A stationary light in a dark room appears to wander: this is the _____ effect.
10. After staring at a waterfall, stationary rocks seem to drift upwards: this is the motion _____.

**Answers:** (1) illusion, (2) permanent, (3) scaling, (4) carpentered, (5) Ames, (6) Titchener, (7) hallucination, (8) entendu, (9) autokinetic, (10) aftereffect.

## 4.7 Pattern and Object Recognition

LOWER YIELD · 2 QS

You recognise your mother's voice on a crackling phone line, a friend's face across a crowded platform at Bengaluru City station, and the letter A whether it is printed in a newspaper, scrawled on a blackboard or carved in stone. You do it in a fraction of a second and you cannot say how. **Pattern recognition** is the process of identifying a sensory pattern by matching it with a representation stored in long-term memory; when the pattern is a three-dimensional thing in the world, psychologists speak of object recognition. The difficulty is variability. The same letter or object almost never produces the same retinal image twice: it changes with size, distance, angle, lighting, partial hiding and, for letters, the hand or font that made it. Any theory of recognition has to explain how a single stored representation can match all of these. This section follows the main attempts, roughly in the order in which they were proposed.

### Template matching

The simplest idea is that memory holds a template for every known pattern: an exact, miniature copy, like a stencil. In **template matching**, the incoming pattern is compared with the stored templates point by point, and it is recognised as whichever template it overlaps best. Machines work this way when the input is standardised. The numbers printed along the bottom of an Indian bank cheque are in a special magnetic ink and a fixed typeface so that a reader in the clearing house can match them against templates; barcode scanners work on the same principle.

For human recognition the theory fails quickly. A template for A matches only an A of the same size, shape, slant and position. To recognise every A you would need a separate tem-

plate for every variant, an impossible number, and you would still be stuck the first time you met a new handwriting. One could add a preliminary stage that rotates, shrinks or centres the input before matching (normalisation), but to know how to rotate it the system would already have to know what it was. And templates cannot explain why the same shape is read differently in different contexts, as you will see at the end of this section. Template matching survives as a component of some modern theories, but no one now believes it is the whole story.

### Feature analysis and the Pandemonium model

A more flexible idea is that patterns are recognised from their parts. According to **feature analysis** (feature theories), every pattern can be described as a set of **distinctive features**, elementary components such as vertical lines, horizontal lines, oblique lines, curves, angles and intersections. An A is two obliques meeting at the top plus a horizontal bar; an H is two verticals plus a horizontal. The perceiver detects the features present in the input and compares this list with the feature lists stored for each pattern. Because features stay the same when a letter changes size or style, the theory handles variability far better than templates do.

Two kinds of evidence supported it. When people are shown letters very briefly and asked to identify them, the letters they confuse are those that share many features: P with R, E with F, C with G, while O and X are almost never confused. Eleanor Gibson and her colleagues used such confusions in the 1960s to work out a set of distinctive features for the letters of the alphabet. Ulric Neisser showed in 1964 that it is much faster to find a Z hidden in a list of rounded letters (O, Q, D, C) than in a list of angular ones (X, V, M, W), because in the first list the Z is the only item with straight-line features, while in the second every item shares them.

The best-known feature model was proposed in 1959 by the computer scientist **Oliver Selfridge** (1926 to 2008), one of the founders of artificial intelligence. He called it **Pandemonium**, after the capital of hell in John Milton's *Paradise Lost*, because it imagined recognition as the work of crowds of shrieking "demons", each a simple processing unit (Figure 4.7). *Image demons* record the incoming pattern. *Feature demons*, each responsible for one feature, inspect the image and shout when they find their feature. *Cognitive demons*, one for each letter, listen to the feature demons and shout in proportion to how many of their own features are present. A *decision demon* listens to the cognitive demons and picks the loudest. Presented with R, the demons for P and B shout fairly loudly, because they share many features with R, but the R demon shouts loudest of all. Two features of the model are important. All the demons at each level work at the same time (parallel processing), and the model could learn by adjusting how much weight each cognitive demon gave to each feature, which made it an ancestor of today's neural network models. Neisser's *Cognitive Psychology* (1967) made it famous among psychologists.

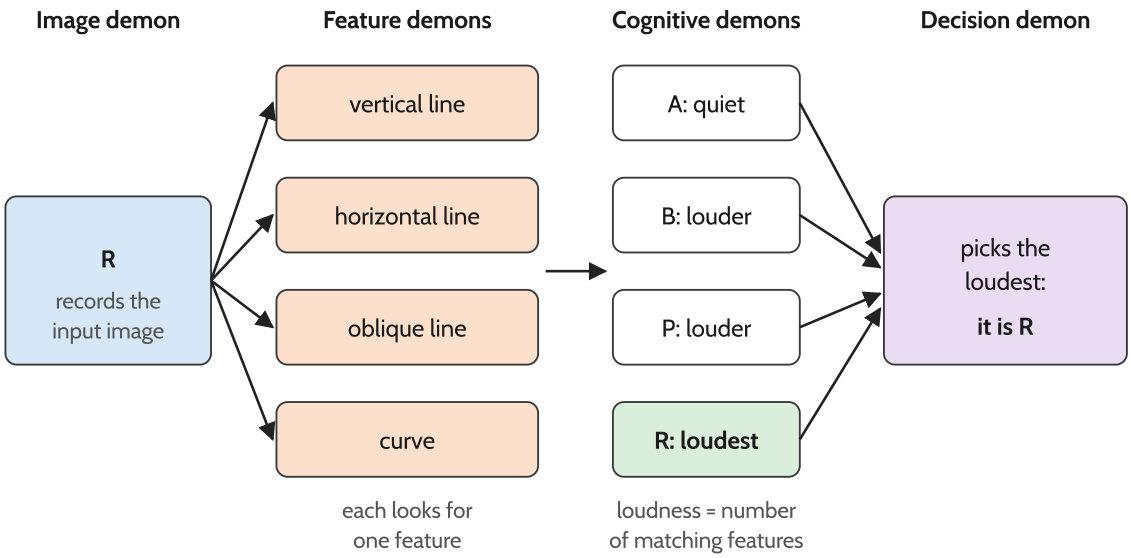


Figure 4.7 Selfridge's Pandemonium (1959) recognising the letter R. Feature demons report the features present; every cognitive demon shouts in proportion to its matching features, so B and P shout too, but R shouts loudest and the decision demon chooses it. Notice that all the demons at each stage work in parallel.

Feature theories received spectacular support from the brain. In the late 1950s, at Johns Hopkins University, the neurophysiologists **David Hubel** (1926 to 2013) and **Torsten Wiesel** (born 1924) began recording from single neurons in the visual cortex of cats.

### KEY STUDY: HUBEL AND WIESEL'S FEATURE DETECTORS (1959 ONWARDS)

**Question.** What kinds of stimuli make individual neurons in the primary visual cortex fire? **Method.** Hubel and Wiesel lowered fine microelectrodes into the striate (primary visual) cortex of anaesthetised cats and recorded the activity of one neuron at a time while projecting simple patterns of light onto a screen in front of the animal. **Results.** For hours, the spots of light that worked so well for retinal cells produced almost nothing. Then, as they slid a glass slide into the projector, one cell began to fire furiously: the faint shadow of the slide's straight edge, sweeping across the screen at a particular angle, was the stimulus it wanted. Systematic work over the next years distinguished **simple cells**, which respond best to a line or edge of a particular orientation at a particular place in the visual field; **complex cells**, which respond to a line of a particular orientation anywhere within a larger field, often only when it moves in a particular direction; and hypercomplex (end-stopped) cells, which respond best to lines of a particular length or to corners. **Conclusion.** The visual cortex contains **feature detectors**, neurons that respond selectively to particular features of a stimulus, arranged so that simple features are combined into more complex ones as information moves through the system. Hubel and Wiesel shared the 1981 Nobel Prize in Physiology or Medicine with Roger Sperry. **Criticism.** Later work showed that such cells are better described as filters tuned to patterns of light and dark at different scales than as detectors of "lines" in the everyday sense, and that their responses are shaped by context and attention. The anatomy is covered in the Physiological book.

In the same year, 1959, Jerome Lettvin and his colleagues published "What the frog's eye tells the frog's brain", describing cells in the frog's visual system that responded best to small dark

moving spots, nicknamed bug detectors; this is the reason a frog snaps at a moving speck but ignores a dead fly. Further along the primate visual system, cells in the inferior temporal cortex respond to complex shapes such as hands and faces, and in 2005 Rodrigo Quian Quiroga and his colleagues found single neurons in the human medial temporal lobe that fired to many different photographs of one particular film star and not to other people. The brain seems to build complex detectors out of simple ones, much as Pandemonium proposed.

Feature theories nevertheless have serious limits. A list of features says nothing about how the features are arranged, yet T, L and + are all one vertical and one horizontal line: the relations between features matter as much as the features themselves, so the stored description must be a structural description, not a list. Features also struggle with three-dimensional objects seen from different angles, and they cannot explain the effects of context that you will meet below. And there is evidence that the whole is often processed before its parts. In 1977 David Navon showed people large letters made up of small letters (a big H made of small S's): they identified the large letter faster than the small ones, and the large letter interfered with naming the small letters more than the reverse. This global precedence is just what the Gestalt psychologists would have expected.

### Prototypes

A third approach replaces exact copies with averages. According to **prototype theory**, memory stores for each category a **prototype**: an idealised, abstract representation of the typical member, the central tendency of all the examples one has met, which need not correspond to any example actually seen. An incoming pattern is recognised as belonging to the category whose prototype it most resembles; it does not need to match exactly, only closely enough. Your prototype of a dog is a sort of average dog, and you recognise a new breed as a dog because it is reasonably similar to it. The idea is most closely associated with **Eleanor Rosch** (born 1938), whose work in the 1970s on natural categories showed that some members of a category are more typical than others (a robin is a better example of a bird than a penguin) and are recognised faster (concepts and categories are treated fully in Chapter 8).

The most elegant evidence came from **Michael Posner** (born 1936) and Steven Keele in 1968. They created random patterns of dots as prototypes, then made distorted versions of each by moving the dots slightly. Participants learned to sort the distortions into categories but never saw the prototypes themselves. In a later test, the never-seen prototypes were classified as accurately as the distortions that had actually been studied, and more accurately than new distortions, and the advantage held up better over time. People had abstracted the prototype from its variants. Prototype theory thus explains recognition of new examples and of typicality, which templates cannot. Its weakness is that it throws away information that people do seem to keep: we know that dogs vary enormously in size and that some birds cannot fly. A rival, exemplar theory, holds that we store many individual examples and compare new items with all of them.

EXAM TRAP

"An idealised representative object of a category" is a **prototype**. A **template** is an exact stored copy that must match point for point; a **feature** is an elementary part such as a line or curve; **geons** are Biederman's three-dimensional building-block shapes. **Pandemonium** is Selfridge's 1959 feature model with image, feature, cognitive and decision demons. Hubel and Wiesel's feature detectors were found first in the **cat**; the frog's "bug detectors" are Lettvin's.

## Recognition-by-components: Biederman's geons

Templates, features and prototypes were mostly developed for flat patterns such as letters. How do we recognise solid objects seen from any angle? The most influential answer is the **recognition-by-components theory** (RBC) proposed in 1987 by the American psychologist **Irving Biederman**. Its central claim is that every object can be described as an arrangement of a small number of basic three-dimensional shapes, which Biederman called **geons** (short for geometric ions): blocks, cylinders, spheres, cones, wedges, arcs and so on. He estimated that about 36 geons are enough. That sounds far too few, until you remember that about 44 phonemes are enough to build every word of spoken English. What matters is the combination and arrangement. A cup is an arc attached to the side of a cylinder; a bucket is the same two geons with the arc attached to the top.

How are geons found in the image? Biederman proposed that the visual system first extracts the edges of the object and then divides it into parts at regions of deep concavity, the places where the contour bends sharply inwards, such as where the handle of a cup meets its body. Each part is then identified as a geon from **non-accidental properties**, properties of edges in the image that are very unlikely to arise by accident from a particular viewpoint and that therefore stay the same from almost any angle. He listed five: curvature, parallel lines, cotermination (edges ending at a common point), symmetry and collinearity (points lying along a straight line). A straight edge in the image almost always means a straight edge in the object; parallel edges almost always mean parallel edges. Because geons can be identified from almost any viewpoint, the theory predicts that recognition should be largely viewpoint-invariant: an object should be recognised equally easily from most angles, unless a key geon is hidden.

The theory made a striking prediction about damaged pictures. Biederman deleted the same amount of contour from line drawings of objects in two ways. When the deletions were in the middle of line segments, people still recognised the objects easily, because the concavities and non-accidental properties survived. When the deletions removed the vertices and concavities, the objects became almost impossible to identify. Biederman and Peter Gerhardstein (1993) also found that seeing an object primed later naming of the same object almost as well when it was rotated by 135 degrees as when the view was identical, which fits viewpoint invariance.

The evaluation is mixed. RBC explains how we tell a cup from a bucket or a dog from a chair, but not how we tell one breed of dog, or one face, from another, because those discriminations depend on fine differences in shape and surface that geons ignore. Recognition is often viewpoint-dependent after all: Michael Tarr and Heinrich Bülthoff (1995), using novel objects learned from particular views, found that recognition was faster and more accurate from the familiar views. Colour and surface texture help recognition more than a pure edge theory al-

lows. And like Marr's theory, below, RBC is almost entirely bottom-up, with little role for expectation and context.

### Marr's computational theory

Biederman built on the work of the British neuroscientist **David Marr** (1945 to 1980), who worked at the Massachusetts Institute of Technology and died of leukaemia at the age of 35; his book *Vision* was published after his death, in 1982. Marr made two lasting contributions. The first was a way of thinking. Any information-processing system, he argued, must be understood at three levels: the computational level (what problem is being solved, and why), the algorithmic or representational level (what representations and procedures solve it) and the implementational level (how the procedures are carried out in neurons or silicon). This is the computational approach you met by name in Section 4.1.

The second was a theory of how vision proceeds, through a series of increasingly rich representations:

- The **primal sketch** is a two-dimensional description of the main changes in light intensity in the image: edges, bars, blobs and line endings. Marr divided it into a raw primal sketch (the intensity changes themselves) and a full primal sketch (in which these are grouped, by Gestalt-like principles, into contours and regions).
- The **2.5-D sketch** (two-and-a-half-dimensional sketch) describes the depth and orientation of every visible surface, worked out from cues such as binocular disparity, motion, shading and texture (Section 4.3). It is *viewer-centred*: it describes the surfaces as they face this observer from this viewpoint, and it says nothing about the hidden back of the object.
- The **3-D model representation** describes the object's three-dimensional shape and the arrangement of its parts in an *object-centred* frame, independent of the observer's viewpoint, typically as a hierarchy of cylinder-like components arranged along axes. Only at this stage can the object be matched with stored descriptions and recognised from any angle.

Marr's achievement was to show how much computation recognition requires, and to inspire a generation of computational models. His theory, however, is overwhelmingly bottom-up (Marr himself admitted that top-down processing is "sometimes used and necessary" but largely left it out), and no one has yet shown how the fully viewpoint-independent 3-D models he proposed could actually be computed. Table 4.8 compares the five approaches.

Table 4.8 Theories of pattern and object recognition.

| Theory | Key name and year | What is stored | Strength | Weakness |
|---|---|---|---|---|
| Template matching | (Early machine models) | Exact copies | Works for standardised input | Cannot cope with variation |
| Feature analysis | Selfridge, 1959; Hubel and Wiesel, 1959 | Lists of distinctive features | Handles size and style; neural support | Ignores relations between features and context |
| Prototype | Posner and Keele, 1968; Rosch, 1970s | An idealised average | Explains new examples and typicality | Loses information about variation |
| Recognition-by-components | Biederman, 1987 | Arrangements of about 36 geons | Viewpoint invariance; damaged drawings | Poor at fine discriminations; bottom-up only |

| Theory | Key name and year | What is stored | Strength | Weakness |
|---|---|---|---|---|
| Computational | Marr, 1982 | 3-D model descriptions | Precise stages: primal sketch, 2.5-D sketch, 3-D model | Neglects top-down processing |

EXAM TRAP

The theory that objects are recognised from **geons** is Biederman's **recognition-by-components** theory (1987), not Pandemonium, not prototype theory and not template matching. Marr's stages come in the order **primal sketch, 2.5-D sketch, 3-D model**; the 2.5-D sketch is **viewer-centred** and the 3-D model is **object-centred**. Marr's book *Vision* appeared in **1982**, two years after his death.

## Faces: a special case

Faces pose the hardest recognition problem of all. Every face has the same parts in the same arrangement, two eyes above a nose above a mouth, so faces cannot be told apart by listing their features or geons. What distinguishes one face from another is the fine spacing and shape of the parts, their configuration, and there is good evidence that faces are processed holistically, as wholes, more than other objects are. In 1969 Robert Yin showed that turning photographs upside down impairs the recognition of faces much more than the recognition of houses or aeroplanes, the **face inversion effect**; inversion seems to disrupt configural processing. In the Thatcher illusion, described by Peter Thompson in 1980, a photograph of Margaret Thatcher with the eyes and mouth cut out and turned upside down looks grotesque when the face is upright but almost normal when the whole picture is inverted. Brain imaging by Nancy Kanwisher and her colleagues (1997) identified a region in the fusiform gyrus of the temporal lobe, the **fusiform face area**, that responds much more strongly to faces than to other objects.

The clearest evidence that face recognition is special comes from people who have lost it. **Prosopagnosia** (face blindness), named in 1947 by the German neurologist **Joachim Bodamer** (1910 to 1985) from the Greek for "face" and "not knowing", is a severe difficulty in recognising familiar faces, including one's own in a mirror, with much less impairment in recognising other objects. Acquired prosopagnosia follows damage to the occipitotemporal region, including the fusiform gyrus. Developmental prosopagnosia occurs without any obvious brain damage and may affect up to about 2 percent of people; Oliver Sacks, whom you met in Section 4.3, had it, and the American writer Heather Sellers did not realise until well into adulthood why she sometimes failed to recognise her own mother and husband. People with prosopagnosia recognise others by their voices, hairstyles, clothes and walk. Some of them show covert recognition: their skin conductance rises when they see a familiar face that they insist they do not know, a dissociation between recognition and awareness that anticipates Section 4.9.

CLINICAL LINK

In Capgras delusion, a patient insists that a spouse or parent has been replaced by an identical impostor. One influential explanation, proposed by Hadyn Ellis and Andrew Young in 1990, is that it is the mirror image of covert recognition in prosopagnosia: conscious recognition of the face is intact, but the emotional response that normally accompanies a familiar face is missing, and the patient explains the strange lack of feeling with a delusion. The disorders of face perception show how recognition, emotional familiarity and belief can come apart (Disorders and Physiological books).

### Top-down help: the word superiority effect and context

Every theory so far has been mainly bottom-up, building recognition from the stimulus. Yet recognition is also driven by what we know. The clearest demonstration is with letters and words. As long ago as 1886, **James McKeen Cattell** (1860 to 1944) found that in a very brief exposure readers could report more letters when the letters formed a word than when they were a random string. The modern version was devised by **Gerald Reicher** in 1969 and refined by Daniel Wheeler in 1970. A string of letters, such as WORD, a single letter such as D, or a non-word such as ORWD, is flashed for a fraction of a second and then covered by a pattern mask. The participant must say which of two letters appeared in a given position, for example D or K in the fourth position. Both choices are chosen so that they make a word (WORD or WORK), so guessing from knowledge of words cannot help. Even so, the letter is identified more accurately when it is part of a word than when it is part of a non-word, or even when it is shown alone. This is the **word superiority effect**: a letter is perceived more accurately in a word than in isolation or in a meaningless string.

How can a whole word be easier than one of its letters? In 1981 James McClelland and David Rumelhart explained it with their interactive activation model, in which information flows both ways between levels: features activate letters, letters activate words, and activated words send activation back down to their own letters, strengthening them. The letter D in WORD gets support from above; the D in ORWD gets none. The model is a modern, more precise descendant of Pandemonium, with the crucial addition of top-down connections.

Context works the same way beyond the page. Selfridge himself drew attention to the phrase THE CAT written with an identical, ambiguous character in the middle of both words, half H and half A: everyone reads it as THE CAT, without noticing that the same shape has been read as two different letters. Stephen Palmer showed in 1975 that after looking at a kitchen scene, people identified a briefly flashed loaf of bread more accurately than a mailbox or a drum of similar shape. In 1970 Richard Warren played listeners a recorded sentence in which one sound in the word "legislatures" had been replaced by a cough; nearly all of them heard the missing sound clearly and could not say where the cough had occurred (phonemic restoration). In all of these, bottom-up information from the stimulus and top-down information from knowledge and context are combined, just as Neisser's perceptual cycle in Section 4.1 proposed. The next section looks more closely at the top-down side: at how expectations, needs, feelings, experience and culture shape what we see.

EXAM TRAP

The **word superiority effect** (Reicher, 1969) is better identification of a **letter** when it appears in a **word** than when it appears alone or in a non-word; it is evidence for **top-down** processing. **Prosopagnosia** is face blindness, linked to the **fusiform** region; it is not a loss of all object recognition (that is visual object agnosia) and not a memory disorder. The **face inversion effect** is Yin (1969); the Thatcher illusion is Thompson (1980).

## 4.8 Perceptual Set, Learning and Culture

LOWER YIELD · 1 Q

If perception were simply a matter of recording the stimulus, everyone would see the same thing in the same picture. They do not. A hungry person, a frightened person, a person who has just been looking at pictures of animals and a person who grew up in a forest may all look at the same figure and see different things. This section examines the perceiver's contribution to perception, which Section 4.1 described in general terms as top-down processing.

### Perceptual set: seeing what we expect

A **perceptual set** is a readiness to perceive a stimulus in a particular way: in the definition given in 1955 by the American psychologist **Floyd Henry Allport** (1890 to 1979), "a perceptual bias or predisposition or readiness to perceive particular features of a stimulus". A set can be created by expectation, context, instructions, recent experience, needs, emotions, values and culture. It works in two ways: it makes us notice some things rather than others (selection, closely linked to selective attention, Chapter 5), and it makes us interpret ambiguous input one way rather than another (interpretation).

Look at Figure 4.8. The shape in the middle is identical whichever way you read it, yet read across it is the letter B and read downwards it is the number 13. **Jerome Seymour Bruner** (1915 to 2016) and Leigh Minturn used exactly this ambiguous figure in 1955: participants who had just seen a series of letters drew it as a B, and those who had just seen numbers drew it as 13. This is the **context effect**, the influence of surrounding stimuli and of the situation on the perception of a target.

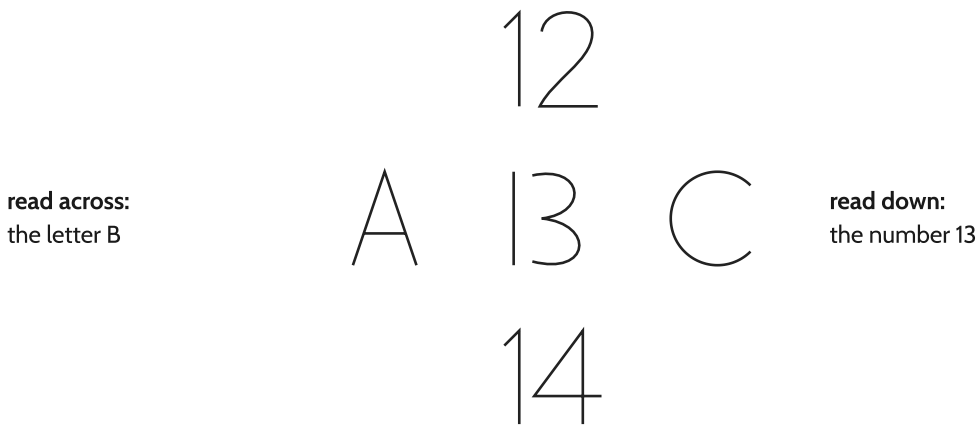


Figure 4.8 The context effect. The central shape is physically identical in both readings: between A and C it is seen as the letter B, between 12 and 14 as the number 13 (after Bruner and Minturn, 1955). The stimulus does not decide; the context does.

Recent experience creates sets just as powerfully. In 1961 B. R. Bugelski and Delia Alampay showed participants an ambiguous drawing that can be seen either as a rat or as a bald man wearing glasses. Those who had first seen a series of pictures of animals overwhelmingly saw a rat; those who had seen no such series, or pictures of people, mostly saw a man. As early as 1935 Robert Leeper had shown that a brief look at an unambiguous version of the "wife and mother-in-law" figure (Section 4.5) determined which woman people saw in the ambiguous one. The same set makes proofreading so hard: read "Paris in the the spring" quickly and most people miss the repeated word, because they see what they expect a sentence to say.

Sets can cross the senses. In 1976 Harry McGurk and John MacDonald dubbed a recording of a voice saying "ba" onto a film of a face saying "ga". Viewers heard "da", a fusion of the two, and went on hearing it even when they knew the trick; with their eyes shut they heard "ba". The **McGurk effect** shows that speech perception integrates sight and sound automatically, and that what we see changes what we hear.

CLINICAL LINK

Clinicians have perceptual sets too. In 1973 David Rosenhan reported that eight healthy "pseudopatients", admitted to psychiatric hospitals after claiming to hear voices, behaved normally from then on, yet staff interpreted their ordinary behaviour through the diagnosis: one nurse recorded that a patient "engages in writing behaviour" when he was taking notes. Later investigators have questioned how the study was carried out, but the lesson stands. Once a label is applied, it creates a set that shapes what the clinician notices, which is why structured assessment and deliberately considering alternative explanations matter. The same danger, called diagnostic overshadowing, occurs when a physical illness in a person with a psychiatric diagnosis is wrongly put down to the psychiatric condition.

## Needs, values and emotions: the New Look ●●

In the late 1940s Bruner and his colleagues launched what became known as the **New Look** in perception, the claim that perception is shaped not only by expectation but by the perceiver's

needs, values and emotions. The evidence came from a run of ingenious experiments. In 1948 David McClelland and John Atkinson showed dim or even blank slides to sailors at a naval training base who had gone one, four or sixteen hours without food and asked them what they could see; the longer they had gone hungry, the more often they reported food-related objects. Values had similar effects. In 1948 Leo Postman, Bruner and Elliott McGinnies measured the values of their participants with the Allport-Vernon Study of Values and found that words related to a person's strongest values (for example, religious words for people with strong religious values) were recognised at shorter exposure times than words related to weak values.

### KEY STUDY: BRUNER AND GOODMAN'S COINS (1947)

**Question.** Do needs and values change the perceived size of objects? **Method.** Bruner and **Cecile Goodman** tested ten-year-old children in Boston. The experimental group, made up of children from a well-off progressive school and children from a settlement house in a poor district, adjusted the size of a circular patch of light until it matched the size of coins (1, 5, 10, 25 and 50 cent pieces) held in the palm of the hand. A control group made the same judgements for cardboard discs of the same sizes. **Results.** Coins were judged larger than they really were, and on the whole the more valuable the coin, the greater the overestimation; cardboard discs were judged much more accurately. Poor children overestimated the coins considerably more than rich children did. **Conclusion.** The value of an object and the perceiver's need for it can influence its perceived size: economic deprivation affects perception. **Criticism.** The samples were small, and the rich and poor children differed in many ways besides income, including familiarity with handling coins. In 1949 Launor Carter and Kermit Schooler found the rich-poor difference when coins were judged from memory but not when they were present, which suggests that the effect may lie partly in memory and judgement rather than in perception itself.

The most controversial New Look claim was **perceptual defence**, the idea that we resist perceiving stimuli that are threatening or emotionally disturbing. In 1949 **Elliott McGinnies** flashed words in a tachistoscope (a device for presenting visual stimuli for precisely controlled, very short times) at gradually longer exposures until participants could read them. Emotionally loaded taboo words needed longer exposures to be recognised than neutral words. More intriguingly, the galvanic skin response, a measure of emotional arousal, rose for the taboo words *before* participants reported seeing them, as if some part of the mind had read the word and decided to keep it out of awareness. Critics were quick to respond. Davis Howes and Richard Solomon (1950) pointed out that taboo words are less frequent in print, and less frequent words need longer exposures anyway; and participants, especially students reporting to a stranger in 1949, may simply have hesitated to say such words aloud until they were certain. The opposite tendency, faster recognition of threatening stimuli, is called perceptual vigilance, and it is now the better supported of the two. In 1951 **Richard Lazarus** (1922 to 2002) and Robert McCleary paired some nonsense syllables with electric shocks; later, when the syllables were flashed too briefly to be identified correctly, the shock-paired syllables still produced larger skin-conductance responses. They called this discrimination without awareness **subception**, a bridge to the subliminal perception of Section 4.9.

Emotions and bodily states shape perception in more everyday ways too. People who are anxious are more likely to interpret ambiguous stimuli as threatening: asked to write down spoken words that sound the same but have different meanings, such as die and dye, anxious

people more often write the threatening spelling. Some studies have reported that hills look steeper to people who are tired or carrying a heavy backpack, although other researchers argue that such effects reflect what participants think the experimenter wants rather than a change in what they see. The safest conclusion is that needs, values and emotions certainly influence attention, interpretation and judgement, and probably influence perception itself, but less dramatically than the New Look first claimed.

EXAM TRAP

"Who demonstrated that economic deprivation affects perception?" **Bruner and Goodman** (1947, the coin study). **Segall** and **Herskovits** (with Campbell) studied culture and illusions (the carpentered world), not poverty. K. D. Broota and H. C. Ganguli are Indian psychologists who wrote on cultural differences in perceptual selectivity in the 1970s, but the classic study of need and perceived size is not theirs. **Perceptual defence** is **McGinnies** (1949); **subception** is **Lazarus and McCleary** (1951); the **New Look** is Bruner's movement. A perceptual set is a **readiness to perceive** in a particular way (Allport, 1955), and the B or 13 figure is **Bruner and Minturn** (1955).

## Perceptual learning

Expectation and need change perception from moment to moment. Experience changes it for good. **Perceptual learning** is a relatively lasting improvement in the ability to perceive, especially to discriminate between stimuli, as a result of experience. A tea taster in an Assam garden distinguishes grades of leaf that taste identical to a visitor; a radiologist sees a small tumour in a chest X-ray that a medical student cannot find even when it is pointed out; a cricket umpire judges in an instant whether the ball pitched in line with the stumps; a student of Hindustani music gradually comes to hear the fine differences of pitch between the same note in two different ragas.

What changes? In 1955 James and Eleanor Gibson posed the question in a paper titled "Perceptual learning: differentiation or enrichment?" On the enrichment view, held by most psychologists since Berkeley, learning adds associations and memories to an impoverished sensation, so that we come to see more than is there. The Gibsons argued for differentiation: learning makes perception more specific, so that we come to detect more of what is there, picking up distinctive features and variables of the stimulus that we had not noticed before. In their experiment participants were shown a meaningless "scribble", a curly line, and then a long series of similar scribbles that varied in the number of coils, their compression and their orientation, with copies of the original mixed in; their task was to say which were identical to the first. At first they accepted many of the variants as the same; with repeated runs, and without any feedback about which answers were right, they came to pick out exactly the right items, and adults learned faster than children. Eleanor Gibson developed the idea in her book *Principles of Perceptual Learning and Development* (1969): perceptual learning is the learning of distinctive features, of invariants and of higher-order structure, and it makes attention more economical, so that the expert looks at less but sees more. It is also why, as Section 4.2 warned, what counts as "similar" or "familiar" for the Gestalt laws depends partly on experience.

## Perceptual adaptation: Stratton's upside-down world

How flexible is the adult visual system? In 1896 the American psychologist **George Malcolm Stratton** (1865 to 1957), who had trained in Wundt's laboratory, put a lens system over one eye that turned the visual world upside down and reversed left and right, and covered the other eye. In a first experiment he wore it for about three days; in 1897 he wore it for eight days, removing it only in darkness or with his eyes closed. At first everything was chaos. He reached for things in the wrong direction, stumbled, felt nauseated, and found that the world swung wildly whenever he moved his head. Day by day his movements became more accurate and the scene less disturbing, and by the end there were periods when things seemed to be in their proper places and he could move about with some ease. When he finally removed the lenses, the normal world looked strange for a while, and his movements were briefly wrong in the opposite direction.

Stratton had demonstrated **perceptual adaptation**: the ability of the perceptual system to adjust to an artificially displaced, distorted or inverted visual input. Later researchers, notably Ivo Kohler at Innsbruck in the 1950s, had participants wear inverting or distorting goggles for weeks; they came to perform complex skilled activities, and the distortions became less noticeable. With prisms that shift the visual world a few degrees to one side, adaptation is much faster: people's reaching errors shrink within minutes, and when the prisms are removed they show a negative aftereffect, reaching too far the other way, which proves that something in the system really changed. How complete the visual change is remains debated. Much of the adaptation seems to lie in the coordination between vision and movement, the felt position of the body, rather than in the picture itself turning the right way up.

Movement, it turns out, is essential to that coordination. **Richard Held** (1922 to 2016) and Alan Hein demonstrated this in a classic experiment of 1963.

### KEY STUDY: HELD AND HEIN'S KITTEN CAROUSEL (1963)

**Question.** Is visual experience enough for normal visually guided behaviour to develop, or must the animal produce its own movement? **Method.** Kittens were reared in darkness and then spent a few hours a day in pairs in a "kitten carousel": a cylinder with vertical stripes on its walls. One kitten walked around freely and actively, pulling a central arm; the other rode passively in a gondola attached to the other end of the arm, so that it received the same visual stimulation but did not produce its own movement. **Results.** When tested, the active kittens behaved normally: they put out their paws when carried towards a surface, blinked at approaching objects and avoided the deep side of a visual cliff (Section 4.3). The passive kittens did not, until they were allowed to move about freely for some days. **Conclusion.** The development of visually guided behaviour requires self-produced movement; visual input that is not linked to one's own actions is not enough. **Criticism.** The passive kittens' movements were restricted in ways other than self-production, and some later studies found a smaller difference; but the result fits the visual cliff finding that fear of heights appears with self-produced locomotion.

Experience matters most early in life. When Austin Riesen reared chimpanzees in darkness in the 1940s, they later had great difficulty learning simple visual discriminations, although part of the problem turned out to be damage to the retina from lack of light. When Colin Blakemore and Grahame Cooper (1970) raised kittens for their first months in a cylinder painted only with vertical stripes, the cats later behaved as if blind to horizontal edges, and

neurons in their visual cortex responded almost only to vertical orientations. Hubel and Wiesel showed that closing one eye of a kitten during the first few months of life leaves the cortex almost unresponsive to that eye, while the same closure in an adult cat has little effect. There is a sensitive period early in life when visual experience shapes the developing visual brain (Developmental and Physiological books).

> EXAM TRAP
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> **Stratton** (1896, 1897) wore **inverting lenses**: perceptual **adaptation**. **Held and Hein** (1963) used the **kitten carousel**: active versus passive movement. **Blakemore and Cooper** (1970) reared kittens with **vertical stripes** only. **Gibson and Gibson** (1955) argued that perceptual learning is **differentiation**, not enrichment. Do not confuse perceptual adaptation (adjusting to distorted input) with sensory adaptation (receptors becoming less responsive to an unchanging stimulus, Chapter 3).

### Restored sight: S.B. and his successors

The most direct test of what experience contributes comes from people who gain sight after a lifetime of blindness. The question was put in 1688 by the Irish scientist **William Molyneux** (1656 to 1698) in a letter to the philosopher John Locke, who printed it in the second edition of his *Essay Concerning Human Understanding* (1694). **Molyneux's question** asks: if a man born blind, who has learned to tell a cube from a sphere by touch, were suddenly made to see, could he tell which was which by sight alone, before touching them? Molyneux and Locke both answered no. In 1932 Marius von Senden collected the published reports of people operated on for congenital cataracts and concluded that most could at first see colours and tell figure from ground, but could not recognise even simple shapes by sight, and learned to do so only slowly and with great effort.

The case you met at the start of this chapter was reported in 1963 by Richard Gregory and **Jean Wallace** in their monograph *Recovery from Early Blindness*. S.B.'s story fits von Senden's conclusion and adds a twist: whatever he had known by touch transferred to vision quickly. He read the time from a clock because he had felt the hands of his watch; he read capital letters but not lower-case ones because he had learned only capitals by touch; he understood the lathe only when he had felt it. His perception of distance was poor, as his belief that he could touch the ground from his high window showed. And Gregory reported that he was less affected than normally sighted people by some of the classic illusions, and that the Necker cube did not reverse in depth for him, as if he lacked the stored depth interpretations on which those effects depend.

Modern cases tell a similar story. Mike May, an American blinded in a chemical accident at the age of three and a half, had his sight restored in one eye at 46. When Ione Fine and her colleagues tested him in 2003, he could see colour and motion well, but even years later he struggled to recognise faces and objects from their shape alone and to see depth in flat drawings. Three-dimensional shape, faces and pictorial depth, the most learned parts of perception, were the parts that did not come back.

INDIAN CONTEXT

India has made Molyneux's question testable on a large scale. Project Prakash ("light" in Sanskrit), founded in the mid-2000s by the Indian-born MIT neuroscientist **Pawan Sinha**, has provided cataract surgery to many blind children and young people in India, most of them from poor rural families, and studied how their vision develops. In 2011 Richard Held, Sinha and their colleagues reported that newly sighted participants could not at first match a shape they had felt with the same shape seen, the answer "no" that Molyneux and Locke had predicted, but that within a few days they learned to do so. Other Prakash studies found that the newly sighted at first had great difficulty separating overlapping shapes, and that motion helped them most to bind parts into objects: common fate (Section 4.2) seems to be one of the first grouping principles available.

## Culture and perception

If experience shapes perception, people who grow up in different physical and cultural environments should perceive differently. You have already met the carpentered-world evidence for illusions (Section 4.5) and the Zulu children who were less fooled by the Ames window. Several other lines of evidence point the same way.

The first concerns distance. The British anthropologist **Colin Turnbull** (1924 to 1994), in *The Forest People* (1961), described a journey out of the dense Ituri rainforest of the Congo with Kenge, a young BaMbuti man who had spent his whole life in the forest, where one can rarely see more than a few dozen metres. On the open plains Kenge saw a herd of buffalo several kilometres away and asked what kind of insects they were. As they drove towards the herd, the "insects" grew larger, and Kenge, frightened and convinced that witchcraft was at work, could not understand how. Lacking experience of judging great distances, he had no basis for size constancy at that range (Section 4.4).

The second concerns pictures. In 1960 the South African psychologist **William Hudson** showed people a set of simple line drawings in which a hunter aims his spear, with an antelope in the middle distance and an elephant drawn small on a hill far behind it. Most people who had been to school, and who were used to pictures, saw the hunter aiming at the nearby antelope; many unschooled participants said he was aiming at the elephant, interpreting the drawing as flat, without reading its pictorial depth cues. Later, Jan Deregowski found that some African participants preferred "split" drawings of an elephant showing it as if flattened out from above, with all four legs visible, to the perspective drawings that Western viewers prefer. Reading depth in pictures, it seems, is a skill learned through exposure to a pictorial culture.

The third concerns what people attend to. In 2001 Takahiko Masuda and Richard Nisbett showed Japanese and American students short animated underwater scenes. Asked to describe them, the Americans tended to begin with the large, fast-moving focal fish; the Japanese more often began with the background ("it looked like a pond") and mentioned the relations between objects and their surroundings. When the focal fish was later shown against a new background, Japanese participants' recognition suffered more. Nisbett argued that East Asian cultures encourage a more holistic, context-sensitive way of seeing and Western cultures a more analytic, object-focused one.

A related individual difference, with cultural roots, is field dependence. In a series of studies beginning in the 1940s, the American psychologist **Herman Witkin** (1916 to 1979) seated peo-

ple in a dark room and asked them to set a luminous rod to the true vertical while it was surrounded by a tilted luminous frame (the rod-and-frame test), or to find a simple shape hidden within a complex pattern (the embedded figures test). Some people's judgements were strongly pulled by the surrounding frame or pattern; others ignored it and relied on the gravitational feel of their own bodies or on the parts themselves. Witkin called this cognitive style **field dependence-independence**: field-dependent people perceive in a more global way, strongly influenced by the surrounding field, while field-independent people perceive analytically, separating items from their context. The Canadian cross-cultural psychologist **John Berry** (born 1939) found in 1966 that Inuit hunters of Baffin Island, whose survival depended on picking out small details in a featureless snowy landscape, were on average more field-independent than Temne farmers of Sierra Leone, whose society stressed conformity; his ecocultural approach holds that each group's ecology and economy shape the perceptual skills it encourages.

INDIAN CONTEXT

Berry's ecocultural approach has been tested extensively in India. Ramesh C. Mishra, Durganand Sinha and John Berry studied Adivasi communities in Bihar, comparing the Birhor, traditionally nomadic hunters and gatherers, with neighbouring groups who had settled into agriculture, and reported their findings in 1996 in the book *Ecology, Acculturation and Psychological Adaptation*. Their broad conclusion was that a group's ecology and way of life, its degree of acculturation (contact with town life) and schooling were all related to performance on perceptual and cognitive tasks, as the ecocultural model predicts.

EXAM TRAP

**Field dependence-independence** is **Witkin**'s perceptual or cognitive style (rod-and-frame test, embedded figures test), not a procedure for studying subliminal perception; that is **semantic priming** (Section 4.9). **Locus of control** is Julian Rotter's personality variable (Personality book), and **signal detection** ability is measured with the methods of Chapter 3. Kenge and the buffalo is **Turnbull** (1961); the hunter, antelope and elephant pictures are **Hudson** (1960); the underwater scenes are **Masuda and Nisbett** (2001).

## 4.9 Perception Without Awareness, and Extrasensory Perception

LOWER YIELD · 2 QS

Throughout this chapter perception has meant conscious seeing and hearing: the vase you see, the moon that looks large, the letter you read. This last section takes up two claims that stretch that idea in opposite directions. The first is that a stimulus can be perceived, and can influence what you think and do, without your ever being aware of it. The second is that information can be perceived without reaching any sense organ at all. The first claim, after a shaky start, is now well supported. The second, after nearly a century of research, is not.

### Subliminal perception and the Vicary hoax ●●

Chapter 3 introduced the absolute threshold, or limen, and the word subliminal for a stimulus below it. **Subliminal perception** is the registration and processing of a stimulus that is

presented too briefly or too weakly to be consciously perceived, shown by its effects on later behaviour, judgements or physiological responses. Its opposite, a stimulus strong enough to be consciously seen or heard, is supraliminal.

The idea burst into public view in 1957. A market researcher named **James Vicary** (1915 to 1977) announced that during screenings of the film *Picnic* at a cinema in Fort Lee, New Jersey, he had flashed the messages "Eat Popcorn" and "Drink Coca-Cola" on the screen every few seconds, each for a tiny fraction of a second, far too briefly to be noticed. He claimed that over six weeks the cinema's sales of Coca-Cola had risen by about 18 percent and of popcorn by nearly 58 percent. The announcement caused a sensation and a moral panic about "hidden persuaders"; broadcasters in several countries banned subliminal advertising. But no one could replicate the result, Vicary never published any details, and in 1962 he admitted in a magazine interview that the study had been a gimmick and that he had collected too little data, if any, to mean anything. The Vicary study is now regarded as a hoax.

The hoax did not settle whether subliminal perception exists; it only showed that this particular claim was false. Answering the real question turned out to require care about what "unaware" means. In 1984 Jim Cheesman and Philip Merikle distinguished two thresholds. A stimulus is below the *subjective* threshold when the person says they saw nothing, and below the *objective* threshold when they cannot even guess its identity better than chance in a forced-choice test. Many early demonstrations used stimuli that were below the subjective threshold only; participants who said they saw nothing might simply have been cautious about saying so, a matter of response criterion in the sense of signal detection theory (Chapter 3).

The most influential evidence used **priming**, the effect of an earlier stimulus (the prime) on the processing of a later one (the target). In **semantic priming**, a prime makes it easier to recognise a target that is related to it in meaning: in 1971 David Meyer and Roger Schvaneveldt found that people decide that BUTTER is a word faster if they have just seen BREAD than if they have just seen NURSE. Semantic priming became the standard procedure for studying subliminal perception when the British psychologist **Anthony Marcel** reported in 1983 that primes flashed so briefly and so heavily masked that participants could not tell whether any word had been presented still speeded responses to related target words. A word that was never consciously seen had been processed for its meaning. Later work with stricter tests of awareness has confirmed that masked primes can affect responses, although the effects are small and short-lived.

Other studies showed effects on feelings and choices. In 1980 William Kunst-Wilson and Robert Zajonc flashed irregular octagons for one millisecond each. Later, shown pairs of octagons, participants could not tell which ones they had seen (recognition was at chance), yet they reliably said that they *liked* the previously shown ones better: the mere exposure effect without awareness. In 2006 Johan Karremans and his colleagues flashed the brand name of a soft drink, Lipton Ice, below awareness; afterwards, participants who were thirsty were more likely to choose that drink, while participants who were not thirsty were unaffected. Subliminal influence, in other words, can nudge a goal the person already has; it cannot create one. And the subliminal self-help tapes sold to improve memory or self-esteem fail completely. In 1991 Anthony Greenwald and his colleagues gave volunteers tapes labelled "memory" or "self-esteem", but secretly swapped the labels on half of them. After a month, people

believed they had improved in whatever the label said, but objective tests showed no improvement of either kind: a placebo effect of the label, not of the hidden message.

> EXAM TRAP
>
> The procedure called **semantic priming** (masked priming) is used to study **subliminal perception**. It is not a method for studying signal detection ability (Chapter 3), field dependence-independence (Witkin, Section 4.8) or locus of control (Rotter, Personality book). **Vicary**'s 1957 popcorn and Coca-Cola study was a **hoax**, admitted in 1962; genuine subliminal effects are real but **small, brief and dependent on existing goals**. Subliminal means **below** the threshold of awareness; supraliminal means above it.

### Blindsight and other dissociations ●●

The most dramatic evidence for perception without awareness comes from patients. In the early 1970s the British neuropsychologist **Lawrence Weiskrantz** (1926 to 2018) and his colleagues at Oxford studied D.B., a man who had had part of the primary visual cortex of his right hemisphere removed in an operation to cure severe migraines caused by an abnormal tangle of blood vessels. As expected, he was blind in much of his left visual field: shown a stimulus there, he said he saw nothing at all. But when the researchers insisted that he guess, pointing to where a light might be, saying whether a line was vertical or horizontal, or whether something was moving, he was correct far more often than chance would allow. When he was shown a film of his own accurate performance, he was openly astonished. Weiskrantz named the condition **blindsight** in 1974: the ability of people with damage to the primary visual cortex to respond accurately to visual stimuli in the blind part of the field while denying any conscious experience of seeing them.

How is it possible? Not all visual information goes through the primary visual cortex. Smaller pathways run from the eye to the superior colliculus in the midbrain and onwards to the cortex, and some fibres from the thalamus reach motion-sensitive areas such as MT directly. These routes can support detection, localisation and the sensing of movement, and they feed the dorsal, vision-for-action stream of Section 4.1, which operates largely outside awareness. Some patients even show "affective blindsight", responding differently to happy and fearful faces they cannot see. Sceptics first suggested that blindsight patients were really responding to stray light scattered onto the intact part of the retina, but D.B. could not detect stimuli presented to his natural blind spot, where stray light would have worked just as well. A more serious doubt is that some patients have faint, degraded visual experiences in the "blind" field and are simply very cautious about reporting them; this is the subjective threshold problem again. Even so, blindsight is accepted as a real dissociation between visual processing and visual awareness.

You have now met several such dissociations in this chapter: D.F., who could not see the orientation of a slot but could post a card through it (Section 4.1); the hand that is not fooled by the Ebbinghaus illusion (Section 4.5); prosopagnosic patients whose skin conductance recognises faces they cannot name (Section 4.7). Attention produces dissociations of its own in healthy people. In inattentional blindness, an unexpected object in plain view goes unseen because attention is elsewhere; in change blindness, large changes to a scene go unnoticed when they coincide with a blink, a cut or a flicker. These are covered fully in Chapter 5, together with the wider question of what consciousness is.

EXAM TRAP

**Blindsight** (Weiskrantz, patient D.B., 1974) follows damage to the **primary visual cortex**; the patient denies seeing yet performs above chance on forced-choice tests. Failing to notice significant changes in a scene, even obvious ones, is **change blindness**; failing to see an unexpected object while attending to something else is **inattentional blindness**; **selective attention** is the focusing of processing on some stimuli at the expense of others; **perceptual constancy** (Section 4.4) is a success of perception, not a failure to notice (Chapter 5).

## Extrasensory perception

**Extrasensory perception** (ESP) is the claimed perception of information without the use of any known sensory channel. It is quite different from subliminal perception: a subliminal stimulus does reach the senses, only too weakly to be noticed, whereas ESP claims that information arrives with no sensory input at all. Three forms of ESP are traditionally distinguished, and a fourth phenomenon is usually studied alongside them (Table 4.9 collects the vocabulary of the field).

- **Telepathy** is the direct transfer of thoughts or feelings from one mind to another, mind reading without speech, gesture or any other sensory signal.
- **Clairvoyance** is the perception of objects or events that are hidden from the senses or far away, without any other mind knowing them: naming a card lying face down in a sealed envelope, or "seeing" a fire in a distant city.
- **Precognition** is the perception of events before they happen; its rarely mentioned partner, retrocognition, is paranormal knowledge of past events.
- **Psychokinesis** (PK) is the claimed ability to influence physical objects or events by mental effort alone, such as influencing the fall of dice. It is a claimed power of action, not of perception, so strictly it is not a form of ESP.

ESP and psychokinesis together are called psi phenomena, and **parapsychology** is the field that studies them. Its scientific history begins with **Joseph Banks Rhine** (1895 to 1980), a botanist by training who came to Duke University in North Carolina in 1927, with his wife and colleague Louisa Rhine, to work with the psychologist William McDougall. Rhine's book *Extra-Sensory Perception* (1934) made the term famous, and the Duke Parapsychology Laboratory became the centre of the field. Rhine's chief tool was a pack of **Zener cards**, designed by his Duke colleague, the perception psychologist **Karl Zener** (1903 to 1964). A pack has 25 cards, five each of five symbols: a circle, a cross, three wavy lines, a square and a star. In a clairvoyance test the participant guesses the order of a shuffled pack; in a telepathy test a "sender" looks at each card in turn; in a precognition test the guesses are made before the pack is shuffled. Pure chance gives an average of 5 correct guesses in 25 (20 percent), and Rhine reported participants who averaged well above that over thousands of trials.

Critics found problems with almost every aspect of the early work. The symbols could sometimes be seen faintly through the backs of cheap cards, or reflected in the sender's glasses or cornea; senders could give unintentional cues by their breathing or timing; shuffling was sometimes inadequate; recording errors tended to favour hits; and successful series were more likely to be reported than unsuccessful ones. Laboratories elsewhere generally failed to replicate Rhine's results. The British researcher S. G. Soal produced some of the strongest

card-guessing data of the 1940s, but statistical analysis by Betty Markwick in 1978 showed that his records had been manipulated.

Modern parapsychologists turned to the **Ganzfeld procedure**, developed in the 1970s by **Charles Honorton** (1946 to 1992) and others. *Ganzfeld* is German for "whole field". The receiver relaxes in a reclining chair with halved table-tennis balls taped over the eyes, a soft red light shining on them and steady hissing noise in headphones, which produces a uniform, featureless sensory field; the idea is that reducing ordinary sensory input makes any faint psi signal easier to notice. In another room, a sender concentrates on a randomly chosen target picture or video clip. Afterwards the receiver is shown four pictures, the target and three decoys, and picks the one that best matches their impressions; chance success is 25 percent. In 1994 the social psychologist **Daryl Bem** (born 1938) and Honorton published, in a leading journal, an analysis of computer-controlled Ganzfeld studies reporting a hit rate of about 32 percent. But in 1999 Julie Milton and Richard Wiseman analysed 30 newer Ganzfeld studies from several laboratories and found no significant effect, and the debate has swung back and forth since. In 2011 Bem reported nine experiments that appeared to show precognition, for example that people could guess, better than chance, behind which of two curtains an erotic picture would appear before the computer had chosen its position. Several independent teams repeated his procedures and found nothing, and the episode became one of the triggers of psychology's "replication crisis", the realisation that many published effects in psychology were not reproducible (Research Methods book).

Why do most scientists remain sceptical?

- The effects do not replicate reliably. Positive results in one laboratory are followed by null results in others, and the effects tend to shrink as experimental controls become stricter.
- There is no plausible mechanism. No known form of energy carries information from mind to mind or from the future to the present, and claimed psi effects do not weaken with distance or with time as physical signals do.
- The history of the field contains sensory leakage, poor randomisation, selective reporting (the file-drawer problem, in which unsuccessful studies are never published) and outright fraud.
- Extraordinary claims require extraordinary evidence. A claim that would overturn much of established physics needs far stronger and more consistent evidence than a claim about, say, the effect of hunger on perception. A large prize offered for decades by the stage magician and sceptic James Randi for any demonstration of paranormal ability under agreed test conditions was never won.
- Belief can be explained without psi. People remember the one dream that came true and forget the thousands that did not; they underestimate how often coincidences occur by chance (in a group of just 23 people, the chance that two share a birthday is better than even); and they notice evidence that confirms what they already believe.

Table 4.9 The vocabulary of parapsychology.

| Term | Meaning | Typical test or example |
|---|---|---|
| Extrasensory perception (ESP) | Perception without any known sensory channel | Umbrella term for the next four rows except psychokinesis |
| Telepathy | Mind-to-mind transfer of thoughts | Sender looks at Zener cards; receiver guesses |

| Term | Meaning | Typical test or example |
|---|---|---|
| Clairvoyance | Perceiving hidden or distant objects or events directly | Guessing the order of a shuffled, unseen pack |
| Precognition | Perceiving future events | Guessing the order before the pack is shuffled; Bem, 2011 |
| Retrocognition | Paranormal knowledge of past events | "Seeing" a past event one could not have known |
| Psychokinesis (PK) | Influencing matter by mind alone; action, not perception | Trying to influence the fall of dice |
| Psi | ESP and psychokinesis together | The subject matter of parapsychology |
| Zener cards | 25 cards, 5 each of 5 symbols; chance = 5 hits | Rhine's Duke studies, 1930s |
| Ganzfeld procedure | Receiver in a uniform sensory field picks 1 of 4 targets; chance = 25 percent | Honorton, 1970s; Bem and Honorton, 1994 |

### INDIAN CONTEXT

Claims of extraordinary perception have a long history in India. The third chapter of Patanjali's *Yoga Sutras*, the Vibhuti Pada, describes *siddhis*, powers said to arise from advanced meditation, including knowledge of other minds and of the past and future. Modern parapsychology also put down roots here: a department of psychology and parapsychology was established at Andhra University in Visakhapatnam, associated with K. Ramakrishna Rao, who had worked with Rhine at Duke and later led the institute Rhine founded. The scientific standard is the same everywhere, however: a claimed ability must be demonstrated under controlled conditions that rule out sensory cues and chance, and replicated by independent investigators.

### CLINICAL LINK

Belief in telepathy, clairvoyance or a "sixth sense" is common in the general population and, on its own, says nothing about mental health; cultural context always matters. But odd beliefs and magical thinking that go well beyond the person's culture, such as a conviction that one can read other people's minds or that one's thoughts can cause distant events, are part of the description of schizotypal personality disorder, and magical thinking also appears in obsessive-compulsive disorder. Telling a culturally shared belief from a clinically significant one is a matter of careful assessment (Disorders book).

### EXAM TRAP

**Telepathy** is mind to mind; **clairvoyance** is perceiving objects or events directly, with no other mind involved; **precognition** is perceiving the future; **psychokinesis** is mind over matter and is **not** a form of ESP, although it belongs to psi. **J. B. Rhine** (Duke, *Extra-Sensory Perception*, 1934) used **Zener cards** (five symbols; chance is 5 out of 25). The **Ganzfeld** (Honorton) is a mild sensory-deprivation procedure with a chance rate of 25 percent. ESP is not subliminal perception: subliminal stimuli reach the senses, ESP claims no sensory input at all.

Perception, then, extends below awareness in ways that are now well established, but not, on present evidence, beyond the senses. Everything you perceive arrives through the sense organs described in Chapter 3 and is organised and interpreted by the processes described in this chapter; what you then select from it, and how much of it reaches consciousness, is the subject of Chapter 5.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Selfridge's 1959 feature model with image, feature, cognitive and decision demons is called _____.
2. Hubel and Wiesel called neurons that respond to a line of particular orientation at a particular place _____ cells.
3. An idealised, average representation of a category is a _____.
4. Biederman's recognition-by-components theory describes objects as arrangements of about 36 _____.
5. In Marr's theory the viewer-centred description of visible surfaces is the _____ sketch.
6. Severe difficulty recognising familiar faces is called _____.
7. A letter is identified better in a word than alone: this is the word _____ effect.
8. McGinnies's finding that taboo words need longer exposures to be recognised was called perceptual _____.
9. Stratton wore lenses that turned the visual world upside down, demonstrating perceptual _____.
10. Responding accurately to stimuli in a field one claims not to see, after damage to the primary visual cortex, is called _____.

**Answers:** (1) Pandemonium, (2) simple, (3) prototype, (4) geons, (5) 2.5-D, (6) prosopagnosia, (7) superiority, (8) defence, (9) adaptation, (10) blindsight.

## The case of S.B., revisited

You can now see why S.B.'s first months of sight were so puzzling, and why they mattered so much to Richard Gregory. The bottom-up machinery of vision worked from the start: S.B. saw light, colour and movement, and he could pick out a figure from its ground. What he lacked was everything this chapter has shown to depend on stored knowledge. With no experience of seeing distance, he had no size constancy across the space below his window (Section 4.4); with no visual templates, features or prototypes for most objects, he could not recognise them (Section 4.7); with no stored depth interpretations of flat drawings, some illusions and the reversing Necker cube had little hold on him (Section 4.5). Where knowledge existed, it transferred at once, even from another sense. He read a clock because he had felt a watch, and he saw the lathe once his hands had built the schema that Neisser's perceptual cycle says must direct looking (Section 4.1). "Now that I've felt it I can see" is as clear a statement of top-down processing as the literature contains.

The story did not end happily. Gregory and Wallace reported that as the months passed S.B. became increasingly disappointed and depressed. The visual world he had imagined turned

out to be drab, the flaking paint and blemishes of things troubled him, and he often preferred to sit in the evenings without turning on the light. He died in 1960, less than two years after the operation. Later cases such as Mike May's and the children of Project Prakash show both sides of the lesson: perception in the adult brain is slow to rebuild after a lifetime without sight, yet young brains given sight can learn to see with remarkable speed. Perception, in short, is not a window on the world but a skill, built from the meeting of what arrives at the senses and what the brain already knows.

## Chapter summary

**4.1 What Perception Is.** Sensation is the detection and transduction of physical energy; perception is the selection, organisation and interpretation of the resulting signals. Because many states of the world can produce the same retinal image (the inverse problem), perception must combine bottom-up (data-driven) and top-down (conceptually driven) processing. Constructivist theories (Helmholtz's unconscious inference and likelihood principle, 1867; Gregory's perceptual hypotheses, 1966 and 1970, illustrated by the hollow-face illusion) treat perception as inference, and the Bayesian approach treats it as the combination of prior beliefs with sensory evidence (Ramachandran's light-from-above prior, 1988). J. J. Gibson's direct, ecological theory (1950, 1966, 1979) holds that the optic array, with its invariants, optic flow and affordances, specifies the world without inference; it is the classic bottom-up theory. Neisser's perceptual cycle (1976) links schema, exploration and object in a loop, and Milner and Goodale's two visual streams (1995) suggest that the constructivist and direct theories describe vision-for-perception and vision-for-action respectively.

**4.2 Perceptual Organisation: The Gestalt Laws.** Gestalt psychology (Wertheimer, Koffka, Köhler, from 1912) held that the whole is different from the sum of its parts. The field is first split into figure and ground (Rubin, 1915): the figure is thing-like, bounded and in front; the ground is shapeless and behind, and reversible figures such as Rubin's vase show that organisation is done by the perceiver. Elements are grouped by proximity, similarity, closure, good continuation, common fate and symmetry, and by the modern laws of common region (Palmer, 1992) and connectedness (Palmer and Rock, 1994), all governed by the master law of Prägnanz, the tendency to see the simplest, most stable organisation. Kanizsa's illusory contours (1955) show the laws creating edges that are not there. The laws describe rather than explain, and grouping is partly learned.

**4.3 Depth and Distance Perception.** Berkeley (1709) posed the problem of how a flat retina yields depth. The binocular cues are retinal disparity, which produces stereopsis (Wheatstone's stereoscope, 1838; Julesz's random-dot stereograms, 1960), and convergence. The monocular pictorial cues are relative and familiar size, interposition, linear perspective, texture gradient, aerial perspective, height in the visual field and shading; motion parallax is a monocular cue that needs movement; accommodation and convergence are oculomotor cues. The visual cliff (Gibson and Walk, 1960) showed that infants of 6 to 14 months avoid an apparent drop and that newborn animals of walking species do too; Campos (1970) found that younger infants detect depth before they fear it, and fear grows with self-produced locomotion.

**4.4 Perceptual Constancy.** Perceptual constancy is the perception of stable object properties despite changing sensory input. Size constancy depends on taking distance into account (size-distance invariance), as Holway and Boring (1941) showed by removing distance cues, and

Emmert's law (1881) shows the same relation for afterimages. Shape constancy is a compromise between retinal and real shape (Thouless's phenomenal regression, 1931); lightness constancy follows Wallach's ratio principle (1948), with lateral inhibition producing simultaneous contrast, Mach bands and the Hermann grid; colour constancy discounts the illuminant (Land's retinex theory); and location constancy uses the corollary discharge (Sperry; von Holst, 1950).

**4.5 Illusions.** An illusion is a systematic misinterpretation of a real stimulus, a distortion of perception rather than sensation; personal illusions vary between people, while universal (permanent) illusions are shared. The Müller-Lyer illusion (1889) is best explained by Gregory's inappropriate constancy scaling (1963), and the carpentered-world hypothesis (Segall, Campbell and Herskovits, 1963, 1966) links susceptibility to experience with rectangular buildings. The Ponzo illusion (1911), the Ames room (1946) and the moon illusion (apparent distance hypothesis; Kaufman and Rock, 1962) are further cases of misapplied size-distance scaling, while the Ebbinghaus (Titchener) illusion reflects size contrast and the Zöllner, Poggendorff and Hering illusions distort orientation. The Necker cube (1832) and other ambiguous figures alternate between interpretations; the Penrose triangle (1958) is an impossible figure. Illusions must be distinguished from sensory distortions, hallucinations (perceptions without an adequate stimulus), imagery and delusions (false fixed beliefs), and from memory illusions such as déjà vu and déjà entendu.

**4.6 Motion Perception.** Real motion is detected by the image-retina and eye-head systems, using direction-selective neurons and area MT (V5), whose damage causes akinetopsia (patient L.M., 1983); Johansson's point-light displays (1973) reveal biological motion. Apparent motion includes Wertheimer's phi phenomenon (1912), strictly distinguished from beta movement, and the stroboscopic motion of films. Induced motion (Duncker, 1929) makes a stationary object appear to move within a moving frame, and vection is illusory self-motion. The autokinetic effect, apparent movement of a stationary light in darkness, was used by Sherif (1935) to study norm formation. The motion aftereffect, or waterfall illusion (Addams, 1834), results from adaptation of direction-selective neurons.

**4.7 Pattern and Object Recognition.** Template matching cannot cope with variability. Feature analysis, supported by letter confusions, visual search and Hubel and Wiesel's simple and complex cells (from 1959), is embodied in Selfridge's Pandemonium (1959). Prototype theory (Posner and Keele, 1968; Rosch) stores idealised averages. Biederman's recognition-by-components theory (1987) describes objects as arrangements of about 36 geons identified from non-accidental properties, and Marr's computational theory (1982) proceeds from the primal sketch through the viewer-centred 2.5-D sketch to the object-centred 3-D model. Faces are processed holistically (face inversion effect, fusiform face area), and their recognition can be lost in prosopagnosia (Bodamer, 1947). The word superiority effect (Reicher, 1969) and other context effects show top-down processing.

**4.8 Perceptual Set, Learning and Culture.** A perceptual set is a readiness to perceive in a particular way (Allport, 1955), created by context (Bruner and Minturn's B or 13, 1955), recent experience (Bugelski and Alampay, 1961) and other senses (the McGurk effect, 1976). The New Look showed effects of need and value: hungry sailors saw food (McClelland and Atkinson, 1948), and poor children overestimated coin sizes more than rich children (Bruner and Goodman, 1947). Perceptual defence (McGinnies, 1949) was challenged on grounds of word frequency and response suppression; subception (Lazarus and McCleary, 1951) showed discrimination without awareness. Perceptual learning is differentiation of distinctive features (Gibson and Gibson, 1955). Stratton's inverting lenses (1896, 1897) demonstrated perceptual adaptation, Held and

Hein (1963) showed the need for self-produced movement, and restored-sight cases (Molyneux's question; S.B.; Project Prakash) show which aspects of perception depend on experience. Culture shapes perception of distance (Turnbull's Kenge), pictorial depth (Hudson, 1960), attention to context (Masuda and Nisbett, 2001) and field dependence-independence (Witkin; Berry, 1966).

**4.9 Perception Without Awareness, and Extrasensory Perception.** Subliminal perception is the processing of stimuli below the threshold of awareness. Vicary's 1957 popcorn study was a hoax, but masked semantic priming (Marcel, 1983), mere exposure without recognition (Kunst-Wilson and Zajonc, 1980) and goal-dependent brand effects (Karremans and colleagues, 2006) show small, short-lived genuine effects, while subliminal self-help tapes are placebos. Blindsight (Weiskrantz, patient D.B., 1974) shows visual processing without awareness after damage to the primary visual cortex. Extrasensory perception comprises telepathy, clairvoyance and precognition, studied with psychokinesis under the name parapsychology; Rhine's Zener-card studies at Duke and the Ganzfeld procedure (Honorton; Bem and Honorton, 1994) have not produced replicable evidence, and most scientists remain sceptical because of failed replications, the absence of a mechanism and the history of methodological flaws.

## Who's who

Table 4.10 The people in this chapter.

| Name | When | Examined for |
|---|---|---|
| Hermann von Helmholtz | 1821 to 1894 | Unconscious inference and the likelihood principle; *Handbook of Physiological Optics* (1867) |
| Richard Langton Gregory | 1923 to 2010 | Constructivist theory, perceptual hypotheses, hollow face; inappropriate constancy scaling (1963); image-retina and eye-head systems; S.B. (1963) |
| Thomas Bayes | about 1701 to 1761 | Bayes's theorem; the Bayesian approach (priors plus sensory evidence) |
| Vilayanur S. Ramachandran | born 1951 | Light-from-above prior in shaded discs (1988) |
| James Jerome Gibson | 1904 to 1979 | Direct (ecological) perception; optic array, invariants, optic flow, affordances (1950, 1966, 1979); the bottom-up theorist; differentiation (1955) |
| Ulric Neisser | 1928 to 2012 | *Cognitive Psychology* (1967); perceptual cycle (1976); visual search for Z (1964) |
| David Milner | 1995 | Two visual streams, vision-for-perception and vision-for-action (with Goodale); patient D.F. |
| Melvyn Goodale | 1995 | Two visual streams; Ebbinghaus illusion "deceives the eye but not the hand" (1995) |
| Max Wertheimer | 1880 to 1943 | Phi phenomenon (1912), founding Gestalt psychology; laws of grouping (1923) |
| Kurt Koffka | 1886 to 1941 | Gestalt co-founder; observer in Wertheimer's 1912 experiments |
| Wolfgang Köhler | 1887 to 1967 | Gestalt co-founder; isomorphism |
| Edgar Rubin | 1886 to 1951 | Figure-ground (1915); Rubin's vase |
| Stephen Palmer | 1975, 1992 | Law of common region (1992); connectedness (with Rock, 1994); kitchen-scene context effect (1975) |
| Irvin Rock | 1922 to 1995 | Connectedness (1994); moon illusion with Kaufman (1962); children and reversible figures (1994) |

| Name | When | Examined for |
|---|---|---|
| Gaetano Kanizsa | 1913 to 1993 | Illusory contours, the Kanizsa triangle (1955) |
| George Berkeley | 1685 to 1753 | *An Essay Towards a New Theory of Vision* (1709): distance is learned |
| Charles Wheatstone | 1802 to 1875 | The stereoscope (1838) |
| Béla Julesz | 1928 to 2003 | Random-dot stereograms (1960) |
| Eleanor Jack Gibson | 1910 to 2002 | Visual cliff (with Walk, 1960); perceptual learning as differentiation (1955, 1969); distinctive features of letters |
| Richard Walk | 1920 to 1999 | Visual cliff (1960) |
| Joseph Campos | 1970 | Heart-rate slowing on the deep side: depth detected before it is feared |
| Susan Barry | stereopsis in her late forties | Neurobiologist who gained stereoscopic vision in adulthood |
| Oliver Sacks | 1933 to 2015 | Neurologist and writer; described Susan Barry; had developmental prosopagnosia |
| Alfred Holway | 1941 | Size constancy depends on distance cues (with Boring) |
| Edwin Garrigues Boring | 1886 to 1968 | Holway and Boring (1941); brought "My wife and my mother-in-law" into psychology (1930) |
| Emil Emmert | 1844 to 1911 | Emmert's law of afterimage size (1881) |
| Robert Thouless | 1894 to 1984 | Phenomenal regression to the real object (1931) |
| Hans Wallach | 1904 to 1998 | Ratio principle of lightness constancy (1948) |
| Ernst Mach | 1838 to 1916 | Mach bands (1865) |
| Ludimar Hermann | 1838 to 1914 | Hermann grid (1870) |
| Edwin Land | 1909 to 1991 | Retinex theory of colour constancy; inventor of the Polaroid camera |
| Roger Sperry | 1913 to 1994 | Corollary discharge (1950); shared the 1981 Nobel Prize with Hubel and Wiesel |
| Erich von Holst | 1908 to 1962 | Efference copy (1950) |
| Franz Carl Müller-Lyer | 1857 to 1916 | Müller-Lyer illusion (1889) |
| Marshall Segall | 1963, 1966 | Carpentered-world hypothesis (with Campbell and Herskovits) |
| Donald Thomas Campbell | 1916 to 1996 | Carpentered-world studies; arbitrary autokinetic norms across generations (with Jacobs, 1961) |
| Melville Jean Herskovits | 1895 to 1963 | Anthropologist in the carpentered-world studies; *The Influence of Culture on Visual Perception* (1966) |
| Mario Ponzo | 1882 to 1960 | Ponzo illusion (1911) |
| Adelbert Ames Jr. | 1880 to 1955 | Ames room (1946) and rotating trapezoidal window; transactionalism |
| Gordon Willard Allport | 1897 to 1967 | Zulu children and the Ames window (with Pettigrew, 1957) |
| Ibn al-Haytham (Alhazen) | about 965 to about 1040 | Apparent distance explanation of the moon illusion |
| Lloyd Kaufman | 1962 | Moon illusion experiments supporting apparent distance (with Rock) |
| Hermann Ebbinghaus | 1850 to 1909 | Ebbinghaus illusion; pioneer of memory research |
| Edward Bradford Titchener | 1867 to 1927 | Popularised the Ebbinghaus figure (Titchener circles) in his 1901 manual |

| Name | When | Examined for |
|---|---|---|
| Johann Karl Friedrich Zöllner | 1834 to 1882 | Zöllner illusion (1860) |
| Johann Christian Poggendorff | 1796 to 1877 | Poggendorff illusion (1860) |
| Ewald Hering | 1834 to 1918 | Hering illusion (1861); opponent-process theory of colour |
| Louis Albert Necker | 1786 to 1861 | Necker cube (1832) |
| Lionel Penrose | 1898 to 1972 | Penrose triangle and staircase (1958, with his son) |
| Roger Penrose | born 1931 | Penrose triangle and staircase (1958); Nobel Prize in Physics (2020) |
| Gunnar Johansson | 1911 to 1998 | Biological motion from point-light displays (1973) |
| Karl Duncker | 1903 to 1940 | Induced motion (1929); also problem solving (Chapter 8) |
| Muzafer Sherif | 1906 to 1988 | Norm formation with the autokinetic effect (1935; *The Psychology of Social Norms*, 1936) |
| Oliver Selfridge | 1926 to 2008 | Pandemonium model (1959); the THE CAT demonstration |
| David Hubel | 1926 to 2013 | Feature detectors: simple and complex cells (from 1959); Nobel Prize (1981) |
| Torsten Wiesel | born 1924 | Feature detectors with Hubel; monocular deprivation in kittens; Nobel Prize (1981) |
| Eleanor Rosch | born 1938 | Prototypes and natural categories (1970s) |
| Michael Posner | born 1936 | Abstraction of unseen prototypes from dot patterns (with Keele, 1968) |
| Irving Biederman | 1987 | Recognition-by-components theory; about 36 geons; non-accidental properties |
| David Marr | 1945 to 1980 | Computational theory; primal sketch, 2.5-D sketch, 3-D model; *Vision* (1982) |
| Joachim Bodamer | 1910 to 1985 | Named prosopagnosia (1947) |
| James McKeen Cattell | 1860 to 1944 | Letters reported better in words than in random strings (1886) |
| Gerald Reicher | 1969 | Word superiority effect |
| Floyd Henry Allport | 1890 to 1979 | Definition of perceptual set (1955) |
| Jerome Seymour Bruner | 1915 to 2016 | New Look; coin study (with Goodman, 1947); B or 13 (with Minturn, 1955) |
| Cecile Goodman | 1947 | Coin study: economic need and perceived size (with Bruner) |
| Elliott McGinnies | 1949 | Perceptual defence for taboo words |
| Richard Lazarus | 1922 to 2002 | Subception (with McCleary, 1951) |
| George Malcolm Stratton | 1865 to 1957 | Inverting lenses (1896, 1897): perceptual adaptation |
| Richard Held | 1922 to 2016 | Kitten carousel (with Hein, 1963); Prakash test of Molyneux's question (2011) |
| William Molyneux | 1656 to 1698 | Molyneux's question (1688) |
| Jean Wallace | 1963 | Studied S.B. with Gregory (*Recovery from Early Blindness*) |
| Pawan Sinha | 2000s to present | Project Prakash: sight restoration and visual learning in India |
| Colin Turnbull | 1924 to 1994 | Kenge and the "insect" buffalo (*The Forest People*, 1961) |
| William Hudson | 1960 | Cultural differences in pictorial depth perception |

| Name | When | Examined for |
|---|---|---|
| Herman Witkin | 1916 to 1979 | Field dependence-independence; rod-and-frame and embedded figures tests |
| John Berry | born 1939 | Ecocultural model; Inuit and Temne field dependence (1966) |
| James Vicary | 1915 to 1977 | Subliminal advertising hoax (1957, admitted 1962) |
| Anthony Marcel | 1983 | Masked semantic priming without awareness |
| Lawrence Weiskrantz | 1926 to 2018 | Blindsight; patient D.B. (1974) |
| Joseph Banks Rhine | 1895 to 1980 | Parapsychology at Duke; *Extra-Sensory Perception* (1934) |
| Karl Zener | 1903 to 1964 | Zener cards |
| Charles Honorton | 1946 to 1992 | Ganzfeld procedure (1970s) |
| Daryl Bem | born 1938 | Ganzfeld analysis with Honorton (1994); precognition studies (2011) |

## Key terms

sensation, perception, inverse problem, bottom-up processing, top-down processing, unconscious inference, likelihood principle, constructivist approach, perceptual hypothesis, hollow-face illusion, Bayesian approach, prior, likelihood, direct perception, ecological approach, optic array, invariants, optic flow, focus of expansion, affordances, perceptual cycle, schema, vision-for-perception, vision-for-action, information-processing approach, computational approach, figure, ground, figure-ground, Rubin's vase, reversible figure, laws of perceptual organisation, law of proximity, law of similarity, law of closure, law of good continuation, law of common fate, law of symmetry, law of common region, connectedness, law of Prägnanz, illusory contours, amodal completion, depth cues, depth perception, binocular cues, retinal disparity, horopter, stereopsis, stereoscope, convergence, monocular cues, pictorial cues, relative size, familiar size, interposition, linear perspective, texture gradient, aerial perspective, height in the visual field, shading, motion parallax, accommodation, oculomotor cue, visual cliff, perceptual constancy, size constancy, size-distance invariance hypothesis, Emmert's law, shape constancy, phenomenal regression to the real object, brightness constancy, ratio principle, simultaneous contrast, lateral inhibition, Mach bands, Hermann grid, colour constancy, location constancy, corollary discharge, illusion, personal illusions, universal illusions, Müller-Lyer illusion, inappropriate constancy scaling, carpentered-world hypothesis, Ponzo illusion, Ames room, moon illusion, apparent distance hypothesis, Ebbinghaus illusion, Zöllner illusion, Poggendorff illusion, Hering illusion, ambiguous figure, Necker cube, impossible figure, hallucination, delusion, sensory distortions, sensory deceptions, completion illusion, affect illusion, pareidolia, déjà vu, déjà entendu, jamais vu, real motion, image-retina system, eye-head system, akinetopsia, biological motion, apparent motion, phi phenomenon, beta movement, stroboscopic motion, induced motion, vection, autokinetic effect, motion aftereffect, pattern recognition, template matching, feature analysis, distinctive features, Pandemonium, simple cells, complex cells, feature detectors, prototype theory, prototype, recognition-by-components theory, geons, non-accidental properties, primal sketch, 2.5-D sketch, 3-D model representation, face inversion effect, fusiform face area, prosopagnosia, word superiority effect, perceptual set, context effect, McGurk effect, New Look, perceptual defence, subception, perceptual learning, perceptual adaptation, Molyneux's question, field dependence-independence, subliminal perception, priming, semantic priming, blindsight, extrasensory perception, telepathy, clairvoyance, precognition, psychokinesis, parapsychology, Zener cards, Ganzfeld procedure

## From the past papers

1. Which approach to perception emphasises the role of unconscious inference and probabilistic reasoning in interpreting sensory information? (RML 2023)

   (a) Computational approach (b) Constructivist approach (c) Direct perception approach (d) Information processing approach

   **Answer: (b) Constructivist approach.** Unconscious inference is Helmholtz's idea and the core of Gregory's constructivism; its probabilistic, Bayesian form treats the percept as the most likely interpretation. The computational approach (Marr) asks what each stage of vision computes, direct perception (Gibson) denies that inference is needed, and the information-processing approach describes perception as a series of stages without stressing inference.

2. According to the Bayesian approach to perception, our perceptual experiences are influenced by: (RML 2023)

   (a) Genetic predispositions (b) Cultural background (c) Prior beliefs and sensory evidence (d) Emotional states

   **Answer: (c) Prior beliefs and sensory evidence.** Bayes's theorem combines the prior (what is likely, from experience or evolution) with the likelihood (how well each possibility fits the current evidence). Genes, culture and emotion can shape the priors, but none of them is the definition.

3. Which psychologist is associated with the concept of "affordances", which refers to the opportunities for action provided by the environment? (RML 2023)

   (a) Hermann von Helmholtz (b) James J. Gibson (c) Richard Gregory (d) Ulric Neisser

   **Answer: (b) James J. Gibson.** Affordances are central to Gibson's ecological theory (1979): a chair affords sitting, a handle affords grasping. Helmholtz and Gregory are the inference (top-down) theorists, and Neisser proposed the perceptual cycle.

4. According to Wundt, "apperceptive mass" constitutes _____. (IHBAS 2025)

   (a) Compound of elemental processes bound by association (b) Sensation and pure perception (c) Pure perception and preexisting ideas (d) Sensation plus unconscious inference

   **Answer: (c) Pure perception and preexisting ideas.** The apperceptive mass (Herbart's term, taken over by Wundt) is the body of ideas a person already has, into which a new perception is absorbed and so made meaningful. Option (a) describes associationist compounds of elements, (b) leaves out the existing ideas, and (d) mixes in Helmholtz's unconscious inference.

5. The most important school of psychology which has contributed a lot towards perception is: (Amity 2023)

   (a) Psychoanalysis (b) Behaviour school (c) Structural school (d) Gestalt psychology

   **Answer: (d) Gestalt psychology.** Wertheimer, Koffka and Köhler began with the phi phenomenon (1912) and gave perception figure-ground organisation, the grouping laws and Prägnanz. Structuralism analysed sensations into elements, behaviourism set perception aside as unobservable, and psychoanalysis dealt with the unconscious, not perception.

6. According to the law of _____, items which are close together in space and time tend to be perceived as belonging together or forming an organised group. (IHBAS 2024)

   (a) Proximity (b) Continuity (c) Similarity (d) Closure

   **Answer: (a) Proximity.** Nearness in space or time is proximity. Continuity concerns elements on a smooth path, similarity concerns elements that look alike, and closure is the completion of gaps.

7. The tendency to complete a figure is called: (RML 2024)
   (a) Continuation (b) Continuity (c) Closure (d) Similarity
   **Answer: (c) Closure.** A circle drawn with gaps is seen as a complete circle. Continuation and continuity are two names for the same law of good continuation, which is about smooth paths, not gaps; similarity groups look-alike elements.

8. Any figure or form not perceived in parts but organised in perception in the best possible whole form is known as: (RML 2020)
   (a) Law of proximity (b) Law of Prägnanz (c) Law of figure and background (d) Law of continuity
   **Answer: (b) Law of Prägnanz.** Prägnanz, the law of good figure or simplicity, says we see the simplest, most stable and complete organisation possible; it is the master law of which the others are special cases. Figure-ground and the other laws describe particular kinds of organisation.

9. Determining a rule of structure from incomplete evidence and then identifying items that fulfil the rule would be a demonstration of: (Amity 2023)
   (a) Interpolation (b) Extrapolation (c) Interposition (d) Structuring
   **Answer: (d) Structuring.** Imposing an organising rule on incomplete input and then picking out what fits it is structuring. Interpolation fills in between known points, extrapolation extends a trend beyond them, and interposition is a monocular depth cue (overlap).

10. Among monocular cues for depth perception is: (JEMAS 2025)
    (a) Convergence (b) Retinal disparity (c) Assimilation (d) Interposition
    **Answer: (d) Interposition.** An object that partly hides another is seen as nearer, and one eye is enough to see it. Convergence and retinal disparity are the two binocular cues; assimilation is not a depth cue at all (it is Piaget's term, or the opposite of contrast).

11. Retinal disparity, which enables three-dimensional depth perception, is classified as which type of visual depth cue? (Real PYQ 2025)
    (a) Monocular static cue (b) Binocular cue (c) Motion parallax cue (d) Pictorial depth cue
    **Answer: (b) Binocular cue.** Disparity is the difference between the two eyes' images, so it needs two eyes by definition and gives stereopsis. Pictorial cues are monocular and static; motion parallax is monocular and needs movement.

12. Highway engineers sometimes paint stripes on freeway off-ramps to enhance the perception of depth. This uses the cue of: (RML 2024)
    (a) Aerial perspective (b) Parallax (c) Overlap (d) Texture gradients
    **Answer: (d) Texture gradients.** Stripes that get closer together exaggerate the normal crowding of texture with distance and heighten the sense of speed. Aerial perspective is haze, parallax needs relative movement of objects at different distances, and overlap is interposition.

13. The density of stones increases as we look farther away. This phenomenon is due to: (RML 2024)
    (a) Convergence (b) Aerial perspective (c) Interposition (d) Texture gradient
    **Answer: (d) Texture gradient.** Texture elements look smaller, finer and more densely packed with distance. Convergence is a binocular eye-muscle cue, aerial perspective is haziness of distant objects, and interposition is one object blocking another.

14. What is the approximate visual field covered by each human eye? (RML 2024)

    (a) 130 degrees (b) 160 degrees (c) 180 degrees (d) 220 degrees

    **Answer: (c) 180 degrees, as keyed.** Be careful: the better textbook figure for one eye is about 160 degrees horizontally (about 60 degrees towards the nose and 100 towards the temple); 180 to 200 degrees is the field of both eyes together, with about 120 degrees of binocular overlap. Choose 180 for this exact item, but recognise 160 if a question is precise about a single eye.

15. When you stretch your right hand farther away from your body, you still perceive it to be of the same size as your left hand. This is due to: (IHBAS 2020)

    (a) Size constancy (b) Shape constancy (c) Motion parallax (d) Depth perception

    **Answer: (a) Size constancy.** The far hand's retinal image is smaller, but the brain scales it by distance (size-distance invariance). Shape constancy concerns changes of orientation, motion parallax is a depth cue from movement, and depth perception is what size constancy uses, not the name of the effect.

16. Illusions are: (JEMAS 2025)

    (a) misinterpretation of stimuli arising from an external object (b) real perceptual object perceived in a distorted way (c) vivid mental images (d) perception without an adequate external stimulus

    **Answer: (a) Misinterpretation of stimuli arising from an external object.** This is the standard psychopathology definition of an illusion. Option (b) describes what Sims calls a sensory distortion (such as micropsia), (c) describes imagery, and (d) is the definition of a hallucination.

17. Illusions are: (Amity 2023)

    (a) Distortions of existing stimuli (b) The same as hallucinations (c) The result of innate mechanisms (d) Not based on external reality

    **Answer: (a) Distortions of existing stimuli.** In an illusion a real stimulus is misperceived. A hallucination has no adequate external stimulus, so (b) and (d) describe hallucinations; (c) fails because culture and learning shape susceptibility, as the carpentered-world studies show.

18. Optical illusions result from distortion in: (Amity 2023)

    (a) Translation (b) Sensation (c) Perception (d) Adaptation

    **Answer: (c) Perception.** The retinal image of the Müller-Lyer figure is accurate and the receptors transduce it correctly; the error lies in the interpretation. Sensation, transduction ("translation") and sensory adaptation work normally.

19. The rail tracks appearing to converge to all of us is an example of: (RML 2024)

    (a) Permanent illusion (b) Personal illusion (c) Visual illusion (d) Müller-Lyer illusion

    **Answer: (a) Permanent illusion, as keyed.** In the personal versus permanent (universal) classification, an illusion that everyone experiences every time is permanent; a personal illusion, such as a rope seen as a snake, varies between people and occasions. It is also, broadly, a visual illusion, but the item is testing the classification; it is not the Müller-Lyer illusion, and the convergence itself is the depth cue of linear perspective.

20. It has been found that the Müller-Lyer illusion exists in certain cultures in which there are: (RML 2022)

    (a) Lot of trees (b) Buildings with corners (c) Structures without many corners and angles (d) More men than women

    **Answer: (b) Buildings with corners.** Segall, Campbell and Herskovits's carpentered-world hypothesis (1963, 1966) holds that experience of rectangular buildings teaches people to read angles as corners in depth, increasing susceptibility. People living among round huts and few straight edges, option (c), were less susceptible; trees and the sex ratio are irrelevant.

21. The Müller-Lyer illusion is best explained by: (Recall set)

    (a) Misapplied size constancy (inappropriate constancy scaling) (b) Binocular disparity between the two eyes' images (c) Lateral inhibition between neighbouring retinal cells (d) Figure-ground reversal

    **Answer: (a) Misapplied size constancy.** Gregory (1963) argued that the fins and arrowheads are read as inside and outside corners, so one shaft is coded as farther away and scaled up. The illusion works with one eye and on a flat page, ruling out disparity; lateral inhibition explains Mach bands and the Hermann grid; figure-ground reversal is Rubin's vase.

22. Déjà entendu refers to: (IHBAS 2025)

    (a) Illusion of having seen something before (b) Memory of past events with clarity (c) Illusion of having heard something previously (d) Repetitive involuntary speech

    **Answer: (c) Illusion of having heard something previously.** Entendu means "heard". Having seen something before is déjà vu, (a); an unusually clear memory of past events is hypermnesia, (b); involuntary repetition of speech, (d), is a speech disorder such as palilalia.

23. Who demonstrated that economic deprivation affects perception? (IHBAS 2024)

    (a) Broota and Ganguli (b) Bruner and Goodman (c) Segall (d) Herskovits

    **Answer: (b) Bruner and Goodman.** In their 1947 coin study, poor ten-year-olds overestimated the sizes of coins more than rich children did. Segall and Herskovits are the carpentered-world researchers, who studied culture and illusions, and Broota and Ganguli are Indian psychologists who are not associated with this classic finding.

24. The procedure called "semantic priming" is used to study which of the following phenomena? (RML 2022)

    (a) Signal detection ability (b) Subliminal perception (c) Field dependence-independence (d) Locus of control

    **Answer: (b) Subliminal perception.** Masked primes too brief to be consciously seen still speed responses to related words (Marcel, 1983), showing processing without awareness. Signal detection ability is measured by d-prime in yes/no detection tasks, field dependence-independence by Witkin's rod-and-frame and embedded figures tests, and locus of control (Rotter) by personality questionnaires.

25. What term describes the phenomenon where individuals fail to notice significant changes in their environment, even when those changes are obvious? (RML 2023)

    (a) Change blindness (b) Inattentional blindness (c) Perceptual constancy (d) Selective attention

    **Answer: (a) Change blindness.** The stem stresses failing to notice *changes*, typically when they coincide with a blink, cut or flicker (Chapter 5). Inattentional blindness is failing to see an unexpected object while attending to something else; perceptual constancy is the stable perception of objects despite changing input; selective attention is the focusing of processing on some stimuli rather than others.

## High-yield sheet

*The facts this chapter's questions ask most, most-asked first. Last read before the exam.*

1 The only **binocular** depth cues are **retinal disparity** (which gives stereopsis) and **convergence**; interposition, accommodation, motion parallax and all the pictorial cues are **monocular**. ASKED 3 TIMES

2 By the Gestalt law of **proximity**, elements close together in **space or time** are perceived as belonging together. ASKED 3 TIMES

3 Stones or pebbles that look smaller and more densely packed with distance, and stripes painted closer and closer together on an off-ramp, are the monocular cue of **texture gradient** (Gibson's favourite cue). ASKED TWICE

4 An **illusion** is a misinterpretation of a real external stimulus (a distortion of an existing stimulus); a **hallucination** is a perception without an adequate external stimulus; a **delusion** is a false, fixed belief. ASKED TWICE

5 The **Müller-Lyer illusion** (Franz Carl Müller-Lyer, 1889) is best explained by Richard Gregory's **inappropriate constancy scaling** (misapplied size constancy, 1963), and it is strongest in cultures with **buildings with corners** (the carpentered-world hypothesis of Segall, Campbell and Herskovits, 1963 and 1966). ASKED TWICE

6 The **visual cliff** (Eleanor Gibson and Richard Walk, 1960) studied **depth perception** and whether it is innate: 36 infants aged 6 to 14 months; only 3 crept onto the deep side, so depth perception is present **by crawling age**. ASKED 3 TIMES

7 **James J. Gibson** proposed the **bottom-up**, direct (ecological) theory of perception and coined **affordances**, the possibilities for action an object offers. ASKED TWICE

8 The **constructivist** approach (Helmholtz's unconscious inference, 1867; Gregory's perceptual hypotheses, 1966 and 1970) treats perception as inference from sensory data plus knowledge. ASKED ONCE

9 In the **Bayesian** approach, the percept combines **prior beliefs** with **sensory evidence** (the likelihood). ASKED ONCE

10 The tendency to **complete** a figure with gaps is **closure**; continuity (good continuation) is grouping along smooth paths. ASKED ONCE

11 The law of **Prägnanz** (good figure, simplicity) is the master Gestalt law: a form is organised in the best, simplest possible whole. ASKED ONCE

12 **Gestalt psychology** (Max Wertheimer, Kurt Koffka, Wolfgang Köhler; founded in 1912 with the phi phenomenon) is the school that contributed most to the study of perception. ASKED ONCE

13 Your outstretched hand looks the same size as your near hand because of **size constancy**: the brain scales retinal size by distance (size-distance invariance; Holway and Boring, 1941). ASKED ONCE

14 Optical illusions are distortions of **perception**, not of sensation: the retinal image is accurate and the receptors work normally. ASKED ONCE

15 Railway tracks that seem to converge are a **permanent (universal) illusion**, seen by everyone every time; a **personal illusion**, such as a rope seen as a snake, varies with the person and occasion. ASKED ONCE

16 Each eye's visual field is about **160 degrees** horizontally, both eyes together about 200 degrees, with about 120 degrees of binocular overlap; the RML 2024 key gave **180 degrees** for one eye. ASKED ONCE

17 **Bruner and Goodman** (1947) showed that economic deprivation affects perception: poor ten-year-olds overestimated coin sizes more than rich children did. ASKED ONCE

18 **Semantic priming** with masked primes (Marcel, 1983) is the procedure used to study **subliminal perception**; Vicary's 1957 popcorn and Coca-Cola study was a hoax, admitted in 1962. ASKED ONCE

19 **Change blindness** is failing to notice large changes in a scene; **inattentional blindness** is failing to see an unexpected object while attending to something else. ASKED ONCE

20 **Déjà entendu** is the illusion of having **heard** something before; déjà vu is "already seen" and jamais vu is a familiar situation feeling strange. ASKED ONCE

21 The **apperceptive mass** (Herbart's term, taken over by Wundt) is pure perception joined to **preexisting ideas**, the store of ideas into which a new perception is absorbed. ASKED ONCE

22 Working out a rule from incomplete evidence and picking the items that fit it is **structuring**; interpolation fills in between known points, extrapolation extends beyond them, interposition is a depth cue. ASKED ONCE

23 The **figure** (Edgar Rubin, 1915) is thing-like, clearly outlined, usually smaller and seen in front; the **ground** is diffuse and seems spatially distant. ASKED ONCE

24 **Lateral inhibition** enhances edge perception and explains Mach bands and the Hermann grid. ASKED ONCE

25 A **prototype** is an idealised, representative member of a category (Eleanor Rosch; Posner and Keele, 1968), unlike a template, an exact stored copy. ASKED ONCE

26 **Geons** (about 36) belong to Irving Biederman's **recognition-by-components** theory (1987). ASKED ONCE

27 **Saccades** (strictly, microsaccades) are tiny rapid eye movements that ensure **constant perception** by stopping the retinal image from fading. ASKED ONCE

CHAPTER 5

# Attention and Consciousness: Selective Attention, Sleep, Dreams, Hypnosis and Altered States

IN THIS CHAPTER

5.1 ● Attention: what it is, how it behaves and what captures it: define attention and explain its characteristics (selectivity, shifting, span, fluctuation and division) and its external and internal determinants.

5.2 ●● Selective listening: filters, attenuators and late selection: describe the cocktail party problem and compare the models of Broadbent, Treisman, Deutsch and Deutsch, Norman, Johnston and Heinz, and Lavie.

5.3 ● Visual attention: spotlights, binding and blindness to the obvious: explain the spotlight and zoom-lens ideas, Posner's cueing task, feature integration theory, inattentional and change blindness, the attentional blink and inhibition of return.

5.4 ●●● Divided attention, automaticity, the Stroop effect and vigilance: explain capacity and multiple-resource theories, automatic versus controlled processing, what the Stroop effect shows and how sustained attention declines.

5.5 ●●● Consciousness: definitions, levels and the brain: define consciousness, describe James's stream of consciousness and Freud's levels, explain the nonconscious and the hard problem, and define altered states.

5.6 ● Biological rhythms: the body's clocks: describe circadian, ultradian and infradian rhythms, the suprachiasmatic nucleus and melatonin, and the problems of jet lag and shift work.

5.7 ●●● The stages of sleep: identify the EEG waves and the stages N1, N2, N3 and REM, and explain the 90-minute cycle and how sleep changes across the night and the lifespan.

5.8 ●● Why we sleep, what deprivation does and when sleep goes wrong: compare restorative, adaptive and memory theories of sleep, describe deprivation and REM rebound, and survey the main sleep disorders.

5.9 ●● Dreams: compare Freud's, Jung's, activation-synthesis, information-processing and threat-simulation theories, and explain lucid dreams and hypnagogic and hypnopompic imagery.

5.10 ● Hypnosis: trace the history from Mesmer to Freud, separate what hypnosis can and cannot do, and compare neodissociation and social-cognitive theories.

**5.11** • Meditation, sensory deprivation and psychoactive drugs: distinguish types of meditation, describe sensory deprivation research, and classify psychoactive drugs with their effects.

**Weightage.** 69 of the 819 questions mapped in this book (8.4%) test this chapter, 28 of them from real past papers. 3 of its 11 sections are High yield.

In the early 1950s **Eugene Aserinsky** (1921 to 1998) was a graduate student in physiology at the University of Chicago, nearly thirty years old, short of money and living with his wife and young son in a converted army barracks. His supervisor, **Nathaniel Kleitman** (1895 to 1999), was the most senior sleep researcher in the world, and he had given Aserinsky a modest job: watch the eyes of sleepers and see what they do. Most scientists of the day thought that sleep was simply the brain switching off, a long quiet night of nothing, so the project did not look promising. Aserinsky found an old, temperamental recording machine in a basement, repaired it as best he could and began taping electrodes around the eyes of anyone who would sleep in his laboratory.

One of his first volunteers was his eight-year-old son, Armond. As the boy slept, the pens of the machine scratched out slow, lazy lines, exactly as expected. Then, deep into the night, the pens began to jerk sharply from side to side, the tracing of eyes darting about as if the sleeper were wide awake and watching something. Aserinsky assumed that his old machine had failed again. But the boy was asleep, and the jerking pens came back, in bursts, at regular intervals through the night. They came back in adult volunteers too.

Aserinsky then did the obvious thing that nobody had done before: he woke people up during these bursts of eye movement and asked them what had been going on in their minds. Most of them described vivid, story-like dreams. Woken at other times, they usually reported nothing much. In 1953, Aserinsky and Kleitman published their discovery in the journal *Science*: sleep contains recurring periods of rapid eye movement, and these periods are when we dream.

That short paper did something larger than add a fact to physiology. It showed that consciousness does not simply switch on in the morning and off at night. The mind changes state, many times a night, and each state has its own brain activity and its own kind of experience. This chapter is about those states. It begins with attention, the process that decides which tiny part of the world reaches awareness at all, and then moves to consciousness itself, the body's daily clocks, sleep, dreams, hypnosis, meditation and the changes that drugs produce. Keep Armond's dancing eyes in mind; you will meet them again in Section 5.7.

## 5.1 Attention: what it is, how it behaves and what captures it

LOWER YIELD · 1 Q

Sit for a moment in a busy chai stall near the Mysuru bus stand. Buses roar past, the stove hisses, two men argue about cricket, a radio plays a film song, your tea is too hot, your shoe pinches and a crow is calling from the roof. All of this reaches your sense organs at once, yet at any instant you are clearly aware of only a small part of it. If your friend starts telling you about tomorrow's exam, the cricket argument fades into a murmur. Then someone at the next table says your name, and suddenly you are listening to them instead. The process that is doing this choosing is attention.

## Defining attention

The most quoted definition comes from **William James** (1842 to 1910), the American philosopher-psychologist, in his *Principles of Psychology* of 1890: "Everyone knows what attention is. It is the taking possession by the mind, in clear and vivid form, of one out of what seem several simultaneously possible objects or trains of thought. Focalization, concentration, of consciousness are of its essence. It implies withdrawal from some things in order to deal effectively with others." Notice the two halves of that definition. **Attention** is a process of *selection* (one object out of several) and of *concentration* (that object is held clearly and vividly while others are withdrawn from). Modern textbooks say much the same thing in plainer words: attention is the process by which we select some information from the vast amount available to the senses and to memory, and concentrate mental resources on it.

Why do we need such a process at all? Because the brain's capacity to process information fully is limited. You can register a great deal at the level of the sense organs, but you cannot analyse it all for meaning, remember it all or act on it all. Attention is the gatekeeper between the flood of sensation and the narrow channel of awareness, which is why Indian exam notes often summarise the sequence as sensation, then attention, then perception. (Sensation and perception have their own chapters, Chapters 3 and 4.)

Psychologists have classified attention in several ways. A classic classification, set out by **Edward Bradford Titchener** (1867 to 1927), separates **involuntary attention** (his "primary" attention), which is pulled by the stimulus without effort (a sudden bang), from **voluntary attention** (his "secondary" attention), which you direct deliberately and with effort (forcing yourself to read a dull chapter), and adds a third, **habitual attention** (his "derived primary" attention), which began as voluntary but has become effortless through long practice and interest (the cricket fan who notices a score flashing on a distant shop television without trying). A clinical classification, proposed by **McKay Moore Sohlberg** and **Catherine Mateer** in 1987 for rehabilitating brain-injured patients, describes five levels: *focused* attention (responding to a single stimulus at all), *sustained* attention (keeping that response going over time), *selective* attention (holding focus in the presence of distractors), *alternating* attention (switching back and forth between tasks) and *divided* attention (responding to two tasks at the same moment). You will meet each of these again in this chapter.

## The characteristics of attention

Classical general psychology describes attention by five characteristics. Each is a simple observation about how attention behaves, and each is a favourite source of exam questions.

- **Selectivity.** Attention picks out a few stimuli and neglects the rest. The mother asleep through traffic noise but awake at her baby's faintest whimper shows selectivity at work even during sleep.
- **Shifting.** Attention moves from one object to another, often several times a second, as when your eyes and mind move between the teacher, the blackboard and your notes. Shifting can be voluntary or forced by a new stimulus.
- **Span.** Attention can take in only a limited number of items in a single glance. This limit is called the span of attention, and it is examined in detail below.
- **Fluctuation.** Even when you try to keep attention on one unchanging stimulus, its clarity waxes and wanes. This is different from shifting: in fluctuation the object stays the same but your hold on it rises and falls.

- **Division.** Under some conditions attention can be shared between two activities at once, such as walking and talking. How far it can be divided, and why it often cannot, is the subject of Section 5.4.

### The span of attention ●●

The **span of attention** (also called the span of apprehension) is the number of separate objects that can be clearly perceived in a single brief exposure, too short for the eyes to move or for counting. The question is old. The Scottish philosopher **Sir William Hamilton** (1788 to 1856) remarked in his lectures that if you throw a handful of marbles on the floor, you will find it difficult to see more than six, or seven at most, at one view without confusion.

KEY STUDY: JEVONS AND THE BEANS (1871)

The English economist and logician **William Stanley Jevons** (1835 to 1882) turned Hamilton's remark into an experiment, with himself as the only participant. He threw random handfuls of black beans into a shallow box, glanced at them for an instant, called out an estimate of how many there were, and then counted them. Over about a thousand trials he was never wrong with three or four beans; errors began at five, and his accuracy fell steadily as the number rose. He published the result in *Nature* in 1871. Later laboratory work used the **tachistoscope**, an apparatus that flashes a display for a fraction of a second, and arrived at the figure that general psychology textbooks still give: for unrelated items, the adult span of attention is about six to eight, usually summarised as "about seven". For perfectly accurate, effortless enumeration the limit is lower, about four; this rapid, error-free judgement of small numbers was named **subitising** by E. L. Kaufman and colleagues in 1949.

The span of attention explains a lot of everyday design. An Indian vehicle registration number is not printed as one long string such as KA09AB1234. It is broken into short groups (a two-letter state code, a two-digit district code, a letter series and a number of no more than four digits), so that a witness to an accident can take in each group at a glance. The same reason lies behind short telephone groupings and four-digit PINs. The closely related limit on how much we can hold in mind for a few seconds, George Miller's "magical number seven, plus or minus two" of 1956, belongs to short-term memory and is taught in Chapter 7.

EXAM TRAP

When a question asks why number plates carry only about four digits with a few letters, the answer is **span of attention**, not selective, divided or sustained attention. Do not confuse the span of attention (how many items can be apprehended in one glance; Jevons, tachistoscope studies) with the memory span of Miller (how many items can be held and repeated back). The numbers are similar; the processes are not.

### Fluctuation of attention

Hold a watch at the distance where its ticking is just audible and listen carefully. You will notice that the ticking seems to fade out and come back, although the watch has not changed. The classical laboratory demonstration of this **fluctuation of attention** used the **Masson disc**, a white disc carrying a broken line of small grey segments that, when the disc is spun, forms faint grey rings at the threshold of visibility. A person who fixes the disc steadily re-

ports that the faint rings appear and disappear, again and again. Ambiguous figures, such as Rubin's vase-faces figure (see Chapter 4), show something similar: attention cannot be held on one organisation for long and keeps flipping between them. Part of the fluctuation has peripheral causes (small eye movements and the adaptation of receptors), and part is central. The broader lesson is that attention is never static. It needs constant renewal, and when a task gives it nothing new, it drifts. That drift, as you will see in Section 5.5, is where daydreams come from.

## The determinants of attention

What decides which stimulus wins? The **determinants of attention** are traditionally divided into two groups. **External determinants** (also called objective factors) are properties of the stimulus itself, and they explain why advertisers do what they do. **Internal determinants** (subjective factors) are properties of the person: motives, interests, expectations and attitudes. Table 5.1 sets them out.

Table 5.1 The determinants of attention.

| Determinant | What it means | Everyday example |
|---|---|---|
| Intensity | Stronger stimuli capture attention more than weaker ones | A loud horn; a bright floodlight |
| Size | Larger stimuli are noticed before smaller ones | A full-page newspaper advertisement |
| Contrast | A stimulus that differs from its background stands out | One person in white at a wedding of bright silks |
| Change | A sudden change, including sudden silence, is noticed | The fan stopping during a power cut |
| Movement | Moving stimuli capture attention over still ones | Blinking and scrolling shop signs |
| Novelty | New or unusual stimuli attract attention | A camel walking down a Bengaluru road |
| Repetition | A repeated stimulus is more likely to be noticed (up to a point) | A jingle repeated in every IPL break |
| Motives | Needs direct attention toward what satisfies them | A hungry person notices every food stall |
| Interests | We attend to what we care about | A cricket fan spots a score from across a room |
| Mental set | Readiness to perceive a particular stimulus | A runner waiting for the starter's gun |
| Attitudes and emotions | Beliefs and feelings bias what we notice | An anxious student notices every frown |

Two of these deserve a further word. **Mental set** is a temporary readiness to perceive or respond in a particular way, created by instructions, expectation or recent experience. A student waiting for exam results online sees her roll number leap out of a long list because she is set for it. Repetition, meanwhile, cuts both ways. A stimulus repeated a few times gains attention, but a stimulus repeated endlessly without change loses it, because the nervous system habituates to it. This is why advertisers keep changing the details of a campaign while keeping the brand the same, and why external and internal determinants usually work together: the size and colour of the hoarding get your eyes there, and your hunger or interest decides whether you stay.

EXAM TRAP

Learn which list each determinant belongs to. Intensity, size, contrast, change, movement, novelty, repetition, colour and position are **external** (objective). Motives, interests, mental set, attitudes, emotions and habit are **internal** (subjective). "Fluctuation" and "span" are not determinants at all; they are characteristics of attention. Also keep shifting (attention moves to a different object) apart from fluctuation (attention to the same object waxes and wanes).

## 5.2 Selective listening: filters, attenuators and late selection

MEDIUM YIELD · 5 QS

The chai stall showed you a puzzle. You can follow your friend's voice through a wall of noise, and yet your name, spoken by a stranger, still gets through. How can the mind be ignoring the other voices and also listening to them? This puzzle drove the first great theoretical debate of cognitive psychology, and it was fought with headphones.

### The cocktail party problem and dichotic listening ●●

The British engineer **Colin Cherry** (1914 to 1979), working on problems of communication, named the puzzle in 1953. He called it the **cocktail party problem**: how do we manage to follow one conversation when many people are talking at once? The everyday experience of hearing your own name across a crowded room, even though you were not listening to that conversation, is called the **cocktail party phenomenon** (or cocktail party effect).

KEY STUDY: CHERRY'S SHADOWING EXPERIMENTS (1953)

Cherry invented the method that dominated attention research for twenty years. In **dichotic listening**, a person wears headphones and a different message is played to each ear. To make sure they really attend to one ear, they must repeat the attended message aloud, word for word, as it arrives, a technique called **shadowing**. Afterwards they are asked what they noticed in the other, unattended ear. The results were striking. People could report almost nothing about the *content* of the unattended message. They did not notice when it switched from English to German, or when the speech was played backwards. But they almost always noticed *physical* changes: a switch from a male to a female voice, or a pure tone in place of speech. Cherry concluded that the unattended message is processed only for its physical features, not for meaning.

A few years later the British psychologist **Neville Moray** added two findings that pulled in opposite directions. First, in 1959 he showed that a short list of words repeated 35 times in the unattended ear left no trace in memory: people did not recognise the words afterwards. That supported Cherry. Second, about a third of his participants noticed when an instruction in the unattended ear was preceded by their own name. Your name, it seems, can break through a channel you are not listening to. This is the laboratory version of the cocktail party phenomenon, and any theory of attention has to explain both halves of Moray's result.

## Broadbent's filter model (1958): early selection

The first theory came from **Donald Eric Broadbent** (1926 to 1993), a British psychologist who had served in the Royal Air Force and studied the problems of pilots and air traffic controllers swamped with radio messages. In his book *Perception and Communication* (1958) he proposed the **filter model** of attention. This was the first major theory to treat the mind explicitly as an information-processing system, a sequence of stages through which information flows, like messages through a telephone exchange, and it is often described as the starting point of the information-processing approach in cognitive psychology.

Broadbent's model has three parts (see the top row of Figure 5.1). All incoming messages first enter a **sensory buffer**, a brief store that holds them for a second or two. Next comes a **selective filter**, which lets through only one message, chosen on the basis of a *physical* characteristic such as which ear it arrives at, the pitch of the voice or its loudness. Everything else is blocked. Finally, the selected message passes into a **limited-capacity channel** where it is analysed for meaning and can enter awareness and memory. Because the filter works on physical features before any analysis of meaning, Broadbent's is the classic **early selection** model, and because the filter passes one channel and blocks the other completely, like a railway switch, it is sometimes called an all-or-none or "switch" model. The filter is also the first **bottleneck** in attention theory: a narrow point in the system through which only a little information can pass at once.

Broadbent supported his model with the **split-span** experiment (1954). Three pairs of digits were played simultaneously, one digit of each pair to each ear (for example 7 to the left ear and 2 to the right, then 4 and 9, then 3 and 6). People found it much easier to recall the digits ear by ear (743, then 296) than pair by pair (72, 49, 36). Broadbent's explanation was that the filter is set to one ear at a time; recalling by ear means switching the filter only once, while recalling by pairs means switching it back and forth.

The model fitted Cherry's findings well, but it could not survive long. If the unattended message is blocked before meaning is analysed, how could Moray's participants hear their own names? And there was more trouble. **Jeffrey Gray** and **A. A. I. Wedderburn** (1960) played "Dear 7 Jane" to one ear and "9 Aunt 6" to the other; people reported "Dear Aunt Jane", grouping by meaning across the ears, not by ear as the filter predicts. **Anne Marie Treisman** (1935 to 2018) found in 1960 that when the story being shadowed suddenly switched to the other ear, people often followed the story into the "wrong" ear for a word or two before catching themselves: meaning in the unattended ear was guiding them. And in 1972 **Robert Corteen** and **B. Wood** conditioned people to expect a mild electric shock when they heard certain city names; later, when those names (and even other city names never paired with shock) were played in the unattended ear, the participants showed a galvanic skin response, although they could not report hearing the words. Something in the unattended channel was being analysed for meaning.

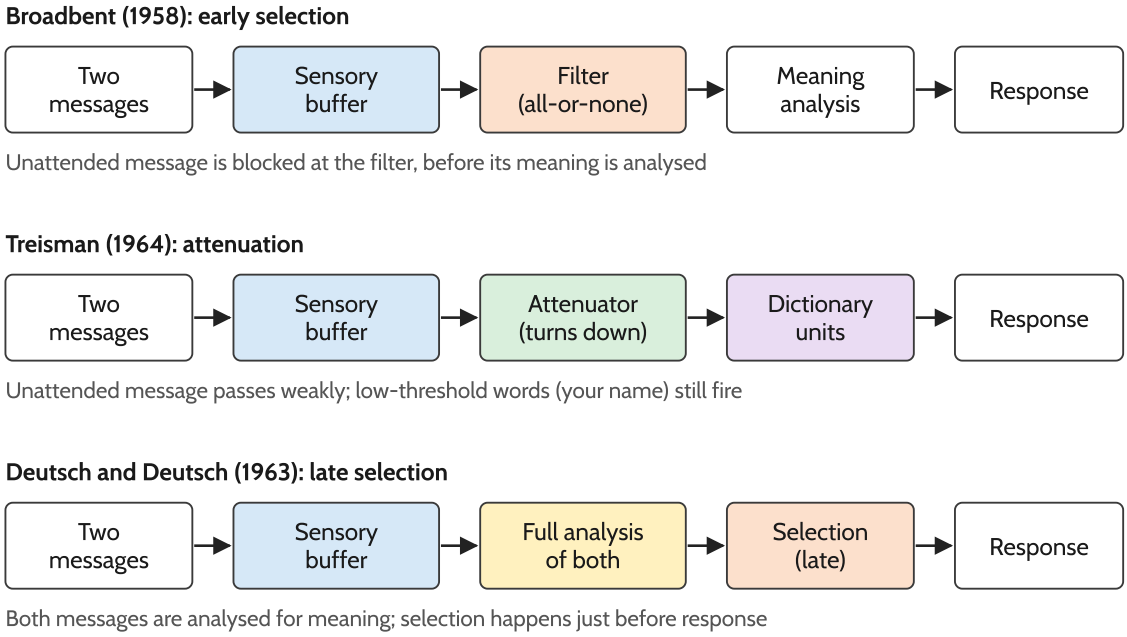


Figure 5.1 Three answers to "where is the bottleneck?". Broadbent places a complete block early, Treisman turns the unattended message down instead of off, and Deutsch and Deutsch let everything be analysed for meaning and select only at the response.

## Treisman's attenuation model (1964) ●●

Treisman's answer, set out fully in 1964, kept Broadbent's idea of an early stage that works on physical features but changed what that stage does. In her **attenuation model**, the filter is not a switch but a volume control, an **attenuator**. The attended message passes at full strength; unattended messages are not blocked but turned down, so that a weakened version still travels on to the next stage (middle row of Figure 5.1).

The next stage is a mental dictionary of **dictionary units**, one for each word you know, and each unit has its own **threshold**, the amount of activation it needs before it fires and the word reaches awareness. Thresholds are not equal. Words that are personally important (your own name, "fire", "help") have permanently low thresholds, so even an attenuated signal is enough to fire them. Words that are made probable by the context of the attended message have their thresholds temporarily lowered too, which is why Treisman's participants followed the story into the wrong ear: the next word of the story was expected, so its unit fired even from the weakened channel. Treisman also proposed that analysis runs through a hierarchy, from physical cues to syllable patterns to individual words to grammar and meaning, and that when capacity runs short, the later stages of analysis are simply not carried out for the unattended message.

The attenuation model explains everything that troubled Broadbent: Moray's own-name effect, Gray and Wedderburn's grouping by meaning, Corteen and Wood's skin responses and Treisman's own ear-switching. That is why it is the standard answer when a question asks which model best explains the cocktail party phenomenon. It is also why Treisman's theory is called a filter-attenuation theory: it is a modified filter theory, still early selection in spirit, but leaky.

EXAM TRAP

"Filter-attenuation theory" is **Treisman**, not Broadbent (pure filter), Johnston or Heinz (multimode theory). The attenuation model is dated **1964** (her ear-switching experiment was 1960). Some Indian notes give 1980 for the attenuation model; 1980 is the year of a *different* Treisman theory, feature integration theory (Treisman and Gelade), taught in Section 5.3. If a question asks which model "better explains the cocktail party effect", the answer is the attenuator model; Broadbent's switch model cannot explain why your name gets through.

### Late selection: Deutsch and Deutsch (1963) and Norman (1968)

At the opposite extreme, **J. Anthony Deutsch** and **Diana Deutsch** argued in 1963 that there is no early filter at all. In their **late selection** model, every incoming message is fully analysed, including for meaning, automatically and without capacity limits. The bottleneck comes only at the end, when the system must decide which analysed message to respond to, remember or become aware of (bottom row of Figure 5.1). Selection is based on *importance*: the most pertinent message wins the response. On this view you do understand the cricket argument at the next table; you simply do not respond to it, and you will not remember it, unless it becomes more important than your friend's voice.

**Donald Arthur Norman** (born 1935) refined the late-selection view in 1968. He proposed that what gets selected depends on two things added together: the strength of the sensory input (how loud and clear it is) and its **pertinence** (how relevant it is to current goals and expectations). A faint word that is highly pertinent can beat a loud word that is irrelevant. Norman's idea gives a late-selection account of the own-name effect: your name is so pertinent that it is selected even though the voice saying it is faint.

So which is right, early or late? The evidence pointed both ways. Treisman and Riley (1969) found that people detected target words far more often in the shadowed message than in the unattended one, which is awkward for a theory in which both are fully analysed. Yet the skin-response and own-name findings show some analysis of meaning in the unattended channel. The debate was eventually resolved not by declaring a winner but by asking a better question: when does selection happen early, and when late?

### Flexible selection: Johnston and Heinz (1978) and Lavie's perceptual load theory

**William A. Johnston** and **Steven P. Heinz** proposed in 1978 that the bottleneck has no fixed position. In their **multimode theory** (flexible multimode model), selection can occur at any of three stages: at the stage of sensory representation (early, on physical features), at the stage of semantic representation (late, on meaning), or after both have been built and are entering consciousness. The listener chooses the mode, but there is a price: the later the selection, the more mental effort and capacity it consumes. Johnston and Heinz showed that when two messages differed physically (one male voice, one female), people selected early and had capacity left for a secondary task; when the messages could be told apart only by meaning, selection had to be late and secondary-task performance suffered. Attention is, in their phrase, flexible but costly.

The most influential modern resolution is the **perceptual load theory** of **Nilli Lavie**, developed in the mid-1990s (with Yehoshua Tsal in 1994, and in her own paper of 1995). Lavie argued that perception has limited capacity and that it runs automatically until that capacity is

full. When the main task carries a *high perceptual load* (many items, hard discriminations), it uses up all the capacity, so nothing is left over for distractors and they are not processed at all: selection looks early. When the main task is easy (*low perceptual load*), spare capacity spills over automatically onto the distractors, which are then processed and can interfere: selection looks late. The early-late debate, on this view, was a debate about task difficulty. A practical prediction follows: if you want to stop being distracted, a harder, more absorbing task helps more than willpower does. (Lavie added that a high load on working memory, a *cognitive* load, has the opposite effect and makes distraction worse, because it weakens the control needed to ignore distractors.) Table 5.2 sets the six theories of selective listening side by side.

Table 5.2 Theories of selective auditory attention.

| Theory | Theorist and year | Where selection happens | What happens to the unattended message |
|---|---|---|---|
| Filter (switch) model | Broadbent, 1958 | Early, on physical features | Blocked completely; meaning never analysed |
| Attenuation model | Treisman, 1964 | Early but leaky | Weakened; important or expected words still reach awareness |
| Late selection | Deutsch and Deutsch, 1963 | Late, at the response | Fully analysed for meaning, then not responded to |
| Pertinence model | Norman, 1968 | Late | Selected if sensory strength plus pertinence is high enough |
| Multimode theory | Johnston and Heinz, 1978 | Flexible: early, late or after both | Depends on the mode chosen; late selection costs more effort |
| Perceptual load theory | Lavie, 1994 to 1995 | Depends on load | Ignored under high load; processed under low load |

EXAM TRAP

Three dating traps cluster here. Deutsch and Deutsch's late selection (**1963**) came *before* Treisman's full attenuation model (**1964**), although textbooks usually teach Treisman first. Broadbent (**1958**) is the answer to "the first major theoretical effort in information-processing psychology". And "bottleneck" in an attention question refers to the narrow stage in processing; it is not a name for the cocktail party phenomenon (the "bottleneck effect" in biology is something else entirely, a sharp fall in a population's genetic variety).

## 5.3 Visual attention: spotlights, binding and blindness to the obvious

LOWER YIELD · 1 Q

The early debate was conducted with the ears, but vision is our dominant sense, and from the 1970s most attention research moved to the eyes. The questions changed shape. What is visual attention *like*: a beam that sweeps across the scene, or something more flexible? What is it *for*, beyond picking one thing out? And how much of the visual world do we actually see when we are not attending to it? The answer to the last question turned out to be: far less than you think.

### The spotlight and the zoom lens

You can attend to something without looking straight at it. Fix your eyes on this word, and without moving them, notice the margin of the page. Moving attention with the eyes is **overt attention**; moving attention while the eyes stay still is **covert attention**. **Michael I. Posner** (born 1936), whose experiments are described below, likened covert attention to a **spotlight**: a beam of limited width that illuminates one region of space, so that whatever falls inside it is processed faster and better, and that can be moved from place to place independently of the eyes. William James had anticipated the idea in 1890 when he described consciousness as having a bright *focus*, surrounded by a dimmer *margin* or *fringe*.

A real spotlight has a fixed beam. Charles Eriksen and James St James argued in 1986 that visual attention is more like a **zoom lens**: you can widen the attended area to take in a whole scene or narrow it to a single letter, but there is a trade-off. The wider the area, the lower the resolution; the same amount of processing power is spread more thinly. A batsman watching the bowler's hand narrows his zoom lens to a tiny patch of space; a fielder at deep midwicket keeps it wide to watch the whole ground. Later work found that under some conditions attention can even be split between two separate locations, like two spotlights, so both metaphors are best treated as approximations of a flexible system.

### Posner's cueing task: endogenous and exogenous attention

#### KEY STUDY: POSNER'S SPATIAL CUEING PARADIGM (1980)

Posner asked people to keep their eyes on a central point and press a key as soon as a light appeared in a box to the left or right. Before the light, a cue appeared. Sometimes it was a *central* cue, an arrow at fixation pointing left or right; sometimes it was a *peripheral* cue, a brief brightening of one of the boxes. Most cues were **valid** (the target appeared where the cue indicated), some were **invalid** (the target appeared on the other side) and on some trials a **neutral** cue gave no information. Responses were fastest after valid cues, slowest after invalid cues and in between after neutral cues. Because the eyes never moved, the benefit and the cost could only mean that attention itself had moved, covertly, to the cued location.

The two kinds of cue revealed two systems. An arrow at fixation must be interpreted and used deliberately; this is **endogenous attention** (also called voluntary, goal-directed or top-down attention), controlled by the person's intentions. It takes a few hundred milliseconds to deploy and, crucially, people ignore central arrows if they learn the arrows are usually wrong. A flash in the periphery grabs attention whether you want it to or not; this is **exogenous attention** (reflexive, stimulus-driven or bottom-up), and it is fast and automatic, pulling attention even when the flash predicts nothing. You use endogenous attention when you search a crowded railway platform for your friend's red bag; exogenous attention takes over when a pigeon suddenly flaps up beside you. Later imaging work by Maurizio Corbetta and Gordon Shulman (2002) mapped these onto a dorsal fronto-parietal network for goal-directed attention and a right-hemisphere ventral network that acts as a "circuit breaker", redirecting attention to unexpected events. Posner himself, with Steven Petersen in 1990, went on to describe three attention networks: alerting (getting ready), orienting (moving attention to a location) and executive control (resolving conflict).

The peripheral flash has one more surprise. If the target appears very soon after an uninformative flash (within about a quarter of a second), responses at the flashed location are faster. If it appears later than about 300 milliseconds, responses at that location become *slower* than elsewhere. Attention, having been pulled to the spot and found nothing, is now biased against going back. Posner and Yoav Cohen described this **inhibition of return** in 1984. Its likely function is to make visual search efficient: once you have checked a location, you are discouraged from wasting time checking it again, rather like a person searching a house who does not reopen a cupboard she has just looked in.

### Feature integration theory (Treisman and Gelade, 1980)

Look at a ripe red tomato. Its redness is analysed by colour-sensitive neurons, its roundness by shape-sensitive neurons, its position by yet other circuits, in different parts of the visual brain. Yet you see one red, round object in one place, not a floating redness next to a floating roundness. How are the separate features put back together? This is the **binding problem**, and Anne Treisman's second great theory, developed with **Garry Gelade** in 1980, proposed that attention is the solution.

**Feature integration theory** divides visual processing into two stages. In the first, **preattentive stage**, simple features (colour, orientation, size, curvature, motion) are registered automatically, in parallel, across the whole visual field, each in its own feature map. No attention is needed and no effort is felt. In the second, **focused attention stage**, attention is directed to one location at a time and acts as a kind of glue: it combines the features found at that location into a single object. Because the glue can be applied to only one location at a time, this stage is serial.

The theory makes a clear prediction about visual search, tested by Treisman and Gelade. Suppose you must find a target among distractors. In **feature search** the target differs from every distractor by a single feature: a red letter among green letters, a vertical line among horizontals. The target is found at the preattentive stage and seems to leap out of the display, a phenomenon called **pop-out**, and search time hardly changes whether there are 5 distractors or 30. In **conjunction search** the target is defined only by a combination of features, say a green T among green X's and brown T's. No single feature map can find it; attention must visit items one by one, binding features at each, so search time rises in a straight line with the number of items. Figure 5.2 shows the pattern. Treisman and Gelade also found that when the target was absent, the conjunction slope was about twice as steep as when it was present, exactly what a serial, self-terminating search predicts: on average you find a present target halfway through, but you must check every item to be sure it is absent.

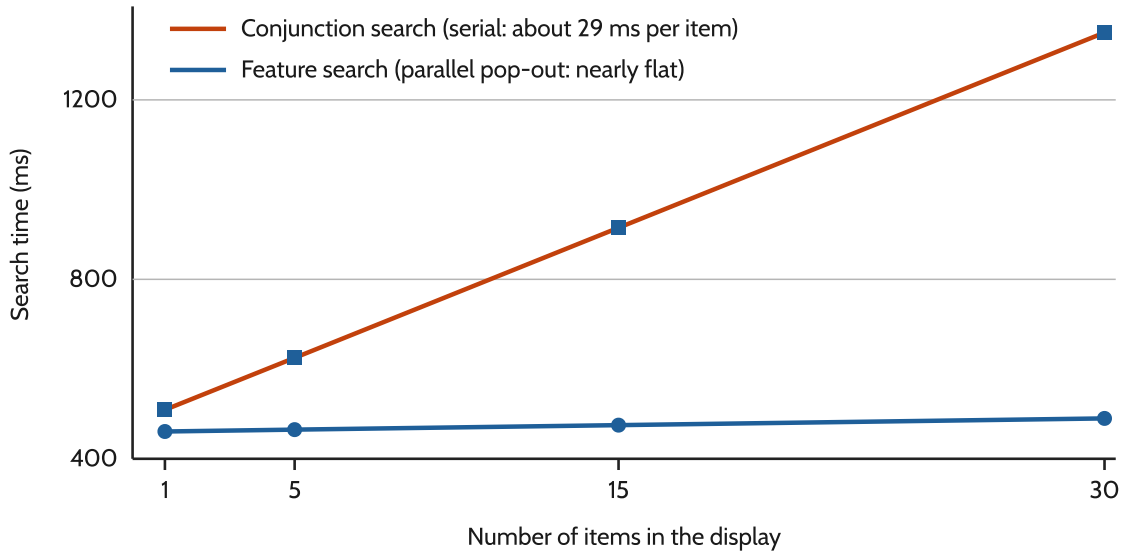


Figure 5.2 Feature versus conjunction search (target-present trials, approximate values in the pattern reported by Treisman and Gelade, 1980). A target defined by one feature pops out, so search time stays almost flat as the display grows; a target defined by a conjunction of features needs serial attention to each item, so search time climbs steadily.

If attention is the glue, then features that are not attended should sometimes be glued wrongly. They are. Treisman and Hilary Schmidt (1982) flashed displays of coloured letters very briefly while people's attention was occupied by a second task. Participants often reported seeing, with confidence, combinations that were not there, such as a red X when the display contained a red T and a green X. These **illusory conjunctions** are strong evidence that features are registered separately and combined only with attention. Patients with damage to both parietal lobes (Balint's syndrome), who have severe difficulty directing spatial attention, produce illusory conjunctions even with long, leisurely displays. Later research qualified the theory (Jeremy Wolfe's guided search model showed that feature information can guide the serial stage, so many conjunction searches are faster than pure serial search), but the distinction between parallel feature detection and attention-dependent binding remains one of the most important ideas in the field.

> **EXAM TRAP**
>
> Anne Treisman gave us **two** theories: the attenuation model of auditory selection (1964) and feature integration theory of visual binding (1980, with Gelade). Questions mixing them up are common. In feature integration theory, the **preattentive** stage is automatic and parallel (features), the **focused attention** stage is serial and binds features into objects; pop-out belongs to feature search, linear search slopes to conjunction search, and illusory conjunctions are the error that occurs when attention is unavailable.

## Blind to the obvious: inattentional blindness and change blindness ●●

Most people believe that they see everything in front of their eyes and that anything unusual would immediately catch their attention. Two families of experiments show that this belief is false.

KEY STUDY: THE INVISIBLE GORILLA (SIMONS AND CHABRIS, 1999)

**Daniel J. Simons** and **Christopher F. Chabris**, then at Harvard University, filmed two teams of students, one in white shirts and one in black, passing basketballs among themselves. Viewers were asked to count the passes made by the white team, a task that demands close attention. Partway through the clip a woman in a full gorilla suit walks into the middle of the players, turns to the camera, thumps her chest and walks off; she is in full view for several seconds. Around half of the viewers did not see her at all, and many refused to believe she had been there until the clip was replayed. When viewers counted the black team's passes instead, far more of them (83 per cent) noticed the black gorilla, because her colour now matched what they were attending to.

The gorilla study is the best-known demonstration of **inattentional blindness**: the failure to notice a fully visible but unexpected object or event because attention is engaged elsewhere. The term was introduced by **Arien Mack** and **Irvin Rock**, whose 1998 book reported that many people failed to see an unexpected shape flashed near the point they were fixating while they judged which arm of a cross was longer. The lesson is that seeing is not just a matter of light reaching the retina; without attention, an object may never reach awareness at all.

**Change blindness** is a related but different failure: the failure to notice that something in a scene has *changed*, even a large change, when the change happens during a brief interruption such as an eye movement, a blink, a cut in a film or a flicker. Ronald Rensink, Kevin O'Regan and James Clark (1997) showed that people can stare for many seconds at two alternating photographs, separated by a brief blank, without spotting that a large object (a jet engine on an aeroplane, for example) appears in one and not the other; this is the *flicker paradigm*. In a famous field version by Simons and Daniel Levin (1998), an experimenter stopped pedestrians on a university campus to ask for directions; while they talked, two workmen carried a door between them, and behind the door the experimenter was replaced by a different person in different clothes. About half the pedestrians carried on giving directions to the new person without noticing the switch. Continuity errors in films, which audiences almost never spot, are everyday change blindness.

Keep the two apart. In inattentional blindness, an unexpected object is *present* and is not seen at all. In change blindness, the person may well see both versions of the scene but fails to notice the *difference* between them, because noticing a change requires attending to the changed item and comparing it across time. Neither should be confused with **perceptual constancy**, the tendency to see objects as having stable size, shape and colour although their image on the retina changes (taught in Chapter 4); constancy is a success of perception, not a failure of attention.

EXAM TRAP

"Failing to notice significant changes in the environment, even obvious ones" is **change blindness** (Rensink and colleagues; the Simons and Levin door study). Failing to see an unexpected object that is in plain view while busy with another task is **inattentional blindness** (Mack and Rock; the gorilla). If the question stresses *change*, choose change blindness; if it stresses an *unexpected object*, choose inattentional blindness.

### The attentional blink

Attention is limited not only in space but in time. In 1992 Jane Raymond, Kimron Shapiro and Karen Arnell used **rapid serial visual presentation**: letters flashed one after another in the same place at about ten per second. Participants had to identify a first target (say, a white letter among black ones) and then report whether a second target (say, the letter X) appeared later in the stream. If the second target came within roughly 200 to 500 milliseconds after the first, it was very often missed; before and after that window it was seen well. This brief "blindness" after a target is called the **attentional blink**. The usual explanation is that attention is still busy consolidating the first target into memory when the second arrives, so the second decays before it can be processed. (Curiously, a second target that comes immediately after the first, with no item between, is often spared, apparently because it slips into the same attentional episode.)

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. The number of objects that can be apprehended in a single brief glance is the _____ of attention.
2. Jevons estimated handfuls of _____ thrown into a box and published his results in 1871.
3. Intensity, contrast and movement are _____ determinants of attention; motives and mental set are internal determinants.
4. In dichotic listening, repeating the attended message aloud is called _____.
5. Broadbent's 1958 model, in which unattended messages are blocked on the basis of physical features, is an _____ selection model.
6. In Treisman's model, unattended messages are weakened by an _____ rather than blocked.
7. Deutsch and Deutsch (1963) argued that all messages are analysed for meaning and selection occurs _____.
8. Responses slowed at a recently cued location after about 300 milliseconds show _____ of return.
9. In feature integration theory, a target that differs by a single feature shows _____ and is found in parallel.
10. Failing to see a gorilla while counting basketball passes is an example of _____ blindness.

**Answers:** (1) span, (2) beans, (3) external, (4) shadowing, (5) early, (6) attenuator, (7) late (at the response stage), (8) inhibition, (9) pop-out, (10) inattentional.

## 5.4 Divided attention, automaticity, the Stroop effect and vigilance

HIGH YIELD · 10 QS

Selective attention asks how we keep one thing in and other things out. **Divided attention** asks the reverse question: how far can we do two things at once? You can walk and talk, and an experienced driver can drive and hold a conversation. But a learner driver cannot, and nobody can hold two conversations at the same time. Why do some pairs of tasks combine easily

and others not at all? Three answers have been proposed: that we have one limited pool of capacity, that we have several pools, and that practice makes some processes stop needing capacity at all.

### Kahneman's capacity model (1973)

In his book *Attention and Effort* (1973), **Daniel Kahneman** (1934 to 2024), later a Nobel laureate in economics, replaced the bottleneck with a different picture. Attention, he proposed, is a single, limited pool of mental effort, or **capacity**, that can be shared out among activities. This is the **capacity model** (also called resource or central capacity theory). Two tasks can be performed together as long as their combined demand does not exceed the capacity available; when it does, performance on one or both suffers. The amount of capacity is not fixed: it rises and falls with arousal, so that a drowsy person has less to share out and a moderately alert one has more. How the capacity is divided depends on an **allocation policy**, which is shaped by four influences: *enduring dispositions* (built-in tendencies to attend to novel or moving stimuli, or to one's own name), *momentary intentions* (what you are trying to do right now), the *evaluation of demands* (the system's judgement of how much each task needs) and the current level of arousal. Kahneman measured effort through the size of the pupil, which widens as a mental task becomes harder.

The capacity model explained why difficult tasks interfere more than easy ones, and why practice, which makes tasks easier, makes them easier to combine. But it had a weakness: it predicts that two equally difficult tasks should interfere equally with a third, and they do not. What matters is how *similar* they are.

### Multiple resource theory and dual-task interference

The study of what happens when people perform two tasks together is called the **dual-task paradigm**, and its main finding is that interference depends heavily on similarity. In a striking demonstration, Alan Allport, Barbara Antonis and Patricia Reynolds (1972) found that skilled pianists could sight-read music while shadowing speech with little loss on either task, although shadowing two spoken messages would have been impossible. **Christopher Wickens** captured this in his **multiple resource theory** (1984, refined in later work). Instead of a single pool, Wickens proposed several separate pools of resources, defined along a few dimensions: the *stage* of processing (perceptual and cognitive processing versus responding), the *modality* of input (visual versus auditory), the *code* of processing (spatial versus verbal) and, within vision, focal versus peripheral (ambient) vision. Two tasks that draw on different pools can be time-shared well; two tasks that compete for the same pool interfere. Driving (visual, spatial, manual) combines reasonably well with listening to the radio (auditory, verbal), and badly with reading a text message (visual, verbal, manual).

Practice also changes what can be combined. In a famous study, Elizabeth Spelke, William Hirst and Ulric Neisser (1976) trained two students for several months to read short stories while writing down words dictated to them. At first their reading speed and comprehension collapsed; after weeks of practice they read as fast and understood as well as when reading alone, while writing to dictation. Practice had apparently turned one of the tasks into something that no longer needed much capacity.

### Multitasking and the phone in the car

What people call multitasking is, for most tasks, rapid switching between them. Every switch has a **switch cost**: responses are slower and more error-prone just after a switch than when the same task is repeated, as Robert Rogers and Stephen Monsell showed in 1995. The real-world consequences are serious. David Strayer and William Johnston (2001) had people perform a simulated driving task while talking on a mobile phone; they missed nearly twice as many red lights and braked more slowly than when driving alone, and a hands-free phone was no better than a hand-held one. Listening to the radio or a recorded book did not have this effect. The problem is the conversation, not the hands. Later work by Strayer's group found that drivers talking on phones were impaired about as much as drivers at the legal limit for alcohol, and that they looked straight at roadside objects and then failed to remember them: inattentional blindness at the wheel. A two-wheeler rider in Bengaluru with a phone wedged inside his helmet is not doing two things well; he is doing two things badly.

### Automatic versus controlled processing ●●

Spelke's students point to the third answer. Some mental processes demand attention and effort; others run off by themselves. In 1975 Michael Posner and Charles Snyder proposed three criteria for an **automatic process**: it occurs without intention, it occurs without conscious awareness, and it does not interfere with other ongoing mental activity. Two years later **Richard M. Shiffrin** and **Walter Schneider** (1977) made the distinction the basis of a major theory. **Controlled processes** are slow, serial, effortful, limited by capacity, open to awareness and flexible: you can change them at once when the situation changes. Automatic processes are fast, run in parallel, need little or no capacity, are hard to stop once triggered and are inflexible. Table 5.3 lists the contrasts.

#### KEY STUDY: SHIFFRIN AND SCHNEIDER'S SEARCH EXPERIMENTS (1977)

Participants memorised a set of up to four characters and then searched rapidly presented displays for any of them. In the **consistent mapping** condition, targets and distractors never swapped roles (for example, targets were always consonants and distractors always digits). In the **varied mapping** condition, the same characters served as targets on some trials and distractors on others. With consistent mapping, search speed hardly depended on how many items had to be compared: search had become automatic and parallel. With varied mapping, every extra item added time: search was controlled and serial. When participants were trained for 2,100 trials with one set of letters as targets and another as distractors, search became automatic; when the roles were then reversed, it took nearly 1,000 trials before performance recovered even to its starting level. Automatic processes, once learned, are fast but stubborn.

Table 5.3 Automatic versus controlled processing (Posner and Snyder, 1975; Shiffrin and Schneider, 1977).

| Feature | Automatic processing | Controlled processing |
|---|---|---|
| Effort and capacity | Little or none | Effortful; limited by capacity |
| Speed and organisation | Fast, parallel | Slow, serial |
| Intention | Triggered without intention; hard to stop | Deliberate; can be started and stopped |
| Awareness | Usually outside awareness | Conscious |

| Feature | Automatic processing | Controlled processing |
|---|---|---|
| Flexibility | Rigid; hard to modify once learned | Flexible; adapts to new situations |
| How acquired | Extensive consistent practice | Needed for novel or variable tasks |
| Examples | Reading a word, an expert's gear change | A learner driver, solving a new problem |

The distinction reaches well beyond the laboratory. **Patricia Devine** (1989) applied it to prejudice: because cultural stereotypes are learned by everyone in a society, they are activated *automatically* in high- and low-prejudiced people alike when a member of the group is encountered; what distinguishes low-prejudiced people is that they use *controlled* processing to override the activated stereotype and replace it with their own egalitarian beliefs. (The social psychology of prejudice is taught in the Social Psychology book.) And the stubbornness of automatic processes produces the most famous interference effect in psychology.

## The Stroop effect (1935) ●●●

KEY STUDY: STROOP'S COLOUR-WORD EXPERIMENTS (1935)

**John Ridley Stroop** (1897 to 1973), an American psychologist, published "Studies of interference in serial verbal reactions" in 1935, based on his doctoral research at George Peabody College. In one experiment, people simply read aloud colour words printed in conflicting ink colours (the word RED printed in green ink) and were barely slower than when reading the same words printed in black. In another, they had to *name the ink colour* of such words, saying "green" to RED printed in green, and here performance collapsed: naming the ink colours of 100 conflicting words took about 74 per cent longer than naming the colours of plain coloured patches, and errors rose. A third experiment showed that days of practice reduced the interference but did not abolish it. Stroop himself soon left experimental work to teach the Bible, never knowing that his dissertation would become one of the most cited studies in psychology.

The **Stroop effect** is the slowing of colour naming (and the rise in errors) when the word and the ink colour conflict. Notice its asymmetry. Words interfere strongly with naming colours, but colours hardly interfere with reading words. Why? The most widely accepted explanation is in terms of automaticity. For a literate adult, reading is so overlearned that it is automatic: you cannot look at a familiar word and not read it. Naming an ink colour, by contrast, is a controlled process that needs attention. When the two conflict, the automatically produced word ("red") competes with the deliberately produced colour name ("green") at the stage of choosing a response, and resolving that conflict takes time. A related account, the *relative speed of processing* explanation, says that words are simply processed faster than colours, so the word's name arrives at the response stage first and must be suppressed. Later connectionist models (Jonathan Cohen, Kevin Dunbar and James McClelland, 1990) recast both ideas in terms of the strength of the pathways involved: the word-reading pathway, trained by years of practice, is stronger.

What, then, does the Stroop effect demonstrate? Above all, a limit on **selective attention**: we cannot completely filter out a stimulus dimension whose processing is automatic, even when we are told to ignore it and it hurts our performance to attend to it. Within a single stimulus, one attribute (the word) dominates the other (the ink colour) in controlling the response. It is

also a measure of **interference** in reaction time due to conflicting information, and of the ability to inhibit a dominant response, which is why the Stroop task is one of the standard tests of the *inhibition* component of executive function and why it is used clinically to assess frontal lobe and attention problems. The **emotional Stroop task** applies the same logic: people name the ink colour of emotional and neutral words, and anxious individuals are slower with words related to their worries (a person with a spider phobia slows down on "web" and "hairy"), revealing an attentional bias toward threat.

EXAM TRAP

The Stroop effect is an **attention** phenomenon: specifically, a failure of selective attention caused by interference from automatic word reading. It is not primarily about long-term memory retrieval, working memory, procedural memory, perception or language development, which are the usual distractors, and it does not show improved recall for related items or better accuracy under divided attention. One past paper (RML 2024) asks what the Stroop effect is "an account of" and keys **stimulus dominance**: one dimension of the stimulus (the printed word) dominates the other (its ink colour). "Semantic dominance" is the tempting wrong option because the intruding word is meaningful, but the keyed term describes one attribute of the stimulus overpowering another; "instructional" and "conceptual" dominance are not terms used for this effect.

## Vigilance and sustained attention ●●●

Some jobs require attention not to many things at once but to one thing for a very long time, while waiting for something rare to happen. Air traffic controllers watching radar screens, radar operators, airport baggage screeners, lifeguards and nurses monitoring intensive-care displays all depend on **sustained attention**, the ability to keep attention on a task over a prolonged period; when the events to be detected are infrequent and unpredictable, it is usually called **vigilance**.

KEY STUDY: THE MACKWORTH CLOCK (1948)

During the Second World War, the Royal Air Force noticed that radar operators hunting for enemy submarines missed more signals as their watches went on. **Norman Mackworth** built a laboratory model of the problem. A pointer moved around a blank clock face in small regular jumps; occasionally, and unpredictably, it made a double jump, and the observer had to press a key whenever this happened. Over a two-hour watch, detection fell sharply, with the largest drop within the first half hour; a short break or a telephone message restored it. Mackworth had demonstrated the **vigilance decrement**: the decline in detection over time on a monotonous monitoring task.

Two kinds of explanation have been offered for the decrement. One is that boring, understimulating tasks lower arousal, so the observer becomes less alert. The other, now more favoured, is that vigilance is hard mental work that steadily depletes attentional resources; people rate vigilance tasks as surprisingly effortful and stressful. Signal detection analysis (Chapter 3) adds that part of the decrement is a shift in criterion: tired observers become more cautious about saying "yes, there it is". Clinically, sustained attention is measured with the **Continuous Performance Test** (respond to a rare target letter in a long stream), and it is one of the first functions to suffer from sleep loss, as you will see in Section 5.8. Brain record-

ings provide another window: when a rare target appears among frequent non-targets (the *oddball* task), the brain produces a positive event-related potential about 300 milliseconds later, the P300, which indexes the allocation of attention to the target and is reduced in conditions such as schizophrenia and Alzheimer's disease.

EXAM TRAP

Air traffic controllers and radar readers are the textbook example of **sustained attention (vigilance)**, not selective attention (choosing one input among several), divided attention (two tasks at once) or "perceptual attention" (not a standard category). The classic experiment is Mackworth's clock test (1948).

CLINICAL LINK

Disorders of attention are among the commonest problems a clinical psychologist assesses. In **attention-deficit/hyperactivity disorder** (ADHD), inattention, hyperactivity and impulsivity begin in childhood (the disorder itself is covered in the Disorders book). In **hemispatial neglect**, usually after damage to the right parietal lobe, patients ignore the left side of space: they eat from only the right half of the plate, shave half the face and bisect a horizontal line far to the right, although their vision is intact. **Delirium** is defined by an acute, fluctuating disturbance of attention and awareness, and that attentional disturbance, together with its acute onset, fluctuating course and usually identifiable (often reversible) medical cause, is the most reliable feature separating it from dementia, which begins insidiously and progresses slowly, with attention relatively preserved early. Attention is measured with tests such as the Continuous Performance Test, the Test of Everyday Attention, the Stroop and the Trail Making Test (taught in the Psychological Assessment book).

## 5.5 Consciousness: definitions, levels and the brain HIGH YIELD · 18 QS

Everything in the last four sections was about how information gets selected. But selected for what? The usual answer is: for awareness. The gorilla that was not attended to was not experienced; the word that broke through the attenuator was. Attention is, very often, the gateway to consciousness. It is time to ask what lies on the other side of the gate.

### What consciousness is

Consciousness is one of those words that everyone understands until asked to define it. A workable definition, used in most introductory texts, is that **consciousness** is our awareness of ourselves and of our environment: of our sensations, perceptions, thoughts, feelings and memories at any given moment. It helps to separate two meanings. The first is consciousness as a *level* or state of wakefulness, running from coma through drowsiness to alert wakefulness; this is what a doctor checks at a road accident. The second is consciousness as *content*: what, in particular, you are aware of right now. The two can come apart. A patient in a **vegetative state** opens the eyes and has sleep-wake cycles (wakefulness is present) but shows no sign of awareness, whereas a patient with **locked-in syndrome**, after damage to the brainstem, is fully aware but almost completely paralysed, able to communicate only with eye movements.

The level of consciousness is measured clinically with the **Glasgow Coma Scale**, published by the neurosurgeons Graham Teasdale and Bryan Jennett in 1974. It scores three responses: eye opening (1 to 4), verbal response (1 to 5) and motor response (1 to 6), giving a total from 3 (deep coma) to 15 (fully alert). A score of 8 or less is generally taken to mean coma. The scale measures consciousness, not attention, memory or intelligence, and it is used every day in emergency rooms across India.

### William James and the stream of consciousness ●●●

The founders of experimental psychology tried to study consciousness by breaking it into elements. Wilhelm Wundt and Edward Titchener trained observers to introspect, to report the sensations, images and feelings that made up an experience, much as a chemist breaks a compound into its elements. Titchener described introspection as a highly specialised form of self-observation, and his structuralism set out to describe the components of consciousness, their combinations and their connections (Chapter 1). William James thought this was a mistake. Consciousness, he wrote in 1890, "does not appear to itself chopped up in bits ... It is nothing jointed; it flows. A 'river' or a 'stream' are the metaphors by which it is most naturally described. In talking of it hereafter, let us call it the stream of thought, of consciousness, or of subjective life." The phrase **stream of consciousness** has belonged to James ever since, and it later gave its name to a style of novel-writing that imitates the flow of thought.

James listed five characteristics of this stream. Every thought is part of a *personal* consciousness: it belongs to someone. Within each person, consciousness is *always changing*: no state, once gone, can recur exactly. It is *sensibly continuous*: even after sleep, you know that your thoughts this morning belong to the same self as last night's. It deals with *objects independent of itself*: thought is always about something. And it is *selective*: it is interested in some parts of its objects to the exclusion of others, which is James's link between consciousness and attention.

One feature of the stream of consciousness has received particularly intense research attention: **daydreaming**, the spontaneous drift of thought away from the task and the surroundings toward fantasies, memories and plans. From the 1960s the American psychologist **Jerome L. Singer** (1924 to 2010) showed that daydreaming is not a sign of laziness or disorder but a normal, almost universal part of mental life, and that it can serve planning, creativity and self-reflection. More recently, Matthew Killingsworth and Daniel Gilbert (2010) used a smartphone app to interrupt thousands of people at random moments; minds were wandering in nearly half of the samples, and people were less happy when their minds wandered than when they were focused, whatever they were doing. Brain imaging has linked mind wandering to the *default mode network*, a set of midline brain regions that become more active when we are not engaged in an external task.

> **EXAM TRAP**
>
> "Stream of consciousness" was coined by **William James** (*Principles of Psychology*, 1890), not by Jung, Wundt or Skinner. Wundt studied the elements of consciousness by introspection; James opposed this with the image of a flowing stream, which is why he is linked with functionalism. The feature of the stream that has received the most research attention is **daydreaming**, not gender differences, brain-wave patterns or eye movements.

## Levels of consciousness: Freud's iceberg and the modern unconscious ●●●

James described the stream as it appears to the person having it. **Sigmund Freud** (1856 to 1939) insisted that most of mental life never appears in the stream at all. In *The Interpretation of Dreams* (1900) he set out what is now called his **topographic model** of the mind, which divides it into three regions, usually drawn as an iceberg (Figure 5.3). The **conscious** is whatever you are aware of at this moment, a small tip above the water. The **preconscious** holds material that is not in awareness right now but can be recalled at will without difficulty: your mobile number, what you ate yesterday, the name of your school. The **unconscious**, by far the largest region, holds wishes, impulses and memories that are actively kept out of awareness by repression because they would cause anxiety; they cannot be recalled at will and are known only indirectly, through dreams, slips of the tongue, symptoms and the free associations of the analytic patient. Freud's principle of **psychic determinism** held that nothing in mental life is accidental: even a slip of the tongue (a *parapraxis*, popularly a "Freudian slip") is caused by unconscious motives. In 1923 Freud added a second, structural model of id, ego and superego (the id, wholly unconscious, works on the pleasure principle; the ego works on the reality principle; the superego is the internalised conscience); that model, and psychoanalysis as a theory of personality, belong to the Personality book. The picture of behaviour driven by unconscious inner forces that lie largely beyond the person's control, and shaped by early childhood experience, is the heart of the psychodynamic perspective (Chapter 2).

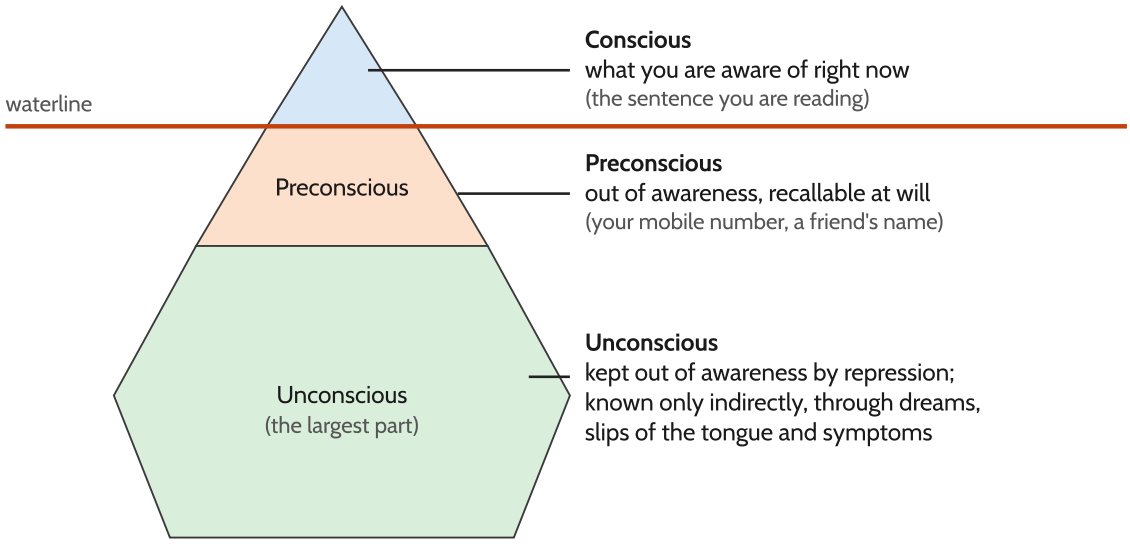


Figure 5.3 Freud's topographic model drawn as an iceberg. Only the tip is conscious; the preconscious lies just below the surface and can be brought up at will; the much larger unconscious is held down by repression and shows itself only indirectly.

Freud's specific claims about repression and the Oedipus complex have fared poorly in research, but the general idea that much mental activity goes on outside awareness has fared very well; it is the part of his thought that modern research most clearly supports. Modern psychologists, however, usually distinguish the dynamic, motivated unconscious of Freud from two other things. The **nonconscious** refers to processes that are never accessible to awareness at all, however hard you try: the regulation of blood pressure and body temperature, or the early stages of visual processing that compute depth from the two eyes. The **cog-**

**nitive unconscious**, a term popularised by John Kihlstrom in 1987, refers to the mental processes that operate outside awareness but are not repressed, simply automatic. You met examples in Section 5.2 (Corteen and Wood's skin responses to unattended words) and Section 5.4 (automatic reading in the Stroop task). Others include priming and implicit memory (Chapter 7), and three findings deserve special mention:

- **Blindsight. Lawrence Weiskrantz** (1926 to 2018) studied patients with damage to the primary visual cortex who insist that they see nothing in part of the visual field, yet, when forced to guess, point to objects there and discriminate their orientation far better than chance. Visual information is processed and used without being consciously seen.
- **The "low road" to fear.** Joseph LeDoux showed that a fast pathway from the thalamus to the amygdala, bypassing the cortex, allows rapid, crude detection of threat and the start of a defensive response before the threat is consciously recognised (Chapter 11).
- **Implicit attitudes.** The Implicit Association Test (Anthony Greenwald and colleagues, 1998) measures how quickly people pair concepts (for example, social groups with good or bad words) and is used to reveal automatic, often unacknowledged stereotyping (taught in the Social Psychology book).

> EXAM TRAP
>
> Accessible memories that can be recalled at will are **preconscious**, not unconscious. The unconscious cannot be reached at will; that is its defining feature. **Psychic determinism** is Freud's principle that all mental events are caused, not a level of consciousness. "Subconscious" is a popular word that Freud did not use as a technical term; in exam options prefer conscious, preconscious and unconscious. The model of three levels is the **topographic** model (1900); id, ego and superego form the later **structural** model (1923).

### Consciousness and the brain, and the hard problem

Where does consciousness come from in the brain? The first clue about the *level* of consciousness came in 1949, when Giuseppe Moruzzi and Horace Magoun found that electrically stimulating the reticular formation in the brainstem of a sleeping or anaesthetised cat immediately changed its EEG to the pattern of alert wakefulness. The **reticular activating system**, a network running from the brainstem through the thalamus to the cortex, maintains arousal; damage to it produces coma. The *contents* of consciousness, in contrast, seem to depend on widespread, coordinated activity across the cortex and thalamus. Francis Crick and Christof Koch made the search for the **neural correlates of consciousness**, the minimal brain activity sufficient for a specific conscious experience, a respectable scientific programme from 1990, and Bernard Baars's **global workspace theory** (1988) proposed that information becomes conscious when it is "broadcast" widely across the brain, so that many specialised systems can use it at once, rather like an actor stepping into the spotlight on a stage watched by a large audience. (The split-brain studies of Roger Sperry and Michael Gazzaniga, which bear on the unity of consciousness, are taught in the Physiological Psychology book.)

Even a complete map of these correlates would leave one question open. In 1995 the Australian philosopher **David Chalmers** (born 1966) distinguished the "easy problems" of consciousness, explaining how the brain discriminates stimuli, integrates information, focuses attention and reports on its states, from the **hard problem of consciousness**: why any of this processing is accompanied by subjective experience at all. Why does the activity of

neurons responding to a certain wavelength *feel like* the redness of a tomato, instead of going on "in the dark"? The easy problems are hard in practice but are problems of mechanism; the hard problem asks why mechanism should produce experience, and there is no agreed answer.

### Altered states of consciousness ●●●

Your ordinary waking consciousness is not the only state your mind can be in. An **altered state of consciousness** (ASC) is a temporary, qualitative change in the overall pattern of mental functioning, such that the person feels that consciousness is radically different from its ordinary state. The definition is associated with the American psychologist **Charles Tart** (born 1937), whose edited book *Altered States of Consciousness* (1969) made the term standard; the psychiatrist Arnold Ludwig (1966) listed the features such states tend to share, including disturbed thinking, a changed sense of time, loss of control, changes in emotional expression and body image, perceptual distortion, a sense that experiences are deeply meaningful or indescribable, and increased suggestibility. Some altered states occur naturally every day (sleep and dreaming); others are induced psychologically (hypnosis, meditation, sensory deprivation, rhythmic drumming or chanting) or physiologically (fever, lack of oxygen, fasting and psychoactive drugs). The rest of this chapter examines the most important of them, beginning with the one you enter every night. But sleep is governed by a clock, so we start with the clock.

> **INDIAN CONTEXT**
>
> Indian philosophy analysed states of consciousness long before Western psychology. The *Mandukya Upanishad* describes four states. In **jagrat** (waking) consciousness is directed outward to the physical world; in **svapna** (dreaming) it is directed inward to subtle images; in **sushupti** (deep, dreamless sleep) there is neither object nor desire, only an undifferentiated rest (the self in this state is called *prajna*). The fourth state, **turiya** (literally "the fourth"), is not a further condition alongside the others but pure consciousness itself, the silent witness present in all three and identified with the *Atman*, the true self or soul, understood as eternal, unchanging consciousness (as distinct from the perishable body and the passing states of mind). Vedanta texts often speak of *avastha-traya*, "the three states", with turiya as what lies beyond them. Indian psychology, whose primary focus is the integration of these Indian philosophical traditions with psychological concepts, is taught in Chapter 2.

## 5.6 Biological rhythms: the body's clocks LOWER YIELD · 4 QS

If you have ever travelled overnight by train from Mysuru to Delhi and back, you know that the body has opinions about time. You feel sleepy at certain hours whatever you are doing, and alert at others. These opinions come from biological rhythms, regular cyclical changes in physiology and behaviour, and the most important of them is set by a tiny clock deep in the brain.

### Circadian rhythms

A **circadian rhythm** is a biological cycle of about 24 hours (from the Latin *circa diem*, "about a day"; the term was coined by the chronobiologist Franz Halberg in 1959). The sleep-wake cy-

cle is the obvious example, but almost every bodily process has one. Body temperature falls through the night and is lowest in the early morning hours, which is why night-shift workers feel coldest and most error-prone around four or five in the morning; it rises through the day. The stress hormone cortisol peaks shortly after waking. The hormone melatonin rises in the evening and is highest in the middle of the night.

Are these rhythms just responses to the day outside, or does the body keep time by itself? The test is to remove every time cue. Volunteers have lived for weeks in caves and underground bunkers with no daylight, no clocks and no news, eating and sleeping whenever they wished. Their rhythms persisted, but drifted: under constant conditions human rhythms run with a period slightly longer than 24 hours (about 24.2 hours in careful modern studies), so that each "day" they went to bed a little later. A rhythm that runs on in the absence of time cues is called **free-running**, and its persistence proves that there is an internal clock. Under normal conditions the clock is reset every day by environmental cues called **zeitgebers** (German, "time givers"). Light is by far the most powerful; meals, exercise and social activity are weaker zeitgebers. The process of being kept in step with the 24-hour day is called **entrainment**.

### The suprachiasmatic nucleus and melatonin ●●●

The master clock is the **suprachiasmatic nucleus** (SCN), a pair of small clusters of neurons in the hypothalamus, sitting just above the optic chiasm (hence "supra-chiasmatic"). The evidence that it is the clock is unusually complete. When the SCN is destroyed in rats and hamsters, circadian rhythms of sleep, activity, eating and drinking disappear: the animals sleep about the same total amount, but at random times of day and night. SCN neurons removed from the brain and kept alive in a dish continue to fire in a 24-hour rhythm. And when Martin Ralph and colleagues (1990) transplanted the SCN from hamsters with a mutant short-period clock into normal hamsters whose own SCN had been destroyed, the recipients took on the *donor's* rhythm. The clock itself is molecular: in each SCN cell, "clock genes" switch each other on and off in a feedback loop that takes about a day to complete, a discovery recognised by the Nobel Prize in Physiology or Medicine in 2017.

How does light reach the clock? Not through the ordinary visual pathway. A special class of retinal ganglion cells containing the photopigment *melanopsin* responds to overall brightness and sends signals directly to the SCN along the **retinohypothalamic tract**. (Some blind people whose rods and cones are destroyed still have entrained rhythms because these cells survive.) The SCN, in turn, controls the **pineal gland**, which secretes **melatonin** in darkness; light suppresses it. Melatonin, sometimes called the hormone of darkness, signals "night" to the rest of the body and promotes sleepiness, and melatonin taken at bedtime in the new location can help reset the clock after long flights.

EXAM TRAP

The SCN is the **master circadian pacemaker**, entrained by light through the retinohypothalamic tract; SCN lesions disrupt **circadian rhythms**, not total sleep amount. Do not confuse it with neighbouring hypothalamic regions that are popular distractors: the lateral and ventromedial hypothalamus (hunger and satiety, Chapter 10), the preoptic area (temperature regulation and, in its ventrolateral part, sleep onset; Section 5.7), the paraventricular nucleus (stress hormones), the hypothalamic control of anterior pituitary hormones, and the mammillary bodies (memory). Memory consolidation belongs to the hippocampus, language to Broca's and Wernicke's areas, and long-term memory is not "stored" in the SCN.

### Jet lag, shift work, larks and owls

The clock works well as long as life keeps to a regular schedule. Problems arise when the zeitgebers change suddenly (**jet lag**, after flying across time zones) or when work forces us to be awake at the wrong phase of the clock (**shift work**). Both produce sleep disturbance, fatigue, poor concentration and digestive upset, because the internal clock and the external day are out of step, and because different bodily rhythms re-adjust at different speeds. Flying east is harder than flying west. Flying east from London to Bengaluru shortens your day, so you must fall asleep and wake earlier than your clock wants (a *phase advance*); flying west lengthens it (a *phase delay*), and since the free-running clock naturally runs a little longer than 24 hours, delaying it is easier. Full adjustment takes roughly a day for each time zone crossed. For the same reason, rotating shifts are easier to tolerate when they move forward, from morning to evening to night, than backward; Charles Czeisler and colleagues showed in 1982 that switching a group of industrial workers to forward rotation improved their health and productivity.

People also differ in the phase of their clocks. "Larks" (morning types) wake early and peak early; "owls" (evening types) are sluggish in the morning and alert at night. These differences, called chronotypes, are partly genetic, and they shift with age: adolescents drift strongly toward owl-ness, which is why a 7 a.m. coaching class is a biological ordeal for many sixteen-year-olds, while older adults drift back toward larkishness.

Sleepiness, finally, is not set by the clock alone. The Swiss researcher Alexander Borbély proposed in 1982 the influential **two-process model** of sleep regulation. *Process S* (sleep pressure) is homeostatic: it builds up the longer you are awake (partly through the accumulation of the chemical adenosine in the brain) and dissipates during sleep. *Process C* is the circadian rhythm from the SCN, which promotes wakefulness during the biological day. You fall asleep easily when sleep pressure is high and the circadian alerting signal is low. Caffeine works by blocking adenosine receptors, hiding sleep pressure for a few hours without removing it.

### Ultradian and infradian rhythms

Not all rhythms are daily. **Ultradian rhythms** are cycles shorter than a day, repeating more than once in 24 hours. The best known is the roughly 90-minute cycle of sleep stages that you will meet in the next section; Kleitman suggested that a similar 90-minute *basic rest-activity cycle* continues through the day as well, as fluctuations in alertness, and pulses of many hormones follow ultradian rhythms. **Infradian rhythms** are cycles longer than a day: the human menstrual cycle of about 28 days is the classic example. Cycles of about a year, called circannual rhythms, include hibernation, bird migration and, in humans, **seasonal affective**

**disorder**, a form of depression that recurs in the dark months of winter and can be treated with bright-light therapy. Table 5.4 summarises the four kinds of rhythm.

Table 5.4 Biological rhythms.

| Rhythm | Period | Examples |
|---|---|---|
| Ultradian | Shorter than 24 hours | The 90-minute NREM-REM sleep cycle; basic rest-activity cycle; hormone pulses |
| Circadian | About 24 hours | Sleep-wake cycle; body temperature; cortisol; melatonin |
| Infradian | Longer than 24 hours | Menstrual cycle (about 28 days) |
| Circannual (a kind of infradian) | About a year | Hibernation, migration, seasonal affective disorder |

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Kahneman's 1973 book, which treated attention as a limited pool of mental effort, was called *Attention and* _____.
2. Wickens argued that tasks drawing on separate pools of resources interfere little; his theory is called _____ resource theory.
3. In Shiffrin and Schneider's experiments, automatic search developed only with _____ mapping.
4. In the Stroop task, naming the ink colour is slowed by the automatic process of _____ the word.
5. The decline in detection over a long monitoring task, first shown with the Mackworth clock, is the vigilance _____.
6. Memories that are not in awareness but can be recalled at will are, in Freud's terms, _____.
7. Chalmers called the question of why brain processing is accompanied by subjective experience the _____ problem.
8. In the Mandukya Upanishad, the fourth state of pure consciousness is _____.
9. Environmental time cues that entrain circadian rhythms are called _____.
10. Light reaches the suprachiasmatic nucleus through the _____ tract.

**Answers:** (1) Effort, (2) multiple, (3) consistent, (4) reading, (5) decrement, (6) preconscious, (7) hard, (8) turiya, (9) zeitgebers, (10) retinohypothalamic.

## 5.7 The stages of sleep HIGH YIELD · 11 QS

We spend about a third of our lives asleep. For most of history that third was a blank: you closed your eyes, and some hours later you opened them. Sleep could not be studied because the sleeper could not report on it. What changed this was a machine that could listen to the brain.

## Listening to the sleeping brain: the EEG ●●

In 1929 the German psychiatrist **Hans Berger** (1873 to 1941) published the first recordings of electrical activity from the human scalp, the **electroencephalogram** (EEG). The EEG picks up the summed activity of millions of cortical neurons. When those neurons fire in an unsynchronised way, each doing its own work, their activity largely cancels out and the EEG shows small, fast waves; when large numbers fire together in rhythm, the waves become large and slow. Waves are described by their *frequency* (cycles per second, or hertz, Hz) and their *amplitude* (height, which reflects voltage). The first rhythm Berger described, a regular wave of about 10 Hz seen when a person rests with eyes closed, is still sometimes called the Berger rhythm; we now call it alpha.

A modern sleep laboratory records three measures together, a procedure called **polysomnography**: the EEG, the **electrooculogram** (EOG), which records eye movements from electrodes beside the eyes, and the **electromyogram** (EMG), which records muscle tension, usually from under the chin. Table 5.5 and Figure 5.4 summarise the EEG patterns you need to know.

Table 5.5 EEG waves and the states they mark.

| Wave | Frequency | Amplitude | State it marks |
|---|---|---|---|
| Beta | About 13 to 30 Hz | Low | Alert wakefulness, active thinking, anxiety; also seen in REM sleep |
| Alpha | About 8 to 12 Hz | Low to moderate, regular | Relaxed wakefulness with eyes closed (clearest over the back of the head) |
| Theta | About 4 to 7 Hz | Moderate | Drowsiness and stage N1; also present in REM sleep |
| Delta | About 0.5 to 4 Hz | High | Deep slow-wave sleep (N3) |
| Sleep spindles | Bursts of about 11 to 15 Hz lasting 0.5 to 2 s | Waxing and waning | Stage N2 |
| K-complexes | A single large sharp wave | High | Stage N2 (often in response to a sound) |
| Sawtooth waves | Notched theta-range waves | Low | REM sleep |

The pattern of small, fast, irregular waves typical of alert wakefulness is called **desynchronised** EEG (also "activated" EEG); the pattern of large, slow, regular waves typical of deep sleep is called **synchronised** EEG. Hold on to these two terms. The single most surprising fact about sleep is that one of its stages has a desynchronised EEG.

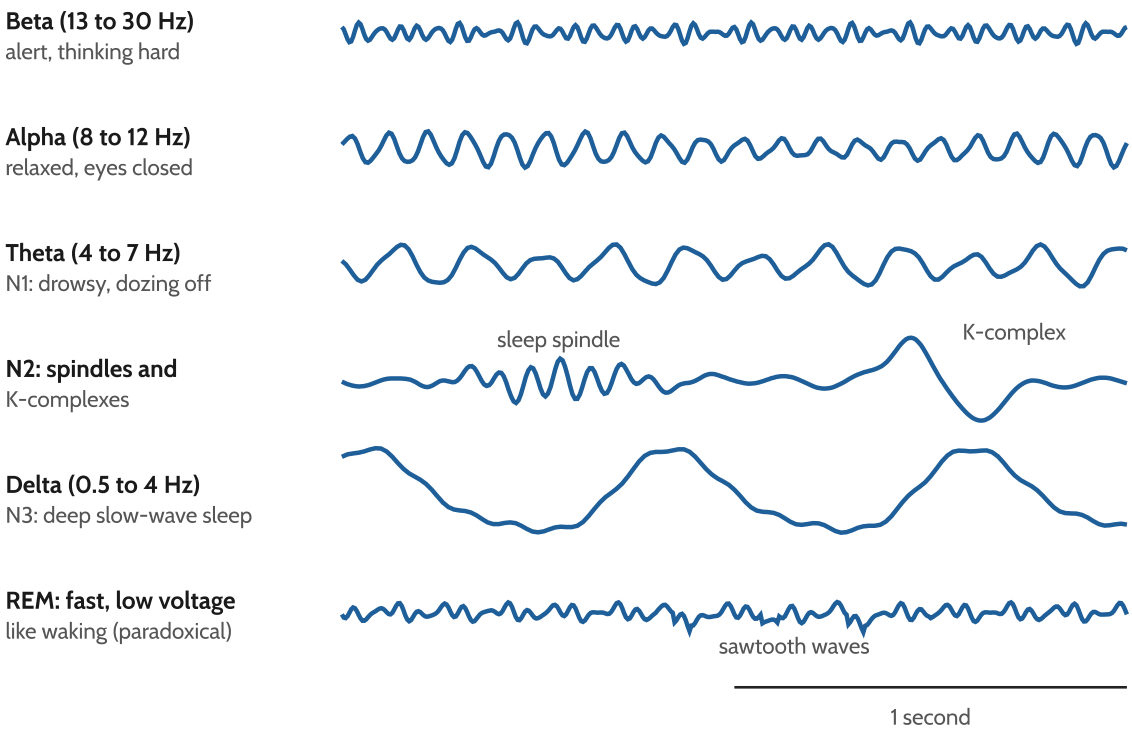


Figure 5.4 Stylised EEG traces from alert waking to deep sleep (each trace about 2 seconds). As sleep deepens, the waves become slower and larger; in REM sleep the EEG returns to a fast, low-voltage, waking-like pattern.

## The discovery of REM sleep

Now return to Eugene Aserinsky and the jerking pens. What Aserinsky and Kleitman reported in 1953 was that the bursts of rapid eye movement coincided with periods of low-voltage, fast EEG, like that of a waking person, and that people woken from these periods usually reported dreams. Their student **William Charles Dement** (1928 to 2020), then a medical student and later the founder of the world's first sleep disorders clinic at Stanford, joined the work, and in 1957 Dement and Kleitman described the full pattern: the stages of sleep follow one another in a regular cycle of about 90 minutes, repeated through the night, with a period of rapid-eye-movement sleep at the end of each cycle. In their studies, about 80 per cent of awakenings from REM periods produced vivid, story-like dream reports, compared with only a small fraction of awakenings from other stages. In 1959 the French researcher **Michel Jouvet** (1925 to 2017) found that in cats this state is accompanied by a complete loss of muscle tone and called it **paradoxical sleep**, because the brain looks awake while the body is more deeply relaxed than at any other time.

Sleep researchers therefore divide sleep into two basic kinds. **REM sleep** (rapid eye movement sleep) is the paradoxical stage. **NREM sleep** (non-REM sleep) is everything else, and it is subdivided into stages by its EEG. The standard scoring manual of Allan Rechtschaffen and Anthony Kales (1968) recognised four NREM stages, numbered 1 to 4, with stages 3 and 4 together called slow-wave sleep. In 2007 the American Academy of Sleep Medicine merged stages 3 and 4 into a single stage, so that modern texts (including Pinel) describe three NREM stages, **N1**, **N2** and **N3**, plus REM (sometimes labelled R). Many exam questions still use the older numbering, so you must know both.

### A night's journey through the stages ●●●

Picture a student going to bed after a long day of revision. As she lies with her eyes closed and relaxes, her EEG shows regular alpha waves. Then, over a few minutes, she passes through the stages of sleep.

**Stage N1** is the brief transition from waking to sleep, lasting only a few minutes. Alpha waves give way to slower, irregular theta waves of low amplitude, and the eyes may roll slowly. She is easy to wake, and if woken she may insist she was not asleep. This is the stage of the **hypnic jerk**, the sudden muscle twitch that sometimes jolts us awake with a feeling of falling, and of dreamlike hypnagogic images (Section 5.9). N1 makes up only about 5 per cent of a night's sleep.

**Stage N2** is light but definite sleep. The EEG is mostly theta, punctuated by two signature waveforms: **sleep spindles**, short waxing and waning bursts of 11 to 15 Hz waves, and **K-complexes**, single large sharp waves, a large negative (upward) deflection followed at once by a large positive (downward) one, which often appear when a sound occurs during sleep. Heart rate and breathing slow, body temperature falls and the eyes are still. N2 is the most common stage, about half of an adult's total sleep.

**Stage N3** is deep sleep. The EEG is dominated by delta waves, the largest and slowest of all, which is why N3 is called **slow-wave sleep** (SWS). In the older system, stage 3 had 20 to 50 per cent delta and stage 4 more than 50 per cent; together they formed slow-wave sleep. This is the hardest stage to wake from: a person roused from N3 is groggy and confused for several minutes. Breathing and heart rate are slow and regular, the release of growth hormone is at its daily peak, and the brain uses less energy than at any other time. N3 is also when sleepwalking and sleep terrors occur.

After N3 the sleeper climbs back up through N2, and then, about 70 to 90 minutes after falling asleep, something remarkable happens. The EEG becomes fast, low-voltage and desynchronised, much like that of an alert, waking person, with bursts of rapid eye movements under the closed lids. This is REM sleep. Its features are a list of paradoxes (Table 5.6 sets them against those of NREM sleep):

- The brain is highly active: its oxygen use and blood flow in many regions rise to waking levels, and the EEG shows beta and theta waves with characteristic sawtooth waves.
- The eyes dart rapidly from side to side in bursts.
- The large skeletal muscles of the body core are almost completely paralysed, a state called **REM atonia**, produced by brainstem circuits that actively inhibit motor neurons in the spinal cord, although small muscles of the fingers and face may twitch.
- Heart rate, blood pressure and breathing become irregular, and there is genital arousal (penile erection or clitoral engorgement) in nearly every REM period, whatever the content of the dream; even infants show it.
- Most vivid, emotional, story-like dreams occur in REM sleep.
- Despite the waking-like EEG, a person in REM sleep is hard to wake by meaningless sounds, although meaningful ones (their own name, a baby's cry) can wake them easily.

Table 5.6 NREM sleep and REM sleep compared.

| Feature | NREM sleep (especially N3) | REM sleep |
|---|---|---|

| Feature | NREM sleep (especially N3) | REM sleep |
|---|---|---|
| EEG | Synchronised; spindles and K-complexes (N2), delta waves (N3) | Desynchronised, fast, low voltage, waking-like; sawtooth waves |
| Eye movements | Few or slow rolling (N1) | Rapid bursts |
| Muscle tone | Present, though reduced | Atonia of core muscles; occasional twitches |
| Heart rate and breathing | Slow and regular | Irregular, variable |
| Brain activity | Lowest energy use of the day | Near waking levels |
| Mental activity | Brief, thought-like fragments, less often recalled | Vivid, emotional, story-like dreams |
| Parasomnias | Sleepwalking, sleep terrors, bedwetting | Nightmares, REM sleep behaviour disorder |
| Share of adult sleep | About 75 to 80 per cent | About 20 to 25 per cent |

EXAM TRAP

REM sleep is **not** characterised by EEG synchrony or delta waves; it has a desynchronised, low-voltage, mixed-frequency EEG. Delta waves and the growth hormone surge belong to N3; spindles and K-complexes belong to N2; dominant alpha over the back of the head belongs to relaxed wakefulness with eyes closed, and beta to alert wakefulness. If a question lists rapid eye movements, atonia, dreams and genital tumescence and asks for the exception, the odd one out is anything describing slow, synchronised waves.

## The 90-minute cycle and its changes across the night ●●

One sequence of NREM stages followed by a REM period is a **sleep cycle**. Each cycle lasts about 90 minutes (anywhere from about 70 to 110 minutes), so a typical adult night of seven to eight hours contains **four to six cycles**. But the cycles are not identical, and the way they change is a favourite exam point (Figure 5.5). Early in the night, cycles contain long periods of N3 and very short REM periods; the first REM period may last only about 10 minutes. As the night goes on, N3 shrinks and often disappears altogether in the last cycles, while REM periods lengthen to half an hour or more. Slow-wave sleep is concentrated in the first third of the night; REM sleep in the last third. This is why you are most likely to be woken from a dream by your morning alarm, and why sleepwalking and sleep terrors (N3 events) happen within a few hours of falling asleep. Brief awakenings, usually forgotten by morning, are normal.

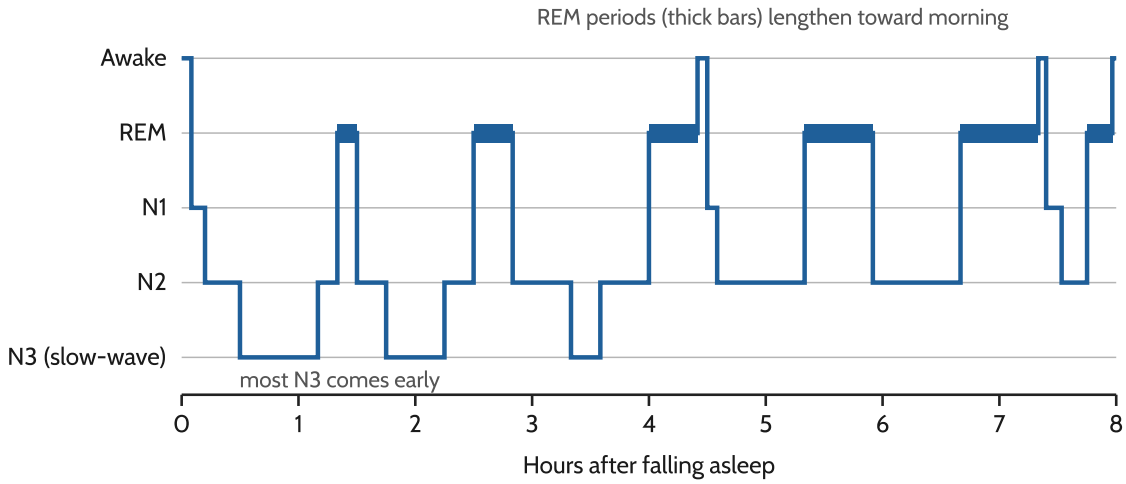


Figure 5.5 A typical night of young adult sleep (a hypnogram). Cycles of about 90 minutes repeat four to six times; N3 is concentrated early in the night, and REM periods become longer toward morning.

EXAM TRAP

How many NREM-REM cycles in a night? The standard answer is **four to six** (about 90 minutes each over seven to eight hours), and that is the keyed answer when it is offered (RML 2024). Some sources give "four to five" or "about five", and one past paper (RML 2022) keys **4 to 7** because 4 to 6 was not among its options; choose the option that contains four to six and excludes the extremes (2 to 4 is too few, 8 to 12 too many). Also remember the direction of change: N3 decreases and REM increases across the night, not the reverse.

## Sleep across the lifespan

Sleep changes dramatically with age, as **Howard Roffwarg** and his colleagues Joseph Muzio and William Dement showed in a classic paper in 1966 (Figure 5.6). A newborn baby sleeps about 16 hours a day, and about half of that is REM sleep; newborns also often enter REM directly at sleep onset. Total sleep and the proportion of REM fall steeply through infancy and childhood. By adolescence REM makes up about 20 to 25 per cent of sleep, and it stays near that level through adulthood, declining slightly in old age. Roffwarg proposed that the huge amount of REM sleep in infancy provides internally generated stimulation that helps the developing brain mature, an idea known as the ontogenetic hypothesis. In older adults total sleep falls to about six hours, slow-wave sleep shrinks greatly (in many elderly people N3 almost disappears), sleep becomes lighter and more broken by awakenings, and its timing shifts earlier in the evening.

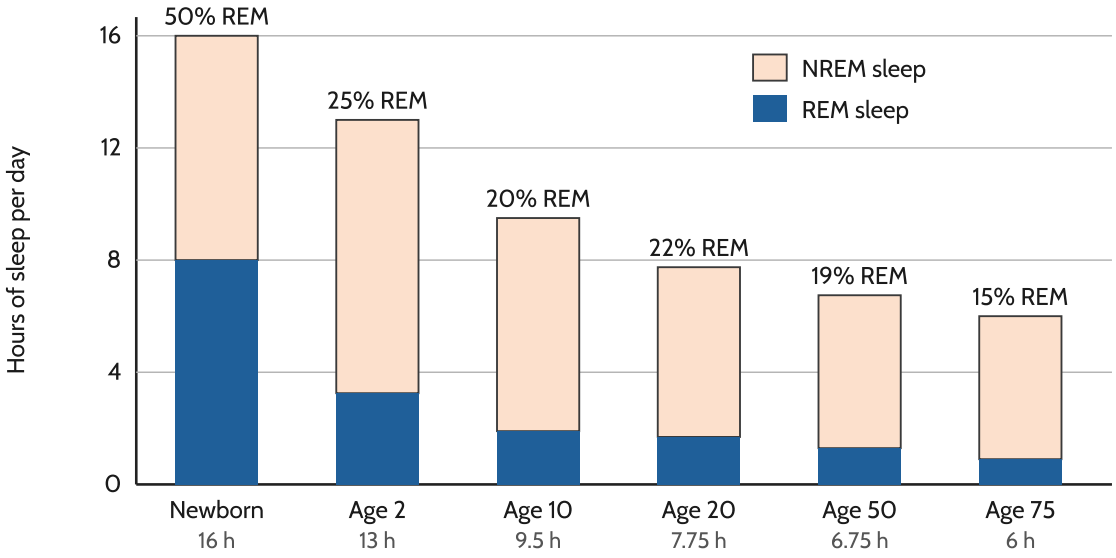


Figure 5.6 Total daily sleep and REM sleep across the lifespan (approximate values redrawn after Roffwarg, Muzio and Dement, 1966). Both total sleep and the share of REM fall steeply in early life; REM is half of a newborn's sleep but about a fifth of an adult's.

## What switches sleep on and off

You read in Section 5.5 that the reticular activating system keeps the cortex awake. Sleep is not simply this system running down. It is actively switched on. A group of inhibitory neurons in the **ventrolateral preoptic area** (VLPO) of the hypothalamus, which use the transmitter GABA, become active at sleep onset and shut down the arousal systems of the brainstem and hypothalamus; the arousal systems, in turn, inhibit the VLPO. Because each side suppresses the other, the brain flips quickly between sleep and wakefulness, like a light switch, instead of drifting between them; Clifford Saper and his colleagues called this the "flip-flop switch". The switch is stabilised by **orexin** (also called hypocretin), a peptide made by neurons in the lateral hypothalamus that keeps the system firmly in the waking position; its loss causes narcolepsy (Section 5.8).

REM sleep is generated in the brainstem, particularly the pons. "REM-on" neurons in a region called the sublaterodorsal nucleus trigger REM sleep and, through circuits in the medulla, the atonia that goes with it; "REM-off" neurons in the ventrolateral periaqueductal grey hold REM in check. During REM, waves of activity called PGO waves (ponto-geniculo-occipital) travel from the pons through the visual thalamus to the occipital cortex, a fact that will matter when we come to theories of dreaming.

EXAM TRAP

Slow-wave (NREM) sleep begins when neurons in the **ventrolateral preoptic area** become active. The **suprachiasmatic nucleus** is the circadian clock (it decides *when* sleep is likely, not how it is switched on). The **sublaterodorsal nucleus** is a REM-on region, and the **ventrolateral periaqueductal grey** is a REM-off region; both concern REM, not slow-wave sleep.

## 5.8 Why we sleep, what deprivation does and when sleep goes wrong

MEDIUM YIELD · 4 QS

Sleep is costly. An animal asleep is not eating, mating or watching for predators. Yet every mammal and bird that has been studied sleeps, and even fruit flies and roundworms have sleep-like states. Something so universal and so risky must do something important. But what?

### Theories of why we sleep

Pinel groups the answers into two kinds. **Recuperation theories** (also called restorative theories), associated with the British psychiatrist Ian Oswald among others, hold that being awake disrupts the body's internal balance in some way and that sleep is needed to restore it: to replenish energy, repair tissue, clear waste products and restore the brain. There is real evidence for restoration. Growth hormone is released mainly during slow-wave sleep. Slow-wave sleep increases after unusually tiring days and after sleep loss. And in 2013 Lulu Xie, Maiken Nedergaard and colleagues showed in mice that the spaces between brain cells widen during sleep, allowing cerebrospinal fluid to flush out waste products, including the protein beta-amyloid that accumulates in Alzheimer's disease, a cleaning system called the glymphatic system.

**Adaptation theories** (Pinel's term; also called adaptive, evolutionary or circadian theories), associated with Wilse Webb and Ray Meddis in the 1970s, hold that sleep is less a response to the wear of waking than a behaviour that evolution timed to keep animals still and safe during the part of the day when activity would be useless or dangerous, and to conserve energy. They explain well why different species sleep such different amounts. Large grazing animals, which must eat for most of the day and cannot hide from predators, sleep only a few hours; animals that are safe in burrows or at the top of the food chain, and that can meet their needs quickly, sleep much longer. They also explain why sleep is so tightly tied to the circadian clock.

A third family of theories stresses **memory consolidation**. As far back as 1924, John Jenkins and Karl Dallenbach found that people forgot less of what they had learned if they slept after learning than if they stayed awake (Chapter 7). Modern research suggests that slow-wave sleep helps to consolidate factual, declarative memories, partly by "replaying" the day's patterns of hippocampal activity, while REM sleep helps to consolidate procedural skills and emotional memories; Avi Karni and colleagues (1994) found that improvement on a visual skill overnight disappeared when participants were deprived of REM sleep. Giulio Tononi and Chiara Cirelli have added that sleep may prune back the synaptic strengthening of the day, keeping the brain efficient. These theories are not rivals so much as parts of one answer: sleep probably does several jobs, and its timing has been fitted to each species' way of life.

## Sleep deprivation

**KEY STUDY: RANDY GARDNER'S 264 HOURS (1964)**

In the winter of 1963 to 1964, Randy Gardner, a 17-year-old high-school student in San Diego, set out to break the world record for staying awake as a school science project, kept awake by two friends. William Dement heard about it and joined them. Gardner stayed awake for 264 hours, eleven days. He grew irritable, his speech became slurred, his memory and concentration lapsed, and he had brief perceptual distortions, but he never became psychotic; near the end he held a press conference and conducted himself well. Afterwards he slept for about 14 hours and then returned to his normal schedule. He did not need to recover all the lost sleep, a typical finding.

Laboratory studies give a more systematic picture. Even moderate sleep loss produces three consistent effects: increased sleepiness, more negative mood and, above all, poor performance on tasks of *sustained attention*, the dull vigilance tasks of Section 5.4. Sleep-deprived people suffer **microsleeps**, episodes of a few seconds in which the eyelids droop, the EEG shows sleep and the person stops responding, even while sitting or standing; at the wheel of a car on a highway, a microsleep can be fatal. Chronic partial deprivation is also harmful: Hans Van Dongen and colleagues (2003) found that people restricted to six hours a night for two weeks performed as badly on attention tests as people kept awake for one or two whole nights, yet they rated themselves as only slightly sleepy. Total deprivation for weeks is lethal in rats (Allan Rechtschaffen's studies in the 1980s), and a rare inherited human prion disease that destroys the thalamus and abolishes sleep, fatal familial insomnia, is invariably fatal. Interpreting human deprivation studies requires caution, however, because sleep loss is so often confounded with stress.

### REM deprivation and REM rebound

In 1960 Dement woke volunteers every time they entered REM sleep, night after night, and woke a control group equally often during NREM sleep. Two things happened. First, the REM-deprived volunteers tried to enter REM more and more often each night, so that they had to be woken ever more frequently. Second, when they were finally allowed to sleep undisturbed, they spent much more time than usual in REM sleep. This compensatory increase is called **REM rebound**. Dement at first thought REM deprivation also caused anxiety and irritability, but later studies did not confirm serious psychological harm; people taking antidepressants that suppress REM sleep for months show no obvious ill effects. REM rebound is clinically familiar too: alcohol and many sleeping pills suppress REM, and when they are stopped, REM rebounds with intense, often disturbing dreams, which can tempt the person to go back to the drug.

### Sleep disorders: an overview ●●●

Sleep disorders fall into a few broad groups: **insomnia** (difficulty falling asleep or staying asleep, or unrefreshing sleep), **hypersomnia** (excessive sleepiness), breathing-related disorders and the **parasomnias**, abnormal behaviours or experiences that occur during sleep or the transitions into and out of it. Table 5.7 gives a brief overview; diagnostic criteria and treatment are taught in the Psychiatric Disorders book.

Table 5.7 The main sleep disorders in brief.

| Disorder | Key features | Stage or mechanism |
|---|---|---|
| Insomnia | Trouble falling or staying asleep; the commonest sleep complaint; sleeping pills can cause tolerance and rebound insomnia | Often linked to stress, anxiety, depression and poor sleep habits |
| Narcolepsy | Irresistible daytime sleep attacks, cataplexy, sleep paralysis, hypnagogic hallucinations | REM intrudes into waking; loss of orexin (hypocretin) neurons |
| Obstructive sleep apnoea | Breathing stops repeatedly as the airway collapses; loud snoring, gasping, daytime sleepiness | Airway obstruction; commoner with obesity; central apnoea is a failure of the brain's drive to breathe |
| Sleepwalking (somnambulism) | Sitting up, walking or doing complex acts while asleep; hard to wake; no memory next day | NREM parasomnia from N3, early in the night; commoner in children |
| Sleep terrors (night terrors) | Sudden scream, intense fear and racing heart; inconsolable; little or no dream recall | NREM parasomnia from N3, early in the night |
| Nightmares | Frightening dreams that wake the sleeper, who is quickly alert and recalls the dream vividly | REM sleep, later in the night |
| REM sleep behaviour disorder | Acting out dreams, often violently (punching, kicking); mostly older men | Loss of normal REM atonia; often precedes Parkinson's disease |
| Restless legs syndrome | Uncomfortable sensations in the legs and an urge to move them, worse at rest and in the evening | A sensorimotor disorder that delays sleep onset |

A few of these deserve a closer look. **Narcolepsy**, named by the French physician Jean-Baptiste-Édouard Gélineau in 1880, is a disorder of hypersomnia in which elements of REM sleep intrude into waking life. Its sufferers fall asleep at inappropriate moments, in the middle of a conversation or a meal, and when they do they often go straight into REM sleep. The most dramatic symptom is **cataplexy**, a sudden loss of muscle tone during full wakefulness, usually triggered by strong emotion such as laughter or anger: the knees buckle or the person collapses, fully conscious, for a minute or two. It is, in effect, REM atonia switched on while awake. **Sleep paralysis** (the inability to move for a few moments just as one falls asleep or wakes up) and hypnagogic hallucinations complete the classic picture, although both also occur occasionally in healthy people. The discovery in 1999 and 2000 that narcoleptic dogs carry a mutation in the orexin receptor gene and that human patients have lost most of their orexin-producing neurons explained the disorder.

**Sleep apnoea** often masquerades as insomnia or as daytime fatigue: the sleeper wakes briefly dozens or hundreds of times a night to restart breathing, usually without remembering it. **REM sleep behaviour disorder**, first described by Carlos Schenck, Mark Mahowald and colleagues in 1986, is the mirror image of cataplexy: the atonia that should accompany REM sleep fails, so the sleeper acts out the dream, sometimes injuring a bed partner. And the contrast between sleep terrors and nightmares is a classic exam point: the first comes from deep NREM sleep early in the night with little recall, the second from REM sleep late in the night with vivid recall.

EXAM TRAP

**Somnambulism** means sitting, walking or performing complex behaviour while asleep. Extreme fear, agitation and screaming while asleep describes **sleep terrors**; acting out dreams (including nightmares) describes **REM sleep behaviour disorder**; uncomfortable sensations in the legs causing movement and lost sleep describe **restless legs syndrome**. Sleepwalking and sleep terrors arise from **N3** (slow-wave) sleep, not REM: a person in REM sleep is paralysed and cannot walk. Sleep talking (somniloquy) can occur in any stage.

CLINICAL LINK

The first-line treatment for chronic insomnia is not a sleeping pill but cognitive behavioural therapy for insomnia (CBT-I), which outperforms medication in the long run. Its techniques are examined by name. **Stimulus control** (Richard Bootzin, 1972) re-associates the bed with sleep: go to bed only when sleepy, use the bed only for sleep, get up if not asleep within about 20 minutes, and rise at the same time every day. **Sleep restriction** (Arthur Spielman) limits the time in bed to the time actually slept, building sleep pressure, and then extends it gradually. **Paradoxical intention**, a technique from Viktor Frankl's logotherapy, asks the patient to lie in bed in the dark and try as hard as possible to stay *awake*; this removes the performance anxiety of "trying to sleep", which is what keeps many insomniacs awake. **Dereflection**, Frankl's other technique, is different: it redirects attention away from oneself and the symptom toward other meaningful goals. A man told to try to stay awake is receiving paradoxical intention.

## 5.9 Dreams MEDIUM YIELD · 8 QS

Every culture has wondered about dreams. In the ancient world they were messages from gods or ancestors; in the Indian epics they foretell victory and disaster; in twentieth-century Vienna they became the key to the unconscious. Sleep research has answered some old questions and sharpened others. Before we look at the theories, let us be clear about what dreams are actually like.

### What dreams are like

A **dream** is a sequence of images, feelings and thoughts experienced during sleep, usually as a story in which the dreamer takes part. Vivid, emotional, narrative dreams are most common in REM sleep, but mental activity also occurs in NREM sleep, mostly as shorter, more thought-like fragments; late in the night, NREM dreams can be as story-like as REM dreams. Everyone dreams every night, including people who insist they never do: "non-dreamers" have normal amounts of REM sleep and report dreams when woken from it.

Early laboratory work tested common beliefs about dreams. Do dreams flash by in an instant? No: in 1957 Dement and Kleitman woke people either 5 or 15 minutes after a REM period began and asked which it had been, judging only from the length of their dream; they were correct in 92 of 111 cases, so dreams unfold in roughly real time. Can outside stimuli enter dreams? Yes: when Dement and Edward Wolpert (1958) sprayed water on sleepers in REM sleep and woke them a few seconds later, the water had been worked into the dream in 14 of 33 cases (a leaking roof, a sudden downpour on a stage). What do people dream about? Calvin Hall, who collected tens of thousands of dream reports, and Robert Van de Castle (1966), who

devised a coding system for them, found that most dreams are surprisingly ordinary: familiar settings and people, everyday activities and the dreamer's current concerns. Negative emotions such as anxiety and fear are more common than positive ones, and being chased, falling and failing an exam are among the commonest themes worldwide. Fragments of the previous day's events, which Freud called the *day residue*, appear often.

Why should a brain produce such experiences at all? The theories fall into a few families.

### Freud: dreams as disguised wishes

Freud's *The Interpretation of Dreams* appeared in November 1899, but the publisher dated it 1900 to mark the new century, and 1900 is the date to remember. Freud called the interpretation of dreams "the royal road to a knowledge of the unconscious activities of the mind". His central claim was that every dream is a **wish fulfilment**: the disguised fulfilment of a repressed wish, often sexual or aggressive and often dating from childhood. During sleep the ego's guard is lowered, and these wishes press toward expression. If they appeared undisguised, the dreamer would be so disturbed that he would wake, so an internal censor disguises them. In this sense, Freud said, dreams are the guardians of sleep.

Freud therefore distinguished two levels of every dream. The **manifest content** is the dream as the dreamer remembers and reports it: the story, the people, the places. The **latent content** is its hidden meaning, the unconscious wishes and thoughts that gave rise to it. The mental process that turns latent content into manifest content is the **dream work**, and it uses four main mechanisms:

- **Condensation**: several latent ideas are compressed into a single image. A person in a dream who has your teacher's face, your father's voice and your uncle's walk condenses feelings about all three.
- **Displacement**: the emotional significance of an important but threatening object is shifted onto a trivial, harmless one. Anger at a parent may appear as anger at a stray dog.
- **Symbolisation** (Freud's "considerations of representability"): abstract thoughts are turned into concrete, mostly visual images and symbols, so that a wish to be free of an obligation might appear as a flight from a locked room.
- **Secondary revision** (secondary elaboration): on waking and in recall, the mind tidies the confused material into a more coherent, logical-seeming story, hiding the latent meaning further.

To interpret a dream, the analyst asks the patient to free associate to each element of the manifest content, retracing the dream work backwards toward the latent thoughts. The theory has been enormously influential in culture and in psychotherapy, but its scientific standing is weak. There is no convincing evidence that dreams systematically disguise wishes; many dreams (the repeated nightmares of trauma survivors, for example) are hard to see as wishes at all, as Freud himself came to accept in 1920; and the brain science on which he built the theory is now obsolete. What has survived is the broader idea that dreams draw on our emotional concerns.

EXAM TRAP

*The Interpretation of Dreams*: **1900** (printed in November 1899 but dated 1900). Manifest content is what is remembered; latent content is the hidden meaning. The four mechanisms of the dream work are condensation, displacement, symbolisation (representation) and secondary revision; "sublimation" and "projection" are defence mechanisms, not dream-work mechanisms, and "day residue" is the raw material, not a mechanism. "Royal road to the unconscious" refers to dreams, not to free association or slips of the tongue.

### Jung: dreams as messages from the psyche ●●●

**Carl Gustav Jung** (1875 to 1961), once Freud's closest colleague, broke with him in 1913 partly over dreams. For Jung a dream is not a disguise but a natural, spontaneous self-portrait of the psyche, speaking in the language of symbols. Its chief function is *compensatory*: it balances a one-sided conscious attitude, so that a man who is outwardly all confidence may dream of being lost and helpless. Dreams can also be *prospective*, anticipating future development. Here Jung's disagreement with Freud over the unconscious matters. Jung held that the unconscious exists on two levels: a *personal* (individual) unconscious of the person's own forgotten and repressed experiences, and beneath it the **collective unconscious**, the inherited layer of the psyche shared by all humanity, which he accounted for in evolutionary terms, through the similar brain structure inherited by all human races. Some dreams draw on the collective unconscious and present universal images, the archetypes (such as the persona, the social mask; the shadow, the disowned dark side; the anima, the feminine side of a man; the animus, the masculine side of a woman; and the wise old man), which Jung found repeated in myths and religions across cultures. Early in his career, at the Burghölzli clinic in Zurich, Jung had also developed the *word association test*, in which hesitations and odd replies to stimulus words reveal unconscious emotional complexes. Instead of free association, Jung used *amplification*, enriching a dream symbol by comparing it with parallel images from myth, folklore and religion. (Jung's theory of personality is taught in the Personality book.)

### Activation-synthesis: dreams from the brainstem

The discovery of REM sleep made a purely physiological theory possible. In 1977 the Harvard psychiatrists **J. Allan Hobson** (1933 to 2021) and **Robert McCarley** proposed the **activation-synthesis hypothesis**. During REM sleep, circuits in the pons fire spontaneously and send waves of activation (the PGO waves of Section 5.7) up to the forebrain, stimulating visual, motor and emotional areas more or less at random. That is the *activation*. The cortex, which is built to make sense of input, does its best with this chaotic barrage and weaves it into a story, drawing on the dreamer's memories and concerns. That is the *synthesis*. Think of how you see faces and animals in drifting clouds: the clouds are random, but your brain imposes a meaning on them.

The hypothesis explains several features of dreams neatly. Dreams are intensely visual because the visual system is activated. They are bizarre, with sudden scene shifts, because the input is random. We dream of running yet being unable to move because motor areas are activated while the body is paralysed. And dreams are quickly forgotten because the brain chemicals that support memory formation (noradrenaline and serotonin) are almost absent in REM sleep. On this view dreams are not disguised messages; any meaning is added by the

synthesising brain, although that meaning can still reflect the dreamer's personality. The hypothesis has been criticised: the neuropsychologist Mark Solms showed that patients with certain forebrain lesions stop dreaming although their REM sleep is intact, so dreaming depends on the forebrain, not just the pons, and dreaming also occurs outside REM. Hobson himself later revised the theory, and in 2009 proposed that dreaming is a kind of "protoconsciousness", a virtual-reality rehearsal that helps develop and maintain waking consciousness.

### Information-processing, memory and threat-simulation theories

**Information-processing theories** see dreaming as a by-product or part of the brain's overnight handling of the day's experiences. In one early, provocative version, **Francis Crick** (1916 to 2004) and Graeme Mitchison (1983) proposed "reverse learning": REM sleep weakens unwanted, parasitic connections formed during the day, so that, in their phrase, we dream in order to forget. More influential today are **memory-consolidation** views, which link dreams to the role of sleep in strengthening and integrating memories (Section 5.8): new learning is replayed and connected with older memories, and dream content reflects fragments of this process. When Robert Stickgold and colleagues (2000) had people play the computer game Tetris for hours, many saw falling blocks as they drifted off to sleep, including amnesic patients who could not remember having played. Rosalind Cartwright, studying people going through divorce, argued that dreams also help regulate mood, working through emotional problems overnight. The **neurocognitive** or continuity view of William Domhoff holds that dreams express our waking concerns and conceptions, using the same cognitive systems that we use when awake; dreams develop gradually in childhood as these systems mature.

The Finnish psychologist **Antti Revonsuo** took an evolutionary approach in his **threat-simulation theory** (2000). Dreams, he argued, evolved as a virtual training ground in which our ancestors rehearsed responses to threats (being chased, attacked, injured or rejected) safely during sleep, making them better at handling such threats when awake. In support, threatening events are greatly over-represented in dreams relative to waking life, and children who have lived through real danger dream of threats more often. Critics note that many dreams contain no threat, and that the rehearsal would be useless if it is forgotten by morning. Table 5.8 brings the theories of dreaming together.

Table 5.8 Theories of dreaming.

| Theory | Proposer and year | Central idea |
|---|---|---|
| Wish fulfilment | Freud, 1900 | Dreams are disguised fulfilments of repressed wishes; manifest content hides latent content |
| Compensation and archetypes | Jung | Dreams are undisguised symbolic messages that balance conscious attitudes and may show archetypes |
| Activation-synthesis | Hobson and McCarley, 1977 | The cortex synthesises a story from random brainstem activation during REM sleep |
| Reverse learning | Crick and Mitchison, 1983 | REM sleep erases unwanted connections; we dream in order to forget |
| Information processing and memory consolidation | Stickgold and others | Dreams reflect the overnight processing and consolidation of recent experience |
| Continuity (neurocognitive) | Hall; Domhoff | Dreams express waking concerns using waking cognitive systems |

| Theory | Proposer and year | Central idea |
|---|---|---|
| Threat simulation | Revonsuo, 2000 | Dreams evolved to rehearse responses to threats |

## Lucid dreams

Sometimes a dreamer realises, within the dream, that he is dreaming, and may even take control of what happens next, choosing to fly or to turn and face a pursuer. This is a **lucid dream**, a term coined by the Dutch psychiatrist Frederik van Eeden in 1913. For a long time scientists doubted that such dreams were real rather than reports made up after waking. The doubt was settled by an ingenious method. Keith Hearne in Britain in the 1970s, and **Stephen LaBerge** at Stanford in the early 1980s, asked lucid dreamers to make a pre-arranged pattern of eye movements (left, right, left, right) as soon as they became lucid. Because the eye muscles are not paralysed in REM sleep, the signals appeared clearly on the EOG, in the middle of unmistakable REM sleep. Many people have at least one lucid dream in their lives, and a few have them regularly; lucid dreaming techniques have been tried as a treatment for recurring nightmares.

## Hypnagogic and hypnopompic imagery

The borders of sleep have experiences of their own. **Hypnagogic imagery** occurs while falling asleep (in N1): flashes of light, geometric patterns, faces, landscapes, snatches of speech or music, or the sensation of falling that may end in a hypnic jerk. The Tetris images described above are hypnagogic. **Hypnopompic imagery** is similar imagery experienced while waking up, sometimes so vivid that a person sees a figure standing by the bed for a moment after opening the eyes. (The terms are credited to the French scholar Alfred Maury and the British writer Frederic Myers respectively.) Such images are common and normal, and they are distinguished from the hallucinations of psychosis by occurring only at the edges of sleep. When they combine with sleep paralysis, the experience can be terrifying, typically a sense of an evil presence pressing on the chest; many cultures interpret it as an attack by a spirit or ghost. Frequent, intense episodes, together with daytime sleep attacks, should raise the question of narcolepsy.

> EXAM TRAP
>
> **Hypnagogic** = falling asleep (think "going" to sleep); **hypnopompic** = waking up. The activation-synthesis hypothesis is **Hobson and McCarley (1977)**, not Freud or Crick; threat simulation is **Revonsuo (2000)**; "we dream in order to forget" is Crick and Mitchison's reverse learning. Lucid dreaming was verified by pre-arranged **eye-movement** signals during REM sleep (LaBerge), because the eyes are the one set of muscles free to move.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Recording EEG, EOG and EMG together during sleep is called _____.
2. Relaxed wakefulness with the eyes closed is marked by _____ waves.
3. Sleep spindles and K-complexes are the signature waveforms of stage _____.
4. Because its EEG looks awake while the body is paralysed, REM sleep is also called _____ sleep.
5. A typical night contains about _____ NREM-REM cycles of about 90 minutes each.
6. About half of a newborn's sleep is _____ sleep.
7. The extra REM sleep that follows a period of REM deprivation is called REM _____.
8. Sudden loss of muscle tone during waking, triggered by strong emotion, is _____, a symptom of narcolepsy.
9. In Freud's theory, the hidden meaning of a dream is its _____ content.
10. Images experienced while falling asleep are called _____ images.

**Answers:** (1) polysomnography, (2) alpha, (3) N2, (4) paradoxical, (5) four to six, (6) REM, (7) rebound, (8) cataplexy, (9) latent, (10) hypnagogic.

## 5.10 Hypnosis

LOWER YIELD · 4 QS

Few topics in psychology are surrounded by so much showmanship. On stage, the hypnotist swings a watch, the volunteers' eyes close, and moments later grown adults are clucking like hens. Is this a special state of consciousness, or are the volunteers simply playing along? Psychologists have argued about that question for two centuries, and the argument is a good model of how science sorts real effects from wishful thinking.

**Hypnosis** is best defined as a social interaction in which one person, the hypnotist, offers suggestions for changes in perception, feeling, memory or behaviour, and the other person, the participant, experiences those changes as real and as happening by themselves, without deliberate effort. It usually begins with an **induction**: the participant is asked to focus attention (on the hypnotist's voice or a point on the wall), to relax and to let go of critical evaluation. The essential ingredient, the crux of the whole phenomenon, is **suggestibility**: a heightened readiness to accept and respond to suggestions. A **suggestion** is a communication that invites an automatic, non-volitional response, such as "your arm is becoming lighter and is rising by itself". Notice what is *not* required: neither sleep nor a special swinging watch, and not even a formal induction, since suggestible people respond to many suggestions without one.

EXAM TRAP

The main crux of hypnosis is **suggestibility**. The "analytical mind" and "logical mind" are what hypnosis is said to quieten, not what it rests on. Hypnosis is not sleep: the EEG of a hypnotised person is a waking EEG, and people under hypnosis can hear, respond and, in general, refuse.

## From animal magnetism to the Nancy school: a short history ●●

The story begins with the Viennese physician **Franz Anton Mesmer** (1734 to 1815). From the mid-1770s, Mesmer claimed that health depended on the free flow of an invisible universal fluid through the body, which he called **animal magnetism**, and that a physician could redirect it to cure disease. In Paris from 1778 he treated fashionable patients in groups around a tub (the *baquet*) filled with "magnetised" water and iron rods; many fell into convulsive "crises" and declared themselves cured. In 1784 King Louis XVI appointed a royal commission, which included the American scientist Benjamin Franklin and the chemist Antoine Lavoisier. In some of the first blind experiments in history, they showed that people reacted dramatically to objects they merely believed had been magnetised, and not at all to objects secretly magnetised. The commission concluded that the effects were real but were caused by imagination, not by any fluid. Mesmer was discredited, but "mesmerism" lived on. In the same year one of his followers, the Marquis de Puységur, found that some patients entered not a convulsive crisis but a calm, sleep-like trance in which they spoke and obeyed instructions and afterwards remembered nothing; he called it *artificial somnambulism*.

The name we use came from the Scottish surgeon **James Braid** (1795 to 1860), practising in Manchester. After watching a travelling mesmerist's demonstration in 1841, Braid concluded that there was no magnetic fluid; the trance was a physiological and psychological effect of prolonged, fixed attention, which he produced by having patients stare at a bright object held above their eyes. He called the state "neuro-hypnotism" (nervous sleep), from Hypnos, the Greek god of sleep, and the shortened form **hypnotism** (hypnosis) stuck; his book *Neurypnology* appeared in 1843. Braid later realised that the state was not sleep at all and tried to rename it "monoideism", concentration on a single idea, but it was too late to change the name.

> **INDIAN CONTEXT**
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> Some of the most remarkable early uses of mesmerism took place in Bengal. James Esdaile (1808 to 1859), a Scottish surgeon with the East India Company at Hooghly, began in 1845 to operate on patients in a mesmeric trance, before chemical anaesthesia had reached India. Over the next few years he reported hundreds of operations, including amputations and the removal of large tumours, performed with little apparent pain and with a lower death rate than usual; the government gave him a small mesmeric hospital in Calcutta, and he described his work in *Mesmerism in India* (1846). Once ether and chloroform became available, mesmeric surgery quickly disappeared, but Esdaile's reports remain an early documented case of hypnotic analgesia.

In the second half of the nineteenth century, hypnosis became the centre of a famous French controversy. At the Salpêtrière hospital in Paris, the great neurologist **Jean-Martin Charcot** (1825 to 1893) studied hypnosis in his patients with hysteria and concluded that hypnotisability was itself a sign of a diseased nervous system, a pathological state that passed through fixed stages and was found mainly in hysterics. His prestige made hypnosis respectable for scientific study. Against him stood the **Nancy school**: the country doctor **Ambroise-Auguste Liébeault** (1823 to 1904), who had treated poor patients with hypnosis for years, and the professor of medicine **Hippolyte Bernheim** (1840 to 1919), who joined him. They argued that hypnosis is a normal phenomenon, that most healthy people can be hypnotised, and that it

works entirely through suggestion. The Nancy view won the argument, and it is essentially the modern view.

The young **Sigmund Freud** learned from both sides: he studied with Charcot in Paris in 1885 to 1886 and visited Bernheim at Nancy in 1889. With the Viennese physician **Josef Breuer** (1842 to 1925), he used hypnosis to help patients with hysteria recall and emotionally discharge forgotten traumatic experiences, the "cathartic method" first developed with Breuer's patient Anna O. Breuer and Freud reported their ideas on the psychical mechanism of hysterical phenomena in 1893 and in *Studies on Hysteria* in 1895. Freud soon abandoned hypnosis, because many patients could not be hypnotised and the improvements often did not last, and replaced it with **free association**, in which the patient, relaxing on a couch, reports whatever comes to mind without censoring it; this became the technique on which psychoanalysis was built. A generation later, the American learning theorist Clark Hull's *Hypnosis and Suggestibility* (1933) brought hypnosis into the experimental laboratory.

> EXAM TRAP
>
> **Mesmer**: animal magnetism (1770s and 1780s), not the word hypnosis. **Braid**: coined hypnotism (1840s; *Neurypnology*, 1843). **Charcot**: hypnosis as pathology linked to hysteria (Salpêtrière, Paris). **Liébeault and Bernheim**: the Nancy school, hypnosis as normal suggestion. **Freud and Breuer**: the mechanism of hysteria (1893) and *Studies on Hysteria* (1895); Freud, not Bernheim, is the father of psychoanalysis, and free association replaced hypnosis.

## Who can be hypnotised? Hypnotic susceptibility

People differ greatly in how readily they respond to hypnotic suggestions, a trait called **hypnotic susceptibility** (or hypnotisability). It is measured by standardised scales in which a standard induction is followed by a graded series of suggestions, from easy to hard, and the person's score is the number passed. The best known are the **Stanford Hypnotic Susceptibility Scales**, devised by André Weitzenhoffer and **Ernest Ropiequet Hilgard** (1904 to 2001) at Stanford University (Forms A and B in 1959, Form C in 1962), each with 12 items. An easy item is that an outstretched arm becomes heavy and drops; harder items include hallucinating a buzzing fly, posthypnotic amnesia and failing to see an object that is in plain view. The Harvard Group Scale (Ronald Shor and Emily Orne, 1962) adapts the test for groups.

Scores form a roughly bell-shaped distribution: perhaps 10 to 15 per cent of people are highly hypnotisable, a similar proportion respond to almost nothing, and most fall in between. Susceptibility is remarkably stable; in one study it correlated about 0.7 with the same person's score 25 years later. It is not related to gullibility, weakness of will or most personality traits, but it is related to **absorption**, the tendency to become deeply immersed in imaginative experiences such as a novel, a film or a daydream (Auke Tellegen and Gilbert Atkinson, 1974).

## What hypnosis can and cannot do

Separating the real effects of hypnosis from the myths is one of the most practically useful things a psychologist can do (Table 5.9).

Table 5.9 What hypnosis can and cannot do.

| Hypnosis can | Hypnosis cannot |
|---|---|

| Hypnosis can | Hypnosis cannot |
|---|---|
| Reduce pain substantially (hypnotic analgesia), used in surgery, dentistry, childbirth and burns care | Give superhuman strength or endurance; motivated people who are not hypnotised can match the feats |
| Produce posthypnotic suggestions, which are carried out on a cue after hypnosis has ended | Reliably recover accurate memories; it increases false recall and confidence together |
| Produce posthypnotic amnesia, which is reversed by a pre-arranged cue | Truly regress adults to childhood ways of thinking, or to "past lives" |
| Alter perception, including positive and negative hallucinations in highly hypnotisable people | Reliably force people to act against their deeply held values |
| Help as an addition to therapy for pain, irritable bowel syndrome, habit control and anxiety | Work equally on everyone; responsiveness depends on susceptibility |

Hypnotic **analgesia** is the most robust effect: in highly hypnotisable people it can rival morphine, and in the 1990s Pierre Rainville and colleagues showed that suggestions changing how *unpleasant* a painful stimulus felt changed activity in the anterior cingulate cortex, a brain region involved in the emotional side of pain. A **posthypnotic suggestion** is an instruction given during hypnosis to be carried out later, after hypnosis has ended, in response to a cue: a participant told that she will open a window when the experimenter touches his tie may do so, and then explain, quite sincerely, that the room felt stuffy. **Posthypnotic amnesia**, forgetting what happened during hypnosis until a cue is given, is instructive because the memories return intact when the cue is given; they were stored all along, and only retrieval was blocked.

The memory myths are the most dangerous. Under hypnosis people do report more "memories", but the extra items are as likely to be false as true, and, crucially, hypnosis makes people more *confident* in everything they recall, a problem called memory hardening. Jean-Roch Laurence and Campbell Perry (1983) suggested to hypnotised people that they had been woken by loud noises on a particular night the previous week; about half later "remembered" the noises, and some insisted on them even after being told of the suggestion. For this reason, many courts treat hypnotically refreshed testimony with great caution or refuse it altogether, and "recovered memories" of abuse produced under hypnosis cannot be trusted without independent evidence. Hypnotic age regression does not make adults think like children; it makes them act as they believe children act.

## Theories of hypnosis: a special state or a social role?

The theoretical debate is between those who think hypnosis involves a special division of consciousness and those who think it is ordinary social and cognitive behaviour.

Ernest Hilgard's **neodissociation theory**, set out in his book *Divided Consciousness* (1977), revived an idea of the French psychiatrist Pierre Janet. Hilgard proposed that the mind contains several subsystems of control, normally coordinated by a central executive. Hypnotic suggestion splits consciousness into separate streams: one part of the mind carries out the suggestion and has experiences that are walled off from the other, which remains unaware of them. The evidence that made the theory famous came by accident. During a classroom demonstration of hypnotic deafness, a blind student showed no response to loud sounds. A member of the class asked whether "some part" of him might be hearing, so Hilgard quietly told the student that if some part of him could hear, his index finger should rise. It rose.

Hilgard called this concealed, aware part of the mind the **hidden observer**. In later experiments using the *cold pressor* test, in which a hand is held in ice water, hypnotised participants given analgesia suggestions reported very little pain aloud, while their hidden observer, reporting through automatic writing or key presses, reported pain almost as high as in the normal waking state. The pain was registered, Hilgard concluded, but kept out of the main stream of awareness by an amnesia-like barrier.

The **social-cognitive theory** of hypnosis (also called sociocognitive or role theory) denies that any special state is needed. Its roots lie in Theodore Sarbin's role theory of the 1950s: hypnotised people are enacting the role of "hypnotised subject" as their culture understands it, not faking but absorbed in the role, as an actor can be absorbed in a part. **Theodore Xenophon Barber** (1927 to 2005) showed in the 1960s that ordinary participants given "task-motivating" instructions (to try their hardest to imagine and experience what is suggested) responded to suggestions almost as well as hypnotised participants. **Nicholas Spanos** (1942 to 1994) argued that hypnotic responses are goal-directed actions shaped by expectations and by how the situation is interpreted; in a direct challenge to Hilgard, he showed that the hidden observer reported more pain or less pain depending on what participants had been told the hidden observer would experience, suggesting that the hidden observer was itself a product of suggestion. Irving Kirsch added that the expectation of responding is itself a powerful cause of responding. Martin Orne, for his part, used the "real-simulator" design, comparing hypnotised participants with people instructed to fake hypnosis, and found subtle differences, such as "trance logic", the untroubled acceptance of logical contradictions (seeing a hallucinated person and the real person at the same time).

Today the two camps have moved closer. Almost everyone agrees that hypnosis is not sleep, that it is not a magical surrender of will, and that expectation and suggestion drive it. But brain imaging shows that, in highly hypnotisable people, suggestion produces real changes in brain activity: Stephen Kosslyn and colleagues (2000) found that the suggestion to see colour in a grey pattern activated colour-processing regions, and Amir Raz, Michael Posner and colleagues (2002) found that a posthypnotic suggestion that words were meaningless symbols greatly reduced the Stroop effect, an effect that, as you saw in Section 5.4, people cannot normally switch off. Table 5.10 summarises the two positions.

Table 5.10 Two views of hypnosis.

| Question | Neodissociation (Hilgard) | Social-cognitive (Sarbin, Barber, Spanos) |
|---|---|---|
| Is hypnosis a special state? | Yes: a division of consciousness | No: ordinary processes of expectation and role enactment |
| Key evidence | The hidden observer; cold pressor analgesia | Task-motivated controls perform as well; hidden observer reports follow instructions |
| Why responses feel involuntary | The executive is unaware of the split-off control | Participants interpret their own goal-directed actions as happening by themselves |

CLINICAL LINK

Hypnosis has respectable clinical uses, mostly as an addition to other treatment: for chronic and procedural pain, irritable bowel syndrome, smoking and weight control, and anxiety; adding hypnotic techniques to cognitive behavioural therapy has been found to improve outcomes. The idea of dissociation, central to Hilgard's theory, also links hypnosis to the dissociative disorders, such as dissociative identity disorder, which requires two or more distinct personality states with recurrent gaps in memory (covered in the Disorders book). And the history of hypnosis is a reminder of the difference between symptoms produced unconsciously, as in conversion disorder, and symptoms faked deliberately for gain, as in malingering.

## 5.11 Meditation, sensory deprivation and psychoactive drugs

LOWER YIELD · 3 QS

Hypnosis showed that a few words from another person can change experience. The last altered states in this chapter are produced in other ways: by deliberately training one's own attention, by taking stimulation away, and by putting chemicals into the brain.

### Meditation ●●

**Meditation** is a family of practices that train attention and awareness, usually with the aim of calming the mind and bringing mental processes under greater voluntary control. The many techniques of the world's traditions fall into two broad kinds, which differ in what they do with attention.

In **concentrative meditation** (now often called focused-attention meditation), attention is fixed on a single object: the breath, a repeated word or sound (a **mantra**), a candle flame, an image or a prayer. Whenever the mind wanders, as it constantly will, the meditator notices this and gently brings attention back. The best-known modern form is **transcendental meditation** (TM), introduced by **Maharishi Mahesh Yogi** (1918 to 2008) in India in the mid-1950s and spread worldwide from 1958, especially after the Beatles stayed at his ashram in Rishikesh in 1968. In TM the meditator sits comfortably and silently repeats a personal mantra for about 20 minutes, twice a day. In **mindfulness meditation** (also called opening-up or open-monitoring meditation), attention is not narrowed but widened: the meditator observes whatever arises, thoughts, feelings, bodily sensations and sounds, with calm, non-judging awareness, noticing each and letting it pass without holding on to it or pushing it away. Its roots are in Buddhist *vipassana* ("insight") practice. Concentrative meditation is like narrowing the zoom lens of Section 5.3 to a single point; mindfulness is like opening it wide while keeping it steady.

What does meditation do? In 1970 Robert Keith Wallace reported that during TM, oxygen consumption, heart rate and breathing rate fell, skin resistance rose (a sign of relaxation) and alpha waves increased. The Harvard cardiologist **Herbert Benson** (1935 to 2022) then showed that the same changes, which he called the **relaxation response** (1975), could be produced by any similar technique (sitting quietly, relaxing the muscles and repeating a neutral word) and were not unique to TM. A sceptical review by David Holmes (1984) went further, concluding that meditation reduced physiological arousal no more than simply resting with the eyes closed. Later and better-controlled work has been kinder. **Jon Kabat-Zinn** (born 1944) devel-

oped **mindfulness-based stress reduction** (MBSR) in 1979 for patients with chronic pain and stress, and Zindel Segal, Mark Williams and John Teasdale combined mindfulness with cognitive therapy in **mindfulness-based cognitive therapy** (MBCT, 2002), which substantially reduces relapse in people who have had three or more episodes of depression. Imaging studies have reported structural and functional brain changes in long-term meditators, for example greater cortical thickness in attention-related regions (Sara Lazar and colleagues, 2005).

Meditation research also teaches an important lesson about evidence. Suppose researchers survey 10,000 randomly selected adults in ten countries and find that people who meditate regularly report less stress than people who rarely or never meditate. Can we conclude that meditation lowers stress? Not necessarily. Random *selection* makes the sample representative, and a large sample makes the estimate precise, but neither makes the study an experiment. Without random *assignment* of people to meditate or not, the finding is correlational: perhaps calmer people find it easier to keep up a meditation habit, or perhaps a third factor, such as a healthier lifestyle or more free time, produces both. Only an experiment can show that meditation causes the reduction (the logic is taught in the Research Methodology book).

INDIAN CONTEXT

Meditation is among India's oldest contributions to psychology. In Patanjali's *Yoga Sutras*, the eight limbs of yoga (*ashtanga*) move from ethical restraints and observances (*yama*, *niyama*), posture (*asana*) and breath control (*pranayama*) to withdrawal of the senses from their objects (*pratyahara*), concentration (*dharana*), meditation (*dhyana*) and absorption (*samadhi*). *Dharana* and *dhyana* are close to what psychologists call concentrative meditation, while Buddhist *vipassana* is the source of mindfulness practice. Sufism describes its process of psycho-spiritual transformation as a comparable journey, from our mundane surface awareness to the depths and heights of self-consciousness, using practices such as *dhikr*, the rhythmic repetition of divine names. (Indian psychology is taught in Chapter 2.)

## Sensory deprivation

Meditation reduces the *variety* of what we attend to. What happens if the world's input is taken away almost entirely? In the early 1950s, in Donald Hebb's laboratory at McGill University in Montreal, William Bexton, Woodburn Heron and Thomas Scott (1954) tried to find out.

KEY STUDY: THE MCGILL SENSORY DEPRIVATION EXPERIMENTS (1954)

Male students were paid 20 dollars a day, generous money at the time, to do nothing at all. Each lay on a comfortable bed in a lit cubicle for as long as he could bear, wearing translucent goggles that let in diffuse light but no patterns, cotton gloves and long cardboard cuffs that limited touch, with his head on a U-shaped foam pillow and the steady hum of an air conditioner masking other sounds. He came out only to eat and use the toilet. Although the money was good, most volunteers quit within two or three days. They found it hard to think clearly or concentrate, did worse on simple cognitive tests, became irritable and restless, and many experienced hallucinations, beginning with dots and geometric patterns and developing into whole scenes; Heron later described one volunteer who saw a procession of squirrels with sacks over their shoulders marching across a snowfield. They also became more easily persuaded by recorded talks about the supernatural.

The McGill studies, sometimes described as research into the "pathology of boredom", showed that the waking brain needs a steady flow of varied stimulation to function normally, an idea that fed into arousal theories of motivation (Chapter 10). Later researchers softened the picture. The neuroscientist John Lilly developed the flotation tank, in which a person floats in skin-temperature salt water in darkness and silence, and the psychologist Peter Suedfeld showed that short sessions of **sensory deprivation**, which he called the restricted environmental stimulation technique (REST), are often experienced as deeply relaxing and can help people stop smoking. The clinical relevance is real: hallucinations are common in people with severe visual loss (the Charles Bonnet syndrome), and the monotony of an intensive care unit contributes to confusion in some patients.

## Psychoactive drugs ●●

The most common way people deliberately alter consciousness is chemical: the morning coffee, the evening drink, the cigarette, the cup of *bhang* at Holi. A **psychoactive drug** is a chemical substance that changes perception, mood, thought or consciousness by acting on the nervous system, mostly at synapses: some mimic a neurotransmitter, some block its receptors, and some change how much of it is released or how quickly it is removed. (The synaptic mechanisms are taught in the Physiological Psychology book.)

Three terms describe what happens with repeated use. **Tolerance** is the decreasing effect of the same dose with repeated use, so that larger doses are needed to get the original effect; tolerance to one drug often brings tolerance to similar drugs (cross-tolerance, as between alcohol and benzodiazepines). **Dependence** is a state in which the person needs the drug to function normally. *Physical dependence* means that the body has adapted to the drug's presence, so that stopping it produces withdrawal; *psychological dependence* is a compulsive craving to use the drug for its emotional effects. **Withdrawal** is the set of unpleasant symptoms that follows stopping a drug on which one is physically dependent, and the symptoms are usually the opposite of the drug's effects: anxiety and tremor after alcohol, fatigue and depression after stimulants, and pain, diarrhoea and gooseflesh after opiates. Table 5.11 summarises the main classes.

Table 5.11 The main classes of psychoactive drugs.

| Class | Examples | Main effects | Notes |
|---|---|---|---|
| Depressants | Alcohol; barbiturates; benzodiazepines (Valium, Xanax, Rohypnol, Halcion) | Slow nervous system activity; relaxation, disinhibition, sedation, sleep | Enhance GABA; dangerous in combination; suppress REM sleep |
| Stimulants | Caffeine, nicotine, amphetamines, cocaine, MDMA (ecstasy) | Increase arousal, alertness and energy; euphoria; reduced appetite | Cocaine blocks dopamine reuptake; a "crash" follows |
| Opiates (opioids) | Opium, morphine, heroin, codeine, fentanyl | Pain relief, euphoria, drowsiness, constricted pupils | Mimic endorphins; overdose stops breathing; strong dependence |
| Hallucinogens (psychedelics) | LSD, psilocybin, mescaline | Distorted perception, hallucinations, altered sense of self and time | Mostly act on serotonin receptors; "bad trips" and flashbacks |
| Cannabis | Marijuana, ganja, charas, bhang (THC) | Relaxation, mild euphoria, altered time sense, impaired memory and coordination | Mixed effects; raises psychosis risk in vulnerable users |

**Depressants** slow the activity of the nervous system. Alcohol, the most widely used, acts largely by enhancing the inhibitory transmitter GABA; at low doses it disinhibits (which is why it can look like a stimulant), at higher doses it impairs judgement, coordination and memory, and at very high doses it can be fatal. Claude Steele and Robert Josephs (1990) described its effect on thought as *alcohol myopia*: alcohol narrows attention to the most salient cues of the moment, so that a drunk person responds to immediate provocations and forgets distant consequences. The **barbiturates** (such as phenobarbital and amobarbital) were once widely prescribed as sedatives and sleeping pills, but the margin between an effective and a fatal dose is narrow, and they have largely been replaced by the **benzodiazepines**, a class that includes diazepam (sold as Valium), alprazolam (Xanax), flunitrazepam (Rohypnol) and triazolam (Halcion). Benzodiazepines are used to reduce anxiety (as anxiolytics), to sedate and to induce sleep (as hypnotics); they are much safer than barbiturates on their own but still cause tolerance and dependence, and they are dangerous when combined with alcohol.

**Stimulants** increase nervous system activity. Caffeine works by blocking adenosine receptors (Section 5.6); nicotine acts on acetylcholine receptors; amphetamines increase the release of dopamine and noradrenaline; cocaine blocks their reuptake, prolonging their action, and is followed by a "crash" of fatigue and low mood. MDMA (ecstasy) is a stimulant with mild hallucinogenic effects that releases large amounts of serotonin. **Opiates**, derived from the opium poppy (morphine, codeine and heroin) and their synthetic relatives (opioids such as fentanyl), relieve pain and produce euphoria because they mimic the brain's own opioids, the endorphins and enkephalins, at opiate receptors; they produce strong physical dependence, and overdose kills by stopping breathing. **Hallucinogens** (psychedelics) distort perception and thought. The best known, LSD (lysergic acid diethylamide), was first synthesised by the Swiss chemist **Albert Hofmann** (1906 to 2008), who discovered its extraordinary effects by accident in 1943; psilocybin (from certain mushrooms) and mescaline (from the peyote cactus) have similar effects, largely through serotonin receptors. **Cannabis** (marijuana, and in India ganja, charas and bhang) is usually placed in a class of its own because its effects are mixed: its active ingredient, THC (tetrahydrocannabinol), acts on the brain's cannabinoid receptors to produce relaxation and mild euphoria, heightened sensory experience, a slowed sense of time, increased appetite and impaired short-term memory and coordination; at high doses it can produce hallucinations, and heavy use in adolescence raises the risk of psychosis in vulnerable people.

> EXAM TRAP
>
> Valium, Xanax, Rohypnol and Halcion are all **benzodiazepines**. They are also sedatives and hypnotics in function, which makes those options tempting, but the question asks for the drug class they share, and "sedative" and "hypnotic" describe uses that barbiturates and other drugs share too. They are **not barbiturates**; examples of barbiturates are phenobarbital, amobarbital (sodium amytal) and secobarbital. Alcohol is a depressant, not a stimulant, despite its early disinhibiting effect; MDMA is usually classed as a stimulant with hallucinogenic properties.

CLINICAL LINK

Tolerance, withdrawal and craving are the building blocks of substance use disorders, which are among the commonest problems in Indian psychiatric practice (alcohol above all, then cannabis, opioids and tobacco); the diagnostic criteria and treatments are covered in the Psychiatric Disorders book. Two facts from this chapter recur in the clinic: benzodiazepines and alcohol suppress REM sleep, so stopping them brings vivid dreams and broken sleep that tempt relapse, and long-term sleeping-pill use can itself cause insomnia, a reason that CBT-I is preferred.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Mesmer claimed to cure illness by redirecting an invisible fluid he called animal _____.
2. The word hypnotism was coined by the Scottish surgeon James _____.
3. Liébeault and Bernheim led the _____ school, which held that hypnosis is normal suggestion.
4. The most widely used test of hypnotisability is the _____ Hypnotic Susceptibility Scale.
5. Hilgard called the concealed part of the mind that registers pain during hypnotic analgesia the hidden _____.
6. Barber and Spanos explained hypnosis with a social-_____ theory rather than a special state.
7. Meditation in which attention is fixed on a single object, such as a mantra, is called _____ meditation.
8. Benson called the calming physiological changes produced by meditation-like techniques the _____ response.
9. The need for larger doses to obtain the same effect is called _____.
10. Valium and Xanax belong to the drug class called _____.

**Answers:** (1) magnetism, (2) Braid, (3) Nancy, (4) Stanford, (5) observer, (6) cognitive, (7) concentrative, (8) relaxation, (9) tolerance, (10) benzodiazepines.

## The case of Armond's dancing eyes, revisited

Think back to the night in Chicago when the pens of Eugene Aserinsky's old machine began to jerk while his son slept. You can now say exactly what he was seeing. About 70 to 90 minutes after falling asleep, Armond had passed down through N1, N2 and the delta waves of N3, climbed back up, and entered his first period of REM sleep: a desynchronised, waking-like EEG, bursts of rapid eye movements, the large muscles of his body held in atonia by circuits in his brainstem, and, very probably, a dream. Through the rest of the night the pattern returned every 90 minutes or so, four to six times, the REM periods growing longer toward morning. At his age, about a fifth to a quarter of his sleep was REM; as a newborn it had been half.

The discovery that Aserinsky first suspected was a machine fault also changed how psychologists think about consciousness. Sleep is not the absence of mind but a sequence of states, switched on by the brain's own circuits and timed by a clock in the hypothalamus. Waking consciousness, too, turned out to be far smaller and more selective than it feels: a narrow stream chosen by attention from a flood of input, while much of the mind's work goes on unseen. Hypnosis, meditation, sensory deprivation and drugs show how far that stream can be bent. Aserinsky's real contribution, in the end, was to wake people up and ask them what they had been experiencing, which is the question this whole chapter has been asking.

## Chapter summary

**5.1 Attention: what it is, how it behaves and what captures it.** William James (1890) defined attention as the mind taking possession of one of several possible objects, with focalisation and concentration as its essence. Attention may be involuntary, voluntary or habitual, and Sohlberg and Mateer's clinical model distinguishes focused, sustained, selective, alternating and divided attention. Its characteristics are selectivity, shifting, a limited span, fluctuation and limited division. The span of attention, studied by Jevons (1871) with beans and later with the tachistoscope, is about six to eight unrelated items (about four for errorless subitising), which is why number plates use short groups. Fluctuation is shown by the Masson disc. Attention is determined by external factors (intensity, size, contrast, change, movement, novelty, repetition) and internal factors (motives, interests, mental set, attitudes).

**5.2 Selective listening: filters, attenuators and late selection.** Cherry (1953) named the cocktail party problem and used dichotic listening with shadowing; listeners noticed physical but not semantic changes in the unattended ear, although Moray (1959) found that about a third heard their own names. Broadbent's filter model (1958), the first information-processing theory, placed an all-or-none filter early. Treisman's attenuation model (1964) turned unattended messages down rather than off, with dictionary units of varying thresholds, and best explains the cocktail party phenomenon. Deutsch and Deutsch (1963) and Norman (1968) placed selection late. Johnston and Heinz (1978) made selection flexible but costly, and Lavie's perceptual load theory showed that high load produces early selection and low load late selection.

**5.3 Visual attention: spotlights, binding and blindness to the obvious.** Covert visual attention behaves like a spotlight or a zoom lens (Eriksen and St James, 1986). Posner's cueing task (1980) separated endogenous (voluntary, central cues) from exogenous (automatic, peripheral cues) attention and revealed inhibition of return. Treisman and Gelade's feature integration theory (1980) proposed a parallel preattentive stage for features and a serial focused-attention stage that binds them, explaining pop-out, linear conjunction search slopes and illusory conjunctions. Inattentional blindness (Mack and Rock; Simons and Chabris's gorilla, 1999), change blindness (Rensink and colleagues; Simons and Levin) and the attentional blink (Raymond and colleagues, 1992) show how much goes unseen without attention.

**5.4 Divided attention, automaticity, the Stroop effect and vigilance.** Kahneman's capacity model (1973) treats attention as a single pool of effort allocated by policy; Wickens's multiple resource theory explains why similar tasks interfere more than different ones. Phone use impairs driving even when hands-free. Posner and Snyder (1975) and Shiffrin and Schneider (1977) distinguished fast, parallel, effortless, inflexible automatic processes from slow, serial, effortful, flexible controlled ones. The Stroop effect (1935) shows that automatic word reading interferes with

colour naming, a failure of selective attention and a measure of inhibition. Vigilance, or sustained attention, declines over time (Mackworth's clock, 1948), as in radar and air traffic control work. ADHD, neglect and delirium are clinical disorders of attention.

**5.5 Consciousness: definitions, levels and the brain.** Consciousness is awareness of ourselves and our environment, with a level (measured by the Glasgow Coma Scale) and a content. James (1890) described a personal, changing, continuous, object-directed and selective stream of consciousness, whose most studied feature is daydreaming. Freud's topographic model divides the mind into conscious, preconscious (recallable at will) and unconscious (repressed) levels; modern psychology adds the nonconscious and the cognitive unconscious, supported by blindsight, fast threat processing and implicit attitudes. The reticular activating system maintains arousal; Chalmers's hard problem asks why processing is experienced at all. Altered states of consciousness (Tart, Ludwig) include sleep, hypnosis, meditation and drug states; the Mandukya Upanishad describes waking, dreaming, deep sleep and turiya.

**5.6 Biological rhythms: the body's clocks.** Circadian rhythms of about 24 hours free-run at about 24.2 hours without time cues and are entrained by zeitgebers, chiefly light. The suprachiasmatic nucleus of the hypothalamus is the master clock, receiving light through the retinohypothalamic tract and controlling melatonin release from the pineal gland. Jet lag is worse flying east (phase advance), and shift rotations work better forward. Borbély's two-process model combines homeostatic sleep pressure with the circadian signal. Ultradian rhythms (such as the 90-minute sleep cycle) are shorter than a day; infradian rhythms (such as the menstrual cycle) are longer.

**5.7 The stages of sleep.** Polysomnography records EEG, EOG and EMG. Beta marks alert waking, alpha relaxed waking with eyes closed, theta N1, spindles and K-complexes N2, and delta N3 (slow-wave sleep; stages 3 and 4 in the older system). REM sleep, discovered by Aserinsky and Kleitman (1953) and called paradoxical by Jouvet, combines a desynchronised EEG, rapid eye movements, atonia, autonomic variability, genital arousal and vivid dreams. Cycles of about 90 minutes repeat four to six times a night, with N3 early and REM late. REM forms half of a newborn's sleep and about a fifth of an adult's (Roffwarg and colleagues, 1966). The VLPO switches sleep on; orexin stabilises waking; the pons generates REM.

**5.8 Why we sleep, what deprivation does and when sleep goes wrong.** Recuperation theories stress restoration and waste clearance, adaptive theories stress safety and energy conservation, and other theories stress memory consolidation. Sleep deprivation (Randy Gardner's 264 hours) mainly impairs mood and sustained attention and produces microsleeps; REM deprivation produces REM rebound. Sleep disorders include insomnia, narcolepsy with cataplexy, sleep apnoea, NREM parasomnias (sleepwalking and sleep terrors, from N3), REM parasomnias (nightmares and REM sleep behaviour disorder) and restless legs syndrome. CBT-I uses stimulus control, sleep restriction and paradoxical intention.

**5.9 Dreams.** Dreams are most vivid in REM sleep, unfold in roughly real time, incorporate outside stimuli and usually reflect everyday concerns. Freud (1900) saw dreams as disguised wish fulfilments in which the dream work (condensation, displacement, symbolisation, secondary revision) turns latent into manifest content; Jung saw them as compensatory symbolic messages that can carry archetypes. Hobson and McCarley's activation-synthesis hypothesis (1977) explains dreams as the cortex's synthesis of random brainstem activation; information-processing, memory-consolidation and continuity theories link dreams to the handling of experience; Revonsuo's threat-simulation theory (2000) gives them an evolutionary function. Lucid

dreams were verified with eye signals; hypnagogic imagery occurs falling asleep and hypnopompic imagery waking up.

**5.10 Hypnosis.** Hypnosis is a social interaction based on suggestibility. Its history runs from Mesmer's animal magnetism and the 1784 commission through Braid, who coined hypnotism, Charcot's view of it as pathology, the Nancy school's view of it as normal suggestion, and Freud and Breuer's cathartic method. Susceptibility, measured by the Stanford scales, is stable and linked to absorption. Hypnosis can relieve pain and produce posthypnotic suggestions and reversible amnesia, but it cannot give superhuman strength or reliably recover memories. Hilgard's neodissociation theory and hidden observer contrast with the social-cognitive theories of Sarbin, Barber and Spanos.

**5.11 Meditation, sensory deprivation and psychoactive drugs.** Concentrative meditation (including TM) focuses attention on one object; mindfulness meditation opens it to whatever arises. Meditation produces Benson's relaxation response, and MBSR and MBCT have clinical value, though correlational surveys cannot prove that meditation causes benefits. The McGill sensory deprivation studies (1954) produced impaired thinking and hallucinations, while brief REST can relax. Psychoactive drugs (depressants including alcohol and benzodiazepines, stimulants, opiates, hallucinogens and cannabis) alter consciousness through synaptic actions and produce tolerance, dependence and withdrawal.

## Who's who

Table 5.12 The people in this chapter.

| Name | When | Examined for |
|---|---|---|
| Eugene Aserinsky | 1921 to 1998; 1953 | Discovered REM sleep with Kleitman (1953) |
| Nathaniel Kleitman | 1895 to 1999 | Pioneer of sleep research; REM discovery (1953); sleep cycle with Dement (1957); basic rest-activity cycle |
| William James | 1842 to 1910; 1890 | Classic definition of attention; stream of consciousness (*Principles of Psychology*, 1890) |
| Edward Bradford Titchener | 1867 to 1927 | Involuntary, voluntary and habitual attention; structuralist introspection |
| McKay Moore Sohlberg and Catherine Mateer | 1987 | Clinical model: focused, sustained, selective, alternating and divided attention |
| Sir William Hamilton | 1788 to 1856 | Marbles remark: six or seven objects at one view |
| William Stanley Jevons | 1835 to 1882; 1871 | Bean experiment on the span of attention |
| E. L. Kaufman and colleagues | 1949 | Named subitising |
| George Miller | 1956 | Magical number seven, plus or minus two (memory span; Chapter 7) |
| Colin Cherry | 1914 to 1979; 1953 | Cocktail party problem; dichotic listening and shadowing |
| Neville Moray | 1959 | Own name heard in the unattended ear; no memory for words repeated 35 times |
| Donald Eric Broadbent | 1926 to 1993; 1958 | Filter model (early selection); split-span; first information-processing theory |

| Name | When | Examined for |
|---|---|---|
| Jeffrey Gray and A. A. I. Wedderburn | 1960 | "Dear Aunt Jane": grouping by meaning across ears |
| Anne Marie Treisman | 1935 to 2018 | Attenuation model (1964); feature integration theory (1980); ear-switching study (1960) |
| Robert Corteen and B. Wood | 1972 | Skin responses to shock-linked words in the unattended ear |
| J. Anthony Deutsch and Diana Deutsch | 1963 | Late selection model |
| Donald Arthur Norman | born 1935; 1968 | Pertinence model of late selection |
| William A. Johnston and Steven P. Heinz | 1978 | Multimode (flexible) theory of selection |
| Nilli Lavie (with Yehoshua Tsal) | 1994 to 1995 | Perceptual load theory |
| Michael I. Posner | born 1936 | Spotlight; cueing task (1980); endogenous and exogenous attention; inhibition of return (with Yoav Cohen, 1984); automaticity criteria (with Charles Snyder, 1975); attention networks (with Steven Petersen, 1990) |
| Charles Eriksen and James St James | 1986 | Zoom-lens model of visual attention |
| Maurizio Corbetta and Gordon Shulman | 2002 | Dorsal and ventral attention networks |
| Garry Gelade | 1980 | Feature integration theory (with Treisman) |
| Hilary Schmidt | 1982 | Illusory conjunctions (with Treisman) |
| Jeremy Wolfe | 1989 onward | Guided search model |
| Daniel J. Simons and Christopher F. Chabris | 1999 | The invisible gorilla (inattentional blindness) |
| Arien Mack and Irvin Rock | 1998 | Named inattentional blindness |
| Ronald Rensink, Kevin O'Regan and James Clark | 1997 | Flicker paradigm (change blindness) |
| Daniel Levin | 1998 | Door study of change blindness (with Simons) |
| Jane Raymond, Kimron Shapiro and Karen Arnell | 1992 | Attentional blink |
| Daniel Kahneman | 1934 to 2024; 1973 | Capacity model (*Attention and Effort*) |
| Christopher Wickens | 1984 | Multiple resource theory |
| Alan Allport, Barbara Antonis and Patricia Reynolds | 1972 | Pianists sight-reading while shadowing speech |
| Elizabeth Spelke, William Hirst and Ulric Neisser | 1976 | Practice in reading while writing to dictation |
| Robert Rogers and Stephen Monsell | 1995 | Task-switch costs |
| David Strayer and William Johnston | 2001 | Phone conversations impair simulated driving |

| Name | When | Examined for |
|---|---|---|
| Richard M. Shiffrin and Walter Schneider | 1977 | Automatic versus controlled processing; consistent and varied mapping |
| Patricia Devine | 1989 | Automatic and controlled components of stereotyping |
| John Ridley Stroop | 1897 to 1973; 1935 | The Stroop effect |
| Jonathan Cohen, Kevin Dunbar and James McClelland | 1990 | Connectionist model of the Stroop effect |
| Norman Mackworth | 1948 | Clock test; vigilance decrement |
| Graham Teasdale and Bryan Jennett | 1974 | Glasgow Coma Scale |
| Wilhelm Wundt | 1832 to 1920 | Introspective analysis of consciousness into elements (Chapter 1) |
| Jerome L. Singer | 1924 to 2010 | Research on daydreaming |
| Matthew Killingsworth and Daniel Gilbert | 2010 | Minds wander in nearly half of waking moments |
| Sigmund Freud | 1856 to 1939 | Topographic model and dream theory (1900); learned hypnosis from Charcot and Bernheim; cathartic method with Breuer |
| John Kihlstrom | 1987 | The cognitive unconscious |
| Lawrence Weiskrantz | 1926 to 2018 | Blindsight |
| Joseph LeDoux | 1990s | Fast thalamus-to-amygdala "low road" for threat |
| Anthony Greenwald and colleagues | 1998 | Implicit Association Test |
| Giuseppe Moruzzi and Horace Magoun | 1949 | Reticular activating system |
| Francis Crick | 1916 to 2004 | Neural correlates of consciousness (with Christof Koch, 1990); reverse-learning theory of dreams (with Graeme Mitchison, 1983) |
| Bernard Baars | 1988 | Global workspace theory |
| Roger Sperry and Michael Gazzaniga | 1960s | Split-brain research (Physiological Psychology book) |
| David Chalmers | born 1966; 1995 | The hard problem of consciousness |
| Charles Tart | born 1937; 1969 | Altered states of consciousness |
| Arnold Ludwig | 1966 | Features of altered states |
| Franz Halberg | 1959 | Coined "circadian" |
| Martin Ralph and colleagues | 1990 | SCN transplants carry the donor's rhythm |
| Charles Czeisler and colleagues | 1982 | Forward-rotating shift schedules |
| Alexander Borbély | 1982 | Two-process model of sleep regulation |
| Hans Berger | 1873 to 1941; 1929 | First human EEG; alpha (Berger) rhythm |

| Name | When | Examined for |
|---|---|---|
| William Charles Dement | 1928 to 2020 | Sleep cycle (1957, with Kleitman); REM deprivation and rebound (1960); studied Randy Gardner |
| Michel Jouvet | 1925 to 2017; 1959 | Named paradoxical sleep |
| Allan Rechtschaffen and Anthony Kales | 1968 | Sleep-stage scoring manual (stages 1 to 4); Rechtschaffen's sleep deprivation studies in rats |
| Howard Roffwarg (with Joseph Muzio and William Dement) | 1966 | Sleep and REM across the lifespan; ontogenetic hypothesis |
| Clifford Saper | 2000s | The VLPO "flip-flop switch" |
| Ian Oswald | 1960s | Recuperation (restorative) theory of sleep |
| Wilse Webb and Ray Meddis | 1970s | Adaptation (adaptive) theories of sleep |
| Lulu Xie, Maiken Nedergaard and colleagues | 2013 | Glymphatic waste clearance during sleep |
| John Jenkins and Karl Dallenbach | 1924 | Less forgetting after sleep than after waking |
| Avi Karni and colleagues | 1994 | REM deprivation blocks overnight skill improvement |
| Giulio Tononi and Chiara Cirelli | 2000s | Synaptic homeostasis theory of sleep |
| Randy Gardner | 1964 | Stayed awake for 264 hours |
| Hans Van Dongen and colleagues | 2003 | Effects of chronic partial sleep restriction |
| Jean-Baptiste-Édouard Gélineau | 1880 | Named narcolepsy |
| Carlos Schenck and Mark Mahowald | 1986 | Described REM sleep behaviour disorder |
| Richard Bootzin | 1972 | Stimulus control therapy for insomnia |
| Arthur Spielman | 1980s | Sleep restriction therapy |
| Viktor Frankl | 1905 to 1997 | Paradoxical intention and dereflection (logotherapy) |
| Edward Wolpert | 1958 | External stimuli incorporated into dreams (with Dement) |
| Calvin Hall and Robert Van de Castle | 1966 | Content analysis of dreams |
| Carl Gustav Jung | 1875 to 1961 | Compensatory dreams; collective unconscious and archetypes; amplification |
| J. Allan Hobson and Robert McCarley | 1977 | Activation-synthesis hypothesis; Hobson's protoconsciousness hypothesis (2009) |
| Mark Solms | 1990s | Forebrain lesions abolish dreaming with REM intact |
| Robert Stickgold | 2000 | Tetris imagery at sleep onset; sleep and memory |
| Rosalind Cartwright | 1990s | Dreams and mood regulation |
| William Domhoff | 2000s | Continuity (neurocognitive) theory of dreams |
| Antti Revonsuo | 2000 | Threat-simulation theory of dreams |

| Name | When | Examined for |
|---|---|---|
| Frederik van Eeden | 1913 | Coined "lucid dream" |
| Keith Hearne and Stephen LaBerge | 1970s to 1980s | Verified lucid dreaming with eye-movement signals |
| Alfred Maury and Frederic Myers | 1848 (Maury); early 1900s (Myers) | Terms hypnagogic and hypnopompic |
| Franz Anton Mesmer | 1734 to 1815 | Animal magnetism |
| Benjamin Franklin and Antoine Lavoisier | 1784 | Royal commission that attributed mesmerism to imagination |
| Marquis de Puységur | 1784 | Artificial somnambulism |
| James Braid | 1795 to 1860 | Coined hypnotism; *Neurypnology* (1843) |
| James Esdaile | 1808 to 1859 | Mesmeric surgery in Bengal |
| Jean-Martin Charcot | 1825 to 1893 | Hypnosis as pathology linked to hysteria (Salpêtrière) |
| Ambroise-Auguste Liébeault | 1823 to 1904 | Founder of the Nancy school |
| Hippolyte Bernheim | 1840 to 1919 | Nancy school: hypnosis as normal suggestion |
| Josef Breuer | 1842 to 1925 | Cathartic method; Anna O; *Studies on Hysteria* (1895, with Freud) |
| Clark Hull | 1933 | *Hypnosis and Suggestibility* (experimental study of hypnosis) |
| Ernest Ropiequet Hilgard | 1904 to 2001 | Stanford scales (with André Weitzenhoffer, 1959); neodissociation theory and the hidden observer (1977) |
| Ronald Shor and Emily Orne | 1962 | Harvard Group Scale of Hypnotic Susceptibility |
| Auke Tellegen and Gilbert Atkinson | 1974 | Absorption and hypnotisability |
| Pierre Rainville and colleagues | 1997 | Hypnotic suggestion and pain unpleasantness in the anterior cingulate |
| Jean-Roch Laurence and Campbell Perry | 1983 | Hypnotically created pseudomemories |
| Pierre Janet | 1859 to 1947 | Dissociation (forerunner of neodissociation) |
| Theodore Sarbin | 1950s | Role theory of hypnosis |
| Theodore Xenophon Barber | 1927 to 2005 | Task-motivational instructions; social-cognitive view |
| Nicholas Spanos | 1942 to 1994 | Social-cognitive theory; challenged the hidden observer |
| Irving Kirsch | 1985 onward | Response expectancy |
| Martin Orne | 1950s to 1970s | Real-simulator design; trance logic |
| Stephen Kosslyn and colleagues | 2000 | Hypnotic colour hallucination changes colour-area activity |
| Amir Raz and colleagues | 2002 | Hypnotic suggestion reduces the Stroop effect |
| Maharishi Mahesh Yogi | 1918 to 2008 | Transcendental meditation |
| Robert Keith Wallace | 1970 | Physiological effects of TM |

| Name | When | Examined for |
|---|---|---|
| Herbert Benson | 1935 to 2022; 1975 | The relaxation response |
| David Holmes | 1984 | Sceptical review of meditation and arousal |
| Jon Kabat-Zinn | born 1944; 1979 | Mindfulness-based stress reduction |
| Zindel Segal, Mark Williams and John Teasdale | 2002 | Mindfulness-based cognitive therapy |
| Sara Lazar and colleagues | 2005 | Cortical thickness in long-term meditators |
| Patanjali | ancient | *Yoga Sutras*: the eight limbs, dharana, dhyana and samadhi |
| Donald Hebb | 1950s | Laboratory where the McGill deprivation studies were run |
| William Bexton, Woodburn Heron and Thomas Scott | 1954 | Sensory deprivation experiments |
| John Lilly | 1950s | Flotation tank |
| Peter Suedfeld | 1970s to 1980s | Restricted environmental stimulation technique (REST) |
| Albert Hofmann | 1906 to 2008; 1943 | Synthesised LSD and discovered its effects |
| Claude Steele and Robert Josephs | 1990 | Alcohol myopia |

## Key terms

attention, involuntary attention, voluntary attention, habitual attention, span of attention, tachistoscope, subitising, fluctuation of attention, Masson disc, determinants of attention, external determinants, internal determinants, mental set, cocktail party problem, cocktail party phenomenon, dichotic listening, shadowing, filter model, early selection, bottleneck, split-span, attenuation model, attenuator, dictionary units, threshold, late selection, pertinence, multimode theory, perceptual load theory, overt attention, covert attention, spotlight, zoom lens, endogenous attention, exogenous attention, inhibition of return, binding problem, feature integration theory, preattentive stage, focused attention stage, feature search, pop-out, conjunction search, illusory conjunctions, inattentional blindness, change blindness, perceptual constancy, attentional blink, divided attention, capacity, capacity model, dual-task paradigm, multiple resource theory, switch cost, automatic process, controlled processes, Stroop effect, emotional Stroop task, sustained attention, vigilance, vigilance decrement, consciousness, Glasgow Coma Scale, stream of consciousness, daydreaming, topographic model, conscious, preconscious, unconscious, psychic determinism, nonconscious, cognitive unconscious, reticular activating system, hard problem of consciousness, altered state of consciousness, turiya, circadian rhythm, free-running, zeitgebers, suprachiasmatic nucleus, retinohypothalamic tract, pineal gland, melatonin, jet lag, two-process model, ultradian rhythms, infradian rhythms, electroencephalogram, polysomnography, electrooculogram, electromyogram, desynchronised, synchronised, paradoxical sleep, REM sleep, NREM sleep, hypnic jerk, sleep spindles, K-complexes, slow-wave sleep, REM atonia, sleep cycle, ventrolateral preoptic area, orexin, recuperation theories, adaptation theories, memory consolidation, microsleeps, REM rebound, insomnia, hypersomnia, parasomnias, narcolepsy, cataplexy, sleep paralysis, sleep apnoea, REM sleep behaviour disorder, dream, wish fulfilment, manifest content, latent content, dream work, condensation, displacement, symbolisation, secondary revision, acti-

vation-synthesis hypothesis, information-processing theories, threat-simulation theory, lucid dream, hypnagogic imagery, hypnopompic imagery, hypnosis, suggestibility, suggestion, animal magnetism, Nancy school, hypnotic susceptibility, Stanford Hypnotic Susceptibility Scales, absorption, posthypnotic suggestion, posthypnotic amnesia, neodissociation theory, hidden observer, social-cognitive theory, meditation, concentrative meditation, mantra, transcendental meditation, mindfulness meditation, relaxation response, sensory deprivation, psychoactive drug, tolerance, dependence, withdrawal, depressants, benzodiazepines, stimulants, opiates, hallucinogens, cannabis

## From the past papers

1. The Stroop effect demonstrates interference in: (JEMAS 2025)

   (a) Long-term memory retrieval (b) Working memory (c) Selective attention (d) Procedural memory

   **Answer: (c) Selective attention.** Automatic word reading cannot be filtered out while naming the ink colour, so the task shows a failure of selective attention. The memory options are distractors; no stored memory is being retrieved or rehearsed.

2. Which psychological process is the Stroop effect most closely associated with? (IHBAS 2024)

   (a) Memory retrieval (b) Attention (c) Perception (d) Language development

   **Answer: (b) Attention.** The colours and words are perceived perfectly well; the difficulty lies in selecting the relevant dimension and inhibiting the automatic one, which is attention. Reading skill matters, but the effect is not about how language develops.

3. In cognitive psychology, the "Stroop effect" primarily demonstrates which cognitive phenomenon? (RML 2025)

   (a) The interference that occurs in reaction time due to conflicting information processing (b) Improved recall when semantically related information is presented (c) The increased accuracy of perception under divided attention tasks (d) The predictive power of schema theory on episodic memory

   **Answer: (a).** Conflicting word and colour information slows reaction time. Option (b) describes semantic priming or organisation effects in memory, (c) is false (dividing attention reduces accuracy) and (d) belongs to memory research.

4. "Stroop effect" is an account of: (RML 2024)

   (a) Stimulus dominance (b) Semantic dominance (c) Instructional dominance (d) Conceptual dominance

   **Answer: (a) Stimulus dominance.** One dimension of the stimulus, the printed word, dominates the other, its ink colour, in controlling the response. "Semantic dominance" tempts because the word is meaningful, but the keyed answer describes one stimulus attribute overpowering another; the other two are not terms used for this effect.

5. Air traffic controllers and radar readers provide us with a good example of which of the following phenomena? (RML 2024)

   (a) Selective attention (b) Sustained attention (c) Divided attention (d) Perceptual attention

   **Answer: (b) Sustained attention.** Watching for rare, unpredictable signals over long periods is vigilance, studied with the Mackworth clock. Selective attention picks one input among several; divided attention handles two tasks at once.

6. Why do the number plates of motorbikes and cars carry only four-digit numbers coupled with some letters? (RML 2024)

   (a) Selected attention (b) Span of attention (c) Divided attention (d) Sustained attention

   **Answer: (b) Span of attention.** Only a limited number of items (about six to eight, fewer for accuracy) can be apprehended at a glance, so plates use short groups that an observer can take in quickly.

7. Who developed the filter-attenuation theory? (RML 2024)

   (a) Broadbent (b) Treisman (c) Johnston (d) Heinz

   **Answer: (b) Treisman.** Treisman (1964) modified Broadbent's all-or-none filter into an attenuator. Broadbent proposed the original filter (1958); Johnston and Heinz (1978) proposed the multi-mode theory.

8. You are at a party and someone calls out your name. Instantly, in the middle of a crowd, you can hear it. This phenomenon is called: (Amity 2023)

   (a) Bottleneck effect (b) Cocktail phenomenon (c) Divided attention (d) Sustained attention

   **Answer: (b) Cocktail party phenomenon.** Named after Cherry (1953) and shown in the lab by Moray (1959); Treisman's low-threshold dictionary units explain it. "Bottleneck" refers to the narrow stage in attention models, not to this effect.

9. What term describes the phenomenon where individuals fail to notice significant changes in their environment, even when those changes are obvious? (RML 2023)

   (a) Change blindness (b) Inattentional blindness (c) Perceptual constancy (d) Selective attention

   **Answer: (a) Change blindness.** The stress on *changes* points to change blindness (Rensink and colleagues; Simons and Levin's door study). Inattentional blindness is missing an unexpected object, perceptual constancy is the stable perception of objects, and selective attention is the general process.

10. Low-prejudiced people engage in: (IHBAS 2020)

    (a) Automatic information processing resulting in involuntary stereotyping (b) Controlled information processing resulting in refuting stereotype (c) Controlled information processing resulting in involuntary stereotyping (d) Automatic information processing resulting in voluntary stereotyping

    **Answer: (b).** In Devine's (1989) model, stereotypes are activated automatically in everyone, but low-prejudiced people use controlled, effortful processing to override them. Controlled processing cannot produce "involuntary" stereotyping, and automatic processing is not "voluntary".

11. The term "stream of consciousness", referring to the flow of thoughts in the human mind, was coined by which psychologist? (IHBAS 2024)

    (a) Carl Jung (b) Wilhelm Wundt (c) William James (d) B.F. Skinner

    **Answer: (c) William James**, in *Principles of Psychology* (1890). Wundt analysed consciousness into elements, Jung described the collective unconscious and Skinner rejected the study of consciousness altogether.

12. According to Freud, the ___ mind includes accessible memories, which can be recalled at will. (RML 2020)

    (a) Conscious (b) Preconscious (c) Unconscious (d) Psychic determinism

    **Answer: (b) Preconscious.** The conscious holds only what is in awareness now; the unconscious cannot be recalled at will; psychic determinism is a principle, not a level of mind.

13. The Mandukya Upanishad describes four states of consciousness (avasthas). The "fourth", pure consciousness beyond waking, dream and deep sleep, is called: (Amity recall)

    (a) Jagrat (b) Swapna (c) Sushupti (d) Turiya

    **Answer: (d) Turiya.** Jagrat is waking, swapna (svapna) dreaming and sushupti dreamless deep sleep; turiya, "the fourth", is the witnessing consciousness present in all three.

14. The suprachiasmatic nucleus (SCN) is crucial for: (IHBAS 2025)

    (a) Memory consolidation (b) Circadian rhythm regulation (c) Language processing (d) Pain perception

    **Answer: (b) Circadian rhythm regulation.** The SCN is the master clock, entrained by light through the retinohypothalamic tract. Memory consolidation depends on the hippocampus and language on Broca's and Wernicke's areas.

15. In a typical major sleep period, how many cycles of NREM-REM are there? (RML 2024)

    (a) 2 to 4 (b) 4 to 6 (c) 6 to 10 (d) 10 to 12

    **Answer: (b) 4 to 6.** Cycles last about 90 minutes, so seven to eight hours hold four to six. Two to four is too few for a full night; six or more would need cycles far shorter than 90 minutes.

16. During a normal night's sleep, how many times do we pass through the different stages of sleep? (RML 2022)

    (a) 8 to 11 (b) 5 to 8 (c) 3 to 5 (d) 4 to 7

    **Answer: (d) 4 to 7.** The standard figure is four to six cycles; of the options given, 4 to 7 is the range that contains it. Eight to eleven is far too many, and 3 to 5 and 5 to 8 each miss part of the usual range.

17. Which of the following characterises slow-wave sleep? (Amity 2023)

    (a) Stage 1 to 4 and REM sleep (b) Stage 3 and 4 (c) NREM and REM sleep (d) Stage 1 to 4

    **Answer: (b) Stages 3 and 4.** In the older Rechtschaffen and Kales system, stages 3 and 4 (dominated by delta waves) were together called slow-wave sleep; since 2007 they are merged as N3. Stages 1 and 2 are light sleep and REM has a fast, desynchronised EEG.

18. Which one of the following characterises somnambulism? (RML 2022)

    (a) Sitting, walking or performing complex behaviour while asleep (b) Extreme fear, agitation, screaming while asleep (c) Acting out nightmares (d) Uncomfortable sensations in the legs causing movement and loss of sleep

    **Answer: (a).** Somnambulism is sleepwalking, an NREM parasomnia from N3. Option (b) is sleep terrors, (c) is REM sleep behaviour disorder and (d) is restless legs syndrome.

19. The main crux of hypnosis is: (RML 2020)

    (a) Analytical mind (b) Logical mind (c) Suggestibility (d) None

    **Answer: (c) Suggestibility.** Hypnosis works through heightened responsiveness to suggestion, measured by scales such as the Stanford. The analytical and logical faculties are what hypnotic induction quietens, not its basis.

20. Researchers studying meditation contacted over 10,000 randomly selected adults in 10 nations. Those who meditated regularly were less likely to suffer from stress than adults who meditated less regularly or not at all. Can we conclude that meditation lowers stress? Why? (RML 2021)

    (a) Yes, because the sample size was large (b) Yes, because the adults were randomly selected (c) Yes, because the adults were randomly assigned (d) Not necessarily, because this wasn't an experiment

    **Answer: (d).** The study is correlational: calmer people may meditate more, or a third factor may cause both. A large, randomly selected sample improves generalisability, not causal inference, and nobody was randomly assigned, so (c) is simply untrue.

21. Valium, Xanax, Rohypnol and Halcion are: (Amity 2023)

    (a) Benzodiazepines (b) Barbiturates (c) Sedatives (d) Hypnotics

    **Answer: (a) Benzodiazepines** (diazepam, alprazolam, flunitrazepam and triazolam). They do act as sedatives and hypnotics, but the class they share is benzodiazepines; barbiturates are an older, more dangerous class such as phenobarbital.

22. A man with sleep-onset insomnia is told to lie in bed with the lights off and try as hard as he can to stay awake. This technique is: (NIMHANS 2026 recall)

    (a) Paradoxical intention (b) Stimulus control (c) Sleep restriction (d) Dereflection

    **Answer: (a) Paradoxical intention**, which removes the performance anxiety of trying to sleep. Stimulus control re-links the bed with sleep, sleep restriction limits time in bed to build sleep pressure, and dereflection (also Frankl's) turns attention away from the symptom altogether.

## High-yield sheet

*The facts this chapter's questions ask most, most-asked first. Last read before the exam.*

1 The **Stroop effect** (John Ridley Stroop, 1935): naming the ink colour of a conflicting colour word (RED printed in green) is slowed because automatic word reading interferes. It is a failure of **selective attention** (interference in reaction time; keyed once as "stimulus dominance"), not a memory effect. ASKED 5 TIMES

2 The **suprachiasmatic nucleus** (SCN) of the hypothalamus is the master circadian pacemaker, entrained by light through the retinohypothalamic tract; SCN lesions abolish circadian rhythms, not total sleep. Hunger is the lateral and ventromedial hypothalamus, stress hormones the paraventricular nucleus, memory the hippocampus and mammillary bodies. ASKED 4 TIMES

3 **Carl Jung** split the unconscious into a personal (individual) unconscious and a **collective unconscious**: an inherited layer shared by all humanity, full of archetypes, which he explained by the similar brain structure of all human races. ASKED 4 TIMES

4 The **psychodynamic** perspective (Freud) holds that behaviour is driven by unconscious inner forces largely beyond the person's control and shaped by early childhood experience. ASKED 3 TIMES

5 **REM (paradoxical) sleep**: desynchronised, fast, low-voltage EEG like waking, rapid eye movements, atonia of the core muscles, genital arousal and vivid, story-like dreams. EEG synchrony and delta waves belong to N3; spindles and K-complexes to N2. ASKED 4 TIMES

6 **Sigmund Freud**, the father of psychoanalysis, published the mechanism of hysteria with Josef Breuer in 1893 (*Studies on Hysteria*, 1895), replaced hypnosis with free association, and published *The Interpretation of Dreams* in **1900** (printed November 1899). ASKED 4 TIMES

7 Freud's topographic model (1900): the **conscious** (aware now), the **preconscious** (not in awareness but recallable at will) and the **unconscious** (repressed; seen in dreams and Freudian slips). The unconscious mind is the part of Freud's theory that research most supports. ASKED 3 TIMES

8 **Sleepwalking** (somnambulism: sitting, walking or complex acts while asleep) and sleep terrors arise from N3 early in the night. Nightmares and **REM sleep behaviour disorder** (loss of REM atonia, so the sleeper acts out dreams) belong to REM. ASKED 3 TIMES

9 A night holds **four to six NREM-REM cycles** of about 90 minutes (one paper keys "4 to 7" because 4 to 6 was not offered). N3 dominates the early cycles; REM periods lengthen toward morning. ASKED TWICE

10 The Mandukya Upanishad's four states: jagrat (waking), svapna (dreaming), sushupti (deep sleep) and **turiya**, "the fourth", pure consciousness identified with the **Atman**, the true self or eternal consciousness. ASKED TWICE

11 **Anne Treisman**'s attenuation (filter-attenuation) model, 1964: unattended messages are turned down, not blocked, so low-threshold dictionary units such as your own name still fire. It best explains the cocktail party effect. ASKED TWICE

12 The **cocktail party phenomenon** (hearing your name across a noisy room) was named by Colin Cherry (1953) using dichotic listening and shadowing; listeners noticed physical changes (male to female voice) in the unattended ear, not its meaning. Moray (1959) showed the own-name effect. ASKED TWICE

13 **Slow-wave sleep** is stages 3 and 4 of the older Rechtschaffen and Kales system (merged as N3 in 2007), dominated by high-amplitude delta waves (about 0.5 to 4 Hz). ASKED TWICE

14 **William James** coined "stream of consciousness" in *Principles of Psychology* (1890); the feature of the stream most studied by researchers is daydreaming. ASKED TWICE

15 **Delirium** is separated from dementia by acute onset, a fluctuating course and disturbed attention and awareness; dementia begins insidiously, with attention preserved early. ASKED TWICE

16 EEG: **beta** (13 to 30 Hz) alert waking; **alpha** (8 to 12 Hz) relaxed waking with eyes closed; theta N1; spindles and K-complexes N2; delta N3. ASKED TWICE

17 Freud's structural model (1923): the **id** works on the pleasure principle, the **ego** on the reality principle. ASKED TWICE

18 Jung's **anima** is the feminine side of a man; the **animus** is the masculine side of a woman. ASKED TWICE

19 **Titchener** called introspection a highly specialised form of self-observation; structuralism describes the components, combinations and connections of consciousness. ASKED TWICE

20 Air traffic controllers and radar readers are the classic example of **sustained attention** (vigilance); its decline over a watch is the vigilance decrement (Mackworth clock, 1948). ASKED ONCE

21 Number plates carry about four digits plus letters because of the **span of attention**: about six to eight unrelated items in one glance (Jevons's beans, 1871). ASKED ONCE

22 **Change blindness** is failing to notice obvious changes in a scene (Rensink, O'Regan and Clark, 1997; Simons and Levin's door study); missing an unexpected object in plain view is inattentional blindness (the gorilla, Simons and Chabris, 1999). ASKED ONCE

23 The crux of **hypnosis** is suggestibility; hypnosis is not sleep. ASKED ONCE

24 Valium (diazepam), Xanax (alprazolam), Rohypnol (flunitrazepam) and Halcion (triazolam) are **benzodiazepines**, depressants that enhance GABA; they are not barbiturates. ASKED ONCE

25 A survey linking regular **meditation** with less stress does not show that meditation lowers stress: a large, randomly selected sample is still not an experiment, because nobody was randomly assigned. ASKED ONCE

26 The primary focus of **Indian psychology** is integrating Indian philosophical traditions with psychological concepts. ASKED ONCE

27 **Patricia Devine** (1989): stereotypes are activated automatically in everyone; low-prejudiced people use controlled processing to refute them. ASKED ONCE

28 The **Implicit Association Test** (Anthony Greenwald and colleagues, 1998) measures unconscious stereotyping. ASKED ONCE

29 Telling an insomniac to lie in the dark and try hard to stay awake is Viktor Frankl's **paradoxical intention**; stimulus control (Bootzin) re-links bed and sleep, sleep restriction limits time in bed. ASKED ONCE

30 **Donald Broadbent**'s filter model (1958) was the first major theory of information-processing psychology: early, all-or-none selection on physical features. ASKED ONCE

CHAPTER 6

# Learning: Conditioning, Cognition and Observation

IN THIS CHAPTER

6.1 ●● What learning is, and what it is not: define learning, separate it from maturation, fatigue and instinct, and explain habituation, sensitisation and imprinting.

6.2 ●● Pavlov and the language of classical conditioning: identify the UCS, UCR, CS and CR in any example and say which timing arrangement conditions best.

6.3 ●● The phenomena of classical conditioning: describe acquisition, extinction, spontaneous recovery, generalisation, discrimination, experimental neurosis, higher-order conditioning and sensory preconditioning.

6.4 ●●● Fear, prediction and preparedness: explain Little Albert and Peter, Rescorla's contingency, blocking, the Rescorla-Wagner model, taste-aversion learning and the conditioning therapies.

6.5 ●●● Thorndike, Skinner and the consequences of behaviour: apply the law of effect and the four kinds of consequence, and explain reinforcers, avoidance, punishment, shaping and extinction.

6.6 ●●● Schedules of reinforcement and other operant phenomena: predict the response pattern of each schedule and explain the Premack principle, superstition, learned helplessness, instinctive drift, matching, contrast and biofeedback.

6.7 ●● Cognitive learning: explain latent learning, cognitive maps, insight, learning sets and the stages of concept formation.

6.8 ●●● Observational learning: describe the Bobo doll studies, Bandura's four processes, self-efficacy and reciprocal determinism.

6.9 ● Other learning theorists: summarise the systems of Hull, Guthrie, Estes, Mowrer and Gagné.

6.10 ● Learning in practice: read learning curves and plateaus, and explain overlearning, spaced practice, whole and part learning, feedback, transfer of training and programmed instruction.

Weightage. 144 of the 819 questions mapped in this book (17.6%) test this chapter, 78 of them from real past papers. 4 of its 10 sections are High yield.

The baby in the old film is calm, almost stolid. In the published report he is simply "Albert B.", the son of a wet nurse at the Harriet Lane Home for Invalid Children, a hospital attached to Johns Hopkins University in Baltimore. When Albert was about nine months old, two psychologists, John B. Watson and his graduate student Rosalie Rayner, brought him a parade of things to look at: a white rat, a rabbit, a dog, a monkey, masks with and without hair, cotton wool, even a burning newspaper. Albert reached for some of them and stared at others. He was afraid of none of them.

Only one thing reliably upset him. When a long steel bar was struck sharply with a hammer just behind his head, Albert startled violently; at the second blow his lips puckered and trembled; at the third he broke into a sudden fit of crying. Watson and Rayner had found what they were looking for: a stimulus that produced fear without any learning at all.

When Albert was eleven months old the experiment proper began. The white rat was taken from its basket and offered to him, and just as his hand touched it the bar was struck behind his head. Albert jumped violently and fell forward, burying his face in the mattress. The pairing was repeated once more that day, and he began to whimper. A week later the rat and the noise were paired five more times, seven pairings in all. Then the rat was shown alone, in silence. The instant he saw it, Albert began to cry, toppled over, raised himself on all fours and crawled away so fast that he was caught only just before he reached the edge of the table.

Five days later he was tested again. Now he whimpered and withdrew from things that had never been paired with the noise: a rabbit, a dog, a sealskin coat, cotton wool, a Santa Claus mask. His wooden blocks, offered between the tests, he played with happily. Soon afterwards Albert left the hospital, and the fears that had been built into him were never removed. A healthy, placid baby had been taught to be afraid, and the fear had spread well beyond the animal that started it. How could this happen in a week? Why did the fear spread to the rabbit and the fur coat but not to the blocks? Could the same process be run backwards to cure a fear? Keep Albert in mind. These questions run through the whole chapter, and we will return to him at the end.

## 6.1 What Learning Is, and What It Is Not MEDIUM YIELD · 6 QS

Almost everything that makes you a particular person rather than a generic human being was learned: your language, your taste in chai, your fear of a certain teacher, your skill at a cricket cover drive, the way your stomach tightens when you walk into an exam hall. Psychologists have studied learning more intensively than almost any other topic, and for the first half of the twentieth century the study of learning very nearly *was* experimental psychology. Before we look at how learning happens, we need to be precise about what it is.

### A working definition

The definition most textbooks use descends from **Gregory Kimble**, who in 1961 revised a classic text on conditioning. In modern form: **learning** is a relatively permanent change in be-

haviour, or in the potential for behaviour, that results from experience. Each phrase does work.

- *Relatively permanent* excludes passing states. If you perform badly because you are sleepy, nothing has been learned; when you wake up, the change is gone. "Relatively" is there because learned things can be forgotten or extinguished; learning is durable, not eternal.
- *Behaviour or the potential for behaviour* acknowledges that learning can be silent. You may learn the route to a new friend's house in Kuvempunagar while riding pillion and show no sign of it until the day you have to drive there yourself.
- *Results from experience* excludes changes caused by growth, injury, drugs or disease. Kimble's own wording said "reinforced practice"; later writers dropped the word "reinforced" because, as you will see in Sections 6.7 and 6.8, a great deal is learned without any reward at all.

The second point leads to one of the most important distinctions in this chapter. **Performance** is what an organism actually does on a given occasion; learning is the change in knowledge or capacity that we infer from performance. The two can come apart. A student who knows the answer but freezes in the exam has learned but not performed. As you will see, Edward Tolman's rats (Section 6.7) and Albert Bandura's children (Section 6.8) both showed that learning can occur without being displayed until there is a reason to display it.

### Changes that are not learning ●●

Several kinds of behavioural change look like learning but are not, and examiners like to use them as distractors.

**Maturation** is change that unfolds according to a biological timetable, largely independent of specific experience. A baby does not learn to walk in the way she learns to ride a bicycle; she walks when her muscles, bones and nervous system are ready, and practice before that point achieves little. Fatigue, illness, injury and drugs also change behaviour, but temporarily or without experience being the cause. So do motivational states: a hungry rat runs faster than a satiated one, but it has not learned anything new.

Finally, much behaviour is innate. Ethologists, biologists who study animal behaviour in natural settings, describe the **fixed action pattern**: an unlearned, stereotyped sequence of behaviour that is the same in every member of a species and, once triggered, runs to completion. The trigger is a **sign stimulus** (also called a releaser), a simple feature of the environment to which the animal is tuned from birth. **Nikolaas (Niko) Tinbergen** (1907 to 1988) found that a male three-spined stickleback in breeding season attacks crude models of fish as long as they have a red underside, the sign stimulus of a rival male, while ignoring a realistic model with no red. A greylag goose that sees an egg outside its nest stretches out and rolls it back with its bill; if the egg is removed halfway, the goose completes the rolling movement anyway. Fixed action patterns are instinctive; they are the result of evolution, not of the individual's experience.

> **EXAM TRAP**
>
> "Fixed action patterns result from ..." The answer is innate (unlearned) behaviour, not classical or operant conditioning. Also watch the word "permanent": learning is *relatively* permanent. And do not confuse habituation (below) with **sensory adaptation**, the reduced firing of receptors under constant stimulation (Chapter 3). Adaptation happens in the sense organ and is not reversed by an unrelated stimulus; habituation happens in the nervous system and is.

## The simplest learning: habituation and sensitisation ●●

The simplest forms of learning involve only one stimulus. Because nothing is being associated with anything else, they are called **non-associative learning**. There are two.

**Habituation** is a decrease in response to a stimulus that is presented repeatedly and turns out to be harmless. The first night in a hostel room near a railway line, every passing train wakes you; by the end of the week you sleep through them. Habituation is adaptive: an organism that kept reacting to every irrelevant event would have no attention left for the ones that matter. It has three diagnostic features. It is stimulus-specific (you still wake if your phone rings). It shows spontaneous recovery (after a long gap without trains, the first train wakes you again). And it shows **dishabituation**: if a new, strong stimulus occurs, the habituated response returns to the old stimulus. Dishabituation is the clearest proof that habituation is a form of learning and not simple fatigue of the muscles or receptors.

**Sensitisation** is nearly the opposite: an increase in response, usually to a wide range of stimuli, after exposure to a strong or noxious stimulus. After a near-miss with a bus on the Mysuru ring road, every horn makes you jump for the rest of the day. Unlike habituation, sensitisation is not specific to the stimulus that caused it. The dual-process theory proposed by **Philip Groves** and **Richard F. Thompson** in 1970 holds that every stimulus sets off both processes at once, and what you observe is their sum.

These two simple forms of learning were the first to be traced to the level of individual neurons. **Eric Kandel** (born 1929) chose the sea slug *Aplysia*, whose nervous system has only about 20,000 neurons, many of them large enough to record from one at a time. Touching the slug's siphon makes it withdraw its gill. With repeated gentle touches the withdrawal habituates, because the sensory neurons release less transmitter onto the motor neurons. A shock to the tail sensitises the reflex, because interneurons release serotonin onto the sensory neurons' terminals and increase their transmitter release. This work earned Kandel a share of the Nobel Prize in 2000 (the cellular details are in the Physiological Psychology book).

Habituation is also one of developmental psychology's most useful tools. Infants cannot tell you what they can distinguish, but they can get bored. **Robert Fantz** showed in 1964 that babies look less and less at a pattern shown repeatedly and look longer at a new one. If an infant habituated to one picture recovers its looking when a second picture appears, the infant must be able to tell the two apart. The habituation-dishabituation paradigm is therefore used to study discrimination and categorisation in babies long before they can speak.

## Imprinting: learning on a biological timetable ●●

Between instinct and ordinary learning lies **imprinting**, the rapid formation of a lasting attachment to the first suitable moving object a young animal encounters. The phenomenon

was described in chicks by **Douglas Spalding** in 1873, but it is linked forever with the Austrian ethologist **Konrad Lorenz** (1903 to 1989). In a 1935 paper Lorenz reported that greylag goslings hatched in an incubator, whose first sight was Lorenz himself, followed him about as if he were their mother, and showed no interest in their real mother. Photographs of a line of goslings marching behind the bearded scientist made the finding famous.

Imprinting differs from ordinary learning in four ways. It happens fast, often within minutes of exposure. It needs no reward: the gosling is not fed for following. It is relatively irreversible: Lorenz's geese continued to prefer humans, and some later directed courtship at them (sexual imprinting). And it can occur only during a **critical period**, a limited window of development in which a particular kind of learning is possible. **Eckhard Hess** found the window peaked about 13 to 16 hours after hatching in mallard ducklings. Later research showed the window is less rigid than Lorenz believed, and many researchers now prefer the term sensitive period. Lorenz shared the 1973 Nobel Prize with Tinbergen and Karl von Frisch, and John Bowlby borrowed the idea of imprinting for his theory of human attachment (Developmental Psychology book).

> EXAM TRAP
>
> Imprinting is rapid, largely irreversible, confined to a critical period and independent of reward. Options describing it as "slow and reversible across the lifespan", "operant conditioning with food reward" or "a purely visual recognition system unrelated to attachment" are all wrong. The classic species is the greylag goose (Lorenz); the 13 to 16 hour peak comes from Hess's mallard ducklings.

### The map of the chapter ●●

Habituation and sensitisation concern a single stimulus. Most learning, by contrast, is **associative learning**: learning that two events go together. Two broad kinds dominate. In classical conditioning the organism learns that one stimulus predicts another (the rat predicts the noise). In operant conditioning it learns that a response produces a consequence (pressing a lever produces food). Most of this research was done on animals, within *comparative psychology*, the branch of psychology that studies animal behaviour and compares species (its history is in Chapter 2). Later, cognitive psychologists showed that animals and people also learn maps, insights and concepts (Section 6.7), and Albert Bandura showed that people learn a great deal simply by watching others (Section 6.8). We begin, as psychology did, with a Russian physiologist and his dogs.

## 6.2 Pavlov and the Language of Classical Conditioning MEDIUM YIELD · 7 QS

### A physiologist and his "psychic secretions" ●

**Ivan Petrovich Pavlov** (1849 to 1936) did not set out to study learning. He was a physiologist, and in 1904 he received the Nobel Prize in Physiology or Medicine for his research on digestion. To measure digestive secretions in living animals he developed surgical techniques of great elegance; for the salivary work, the duct of a dog's salivary gland was redirected through a small opening (a fistula) in the cheek so that every drop of saliva could be collected through a tube and counted.

In the late 1890s Pavlov and his co-workers noticed something that spoiled their measurements. Dogs that had been in the laboratory for a while began to salivate before food reached their mouths: at the sight of the food, at the sight of the dish, even at the sound of the attendant's footsteps. Pavlov at first called these "psychic secretions". Instead of treating them as a nuisance, he realised that they offered an objective way to study how the brain forms new connections, and from about 1897 he and his students studied them systematically. He described the work publicly at a congress in Madrid in 1903, and his lectures appeared in English in 1927 as *Conditioned Reflexes*.

The procedure was simple. A hungry dog stood in a harness in a quiet room. A neutral stimulus, often the ticking of a metronome, a buzzer or a bell, was presented, and a few seconds later meat powder was delivered to the dog's mouth. After a number of such pairings the metronome alone made the dog salivate. Pavlov called the new reflex "conditional", because it depended on the conditions of the dog's experience; the English translation "conditioned" stuck.

Pavlov was not quite the only discoverer. In 1902 the American graduate student **Edwin Burket Twitmyer**, studying the knee-jerk reflex, noticed that his participants' legs began to jerk when they heard the bell that warned of the hammer tap, even without the tap. He reported it to the American Psychological Association in 1904, where it was ignored. Discovery in science belongs partly to the person who sees what a finding means.

### The five terms ●●●

**Classical conditioning** (also called Pavlovian or respondent conditioning) is learning in which a neutral stimulus, through repeated pairing with a stimulus that already produces a response, comes to produce a similar response itself. Its vocabulary has five terms, and exam questions test them constantly.

- The **unconditioned stimulus** (UCS, or US) is a stimulus that produces a response without any prior learning: food in the mouth.
- The **unconditioned response** (UCR, or UR) is the unlearned, reflexive response to the UCS: salivation to food.
- The **neutral stimulus** (NS) is a stimulus that, before conditioning, does not produce the response of interest. It is not completely without effect: the first time the metronome ticks, the dog pricks up its ears and turns towards it, the **orienting response** that Pavlov called the "what-is-it?" reflex. But it does not produce salivation.
- The **conditioned stimulus** (CS) is the formerly neutral stimulus after it has been paired with the UCS and has come to produce a response.
- The **conditioned response** (CR) is the learned response to the CS: salivation to the metronome.

Notice two things. First, the same physical stimulus changes its name: the metronome is a neutral stimulus before conditioning and a conditioned stimulus after it. Second, the same physical response also changes its name according to what produces it. Salivation to food is a UCR; salivation to the metronome is a CR. The CR is usually similar to the UCR but not identical: Pavlov's dogs produced less saliva to the metronome than to food, and with a different chemical composition. What makes the whole thing work is *association*: the pairing of two stimuli close together in time. Figure 6.1 shows the three stages.

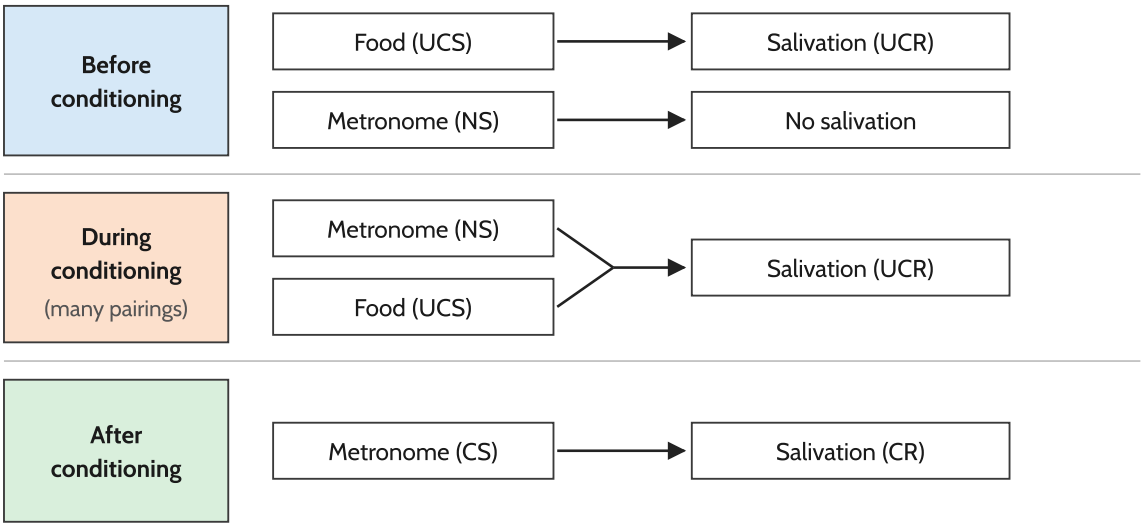


**Figure 6.1** The three stages of classical conditioning. Notice that the metronome changes its label from NS to CS, and that salivation is called a UCR when food produces it but a CR when the metronome produces it.

The best way to master the five terms is to push real cases through four questions. What produced the response before any learning took place? That is the UCS. What was that unlearned response? The UCR. What stimulus was present, neutral at first, and paired with the UCS? The CS. What does that stimulus now produce on its own? The CR. Table 6.1 applies the four questions to five situations, including Albert's.

**Table 6.1** Five classical conditioning situations broken into their parts.

| Situation | UCS | UCR | CS | CR |
|---|---|---|---|---|
| Pavlov's dog | Meat powder | Salivation | Metronome | Salivation to the metronome |
| Little Albert | Loud clang | Startle and fear | White rat | Fear of the rat |
| Child at the clinic | Injection pain | Crying, fear | Doctor's white coat | Crying at a white coat |
| Student who lost a match | Defeat and humiliation | Distress | The sports ground | Anxiety at the ground |
| Cancer patient | Chemotherapy drug | Nausea | Hospital smell | Nausea on arrival (anticipatory nausea) |

The last two rows are worth pausing over. Nobody chooses to become anxious at a cricket ground or nauseated at the smell of a hospital corridor; the response is elicited, automatic and emotional. That is the signature of classical conditioning: it attaches *involuntary* responses, especially emotional and bodily ones, to new stimuli. Your racing heart when you see the exam timetable was learned the same way Pavlov's dogs learned to drool.

EXAM TRAP

"In Pavlov's experiment the bell was ..." The answer is: a neutral stimulus *before* it was associated with food and a conditioned stimulus *after*. It is never an unconditioned stimulus, and never an "operant stimulus". Likewise, a neutral stimulus becomes a CS "when it is repeatedly paired with an unconditioned stimulus", not when it is "reinforced with a reward" (that is operant language) or "occurs randomly". If a question asks for the "fundamental element" of classical conditioning, the answer is association. Some question writers loosely say the pairing brings out an "unconditioned behaviour"; read past the slip and answer classical conditioning.

### Timing: which arrangement conditions best? ●●

Pavlov and his successors quickly found that the order and spacing of the CS and UCS matter enormously. Four arrangements are distinguished (Figure 6.2).

In **delay conditioning**, the CS comes on first and stays on until the UCS arrives, so the two overlap. The metronome starts, keeps ticking, and the food arrives while it is still ticking. This is the most effective arrangement and the standard laboratory procedure. The word "delay" refers to the delay between CS onset and UCS onset, not to any weakness. For fast reflexes such as the eyeblink, conditioning is best when the CS leads by about half a second; for slower responses, longer intervals work.

In **trace conditioning**, the CS comes on and goes off before the UCS begins, leaving a gap. The organism must keep a memory "trace" of the CS across the gap. Trace conditioning works if the gap is short, but it is weaker than delay conditioning and gets weaker as the gap lengthens. Because the CS precedes the UCS in both delay and trace conditioning, both are sometimes grouped together as forward conditioning.

In **simultaneous conditioning**, the CS and UCS begin and end together. Conditioning is weak, because the CS gives no advance warning; the UCS is already there.

In **backward conditioning**, the UCS comes first and the CS follows. This produces little or no excitatory conditioning. If anything, the CS comes to signal that the UCS is over, so it may become a signal for safety (a conditioned inhibitor).

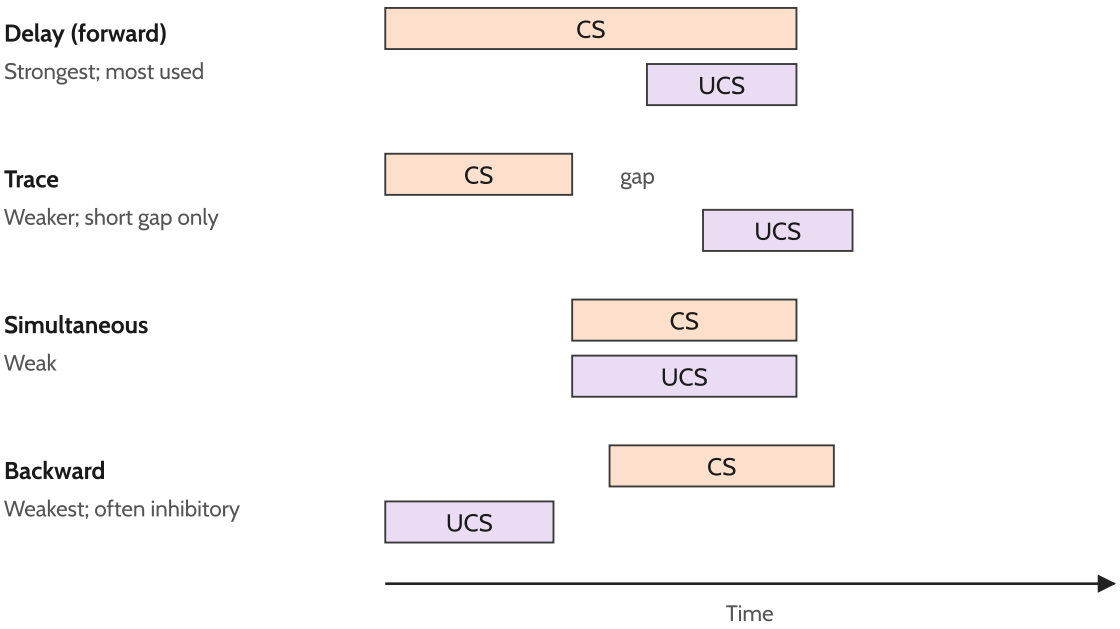


Figure 6.2 The four timing arrangements. In each row the upper bar shows when the CS is on and the lower bar when the UCS is on. Conditioning is strongest when the CS comes first and is still present when the UCS arrives (delay conditioning).

Why should the order matter so much? Because, as Section 6.4 will show, classical conditioning is not really about two things happening together; it is about one thing *predicting* another. A stimulus that comes before the UCS can serve as a warning. A stimulus that comes with it or after it predicts nothing.

> **EXAM TRAP**
>
> "Which is most effective in conditioning?" The answer is delayed (delay) conditioning. Students who think "delay" means "slow" pick trace conditioning instead; trace is the one with the gap. Backward conditioning is the least effective. The one famous exception to the short-interval rule is taste aversion, which forms even when illness follows the taste by hours (Section 6.4).

## 6.3 The Phenomena of Classical Conditioning

MEDIUM YIELD · 6 QS

Once Pavlov had a reliable way to create a conditioned reflex, he and his students spent three decades finding out what could be done to it. They discovered that a conditioned response can be strengthened, weakened, revived, spread to new stimuli, narrowed to one stimulus and even used to build further conditioned responses. These phenomena are the grammar of classical conditioning, and most of them reappear, under the same names, in operant conditioning.

### Acquisition, extinction and the return of the response ●●●

**Acquisition** is the stage in which the CS-UCS pairing is repeated and the CR grows stronger. Plotted against trials, it rises steeply at first and then levels off: each pairing adds less than

the one before (Figure 6.3, left). Acquisition is faster when the UCS is intense, when the CS is noticeable, when the timing is right (delay conditioning with a short interval) and, as Section 6.4 will show, when the CS reliably predicts the UCS.

What happens when the UCS stops coming? If the metronome is sounded again and again with no food, the dog salivates less and less until it stops. This is **extinction**: the gradual weakening and disappearance of a conditioned response when the CS is repeatedly presented without the UCS. Pavlov called the UCS the "reinforcement" of the conditioned reflex, so in his language extinction occurs when reinforcement is withdrawn. Do not let that word mislead you into thinking of operant conditioning; in a classical-conditioning question, "withdrawing reinforcement" simply means omitting the UCS.

It would be natural to assume that extinction erases what was learned. Pavlov showed that it does not, and his evidence is still the core of every modern account.

- **Spontaneous recovery**. If the extinguished animal is given a rest, say a day, and the CS is presented again, the CR reappears, weaker than before, with no further pairings (Figure 6.3, right). Something was suppressing the response, and the suppression faded with time.
- **Disinhibition**. If a novel, unrelated stimulus (a sudden noise) occurs during extinction, the extinguished CR temporarily comes back. Pavlov reasoned that extinction is an active inhibitory process and that the distraction disrupted the inhibition.
- *Rapid reacquisition* (savings). If CS-UCS pairings are resumed after extinction, the CR returns much faster than it was first acquired. This is the classical equivalent of finding that you relearn a forgotten poem in fewer readings than it took the first time.
- **Reinstatement**. After extinction, a few presentations of the UCS on its own, with no CS, are enough to bring the CR back to the CS. Robert Rescorla and Donald Heth demonstrated this in 1975 (you will meet Rescorla again in Section 6.4).
- **Renewal**. Extinction is tied to the context in which it happens. If fear is conditioned in one room (context A) and extinguished in another (context B), the fear returns when the animal is put back in room A, or into any new room. **Mark Bouton** has shown in many experiments that what extinction produces is a new, context-dependent memory ("the CS is safe *here*") that competes with the old one.

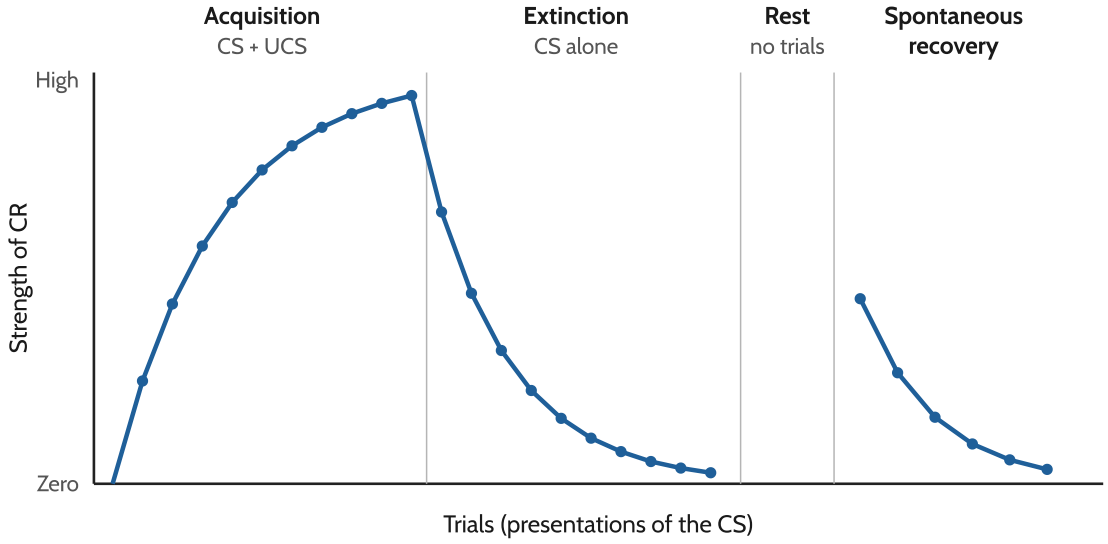


Figure 6.3 Acquisition, extinction and spontaneous recovery. Each pairing adds less strength than the last (a negatively accelerated curve). After extinction and a rest, the CR returns without any new pairings, which shows that extinction had suppressed the response rather than erased it.

CLINICAL LINK

Spontaneous recovery, reinstatement and renewal are the laboratory faces of relapse. A client whose fear of lifts was extinguished in the therapist's building may find the fear back when she steps into the lift at her own office (renewal), or after an unrelated frightening event (reinstatement). This is why modern exposure therapy is carried out in several settings and why therapists warn clients that a brief return of fear is expected and is not failure.

EXAM TRAP

Keep the four "comeback" terms apart. **Spontaneous recovery** needs only the passage of time. **Disinhibition** needs a novel distracting stimulus. **Reinstatement** needs a presentation of the UCS alone. **Reconditioning** (rapid reacquisition) needs new CS-UCS pairings. **Sensitisation** is not a comeback at all; it is the non-associative increase in responding from Section 6.1. "If the UCS is subsequently omitted, the CR will undergo …" Extinction.

## Generalisation and discrimination ●●

Pavlov's dogs, conditioned to one tone, also salivated to other tones, the more so the more similar they were to the original. This is **stimulus generalisation**: the tendency for a CR to be produced by stimuli similar to the CS. If you plot the strength of the response against the similarity of the test stimulus to the original CS, you get a hump-shaped **generalisation gradient**, highest at the original CS and falling off on either side. In 1937 **Carl Hovland** demonstrated such a gradient in humans: participants conditioned to show a skin-conductance response to one tone responded less and less to tones further away from it in pitch. Albert's fear of the rabbit, the dog and the sealskin coat was generalisation from the white rat; things that were furry and rat-like produced fear, while wooden blocks, which shared nothing with the rat, did not.

Generalisation is adaptive. A gazelle that has been chased by one lion does well to be wary of a lion of slightly different size or colour. But generalisation that goes too far is costly, and so organisms also learn **stimulus discrimination**: the learned ability to respond to the CS but not to similar stimuli that have never been followed by the UCS. Pavlov trained it by differential conditioning. One tone (called CS+) was always followed by food; a similar tone (CS-) never was. At first the dog salivated to both; gradually it salivated only to CS+. A child who was frightened by a large black dog but still plays happily with her grandmother's small tabby cat, or a child who avoids one furry black cat but cuddles another, is showing discrimination.

KEY STUDY: EXPERIMENTAL NEUROSIS IN PAVLOV'S LABORATORY (1921)

**Question.** What happens when a discrimination becomes impossible? **Method.** In Pavlov's laboratory, **Shenger-Krestovnikova** projected a circle and an ellipse onto a screen in front of a dog. The circle was always followed by food; the ellipse never was. Once the dog discriminated them, the ellipse was made rounder step by step, until the ratio of its two axes was 9 to 8, almost a circle. **Result.** The dog's discrimination broke down; worse, the formerly placid animal began to squeal and wriggle in its stand, bit through the tubes and barked violently whenever it was brought to the room, and even easy discriminations it had previously mastered were lost. **Conclusion.** Pavlov called this state **experimental neurosis**: a lasting emotional disturbance produced by an unresolvable conflict between excitation and inhibition. **Significance.** It was the first laboratory model of anxiety, and it pointed toward the idea, developed later by Joseph Wolpe, that neurotic anxiety is learned and can therefore be unlearned.

A second, accidental "experiment" convinced Pavlov that his dogs could be traumatised. In September 1924 the River Neva flooded Leningrad, and the dogs in the laboratory's kennels were trapped as the water rose almost to the tops of their cages before they were rescued. Afterwards several of the dogs had lost their carefully trained conditioned reflexes, and some showed lasting fear: a trickle of water allowed to run under the door of the room was enough to throw them back into agitation. Pavlov devoted much of his final decade to these "neuroses" and to applying conditioning ideas to psychiatry.

EXAM TRAP

Generalisation is responding to *similar* stimuli; discrimination is responding *differently* to similar stimuli. A child who avoids a furry black cat but plays with a short-haired tabby is showing discrimination, not extinction (the fear of the black cat has not gone), not successive approximation (that is shaping, Section 6.5), and not negative transfer (old learning interfering with new learning, Section 6.10).

## Building on conditioning: higher-order conditioning and sensory preconditioning

Food, pain and loud noises are rare in daily life, yet adults have strong emotional reactions to words, logos, places and faces that have never been directly paired with any of them. How? Pavlov's laboratory found one answer. Once a metronome has become a strong CS for salivation, it can itself be used in place of food to condition a second stimulus. Pair a black square with the metronome, with no food anywhere, and after some trials the black square alone produces salivation. This is **higher-order conditioning** (second-order conditioning when there are two links): an established CS serves as the UCS for conditioning a new stimulus.

Third-order conditioning is very weak or absent, partly because during second-order training the first CS is repeatedly presented without food and so begins to extinguish.

Higher-order conditioning helps explain why a word like "exam" can make a student's heart race. The word was never itself painful, but it was paired, over years, with the anxious waiting and the stern invigilator that had themselves become conditioned stimuli for fear.

A related procedure reverses the order of training. In **sensory preconditioning**, two neutral stimuli are first paired with each other when neither means anything. Only afterwards is one of them paired with a UCS. The other then produces the CR too, even though it was never paired with the UCS or with a CS. **W. J. Brogden** demonstrated this in dogs in 1939: a light and a bell were presented together many times; then the bell alone was paired with a shock to the leg until it produced leg flexion; finally the light alone, never paired with shock, also produced some flexion. The finding mattered because it showed that animals learn associations between neutral stimuli that have no consequences at the time; the learning stays silent until later events reveal it, a point Edward Tolman would make about rats in mazes (Section 6.7). Table 6.2 sets the two procedures side by side.

Table 6.2 Higher-order conditioning versus sensory preconditioning (S1 and S2 are two neutral stimuli).

| Procedure | Phase 1 | Phase 2 | Test |
|---|---|---|---|
| Higher-order conditioning (Pavlov's laboratory) | S1 + UCS, so S1 becomes a CS | S2 + S1 (no UCS) | S2 alone produces the CR |
| Sensory preconditioning (Brogden, 1939) | S2 + S1 (both neutral) | S1 + UCS | S2 alone produces the CR |

A final point completes the picture. Not every CS signals that something is coming; some signal that something is *not* coming. If a light is always followed by shock except when a tone accompanies it, the tone comes to inhibit fear. Such a stimulus is a **conditioned inhibitor**. Conditioned inhibitors are the laboratory version of safety signals, such as a trusted friend whose presence lets an anxious person enter a crowded mall.

> **EXAM TRAP**
>
> "Using an already-established CS to condition a new neutral stimulus" is higher-order (second-order) conditioning. It is not backward conditioning (UCS before CS), not stimulus discrimination, and not counterconditioning (replacing one response to a stimulus with an incompatible response, Section 6.4). If the two neutral stimuli were paired *before* any UCS appeared, it is sensory preconditioning.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. A decrease in responding to a repeated, harmless stimulus is called _____; its near opposite is _____.
2. The return of a habituated response when a novel strong stimulus occurs is called _____.
3. Lorenz's goslings followed the first moving object they saw during a limited window called the _____ period.
4. In Pavlov's work, the metronome was a _____ stimulus before conditioning and a _____ stimulus after it.
5. The most effective timing arrangement, in which the CS begins first and overlaps the UCS, is _____ conditioning.
6. When an extinguished CR reappears after a rest, with no new pairings, the phenomenon is called _____.
7. Shenger-Krestovnikova's dog, unable to tell a circle from a near-circular ellipse, developed experimental _____.
8. Pairing two neutral stimuli first and only then pairing one of them with a UCS is called _____.
9. Fear that returns when a client goes back to the context in which it was originally learned is an example of _____.

**Answers:** (1) habituation, sensitisation, (2) dishabituation, (3) critical, (4) neutral, conditioned, (5) delay (forward), (6) spontaneous recovery, (7) neurosis, (8) sensory preconditioning, (9) renewal.

## 6.4 Fear, Prediction and Preparedness HIGH YIELD · 29 QS

For its first half-century, classical conditioning was thought of as a simple, mechanical process: pair two stimuli often enough, close enough together in time, and the first will come to produce the response to the second. This section tells how that picture was enriched in three directions. Conditioning was applied to human emotion, first to create fears and then to remove them. It was shown to depend on prediction, not mere pairing. And it was shown to be shaped by evolution, so that some associations are learned in a single trial and others hardly at all.

### Little Albert and conditioned emotional responses ●●

**John Broadus Watson** (1878 to 1958) had launched behaviourism in 1913 with the claim that psychology should study only observable behaviour (Chapter 2). He believed that most adult emotional life consisted of conditioned versions of a few innate emotional reactions, and the study you read about at the opening of this chapter, published with **Rosalie Rayner** in 1920 under the title "Conditioned emotional reactions", was meant to prove it. In the terms of Table 6.1, the loud clang was the UCS, the startle and fear it produced was the UCR, the white rat became the CS, and Albert's fear of the rat was the CR. His fear of the rabbit, the dog, the coat and the mask was stimulus generalisation.

A fear or other emotional reaction acquired through classical conditioning is called a **conditioned emotional response**. In animal laboratories it is usually measured indirectly. In a

procedure introduced by William Estes and B. F. Skinner in 1941, rats press a lever for food; a tone is then paired with a mild shock; afterwards, whenever the tone sounds, the rats stop pressing. The degree of this conditioned suppression measures how much fear the tone produces, and you will meet the procedure again in Kamin's experiments below. Fear conditioning depends critically on the amygdala, a structure in the temporal lobe that receives information about the CS and organises the bodily fear response (**Joseph LeDoux**'s work on these circuits is covered in the Physiological Psychology book).

The Albert study made Watson's case vividly, but it is a weak experiment by modern standards. There was one participant, no control condition, and the fear was measured by informal description; the fear also faded between sessions and had to be "freshened" with extra pairings. Ethically it could not be done today: the researchers deliberately frightened an infant and made no attempt to remove the fear. As you will see in the revisit at the end of the chapter, even Albert's identity is disputed.

### Peter, counterconditioning and the conditioning therapies ●●●

If fear can be conditioned, can it be unconditioned? Four years after Albert, **Mary Cover Jones** (1897 to 1987), working with Watson's advice, answered yes. Her 1924 report "A laboratory study of fear: The case of Peter" described a boy of almost three who was intensely afraid of rabbits, and also, by generalisation, of a fur coat, feathers and cotton wool, much as Albert had been. Jones used two methods. In one, Peter played alongside other children who were not afraid of the rabbit. In the other, while Peter sat eating a favourite food, the rabbit in its cage was brought a little closer each day, never so close that he became upset. Over many sessions Peter came to tolerate the rabbit, then to touch it, and finally to let it nibble his fingers. His fears of the related objects declined as well.

The method is **counterconditioning**: replacing a conditioned response to a stimulus with a new response that is incompatible with it. Eating and contentment cannot easily coexist with terror, so the rabbit, paired with eating, gradually came to signal something that was no longer threatening. Jones is often called the mother of behaviour therapy.

The idea was developed into a systematic treatment three decades later by the South African psychiatrist **Joseph Wolpe** (1915 to 1997). Wolpe first made cats anxious by giving them mild shocks in an experimental cage; afterwards the cats were too anxious to eat in it. He then fed them first in a room quite unlike the cage, where their anxiety was weak enough for them to eat, and then in rooms that resembled the cage more and more closely, until eating, a response incompatible with anxiety, had overcome the anxiety even in the cage itself. He borrowed the physiologist **Charles Scott Sherrington**'s term for the principle: **reciprocal inhibition**, the idea that if a response incompatible with anxiety can be made to occur in the presence of an anxiety-provoking stimulus, it will weaken the bond between that stimulus and anxiety. In *Psychotherapy by Reciprocal Inhibition* (1958) he described the treatment that grew out of it, **systematic desensitisation**, which has three steps.

1. *Relaxation training*. The client learns deep muscle relaxation, usually a shortened form of the progressive relaxation developed by the American physician **Edmund Jacobson** (1888 to 1983), in which muscle groups are tensed and then released one by one. Relaxation is the response that will compete with anxiety.
2. *Constructing an anxiety hierarchy*. Client and therapist list situations involving the feared object, from the mildest to the most frightening, and the client rates each on the

Subjective Units of Disturbance Scale (SUDS), a 0 to 100 rating of felt anxiety that Wolpe introduced in *The Practice of Behavior Therapy* (1969).

3. *Desensitisation proper*. While deeply relaxed, the client imagines (or, in vivo, actually encounters) the lowest item. Only when it no longer produces anxiety does the therapist move up to the next item, and so on to the top.

Systematic desensitisation is one of several treatments that grew directly out of classical conditioning. In **flooding**, the client is exposed to the most feared situation at full intensity, without gradual steps or relaxation, and is kept there until the anxiety subsides on its own; a person afraid of heights stands on a high balcony and does not leave. Flooding works by extinction: the CS (height) is presented without any UCS (falling), and escape, which would otherwise prevent extinction, is blocked. Implosive therapy, developed by **Thomas Stampfl** in the 1960s, is flooding carried out in imagination with deliberately exaggerated scenes. In aversion therapy the logic is reversed: a stimulus linked to an unwanted behaviour (the taste of alcohol) is paired with an aversive UCS (a nausea-producing drug) so that the stimulus becomes unpleasant. In *covert sensitisation*, introduced by **Joseph Cautela** in 1967, both the behaviour and the unpleasant consequence are only imagined: the client pictures reaching for a cigarette and then vividly imagines being violently sick. And in exposure and response prevention, the standard treatment for obsessive-compulsive disorder developed by **Victor Meyer** in 1966, the client is exposed to the obsessional cue (touching a "contaminated" doorknob) and prevented from performing the compulsion (washing). The anxiety rises, peaks and falls; the treatment works through habituation and extinction. The full clinical details of all these treatments are in the Clinical Psychology book; Table 6.10 in Section 6.6 sorts them by the learning principle on which each rests.

> EXAM TRAP
>
> Systematic desensitisation is **Wolpe's**, and its principle is **reciprocal inhibition** (counter-conditioning), not operant extinction, not habituation, not the law of effect and not catharsis. Relaxation comes from **Jacobson**. The anxiety ratings are **SUDS**. Graded steps plus relaxation means desensitisation; full intensity with no escape and no relaxation means flooding; imagined unpleasant consequences of an imagined behaviour means covert sensitisation (Cautela). Exposure and response prevention rests on **habituation and extinction**. And do not confuse Wolpe's reciprocal *inhibition* with Bandura's reciprocal *determinism* (Section 6.8). **Symptom substitution**, sometimes offered as an option, is not a technique at all: it is the psychoanalytic prediction that removing a symptom without resolving its supposed underlying cause will simply produce a new symptom, a criticism of behaviour therapy that outcome research on desensitisation has generally not supported.

## Contiguity is not enough

From Pavlov to the 1960s, almost everyone assumed that the key to conditioning was *contiguity*, the occurrence of the CS and UCS close together in time. Four findings from the late 1950s and 1960s, summarised in Table 6.3, showed that contiguity is not sufficient. What matters is whether the CS gives the animal new information about the UCS.

The first came from **Robert Rescorla** (1940 to 2020) in 1968. He gave groups of rats the same number of tone-shock pairings, but varied how often shocks also occurred when the tone was absent. When shocks came only with the tone, the rats developed strong fear of the tone.

When shocks were just as likely without the tone as with it, the rats showed no fear of the tone at all, even though they had experienced exactly as many tone-shock pairings. Rescorla concluded that conditioning depends on **contingency**: the degree to which the CS predicts the UCS, that is, how much more likely the UCS is in the presence of the CS than in its absence. A positive contingency produces excitatory conditioning; zero contingency produces none; a negative contingency (UCS less likely when the CS is present) produces conditioned inhibition.

The second finding, **blocking**, was reported by **Leon Kamin** (1927 to 2017) in 1969. Kamin first conditioned rats to fear a noise by pairing it with shock. Then he presented the noise together with a light, followed by the same shock. Finally he tested the light alone. Rats that had had the first phase showed almost no fear of the light; control rats that had experienced only the noise-plus-light pairings feared the light strongly. Prior conditioning of the noise had blocked conditioning of the light. The light was perfectly contiguous with the shock, but it was redundant: the noise already predicted the shock, so the light added no information.

The third, **overshadowing**, had been noted by Pavlov himself. When two stimuli are presented together as a compound CS, the more salient one (the louder, brighter or more noticeable) acquires most of the conditioning and the weaker one acquires little, even though either would have been conditioned well if presented alone.

The fourth, **latent inhibition**, was described by **Robert Lubow** and **A. U. Moore** in 1959. If a stimulus is presented many times on its own before conditioning begins, it is later much harder to condition. The animal has learned that the stimulus is irrelevant. Latent inhibition explains a useful clinical fact: a child who has had many uneventful visits to the dentist is less likely to develop a dental phobia after one painful visit than a child whose first visit is the painful one.

Table 6.3 Four findings showing that pairing alone does not produce conditioning.

| Finding | Who, when | What was done | What it shows |
|---|---|---|---|
| Contingency | Rescorla, 1968 | Equal CS-UCS pairings; UCS also given without the CS | No conditioning unless the CS predicts the UCS |
| Blocking | Kamin, 1969 | Noise trained first, then noise plus light | A redundant CS is not conditioned |
| Overshadowing | Pavlov | Strong and weak stimuli presented as a compound | The more salient stimulus takes most of the learning |
| Latent inhibition | Lubow and Moore, 1959 | CS presented alone many times before conditioning | A familiar, meaningless stimulus conditions slowly |

## The Rescorla-Wagner model: learning from surprise ●●

In 1972 Rescorla and **Allan Wagner** (1934 to 2018) proposed a simple mathematical model that captures most of these findings. The **Rescorla-Wagner model** says that on every trial the associative strength of a CS changes in proportion to how surprising the outcome is. In words: the animal learns only when what happens differs from what it expected. In symbols:

$$\Delta V = \alpha\beta(\lambda - \Sigma V)$$

Here V is the associative strength of the CS (roughly, how strongly it predicts the UCS); $\Delta V$ is the change in that strength on this trial; $\alpha$ is the salience of the CS (how noticeable it is); $\beta$ is a learning-rate value set by the UCS; $\lambda$ (lambda) is the maximum associative strength the UCS

can support, which is set to zero on trials when the UCS is absent; and ΣV is the sum of the associative strengths of all the stimuli present on the trial, which is the animal's total expectation. The term in brackets, (λ − ΣV), is the surprise, or **prediction error**.

Push some numbers through it. Let αβ = 0.3 and λ = 100. On trial 1 the CS has no strength, so ΔV = 0.3 × (100 − 0) = 30, and V becomes 30. On trial 2, ΔV = 0.3 × (100 − 30) = 21, so V becomes 51. On trial 3, ΔV = 0.3 × 49 = 14.7, so V becomes 65.7. Each increment is smaller than the last because the UCS is becoming less surprising; that is why acquisition curves are negatively accelerated (Figure 6.3). Now blocking: after many noise-shock trials the noise has V close to 100. When the light is added, ΣV = 100 (noise) + 0 (light) = 100, so the prediction error is 100 − 100 = 0, and the light gains nothing. Extinction follows too: on a CS-alone trial λ = 0, so ΔV = 0.3 × (0 − 65.7), a negative change.

The model is not perfect. It treats extinction as unlearning, which cannot explain spontaneous recovery or renewal, and it predicts no effect of presenting a CS alone before conditioning, so it cannot explain latent inhibition. Later models added attention to deal with these problems. But its central idea proved extraordinarily fruitful. In the 1990s **Wolfram Schultz** recorded from dopamine neurons in the ventral tegmental area and substantia nigra of monkeys. The neurons fired in a burst to an unexpected reward; once a light reliably predicted the reward, they fired to the light and not to the reward; and if the predicted reward failed to arrive, their firing dipped below baseline at the moment it should have come. The neurons were signalling prediction error, the quantity in the Rescorla-Wagner brackets. The same idea now drives the reinforcement-learning algorithms used in artificial intelligence.

> EXAM TRAP
>
> The brain's reward "prediction error" signal is carried by **dopamine neurons of the ventral tegmental area** (and substantia nigra), not by serotonin release, GABA inhibition or acetylcholine release. Blocking is Kamin's; contingency is Rescorla's; the model is Rescorla and Wagner's (1972); latent inhibition is Lubow and Moore's. Latent *inhibition* (pre-exposure slows conditioning) is a different thing from latent *learning* (Tolman, Section 6.7).

### What is learned: a substitute stimulus or a signal?

Pavlov held a **stimulus substitution theory**: through pairing, the CS comes to activate the same brain centre as the UCS, so it becomes a substitute for the UCS and produces the same response. The theory predicts that the CR should be a copy of the UCR, and often it is not. Rats shocked through the floor jump and run (the UCR), but to a tone that predicts shock they freeze (the CR). The most striking evidence comes from drug tolerance. **Shepard Siegel** showed that the cues that regularly accompany a drug (the room, the syringe, the ritual) come to produce a *compensatory* response, opposite to the drug's effect, that prepares the body for it. This is part of why tolerance develops. It also explains a grim finding: rats given a large dose of heroin in a familiar drug context survived more often than rats given the same dose in an unfamiliar context, where the protective compensatory response was not triggered. Many human "overdoses" occur at doses the person usually tolerates, but in an unusual setting.

The modern view, which Rescorla championed, is an information or expectancy view: in classical conditioning the organism learns that one event predicts another, and the CR is whatever response best prepares it for the predicted event. On this view the CS is a signal,

not a substitute. The idea that conditioning depends on prediction and surprise fits the evidence from contingency, blocking and the dopamine neurons.

### Biological constraints: Garcia's rats and the idea of preparedness ●●

Behind classical learning theory lay an assumption that was rarely stated because it seemed obvious. It was called **equipotentiality**: the view that any stimulus can be associated with any response or UCS with equal ease, so that the laws of learning are the same whatever stimuli and responses are used. Pavlov's dogs and the metronome were assumed to be a convenient example of a universal process. In the 1960s an American psychologist overturned this assumption.

#### KEY STUDY: GARCIA AND KOELLING'S "BRIGHT, NOISY, TASTY WATER" (1966)

**Question.** Are all stimuli equally easy to associate with all consequences? **Method. John Garcia** (1917 to 2012) and **Robert Koelling** let thirsty rats drink from a spout. The water was flavoured with saccharin (taste), and every lick also produced a flash of light and a click (bright, noisy). Some rats were then made ill (by X-irradiation or a toxic drug); others received electric shocks to the feet. Later each rat was tested with tasty water (flavour, no light or click) and with bright-noisy water (plain water with light and click). **Result.** Rats made ill avoided the tasty water but drank the bright-noisy water freely; rats shocked avoided the bright-noisy water but drank the tasty water freely. **Conclusion.** Rats readily associate taste with illness and sights and sounds with external pain, but not the reverse. The cue and the consequence must "belong" together. **Criticism and fate.** The findings so contradicted accepted theory that Garcia had great difficulty getting them published; they are now among the most cited in the field.

Garcia's broader work on **conditioned taste aversion**, the avoidance of a taste that has been followed by illness, broke three accepted principles at once. It showed one-trial learning: a single pairing of a new taste with illness is enough, whereas a metronome needs many pairings with food. It showed long-delay learning: the aversion forms even when the illness follows the taste by several hours, which flatly contradicts the rule that the CS and UCS must be close in time. And it refuted equipotentiality, as the key study shows. The adaptive logic is obvious once stated: in the wild, poisoned food makes an animal ill hours after eating, and an animal that could learn from a single such experience would survive to reproduce.

**Martin Seligman** (born 1942) generalised the point in 1970 as the principle of **preparedness**: through evolution, each species is prepared to learn some associations easily, unprepared for others (they can be learned with effort), and contraprepared for still others (they are learned with great difficulty or not at all). In 1971 he applied the idea to human phobias. People commonly develop phobias of snakes, spiders, heights, the dark and enclosed spaces, which were dangers to our ancestors, but rarely of guns, electric sockets or cars, which are far more dangerous today. Preparedness may also explain why Albert's fear attached so readily to a furry animal: when **Elsie Bregman** tried, in 1934, to condition infants to fear wooden blocks and cloth objects, she largely failed.

Taste-aversion learning has practical uses. **Ilene Bernstein** showed in 1978 that children receiving nausea-producing chemotherapy developed aversions to an unusual ice cream eaten shortly before treatment; later work found that giving patients a novel "scapegoat" food before chemotherapy can protect their usual diet from becoming aversive. Ranchers in the

western United States have reduced coyote attacks on sheep by leaving out lamb meat laced with lithium chloride, which makes the coyotes ill and gives them an aversion to lamb. Aversion therapy for alcohol use, described above, uses the same principle.

Advertisers apply classical conditioning constantly. A brand (the CS) is shown again and again with a beloved film star, a cricket hero, attractive people or pleasant music (the UCS), in the hope that the warm feelings these produce will transfer to the brand. The transfer of liking or disliking through pairing is called evaluative conditioning. In a well-known 1982 experiment, the marketing researcher **Gerald Gorn** found that students who saw a pen of one colour while hearing music they liked were more likely to choose a pen of that colour afterwards.

> EXAM TRAP
>
> Taste aversion is an example of **classical conditioning** (taste = CS, illness = UCS, nausea = UCR, aversion = CR), not operant conditioning, not second-order conditioning, and not a "spurious association" (a statistical term for a coincidental link). The Garcia effect best illustrates **biological preparedness (constraints on learning)**. It contradicts two textbook rules: that all stimuli are equally easy to associate (equipotentiality) and that conditioning needs near-simultaneous events (contiguity).

## 6.5 Thorndike, Skinner and the Consequences of Behaviour

HIGH YIELD · 41 QS

Classical conditioning explains how stimuli come to trigger reflexive and emotional responses. But most of what you do is not triggered in that way. You choose to study, to message a friend, to order masala dosa rather than idli. These actions are not elicited by a stimulus; they are emitted, and whether you repeat them depends on what happened the last time you did them. The study of how consequences shape behaviour began, a few years before Pavlov's first reports, with an American graduate student and a houseful of cats.

### Thorndike's puzzle boxes and the law of effect ●●

**Edward Lee Thorndike** (1874 to 1949) began his animal work as a student of William James at Harvard, training chicks to run through mazes; when his landlady objected, James let him keep the chicks in the cellar of his own house. Thorndike then moved to Columbia University, and his doctoral dissertation, *Animal Intelligence* (1898), became one of the founding documents of comparative psychology.

His best-known apparatus was the puzzle box: a slatted wooden crate from which a hungry cat could escape, to reach a piece of fish placed outside, only by performing a particular action such as pulling a loop of string, pressing a lever or stepping on a platform. Placed in the box, the cat clawed at the bars, squeezed at the gaps, bit and scrambled, until by accident it operated the release and got out. On the next trial it went through much the same frantic routine, but it hit the release a little sooner. Over many trials the useless movements dropped out and the escape time fell, but gradually and irregularly, not suddenly. When Thorndike plotted escape time against trials he produced some of the first learning curves in psychology, and their gradual slope persuaded him that the cats were not reasoning their way out. He

called the process **trial-and-error learning** (he also called it learning by "trial and accidental success").

To explain it, Thorndike proposed the **law of effect**, stated in its classic form in 1911: of several responses made to the same situation, those that are accompanied or closely followed by satisfaction to the animal will, other things being equal, be more firmly connected with the situation, so that when the situation recurs they will be more likely to recur; those followed by discomfort will have their connections with the situation weakened. Thorndike defined "satisfaction" behaviourally, as a state of affairs the animal does nothing to avoid and often acts to attain. In plain words: behaviour followed by pleasant consequences is more likely to be repeated, and behaviour followed by unpleasant consequences is less likely to be repeated.

Thorndike thought of learning as the strengthening and weakening of connections (bonds) between situations and responses, and his theory is therefore called **connectionism**. (It should not be confused with the modern connectionist, or neural-network, models of cognition described in Chapter 8.) He added two further primary laws. The **law of exercise** (also called the law of use and disuse) held that connections are strengthened by being used and weakened by disuse. The **law of readiness** held that when an organism is prepared to act, acting is satisfying and being prevented from acting is annoying, while being forced to act when unprepared is annoying. He also proposed five subsidiary laws, summarised in Table 6.4.

Table 6.4 Thorndike's laws of learning.

| Law | What it says |
|---|---|
| Law of effect (primary) | Responses followed by satisfaction are strengthened; those followed by discomfort are weakened. |
| Law of exercise (primary) | Use strengthens a connection; disuse weakens it. Later abandoned. |
| Law of readiness (primary) | Acting when ready is satisfying; being blocked when ready, or forced when unready, is annoying. |
| Multiple response | Faced with a problem, the learner tries response after response until one succeeds. |
| Set or attitude | What the learner brings to the situation (hunger, mood, expectation) shapes what is learned. |
| Prepotency of elements | The learner responds selectively to the dominant (prepotent) features of a situation. |
| Response by analogy | A new situation is met with the response learned for a similar old one. |
| Associative shifting | A response can be transferred to a new stimulus by gradually changing the situation. |

Thorndike was an honest scientist and revised his laws when his own data demanded it. In the early 1930s he reported that repetition alone does not improve performance: people who drew line after line, blindfolded, trying to make each one three inches long, did not get better if they were never told how close they had come. He therefore abandoned the law of exercise. He also concluded that punishment does not weaken connections as strongly as reward strengthens them, so that only the "satisfaction" half of the law of effect survived. This revised version is often called the truncated law of effect.

EXAM TRAP

The law of effect is **Thorndike's**, not Skinner's, Pavlov's or Watson's. It states that behaviours followed by pleasant consequences are more likely to be repeated. Do not confuse it with learning by observing models (Bandura), sudden restructuring of a problem (Köhler's insight) or unconscious drives (Freud). Thorndike's process is trial-and-error learning; if a question asks for the general name of "a process in which a response is made more frequent by reinforcement", the answer is operant conditioning.

## Skinner and operant behaviour ●●●

**Burrhus Frederic Skinner** (1904 to 1990) set out to be a novelist, gave it up, and arrived at Harvard as a psychology graduate student in 1928 with Watson's behaviourism as his inspiration. In *The Behavior of Organisms* (1938) he set out the system that dominated learning research for the next thirty years. He also wrote a utopian novel, *Walden Two* (1948), and the books *Science and Human Behavior* (1953), *Verbal Behavior* (1957) and *Beyond Freedom and Dignity* (1971).

Skinner drew a sharp distinction between two kinds of behaviour. **Respondent behaviour** is elicited by a known stimulus: salivation to food, a blink to a puff of air. It is the behaviour Pavlov studied, and Skinner called its conditioning Type S. **Operant behaviour** is emitted by the organism, without any obvious eliciting stimulus, and it operates on the environment to produce consequences: a rat presses a lever, a child raises her hand, you unlock your phone. Skinner coined the term "operant" in 1937 and called its conditioning Type R. **Operant conditioning** is learning in which the probability of a response is changed by its consequences.

To study operant behaviour Skinner built the apparatus now known as the **Skinner box** (he preferred "operant chamber"): a small enclosure with a lever for a rat, or a lit plastic disc (a key) for a pigeon to peck, a mechanism that delivers food pellets or grain, and often lights, a loudspeaker and a floor that can deliver mild shock. Unlike Thorndike's cat, which had to be put back in the box after every escape, Skinner's rat could respond whenever it liked, as often as it liked. Skinner therefore measured the rate of responding, and he recorded it with an elegant device, the **cumulative recorder**. A roll of paper moves steadily under a pen; each response moves the pen one small step upward, and the pen never comes down. The result is a line whose slope shows the rate: a steep line means rapid responding, a flat line means none, and small diagonal ticks mark the moments of reinforcement. You will see four such records in Figure 6.4.

Skinner's philosophy is called radical behaviourism. It holds that private events such as thoughts and feelings are themselves behaviour, governed by the same laws as public behaviour, and that explanations in terms of mental causes ("he studied because he wanted to pass") are incomplete until we identify the environmental history that produced the wanting. Thorndike's procedure, with separate trials, is often called **instrumental conditioning**, and Skinner's free-responding procedure operant conditioning, but the two terms are used almost interchangeably today.

Operant behaviour does not occur in a vacuum. A rat whose lever presses produce food only when a light is on soon presses mainly when the light is on. A stimulus in whose presence a response is reinforced is a **discriminative stimulus** (written $S^D$); one in whose presence it is not reinforced is an S-delta ($S^\Delta$). The discriminative stimulus does not elicit the response, as a

CS does; it "sets the occasion" for it. The green signal at a crossing, the "Open" board outside a chai stall and the presence of a strict teacher are all discriminative stimuli. Skinner described operant learning as a **three-term contingency**: discriminative stimulus, response, consequence, which clinicians call the A-B-C (antecedent, behaviour, consequence). When behaviour reliably occurs in the presence of a discriminative stimulus and not in its absence, it is said to be under stimulus control. Operant responses show generalisation and discrimination just as conditioned reflexes do.

CLINICAL LINK

A **functional analysis** (or functional behavioural assessment) is the operant method of finding out why a problem behaviour persists. The clinician records each episode with its antecedents and consequences, asking what the behaviour *gets* the person (attention, an object, sensory stimulation) or what it lets the person *escape* (a demand, a feared situation). Applied behaviour analysis (ABA), the use of operant principles to change socially important behaviour, is built on this method, so "behaviour analysis is based on" operant conditioning. A practical wrinkle: when clients start recording their own behaviour (counting cigarettes, logging snacks), the behaviour often changes before any treatment begins. This is called **reactivity**.

## Four kinds of consequence ●●●

The heart of operant conditioning is a simple classification that confuses more students than any other idea in psychology. It rests on two separate questions about any consequence.

First: does the behaviour become more likely or less likely in future? If more likely, the consequence is **reinforcement**; if less likely, it is **punishment**. Reinforcement and punishment are defined entirely by their effect on behaviour, not by anyone's intentions or by whether the consequence seems pleasant.

Second: was a stimulus added or removed? If added, the consequence is called positive; if removed, negative. "Positive" and "negative" here mean plus and minus, exactly as in arithmetic. They do not mean good and bad.

Crossing the two questions gives four combinations (Table 6.5). In **positive reinforcement**, a stimulus is added and the behaviour increases (praise after a correct answer). In **negative reinforcement**, a stimulus is removed and the behaviour increases; the removed stimulus is usually aversive (a headache that goes away after you take a pill). In **positive punishment**, a stimulus is added and the behaviour decreases (a scolding after you are rude). In **negative punishment**, a stimulus is removed and the behaviour decreases; the removed stimulus is usually something the person values (a mobile phone confiscated for a week).

Table 6.5 The four kinds of operant consequence.

| Effect on behaviour | Stimulus added (positive) | Stimulus removed (negative) |
|---|---|---|
| Behaviour increases (reinforcement) | Positive reinforcement: something wanted is given | Negative reinforcement: something unpleasant is taken away |
| Behaviour decreases (punishment) | Positive punishment: something unpleasant is given | Negative punishment: something wanted is taken away (time-out, response cost) |

The only reliable way to classify a real example is to ask the two questions in order, every time. Table 6.6 does this for eleven cases. Work through them with the right-hand column covered.

**Table 6.6** Classifying consequences: ask "added or removed?" and then "does the behaviour go up or down?"

| Situation | Added or removed? | Behaviour | Consequence |
|---|---|---|---|
| A child gets a chocolate after finishing homework | Chocolate added | Homework up | Positive reinforcement |
| You express an opinion and a friend warmly agrees | Approval added | Opinion strengthened | Positive reinforcement |
| You buckle your seatbelt and the beeping stops | Beeping removed | Buckling up | Negative reinforcement |
| A rat presses a lever that switches off a shock already on | Shock removed | Pressing up | Negative reinforcement (escape) |
| You take a painkiller and the headache goes | Pain removed | Pill-taking up | Negative reinforcement |
| A toddler touches a hot tawa | Burn added | Touching down | Positive punishment |
| A driver is fined for jumping a red light | Money removed | Jumping down | Negative punishment (response cost) |
| A boy who hits his sister is sent from the game for five minutes | Play removed | Hitting down | Negative punishment (time-out) |
| A teacher scolds a pupil for talking, and the talking increases | Attention added | Talking up | Positive reinforcement, not punishment |
| A child's tantrum in a shop ends when she is bought a sweet | Sweet added (for the child) | Tantrums up | Positive reinforcement of the child |
| The same parent gives in, and the screaming stops | Screaming removed (for the parent) | Giving in up | Negative reinforcement of the parent |

The last two rows describe one of the most common traps in family life. Each person's behaviour reinforces the other's: the child learns that screaming produces sweets, and the parent learns that sweets stop screaming. Both behaviours grow. The scolding example makes a related point: a consequence intended as punishment is actually reinforcement if the behaviour increases, which is what often happens when a child starved of attention finds that misbehaviour reliably earns it.

EXAM TRAP

**Negative reinforcement is not punishment.** All reinforcement, positive or negative, *increases* behaviour; all punishment decreases it. A rat pressing a lever to turn off a shock is negative reinforcement (escape conditioning). Time-out and response cost are **negative punishment**. One real paper asked "which consequence increases the likelihood of a behaviour being repeated?" and offered both positive reinforcement and negative reinforcement; the keyed answer was positive reinforcement, but negative reinforcement increases behaviour too. If you meet such an item, choose positive reinforcement (the intended "reward" answer) and remember that the item, not the principle, is at fault. Operant conditioning motivates behaviour through reinforcement and punishment, not through modelling (Bandura), classical conditioning or cognitive dissonance (Festinger).

### Primary, secondary and generalised reinforcers ●●●

Why does a rat work for food, a pigeon for grain, and you for a pay slip? Food is a **primary reinforcer** (or unconditioned reinforcer): it is reinforcing without any learning, because it satisfies a biological need. Water, warmth, sexual contact and relief from pain are other examples. Money is different. A five-hundred-rupee note is just printed paper; it is reinforcing only because it has been repeatedly associated with things that are reinforcing in themselves. A stimulus that acquires its reinforcing power through association with a primary reinforcer is a **secondary reinforcer** (or conditioned reinforcer). Grades, praise, a smile, the "click" a dog trainer sounds just before giving a treat, and the ping of a notification are all secondary reinforcers. Notice that the process by which a neutral stimulus becomes a secondary reinforcer is itself classical conditioning: the click is paired with food until it acquires some of food's value.

The power of secondary reinforcers was shown dramatically by **John B. Wolfe** in 1936. Chimpanzees learned to insert poker chips into a vending machine (nicknamed the "Chimp-O-Mat") that dispensed grapes, and then to work at a heavy lever-pulling task to earn the chips. They would work for chips even when the machine was not immediately available, and some hoarded them. A secondary reinforcer that has been paired with many different primary reinforcers, so that it works whatever the person currently needs, is a **generalised reinforcer**. Money is the obvious example; Skinner also listed attention, approval and affection.

These ideas were turned into a treatment method in the early 1960s by **Teodoro Ayllon** and **Nathan Azrin** at Anna State Hospital in Illinois. On a ward of long-stay psychiatric patients, many of whom had been diagnosed with schizophrenia and had become withdrawn and inactive, patients earned tokens for clearly defined behaviours: self-care, making their beds, working on ward jobs. The tokens could be exchanged for backup reinforcers such as a private room, walks in the grounds, extra time with staff, snacks and cigarettes. Participation in the target behaviours rose sharply; when tokens were given regardless of behaviour, participation fell; when the contingency was restored, it rose again. Their book *The Token Economy* (1968) gave the method its name. A **token economy** is a behaviour-modification system in which tokens (secondary reinforcers) are earned for target behaviours and exchanged for backup reinforcers. It is an application of operant conditioning, and it has since been used in classrooms, homes for children, prisons and rehabilitation units.

EXAM TRAP

Who developed the token economy? The historically correct answer is **Ayllon and Azrin** (Anna State Hospital; book 1968). One real paper (Amity 2023) keyed **Skinner**, presumably because the method rests on Skinner's operant principles. If both appear, Ayllon and Azrin is the accurate answer; Skinner supplied the principles, not the ward programme. Token economy is based on **operant** conditioning (secondary reinforcement), never on classical conditioning, observational learning, insight or free association. A teacher's stickers exchangeable for a homework exemption are best called **tokens**; they are secondary, not primary, reinforcers. Tokens can also be generalised reinforcers when they buy many different backups, but "token" is the specific answer. A "prepotent response" is simply a dominant, highly probable response (the word echoes Thorndike's law of prepotency of elements), not a kind of reinforcer.

REMEMBER

Harry Harlow's famous monkeys (1958) were a test of the secondary-reinforcer idea. Many learning theorists held that infants love their mothers because mothers are paired with feeding, so that the mother is a secondary reinforcer (the "cupboard love" or secondary-drive theory). Harlow's infant rhesus monkeys clung to a soft cloth surrogate that gave no milk in preference to a wire surrogate that fed them, and ran to the cloth mother when frightened. Attachment rests on contact comfort, not on conditioned reinforcement from feeding. (The attachment research belongs to the Developmental Psychology book; Harlow's learning-set research is in Section 6.7.)

## Escape, avoidance and two-factor theory ●●●

Negative reinforcement comes in two forms. In **escape learning**, a response terminates an aversive stimulus that is already present: the rat presses a lever to turn off a shock that is on; you step out of the midday sun into the shade. In **avoidance learning**, a response prevents an aversive stimulus from occurring at all: you leave home early to avoid the traffic jam; you carry an umbrella in the monsoon.

Avoidance is usually studied in a shuttle box, a two-compartment box with a low barrier between the halves. A warning signal, such as a light, comes on, and a few seconds later the floor on that side is electrified. At first the animal jumps the barrier only after the shock begins (escape). Soon it jumps as soon as the light comes on, before any shock (avoidance). Avoidance responses are remarkably durable. In 1953 **Richard Solomon** and **Lyman Wynne** found that dogs which had learned to avoid a strong shock went on jumping, quickly and calmly, for hundreds of trials after the shock had been disconnected.

Avoidance posed a puzzle. In escape, the reinforcer is obvious: the shock goes off. But in successful avoidance nothing happens; the shock never arrives. How can the absence of an event reinforce anything? **Orval Hobart Mowrer** (1907 to 1982) answered in 1947 with his **two-factor theory** (two-process theory) of avoidance. In the first factor, classical conditioning, the warning signal is paired with shock and comes to elicit fear. In the second factor, operant conditioning, jumping the barrier takes the animal away from the warning signal, and the reduction in fear negatively reinforces the jump. The animal is not avoiding the shock; it is escaping the fear. The theory also explains why avoidance resists extinction: because the ani-

mal always leaves before the shock would have come, it never discovers that the shock has stopped.

Two-factor theory became the standard behavioural account of anxiety disorders. A woman who was once trapped in a lift learns to fear lifts (classical); taking the stairs reduces the fear and is negatively reinforced (operant); and because she never rides a lift again, the fear never extinguishes. Obsessive-compulsive disorder fits the same mould. An intrusive thought of contamination produces anxiety; washing the hands reduces the anxiety; the washing is therefore negatively reinforced, and from the standpoint of learning theory compulsions are escape and avoidance behaviours (often called safety behaviours). This is exactly why exposure and response prevention (Section 6.4) works: by blocking the escape response, it allows the anxiety to extinguish and habituate.

> EXAM TRAP
>
> Escape **terminates** an aversive stimulus that is already present; avoidance **prevents** its onset. Both are negative reinforcement, not positive reinforcement. Compulsive acts in OCD are best described as **escape (avoidance) behaviours** maintained by anxiety reduction; one real paper keyed "escape", and "conditioning" is too general to be the best answer. Handwashing in OCD is an avoidance response or safety behaviour, not a conditioned stimulus, not an unconditioned response, and not behaviour maintained by positive reinforcement. "Two-factor theory" in a learning question means Mowrer's theory of avoidance (do not confuse it with Schachter's two-factor theory of emotion, Chapter 11).

### Punishment and its problems ●●●

Punishment is the most widely used and most poorly understood way of changing behaviour. Two forms are used deliberately in clinics and classrooms. **Time-out** (short for time-out from positive reinforcement) removes the person for a short period from a setting where reinforcement is available: the child who hits is taken out of the game for a few minutes. **Response cost** removes a specific amount of a reinforcer the person already has, after each instance of the unwanted behaviour: a fine, the loss of tokens, the loss of an hour of television. Both are negative punishment. A third method, overcorrection, requires the person to restore the environment to a better state than before and to practise the correct behaviour repeatedly.

Does punishment work? **William Kaye Estes** (1919 to 2011) gave an early and influential answer in 1944. Rats that had learned to press a lever for food were put on extinction, and some were also given a brief shock whenever they pressed. The punished rats stopped pressing quickly. But when the punishment ended they started pressing again, and by the end of extinction they had made about as many responses in total as unpunished rats. Punishment had suppressed the response, not erased it. Later work by Azrin and others showed that punishment can produce lasting suppression, but only under demanding conditions: it must be immediate, it must follow every instance of the behaviour, it must be intense enough from the start (animals and children adapt to punishment that is gradually increased), and it works best when an alternative behaviour is available and reinforced.

Even when it works, punishment has well-documented side effects. It tells the person what not to do but not what to do. It produces fear, and by classical conditioning the fear spreads to the punisher and the setting, so that the child learns to avoid the parent or the school. It

can provoke aggression, and physical punishment models aggression for the child to imitate (Section 6.8). It often suppresses the behaviour only while the punisher is present, so that the punisher becomes a discriminative stimulus for *not* misbehaving. And it is negatively reinforcing for the punisher, because it stops the annoying behaviour at once, so punishment tends to escalate. Skinner argued throughout his career that society should rely on reinforcement instead.

> **INDIAN CONTEXT**
>
> Section 17 of India's Right of Children to Free and Compulsory Education Act, 2009 prohibits physical punishment and mental harassment of children in schools. The behavioural evidence summarised above, that punishment suppresses rather than teaches and carries fear, avoidance and aggression with it, is part of the scientific case for such laws. Clinical psychologists working with families routinely teach parents to replace smacking with time-out, response cost and, above all, reinforcement of the behaviour they want.

> **EXAM TRAP**
>
> Removing a desirable stimulus after an undesirable behaviour to reduce it is **response cost** (a form of negative punishment). It is not negative reinforcement (removal that *increases* behaviour), not positive punishment (something unpleasant *added*), and not extinction (simply no longer delivering the reinforcer that maintained the behaviour). Time-out is technically **negative punishment**.

### Shaping, chaining, prompting and fading ●●●

Reinforcement can strengthen only a behaviour that occurs. How, then, do you teach a rat to press a lever it has never touched, or a child with autism to say a word he has never said? Skinner's answer was **shaping**, also called the method of successive approximations: reinforcing behaviours that are closer and closer to the target behaviour. To shape lever pressing, the trainer first reinforces the rat for facing the lever, then only for moving towards it, then only for rearing near it, then only for touching it, and finally only for pressing it. At each stage the previous approximation is no longer reinforced (it is put on extinction) while the closer one is. Skinner could shape a pigeon to turn a full circle in a few minutes this way.

Shaping can work on any dimension of a response: its form, its speed or rate, its force or intensity, its duration, or how quickly it begins. And because it relies only on reinforcing what the learner does, it needs no verbal instructions at all, which is why it is used to train animals, infants and people with severe intellectual disability or autism. A speech therapist who first reinforces any sound, then a sound beginning with "b", then "ba", and finally "ball" is shaping.

Many skills are sequences rather than single responses. A **chaining** procedure teaches such a sequence by linking responses so that each one produces the discriminative stimulus for the next, with reinforcement coming at the end. Each link's stimulus is therefore both a conditioned reinforcer for the response before it and a signal for the response after it. Chains are often taught backwards (backward chaining): when teaching a child to put on a shirt, the therapist first teaches the last step (pulling it down) so that the child always ends with success, then the step before, and so on. A rat named Barnabus at Brown University in the early

1960s was trained by backward chaining to climb a spiral staircase, cross a drawbridge, climb a ladder, pull a toy car to him, ride it through a tunnel and finally press a lever for food.

Two further techniques help get behaviour started. A **prompt** is an extra stimulus that makes the correct response more likely: a verbal hint, a gesture, a demonstration or physical guidance of the learner's hand. **Fading** is the gradual removal of the prompt so that, in the end, the natural discriminative stimulus alone controls the behaviour.

EXAM TRAP

Reinforcing successive approximations is **shaping**. Linking a sequence of responses is **chaining**; adding an extra cue is **prompting**; gradually withdrawing the cue is **fading**. Shaping is not "removing a reinforcer to weaken behaviour", not "pairing two neutral stimuli", not "punishing every incorrect response", not "pigeons pressing a button for food" (that describes operant conditioning in general) and not "any behaviour with pleasant consequences is repeated" (the law of effect). One unusual real item (RML 2024) asked on which "principles" shaping is based and keyed **speed, power and non-verbal**, excluding "performance". The logic, as far as it can be reconstructed, is that shaping works on the speed and force of responses and needs no verbal instruction; learn the definition first, and treat this item as a one-off.

## Operant extinction ●●●

When a response that has been reinforced stops being reinforced, it declines. This is operant extinction, and it follows a characteristic course. At first, responding often gets worse before it gets better: the rate rises, the responses become more forceful and varied, and emotional behaviour such as aggression may appear. This temporary surge is the **extinction burst**. You have shown one yourself if you have ever pressed a lift button again and again, harder and harder, when the lift did not come. Then responding declines until it returns to its **operant level**, the baseline rate at which the response occurred before it was ever reinforced. (Because many responses occur occasionally even without reinforcement, extinction brings the rate back to this earlier level, which is not necessarily zero.) After a rest, the extinguished response shows spontaneous recovery, just as in classical conditioning.

KEY STUDY: EXTINGUISHING A TODDLER'S BEDTIME TANTRUMS (WILLIAMS, 1959)

**Problem.** A 21-month-old boy, recovering from a long illness during which he had received constant attention, screamed and fussed whenever he was put to bed unless a parent stayed in the room until he fell asleep. **Method. Carl Williams** reasoned that the tantrums were maintained by parental attention, so he advised the parents to put the boy to bed in a relaxed way, leave the room, and not go back in however much he cried. **Result.** The first night the boy screamed for 45 minutes; the second night he did not cry at all; within ten occasions he was smiling as his parents left. About a week later, when an aunt put him to bed, he cried and she went back in; the tantrums returned (spontaneous recovery, reinforced), and had to be extinguished a second time, after which none occurred for the following two years. **Conclusion.** Ignoring an attention-maintained behaviour is extinction, and it works if it is consistent. A single "giving in" during the extinction burst puts the behaviour on intermittent reinforcement, which, as Section 6.6 will show, makes it even harder to extinguish.

EXAM TRAP

"A parent deliberately ignores a child's temper tantrums to discourage them." The correct answer is **extinction**. The source key for this real item (Amity 2023, repeated in JSS practice sets) marks "operant escape", which is wrong: escape is a response that *terminates* an aversive stimulus already present (negative reinforcement), and nothing in the parent's ignoring fits that description. Learn the item as extinction. When reinforcement of a previously reinforced behaviour stops and the behaviour fades, the term is extinction, not generalisation, discrimination or shaping. "Habit weakening through the elimination of any reward" is likewise extinction, not negative reinforcement, reciprocal inhibition or flooding.

## 6.6 Schedules of Reinforcement and Other Operant Phenomena

HIGH YIELD · 25 QS

In the 1930s, Skinner later recalled, he was running short of food pellets, which he made by hand, and he began reinforcing his rats only once a minute instead of after every press to make his supply last. To his surprise the rats kept pressing, and at steady rates. That economy launched one of the most productive research programmes in psychology. With **Charles Ferster** he spent years mapping how different rules for delivering reinforcement shape the pattern of responding, and their book *Schedules of Reinforcement* (1957) contains hundreds of cumulative records. The basic findings are few, highly examined and easy to remember once you see the records.

### Continuous and partial reinforcement ●●

A **schedule of reinforcement** is the rule that determines which responses are reinforced. Under **continuous reinforcement** (CRF), every correct response is reinforced. Under **partial reinforcement** (also called intermittent reinforcement), only some responses are. Continuous reinforcement produces the fastest learning, so it is the best way to establish a new behaviour. But behaviour that has been continuously reinforced also extinguishes fastest. When the vending machine that always works suddenly swallows your coin, you stop putting coins in almost at once; when the slot machine that rarely pays fails again, you keep playing.

This greater persistence after partial reinforcement is the **partial reinforcement extinction effect**. It was demonstrated in humans by **Lloyd Humphreys** in 1939, using conditioned eyeblinks: people for whom the puff of air had followed the signal on only half the trials kept blinking to the signal far longer in extinction than people for whom it had followed every time. The simplest explanation is that it is harder to tell that reinforcement has stopped when it was intermittent to begin with. The practical rule follows: use continuous reinforcement to teach a behaviour, then shift to partial reinforcement to maintain it.

### The four basic schedules ●●●

Partial schedules are built from two choices. Reinforcement can depend on the *number* of responses (a ratio schedule) or on the *time* that has passed (an interval schedule, in which the first response after the interval is reinforced). And the requirement can be the same every time (fixed) or can vary unpredictably around an average (variable). Crossing the two choices

gives four schedules, and each produces its own signature on the cumulative record (Figure 6.4); Table 6.7 summarises them, with continuous reinforcement for comparison.

- On a **fixed-ratio schedule** (FR), reinforcement follows a fixed number of responses: FR-10 means every tenth response. Responding is high and steady, because the faster the animal responds, the sooner it is paid, but each reinforcer is followed by a short break, the **post-reinforcement pause**. The pause lengthens as the ratio grows, and if the ratio is raised too fast responding can break down (ratio strain). Piece-rate work, such as a tailor paid for every ten blouses stitched, and "buy nine coffees, get the tenth free" cards are fixed-ratio schedules.
- On a **variable-ratio schedule** (VR), reinforcement follows an unpredictable number of responses that averages a set value. The very next response might pay off, so there is no reason to pause. VR produces the highest and steadiest rate of responding and the greatest resistance to extinction of all the schedules. Slot machines, lottery tickets and a salesperson's cold calls are the classic examples, and they explain why gambling is so hard to stop.
- On a **fixed-interval schedule** (FI), the first response after a fixed period is reinforced: FI-60 seconds. Responses during the interval earn nothing, so animals learn to pause after each reinforcer and then respond faster and faster as the end of the interval approaches. The cumulative record shows a curve called the FI scallop. Overall rates are relatively low. You are on something like an FI schedule when a dish is due out of the oven in thirty minutes: you hardly check at first and check more and more often as the time approaches. Students who study little early in term and frantically before a fixed examination date show the same scallop.
- On a **variable-interval schedule** (VI), the first response after an unpredictable period, averaging a set value, is reinforced. Because reinforcement could become available at any moment, responding is moderate but very steady, with no scallop, and it is quite resistant to extinction. Checking your phone for a reply to a message, fishing, and preparing for a teacher who sets surprise quizzes are VI schedules.

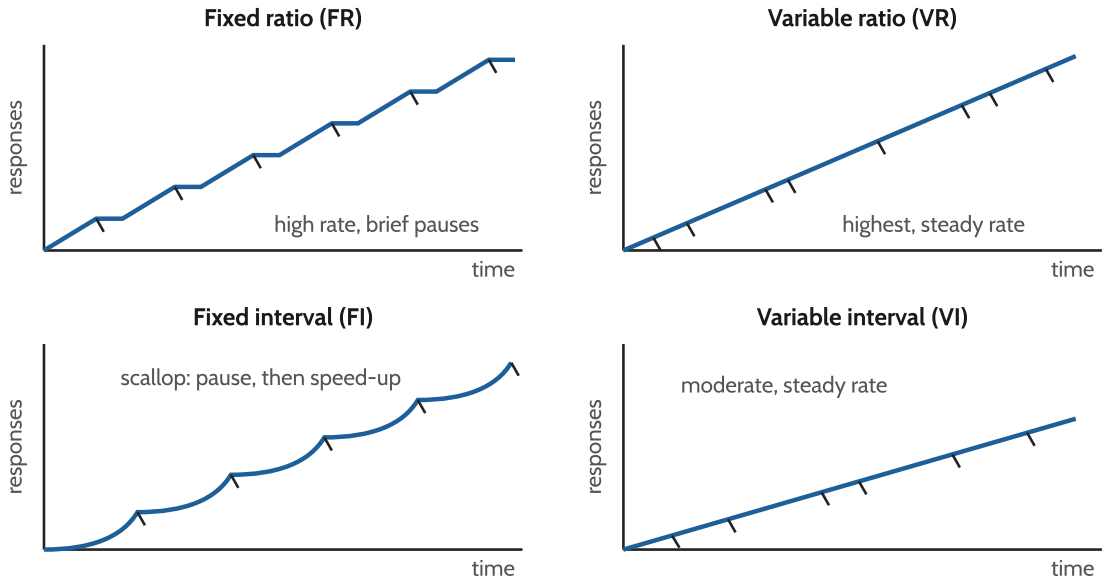


**Figure 6.4** Stylised cumulative records for the four basic schedules. The steeper the line, the faster the responding; short diagonal ticks mark reinforcers. Notice the flat steps after each reinforcer on FR, the unbroken steep line on VR, the scallops on FI and the steady, shallower line on VI.

**Table 6.7** Schedules of reinforcement at a glance.

| Schedule | Rule | Response pattern | Everyday example |
|---|---|---|---|
| Continuous (CRF) | Every response reinforced | Fast learning; fastest extinction | A vending machine that always works |
| Fixed ratio (FR) | After a set number of responses | High, steady; pause after each reinforcer | Piece-rate pay; loyalty cards |
| Variable ratio (VR) | After an unpredictable number | Highest, steadiest; most resistant to extinction | Slot machines; lottery; sales calls |
| Fixed interval (FI) | First response after a set time | Scallop: slow after reinforcer, faster near the end | Checking a dish due out of the oven |
| Variable interval (VI) | First response after an unpredictable time | Moderate, steady; resistant to extinction | Checking for a reply; fishing; surprise quizzes |

Two general rules tie the table together. Ratio schedules produce higher rates than interval schedules, because on ratio schedules responding faster brings reinforcement sooner, whereas on interval schedules it does not. And variable schedules produce steadier responding and greater resistance to extinction than fixed ones, because the learner can never predict when the next reinforcer is due, and therefore cannot easily tell when reinforcement has stopped. Put the two rules together and VR comes out on top for both rate and persistence.

EXAM TRAP

Cue words decide most schedule questions: "number of responses" means ratio; "time" or "minutes" means interval; "unpredictable" means variable. Highest rate and most resistant to extinction: **VR**. Scallop: **FI**. High rate with brief pauses after reinforcement: **FR**. Reinforcement after an unpredictable amount of time: **VI**. Reinforcement after every correct response: **continuous**. Continuous reinforcement is the *least* resistant to extinction. In one real item, Sheena rewards herself with a chocolate "for every 45 minutes" of study; the keyed answer is **fixed interval**, because the requirement is a constant amount of time. (Strictly, a reward that arrives after a fixed time regardless of responding is a fixed-time arrangement, and the same is true of a monthly salary, the textbook "FI" example; but among the four options, FI is correct.)

## Differential reinforcement

Clinicians rarely want simply to strengthen a behaviour; more often they want to weaken a problem behaviour without the side effects of punishment. **Differential reinforcement** does this by reinforcing some responses and not others. The main variants are listed in Table 6.8. In each, the problem behaviour is put on extinction while something else is reinforced.

Table 6.8 Forms of differential reinforcement.

| Procedure | What is reinforced | Example |
|---|---|---|
| DRO: differential reinforcement of other behaviour | The absence of the target behaviour for a set interval (also called omission training) | A sticker for every ten minutes without hitting |
| DRI: differential reinforcement of incompatible behaviour | A behaviour that cannot occur at the same time as the problem | Praise for hands in lap, which rules out hand-flapping |
| DRA: differential reinforcement of alternative behaviour | A more appropriate behaviour serving the same function | Reinforcing "Can I have a break?" instead of throwing books |
| DRL: differential reinforcement of low rates | The behaviour, but only when it occurs at a low rate | Praising a pupil who asks no more than two questions per class |
| DRH: differential reinforcement of high rates | The behaviour, but only when it occurs at a high rate | A bonus only for typing above a target speed |

## The Premack principle ●●●

What can serve as a reinforcer? The traditional answer was a list of stimuli: food, water, praise, money. In 1959 **David Premack** (1925 to 2015) proposed something more radical. Reinforcers, he argued, are not stimuli but activities, and what makes an activity reinforcing is its probability relative to other activities. The **Premack principle** states that a more probable (preferred) behaviour can be used to reinforce a less probable (less preferred) behaviour. You find out how probable each behaviour is by watching how much time the person spends on it when free to choose.

Premack's experiments showed that reinforcement is relative, not fixed. Some children preferred eating sweets to playing a pinball machine, and for them sweets reinforced pinball playing; for children who preferred pinball, the opportunity to play reinforced eating sweets. In rats, access to water reinforced running in a wheel when the rats were thirsty, but when the rats had free water and had been prevented from running, the opportunity to run reinforced drinking. The everyday version is called grandma's rule: "first eat your vegetables,

then you can have dessert"; "first finish your homework, then you can play video games." For a child who rarely does homework, letting him play video games only after he finishes it is more effective than vague praise, and it costs nothing.

A later refinement, the response-deprivation hypothesis of **William Timberlake** and **James Allison** (1974), showed that even a low-probability activity can serve as a reinforcer if the person has been restricted to less of it than they would normally choose. (Premack is also remembered for a quite different idea: in 1978, with **Guy Woodruff**, he asked whether a chimpanzee has a "theory of mind", a question covered in the Developmental Psychology book.)

> EXAM TRAP
>
> The Premack principle says **high-probability behaviour reinforces low-probability behaviour**, never the reverse. It does not say that reinforcement works best when intermittent (that is the partial reinforcement effect), that behaviour followed by satisfaction recurs (the law of effect), that complex behaviour is built from successive approximations (shaping), that punishment beats reinforcement, or that continuous reinforcement is most resistant to extinction (it is the least). Some question banks date the principle 1965, the year of Premack's longer statement of it; his first statement was in 1959.

## Superstitious behaviour

Reinforcement strengthens whatever response happens to precede it, whether or not the response produced it. In a 1948 paper titled "'Superstition' in the pigeon", Skinner gave hungry pigeons food every 15 seconds, whatever they were doing. Six of his eight birds developed odd, repetitive rituals: one turned anticlockwise two or three times between feedings, another kept thrusting its head into an upper corner of the cage, two swung their heads from side to side like a pendulum. Each bird, Skinner argued, had happened to be doing something when food arrived; that act was reinforced, became more likely, was therefore more likely to be under way when the next food came, and so on. **Superstitious behaviour** is behaviour strengthened by accidental (adventitious) reinforcement, with no real relationship between the act and the reinforcer. A batsman who will not change his "lucky" pads during a run of good scores, or a student who takes the same pen to every exam, is showing it.

The interpretation has been challenged. When **John Staddon** and **Virginia Simmelhag** repeated the experiment in 1971 and watched more closely, they found that the behaviours just before food were mostly the same species-typical, food-related acts in every bird (pecking near the food hopper), which suggests that much of what Skinner saw reflected Pavlovian anticipation of food rather than accidental operant reinforcement.

> EXAM TRAP
>
> Strictly, Skinner's pigeons were on a fixed-*time* schedule (food every 15 seconds, regardless of behaviour), but many textbooks describe it as fixed-interval reinforcement. Either way, the answer to "what did the superstition experiment demonstrate?" is accidental (adventitious) reinforcement.

## Learned helplessness ●●●

What happens to an animal that learns that nothing it does makes any difference? The answer came from a discovery that its investigators did not expect, and it became one of the most influential models in clinical psychology.

### KEY STUDY: LEARNED HELPLESSNESS IN DOGS (OVERMIER AND SELIGMAN, 1967; SELIGMAN AND MAIER, 1967)

**Question.** Does experience with uncontrollable aversive events change later learning? **Method.** In the "triadic" design used by Martin Seligman and **Steven Maier**, dogs were first placed in harnesses in three groups. Dogs in the escape group received shocks that they could switch off by pressing a panel with their heads. Each dog in the yoked group was wired to an escape dog and received exactly the same shocks, at the same times and of the same duration, but nothing it did affected them. A third group received no shocks. A day later all the dogs were placed in a shuttle box where they could escape or avoid shock simply by jumping a low barrier. **Result.** The escape dogs and the unshocked dogs quickly learned to jump. Most of the yoked dogs did not: after a brief struggle they lay down and whined, taking shock after shock without trying to escape, and even when one did jump by chance it seldom repeated the response. **Conclusion.** The yoked dogs had learned that their responses and the shocks were independent. This expectation produced three deficits: motivational (they stopped trying), cognitive (they had difficulty learning that a response now worked) and emotional (they appeared depressed). Seligman called this **learned helplessness**: passive acceptance of aversive events, even avoidable ones, that follows experience with uncontrollable aversive events. **Significance.** Because the shocks were identical for escape and yoked dogs, the damage was caused not by the shocks but by the lack of control.

The first published report was by **J. Bruce Overmier** and Seligman in 1967, with Seligman and Maier's follow-up in the same year. Later experiments by **Donald Hiroto** (1974) produced the same pattern in people: university students exposed to loud noise they could not turn off later failed to learn a simple hand movement that would have turned it off.

Seligman proposed learned helplessness as a model of depression, because depressed people, like the yoked dogs, often stop trying, find it hard to see that their actions can work, and feel low. The model had a problem: people do not become depressed after every uncontrollable event. In 1978 **Lyn Abramson**, Seligman and **John Teasdale** reformulated it in cognitive terms: what matters is how people explain uncontrollable events to themselves. Depression is most likely in people who attribute bad events to causes that are internal ("it's me"), stable ("it will always be like this") and global ("it affects everything I do"). This attributional reformulation, and the hopelessness theory that followed it, are covered with depression in the Psychiatric Disorders book. Fifty years after the original studies, Maier and Seligman (2016) revised their own account using neuroscience: passivity, they now argue, is the brain's default reaction to prolonged aversive events, and what animals learn in the escape condition is that they have control, a discovery that recruits the prefrontal cortex to damp down the stress response.

EXAM TRAP

Learned helplessness was offered by **Martin Seligman and Steven Maier** (with Bruce Overmier), not by Aaron Beck (cognitive triad), Harry Stack Sullivan (interpersonal theory) or Daniel Kahneman (heuristics). It is passive acceptance of later aversive events after repeated *unavoidable* ones. It is not escaping an aversive stimulus, not watching a model rewarded for aggression, and not extinction from loss of reinforcement. One real stem places it "in social learning theory"; the concept actually comes from animal learning research and was later reformulated cognitively, but the answer is unchanged.

### Instinctive drift and other limits on operant conditioning

Just as Garcia showed that classical conditioning is constrained by biology, two of Skinner's own students showed the same for operant conditioning. **Keller Breland** and **Marian Breland** set up a business training animals for zoos, fairs and television advertisements, and by the late 1950s they had trained thousands of animals of dozens of species. In 1961 they published a paper wittily titled "The misbehavior of organisms", a play on the title of Skinner's 1938 book. They had trained a raccoon to pick up a coin and drop it into a metal box for food. With one coin all went well. Given two coins, the raccoon began rubbing them together, dipping them into the box and pulling them out again, for minutes at a time, exactly as raccoons rub and wash food, and it never stopped doing so despite losing reinforcement. Pigs trained to carry large wooden coins to a "piggy bank" began, after some weeks, to drop them, root them along the ground with their snouts and toss them in the air, as pigs root for food. The Brelands called this **instinctive drift**: the tendency of a learned behaviour to drift towards the instinctive, species-typical behaviour associated with the reinforcer. Their conclusion was blunt: operant conditioning cannot be understood without understanding the animal's instinctive repertoire.

A related finding is autoshaping. In 1968 **Paul Brown** and **Herbert Jenkins** found that if a pigeon's key is simply lit for a few seconds before food is delivered, the pigeon begins pecking the key even though pecking has no effect on the food. A Pavlovian pairing of light and food produced what looks like an operant response. Classical and operant conditioning are separable in principle, but in living animals they constantly interact.

### Choice: the matching law

In everyday life you are rarely choosing whether to respond; you are choosing which of several reinforced activities to do. **Richard Herrnstein** (1930 to 1994) studied this in 1961 by giving pigeons two keys, each reinforced on its own variable-interval schedule. He found that the proportion of pecks a pigeon made on each key matched the proportion of reinforcers it obtained from that key. This is the **matching law**: $B_1/(B_1 + B_2) = R_1/(R_1 + R_2)$, where B is the rate of behaviour on each alternative and R is the rate of reinforcement it produces. If the left key yields 40 reinforcers an hour and the right key 20, the pigeon makes about two-thirds of its pecks on the left. People behave similarly: the time we spend with different friends, or on different activities, tends to be proportional to the rewards each provides.

### Contrast effects: the Crespi effect ●●

Does a bigger reward produce better learning, or just more effort? In 1942 **Leo Crespi** trained rats to run down a runway for food, some for a large amount and some for a small amount,

and then shifted all of them to the same intermediate amount. The rats shifted down from a large reward immediately ran *slower* than rats that had received the intermediate amount all along. The rats shifted up from a small reward immediately ran *faster* than those rats. Crespi called these the depression and elation effects; in 1949 **David Zeaman** renamed them negative contrast and positive contrast, and today they are known together as **contrast effects**, or the Crespi effect. Two conclusions followed. First, the value of a reinforcer is relative to what the organism is used to: a mark of 85 delights a student who usually gets 70 and dismays one who usually gets 95. Second, the changes in running speed were too abrupt to be changes in learning; they were changes in motivation. Reward size affects performance through incentive, a point that forced Clark Hull to add incentive motivation to his equations (Section 6.9).

> EXAM TRAP
>
> The Crespi effect refers to a **change in motivation (performance) when the magnitude of reinforcement changes**: an increase in reward produces a disproportionate increase in performance (positive contrast) and a decrease produces a disproportionate decrease (negative contrast). The keyed answer in IHBAS 2025 described the positive half. It is not retroactive inhibition (forgetting caused by later learning, Chapter 7), not extinction and not insight.

### Biofeedback: operant control of the body ●●●

For decades it was assumed that only skeletal muscles could be operantly conditioned, while heart rate, blood pressure and other autonomic responses could only be classically conditioned. In the late 1960s **Neal Miller** (1909 to 2002) and his student **Leo DiCara** reported that rats paralysed with curare, so that they could not use their muscles to influence their hearts indirectly, could learn to raise or lower their heart rates when rewarded with electrical stimulation of the brain's reward areas. Later attempts, including Miller's own, failed to replicate the strongest results, but the idea that autonomic functions can be brought under voluntary control with feedback had already launched a clinical field.

**Biofeedback** is a technique in which a bodily process that is normally outside awareness is measured by an instrument and displayed back to the person, as a tone, a moving line or a number, so that he or she can learn to control it. Each change in the desired direction is reinforced by the feedback itself, so biofeedback is operant conditioning combined with knowledge of results (Section 6.10). Electromyographic (EMG) biofeedback of muscle tension in the forehead and neck is used for tension-type headache and chronic pain; thermal biofeedback of finger temperature is used for migraine; electroencephalographic biofeedback (neurofeedback) has been used for attention-deficit/hyperactivity disorder and epilepsy; and feedback of skin conductance or heart-rate variability is used in anxiety and stress management. The idea is older than the name. In 1948 the gynaecologist **Arnold Kegel** taught women pelvic-floor exercises with a perineometer, a pressure gauge that let a patient watch her own muscle squeeze on a dial, an early form of biofeedback.

EXAM TRAP

"The technique based on the idea that the autonomic nervous system can come under voluntary control through operant conditioning" is **biofeedback**, not hypnosis, sublimation (a Freudian defence mechanism) or systematic desensitisation (Wolpe's counterconditioning). EMG biofeedback is most useful for tension-type headache. Kegel's perineometer was an early form of biofeedback, not aversion therapy, a token economy or desensitisation.

## Classical and operant conditioning compared ●●●

You now have both forms of associative learning in hand. Table 6.9 summarises their differences, and Table 6.10 shows how each became a family of clinical techniques. Keep in mind that the two processes, though distinct in principle, usually operate together: Albert's fear of the rat was classical, but if he had been allowed to crawl away and feel relief, his avoidance would have been operant, which is exactly Mowrer's two-factor theory.

Table 6.9 Classical versus operant conditioning.

| Feature | Classical conditioning | Operant conditioning |
|---|---|---|
| Key figures | Pavlov, Watson | Thorndike, Skinner |
| Other names | Pavlovian, respondent, Type S | Instrumental, Type R |
| What is associated | Two stimuli (CS and UCS) | A response and its consequence |
| Kind of behaviour | Elicited, reflexive, largely involuntary | Emitted, voluntary, goal-directed |
| Learner's role | Relatively passive | Active; behaviour operates on the environment |
| Order of events | Stimulus (CS) comes before the response | Consequence comes after the response |
| Typical responses | Emotional and glandular: fear, salivation, nausea | Skeletal: pressing, walking, speaking, studying |
| Extinction | CS repeatedly presented without the UCS | Response no longer reinforced |

Table 6.10 Learning principles behind common behaviour-therapy techniques.

| Technique | Developed by | Principle | Family |
|---|---|---|---|
| Systematic desensitisation | Wolpe (1958) | Counterconditioning by reciprocal inhibition | Classical |
| Flooding; implosive therapy | Implosion: Stampfl | Extinction with escape prevented | Classical |
| Aversion therapy | Various | Pairing a cue with an aversive UCS | Classical |
| Covert sensitisation | Cautela (1967) | Imagined behaviour paired with imagined aversive outcome | Classical |
| Exposure and response prevention | Meyer (1966) | Habituation and extinction; escape blocked | Classical, with an operant element |
| Bell-and-pad for bed-wetting | Mowrer and Mowrer (1938) | Full-bladder sensations become a signal to wake | Classical |
| Token economy | Ayllon and Azrin (1968) | Secondary (generalised) reinforcement | Operant |
| Shaping, chaining, time-out, response cost, DRO | Skinner's principles, applied behaviour analysis | Reinforcement, punishment, extinction | Operant |
| Biofeedback | Miller; Kegel (1948) | Feedback reinforces bodily change | Operant |

| Technique | Developed by | Principle | Family |
|---|---|---|---|
| Modelling, participant modelling | Bandura | Observational learning | Social learning |

EXAM TRAP

"Which is NOT a treatment developed from classical conditioning?" **Token economy** (operant). Flooding, systematic desensitisation and aversion therapy are all classical. The main difference between the two kinds of conditioning is that **classical pairs two stimuli while operant relies on consequences (reinforcement and punishment).** It is not that "classical relies on reinforcement and operant does not" (Pavlov did call the UCS a reinforcement, but the statement is backwards), nor that only operant involves cognition (Rescorla showed classical conditioning involves prediction), nor that they are "entirely different" (they interact constantly). The hallmark of behaviour therapy is the use of classical and operant conditioning techniques to eliminate unwanted behaviour, not cognitive restructuring, dream analysis or free association, and not the search for underlying causes or insight.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Rescorla showed that conditioning depends on _____, not mere contiguity.
2. Kamin found that prior conditioning of a noise prevented conditioning of an added light, a phenomenon called _____.
3. Garcia's taste-aversion work refuted the principle of _____, the idea that all stimuli are equally easy to associate.
4. Thorndike's principle that satisfying consequences strengthen a response is the law of _____.
5. Removing an aversive stimulus to increase a behaviour is _____ reinforcement.
6. Taking a child out of a game for five minutes after he hits someone is _____, a form of negative punishment.
7. Mowrer explained avoidance with his _____ theory.
8. Reinforcing successive approximations to a target behaviour is called _____.
9. The schedule that produces the highest rate of responding and the greatest resistance to extinction is the _____ schedule.
10. Seligman and Maier's yoked dogs, given inescapable shock, later showed learned _____.

**Answers:** (1) contingency, (2) blocking, (3) equipotentiality, (4) effect, (5) negative, (6) time-out, (7) two-factor (two-process), (8) shaping, (9) variable-ratio, (10) helplessness.

## 6.7 Cognitive Learning

MEDIUM YIELD · 9 QS

By the 1930s the dominant view in American psychology was that learning consists of stimulus-response connections stamped in by reinforcement, with nothing in between worth talking about. Three lines of research, in rats, apes and monkeys, showed that this view left out something essential: what animals learn is often knowledge, a representation of how the world is arranged, which they use flexibly when the occasion demands. This section de-

scribes those three lines of research and ends with a brief account of how concepts are formed.

### Tolman: latent learning and cognitive maps ●●●

**Edward Chace Tolman** (1886 to 1959) was a behaviourist in method: he studied rats in mazes and measured only what they did. But he insisted that behaviour is purposive, directed towards goals, and that it should be described at the level of whole, meaningful acts ("running to the food box") rather than muscle twitches. His approach, set out in *Purposive Behavior in Animals and Men* (1932), is called purposive behaviourism. Tolman argued that between the stimulus and the response there are intervening variables, such as expectancies and knowledge of the environment, that cannot be seen but can be inferred from behaviour. (A similar point had been made by **Robert Sessions Woodworth**, who proposed replacing the S-R formula with S-O-R, stimulus-organism-response, to show that what the organism brings to the situation shapes its response.)

#### KEY STUDY: TOLMAN AND HONZIK'S LATENT LEARNING EXPERIMENT (1930)

**Question.** Do rats learn a maze only when they are reinforced for running it? **Method.** Tolman and **Charles Honzik** ran three groups of hungry rats through a complex maze with fourteen choice points, once a day for seventeen days. One group always found food in the goal box. A second group never found food; they were simply removed from the goal box when they reached it. A third group found no food for the first ten days, but from day 11 onwards found food in the goal box. **Result.** The always-rewarded rats steadily reduced their errors. The never-rewarded rats improved only slightly. The third group made as many errors as the never-rewarded rats for ten days, but the day after food was introduced their errors dropped dramatically, to the level of the always-rewarded group and even slightly below it (Figure 6.5). **Conclusion.** The third group must have been learning the layout of the maze all along, during the unrewarded days, without showing it. The reward did not create the learning; it gave the rats a reason to display it. **Significance.** This is the clearest demonstration of the difference between learning and performance. (Hugh Blodgett had reported a similar result in 1929.)

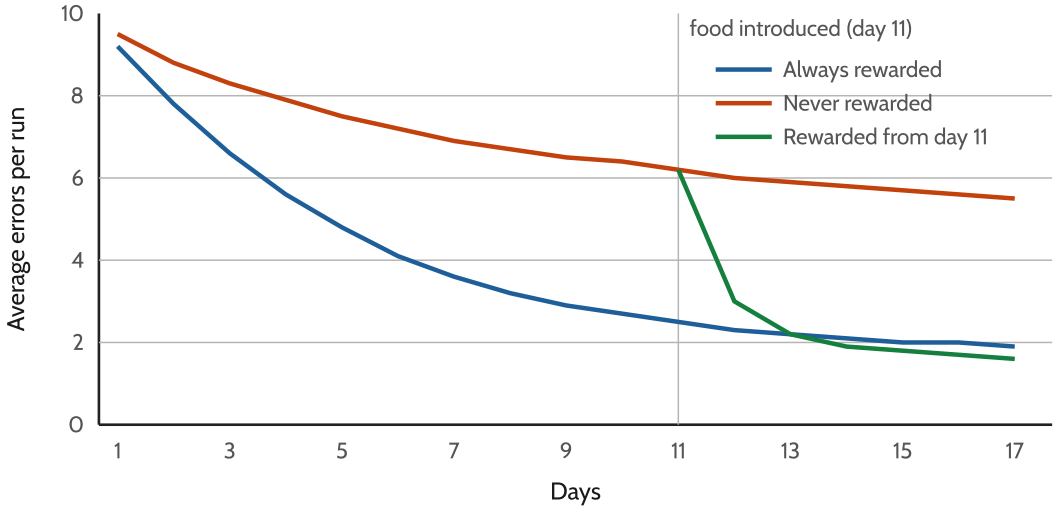


Figure 6.5 Latent learning, stylised after Tolman and Honzik (1930). The group rewarded only from day 11 (dotted line) performed like the never-rewarded group until food appeared, then immediately matched the always-rewarded group. The learning had been there, unseen, before the reward.

**Latent learning** is learning that occurs without obvious reinforcement and remains hidden until there is an incentive to demonstrate it. What had the rats learned? Tolman's answer, developed in a famous 1948 paper, "Cognitive maps in rats and men", was that they had formed a **cognitive map**: a mental representation of the spatial layout of the environment. Evidence came from experiments on **place learning**. In 1946 Tolman and his colleagues trained one group of rats always to find food in the same place in a cross-shaped maze, whichever arm they started from, so that sometimes they had to turn left and sometimes right; another group always had to make the same turn, so the food's location changed. The rats that had to learn a place learned faster than those that had to learn a response, which a pure S-R theory cannot explain. When rats that had learned a roundabout route to food were later offered many straight paths radiating in different directions, many chose the path that pointed most directly towards where the food had been, as if they knew where it was, not just how to get there.

Tolman also argued that animals learn *expectancies*: that a particular goal will follow a particular action. A vivid demonstration had come from **Otto Tinklepaugh** in 1928. A monkey watched a banana being hidden under a cup; when it lifted the cup and found that the banana had been secretly replaced by lettuce, a food it normally ate happily, it refused the lettuce, searched around and shrieked at the observers. It had expected a banana. Because what is learned in Tolman's view is knowledge of "what leads to what", his position is variously called sign learning theory, expectancy theory and place learning theory.

EXAM TRAP

Latent learning and the cognitive map belong to **Tolman**, not Bandura, Skinner, Köhler or Harlow. "A sudden increase in performance upon reinforcement, despite never being reinforced on previous trials" is latent learning, not observational learning, discrimination learning or vicarious reinforcement. Tolman's rats do not show shaping, pure S-R learning or spontaneous recovery. An option naming "Gould" refers to the biologist James Gould, who studied honeybee navigation and asked whether insects might have map-like knowledge; the concept of the cognitive map is Tolman's (1948).

## Köhler: insight ●●●

On the other side of the Atlantic, a different challenge to trial-and-error learning was taking shape. **Wolfgang Köhler** (1887 to 1967) was, with Max Wertheimer and Kurt Koffka, one of the three founders of Gestalt psychology, which holds that experience is organised into wholes that are more than the sum of their parts (Chapters 2 and 4). From 1913 to 1920 Köhler directed a research station for apes on Tenerife in the Canary Islands, where the First World War kept him far longer than planned. There he set chimpanzees problems that could not be solved by any single learned response.

In the simplest problems a banana lay on the ground outside the cage, out of reach, and a stick lay inside; the chimpanzees soon used the stick to rake the banana in. In the famous two-stick problem, the banana was farther away, and neither of two sticks was long enough on its own; one stick was hollow, so the other could be fitted into it. The most gifted chimpanzee, Sultan, tried with one stick and then the other, failed, and gave up. Later, while playing idly with the two sticks, he happened to hold them in a line, pushed the thinner one into the hollow end of the thicker, and at once ran to the bars and raked in the banana. In the box problems a banana was hung from the roof of the enclosure, too high to reach by jumping. The chimpanzees jumped in vain, paced about, and then, often quite suddenly, dragged a box under the banana and climbed on it; some later learned to stack two or three boxes. Köhler described this work in a book published in German in 1917 and in English in 1925 as *The Mentality of Apes*.

**Insight learning** is the sudden grasp of a problem's solution through perceiving the relationships among its elements, a reorganisation of the perceptual field. Köhler identified its features. The change from no solution to solution is sudden, not gradual like Thorndike's curves. It is often preceded by a period of apparent inactivity in which the animal surveys the situation. Once found, the solution is carried out smoothly and without errors. It is remembered well. And it transfers readily to similar problems. It also depends on all the elements of the problem being in view at once: the chimpanzees rarely used a stick that was behind them. The subjective feeling of sudden understanding that accompanies insight is called the **aha experience** (the German "Aha-Erlebnis", a term usually credited to the psychologist **Karl Bühler**).

Insight was a powerful argument against the view that all learning is the gradual strengthening of S-R connections. But it has a qualification. In 1945 **Herbert Birch** found that young chimpanzees who had never handled sticks did not solve stick problems insightfully; after a few days of playing with sticks, they did. Insight, it seems, recombines things that were learned earlier. Harlow's work, described next, explains how.

EXAM TRAP

Sultan, sticks and stacked boxes mean **insight learning** (Köhler), not trial-and-error, classical conditioning or latent learning. The sudden realisation of a solution is the **"aha" experience**, not the "light bulb", "momentary" or "learning". Do not confuse Köhler's insight with *insight* in psychiatry, the patient's awareness of his own illness, which is rated in the mental status examination on a six-grade scale (with intellectual insight usually placed at grade 5; Clinical Psychology book), or with insight as a goal of psychoanalysis.

### Harlow: learning to learn

**Harry Harlow** (1905 to 1981), working with rhesus monkeys at the University of Wisconsin, asked what happens when an animal is given not one problem but hundreds of similar ones. In each discrimination problem a monkey was shown two objects it had never seen before, such as a cup and a wooden block, one of which always covered a raisin or a piece of food. The monkey had a few trials to discover which object paid. Then a new pair of objects was introduced, and so on through several hundred problems.

On the first few problems the monkeys learned slowly and gradually, in true trial-and-error fashion, and on the second trial of each new problem they were right only about half the time, no better than chance. But each problem was solved a little faster than the last. By the end, the monkeys were right on the second trial of a new problem more than 95 percent of the time: one trial was enough to tell them which object to choose. They had learned a general strategy, which can be described as "if the object you picked paid, stay with it; if not, switch" (win-stay, lose-shift). Harlow, in a 1949 paper, called this a **learning set**: an improvement in the ability to learn new problems of a given type that results from experience with many problems of that type, or, more simply, learning to learn.

Learning sets reconcile Thorndike and Köhler. A naive animal facing a new kind of problem learns by trial and error; an experienced one, armed with learning sets, appears to solve problems by insight. What looks like sudden understanding in an adult may rest on a long history of earlier, gradual learning. Students do the same: the first algebra problem of a new type takes an hour, the twentieth takes a minute.

### Concept formation ●●

A **concept** is a mental category that groups objects, events or ideas sharing common properties: "dog", "fruit", "democracy". Concepts let us respond to new instances without learning about each one separately. The full cognitive treatment of concepts, including the strategies that Jerome Bruner, Jacqueline Goodnow and George Austin described in 1956 and the later prototype view, is in Chapter 8. Here we need only the stage account of **concept formation** that is taught in general and educational psychology, in which a concept is built up in four steps.

1. *Observation*. The learner encounters many instances. A two-year-old in Mysuru sees the neighbour's Labrador, the street dogs outside the temple, a Pomeranian on television.
2. *Discrimination*. The learner notices how instances differ from non-instances: cows and cats, though they also have four legs and fur, are not dogs.
3. *Generalisation*. The learner extends the response (the word "dog") to all the instances that share the essential features, however different their size and colour.

4. *Abstraction*. The learner extracts the common defining properties from the particular instances, so that the concept exists independently of any one dog and can be applied to breeds never seen before. The concept is then named and can be defined.

Notice that the stages parallel the conditioning processes of Section 6.3: generalisation and discrimination, which Pavlov studied in dogs, are the raw materials from which human concepts are built.

> EXAM TRAP
>
> The keyed order (JEMAS 2025, repeated in JSS practice sets) is **Observation, Discrimination, Generalisation, Abstraction**. Options that start with abstraction, or that put generalisation before discrimination, are wrong; so is any option containing **assimilation**, which is Piaget's term for fitting new experience into existing schemas (Developmental Psychology book), not a stage of concept formation. Some educational psychology texts list the steps as observation, comparison, abstraction and generalisation, followed by naming; if you see that version, accept it too, because "comparison" is discrimination under another name.

## 6.8 Observational Learning HIGH YIELD · 17 QS

Imagine learning to drive purely by operant conditioning: you would press pedals and turn the wheel at random, and be reinforced whenever the car happened to move safely forward. You would not survive the first lesson. Most complex human skills, from tying a turban to rolling a round chapati to performing a cover drive, are learned largely by watching someone else do them first. So are many attitudes, fears, habits of speech and ways of handling anger. **Observational learning** (also called social learning or modelling) is learning that occurs by watching the behaviour of others, called models, and its consequences.

### Before Bandura: imitation as learned behaviour

The first learning theorists to take imitation seriously were Neal Miller and **John Dollard** (1900 to 1980) at Yale. In *Social Learning and Imitation* (1941) they argued that imitation is itself an operant response that is learned because it is reinforced. A younger child who follows an older sibling to where the father hands out sweets is rewarded for copying, so copying becomes a habit; they called this matched-dependent behaviour. In *Personality and Psychotherapy* (1950) they went further and translated Freud's ideas into the language of Clark Hull's learning theory, describing behaviour in terms of drive, cue, response and reward. This work is usually called early social learning theory. Its limitation was that, on their account, imitation still required reinforcement of the observer.

### The Bobo doll studies ●●

**Albert Bandura** (1925 to 2021), born in a small town in Alberta, Canada, and for most of his career a professor at Stanford University, showed that observers can learn new behaviour without performing it and without any reinforcement at all.

**KEY STUDY: BANDURA, ROSS AND ROSS'S BOBO DOLL EXPERIMENT (1961)**

**Question.** Will children imitate aggression they have merely watched, without being reinforced? **Method.** Bandura, **Dorothea Ross** and **Sheila Ross** studied 72 children, aged about three to six, from the Stanford University nursery school. In one condition each child, busy making pictures with potato prints and stickers in a corner of a room, watched an adult model attack a large inflatable "Bobo" clown doll in unusual ways: sitting on it and punching it on the nose, striking it with a mallet, tossing it in the air, kicking it about the room, while saying things like "Sock him in the nose!" and "Pow!". In a second condition the model played quietly with tinker toys and ignored Bobo; a third group saw no model. Each child was then mildly frustrated (shown some very attractive toys and told they were being kept for other children) and left in a room containing various toys, including Bobo and a mallet. **Result.** Children who had watched the aggressive model reproduced many of the model's specific, novel acts and phrases, far more than children in the other two groups. Boys showed more physical aggression than girls, and the male model was imitated more for physical aggression. **Conclusion.** New behaviour can be acquired simply by observation, with no reinforcement to either the model or the child. **Criticism.** A Bobo doll is designed to be hit, so the study may show play rather than aggression towards people; the children may have sensed what the adults expected; and only short-term effects were measured.

Two follow-up studies extended the finding. In 1963 Bandura and the Rosses showed that a filmed model, and even a cartoon-like model dressed as a cat, produced as much imitative aggression as a live model, a result that made Bandura a central figure in the debate about violence on television (Social Psychology book). And in 1965 Bandura showed children a film in which the model's aggression was rewarded, punished, or had no consequences. Children who saw the model punished imitated less aggression than the others. But when the experimenter then offered every child an attractive incentive (stickers and fruit juice) for showing what the model had done, the differences disappeared: children in all three groups could reproduce the aggressive acts equally well. The consequences to the model had affected *performance*, not *learning*. This is the same learning-performance distinction that Tolman's rats had shown.

The 1965 study also illustrates two key terms. **Vicarious reinforcement** is an increase in an observer's tendency to perform a behaviour after seeing a model rewarded for it; vicarious punishment is a decrease after seeing a model punished. A younger brother who watches his elder sister get scolded for answering back learns, without being scolded himself, not to answer back, at least when the parents are watching.

## The four processes of observational learning ●●●

Bandura analysed observational learning into four component processes, all of which must operate for an observed behaviour to be reproduced (Figure 6.6, left).

- *Attention*. The observer must notice the model and the relevant features of what the model does. We attend more to models who are attractive, competent, high in status, similar to ourselves, and whose behaviour is distinctive or rewarded.
- *Retention*. The observer must store what was seen, in images and in words ("first fold, then tuck"), and rehearse it, because the behaviour may not be performed until much later.
- *Motor reproduction* (or production). The observer must be physically able to turn the memory into action, and usually refines the performance with practice and feedback.

You can watch Virat Kohli for years and remember every detail of his cover drive without being able to play it.

- *Motivation*. Finally, the observer must have a reason to perform the behaviour: direct reinforcement, vicarious reinforcement, or self-reinforcement (the satisfaction of meeting one's own standards).

The first three processes govern learning; the fourth governs performance. That is why direct reinforcement is not a necessary condition of observational *learning*, although it strongly affects whether the learned behaviour is ever shown. It also explains why observational learning helps someone acquire adaptive behaviour only when the model actually demonstrates the target behaviour: a model who acts angrily, playfully or randomly gives the observer nothing useful to attend to and retain.

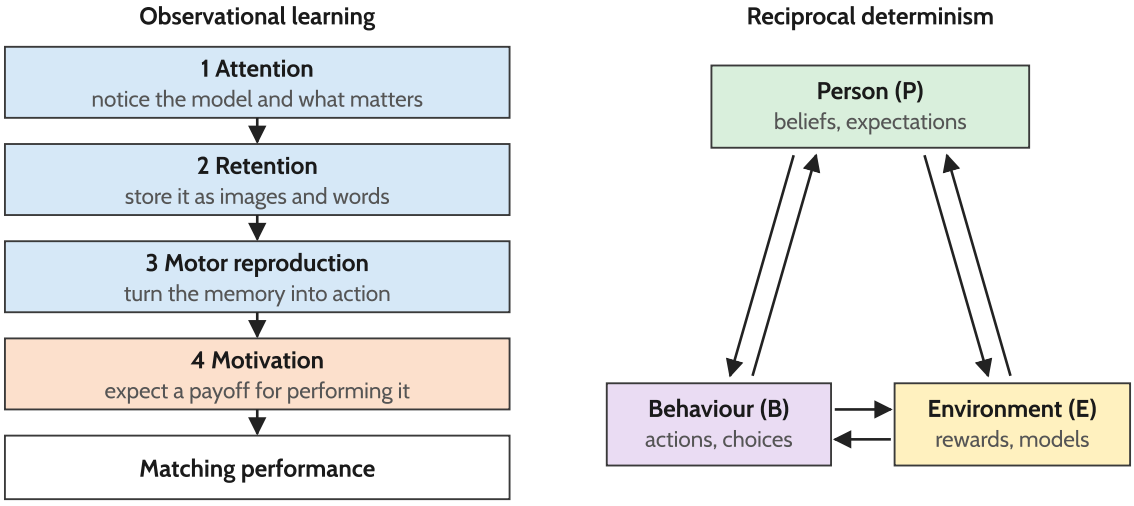


Figure 6.6 Left: Bandura's four processes. The first three govern whether a behaviour is learned; the fourth (shaded differently) governs whether it is performed. Right: reciprocal determinism. Personal factors, behaviour and environment each influence, and are influenced by, the other two.

EXAM TRAP

The four processes are **attention, retention, motor reproduction (production) and motivation**. They are not "sensory, perceptual, cognitive, affective"; not "encoding, storage, retrieval, evaluation" (the stages of memory, Chapter 7); and not "stimulus, response, reinforcement, extinction" (operant conditioning). "Which is NOT necessary for observational learning?" **Reinforcement**. The view that reinforcement is not required to learn a behaviour by observation is **Bandura's**, not Skinner's, Tolman's or Miller and Dollard's (who held that imitation must be reinforced). The Bobo doll studies demonstrate observational learning of aggression through modelling, not operant shaping, classical conditioning of fear, negative reinforcement or the effects of sensory deprivation.

## From social learning to social cognitive theory ●●●

Bandura's early theory, set out with his student **Richard Walters** in *Social Learning and Personality Development* (1963), is often called later social learning theory, to distinguish it from Dollard and Miller's. Over the following decades Bandura gave cognition an ever larger

role, and in *Social Foundations of Thought and Action* (1986) he renamed his position social cognitive theory. Three ideas from this period are heavily examined.

The first is **reciprocal determinism** (Bandura, 1978), which Bandura later called triadic reciprocal causation. Strict behaviourists such as Skinner held a one-way view: the environment determines behaviour. Bandura argued that three sets of factors, the person (beliefs, expectations, self-efficacy), behaviour and the environment, continuously influence one another in both directions (Figure 6.6, right). A student who believes she can master statistics (person) attends classes and practises (behaviour); her practice earns praise and a seat in the advanced group (environment); the new environment strengthens her belief, and so on round the triangle.

The second is **self-regulation**. Bandura noted that most human behaviour occurs in the absence of any immediate external reward or punishment. People set their own standards, observe their own behaviour (self-observation), compare it with the standards (the judgemental process) and reward or criticise themselves accordingly (self-reaction). Much of what keeps a student at her desk at eleven at night is self-regulation, not anyone else's reinforcement.

The third is **self-efficacy**, introduced in a 1977 paper: a person's belief in his or her capability to organise and carry out the actions needed to achieve a particular goal or deal with a particular situation. Self-efficacy is specific to a task: you may have high self-efficacy for public speaking and low self-efficacy for mathematics. It is also different from an outcome expectation, the belief that a behaviour will lead to a result: you may be sure that regular exercise would make you fitter (outcome expectation) and doubt that you could keep it up (efficacy expectation). Bandura identified four sources of self-efficacy: performance attainments (past successes, the most powerful source), vicarious experiences (seeing similar others succeed), verbal persuasion (being told you can do it) and physiological and emotional states (reading a calm body as a sign of competence). Within social cognitive theory, self-efficacy is the most important motivational construct: people with high self-efficacy choose harder tasks, try harder and persist longer.

CLINICAL LINK

Modelling is a therapy technique in its own right. In 1969 Bandura and his colleagues treated people with snake phobia by participant modelling: the therapist first handled the snake while the client watched, then guided the client through closer and closer contact. It worked better than watching a film of models or than systematic desensitisation, and Bandura argued that every successful treatment works by raising the client's self-efficacy. Social skills training, assertiveness training and much of cognitive behaviour therapy use modelling and rehearsal (Clinical Psychology book).

### REMEMBER: ROTTER'S SOCIAL LEARNING THEORY

**Julian Rotter** (1916 to 2014) developed a separate social learning theory in *Social Learning and Clinical Psychology* (1954). For Rotter, the **behaviour potential** (BP) is the likelihood of a particular behaviour occurring in a given situation, calculated in relation to a particular reinforcement or set of reinforcements. It depends on the **expectancy** (E), the person's subjective probability that the behaviour will lead to the reinforcement, and on the **reinforcement value** (RV), how much the person wants that reinforcement: BP = f(E and RV), within a given psychological situation. Generalised expectancies about whether outcomes depend on one's own actions became his concept of **locus of control** (1966), internal or external. Social learning frameworks in general explain behaviour by cognitive variables such as expectancies, situational demands and incentives, rather than by reinforcement history alone. (Rotter's theory is treated fully in the Personality book.)

### EXAM TRAP

**Reciprocal determinism** (Bandura): cognition, behaviour and environment influence one another bidirectionally; the "reciprocal causation model" belongs to **social cognitive theory**, not to two-factor theory (Mowrer), drive-reduction theory (Hull) or constructivist theory (Piaget, Vygotsky). Do not confuse it with reciprocal **inhibition** (Wolpe) or reciprocal **altruism** (the evolutionary biologist Robert Trivers). **Self-efficacy** is belief in one's capability for a specific task; it is not overall self-worth (self-esteem), not attributing outcomes to fate or chance (external locus of control, Rotter) and not the striving to overcome inferiority (Adler). The theorist who developed self-efficacy is Bandura, not Beck, Ellis or Skinner. Bandura's claim that most behaviour comes from **self-regulation** is the right answer when the alternatives are negative reinforcement, "reciprocal dualism" or "behavioural determinism". In a match-the-list question: Dollard and Miller = early social learning; Bandura and Walters = later social learning; Skinner = radical behaviourism; Maslow = self-actualisation. Behaviour that is emitted and operates on the environment is **operant behaviour**, not Jung's principle of opposites, Rotter's behaviour potential or Allport's personal dispositions.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Rats that found food only from day 11 suddenly matched the always-rewarded rats, showing _____ learning.
2. Tolman called the rats' mental representation of a maze a cognitive _____.
3. Köhler's chimpanzee _____ joined two sticks to reach a banana.
4. Harlow's monkeys, after hundreds of discrimination problems, formed a learning _____.
5. The keyed stages of concept formation are observation, _____, generalisation and abstraction.
6. Miller and Dollard held that imitation must itself be _____.
7. Bandura's four processes are attention, _____, motor reproduction and motivation.
8. Seeing a model rewarded and then imitating the behaviour shows _____ reinforcement.
9. A person's belief in his capability to perform a specific task is called _____.
10. Bandura's idea that person, behaviour and environment influence one another is called reciprocal _____.

**Answers:** (1) latent, (2) map, (3) Sultan, (4) set, (5) discrimination, (6) reinforced, (7) retention, (8) vicarious, (9) self-efficacy, (10) determinism.

## 6.9 Other Learning Theorists

LOWER YIELD · 1 Q

Between about 1930 and 1960, learning was the battleground on which American psychology fought out its biggest theoretical arguments. Pavlov, Thorndike, Skinner, Tolman and Bandura, whom you have already met, were not alone on that field. A handful of other theorists built complete systems that tried to explain all learning with a few principles. None of these "grand theories" survived intact, but their ideas and their vocabulary run all through later psychology, and their names run all through examination papers. The standard guide to them was *Theories of Learning* (1948) by **Ernest Ropiequet Hilgard** (1904 to 2001), a book that went through five editions (the later ones with Gordon Bower) and taught generations of students to compare the theories by putting the same few questions to each. (Hilgard is also remembered for his work on hypnosis, Chapter 5.) This section begins with those questions, then takes the theorists one at a time.

### Two questions that divided the theorists ●●

The first question was *what is learned*. For the **stimulus-response theories** (S-R theories), what is learned is a connection between a stimulus and a response. Thorndike's bonds, Watson's conditioned reflexes and the theories of Guthrie and Hull below all belong here. They explain learning without any reference to the mind: the organism does not come to know anything; it comes to respond in a new way. Textbooks often call this the stimulus-response behaviouristic approach, and it is the approach that "emphasises learning through stimulus-response associations, mainly via conditioning". For the cognitive theorists, Tolman and the Gestalt psychologists such as Köhler, what is learned is knowledge: an ex-

pectancy that one event leads to another, a map of a maze, an insight into how the parts of a problem fit together. The organism then uses that knowledge flexibly. Tolman's place-learning experiment (Section 6.7) was the classic test case. An S-R theorist says the rat learned a turning response; a cognitive theorist says it learned where the food is.

The second question was *what makes learning happen*. Contiguity theorists answered that a stimulus and a response need only occur together. Pavlov's account of conditioning rested on contiguity, and Watson, who regarded Thorndike's talk of "satisfaction" as unscientific, explained learning by two laws of his own: the law of frequency (the response made most often in a situation is the one learned) and the law of recency (the response made most recently is the one that will occur next time). Reinforcement theorists, from Thorndike through Hull to Skinner, answered that a response must be followed by a consequence. Tolman took a third position: learning happens through exploration, with or without reward (latent learning), but reward decides whether what was learned is performed.

Skinner, characteristically, refused to play the game. In a 1950 paper titled "Are theories of learning necessary?" he argued that explanations in terms of unobserved events at some other level, whether neural, mental or mathematical, were unnecessary and distracting; a science of behaviour should simply describe how behaviour changes with its environment. That is why Skinner's system is often called descriptive behaviourism. The theorists in the rest of this section disagreed with him. They believed that good theory, the more precise the better, was exactly what the study of learning needed.

> EXAM TRAP
>
> "Which approach emphasises learning through stimulus-response associations, mainly via conditioning?" The answer is the **stimulus-response (S-R) behaviouristic approach**. The cognitive approach emphasises knowledge, expectancy and insight; the psychoanalytic approach emphasises unconscious conflict (Freud); the existential approach emphasises freedom, choice and meaning. Within learning theory, keep the two dimensions apart: Guthrie is an S-R *contiguity* theorist, Hull an S-R *reinforcement* theorist, and Tolman a *cognitive* theorist.

### Guthrie: we learn what we do

**Edwin Ray Guthrie** (1886 to 1959) spent his whole career at the University of Washington in Seattle. Trained first in mathematics and philosophy, he wrote clearly, with homely examples from farms, schools and family life, and did few experiments himself. His book *The Psychology of Learning* (1935, revised 1952) offered the simplest theory of learning ever proposed. It had one law, the law of **contiguity**: "A combination of stimuli which has accompanied a movement will on its recurrence tend to be followed by that movement." Nothing about satisfaction, reward or drive. If a stimulus pattern and a movement occur together, they become associated, and that is all learning is.

Two corollaries follow. The first is **one-trial learning**: in Guthrie's words, a stimulus pattern gains its full associative strength on the occasion of its first pairing with a response. Learning does not grow bit by bit; each association is made all at once. The second is the recency principle: whatever an organism did last in a situation is what it will do when the situation recurs, because that was the last movement attached to those stimuli. "We learn only what we do," Guthrie said, and teachers have quoted him ever since. A student who reads a chapter pas-

sively learns to read passively; if you want her to be able to explain the ideas, she must practise explaining them.

If learning is complete in one trial, why does practice help? Guthrie's answer rested on a distinction between movements and acts. A *movement* is a specific muscle contraction; an *act* is a class of movements defined by what it achieves, such as playing a cover drive or parking a scooter. An act is made up of thousands of movements, and each movement must be attached to its own stimuli, including the **movement-produced stimuli** (the sensations from the muscles and joints) generated by the movement before it, which link the movements into a chain. An act must also be performed in many different situations: against fast bowling and slow, on a green pitch and a dry one. Practice attaches the act to more and more of these stimulus combinations. The learning curve for the act is gradual even though every single association is formed in one trial.

What, then, does a reward do? For Guthrie, nothing special. A reward simply changes the stimulus situation. When the cat in the puzzle box operates the latch, the door opens and the cat leaves; the last movement it made in the box is therefore preserved, because the cat has no opportunity to do anything else in the presence of those stimuli. If the cat stayed in the box after escaping, it would go on to make other movements, and those would be the ones attached to the box. Forgetting, likewise, is not the fading of an association with time; it occurs only when a new response becomes attached to the old stimuli. Guthrie was one of the first to argue that forgetting is caused by interference rather than decay (Chapter 7).

### KEY STUDY: GUTHRIE AND HORTON'S CATS IN A PUZZLE BOX (1946)

**Question.** If the last movement before escape is what is learned, will each cat repeat its own particular movement? **Method.** Guthrie and **George P. Horton** built a puzzle box with a glass front and a slender pole standing in the middle of the floor. Tilting the pole in any direction opened the door, and a camera triggered by the pole photographed each cat at the instant it escaped. **Result.** Each cat settled into its own stereotyped way of moving the pole, backing into it, pushing it with a paw, or rubbing against it with its head or flank, and repeated that movement with remarkable consistency, escape after escape, while different cats used different movements. **Conclusion.** Guthrie and Horton took the photographs as strong support for contiguity and recency: what each cat learned was simply the movement it had happened to make at the moment of escape. **Criticism.** In 1979 **Bruce Moore** and **Susan Stuttard** showed that cats in such a box rub the pole with their heads and flanks whenever a person is in view, even when rubbing never opens the door. Head-rubbing is a feline greeting. Much of what the photographs showed was species-typical behaviour released by the watching experimenters, the same lesson the Brelands drew from instinctive drift (Section 6.6).

Guthrie's view of punishment followed directly from his one law. Punishment works, he said, only if it makes the animal do something incompatible with the punished response in the presence of the stimuli that produced it. Slap a dog on the nose as it runs at a passing car, and it recoils, so recoiling becomes attached to the sight of cars; slap it on the rump, and you may simply drive it forward, strengthening the chase. What matters is not the pain but what the pain makes the animal do.

### Guthrie's methods of breaking habits

Guthrie's most practical contribution, and the one most often examined, was his advice on breaking habits. If a habit is a response attached to cues, it cannot be removed by wishing; the only way to break it is to attach a different response to the same cues. He described three methods (Table 6.11).

- In the **threshold method**, the stimulus is introduced at an intensity so weak that it does not produce the unwanted response, and is then strengthened so gradually that it always stays below the threshold for that response. Guthrie's example was breaking a horse to the saddle: first a light blanket is laid on its back, then a slightly heavier one, then a light saddle, and so on, each so mild that the horse never bucks. By the time the full saddle and rider arrive, standing quietly has been attached to the feel of weight on its back.
- In the **fatigue method**, the stimulus is presented again and again, and the unwanted response is allowed or even made to occur until the organism is so tired that it stops and does something else in the presence of the stimulus. The "bronco buster" rides a bucking horse until it is too exhausted to buck. A child who keeps playing with matches can be made to strike match after match until he is heartily sick of it.
- In the **incompatible response method**, the stimulus for the unwanted response is presented together with other stimuli that produce a response incompatible with it, so that the incompatible response becomes attached to the old cue. A small child frightened of a toy panda is given the panda while she sits relaxed on her mother's lap; the warmth and calm that her mother produces cannot coexist with fear, and calm becomes attached to the panda.

Guthrie added a fourth piece of advice that is sometimes called sidetracking. Find the cues that start the unwanted habit and practise a different response in their presence. A man who smokes with his morning tea should not try to resist the cigarette at noon, when the urge is weak; he should change what he does with his tea. Guthrie's best-known anecdote makes the same point. A mother complained that her ten-year-old daughter had for two years thrown her hat and coat on the floor every time she came home, despite constant scolding. On a psychologist's advice, the next time it happened the mother made the girl put the coat and hat back on, go outside, come in again and hang them up. A few repetitions cured the habit. Scolding had attached picking up the coat to the mother's voice; the new routine attached it to the cue that really mattered, walking in through the door.

Table 6.11 Guthrie's three methods of breaking habits and their descendants in behaviour therapy.

| Method | What you do | Guthrie's example | Modern parallel |
|---|---|---|---|
| Threshold | Present the cue so weakly that the habit is not triggered; increase it very gradually | Blanket, then saddle, then rider on a young horse | Graded exposure; systematic desensitisation |
| Fatigue | Make the unwanted response occur over and over until the learner is exhausted | Riding a bucking horse until it stops; striking match after match | Negative practice (Dunlap); stimulus satiation (Ayllon) |
| Incompatible response | Present the cue while a response incompatible with the habit is occurring | The feared panda given to a child on her mother's lap | Counterconditioning; reciprocal inhibition (Jones, Wolpe) |

The parallels in the last column are close. The threshold method is, in essence, the graded hierarchy of systematic desensitisation, and the incompatible response method is Mary Cover Jones's treatment of Peter and Wolpe's reciprocal inhibition (Section 6.4). The fatigue method resembles two later techniques. In negative practice, introduced by **Knight Dunlap** in *Habits: Their Making and Unmaking* (1932), the person deliberately and repeatedly performs the unwanted habit, as when a typist who habitually types "hte" for "the" practises typing "hte" on purpose. In stimulus satiation, the reinforcer for a habit is supplied in such excess that it loses its value: in 1963 Teodoro Ayllon treated a long-stay psychiatric patient who hoarded towels by giving her towels freely, until several hundred were piled in her room and she began carrying them out herself, after which the hoarding stopped.

> EXAM TRAP
>
> Guthrie's theory is **contiguity** (stimulus and movement occurring together), with **one-trial learning** and the **recency principle**; reinforcement is not needed. Do not confuse Guthrie's one-trial learning (true, he said, of every association) with the one-trial learning of Garcia's taste aversion (a special, biologically prepared case, Section 6.4). His three methods of breaking habits are **threshold** (gradual increase below the threshold), **fatigue** (repeat the response to exhaustion) and **incompatible response** (pair the cue with a competing response). A question that describes introducing a saddle a little at a time wants the threshold method, not the fatigue method. Guthrie's book is *The Psychology of Learning* (1935); his cat study was with Horton (1946).

### Hull: a mathematical theory of habit

If Guthrie built the smallest theory of learning, **Clark Leonard Hull** (1884 to 1952) built the largest. Before turning to learning, Hull had written books on aptitude testing (1928) and on hypnosis and suggestibility (1933). At Yale University's Institute of Human Relations from 1929, he set out to make psychology as rigorous as Newton's physics, using the hypothetico-deductive method described in Chapter 2: state a set of postulates, deduce precise predictions from them, test the predictions in experiments and revise the postulates when they fail. *Principles of Behavior* (1943) presented sixteen postulates; *Essentials of Behavior* (1951) and *A Behavior System* (1952), published in the year of his death, revised them. In the 1940s and 1950s Hull was one of the most cited psychologists in experimental research, and the motivational side of his theory, drive reduction, is described in Chapter 10. Here we need its learning side.

Hull began from survival. An organism has biological needs, for food, water, air and the avoidance of injury. A need produces a *drive* (D), an aroused state that energises behaviour. Whatever response reduces the drive is reinforced. This is **drive-reduction theory**, and for Hull it was the definition of reinforcement: a reinforcer is anything that reduces a drive (or, in later versions, the stimuli that accompany a drive). Primary reinforcement reduces a biological drive directly; secondary reinforcement comes from stimuli that have been associated with drive reduction, the click before the food or the coin before the meal.

Each time a response is followed by reinforcement in the presence of a stimulus, the connection between them is strengthened. Hull called the strength of this learned connection **habit strength**, written ${}_{S}H_{R}$. Habit strength grows with the number of reinforced trials, and it grows in a particular way: each reinforcement adds a fixed fraction of the habit strength still to be gained. The early trials therefore add a great deal and the later ones less and less, which

is why Hull's theory predicts the negatively accelerated learning curve you saw in Figure 6.3. Once formed, habit strength is permanent; it can be opposed by inhibition, but never unlearned.

Habit alone does not produce behaviour. A rat that knows the maze perfectly but has just eaten sits in the start box. Hull therefore distinguished the learned connection from the tendency to act on it, which he called reaction potential (${}_{S}E_{R}$). In 1943 the relation was a simple product:

$${}_{S}E_{R} = {}_{S}H_{R} \times D$$

The multiplication sign is the heart of the theory. If habit strength is zero (the rat has never been trained), there is no response however hungry the rat is; if drive is zero (the rat is satiated), there is no response however well it is trained. Push numbers through it. A rat with habit strength 0.8 (on a scale from 0 to 1) and moderate hunger (D = 0.5) has a reaction potential of 0.4; double its hunger to D = 1.0 and the reaction potential doubles to 0.8, though it has learned nothing new; feed it to satiation (D = 0) and the reaction potential falls to zero, though it has forgotten nothing.

The experiments of the 1940s forced additions. Crespi's rats (Section 6.6) changed their running speed abruptly when the size of the reward changed, far too quickly for habit strength to have changed, so Hull added **incentive motivation** (K), the motivating effect of the size and quality of the reward. He added stimulus-intensity dynamism (V), the fact that a stronger stimulus evokes a learned response more readily, and he subtracted two kinds of inhibition. In its final form:

$$\text{Effective reaction potential} = ({}_{S}H_{R} \times D \times K \times V) - (I_{R} + {}_{S}I_{R})$$

Some versions also include J, a factor for delay of reinforcement (Chapter 10, Table 10.4, lists all the symbols). Two last refinements made the theory probabilistic. The effective reaction potential fluctuates randomly from moment to moment (oscillation, ${}_{S}O_{R}$), which is why a well-trained rat sometimes fails to respond; and it must exceed a reaction threshold (${}_{S}L_{R}$) before any response occurs. Hull then measured reaction potential by four observable outcomes: how quickly the response begins (latency), how vigorous it is (amplitude), how likely it is, and how many unreinforced trials it takes to extinguish.

The two inhibition terms deserve a closer look, because they explain three phenomena at once. Every response, Hull proposed, produces **reactive inhibition** ($I_{R}$), a fatigue-like negative drive that builds up with repeated responding and dissipates with rest. Resting reduces this fatigue-like drive, and drive reduction is reinforcing, so the response of *not* responding becomes conditioned to the situation; this learned habit of not responding is conditioned inhibition (${}_{S}I_{R}$). During extinction, with no reinforcement to offset them, the two kinds of inhibition accumulate until they cancel the reaction potential and responding stops. After a rest, reactive inhibition dissipates but conditioned inhibition remains, so the response returns, but only partly: spontaneous recovery. And because reactive inhibition piles up when trials are crowded together and dissipates when they are spaced out, Hull's theory predicts that distributed practice should beat massed practice and that performance should improve after a rest, two findings you will meet in Section 6.10.

### Hull's answers to purpose, and Spence's revisions

Tolman's rats seemed to act with purpose, to expect food and to know where it was. Hull set out to explain purpose without leaving S-R theory, and three of his concepts are still examined.

The **goal gradient** hypothesis (1932) states that the strength of a response increases the nearer it is to the goal. Rats run faster as they approach the goal box, and in a complex maze they eliminate the blind alleys nearest the goal first and those nearest the start last. The effect is not confined to rats. In a 2006 study by **Ran Kivetz** and colleagues, customers given a coffee-shop loyalty card bought coffee more and more often as they approached the free cup, an everyday goal gradient.

The **fractional anticipatory goal response** ($r_G$) was Hull's mechanical version of an expectation. When a hungry rat eats in the goal box, its eating (licking, chewing, salivating) is the goal response. Small fragments of that response become classically conditioned to the stimuli of the maze leading up to the goal, so that the rat makes tiny anticipatory licking and chewing movements as it runs. These fractional responses produce their own internal stimuli ($s_G$), which guide the rat towards the goal. What Tolman called an expectancy of food, Hull called an $r_G$-$s_G$ mechanism.

The **habit-family hierarchy** explained flexibility. Any goal can usually be reached by several routes, and each route has its own habit strength, so they form a family ranked from strongest to weakest. If the preferred route is blocked, the organism takes the next one in the hierarchy. Where Tolman saw a rat consulting a cognitive map, Hull saw a family of habits of different strengths.

Hull's most important student, **Kenneth Wartinbee Spence** (1907 to 1967) of the University of Iowa, refined the system so thoroughly that it is often called Hull-Spence theory. It was Spence who argued for incentive motivation and explained it through the fractional anticipatory goal response: a bigger reward produces a bigger $r_G$, and so more vigour. In his later work Spence concluded that habit strength grows with the number of trials whether or not they are reinforced, and that reinforcement works through incentive. His version was, in effect, a contiguity theory of learning combined with a reinforcement theory of performance. In 1937 Spence also gave a celebrated S-R explanation of transposition, the Gestalt psychologists' favourite evidence for relational learning (Section 6.10).

CLINICAL LINK

If drive multiplies habit, then anxious people, whose general level of drive is high, should condition faster on simple tasks. To test this, **Janet Taylor Spence** (1923 to 2015), who trained in Spence's laboratory at Iowa and later married him, selected items from the Minnesota Multiphasic Personality Inventory to build the Taylor Manifest Anxiety Scale (1953). High scorers did acquire conditioned eyeblinks faster than low scorers. On complex tasks with many competing responses, however, high drive strengthened the wrong responses too, and anxious participants did worse. The finding helps explain why a moderately anxious student may race through simple revision but freeze on a hard, unfamiliar problem (the Yerkes-Dodson law, Chapter 10, makes a related point).

How should Hull be judged? His system collapsed under its own weight. Each new finding required a new constant or a new term, and the predictions became so hedged that they could

hardly be tested. Drive reduction itself failed as a universal account of reinforcement: rats learned to run for saccharin, which reduces no need, and monkeys worked for the chance to look out of a window (Chapter 10). Latent learning showed that learning can occur without reinforcement at all. Yet Hull's influence was enormous. He showed what a precise, testable theory of behaviour would look like; his intervening variables (habit, drive, incentive) became the common language of motivation research; and his students and colleagues, among them Spence, Neal Miller, John Dollard, Hobart Mowrer, Carl Hovland and Robert Sears, carried his ideas into personality, social psychology, attitude change and therapy. Wolpe's reciprocal inhibition was framed in Hull's terms, and Dollard and Miller's translation of Freud (Section 6.8) used his vocabulary of drive, cue, response and reward.

> EXAM TRAP
>
> **Drive-reduction theory** is **Hull's** (*Principles of Behavior*, 1943); the word "drive" was introduced earlier by Woodworth (1918), and homeostasis is Cannon's term. Hull's method was **hypothetico-deductive**. In ${}_{S}E_{R} = {}_{S}H_{R} \times D$, E is reaction potential, H habit strength, D drive; K (incentive motivation) was added after Crespi's contrast effects, and V is stimulus-intensity dynamism. **Reactive inhibition** is Hull's explanation of why spaced practice beats massed practice. The **goal gradient** and the **fractional anticipatory goal response** are Hull's; the **cognitive map** is Tolman's. And the "reciprocal causation model" belongs to Bandura's social cognitive theory, not to drive-reduction theory.

### Estes: stimulus sampling theory

You have already met William Kaye Estes twice: in 1941, with Skinner, he invented the conditioned-suppression measure of fear (Section 6.4), and in 1944 he showed that punishment suppresses a response without erasing it (Section 6.5). Estes had taken his doctorate under Skinner at the University of Minnesota, but in 1950, in a paper titled "Toward a statistical theory of learning", he did something his teacher would never have done. He built a mathematical model of learning, and he built it on Guthrie's ideas.

Estes's **stimulus sampling theory** starts by treating any stimulus situation, such as a Skinner box or a classroom, as a large population of small stimulus elements: a particular sound, a patch of light, a smell, a sensation from the body. On any trial the learner samples, or notices, only a fraction of these elements, a fraction Estes called θ (theta). Following Guthrie, each sampled element becomes connected, completely and at once, to whatever response occurs on that trial (an all-or-none, one-trial process). The probability that the learner will make the response on the next trial is simply the proportion of all the elements in the situation that are now connected to it.

From these assumptions Estes derived a learning curve. If $P_1$ is the probability of the response on the first trial, then on trial n:

$$P_n = 1 - (1 - P_1)(1 - \theta)^{n-1}$$

Work through it with $\theta = 0.2$ and a response that never occurs at first ($P_1 = 0$). On trial 2 the probability is $1 - 0.8 = 0.20$; on trial 3, $1 - 0.8 \times 0.8 = 0.36$; on trial 4, $1 - 0.512 = 0.49$; on trial 5, about 0.59. Each trial conditions a fifth of the elements not yet conditioned, so the gains shrink as fewer unconditioned elements remain. The result is the familiar negatively accelerated curve, yet every individual element was conditioned in a single step. Notice the family resemblance: Hull's habit strength, Estes's θ and the αβ of the Rescorla-Wagner model (Sec-

tion 6.4) all describe learning as the capture of a constant fraction of what is still to be learned.

The theory explained other phenomena with the same economy. Generalisation occurs because two similar situations share many elements, so a response conditioned to one is partly conditioned to the other. Spontaneous recovery, Estes argued in 1955, occurs because the elements available to the learner fluctuate from session to session: elements that were conditioned during acquisition but happened not to be sampled during extinction drift back into the available set after a rest, carrying the response with them. And the theory predicted a striking pattern called **probability matching**. Ask people to guess, trial after trial, whether a light will come on, when in fact it comes on at random on 70 per cent of trials, and most of them end up predicting "on" on about 70 per cent of trials, rather than on every trial, which would maximise their score. The first hints of this pattern had appeared in a 1939 experiment by Lloyd Humphreys, and Estes's model predicted it.

Stimulus sampling theory, together with the linear models of **Robert R. Bush** and **Frederick Mosteller**, launched mathematical learning theory. Estes himself later moved into cognitive psychology, building models of memory and categorisation, and he is one of the few learning theorists whose work bridges the behaviourist and cognitive eras. He made one further point that every reader of learning curves should remember. In 1956 he showed that a curve averaged over many learners can look smooth and gradual even when each individual learner improves in sudden jumps (Section 6.10).

> EXAM TRAP
>
> **Stimulus sampling theory** is **Estes's** (1950), a statistical (mathematical) learning theory that formalised Guthrie's contiguity and one-trial ideas; it is not Skinner's. Its key quantity, θ, is the proportion of stimulus elements sampled on a trial. Estes is also examined for conditioned suppression (with Skinner, 1941) and for showing that punishment suppresses rather than eliminates responding (1944).

### Mowrer: fear, hope and the neurotic paradox

Orval Hobart Mowrer, whose two-factor theory of avoidance you met in Section 6.5, was another product of Hull's Yale institute, and his career moved through almost every question in this chapter. His first lasting contribution was practical. In 1938, with his wife **Willie Mae Mowrer**, he described the **bell-and-pad method** for treating nocturnal enuresis (bed-wetting). The child sleeps on a pad that completes an electrical circuit at the first drop of urine and rings a bell. In classical-conditioning terms the bell is a UCS that produces waking and contraction of the bladder sphincter (the UCR); the sensations of a full bladder, which immediately precede the bell, are the CS. After a number of pairings the full bladder alone wakes the child or inhibits urination. (Some writers describe the later stages as avoidance learning, since the child learns to prevent the bell.) The enuresis alarm descended from it is still one of the most effective treatments available.

In 1947 Mowrer proposed that there are two fundamentally different kinds of learning. **Sign learning** is classical conditioning: signs (stimuli) become attached to emotional responses, especially fear, which are involuntary and governed by the autonomic nervous system. **Solution learning** is instrumental learning: skeletal, voluntary responses that solve prob-

lems are strengthened because they reduce a drive. Two-factor theory is the combination of the two, fear learned by sign learning and avoidance learned by solution learning.

A year later Mowrer used the theory to attack what he called the **neurotic paradox**: why do people go on behaving in ways that are obviously self-defeating, avoiding what they fear at the cost of jobs, friendships and happiness? A simple law of effect would predict that behaviour with bad consequences should die out. Mowrer's answer was timing. The avoidance response brings immediate relief from anxiety, while its costs are delayed and diffuse, and immediate consequences control behaviour far more strongly than delayed ones. Avoidance also prevents extinction, because the person never stays in the feared situation long enough to discover that nothing terrible happens. The man who leaves every party after ten minutes feels better each time he leaves, and never learns that he could have survived the evening.

Two-factor theory was soon challenged. Solomon's dogs (Section 6.5) went on jumping long after they showed any outward signs of fear, which raised the question of what was being reduced. In 1966 Richard Herrnstein and **Philip Hineline** showed that rats would learn to press a lever that merely lowered the overall frequency of shocks arriving at unpredictable moments, with no warning signal to be afraid of; the avoidance response seemed to be reinforced directly by the reduction in shock. And in 1970 **Robert Bolles** pointed out that rats learn some avoidance responses in a trial or two (running away, jumping out of a box) but need hundreds of trials to learn others (pressing a lever). Avoidance is learned quickly, Bolles argued, only when the required response is one of the animal's **species-specific defence reactions**: fleeing, freezing or fighting. Biology constrains avoidance learning just as it constrains taste aversion and operant training.

Mowrer, meanwhile, had changed his mind. In *Learning Theory and Behavior* (1960) he concluded that all learning is sign learning. What organisms learn, he now argued, is to attach emotions to stimuli, including the stimuli produced by their own movements. Four emotions do all the work, depending on whether a stimulus signals the beginning or the end of something painful or something pleasant (Table 6.12). When the feedback from a response has become attached to hope, the organism continues the response; when it has become attached to fear, the organism stops. Even reward and punishment, on this view, work through classically conditioned emotions.

Table 6.12 Mowrer's four emotions (1960): what a stimulus comes to signal decides what it comes to make us feel.

| The stimulus signals | Painful event | Pleasant event (reward) |
|---|---|---|
| Onset (it is coming) | Fear | Hope |
| Termination (it is ending or withheld) | Relief | Disappointment |

Mowrer applied learning theory to language too. He suggested that a talking bird, or a babbling infant, first reproduces the sounds of its caregiver's voice because those sounds have been paired with food and comfort and so have become secondary reinforcers; producing them is then self-rewarding. He called this the "autism theory" of speech, using "autism" in its old sense of self (it has nothing to do with autism spectrum disorder). Later still he broke sharply with Freud, arguing that neurotic suffering comes from real guilt over real wrongdoing rather than from repressed impulses, and he founded the confession-based integrity groups of the 1960s (Clinical Psychology book).

EXAM TRAP

"Two-factor theory" in a learning question is **Mowrer's** theory of avoidance (1947): classical conditioning of fear plus operant (negative) reinforcement of the avoidance response by fear reduction. It is not Schachter and Singer's two-factor theory of emotion (Chapter 11) and not Herzberg's two-factor theory of work motivation. **Sign learning versus solution learning** is Mowrer's distinction; but "sign learning theory" is also a name for **Tolman's** expectancy theory (Section 6.7), in which "signs" are cues to what leads to what. Read the options. The **bell-and-pad** for enuresis is Mowrer and Mowrer (1938), a classical-conditioning treatment. The **neurotic paradox** (1948) is Mowrer's question of why self-defeating avoidance persists.

## Gagné: the conditions of learning

The theorists so far wanted one theory for all learning. **Robert Mills Gagné** (1916 to 2002), an American educational psychologist who had spent much of the 1940s and 1950s in military research on training pilots and technicians, drew the opposite conclusion. In *The Conditions of Learning* (1965) he argued that there is not one kind of learning but several, and that each requires its own conditions: internal conditions (what the learner must already know or be able to do) and external conditions (what the teacher or the situation must provide). His central claim was that the kinds of learning form a hierarchy. Each type builds on the ones below it, so that a higher type cannot be learned until its prerequisites have been mastered, a view he called cumulative learning. Table 6.13 shows the eight types in order, from the simplest to the most complex.

Table 6.13 Gagné's eight types of learning (1965), from simplest to most complex.

| Type | What is learned | Example | Echoes |
|---|---|---|---|
| 1 Signal learning | An involuntary, emotional response to a signal | A child flinches at the sight of the syringe | Pavlov |
| 2 Stimulus-response learning | A precise voluntary response to a stimulus, through reinforcement | A puppy learns to sit when told "sit" | Thorndike, Skinner |
| 3 Chaining | A sequence of motor S-R links | Starting a scooter: key, choke, kick | Skinner's chaining |
| 4 Verbal association | Chains of words | Learning that "neeru" means water; reciting tables | Verbal learning |
| 5 Multiple discrimination | Different responses to many similar stimuli | Telling each of thirty pupils apart by name | Discrimination |
| 6 Concept learning | A common response to a class of objects | "Triangle", "mammal" | Concept formation (6.7) |
| 7 Principle (rule) learning | A chain of two or more concepts | "Metals expand when heated" | Rules |
| 8 Problem solving | Combining principles into a new, higher-order principle | Freeing a tight steel lid by running hot water over it | Thinking (Chapter 8) |

The hierarchy has a practical message for teachers. If a pupil cannot solve problems about expanding metals, the fault may lie further down: perhaps he never learned the principle, or never mastered the concepts of "metal" and "expansion" from which it is built. Diagnose the missing prerequisite and teach it first.

In later editions of the book Gagné recast his scheme as five varieties of learned capability, or learning outcomes: verbal information (knowing that), intellectual skills (knowing how, which include discriminations, concepts, rules and higher-order rules), cognitive strategies (ways of managing one's own learning and thinking), motor skills and attitudes. He also proposed nine "events of instruction" that a good lesson should contain, in order: gain the learners' attention; tell them the objective; stimulate recall of prerequisite learning; present the material; provide learning guidance; elicit performance; provide feedback; assess performance; and enhance retention and transfer. Several of these, feedback and transfer especially, are the subject of Section 6.10. Gagné also distinguished lateral from vertical transfer, a distinction you will meet there.

> **EXAM TRAP**
>
> Gagné's eight types run **signal learning, stimulus-response learning, chaining, verbal association, multiple discrimination, concept learning, principle (rule) learning, problem solving**. Signal learning (the lowest) is classical conditioning; problem solving is the highest. Concept learning comes *before* principle learning, because principles are chains of concepts. Do not mix the eight types with his five learning outcomes (verbal information, intellectual skills, cognitive strategies, motor skills, attitudes) or his nine events of instruction. The book is *The Conditions of Learning* (1965).

## The theories side by side

By the 1960s the age of grand theories was over. The questions they had argued about did not go away; they were absorbed into cognitive psychology, which took Tolman's side on what is learned, and into the neuroscience of learning, which found that both S-R habits and cognitive maps exist, in different brain systems (Chapter 7 and the Physiological Psychology book). Table 6.14 places the main theorists of this chapter in their families, a useful revision tool because examiners love to ask which theorist belongs to which idea.

Table 6.14 The main learning theorists at a glance.

| Theorist | Family | Central idea | Key work |
|---|---|---|---|
| Pavlov | Classical conditioning | The conditioned reflex; CS-UCS pairing | *Conditioned Reflexes* (English 1927) |
| Watson | S-R contiguity | Frequency and recency; conditioned emotions | Behaviourist manifesto (1913) |
| Thorndike | S-R reinforcement | Law of effect; connectionism | *Animal Intelligence* (1898, 1911) |
| Guthrie | S-R contiguity | One-trial learning; recency; habit breaking | *The Psychology of Learning* (1935) |
| Hull | S-R reinforcement | Drive reduction; habit strength × drive | *Principles of Behavior* (1943) |
| Skinner | Operant, descriptive | Contingencies and schedules of reinforcement | *The Behavior of Organisms* (1938) |
| Tolman | Cognitive (purposive) | Expectancy; cognitive map; latent learning | *Purposive Behavior in Animals and Men* (1932) |
| Köhler | Cognitive (Gestalt) | Insight | *The Mentality of Apes* (English 1925) |
| Estes | Mathematical | Stimulus sampling | "Toward a statistical theory of learning" (1950) |

| Theorist | Family | Central idea | Key work |
|---|---|---|---|
| Mowrer | Two-factor | Sign and solution learning; fear and hope | *Learning Theory and Behavior* (1960) |
| Bandura | Social cognitive | Observational learning; self-efficacy | *Social Foundations of Thought and Action* (1986) |
| Gagné | Instructional | Eight types in a hierarchy | *The Conditions of Learning* (1965) |

One family is missing from the table because it grew up in developmental and educational psychology rather than in the animal laboratory. **Constructivism** is the view that learners do not passively receive knowledge or have responses stamped in, but actively construct their own understanding by fitting new experience to what they already know and revising what they know when it does not fit. Jean Piaget's theory of cognitive development, Lev Vygotsky's account of learning through social interaction, and Jerome Bruner's discovery learning are its main forms (Developmental Psychology book). Constructivist theories share Tolman's and Köhler's conviction that what is learned is knowledge, and they share Bandura's conviction that the learner is an agent, not a passive recipient.

> EXAM TRAP
>
> Match the theory to its owner. **Contiguity and one-trial learning**: Guthrie. **Drive reduction and habit strength**: Hull. **Incentive motivation (K)**: Hull and Spence. **Stimulus sampling**: Estes. **Two-factor theory of avoidance**: Mowrer. **Eight types of learning**: Gagné. **Reciprocal causation (reciprocal determinism)**: Bandura's social cognitive theory, not two-factor theory, drive-reduction theory or constructivist theory. **Constructivism**: Piaget, Vygotsky and Bruner. The **law of effect** is Thorndike's and the **matching law** Herrnstein's; Watson's own laws of learning were **frequency and recency**.

## 6.10 Learning in Practice LOWER YIELD · 3 QS

The theories in this chapter were built in laboratories, with rats, pigeons, dogs and cats. But the people who most need to know about learning are not in laboratories. A student with four weeks left before an entrance examination, a nursing trainee learning to give injections, a child learning to read, a batsman remodelling his technique: all of them face the same practical questions. How quickly will I improve, and why have I stopped improving? How much should I practise, and in one long session or many short ones? Should I learn the whole thing at once or in parts? How much does feedback matter? Will what I learn here help me there? Can a machine teach me? Psychologists have been studying these questions since the 1890s, and the answers are among the most useful in this book.

### Learning curves

A **learning curve** is a graph of performance plotted against the amount of practice, measured in trials, sessions or time. Ebbinghaus's curves of 1885 (Chapter 7) and Thorndike's escape-time curves for his cats (Section 6.5) were among the first. Performance can be measured in many ways: correct responses, words typed per minute, runs scored, or, the other way up, errors made and seconds taken. When errors or time are plotted, a curve of improvement falls instead of rising, as in Tolman's maze data (Figure 6.5). What matters is the shape

of the improvement, not which way the line slopes. Three shapes are distinguished (Figure 6.7, left).

A **negatively accelerated curve** (a decelerating curve) rises steeply at first and then more and more slowly, levelling off as performance approaches its limit. Each unit of practice adds less than the one before, a law of diminishing returns. This is the most common shape, especially for simple tasks and for learners who already have the relevant background. Someone learning to type sees huge gains in the first week and tiny ones in the tenth. You have already met three explanations of the shape: in Hull's theory, Estes's theory and the Rescorla-Wagner model, each trial captures a constant fraction of what remains to be learned, and as less remains, each gain shrinks. There is also a simpler reason: learners tend to master the easy parts of a task first, leaving the hard parts for later.

A **positively accelerated curve** (an accelerating curve) does the opposite: progress is slow at first and then speeds up. It appears when a task is difficult or unfamiliar and the learner must first acquire basic components before the real gains can come. An adult beginning a new language spends weeks on sounds and vocabulary with little to show for it; once enough words and grammar are in place, comprehension suddenly takes off. Positively accelerated curves are common in the early stages of complex learning and in learners with little prior knowledge.

An **S-shaped curve** (a sigmoid curve) combines the two: a slow start while the learner gets oriented, a rapid middle phase, and a slowing as performance nears its limit. It is the typical curve when a complex skill, such as driving a car or playing the tabla, is learned from scratch and followed to mastery. Straight-line (linear) curves, in which every trial adds the same amount, are rare.

One caution, which comes from William Estes (Section 6.9). In 1956 he showed that a curve averaged over many learners can look smooth and gradual even when each individual learner improves in sudden jumps. If ten students each have their "aha" at a different session, the average of their scores rises gently, though none of them improved gently. A smooth curve in a textbook is therefore not proof that each learner improved bit by bit; Köhler's chimpanzees, plotted one at a time, would have produced steps.

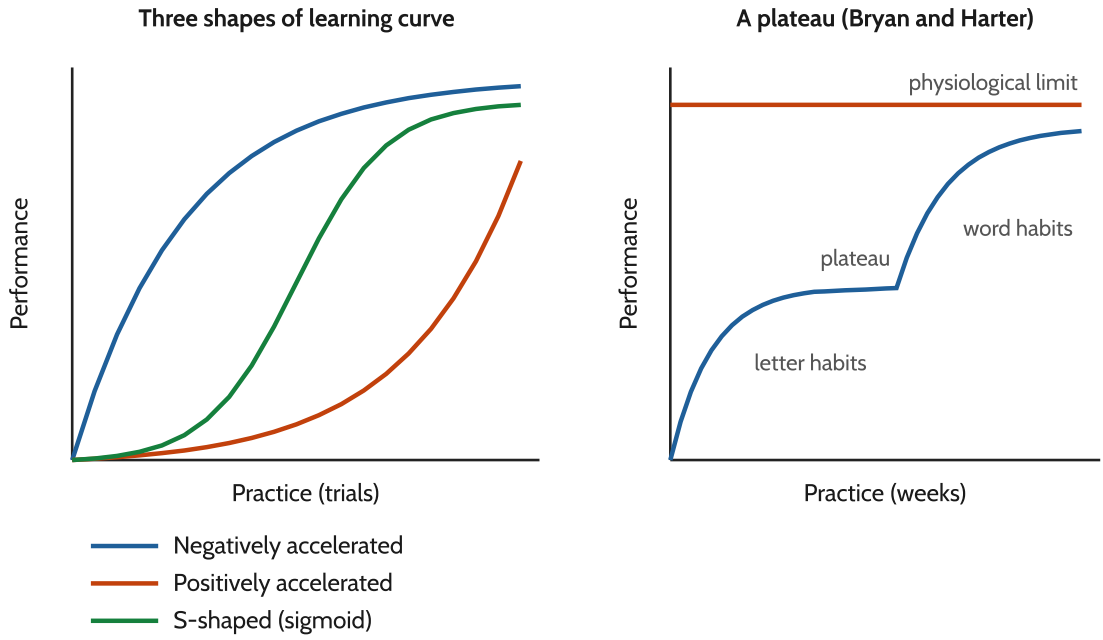


Figure 6.7 Left: the three shapes of learning curve, drawn from their mathematical functions. Right: a stylised curve with a plateau, after Bryan and Harter's telegraphers. Improvement stalls while letter-by-letter habits become automatic, then resumes as word habits form; a plateau is a pause in the middle of learning, not the final physiological limit.

## Plateaus and the limits of improvement

In the 1890s, at Indiana University, **William Lowe Bryan** (1860 to 1955) and **Noble Harter** studied people learning the telegraphic Morse code, measuring each week how many letters per minute they could send and receive. Their reports (1897 and 1899) contain the most famous learning curves in the history of psychology. Sending improved smoothly. Receiving improved rapidly for some weeks, then stopped improving for a long stretch, sometimes for several weeks, and then began to rise again, eventually to the level of an expert operator (Figure 6.7, right).

Such a flat stretch in the middle of a learning curve, preceded and followed by improvement, is called a **plateau**. Bryan and Harter explained it by a *hierarchy of habits*. A beginner receives code letter by letter. Only when letter habits have become automatic can the learner's attention move on to grasping whole words as units, and later whole phrases. The plateau is the period during which the lower-order habits are being perfected but are not yet automatic enough to free attention for the higher-order ones. When word habits take over, improvement resumes.

Later writers listed several other causes of plateaus, which suggest their remedies:

- A shift from a lower-order to a higher-order method of working, as in the telegraphers, during which neither method is yet efficient.
- Loss of motivation, boredom or fatigue after the excitement of early rapid progress has worn off.
- Faulty methods or bad habits that limit performance until they are corrected, such as typing with two fingers.

- A particularly difficult stage of the task, or a period in which recently learned material is being consolidated.

Plateaus can therefore often be shortened: by rest, by changing the method of practice, by setting new goals, by giving precise feedback, and by encouraging the learner, who may be on the verge of a breakthrough without knowing it. They should not be overstated, though. When later researchers looked again at telegraphy and other skills, many found no plateau at all, and in 1958 Fred Keller titled his review of the evidence "The phantom plateau". A plateau is common but not inevitable.

A plateau must also be distinguished from the **physiological limit**, the final ceiling of performance set by the limits of the body and nervous system, beyond which no amount of practice brings improvement. A plateau is temporary and in the middle of the curve; the physiological limit is permanent and at the end. In practice, most people stop practising long before they reach their physiological limit; what they reach is the level that satisfies them.

> EXAM TRAP
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> **Negatively accelerated** means rapid early gains and slowing later (the commonest curve); **positively accelerated** means slow early gains and faster later; **S-shaped** combines them. A **plateau** is a temporary flat period of no apparent progress in the middle of learning (Bryan and Harter, telegraphy, 1897 and 1899), explained by a **hierarchy of habits**; it is not the physiological limit (the final ceiling), not extinction (the weakening of a response when reinforcement stops) and not forgetting. Do not confuse a learning plateau with the plateau phase of Masters and Johnson's sexual response cycle (Physiological Psychology book).

## From clumsy to automatic: the stages of skill learning

Bryan and Harter's telegraphers were acquiring a skill, and the course of skill learning in general was described by **Paul Fitts** (1912 to 1965) and **Michael Posner** (born 1936) in *Human Performance* (1967). They proposed three stages. In the *cognitive stage* the learner works out what to do, often talking himself through it ("clutch in, first gear, release slowly"); performance is slow, effortful, inconsistent and full of errors, and it needs all of his attention. In the *associative stage* the components are linked into smooth sequences, errors fall, and the learner begins to notice and correct his own mistakes. In the *autonomous stage* the skill runs almost automatically: it is fast, accurate and consistent, needs little attention, and leaves the mind free for other things, so that an experienced rider can weave a scooter through Mysuru traffic while carrying on a conversation. The cost of automaticity is that the expert often cannot say how she does what she does, and finds bad habits hard to change.

The stages map onto brain systems described in the Physiological Psychology book. Early, effortful learning depends heavily on the prefrontal cortex. As a skill becomes automatic, control shifts to systems for procedural learning: the cerebellum, which supports error correction and the fine coordination of movement, and the basal ganglia, which support the selection of actions and the formation of habits (procedural memory is in Chapter 7).

## Overlearning

When have you learned something? A natural answer is: when you can do it once without error. **Overlearning** is practice that continues beyond that point. The classic study was published in 1929 by **William C. F. Krueger**. Participants learned lists of single-syllable nouns

until they could recite a list once without error. Some stopped there; others continued for half as many trials again as they had needed (50 per cent overlearning), or for as many trials again (100 per cent overlearning). Retention was tested after intervals of up to 28 days. Fifty per cent overlearning produced a clear improvement in retention at every interval. Doubling the extra practice to 100 per cent produced a further, smaller gain, so that the benefit per extra trial fell, another example of diminishing returns.

Push a case through it. If you need 12 readings to recite a 20-line poem for the first time, 50 per cent overlearning means 6 more readings, and 100 per cent overlearning means 12 more. The first 6 extra readings are well worth the time; the next 6 are worth less. Overlearned material resists forgetting, resists interference from later learning, and, importantly for anyone who has to perform under pressure, resists disruption by stress and anxiety. This is why airline pilots, surgeons and emergency teams drill procedures they already know, and why a dancer rehearses an item long after she can perform it correctly. (Ebbinghaus's savings method, which first revealed the benefits of extra repetitions, is in Chapter 7.)

### Massed versus distributed practice

Suppose you have twenty hours to give to a subject. You could spend them in one long weekend, or as ten two-hour sessions spread over a fortnight. In **massed practice**, practice sessions are crowded together with little or no rest between them; in **distributed practice** (spaced practice), the same amount of practice is spread out over time with rest intervals. For both verbal learning and motor skills, distributed practice usually produces better learning and, above all, better long-term retention. The advantage is called the spacing effect, and it is one of the most reliable findings in all of psychology; a 2006 review by **Nicholas J. Cepeda** and colleagues analysed several hundred experiments on it.

Ebbinghaus was the first to show it, in 1885. Twelve years later **Adolf Jost** stated what is still called Jost's law (1897): if two associations are equally strong but of different ages, a further repetition strengthens the older one more, and the older one is forgotten more slowly. Spaced repetitions, which give each association time to age between repetitions, therefore gain more from each repetition than massed ones. A striking applied study came from **Alan Baddeley** and D. J. A. Longman in 1978. British postal workers learning to type postcodes on a new keyboard were trained in one or two daily sessions lasting one or two hours each. Those given a single one-hour session a day learned the keyboard in fewer total hours of practice than those on the more massed schedules, though it took them more days; and, interestingly, the workers preferred the massed schedules, which felt more productive.

Why does spacing help? Hull's answer was reactive inhibition (Section 6.9): a fatigue-like inhibition builds up when responses are crowded together and dissipates during rest. Hull's theory also explains **reminiscence**, the improvement in performance after a rest with no further practice, which **Philip Boswood Ballard** reported in 1913 when schoolchildren recalled more of a poem two or three days after learning it than immediately afterwards. In motor skills, reminiscence is often seen when a learner returns to massed practice after a break and performs better than before the break. Other explanations stress memory: spaced sessions allow consolidation, they encode the material in more varied contexts, and they force effortful retrieval at each new session, a "desirable difficulty" in **Robert A. Bjork**'s phrase, which strengthens memory more than easy repetition does.

Massed practice is not always worse. It is efficient for short, simple tasks; for tasks that require a long warm-up before useful work can begin; and for problems whose solution depends on keeping the whole problem in mind at once, which a break would interrupt. Cramming the night before can also raise performance the next morning. What massed practice does badly is long-term retention: the crammed material is largely gone a month later.

> EXAM TRAP
>
> **Overlearning** is practice beyond the first errorless performance (Krueger, 1929: 50 per cent overlearning clearly helps; 100 per cent adds less), not learning so much that performance suffers. **Distributed (spaced) practice** generally beats **massed practice**, especially for long-term retention; Hull's explanation is **reactive inhibition**. **Reminiscence** (Ballard, 1913) is improvement after rest without practice; it is not spontaneous recovery, which is the return of an *extinguished* conditioned response. Jost's law (1897) says a repetition benefits an older association more than a newer one of equal strength.

### Whole versus part learning

You have to learn a 40-line poem for a recitation, or a long Carnatic composition, or a new bowling action. Should you work through the whole thing each time, or break it into parts? In the **whole method**, the material is practised as a single unit from beginning to end on every trial. In the **part method**, it is divided into sections, each section is learned separately, and the sections are then joined. A compromise, the **progressive part method**, learns the first part, then the first and second together, then the first three, and so on, so that the joins are practised as the learning proceeds.

Studies from the early 1900s onwards found that neither method always wins; the answer depends on the material and the learner.

- The whole method tends to be better when the material is meaningful, well organised and of moderate length, so that its unity helps (a poem with a clear story); when the learner is older, more able and well motivated; and when practice is distributed. It avoids the main weakness of the part method, which is that the parts must later be linked, and the learner tends to stumble at the junctions.
- The part method tends to be better when the material is very long, difficult or made of loosely connected units (a list of dates, the bones of the hand); and for younger or less able learners. It also gives an early sense of success, which sustains motivation, whereas with the whole method progress may be invisible for a long time.

The same principle applies to motor skills. A highly integrated movement, in which each component depends on the others, such as a bowling action, a golf swing or a swimming stroke, is usually best practised whole, perhaps slowed down, because practising its pieces separately destroys their timing. A skill made of fairly independent components, such as a dance with distinct sections or the separate checks of a pre-flight routine, can be practised in parts. Many teachers use a whole-part-whole sequence: show and attempt the whole, practise the difficult parts, then put the whole together again.

EXAM TRAP

There is no single "best" method. **Whole** learning favours short or moderate, meaningful, unified material and mature, able learners; **part** learning favours long, difficult, unconnected material and younger learners; the **progressive part** method adds each new part to those already learned. Do not confuse whole and part *learning* (how material is divided for practice) with the whole-report and partial-report methods of Sperling's sensory-memory experiments (Chapter 7).

## Feedback and goals ●●

You saw in Section 6.5 that Thorndike's blindfolded participants, drawing line after line and trying to make each one three inches long, did not improve at all when they were never told how close they had come. Practice alone, without information about its results, taught them nothing. **Knowledge of results** (KR) is information given to the learner about the outcome of his or her performance; in everyday language it is feedback. In 1932 **Margery Trowbridge** and **Hulsey Cason** repeated the line-drawing task with groups that received different kinds of feedback after each line: none, an irrelevant nonsense syllable, a simple "right" or "wrong", or the exact size and direction of the error. The group given no feedback did not improve; the "right or wrong" group improved; the group told exactly how far off they were improved most.

Research since then has added several practical rules. Feedback works best when it is precise rather than vague ("two centimetres too long" beats "wrong"), and when it arrives soon after the performance, while the learner can still relate it to what he did. Feedback after every single trial helps performance during practice, but learners who receive it too often can come to depend on it and perform worse when it is withdrawn, so skilled coaches give frequent feedback early and gradually reduce it. Teachers of motor skills distinguish knowledge of results, which concerns the outcome (the ball went for four), from knowledge of performance, which concerns the movement itself (your front elbow dropped). Feedback can also come from the task itself (intrinsic feedback, the feel of a clean strike) or from an outside source (extrinsic feedback, the coach's comment). Feedback works in two ways at once: it is information, telling the learner what to change, and it is a reinforcer and motivator, telling him that his efforts are paying off. Biofeedback (Section 6.6) and programmed instruction (below) are both built on it.

Feedback has a natural partner. In 1968 the American psychologist **Edwin Locke** (born 1938) proposed, and in later work with **Gary Latham** demonstrated in hundreds of studies, that people who set specific, difficult but attainable goals perform better than people who are told to "do your best" or who set no goal at all. A specific goal directs attention to what matters, mobilises effort, increases persistence and prompts the learner to search for better strategies. Goals and feedback need each other: a goal without feedback leaves the learner unable to tell whether he is on course, and feedback without a goal gives him nothing to compare it with. So practice is most effective when the learner has a specific goal and receives knowledge of results about progress towards it. (Goal-setting theory as a theory of motivation is in Chapter 10; self-efficacy, which decides whether a person will commit to a hard goal, is in Section 6.8.)

EXAM TRAP

"Learning always leads to efficacy, provided …" The keyed answer is that **the learner has a specific goal** (Locke and Latham's goal setting). Ability, aptitude and knowledge help, but on their own they do not make practice purposeful. Knowledge of results is feedback about the outcome; without it, as in Thorndike's line drawing, practice brings no improvement. In a question about what **social cognitive theory** stresses in motivation, the answer is **self-efficacy**; "goal difficulty" belongs to Locke's goal-setting theory, and "external rewards only" and "biological drives" to behaviourist and drive theories.

CLINICAL LINK

The principles of this section are the backbone of teaching children with learning difficulties. Effective programmes present material through several senses at once, a **multisensory** approach in which the child sees, hears, says and traces a word, as in the method **Grace Fernald** described in 1943 (visual, auditory, kinaesthetic and tactile, VAKT) and the Orton-Gillingham method; they break skills into small steps, give immediate feedback and reinforcement, and build in regular, spaced repetition and overlearning. Relying solely on written instructions, refusing accommodations, or drilling rote memorisation alone are the opposite of good practice. (Specific learning disorder itself is covered in the Psychiatric Disorders book.)

## Transfer of training

Nobody learns anything for use in only one situation. You study psychology at a desk in order to use it in a hospital ward; a pilot trains in a simulator in order to fly an aircraft. **Transfer of training** is the effect that learning one task has on the learning or performance of another. It can take three forms. In **positive transfer**, earlier learning makes later learning easier: someone who rides a bicycle learns to ride a motorcycle faster, and a student who reads Hindi in the Devanagari script finds written Marathi easy. In **negative transfer**, earlier learning makes later learning harder, usually because the same stimuli now require different responses: a driver from India, where traffic keeps left, keeps reaching for the wrong side of the road in a country where traffic keeps right, and a driver who switches from a Japanese car to a European one switches on the wipers every time he means to signal a turn. In **zero transfer**, the first task has no effect on the second: learning to swim neither helps nor hinders learning Sanskrit grammar.

Transfer is measured with a simple design. An experimental group learns Task A and then Task B; a control group learns only Task B (often after resting or doing an unrelated task for the same time). If the experimental group learns Task B faster than the control group, transfer was positive; if slower, negative; if the same, zero. Several special kinds of transfer have names. In **bilateral transfer**, a skill practised with one side of the body improves on the other: people who practise mirror drawing with the right hand also draw better with the left. Gagné distinguished *lateral transfer*, the application of learning to a new situation at the same level of complexity (calculating the area of the classroom floor after learning to calculate areas of rectangles in a textbook), from *vertical transfer*, in which lower-level learning makes higher-level learning possible (concepts making principles possible, as in his hierarchy). And Harlow's learning sets (Section 6.7) show non-specific transfer: learning *how* to learn a kind of problem carries over even when no specific content is shared.

Why does transfer occur, and when? For two thousand years educators had a confident answer. According to the doctrine of **formal discipline**, based on the old faculty psychology, the mind consists of general faculties, such as memory, reasoning, attention and will, that are strengthened by exercise as muscles are. On this view it hardly matters what pupils study, so long as it is hard: Latin, Greek and geometry were taught not for their content but because they were supposed to train the mind for everything else. Three pieces of evidence demolished the doctrine.

- **William James** (1842 to 1910) tested it on himself and reported the result in *The Principles of Psychology* (1890). He learned 158 lines of Victor Hugo's poem *Satyr*, which took him 131 and five-sixths minutes. He then spent twenty-odd minutes a day for 38 days learning the first book of Milton's *Paradise Lost*, to train his "memory faculty". Finally he learned another 158 lines of *Satyr*. It took him 151 and a half minutes: if anything, he was slower. (James had a few friends repeat the experiment, with mixed results, but the point was made.)
- In 1901 Thorndike and Robert Sessions Woodworth trained people to estimate the areas of rectangles of certain sizes. Accuracy improved for the trained shapes and sizes, but transferred very little to estimating the areas of other shapes, or to related judgements of length and weight.
- In 1924 Thorndike compared the gains in reasoning test scores of more than 8,000 American high-school pupils over a school year according to the subjects they had studied. The subjects studied made very little difference: the gains that could be credited to Latin or geometry were tiny, and no larger than those from several "practical" subjects. The pupils who took the classical subjects did score higher, but mainly because they had been abler to begin with.

In place of formal discipline, Thorndike proposed his **identical elements theory**: transfer occurs only to the extent that the two tasks share identical elements, whether facts, procedures, stimulus-response connections, attitudes or methods of work. Learning to drive a car transfers to driving a van because nearly all the elements are the same; studying Latin helps with learning Latin-derived English vocabulary but not with reasoning in general. The theory had a large practical effect: if transfer depends on shared content, schools should teach directly what pupils will need, and American curricula became more practical as a result.

A second view emphasised understanding. In 1908 **Charles Hubbard Judd** (1873 to 1946) had two groups of boys throw darts at a target placed under about 12 inches of water. One group had first been taught the principle of refraction, how water bends light so that an object appears displaced; the other had not. Both groups did equally well at first, since practice was what counted. But when the target was moved to a depth of only 4 inches, the boys who understood refraction adjusted quickly, while the others were confused and made many errors. Judd's *generalisation theory* of transfer holds that what transfers is a general principle or rule that the learner has understood and can apply to new situations. Good teaching, on this view, emphasises principles rather than isolated facts.

The Gestalt psychologists offered a third view through the phenomenon of **transposition**: the transfer of a learned relationship to new stimuli. In 1918 Köhler trained chickens to peck grain from the darker of two grey papers and not from the lighter. He then tested them with the darker grey and a new, still darker one. Most of the chickens pecked at the new paper, the one they had never been rewarded on, because it was now "the darker one". They had learned

a relation, not a response to a particular grey. Spence's 1937 S-R reply showed that overlapping generalisation gradients of excitation and inhibition around the two training stimuli could produce the same choice, and the argument between relational and S-R accounts of transposition ran for decades.

Finally, **Charles Egerton Osgood** (1916 to 1991) drew the findings together in 1949. Whether transfer is positive or negative, he showed, depends on the similarity of both the stimuli and the responses in the two tasks. When the stimuli are similar and the responses are the same, transfer is positive, and the more similar the stimuli the more positive it is. When the stimuli are similar but the responses must be different, especially opposite, transfer is negative, and the more similar the stimuli the worse the interference. The driver crossing from left-hand to right-hand traffic faces nearly identical stimuli (roads, junctions, cars) that demand opposite responses, the worst combination of all. Table 6.15 summarises the theories.

Table 6.15 Theories of transfer of training.

| Theory | Proposed by | Transfer occurs because | Key evidence |
|---|---|---|---|
| Formal discipline | Faculty psychology (traditional) | Exercise strengthens general mental faculties | Refuted by James (1890) and Thorndike (1901, 1924) |
| Identical elements | Thorndike (with Woodworth, 1901) | The tasks share identical elements | Area-estimation training transfers little |
| Generalisation | Judd (1908) | A general principle is understood and applied | Darts under water and refraction |
| Transposition | Köhler and the Gestalt school (1918) | A relationship is learned and applied | Chickens choose "the darker one" |
| Transfer surface | Osgood (1949) | Stimulus and response similarity jointly decide direction | Similar stimuli with new responses give negative transfer |

EXAM TRAP

**Identical elements** is Thorndike's theory (with Woodworth, 1901); **generalisation** of principles is Judd's (1908); **transposition** is the Gestalt account (Köhler's chickens, 1918); **formal discipline** is the faculty-psychology doctrine that Latin and geometry "train the mind", refuted by James and Thorndike. **Negative transfer** is old learning hindering new learning; in memory research the matching term for old learning disrupting the recall of new material is **proactive interference** (Chapter 7). A child who stops fearing one cat but still fears another is showing discrimination, not negative transfer (Section 6.3). Bilateral transfer is transfer from one side of the body to the other.

## Programmed instruction and teaching machines

If learning depends on active responding, immediate feedback and practice at the learner's own pace, then an ordinary classroom, with forty pupils listening to one teacher and homework returned a week later, breaks almost every rule. Two psychologists tried to fix this with machines.

The first was **Sidney Leavitt Pressey** (1888 to 1979) of Ohio State University. In the mid-1920s he built a small device, described in 1926, that presented a multiple-choice question in a window. The student pressed one of four keys. In its teaching mode the machine would not move on to the next question until the right key had been pressed, so that every error was correct-

ed at once, and it kept count of errors; it could even be fitted with an attachment that dropped a sweet when the student reached a set score. Pressey predicted an "industrial revolution" in education. It did not come, and his **teaching machine**, a device that presents material, requires a response and gives immediate knowledge of results, was largely forgotten.

The second was Skinner. On 11 November 1953 he visited his younger daughter's fourth-grade arithmetic class on a parents' day and was appalled: every child was being taught at the same pace, whatever his understanding, and no child learned whether an answer was right until long afterwards. From an operant point of view, the reinforcement was too rare and too delayed to shape anything. Within days he had built a prototype machine, and in "The science of learning and the art of teaching" (1954) and "Teaching machines" (1958) he set out the method that came to be called **programmed instruction**. Its principles are those of shaping, applied to knowledge.

- *Small steps*. The material is broken into a long sequence of small units called frames, each a little harder than the last.
- *Active responding*. The learner must respond to every frame, usually by writing a word or completing a sentence.
- *Immediate knowledge of results*. The correct answer is revealed at once, so that each right answer is reinforced and each wrong one corrected before it can be practised.
- *Self-pacing*. Each learner moves through the programme at his own speed.
- *A low error rate*. The steps are made so small, with prompts that are gradually faded, that the learner almost always succeeds; Skinner believed that errors are themselves practised and should be avoided.

Skinner's design is the **linear programme**: every learner works through the same fixed sequence of frames, writing his own (constructed) answers. Around 1960 **Norman Crowder** developed the alternative **branching programme** (also called intrinsic programming). Each frame presents a larger amount of material followed by a multiple-choice question, and the answer the learner chooses decides where he goes next: a correct answer leads on to new material, while each wrong answer leads to a remedial frame that explains that particular mistake before returning him to the main line. Crowder treated errors as useful diagnostic information rather than something to be avoided; his programmes were often printed as "scrambled textbooks" in which the learner's answer sent him to a different page.

Programmed instruction was a sensation in the early 1960s and then faded, partly because many programmes were poorly written and boring, with steps too small for able learners. But its principles survived and spread. Computer-assisted instruction, beginning with the PLATO system at the University of Illinois in 1960, put programmes on screens. **Fred Simmons Keller** (1899 to 1996), Skinner's friend since their graduate-student days, developed the Personalized System of Instruction in the 1960s, described in his 1968 paper "Good-bye, teacher...", in which university students work through a course in small written units at their own pace and must pass a test on each unit before moving on. The same logic underlies mastery learning and the adaptive learning applications on today's phones, which give small steps, instant feedback and spaced review. Programmed instruction works best for factual material and well-defined skills; it is less suited to teaching open-ended reasoning, discussion and creativity.

EXAM TRAP

The first teaching machine was **Pressey's** (1920s); **Skinner** popularised teaching machines and programmed instruction (1954, 1958), based on **operant conditioning** (shaping, active responding and immediate reinforcement), not on classical conditioning, insight or observational learning. **Linear** programmes (fixed sequence, constructed answers, very few errors) are Skinner's; **branching** (intrinsic) programmes (multiple choice, remedial frames for wrong answers) are **Crowder's**. The Personalized System of Instruction is **Keller's** (1968).

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Guthrie's single law of learning is the law of _____, with learning completed in one trial.
2. Getting a young horse used to a saddle by starting with a light blanket and increasing the weight very gradually is Guthrie's _____ method.
3. Hull held that reinforcement consists of _____ reduction.
4. Hull's fatigue-like inhibition, which builds up with responding and dissipates with rest, is called _____ inhibition.
5. Estes's mathematical model, in which a fraction θ of stimulus elements is sampled on each trial, is _____ theory.
6. The lowest of Gagné's eight types of learning, equivalent to classical conditioning, is _____ learning.
7. A temporary flat stretch in the middle of a learning curve, as in Bryan and Harter's telegraphers, is called a _____.
8. Krueger found that _____ per cent overlearning produced a clear gain in retention, with smaller gains beyond it.
9. Thorndike's view that transfer depends on what two tasks share is the theory of _____ elements.
10. Programmed instruction in which wrong answers lead to remedial frames is called _____ programming, developed by Crowder.

**Answers:** (1) contiguity, (2) threshold, (3) drive, (4) reactive, (5) stimulus sampling, (6) signal, (7) plateau, (8) 50, (9) identical, (10) branching (intrinsic).

## The case of Little Albert, revisited

The chapter opened with three questions about Albert, and you can now answer all of them. How could a placid baby be taught to fear a white rat in a week? Because seven pairings of a neutral stimulus (the rat) with an intense unconditioned stimulus (the clang of the steel bar) were enough to make the rat a conditioned stimulus for fear, especially since the rat appeared just before the noise and was still in view when it came, the most effective delay arrangement. Seligman's idea of preparedness adds a further reason: a furry, moving animal is just the kind of stimulus human beings are prepared to fear, which may be why Elsie Bregman failed when she tried to condition infants to fear wooden blocks. Why did the fear spread to the rabbit, the dog, the sealskin coat, the cotton wool and the Santa Claus mask,

but not to the blocks? Because of stimulus generalisation: the more a stimulus resembled the furry white rat, the more fear it produced, while the blocks shared nothing with it. And could the process be run backwards? Mary Cover Jones showed four years later, with Peter and the rabbit, that it could, by counterconditioning, and Wolpe turned her method into systematic desensitisation. A modern clinician would add two warnings. Had Albert been allowed to crawl away and feel relief, his avoidance would have been negatively reinforced, which, on Mowrer's two-factor theory, would have protected his fear from extinction. And even a fear that has been successfully extinguished can return through spontaneous recovery, reinstatement or renewal.

One question Watson and Rayner never answered is who Albert was. In 2009 **Hall P. Beck** and his colleagues, after years of detective work in birth records and census returns, identified him as Douglas Merritte, the son of a wet nurse at the Harriet Lane Home, who died of hydrocephalus in 1925 at the age of six. In 2012 **Alan J. Fridlund** and colleagues went further and argued, from the film and from Douglas's medical records, that he had been neurologically impaired from birth, which would mean that Watson's "healthy" and "stolid" baby was nothing of the kind. Then, in 2014, **Russell A. Powell**, **Nancy Digdon** and their colleagues pointed to another candidate, William Albert Barger, the son of another wet nurse at the same home, whose age, weight and name fitted Watson's records better. Barger, known in later life as Albert Martin, lived until 2007, apparently without ever knowing that he might have been the most famous baby in psychology; a relative recalled that he had always disliked dogs, though nobody can say whether that had anything to do with a white rat and a steel bar. The debate is not settled. What is settled is the lesson of the experiment itself: fears can be learned, they spread to similar things, and they can be unlearned, which is the foundation on which the behaviour therapies described in this chapter were built. No ethics committee today would allow the experiment, and its story is one of the reasons such committees exist.

## Chapter summary

6.1 What learning is, and what it is not. Learning is a relatively permanent change in behaviour, or in the potential for behaviour, that results from experience (Kimble). It differs from performance, which is what an organism actually does, and from changes caused by maturation, fatigue, drugs, injury or motivation. Fixed action patterns, triggered by sign stimuli (Tinbergen), are innate. The simplest, non-associative learning involves a single stimulus: habituation is a decrease in response to a repeated harmless stimulus, shown to be learning by its stimulus specificity, spontaneous recovery and dishabituation; sensitisation is an increase in responding after a strong stimulus. Kandel traced both to synaptic changes in *Aplysia*, and Fantz used habituation to study infant perception. Imprinting (Spalding, Lorenz) is rapid, largely irreversible, unrewarded learning confined to a critical period, peaking at 13 to 16 hours in Hess's ducklings.

6.2 Pavlov and the language of classical conditioning. Pavlov, a Nobel Prize-winning physiologist of digestion, noticed "psychic secretions" and from about 1897 studied them as conditioned reflexes; Twitmyer had seen the same thing in the knee jerk. In classical conditioning a neutral stimulus, paired repeatedly with an unconditioned stimulus (UCS) that produces an unconditioned response (UCR), becomes a conditioned stimulus (CS) that produces a conditioned response (CR). The CR resembles the UCR but is not identical. Conditioning is strongest in delay (forward) conditioning, where the CS begins first and overlaps the UCS, weaker in trace condition-

ing, weak in simultaneous conditioning and weakest, often inhibitory, in backward conditioning.

**6.3 The phenomena of classical conditioning.** Acquisition produces a negatively accelerated rise in the CR; extinction, the omission of the UCS, weakens it. Spontaneous recovery, disinhibition, rapid reacquisition, reinstatement (Rescorla and Heth) and renewal (Bouton) show that extinction suppresses rather than erases learning, which is why fears relapse. Stimulus generalisation produces a generalisation gradient (Hovland); stimulus discrimination is learned by differential conditioning, and an impossible discrimination produced experimental neurosis in Shenger-Krestovnikova's dog. In higher-order conditioning an established CS acts as the UCS for a new stimulus; in sensory preconditioning (Brogden) two neutral stimuli are paired before one of them is conditioned. A conditioned inhibitor signals that the UCS will not occur.

**6.4 Fear, prediction and preparedness.** Watson and Rayner's Little Albert (1920) showed that fear can be classically conditioned and generalised; Mary Cover Jones's Peter (1924) showed that it can be removed by counterconditioning, the principle that Wolpe developed into reciprocal inhibition and systematic desensitisation (relaxation from Jacobson, a hierarchy rated in SUDS, then graded exposure). Flooding, implosion (Stampfl), aversion therapy, covert sensitisation (Cautela) and exposure and response prevention (Meyer) also rest on conditioning. Rescorla's contingency, Kamin's blocking, overshadowing and Lubow and Moore's latent inhibition showed that a CS must predict the UCS; the Rescorla-Wagner model (1972) formalised learning as driven by prediction error, which dopamine neurons signal (Schultz). The CS acts as a signal rather than a substitute (Siegel's compensatory responses). Garcia and Koelling's taste aversion (1966), learned in one trial and across long delays, refuted equipotentiality and led to Seligman's preparedness (1970), with applications in chemotherapy, wildlife management and advertising.

**6.5 Thorndike, Skinner and the consequences of behaviour.** Thorndike's cats in puzzle boxes learned by trial and error, and his law of effect (1911) states that satisfying consequences strengthen responses; his connectionism also included the laws of exercise and readiness, and he later abandoned the law of exercise. Skinner distinguished elicited respondent behaviour from emitted operant behaviour, studied it in the Skinner box with the cumulative recorder, and described the three-term contingency of discriminative stimulus, response and consequence. Reinforcement increases behaviour and punishment decreases it; "positive" adds a stimulus and "negative" removes one, so negative reinforcement is not punishment. Secondary and generalised reinforcers (Wolfe's chimpanzees) underlie the token economy (Ayllon and Azrin). Escape terminates and avoidance prevents an aversive stimulus, and Mowrer's two-factor theory explains avoidance and compulsions. Punishment (including time-out and response cost) suppresses rather than teaches and has side effects (Estes). Shaping reinforces successive approximations; chaining, prompting and fading build complex behaviour; extinction produces an extinction burst and a return to the operant level.

**6.6 Schedules of reinforcement and other operant phenomena.** Continuous reinforcement teaches fastest and extinguishes fastest; partial reinforcement produces the partial reinforcement extinction effect (Humphreys). Fixed-ratio schedules give high rates with post-reinforcement pauses, variable-ratio schedules the highest, steadiest rates and greatest resistance to extinction, fixed-interval schedules a scallop, and variable-interval schedules moderate, steady rates (Ferster and Skinner). Differential reinforcement (DRO, DRI, DRA, DRL, DRH) weakens problem behaviour without punishment. Premack showed that a more probable behaviour reinforces a less probable one. Accidental reinforcement produces superstitious behaviour (Skin-

ner, 1948). Uncontrollable shock produces learned helplessness (Overmier, Seligman and Maier, 1967), a model of depression later reformulated in attributional terms. The Brelands' instinctive drift and Brown and Jenkins's autoshaping show biological limits on operant learning; Herrnstein's matching law describes choice; Crespi's contrast effects show that reward size affects motivation and performance; and biofeedback (Miller, Kegel) brings bodily processes under operant control. Classical and operant conditioning differ in what is associated and in the kind of behaviour involved, but constantly interact.

**6.7 Cognitive learning.** Tolman's purposive behaviourism held that animals learn expectancies and cognitive maps; in Tolman and Honzik's latent learning experiment (1930) rats rewarded only from day 11 immediately matched always-rewarded rats, showing that reinforcement affects performance rather than learning, and place learning and Tinklepaugh's monkey supported the idea of expectancy. Köhler's chimpanzees, Sultan above all, solved stick and box problems by insight, a sudden perceptual reorganisation accompanied by the "aha" experience, though Birch showed that insight depends on earlier experience. Harlow's monkeys formed learning sets, learning to learn, which reconcile trial-and-error and insight. Concepts are formed in the stages of observation, discrimination, generalisation and abstraction.

**6.8 Observational learning.** Miller and Dollard (1941) treated imitation as reinforced behaviour. Bandura's Bobo doll studies (1961, 1963, 1965) showed that children learn new aggressive acts simply by watching a live or filmed model, and that the model's consequences affect performance rather than learning. Observational learning requires attention, retention, motor reproduction and motivation; reinforcement, direct or vicarious, governs only the last. Bandura's social cognitive theory added reciprocal determinism (person, behaviour and environment influencing one another), self-regulation and self-efficacy (1977), with four sources led by performance attainments. Rotter's separate social learning theory explains behaviour potential through expectancy and reinforcement value.

**6.9 Other learning theorists.** The grand theories, surveyed in Hilgard's *Theories of Learning* (1948), divided over whether learning consists of S-R connections or knowledge and whether it requires contiguity or reinforcement. Guthrie's contiguity theory held that learning is complete in one trial and that practice attaches acts to more stimuli; he proposed the threshold, fatigue and incompatible-response methods of breaking habits, and his cat studies with Horton were later reinterpreted as greeting behaviour. Hull's hypothetico-deductive theory made reinforcement equal to drive reduction and reaction potential equal to habit strength multiplied by drive, later with incentive motivation, stimulus-intensity dynamism and reactive and conditioned inhibition; his goal gradient, fractional anticipatory goal response and habit-family hierarchy were S-R answers to Tolman, and Spence extended the system. Estes's stimulus sampling theory (1950) gave learning a statistical form and predicted probability matching. Mowrer developed the bell-and-pad method, two-factor theory with its sign and solution learning, the neurotic paradox, and later a theory of fear, relief, hope and disappointment; Herrnstein and Hineline and Bolles challenged two-factor theory. Gagné's *The Conditions of Learning* (1965) arranged eight types of learning in a hierarchy from signal learning to problem solving. Constructivism (Piaget, Vygotsky, Bruner) sees learners as active builders of knowledge.

**6.10 Learning in practice.** Learning curves may be negatively accelerated (the commonest), positively accelerated or S-shaped, and group averages can hide sudden individual jumps (Estes). Bryan and Harter's telegraphers showed a plateau, explained by a hierarchy of habits; a plateau is temporary, unlike the physiological limit. Skills pass through cognitive, associative and au-

tonomous stages (Fitts and Posner). Overlearning improves retention (Krueger, 1929). Distributed practice usually beats massed practice (Ebbinghaus, Jost, Baddeley and Longman), which Hull explained by reactive inhibition; reminiscence (Ballard) is improvement after rest. Whole learning suits meaningful, unified material and part learning long, difficult material. Knowledge of results is essential (Trowbridge and Cason), and specific, difficult goals improve performance (Locke and Latham). Transfer of training may be positive, negative or zero; formal discipline was refuted by James and Thorndike, and transfer has been explained by identical elements (Thorndike), generalisation of principles (Judd), transposition (Köhler) and the similarity of stimuli and responses (Osgood). Programmed instruction, from Pressey's teaching machine to Skinner's linear and Crowder's branching programmes and Keller's Personalized System of Instruction, applies shaping and immediate feedback to teaching.

## Who's who

Table 6.16 The people in this chapter.

| Name | When | Examined for |
|---|---|---|
| Lyn Abramson | 1978 | Attributional reformulation of learned helplessness (with Seligman and Teasdale) |
| James Allison | 1974 | Response-deprivation hypothesis (with Timberlake) |
| Teodoro Ayllon | 1960s; book 1968 | Token economy at Anna State Hospital (with Azrin); stimulus satiation (1963) |
| Nathan Azrin | 1960s; book 1968 | Token economy (with Ayllon); conditions for effective punishment |
| Alan Baddeley | 1978 | Spaced beats massed practice: postal workers learning a keyboard (with Longman) |
| Philip Boswood Ballard | 1913 | Reminiscence: better recall after a rest |
| Albert Bandura | 1925 to 2021 | Bobo doll studies (1961, 1963, 1965); four processes; self-efficacy (1977); reciprocal determinism |
| Hall P. Beck | 2009 | Identified Little Albert as Douglas Merritte (with colleagues) |
| Ilene Bernstein | 1978 | Taste aversions in children receiving chemotherapy |
| Herbert Birch | 1945 | Insight depends on earlier experience with sticks |
| Robert A. Bjork | 1990s onwards | "Desirable difficulties": why spaced, effortful practice strengthens memory |
| Robert Bolles | 1970 | Species-specific defence reactions in avoidance learning |
| Mark Bouton | 1979 onwards | Renewal; extinction as new, context-dependent learning |
| Elsie Bregman | 1934 | Failed to condition infants' fear of blocks and cloth (preparedness) |
| Keller Breland | 1961 | Instinctive drift, "The misbehavior of organisms" (with Marian Breland) |
| Marian Breland | 1961 | Instinctive drift (with Keller Breland) |
| W. J. Brogden | 1939 | Sensory preconditioning |
| Paul Brown | 1968 | Autoshaping in pigeons (with Jenkins) |
| William Lowe Bryan | 1860 to 1955 | Telegraphy learning curves and the plateau (with Harter, 1897, 1899) |
| Karl Bühler | 1879 to 1963 | The "aha" experience (Aha-Erlebnis) |

| Name | When | Examined for |
|---|---|---|
| Robert R. Bush | 1950s | Linear models of learning (with Mosteller) |
| Hulsey Cason | 1932 | Precise knowledge of results in line drawing (with Trowbridge) |
| Joseph Cautela | 1967 | Covert sensitisation |
| Nicholas J. Cepeda | 2006 | Review of several hundred experiments on the spacing effect |
| Leo Crespi | 1942 | Contrast effects (the Crespi effect) |
| Norman Crowder | About 1960 | Branching (intrinsic) programmed instruction |
| Leo DiCara | Late 1960s | Heart-rate conditioning in curarised rats (with Miller) |
| Nancy Digdon | 2014 | Proposed William Albert Barger as Little Albert (with Powell) |
| John Dollard | 1900 to 1980 | Imitation as reinforced behaviour (1941); early social learning theory (1950, with Miller) |
| Knight Dunlap | 1932 | Negative practice for breaking habits |
| William Kaye Estes | 1919 to 2011 | Conditioned suppression (1941); punishment suppresses (1944); stimulus sampling theory (1950) |
| Robert Fantz | 1964 | Habituation-dishabituation method in infants |
| Grace Fernald | 1943 | Multisensory (VAKT) teaching for learning difficulties |
| Charles Ferster | 1957 | *Schedules of Reinforcement* (with Skinner) |
| Paul Fitts | 1912 to 1965 | Cognitive, associative and autonomous stages of skill learning (with Posner, 1967) |
| Alan J. Fridlund | 2012 | Argued that Little Albert was neurologically impaired |
| Robert Mills Gagné | 1916 to 2002 | Eight types of learning; *The Conditions of Learning* (1965); nine events of instruction |
| John Garcia | 1917 to 2012 | Taste aversion; Garcia and Koelling (1966); equipotentiality refuted |
| Gerald Gorn | 1982 | Classical conditioning in advertising (music and pen colour) |
| Philip Groves | 1970 | Dual-process theory of habituation and sensitisation (with Thompson) |
| Edwin Ray Guthrie | 1886 to 1959 | Contiguity; one-trial learning; threshold, fatigue and incompatible-response methods |
| Harry Harlow | 1905 to 1981 | Learning sets (1949); contact comfort (1958) |
| Noble Harter | 1897, 1899 | The plateau in telegraphy learning (with Bryan) |
| Richard Herrnstein | 1930 to 1994 | Matching law (1961); avoidance as shock-frequency reduction (1966) |
| Eckhard Hess | 1950s | Imprinting window peaking at 13 to 16 hours in mallard ducklings |
| Ernest Ropiequet Hilgard | 1904 to 2001 | *Theories of Learning* (1948); hypnosis research |
| Philip Hineline | 1966 | Unsignalled avoidance learning (with Herrnstein) |
| Donald Hiroto | 1974 | Learned helplessness in humans (inescapable noise) |
| Charles Honzik | 1930 | Latent learning experiment (with Tolman) |
| George P. Horton | 1946 | *Cats in a Puzzle Box* (with Guthrie) |
| Carl Hovland | 1912 to 1961 | Generalisation gradient in humans (1937) |
| Clark Leonard Hull | 1884 to 1952 | Drive reduction; habit strength; reaction potential = habit × drive; *Principles of Behavior* (1943) |

| Name | When | Examined for |
|---|---|---|
| Lloyd Humphreys | 1939 | Partial reinforcement extinction effect in human eyeblink conditioning |
| Edmund Jacobson | 1888 to 1983 | Progressive muscle relaxation |
| William James | 1842 to 1910 | Memory-training self-experiment against formal discipline (1890) |
| Herbert Jenkins | 1968 | Autoshaping (with Brown) |
| Mary Cover Jones | 1897 to 1987 | Peter and the rabbit (1924); counterconditioning |
| Adolf Jost | 1897 | Jost's law: repetition benefits an older association more |
| Charles Hubbard Judd | 1873 to 1946 | Generalisation theory of transfer: darts under water (1908) |
| Leon Kamin | 1927 to 2017 | Blocking (1969) |
| Eric Kandel | Born 1929 | Habituation and sensitisation in *Aplysia*; Nobel Prize 2000 |
| Arnold Kegel | 1948 | Perineometer, an early form of biofeedback |
| Fred Simmons Keller | 1899 to 1996 | "The phantom plateau" (1958); Personalized System of Instruction (1968) |
| Gregory Kimble | 1961 | The standard definition of learning |
| Ran Kivetz | 2006 | Goal gradient in coffee-shop loyalty cards |
| Robert Koelling | 1966 | "Bright, noisy, tasty water" (with Garcia) |
| Wolfgang Köhler | 1887 to 1967 | Insight in chimpanzees (Sultan); *The Mentality of Apes*; transposition in chickens (1918) |
| William C. F. Krueger | 1929 | Overlearning: 50 per cent overlearning improves retention |
| Gary Latham | 1970s onwards | Goal setting (with Locke) |
| Joseph LeDoux | 1990s | Amygdala circuits of fear conditioning |
| Edwin Locke | Born 1938 | Goal setting: specific, difficult goals improve performance (1968) |
| Konrad Lorenz | 1903 to 1989 | Imprinting in greylag geese (1935); Nobel Prize 1973 |
| Robert Lubow | 1959 | Latent inhibition (with Moore) |
| Steven Maier | 1967 | Learned helplessness (with Seligman); neuroscientific revision (2016) |
| Victor Meyer | 1966 | Exposure and response prevention for obsessive-compulsive disorder |
| Neal Miller | 1909 to 2002 | Imitation (with Dollard, 1941); visceral learning and biofeedback |
| A. U. Moore | 1959 | Latent inhibition (with Lubow) |
| Bruce Moore | 1979 | Showed that Guthrie's cats were greeting, not learning (with Stuttard) |
| Frederick Mosteller | 1950s | Linear models of learning (with Bush) |
| Orval Hobart Mowrer | 1907 to 1982 | Two-factor theory (1947); bell-and-pad (1938); neurotic paradox (1948); sign and solution learning |
| Willie Mae Mowrer | 1938 | Bell-and-pad treatment of enuresis (with O. H. Mowrer) |
| Charles Egerton Osgood | 1916 to 1991 | Transfer surface: stimulus and response similarity (1949) |
| J. Bruce Overmier | 1967 | First report of learned helplessness (with Seligman) |
| Ivan Petrovich Pavlov | 1849 to 1936 | Classical conditioning; Nobel Prize 1904; *Conditioned Reflexes* (English 1927) |
| Michael Posner | Born 1936 | Stages of skill learning (with Fitts, 1967) |

| Name | When | Examined for |
|---|---|---|
| Russell A. Powell | 2014 | Proposed William Albert Barger as Little Albert (with Digdon) |
| David Premack | 1925 to 2015 | Premack principle (1959); theory of mind (1978) |
| Sidney Leavitt Pressey | 1888 to 1979 | The first teaching machine (1920s) |
| Rosalie Rayner | 1920 | Little Albert (with Watson) |
| Robert Rescorla | 1940 to 2020 | Contingency (1968); Rescorla-Wagner model (1972); reinstatement (1975) |
| Dorothea Ross | 1961 | Bobo doll studies (with Bandura and Sheila Ross) |
| Sheila Ross | 1961 | Bobo doll studies (with Bandura and Dorothea Ross) |
| Julian Rotter | 1916 to 2014 | Social learning theory (1954): behaviour potential, expectancy, reinforcement value; locus of control (1966) |
| Wolfram Schultz | 1990s | Dopamine neurons signal reward prediction error |
| Martin Seligman | Born 1942 | Learned helplessness (1967); preparedness (1970) |
| Shenger-Krestovnikova | 1921 | Experimental neurosis (circle and ellipse) |
| Charles Scott Sherrington | 1857 to 1952 | Coined "reciprocal inhibition", the term Wolpe borrowed |
| Shepard Siegel | 1970s and 1980s | Conditioned compensatory responses, drug tolerance and overdose |
| Virginia Simmelhag | 1971 | Reinterpreted Skinner's superstition (with Staddon) |
| Burrhus Frederic Skinner | 1904 to 1990 | Operant conditioning; Skinner box; schedules; superstition (1948); teaching machines (1954, 1958) |
| Richard Solomon | 1953 | Persistence of avoidance responding (with Wynne) |
| Douglas Spalding | 1873 | First description of imprinting (chicks) |
| Janet Taylor Spence | 1923 to 2015 | Taylor Manifest Anxiety Scale (1953); anxiety as drive |
| Kenneth Wartinbee Spence | 1907 to 1967 | Hull-Spence theory; incentive motivation; transposition (1937) |
| John Staddon | 1971 | Reinterpreted Skinner's superstition (with Simmelhag) |
| Thomas Stampfl | 1960s | Implosive therapy |
| Susan Stuttard | 1979 | Reinterpretation of Guthrie's cats (with Bruce Moore) |
| John Teasdale | 1978 | Attributional reformulation of learned helplessness |
| Richard F. Thompson | 1970 | Dual-process theory of habituation (with Groves) |
| Edward Lee Thorndike | 1874 to 1949 | Puzzle box; law of effect (1898, 1911); connectionism; identical elements theory |
| William Timberlake | 1974 | Response-deprivation hypothesis (with Allison) |
| Nikolaas (Niko) Tinbergen | 1907 to 1988 | Fixed action patterns and sign stimuli (stickleback); Nobel Prize 1973 |
| Otto Tinklepaugh | 1928 | Expectancy: the monkey that rejected lettuce in place of a banana |
| Edward Chace Tolman | 1886 to 1959 | Purposive behaviourism (1932); latent learning (1930); cognitive maps (1948) |
| Margery Trowbridge | 1932 | Precise knowledge of results works best (with Cason) |
| Edwin Burket Twitmyer | 1902 | Independent discovery of the conditioned knee jerk |
| Allan Wagner | 1934 to 2018 | Rescorla-Wagner model (1972) |

| Name | When | Examined for |
|---|---|---|
| Richard Walters | 1963 | Later social learning theory (with Bandura) |
| John Broadus Watson | 1878 to 1958 | Behaviourism (1913); Little Albert (1920); laws of frequency and recency |
| Carl Williams | 1959 | Extinction of a toddler's bedtime tantrums |
| John B. Wolfe | 1936 | Token rewards for chimpanzees (the "Chimp-O-Mat") |
| Joseph Wolpe | 1915 to 1997 | Reciprocal inhibition; systematic desensitisation (1958); SUDS |
| Guy Woodruff | 1978 | Theory of mind in chimpanzees (with Premack) |
| Robert Sessions Woodworth | 1869 to 1962 | S-O-R formula; transfer studies with Thorndike (1901) |
| Lyman Wynne | 1953 | Persistence of avoidance responding (with Solomon) |
| David Zeaman | 1949 | Named positive and negative contrast |

## Key terms

learning, performance, maturation, fixed action pattern, sign stimulus, non-associative learning, habituation, dishabituation, sensitisation, imprinting, critical period, associative learning, classical conditioning, unconditioned stimulus, unconditioned response, neutral stimulus, orienting response, conditioned stimulus, conditioned response, delay conditioning, trace conditioning, simultaneous conditioning, backward conditioning, acquisition, extinction, spontaneous recovery, disinhibition, reinstatement, renewal, stimulus generalisation, generalisation gradient, stimulus discrimination, experimental neurosis, higher-order conditioning, sensory preconditioning, conditioned inhibitor, conditioned emotional response, counterconditioning, reciprocal inhibition, systematic desensitisation, flooding, contingency, blocking, overshadowing, latent inhibition, Rescorla-Wagner model, prediction error, stimulus substitution theory, equipotentiality, conditioned taste aversion, preparedness, trial-and-error learning, law of effect, connectionism, law of exercise, law of readiness, respondent behaviour, operant behaviour, operant conditioning, Skinner box, cumulative recorder, instrumental conditioning, discriminative stimulus, three-term contingency, reinforcement, punishment, positive reinforcement, negative reinforcement, positive punishment, negative punishment, primary reinforcer, secondary reinforcer, generalised reinforcer, token economy, escape learning, avoidance learning, two-factor theory, time-out, response cost, shaping, chaining, prompt, fading, extinction burst, operant level, schedule of reinforcement, continuous reinforcement, partial reinforcement, partial reinforcement extinction effect, fixed-ratio schedule, post-reinforcement pause, variable-ratio schedule, fixed-interval schedule, variable-interval schedule, differential reinforcement, Premack principle, superstitious behaviour, learned helplessness, instinctive drift, matching law, contrast effects, biofeedback, latent learning, cognitive map, place learning, insight learning, aha experience, learning set, concept, concept formation, observational learning, vicarious reinforcement, reciprocal determinism, self-regulation, self-efficacy, stimulus-response theories, contiguity, one-trial learning, movement-produced stimuli, threshold method, fatigue method, incompatible response method, drive-reduction theory, habit strength, incentive motivation, reactive inhibition, goal gradient, fractional anticipatory goal response, habit-family hierarchy, stimulus sampling theory, probability matching, bell-and-pad method, sign learning, solution learning, neurotic paradox, species-specific defence reactions, constructivism, learning curve, negatively accelerated curve, positively accelerated curve, S-shaped curve, plateau, physiological limit, overlearning, massed practice, distributed practice, reminiscence, whole method, part method, progressive part method, knowledge of re-

sults, transfer of training, positive transfer, negative transfer, zero transfer, bilateral transfer, formal discipline, identical elements theory, transposition, teaching machine, programmed instruction, linear programme, branching programme

## From the past papers

1. The phenomenon that is nearly opposite of habituation is termed: (RML 2024)
   (a) Sensitisation (b) Stabilization (c) Inhabitation (d) Habitation

   **Answer: (a) Sensitisation.** Habituation is a decrease in responding to a repeated, harmless stimulus; sensitisation is an increase in responding, usually to many stimuli, after a strong or noxious one (Section 6.1). The other three are not learning terms; "inhabitation" and "habitation" are sound-alike distractors.

2. In Pavlov's experiment with dogs, the sound of the bell/metronome was the: (RML 2020)
   (a) operant stimulus after food was associated (b) neutral stimulus before the food was associated and conditioned stimulus after food was associated (c) unconditioned stimulus before the food was associated (d) conditioned stimulus before the food was associated and neutral stimulus after food was associated

   **Answer: (b).** Before pairing, the metronome does not produce salivation, so it is a neutral stimulus; after pairing with food, it is a conditioned stimulus. Food, not the bell, is the unconditioned stimulus; "operant stimulus" is not a term of classical conditioning; and (d) reverses the order.

3. Which of the following is most effective in conditioning? (Amity 2023)
   (a) Backward Conditioning (b) Delayed Conditioning (c) Simultaneous Conditioning (d) Trace Conditioning

   **Answer: (b) Delayed (delay) conditioning.** The CS begins first and is still present when the UCS arrives, so it serves as a warning. Trace conditioning leaves a gap and is weaker; simultaneous conditioning gives no advance warning; backward conditioning is the least effective and often makes the CS a signal of safety.

4. A child has learned to avoid a furry, black cat. However, she still plays with her grandmother's short-haired tabby. Her response demonstrates: (RML 2024)
   (a) Discrimination (b) Extinction (c) Successive approximation (d) Negative transfer

   **Answer: (a) Discrimination.** She responds differently to similar stimuli. Her fear of the black cat has not gone, so it is not extinction; successive approximation is shaping; negative transfer is earlier learning hindering new learning (Section 6.10).

5. In reinforcement learning, the "prediction error" signal in the brain is primarily associated with: (JEMAS 2025)
   (a) Serotonin release (b) Dopamine neurons in the ventral tegmental area (c) GABA inhibition (d) Acetylcholine release

   **Answer: (b).** Schultz found that dopamine neurons of the ventral tegmental area and substantia nigra fire to an unexpected reward, shift their firing to a cue that predicts it, and dip when a predicted reward fails to arrive: the prediction error of the Rescorla-Wagner model (Section 6.4). Serotonin, GABA and acetylcholine do not carry this signal.

6. Taste aversion is an example of: (RML 2022)
   (a) Classical Conditioning (b) Second-order conditioning (c) Operant conditioning (d) Spurious association
   **Answer: (a) Classical conditioning.** The taste is the CS and the illness the UCS, paired once, often hours apart. It is not second-order conditioning (no established CS is used to condition a new one), not operant (no response is reinforced) and not a spurious association, which is a statistical term.

7. Systematic desensitization technique is based upon the idea of: (IHBAS 2025)
   (a) Reciprocal inhibition (b) Law of effect (c) Habituation (d) Adaptation to stress
   **Answer: (a) Reciprocal inhibition.** Wolpe paired each step of an anxiety hierarchy with relaxation, a response incompatible with anxiety (Section 6.4). The law of effect is Thorndike's operant principle; habituation underlies exposure and response prevention; adaptation to stress belongs to stress theory (Chapter 11).

8. Exposure and response prevention procedure is based on behaviour therapy principal of: (IHBAS 2024)
   (a) Counter conditioning (b) Habituation and extinction (c) Reciprocal inhibition (d) Graded exposure
   **Answer: (b) Habituation and extinction.** The client meets the obsessional cue and is prevented from performing the compulsion, so the anxiety rises, peaks and falls without the escape response. No incompatible response is trained, so it is neither counterconditioning nor reciprocal inhibition (those describe systematic desensitisation), and graded exposure is a way of presenting stimuli, not the learning principle behind the change.

9. The "law of effect," formulated by Edward Thorndike, states that: (Real PYQ 2024)
   (a) Behaviors followed by pleasurable consequences are more likely to be repeated. (b) Organisms learn by observing and imitating models. (c) Insight occurs when the elements of a problem are suddenly restructured. (d) All behavior is motivated by unconscious drives.
   **Answer: (a).** Responses followed by satisfaction are strengthened (Section 6.5). Option (b) is Bandura's observational learning, (c) is Köhler's insight and (d) is Freud.

10. The behavioral technique that involves removing a desirable stimulus following an undesirable behavior in order to reduce the frequency of that behavior is called: (Real PYQ 2025)
    (a) Negative reinforcement (b) Response cost (c) Positive punishment (d) Extinction
    **Answer: (b) Response cost,** a form of negative punishment. Negative reinforcement removes something to *increase* behaviour; positive punishment *adds* something unpleasant; extinction simply stops delivering the reinforcer that maintained the behaviour.

11. The first-grade teacher gives students stickers when they perform well. If they earn five stickers in one day they are exempt from homework. The stickers in this example could also be called: (RML 2024)
    (a) Primary reinforcer (b) Token (c) Generalized reinforcer (d) Prepotent responses
    **Answer: (b) Token.** Stickers exchanged for a backup reinforcer are tokens, which are secondary reinforcers, not primary ones. A generalised reinforcer buys many different backups (money), whereas these buy one; a prepotent response is a dominant response, not a reinforcer.

12. A parent deliberately ignores a child's temper tantrums in an attempt to discourage them. The parent's strategy is in accordance with: (Amity 2023)

    (a) Negative reinforcement (b) Extinction (c) Operant escape (d) Operant avoidance

    **Answer: (b) Extinction.** The source key marks (c), but that is wrong: ignoring withholds the attention that reinforced the tantrums, which is extinction (Williams, 1959; Section 6.5). Escape terminates, and avoidance prevents, an aversive stimulus; negative reinforcement would increase behaviour.

13. A client with OCD washes their hands excessively to neutralize the anxiety caused by intrusive thoughts of contamination. This hand-washing behavior is BEST conceptualized as a(n): (Real PYQ 2024)

    (a) Conditioned stimulus (b) Unconditioned response (c) Avoidance response / Safety behavior (d) Operant behavior maintained by positive reinforcement

    **Answer: (c).** Washing reduces anxiety and is therefore negatively reinforced, the second factor of Mowrer's theory (Section 6.5). It is a learned response, not a stimulus and not an unconditioned response, and it is maintained by the removal of anxiety, not by positive reinforcement.

14. Sheena is studying for her final exam in her psychology class. For every 45 minutes that she studies, she gives herself a mini chocolate, then goes back to studying. What Schedule of reinforcement is Sheena using? (RML 2021)

    (a) Variable ratio schedule (b) Variable interval schedule (c) Fixed interval schedule (d) Fixed ratio schedule

    **Answer: (c) Fixed interval.** The requirement is a constant amount of time, not a number of responses (ratio) and not an unpredictable period (variable). Strictly, reward after a fixed time regardless of responding is a fixed-time arrangement, but among these options fixed interval is correct (Section 6.6).

15. Among the following schedules of reinforcement, which typically produces the highest rate of response and is most resistant to extinction? (Real PYQ 2024)

    (a) Fixed Ratio (b) Variable Ratio (c) Fixed Interval (d) Variable Interval

    **Answer: (b) Variable ratio.** Ratio schedules reward fast responding, and unpredictability removes any reason to pause and makes it hard to detect that reinforcement has stopped, as with slot machines. Fixed ratio produces pauses after reinforcement, fixed interval a scallop and variable interval a moderate rate.

16. According to the Premack Principle, a high-probability behavior (for example, playing video games) can be used to: (Real PYQ 2024)

    (a) Reinforce a low-probability behavior (for example, completing homework). (b) Punish a low-probability behavior. (c) Reinforce another high-probability behavior. (d) Extinguish a maladaptive behavior.

    **Answer: (a).** A more preferred activity, made available only after a less preferred one, reinforces the less preferred one: grandma's rule (Section 6.6). The principle concerns reinforcement, not punishment or extinction, and runs from high to low probability, not between two high-probability activities.

17. Learned helplessness phenomenon is offered by: (JEMAS 2025)

    (a) Aaron T. Beck (b) Martin Seligman and Steven F. Maier (c) Harry Stack Sullivan (d) Daniel Kahneman

    **Answer: (b).** Seligman and Maier's yoked dogs, given inescapable shock, later failed to escape avoidable shock (Section 6.6). Beck is known for the cognitive triad of depression, Sullivan for interpersonal theory and Kahneman for heuristics and biases.

18. The Crespi effect in learning experiments refers to: (IHBAS 2025)

    (a) Loss of previously learned behavior due to retroactive inhibition (b) Increase in motivation when the magnitude of reinforcement increases (c) Decrease in response strength after extinction training (d) Sudden improvement in performance due to insight

    **Answer: (b).** Crespi's rats shifted from a small to a larger reward ran faster than rats always given the larger reward (positive contrast), and the reverse shift produced negative contrast; the changes were too abrupt to be learning, so they reflect motivation (Section 6.6). Option (a) is memory interference, (c) is extinction and (d) is Köhler's insight.

19. The technique that is based on the idea that the autonomic nervous system can come under voluntary control through operant conditioning is known as: (JEMAS 2025)

    (a) Biofeedback (b) Hypnosis (c) Sublimation (d) Systematic desensitization

    **Answer: (a) Biofeedback.** A bodily process is measured and displayed so that changes in the desired direction are reinforced by the feedback (Miller; Section 6.6). Hypnosis is an altered state (Chapter 5), sublimation a Freudian defence mechanism, and systematic desensitisation a classical counterconditioning technique.

20. What is the main difference between classical and operant conditioning? (IHBAS 2024)

    (a) Classical conditioning relies on reinforcement; operant conditioning does not (b) Classical conditioning pairs two stimuli; operant conditioning relies on reinforcement and punishment (c) Operant conditioning involves cognitive processes; classical conditioning does not (d) Both are entirely different learning processes

    **Answer: (b).** Classical conditioning associates two stimuli; operant conditioning associates a response with its consequences (Table 6.9). Option (a) is backwards, (c) is false because classical conditioning depends on prediction (Rescorla), and (d) ignores how constantly the two interact (two-factor theory, autoshaping).

21. Latent learning was demonstrated by: (IHBAS 2025)

    (a) Tolman (b) Bandura (c) Skinner (d) Kohler

    **Answer: (a) Tolman,** with Honzik (1930): rats that found food only from day 11 immediately performed as well as rats rewarded throughout (Section 6.7). Bandura is observational learning, Skinner operant conditioning and Köhler insight.

22. Concept formation stages include the following: (JEMAS 2025)

    (a) Observation, Discrimination, Generalization, Abstraction (b) Abstraction, Generalisation, Discrimination, Assimilation (c) Observation, Generalisation, Discrimination, Abstraction (d) Observation, Assimilation, Generalisation, Discrimination

    **Answer: (a).** The learner observes instances, discriminates them from non-instances, generalises to all instances sharing the essential features, and abstracts the defining properties (Section 6.7). Option (c) puts generalisation before discrimination, and assimilation, in (b) and (d), is Piaget's term, not a stage of concept formation.

23. According to Albert Bandura, which of the following is NOT a necessary component process for observational learning to occur? (Real PYQ 2024)

    (a) Attention (b) Motivation (c) Reinforcement (d) Reproduction

    **Answer: (c) Reinforcement.** The four processes are attention, retention, motor reproduction and motivation; the 1965 Bobo doll study showed that children learn a model's behaviour whether or not anyone is reinforced, and reinforcement affects only whether the behaviour is performed (Section 6.8).

24. Reciprocal causation model has been described under: (IHBAS 2020)

    (a) Social cognitive theory (b) Two-factor theory (c) Drive reduction theory (d) Constructivist theory

    **Answer: (a) Social cognitive theory.** Bandura's triadic reciprocal causation (reciprocal determinism) holds that person, behaviour and environment influence one another (Section 6.8). Two-factor theory is Mowrer's account of avoidance, drive-reduction theory is Hull's, and constructivism (Piaget, Vygotsky, Bruner) concerns how learners build knowledge (Section 6.9).

25. Learning always leads to efficacy, provided presence of a significant factor which is: (JEMAS 2025)

    (a) The learner has ability. (b) The learner has aptitude. (c) The learner has knowledge. (d) The learner has a specific goal.

    **Answer: (d).** Locke and Latham showed that specific, difficult goals, together with knowledge of results, direct attention, mobilise effort and increase persistence, so that practice becomes purposeful (Section 6.10). Ability, aptitude and knowledge help, but on their own they do not turn learning into effective performance.

## High-yield sheet

*The facts this chapter's questions ask most, most-asked first. Last read before the exam.*

1 **Systematic desensitisation** (Joseph Wolpe, *Psychotherapy by Reciprocal Inhibition*, 1958) rests on **reciprocal inhibition**: relaxation, a response incompatible with anxiety, is paired step by step with a graded anxiety hierarchy. It is counterconditioning, not operant extinction, habituation or the law of effect. ASKED 17 TIMES

2 The **token economy** is **operant** conditioning: tokens are secondary (conditioned) reinforcers exchanged for backup reinforcers. It was developed on a psychiatric ward by **Teodoro Ayllon and Nathan Azrin** (*The Token Economy*, 1968), and it is the listed therapy NOT derived from classical conditioning. Stickers that earn a reward are tokens. ASKED 10 TIMES

3 **Schedules** (Ferster and Skinner, 1957): variable ratio gives the highest rate and greatest resistance to extinction; fixed ratio, high rates with a post-reinforcement pause; fixed interval, the scallop (a reward every 45 minutes of study is keyed fixed interval); variable interval, moderate steady rates; continuous reinforcement, every response reinforced, learns fastest and extinguishes fastest. ASKED 7 TIMES

4 **Extinction**: a conditioned response weakens when the CS is repeatedly presented without the UCS (classical), or a response fades when it is no longer reinforced (operant), falling back to its earlier operant level. A parent who ignores tantrums is using extinction; the Amity 2023 key "operant escape" is wrong. ASKED 7 TIMES

5 In **Pavlov's** experiment the bell or metronome is a **neutral stimulus before** pairing with food and a **conditioned stimulus after**; a neutral stimulus becomes a CS by repeated pairing with an unconditioned stimulus, and the fundamental element is **association**. Fear of the ground where you lost a match is classical conditioning. ASKED 6 TIMES

6 **Operant behaviour** is emitted and operates on the environment (Skinner), and a response made more frequent by reinforcement is operant conditioning. **Reinforcement** always increases behaviour and punishment decreases it; positive adds and negative removes, so switching off an ongoing shock is **negative reinforcement** (escape), not punishment. ASKED 6 TIMES

7 **Shaping** (Skinner) is reinforcing **successive approximations** to a target behaviour; chaining links a sequence of responses, prompting adds a cue and fading withdraws it. ASKED 5 TIMES

8 **Exposure and response prevention** (Victor Meyer, 1966) rests on **habituation and extinction**; **flooding** is prolonged full-intensity exposure without relaxation; **covert sensitisation** (Joseph Cautela, 1967) pairs an imagined behaviour with an imagined unpleasant consequence; **counterconditioning** teaches that a feared stimulus is no longer threatening. ASKED 4 TIMES

9 **Latent learning**: Edward Tolman and Charles Honzik (1930) found that rats unrewarded for ten days and then fed from day 11 almost at once matched rats rewarded throughout. Tolman called what they had learned a **cognitive map** (1948). ASKED 5 TIMES

10 **Premack principle** (David Premack, 1959): a **high-probability** (preferred) behaviour reinforces a **low-probability** one, never the reverse (grandma's rule: homework first, then video games). ASKED 5 TIMES

11 **Learned helplessness**: **Martin Seligman and Steven Maier** (1967; first reported by Overmier and Seligman, 1967) found that dogs given inescapable shock later failed to escape avoidable shock; it became a model of depression. ASKED 4 TIMES

12 **Bandura's four processes** of observational learning are **attention, retention, motor reproduction and motivation**. **Reinforcement is not necessary** for learning by observation (it governs performance), and the model must demonstrate the target behaviour. ASKED 4 TIMES

13 **Escape** terminates an aversive stimulus already present; **avoidance** prevents its onset; both are negative reinforcement. On Mowrer's two-factor theory (1947), OCD compulsions such as hand-washing are escape (avoidance) behaviours, or safety behaviours. ASKED 3 TIMES

14 **Behaviour analysis** (applied behaviour analysis) and **functional analysis** are based on **operant conditioning**: they examine a behaviour's antecedents and consequences. ASKED 3 TIMES

15 **Classical conditioning pairs two stimuli; operant conditioning relies on consequences** (reinforcement and punishment). The hallmark of behaviour therapy is the use of classical and operant conditioning techniques to eliminate unwanted behaviour. ASKED 3 TIMES

16 **Reciprocal determinism** (triadic reciprocal causation) belongs to **Bandura's social cognitive theory**: person, behaviour and environment influence one another in both directions. It is not Wolpe's reciprocal inhibition. ASKED 3 TIMES

17 **Time-out** and **response cost** (removing a desirable stimulus after an unwanted behaviour) are both **negative punishment**. ASKED 3 TIMES

18 **Biofeedback** brings autonomic and other bodily processes under voluntary control through operant conditioning with instrument feedback (Neal Miller); EMG biofeedback suits tension-type headache, and Arnold Kegel's perineometer (1948) was an early form. ASKED 3 TIMES

19 **Insight learning** (Wolfgang Köhler, Tenerife, 1913 to 1920): Sultan joined two sticks and the chimpanzees stacked boxes to reach bananas; the sudden solution is the **"aha" experience**. Gestalt psychology holds that the whole is greater than the sum of its parts. ASKED 3 TIMES

20 **Harry Harlow** (1958): infant rhesus monkeys clung to a cloth surrogate rather than a wire one that fed them, so attachment rests on **contact comfort**, not on conditioned reinforcement from feeding. ASKED 4 TIMES

21 **Self-efficacy** (Albert Bandura, 1977) is a person's belief in his capability to carry out the actions a specific task requires; it is the central motivational construct of social cognitive theory. ASKED 3 TIMES

22 **Law of effect** (Edward Lee Thorndike; puzzle-box cats, 1898; classic statement, 1911): responses followed by satisfying consequences become more likely to recur. ASKED 3 TIMES

23 **Bobo doll** (Albert Bandura, Dorothea Ross and Sheila Ross, 1961): children who watched an adult attack the doll imitated its specific aggressive acts with no reinforcement at all, which is **observational learning**. ASKED 3 TIMES

24 The relaxation in desensitisation is **Edmund Jacobson's** progressive muscle relaxation, and the client rates each hierarchy scene on Wolpe's 0 to 100 **Subjective Units of Disturbance Scale (SUDS)**. ASKED TWICE

25 **Taste aversion** (John Garcia and Robert Koelling, 1966) is **classical conditioning** learned in one trial across long delays; it shows **biological preparedness** (constraints on learning) and refutes equipotentiality. ASKED TWICE

26 **Sensitisation** (increased responding after a strong stimulus) is the near opposite of **habituation** (a decreasing response to a repeated harmless stimulus); habituation and dishabituation reveal what infants can discriminate before they can speak. ASKED TWICE

27 **Little Albert** (John B. Watson and Rosalie Rayner, 1920): seven pairings of a white rat with a loud noise produced classically conditioned fear that generalised to furry objects. ASKED TWICE

28 **Imprinting** (Konrad Lorenz, 1935, greylag goslings) is rapid, largely irreversible attachment to the first moving object, confined to a critical period. ASKED TWICE

29 **Delay (forward) conditioning**, in which the CS begins first and overlaps the UCS, is the most effective timing arrangement; backward conditioning is the least effective. ASKED ONCE

30 The keyed stages of **concept formation** are **observation, discrimination, generalisation, abstraction.** ASKED ONCE

CHAPTER 7

# Memory: Encoding, Storage, Retrieval and Forgetting

IN THIS CHAPTER

7.1 ●● What memory is, and how Ebbinghaus measured it: define encoding, storage and retrieval, describe Ebbinghaus's method and findings, and compare recall, recognition, relearning and reconstruction as measures of retention.

7.2 ●●● The multi-store model: describe sensory, short-term and long-term memory in the Atkinson and Shiffrin model, with their capacity, duration and coding, and evaluate the evidence for and against separate stores.

7.3 ●●● Working memory: explain the four components of Baddeley and Hitch's model and the dual-task evidence for each, and distinguish working memory from short-term memory.

7.4 ●● Levels of processing and the quality of encoding: explain why deep, elaborate and self-related processing produces durable memories, and why the levels idea was criticised.

7.5 ●●● The systems of long-term memory: distinguish episodic, semantic and procedural memory, explicit and implicit memory, and describe autobiographical, prospective, flashbulb and metamemory phenomena.

7.6 ●● How knowledge is organised: describe hierarchical network, spreading activation and connectionist models, and the role of schemas and scripts.

7.7 ●● Retrieval: cues, context and state: explain the encoding specificity principle and context-, state- and mood-dependent memory.

7.8 ●●● Forgetting: evaluate decay, interference, retrieval failure, motivated forgetting and consolidation failure, and describe the classic effects on retention.

7.9 ●● Memory as reconstruction: describe Bartlett's and Loftus's research, false memories, source monitoring and the problems of eyewitness testimony.

7.10 ●●● Amnesia and the biology of memory: distinguish anterograde and retrograde amnesia, describe the classic amnesic patients and Korsakoff's syndrome, and outline consolidation and the search for the engram.

7.11 ● Improving memory: explain how mnemonics, study systems, spacing, testing and sleep exploit the principles in this chapter.

Weightage. 111 of the 819 questions mapped in this book (13.6%) test this chapter, 72 of them from real past papers. 5 of its 11 sections are High yield.

In March 1985, Clive Wearing was at the top of his profession. He was a British musician and musicologist, a producer of music programmes for BBC radio and an authority on the Renaissance composer Orlande de Lassus. Then he fell ill with what looked like a bad flu and a headache that would not go away. It was herpes simplex encephalitis: the same virus that causes ordinary cold sores had invaded his brain, and within days it had destroyed the hippocampus on both sides and damaged large parts of his temporal and frontal lobes.

Clive survived, but the man who woke up was trapped inside a moment that lasted only seconds. He could hold a conversation as long as it kept moving, and he could follow a sentence to its end. But a short while after any experience ended, it was gone, and with it any sense that it had ever happened. He felt, continuously, as if he had just regained consciousness for the first time. His diary filled with page after page of entries, each giving the time and announcing that he was now, at last, truly awake; every earlier entry, which he had no memory of writing, was angrily crossed out. When his wife Deborah stepped out of the room for a few minutes and came back, he greeted her with tears of joy, as if they had been apart for years.

The damage reached backwards as well as forwards. Almost every particular episode of his earlier life had become inaccessible to him. He knew certain facts, that he was a musician and that he was married to Deborah, but he could bring back almost no specific event from the decades he had lived.

And yet some things survived untouched. Seat Clive at a piano or in front of a choir and he could still sight-read, play and conduct with his old skill; the music carried him from one bar to the next for as long as it lasted. Nor did he ever lose his love for Deborah or his recognition of her, although he could not remember a single conversation with her from one visit to the next.

Clive Wearing's case raises nearly every question this chapter tries to answer. How can a person hold a thought for a few seconds but not for a minute? How can someone lose his whole personal past and still remember how to play the piano? Is there one memory or many? Where do memories go when we forget them, and why do some of them come back changed? Keep Clive in mind; you will meet him again at the end of the chapter.

## 7.1 What Memory Is, and How Ebbinghaus Measured It

MEDIUM YIELD · 5 QS

You use memory in everything you do. Reading this sentence requires you to hold its first words in mind until you reach its last ones; understanding it requires the meanings of words you learned years ago; and if you can answer a question about it next week, it is because something of it was kept. **memory** is the capacity to retain information over time and to use it later. Notice that this definition says nothing about consciousness: as Clive Wearing's piano playing shows, much of what we retain is expressed in what we can do rather than in what we can report.

## Three processes: encoding, storage and retrieval ●●

Psychologists find it useful to break remembering into three processes, an idea that grew out of the information-processing approach of the 1950s and 1960s, which treated the mind as a system that takes in, transforms, stores and outputs information, much as a computer does. The study of mental processes such as attention, memory and problem solving through this information-processor metaphor is cognitive psychology, whose rise is told in Chapter 2. **encoding** is the transformation of incoming information into a form, or code, that the memory system can hold. **storage** is the maintenance of encoded information over time. **retrieval** is the recovery of stored information when it is needed. A library is a fair analogy: books must be acquired and catalogued (encoding), kept on the shelves (storage), and found again when a reader asks for them (retrieval). A book can be missing because it was never bought, because it was lost from the shelf, or because it was shelved under the wrong number and cannot be found; in the same way, a failure to remember can be a failure at any of the three stages.

The first stage depends heavily on attention. What you do not attend to is usually never encoded at all. In 1979 Raymond Nickerson and Marilyn Adams asked American adults to pick out the correct drawing of a one-cent coin from fifteen versions; fewer than half chose correctly, although every one of them had handled thousands of such coins. You can try the same test on yourself with a ten-rupee coin. The failure is not forgetting in the ordinary sense but **encoding failure**: the details were never encoded, because nothing in daily life requires you to attend to them, and what was never stored cannot later be retrieved. Attention itself is the subject of Chapter 5.

## Ebbinghaus and the first experiments on memory ●●

For most of the nineteenth century, memory was a subject for philosophers. Even Wilhelm Wundt, who founded the first psychology laboratory in 1879, doubted that the higher mental processes could be studied by experiment. The person who proved otherwise was **Hermann Ebbinghaus** (1850 to 1909), a young German scholar working alone, without a laboratory, who had been inspired by Gustav Fechner's demonstration that sensations could be measured (Chapter 3). If sensations could be measured, Ebbinghaus reasoned, why not memories?

His first problem was material. Real words and poems come loaded with associations: a word you already know is not learned from scratch, and some words are far easier than others. So Ebbinghaus invented the **nonsense syllable**, a pronounceable but meaningless consonant-vowel-consonant combination such as DAX, BUP or ZOL. He built a pool of about 2,300 of them, drew lists at random, and, acting as his own and only participant, read each list aloud at a steady rate until he could recite it without error. Then, after a delay that ranged from twenty minutes to a month, he learned the same list again to the same standard.

His measure of what remained was ingenious. Even when he could not recall a single syllable, relearning a list was faster than learning it had been the first time. The **savings method** expresses this as a percentage: savings equals the time (or number of trials) for original learning minus the time for relearning, divided by the time for original learning, multiplied by 100. Here is a worked example. Suppose a list took 20 readings to learn on Monday and 12 readings to relearn on Friday. The savings are (20 minus 12) divided by 20, multiplied by 100, which is 40 per cent. Forty per cent of the original work did not have to be repeated, so some-

thing of Monday's learning survived, even if Friday's attempt at recall produced nothing at all.

### KEY STUDY: EBBINGHAUS, MEMORY (1885)

Ebbinghaus published his results in 1885 as *Über das Gedächtnis* (On Memory; translated into English in 1913). His central result was the **forgetting curve**: savings fell steeply at first and then levelled off. After twenty minutes he had already lost about 42 per cent of the original learning; after an hour, more than half; after a day, about two thirds. But after that the loss slowed dramatically, and a month later about a fifth of the learning still survived (Figure 7.1). He also found that a list of about seven syllables could be repeated after a single reading but that the number of readings needed rose steeply for longer lists; that extra repetitions beyond the point of mastery increased later savings; that repetitions spread over several days were more efficient than the same repetitions crowded into one sitting; that syllables at the beginning and end of a list were learned before those in the middle; and that meaningful material (stanzas of poetry) needed far fewer readings than nonsense syllables.

The criticisms are obvious and Ebbinghaus knew most of them: one participant, who was also the experimenter; artificial material stripped of meaning; and syllables that were not, in fact, equally meaningless (later researchers measured the "association value" of each syllable). Frederic Bartlett would later argue that by removing meaning Ebbinghaus had removed the most important thing about human memory (Section 7.9). Even so, his curves have been replicated many times, and his work showed that memory could be studied with the same precision as sensation.

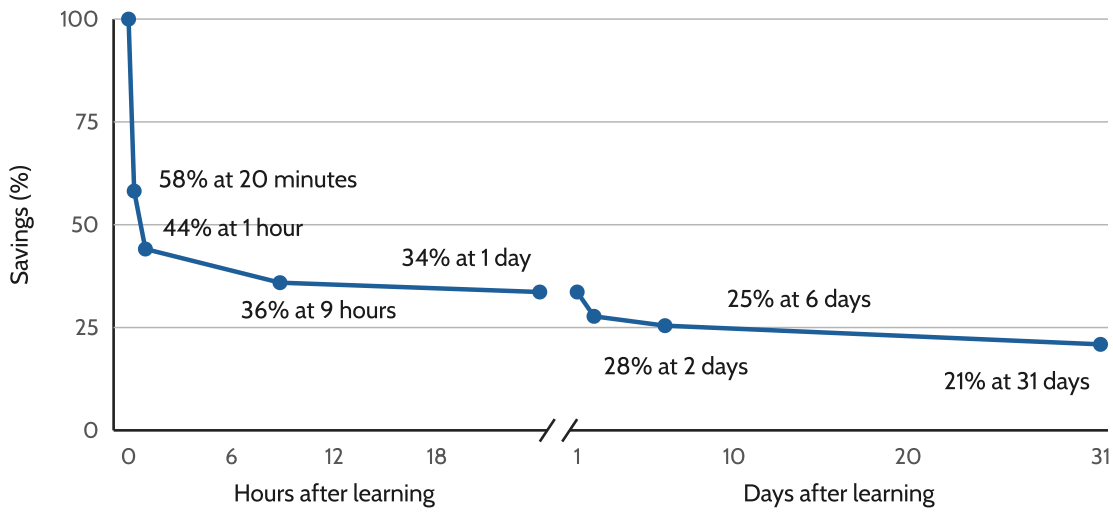



Figure 7.1 Ebbinghaus's forgetting curve, plotted from his 1885 savings data. The time axis is broken: hours on the left, days on the right. Notice how much is lost in the first hour and how little is lost between the sixth and the thirty-first day.

Two features of Figure 7.1 are worth fixing in your mind, because examiners return to them. First, the curve is negatively accelerated: forgetting is fastest immediately after learning and slows as time passes. Second, it never reaches zero; Ebbinghaus's savings method detected traces of learning long after recall had failed. Ebbinghaus also left psychology one of its best-

known sentences, from a textbook of 1908: "Psychology has a long past, but only a short history."

> **EXAM TRAP**
>
> A question asks what Hermann Ebbinghaus, Francis Galton and Sigmund Freud have in common. All three worked largely independently and were pioneers, and each devised a new method for studying the mind (Ebbinghaus the nonsense syllable and savings, Galton questionnaires and word association, Freud free association). None of them founded a formal psychology laboratory in the sense that Wundt did (Galton's Anthropometric Laboratory, opened in London in 1884, measured physical and sensory traits of the public rather than running psychological experiments). The RML 2020 key gives "each worked independently and were pioneers" alone, while the Panwar bank keys "(a) and (c)". The difference turns on the word "observational": Ebbinghaus's method was experimental, not observational, which is the likely reason the RML key stops at (a). Read the options carefully and choose the statement that is true of all three without qualification.

### Measuring retention: recall, recognition, relearning and reconstruction ●●

Ebbinghaus measured savings, but memory can be measured in several ways, and the same memory can look strong on one test and absent on another. All of the methods below are explicit tests: the person knows that memory is being tested and tries to remember. (Implicit tests, where memory shows itself without any attempt to remember, are described in Section 7.5.)

**recall** requires the person to reproduce information with few or no hints. It comes in three forms. In **free recall** the person reports the items in any order, as when you list the films you saw last year. In **serial recall** the items must be reproduced in the order presented, as when you repeat a phone number. In **cued recall** a hint is given, such as a category name or the first letter. A special and historically important form of cued recall is **paired-associate learning**, introduced by **Mary Whiton Calkins** (1863 to 1930) in 1894: the person learns pairs of items (HOUSE and 47, or a Kannada word and its English meaning) and is later given the first member of each pair as a cue for the second. Learning the vocabulary of a new language is paired-associate learning in everyday life.

**recognition** requires the person only to identify previously encountered items among new ones, called distractors or lures. It may take a yes or no form ("Was this face in the set?") or a forced-choice form, like the multiple-choice questions in your exam. A police line-up is a recognition test. Recognition is usually easier than recall, because the item itself is supplied and the person need only judge its familiarity.

**relearning** is Ebbinghaus's savings method: the person learns the material again, and the reduction in time or trials is the measure of what was retained. It is the most sensitive of the measures, because it detects traces too weak to support either recall or recognition. Finally, the **reconstruction method** presents items in a particular order, then hands them back shuffled and asks the person to put them back into the original order. The score reflects memory for order rather than for the items themselves, which are all in front of the person. Table 7.1 sets the methods side by side.

Table 7.1 Measures of explicit retention, roughly from least to most sensitive: the recall tests are the least sensitive and relearning the most.

| Method | What the person does | Everyday example |
|---|---|---|
| Free recall | Produces the items in any order, with no cues | Writing an essay answer from memory |
| Serial recall | Produces the items in their original order | Repeating a mobile number |
| Cued recall (including paired associates) | Produces an item when given a hint or partner item | Giving the English meaning of a foreign word |
| Reconstruction | Restores the original order of items that are supplied | Putting the steps of a recipe back in sequence |
| Recognition | Picks out old items from among new ones | A multiple-choice question; a police line-up |
| Relearning (savings) | Learns the material again; time saved is measured | Revising a subject studied years ago |

EXAM TRAP

The "reconstruction method" (restoring the original order of supplied items) is a measurement technique. "Reconstructive memory" (Bartlett's idea that remembering rebuilds the past from schemas, Section 7.9) is a theory. A question describing items presented in order, then scrambled, then put back in order by the participant is asking for the reconstruction method, not recall, recognition or relearning. And when asked which measure is most sensitive, the answer is relearning (savings); recall, with no cues to help, is the least sensitive.

## 7.2 The Multi-Store Model HIGH YIELD · 17 QS

Ebbinghaus studied memory as a single thing that weakens over time. The idea that memory might be made of several different stores is older than experiments on it. **William James** (1842 to 1910), in his *Principles of Psychology* (1890), distinguished primary memory, the small amount of information that is still present in consciousness (the sentence you are reading), from secondary memory, the vast store of knowledge that must be brought back into consciousness. In 1965 Nancy Waugh and Donald Norman revived James's terms in an experimental model. Three years later came the version that every textbook teaches.

### Atkinson and Shiffrin's modal model (1968)

In 1968 **Richard C. Atkinson** and **Richard M. Shiffrin** proposed the **multi-store model** of memory. Because so many theorists of the period proposed something similar, it is also called the modal model (the most typical model). It has two kinds of parts. The structural features are three stores through which information passes: sensory memory, short-term memory and long-term memory. The **control processes** are strategies that the person can choose to use, such as attending, rehearsing, coding information in a particular way, and searching memory during retrieval. As Figure 7.2 shows, information from the environment enters the sensory stores; what is attended to passes into short-term memory; what is rehearsed there is transferred to long-term memory; and information can be retrieved from long-term memory back into short-term memory when it is needed.

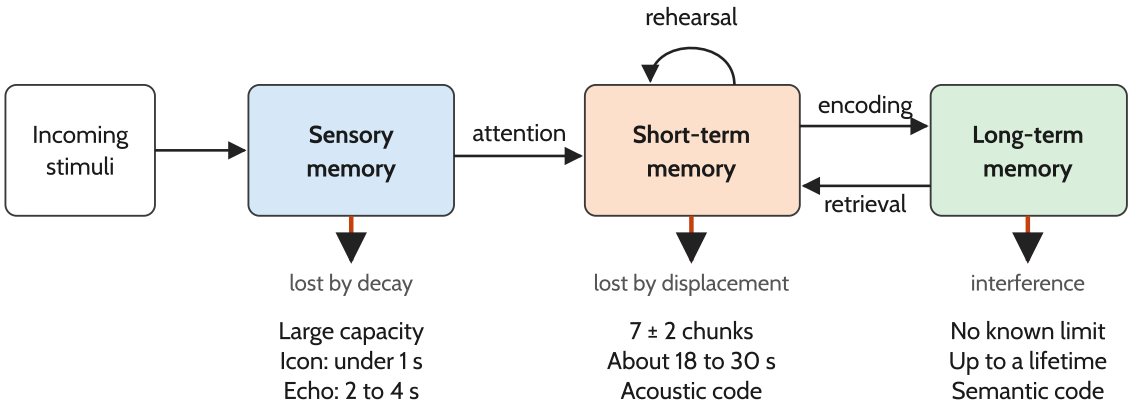



Figure 7.2 The multi-store (modal) model of Atkinson and Shiffrin (1968). Solid arrows show the flow of information; dashed arrows show how information is lost from each store. The figures beneath each store are the classic textbook values.

## Sensory memory ●●●

Wave a lit sparkler in a circle on Diwali night and you see a circle of light, not a moving dot. The circle exists only in your memory: the image of each position persists for a fraction of a second after the sparkler has moved on. **sensory memory** (also called the sensory register) is a very brief store that holds a large amount of raw, unprocessed sensory information, one store for each sense, just long enough for attention to select some of it for further processing. Its contents are pre-categorical: they are a record of the stimulus, not yet of its meaning. The visual sensory store is **iconic memory** and the auditory store is **echoic memory**; both terms were coined by **Ulric Neisser** (1928 to 2012) in his book *Cognitive Psychology* (1967), which gave the new field its name. A tactile store, sometimes called haptic memory, has been studied far less.

### KEY STUDY: SPERLING'S PARTIAL-REPORT EXPERIMENTS (1960)

When people see an array of twelve letters (three rows of four) flashed for 50 milliseconds, they can report only about four or five of them. Yet they insist that they saw more, and that the rest faded while they were reporting. **George Sperling** (born 1934) tested this claim with the **partial-report procedure**. Immediately after the flash, a tone told the participant which row to report: a high tone for the top row, a medium tone for the middle row, a low tone for the bottom row. Because the participant could not know in advance which row would be cued, the proportion of the cued row reported must be true of the whole array. Participants reported about three of the four letters in whichever row was cued, which means that about nine of the twelve letters, roughly 75 per cent, were available at the moment of the tone, about twice as many as the whole-report method suggested.

Then Sperling delayed the tone. With every fraction of a second of delay the partial-report advantage shrank, and by about one second it had vanished: performance fell to the level of whole report. Sperling concluded that a rich visual image persists for a few hundred milliseconds after the stimulus ends and fades rapidly, so that whole report underestimates what was seen simply because the image decays while the first letters are being named. A bright visual stimulus that follows the array (a mask) wipes the icon out even faster.

Iconic memory therefore lasts roughly 250 to 500 milliseconds (a quarter to half a second, a little longer in some conditions). Echoic memory lasts longer. In 1972 Christopher Darwin, Michael Turvey and Robert Crowder ran an auditory version of Sperling's experiment, the "three-eared man" study: participants heard three different lists of letters and digits at once over headphones, one in the left ear, one in the right ear and one in both ears (which is heard as coming from the middle of the head). A visual cue then indicated which "ear" to report. A partial-report advantage appeared again, and it lasted for up to about four seconds. Echoic memory is generally put at about two to four seconds. You use it every time someone speaks to you while your attention is elsewhere: you say "What did you say?" and then realise you can replay the last few words yourself before they repeat them. The longer life of the echo makes sense, because speech unfolds over time and the end of a word can only be understood together with its beginning.

> **EXAM TRAP**
>
> Two past papers ask about "the largest capacity" and get different answers. If a question says "largest capacity but shortest duration" (JEMAS 2025), the answer is sensory memory: it registers almost everything that reaches the senses, far more than short-term memory can hold, but for under a second (icon) or a few seconds (echo). If a question asks only which store has "the largest capacity" in the Atkinson and Shiffrin model, the answer is long-term memory, whose capacity has no known limit. Read the second half of the stem. Also note that "icon and echo" belong to sensory memory, never to short-term or working memory; that iconic memory is shorter than echoic memory; and that when a question says "lasts only a few seconds" and offers sensory memory, sensory memory is the intended answer, not eidetic imagery (Section 7.5), which is a rare, long-lasting visual image.

## Short-term memory: capacity and chunking ●●●

Information that is attended to passes into **short-term memory** (STM), a store of very limited capacity that holds information for a matter of seconds unless it is rehearsed. Its capacity is measured by the **memory span**: the longest sequence of items that a person can repeat back in the correct order immediately after hearing it. Joseph Jacobs measured digit span systematically as early as 1887, and digit span remains part of every major intelligence and memory scale (Psychological Assessment book).

In 1956 **George Armitage Miller** (1920 to 2012) published a paper with one of the most famous titles in psychology, "The magical number seven, plus or minus two: Some limits on our capacity for processing information." Across many tasks, he noted, people could hold about seven items (between five and nine) in immediate memory. But Miller's deeper point was that the limit applies to meaningful units, not to raw items. He called these units chunks, and the process of grouping items into larger meaningful units **chunking**. Try to hold this string: I N D I A I I T I S R O. Twelve letters is beyond most people's span. Read it as INDIA, IIT, ISRO and it becomes three chunks, well within span. You chunk your mobile number into groups (98450 12345), and your twelve-digit Aadhaar number into three groups of four, for the same reason.

Chunking depends on knowledge already stored in long-term memory: IIT is a chunk only for someone who knows what it is. The most dramatic demonstration came from a student known as S.F., studied by Anders Ericsson, William Chase and Steve Faloon (1980). S.F. began with an ordinary digit span of about seven. Over more than 200 hours of practice spread

across about twenty months he raised it to around 80 digits, by recoding groups of digits as running times (he was a keen club runner, so 3492 became "3 minutes 49.2 seconds, a near world-record mile"). His span for letters, which he did not practise, stayed at about six. His short-term store had not grown; he had learned to chunk. In the same way, chess masters glancing at a board from a real game for five seconds can reconstruct it almost perfectly, but are no better than novices when the pieces are placed at random (Chase and Simon, 1973).

When rehearsal and chunking are prevented, the true capacity of short-term memory appears to be smaller than seven. In 2001 Nelson Cowan reviewed this evidence and argued for "the magical number 4": about four chunks, plus or minus one. Modern textbooks therefore give both figures: Miller's seven plus or minus two for memory span, and Cowan's four for the pure capacity of the store.

### Short-term memory: duration, forgetting and coding

How long does information last in short-term memory if you are prevented from rehearsing it? Two studies answered almost simultaneously: John Brown in Britain in 1958 and, in a paper that every student learns, Lloyd and Margaret Peterson in the United States in 1959.

**KEY STUDY: THE BROWN-PETERSON TECHNIQUE (1958, 1959)**

Lloyd Peterson and Margaret Peterson read a consonant trigram (such as CHJ) to each participant, followed at once by a three-digit number. The participant had to count backwards by threes from that number (506, 503, 500 ...) until a signal, then recall the trigram. Counting backwards prevented rehearsal. The retention interval varied from 3 to 18 seconds. Recall fell rapidly: from about 80 per cent after 3 seconds to about 10 per cent after 18 seconds. This method of filling the retention interval with a distracting task became known as the **Brown-Peterson technique**.

The Petersons interpreted the result as decay: an unrehearsed trace simply fades in under half a minute, which is the origin of the textbook figure of "about 18 to 30 seconds" for the duration of short-term memory. But the decay interpretation did not survive. In 1962 Geoffrey Keppel and Benton Underwood looked at the very first trial each participant received and found almost no forgetting at all, even after 18 seconds. Forgetting grew only over successive trials, as trigrams from earlier trials piled up and competed with the current one. That is proactive interference (Section 7.8), not decay.

Further evidence pointed the same way. Delos Wickens and his colleagues showed in the 1960s and early 1970s that if several Brown-Peterson trials use items from one category (say, fruits), recall declines from trial to trial; but if the category is then switched (to professions), recall jumps back up. This **release from proactive interference** shows that the earlier items were interfering with the new ones, and, incidentally, that items in short-term memory carry some meaning. Waugh and Norman (1965) presented digits at fast or slow rates and found that forgetting depended on how many items came after the target, not on how much time had passed: new items displace old ones from a store with few slots. The modern view is that short-term forgetting is mostly due to interference and displacement, with decay playing a role, if at all, only over the first few seconds.

What form does information take in short-term memory? In 1964 Reuben Conrad showed people strings of letters visually and recorded their recall errors. The errors were acoustic: people wrote V when the letter had been B, or S when it had been X, letters that sound alike

but look different. Even visual material, it seemed, was held as sound. Two years later **Alan Baddeley** (born 1934) gave people short lists of words to recall immediately. Lists of words that sounded alike (man, mad, cap, can, map) were recalled very poorly; lists of words with similar meanings (big, huge, large, broad, great) caused little difficulty. When recall was delayed by twenty minutes, so that it depended on long-term memory, the pattern reversed: similarity of meaning now caused confusion, and similarity of sound did not. The classic conclusion is that **acoustic coding** dominates in short-term memory and semantic coding in long-term memory. "Dominates" is the right word: short-term memory can also hold visual codes for a second or two, and Wickens's release effect shows that meaning is present too.

## Long-term memory ●

**long-term memory** (LTM) is the relatively permanent store of information. Its capacity has no known limit: nobody has ever been turned away from learning a new face because the store was full. Its duration ranges from minutes to a lifetime. Harry Bahrick and his colleagues (1975) tested nearly four hundred American adults on the names and faces of their high-school classmates; even 34 years after graduation, people recognised about 90 per cent of their classmates' faces, although their ability to recall names freely was far poorer and declined steadily. In a later study, Bahrick (1984) found that much of the Spanish people had learned at school was still there fifty years later; he called this stable residue permastore. Information in long-term memory is coded mainly by meaning. Jacqueline Sachs (1967) found that people who had heard a passage could not tell, after a short delay, whether a test sentence matched the exact wording they had heard, but readily detected any change in its meaning: we keep the gist and lose the words. Long-term memory also holds sounds (a song, a voice), images (a face) and movements (how to ride a bicycle), so semantic coding is the typical, not the only, code. Table 7.2 compares the three stores.

Table 7.2 The three stores of the multi-store model compared.

| Feature | Sensory memory | Short-term memory | Long-term memory |
|---|---|---|---|
| Capacity | Large (most of what reaches the senses) | About 7 ± 2 items (Miller); about 4 chunks (Cowan) | No known limit |
| Duration | Icon about 0.25 to 0.5 s; echo about 2 to 4 s | About 18 to 30 s without rehearsal | Minutes to a lifetime |
| Main code | Raw sensory (modality-specific) | Acoustic (Conrad; Baddeley) | Semantic (Baddeley; Sachs) |
| Main cause of loss | Decay and masking | Displacement and interference | Interference and retrieval failure |
| Classic study | Sperling (1960); Darwin, Turvey and Crowder (1972) | Miller (1956); Peterson and Peterson (1959) | Bahrick and colleagues (1975) |

## Evidence for separate stores: the serial position curve and amnesic patients ●●●

If you hear a list of fifteen words and then write down as many as you can in any order, which words will you remember? Not the middle ones. Plot the probability of recall against each word's position in the list and you get a U-shaped curve, the **serial position effect**, also called the primacy-recency effect (Figure 7.3). The superior recall of the first few items is the **primacy effect**; the superior recall of the last few is the **recency effect**. Bennet Murdock (1962) showed how reliable this curve is in free recall across lists of different lengths and pre-

sentation rates. The multi-store model explains it neatly. The first items receive more rehearsal than later ones, because each of them is rehearsed alone and then together with the next few, so they are more likely to be transferred to long-term memory: primacy is a long-term memory effect. The last items are still sitting in short-term memory at the moment of recall and are simply read out first: recency is a short-term memory effect. Dewey Rundus (1971) confirmed the first claim by asking people to rehearse aloud; the early items were rehearsed far more often than the rest, and the number of rehearsals predicted recall.

**KEY STUDY: GLANZER AND CUNITZ (1966)**

If primacy and recency come from different stores, it should be possible to change one without touching the other. Murray Glanzer and Anita Cunitz did exactly that. When recall was delayed by 30 seconds filled with counting (which empties short-term memory), the recency effect disappeared, but the primacy effect was unchanged. When words were presented more slowly (allowing more rehearsal), recall of early and middle items improved, but recency was unaffected. Two variables, two stores: a classic dissociation.

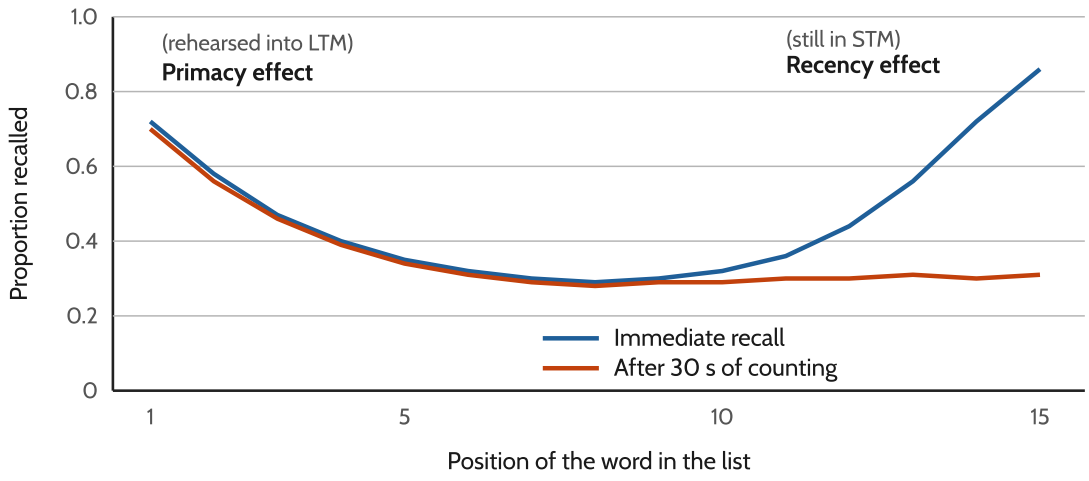



Figure 7.3 The serial position curve in free recall (schematic, with the pattern reported by Glanzer and Cunitz, 1966). Immediate recall (solid line) shows both primacy and recency. A 30-second filled delay (dashed line) removes the recency effect but leaves primacy intact.

The second line of evidence comes from brain damage. Patients with the classic amnesic syndrome, such as H.M. (Section 7.10) and Clive Wearing, have normal digit spans and normal recency effects, but cannot transfer new information into long-term memory: short-term memory intact, long-term learning destroyed. If the stores are truly separate, the opposite pattern should also exist. It does. In 1970 Tim Shallice and Elizabeth Warrington described K.F., a man who had injured the left side of his brain in a motorcycle accident. His digit span was only about two items, and his recency effect was almost absent, yet his long-term learning was essentially normal. When one patient shows damage to function A with function B intact, and another shows the reverse, the result is a **double dissociation**, the strongest kind of neuropsychological evidence that two functions depend on different systems.

EXAM TRAP

Who first observed the primacy-recency effect? Ebbinghaus noted in 1885 that the beginning and end of a list were learned first, and many textbooks credit him with the first observation of the serial position effect. The RML 2024 key, however, gives "McCray and Hunter", which refers to J. W. McCrary and W. S. Hunter, who published influential serial position curves for serial learning in 1953. If Ebbinghaus and McCrary and Hunter both appear, remember that the official key for that paper chose McCrary and Hunter, and that the historically earlier observation is Ebbinghaus's. Two more confusables: primacy and recency effects in impression formation and persuasion belong to the Social book, and primacy and recency as order effects in a psychophysics experiment are controlled by counterbalancing (Research Methodology book).

## Criticisms of the multi-store model ●●

The modal model gave memory research a common map, and the distinction between a small, fragile short-term store and a vast, durable long-term store has survived. But nearly every detail has been challenged. First, the stores are not unitary. Short-term memory turned out to have separate verbal and visual parts and an attentional controller, which is why it was replaced by the working memory model (Section 7.3); long-term memory turned out to contain several systems (Section 7.5). Second, rehearsal is not the royal road to long-term memory that the model assumed: simply repeating an item can keep it in short-term memory for as long as you like without making it much more memorable later (Section 7.4). Third, K.F. is a problem as well as a support: if information could reach long-term memory only by passing through short-term memory, his damaged short-term store should have crippled his long-term learning, and it did not. Fourth, the order of the stores is wrong in an important way: we can chunk IIT in short-term memory only because long-term memory has already been consulted, so long-term knowledge acts on information before and during its stay in short-term memory, not merely after it. Finally, the model treats short-term memory as a passive waiting room, when in daily life it is where we calculate, reason and understand.

EXAM TRAP

"Which of the following is NOT a store in the stage model of memory?" The stage (multi-store) model has three stores: sensory, short-term and long-term. Working memory is not one of them; it is the separate model of Baddeley and Hitch (1974). A related question asks which store information must pass through before reaching long-term memory in the Atkinson and Shiffrin model: short-term memory, which some practice papers write as "short-term (working) memory" because Atkinson and Shiffrin themselves described the short-term store as a working memory.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. The three processes of memory are encoding, storage and _____.
2. Ebbinghaus invented the _____ syllable and measured retention by the _____ method.
3. A list that took 30 trials to learn and 18 trials to relearn shows savings of _____ per cent.
4. Learning pairs of items, with the first item later given as a cue for the second, is called _____ learning; it was introduced by Mary Whiton Calkins.
5. Sperling used the _____ procedure to show that about nine of twelve letters are briefly available in iconic memory.
6. Miller's 1956 estimate of short-term memory span was seven plus or minus two; Cowan's later estimate of pure capacity was about _____ chunks.
7. Keppel and Underwood showed that forgetting in the Brown-Peterson task was largely due to _____ interference rather than decay.
8. A 30-second filled delay before recall abolishes the _____ effect but leaves the primacy effect intact.
9. Patient K.F. had a very small digit span but normal long-term learning; with H.M. he provides a _____ dissociation.

**Answers:** (1) retrieval, (2) nonsense, savings, (3) 40, (4) paired-associate, (5) partial-report, (6) four, (7) proactive, (8) recency, (9) double.

## 7.3 Working Memory

HIGH YIELD · 15 QS

Try this: multiply 24 by 7 in your head. You must hold the numbers, carry out the steps, store the partial result (140) while you calculate the next one (28), and then combine them. Short-term memory in the Atkinson and Shiffrin sense, a passive box where items wait to be rehearsed or lost, does not capture what your mind just did. In 1974 Alan Baddeley and **Graham Hitch** proposed that short-term memory is better thought of as a **working memory**: a limited-capacity system for temporarily holding and manipulating information while we carry out complex tasks such as reasoning, comprehension, mental arithmetic and learning.

### The dual-task method and the case against a single store ●●

Baddeley and Hitch's main tool was the **dual-task method**: give people two tasks at once and see whether they interfere. The logic is simple. If two tasks need the same limited resource, doing them together will impair one or both; if they draw on different resources, they can be combined with little cost. In their key experiments, participants held a list of up to six digits (a full short-term memory load, by the multi-store account) while solving reasoning problems, such as deciding whether "A follows B" correctly describes the pair BA. If short-term memory were a single store, filling it with six digits should have wrecked the reasoning. It did not. Reasoning slowed somewhat, but accuracy held up. Baddeley and Hitch concluded that short-term memory has several components: a store that was occupied by the digits, and something else, which was still available to do the reasoning.

The model that resulted originally had three components, a controlling central executive and two "slave" systems, the phonological loop and the visuospatial sketchpad. In 2000 Baddeley added a fourth, the episodic buffer. Figure 7.4 shows the model in its current form.

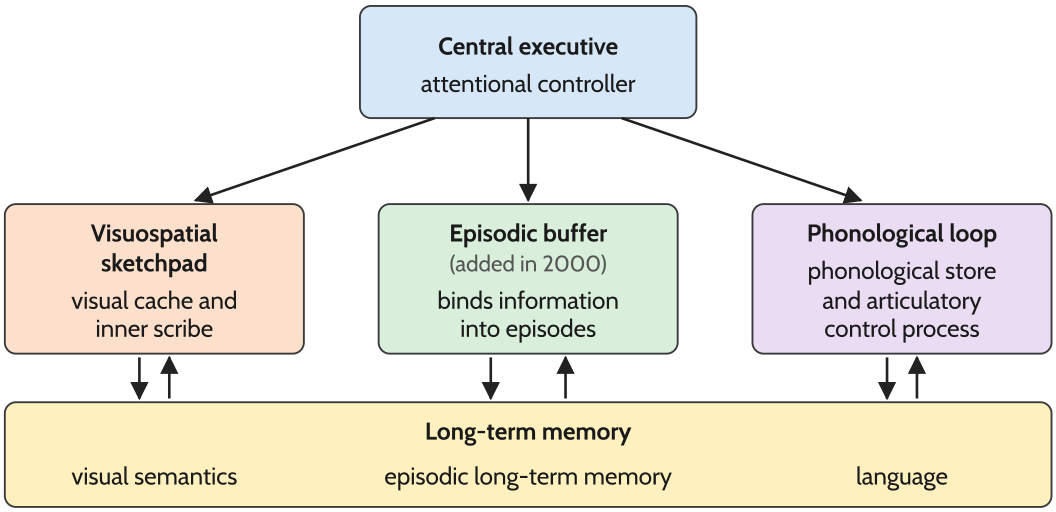



Figure 7.4 Baddeley's working memory model (Baddeley and Hitch, 1974; episodic buffer added by Baddeley, 2000). The central executive directs attention but stores nothing; the three subsidiary systems hold information and exchange it with long-term memory.

## The phonological loop ●●

The **phonological loop** holds and rehearses speech-based, auditory-verbal information. It has two parts. The phonological store, sometimes called the inner ear, holds speech sounds for about two seconds before they fade. The articulatory control process, the inner voice, refreshes the contents of the store by silent rehearsal (the way you repeat a phone number under your breath) and converts written material into a speech code so that it can enter the store. Four findings support this account.

- The **phonological similarity effect**: lists of similar-sounding items (B, G, V, P, T) are recalled worse than dissimilar ones (F, W, K, R, Y), because similar sounds are confused in the store. This is Conrad's and Baddeley's finding from Section 7.2, now given a home.
- The **word-length effect**: memory span is greater for short words than long ones. Baddeley, Neil Thomson and Mary Buchanan (1975) found that people could recall about as many words as they could say in roughly two seconds, so span for words like "wit, sum, harm" is larger than for "university, opportunity, constitutional". Even words with the same number of syllables but longer spoken durations (Friday, harpoon) are recalled worse than quick ones (bishop, wicket). Digit span depends on language for the same reason: Nick Ellis and Richard Hennelly (1980) showed that Welsh-English bilingual children had smaller digit spans in Welsh, whose digit names take longer to say, than in English.
- **articulatory suppression**: if people repeat an irrelevant word aloud ("the, the, the") while studying a list, the articulatory process is occupied. Suppression abolishes the word-length effect, and with visually presented items it also abolishes the phonological similarity effect, because written items can no longer be recoded into sound and so never

reach the store. With spoken items, the similarity effect survives, because heard speech enters the store directly.

- The irrelevant speech effect: background speech, even in a language the listener does not know, impairs serial recall of visually presented digits, whereas non-speech noise of the same loudness does not (Pierre Salamé and Baddeley, 1982). Speech gains automatic access to the phonological store.

What is the loop for? Not, mainly, for remembering phone numbers. Baddeley and his colleagues argue that it evolved as a language-learning device: to learn a new word, you must hold its unfamiliar sound sequence long enough to build a long-term representation. A patient known as P.V., whose phonological store was damaged, could learn pairs of words in her own language normally but was almost unable to learn words in Russian, which offered nothing familiar to lean on.

### The visuospatial sketchpad ●

The **visuospatial sketchpad** holds and manipulates visual and spatial information: the route from your hostel to the exam hall, the layout of your room in the dark, an image you rotate in your mind. Robert Logie (1995) divided it into a visual cache, which stores form and colour, and an inner scribe, which handles spatial and movement information and rehearses the contents of the cache. When Roger Shepard and Jacqueline Metzler (1971) showed that the time needed to decide whether two drawings show the same object increases steadily with the angle through which one must be mentally rotated, they were measuring the sketchpad at work; any question about the component of working memory that "mentally rotates an object" is asking for the visuospatial sketchpad.

The dual-task evidence for a separate sketchpad is elegant. Lee Brooks (1968) asked people to imagine a block capital F and to move around its corners in their mind, classifying each corner as "yes" if it was on the extreme top or bottom and "no" otherwise. They could respond either by saying "yes" or "no" or by pointing to Y and N letters scattered over a sheet of paper. Pointing, which needs spatial control, made the imagery task much slower than speaking. With a verbal task (classifying the words of a remembered sentence), the reverse was true: speaking was slower than pointing. Spatial responses interfere with spatial memory and verbal responses with verbal memory, which is what separate visuospatial and phonological systems predict. Clinically, visuospatial span is measured with the Corsi block-tapping test, in which the examiner taps a sequence of blocks and the patient copies it.

### The central executive ●●●

The **central executive** is the most important and least understood component. It is an attentional control system with no storage of its own. Baddeley (1996) listed its functions as focusing attention, dividing attention between tasks, switching attention, and linking working memory with long-term memory. He modelled it on Donald Norman and Tim Shallice's supervisory attentional system, which takes over from habit whenever a situation is new, difficult or dangerous. Later work by Akira Miyake and colleagues (2000) separated three related executive functions: inhibition of unwanted responses, shifting between tasks, and updating the contents of working memory. The best-known measure of inhibition is the Stroop task (John Ridley Stroop, 1935): naming the ink colour of a colour word printed in a conflicting colour (RED in blue ink) is slow because automatic reading competes with the controlled task

of colour naming. The Stroop effect is therefore interference in selective attention, not the interference between memories that causes forgetting (Section 7.8); it is taught fully in Chapter 5.

The classic way to load the central executive is random generation. Try saying a random sequence of digits for a minute: you must constantly monitor what you have said, suppress the habitual sequences (1, 2, 3 or your phone number) and decide what to say next. Random number or letter generation disrupts reasoning, planning and the selection of chess moves, and the more demanding the main task, the more random generation suffers. Damage to the frontal lobes, especially the dorsolateral prefrontal cortex, produces what Baddeley and Barbara Wilson (1988) called the dysexecutive syndrome: poor planning, distractibility and perseveration, with memory stores themselves intact (Physiological book). Table 7.3 sums up which secondary task loads which component.

Table 7.3 Secondary tasks used in dual-task experiments and the component each one occupies.

| Secondary task | What the person does | Component loaded | Typical effect |
|---|---|---|---|
| Articulatory suppression | Repeats a word aloud ("the, the, the") | Phonological loop (articulatory control process) | Removes the word-length effect |
| Irrelevant speech | Ignores background speech | Phonological store | Impairs serial recall of digits |
| Spatial tapping or tracking | Taps a pattern or follows a moving target | Visuospatial sketchpad (inner scribe) | Impairs spatial imagery |
| Random number generation | Produces a random sequence of digits | Central executive | Impairs reasoning and planning |

EXAM TRAP

"Which secondary task affects the central executive?" Random number generation (RML 2022). The other options in that question each load a different component: continually repeating a word is articulatory suppression (phonological loop), remembering a spatial configuration loads the visuospatial sketchpad, and listening to irrelevant speech reaches the phonological store. Note too that the central executive stores nothing; if an option says it "holds verbal information", it is wrong.

## The episodic buffer ●●●

By the late 1990s the three-part model had a problem. People can repeat back a meaningful sentence of fifteen or sixteen words, far more than the five or six unrelated words the phonological loop can hold. Some densely amnesic patients can recall a short prose passage immediately, at normal levels, although they will have forgotten it entirely an hour later. Where is all this information held, and where are words, images and meaning combined into a single experience? In 2000 Baddeley proposed the **episodic buffer**: a limited-capacity store, holding perhaps four chunks, that uses a multidimensional code to bind information from the phonological loop, the visuospatial sketchpad and long-term memory into integrated episodes. It is "episodic" because it holds integrated scenes or episodes, and a "buffer" because it serves as an interface between systems that use different codes. It is controlled by the central executive and is thought to be the place where we are consciously aware of the bound result.

EXAM TRAP

The four components of working memory are the central executive, the phonological loop, the visuospatial sketchpad and the episodic buffer. Distractors invent plausible-sounding parts: "visual executive loop", "verbal sketchpad", "semantic buffer". The phonological loop handles auditory-verbal information; the sketchpad handles visual and spatial information (including mental rotation); the episodic buffer integrates information into coherent episodes and was the one added in 2000. The original 1974 model had three components.

### Working memory versus short-term memory ●●

Are working memory and short-term memory just two names for the same thing? Not quite. Short-term memory refers to the temporary storage of information, measured by simple span tasks: hear a list, repeat it back. Working memory refers to storage combined with processing, measured by complex span tasks. In the reading span task of Meredyth Daneman and Patricia Carpenter (1980), for example, people read a series of sentences aloud and then try to recall the last word of each one; they must store words while processing sentences. Simple digit span repeated forwards is the classic short-term memory test; digit span repeated backwards requires the person to hold the sequence and reverse it, so it is treated as a working memory test. Working memory capacity, measured by complex span, predicts reading comprehension, reasoning and fluid intelligence far better than simple span does (Chapter 9). In short, short-term memory is a component of the larger working memory system; the multi-store model's single box became Baddeley's set of interacting parts.

CLINICAL LINK

Working memory tasks are central to cognitive assessment. The Wechsler intelligence scales combine forward, backward and sequenced digit span into a working memory index, and the Wechsler Memory Scale, Fourth Edition (WMS-IV), reports five index scores: Auditory Memory, Visual Memory, Visual Working Memory, Immediate Memory and Delayed Memory. Working memory deficits are prominent in attention-deficit/hyperactivity disorder, schizophrenia and frontal lobe injury, and tests of executive function such as the Wisconsin Card Sorting Test tap the central executive. (Test details are in the Psychological Assessment book.)

## 7.4 Levels of Processing and the Quality of Encoding MEDIUM YIELD · 4 QS

The multi-store model said that what gets into long-term memory depends on how long an item is rehearsed in short-term memory. Think about your own studying for a moment. Have you ever read a page five times and remembered nothing, and heard a single striking story once and remembered it for years? If so, you already suspect that the amount of rehearsal is not the whole story. In 1972 **Fergus Craik** and **Robert Lockhart** turned that suspicion into a theory.

### Craik and Lockhart's framework (1972) ●●

Craik and Lockhart argued that memory is not a series of boxes but a by-product of the way information is processed. Incoming information can be analysed at different depths. At the shallowest level, structural (or orthographic) processing, we attend to physical appearance:

is the word printed in capital letters? At an intermediate level, phonemic processing, we attend to sound: does the word rhyme with "weight"? At the deepest level, semantic processing, we attend to meaning: is it a kind of fish? Would it fit the sentence "He met a ___ in the street"? According to the **levels of processing** framework, deeper processing produces more elaborate, longer-lasting and stronger memory traces. Memory is what is left behind by the analyses that perception and comprehension perform, so the crucial question is not "How long did you rehearse it?" but "What did you do with it?"

An important consequence is that the intention to learn matters very little. Thomas Hyde and James Jenkins (1973) gave people word lists with different orienting tasks: some rated each word for pleasantness (a semantic task), others checked whether the word contained the letters E or G (a structural task). Some participants in each condition were warned of a later memory test; others were not. Recall depended strongly on the orienting task and hardly at all on the warning: people who rated pleasantness without trying to learn remembered about as well as people who were deliberately trying to learn. What you do with information determines what you remember, whether or not you mean to remember it.

#### KEY STUDY: CRAIK AND TULVING (1975)

Fergus Craik and Endel Tulving presented words, each preceded by a question that required one of three levels of processing: a structural question ("Is the word in capital letters?"), a phonemic question ("Does the word rhyme with WEIGHT?") or a semantic question ("Would the word fit the sentence 'He met a ___ in the street'?"). Participants did not know that a memory test would follow. On a later surprise recognition test, memory rose steadily from structural to phonemic to semantic questions, with semantic processing producing performance more than three times as good as structural processing. Words that received a "yes" answer were remembered better than "no" words, especially at the semantic level, because a "yes" word fits into a meaningful structure.

Was depth just a disguise for time? Semantic questions take a little longer to answer. Craik and Tulving checked by devising a slow but shallow task (judging the pattern of consonants and vowels in a word). It took longer than the semantic task and still produced poorer memory. In a further experiment, words judged against complex sentence frames ("The great bird swooped down and carried off the struggling ___") were recalled about twice as well as words judged against simple frames ("She cooked the ___"): within the same level, more elaborate processing helped too.

### Maintenance versus elaborative rehearsal

Craik and Lockhart drew a distinction that every student should use. **maintenance rehearsal** (which they called Type I processing) is the simple repetition of information at the same level, like repeating a phone number until you dial it. It keeps information available in short-term memory but adds little to long-term memory. **elaborative rehearsal** (Type II processing) links the information to other knowledge: thinking about what it means, how it relates to what you already know, what example illustrates it. It is elaborative rehearsal that builds durable memories. Craik and Michael Watkins (1973) demonstrated the difference neatly. Participants heard lists of words and had to keep in mind the most recent word beginning with a target letter, say G, so that some G-words were held in mind for a long time and others only briefly. On a surprise final recall test, the time a word had spent being main-

tained made almost no difference to whether it was remembered. Repetition kept words in mind; it did not make them memorable.

Two further ideas extend the levels framework. Elaboration refers to the richness of processing within a level, as the complex sentences in Craik and Tulving's study showed. Distinctiveness refers to how different a trace is from other traces; a distinctive memory has fewer competitors at retrieval. Michael Eysenck and Christine Eysenck (1980) asked people to pronounce words with silent letters as they are spelled (the "b" in comb). This was shallow, phonemic processing, yet because it was distinctive it produced recognition as good as semantic processing. (The classic demonstration of distinctiveness, the von Restorff effect, is described in Section 7.8.)

### The self-reference, generation and dual-coding effects ●●

If semantic processing beats phonemic processing, is there anything deeper than meaning? In 1977 T. B. Rogers, Nicholas Kuiper and W. S. Kirker added a fourth orienting question to the classic three: "Does this word describe you?" Adjectives judged in relation to the self were recalled about twice as well as adjectives judged for meaning alone. This **self-reference effect**, the finding that information is remembered better when it is related to oneself, has been confirmed many times. The usual explanation is that the self is the richest, most organised and most often used body of knowledge each of us has, so linking a word to it provides both elaboration and organisation. Interestingly, in some collectivist cultures, relating words to one's mother produces almost as large an advantage as relating them to oneself, which suggests that what matters is an elaborate, well-organised knowledge structure rather than the self as such.

The **generation effect** is the finding that information you produce yourself is remembered better than information you merely read. Norman Slamecka and Peter Graf (1978) had people either read word pairs (HOT and COLD) or generate the second word from a rule and a cue (an opposite of HOT beginning with C). Generated words were remembered better. The lesson for study is direct: an answer you work out for yourself is worth more than an answer you highlight.

A third principle concerns images. **Allan Paivio** (1925 to 2016) proposed his **dual coding theory** in 1971: information can be represented in a verbal code and in an imagery code (in his later terms, by logogens and imagens). Concrete words such as "elephant" are easily coded both ways, so they have two routes to retrieval; abstract words such as "truth" are coded mainly verbally. Dual coding explains why concrete words are remembered better than abstract ones and why pictures are remembered better than words (the picture superiority effect), and it underlies the imagery mnemonics of Section 7.11.

EXAM TRAP

The self-reference effect was demonstrated by Rogers, Kuiper and Kirker (1977); exam keys often write "Rogers and Kuiper". It is not Craik and Lockhart's, although it extends their framework. The other names offered in the RML 2024 question have their own claims: Allan Paivio is dual coding theory; Michael Posner is known for attention research and the letter-matching task (Chapter 5). And the "self-reference effect" is a memory phenomenon; the "sum of a person's impressions of himself", "the beliefs we hold about what we are like" and "the knowledge that we exist as separate individuals" describe self-concept, self-schema and self-awareness (Social and Personality books).

### Criticisms and transfer-appropriate processing

The levels framework was enormously productive, but three criticisms stuck. The first is circularity. How do we know that a task involves deep processing? Because it produces good memory. Why does it produce good memory? Because it involves deep processing. Without an independent measure of depth, the theory cannot be tested properly; Alan Baddeley made this point forcefully in a 1978 critique. The second is that depth, elaboration and distinctiveness were never clearly separated or measured. The third, and most important, is that the framework looked only at encoding and ignored the conditions of retrieval.

The third criticism came with an experiment by Donald Morris, John Bransford and Jeffery Franks (1977). Participants answered semantic or rhyme questions about words, as in the levels experiments. Then half of them took a standard recognition test, and half took a rhyming recognition test, in which they had to pick out new words that rhymed with words they had studied. On the standard test, semantic processing won, as usual. On the rhyming test, rhyme processing won. Deep processing is not better in any absolute sense; what matters is the match between the processing done at study and the processing demanded by the test. Morris and colleagues called this **transfer-appropriate processing**. It is a close cousin of Tulving's encoding specificity principle (Section 7.7), and it explains why practising with past papers in exam format is so valuable: the processing you rehearse is the processing the exam will require.

REMEMBER

Levels of processing: Craik and Lockhart (1972). The three levels from shallow to deep: structural, phonemic, semantic. Best retention: deep, semantic, elaborate processing. Maintenance rehearsal keeps; elaborative rehearsal stores. Main criticism: circularity. Main rival: transfer-appropriate processing (Morris, Bransford and Franks, 1977).

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. The working memory model was proposed by Baddeley and _____ in 1974.
2. Repeating "the, the, the" while studying a list is called articulatory _____; it occupies the phonological loop.
3. Memory span is greater for short words than for long ones; this is the _____ effect.
4. Mentally rotating an object is a job for the visuospatial _____.
5. Random number generation is used to load the _____.
6. The component added to the working memory model in 2000 is the _____.
7. Repeating digits in reverse order (backward digit span) is treated as a test of _____ memory.
8. In Craik and Lockhart's framework, the three levels from shallow to deep are structural, phonemic and _____.
9. Simple repetition that keeps information in mind without linking it to meaning is _____ rehearsal.
10. Better memory when study processing matches test processing is called _____ processing.

**Answers:** (1) Hitch, (2) suppression, (3) word-length, (4) sketchpad, (5) central executive, (6) episodic buffer, (7) working, (8) semantic, (9) maintenance, (10) transfer-appropriate.

## 7.5 The Systems of Long-Term Memory HIGH YIELD · 15 QS

Clive Wearing could not tell you a single thing he had done the day before, yet he knew that he was a musician, knew what a piano was, and could play one. If long-term memory were a single store, this pattern would be impossible. The modern view, built from just such cases, is that long-term memory is a family of systems that store different kinds of information, operate by different rules and depend on different parts of the brain.

### Episodic, semantic and procedural memory ●●●

In 1972 the Estonian-born Canadian psychologist **Endel Tulving** (1927 to 2023) proposed a distinction that is now part of everyday psychology. **episodic memory** is memory for personally experienced events, each located in a particular time and place: what happened, where it happened and when. Remembering your first day of college, or what you ate for dinner last night, is episodic memory. **semantic memory** is general knowledge about the world: facts, concepts, word meanings and rules, held without any memory of when or where they were learned. Knowing that Mysuru is in Karnataka, that a sparrow is a bird and that "anterograde" means forward-acting is semantic memory. Semantic memory is often described as a vast network of associated facts and concepts that make up our knowledge of the world (Section 7.6). Because its contents are generalised across many experiences rather than tied to one, some textbooks call it **generic memory**.

Tulving later (1985) tied each kind of memory to a kind of consciousness. Episodic remembering involves **autonoetic consciousness** ("self-knowing"), the feeling of mentally travelling back in time and re-experiencing an event as one's own. Semantic memory involves noetic

consciousness ("knowing"), an awareness of facts without reliving anything. Procedural memory involves anoetic ("non-knowing") consciousness, with no awareness of the memory at all. Tulving turned this into a simple test, the remember-know procedure: when you recognise a word from a list, do you "remember" seeing it (you can recall the moment, what you thought) or do you just "know" it was there? Remember responses reflect episodic recollection; know responses reflect familiarity.

The third major kind of long-term memory is **procedural memory**, memory for how to do things: riding a bicycle, tying shoelaces, typing, playing the piano. The philosopher Gilbert Ryle (1949) had already distinguished "knowing that" from "knowing how", and in 1980 Neal Cohen and Larry Squire used the same contrast to separate declarative from procedural memory. Procedural memories are acquired gradually through practice, are expressed in performance rather than words, and are remarkably resistant both to forgetting and to brain damage that devastates episodic memory.

The strongest evidence that episodic and semantic memory are separable is again neuropsychological. Tulving's patient K.C. (Kent Cochrane, 1951 to 2014) suffered a motorcycle accident in 1981 that damaged his medial temporal lobes and other regions. Afterwards his general knowledge, language and intelligence were normal; he knew facts about his own life, such as the address of the family's summer cottage. Yet he could not recall a single event that he had personally experienced, at any point in his life. K.C. was the man who could not travel back in time. The opposite pattern appears in semantic dementia, a degenerative disease of the anterior temporal lobes in which patients progressively lose the meanings of words and concepts while their memory for recent personal events is relatively spared. Table 7.4 summarises Tulving's three kinds of long-term memory.

Table 7.4 Tulving's three kinds of long-term memory.

| Kind | Content | Type of awareness | Example | Key evidence |
|---|---|---|---|---|
| Episodic | Personal events, located in time and place | Autonoetic (re-experiencing) | Your first day of college | K.C. lost it; semantic spared |
| Semantic | General facts and concepts | Noetic (knowing) | Mysuru is in Karnataka | Lost in semantic dementia |
| Procedural | Skills and habits | Anoetic (no awareness) | Riding a bicycle | Spared in H.M. and Clive Wearing |

## Explicit and implicit memory ●●

A second way of dividing long-term memory cuts across Tulving's. **explicit memory** involves conscious recollection of information or of the situation in which it was learned; it is tested by recall and recognition. **implicit memory** is an influence of past experience on behaviour, feelings or thoughts without any conscious recollection of that experience. The terms were introduced by Peter Graf and **Daniel Schacter** in 1985. Larry Squire's preferred terms, declarative and nondeclarative memory, mean almost the same thing: declarative memory can be "declared" (stated in words); nondeclarative memory is expressed through performance.

Implicit memory was discovered through amnesia. In 1911 the Swiss psychologist Édouard Claparède hid a pin in his hand and pricked an amnesic patient as they shook hands. Next time, she refused to shake his hand, although she had no memory of the pinprick and could not say why; when pressed, she offered only that "sometimes pins are hidden in people's

hands". In the 1960s and 1970s Elizabeth Warrington and Lawrence Weiskrantz found that amnesic patients who could not recall or recognise a list of words nevertheless identified fragmented versions of those words more easily, and completed word stems with them, just as normal people did. Graf, Squire and George Mandler (1984) sharpened the contrast: given the stem MOT___, amnesic patients told to recall a studied word performed poorly, but told simply to write the first word that came to mind, they produced the studied word (MOTEL) as often as healthy participants. The same memory trace, tested two ways, was absent explicitly and intact implicitly.

H.M., the most studied amnesic patient (Section 7.10), showed the same dissociation for skills. Asked to trace a star while watching his hand only in a mirror, he improved steadily over three days of practice, just like anyone else, while denying each day that he had ever done the task before.

Squire gathered these findings into the taxonomy shown in Figure 7.5. Declarative memory (episodic and semantic) depends on the hippocampus and the surrounding medial temporal lobe. Nondeclarative memory is a collection of quite different abilities: procedural skills and habits (dependent on the striatum, part of the basal ganglia), priming (dependent on the neocortex), simple classical conditioning (the amygdala for emotional responses, the cerebellum for skeletal responses such as the conditioned eyeblink), and non-associative learning such as habituation and sensitisation (dependent on reflex pathways; Chapter 6).

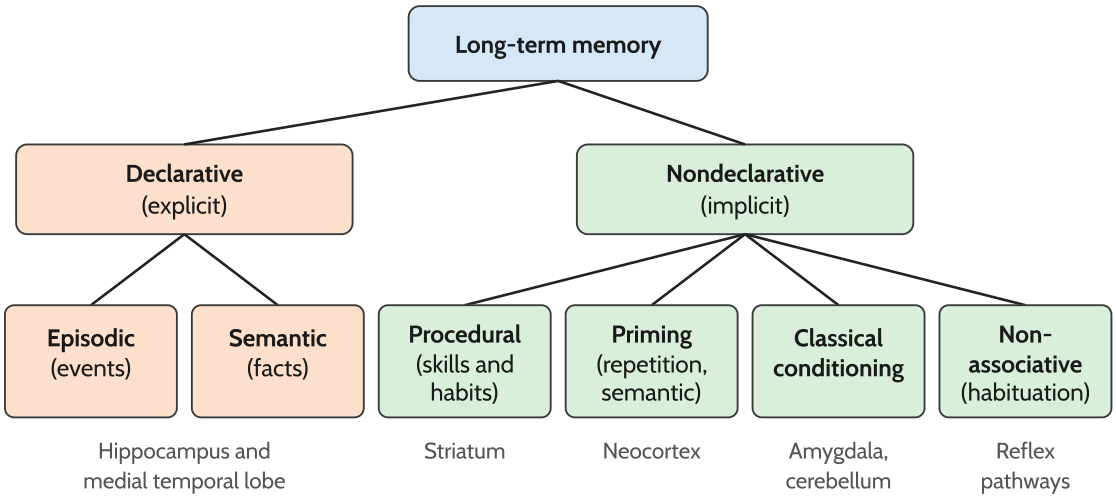



**Figure 7.5** Squire's taxonomy of long-term memory, with the brain structures most closely associated with each kind (bottom row). Amnesic patients lose the left branch and keep the right.

> **EXAM TRAP**
>
> Implicit memory does NOT involve conscious recollection; that is the definition of explicit memory. A "which statement does not accurately reflect these terms" question will usually hide this swap ("implicit memory involves a conscious recollection of the original events"). Also note the matched pairs: explicit = declarative = conscious recollection = recall and recognition; implicit = nondeclarative = performance without awareness = priming, skills, conditioning. Procedural memory is one kind of implicit memory, not a synonym for it.

## Priming ●●

**priming** is a change in the speed, accuracy or likelihood of processing a stimulus as a result of earlier exposure to the same or a related stimulus, without any intention to remember. In **repetition priming** (also called direct or perceptual priming), the earlier stimulus is the same item: having read "assassin" on a list, you are more likely to complete the fragment _ss_ss_n correctly later, even if you cannot recognise "assassin" as a word from the list (Tulving, Schacter and Stark, 1982). In **semantic priming**, the earlier stimulus is related in meaning. David Meyer and Roger Schvaneveldt (1971) asked people to decide as quickly as possible whether strings of letters were words (a lexical decision task). People recognised BUTTER as a word faster when it followed BREAD than when it followed NURSE. The prime activated a region of semantic memory, and activation spread to related concepts (Section 7.6).

Priming can occur even when the prime is not consciously seen. In masked priming, a word is flashed for a few milliseconds and immediately covered by a pattern mask, so that people cannot report it; yet it still speeds responses to related words. Anthony Marcel (1983) used this method to argue for semantic processing without awareness, and masked semantic priming became one of the main techniques for studying subliminal perception (Chapter 5).

> EXAM TRAP
>
> An RML 2022 question asks what "semantic priming" is used to study and keys "subliminal perception". Semantic priming is, first of all, a tool for studying the organisation of semantic memory and spreading activation. Its masked (subliminal) form is the standard way of showing that meaning can be processed without awareness, which is why the key chose subliminal perception from options that otherwise belong to other subjects (signal detection is psychophysics, field dependence is a cognitive style, locus of control is personality). If "semantic memory organisation" were offered, it would be the better answer.

## Autobiographical memory and the reminiscence bump ●●

**autobiographical memory** is memory for the events and facts of one's own life. It contains specific episodes (the day you received your Class 12 results) and personal semantic facts (the name of your school, your father's occupation), organised around the self and its goals. It is the memory that makes you a person with a history.

Francis Galton (1879) pioneered its study with the cue-word technique, revived by Herbert Crovitz and Harold Schiffman (1974): give a person a word such as "river" or "fire" and ask for the first specific personal memory it brings to mind, then ask how old they were at the time. When adults over about forty are tested this way and their memories are plotted against age at the time of the event, a characteristic curve appears. Hardly any memories come from the first three or four years (infantile amnesia, Section 7.10). Many come from the last few years (ordinary recency). And there is a surprising hump of extra memories from the ages of about 10 to 30: the **reminiscence bump**, described by David Rubin, Scott Wetzler and Robert Nebes (1986). Explanations include the many "firsts" of adolescence and early adulthood (first love, first job, leaving home), which are novel and distinctive; the formation of a stable identity in those years, so that events become anchors of the self; the peak of cognitive efficiency; and a cultural life script, which tells us that the major events of a life (marriage, career, children) happen in that period, so that we search our memories accordingly.

A handful of people have astonishingly complete autobiographical memories. The first described, A.J. (Elizabeth Parker, Larry Cahill and James McGaugh, 2006), could name the day of the week and the events of almost any date from her adolescence onwards. This highly superior autobiographical memory, or hyperthymesia, is not a general super-memory: such people are ordinary at memorising lists.

### Prospective versus retrospective memory

Almost everything so far concerns **retrospective memory**, remembering information or events from the past. **prospective memory** is remembering to carry out an intended action in the future: to take a tablet at 8 p.m., to buy milk on the way home, to pass on a message when you see a friend. Gilles Einstein and Mark McDaniel (1990) distinguished event-based prospective memory, triggered by an external cue (seeing your friend reminds you of the message), from time-based prospective memory, which must be self-initiated at the right time (remembering at 8 p.m. that it is time for the tablet). Time-based tasks are generally harder because nothing in the environment prompts them. Prospective memory failures are the commonest memory complaints of everyday life, and in clinical work they matter enormously: a patient who forgets to take medication is a patient whose treatment fails.

### Flashbulb memories ●●●

Many people in India can tell you exactly where they were when they heard about the attacks on Mumbai in November 2008, or when they watched the final of the 2011 Cricket World Cup. In 1977 **Roger Brown** (1925 to 1997) and James Kulik named such memories **flashbulb memory**: vivid, detailed and long-lasting memories of the circumstances in which a person learned of a surprising, emotionally significant event. Their main example was the assassination of President John F. Kennedy in 1963. People typically recalled where they were, what they were doing, who told them, how they felt, how others felt and what happened next. Brown and Kulik proposed that a special neural mechanism, which they called "Now Print!" (borrowing the phrase from the neuroscientist Robert Livingston), was triggered by events that were surprising and consequential and fixed the whole scene like a photograph.

#### KEY STUDY: NEISSER AND HARSCH AND THE CHALLENGER DISASTER (1992)

On 28 January 1986 the space shuttle Challenger exploded shortly after launch. The next morning, Ulric Neisser and Nicole Harsch asked students at Emory University to write down how they had heard the news. About two and a half years later they found 44 of these students and asked them the same questions. The memories had changed dramatically. A typical student who had first written that she heard the news in a religion class, when people started talking, later "remembered" that she had been in her dormitory with her roommate when the news came on television. About a quarter of the students were wrong about every major detail, and few were right about all of them. Yet confidence ratings were high. Some students, shown their own handwritten accounts from the morning after, insisted that their current memory must be the correct one.

The same pattern appeared after the attacks of 11 September 2001. On 12 September, Jennifer Talarico and David Rubin (2003) asked 54 students at Duke University to record how they had heard the news, and also to record an ordinary memory from the preceding days, such as a party or a study session. They retested each student 1, 6 or 32 weeks later. Both kinds of mem-

ory lost consistency at the same rate: details changed, dropped out or were added. What differed was the experience of remembering. Flashbulb memories stayed far more vivid, and students' belief in their accuracy stayed high, while their accuracy fell. Kathy Pezdek (2003) found that 73 per cent of American students said they had seen the television footage of the first plane hitting the World Trade Center on 11 September itself; no such footage was broadcast that day.

So flashbulb memories are special in their vividness and in the confidence they inspire, not in their accuracy. No special "Now Print!" mechanism is needed to explain them. Two ordinary processes do the work: emotional arousal at encoding (which, through the amygdala and stress hormones, strengthens the central features of an experience) and repeated rehearsal and retelling afterwards, through conversation and news coverage, which keeps the story alive while slowly reshaping it.

> EXAM TRAP
>
> Flashbulb memories: Brown and Kulik (1977). Vividness and confidence are very high and stay high; accuracy and consistency decline like those of ordinary memories (Neisser and Harsch, 1992; Talarico and Rubin, 2003, whose title was "Confidence, not consistency, characterizes flashbulb memories"). The current explanation is emotional arousal plus rehearsal, not a special Now Print! mechanism, not freedom from interference and not eidetic imagery. Do not confuse "flashbulb memory" (an emotional autobiographical memory) with "eidetic imagery" (a lingering visual image) or with an "engram" (the physical memory trace, Section 7.10).

### Eidetic imagery and exceptional memory

Is there such a thing as photographic memory? The nearest thing that has been reliably demonstrated is **eidetic imagery**: a visual image that persists for half a minute or more after a picture is removed, vivid enough that the person can scan it with their eyes and describe details they did not notice while the picture was present. Ralph Haber and Ruth Haber (1964) found it in a small minority of primary school children (estimates range from about 2 to 10 per cent); it is very rare in adults. Even eidetic images are not photographs: they fade, they contain errors, and the children who have them do not have better memories in general.

The most famous mnemonist in psychology, the Russian journalist Solomon Shereshevsky, studied for thirty years by **Alexander Romanovich Luria** (1902 to 1977) and described in *The Mind of a Mnemonist* (1968), could reproduce long tables of numbers and lists of words, forwards or backwards, even years later. He did it not with an eidetic image but with intense synaesthesia (words evoked colours, tastes and textures) and by placing images along an imagined street, the method of loci (Section 7.11). His problem was the opposite of ours: he found it hard to forget, and hard to grasp the general meaning of a passage through the crowd of images it evoked.

### Metamemory: feeling of knowing and the tip of the tongue •••

**metamemory** is knowledge about one's own memory and the ability to monitor and control it: knowing that you remember lists better if you organise them, judging whether you have learned a chapter well enough, sensing that you know an answer you cannot yet retrieve. The term was introduced by **John Flavell** (1928 to 2025) in 1971, and metamemory is part of his

broader concept of metacognition, thinking about one's own thinking. A person who is very good at planning, monitoring and controlling every phase of the memory system, from what to attend to, to how to rehearse, to how to search memory, is showing good metacognition.

Metamemory has two sides: monitoring (judging the state of one's memory) and control (acting on that judgement, for example by restudying what feels weak). Monitoring is often surprisingly accurate. J. T. Hart (1965) asked people general-knowledge questions; when they could not recall an answer, he asked them to rate their **feeling of knowing**, that is, how likely they were to recognise it among alternatives. Their ratings predicted later recognition well above chance. But monitoring can also mislead: rereading a chapter makes it feel familiar and fluent, which students mistake for having learned it, an illusion of competence that explains why rereading is such a poor study method (Section 7.11).

KEY STUDY: BROWN AND MCNEILL, THE TIP OF THE TONGUE (1966)

William James described it in 1890: a gap that is "intensely active", in which the sought name beckons and then retreats. Roger Brown and David McNeill set out to catch the **tip-of-the-tongue state** in the laboratory. They read out dictionary definitions of rare words, such as "a navigational instrument used in measuring angular distances, especially the altitude of the sun, moon and stars at sea" (sextant). Sometimes participants knew the word at once and sometimes they had no idea, but quite often they were in agony: sure they knew it, unable to produce it. In those states they could guess the first letter and the number of syllables well above chance, and they produced words with similar sounds (secant, sextet) and similar meanings (astrolabe, compass). The memory was clearly stored, and partly accessible; the failure was one of retrieval.

Tip-of-the-tongue states become more frequent with age, are more common for proper names than for other words, and are more frequent in bilinguals. They are the best everyday evidence that much forgetting is a failure of retrieval rather than of storage, a theme taken up in Sections 7.7 and 7.8. They are not aphasia: in Wernicke's aphasia (fluent but meaningless speech with poor comprehension) and Broca's aphasia (laboured, non-fluent speech), language itself is damaged (Physiological book), whereas the tip-of-the-tongue state is a temporary failure to retrieve one well-known word.

## 7.6 How Knowledge Is Organised: Networks, Schemas and Scripts

MEDIUM YIELD · 4 QS

Semantic memory holds an enormous amount of knowledge, yet you can answer "Is a canary a bird?" in about a second. That speed is only possible if knowledge is organised. How? Psychologists have proposed several kinds of organisation, from tidy hierarchies to sprawling webs to patterns of connection strength spread across a whole network.

### The hierarchical network model (Collins and Quillian, 1969)

Allan Collins and Ross Quillian (1969) proposed that semantic memory is a network of nodes, each representing a concept, linked into a hierarchy (Figure 7.6). Properties are attached to the nodes they belong to. The **hierarchical network model** adds one more assumption, the principle of **cognitive economy**: each property is stored only once, at the highest level in the

hierarchy to which it applies. "Has skin" is stored with ANIMAL, not separately with every animal; "can fly" is stored with BIRD; "can sing" with CANARY. Exceptions are stored at the lower node ("cannot fly" with OSTRICH).

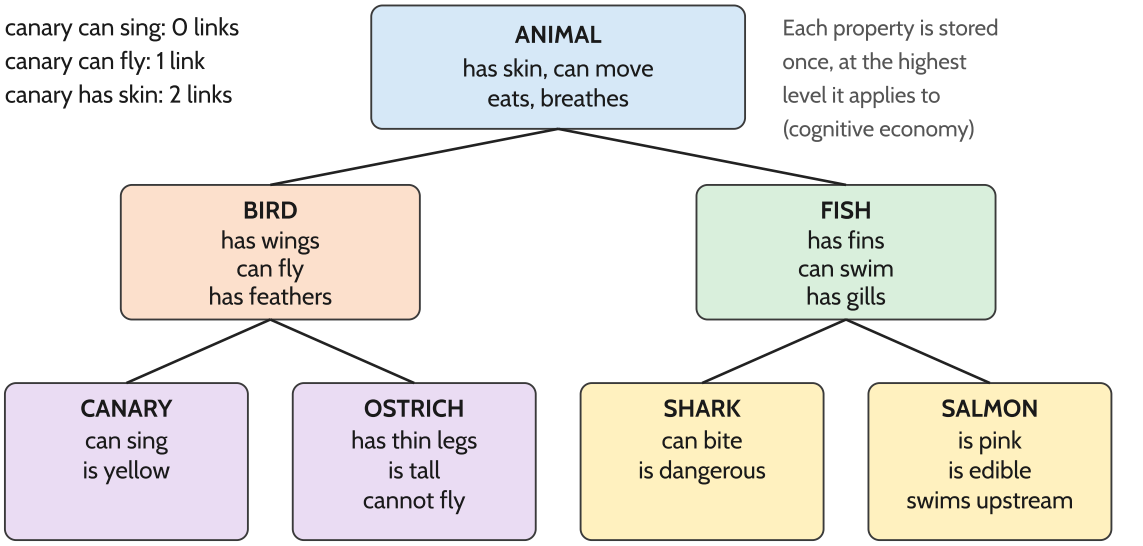



Figure 7.6 Part of the hierarchical network of Collins and Quillian (1969). To verify "A canary has skin", the search must travel up two links to ANIMAL, so it takes longer than verifying "A canary can sing", which is stored at the CANARY node itself.

The model makes a precise prediction. To verify a sentence, you must find the relevant property by travelling along the links, and each link takes time. Collins and Quillian measured how long people took to verify sentences in a sentence verification task. "A canary can sing" (property at the same node, zero links) was verified fastest, "A canary can fly" (one link up) more slowly, and "A canary has skin" (two links up) more slowly still. Category sentences behaved the same way: "A canary is a bird" was faster than "A canary is an animal". Each level added a roughly constant increment of time, just as the model predicted.

But the model ran into trouble. First, the typicality effect: Lance Rips, Edward Shoben and Edward Smith (1973) found that "A robin is a bird" is verified faster than "A penguin is a bird", although both are one link apart. Some members of a category are better examples than others (the idea of prototypes, developed by Eleanor Rosch, is taught in Chapter 8). Second, the hierarchy is sometimes violated: a sentence such as "A dog is an animal" is often verified faster than "A dog is a mammal", although "mammal" should be one link closer. Third, Carol Conrad (1972) showed that the apparent effect of hierarchical distance was partly an effect of familiarity: properties such as "can sing" are more often thought about in connection with canaries than "has skin" is. These problems led Smith, Shoben and Rips (1974) to propose a feature comparison model, in which concepts are lists of defining and characteristic features and verification compares feature lists, and led Collins to a more flexible network.

### Spreading activation (Collins and Loftus, 1975)

In 1975 Allan Collins and **Elizabeth Loftus** (born 1944) abandoned the strict hierarchy. In their **spreading activation** model, semantic memory is a web of concept nodes, and the length of each link represents how closely the two concepts are related: FIRE ENGINE is close

to RED and TRUCK, further from CHERRIES. When a concept is processed, activation spreads outwards along its links, weakening with distance and over time, and partially activates related concepts. A concept that is already partly active is easier to retrieve. This explains semantic priming at once: reading BREAD sends activation to BUTTER, which is then recognised faster. It also explains typicality, because typical members (robin) are closely linked to the category node (BIRD) and atypical members (penguin) are distant. The model's flexibility is also its weakness: with link lengths set to fit the data, it is hard to imagine a result it could not explain.

### Propositions and connectionist networks ●●

Knowledge is not only a set of concepts; it is a set of relations among them. A **proposition** is the smallest unit of knowledge that can be judged true or false, such as "Ravi studies memory" or "the exam is in October". Propositional network models, such as John R. Anderson's adaptive control of thought (ACT) theory and its successors, represent knowledge as networks of interlinked propositions. Propositions are what we keep when we keep the gist of a sentence and lose its words, as in Sachs's study. (In a question about the "meaningful units" that define working memory capacity, however, the answer is chunks, not propositions, features or scripts.)

A more radical alternative came in 1986, when David Rumelhart, James McClelland and the PDP Research Group published *Parallel Distributed Processing*. **connectionism**, also called parallel distributed processing (PDP), models memory as a network of simple, neuron-like units (nodes) connected by links of varying strength, or weight. Unlike the earlier semantic networks, a concept is not stored in one node. It is a pattern of activation distributed across many units, and the knowledge of the network lies in its connection weights. Processing is parallel: many units activate one another at the same time. Learning consists of adjusting the weights after experience, a modern version of Donald Hebb's (1949) idea that connections between neurons that fire together are strengthened. Connectionist networks have three properties that resemble human memory. They are content-addressable (any part of a memory can retrieve the rest, as a smell can bring back a whole day). They degrade gracefully (damage to some units blurs memories rather than deleting particular ones, rather like the effect of diffuse brain damage, and recalling Karl Lashley's search for the engram, Section 7.10). And they generalise automatically from examples to new cases.

> **EXAM TRAP**
>
> "Which model proposes that memory is organised in a network of interconnected nodes?" (IHBAS 2024) keys the connectionist model, although "network model" is also offered. Both Collins and Quillian's semantic network and connectionist models use nodes and links. The distinguishing features of connectionism (parallel distributed processing, Rumelhart and McClelland, 1986) are distributed representation, simultaneous (parallel) activation of many interconnected nodes, and knowledge stored in connection weights. When a question stresses "parallel", "distributed", "simultaneous activation" or "neural network", choose connectionist or PDP; when it stresses a hierarchy, cognitive economy or sentence verification time, choose Collins and Quillian's hierarchical network model; when it stresses activation spreading to related concepts, choose Collins and Loftus.

## Schemas and scripts ●●

Read this passage once: "The procedure is actually quite simple. First you arrange things into different groups. Of course, one pile may be sufficient depending on how much there is to do. If you have to go somewhere else due to lack of facilities, that is the next step..." John Bransford and Marcia Johnson (1972) gave passages like this to students. Without a title it is nearly incomprehensible and is poorly recalled. Told beforehand that it is about washing clothes, students found it easy to understand and recalled about twice as much. (A title given after reading did not help.) The title activated a schema.

A **schema** is an organised body of knowledge, built from experience, about a concept, object, person, event or situation, which guides how new information is perceived, interpreted, encoded and remembered. The term entered memory research through Frederic Bartlett (1932; Section 7.9). Schemas are economical: they let us fill gaps, make predictions and understand quickly. The price is distortion: we remember what fits the schema, add what the schema leads us to expect, and drop or alter what does not fit. William Brewer and James Treyens (1981) asked students to wait for about half a minute in a room they were told was a graduate student's office. Later asked to recall the contents of the room, many "remembered" books, which an office ought to contain but this one did not, while missing unusual objects that did not fit the office schema. The "schema approach" to memory and social understanding is simply the study of how such organised knowledge shapes what we notice and recall.

A **script** is a schema for a familiar, routine sequence of events. The concept was introduced by Roger Schank and Robert Abelson in 1977, whose best-known example is the restaurant script: enter, wait to be seated, read the menu, order, eat, pay, leave. An Indian example is the script for a wedding or for taking a train journey: book the ticket, check the platform, find your coach and berth, show the ticket to the conductor. Gordon Bower, John Black and Terrence Turner (1979) found that when people read stories about routine activities, they later "remembered" script actions that had not been mentioned. Scripts belong to generic (semantic) memory: they are abstracted from many similar episodes, which is why the developmental psychologist Katherine Nelson found that young children can tell you what happens at a birthday party in general well before they can recount a particular party.

> **EXAM TRAP**
>
> "Which type of memory produces scripts of familiar routines?" Generic memory (RML 2025), the same thing as general, semantic knowledge abstracted from many episodes. Episodic and autobiographical memory hold particular episodes; a script is precisely what is left when the particulars are stripped away. And "how people organise information in memory and use it to understand behaviour" describes the schema approach. Schema theory is Bartlett's (1932), not Wundt's, Titchener's or Watson's.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Tulving distinguished episodic memory from _____ memory in 1972.
2. The re-experiencing that accompanies episodic recall is called _____ consciousness.
3. Tulving's patient _____ lost all episodic memory for his life but kept his semantic knowledge.
4. Memory without conscious recollection, such as priming, is called _____ memory.
5. Recognising BUTTER faster after BREAD than after NURSE is an example of _____ priming.
6. The excess of memories from ages 10 to 30 in older adults is the _____.
7. Remembering to take a tablet at 8 p.m. is _____ memory.
8. Flashbulb memories were named by Brown and _____ in 1977; their accuracy declines, but confidence stays high.
9. Brown and McNeill studied the _____ state, which shows that the failure is one of retrieval.
10. In Collins and Quillian's model, storing each property only at the highest applicable node is called _____.

**Answers:** (1) semantic, (2) autonoetic, (3) K.C., (4) implicit, (5) semantic, (6) reminiscence bump, (7) prospective, (8) Kulik, (9) tip-of-the-tongue, (10) cognitive economy.

## 7.7 Retrieval: Cues, Context and State

MEDIUM YIELD · 3 QS

You walk into the kitchen and cannot remember what you came for. You go back to your desk, and the moment you sit down, it comes back: the glass of water. The memory was there all along; what changed was the setting in which you searched for it. This section is about the conditions that make stored information accessible.

### Availability, accessibility and retrieval cues

Endel Tulving and Zena Pearlstone (1966) gave people long lists of words drawn from several categories (animals, fruits, professions and so on). At test, one group simply recalled as many words as they could; the other group was given the category names as prompts. The prompted group recalled far more words. The words the first group failed to recall had not been lost: they were available (stored) but not accessible (retrievable without help). A **retrieval cue** is any stimulus, internal or external, that helps us gain access to a stored memory: a category name, a smell, a place, a question, a mood.

The availability-accessibility distinction also explains why recognition is usually easier than recall. According to the generate-recognise (two-stage) account, recall involves two steps: generating possible candidates from memory, then recognising which of them are correct. A recognition test supplies the candidates, so only the second step is needed. A multiple-choice question is easier than the same question with blanks because the options themselves act as powerful retrieval cues.

### The encoding specificity principle ●●

Which cues work best? In 1973 Endel Tulving and Donald Thomson gave the answer that now organises the whole field. The **encoding specificity principle** states that a retrieval cue is effective to the extent that the information it contains was encoded together with the memory at the time of learning. Memory traces are not stored in isolation; they are stored together with the context, thoughts and associations present at encoding, and retrieval succeeds when the cue reinstates enough of that encoded package.

Tulving and Thomson's evidence was striking. Participants studied target words paired with weak associates (glue and CHAIR). At test, the weak associate (glue) was a better cue for CHAIR than a strong associate such as "table", even though "table" is far more closely linked to "chair" in general knowledge, because "table" had not been part of the encoded event. In a further twist, participants sometimes failed to recognise a target word when it was presented on its own, yet recalled it moments later when given its weak study partner. This recognition failure of recallable words embarrasses the generate-recognise account, which predicts that anything recallable must also be recognisable, and it shows how strongly memory depends on the match between encoding and retrieval. Transfer-appropriate processing (Section 7.4) is the same idea applied to types of processing.

### Context-dependent memory ●●

The external environment is part of what gets encoded. **context-dependent memory** is the finding that recall is better when the physical context at retrieval matches the context at encoding.

#### KEY STUDY: GODDEN AND BADDELEY'S DIVERS (1975)

Duncan Godden and Alan Baddeley worked with members of a university diving club. The divers listened to lists of unrelated words either on the beach or several metres under the sea (accounts of the exact depth differ), and later recalled them either in the same environment or in the other one. Words learned underwater were best recalled underwater; words learned on land were best recalled on land. Changing the context reduced recall by roughly a third. The study has become the classic demonstration of context-dependent memory. Five years later, however, the same researchers found that changing the environment had no effect on recognition (Godden and Baddeley, 1980). Context helps mainly when we must generate the memory ourselves; when the test provides the item, the item itself is a strong enough cue.

Context effects in everyday settings are real but often small, because rich internal cues (the meaning of the question) usually outweigh the room. One study of real university examinations found no difference between students tested in their usual lecture room and those tested elsewhere. A useful finding for exam candidates is that the context does not have to be physically present: Steven Smith (1979) found that simply imagining the room in which material was learned restored much of the benefit. Mentally reinstating the context is also the first technique of the cognitive interview used with eyewitnesses (Section 7.9).

## State-dependent memory ●●

The internal environment matters too. **state-dependent memory** is the finding that recall is better when a person's physiological or drug-induced state at retrieval matches their state at encoding. Donald Goodwin and his colleagues (1969) had volunteers learn material either sober or after drinking alcohol, and test it either sober or intoxicated. As expected, alcohol at learning impaired memory overall; but material learned while intoxicated was recalled better when the person was intoxicated again than when sober. Clinical reports describe heavy drinkers who hide money or bottles while drunk, cannot find them when sober, and remember the hiding place when they next get drunk. Similar effects have been found with marijuana, nicotine and other drugs. As with context, state-dependent effects are strongest in free recall and weak or absent in cued recall and recognition, where better cues are available (Eric Eich, 1980).

A caution about alcohol "blackouts": heavy intoxication can block the formation of new long-term memories altogether (an alcohol-induced anterograde amnesia), and no change of state can bring back what was never stored. Only memories that were encoded but became inaccessible on sobering up show the state-dependent pattern.

## Mood-dependent and mood-congruent memory

Mood is an internal state as well, and it produces two different effects that students often confuse. In mood-dependent memory, recall is better when one's mood at retrieval matches one's mood at encoding, whatever the emotional tone of the material. In **mood-congruent memory**, people tend to learn and recall material whose emotional tone matches their current mood: when sad, sad memories come to mind more easily; when happy, happy ones. **Gordon Bower** (1932 to 2020) reported both effects in 1981, using hypnosis to induce happy or sad moods; later studies found mood-congruent memory to be the more robust of the two. Table 7.5 sets all of these matching effects side by side.

CLINICAL LINK

Mood-congruent memory helps explain why depression feeds itself. A depressed person recalls failures, losses and rejections more readily than successes, which deepens the low mood, which makes negative memories still more accessible. Depressed patients also tend to produce overgeneral autobiographical memories ("I always mess things up") instead of specific episodes when given cue words, a pattern linked to poorer recovery. Cognitive therapy's insistence on examining specific situations and evidence is, among other things, a way of breaking this retrieval cycle (Clinical book).

Table 7.5 The match between encoding and retrieval: five related ideas.

| Idea | What must match | Classic study |
|---|---|---|
| Encoding specificity principle | Any cue present at encoding (the general principle) | Tulving and Thomson (1973) |
| Context-dependent memory | The external, physical setting | Godden and Baddeley (1975), divers |
| State-dependent memory | The internal physiological or drug state | Goodwin and colleagues (1969), alcohol |
| Mood-dependent memory | The mood at encoding and at retrieval | Bower (1981) |

| Idea | What must match | Classic study |
|---|---|---|
| Mood-congruent memory | The current mood and the emotional tone of the material | Bower (1981); depression research |

EXAM TRAP

Rahul does better in the exam held in his usual classroom than in the one held in a new building (IHBAS 2024). The keyed answer is the principle of encoding specificity, because "context-dependent memory" was not among the options; context-dependent memory is the specific case and encoding specificity the general principle, so either is correct depending on what is offered. "The principle of congruity" is a distractor; the nearest real memory phenomenon is mood-congruent memory, which concerns the match between mood and material, not between rooms. If a question asks which phenomenon says recall is better when retrieval context matches encoding context and offers both context-dependent and state-dependent memory, choose context-dependent for external surroundings and state-dependent for internal states such as alcohol. Alcohol-induced amnesia that lifts on re-intoxication is the classic example of state-dependent memory.

## 7.8 Forgetting HIGH YIELD · 10 QS

Ebbinghaus showed how fast we forget; he did not show why. Psychologists have offered five main explanations: decay, interference, retrieval failure, motivated forgetting and consolidation failure. Before we examine them, one point is worth making. Forgetting is not simply a defect. A memory that kept everything would drown the important in the trivial, as Shereshevsky found (Section 7.5). By letting unused information fade, forgetting clears the way for current information, reduces interference and helps us abstract general knowledge from particular cases. Retention and forgetting are two sides of one coin: the more that is retained, the less that is forgotten, never the reverse.

### Decay (trace) theory ●●

The oldest and most intuitive explanation is **decay theory**, also called trace decay theory: learning leaves a physical trace (an engram) in the nervous system, and if the trace is not used, it fades with the passage of time, like a footpath that grows over when nobody walks on it. In psychology, the idea is traditionally credited to **Edward Lee Thorndike** (1874 to 1949), whose law of disuse (the companion of his law of use, together forming the law of exercise; Chapter 6) held that connections not exercised grow weaker. Decay fits the shape of Ebbinghaus's curve and the rapid loss from sensory memory.

The trouble is that time never passes empty. While time passes, other things are learned. In 1932 John McGeoch made the classic objection: iron rusts over time, but time does not cause rust; oxidation does. In the same way, forgetting happens over time, but what causes it might be the events that fill the time. The decisive test is to let time pass while reducing those events, and the classic attempt is one of the most quoted studies in memory research.

**KEY STUDY: JENKINS AND DALLENBACH, SLEEP VERSUS WAKING (1924)**

John Jenkins and Karl Dallenbach had two students learn lists of ten nonsense syllables, either late at night just before going to sleep or in the morning before their usual activities. Recall was tested after 1, 2, 4 or 8 hours. After a night's sleep, forgetting was modest and levelled off after about two hours; after eight hours, more than half of the syllables were still recalled. After the same time spent awake, forgetting continued steadily, and after eight hours only about one syllable in ten survived. Since the same amount of time had passed in both conditions, decay could not explain the difference. Jenkins and Dallenbach concluded that forgetting depends less on the passage of time than on "the interference, inhibition or obliteration of the old by the new".

Later studies pointed the same way. Cockroaches that were immobilised after learning forgot less than those left free to move about (Minami and Dallenbach, 1946). Baddeley and Hitch (1977) asked rugby players to recall the teams they had played that season; how much they forgot depended on how many games they had played since, not on how much time had passed. One qualification is important: we now know that sleep does more than protect memories from interference; it actively helps to consolidate them (Section 7.10).

**EXAM TRAP**

"Who developed the decay theory?" The RML 2024 key is Thorndike, through his law of disuse. Ebbinghaus described the time course of forgetting but did not propose decay as its cause; Tulving is cue-dependent forgetting; Melton and Irwin are the two-factor theory of interference. In short-term memory, the Brown-Peterson studies were first read as evidence for decay and then reinterpreted as interference (Section 7.2).

## Interference theory ●●●

**interference theory** holds that forgetting occurs because other memories compete with or disrupt the one we are trying to retrieve. John Bergström reported the first experiments on interference in the early 1890s, using a card-sorting task in which learning new sorting locations disrupted the old ones. In 1900 Georg Elias Müller and Alfons Pilzecker showed that an activity interpolated between learning and recall impaired memory, and named the effect retroactive inhibition. In the 1930s McGeoch made interference the leading theory of forgetting. There are two directions of interference, and they are among the most examined ideas in this chapter.

**proactive interference** (PI) occurs when earlier learning disrupts the learning or recall of newer information: the old interferes forward with the new. **retroactive interference** (RI) occurs when newer learning disrupts the recall of older information: the new interferes backward with the old. A simple way to keep them straight: PRO means forward, so in proactive interference old memories push forward and disrupt new ones; RETRO means backward, so in retroactive interference new memories reach back and disrupt old ones. In four words: PRO = old disrupts new; RETRO = new disrupts old.

Work through these cases. You change your mobile number and keep giving people your old one: the old number is disrupting the new one, so this is proactive interference. Years later, you cannot remember your old number at all, because the new one has replaced it: retroactive interference. A student studies psychology, then sociology, and finds the psychology getting in the way when she is tested on sociology: the earlier learning disrupts the later, so

proactive interference. If instead she is tested on psychology and finds sociology concepts intruding, that is retroactive interference. An English speaker learning Spanish who keeps applying English grammar is suffering from the earlier language disrupting the newer one: proactive interference.

In the laboratory, interference is studied with paired-associate lists in which the same stimulus terms are paired with different responses: list A-B (for example, DOG-table) followed by list A-C (DOG-river). Table 7.6 shows the two designs. Notice that the difference between them lies only in which list is tested.

Table 7.6 Experimental designs for retroactive and proactive interference.

| Design | Group | Stage 1 | Stage 2 | Test |
|---|---|---|---|---|
| Retroactive | Experimental | Learn A-B | Learn A-C | Recall A-B |
| Retroactive | Control | Learn A-B | Rest (unrelated activity) | Recall A-B |
| Proactive | Experimental | Learn A-B | Learn A-C | Recall A-C |
| Proactive | Control | Rest (unrelated activity) | Learn A-C | Recall A-C |

Three classic findings fill in the theory. First, interference increases with similarity: John McGeoch and William McDonald (1931) found that an interpolated list of synonyms of the original words produced much more retroactive interference than unrelated words or numbers. Second, proactive interference is the larger cause of forgetting in the laboratory. Benton Underwood (1957) noticed that participants in experiments who had already learned many lists forgot a new list far more quickly over 24 hours than naive participants, and that the forgetting rose steeply with the number of earlier lists. Ebbinghaus, who had learned thousands of lists, was the extreme case: his own forgetting curve was probably steeper than that of a naive learner. Third, interference has two components. Arthur Melton and J. M. Irwin (1940) proposed a two-factor theory: response competition (the A-B and A-C responses compete at recall) and unlearning of the first list during learning of the second, which they called "factor X". Jean Barnes and Underwood (1959) showed that the first-list responses are indeed weakened during second-list learning, and later work showed that they partly recover with time, rather like spontaneous recovery after extinction in conditioning (Chapter 6).

> **EXAM TRAP**
>
> The RML 2024 question about an English speaker applying English grammar while learning Spanish keys "retroactive interference". That key is wrong. English was learned first and is disrupting the newer learning of Spanish: that is proactive interference by definition, and every standard textbook would classify it so. (Strictly, a learning difficulty caused by earlier habits is also called negative transfer, Chapter 6.) Know the correct answer, and know that this key exists. The reliable test: ask which was learned first and which is being disrupted. Old disrupting new is proactive; new disrupting old is retroactive. "Difficulty recalling information due to interference from new information" (IHBAS 2024) is retroactive interference.

## Retrieval failure and retrieval-induced forgetting ●●

The tip-of-the-tongue state, Tulving and Pearlstone's category cues, and the encoding specificity principle all point to a third explanation: much forgetting is a failure to find informa-

tion that is still stored. Tulving (1974) called this **cue-dependent forgetting** and contrasted it with trace-dependent forgetting, in which the trace itself has been lost. Cue-dependent forgetting happens when the cues available at retrieval do not match those encoded with the memory. On this view, much of what we call forgetting is temporary, which is why a smell, a song or a return to a childhood home can bring back memories we believed long gone.

Retrieval can also cause forgetting. Michael Anderson, Robert Bjork and Elizabeth Bjork (1994) had people study category-word pairs (FRUIT-orange, FRUIT-banana, DRINK-vodka) and then practise retrieving half of the words from half of the categories (FRUIT-or___). On a final test, the practised words were, unsurprisingly, recalled best. The striking result concerned the unpractised words from practised categories (FRUIT-banana): they were recalled worse than words from categories that had not been practised at all. Retrieving some members of a category had suppressed their competitors. This is **retrieval-induced forgetting**, and it is thought to reflect inhibition of competing memories during retrieval. It has a practical sting: revising only some topics from a chapter may make the unrevised topics from the same chapter harder to recall than if you had revised none of it.

> EXAM TRAP
>
> An RML 2020 question defines retrieval-induced forgetting as "people remembered practised information better than unpractised information". That is the first half of the finding. The defining half is that unpractised items related to the practised ones are remembered worse than baseline items from unpractised categories: retrieving some memories makes related memories harder to retrieve. Choose the keyed option when it is the only one about practice, but be ready for a better-worded version. The options about names and faces are irrelevant distractors.

### Motivated forgetting and repression ●●

Sigmund Freud proposed that some forgetting is motivated: memories that would provoke anxiety, guilt or shame are actively kept out of consciousness. **repression** is the unconscious process by which threatening or anxiety-provoking memories and impulses are excluded from awareness; according to Freud, the repressed material is not lost but continues to exert effects, for example in symptoms and slips of the tongue. Repression must be distinguished from suppression, the deliberate, conscious decision not to think about something. (Freud's other defence mechanisms and his model of conscious, preconscious and unconscious mind belong to the Personality book.)

Laboratory evidence for repression in Freud's sense is weak; after reviewing decades of studies, David Holmes (1990) concluded that there was no convincing controlled evidence for it. Evidence for conscious, motivated forgetting is much better. In directed forgetting experiments, people told to forget some items remember them less well than items they were told to remember. In the think/no-think paradigm of Michael Anderson and Collin Green (2001), people learned word pairs and then, when shown the first word of some pairs, were told to prevent the associated word from coming to mind; the more often they did so, the worse they later recalled those words, even when offered money for correct answers. Brain imaging shows the prefrontal cortex reducing hippocampal activity during such suppression. Motivated forgetting is real; whether the unconscious repression Freud described is real re-

mains disputed, and the dispute became fierce in the recovered memory controversy (Section 7.9).

> **EXAM TRAP**
>
> Repression is unconscious; suppression is conscious. "Very threatening material is unable to gain access to conscious awareness" describes repression. "Motivated forgetting" is the broader category that includes both. Do not confuse Freudian repression with the cognitive term "inhibition", which refers to the active weakening of competing memories (as in retrieval-induced forgetting).

### Consolidation failure

The final explanation looks at the biology. New memories are fragile; they become stable only gradually through **consolidation**, the process by which a memory trace is transformed into a durable form. Müller and Pilzecker introduced the term in 1900, noting that their participants' new associations seemed to "perseverate" and set over time. If consolidation is disrupted, the memory is never fully formed. A blow to the head is the everyday case. American football players who had been concussed could often tell a trainer, immediately afterwards, what play they had been running; a few minutes later they could not, because the memory had not consolidated (Lynch and Yarnell, 1973). Electroconvulsive shock given to rats soon after a learning trial disrupts learning, whereas the same shock given hours later does not (Duncan, 1949), and patients receiving electroconvulsive therapy (ECT) for depression often lose memories for events shortly before the treatment course.

This kind of memory loss shows a temporal gradient: recent memories are more vulnerable than remote ones. The principle was stated by the French psychologist Théodule Ribot in 1881 and is known as **Ribot's law**: in progressive memory loss, the new perishes before the old. It is why an elderly person with dementia may forget breakfast but vividly recall their wedding. Consolidation and its brain basis are taken up again in Section 7.10.

### Classic effects on retention

Several named effects reveal what makes a memory last, and examiners love them because their names are easy to swap.

In the 1920s, the Gestalt psychologist Kurt Lewin noticed that a Berlin waiter remembered every detail of unpaid orders but forgot them as soon as the bill was settled. His student **Bluma Zeigarnik** (1900 to 1988) tested the observation in 1927: she gave people a series of simple tasks and interrupted half of them before completion. Later, people recalled the interrupted tasks nearly twice as often as the completed ones. This **Zeigarnik effect** was explained in Lewinian terms: an unfinished task leaves a state of tension that keeps it active in memory. A year later Maria Ovsiankina showed that people also have a strong tendency to resume an interrupted task when given the chance, even without instructions to do so: the Ovsiankina effect. The Zeigarnik effect is about remembering unfinished tasks; the Ovsiankina effect is about returning to them.

Sometimes memory improves with time. In 1913 Philip Ballard had schoolchildren learn poetry and found that many recalled more lines two or three days later than they had immediately after learning, without any further study. This rise in recall after a delay is **reminiscence**. (A related effect, hypermnesia, is a net increase in recall over repeated tests; Matthew

Erdelyi and Jeffrey Becker described it in 1974.) Do not confuse reminiscence with the reminiscence bump of autobiographical memory (Section 7.5).

In 1933 Hedwig von Restorff showed that an item that stands out from the rest of a list (a number among syllables, a word printed in red among black words) is remembered better than the same item in a homogeneous list. This **von Restorff effect**, also called the isolation effect, is the classic demonstration of distinctiveness (Section 7.4); highlighters exploit it, which is also why highlighting everything makes nothing stand out.

Two final effects matter more than any others for anyone studying for an exam. The **spacing effect** is the finding that study sessions spread out over time (distributed practice) produce better long-term retention than the same total study time crowded together (massed practice). Ebbinghaus observed it in 1885, and a large review by Nicholas Cepeda and colleagues (2006) confirmed it across hundreds of studies; the optimal gap grows with how long you need to remember. The **testing effect** (or retrieval practice effect) is the finding that retrieving information from memory strengthens it more than restudying it. Arthur Gates (1917) found that schoolchildren who spent much of their study time reciting from memory learned more than those who only read. In a modern classic, Henry Roediger and Jeffrey Karpicke (2006) had students study a prose passage either four times, or once followed by three recall tests. Five minutes later, the repeated-study group did slightly better; a week later, the repeated-testing group recalled about 50 per cent more, although the repeated-study group had predicted they would do better. Table 7.7 collects these named effects.

Table 7.7 Classic named effects on retention.

| Effect | Who and when | What happens |
|---|---|---|
| Zeigarnik effect | Bluma Zeigarnik, 1927 | Interrupted tasks are recalled better than completed ones |
| Ovsiankina effect | Maria Ovsiankina, 1928 | People tend to resume interrupted tasks |
| Reminiscence | Philip Ballard, 1913 | Recall rises after a delay, without further study |
| Von Restorff (isolation) effect | Hedwig von Restorff, 1933 | A distinctive item in a list is remembered better |
| Serial position effect | Ebbinghaus, 1885; Murdock, 1962 | First and last items are recalled better than middle ones |
| Spacing effect | Ebbinghaus, 1885 | Distributed practice beats massed practice |
| Testing effect | Gates, 1917; Roediger and Karpicke, 2006 | Retrieval practice beats restudy over long delays |

EXAM TRAP

In the RML practice question on items at the beginning and end of a list, the answer is the serial position effect; von Restorff (a distinctive item), Zeigarnik (an unfinished task) and flashbulb memory (an emotional event) are the distractors. The recency effect is the better recall of the last items; the Flynn effect is the rise in IQ scores across generations (Chapter 9) and the halo effect is a bias in judging people (Social book), both often offered as fillers.

## Schacter's seven sins of memory

In 2001 Daniel Schacter organised the ways memory fails into seven "sins", a framework that ties this section to the next (Table 7.8). The first three are sins of omission: something we want to remember is not available. The last four are sins of commission: something is remembered, but wrongly, or remembered when we would rather forget it. Schacter's point is that each sin is a by-product of a feature that usually serves us well: a memory system that kept every detail and resisted all suggestion would be slower, less flexible and far less able to generalise.

Table 7.8 Schacter's seven sins of memory (2001).

| Sin | Kind | What it is | Example |
|---|---|---|---|
| Transience | Omission | Weakening of memory over time | Forgetting a lecture from last year |
| Absent-mindedness | Omission | Lapses of attention at encoding or when an action is due | Misplacing keys; forgetting to take a tablet |
| Blocking | Omission | Temporary inability to retrieve stored information | The tip-of-the-tongue state |
| Misattribution | Commission | Right memory, wrong source | Cryptomnesia; false recognition |
| Suggestibility | Commission | Memories altered or implanted by suggestion | The misinformation effect |
| Bias | Commission | Present beliefs and feelings reshape the past | Hindsight bias; consistency bias |
| Persistence | Commission | Unwanted memories that will not go away | Intrusive memories after trauma |

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Words that are stored but cannot be retrieved without help are available but not _____.
2. Tulving and Thomson's principle that a cue works only if it was encoded with the memory is the _____ principle.
3. Godden and Baddeley's divers demonstrated _____-dependent memory.
4. Recall of material learned while drunk is better when drunk again; this is _____-dependent memory.
5. Jenkins and Dallenbach found less forgetting after _____ than after the same period awake.
6. Repeatedly dialling your old phone number instead of your new one is _____ interference.
7. Melton and Irwin's two factors in interference are response competition and _____.
8. Recalling interrupted tasks better than completed ones is the _____ effect.
9. Better recall two or three days after learning than immediately after, as in Ballard's study, is called _____.
10. Recent memories are more vulnerable than old ones; this is _____ law.

**Answers:** (1) accessible, (2) encoding specificity, (3) context, (4) state, (5) sleep, (6) proactive, (7) unlearning, (8) Zeigarnik, (9) reminiscence, (10) Ribot's.

## 7.9 Memory as Reconstruction: Distortion, False Memory and Eyewitnesses

MEDIUM YIELD · 5 QS

Most of us think of memory as a recording: an experience is filed away, and remembering plays the file back, perhaps faded but essentially faithful. Four of Schacter's seven sins say otherwise. This section tells the story of the research that replaced the recording metaphor with a different one: remembering is less like playing a video and more like a palaeontologist rebuilding a dinosaur from a few bones and a great deal of knowledge about what dinosaurs ought to look like.

### Bartlett and the War of the Ghosts (1932)

At Cambridge in the years around the First World War, **Frederic Charles Bartlett** (1886 to 1969) set out to study memory in a way that was the opposite of Ebbinghaus's. Instead of meaningless syllables he used meaningful material, and instead of counting items correct he looked at what people did to the material when they remembered it. His most famous material was "The War of the Ghosts", a Native American folk tale from the Pacific north-west recorded by the anthropologist Franz Boas. In it, two young men go down to a river to hunt seals; a canoe of warriors appears out of the fog and invites them to a war party; one goes, is wounded but feels nothing, and realises that his companions are ghosts; he returns home, tells his story, and at sunrise "something black came out of his mouth" and he dies. To English students the story was strange: its logic and its supernatural elements did not fit their expectations.

Bartlett used two methods. In repeated reproduction, the same person retold the story after intervals of minutes, days, months and even years. In serial reproduction, one person's version was given to the next, like a message passed along a line of people. His results, published in *Remembering: A Study in Experimental and Social Psychology* (1932), showed consistent changes. The stories became shorter, as details were omitted. They became more coherent and conventional: puzzling elements were explained or dropped, a process Bartlett called rationalisation, so that "something black came out of his mouth" became the man foaming at the mouth, or his soul leaving his body. Unfamiliar details were transformed into familiar ones: canoes became boats, and hunting seals became fishing. Some details became more prominent and acquired a central place. And the supernatural theme, so foreign to Edwardian Cambridge, tended to disappear.

Bartlett's explanation was that we do not store and replay experiences. We store a general impression together with a few outstanding details, and at retrieval we rebuild the rest using our schema knowledge (Section 7.6) of how such events usually go. Remembering, he wrote, is an "effort after meaning". **reconstructive memory**, the view that remembering actively rebuilds the past from fragments and schemas rather than reproducing it, is Bartlett's lasting legacy. Textbooks often describe the changes in his stories with three terms made famous by Gordon Allport and Leo Postman's study of rumour (1947): levelling (the story becomes shorter and simpler), sharpening (a few details are emphasised) and assimilation (details are distorted to fit the listener's expectations and culture).

Bartlett's methods were informal: he gave no standard instructions, used no statistics and reported selected examples. When Alan Gauld and Geoffrey Stephenson (1967) instructed participants to be as accurate as possible, many of the errors disappeared, suggesting that some

of Bartlett's "distortions" were deliberate guesses rather than true memory changes. But Erik Bergman and Henry Roediger (1999) repeated the War of the Ghosts study under careful control and found that the distortions grew over repeated reproductions, much as Bartlett had said.

> **EXAM TRAP**
>
> "Rationalisation" has two meanings in psychology. In Bartlett's memory research, it is the reshaping of a story to make it sensible to the rememberer. In psychoanalysis, it is a defence mechanism: offering a plausible, acceptable reason for behaviour after the fact, to hide the real motive (Personality book). Questions about schema and reconstruction belong to Bartlett, not to Wundt, Titchener or Watson.

### Loftus and the misinformation effect ●●

If memories are rebuilt at retrieval, information encountered after an event might be built into the memory. In the 1970s Elizabeth Loftus showed that this happens, and her work changed how courts treat eyewitnesses.

> **KEY STUDY: LOFTUS AND PALMER, "SMASHED" VERSUS "HIT" (1974)**
>
> Elizabeth Loftus and John Palmer showed 45 students seven short films of traffic accidents. Afterwards, each student answered questions, one of which was critical: "About how fast were the cars going when they ___ each other?" The verb varied between groups: smashed, collided, bumped, hit or contacted. The average speed estimates followed the verb: about 41 miles per hour for "smashed", 39 for "collided", 38 for "bumped", 34 for "hit" and 32 for "contacted".
>
> Was this just a response bias, or had the memory itself changed? In a second experiment, 150 students saw a film of a multiple-car accident and were asked either the "smashed" question, the "hit" question or no question about speed. A week later, without seeing the film again, they were asked: "Did you see any broken glass?" There was no broken glass in the film. Of those asked the "smashed" question, 32 per cent said yes, compared with 14 per cent of the "hit" group and 12 per cent of the controls. A single word in a question, asked after the event, had altered what people later reported seeing.

The broader phenomenon is the **misinformation effect**: the distortion of memory for an event by misleading information received afterwards, through leading questions, other witnesses' accounts or media reports. In a classic follow-up, Loftus, David Miller and Helen Burns (1978) showed people slides of a car at a stop sign; those later asked a question that presupposed a yield sign were much more likely to "recognise" a slide with a yield sign. Whether misinformation overwrites the original trace or merely creates a competing memory whose source is confused is still debated; Michael McCloskey and Maria Zaragoza (1985) showed that with better test procedures the original memory can often still be found, which supports the source-confusion account.

### Creating false memories

Changing a detail is one thing. Could a whole event that never happened be implanted? Loftus and Jacqueline Pickrell (1995) sent participants booklets describing four events from

their childhood, supposedly supplied by a relative. Three were true. The fourth, being lost in a shopping mall at about the age of five, crying, and being rescued by an elderly person and reunited with the family, was false, although the relative had confirmed that it fitted the family's history. Over two interviews, about a quarter of the participants came to remember the false event, fully or partly, sometimes adding details of their own. Other researchers implanted memories of spilling a bowl of punch on the bride's parents at a wedding (Ira Hyman and colleagues, 1995) and, using a doctored photograph, of a childhood hot-air balloon ride; in that study about half of the participants came to report some memory of the ride (Kimberley Wade and colleagues, 2002). Merely imagining an event repeatedly increases confidence that it happened, an effect called imagination inflation.

A **false memory** is a recollection of an event, or a detail of an event, that did not occur. The simplest way to produce one in the laboratory is the Deese-Roediger-McDermott (DRM) paradigm, introduced by James Deese in 1959 and revived by Henry Roediger and Kathleen McDermott in 1995. Read this list once, cover it, and then write down as many words as you can: bed, rest, awake, tired, dream, wake, snooze, blanket, doze, slumber, snore, nap, peace, yawn, drowsy. Did you write "sleep"? It was not on the list. In Roediger and McDermott's experiments, the related but unpresented "critical lure" was recalled about as often as words that had actually been studied, and people often claimed to "remember" hearing it, with the full sense of re-experiencing that Tulving called autonoetic. Every word on the list activated "sleep" (spreading activation, Section 7.6), and at test that internally generated activation was mistaken for a memory of hearing the word.

### Source monitoring, cryptomnesia and other errors of remembering ●●

Why do we confuse what we heard with what we imagined, or what we saw with what we were told? Marcia Johnson and her colleagues (Johnson, Hashtroudi and Lindsay, 1993) proposed that memories do not carry labels stating where they came from. Instead, we infer the source of a memory from its characteristics: memories of perceived events tend to be rich in sensory and contextual detail, while memories of imagined events contain more traces of the thinking that produced them. **source monitoring** is the set of processes by which we attribute a memory to its origin: when, where, how and from whom it was acquired. A deficit in source monitoring means being unable to recall the relevant contextual elements of an experience and so misattributing it. A special case, studied earlier by Johnson and Carol Raye (1981), is reality monitoring: telling memories of external events from memories of internally generated ones (what you did from what you only thought of doing). Source monitoring is the general framework; reality monitoring is one kind of source judgement.

Source errors explain a whole family of memory errors. In source amnesia, a person remembers a fact but not how they learned it; it is common in older adults and in patients with frontal lobe damage. In **cryptomnesia**, or unconscious plagiarism, a person produces an idea or a tune believing it to be new, when in fact it is a memory of someone else's work. In 1976 a court in the United States ruled that George Harrison's song "My Sweet Lord" had unconsciously copied the melody of an earlier song, "He's So Fine". Alan Brown and Dana Murphy (1989) produced the same effect in the laboratory: after a group generated examples of a category in turn, people later "generated" new examples that had in fact been produced by others. In hindsight bias, described by Baruch Fischhoff (1975), people who know how an event turned out remember their earlier predictions as closer to the outcome than they were: the "I knew it all along" effect.

Several experiences of false familiarity and false recall have names that appear in examinations. **déjà vu** (French for "already seen") is the illusion of recognition: the strong but false feeling that a new situation has been experienced before. It is common and usually harmless, though frequent déjà vu can be an aura of temporal lobe epilepsy. Jamais vu ("never seen") is the reverse: a familiar situation suddenly feels strange and new, as when a common word looks foreign after you have stared at it for a while. **confabulation** is the production of false memories, often detailed and plausible, to fill gaps in memory, without any intention to deceive; it is typical of Korsakoff's syndrome (Section 7.10). Pseudologia fantastica (pathological lying) is different again: elaborate, often grandiose stories told repeatedly, which the teller may half believe (Psychiatric Disorders book).

> EXAM TRAP
>
> "Being unable to recall the relevant elements of an autobiographical experience" is a deficit in source monitoring (RML 2022); reality monitoring is a narrower case (internal versus external origin), and "personal-monitoring" and "social-monitoring" are invented distractors. The "illusion of recognition" is déjà vu; jamais vu is the illusion of unfamiliarity; confabulation fills memory gaps with false but sincere stories; pseudologia fantastica is pathological lying. Hindsight bias is a memory bias, not a memory store or a physical trace.

## Eyewitness testimony ●●

Few things persuade a jury like an eyewitness who points at the defendant and says, "That is the man." Yet research on reconstructive memory implies that eyewitnesses should be treated with caution, and the evidence of DNA exonerations confirms it: in the United States, mistaken eyewitness identification played a part in roughly 70 per cent of the wrongful convictions later overturned by DNA testing. The problem is not usually poor eyesight or a tendency to forget; it is that memory can be confidently wrong.

Many factors affect accuracy. At encoding: extreme stress, brief exposure, poor lighting, and weapon focus, the tendency for a visible weapon to capture attention at the expense of the culprit's face (Loftus, Geoffrey Loftus and Jane Messo, 1987). People are also less accurate at identifying faces of another ethnic group than of their own (the cross-race or own-race effect). After the event: leading questions, conversations with other witnesses and media reports introduce misinformation, and source errors can make a familiar face (seen at the scene, or in a photograph shown by police) feel like the culprit's face. At identification: biased line-ups, suggestive instructions and the investigator's own expectations all increase errors. Confirming feedback after an identification ("Good, you identified our suspect") inflates the witness's confidence, so that by the time of the trial a hesitant witness has become certain (Gary Wells and Amy Bradfield, 1998). Confidence expressed at the first, fair identification is reasonably informative; confidence at trial is not. Young children are especially suggestible when questioned repeatedly with leading questions (Stephen Ceci and Maggie Bruck, 1993).

Psychology has also offered remedies. The cognitive interview developed by Ronald Fisher and Edward Geiselman applies memory principles directly: mentally reinstate the context of the event (context-dependent memory), report everything however trivial, recall the events in different orders, and recall them from different perspectives (each provides different retrieval cues). It reliably increases correct recall without much increase in errors. Line-ups

conducted by an officer who does not know who the suspect is, with fair fillers and a record of the witness's initial confidence, reduce false identifications.

CLINICAL LINK

Clinical and forensic psychologists are often asked about memory in court. The safest summary of the research is that memory is usually accurate in its gist, often inaccurate in its details, readily altered by suggestion, and that confidence and vividness are poor guides to accuracy. A clinician who interviews a child about possible abuse, or an adult about a traumatic event, must therefore use open, non-leading questions, avoid repeated suggestive questioning, and record exactly what was asked and answered.

### The recovered memory controversy

In the late 1980s and 1990s, a growing number of adults in the United States and elsewhere reported recovering, often during therapy, memories of childhood sexual abuse that they had previously had no awareness of. Some therapists believed that such memories had been repressed and that techniques such as hypnosis, guided imagery, dream interpretation and repeated suggestive questioning could bring them back. Some recovered memories led to accusations, family breakdown and criminal trials. Critics, led by Elizabeth Loftus, argued that the very techniques used to "recover" memories were the techniques known to create false ones, and the term "false memory syndrome" was coined by an organisation founded in 1992 by families who said they had been falsely accused. The resulting dispute became known as the memory wars.

The evidence supports a middle position. Genuine abuse can be forgotten for long periods and later remembered: Linda Meyer Williams (1994) interviewed women whose abuse in childhood had been documented in hospital records seventeen years earlier and found that 38 per cent did not report it. (Some of these women may simply have chosen not to disclose it, and ordinary forgetting of early childhood events need not involve repression.) But false memories of traumatic events can also be created, especially through suggestion by an authority figure, imagination and repetition, and hypnosis increases confidence far more than it increases accuracy. A recovered memory is therefore neither automatically true nor automatically false; it needs independent corroboration, and therapists should never try to dig up memories that the client does not report.

## 7.10 Amnesia and the Biology of Memory HIGH YIELD · 31 QS

Much of what we know about the organisation of memory has come from people like Clive Wearing whose memories were damaged. This section describes the main forms of amnesia and the classic cases briefly, and outlines what they reveal about consolidation and the brain; the neural detail belongs to the Physiological book.

### Kinds of amnesia ●●●

**amnesia** is a pathological loss of memory, disproportionate to any other cognitive impairment. Take the moment of the brain injury, illness or surgery as a fixed point on a time line. **anterograde amnesia** (forward-acting) is the inability to form new long-term memories for events after that point: the person cannot learn new names, cannot remember what hap-

pened an hour ago. **retrograde amnesia** (backward-acting) is the loss of memories formed before that point: the person cannot remember events that preceded the injury. Most organic amnesias combine the two, with the anterograde component usually the more disabling. Retrograde amnesia typically follows Ribot's law: the years just before the injury are lost, while remote memories from childhood survive.

Amnesia should be distinguished from the other classic disorders that neuropsychologists name with the prefix "a-": aphasia (disorder of language, often listed with alexia, an acquired loss of the ability to read, and acalculia, an acquired loss of the ability to calculate), apraxia (inability to carry out skilled, learned movements despite intact strength), agnosia (inability to recognise objects despite intact sensation) and dementia (progressive decline across several cognitive functions, including memory, judgement and abstraction).

### H.M.: the man who changed the study of memory ●●●

Henry Gustav Molaison (1926 to 2008), known for over fifty years only by his initials, had suffered from severe epilepsy since childhood. In 1953, when he was 27, the neurosurgeon William Beecher Scoville removed the medial portions of both of his temporal lobes, including most of the hippocampus, the amygdala and the adjacent cortex, in an attempt to stop his seizures. The seizures were greatly reduced. But H.M. was left with a profound anterograde amnesia. Scoville and the psychologist **Brenda Milner** (born 1918) described the case in 1957, and H.M. was studied for the rest of his life, especially by Milner and later by Suzanne Corkin.

H.M.'s case established four principles. First, memory can be separated from other cognitive abilities: his intelligence, language and perception were normal, and his personality largely unchanged. Second, short-term memory is separate from long-term memory: his digit span was normal, and he could hold a conversation, but as soon as his attention moved on, the content was lost. He read the same magazines again and again with fresh interest and could not learn the address of the house his family moved to after his operation. Third, the medial temporal lobes are needed to form new declarative memories but are not where old memories are stored: his memories of childhood were largely intact, and his retrograde amnesia covered only the years before the surgery (originally estimated at about two years; later testing suggested it reached further back). Fourth, implicit memory is separate from explicit memory: as described in Section 7.5, H.M. learned to trace a star in a mirror, improving over three days just like a healthy person, while having no conscious memory of ever having done the task. He also showed normal priming. After his death in 2008, his brain was preserved and sliced into thousands of sections for study.

EXAM TRAP

H.M.'s most profound deficit was anterograde amnesia for declarative (explicit) memory; his procedural memory (mirror tracing) was spared, his short-term memory was normal, and his retrograde amnesia was limited. Options saying he lost procedural memory, semantic memory "only", working memory span, or all old memories are wrong. The structures removed were the medial temporal lobes on both sides, including the hippocampus and amygdala, not the frontal, parietal or occipital lobes. The hippocampus is the structure most often linked to anterograde amnesia and to the consolidation of new declarative memories, not the hypothalamus (motivation, temperature, hunger), the pituitary (hormones) or the cerebellum (motor coordination and motor learning).

### Korsakoff's syndrome and confabulation ●●●

**Korsakoff's syndrome** (also called Korsakoff's psychosis, or Korsakoff's amnesic syndrome) is a memory disorder most commonly seen in people with a long history of heavy alcohol use. It was described by the Russian psychiatrist Sergei Korsakoff in the late 1880s, and it is caused mainly by deficiency of thiamine (vitamin B1), which commonly accompanies chronic alcoholism because of poor diet and poor absorption. It often follows an acute phase called Wernicke's encephalopathy, described by Carl Wernicke in 1881, marked by confusion, unsteady gait (ataxia) and abnormal eye movements such as nystagmus and paralysis of eye muscles; the two together are called the Wernicke-Korsakoff syndrome. The damage centres on the medial diencephalon, especially the mammillary bodies and the mediodorsal (also called dorsomedial) nuclei of the thalamus, with more diffuse damage elsewhere.

The memory disorder has three features. There is a severe anterograde amnesia for explicit memories. There is also an extensive retrograde amnesia, which grows as the disease progresses and can reach back into childhood, with the usual temporal gradient. And there is confabulation: asked what they did yesterday, patients may give a fluent, detailed and entirely false account, often built from fragments of real older memories, with no intent to deceive and no awareness that it is false. Confabulation must be distinguished from malingering (deliberately faking or exaggerating symptoms for gain), from delirium (an acute, fluctuating disturbance of attention and consciousness) and from simple confusion. The neurologist Oliver Sacks described a Korsakoff patient, Jimmie G., who in 1975 still believed he was a nineteen-year-old sailor at the end of the Second World War, "the lost mariner". Treatment with thiamine in the acute Wernicke stage can prevent much of the damage, which is why it matters clinically to recognise the syndrome early.

Other neurological diseases affect memory in contrasting ways. In Alzheimer's disease, episodic memory for recent events is usually the first to fail and semantic memory declines later. In Huntington's and Parkinson's diseases, which damage the basal ganglia, procedural learning is impaired more than declarative memory, the mirror image of H.M. (Physiological and Psychiatric Disorders books).

### Other amnesias

**infantile amnesia**, also called childhood amnesia, is the near-total absence, in adults, of memories from the first three or four years of life; the average earliest memory dates from about three to three and a half years. Freud, who named it, attributed it to repression. Current explanations are developmental: the hippocampus and prefrontal cortex are imma-

ture in infancy; young children lack the language in which later memories are organised and retrieved; a stable sense of self, needed to organise memories as "things that happened to me", emerges only around the second year (Mark Howe and Mary Courage, 1993); and the schemas through which an adult searches memory differ from those of the infant who encoded it.

Transient global amnesia is a sudden, temporary anterograde amnesia, usually in middle-aged or older adults, lasting several hours and never more than a day. The person is alert and knows who they are but asks the same questions over and over; the episode resolves completely, leaving only a gap for the hours it lasted. Post-traumatic amnesia is the period of confusion and failed new learning after a head injury; its length is one of the best predictors of the severity of the injury.

Dissociative amnesia, by contrast, is an inability to recall important autobiographical information, usually of a traumatic or stressful nature, too extensive to be ordinary forgetting and without a medical cause. It may be localised (a specific period), selective (some events of a period), generalised (the person's whole life history), systematised (a category of information) or continuous (everything from a certain point onwards). It may come with dissociative fugue, purposeful travel or wandering with amnesia for identity, and it is distinct from dissociative identity disorder, in which two or more personality states alternate with gaps in memory. These conditions are taught in the Psychiatric Disorders book.

### Consolidation, the hippocampus and the search for the engram ●●●

Where is a memory stored? In 1904 the German biologist Richard Semon coined the word **engram** for the physical trace that learning leaves in the nervous system. The most determined search for it was made by the American neuropsychologist **Karl Spencer Lashley** (1890 to 1958). For three decades Lashley trained rats to run mazes and then removed different parts of their cerebral cortex, hoping to find the place where the maze memory was kept. He never found it. Memory loss depended on how much cortex was removed, not on which part; he called this the principle of mass action. And different parts of the cortex seemed able to take over for one another, the principle of equipotentiality (both set out in his 1929 book *Brain Mechanisms and Intelligence*). In his 1950 paper "In search of the engram" he wrote, only half joking, that reviewing the evidence sometimes made him feel "that the necessary conclusion is that learning just is not possible." His conclusion was that memories are not localised in single spots but are distributed widely across the cortex.

Lashley's student **Donald Olding Hebb** (1904 to 1985) suggested how a distributed memory could form. In *The Organization of Behavior* (1949) he proposed that when one neuron repeatedly helps to fire another, the connection between them is strengthened, so that groups of neurons that are active together become linked into cell assemblies, often summarised as "cells that fire together wire together". A physical mechanism of exactly this kind was found in 1973, when Timothy Bliss and Terje Lømo showed that brief, high-frequency stimulation of a pathway into the rabbit hippocampus produced a lasting increase in the strength of the synapses. This **long-term potentiation** (LTP), which depends on NMDA glutamate receptors, is the leading candidate for the cellular basis of learning and memory (Physiological book). Modern methods have even allowed researchers to tag and reactivate the specific "engram cells" that are active during learning in mice, vindicating Semon's term a century later.

How do these findings fit with H.M.? The standard account, the standard consolidation theory, is that the hippocampus and medial temporal lobe bind together the scattered cortical el-

ements of a new declarative memory and are needed for that memory for some time; through repeated reactivation, much of it during sleep, the memory is gradually reorganised so that it can be retrieved from the neocortex without the hippocampus (systems consolidation). This explains why damage to the hippocampus prevents new learning and erases recent memories while sparing remote ones. Rival accounts hold that vivid episodic memories always depend on the hippocampus (multiple trace theory, Lynn Nadel and Morris Moscovitch, 1997), and research on reconsolidation (Karim Nader, Glenn Schafe and Joseph LeDoux, 2000) has shown that a consolidated memory, when it is reactivated, becomes fragile again for a time and must be restabilised, a biological counterpart of Bartlett's reconstructive memory. Table 7.9 summarises the brain structures that appear in memory questions.

Table 7.9 Brain structures most often linked with memory.

| Structure | Role in memory | Classic evidence |
|---|---|---|
| Hippocampus (medial temporal lobe) | Consolidation of new declarative memories; spatial memory | H.M.; Clive Wearing |
| Amygdala | Emotional memory; fear conditioning; strengthens arousing memories | Flashbulb memory research |
| Mammillary bodies and mediodorsal thalamus | Part of the diencephalic memory circuit | Korsakoff's syndrome |
| Striatum (basal ganglia) | Procedural learning and habits | Huntington's and Parkinson's diseases |
| Cerebellum | Motor learning; conditioned eyeblink | Eyeblink conditioning studies |
| Prefrontal cortex | Working memory, source memory, strategic retrieval | Dysexecutive syndrome; source amnesia |
| Neocortex | Long-term storage of consolidated memories; priming | Lashley; semantic dementia |

EXAM TRAP

Lashley concluded that memories are distributed throughout the cortex, not localised in specific regions, not stored exclusively in the hippocampus, and not confined to the frontal and temporal lobes (Real PYQ 2025). The engram is the physical trace of memory (Semon's term), not a process of forgetting or a type of explicit memory. LTP is a lasting strengthening of synaptic transmission after high-frequency stimulation, associated with memory, not with sleep, thermoregulation or circadian rhythm. And the hippocampus is critical for consolidating new declarative memories but is not the storage site of remote memories.

## 7.11 Improving Memory LOWER YIELD · 2 QS

Everything in this chapter can be turned into advice. Attention determines what is encoded; meaning, elaboration, self-reference and imagery determine how well it is encoded; organisation and distinctiveness make it findable; spacing and retrieval practice make it last; matching study to test conditions makes it accessible when needed; and sleep consolidates it. The techniques below are those principles put to work.

## Mnemonic devices

A **mnemonic** is any technique that improves memory by imposing organisation, meaning or imagery on material that lacks them. The oldest is the **method of loci**. The Roman orator Cicero told how the Greek poet Simonides of Ceos, having stepped out of a banquet hall moments before its roof collapsed, was able to identify the crushed guests by remembering where each had been sitting. Order, Simonides realised, can be held by place. To use the method, you choose a familiar route (your house, the walk from your hostel to the library) and place a vivid image of each item to be remembered at successive locations along it; to recall, you walk the route in imagination and "see" each item where you left it. Shereshevsky used this method, and modern memory champions still do.

The **peg-word system** replaces locations with a memorised list of "pegs" that rhyme with the numbers: one is a bun, two is a shoe, three is a tree, four is a door, five is a hive, and so on. Each item to be learned is linked to its peg by an interactive image: to remember that the third item is "milk", picture milk pouring over a tree. The method gives direct access to any item by number. The **keyword method**, developed for foreign vocabulary by Richard Atkinson (1975; tested with Michael Raugh on Russian vocabulary), links a foreign word to its meaning through an English word that sounds like part of it. The Spanish word "pato" means duck; it sounds like "pot", so you imagine a duck sitting in a pot.

Verbal mnemonics use the first letters of the items. An **acronym** is a word or pronounceable string formed from the first letters of the items, such as VIBGYOR for the colours of the rainbow or OCEAN for the five broad personality traits. An **acrostic** is a sentence whose words begin with the first letters of the items, such as "Every Good Boy Deserves Fruit" for the notes on the lines of the treble stave (E, G, B, D, F). Other techniques draw on principles we have met. Chunking groups items into fewer, larger units. Categorical clustering organises a list by meaning; W. A. Bousfield (1953) showed that people spontaneously recall randomly ordered lists in category clusters. Interactive imagery links items in a single scene, and it is the interaction, not bizarreness, that matters (Keith Wollen and colleagues, 1972). Narrative chaining weaves items into a story: Gordon Bower and Michael Clark (1969) had people learn twelve lists of ten words each; those who made up a story for each list later recalled about 93 per cent of the words, while those who simply studied recalled about 13 per cent. Table 7.10 summarises the devices and the principle each one uses.

Table 7.10 Mnemonic devices and the principle each one uses.

| Mnemonic | How it works | Principle used |
|---|---|---|
| Method of loci | Images placed along a familiar route | Imagery; organisation by place |
| Peg-word system | Items linked to rhyming number pegs | Imagery; ready-made retrieval cues |
| Keyword method | Foreign word linked to a sound-alike image | Imagery; elaboration |
| Acronym | A word made of first letters (VIBGYOR) | Chunking; retrieval cues |
| Acrostic | A sentence made of first letters | Retrieval cues; meaning |
| Narrative chaining | Items woven into a story | Elaboration; organisation |
| Categorical clustering | Items grouped by category | Organisation; category cues |

EXAM TRAP

Acronym = a word made from first letters (VIBGYOR). Acrostic = a sentence made from first letters ("Every Good Boy Deserves Fruit"). The peg-word system uses a memorised rhyming list; the keyword system uses a sound-alike word to link a foreign word with its meaning; the method of loci uses places. A question describing "a sentence framed using the first letter of each item" is asking for the acrostic.

INDIAN CONTEXT

The Vedas were transmitted orally for many centuries before they were written down, with extraordinary fidelity. Reciters learned each text in several forms: the continuous text (samhita patha), word by word (pada patha), in overlapping pairs (krama patha: word 1 and 2, then 2 and 3, then 3 and 4), and in elaborate forward-and-backward weavings (jata and ghana patha). Each recitation pattern acts as a check on the others, so that an error in one would be exposed by the rest. It is one of the most sophisticated memory systems ever devised, and UNESCO recognised the tradition of Vedic chanting as part of the world's intangible heritage in 2003.

## Study systems: SQ3R and PQRST ●●

Two study systems for textbooks build elaboration, self-questioning and retrieval practice into reading. **SQ3R**, devised by the American educational psychologist Francis P. Robinson in 1946, has five steps: Survey (skim headings and summaries to build a framework), Question (turn each heading into a question), Read (read actively to answer the question), Recite (close the book and state the answer from memory) and Review (go over the whole chapter, testing yourself). PQRST, described by Ellen Thomas and H. Alan Robinson (1972) and widely taught in introductory psychology, is a close variant: Preview, Question, Read, Self-recitation and Test. In both, the most important steps are the ones students skip: the questions, which give reading a purpose and a structure, and the recitation, which is retrieval practice.

## Spacing, testing, overlearning and sleep

When John Dunlosky and his colleagues (2013) reviewed the evidence on ten popular study techniques, two were rated as having high utility: practice testing and distributed (spaced) practice. Highlighting, rereading and summarising were rated as having low utility. Rereading feels productive because the material becomes fluent and familiar, but fluency is a poor guide to what can be retrieved later (the illusion of competence, Section 7.5). Retrieval practice works because every successful retrieval strengthens the memory and the route to it, and every failed one shows you exactly what to restudy. Spacing works partly because each return to the material after some forgetting requires effortful retrieval, and partly because material studied in different contexts acquires more varied retrieval cues. Robert Bjork has called such techniques "desirable difficulties": they make learning feel harder and slower, and make it last longer. Mixing different kinds of problems in one practice session (interleaving), rather than practising one kind at a time, is another desirable difficulty: it teaches you to recognise which method a problem needs.

Here is a worked example. Suppose you have two days to master a chapter. Massed study would mean reading it four times on the first day. A spaced-retrieval plan would mean reading it once on the first morning; that evening, writing down everything you can recall about

each section heading before checking; the next morning, answering the chapter's self-test questions without looking; and the next evening, attempting the past-paper questions and restudying only what you got wrong. The total time is similar. The second plan uses testing, spacing, generation and feedback, and on the evidence reviewed in this chapter it will produce far better retention a week later.

Two further aids complete the picture. **overlearning**, continuing to practise after material has been mastered, increases retention: William Krueger (1929) found that 50 per cent overlearning (half as many extra trials as were needed to learn) produced a clear gain, and 100 per cent overlearning a smaller additional gain, a case of diminishing returns. And sleep after learning reduces interference and actively consolidates new memories, as Jenkins and Dallenbach's results first hinted; studying until the small hours at the expense of sleep is a poor bargain. (Massed versus distributed practice and overlearning in skill learning are also discussed in Chapter 6.)

EXAM TRAP

SQ3R (Robinson, 1946): Survey, Question, Read, Recite, Review. PQRST: Preview, Question, Read, Self-recitation, Test. In PQRST the "S" stands for self-recitation, not storing, sequencing or semantics.

CLINICAL LINK

Memory rehabilitation for people with brain injury or dementia applies the same principles in modified form. Because amnesic patients cannot tell their own errors from correct responses, errorless learning (Baddeley and Barbara Wilson, 1994), in which the patient is guided to the correct response from the start rather than encouraged to guess, prevents errors from being learned through intact implicit memory. Spaced retrieval, in which a patient recalls a fact at gradually lengthening intervals, and the method of vanishing cues (Elizabeth Glisky, Daniel Schacter and Endel Tulving, 1986), in which letter cues are withdrawn one at a time, exploit spared implicit learning. External aids such as diaries, alarms and smartphone reminders compensate for failures of prospective memory.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Bartlett's War of the Ghosts study (1932) showed that remembering is an "effort after _____".
2. In Loftus and Palmer's study, witnesses asked about cars that "_____" each other gave the highest speed estimates and reported more broken glass.
3. Recalling the unpresented word "sleep" after studying bed, rest, tired and dream is the _____ paradigm.
4. Unconsciously presenting someone else's idea as your own is called _____.
5. Inability to form new memories after brain damage is _____ amnesia.
6. Korsakoff's syndrome is caused mainly by deficiency of _____ (vitamin B1).
7. Producing false but sincere stories to fill memory gaps is called _____.
8. Lashley's principle that memory loss depends on how much cortex is removed is the principle of _____.
9. A lasting increase in synaptic strength after high-frequency stimulation, found by Bliss and Lømo in 1973, is _____.
10. A sentence whose words begin with the first letters of items to be remembered is an _____.

**Answers:** (1) meaning, (2) smashed, (3) Deese-Roediger-McDermott (DRM), (4) cryptomnesia, (5) anterograde, (6) thiamine, (7) confabulation, (8) mass action, (9) long-term potentiation, (10) acrostic.

## The case of Clive Wearing, revisited

Go back to Clive Wearing, writing in his diary yet again that he is now, at last, truly awake. You can now describe what happened to him in the language of this chapter. Herpes simplex encephalitis destroyed the hippocampus and much of the medial temporal lobe on both sides, the structures that consolidate new declarative memories. His short-term (working) memory still worked, which is why he could follow a sentence or a conversation for as long as it kept moving. But nothing he experienced could be consolidated into durable long-term memory: he had a dense anterograde amnesia like H.M.'s. Alongside it, and probably because the damage spread well beyond the hippocampus, he had a far more extensive retrograde amnesia than H.M. ever had, one that wiped out almost every particular episode of his past. Without episodic memory he had none of the autonoetic consciousness that lets a person travel back in time and feel himself to be the same person who lived through earlier events. Every moment of awareness seemed to him like the first, and so every earlier diary entry looked like a lie.

What survived teaches just as much. Clive kept some semantic knowledge (that he was a musician, that he was married to Deborah), much as Tulving's patient K.C. kept facts about his own life while losing every episode. He kept his procedural memory: sight-reading, playing and conducting, skills that depend on circuits outside the hippocampus, exactly as Squire's taxonomy predicts and as H.M.'s mirror tracing had already shown. And his joy at seeing Deborah, and his love for her, survived without a single remembered conversation to anchor them, a reminder that past experience can shape feeling and behaviour without any conscious recollection. In one life, then, you can see three of this chapter's distinctions: short-

term versus long-term memory, episodic versus semantic memory, and explicit versus implicit memory. It also shows what the chapter has argued from Ebbinghaus to Bartlett: memory is not one thing but several, and it is less a recording of the past than the means by which each of us builds a continuing self.

## Chapter summary

**7.1 What memory is, and how Ebbinghaus measured it.** Memory is the capacity to retain information over time and use it later. It involves three processes: encoding (turning input into a code the system can hold), storage and retrieval. Encoding depends on attention, and details that were never attended to are lost through encoding failure, as Nickerson and Adams's coin study showed. Hermann Ebbinghaus (1885) invented the nonsense syllable, acted as his own participant and measured retention by the savings method: original learning minus relearning, divided by original learning, multiplied by 100. His forgetting curve is negatively accelerated: steep loss in the first hour, then a slow decline that never reaches zero. He also first noted the benefits of overlearning and spaced repetitions, the advantage of the beginning and end of a list, and the ease of meaningful material. Retention can be measured by recall (free, serial and cued, including the paired-associate learning introduced by Calkins in 1894), by recognition, by the reconstruction method (restoring the order of supplied items) and by relearning, which is the most sensitive measure.

**7.2 The multi-store model.** Building on William James's primary and secondary memory (1890), Atkinson and Shiffrin (1968) proposed three stores (sensory, short-term and long-term) worked by control processes such as attention and rehearsal. Sensory memory briefly holds a large amount of pre-categorical information. Sperling's partial-report procedure (1960) showed that about nine of twelve letters are available in iconic memory for roughly a quarter to half a second, and Darwin, Turvey and Crowder (1972) put echoic memory at about two to four seconds. Short-term memory holds seven plus or minus two items (Miller, 1956), or about four chunks (Cowan, 2001), and chunking draws on long-term knowledge. Without rehearsal it lasts about 18 to 30 seconds (Brown, 1958; Peterson and Peterson, 1959), but Keppel and Underwood (1962) showed that the loss came from proactive interference rather than decay. Coding is mainly acoustic in short-term memory (Conrad, 1964; Baddeley, 1966) and semantic in long-term memory, which has no known capacity limit and can last a lifetime (Bahrick). The serial position curve (primacy from long-term memory, recency from short-term memory; Glanzer and Cunitz, 1966) and the double dissociation between H.M. and K.F. support separate stores. Critics point out that neither store is a single unit, that rehearsal is not the main route into long-term memory, and that long-term knowledge acts on information while it is still in short-term memory.

**7.3 Working memory.** Baddeley and Hitch (1974) used the dual-task method to show that short-term memory has several parts. The central executive controls attention but stores nothing: it focuses, divides and switches attention and links working memory with long-term memory. Random number generation loads it, and the Stroop task measures its inhibitory function. The phonological loop (a phonological store and an articulatory control process) holds speech-based information. Four findings support it: the phonological similarity effect, the word-length effect, articulatory suppression and the irrelevant speech effect. It also helps in learning new words. The visuospatial sketchpad (a visual cache and an inner scribe, in Logie's terms) holds visual and spatial information and carries out mental rotation. The episodic buffer,

added in 2000, binds information from the other components and from long-term memory into integrated episodes. Short-term memory is simple storage, measured by simple span; working memory is storage plus processing, measured by complex span and backward digit span, and it predicts comprehension and reasoning.

**7.4 Levels of processing and the quality of encoding.** Craik and Lockhart (1972) argued that memory is a by-product of processing and that deeper processing (structural, then phonemic, then semantic) leaves more durable traces whether or not the person intends to learn (Hyde and Jenkins, 1973). Craik and Tulving (1975) found the best recognition after semantic questions, and more elaborate processing helped even within one level. Maintenance (Type I) rehearsal keeps information available but adds little; elaborative (Type II) rehearsal builds lasting memories. Distinctiveness, the self-reference effect (Rogers, Kuiper and Kirker, 1977), the generation effect (Slamecka and Graf, 1978) and Paivio's dual coding theory (1971) show other ways of strengthening encoding. The framework was criticised as circular. Morris, Bransford and Franks (1977) showed with a rhyming test that what matters is transfer-appropriate processing, the match between processing at study and processing at test.

**7.5 The systems of long-term memory.** Tulving (1972) distinguished episodic memory (personal events located in time and place, recalled with autonoetic consciousness) from semantic, or generic, memory (general knowledge, held with noetic awareness). Procedural memory holds skills. K.C. lost his episodic memory but kept his semantic knowledge. Explicit (declarative) memory involves conscious recollection, while implicit (nondeclarative) memory shows itself in performance without awareness (Graf and Schacter, 1985), as in Claparède's pinprick, word-stem completion by amnesic patients and H.M.'s mirror tracing. Squire's taxonomy links declarative memory to the medial temporal lobe, and links procedural skills, priming, classical conditioning and non-associative learning to other structures. Priming may be repetition priming or semantic priming (Meyer and Schvaneveldt, 1971); its masked form is used to study subliminal perception. Autobiographical memory shows a reminiscence bump for ages 10 to 30. Prospective memory (remembering to act, cued by an event or by time) contrasts with retrospective memory. Flashbulb memories (Brown and Kulik, 1977) are vivid and confidently held but no more accurate than ordinary memories (Neisser and Harsch, 1992; Talarico and Rubin, 2003); emotional arousal and rehearsal explain them, not a "Now Print!" mechanism. Eidetic imagery is rare and found mainly in children. Metamemory (Flavell, 1971) covers the feeling of knowing (Hart, 1965) and the tip-of-the-tongue state (Brown and McNeill, 1966), which is a failure of retrieval, not of storage.

**7.6 How knowledge is organised.** Collins and Quillian's hierarchical network model (1969) stores concepts as nodes in a hierarchy, with each property stored only at the highest level it applies to (cognitive economy), and predicts that verification time rises with the number of links travelled. Typicality effects and familiarity confounds undermined it. They led to the feature comparison model and to Collins and Loftus's spreading activation model (1975), in which activation spreads along links of varying length; this explains semantic priming and typicality. A proposition is the smallest unit of knowledge that can be true or false. Connectionist, or parallel distributed processing, models (Rumelhart and McClelland, 1986) represent knowledge as patterns of simultaneous activation across many interconnected units and store it in connection weights; such networks can retrieve a whole memory from any part of it and degrade gradually when damaged. Schemas are organised bodies of knowledge (Bartlett, 1932) that guide understanding and distort recall (Bransford and Johnson, 1972; Brewer and Treyens, 1981).

Scripts are schemas for routine events (Schank and Abelson, 1977) and belong to generic memory.

**7.7 Retrieval: cues, context and state.** Information may be available but not accessible, and retrieval cues such as category names make it accessible (Tulving and Pearlstone, 1966). Recognition is usually easier than recall because the test supplies the item, as the generate-recognise account says, although the recognition failure of recallable words shows that this is not the whole story. Tulving and Thomson's encoding specificity principle (1973) states that a cue works to the extent that it was encoded with the memory. Its special cases include context-dependent memory (Godden and Baddeley's divers, 1975; merely imagining the original room also helps), state-dependent memory (Goodwin and colleagues, 1969, with alcohol) and mood-dependent memory. Mood-congruent memory, the easier recall of material that matches one's current mood (Bower, 1981), helps to keep depression going. Context and state effects are strongest in free recall and weak or absent in recognition.

**7.8 Forgetting.** Decay theory, traced to Thorndike's law of disuse, holds that unused traces fade. McGeoch (1932) objected that time itself causes nothing, and Jenkins and Dallenbach (1924) found less forgetting after sleep than after the same time awake. Interference theory distinguishes proactive interference (old learning disrupts new) from retroactive interference (new learning disrupts old), and studies both with A-B, A-C designs. Interference grows with similarity, proactive interference is the larger source of laboratory forgetting (Underwood, 1957), and Melton and Irwin (1940) named response competition and unlearning as its two factors. Tulving's cue-dependent forgetting and retrieval-induced forgetting (Anderson, Bjork and Bjork, 1994) show that much forgetting is a failure of access. Freud's repression is unconscious and lacks controlled evidence; suppression, directed forgetting and the think/no-think paradigm show that conscious motivated forgetting is real. Consolidation failure explains the loss of recent memories after concussion or ECT, with the temporal gradient of Ribot's law (1881). The classic named effects are the Zeigarnik effect (1927), the Ovsiankina effect (1928), Ballard's reminiscence (1913), the von Restorff isolation effect (1933), the spacing effect and the testing effect (Roediger and Karpicke, 2006). Schacter's seven sins (2001) sort memory failures into sins of omission and sins of commission. Retention and forgetting are inversely related: the more is retained, the less is forgotten.

**7.9 Memory as reconstruction.** Bartlett (1932) used "The War of the Ghosts" to show that remembering is an "effort after meaning": retold stories became shorter and more conventional, and were rationalised to fit the rememberer's schemas. The changes are often described as levelling, sharpening and assimilation (Allport and Postman, 1947). Loftus and Palmer (1974) showed that the verb "smashed" raised speed estimates and later false reports of broken glass, which is the misinformation effect. Whole false events can be implanted (lost in the mall; Loftus and Pickrell, 1995), and the Deese-Roediger-McDermott paradigm produces confident recall of an unpresented critical lure. Source monitoring (Johnson, Hashtroudi and Lindsay, 1993) explains source amnesia, cryptomnesia and misattribution. Déjà vu is an illusion of recognition, and confabulation is the sincere filling of memory gaps. Eyewitness testimony is undermined by weapon focus, the cross-race effect, leading questions, source errors and inflated confidence; the cognitive interview and fair line-ups reduce errors. In the recovered memory controversy the evidence supports a middle position: real abuse can be forgotten and later remembered, and false memories can be created by suggestion.

**7.10 Amnesia and the biology of memory.** Amnesia is a pathological loss of memory. Anterograde amnesia is the inability to form new memories after a brain injury; retrograde amnesia is the

loss of memories formed before it, usually following Ribot's law. After the removal of both medial temporal lobes, H.M. (Scoville and Milner, 1957) had a severe anterograde amnesia for declarative memory. His short-term memory, intelligence, priming and procedural learning (mirror tracing) were normal, and his retrograde amnesia was limited. Korsakoff's syndrome is caused by thiamine deficiency in chronic alcoholism and is often preceded by Wernicke's encephalopathy (confusion, ataxia and nystagmus). It damages the mammillary bodies and the mediodorsal thalamus and produces anterograde amnesia, an extensive retrograde amnesia and confabulation. Infantile amnesia reflects the development of the brain, language and the self rather than repression. Transient global amnesia and dissociative amnesia (Psychiatric Disorders book) are other forms. Lashley's search for the engram (Semon's term) produced the principles of mass action and equipotentiality and the conclusion that memories are distributed across the cortex; Hebb's cell assemblies (1949) and long-term potentiation (Bliss and Lømo, 1973) suggest the mechanism. The hippocampus consolidates new declarative memories, which in time come to depend on the neocortex.

**7.11 Improving memory.** Mnemonics add organisation, meaning and imagery to material. The main ones are the method of loci (Simonides), the peg-word system, the keyword method (Atkinson, 1975), acronyms (VIBGYOR), acrostics ("Every Good Boy Deserves Fruit"), chunking, categorical clustering, interactive imagery and narrative chaining (Bower and Clark, 1969). Study systems such as SQ3R (Robinson, 1946: Survey, Question, Read, Recite, Review) and PQRST (Preview, Question, Read, Self-recitation, Test) build questioning and retrieval practice into reading. The best-supported techniques are practice testing and distributed practice (Dunlosky and colleagues, 2013). Other aids are desirable difficulties such as interleaving, overlearning (Krueger, 1929), whose returns diminish, and sleep, which consolidates new learning. Rereading and highlighting feel effective but are not. Memory rehabilitation uses errorless learning, spaced retrieval and the method of vanishing cues.

## Who's who

**Table 7.11** The people in this chapter.

| Name | When | Examined for |
|---|---|---|
| Hermann Ebbinghaus | 1850 to 1909; *Memory*, 1885 | Nonsense syllables, savings method, forgetting curve; first noted serial position and spacing effects |
| Mary Whiton Calkins | 1863 to 1930; 1894 | Paired-associate learning |
| William James | 1842 to 1910; 1890 | Primary versus secondary memory; classic description of the tip-of-the-tongue state |
| Richard C. Atkinson | Born 1929; 1968 | Multi-store (modal) model with Shiffrin; keyword method (1975) |
| Richard M. Shiffrin | Born 1942; 1968 | Multi-store (modal) model with Atkinson |
| Ulric Neisser | 1928 to 2012; 1967 | Coined iconic and echoic memory; Challenger flashbulb study with Harsch (1992) |
| George Sperling | Born 1934; 1960 | Partial-report procedure; iconic memory |
| Darwin, Turvey and Crowder | 1972 | "Three-eared man" study; echoic memory of 2 to 4 s |
| George Armitage Miller | 1920 to 2012; 1956 | "Magical number seven, plus or minus two"; chunking |

| Name | When | Examined for |
|---|---|---|
| Nelson Cowan | 2001 | "Magical number four" chunks |
| John Brown; Lloyd and Margaret Peterson | 1958; 1959 | Brown-Peterson technique; short-term memory lasts about 18 to 30 s |
| Keppel and Underwood | 1962 | Short-term forgetting is proactive interference, not decay |
| Reuben Conrad | 1964 | Acoustic confusions: acoustic coding in short-term memory |
| Alan Baddeley | Born 1934; 1966, 1974, 2000 | Acoustic versus semantic coding; working memory with Hitch; episodic buffer; divers study with Godden (1975) |
| Bennet Murdock; Glanzer and Cunitz | 1962; 1966 | Serial position curve; a filled delay abolishes recency but not primacy |
| McCrary and Hunter | 1953 | Serial position curves; keyed in RML 2024 for first observing primacy-recency |
| Shallice and Warrington | 1970 | Patient K.F.: impaired short-term memory, normal long-term learning |
| Graham Hitch | 1974 | Working memory model with Baddeley |
| Robert Logie | 1995 | Visual cache and inner scribe of the sketchpad |
| Fergus Craik | Born 1935; 1972, 1975 | Levels of processing with Lockhart; depth of processing study with Tulving |
| Robert Lockhart | 1972 | Levels of processing with Craik |
| Rogers, Kuiper and Kirker | 1977 | Self-reference effect |
| Slamecka and Graf | 1978 | Generation effect |
| Allan Paivio | 1925 to 2016; 1971 | Dual coding theory (verbal and imagery codes) |
| Morris, Bransford and Franks | 1977 | Transfer-appropriate processing |
| Endel Tulving | 1927 to 2023; 1972, 1973 | Episodic versus semantic memory; autonoetic consciousness; encoding specificity (with Thomson); cue-dependent forgetting; patient K.C. |
| Larry Squire | From 1980 | Declarative versus nondeclarative memory and its taxonomy |
| Daniel Schacter | Born 1952; 1985, 2001 | Explicit versus implicit memory (with Graf); seven sins of memory |
| Meyer and Schvaneveldt | 1971 | Semantic priming in lexical decision (BREAD and BUTTER) |
| Rubin, Wetzler and Nebes | 1986 | Reminiscence bump |
| Einstein and McDaniel | 1990 | Event-based versus time-based prospective memory |
| Roger Brown | 1925 to 1997; 1966, 1977 | Tip-of-the-tongue state (with McNeill); flashbulb memory (with Kulik) |
| Talarico and Rubin | 2003 | 9/11 memories: "confidence, not consistency" |
| Alexander Romanovich Luria | 1902 to 1977; 1968 | *The Mind of a Mnemonist* (Shereshevsky) |
| John Flavell | 1928 to 2025; 1971 | Metamemory; metacognition |
| J. T. Hart | 1965 | Feeling of knowing |
| Collins and Quillian | 1969 | Hierarchical network model; cognitive economy |
| Elizabeth Loftus | Born 1944; 1974, 1975, 1995 | Spreading activation (with Collins); "smashed" versus "hit" (with Palmer); misinformation effect; lost in the mall |

| Name | When | Examined for |
|---|---|---|
| Rumelhart and McClelland | 1986 | Connectionism (parallel distributed processing) |
| Schank and Abelson | 1977 | Scripts (the restaurant script) |
| Godden and Baddeley | 1975 | Context-dependent memory in divers |
| Donald Goodwin and colleagues | 1969 | State-dependent memory with alcohol |
| Gordon Bower | 1932 to 2020; 1981 | Mood-dependent and mood-congruent memory; narrative chaining with Clark (1969) |
| Edward Lee Thorndike | 1874 to 1949 | Law of disuse; keyed as the source of decay theory |
| Jenkins and Dallenbach | 1924 | Less forgetting after sleep than after waking |
| John McGeoch | 1931, 1932 | Interference rather than decay; the rust analogy; similarity increases interference |
| Müller and Pilzecker | 1900 | Retroactive inhibition; the term consolidation |
| Melton and Irwin | 1940 | Two-factor theory of interference: response competition and unlearning |
| Benton Underwood | 1957 | Proactive interference from earlier lists |
| Anderson, Bjork and Bjork | 1994 | Retrieval-induced forgetting |
| Sigmund Freud | 1856 to 1939 | Repression (motivated forgetting); named infantile amnesia |
| Théodule Ribot | 1881 | Ribot's law: the new perishes before the old |
| Bluma Zeigarnik | 1900 to 1988; 1927 | Zeigarnik effect: interrupted tasks are better recalled |
| Maria Ovsiankina | 1928 | Ovsiankina effect: interrupted tasks tend to be resumed |
| Philip Ballard | 1913 | Reminiscence: recall rises after a delay |
| Hedwig von Restorff | 1933 | Isolation effect for distinctive items |
| Roediger and Karpicke | 2006 | Testing (retrieval practice) effect |
| Frederic Charles Bartlett | 1886 to 1969; 1932 | "The War of the Ghosts"; schema; reconstructive memory; "effort after meaning" |
| Deese; Roediger and McDermott | 1959; 1995 | DRM false memory paradigm (critical lure "sleep") |
| Marcia Johnson | 1981, 1993 | Reality monitoring (with Raye); source monitoring |
| William Beecher Scoville | 1953; 1957 | Operated on H.M.; reported the case with Milner |
| Brenda Milner | Born 1918; 1957 | Studies of H.M.; spared mirror-tracing learning |
| Sergei Korsakoff | 1854 to 1900 | Korsakoff's syndrome: amnesia with confabulation |
| Carl Wernicke | 1848 to 1905; 1881 | Wernicke's encephalopathy |
| Richard Semon | 1904 | Coined the term engram |
| Karl Spencer Lashley | 1890 to 1958; 1929, 1950 | "In search of the engram"; mass action; equipotentiality; memory distributed in the cortex |
| Donald Olding Hebb | 1904 to 1985; 1949 | Cell assemblies; "cells that fire together wire together" |
| Bliss and Lømo | 1973 | Long-term potentiation |

| Name | When | Examined for |
|---|---|---|
| Francis P. Robinson | 1946 | SQ3R study method |
| William Krueger | 1929 | Overlearning and its diminishing returns |

## Key terms

memory, encoding, storage, retrieval, encoding failure, nonsense syllable, savings method, forgetting curve, recall, free recall, serial recall, cued recall, paired-associate learning, recognition, relearning, reconstruction method, multi-store model, control processes, sensory memory, iconic memory, echoic memory, partial-report procedure, short-term memory, memory span, chunking, Brown-Peterson technique, release from proactive interference, acoustic coding, long-term memory, serial position effect, primacy effect, recency effect, double dissociation, working memory, dual-task method, phonological loop, phonological similarity effect, word-length effect, articulatory suppression, visuospatial sketchpad, central executive, episodic buffer, levels of processing, maintenance rehearsal, elaborative rehearsal, self-reference effect, generation effect, dual coding theory, transfer-appropriate processing, episodic memory, semantic memory, generic memory, autonoetic consciousness, procedural memory, explicit memory, implicit memory, priming, repetition priming, semantic priming, autobiographical memory, reminiscence bump, retrospective memory, prospective memory, flashbulb memory, eidetic imagery, metamemory, feeling of knowing, tip-of-the-tongue state, hierarchical network model, cognitive economy, spreading activation, proposition, connectionism, schema, script, retrieval cue, encoding specificity principle, context-dependent memory, state-dependent memory, mood-congruent memory, decay theory, interference theory, proactive interference, retroactive interference, cue-dependent forgetting, retrieval-induced forgetting, repression, consolidation, Ribot's law, Zeigarnik effect, reminiscence, von Restorff effect, spacing effect, testing effect, reconstructive memory, misinformation effect, false memory, source monitoring, cryptomnesia, déjà vu, confabulation, amnesia, anterograde amnesia, retrograde amnesia, Korsakoff's syndrome, infantile amnesia, engram, long-term potentiation, mnemonic, method of loci, peg-word system, keyword method, acronym, acrostic, SQ3R, overlearning

## From the past papers

1. In an experiment the stimuli were presented to the subjects in certain order and after the presentation of stimuli their order was disturbed. In the test phase the subjects were needed to set them in the presented order. Which of the following methods was used in this experiment? (RML 2024)

   (a) Reconstruction (b) Relearning (c) Recognition (d) Recall

   **Answer: (a) Reconstruction.** All the items are handed back, and the participant only has to restore their original order, which is the reconstruction method. Recall would require producing the items without them being supplied. Recognition would require picking old items out from among new ones. Relearning measures the time or trials saved when the same material is learned again. Do not confuse the method with Bartlett's theory of reconstructive memory.

2. Which type of memory has the largest capacity but shortest duration? (JEMAS 2025)
   (a) Sensory memory (b) Short-term memory (c) Long-term memory (d) Working memory

   **Answer: (a) Sensory memory.** The sensory register briefly holds almost everything that reaches the senses, for less than a second (icon) or about two to four seconds (echo). Long-term memory also has a vast capacity, but it has the longest duration, not the shortest. Short-term and working memory are small (about seven items, or four chunks) and last seconds rather than fractions of a second.

3. The capacity of working memory is best-described in terms of number of meaningful units of information, called (JEMAS 2025)
   (a) Chunks (b) Features (c) Scripts (d) Proposition

   **Answer: (a) Chunks.** Miller (1956) showed that the span limit applies to chunks, meaningful units built with long-term knowledge, and Cowan (2001) puts pure capacity at about four chunks. Features are the attributes of a concept (as in the feature comparison model). Scripts are long-term schemas for routine events. Propositions are units of meaning that can be true or false; they describe how knowledge is represented, not how much the short-term system holds.

4. Who developed the concept of 'magic number'? (RML 2024)
   (a) Heinz (b) Ebbinghaus (c) Triesman (d) Miller

   **Answer: (d) Miller.** George A. Miller's 1956 paper was titled "The magical number seven, plus or minus two." Anne Treisman (misspelt in the paper) is known for the attenuation model of attention (1964) and feature integration theory (1980), and Heinz for Johnston and Heinz's multimode theory of attention (1978); both belong to Chapter 5. Ebbinghaus noticed that about seven syllables could be repeated after a single reading, but he did not propose the "magic number".

5. Primacy-recency effect was first observed by whom? (RML 2024)
   (a) McCray and Hunter (b) Ebbinghaus (c) Craik and Lockhart (d) Baddeley and Hitch

   **Answer: (a) McCray and Hunter (the official key).** J. W. McCrary and W. S. Hunter published influential serial position curves in 1953, and the key credits them. Historically, though, Ebbinghaus had already noted in 1885 that the beginning and end of a list are learned first, so be ready for a paper that keys (b). Craik and Lockhart proposed levels of processing (1972), and Baddeley and Hitch proposed working memory (1974); neither is connected with the discovery of the serial position effect.

6. Which of the following secondary tasks is thought to affect the central executive in terms of working memory? (RML 2022)
   (a) Continually repeating a word (b) Remembering a spatial configuration (c) Listening to irrelevant speech (d) Random number generation

   **Answer: (d) Random number generation.** Producing a random sequence requires constant monitoring and suppression of habitual sequences, which is exactly the central executive's job. Repeating a word is articulatory suppression, which occupies the phonological loop. A spatial configuration loads the visuospatial sketchpad. Irrelevant speech gains access to the phonological store.

7. In Baddeley's model of working memory, the subsystem responsible for integrating information from the phonological loop, visuospatial sketchpad, and long-term memory into a single, coherent episode is the: (Real PYQ 2024)

   (a) Central Executive (b) Episodic Buffer (c) Visuospatial Sketchpad (d) Phonological Loop

   **Answer: (b) Episodic Buffer.** Baddeley added it in 2000 as a limited-capacity store that binds information from different codes into integrated episodes. The central executive controls attention but stores nothing, so it cannot hold the bound result. The sketchpad and the loop each hold only one kind of information.

8. The concept of "Self-reference effect" was founded by: (RML 2024)

   (a) Craik and Lockhart (b) Posner (c) Paivio (d) Rogers and Kuiper

   **Answer: (d) Rogers and Kuiper.** Rogers, Kuiper and Kirker (1977) added the question "Does this word describe you?" to the levels-of-processing tasks and found that recall was about twice as good as after semantic judgements. Craik and Lockhart created the framework the effect extends, but they did not discover it. Posner is known for attention research, and Paivio for dual coding theory.

9. We know that a common distinction is made between explicit and implicit memory. But which one of the following does NOT accurately reflect one of these terms? (RML 2020)

   (a) Explicit memory involves conscious awareness of the original information or the situation in which the learning occurred (b) Implicit memory refers to an influence on behaviour, feelings or thoughts as a result of prior experience (c) Explicit memory involves recollection of the original information or experience that is subsequently recalled (d) Implicit memory involves a conscious recollection of the original events

   **Answer: (d).** Implicit memory is, by definition, memory without conscious recollection (Graf and Schacter, 1985): the amnesic patient who completes MOT___ with MOTEL without remembering the list. Options (a), (b) and (c) are accurate. The wrong option takes the definition of explicit memory and gives it to implicit memory.

10. Which type of memory produces scripts of familiar routines? (RML 2025)

    (a) Generic memory (b) Episodic memory (c) Autobiographical memory (d) Working memory

    **Answer: (a) Generic memory.** A script (Schank and Abelson, 1977) is general knowledge of how a routine usually goes, abstracted from many episodes, so it belongs to generic (semantic) memory. Episodic and autobiographical memory hold particular events of one's own life, which are exactly the details that a script leaves out. Working memory is a temporary workspace, not a store of routines.

11. A person who is very adept at planning, monitoring, and controlling all phases of the multi-store model must have very good: (IHBAS 2020)

    (a) Attention (b) Inhibition (c) Meta-cognition (d) Knowledge base

    **Answer: (c) Meta-cognition.** Metacognition (including Flavell's metamemory) is knowledge about one's own cognition and the ability to monitor and control it, at every stage from encoding to retrieval. Attention selects what enters the system but does not plan or monitor the whole process. Inhibition is one executive function, the suppression of unwanted responses or memories. A knowledge base is the stored long-term knowledge that makes chunking and elaboration possible: it is what is remembered, not control over remembering.

12. Current research explains the vividness of flashbulb memories mainly by: (NIMHANS 2026 recall)

    (a) A special 'Now Print!' mechanism that stores the scene permanently and exactly (b) The absence of any interference from later events (c) Photographic (eidetic) memory, which only some people have (d) Emotional arousal at encoding plus repeated rehearsal and retelling

    **Answer: (d) Emotional arousal at encoding plus repeated rehearsal and retelling.** "Now Print!" was Brown and Kulik's original 1977 proposal, and it was abandoned when Neisser and Harsch (1992) and Talarico and Rubin (2003) showed that flashbulb memories change as much as ordinary ones. They are not free of interference: later accounts and news coverage reshape them. Eidetic imagery is a rare, lingering visual image found mainly in children, and it has nothing to do with flashbulb memories, which almost everyone forms.

13. Which model proposes that memory is organized in a network of interconnected nodes? (IHBAS 2024)

    (a) Associative model (b) Connectionist model (c) Network model (d) Linkage model

    **Answer: (b) Connectionist model.** Connectionism, or parallel distributed processing (Rumelhart and McClelland, 1986), represents memories as patterns of activation spread across many interconnected nodes, with knowledge stored in connection weights. The question is loosely worded, because Collins and Quillian's hierarchical "network model" also uses nodes and links. That is why the stem's stress on interconnection points to connectionism. "Associative" and "linkage" describe what any network does rather than naming the model in question.

14. Rahul thought that he could do much better in the English exam as it was held in the same classroom where his classes were being held whereas in geography exam he was very nervous as this was conducted in a totally new environment. This perhaps was due to: (IHBAS 2024)

    (a) Examination fear (b) The principle of congruity (c) His anxious personality (d) The principle of encoding specificity

    **Answer: (d) The principle of encoding specificity.** The familiar classroom reinstates the context in which the material was encoded, so it supplies extra retrieval cues (Tulving and Thomson, 1973); context-dependent memory is the specific case. Examination fear and an anxious personality might explain nervousness, but they do not explain why the same student expects to do better in the old room. "The principle of congruity" is not a memory principle; the nearest real idea, mood-congruent memory, concerns the match between mood and material.

15. Alcohol-induced amnesia may be an example of: (RML 2022)

    (a) Dependent memory (b) State-dependent memory (c) Independent memory (d) State-independent memory

    **Answer: (b) State-dependent memory.** Material learned while intoxicated is recalled better when the person is intoxicated again (Goodwin and colleagues, 1969), so the forgetting on sobering up depends on the mismatch of internal states. "Dependent memory", "independent memory" and "state-independent memory" are not established terms; they are distractors built from the words of the right answer. Strictly, a complete blackout in which nothing was ever encoded is anterograde amnesia, and no change of state can bring it back.

16. An English-speaking person may have greater difficulty learning Spanish because of his or her tendency to wish to apply English grammar to the new language. This occurs due to which of the following phenomenon? (RML 2024)

    (a) Proactive reference (b) Retroactive reference (c) Trace decay (d) Retrieval failure

    **Answer: (a) Proactive interference; the official key, (b), is wrong.** English was learned first and is disrupting the newer learning of Spanish. Old disrupting new is proactive interference by definition ("reference" in the options means interference). Retroactive interference would be Spanish making English grammar harder to recall. Trace decay is fading with disuse, and retrieval failure is a missing cue; neither fits an old habit intruding on new learning.

17. Who developed the Decay theory? (RML 2024)

    (a) Ebbinghaus (b) Thorndike (c) Tulving (d) Melton and Irwin

    **Answer: (b) Thorndike.** Thorndike's law of disuse held that connections that are not exercised grow weaker, which is the decay idea in psychology. Ebbinghaus charted the time course of forgetting but did not claim that time causes it. Tulving is associated with cue-dependent forgetting, and Melton and Irwin (1940) with the two-factor theory of interference.

18. Retrieval-induced forgetting refers to which memory-based phenomenon? (RML 2020)

    (a) People remembered practised information better than unpractised information (b) People remembered unpractised information better than practised information (c) People remember names better than faces (d) People remember faces better than names

    **Answer: (a).** In the study by Anderson, Bjork and Bjork (1994), practised items were recalled best. The defining finding, though, is that unpractised items related to the practised ones were recalled worse than items from unpractised categories: retrieving some memories inhibits their competitors. Option (b) reverses the result, and the options about names and faces have nothing to do with the phenomenon.

19. Which statement about forgetting is incorrect? (RML 2025)

    (a) Forgetting helps clear out unnecessary information (b) Forgetting improves memory by prioritizing key concepts (c) Less retention leads to less forgetting (d) High retention correlates with less forgetting

    **Answer: (c) Less retention leads to less forgetting.** Retention and forgetting are two sides of one coin: whatever is not retained has been forgotten, so less retention means more forgetting, as (d) correctly says. Options (a) and (b) describe the adaptive side of forgetting: letting unused detail fade reduces interference and helps us abstract what matters.

20. Being unable to recall the relevant elements of an autobiographical experience from memory is known as a deficit in: (RML 2022)

    (a) Reality-monitoring ability (b) Personal-monitoring ability (c) Social-monitoring ability (d) Source-monitoring ability

    **Answer: (d) Source-monitoring ability.** Source monitoring (Johnson, Hashtroudi and Lindsay, 1993) is the attribution of a memory to its origin from its contextual details. Losing those details produces source errors and source amnesia. Reality monitoring is a narrower case, distinguishing memories of real events from imagined ones. "Personal-monitoring" and "social-monitoring" are invented distractors.

21. Defense attorneys often protest the prosecutor's use of eyewitness testimony because: (Amity 2023)

    (a) Eyewitnesses are more prone to forgetting (b) A confident witness is usually right (c) Source errors may cause an eyewitness to make a false identification (d) Poor eyesight of eyewitnesses tends to invalidate their testimony

    **Answer: (c) Source errors may cause an eyewitness to make a false identification.** A face seen at the scene, in a photograph or in a line-up can feel familiar for the wrong reason, and the witness then confidently picks the wrong person. The problem is not that eyewitnesses forget more than anyone else, and it is rarely poor eyesight. Option (b) is the opposite of the research: confidence, especially when inflated by feedback, is a poor guide to accuracy.

22. According to Karl Lashley's research on memory storage, where did he conclude that memories are primarily stored in the brain? (Real PYQ 2025)

    (a) In specific localized brain regions dedicated to particular memories (b) Distributed throughout the entire brain cortex (c) Exclusively in the hippocampus (d) Only in the frontal and temporal lobes

    **Answer: (b) Distributed throughout the entire brain cortex.** Lashley's lesion studies in rats found that maze memory depended on how much cortex was removed (mass action), not on which part, and that cortical areas could substitute for one another (equipotentiality). He therefore rejected localisation (a) and (d). The hippocampus (c) was linked to memory later by the case of H.M., and even then as the structure for consolidation, not the exclusive store.

23. The famous patient H.M. (Henry Molaison) underwent surgical removal of the medial temporal lobes, including the hippocampus, to treat severe epilepsy. While he retained most of his intellectual abilities and working memory, his primary and most profound deficit was in: (Real PYQ 2024)

    (a) Procedural memory (b) Semantic memory (c) Anterograde amnesia (d) Retrograde amnesia

    **Answer: (c) Anterograde amnesia.** H.M. could not form new declarative memories after his 1953 operation. His procedural learning was spared: he improved at mirror tracing over three days without remembering the task. His old semantic knowledge was intact, so semantic memory was not selectively lost. His retrograde amnesia covered only the years before the surgery and was far milder than his anterograde loss.

24. A patient presents with severe memory deficits, confabulation, and nystagmus, with a history of chronic alcohol use. This clinical picture is classic for: (Real PYQ 2024)

    (a) Alzheimer's Disease (b) Wernicke-Korsakoff Syndrome (c) Vascular Dementia (d) Frontotemporal Dementia

    **Answer: (b) Wernicke-Korsakoff Syndrome.** Thiamine deficiency in chronic alcoholism produces Wernicke's encephalopathy (confusion, ataxia and eye signs such as nystagmus), followed by Korsakoff's amnesia with confabulation. Alzheimer's disease is a gradual dementia of older age, beginning with recent episodic memory, without nystagmus or an alcohol link. Vascular dementia typically declines step by step after strokes, and frontotemporal dementia begins with changes in personality, behaviour or language (Psychiatric Disorders book).

25. In PQRST method, a method of enhancing memory, what does the alphabet 'S' stand for? (RML 2024)

    (a) Storing (b) Self-recitation (c) Sequencing (d) Semantics

    **Answer: (b) Self-recitation.** PQRST stands for Preview, Question, Read, Self-recitation and Test, a close variant of Robinson's SQ3R (Survey, Question, Read, Recite, Review). Self-recitation is retrieval practice, the step that does most for retention. Storing, sequencing and semantics sound like memory terms but are not steps of the method.

## High-yield sheet

*The facts this chapter's questions ask most, most-asked first. Last read before the exam.*

1 **Korsakoff's syndrome** (Korsakoff's psychosis) follows chronic alcoholism and thiamine (vitamin B1) deficiency, often after Wernicke's encephalopathy (confusion, ataxia, nystagmus; together the Wernicke-Korsakoff syndrome). It damages the mammillary bodies and mediodorsal thalamus and produces dense anterograde amnesia, extensive retrograde amnesia and confabulation: sincere, false stories that fill memory gaps, not malingering. ASKED 8 TIMES

2 **H.M.** (Henry Molaison): both medial temporal lobes (hippocampus and amygdala) removed by William Beecher Scoville in 1953; reported by Scoville and Brenda Milner in 1957. Profound anterograde amnesia for declarative memory, with normal short-term memory, limited retrograde amnesia and spared procedural learning (mirror tracing). ASKED 7 TIMES

3 The **hippocampus** (medial temporal lobe) is the structure damaged in anterograde amnesia and is critical for consolidating new declarative memories; it is not the store of remote memories. Amnesia questions offering the hypothalamus, pituitary or cerebellum want the hippocampus. ASKED 6 TIMES

4 **Proactive interference**: old learning disrupts new (dialling your old phone number; English grammar intruding on Spanish, although RML 2024 keyed "retroactive"). **Retroactive interference**: new learning disrupts old. PRO = old disrupts new. ASKED 5 TIMES

5 **Sensory memory** (icon about 250 to 500 ms, echo about 2 to 4 s) has the largest capacity and the shortest duration, and "icon and echo" always mean sensory memory. Asked only for the largest capacity in the Atkinson and Shiffrin model, answer long-term memory (no known limit). ASKED 5 TIMES

6 **Endel Tulving** (1972) separated episodic memory, personal events with their what, where and when (your first day of college), from semantic memory, the network of facts and concepts. Procedural memory holds skills (riding a bicycle); autobiographical memory is memory for specific episodes of one's own life. Patient K.C. lost episodic memory but kept semantic knowledge. ASKED 5 TIMES

7 **Flashbulb memories** (Roger Brown and James Kulik, 1977) are vivid memories of learning of a shocking event. Confidence and vividness stay very high while accuracy and consistency fall (Neisser and Harsch, 1992; Talarico and Rubin, 2003); emotional arousal plus rehearsal explains them, not a "Now Print!" mechanism. ASKED 4 TIMES

8 **Anterograde amnesia** is the inability to form new memories after the injury; **retrograde amnesia** is the loss of memories from before it, usually following Ribot's law; amnesia itself is a pathological loss of memory. ASKED 4 TIMES

9 **Karl Lashley** concluded that memories are distributed throughout the cortex (mass action, equipotentiality). The **engram** (Richard Semon's term, 1904) is the physical memory trace. **Long-term potentiation** (Timothy Bliss and Terje Lømo, 1973) is a lasting strengthening of synapses after high-frequency stimulation, linked with memory. ASKED 4 TIMES

10 **Misinformation effect** (Elizabeth Loftus; Loftus and Palmer, 1974, "smashed" versus "hit"): memories are distorted by leading questions or later misleading information. Source errors cause false eyewitness identifications, and failing to recall the contextual elements of an experience is a deficit in **source monitoring** (Marcia Johnson). ASKED 3 TIMES

11 **George A. Miller** (1956), "The magical number seven, plus or minus two": short-term memory span is 7 ± 2 meaningful units called chunks (Nelson Cowan, 2001: about 4 chunks). ASKED 4 TIMES

12 **Serial position effect** (primacy-recency effect): the first and last items of a list are recalled better than the middle. Primacy reflects rehearsal into long-term memory; recency reflects items still in short-term memory, and a 30-second filled delay removes it (Glanzer and Cunitz, 1966). ASKED 4 TIMES

13 **Encoding specificity principle** (Tulving and Thomson, 1973): recall is best when the cues at retrieval match those at encoding. Context-dependent memory is the same external place (Godden and Baddeley's divers, 1975; Rahul's exam in his usual classroom); state-dependent memory is the same internal state (alcohol). ASKED 3 TIMES

14 **Working memory** (Alan Baddeley and Graham Hitch, 1974) has a central executive, a phonological loop and a visuospatial sketchpad, plus the episodic buffer added in 2000; a "visual executive loop" does not exist. ASKED 3 TIMES

15 Random number generation loads the **central executive**; backward digit span tests working memory; mentally rotating an object is the job of the **visuospatial sketchpad**. ASKED 3 TIMES

16 The **Stroop effect** (John Ridley Stroop, 1935) is interference in selective attention: automatic reading of the word competes with the controlled task of naming the ink colour. ASKED 3 TIMES

17 A **schema** (Frederic Bartlett, 1932, "The War of the Ghosts") is organised knowledge that guides and distorts recall; the schema approach studies how organised knowledge is used to understand behaviour; **scripts** of familiar routines (Schank and Abelson, 1977) belong to generic memory. ASKED 3 TIMES

18 The **episodic buffer**, added by Baddeley in 2000, integrates information from the phonological loop, the visuospatial sketchpad and long-term memory into a single coherent episode. ASKED 3 TIMES

19 The **phonological loop** (phonological store plus articulatory control process) stores and rehearses auditory-verbal information. ASKED 3 TIMES

20 The **tip-of-the-tongue state** (Roger Brown and David McNeill, 1966) is a temporary inability to retrieve a well-known word: a failure of retrieval, not of storage. ASKED 2 TIMES

21 The **self-reference effect**, better recall of information related to oneself, was shown by T. B. Rogers, Nicholas Kuiper and W. S. Kirker (1977); keys write "Rogers and Kuiper". ASKED 2 TIMES

22 **Implicit memory** affects performance without conscious awareness (priming, skills); **explicit memory** involves conscious recollection (Graf and Schacter, 1985). ASKED 2 TIMES

23 The **connectionist** or parallel distributed processing model (Rumelhart and McClelland, 1986) treats memory as a network of interconnected nodes activated simultaneously, with knowledge held in the connection weights. ASKED 2 TIMES

24 In the RML 2025 matching: dementia goes with abstraction, amnesia with memory, aphasia (with alexia and acalculia) with language, and apraxia with skilled movement. ASKED 2 TIMES

25 **Levels of processing** (Fergus Craik and Robert Lockhart, 1972): structural, then phonemic, then semantic; deep, semantic processing gives the best retention. ASKED 2 TIMES

26 **Repression** (Freud) is the unconscious exclusion of threatening memories from awareness; suppression is conscious. ASKED 2 TIMES

27 The stage (multi-store) model of Richard Atkinson and Richard Shiffrin (1968) has three stores, sensory, short-term and long-term; information must pass through short-term memory to reach long-term memory, and working memory is not one of its stores. ASKED 2 TIMES

28 RML 2024 keys **McCrary and Hunter** (1953) for first observing the primacy-recency effect; the earlier observation is Ebbinghaus's (1885). ASKED ONCE

29 **Decay theory** is traced to Edward Lee Thorndike's law of disuse (RML 2024 key); Melton and Irwin (1940) proposed the two-factor theory of interference. ASKED ONCE

30 **PQRST** is Preview, Question, Read, Self-recitation, Test; SQ3R (Francis P. Robinson, 1946) is Survey, Question, Read, Recite, Review. ASKED ONCE

CHAPTER 8

# Thinking, Problem Solving, Reasoning, Decision Making, Creativity and Language

IN THIS CHAPTER

8.1 ● What Is Thinking?: Define thinking, name its elements, explain what the mental-imagery experiments showed, and distinguish the main types of thinking.

8.2 ●● Concepts: Describe how concepts are formed, compare the classical, prototype and exemplar views, and explain how concepts are organised in memory.

8.3 ●● Problem Solving: Explain the problem space, the main problem-solving strategies, insight and incubation, and the obstacles that block a solution.

8.4 ●●● Reasoning: Distinguish deductive, inductive and abductive reasoning and explain why people go wrong on syllogisms, conditional rules and the Wason selection task.

8.5 ●● Judgement and Decision Making: Contrast normative and descriptive models of choice and explain bounded rationality, the classic heuristics and biases, framing and prospect theory.

8.6 ●● Creativity: Define creativity and describe its components, its stages, how it is measured, how it relates to intelligence and the main theories of it.

8.7 ● The Structure of Language: Identify the defining properties and units of language and explain Chomsky's transformational grammar.

8.8 ●● Acquiring Language: Compare learning, nativist and interactionist theories, evaluate the critical period, and trace the milestones of language development.

8.9 ●● Language, Thought and Communication: Evaluate the Sapir-Whorf hypothesis and the Piaget-Vygotsky debate, and survey bilingualism, animal language, speech perception, reading and aphasia.

Weightage. 53 of the 819 questions mapped in this book (6.5%) test this chapter, 28 of them from real past papers. 1 of its 9 sections is High yield.

In 1913 a young German psychologist named **Wolfgang Köhler** (1887 to 1967) sailed to Tenerife, in the Canary Islands, to run a small research station of the Prussian Academy of Sciences. The station kept a colony of chimpanzees, and Köhler, one of the three founders of Gestalt psychology, wanted to answer a question that had been nagging him: could an ape solve a problem by *understanding* it, or only by blundering about until it stumbled on the answer? The First World War broke out the following year and kept him on the island far longer than he had planned, until 1920. The war's loss was psychology's gain.

The brightest of Köhler's chimpanzees was a male called Sultan. In one test a banana lay on the ground outside the bars of Sultan's enclosure, too far to reach by hand. Inside lay two hollow bamboo sticks, neither long enough on its own to reach the fruit. Sultan tried one stick, then the other. He pushed one stick out with the other until its tip touched the banana, but he could not draw the fruit in. Eventually he seemed to give up and sat playing idly with the sticks. Then, according to the record Köhler published, he happened to hold the two sticks in a line, pushed the thinner one into the hollow end of the thicker, jumped up, ran to the bars and raked in the banana with his new, longer tool. When the test was repeated he made the double stick at once, without fumbling.

Other chimpanzees stacked boxes to reach fruit hung from the roof, or fetched a stick from another room to use as a rake. To Köhler, these solutions did not look like the slow, accidental trial and error that Edward Thorndike had described in cats escaping from puzzle boxes in 1898. They appeared suddenly, as a whole, and once found they were kept. Köhler called this *Einsicht*, insight, and described the work in a 1917 book that appeared in English in 1925 as *The Mentality of Apes*.

What was going on inside Sultan's head in the moment before he jumped up? Did he "see" a picture of the joined sticks? Did he have a concept of a tool? Was he reasoning? And here is the question that will follow us to the end of this chapter: Sultan had no words, so was he *thinking* at all? Remember Sultan; you will meet him again at the end of the chapter.

## 8.1 What Is Thinking? LOWER YIELD · 3 QS

Ask ten people what thinking is and most of the answers will go round in a circle ("thinking is when you think about things"). Psychologists define **thinking** (or thought) as the mental manipulation of internal representations of information to form new representations: to understand something, solve a problem, reason, decide or create. The key word is *manipulation*. Seeing the chai stall in front of you is not thinking, and neither is simply remembering what a cup cost yesterday. Working out whether you can afford a second cup and still catch the bus is. Thinking takes what perception and memory supply and transforms it.

A broader word you will meet constantly is *cognition*: all the mental processes by which we acquire, store, transform and use knowledge, from attention and perception through memory to language and thought. Thinking is the part of cognition that does the transforming.

Attention, perception and memory have their own chapters (4 to 7); this chapter is about what the mind does with what they deliver.

## How Thought Came Back Into Psychology ●●

If you want a science of thought you need some way to observe it, and for the first half of the twentieth century psychology could not agree on one. The structuralists, led in America by **Edward Bradford Titchener** (1867 to 1927), held that every thought is built from sensations, images and feelings, and that trained introspection could reveal them. In the first decade of the 1900s a group of researchers at the University of Würzburg, led by **Oswald Külpe** (1862 to 1915), gave trained observers simple tasks (find a word that is a superordinate of "bird", for example) and asked them to describe what passed through their minds. To everyone's surprise the observers often reported thoughts with no image or sensation in them at all. This became known as **imageless thought**. The Würzburg group also noticed that the task itself (the *Aufgabe*) steered thinking without being consciously present; Narziss Ach called this guiding influence the *determining tendency*.

Titchener's observers at Cornell, doing the same kind of introspection, insisted that they always found images. Who was right? There was no way to tell, because introspective reports cannot be checked by anyone else. That deadlock was one reason **John Broadus Watson** (1878 to 1958) threw introspection out of psychology in 1913. For Watson thinking was nothing more than implicit, subvocal speech: tiny movements of the larynx and tongue (you will see in Section 8.9 how that idea was tested and failed).

Thought came back with the *cognitive revolution* of the 1950s and 1960s. **Noam Chomsky** (born 1928) attacked the behaviourist account of language; **Allen Newell** (1927 to 1992) and **Herbert Alexander Simon** (1916 to 2001) built computer programs that solved problems; **Jerome Seymour Bruner** (1915 to 2016) and his colleagues published *A Study of Thinking* (1956); and in 1967 **Ulric Neisser** (1928 to 2012) wrote the textbook *Cognitive Psychology*, which gave the new field its name. Notice the irony. Cognitive psychology returned to the very topics the structuralists had cared about (consciousness, mental imagery, thought) but measured them objectively, with reaction times and error rates, instead of trusting introspective reports. (The history of the schools is told in Chapter 2.)

> EXAM TRAP
>
> **Which modern approach is closest to structuralism?** Cognitive psychology, because it studies the structuralists' topics (consciousness, imagery, thought processes), not behaviourism (which banned them), not functionalism (which asked what the mind is *for*) and not neuroscience (which studies the nervous system that underlies all behaviour rather than continuing the structuralists' programme). **Who pioneered cognitive psychology?** Neisser named the field (1967) and is often called its father; Chomsky, Newell and Simon, Bruner and George Miller drove the revolution. Jean Piaget, who studied how children's thinking develops, is keyed as "a pioneer contributor to cognitive psychology" in one recent paper. Köhler was a forerunner (Gestalt work on insight). Lawrence Kohlberg is the name to reject: he is examined for moral development.

## The Elements of Thought

What is thought made of? Textbooks list five building blocks. To see them at work, imagine you are deciding how to get to an exam hall across Mysuru early on a Monday morning.

- **Images.** You "see" the route in your mind's eye: the crowded junction, the bus stop beside the temple. A mental image is a quasi-perceptual representation of something not present to the senses, and images can be auditory or tactile as well as visual.
- **Concepts.** You think about "buses", "auto-rickshaws" and "traffic jams". A concept is a mental category that groups things sharing common properties (Section 8.2 is devoted to them).
- **Symbols and signs.** The ₹ sign on the fare chart and the number 201 on the bus are symbols: they stand for something purely by convention. A sign, in the older textbook usage, has a natural or causal link to what it signals: dark clouds are a sign of rain, smoke a sign of fire.
- **Propositions.** "The eight o'clock bus is usually late." A proposition is the smallest unit of knowledge that can be judged true or false; it asserts a relation between concepts.
- **Language.** Finally you tell yourself, in words, "Take the auto." Language is the most powerful symbol system we have, because a finite set of words and rules can express an unlimited number of propositions.

Older textbooks sometimes add a sixth element, implicit muscular activity, which reflects Watson's motor theory of thought. Most modern accounts drop it.

## Mental Imagery: Are There Pictures in the Head? ●●

When you imagine the route to the exam hall, is there really something like a picture in your head, or does it only feel that way? Psychologists define **mental imagery** as the mental representation of a stimulus that is not physically present, experienced as if one were perceiving it. For decades the question of what images "are" seemed unanswerable, for exactly the reasons that sank the imageless-thought debate. Then, in 1971, an experiment showed that images could be measured with a stopwatch.

### KEY STUDY: SHEPARD AND METZLER'S MENTAL ROTATION (1971)

**The question.** When you judge whether two objects in different orientations are the same, do you turn one of them in your mind? **The method. Roger Newland Shepard** (1929 to 2022) and Jacqueline Metzler showed eight adults pairs of line drawings of three-dimensional objects, each built from ten cubes joined face to face. The second object was rotated by 0 to 180 degrees relative to the first, either in the picture plane or in depth, and was either the same object or its mirror image. Participants pressed "same" or "different" as fast as they could.

**The result.** The time needed to say "same" increased in a straight line with the angle between the two objects, by roughly one second for every 60 degrees (Figure 8.1), and rotation in depth took no longer than rotation in the picture plane. **The conclusion.** People appear to rotate an analogue representation through intermediate positions, much as one would turn a real object; Shepard called this **mental rotation**. **The criticism.** A straight line shows that the process is gradual and continuous, but it does not prove that the representation is literally a picture.

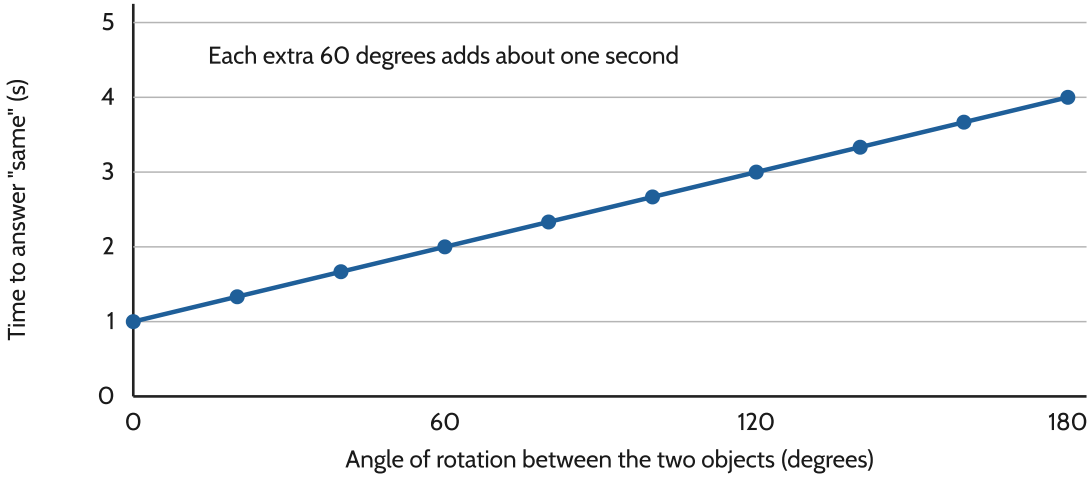



Figure 8.1 The mental rotation function, idealised from Shepard and Metzler (1971). Notice that the line is straight: the further the object must be turned, the longer the judgement takes, just as if a real object were being rotated at a steady speed.

**Stephen Michael Kosslyn** (born 1948) extended the stopwatch approach. In one experiment (Kosslyn, Ball and Reiser, 1978) participants memorised a map of a fictional island with seven landmarks (a hut, a tree, a rock, a well, a lake, a patch of sand and a patch of grass). Later, with eyes closed, they imagined a black speck travelling from one landmark to another. The time taken rose in a straight line with the real distance on the map, as if they were scanning across a picture. In another study, people asked to imagine a rabbit beside an elephant took longer to "see" whether the rabbit had whiskers than people who imagined the same rabbit beside a fly: next to the elephant, the imagined rabbit was small and its details hard to inspect. Kosslyn concluded that images are *depictive* representations that preserve spatial layout, size and distance.

Not everyone was convinced. In 1973 **Zenon Walter Pylyshyn** (1937 to 2022) launched what became known as the *imagery debate*. According to his **propositional theory**, the experience of an image is an epiphenomenon, a by-product that does no work; underneath, knowledge is stored as abstract, language-like propositions ("the hut is north of the lake"). Why, then, does scanning time follow distance? Pylyshyn's answer was *tacit knowledge*: participants know how long it takes to look across a real map and, consciously or not, they reproduce that timing when asked to "scan" an image.

The debate was eventually settled largely by the brain. Imagining an object activates early visual cortex, including the primary visual cortex, and does so in a spatially organised way. Brain damage that disturbs vision tends to disturb imagery in parallel. In a famous demonstration, Edoardo Bisiach and Claudio Luzzatti (1978) asked patients with left-sided neglect after right-hemisphere damage to imagine standing on the cathedral steps in Milan's Piazza del Duomo and describe what they "saw". They listed buildings on the right of the square only. Asked to imagine standing at the opposite end, they now listed the buildings they had left out, and omitted the ones they had just named. A purely propositional store has no left or right side to neglect. Yet images are not simply photographs either: Deborah Chambers and Daniel Reisberg (1985) found that people who formed an image of the ambiguous duck-rabbit figure could not "flip" it in imagery, although they flipped it at once when they drew their im-

age on paper. The modern consensus is that images are depictive, but they come with their interpretation attached.

Another Canadian psychologist, **Allan Paivio** (1925 to 2016), asked a related question about memory. His **dual-coding theory** (1971) proposes two separate but interconnected systems for representing information: a *verbal* system for words and a *nonverbal* or imagery system for pictures and objects. Concrete words such as "elephant" can be coded in both systems, whereas abstract words such as "justice" are coded mainly in the verbal one. Two codes give two routes to retrieval, which explains why concrete words are remembered better than abstract ones (the *concreteness effect*) and pictures better than words (the *picture superiority effect*). Imagery mnemonics such as the method of loci exploit the same advantage (Chapter 7).

> EXAM TRAP
>
> "According to dual-coding theory, which information is more easily remembered?" The answer is **visual** (pictorial, imageable) information, because it can be coded twice. Verbal, auditory and abstract "semantic" material usually gets only the verbal code. Dual coding belongs to **Paivio (1971)**; do not confuse it with Craik and Lockhart's levels of processing (1972) or with Baddeley and Hitch's working memory model (1974). And keep the two imagery theorists straight: **Kosslyn** = depictive images (scanning the island); **Pylyshyn** = propositions and tacit knowledge.

## Types of Thinking

Not all thinking is alike. Some of it is aimed at a goal and checked against reality; some drifts. Some narrows down to one answer; some fans out. The distinctions below recur in every entrance paper, usually as "match the type with its description" or "who coined the term" items. Table 8.1 gathers them.

The most basic contrast is between *directed* and *undirected* thinking. Directed (or realistic) thinking is controlled by a goal and constrained by logic and reality: planning a journey, solving a sum, writing an answer. Undirected or associative thinking lets one idea call up the next by association, with no goal steering it; daydreaming in a warm afternoon lecture is the everyday example, and Freud's free association is its clinical use.

At the far end of undirected thinking lies what the Swiss psychiatrist **Eugen Bleuler** (1857 to 1939) called **autistic thinking** in his 1911 monograph on dementia praecox, the book that made his new terms "schizophrenia" and "autism" famous. Autistic thinking is thinking governed by wishes and needs rather than by reality: fantasy in which you top the merit list, receive the award and silence your critics. Because the word "autistic" invited confusion, Bleuler later offered the alternative *dereistic* thinking (thinking "away from reality"). Jean Piaget borrowed Bleuler's term and placed the young child's egocentric thought halfway between autistic thought and mature directed thought, a point you will meet again in Section 8.9.

The American psychometrician **Joy Paul Guilford** (1897 to 1987) contributed the most famous pair. In **convergent thinking**, thought narrows down to the single best or correct answer, as when you solve a multiple-choice question or a crossword clue. In **divergent thinking**, it fans out to generate many possible answers, as when you list all the uses you can think of for a brick. Divergent thinking is the core of Guilford's account of creativity and of most creativity tests (Section 8.6).

By **critical thinking** psychologists mean the careful evaluation of claims, arguments and evidence before accepting them; the philosopher Robert Ennis's widely used definition is "reasonable reflective thinking focused on deciding what to believe or do". Checking who wrote a health message forwarded on a family WhatsApp group, and on what evidence, is critical thinking. *Creative thinking*, by contrast, produces ideas or products that are both new and useful. Critical thinking evaluates; creative thinking generates. Good problem solvers alternate between them.

The term **reflective thinking** belongs to the American philosopher and educator **John Dewey** (1859 to 1952). In *How We Think* (1910) he defined it as the active, persistent and careful consideration of a belief in the light of the grounds that support it, and described it as beginning with a felt difficulty and moving through defining the problem, suggesting solutions, reasoning out their implications and testing them. (Those five steps are also one of the earliest models of problem solving; see Section 8.3.) Finally, the Gestalt psychologist **Max Wertheimer** (1880 to 1943), in *Productive Thinking* (published in 1945, after his death), distinguished *productive thinking*, which restructures a problem and so produces genuine understanding, from *reproductive thinking*, which merely reapplies a learned method. A child who sees that a parallelogram can be turned into a rectangle by cutting off a triangle at one end and moving it to the other end has thought productively; a child who simply plugs numbers into a memorised formula has thought reproductively.

Table 8.1 Types of thinking and the names attached to them.

| Type | Associated with | What it is | Everyday example |
|---|---|---|---|
| Directed (realistic) | Würzburg school; cognitive psychology | Goal-guided, controlled, checked against reality | Planning a train journey |
| Undirected (associative) | Associationists; Freud's free association | Ideas follow one another by association, no goal | Daydreaming in a lecture |
| Autistic (dereistic) | Eugen Bleuler, 1911 | Wish-fulfilling fantasy, turned away from reality | Imagining topping the merit list |
| Convergent | J. P. Guilford | Narrows to one correct answer | Solving an MCQ |
| Divergent | J. P. Guilford | Generates many varied answers | Listing uses for a brick |
| Critical | Robert Ennis and others | Evaluates claims and evidence | Checking a forwarded health claim |
| Creative | Wallas, Guilford, Torrance | Produces something novel and useful | Inventing a new teaching aid |
| Reflective | John Dewey, 1910 | Persistent, careful weighing of a belief and its grounds | Asking "how do I know this?" |
| Productive versus reproductive | Max Wertheimer, 1945 | Restructuring a problem versus reapplying an old method | Insight versus rote formula |

EXAM TRAP

**Autistic thinking is not autism spectrum disorder.** It is Bleuler's 1911 term for reality-ignoring, wish-driven fantasy, and it appears in thinking chapters, not in the developmental disorders. **Reflective thinking** is Dewey (1910); **productive thinking** is Wertheimer (1945); **convergent and divergent** are Guilford. If an option pairs divergent thinking with "one correct answer", it has the pair reversed.

CLINICAL LINK

In the mental status examination, disturbances of thinking are classically sorted under four headings. The **stream** of thought (its speed and flow: pressure of thought, poverty of thought, thought block); the **form** of thought, meaning how ideas are linked (loosening of associations or derailment, tangentiality, circumstantiality, flight of ideas, "knight's move" thinking, word salad, neologisms), the formal thought disorder characteristic of schizophrenia, of which word salad is the most severe form; the **content** of thought (delusions, overvalued ideas, preoccupations); and the **possession** of thought, the sense that one's thoughts are one's own (obsessions, and thought insertion, withdrawal and broadcasting). A patient whose answer wanders through needless detail but finally reaches the point shows *circumstantiality*, a disorder of form; one who drifts off and never returns shows *tangentiality*. The disorders themselves are covered in the Psychiatric Disorders book.

## 8.2 Concepts MEDIUM YIELD · 4 QS

Imagine a world without concepts. Every dog you met would be a brand-new creature: you would not know whether it might bite, whether it eats grass, or whether it can be stroked. A **concept** is a mental representation of a category, a class of objects, events, ideas or people that share common properties. Concepts give us two enormous savings. They provide *cognitive economy*, because we can store what we know about dogs once rather than separately for every dog; and they support *inference*, because once you classify a new animal as a dog you can predict a great deal about it that you have not observed. Most of the elements of thought described in Section 8.1 depend on concepts.

### How Concepts Are Formed

The term **concept formation** refers to the process by which we learn to group things into categories and to attach the right properties to each. Many textbooks, especially in India, describe it as passing through a series of steps. Follow a two-year-old learning the concept "dog".

1. **Observation.** The child meets many particular instances: the neighbour's dog, the street dog near the gate, a dog in a picture book.
2. **Discrimination.** The child notices differences: dogs bark and cows do not; dogs are smaller than cows; cats have a different face.
3. **Generalisation.** The child notices what the instances have in common (four legs, fur, a tail, barking) and extends the label to new animals that share these features.
4. **Abstraction.** The child pulls out the common features and represents them apart from any particular dog, so that "dog" now means the shared pattern, not Tommy next door.

Many books add a final step of *naming* or labelling, because attaching a word to the abstraction makes the concept stable and shareable. The steps overlap in real life, and mistakes along the way are informative: the toddler who calls every four-legged animal "dog" has generalised too widely (an overextension; see Section 8.8).

The first laboratory study of concept learning was carried out by **Clark Leonard Hull** (1884 to 1952), better known for his learning theory (Chapter 6). In 1920 Hull taught participants nonsense names for Chinese characters. Characters that shared a particular radical (a compo-

nent stroke pattern) all received the same name, although the radical was embedded in otherwise different characters. Participants gradually came to name new characters correctly, often without being able to say what the common element was. Hull concluded that a concept is learned by the gradual, reinforced abstraction of a common element, an associative, step-by-step view.

### The Classical View and Artificial Concepts ●●

The oldest account of concepts goes back to Aristotle. According to the **classical view** (also called the defining-attribute view), every concept is defined by a set of *defining features* that are individually necessary and jointly sufficient. "Triangle" is a closed figure with three straight sides: anything with all three features is a triangle, and anything lacking one of them is not. "Bachelor" is an unmarried adult male. On this view membership is all-or-none, the boundaries are sharp, and every member is as good a member as any other.

The classical view fits well-defined, rule-based categories, and these are exactly the categories that dominated early laboratory work. An **artificial concept** (or formal concept) is one defined by an explicit rule, typically invented by an experimenter, such as "all cards with two red borders". The rules can be combined in three logical ways:

- A **conjunctive concept** requires the joint presence of two or more attributes ("red *and* square"). An eligible voter in India is a conjunctive concept: an Indian citizen *and* aged eighteen or over *and* on the electoral roll.
- A **disjunctive concept** requires one attribute *or* another, or both ("red *or* square"). "Out" in cricket is beautifully disjunctive: bowled, or caught, or leg before wicket, or run out, or stumped, and so on. Two dismissals can share no feature at all and still both be "out".
- A **relational concept** is defined by a relationship between attributes rather than by the attributes themselves: "taller than", "between", "uncle" (a brother of one's parent), or "a card with more figures than borders".

People find conjunctive concepts easiest and disjunctive ones hardest, partly because we expect members of a category to share something.

#### KEY STUDY: BRUNER, GOODNOW AND AUSTIN'S *A STUDY OF THINKING* (1956)

**The question.** When people learn an artificial concept, do they gradually strengthen associations (as Hull thought), or do they actively form and test hypotheses? **The method.** Jerome Bruner, Jacqueline Goodnow and George Austin used an array of 81 cards that varied on four attributes, each with three values: the shape of the figures (cross, circle or square), their colour (red, green or black), the number of figures (one, two or three) and the number of borders (one, two or three). The experimenter had a concept in mind, such as "red circles". In the *selection* paradigm the participant chose cards one at a time and was told whether each was a positive instance; in the *reception* paradigm the experimenter chose the cards.

**The result.** Participants did not behave like associative machines. They formed hypotheses about the rule and used recognisable strategies to test them (Table 8.2). **The conclusion.** Concept attainment is an active process of hypothesis testing. This is the "hypothesis-testing model of concept formation". **The criticism.** The concepts were artificial and rule-defined; the real concepts of everyday life, as Eleanor Rosch was to show, are not like that.

Table 8.2 Bruner, Goodnow and Austin's concept-attainment strategies.

| Strategy | Family | How it works | Cost and benefit |
|---|---|---|---|
| Simultaneous scanning | Scanning | Hold every possible hypothesis in mind and eliminate those each new card rules out | Logically perfect, but overloads memory |
| Successive scanning | Scanning | Test one hypothesis at a time; keep it while it works, switch when it fails | Easy on memory, slow and often repeats choices |
| Conservative focusing | Focusing | Take a positive card as the focus and change one attribute at a time | Most efficient and reliable; each choice is informative |
| Focus gambling | Focusing | Take a positive card and change more than one attribute at a time | Can win fast, but a negative result is ambiguous |

Take an example. Suppose the first positive card shows two red circles with one border. A conservative focuser next picks a card that is identical except that the circles are green. If that card is negative, colour matters (red is part of the concept); if positive, colour is irrelevant. One attribute changes, one question is answered. A focus gambler changes colour and number together, hoping to finish sooner; a positive answer then rules out both attributes at once, but a negative answer leaves it unclear which change caused it. Bruner's group found conservative focusing to be the most commonly used and most efficient strategy. Keep the families straight: **conservative focusing** and focus gambling begin from a positive instance; the two scanning strategies begin from hypotheses.

> EXAM TRAP
>
> "The researcher whose theory explains how we form *artificial* concepts by developing hypotheses" is **Jerome Bruner** (with Goodnow and Austin, 1956). **Eleanor Rosch** is the name for *natural* concepts and prototypes. **Clark Hull** (1920) did study concept learning, but as gradual associative abstraction of a common element, not hypothesis testing. **John Watson** has nothing to do with concepts. Also remember: in cricket, "out" is **disjunctive** (any one of several conditions), a voter is **conjunctive** (all conditions), "uncle" is **relational**.

## Natural Concepts and Prototype Theory ●●

Try to give a classical definition of "game". The philosopher **Ludwig Wittgenstein** (1889 to 1951) tried, in his posthumous *Philosophical Investigations* (1953), and failed. Cricket has teams and a winner; solitaire has no team; ring-a-ring-a-roses has no winner; chess involves skill, snakes and ladders only luck. There is no single feature that all games share. Instead, Wittgenstein said, games are related by a network of overlapping similarities, the way members of a family resemble one another: one has the father's nose and the mother's eyes, another the mother's eyes and the grandmother's chin, yet no single feature runs through them all. He called this **family resemblance**.

**Eleanor Rosch** (born 1938) turned Wittgenstein's insight into experimental psychology in the 1970s. A **natural concept** is a category that people form through ordinary experience with the world (bird, fruit, furniture, vehicle), as opposed to an artificial concept defined by an experimenter's rule. Rosch showed that natural concepts have three properties that the classical view cannot explain.

First, members differ in how typical they are. Asked to rate how good an example of "bird" each animal is, people agree that a robin or a sparrow is an excellent example, a hen a poorer

one and a penguin a poor one. Second, typicality has measurable consequences. In a *sentence verification* task, people say "true" faster to "A robin is a bird" than to "A penguin is a bird", although both are equally true; this is the *typicality effect*. Typical members are also named first when people list category members, and children learn them first. Third, the boundaries of natural concepts are fuzzy. Michael McCloskey and Sam Glucksberg (1978) found that people disagreed about whether a stroke is a disease or a tomato a fruit, and many even changed their own answers when asked again a few weeks later.

Rosch's explanation is **prototype theory**. A **prototype** is the best, most typical representative of a category: an abstract average or central tendency of the members we have encountered, not any particular one of them. Rosch and Carolyn Mervis (1975) showed that an item's typicality depends on its family resemblance score, the number of features it shares with other members of the category. A new object is classified by comparing it with the prototype; the more similar it is, the faster and more confidently it is accepted.

Rosch made a second discovery about how categories are arranged vertically. Rosch and colleagues (1976) distinguished three levels: the *superordinate* level (furniture, animal), the **basic level** (chair, dog) and the *subordinate* level (kitchen chair, Labrador). The basic level is psychologically privileged. It is the most general level at which members share a common shape and are used with similar movements, it is the level children learn first, and it is the level people use to name things: shown pictures of objects, Rosch's participants used basic-level names 1,595 times, subordinate names 14 times and a superordinate name just once. The basic level gives the best trade-off between being informative and being distinctive. Figure 8.2 shows both of Rosch's discoveries side by side. (Experts shift downwards: a birdwatcher says "Indian robin" where you say "bird".)

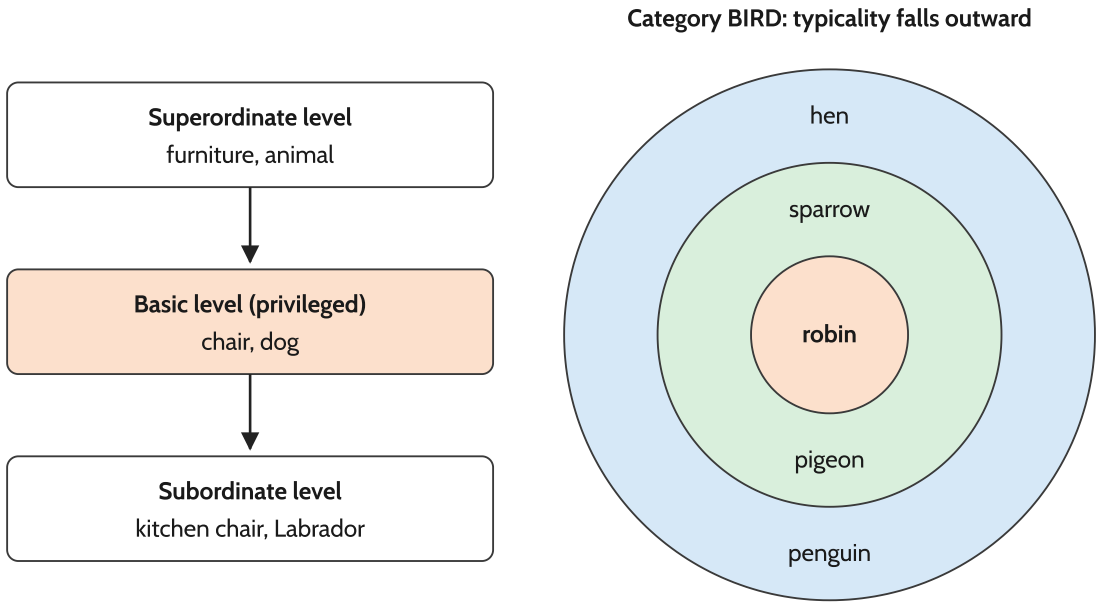



Figure 8.2 Two discoveries of Eleanor Rosch. Left: categories form levels, and the middle (basic) level is the one people learn first and name most readily. Right: members of a natural category differ in typicality, from the prototype-like robin at the centre to the penguin at the fuzzy edge.

EXAM TRAP

"An idealised representative object of a category" is a **prototype**, and a prototype is **abstract and averaged**: it need not be any real member you have met. Do not confuse it with three terms from pattern recognition (Chapter 4): a **template** is an exact stored copy that input must match; **features** are the elementary parts (lines, angles) that feature-analysis theories detect; **geons** are the 36 or so simple three-dimensional shapes in Irving Biederman's recognition-by-components theory (1987). Family resemblance began with **Wittgenstein** (1953) and was tested by **Rosch and Mervis** (1975).

## Exemplar Theory

Prototype theory has a weakness. If all you keep is an average, how do you know that birds vary a great deal in size, or that a small, plain brown bird may still be a sparrow? An alternative, **exemplar theory**, developed by Douglas Medin and Marguerite Schaffer (1978) and later Robert Nosofsky, proposes that we store many specific remembered instances (*exemplars*) of each category, and classify a new item by comparing it with all of them. A new animal is called a dog if it resembles the dogs you remember more than it resembles the cats you remember. Typicality effects still appear, because typical items resemble many stored exemplars; but the theory also explains why we know about a category's variability and its unusual members.

Most researchers now think we use both kinds of representation: prototypes when categories are large and well learned, exemplars when they are small or newly learned. A third approach, the *theory* or *knowledge-based* view, adds that concepts are embedded in our wider explanatory knowledge: you know that a whale is a mammal not because it looks like other mammals (it looks like a fish) but because of what you know about how it breathes and bears its young. Table 8.3 compares the three main views.

Table 8.3 Three views of how concepts are represented.

| View | Key names | A concept is | Explains well | Struggles with |
|---|---|---|---|---|
| Classical (defining features) | Aristotle; Bruner, Goodnow and Austin | A list of necessary and sufficient features | Artificial and technical concepts | Typicality, fuzzy boundaries |
| Prototype | Rosch; Rosch and Mervis | An abstract, averaged best example | Typicality, family resemblance | Knowledge of variability |
| Exemplar | Medin and Schaffer; Nosofsky | A set of stored concrete instances | Variability, atypical members | Memory load, abstract rules |

## How Concepts Are Organised: Networks ●●

Concepts do not float about separately; they are linked. The first influential model of how, the *hierarchical network model* of **Allan Collins** and **Ross Quillian** (1969), pictured knowledge as a **semantic network**: concepts are *nodes*, and labelled links connect them ("canary IS A bird", "bird HAS wings"). To save space (cognitive economy), each property is stored only once, at the highest level to which it applies: "has skin" at the animal node, "can fly" at the bird node, "can sing" at the canary node. The model predicts that verifying a sentence takes longer the more links must be travelled, and so it does: "A canary can sing" is verified faster than "A canary can fly", which is faster than "A canary has skin". But the model could not explain the

typicality effect, or why "A pig is an animal" is verified faster than "A pig is a mammal" (Rips, Shoben and Smith, 1973), which the hierarchy says should be the other way round.

Collins and Elizabeth Loftus (1975) replaced the rigid hierarchy with a *spreading activation* model. Links now vary in strength, short for closely related concepts and long for distant ones, and activating one node spreads activation outward to its neighbours. This explains *semantic priming*: you recognise "nurse" faster just after reading "doctor" than after reading "bread" (Meyer and Schvaneveldt, 1971).

A different kind of network arrived in 1986 with *Parallel Distributed Processing* by David Rumelhart, James McClelland and the PDP Research Group. In **connectionism**, knowledge is not stored in individual nodes that each stand for a concept. A concept is a *pattern of activation* distributed across many simple, neuron-like units, and all knowledge lies in the strengths (weights) of the connections between units, which change gradually with experience. Connectionist networks learn from examples, generalise to new ones, and degrade gracefully when damaged rather than losing whole facts, which makes them attractive models of the brain. Organised knowledge structures for familiar situations are also called *schemas*, and schemas for routine event sequences (going to a restaurant, attending a wedding) are called *scripts* (Schank and Abelson, 1977); their role in memory is discussed in Chapter 7.

> EXAM TRAP
>
> "Which model proposes that memory is organised in a network of interconnected nodes?" A recent paper keyed **connectionist model**. That works, because connectionist (PDP) models are networks of interconnected units. But be careful: Collins and Quillian's **semantic network model** is also literally a network of nodes, so an option reading "semantic network model" would be equally defensible. When "network model" and "connectionist model" both appear, follow the key (connectionist), and remember the difference: semantic networks have one node per concept (localist); connectionist networks spread each concept over many units (distributed). "Associative model" is the older umbrella idea from Aristotle and the British associationists; "linkage model" is not a standard term.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. The Würzburg school reported that some thoughts contain no image; this finding is called _____ thought.
2. Shepard and Metzler found that the time to judge two objects as the same increased _____ with the angle between them.
3. Pylyshyn argued that images are epiphenomenal and that knowledge is stored as abstract _____.
4. According to Paivio's _____ theory, concrete words are remembered well because they have both a verbal and an imagery code.
5. Bleuler's term for wish-fulfilling thinking that ignores reality is _____ thinking.
6. A concept such as "out" in cricket, satisfied by any one of several conditions, is a _____ concept.
7. In Bruner's study, choosing a positive card and changing one attribute at a time is called _____.
8. Wittgenstein's idea that members of a category share overlapping features but no single common one is called _____.
9. Rosch's "chair", as opposed to "furniture" or "kitchen chair", belongs to the _____ level.
10. Theories in which a category is represented by many stored instances are called _____ theories.

**Answers:** (1) imageless, (2) linearly, (3) propositions, (4) dual-coding, (5) autistic (or dereistic), (6) disjunctive, (7) conservative focusing, (8) family resemblance, (9) basic, (10) exemplar.

## 8.3 Problem Solving MEDIUM YIELD · 9 QS

Sultan had a problem, and so do you most days. A *problem* exists whenever you are in one state (the *initial state*), want to be in another (the *goal state*), and the route from one to the other is not obvious. Problem solving is thinking directed at getting from the first to the second. Psychologists have studied it in two very different traditions: the Gestalt tradition of Köhler and Duncker, which asked how a problem is suddenly *seen* in a new way, and the information-processing tradition of Newell and Simon, which asked how a problem is *searched*. You need both.

### Kinds of Problems

A **well-defined problem** has a clearly specified initial state, a clear goal and a clear set of allowable moves: an algebra equation, a sudoku, the Tower of Hanoi. An **ill-defined problem** leaves one or more of these vague: "write a good answer on attachment theory", "choose a career", "help this client feel less hopeless". Most of the problems of real life, and nearly all clinical problems, are ill-defined, and much of the work of solving them lies in defining them. Researchers also distinguish *knowledge-lean* problems, which can be solved with the information given, from *knowledge-rich* problems, which need expertise; and *insight* problems, solved by a sudden change of view, from *non-insight* (analytic) problems, solved step by step.

## The Problem Space

In 1956 Allen Newell and Herbert Simon, working with the programmer Cliff Shaw, unveiled the *Logic Theorist*, a computer program that proved theorems from Whitehead and Russell's *Principia Mathematica*. They followed it in the late 1950s with the *General Problem Solver* (GPS), designed to solve any well-defined problem, and set out the full theory in their 1972 book *Human Problem Solving*. Their method was to have people think aloud while solving puzzles, record every utterance (a *verbal protocol*), and check whether the program went through the same steps.

Their central idea was the **problem space**: an abstract description of all the possible states a problem can be in. It contains the initial state, the goal state, the *operators* (the legal moves that change one state into another), all the intermediate states those moves can produce, and the constraints on which moves are allowed. Solving a problem means *searching* this space for a path from the initial state to the goal. Take the Tower of Hanoi: three discs of decreasing size sit on the left of three pegs; the goal is to move them all to the right peg, moving one disc at a time and never placing a larger disc on a smaller. The shortest solution takes seven moves (in general, $2^n$ minus 1 moves for n discs). For three discs you could map the whole space on a sheet of paper; for a game such as chess, the number of possible paths is astronomically large. Because short-term memory is small and we can do only one thing at a time, Newell and Simon argued, people cannot search such spaces exhaustively. They must take shortcuts.

## Algorithms Versus Heuristics ●●

An **algorithm** is a step-by-step procedure that is guaranteed to produce the solution if it is followed correctly. Long division is an algorithm. So is trying every combination of a three-digit padlock: there are only 1,000 combinations, and if you try them all, one will open the lock. Algorithms are certain but often slow, and for many problems they are impossibly slow.

A **heuristic** is a rule of thumb: a mental shortcut that usually leads to a solution quickly but does not guarantee one. The word comes from the same Greek root as "eureka" (to find), and the mathematician George Pólya popularised it in *How to Solve It* (1945). Suppose you must unscramble MSTAE into an English word. An algorithm would test all 120 possible orderings. A heuristic says: put consonant clusters that often begin English words at the front, and try familiar endings. "ST" looks promising, and STEAM (or MATES, MEATS, TAMES) appears within seconds. The price of speed is that a heuristic can mislead; you will see in Section 8.5 that the same trade-off runs through judgement and decision making.

> EXAM TRAP
>
> "A rule of thumb that allows an efficient search for the solution of a problem" is a **heuristic**, not an algorithm. The defining contrast: an algorithm **guarantees** a solution; a heuristic is **efficient but not guaranteed**, "a general rule that is usually right". Wrong options to reject: "a heuristic always produces a solution" (that is an algorithm), "a heuristic is more accurate than an algorithm" (it is less certain, not more accurate), "heuristics can only be used by computers", "heuristics always produce errors". Education and learning are not problem-solving methods at all.

## Heuristic Strategies

Newell and Simon regarded **means-end analysis** as the most important heuristic of all, and it was the engine of GPS. It has three steps: note the most important *difference* between the current state and the goal state; set a *subgoal* of reducing that difference; and select an operator that achieves the subgoal. Then repeat. Suppose you are in Mysuru and need a certificate attested at an office in Bengaluru by Friday. The biggest difference is distance, so the subgoal is "get to Bengaluru"; the operator is a train. A new difference appears (you are at home, not at the station), creating a sub-subgoal ("get to the station") with its own operator (an auto). Breaking a problem into such subgoals is the heart of the method.

The strategy of **working backward** starts from the goal and works back towards the initial state. It is useful when the goal is clearly specified and there are many possible paths forward from the start. Try this: water lilies on a pond double the area they cover every day, and the pond is completely covered on day 30. On which day was it half covered? Working forward is hopeless; working backward gives the answer at once: day 29. Planning a revision timetable from the exam date backwards is the same strategy.

The heuristic of **hill climbing** is simpler than means-end analysis: at each step you choose the move that takes you closest to the goal state, judged by some simple measure of progress. It is like trying to reach the top of a hill in thick fog by always stepping uphill. It works when progress is steady, but it fails whenever reaching the goal requires a move that takes you temporarily further away, because hill climbing looks only one step ahead and never plans. The classic demonstration is the *Hobbits and Orcs* problem (a version of the missionaries and cannibals puzzle): three hobbits and three orcs must cross a river in a boat that holds two, and the orcs must never outnumber the hobbits on either bank. At one point the only legal move is to bring two creatures back across the river, apparently undoing progress, and that is precisely where solvers stall (Thomas, 1974). Climbers call the trap a false summit; problem solvers call it a *local maximum*.

Solving by **analogy** means using the solution of an earlier, similar problem to solve a new one. It is powerful when people notice the analogy, and the striking finding is how rarely they do.

### KEY STUDY: GICK AND HOLYOAK'S RADIATION PROBLEM (1980)

**The problem.** Karl Duncker had posed it in 1945. A patient has an inoperable stomach tumour. A ray strong enough to destroy the tumour will also destroy the healthy tissue it passes through; a ray weak enough to spare the tissue will not destroy the tumour. What can the doctor do? **The method. Mary Gick** and **Keith Holyoak** first had participants memorise a story about a general who wished to capture a fortress. Every road to the fortress was mined so that a large force would set off the mines, but small groups could pass safely; the general divided his army and had the groups converge on the fortress at the same moment. Then participants attempted the radiation problem.

**The result.** Only about 10 per cent solved the radiation problem without the story. Of those who had memorised the story, about 40 per cent used it spontaneously, but about 80 per cent solved the problem once told that the story might help. (The answer: aim several weak rays at the tumour from different directions so that they converge on it.) **The conclusion.** People can transfer a solution across problems that share *structural* similarity (the underlying principle of convergence), but they often fail to notice an analogy when the *surface* features (armies versus rays) differ. In a later study, Mark Keane (1987) used a story that was superficially similar to the problem (a surgeon using rays on a cancer), and 88 per cent of participants recalled it spontaneously.

The most primitive strategy of all is **trial and error**: trying responses more or less at random and keeping whatever works. **Edward Lee Thorndike**'s cats in their puzzle boxes (1898) escaped this way, their escape times falling gradually over many trials (Chapter 6). Trial and error works when the possible moves are few and mistakes are cheap; it is hopeless for problems with large problem spaces. Table 8.4 summarises the strategies.

Table 8.4 Problem-solving strategies.

| Strategy | What you do | Example | Weakness |
|---|---|---|---|
| Algorithm | Follow a procedure that guarantees the answer | Trying all 1,000 padlock codes | Slow; impossible in large spaces |
| Means-end analysis | Reduce the biggest difference by setting subgoals | Mysuru to Bengaluru by subgoals | Demands memory for subgoals |
| Working backward | Start from the goal and reason back | Water lilies, day 29 | Needs a well-specified goal |
| Hill climbing | Take whichever move gets closest to the goal | Always stepping uphill in fog | Stuck when a backward move is needed |
| Analogy | Reuse the solution of a similar earlier problem | Fortress story to radiation problem | Analogies often go unnoticed |
| Trial and error | Try responses until one works | Thorndike's cats | Inefficient in large spaces |

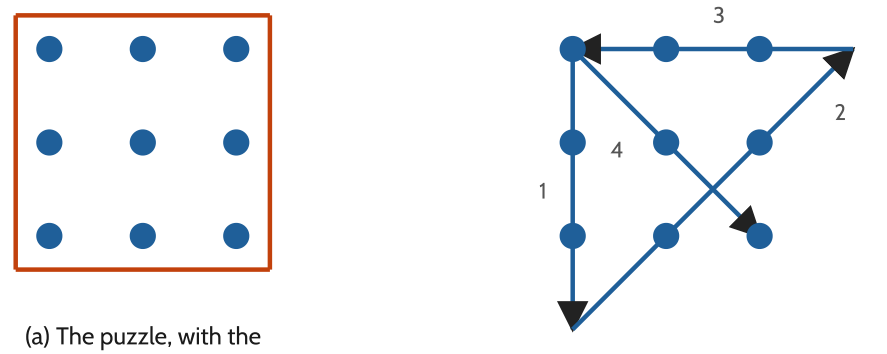
### EXAM TRAP

Match the description to the heuristic. "Choosing the option that moves you closer to the goal state" is **hill climbing**. "Breaking the problem into smaller parts" is subgoaling, the core of **means-end analysis**. "Employing the solution of a similar earlier problem" is **analogy**. "Trying different solutions until one works" is **trial and error**. Hill climbing differs from means-end analysis in having no subgoals: it looks only one step ahead.

## Insight and Incubation

The Gestalt psychologists did not think that problems were solved by search. For Köhler, Wertheimer, **Karl Duncker** (1903 to 1940) and **Norman Maier** (1900 to 1977), many problems are solved by **insight**: a sudden restructuring of the way the problem is represented, so that its elements are seen in a new relationship and the solution becomes obvious, often with an "aha" experience. Köhler listed the marks of insightful solutions in his apes: the transition from helplessness to solution is sudden and complete; the solution, once found, is performed smoothly; it is retained; and it transfers to similar problems. Compare Thorndike's cats, whose times fell gradually and erratically over dozens of trials.

Try the *nine-dot problem* in Figure 8.3 before you read on: join all nine dots with four straight lines without lifting your pen or retracing a line. Norman Maier used it in research in 1930, and most people fail. They assume, without being told, that the lines must stay inside the square the dots form. The solution requires going outside it. Yet even when people are told explicitly that they may go outside the square, only a minority succeed, which tells us that restructuring the problem is necessary but not sufficient: after the insight, there is still a search to do.




(a) The puzzle, with the imagined boundary dashed

(b) The solution leaves the square

Figure 8.3 The nine-dot problem. Most people fail because they impose a boundary that the instructions never mention; lines 1 and 2 must run beyond the square formed by the dots. Insight here means relaxing a self-imposed constraint.

Maier's other famous task, the *two-string problem* (1931), showed that insight can be nudged without awareness. Two strings hang from the ceiling, too far apart to hold both at once, and the task is to tie them together. The solution is to tie a heavy object, such as a pair of pliers, to one string and set it swinging like a pendulum, then catch it while holding the other. When Maier, walking past, casually brushed one string so that it swung, many participants solved the problem shortly afterwards, and most of them denied having noticed his hint.

KEY STUDY: METCALFE AND WIEBE'S WARMTH RATINGS (1987)

**The question.** Is insight really sudden, or does it only feel sudden? **The method. Janet Metcalfe** and David Wiebe gave people insight problems and ordinary analytic problems (such as algebra), and at short, regular intervals asked them to rate how "warm" they felt, that is, how close to the solution. **The result.** On analytic problems warmth rose gradually and steadily, as you would expect from step-by-step search. On insight problems warmth stayed low and flat until just before the solution, then jumped. **The conclusion.** Insight problems are solved by a qualitatively different, sudden process, as the Gestaltists had claimed, and people cannot sense their approach to an insight solution. **The criticism.** Feelings of warmth are self-reports; some later researchers argue for a "business as usual" view in which insight uses ordinary memory and search processes that simply happen to end abruptly.

Köhler's own conclusions also met a challenge. Herbert Birch (1945) found that young chimpanzees with no previous experience of handling sticks rarely solved stick problems; after a few days of playing with sticks, they solved them readily. Insight, it seems, builds on past experience rather than replacing it.

What should you do when you are stuck? Often, the best advice is to stop. Researchers call the period during which a problem is set aside **incubation**; after it, the solution sometimes arrives with surprisingly little effort. The term comes from Graham Wallas's stages of creative thought (Section 8.6). Why might a break help? Perhaps the mind continues to work on the problem unconsciously; perhaps fatigue lifts; but the best-supported explanation is that a break lets you forget a misleading approach that had been blocking you (the *forgetting fixation* hypothesis of Steven Smith and Steven Blankenship). A meta-analysis by Ut Na Sio and Thomas Ormerod (2009) concluded that incubation effects are real but modest, and largest for creative problems that have many possible solutions.

EXAM TRAP

**Two meanings of "insight".** In problem solving, insight (Köhler, Gestalt) is the sudden restructuring of a problem. In the mental status examination, insight means the patient's awareness that he or she is ill, graded on a six-point scale: complete denial (grade 1); slight awareness of being ill and needing help, while denying it at the same time (grade 2); awareness of being ill but blaming others or external or physical causes (grade 3); awareness that the illness is due to something unknown in oneself (grade 4); **intellectual insight**, admitting illness and its psychological roots without applying the knowledge to change behaviour (**grade 5**); and true emotional insight, which does lead to change (grade 6). Read the stem to see which meaning is meant.

## Obstacles to Problem Solving ●●●

If problem solving were only a matter of search and insight, more of us would be good at it. The Gestaltists were especially interested in how past experience, usually our best friend, can get in the way.

The first obstacle, **functional fixedness**, is the tendency to think of an object only in terms of its usual function, which blinds us to other uses that would solve a problem. Karl Duncker's *candle problem*, published in his monograph *On Problem Solving* (English translation 1945), is

the classic demonstration. Participants received a candle, a book of matches and a box of drawing pins, and were asked to fix the candle to the wall so that it would burn without dripping wax onto the table. Most tried to pin the candle directly to the wall or to stick it on with melted wax. The solution is to empty the box, pin it to the wall as a shelf, and stand the candle in it. Far more people solved the problem when the box was presented empty, with the pins beside it, because an empty box does not look so much like a container. Functional fixedness is a case of top-down processing working too well: what we know about an object's typical use dominates what we perceive about its actual properties. Curiously, it seems to be learned. Tim German and Margaret Defeyter (2000) found that five-year-olds, who have not yet learned to think of objects in terms of their designed function, showed no functional fixedness on a version of the task, whereas six- and seven-year-olds did.

A **mental set** is the broader tendency to approach a problem in a way that has worked before, even when a better or simpler method is available. Its classic demonstration is the water-jar experiment of **Abraham Luchins** (1942), whose German term for the phenomenon, *Einstellung* ("setting" or "attitude"), gives it its other name, the Einstellung effect. Work through Table 8.5. You have three jars of given capacities and an unlimited water supply, and you must measure out exactly the required amount.

Table 8.5 Water-jar problems of the kind Luchins used (capacities in litres).

| Problem | Jar A | Jar B | Jar C | Obtain exactly |
|---|---|---|---|---|
| Set problem (one of five) | 21 | 127 | 3 | 100 |
| Critical problem | 23 | 49 | 3 | 20 |
| Extinction problem | 28 | 76 | 3 | 25 |

The set problem is solved by filling B, pouring off enough to fill A once, and then pouring off enough to fill C twice: B minus A minus 2C (127 minus 21 minus 6 is 100). After five problems that all yield to the same formula, people meet the critical problem. B minus A minus 2C still works (49 minus 23 minus 6 is 20), but so does a far simpler method: A minus C (23 minus 3). Most participants who had practised the formula used the long method and never noticed the short one. On the extinction problem the formula no longer works (76 minus 28 minus 6 is 42), and only A minus C does. In one report, 95 per cent of people without the training solved it, against only 36 per cent of those who had acquired the set. Even chess masters show the effect, overlooking a short winning line because a familiar long one comes to mind first.

The other obstacles are easier to describe. **Confirmation bias**, the tendency to seek evidence that supports our current hypothesis rather than evidence that could refute it, keeps us on a wrong track (it is examined with Wason's experiments in Section 8.4). **Irrelevant information** distracts: "Your drawer holds black and brown socks in the ratio 4 to 5; how many must you pull out in the dark to be sure of a matching pair?" The ratio is irrelevant; the answer is three. *Fixation*, the Gestaltists' general word for being stuck on an inappropriate representation, covers both functional fixedness and mental set; and anxiety, fatigue and time pressure all narrow the search.

EXAM TRAP

**Functional fixedness** is about **objects** (the box seen only as a container); **mental set** is about **strategies** (the formula that worked before). Functional fixedness is usually described as a special case of mental set, which is why one recent paper keyed mental set as "the most basic and common obstacle to problem solving". Names: candle problem = **Duncker (1945)**; water jars and Einstellung = **Luchins (1942)**; nine-dot and two-string problems = **Maier**. "Mental barrier" is not a technical term, and trial and error is a strategy, not an obstacle. Seeing a matchbox only as a container, not as a platform, is functional fixedness.

## Stages of Problem Solving and the Expert Advantage

Several writers have broken problem solving into stages. Dewey (1910), as you saw, described a felt difficulty, its definition, suggested solutions, reasoning about them and testing. Pólya's *How to Solve It* (1945) offered four steps for mathematics students: understand the problem, devise a plan, carry out the plan, and look back to check. **Robert Jeffrey Sternberg** (born 1949) describes a *problem-solving cycle* of seven steps: identify the problem, define and represent it, formulate a strategy, organise the information, allocate resources (time, effort, money), monitor progress, and evaluate the solution. It is a cycle because evaluation often reveals a new problem. Whatever the labels, every scheme moves from understanding the problem, through generating and choosing a strategy, to carrying it out and checking the result.

Why are experts better problem solvers? Not, it turns out, because they have better general reasoning. William Chase and Herbert Simon (1973), building on the Dutch psychologist Adriaan de Groot, showed chess positions for five seconds and asked players to reconstruct them. Masters reproduced real game positions far more accurately than novices, but when the pieces were placed at random, their advantage disappeared. Experts store thousands of meaningful patterns, or *chunks*, and see a board as a few familiar configurations rather than as twenty-odd separate pieces. Similarly, Michelene Chi and her colleagues (1981) found that physics experts sorted problems by their underlying principle (conservation of energy, say), whereas novices sorted them by surface features (problems with inclined planes, problems with springs). Experts spend longer representing a problem before starting, and then often work forward from what they know, whereas novices plunge in and work backward from the unknown.

CLINICAL LINK

Thomas D'Zurilla and Marvin Goldfried (1971) turned research on problem solving into a therapy. **Problem-solving therapy** teaches clients a structured sequence: a positive problem orientation, defining and formulating the problem, generating many alternative solutions without judging them, deciding among them, and implementing and verifying the chosen solution. It is used for depression, for people at risk of self-harm (who often show narrow, rigid problem solving under stress), and in primary care. Notice how much of it is the research of this section put to work: defining ill-defined problems, brainstorming to escape mental set, and evaluating outcomes.

## 8.4 Reasoning HIGH YIELD · 9 QS

Reasoning is the process of drawing conclusions from premises or evidence. It overlaps with problem solving (you often reason your way to a solution) but its focus is on the *inference*: what follows from what? Philosophers have studied the rules of correct reasoning since Aristotle. Psychologists ask a different question: how do people actually reason, and why do they so often go wrong?

### Deduction, Induction and Abduction ●●●

The first kind, **deductive reasoning**, moves from general premises to a specific conclusion that must be true if the premises are true. It is sometimes called top-down reasoning. "All autos in Mysuru must charge by the meter. This is a Mysuru auto. Therefore this auto must charge by the meter." The second kind, **inductive reasoning**, moves from specific observations to a general conclusion that is probable but never certain; it is bottom-up. "Every auto I have taken in Mysuru has asked for more than the meter. Therefore Mysuru autos overcharge." The next auto may prove you wrong. Deductive conclusions go no further than the premises; inductive conclusions go beyond them, which is both their power and their risk.

Two terms keep deduction honest. An argument is **valid** when its conclusion follows necessarily from its premises because of its logical form, whatever the content. An argument is **sound** when it is valid *and* its premises are true. "All psychologists are vegetarians; Sihan is a psychologist; therefore Sihan is a vegetarian" is perfectly valid but unsound, because the first premise is false. So the truth of a deductive conclusion is only as good as the truth of the premises (or theory) it starts from.

The American philosopher **Charles Sanders Peirce** (1839 to 1914) described a third kind of inference, **abductive reasoning**: inference to the best explanation. A surprising fact is observed; if some hypothesis were true, the fact would follow naturally; therefore there is reason to suspect the hypothesis is true. Diagnosis is abductive. A clinician sees early waking, weight loss, slowed speech and hopelessness, and infers the disorder that best explains them, while remaining ready to revise the conclusion.

Science combines all three. In the *hypothetico-deductive method*, associated with the philosopher **Karl Raimund Popper** (1902 to 1994), a researcher derives a hypothesis from a theory, deduces a prediction that could be observed, and tests it. If the prediction fails, the hypothesis is refuted; if it succeeds, the hypothesis survives, but it is not proved, because another hypothesis might have made the same prediction. (You will see in a moment that this asymmetry is simply modus tollens versus affirming the consequent.) In research methods, a *deductive* approach starts from theory and tests it against data, as most quantitative research does; an *inductive* approach starts from data, looks for patterns and builds theory, as most qualitative research does (grounded theory is the purest example).

EXAM TRAP

Recent papers ask the research-methods face of reasoning. "Starts with a theory and tests it with observations" = **deductive reasoning**. "Gathers data, looks for patterns, then generates a theory" or "moves from specific observations to broader generalisations" = **inductive reasoning**, which is the reasoning of **theory development** and of **qualitative research**. Abductive = best explanation. The hypothetico-deductive method is the full cycle of hypothesis, deduced prediction and test; when both "deductive reasoning" and "hypothetico-deductive method" appear, the keys have chosen the plain label. One paper asked what the "validity of conclusions" in deduction depends on, keying **the soundness of the initial theory**. Strictly, logical validity depends on form alone, and soundness on true premises; the item uses "validity" loosely to mean "trustworthiness", so choose the soundness option.

## Syllogisms, Belief Bias and the Atmosphere Effect

A **syllogism** is an argument with two premises and a conclusion, introduced by Aristotle. In a *categorical* syllogism each statement uses one of four quantifiers: all, no, some, or some … not. "All B are C; all A are B; therefore all A are C" is valid. "All A are B; all C are B; therefore all A are C" is not: all cats are mammals and all dogs are mammals, but it does not follow that all cats are dogs. People make many errors on syllogisms, and the errors are systematic.

The most famous is **belief bias**: the tendency to accept invalid conclusions that are believable and to reject valid conclusions that are unbelievable. Consider: "All roses are flowers. Some flowers fade quickly. Therefore some roses fade quickly." Most people accept it because the conclusion is plausible, yet it is invalid: the flowers that fade quickly might all be marigolds. Now consider: "All mammals can fly. Elephants are mammals. Therefore elephants can fly." The conclusion is absurd, but the argument is perfectly valid. **Jonathan Evans**, Julie Barston and Paul Pollard (1983) showed that belief and logic both influence judgements and interact: belief has its biggest effect on invalid arguments, where believable conclusions are accepted far more often than unbelievable ones.

An older account of syllogistic error is the **atmosphere effect**, proposed by **Robert Woodworth** and **Saul Sells** in 1935. The "mood" of the premises creates an atmosphere that biases the conclusion people accept: if either premise is particular ("some"), people favour a particular conclusion; if either premise is negative ("no"), they favour a negative one. So "Some A are B; some B are C; therefore some A are C" feels right, although it is invalid. Loren and Jean Chapman (1959) offered a rival explanation, *illicit conversion*: people treat "All A are B" as if it also meant "All B are A".

EXAM TRAP

**Belief bias** (reasoning: judging an argument by the believability of its conclusion) is easy to confuse with three social biases that share the word: **correspondence bias** or the fundamental attribution error (explaining others' behaviour by their dispositions), **self-serving bias** (credit for success, blame situations for failure), and **confirmation bias** (seeking supporting evidence). Atmosphere effect = **Woodworth and Sells (1935)**.

## Conditional Reasoning and the Wason Selection Task

Much everyday reasoning uses conditional statements of the form "if p, then q". The "if" part (p) is the *antecedent*; the "then" part (q) is the *consequent*. Given such a rule and one further

fact, four kinds of argument are possible, and only two are valid. Table 8.6 uses a cricket example: "If it rains, the match is cancelled."

Table 8.6 The four forms of conditional argument.

| Form | Pattern | Cricket example | Valid? |
|---|---|---|---|
| Modus ponens (affirming the antecedent) | If p then q; p; so q | It rained, so the match was cancelled | Valid |
| Modus tollens (denying the consequent) | If p then q; not q; so not p | The match was not cancelled, so it did not rain | Valid |
| Affirming the consequent | If p then q; q; so p | The match was cancelled, so it rained | Invalid (a bad pitch could cancel it) |
| Denying the antecedent | If p then q; not p; so not q | It did not rain, so the match was not cancelled | Invalid (same reason) |

The valid inference called **modus ponens** ("the mode that affirms") is made correctly by nearly everyone. Its partner, **modus tollens** ("the mode that denies"), is equally valid but is made far less often; people hesitate to reason from the absence of the consequent. Meanwhile many people accept the two fallacies, **affirming the consequent** and **denying the antecedent**, as if they were valid. One reason is that everyday language often uses "if" to mean "if and only if": when a parent says "If you finish your homework, you can watch television", the child rightly infers that no homework means no television. Logic does not allow that inference; conversation usually does.

### KEY STUDY: THE WASON SELECTION TASK (1966)

**The task.** The British psychologist **Peter Cathcart Wason** (1924 to 2003) showed people four cards, each with a letter on one side and a number on the other (Figure 8.4), and a rule: "If a card has a vowel on one side, then it has an even number on the other side." Which cards must you turn over to find out whether the rule is true? Decide before reading on.

**The result.** Most people choose E alone, or E and 4. Typically fewer than one person in ten gives the correct answer: E and 7. E must be turned because an odd number behind it would break the rule (this is modus ponens). The 7 must be turned because a vowel behind it would break the rule (modus tollens). The 4 is irrelevant: the rule says nothing about what must be behind an even number, so whatever is there cannot falsify the rule (choosing it is affirming the consequent). **The conclusion.** People look for cards that could *confirm* the rule, or simply cards that *match* the items named in it (Jonathan Evans's *matching bias*), rather than cards that could *falsify* it.

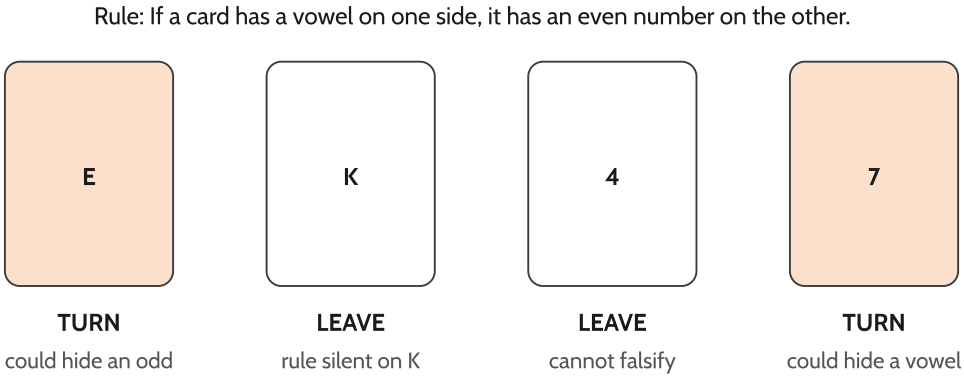



Figure 8.4 The abstract Wason selection task. The two cards with heavy outlines (E and 7) are the only ones that could reveal a counterexample. The popular choice of the 4 card tests for confirmation, which can never prove a general rule.

The story took a twist when the same logical task was given meaningful content. Richard Griggs and James Cox (1982) told participants to imagine they were police officers checking the rule "If a person is drinking beer, then the person must be over 19 years of age". The four cards read "drinking beer", "drinking cola", "16 years of age" and "22 years of age". About three-quarters now chose correctly ("drinking beer" and "16 years"), although the logic was identical to the vowels and numbers. Why? Patricia Cheng and Keith Holyoak (1985) proposed that people reason with *pragmatic reasoning schemas*, rules of thumb for familiar situations such as permissions and obligations ("if you want to do X, you must satisfy Y"). The evolutionary psychologist **Leda Cosmides** (1989) went further: she argued that natural selection equipped us with a specialised *cheater-detection* mechanism for social contracts ("if you take the benefit, you must pay the cost"), and that the task becomes easy whenever it is framed as catching someone who has cheated. Rules about what people *must* or *may* do (deontic rules) are reliably easier than rules about what *is* the case (indicative rules), though researchers still argue about why.

## Mental Models

If people do not reason with formal logic, how do they reason? The most influential answer comes from the British psychologist **Philip Nicholas Johnson-Laird** (born 1936), whose book *Mental Models* appeared in 1983. A **mental model** is an internal representation of a possible state of affairs described by the premises, built to resemble the situation rather than the sentences. Given "The cup is to the right of the plate; the spoon is to the left of the plate", you build one model (spoon, plate, cup) and read the conclusion off it: the cup is to the right of the spoon. Reasoning, on this view, has three steps: construct a model of the premises, draw a tentative conclusion from it, and search for an alternative model in which the premises are true but the conclusion false. If you find none, the conclusion is valid. The theory predicts, correctly, that problems are harder the more models they require, and that people often stop after the first model, which produces errors when other models exist. The rival *mental logic* approach holds that people do have inference rules in their heads, such as modus ponens, but lack others, such as modus tollens, which must be derived in several steps.

## Inductive Reasoning and Hypothesis Testing ●●●

Most of the reasoning that keeps us alive is inductive: generalising from experience, reasoning from one case to a similar one (analogy), and inferring causes from patterns of events. The Scottish philosopher David Hume pointed out in the eighteenth century that induction can never be logically guaranteed. Europeans observed thousands of swans, every one of them white, and concluded that all swans are white; then, in 1697, Dutch explorers found black swans in Western Australia. Concept formation (Section 8.2) is inductive, and so is everyday judgement under uncertainty (Section 8.5).

### KEY STUDY: WASON'S 2-4-6 TASK (1960)

**The method.** Wason told participants that the sequence 2-4-6 fits a rule he had in mind, and asked them to discover the rule by proposing sequences of their own; after each, he said whether it fitted. They were to announce the rule only when sure. **The result.** Most people formed a hypothesis such as "numbers increasing by two" and tested it with sequences like 8-10-12 and 20-22-24, all of which fitted. Confident, they announced their rule, and were told it was wrong. Only about one in five announced the correct rule the first time. The rule was simply "any three increasing numbers". **The conclusion.** People test hypotheses by seeking instances that would confirm them, never trying a sequence such as 1-2-3 or 6-4-2 that could have shown their hypothesis to be wrong. Later writers called this confirmation bias; Wason himself, a few years later, spoke of a "verification bias". **The criticism.** Joshua Klayman and Young-Won Ha (1987) pointed out that testing instances that fit your hypothesis (a *positive test strategy*) is not irrational in itself; it fails on the 2-4-6 task because the true rule is much broader than the hypothesis people start with.

The term **confirmation bias**, now firmly attached to Wason's experiments, refers to the tendency to search for, interpret, favour and remember information in ways that confirm what we already believe, and to neglect evidence that could disconfirm it. It is one of the most robust findings in psychology, and it matters wherever people test beliefs: in science, in courtrooms, in diagnosis and in prejudice. A self-fulfilling prophecy (Robert Merton, 1948) is a different thing. Confirmation bias changes how we *see* the evidence; a self-fulfilling prophecy changes the *world*, because our expectation alters our behaviour, which in turn makes the expected event more likely (a teacher's low expectations lead to less attention and so to poorer results). The social face of these ideas is covered in the Social Psychology book.

EXAM TRAP

**"Which does NOT reflect inductive reasoning: probabilistic conclusions, heuristics, decision making, concept formation?"** The key is **heuristics**. Inductive conclusions are probabilistic, concept formation is induction from instances, and decision making under uncertainty rests on inductive judgements of likelihood; a heuristic is a shortcut strategy rather than a form of inference. The item is debatable (heuristics generate probabilistic judgements), but follow the key. **"Confirmation bias is closely related to which heuristic?"** One paper keyed **representativeness**, reasoning that both involve judging by fit to a preconceived model. No standard textbook ties confirmation bias to one particular heuristic; it is a bias in hypothesis testing (Wason's tasks). Use the key if the item recurs. The **recognition heuristic** (Gerd Gigerenzer's group) means choosing the option you recognise, and is the least related. Finally, "acting in line with one's own expectation so that the expected event becomes more likely" is a **self-fulfilling prophecy**, not confirmation bias (which distorts only how evidence is read), not the base-rate fallacy (Section 8.5) and not "motivational reaction", which is not a technical term.

### Reasoning Across the Lifespan: a Pointer

Reasoning develops. In Jean Piaget's theory (taught fully in the Developmental Psychology book), the *concrete operational* child of about 7 to 11 years can reason logically about concrete, present objects: she understands conservation, *reversibility* (mentally undoing an action) and *transitive inference* (if Asha is taller than Bina and Bina taller than Chitra, Asha is taller than Chitra). From about 11 or 12, in the *formal operational* stage, the adolescent can reason about the abstract and the possible. Formal operations include *hypothetico-deductive reasoning* (forming hypotheses and testing them systematically, as in Piaget and Inhelder's pendulum problem), propositional reasoning about statements rather than objects, *counterfactual* reasoning about states of affairs contrary to fact, and a new capacity for thinking about thinking, which Piaget described as "operations on operations" and which we would now call metacognition.

Some theorists argue that adult thought goes beyond formal operations. What they call **dialectical thinking**, described by Klaus Riegel (1973) and Michael Basseches (1984), is the ability to hold contradictory perspectives together, to see truth as relative to context and changing over time, and to integrate opposing ideas into a new synthesis. It is the hallmark of *post-formal* thought, not of formal operations.

EXAM TRAP

"Formal operational reasoning involves all EXCEPT: hypothetical, counterfactual, dialectical, metacognitive." The answer is **dialectical** (postformal; Riegel, Basseches). **Hypothetico-deductive reasoning** is a capacity that emerges *in* the formal operational stage; it is not itself one of Piaget's four stages, and a concrete operational child "struggles with" it. Reversibility, transitive inference, conservation and reduced egocentrism are concrete operational achievements. Earlier milestones make tempting distractors: **object permanence** (knowing that hidden objects still exist) and **deferred imitation** (copying a model after a delay) belong to the sensorimotor stage (birth to about 2), and **symbolic or pretend play** to the preoperational stage (about 2 to 7), whose child still lacks **conservation**, the understanding that quantity stays the same when only appearance changes (water poured into a taller, thinner glass is not "more"). The four stages themselves are sensorimotor, preoperational, concrete operational and formal operational; "reversibility of formal operations" is a scrambled option, since reversibility is concrete operational.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Newell and Simon described the initial state, goal state, operators and intermediate states of a problem as its _____.
2. A procedure guaranteed to produce a solution is an _____; a rule of thumb that usually works is a _____.
3. The heuristic that always chooses the move bringing you closest to the goal, and so fails when a backward move is needed, is _____.
4. Duncker's candle problem demonstrates _____; Luchins's water jars demonstrate _____.
5. Metcalfe and Wiebe found that "warmth" ratings rose gradually for analytic problems but suddenly for _____ problems.
6. Reasoning from "If p then q" and "not q" to "not p" is called _____.
7. Accepting an invalid argument because its conclusion is believable is called _____.
8. Peirce's "inference to the best explanation" is _____ reasoning.
9. In Wason's 2-4-6 task, people tested only sequences that fitted their hypothesis, showing _____.
10. Riegel's and Basseches's postformal form of reasoning that integrates contradictions is _____ thinking.

**Answers:** (1) problem space, (2) algorithm; heuristic, (3) hill climbing, (4) functional fixedness; mental set (Einstellung), (5) insight, (6) modus tollens, (7) belief bias, (8) abductive, (9) confirmation bias, (10) dialectical.

## 8.5 Judgement and Decision Making MEDIUM YIELD · 4 QS

Should you revise attachment theory or the heuristics tonight? Take the auto or wait for the bus? Accept the first seat you are offered in counselling or hold out for a better college? Two related processes are at work in such choices. *Judgement* is estimating how likely something

is, usually from incomplete information ("How likely is it that this topic will be asked?"). *Decision making* is choosing among options, and it usually depends on judgements. For most of the twentieth century economists assumed that people make both rationally. Psychologists discovered otherwise, and in 2002 one of them won the Nobel Prize in Economics for showing it.

### How We Should Decide: Normative Models

It helps to separate three kinds of model. A *normative* model says how an ideal, fully rational agent *should* decide, using the rules of logic and probability. A *descriptive* model says how people *actually* decide. A *prescriptive* model offers practical advice for bringing real decisions closer to the normative ideal. Most of this section is about the gap between the first two.

The central normative idea is *expected value*: multiply each possible outcome by its probability and add the results. A lottery ticket costs ₹100 and gives a 1 in 1,000 chance of winning ₹50,000. Its expected value is ₹50,000 divided by 1,000, which is ₹50, so on average you lose ₹50 on every ticket. In 1738 the Swiss mathematician Daniel Bernoulli pointed out that people value outcomes not by their money value but by their *utility* (subjective worth), and that utility shows diminishing returns: an extra ₹1,000 means far more to a student than to a millionaire. John von Neumann and Oskar Morgenstern (1944) formalised *expected utility theory*: a rational decision maker, whose preferences are consistent (if she prefers A to B and B to C, she must prefer A to C), always chooses the option with the highest expected utility.

When options differ on several attributes, a fully rational chooser uses a *compensatory* strategy: weigh every attribute of every option, so that a weakness on one attribute can be compensated by a strength on another. Choosing a paying-guest room in Mysuru, you would score each room on rent, distance from college, food and safety, multiply by how much each attribute matters to you, and total the scores. Real people, faced with many options, more often use *non-compensatory* shortcuts. In *elimination by aspects*, described by Amos Tversky in 1972, you pick the most important attribute, eliminate every option that falls below your cut-off on it ("nothing over ₹8,000 a month"), move to the next attribute, and continue until one option is left. A room that is perfect in every other way is gone the moment it fails the first test.

### Bounded Rationality and Satisficing ●●

The first serious challenge to the rational model came from Herbert Simon, the same Simon who, with Newell, modelled problem solving. In a series of papers from 1955 and in his 1957 book *Models of Man*, Simon argued that real decision makers have **bounded rationality**: their rationality is limited by the information available to them, by the limited capacity of their minds to process it, and by the time available to decide. A person who tried to compute the expected utility of every possible option before buying a mobile phone would still be comparing specifications when the phone became obsolete.

Instead, said Simon, people practise **satisficing**, a word he coined in 1956 by blending "satisfy" and "suffice". Satisficing means setting an *aspiration level* (good enough on the features that matter), examining options one at a time, and choosing the first that meets it. You decide that any phone under ₹20,000 with a good camera and a day's battery will do, and you buy the first one that qualifies. Satisficing does not find the best option, but it finds a good one at a sensible cost. Simon received the Nobel Prize in Economics in 1978, largely for this work.

The German psychologist **Gerd Gigerenzer** (born 1947) has developed Simon's idea in a more optimistic direction. Gigerenzer argues that the mind holds an "adaptive toolbox" of *fast and frugal heuristics* that use little information and yet perform well in the environments they evolved for. In the *recognition heuristic*, if you recognise one of two options and not the other, you infer that the recognised one scores higher. Daniel Goldstein and Gigerenzer (2002) asked American students which of two German cities was larger. When the students recognised only one city of a pair, they chose it about 90 per cent of the time. Strikingly, the same students did slightly better on German cities, many of which they had barely heard of, than on American cities, where they usually recognised both names and had to rely on what they knew: a "less-is-more" effect. For Gigerenzer, heuristics are not mere sources of error but tools that are *ecologically rational*.

> EXAM TRAP
>
> **Bounded rationality and satisficing** belong to **Herbert Simon** (1950s; Nobel 1978), not to Kahneman or Tversky, whose names go with heuristics and biases and prospect theory. Howard Gardner is the multiple-intelligences theorist (Chapter 9). Elimination by aspects is **Tversky (1972)**; the recognition heuristic and fast and frugal heuristics are **Gigerenzer**.

## Heuristics and Biases ●●

In 1969 two Israeli psychologists at the Hebrew University of Jerusalem, **Amos Tversky** (1937 to 1996) and **Daniel Kahneman** (1934 to 2024), began a collaboration that would change the study of judgement. Their method was disarmingly simple: they posed short problems to ordinary people and to experts, and catalogued the systematic ways in which the answers departed from the rules of probability. In a famous 1974 paper in *Science*, "Judgment under uncertainty: Heuristics and biases", they argued that people judge probability using a small number of heuristics. These usually serve us well, but they lead to predictable errors, called *biases*. Tversky died in 1996; when the Nobel Prize recognised the work in 2002, it went to Kahneman alone, because the prize is not awarded after death.

The **availability heuristic** (Tversky and Kahneman, 1973) is judging the frequency or probability of an event by how easily examples come to mind. Answer quickly: are there more English words that begin with the letter K, or more words that have K as their third letter? Most people say more begin with K, because words are easy to retrieve by their first letter (kitchen, king, kind) and hard to retrieve by their third (ask, make, like). In fact there are about twice as many with K in third position. Availability usually works because frequent events really are easier to recall, but anything that makes an event memorable distorts it: dramatic, vivid or recent events. After a widely reported train accident, people overestimate the danger of train travel; deaths from plane crashes and terrorism are overestimated, and deaths from diabetes and asthma underestimated. Norbert Schwarz and his colleagues (1991) showed that it is the *ease* of recall, not the number of examples, that matters: people asked to list twelve occasions on which they had been assertive (a hard task) later rated themselves as *less* assertive than people asked to list only six.

The **representativeness heuristic** (Kahneman and Tversky, 1972) is judging the probability that something belongs to a category by how much it resembles the typical member of that category. The trouble is that resemblance ignores other information that matters for probability, above all base rates.

The most important casualty is **base-rate neglect**: the failure to take account of how common something is in the population. Kahneman and Tversky (1973) described "Jack", a 45-year-old man who is conservative, careful and ambitious, shows no interest in politics and enjoys carpentry and mathematical puzzles. Participants were told that Jack's description had been drawn at random from 100 descriptions, either 70 engineers and 30 lawyers or 30 engineers and 70 lawyers. Both groups judged it about 90 per cent likely that Jack was an engineer: the description sounded like an engineer, and the base rate made almost no difference. Now try the *taxi-cab problem* yourself. A cab was involved in a hit-and-run at night. In the city, 85 per cent of cabs are Green and 15 per cent are Blue. A witness says the cab was Blue, and tests show that she identifies colours correctly 80 per cent of the time under those conditions. What is the probability that the cab was Blue? Most people answer 80 per cent. Work it through with 100 accidents instead of percentages. Fifteen involve Blue cabs, and the witness correctly calls 12 of them Blue. Eighty-five involve Green cabs, and she wrongly calls 17 of them Blue. So she says "Blue" 29 times, and is right 12 times: the probability is 12 out of 29, about 41 per cent. The low base rate of Blue cabs pulls the answer far below the witness's accuracy.

KEY STUDY: LINDA THE BANK TELLER (TVERSKY AND KAHNEMAN, 1983)

**The problem.** "Linda is 31 years old, single, outspoken and very bright. She majored in philosophy. As a student, she was deeply concerned with issues of discrimination and social justice, and also participated in anti-nuclear demonstrations." Which is more probable: (a) Linda is a bank teller, or (b) Linda is a bank teller and is active in the feminist movement? **The result.** In one version about 85 per cent of participants chose (b). **Why it is wrong.** The probability of two things both being true can never exceed the probability of either one alone: every feminist bank teller is a bank teller, so (a) must be at least as likely as (b). Choosing (b) is the **conjunction fallacy**. **The explanation.** The description is representative of a feminist, so the conjunction "feels" more probable. **The criticism.** Some participants may read (a) as "a bank teller who is *not* a feminist", and the error shrinks when the problem is posed in frequencies ("Of 100 women like Linda, how many..."), as Gigerenzer has emphasised. But the fallacy survives careful rewording.

Representativeness also explains the **gambler's fallacy**: the belief that after a run of one outcome, the opposite outcome is "due". On 18 August 1913, at the casino in Monte Carlo, the roulette ball fell on black 26 times in a row, and gamblers lost fortunes betting on red in the belief that red had become ever more likely. But a roulette wheel has no memory; each spin is independent. People expect even a short sequence to look like a "representative" random sequence, with reds and blacks well mixed. The same expectation produces *insensitivity to sample size*. A large hospital delivers about 45 babies a day and a small one about 15. Over a year, which hospital records more days on which over 60 per cent of babies born are boys? Most people say the two are about the same; in fact the small hospital will, because small samples stray further from the population average.

The third heuristic is **anchoring and adjustment**: estimating a quantity by starting from an initial value (the anchor) and adjusting from it, usually too little. In Tversky and Kahneman's demonstration (1974), a wheel of fortune, apparently random but rigged, stopped at either 10 or 65 in front of participants, who were then asked what percentage of African countries were members of the United Nations. Median estimates were 25 per cent after seeing 10 and

45 per cent after seeing 65: an obviously irrelevant number dragged the estimates. In another study, people given five seconds to estimate 8 × 7 × 6 × 5 × 4 × 3 × 2 × 1 gave a median answer of 2,250; those shown 1 × 2 × 3 × 4 × 5 × 6 × 7 × 8 gave 512. Both anchored on the first few products and adjusted too little; the true answer is 40,320. The struck-out "MRP" above a sale price in a shop window is an anchor. Table 8.7 gathers the heuristics and the errors each one produces.

Table 8.7 The classic heuristics and their typical biases.

| Heuristic | Proposed by | We judge by | Typical error |
|---|---|---|---|
| Availability | Tversky and Kahneman, 1973 | Ease with which examples come to mind | Overestimating vivid, recent, dramatic events (K-words) |
| Representativeness | Kahneman and Tversky, 1972 | Similarity to the typical case | Base-rate neglect, conjunction fallacy, gambler's fallacy, ignoring sample size |
| Anchoring and adjustment | Tversky and Kahneman, 1974 | Adjusting from a starting value | Estimates stay too close to an arbitrary anchor |
| Recognition | Goldstein and Gigerenzer, 2002 | Whether an option is recognised | Fails when recognition is unrelated to the criterion |
| Affect | Paul Slovic and colleagues | Our feeling about the option | Judging risks low for liked things, high for disliked |

EXAM TRAP

"The representativeness heuristic leads to errors when individuals **ignore base rates and focus solely on similarity**." The other options each describe a different shortcut: relying on readily available information is **availability**; adjusting judgements from initial estimates is **anchoring**; failing to consider alternative explanations is **confirmation bias**. "More words seem to begin with K" is availability. Linda is representativeness (conjunction fallacy). Base-rate neglect, the gambler's fallacy and insensitivity to sample size are all representativeness errors.

## Framing, Prospect Theory and Loss Aversion

According to expected utility theory, the way a choice is described should not matter if the outcomes are the same. Tversky and Kahneman showed that it matters a great deal. The **framing effect** is the finding that logically equivalent descriptions of a choice lead to different decisions.

KEY STUDY: THE ASIAN DISEASE PROBLEM (TVERSKY AND KAHNEMAN, 1981)

**The method.** Participants were told that the United States was preparing for an unusual disease expected to kill 600 people, and that two programmes had been proposed. One group chose between: Programme A, "200 people will be saved", and Programme B, "a one-third probability that 600 people will be saved and a two-thirds probability that no one will be saved". A second group chose between: Programme C, "400 people will die", and Programme D, "a one-third probability that nobody will die and a two-thirds probability that 600 people will die".

**The result.** In the first (gain) frame, 72 per cent chose the certain option A. In the second (loss) frame, 78 per cent chose the risky option D. Yet A and C are identical (200 live, 400 die), and so are B and D. **The conclusion.** When outcomes are framed as gains, people are risk averse; when the same outcomes are framed as losses, people become risk seeking. Rational choice cannot explain this; prospect theory can.

Kahneman and Tversky set out **prospect theory** in 1979 as a descriptive theory of decisions under risk. It has three key ideas. First, people evaluate outcomes as gains or losses relative to a *reference point*, usually the current state or an expectation, not in terms of final wealth. A salary of ₹50,000 feels like a gain if you expected ₹40,000 and like a loss if you expected ₹60,000. Second, the *value function* shows *diminishing sensitivity* in both directions (Figure 8.5): the difference between gaining ₹100 and ₹200 feels larger than the difference between gaining ₹1,100 and ₹1,200. The curve is therefore concave for gains, which makes people risk averse for gains (a sure ₹500 beats a 50 per cent chance of ₹1,000), and convex for losses, which makes them risk seeking for losses (a 50 per cent chance of losing ₹1,000 beats a sure loss of ₹500). Third, the curve is steeper for losses than for gains. This is **loss aversion**: losing ₹1,000 hurts roughly twice as much as winning ₹1,000 pleases, so most people refuse a fifty-fifty bet unless they stand to win about twice what they might lose. Prospect theory adds that people overweight small probabilities, which helps explain why the same person buys both a lottery ticket and an insurance policy.

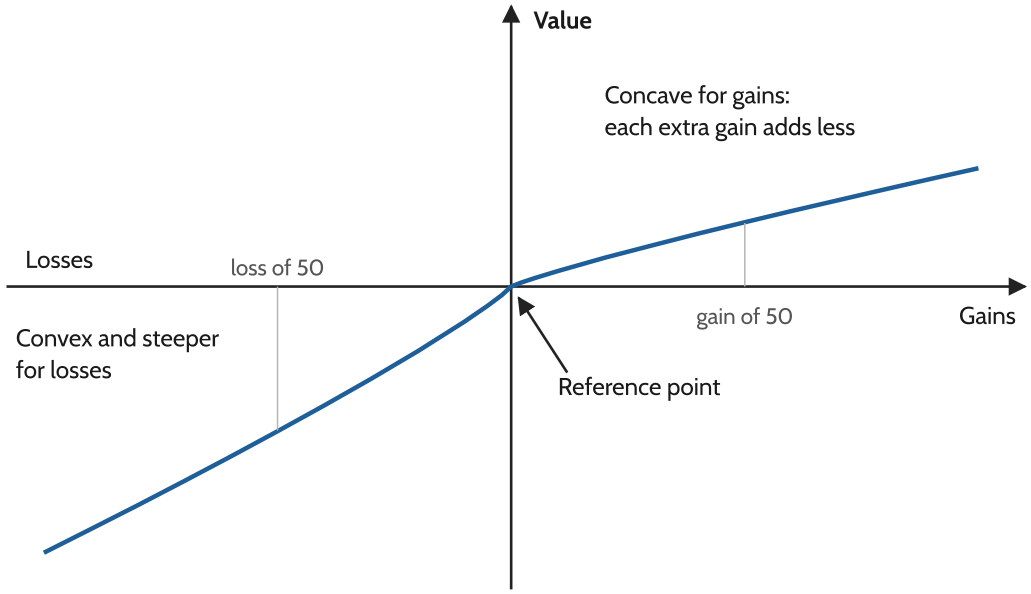



Figure 8.5 The prospect theory value function, plotted with the parameters Tversky and Kahneman estimated in 1992. Compare the two thin vertical lines: a loss of 50 lowers felt value about twice as much as a gain of 50 raises it. That asymmetry is loss aversion.

Loss aversion explains a family of everyday biases. The **endowment effect**, named by the economist **Richard Thaler** in 1980, is the tendency to value something more simply because we own it. In a classic experiment (Kahneman, Knetsch and Thaler, 1990), students who were given a university coffee mug demanded roughly twice as much to sell it as other students were willing to pay to buy one. Giving up the mug is coded as a loss, and losses loom large. **Status quo bias** (William Samuelson and Richard Zeckhauser, 1988) is the preference for keeping things as they are, because any change involves losing something. Defaults exploit it: in countries where citizens are organ donors unless they opt out, far more people are donors than in countries where they must opt in.

The **sunk-cost fallacy** is continuing to invest in a course of action because of what has already been spent on it, although past costs cannot be recovered and should not affect a decision about the future. Hal Arkes and Catherine Blumer (1985) sold season tickets for a university theatre at full price or at a discount, assigned at random. Those who had paid full price attended more plays in the first half of the season, although the money was equally gone in both cases. Staying in a film you dislike because the ticket cost ₹300, or continuing a failing project "because we have already put two years into it", are the same error, captured by the phrase "throwing good money after bad".

## Overconfidence and Hindsight ●●

We are also poor judges of our own knowledge. **Overconfidence** is the tendency for confidence to exceed accuracy. In calibration studies, people answer general-knowledge questions and rate how sure they are; when they say they are 100 per cent certain, they are wrong far more often than that would allow. A close relative is the *planning fallacy* (Kahneman and Tversky, 1979): we underestimate how long our own tasks will take, even when we know that similar tasks took longer in the past. (Check your revision timetable.)

Then there is **hindsight bias**, the "I knew it all along" effect: the tendency, once an outcome is known, to believe that it was more predictable than it really was. **Baruch Fischhoff** (born 1946) called it "creeping determinism". Before President Richard Nixon's visits to China and the Soviet Union in 1972, Fischhoff and Ruth Beyth (1975) asked people to estimate the probabilities of various possible outcomes. After the visits, the same people were asked to recall their estimates. They remembered having given higher probabilities to the events that had actually happened, and lower ones to those that had not. Hindsight bias makes the past look inevitable, which is why it is so hard to learn from it.

> EXAM TRAP
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> "The tendency to assume we would have been better at predicting actual events than is really true" is the **hindsight effect** (Fischhoff, 1975). The **just-world hypothesis** (Melvin Lerner) is the belief that people get what they deserve, a social psychology concept. "Decisiveness" is not a bias. **Confirmation bias** concerns testing beliefs, not misremembering predictions. And keep the decision terms straight: **sunk cost** = past investment; **endowment** = ownership; **status quo** = preference for no change; all three are driven by loss aversion.

## Two Systems of Thought

Try this. A cricket bat and a ball cost ₹1,100 in total. The bat costs ₹1,000 more than the ball. How much does the ball cost? The answer that leaps to mind is ₹100, and it is wrong: if the ball cost ₹100, the bat would cost ₹1,100, and together they would cost ₹1,200. The ball costs ₹50. This is an Indian version of the "bat-and-ball" item from Shane Frederick's *Cognitive Reflection Test* (2005), and it neatly separates two kinds of thinking.

According to **dual-process theory**, judgement and reasoning involve two different kinds of processing. Keith Stanovich and Richard West (2000) gave them the neutral labels that Kahneman later made famous in *Thinking, Fast and Slow* (2011). *System 1* is fast, automatic, effortless, associative and intuitive; it generates impressions and is the home of the heuristics. *System 2* is slow, deliberate, effortful and rule-based; it is limited by working memory and can check, and sometimes override, the answers System 1 proposes. "₹100" is System 1's answer; ₹50 is what System 2 finds if you let it work. Belief bias in syllogisms (Section 8.4) is the same conflict: System 1 judges the believability of the conclusion, System 2 its logic. The "systems" are best thought of as modes of processing, not separate places in the brain.

Feeling plays a larger part in decisions than the rational model admits. Paul Slovic's *affect heuristic* describes judging risks and benefits by how we feel about something: a technology we like seems low in risk and high in benefit. The neurologist Antonio Damasio's *somatic marker hypothesis* (1994) proposes that bodily signals of emotion, learned from past outcomes, steer choices; patients with damage to the ventromedial prefrontal cortex, who can reason perfectly well about a decision, nonetheless make disastrous real-life choices. (Emotion is the subject of Chapter 11.) Decisions made by groups bring further distortions, such as *groupthink* (Irving Janis, 1972), in which a cohesive group's desire for agreement overrides realistic appraisal and leads to poor decisions; these are covered in the Social Psychology book.

CLINICAL LINK

**In the client.** Aaron Beck's *cognitive distortions* are clinical relatives of the biases in this section. *Catastrophising* is expecting that the worst will happen; *all-or-nothing (dichotomous) thinking* sees experiences as wholly good or wholly bad; *selective abstraction* (the mental filter) focuses on negative details and ignores positive ones; *personalisation* takes events too personally; *overgeneralisation* draws sweeping conclusions from one event. **In the clinician.** Clinicians anchor on a first impression, judge by availability after a memorable case, and neglect base rates. If a screening test detects 90 per cent of cases of a disorder that affects 1 per cent of people, and wrongly flags 10 per cent of healthy people, then of 1,000 people screened, 9 true cases and 99 healthy people test positive: only about 8 per cent of positives have the disorder. Loren and Jean Chapman (1967) found that clinicians "saw" correlations between Draw-a-Person signs (such as large eyes) and symptoms (such as suspiciousness) that were not in the data, an *illusory correlation*. Paul Meehl's *Clinical Versus Statistical Prediction* (1954) showed that simple actuarial formulas usually predict at least as well as clinical judgement.

## 8.6 Creativity

MEDIUM YIELD · 5 QS

What do a new raga improvisation, a vaccine designed in record time and a child's drawing of a purple elephant playing the tabla have in common? Each is creative. Most researchers now define **creativity** as the ability to produce work that is both *novel* (original, unexpected) and *appropriate* (useful, or fitted to the constraints of the task), a definition associated with Robert Sternberg and Todd Lubart. Both halves matter. Novelty without appropriateness is merely odd: a random string of words is original but not creative. Appropriateness without novelty is merely competent.

Creativity can be studied from four directions, which Mel Rhodes (1961) called the *four Ps*: the creative *person* (traits and abilities), the creative *process* (the stages of thought), the creative *press* (the environment that supports or stifles it) and the creative *product* (what is made, and how it is judged). Creativity also varies in scale. "Big-C" creativity is eminent, field-changing work (Srinivasa Ramanujan's mathematics, Rabindranath Tagore's poetry); "little-c" creativity is everyday originality (a new recipe, a clever way to rearrange a small room). James Kaufman and Ronald Beghetto (2009) added *mini-c*, the personally meaningful insights of learning, and *Pro-c*, professional-level creativity, to make a *Four C model*.

### Guilford and Divergent Thinking

The scientific study of creativity is usually dated to September 1950, when J. P. Guilford gave his presidential address to the American Psychological Association. Guilford complained that creativity had been shamefully neglected: of some 121,000 titles indexed in *Psychological Abstracts* over the previous 23 years, only 186 bore definitely on creativity, fewer than two in every thousand. In his *Structure of Intellect* model (Chapter 9), Guilford distinguished *convergent production*, the generation of the one correct answer, from *divergent production*, the generation of many varied answers, and identified creativity largely with divergent production. He broke divergent thinking into four abilities, which are still the standard scoring categories of creativity tests:

- The first is **fluency**: the sheer number of relevant ideas produced.

- The second is **flexibility**: the number of different categories the ideas fall into, that is, the ease of shifting from one kind of idea to another.
- The third is **originality**: the statistical unusualness of the ideas, how rarely other people give them.
- The fourth is **elaboration**: the amount of detail used to develop and embellish an idea.

Guilford also listed *sensitivity to problems* (noticing that something needs doing) and *redefinition* (seeing familiar things in new ways) among the creative abilities. Here is how scoring works. Asked to list uses for a brick in two minutes, Ravi writes: build a wall, build a house, build a temple, a doorstop, a paperweight, a pestle for crushing masala, a marker for the stumps in street cricket. His *fluency* is 7. The ideas fall into four categories (construction, weight, kitchen tool, sport), so his *flexibility* is 4. If, in a large group, only Ravi thought of the masala and the stumps, those two earn *originality* credit. Had he written "a doorstop wrapped in an old sari so it will not scratch the floor", he would earn *elaboration* credit as well.

### The Creative Process: Wallas's Four Stages ●●●

The most famous account of the creative process came from **Graham Wallas** (1858 to 1932), an English political scientist and one of the founders of the London School of Economics. In *The Art of Thought* (1926), drawing on the self-reports of the physicist Hermann von Helmholtz and the mathematician Henri Poincaré, he proposed four stages (Table 8.8).

Table 8.8 Wallas's four stages of the creative process (1926).

| Order | Stage | What happens | Conscious? |
|---|---|---|---|
| 1 | Preparation | The problem is defined, information gathered and solutions attempted, often without success | Yes, effortful |
| 2 | Incubation | The problem is set aside; work on it continues, if at all, outside awareness | No |
| 3 | Illumination | The solution appears suddenly, the "aha" moment of insight | Sudden entry into awareness |
| 4 | Verification | The idea is tested, worked out and refined | Yes, effortful |

Poincaré's account of discovering a class of mathematical functions (which he called Fuchsian functions) fits the stages almost perfectly. After days of fruitless work (preparation), he left on a geological excursion and stopped thinking about mathematics (incubation). "At the moment when I put my foot on the step" of a bus, he wrote, the idea came to him, with complete certainty (illumination). Back home, he verified the result at leisure (verification). Wallas also described a fleeting moment just before illumination, a dim feeling that a solution is on its way, which he called *intimation*; some textbooks count it as a fifth stage. Notice that Wallas's model and the Gestalt research of Section 8.3 describe the same phenomena from two angles: incubation and illumination are what problem-solving researchers call incubation and insight.

EXAM TRAP

The order is **preparation, incubation, illumination, verification** (a memory hook: "**P**lease **I**ncubate **I**deas **V**ery carefully"). Second stage = incubation; third = illumination ("achieving insight"); "leaving the problem while considering other things" = incubation; "testing or carrying out the solution" = verification; "formulating the problem and making initial attempts" = preparation. Some books add **intimation** between incubation and illumination, which is why you may read of "five stages". Do not confuse the stages with Guilford's **abilities** (fluency, flexibility, originality, elaboration, sensitivity) or with Sternberg's **analytical, creative and practical** intelligences. A sequence such as "problem-finding, brainstorming, selection, implementation" resembles the later applied creative problem-solving schemes that grew out of Osborn's brainstorming work, not Wallas.

## Measuring Creativity

The most widely used creativity test is the *Torrance Tests of Creative Thinking* (TTCT), published by **Ellis Paul Torrance** (1915 to 2003) in 1966. The TTCT has a *verbal* form, with activities such as Ask-and-Guess (asking questions about a picture and guessing its causes and consequences), Product Improvement (making a toy more fun to play with), Unusual Uses and Just Suppose (imagining the consequences of an improbable event), and a *figural* form, with Picture Construction, Picture Completion and Repeated Figures (turning many identical circles or parallel lines into different pictures). Responses are scored on Guilford's dimensions of fluency, flexibility, originality and elaboration; later editions of the figural test also score abstractness of titles and resistance to premature closure. Torrance followed some of his child participants for decades, and their test scores predicted later creative achievement modestly but genuinely.

**Sarnoff Mednick** (1928 to 2015) approached creativity through association. In his *associative theory* (1962), creativity is the forming of new combinations of remote associative elements that are useful. People differ in their *associative hierarchies*. An uncreative person has a steep hierarchy: one or two strong, conventional associations to a word ("table" calls up "chair", then little else). A creative person has a flat hierarchy: many weaker associations, including remote ones, are almost equally available. Mednick's *Remote Associates Test* (RAT) presents three words and asks for a fourth word linked to all three. What links cottage, Swiss and cake? (Cheese.) Cream, skate and water? (Ice.) Note the irony: although the RAT measures remote association, each item has only one correct answer, so it also demands convergent thinking.

Michael Wallach and Nathan Kogan (1965) showed that the conditions of testing matter. When divergent-thinking tasks were presented to children as relaxed, untimed games rather than as tests, the creativity scores hung together and were largely independent of intelligence scores. Instead of a test, some researchers judge products directly. In Teresa Amabile's *consensual assessment technique*, experts in a domain rate the creativity of products (poems, collages, designs) independently, and their agreement is usually high: creativity, like beauty, is recognised reliably by those who know the field.

## Enhancing Creativity: Lateral Thinking and Brainstorming

The Maltese physician and writer **Edward de Bono** (1933 to 2021) introduced the term **lateral thinking** in *The Use of Lateral Thinking* (1967). Ordinary *vertical* thinking, he said, is logical

and sequential: it digs the same hole deeper. Lateral thinking deliberately moves sideways to dig a hole somewhere else, by challenging assumptions, reversing the problem, using a random word as a starting point, or making a deliberately provocative statement (which he marked with the made-up word "po") to jolt thinking out of its usual tracks. In *Six Thinking Hats* (1985) he proposed that a group take one perspective at a time: white for facts, red for feelings, black for caution, yellow for benefits, green for new ideas and blue for managing the thinking process.

*Brainstorming* was developed by the advertising executive **Alex Faickney Osborn** (1888 to 1966) at his agency from 1939 and popularised in his book *Applied Imagination* (1953). Its four rules are: defer judgement (no criticism during idea generation); go for quantity, because quantity breeds quality; welcome wild ideas; and combine and improve on others' ideas. The rules are sound, but the group format is not. Many studies have compared interacting brainstorming groups with *nominal groups*, the same number of individuals working alone whose ideas are pooled afterwards. The nominal groups usually produce more ideas, and better ones. Michael Diehl and Wolfgang Stroebe (1987) identified the main culprit as *production blocking*: in a group only one person can speak at a time, so others forget or suppress ideas while they wait. Evaluation apprehension and social loafing add to the loss. Writing ideas down silently first ("brainwriting") or brainstorming electronically recovers much of it.

## Creativity and Intelligence

Are creative people simply intelligent people? The usual answer is the **threshold hypothesis**, associated with Guilford, Torrance and others. Below an IQ of about 120, creativity and intelligence are positively correlated: you need a certain level of intelligence to be creative. Above that threshold the correlation largely disappears, and other factors (personality, motivation, knowledge, environment) decide who is creative. Intelligence, in short, is necessary but not sufficient for creativity.

Jacob Getzels and Philip Jackson (1962) compared adolescents at a Chicago school who scored very high on creativity tests but not on IQ tests with others who scored very high on IQ but not on creativity. Both groups did equally well in school achievement, but teachers preferred the high-IQ students; the high-creative students showed more humour, more unconventional goals and more playful, imaginative stories. A 2005 meta-analysis by Kyung Hee Kim put the overall correlation between creativity and intelligence scores at only about 0.17. Creativity and intelligence overlap, but they are not the same thing (theories of intelligence are the subject of Chapter 9).

## Theories of Creativity

Several traditions have offered explanations of creativity. For Freud, creative work is a *sublimation* of instinctual energy into socially valued activity, and the psychoanalyst Ernst Kris spoke of creativity as "regression in the service of the ego", a controlled dipping into primary-process thought. The humanists saw creativity as an expression of growth: Abraham Maslow distinguished the rare "special-talent" creativity of the genius from the "self-actualising" creativity open to anyone, and Carl Rogers argued that psychological safety and freedom foster it. Mednick's associative theory you have already met. Three modern theories are most examined.

The **investment theory** of creativity was proposed by Robert Sternberg and Todd Lubart (1991) and developed in their book *Defying the Crowd* (1995). Creative people, they argued, are like good investors: they "buy low", pursuing ideas that are unknown or out of favour but have growth potential, and they "sell high", moving on once the idea has become accepted and valuable. Doing this requires a *confluence* of six resources: intellectual abilities (the synthetic ability to see problems in new ways, the analytic ability to judge which ideas are worth pursuing, and the practical ability to persuade others of their value), knowledge (enough to know the field, but not so much that it entrenches you), a *legislative* thinking style (a preference for creating one's own rules and ways of doing things), personality (tolerance of ambiguity, willingness to take sensible risks, perseverance), intrinsic motivation, and a supportive environment.

**Teresa Amabile** (born 1950) proposed a *componential model* (1983) with three components: *domain-relevant skills* (knowledge and technique in the field), *creativity-relevant processes* (a cognitive style that breaks set, tolerates ambiguity and persists), and *intrinsic task motivation*. Her "intrinsic motivation principle" holds that people are most creative when they work for the interest and challenge of the task itself. In one study (1985), creative-writing students who had first ranked extrinsic reasons for writing (money, recognition) wrote less creative poems than those who had first ranked intrinsic reasons (enjoyment, self-expression). **Mihaly Csikszentmihalyi** (1934 to 2021) proposed a *systems model* (1988) in which creativity is not inside a person at all but emerges from the interaction of three parts: the *individual*, who produces a variation; the *domain*, the body of symbolic knowledge (mathematics, Carnatic music) in which the variation is made; and the *field*, the gatekeepers (editors, critics, examiners) who decide whether it enters the domain. Table 8.9 summarises the main theories.

Table 8.9 Theories of creativity.

| Theory | Theorist and year | Core idea |
|---|---|---|
| Divergent production | J. P. Guilford, 1950 | Creativity is fluency, flexibility, originality and elaboration of thinking |
| Stages of creative thought | Graham Wallas, 1926 | Preparation, incubation, illumination, verification |
| Associative theory | Sarnoff Mednick, 1962 | Combining remote associations; flat associative hierarchies |
| Investment (confluence) theory | Sternberg and Lubart, 1991 and 1995 | Buy low, sell high; six converging resources |
| Componential model | Teresa Amabile, 1983 | Domain skills, creative processes, intrinsic motivation |
| Systems model | Mihaly Csikszentmihalyi, 1988 | Individual, domain and field together create creativity |
| Four Ps | Mel Rhodes, 1961 | Person, process, press, product |

## The Creative Personality

What are highly creative people like? In the 1950s and 1960s, researchers at the Institute of Personality Assessment and Research at Berkeley, including Frank Barron and Donald MacKinnon, invited eminent writers, scientists and architects for intensive weekends of assessment. MacKinnon's (1962) study of architects, judged creative or less creative by their peers, became the classic. The most creative architects were independent in judgement, self-confident and open to their own feelings and to new experience; they preferred complexity (on Barron's art-preference tests they liked complex, asymmetrical figures that less creative

people disliked), had wide interests and high aesthetic sensitivity, and showed a notable openness to traits stereotyped as feminine. Later work confirms the picture. In Gregory Feist's (1998) meta-analysis, *openness to experience* was the personality trait most consistently linked to creativity in both artists and scientists, along with autonomy, drive and confidence; tolerance of ambiguity and willingness to take risks complete the profile.

CLINICAL LINK

Is there a link between creativity and mental illness? Nancy Andreasen (1987) interviewed writers at the University of Iowa Writers' Workshop and found far higher lifetime rates of mood disorder, particularly bipolar disorder, than in matched controls, and Kay Redfield Jamison's *Touched with Fire* (1993) documented the same pattern in the lives of eminent poets and artists. The samples were small and selective, and the link, where it exists, seems strongest for milder mood elevation and for first-degree relatives rather than for severe illness. For a clinician the practical point is the reverse one: many clients fear that treatment will dull their creativity, and this fear needs to be discussed openly.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Herbert Simon's term for choosing the first option that is good enough is _____.
2. Judging that more words begin with K than have K third, because they come to mind more easily, reflects the _____ heuristic.
3. Judging that Linda is more likely a feminist bank teller than a bank teller is the _____ fallacy.
4. Giving estimates that stay too close to an arbitrary starting number shows _____ and adjustment.
5. In the Asian disease problem, people were risk averse when outcomes were framed as gains and risk seeking when framed as _____.
6. In prospect theory, the value function is steeper for losses than for gains; this is called _____.
7. Fischhoff's "I knew it all along" effect is _____ bias.
8. Guilford's divergent-thinking score for the number of different categories of ideas is _____.
9. Wallas's third stage of creative thought is _____.
10. Sternberg and Lubart's theory that creative people "buy low and sell high" is the _____ theory.

**Answers:** (1) satisficing, (2) availability, (3) conjunction, (4) anchoring, (5) losses, (6) loss aversion, (7) hindsight, (8) flexibility, (9) illumination, (10) investment.

## 8.7 The Structure of Language

LOWER YIELD · 3 QS

You are reading these words without effort, yet you have almost certainly never met this exact sentence before. How is that possible? Language is so familiar that we rarely notice how strange it is. A **language** is a system of arbitrary symbols (spoken words or manual signs) that

are combined according to rules, allowing the members of a community to communicate an unlimited number of messages. Many animals communicate; as you will see in Section 8.9, whether any other animal has language is a much harder question. To answer it we first need to know what makes language language.

### The Defining Properties of Language

In 1960 the American linguist **Charles Francis Hockett** (1916 to 2000) listed thirteen *design features* that together distinguish human language from animal communication systems. The thirteen are: the vocal-auditory channel, broadcast transmission and directional reception, rapid fading, interchangeability (anyone who can hear a message can also send it), total feedback (speakers hear themselves), specialisation, semanticity, arbitrariness, discreteness, displacement, productivity, cultural (traditional) transmission and duality of patterning. With Stuart Altmann he later added prevarication (the ability to lie), reflexiveness (using language to talk about language) and learnability. Five of the features carry most of the weight, and they are the ones you are examined on.

- **Semanticity.** Linguistic symbols carry meaning: they refer to objects, events, actions and ideas.
- **Arbitrariness.** There is no natural connection between the form of a word and what it means. The same animal is "dog" in English, *kutta* in Hindi and *nayi* in Kannada, and nothing about any of these sounds is doglike. The Swiss linguist **Ferdinand de Saussure** (1857 to 1913) made this "arbitrariness of the sign" the first principle of modern linguistics (1916). Onomatopoeic words such as "buzz" are the rare exceptions.
- **Displacement.** Language allows **displacement**, communication about things that are not here and now: yesterday's lecture, next month's results, a city you have never visited, a unicorn that never existed.
- **Generativity** (Hockett's *productivity*): a finite set of words and rules can generate an infinite number of new sentences, which listeners understand at once although they have never heard them.
- **Duality of patterning.** Language is organised on two levels. Meaningless sounds (phonemes) combine into meaningful units (morphemes and words), and these combine into sentences. The same three sounds /k/, /æ/ and /t/ make "cat", "act" and "tack".

A related property, *recursion*, allows a phrase to be embedded inside another phrase of the same kind, without limit: "the girl who met the boy who lost the pen that his father bought". Marc Hauser, Noam Chomsky and Tecumseh Fitch (2002) proposed that recursion is the one feature of language that is uniquely human. The same embedding applied to minds gives *recursive thought* ("I think that you think that I am bluffing"), a form of perspective taking that develops in late childhood and is covered in the Developmental Psychology book. Do not confuse it with Piaget's *reversibility* (mentally undoing an action, Section 8.4), with Katherine Nelson's *referential style* of early word learning (Section 8.8), or with *reconstruction*, the rebuilding of a memory at recall (Chapter 7).

INDIAN CONTEXT

The scientific study of language structure has a long Indian history. Around the fourth century BCE (the dates are uncertain), the grammarian Pāṇini composed the *Aṣṭādhyāyī*, a grammar of Sanskrit in nearly 4,000 compact rules (sūtras) that generate the correct forms of words from roots and affixes. Modern linguists regard it as the earliest generative grammar, a set of finite rules that produces all and only the correct forms. Some nine centuries later, the philosopher Bhartṛhari, in the *Vākyapadīya*, argued that meaning is grasped as an indivisible whole (*sphoṭa*) rather than sound by sound, and that "there is no cognition in the world that does not follow language": an Indian position on the language and thought question of Section 8.9.

## The Units of Language ●●●

Language is a hierarchy of units, each built from the one below (Table 8.10). At the bottom is the **phoneme**, the smallest unit of sound that can distinguish one word from another in a particular language. The /p/ in "pat" and the /b/ in "bat" are different phonemes because swapping them changes the word. A phoneme has no meaning of its own: /p/ means nothing. English has about 40 to 45 phonemes, depending on the dialect, and the world's languages range from about a dozen to more than a hundred. Languages also differ in which sound differences count. In Hindi the unaspirated /k/ of *kal* (yesterday or tomorrow) and the aspirated /kh/ of *khal* (a villain) are different phonemes; in English the same two sounds occur (in "skin" and "kin") but never distinguish words, so English speakers barely hear the difference. The study of a language's sound system is *phonology*.

Phonemes combine into the **morpheme**, the smallest unit of language that carries meaning. "Cat" is one morpheme; "cats" is two (cat plus the plural -s); "unhappiness" is three (un + happy + ness). A *free* morpheme can stand alone as a word (happy, cat, run). A *bound* morpheme cannot (un-, -ness, -s, -ed). Bound morphemes are either *inflectional*, adding grammatical information without changing the word's class (the plural -s, the past tense -ed, -ing), or *derivational*, creating a new word, often of a new class (happy becomes happiness; teach becomes teacher). The study of word structure is *morphology*. Notice that a *syllable* (ba-na-na) is a unit of pronunciation organised around a vowel: it is neither the smallest unit of sound nor the smallest unit of meaning.

Words are combined into phrases and sentences according to **syntax**, the rules of sentence structure. "The dog bit the man" and "The man bit the dog" contain the same words; only syntax tells you who needs the injection. English is a subject-verb-object language; Hindi and Kannada put the verb last (subject-object-verb: *Ram ne seb khaaya*, "Ram the apple ate"). *Grammar* is the umbrella term for a language's phonological, morphological and syntactic rules.

The study of meaning is **semantics**: the meaning of words, and how word meanings combine into the meaning of sentences. The study of how language is used in social context is **pragmatics**: how speakers convey, and listeners infer, more than the literal words say. "Can you pass the salt?" is literally a question about ability, but everyone understands it as a request. The philosopher H. Paul Grice (1975) argued that conversation rests on a *cooperative principle* expressed in four maxims: give as much information as needed and no more (quantity), say what you believe true (quality), be relevant (relation), and be clear (manner). Finally, **prosody** is the melody of speech: its rhythm, stress and intonation. "You're coming?" and

"You're coming." differ only in prosody. Prosody also carries emotion (the warmth, sarcasm or anger in a voice), and this *emotional prosody* depends chiefly on regions of the right hemisphere that mirror the left hemisphere's language areas; damage there produces *aprosodia*, flat or uninterpretable emotional tone (Elliott Ross, 1981).

Table 8.10 The units and levels of language.

| Level | Definition | Example |
|---|---|---|
| Phoneme (phonology) | Smallest unit of sound that distinguishes words; no meaning of its own | /p/ versus /b/ in pat and bat |
| Morpheme (morphology) | Smallest unit that carries meaning; free or bound | un + happy + ness (three) |
| Word (lexicon) | Free-standing unit made of one or more morphemes | teacher |
| Syntax | Rules for combining words into phrases and sentences | Dog bites man versus man bites dog |
| Semantics | Meaning of words and sentences | Knowing that a bachelor is unmarried |
| Pragmatics | Use of language in social context | "Can you pass the salt?" as a request |
| Prosody | Rhythm, stress and intonation of speech | "You're coming?" versus "You're coming." |

EXAM TRAP

"The smallest unit of **meaning**" is the **morpheme**. "The basic units of **sound** used in a spoken language" are **phonemes**. Syllables and consonants are not the answer to either: a consonant is just one kind of phoneme, and a syllable is a unit of pronunciation. Keep the other levels apart by their definitions: rules for combining words into sentences = **syntax**; meaning = **semantics**; culturally appropriate use in context = **pragmatics**; intonation and emotional tone = **prosody**, processed mainly by the **right hemisphere** regions homotopic to the left-hemisphere language areas (not by Broca's area).

## Chomsky's Transformational Grammar

In 1957 a 28-year-old linguist at the Massachusetts Institute of Technology, Noam Chomsky, published a short book, *Syntactic Structures*, that changed both linguistics and psychology. He began with a sentence that has since become famous: "Colorless green ideas sleep furiously." It is meaningless, yet every English speaker recognises it as grammatical, while "Furiously sleep ideas green colorless" is not. So our knowledge of grammar cannot be knowledge of meaning, and it cannot be knowledge of which words usually follow which, because none of these words normally follow one another. It must be knowledge of abstract rules.

Chomsky proposed a *generative grammar*: a finite set of rules that can generate all, and only, the grammatical sentences of a language. *Phrase-structure rules* describe how a sentence breaks into its constituents: a sentence (S) consists of a noun phrase (NP) and a verb phrase (VP); a noun phrase can consist of a determiner and a noun ("the dog"); a verb phrase can consist of a verb and a noun phrase ("chased the cat"). *Transformational rules* then convert one structure into another, turning an active sentence into a passive one or a statement into a question.

In *Aspects of the Theory of Syntax* (1965), Chomsky distinguished two levels of structure. The **surface structure** is the actual form of the sentence as it is spoken or written: the words in their order. The **deep structure** is the underlying, abstract structure that represents the basic grammatical relations, who did what to whom. Transformations map deep structure onto surface structure. Two kinds of example show why both levels are needed:

- One deep structure can appear as different surface structures. "The dog chased the cat" and "The cat was chased by the dog" look different but say the same thing.
- One surface structure can come from different deep structures. "Visiting relatives can be boring" means either that it is boring to visit relatives or that relatives who visit are boring. "The shooting of the hunters was terrible" is ambiguous in the same way: were the hunters shooting badly, or being shot?

Chomsky also distinguished **competence**, a speaker's tacit, unconscious knowledge of the rules of the language, from *performance*, the actual use of language in real situations, which is marred by slips of the tongue, false starts, hesitations and the limits of memory. (The distinction echoes Saussure's earlier contrast between *langue*, the language system, and *parole*, actual speech.) Linguistics, for Chomsky, is the study of competence. In later work, Chomsky replaced deep and surface structure with a theory of universal *principles and parameters* (1981) and then the *Minimalist Program* (1995), but the 1957 and 1965 ideas are the ones psychology absorbed, and the ones you will be asked about.

> EXAM TRAP
>
> **1957** = *Syntactic Structures* (transformational generative grammar); **1959** = Chomsky's review of Skinner's *Verbal Behavior*; **1965** = *Aspects of the Theory of Syntax* (deep and surface structure, competence and performance, the language acquisition device). Ambiguous sentences such as "Visiting relatives can be boring" illustrate **one surface structure with two deep structures**; active and passive pairs illustrate **one deep structure with two surface structures**. Competence is knowledge; performance is use.

## 8.8 Acquiring Language

MEDIUM YIELD · 8 QS

By the age of five, without a single grammar lesson, a typical child has a vocabulary of thousands of words and commands most of the grammar of her language, including rules that her parents could not state if asked. She does this whether her language is Kannada, Tamil, English or Mandarin, and whether her parents are professors or labourers. How? Three families of theory compete to explain it.

### Learning Theory: Skinner's Verbal Behavior

For **Burrhus Frederic Skinner** (1904 to 1990), language was behaviour like any other, acquired through reinforcement, shaping and imitation. In *Verbal Behavior* (1957) he classified "verbal operants" by their controlling conditions. A *mand* is a demand or request reinforced by what it asks for (the child says "milk" and gets milk). A *tact* names something in the environment and is reinforced socially ("car!", "Yes, that's a car!"). An *echoic* repeats what has just been heard. An *intraverbal* responds to someone else's words without repeating them ("How are you?" "Fine."). *Textual* behaviour is reading aloud, and *transcription* is writing down what

is heard (taking dictation). An *autoclitic* modifies the speaker's own verbal behaviour ("I think...", "I'm not sure, but..."). Learning theory explains some things well: children learn the particular language, words and accent they hear, and they learn frequent words earlier.

In 1959 Chomsky published a review of *Verbal Behavior* in the journal *Language* that is often said to have ended behaviourism's claim on language. His arguments were these. Children constantly produce sentences they have never heard and therefore cannot have imitated or been reinforced for. Their characteristic errors ("I goed", "two mouses") are not imitations of anything adults say; they show that the child has extracted a rule and is over-applying it. Parents rarely correct grammar anyway: Roger Brown and Camille Hanlon (1970) found that parents responded to the *truth* of what children said, not its form, approving "Mama isn't boy, he a girl" because it was true. And children everywhere acquire language too fast and too uniformly for trial-and-error learning to explain.

### The Nativist Theory: Chomsky's Language Acquisition Device ●●

Chomsky's alternative was that children are born with an innate, biologically based capacity for language, which he called the **language acquisition device** (LAD). The LAD contains a **universal grammar**: a set of principles and categories common to all human languages (every language has nouns and verbs, consonants and vowels, ways of negating and of asking questions). Exposure to a particular language does not teach the child grammar from scratch; it sets the values of a small number of *parameters*, like switches. English puts the verb before its object; Hindi and Kannada put it after; a child hearing either language needs only to set the switch.

The central argument for nativism is the **poverty of the stimulus**: the speech children hear is fragmentary, full of errors and false starts, and contains almost no information about which sentences are *not* allowed, yet every normal child arrives at the same intricate grammar. The input, Chomsky argued, is too poor to explain the output, so much of the knowledge must be innate. Further evidence came from the regular, universal sequence of milestones (Table 8.12), from deaf children who, brought together without an adult model, create a full sign language of their own (as happened in Nicaragua from the late 1970s), and from the critical period discussed below. **Steven Pinker** (born 1954) popularised the nativist case in *The Language Instinct* (1994).

Critics reply that the LAD is a label rather than an explanation, that nativism ignores the rich social interaction in which language is learned, and that the input is richer than Chomsky supposed. Jenny Saffran, Richard Aslin and Elissa Newport (1996) played eight-month-old infants a continuous stream of nonsense syllables for just two minutes. The infants learned which syllables tended to follow one another, and so where the "word" boundaries lay: powerful *statistical learning* that needs no innate grammar.

### Interactionist Theories ●●

Interactionist theories accept that children bring special capacities to the task but insist that social interaction does essential work. Jerome Bruner, in *Child's Talk* (1983), answered Chomsky's LAD with a **language acquisition support system** (LASS): the structured help that adults provide. Bruner emphasised *formats*, routine, predictable games such as peekaboo and picture-book reading ("What's that?" "A cow!" "Yes, it's a cow!"), within which the child can work out what words refer to. The adult's support is a form of *scaffolding*, a term in-

troduced by David Wood, Bruner and Gail Ross in 1976 for help that is adjusted to the child's level and withdrawn as competence grows.

Part of the support system is the special way adults talk to infants, known as *motherese*, *baby talk* or, more neutrally, **child-directed speech** (infant-directed speech). Compared with speech between adults, it has a higher overall pitch and a wider, exaggerated range of pitch (a sing-song melody), a slower rate, shorter and simpler utterances (a lower mean length of utterance), more repetition, more questions, clearer pronunciation, more frequent and longer pauses, and a focus on the here and now. Infants prefer to listen to it, and its exaggerated contours help them pick out words and phrase boundaries. But it is not strictly necessary: in some cultures adults talk relatively little to infants, and their children still learn to speak.

The developmental psycholinguist **Dan Isaac Slobin** (born 1939), comparing acquisition across dozens of languages, proposed that children bring a *language-making capacity* (LMC) to the task (1985). Unlike the LAD, the LMC is not innate knowledge of grammar. It is a set of cognitive and perceptual abilities, which Slobin called *operating principles*, that are highly specialised for learning language: for example, "pay attention to the ends of words" (where many languages put their grammatical endings) and "avoid exceptions" (which produces overregularisation). Table 8.11 compares the theories.

Table 8.11 Theories of language acquisition.

| Theory | Key theorist | How language is acquired | Key concepts | Main weakness |
|---|---|---|---|---|
| Learning (behaviourist) | B. F. Skinner, 1957 | Reinforcement, shaping, imitation | Mand, tact, echoic, intraverbal, autoclitic | Cannot explain novel sentences or "goed" |
| Nativist | Noam Chomsky, 1957 to 1965 | Innate grammar triggered by input | LAD, universal grammar, poverty of the stimulus | LAD is unspecified; neglects social input |
| Interactionist (social) | Jerome Bruner, 1983 | Innate capacity plus structured adult support | LASS, formats, scaffolding, child-directed speech | Some cultures give little such support |
| Cognitive-perceptual | Dan Slobin, 1985 | General but language-tuned learning strategies | Language-making capacity, operating principles | Principles are descriptive lists |
| Cognitive (constructivist) | Jean Piaget | Language maps onto concepts already formed | Thought precedes language | Underplays language's effect on thought |

EXAM TRAP

Chomsky's innate capacity is the **language acquisition device**, not the phonological loop (part of Baddeley and Hitch's working memory, Chapter 7), not the zone of proximal development (Vygotsky) and not a schema (Piaget, Bartlett). Distinguish the three acronyms: **LAD** = Chomsky (an inborn processor activated by verbal input, containing universal grammar, "rules common to all languages"); **LASS** = Bruner (adult support); **LMC** = Slobin ("a set of cognitive and perceptual abilities highly specialised for language learning"). Motherese has a **shorter** mean length of utterance, **higher** pitch, **wider** pitch range, **slower** rate and **more** pauses than adult-to-adult speech.

## Is There a Critical Period for Language? ●●

In *Biological Foundations of Language* (1967), **Eric Heinz Lenneberg** (1921 to 1975) argued that language must be acquired during a **critical period**, a biologically fixed window of develop-

ment (roughly from the age of two to puberty, in his account) during which the brain is especially receptive to linguistic input and after which full acquisition becomes impossible. He linked the end of the window to the completion of brain maturation and lateralisation. Today most researchers prefer the softer term *sensitive period*: a window during which learning is easiest and most complete, although some learning is still possible afterwards.

The most famous, and saddest, evidence came from a girl known as Genie. In November 1970, social workers in Los Angeles found a 13-year-old girl who had been kept by her father in near-total isolation, strapped to a potty chair or confined in a crib, from about the age of 20 months. She had heard almost no language and could not speak. Over the following years, studied in detail by the linguist **Susan Curtiss** (1977), Genie learned many words and could communicate, but her grammar never developed. Her utterances remained telegraphic ("Applesauce buy store"), with little use of function words or word-order rules. The case supports a sensitive period for grammar in particular, but it is not clean: Genie was also severely abused and malnourished, and nobody knows whether she was developmentally normal to begin with.

KEY STUDY: JOHNSON AND NEWPORT'S IMMIGRANTS (1989)

**The question.** Does the age at which you start learning a second language limit how well you finally learn it? **The method.** Jacqueline Johnson and Elissa Newport tested Korean and Chinese immigrants to the United States who had all lived there for years, but who had arrived at ages ranging from 3 to 39. Participants judged whether each of 276 spoken English sentences was grammatical. **The result.** Those who had arrived between 3 and 7 performed like native speakers. Performance fell steadily with each later age of arrival up to puberty; those who arrived as adults scored lower and varied widely, unrelated to their age at arrival. **The conclusion.** There is a maturational sensitive period for acquiring grammar that closes gradually through childhood. **The criticism.** Age of arrival is confounded with amount and kind of exposure, schooling and motivation.

EXAM TRAP

"Infants who have no exposure to language show severely limited language development that is not reversible" supports **sensitive (critical) periods of development**. Semantics and syntax are levels of language, not developmental principles; nativism is the broader theory that language is innate, and the item asks for the term the evidence supports most directly. Critical period for language = **Lenneberg (1967)**; Genie studied by **Susan Curtiss**.

## Milestones of Language Development ●●

Language begins before the first word. Newborns already prefer their mother's voice to a stranger's (Anthony DeCasper and William Fifer, 1980), having heard it in the womb. Young infants can discriminate the speech sounds of all the world's languages, but their perception narrows to their own language during the first year. Janet Werker and Richard Tees (1984) found that English-learning infants of 6 to 8 months could distinguish two Hindi consonants (a retroflex "ta" and a dental "ta") that English does not use, but by 10 to 12 months most could not; Hindi-learning infants kept the distinction. Infants become specialists in the sounds that matter around them.

Production follows its own timetable (Table 8.12). At about two months infants begin **cooing**, producing vowel-like sounds ("ooo", "aah"), often in response to a smiling face. At about six months **babbling** begins: consonant-vowel combinations, soon repeated in strings ("ba-ba-ba", "da-da-da"). Babbling is at first universal, and early babbling sounds similar whatever language the child hears; but it gradually drifts towards the sounds of the home language. Deaf infants' vocal babbling is delayed and reduced, and deaf infants exposed to sign language "babble" with their hands, producing rhythmic, syllable-like hand movements (Laura-Ann Petitto and Paula Marentette, 1991). Babbling, it seems, is the practice of language's sound units in whatever medium the child meets.

The first recognisable words appear around the first birthday, often for people, animals, food and toys that move or can be acted on. In the *one-word stage* a single word can do the work of a whole sentence: "Milk!" may mean "I want milk", "The milk spilled" or "That is milk". Such a word is called a **holophrase**. **Katherine Nelson** (1973) noticed two styles among children at this stage: *referential* children's first fifty words were mostly names for objects, while *expressive* children used more social formulas ("stop it", "thank you", "I want it"). Vocabulary grows slowly at first and then, around 18 months and about 50 words, accelerates sharply: the *vocabulary spurt* or naming explosion.

Early word meanings rarely match adult ones. In **overextension** a child applies a word too broadly: "dog" for every four-legged animal, "papa" for every man. In *underextension* the child applies it too narrowly: "dog" only for the family dog, "car" only for cars seen moving in the street. And children learn words astonishingly fast. Susan Carey and Elsa Bartlett (1978) casually asked preschoolers to "bring me the chromium tray, not the red one", where the tray was olive green. A week later, some of the children still had a partial idea of what "chromium" meant, after a single exposure. This ability to form a quick first hypothesis about a word's meaning from one encounter is called **fast mapping**.

Between 18 and 24 months children begin to combine two words. **Roger Brown** (1925 to 1997) called this **telegraphic speech**, because, like a telegram, it keeps the content words (nouns, verbs, adjectives) and leaves out function words and grammatical endings: "Mummy go", "more milk", "allgone ball", "Daddy shoe". These short utterances already express a small set of meaning relations: agent and action, action and object, possession, location, recurrence and disappearance. Brown, in *A First Language* (1973), followed three children (Adam, Eve and Sarah) and measured progress by the *mean length of utterance* (MLU) in morphemes, which predicts grammatical development better than age. Grammatical morphemes then appear in a remarkably regular order, beginning with -ing and the plural -s.

From about two to three years, children's grammar shows its most revealing error. Having learned rules such as "add -ed for the past" and "add -s for the plural", children apply them to irregular words they had previously used correctly: "I goed", "two foots", "the mouses". This **overregularisation** means that performance gets worse before it gets better (a *U-shaped* pattern), and it is strong evidence that children are learning rules, not imitating. **Jean Berko** (1958) showed the same thing with her "wug test". She showed children a picture of a small imaginary creature: "This is a wug. Now there is another one. There are two of them. There are two ...?" Preschoolers happily answered "wugs", applying a rule to a word they had never heard. Between three and five, sentences lengthen and become complex, with questions, negatives and embedded clauses; pragmatic skills grow too (four-year-olds simplify their speech when talking to two-year-olds), and in the school years children develop *metalinguistic*

*awareness*, the ability to think about language itself, which shows up in their delight in puns and riddles.

Table 8.12 Milestones of language development (ages are approximate and vary widely).

| Age | Milestone | What the child does | Example |
|---|---|---|---|
| Birth to 2 months | Crying, early perception | Prefers mother's voice; discriminates speech sounds of all languages | Turns to mother's voice |
| About 2 months | Cooing | Vowel-like sounds | "ooo", "aah" |
| About 6 months | Babbling | Repeated consonant-vowel strings, drifting towards home language | "ba-ba-ba" |
| 10 to 12 months | Perceptual narrowing, gestures | Loses non-native contrasts; points | Pointing at a bird |
| About 12 months | One-word (holophrastic) stage | Single words carry whole sentences; overextension, underextension | "Milk!" |
| About 18 months | Vocabulary spurt | Rapid word learning after about 50 words; fast mapping | "What's that?" naming games |
| 18 to 24 months | Two-word, telegraphic speech | Content words without function words | "More milk", "Mummy go" |
| 2 to 3 years | Grammatical morphemes, overregularisation | Applies rules to irregular words | "I goed", "two foots" |
| 3 to 5 years | Complex sentences, pragmatics | Questions, negatives, clauses; adjusts to listener | "Why did the man go?" |
| School years | Metalinguistic awareness | Thinks about language; enjoys puns | Riddles and wordplay |

EXAM TRAP

Two-word utterances such as "more milk" and "mummy go", without function words, appear at about **18 to 24 months** and are called **telegraphic speech** (not the holophrastic stage, which is one word at about 12 months, and not babbling or cooing). Overextension = too broad ("dog" for a cow); underextension = too narrow. Referential versus expressive **styles** are Katherine Nelson's (1973), and "referential style" is a language-development term, not a perspective-taking one.

CLINICAL LINK

Language delay is often the first thing that brings a child to a clinic. Widely used red flags include no babbling or pointing by 12 months, no single words by 16 months, no spontaneous two-word phrases by 24 months, and *any* loss of language or social skills at any age. Hearing should always be checked first. Of the neurodevelopmental conditions, language and communication are most consistently compromised in **autism spectrum disorder**, where delayed speech is often accompanied by echolalia, pronoun reversal ("you want juice" meaning "I want juice") and, even in fluent speakers, impaired pragmatics. Developmental language disorder, by contrast, affects language with relatively intact social communication. In attention-deficit/hyperactivity disorder language difficulties are common but are not a defining feature; specific learning disability is defined by difficulty with academic skills (reading, writing, arithmetic) rather than by delayed speech; and childhood dissociative conditions do not characteristically arrest language development. (The disorders are covered in the Psychiatric Disorders book.)

## 8.9 Language, Thought and Communication

MEDIUM YIELD · 8 QS

We can now return to the question Sultan raised at the start of this chapter. What is the relationship between language and thought? Does the language you speak shape the way you think? Can you think without language at all?

### Does Language Shape Thought? The Sapir-Whorf Hypothesis ●●

**Benjamin Lee Whorf** (1897 to 1941) was a fire-prevention inspector for an insurance company in Connecticut and, in his spare time, a student of linguistics under the anthropologist-linguist **Edward Sapir** (1884 to 1939). In his work he noticed that workers who were careful around drums labelled "gasoline" smoked carelessly near drums labelled "empty", although the "empty" drums were full of explosive vapour. The word, it seemed, had shaped the behaviour. Whorf went on to argue, from his study of the Hopi language of Arizona, that the Hopi conceive of time quite differently from speakers of European languages because their grammar treats it differently. His essays, collected after his death as *Language, Thought, and Reality* (1956), gave his name and Sapir's to the *Sapir-Whorf hypothesis*.

The hypothesis comes in two strengths. The strong version, *linguistic determinism*, holds that language determines thought: we can think only in the categories our language provides, and speakers of very different languages live in different mental worlds. The weak version, **linguistic relativity**, holds that language influences thought: the habits of a language make some distinctions easier to notice, remember or express, without making others unthinkable.

The strong version has not survived. Later fieldwork found that Hopi has plenty of ways of talking about time. The often-repeated claim that "Eskimo" languages have dozens or hundreds of words for snow turned out to be an exaggeration that grew with each retelling. And the most systematic test, on colour, pointed the other way. **Brent Berlin** and **Paul Kay** (1969), in *Basic Color Terms*, examined 98 languages and found that they have between two and eleven basic colour terms, added in a nearly universal order: a language with only two terms has words for dark and light (black and white); a language with three adds red; then come green and yellow; then blue; then brown; and finally purple, pink, orange and grey. Colour

vocabularies, in other words, are constrained by universal features of perception. Eleanor Rosch, publishing as Eleanor Rosch Heider (1972), then studied the Dani of New Guinea, whose language has only two basic colour terms (*mili* for dark, cool colours and *mola* for light, warm ones). The Dani remembered the "focal" colours (the best red, the best green) better than non-focal ones, and learned names for them more easily, just as English speakers do. Lacking a word for a colour did not stop them perceiving and remembering it.

The weak version, however, has strong support. Russian has two basic words for what English calls blue: *goluboy* (light blue) and *siniy* (dark blue). Jonathan Winawer and his colleagues (2007) found that Russian speakers discriminated two blues faster when the shades fell on opposite sides of the goluboy-siniy boundary than when they fell on the same side; English speakers showed no such effect. Speakers of languages that describe space in absolute terms (north, south) rather than relative ones (left, right) keep track of compass directions with remarkable accuracy. You can see the same thing in kinship terms. A Hindi speaker must choose between *chacha* (father's younger brother), *tau* (father's elder brother), *mama* (mother's brother), *mausa* and *phupha*, where English offers only "uncle", so Hindi speakers habitually attend to which side of the family a relative belongs to. English speakers can understand the distinction perfectly well; they simply do not have to make it every time they speak. That is linguistic relativity in its defensible, weak form. Steven Pinker (1994) argues against even this much emphasis on language: we think, he says, in a language-independent "mentalese", which is why translation is possible and why babies and animals can think without words.

> EXAM TRAP
>
> The linguistic relativity hypothesis is associated with **Benjamin Lee Whorf** (and his teacher Edward Sapir). Chomsky (universal grammar), Piaget (thought before language) and Pinker (mentalese, *The Language Instinct*) are all wrong answers, and Pinker is a critic. **Strong** version = determinism; **weak** version = relativity (the one with support). Berlin and Kay's colour universals and Heider's Dani study count **against** the strong version.

## Piaget Versus Vygotsky ●●●

Recall Watson's claim from Section 8.1 that thinking is merely subvocal speech. It was put to a dramatic test in 1947, when the anaesthetist Scott Smith had himself injected with curare, the drug that paralyses all the skeletal muscles, including those of the larynx and tongue (his breathing was maintained artificially). Afterwards he reported that he had remained fully conscious throughout, had understood what was said around him and had been able to think clearly. Thinking does not require movements of the speech muscles. But that does not settle how language and thought are related, and on this the two great developmental theorists disagreed.

For **Jean Piaget** (1896 to 1980), thought comes first. Children build concepts through action on the world, and language is a way of expressing understanding that is already there; a child says "allgone" only after understanding that objects persist and can disappear. In *The Language and Thought of the Child* (1923) Piaget described *egocentric speech*: speech that is not adapted to a listener, such as the "collective monologue" of a group of preschoolers, each talking about his own activity without responding to the others. For Piaget, egocentric speech reflected the egocentrism of young children's thought, a stage between Bleuler's autistic

thought and socialised, directed thought, and it simply faded away, around the age of seven, as thought became socialised.

**Lev Semyonovich Vygotsky** (1896 to 1934), the Russian founder of sociocultural theory, read Piaget closely and disagreed. In *Thinking and Speech* (1934; translated as *Thought and Language* in 1962), he argued that thought and speech have different genetic roots. There is *pre-linguistic thought*, the practical intelligence of infants and of Köhler's chimpanzees; and there is *pre-intellectual speech*, the crying, babbling and first words of the infant, used socially and emotionally but not yet as a tool of thought. At around two years of age the two lines meet: the child, in the words Vygotsky quoted, "discovers that each thing has its name", asks constantly for names, and from then on thought becomes verbal and speech becomes rational.

Speech, for Vygotsky, begins as social communication with others. Children then turn it on themselves as *private speech* (Piaget's egocentric speech) to plan and guide their own actions, and it gradually "goes underground" to become **inner speech**, silent verbal thought that is abbreviated, condensed and dense with personal meaning. So private speech does not simply disappear: it is internalised. Vygotsky supported this with an ingenious experiment. When he introduced obstacles into children's activities (for example, taking away the coloured pencil a child needed for a drawing), the amount of egocentric speech nearly doubled, because the children used it to work out what to do. Later research, notably by Laura Berk, confirms that children use more private speech on harder tasks and that those who do so tend to perform better. Vygotsky summarised development in four stages of speech and thought: a *primitive* or natural stage of pre-intellectual speech and pre-verbal thought; a *naive psychology* stage in which the child uses grammatical forms such as "because" before grasping the logical relations they express; an *external sign* stage of egocentric speech and external aids such as counting on fingers; and an *ingrowth* stage in which these operations are internalised as inner speech and mental calculation.

Vygotsky also studied how true concepts develop. With his colleague Leonid Sakharov he gave children blocks of different colours, shapes, heights and sizes, each marked underneath with one of four nonsense words (*lag*, *bik*, *mur*, *cev*) that stood for combinations of height and size. Young children grouped the blocks into unorganised *heaps*; older children formed *complexes*, groupings based on shifting, concrete links, including *pseudoconcepts* that look like real concepts but rest on visible similarity; only in adolescence do *true concepts* appear, based on abstract, systematic features. Vygotsky linked this achievement to the new tasks that society sets adolescents, to their participation in social institutions such as school and work, and to the growth of self-reflection, the ability to step back and reflect on one's acting self. He distinguished *everyday* (spontaneous) concepts, picked up in daily life, from *scientific* concepts, learned systematically through instruction; the two meet and enrich each other in the zone of proximal development, the gap between what a learner can do alone and what she can do with guidance (Developmental Psychology book). Table 8.13 sets the two theorists side by side.

Table 8.13 Piaget and Vygotsky on language and thought.

| Issue | Piaget | Vygotsky |
|---|---|---|
| Which comes first? | Thought; language depends on and reflects it | Separate roots; after about age 2, thought depends on language |

| Issue | Piaget | Vygotsky |
|---|---|---|
| Egocentric (private) speech | Reflects egocentric thought; serves no function | A tool for self-guidance and self-regulation |
| What happens to it? | Fades away as thought becomes socialised (about age 7) | Is internalised as inner speech (about age 7) |
| Direction of development | From individual to social | From social to individual |
| Key book | *The Language and Thought of the Child*, 1923 | *Thinking and Speech*, 1934 (English 1962) |

EXAM TRAP

The standard contrast: **Vygotsky argued that thought depends on language; Piaget argued that language depends on thought.** Options that swap the names, or that say one studied early and the other later language, are wrong. Private speech, for Vygotsky, guides and regulates behaviour and is **internalised as inner speech**; it is not a regression to autistic thinking, mere imitation, a sign of disorder, or egocentrism that fades only with schooling. In Vygotsky's account of adolescent conceptual thinking, societal tasks, social institutions and self-reflection all matter; **gender differences** play no part. One recent paper asked whether a person's language and thought involve simultaneous, independent, shared or alternate processes, with "simultaneous" as the expected answer: in mature verbal thought, language and thought run together and interdependently. "Independent" describes only their separate early roots, and "shared" is the most defensible alternative if the key is disputed.

## Bilingualism

Most people in the world, and most people in India, speak more than one language. **Bilingualism** is the ability to use two languages; a person fluent in many languages is a *polyglot* or multilingual (a linguist, by contrast, studies language scientifically, whether or not she speaks many, and an orator is simply a skilled public speaker). In *simultaneous* bilingualism, a child learns two languages from birth, as two first languages; in *sequential* bilingualism, a second language is added after the first is established. Uriel Weinreich (1953) distinguished *compound* bilinguals, who learned both languages in the same context and attach two labels to a single set of meanings, from *coordinate* bilinguals, who learned them in separate contexts and have partly separate meaning systems (and *subordinate* bilinguals, who reach the second language through the first). Wallace Lambert distinguished *additive* bilingualism, in which a second language is added while the first remains valued and maintained, from *subtractive* bilingualism, in which the second language gradually replaces the first, as often happens to minority languages.

Early studies claimed that bilingualism harmed children's intelligence, but they compared poor immigrant children with middle-class monolinguals and tested them in their weaker language. When Elizabeth Peal and Wallace Lambert (1962) compared balanced French-English bilingual children in Montreal with monolinguals matched for social class, the bilinguals did better on both verbal and non-verbal tests. Since then Ellen Bialystok and others have reported that bilinguals show greater *metalinguistic awareness* and, in some studies, better executive control, presumably from constantly managing two languages; some studies

even report a later onset of dementia symptoms. These advantages are now debated, and several large studies find small or no executive benefits. The costs are better established and modest: bilingual children reach the main milestones of language development on the usual timetable, but bilinguals tend to have a smaller vocabulary in each language and to retrieve words slightly more slowly. *Code-switching*, moving between languages within a conversation ("Kal exam hai, so I'm going to sleep early"), follows grammatical rules and is a sign of competence, not confusion.

> **INDIAN CONTEXT**
>
> India is one of the most multilingual societies on earth, with 22 languages listed in the Eighth Schedule of the Constitution and hundreds of mother tongues. Multilingualism is the ordinary condition: a Mysuru student may speak Kannada at home, English in class, Hindi with friends from the north and some Tamil or Telugu with neighbours. The *three-language formula*, adopted in the National Policy on Education of 1968, asks schools to teach the regional language, Hindi and English (with another modern Indian language in Hindi-speaking states). For a clinician, this means assessing and treating clients in their dominant language whenever possible, since emotional memories are often encoded in the language in which they were lived.

### Can Animals Learn Language?

Animals certainly communicate. The Austrian zoologist Karl von Frisch, who shared a Nobel Prize in 1973, showed that a honeybee's "waggle dance" tells its hivemates the direction and distance of a food source: a genuine case of displacement, but the system is fixed, cannot express anything new, and is not learned. Vervet monkeys give different alarm calls for leopards, eagles and snakes, and other vervets respond appropriately (running up trees, looking up, or standing up and scanning the ground) even to recordings of the calls (Robert Seyfarth, Dorothy Cheney and Peter Marler, 1980): semanticity, but no combination of calls into new messages.

Could an ape do better if taught? Early attempts to teach chimpanzees to speak failed; Vicki, raised as a child by Keith and Catherine Hayes in the late 1940s, learned to say only about four words (mama, papa, cup and up), because the ape vocal tract cannot produce human speech. So researchers turned to other media. From 1966 **Beatrix Gardner** and **Allen Gardner** taught the chimpanzee Washoe American Sign Language; she learned about 130 signs and combined them, famously signing "water bird" on seeing a swan. David Premack taught a chimpanzee called Sarah to use plastic tokens on a magnetic board, and Duane Rumbaugh taught Lana to press symbols (lexigrams) on a computer keyboard.

Then came the sceptics. **Herbert Terrace** raised a chimpanzee with the pointed name Nim Chimpsky and taught him sign language. When Terrace and his colleagues analysed hours of videotape (1979), they found that most of Nim's signs were prompted by his teachers or were partial imitations of signs the teacher had just made; his utterances did not grow longer in a rule-governed way, and his longest, sixteen signs long, was "Give orange me give eat orange me eat orange give me eat orange give me you". Terrace's answer to the question in the title of his paper, "Can an ape create a sentence?", was no. The most impressive evidence came later from **Sue Savage-Rumbaugh**'s bonobo Kanzi, who learned lexigrams spontaneously as an infant by watching researchers try to train his adoptive mother, and who could carry out novel

spoken requests ("put the soap on the apple") at a level roughly comparable to a child of two to two and a half. The current verdict is that apes can learn symbols with meaning (semanticity, arbitrariness, some displacement), but show little productive grammar and no clear recursion. On Hockett's list, generativity remains the human signature.

## Speech Perception

Listen to someone speaking a language you do not know and you will notice something important: you cannot tell where one word ends and the next begins. The speech signal is continuous; the gaps we "hear" between words in our own language are mostly supplied by our knowledge (the *segmentation problem*). Worse, each phoneme is pronounced differently depending on the sounds around it (*coarticulation*), so there is no fixed acoustic pattern for, say, /d/. How do we perceive speech at all?

Part of the answer is **categorical perception**: the tendency to perceive a continuously varying stimulus as falling into discrete categories, with sharp boundaries between them and poor discrimination within them. **Alvin Liberman** (1917 to 2000) and his colleagues at the Haskins Laboratories (1957) created synthetic syllables that changed in equal acoustic steps from one consonant to another. Consider the difference between /ba/ and /pa/. Both are made by opening the lips; they differ in *voice onset time* (VOT), the delay between the release of the lips and the start of vocal-cord vibration, which is short for /ba/ and longer for /pa/. When VOT is varied smoothly, listeners do not hear a gradual change. They hear /ba/, /ba/, /ba/, and then, at a boundary of about 25 milliseconds in English, an abrupt switch to /pa/ (Figure 8.6). Two sounds 20 milliseconds apart are easy to tell apart if they straddle the boundary and almost impossible if they fall on the same side. Peter Eimas and his colleagues (1971) found the same boundary in infants only one and four months old, which at first suggested a special inborn mechanism for speech; but chinchillas show it too, so it may reflect general properties of mammalian hearing that languages have exploited.

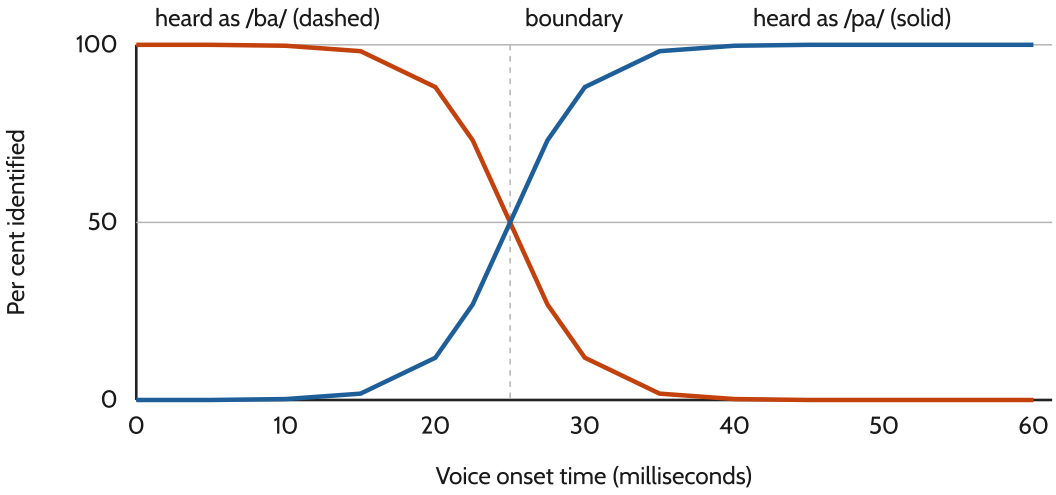


Figure 8.6 Categorical perception of speech. As voice onset time increases in equal steps, listeners' labels do not change gradually: they switch from /ba/ to /pa/ abruptly near 25 milliseconds. Idealised curves of the kind first reported by Liberman and colleagues.

Speech perception also uses the eyes. In the **McGurk effect**, described by **Harry McGurk** and **John MacDonald** (1976), the sound of a person saying "ba" is dubbed onto a video of the same person's lips saying "ga". Viewers hear "da", a compromise between what they hear and what they see; when they close their eyes, they hear "ba" again. The illusion persists even when you know exactly how it works, which shows how automatically sight and sound are combined. Knowledge and context fill in gaps too. In the *phonemic restoration effect*, Richard Warren (1970) replaced the first /s/ in "legislatures" in a recorded sentence with a cough. Listeners "heard" the missing /s/ and could not say where the cough had occurred. Remarkably, a word heard later in the sentence can determine which sound is restored: "the *eel was on the axle" is heard as "wheel", "the *eel was on the orange" as "peel". Speech perception is as much top-down as bottom-up, a point you met for vision in Chapter 4.

### Reading and Aphasia: Pointers ●●

Reading is a recent cultural invention, a few thousand years old, built on top of spoken language. Before children can learn to read in an alphabetic script, they must already have spoken vocabulary and grammar, and above all *phonological awareness*: the knowledge that spoken words are made of smaller sounds (syllables, rhymes and phonemes) that can be manipulated. Phonological awareness in the preschool years is one of the best predictors of later reading. So the cognitive skill that learning to read builds on most directly is spoken language, not theory of mind (the understanding that other people have beliefs and desires of their own), imagination, or the general capacity for mental representation. In the alphasyllabic scripts of India, such as Kannada and Devanagari, the basic written unit is the *akshara* (roughly a consonant with its vowel), and awareness of syllables and aksharas plays the same role. Skilled reading, according to Max Coltheart's *dual-route model*, uses two routes: a lexical route that recognises familiar words as wholes (essential for irregular words such as "yacht" and "colonel") and a non-lexical route that converts letters to sounds by rule (essential for new words and nonwords such as "blint"). Damage to the lexical route produces *surface dyslexia* (irregular words are "regularised", so "pint" is read to rhyme with "mint"); damage to the non-lexical route produces *phonological dyslexia* (familiar words can be read, but nonwords cannot). Developmental dyslexia and specific learning disorders are covered in the Psychiatric Disorders book.

Finally, language can be lost. An acquired disturbance of language caused by brain damage, typically to the left hemisphere after a stroke, is called **aphasia**. It is a disorder of language specifically, not of intelligence, memory, abstract thinking or the muscles of speech (weakness of the speech muscles is *dysarthria*). In 1861 the French physician **Paul Broca** (1824 to 1880) described a patient, Louis Victor Leborgne, nicknamed "Tan" because it was almost the only syllable he could say; at autopsy Broca found damage in the left inferior frontal gyrus, now called Broca's area. In 1874 the German neurologist **Carl Wernicke** (1848 to 1905) described a different syndrome after damage to the posterior part of the left superior temporal gyrus, now Wernicke's area. The two classic aphasias, and the brain's language network, are treated fully in the Physiological Psychology book; the essentials are in the box below.

CLINICAL LINK

**Broca's aphasia** (expressive, non-fluent): effortful, halting, agrammatic speech much like telegraphic speech, with relatively preserved comprehension and impaired repetition; patients are usually aware of their difficulty and frustrated by it. Broca's area is involved in **speech production**. **Wernicke's aphasia** (receptive, fluent): fluent, well-articulated speech with normal prosody but empty of meaning, full of word substitutions (paraphasias) and invented words (neologisms), with poor comprehension and repetition; patients are often unaware of their errors. **Conduction aphasia**: fluent speech and good comprehension but strikingly poor repetition, after damage to the arcuate fasciculus linking the two areas. **Global aphasia** affects everything. Damage to the corresponding **right-hemisphere** regions spares grammar but disturbs emotional prosody, humour, metaphor and the flow of conversation. Three neighbours are not aphasias of speech: damage to the **left angular gyrus** disturbs reading and writing (alexia with agraphia); **pure alexia** (alexia without agraphia), in which the patient can write but cannot read, follows left occipital damage; and inability to recognise faces is **prosopagnosia**, a disorder of visual recognition.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Hockett's design feature that lets us talk about things not present in time or space is _____.
2. The smallest unit of language that carries meaning is the _____; the smallest unit of sound that distinguishes words is the _____.
3. "Visiting relatives can be boring" has one surface structure but two _____ structures.
4. Chomsky's argument that children's linguistic input is too impoverished to explain their grammar is called the _____ of the stimulus.
5. Bruner's answer to the LAD, the structured help adults give, is the _____.
6. Lenneberg's 1967 claim that language must be acquired between about the age of two and puberty concerns a _____ period.
7. A child who says "two foots" after previously saying "feet" is showing _____.
8. The weak form of the Sapir-Whorf hypothesis, that language influences thought, is called linguistic _____.
9. For Vygotsky, private speech is internalised as _____ speech.
10. Hearing "da" when the sound "ba" is dubbed onto lips saying "ga" is the _____ effect.

**Answers:** (1) displacement, (2) morpheme; phoneme, (3) deep, (4) poverty, (5) language acquisition support system (LASS), (6) critical (or sensitive), (7) overregularisation, (8) relativity, (9) inner, (10) McGurk.

## The case of Sultan and the two sticks, revisited

Go back to Tenerife, to the moment before Sultan jumped up. You can now describe what happened in several vocabularies. In Newell and Simon's terms, the banana problem was well defined (a clear initial state, a clear goal), but Sultan's problem space lacked the one oper-

ator that would solve it: joining two sticks into one. Hill climbing could never have found it. Pushing one stick out with the other brought a tip closer to the banana without bringing the banana any closer to Sultan, a local maximum from which no small step led further. In Gestalt terms he solved the problem by insight: the two short sticks were suddenly seen as parts of one long one, and the solution appeared all at once, was performed smoothly and was kept, exactly the marks of insight that Köhler listed. His idle play with the sticks even resembles Wallas's incubation followed by illumination, and Metcalfe and Wiebe's warmth ratings suggest that, had he been able to tell us, he would have felt no nearer the answer a moment before he found it. Herbert Birch's work adds the caution: Sultan had handled sticks before, and insight grows out of experience rather than replacing it.

Was he thinking without words? Everything in this chapter says yes. Of the five elements of thought he plainly had images and concepts (a concept of "stick" flexible enough to include a stick made of two sticks), though not conventional symbols, propositions or language. The curare experiment showed that human thought does not need the muscles of speech, and Vygotsky placed Köhler's chimpanzees at the very start of his account of thought and language, as the clearest case of pre-linguistic thought: practical intelligence that has not yet met speech. What Sultan lacked is what the last three sections of the chapter describe, a generative symbol system that can refer to things not present. Washoe, Nim and Kanzi showed that apes can learn symbols that mean something, but not a grammar that produces endless new sentences. In children, Vygotsky argued, the two lines meet at about the age of two, and from then on thought becomes verbal: we can reason about what is absent, test hypotheses we have never acted out, and tell someone else how we solved a problem. Sultan could do the first thing this chapter describes. He could not do the last: he had no way to tell another chimpanzee how he had done it.

## Chapter summary

**8.1 What Is Thinking?** Thinking is the mental manipulation of internal representations to form new ones, the part of cognition that transforms what perception and memory supply. The Würzburg school of Oswald Külpe reported imageless thought, against Titchener's view that every thought contains images; Watson (1913) reduced thinking to subvocal speech; and the cognitive revolution (Chomsky, Newell and Simon, Bruner, and Neisser's *Cognitive Psychology* of 1967) brought thought back with objective measures. The elements of thought are images, concepts, symbols and signs, propositions and language. Shepard and Metzler (1971) found that the time to compare rotated objects rises linearly with the angle (mental rotation), and Kosslyn's island-scanning studies suggested depictive images; Pylyshyn's propositional theory (1973) explained such results by tacit knowledge, but imagery's use of visual cortex and the imaginal neglect of Bisiach and Luzzatti's patients favour depictive images. Paivio's dual-coding theory (1971) explains why pictures and concrete words are remembered best. Thinking may be directed or undirected, autistic or dereistic (Bleuler, 1911), convergent or divergent (Guilford), critical, creative, reflective (Dewey, 1910), and productive or reproductive (Wertheimer, 1945).

**8.2 Concepts.** A concept is a mental representation of a category; concepts give cognitive economy and support inference. Concept formation is described as observation, discrimination, generalisation and abstraction, often followed by naming; Hull (1920) treated it as gradual associative abstraction. The classical view defines a concept by necessary and sufficient features and

fits artificial concepts, which may be conjunctive (this and that), disjunctive (this or that) or relational. Bruner, Goodnow and Austin (1956) showed that people attain artificial concepts by testing hypotheses, with conservative focusing the most efficient strategy, alongside focus gambling and simultaneous and successive scanning. Natural concepts, as Eleanor Rosch showed, have typicality gradients, fuzzy boundaries and Wittgenstein's family resemblance; prototype theory represents them by an abstract, averaged best example, and the basic level (chair, dog) is learned and named first. Exemplar theory (Medin and Schaffer, 1978) stores many instances instead. Concepts are linked in semantic networks (Collins and Quillian, 1969; spreading activation, Collins and Loftus, 1975) or, in connectionism (Rumelhart and McClelland, 1986), distributed across many interconnected units.

**8.3 Problem Solving.** A problem is a gap between an initial state and a goal state; problems may be well defined or ill defined. Newell and Simon (General Problem Solver; *Human Problem Solving*, 1972) described solving as search through a problem space of states and operators. An algorithm guarantees a solution; a heuristic is an efficient rule of thumb that does not. The main heuristics are means-end analysis (reducing differences by subgoals), working backward, hill climbing (which fails at a local maximum, as in the hobbits and orcs problem) and analogy, which people use readily when told to but rarely notice spontaneously (Gick and Holyoak, 1980); trial and error is the most primitive strategy (Thorndike, 1898). Gestalt psychologists stressed insight, a sudden restructuring of the problem (Köhler's apes, Maier's nine-dot and two-string problems), and Metcalfe and Wiebe's warmth ratings (1987) confirmed that insight solutions arrive suddenly; incubation helps, probably by letting a misleading approach be forgotten. The obstacles are functional fixedness (Duncker's candle problem, 1945), mental set or Einstellung (Luchins's water jars, 1942), confirmation bias and irrelevant information. Stage models run from Dewey and Pólya to Sternberg's problem-solving cycle, and experts excel because they store meaningful chunks and represent problems by deep principles (Chase and Simon, 1973; Chi, 1981).

**8.4 Reasoning.** Deductive reasoning moves from general premises to a conclusion that must be true if the premises are; an argument is valid by its form and sound if its premises are also true. Inductive reasoning moves from specific observations to probable generalisations and is the logic of theory development and qualitative research, while abductive reasoning (Peirce) is inference to the best explanation; Popper's hypothetico-deductive method combines them. In syllogisms people show belief bias (Evans, Barston and Pollard, 1983) and the atmosphere effect (Woodworth and Sells, 1935). Of the four conditional arguments, modus ponens and modus tollens are valid, while affirming the consequent and denying the antecedent are fallacies. In Wason's selection task (1966) few people choose the correct cards (E and 7), but performance improves with familiar permission rules (Griggs and Cox, 1982), explained by pragmatic reasoning schemas (Cheng and Holyoak, 1985) or cheater detection (Cosmides, 1989). Johnson-Laird's mental models theory (1983) holds that we reason by building and checking models of possible situations. Wason's 2-4-6 task (1960) revealed confirmation bias. In Piaget's theory, hypothetico-deductive, counterfactual and metacognitive reasoning emerge with formal operations; dialectical thinking is postformal.

**8.5 Judgement and Decision Making.** Normative models (expected value; expected utility, von Neumann and Morgenstern, 1944) say how a rational agent should decide; descriptive models say how people actually decide. Simon's bounded rationality holds that information, mental capacity and time limit choice, so people satisfice rather than optimise; Tversky's elimination by aspects (1972) is a non-compensatory shortcut, and Gigerenzer's fast and frugal heuristics,

such as the recognition heuristic, can be ecologically rational. Tversky and Kahneman (1974) described the availability heuristic (the K-words), the representativeness heuristic, which produces base-rate neglect, the conjunction fallacy (Linda, 1983), the gambler's fallacy and insensitivity to sample size, and anchoring and adjustment. The framing effect (the Asian disease problem, 1981) shows risk aversion for gains and risk seeking for losses, explained by prospect theory (1979): outcomes are judged from a reference point, sensitivity diminishes, and losses loom about twice as large as gains (loss aversion), which also underlies the endowment effect, status quo bias and the sunk-cost fallacy. Overconfidence, the planning fallacy and Fischhoff's hindsight bias (1975) distort judgements of our own knowledge. Dual-process theory contrasts fast, intuitive System 1 with slow, deliberate System 2, and Beck's cognitive distortions are the clinical cousins of these biases.

**8.6 Creativity.** Creativity is the production of work that is both novel and appropriate, studied through Rhodes's four Ps (person, process, press, product) and at levels from mini-c to Big-C. Guilford's 1950 address launched the field; he identified creativity with divergent production and scored it by fluency, flexibility, originality and elaboration. Wallas (1926) described four stages: preparation, incubation, illumination and verification, with intimation sometimes added. Creativity is measured by the Torrance Tests of Creative Thinking (1966), Mednick's Remote Associates Test (1962), which rests on his associative theory, and Amabile's consensual assessment technique. De Bono's lateral thinking (1967) and Osborn's brainstorming (rules: defer judgement, go for quantity, welcome wild ideas, combine) aim to enhance it, although nominal groups outperform interacting ones because of production blocking. The threshold hypothesis holds that intelligence is necessary but not sufficient for creativity: the two are correlated below an IQ of about 120 but hardly at all above it, and overall the correlation is weak (about 0.17 in Kim's 2005 meta-analysis); Getzels and Jackson (1962) found high-creative and high-IQ adolescents equally successful at school. Sternberg and Lubart's investment theory (buy low, sell high; six converging resources), Amabile's componential model with its intrinsic motivation principle, and Csikszentmihalyi's systems model (individual, domain and field) are the main modern theories. Creative people are above all open to experience, independent, tolerant of ambiguity and willing to take risks.

**8.7 The Structure of Language.** A language is a system of arbitrary symbols combined by rules to produce unlimited messages. Hockett (1960) listed design features, of which semanticity, arbitrariness, displacement, generativity (productivity) and duality of patterning are the most examined; recursion may be uniquely human. The units of language form a hierarchy: phonemes (smallest units of sound that distinguish words), morphemes (smallest units of meaning, free or bound, inflectional or derivational), words, syntax (rules of sentence structure), semantics (meaning), pragmatics (use in social context, including Grice's maxims) and prosody (rhythm, stress and intonation, with emotional prosody served by right-hemisphere regions). Chomsky's *Syntactic Structures* (1957) proposed a transformational generative grammar of phrase-structure and transformational rules, and *Aspects of the Theory of Syntax* (1965) distinguished surface structure from deep structure (ambiguous sentences have one surface structure and two deep structures) and competence (tacit knowledge) from performance (actual use).

**8.8 Acquiring Language.** Skinner's *Verbal Behavior* (1957) explained language by reinforcement of verbal operants (mand, tact, echoic, intraverbal, textual, transcription, autoclitic). Chomsky's 1959 review replied that children produce novel sentences, make rule-based errors and are rarely corrected for grammar, and proposed an innate language acquisition device containing

universal grammar, defended by the poverty of the stimulus; Pinker popularised nativism, while statistical learning (Saffran, Aslin and Newport, 1996) shows the input is richer than Chomsky supposed. Interactionists stress social support: Bruner's language acquisition support system (formats and scaffolding) and child-directed speech, with its higher, wider pitch, slower rate, more pauses and shorter utterances; Slobin's language-making capacity is a set of language-tuned cognitive and perceptual operating principles. Lenneberg (1967) proposed a critical period from about two to puberty, supported, imperfectly, by Genie and by Johnson and Newport's immigrants (1989). The milestones are cooing (about 2 months), babbling (about 6 months), holophrases (about 12 months), the vocabulary spurt with fast mapping (about 18 months), telegraphic speech (18 to 24 months), overregularisation (2 to 3 years), and complex sentences and metalinguistic awareness after that, with overextension and underextension along the way. Language is most consistently compromised in autism spectrum disorder.

**8.9 Language, Thought and Communication.** The Sapir-Whorf hypothesis has a strong version, linguistic determinism, which has failed (Hopi time, Berlin and Kay's colour universals of 1969, Heider's Dani study of 1972), and a weak version, linguistic relativity, which is supported (Russian blues, absolute spatial languages, kinship terms); Pinker argues that we think in a language-independent mentalese. The curare experiment (1947) refuted Watson's motor theory. Piaget held that thought comes first and egocentric speech simply fades; Vygotsky held that thought and speech have separate roots that merge at about two, after which private speech guides behaviour and is internalised as inner speech; he also traced concepts from heaps through complexes to true concepts in adolescence. Bilingualism may be simultaneous or sequential, compound, coordinate or subordinate (Weinreich), additive or subtractive (Lambert); balanced bilinguals can show advantages (Peal and Lambert, 1962), though executive benefits are debated. Bees and vervet monkeys communicate, and apes (Washoe, Sarah, Lana, Nim, Kanzi) learn meaningful symbols but little productive grammar. Speech perception is categorical (Liberman, 1957; voice onset time), audiovisual (the McGurk effect, 1976) and partly top-down (phonemic restoration). Reading builds on spoken language and phonological awareness (dual-route model), and aphasia is an acquired language disorder: Broca's (production) and Wernicke's (comprehension), treated fully in the Physiological Psychology book.

## Who's who

Table 8.14 The people in this chapter.

| Name | When | Examined for |
|---|---|---|
| Teresa Amabile | born 1950 | Componential model of creativity (1983); intrinsic motivation principle; consensual assessment technique |
| Jean Berko | 1958 | The "wug test": preschoolers apply plural and past-tense rules to invented words |
| Brent Berlin and Paul Kay | 1969 | *Basic Color Terms*: colour words appear in a near-universal order, evidence against strong Whorf |
| Eugen Bleuler | 1857 to 1939 | Autistic (dereistic) thinking, 1911; coined "schizophrenia" |
| Paul Broca | 1824 to 1880 | Patient "Tan" (1861); Broca's area; non-fluent (expressive) aphasia; speech production |
| Roger Brown | 1925 to 1997 | Telegraphic speech; *A First Language* (1973); mean length of utterance |

| Name | When | Examined for |
|---|---|---|
| Jerome Seymour Bruner | 1915 to 2016 | *A Study of Thinking* (1956, with Goodnow and Austin): artificial concepts by hypothesis testing, focusing and scanning; LASS (1983); scaffolding (1976) |
| Noam Chomsky | born 1928 | *Syntactic Structures* (1957); review of Skinner (1959); *Aspects* (1965): deep and surface structure, competence and performance; LAD, universal grammar, poverty of the stimulus |
| Allan Collins and Ross Quillian | 1969 | Hierarchical semantic network model |
| Leda Cosmides | born 1957 | Cheater-detection account of the social-contract selection task (1989) |
| Mihaly Csikszentmihalyi | 1934 to 2021 | Systems model of creativity (1988): individual, domain and field |
| Susan Curtiss | 1977 | Studied Genie; evidence for a sensitive period for grammar |
| Edward de Bono | 1933 to 2021 | Lateral thinking (1967); *Six Thinking Hats* (1985) |
| John Dewey | 1859 to 1952 | Reflective thinking; *How We Think* (1910) and its five steps of problem solving |
| Karl Duncker | 1903 to 1940 | Candle problem and functional fixedness; radiation problem; *On Problem Solving* (English 1945) |
| Jonathan Evans | 1983 | Belief bias in syllogisms (with Barston and Pollard); matching bias in the selection task |
| Baruch Fischhoff | born 1946 | Hindsight bias, "creeping determinism" (1975, with Beyth) |
| Beatrix and Allen Gardner | from 1966 | Taught the chimpanzee Washoe American Sign Language |
| Mary Gick and Keith Holyoak | 1980 | Analogical problem solving: the fortress story and the radiation problem |
| Gerd Gigerenzer | born 1947 | Fast and frugal heuristics; recognition heuristic (2002, with Goldstein); ecological rationality |
| Joy Paul Guilford | 1897 to 1987 | Convergent and divergent production; 1950 APA address; fluency, flexibility, originality, elaboration |
| Charles Francis Hockett | 1916 to 2000 | Design features of language (1960) |
| Clark Leonard Hull | 1884 to 1952 | First laboratory study of concept learning (1920): gradual associative abstraction |
| Philip Nicholas Johnson-Laird | born 1936 | Mental models theory of reasoning (1983) |
| Daniel Kahneman | 1934 to 2024 | Heuristics and biases (1974), prospect theory (1979) and framing (1981), all with Tversky; Nobel 2002; *Thinking, Fast and Slow* (2011) |
| Wolfgang Köhler | 1887 to 1967 | Insight in chimpanzees (Sultan) on Tenerife, 1913 to 1920; *The Mentality of Apes* (1917; English 1925) |
| Stephen Michael Kosslyn | born 1948 | Image scanning (the island map, 1978); images as depictive representations |
| Oswald Külpe | 1862 to 1915 | Würzburg school; imageless thought |

| Name | When | Examined for |
|---|---|---|
| Eric Heinz Lenneberg | 1921 to 1975 | Critical period for language; *Biological Foundations of Language* (1967) |
| Alvin Liberman | 1917 to 2000 | Categorical perception of speech sounds (Haskins Laboratories, 1957) |
| Abraham Luchins | 1914 to 2005 | Water-jar problems; mental set (Einstellung), 1942 |
| Norman Maier | 1900 to 1977 | Nine-dot problem (1930) and two-string problem (1931) |
| Harry McGurk and John MacDonald | 1976 | McGurk effect: seen lip movements change the speech sound heard |
| Sarnoff Mednick | 1928 to 2015 | Associative theory of creativity; Remote Associates Test (1962) |
| Janet Metcalfe | 1987 | Warmth ratings (with Wiebe): insight solutions arrive suddenly |
| Ulric Neisser | 1928 to 2012 | *Cognitive Psychology* (1967), which named the field |
| Katherine Nelson | 1973 | Referential versus expressive styles of early word learning |
| Allen Newell | 1927 to 1992 | With Simon: Logic Theorist (1956), General Problem Solver, *Human Problem Solving* (1972); problem space; means-end analysis |
| Alex Faickney Osborn | 1888 to 1966 | Brainstorming; *Applied Imagination* (1953) |
| Allan Paivio | 1925 to 2016 | Dual-coding theory (1971) |
| Charles Sanders Peirce | 1839 to 1914 | Abductive reasoning, inference to the best explanation |
| Jean Piaget | 1896 to 1980 | Thought precedes language; egocentric speech (1923); formal operations; keyed as a pioneer of cognitive psychology |
| Steven Pinker | born 1954 | *The Language Instinct* (1994); "mentalese"; critic of linguistic determinism |
| Karl Raimund Popper | 1902 to 1994 | Hypothetico-deductive method; falsification |
| Zenon Walter Pylyshyn | 1937 to 2022 | Propositional theory and the imagery debate (1973); tacit knowledge |
| Eleanor Rosch | born 1938 | Natural concepts, prototype theory, family resemblance (with Mervis, 1975), basic level (1976); Dani colour study (as Heider, 1972) |
| Edward Sapir | 1884 to 1939 | Whorf's teacher; the Sapir-Whorf hypothesis |
| Ferdinand de Saussure | 1857 to 1913 | Arbitrariness of the linguistic sign; *langue* versus *parole* (1916) |
| Sue Savage-Rumbaugh | 1980s | The bonobo Kanzi: spontaneous lexigram learning and comprehension of spoken requests |
| Roger Newland Shepard | 1929 to 2022 | Mental rotation (1971, with Metzler) |
| Herbert Alexander Simon | 1916 to 2001 | Bounded rationality and satisficing (1950s); Nobel 1978; General Problem Solver with Newell; chess chunks (1973, with Chase) |

| Name | When | Examined for |
|---|---|---|
| Burrhus Frederic Skinner | 1904 to 1990 | *Verbal Behavior* (1957): mand, tact, echoic, intraverbal, autoclitic |
| Dan Isaac Slobin | born 1939 | Language-making capacity and operating principles (1985) |
| Robert Jeffrey Sternberg | born 1949 | Problem-solving cycle; investment theory of creativity (with Lubart, 1991 and 1995) |
| Herbert Terrace | 1979 | Nim Chimpsky; concluded that apes do not create sentences |
| Richard Thaler | born 1945 | Named the endowment effect (1980); coffee-mug study (1990, with Kahneman and Knetsch) |
| Edward Lee Thorndike | 1874 to 1949 | Trial-and-error learning in puzzle-box cats (1898) |
| Edward Bradford Titchener | 1867 to 1927 | Structuralism; held that every thought contains images, against Würzburg |
| Ellis Paul Torrance | 1915 to 2003 | Torrance Tests of Creative Thinking (1966) |
| Amos Tversky | 1937 to 1996 | Heuristics and biases (1974); elimination by aspects (1972); prospect theory (1979); Asian disease (1981); Linda (1983) |
| Lev Semyonovich Vygotsky | 1896 to 1934 | *Thinking and Speech* (1934): separate roots of thought and speech; private speech becomes inner speech; heaps, complexes, true concepts; zone of proximal development |
| Graham Wallas | 1858 to 1932 | *The Art of Thought* (1926): preparation, incubation, illumination, verification |
| Peter Cathcart Wason | 1924 to 2003 | 2-4-6 task (1960), selection task (1966), confirmation bias |
| John Broadus Watson | 1878 to 1958 | Behaviourism (1913); thinking as subvocal speech |
| Carl Wernicke | 1848 to 1905 | Wernicke's area (1874); fluent (receptive) aphasia; speech comprehension |
| Max Wertheimer | 1880 to 1943 | Productive versus reproductive thinking; *Productive Thinking* (1945) |
| Benjamin Lee Whorf | 1897 to 1941 | Linguistic relativity; the Hopi; *Language, Thought, and Reality* (1956) |
| Ludwig Wittgenstein | 1889 to 1951 | Family resemblance; *Philosophical Investigations* (1953) |
| Robert Woodworth and Saul Sells | 1935 | Atmosphere effect in syllogistic reasoning |

## Key terms

thinking, imageless thought, mental imagery, mental rotation, propositional theory, dual-coding theory, autistic thinking, convergent thinking, divergent thinking, critical thinking, reflective thinking, concept, concept formation, classical view, artificial concept, conjunctive concept, disjunctive concept, relational concept, conservative focusing, family resemblance, natural concept, prototype theory, prototype, basic level, exemplar theory, semantic network, connectionism,

problem space, algorithm, heuristic, means-end analysis, working backward, hill climbing, insight, incubation, functional fixedness, mental set, deductive reasoning, inductive reasoning, abductive reasoning, syllogism, belief bias, atmosphere effect, modus ponens, modus tollens, affirming the consequent, denying the antecedent, mental model, confirmation bias, dialectical thinking, bounded rationality, satisficing, availability heuristic, representativeness heuristic, base-rate neglect, conjunction fallacy, gambler's fallacy, anchoring and adjustment, framing effect, prospect theory, loss aversion, endowment effect, sunk-cost fallacy, hindsight bias, dual-process theory, creativity, fluency, flexibility, originality, elaboration, lateral thinking, threshold hypothesis, investment theory, language, displacement, phoneme, morpheme, syntax, semantics, pragmatics, prosody, surface structure, deep structure, competence, language acquisition device, universal grammar, poverty of the stimulus, critical period, holophrase, overextension, telegraphic speech, overregularisation, linguistic relativity, inner speech, categorical perception, McGurk effect, aphasia

## From the past papers

1. Who is a pioneer contributor to Cognitive Psychology? (Amity 2023)

   (a) Jean Piaget (b) Kohler (c) Chomsky (d) Kohlberg

   **Answer: (a) Jean Piaget** (the keyed answer). Piaget's studies of how children's thinking develops, begun in the 1920s, were among the first systematic accounts of mental structures and processes, decades before the cognitive revolution. Chomsky is the strongest alternative, since his attack on behaviourism helped launch that revolution, so the item is debatable; Köhler was a Gestalt forerunner whose insight research predates cognitive psychology; Kohlberg is examined for moral development and is never the answer here.

2. According to the dual-coding theory of memory, which type of information is more easily remembered? (RML 2023)

   (a) Verbal information (b) Visual information (c) Auditory information (d) Semantic information

   **Answer: (b) Visual information.** In Paivio's theory (1971), pictures and concrete, imageable material are coded both in the imagery system and in the verbal system, giving two routes to retrieval (the picture superiority and concreteness effects). Verbal and auditory material, and abstract meaning, usually receive only the verbal code. "Semantic" tempts students who confuse dual coding with Craik and Lockhart's levels of processing.

3. The researcher who formulated the theory which explains how we form artificial concepts rather than natural concepts by developing a hypothesis is: (IHBAS 2020)

   (a) Clark Hull (b) Eleanor Rosch (c) Jerome Bruner (d) John Watson

   **Answer: (c) Jerome Bruner.** With Goodnow and Austin (1956), Bruner showed that people learning artificial, rule-defined concepts form and test hypotheses, using strategies such as conservative focusing. Rosch is the name for natural concepts and prototypes; Hull (1920) treated concept learning as gradual associative abstraction of a common element, not hypothesis testing; Watson did no work on concepts.

4. Which model proposes that memory is organized in a network of interconnected nodes? (IHBAS 2024)

   (a) Associative model (b) Connectionist model (c) Network model (d) Linkage model

   **Answer: (b) Connectionist model** (the keyed answer). Connectionist, or parallel distributed processing, models (Rumelhart and McClelland, 1986) represent knowledge as patterns of activation across a network of interconnected units whose connection weights store what has been learned. "Network model" is the strongest distractor, because Collins and Quillian's semantic network is also a network of nodes, but follow the key. "Associative model" is the older, general idea of linked ideas, and "linkage model" is not a standard term.

5. A rule-of-thumb that allows for an efficient search for the selection of a problem is termed as (RML 2024)

   (a) Algorithm (b) Education (c) Learning (d) Heuristic

   **Answer: (d) Heuristic.** ("Selection" in the stem is a slip for "solution".) A heuristic is an efficient shortcut that usually works but does not guarantee success. An algorithm is the opposite case, a step-by-step procedure that guarantees a solution but may be slow; education and learning are not problem-solving strategies at all.

6. The tendency to think of the use of objects only in terms of their habitual use is known as: (RML 2024)

   (a) Mental barrier (b) Functional fixedness (c) Einstellung (d) Trial and error

   **Answer: (b) Functional fixedness**, demonstrated by Duncker's candle problem (1945), in which the box is seen only as a container. Einstellung (mental set) is the related fixation on a previously successful *strategy* (Luchins's water jars), not on an object's use; "mental barrier" is not a technical term; trial and error is a problem-solving strategy, not an obstacle.

7. Which is the most basic and common obstacle to problem solving? (Amity 2023)

   (a) Confirmation bias (b) Mental set (c) Functional fixedness (d) Fixation

   **Answer: (b) Mental set**, the tendency to keep using an approach that worked before. Functional fixedness is usually treated as a special case of mental set, applied to objects, so it is narrower. Fixation is the Gestaltists' umbrella word for being stuck rather than a specific obstacle, and confirmation bias concerns how hypotheses are tested rather than how a problem is approached.

8. In which method does the researcher start with a theory and then test it with observations? (RML 2023)

   (a) Inductive reasoning (b) Deductive reasoning (c) Abductive reasoning (d) Hypothetico-deductive method

   **Answer: (b) Deductive reasoning**, which runs from the general (a theory) to the specific (a prediction tested against observations). Inductive reasoning runs the other way, from data to theory; abductive reasoning infers the best explanation of a surprising observation. The hypothetico-deductive method also begins with theory, but it is the whole cycle of hypothesis, deduced prediction and test, and the key chooses the plain label.

9. In deductive reasoning, the validity of conclusions depends on the: (RML 2023)

   (a) Breadth of observations (b) Depth of data collected (c) Soundness of the initial theory (d) Use of statistical analysis

   **Answer: (c) Soundness of the initial theory.** A deductive conclusion is only as trustworthy as the premises it starts from: a valid argument from a false premise can reach a false conclusion. (Strictly, logicians say validity depends on form and soundness on true premises, so the item uses "validity" loosely.) Breadth of observations and depth of data are concerns of inductive, data-first research, and statistics is a tool, not the basis of deduction.

10. Which of the following does NOT reflect inductive reasoning? (RML 2024)

    (a) Probabilistic conclusions (b) Heuristics (c) Decision making (d) Process involved in concept formation

    **Answer: (b) Heuristics** (the keyed answer). Inductive conclusions are probabilistic by nature; concept formation abstracts general categories from specific instances; decision making under uncertainty rests on inductive estimates of likelihood. A heuristic is a shortcut strategy rather than a form of inference, although it can produce inductive judgements, which is why the item is debatable.

11. Formal operational reasoning involves all the following EXCEPT: (RML 2022)

    (a) hypothetical reasoning (b) counterfactual reasoning (c) dialectical reasoning (d) metacognitive reasoning

    **Answer: (c) Dialectical reasoning.** Holding contradictory perspectives together and integrating them is postformal thought (Riegel, 1973; Basseches, 1984), beyond Piaget's final stage. Hypothetical (hypothetico-deductive), counterfactual and metacognitive reasoning ("operations on operations") are exactly what emerges in the formal operational stage from about 11 or 12.

12. Who is known for his work on the concept of bounded rationality, which suggests that decision-making is often constrained by factors such as limited information and cognitive resources? (RML 2023)

    (a) Herbert Simon (b) Daniel Kahneman (c) Amos Tversky (d) Howard Gardner

    **Answer: (a) Herbert Simon**, who also coined "satisficing" (1956) and won the Nobel Prize in Economics in 1978. Kahneman and Tversky tempt because they too showed that people depart from rational choice, but their contributions are heuristics and biases and prospect theory; Gardner is the theorist of multiple intelligences.

13. The representativeness heuristic can lead to errors when: (RML 2023)

    (a) Individuals ignore base rates and focus solely on similarity (b) Individuals rely on readily available information (c) Individuals fail to consider alternative explanations (d) Individuals adjust their judgements based on initial estimates

    **Answer: (a).** Representativeness judges probability by resemblance to the typical case, and its classic failure is base-rate neglect, as with Jack the "engineer" and the taxi-cab problem. Option (b) describes the availability heuristic, (c) confirmation bias and (d) anchoring and adjustment.

14. Confirmation bias is closely related to which heuristics? (RML 2023)

    (a) Availability heuristic (b) Representativeness heuristic (c) Anchoring and adjustment heuristic (d) Recognition heuristic

    **Answer: (b) Representativeness heuristic** (the keyed answer). The key reasons that both involve judging by fit to a preconceived model: evidence that "looks like" what we expect is accepted. No standard textbook ties confirmation bias to one heuristic (it is a bias in hypothesis testing, Wason, 1960), so treat this as a recall of the key. Availability concerns ease of retrieval and anchoring an initial value; the recognition heuristic (Gigerenzer) simply favours the option one recognises and is the least related.

15. The tendency to assume that we would have been better at predicting actual events than is really true is termed as: (RML 2024)

    (a) Hindsight effect (b) Just world hypothesis (c) Decisiveness (d) Confirmation

    **Answer: (a) Hindsight effect**, the "I knew it all along" bias shown by Fischhoff and Beyth (1975) with predictions about Nixon's 1972 visits. The just-world hypothesis (Melvin Lerner) is the belief that people get what they deserve; decisiveness is not a bias; confirmation bias concerns how we seek and read evidence for a belief, not how we misremember our own predictions.

16. The correct sequence of the four stages of creative thinking, as proposed by Graham Wallas, is: (Real PYQ 2024)

    (a) Preparation, Illumination, Incubation, Verification (b) Incubation, Preparation, Verification, Illumination (c) Preparation, Incubation, Illumination, Verification (d) Illumination, Preparation, Verification, Incubation

    **Answer: (c)**, from *The Art of Thought* (1926). The other options scramble the logic of the sequence: incubation (setting the problem aside) has to follow the effortful work of preparation and comes before the sudden illumination it prepares, and verification must come last because there has to be an idea to test.

17. Which is the smallest unit of meaning in speech perception? (RML 2024)

    (a) Syllables (b) Consonant (c) Phonemes (d) Morphemes

    **Answer: (d) Morphemes**, such as the two in "cat-s". Phonemes are the smallest units of *sound* that distinguish words, but they carry no meaning of their own; a consonant is just one kind of phoneme; a syllable is a unit of pronunciation built around a vowel.

18. The brain region most critically involved in processing the prosody or emotional tone of speech (speech intonation) is typically: (Real PYQ 2024)

    (a) Left inferior frontal gyrus (Broca's area) (b) Right hemisphere regions homotopic to language areas (c) Primary motor cortex (d) Occipital lobe

    **Answer: (b).** Emotional prosody depends chiefly on right-hemisphere regions that mirror the left hemisphere's language areas, and damage there produces aprosodia (Ross, 1981). Broca's area serves speech production and grammar rather than emotional tone; the primary motor cortex moves the speech muscles without interpreting anything; the occipital lobe is visual.

19. Adults typically talk to babies in a special way. This has been defined as 'Motherese', 'baby talk register' or 'adult-child speech'. Identify one characteristic of this type of adult-child speech. (RML 2021)

    (a) The mean length of utterance is shorter in adult-child speech than it is in adult-adult speech. (b) Adults speak much more quickly to babies than they do to other adults. (c) Adults pause more frequently when they talk to other adults than they do when they talk to babies. (d) Adult-child speech has a lower pitch and a narrower range of pitch.

    **Answer: (a).** Child-directed speech uses shorter, simpler utterances. Each other option reverses a real feature: adults speak more *slowly* to babies, pause *more* often with babies than with adults, and use a *higher* pitch with a *wider* range.

20. When infants and very young children live in environments where they have no exposure to or experience with language, their language development is severely limited. These limitations are not reversible. This supports which of the following terms? (RML 2021)

    (a) Sensitive periods of development (b) Semantics (c) Syntax (d) Nativism

    **Answer: (a) Sensitive periods of development.** Irreversible deficits after early deprivation (Genie; Lenneberg, 1967) are the defining evidence for a sensitive or critical period. Semantics and syntax are levels of language, not developmental principles; nativism is the broader claim that the capacity for language is innate, which such evidence fits but does not name directly.

21. Who emphasised the linguistic relativity hypothesis? (Amity 2023)

    (a) Noam Chomsky (b) Jean Piaget (c) Steven Pinker (d) B.L. Whorf

    **Answer: (d) B. L. Whorf**, Sapir's student, who argued from the Hopi language that language shapes habitual thought. Chomsky stresses the universal grammar common to all languages, the opposite emphasis; Piaget held that thought precedes and shapes language; Pinker is an outspoken critic of Whorf who argues that we think in "mentalese".

22. What is the primary difference between Piaget and Vygotsky's theories of language development? (RML 2021)

    (a) Piaget argued thought depends on language while Vygotsky argued language depends on thought (b) Vygotsky argued thought depends on language while Piaget argued language depends on thought (c) Vygotsky was concerned with early language development while Piaget was concerned with later language development. (d) None of the above

    **Answer: (b).** For Piaget, language expresses concepts already formed through action; for Vygotsky, thought and speech merge at about two, after which thought becomes verbal and depends on language. Option (a) swaps the names, and (c) is false, because both studied the same preschool years (Piaget's egocentric speech and Vygotsky's private speech are the same phenomenon, interpreted differently).

23. In Vygotsky's writings, conceptual thinking is linked to all the following EXCEPT: (RML 2022)

    (a) the distinction between the acting 'I' and the reflecting 'I' (b) the problems and developmental tasks that society creates for adolescents (c) participation in particular societal institutions (d) gender differences

    **Answer: (d) Gender differences.** Vygotsky linked the emergence of true concepts in adolescence to the tasks society sets adolescents, to their participation in social institutions such as school and work, and to self-reflection, the ability to step back from the acting self. Gender plays no part in his account.

24. A person's language and thought involve (JEMAS 2025)

    (a) Simultaneous processes. (b) Independent processes. (c) Shared processes. (d) Alternate process.

    **Answer: (a) Simultaneous processes** (the expected answer; the official key is not available). In mature verbal thought, language and thought run together and depend on each other, as Vygotsky described once their two lines merge at about two. "Independent" describes only their separate early roots (pre-linguistic thought and pre-intellectual speech); "shared" is the most defensible alternative if the key is disputed; "alternate", meaning that they take turns, describes no theory.

25. Aphasia is the disturbance in: (IHBAS 2020)

    (a) Language function due to organic disease (b) Intellectual functioning (c) Abstract thinking (d) Memory

    **Answer: (a).** Aphasia is an acquired disorder of language caused by brain damage, classically to Broca's area (non-fluent aphasia) or Wernicke's area (fluent aphasia) in the left hemisphere. Intelligence, abstract thinking and memory can all be intact; a general loss of intellectual functioning suggests intellectual disability or dementia, and a loss of memory suggests amnesia.

## High-yield sheet

*The facts this chapter's questions ask most, most-asked first. Last read before the exam.*

1 **Inductive reasoning** moves from specific observations to broader, probable generalisations: gather data, find patterns, build theory. It is the reasoning of theory development and of qualitative research. ASKED 4 TIMES

2 **Graham Wallas**, *The Art of Thought* (1926): preparation, incubation, illumination, verification. Incubation (setting the problem aside) is second; illumination (the sudden insight) is third; intimation is sometimes added before illumination. ASKED 5 TIMES

3 **Vygotsky versus Piaget:** Vygotsky held that thought depends on language (the two lines merge at about age two, after which they work together); Piaget held that language depends on thought. For Vygotsky, private speech guides behaviour and is internalised as inner speech. ASKED 4 TIMES

4 **Functional fixedness** is seeing an object only in terms of its usual function (Karl Duncker's candle problem, 1945: the box seen only as a container). It is about objects, mental set is about strategies, and it reflects top-down processing working too well. ASKED 4 TIMES

5 A **heuristic** is a rule of thumb that allows an efficient search and is usually right but does not guarantee a solution; an **algorithm** guarantees a solution but may be slow. ASKED 3 TIMES

6 **Deductive reasoning** starts from a theory or general premises and tests specific predictions against observations; a deductive conclusion is only as trustworthy as the soundness of its premises (the initial theory). ASKED TWICE

7 **Noam Chomsky's language acquisition device** is an innate, biologically based capacity containing universal grammar, argued from the poverty of the stimulus; not the phonological loop, the zone of proximal development or a schema. ASKED 3 TIMES

8 The **morpheme** is the smallest unit of meaning (cat + s is two); the **phoneme** is the smallest unit of sound that distinguishes words and has no meaning of its own. ASKED TWICE

9 **Cognitive psychology** is the modern approach closest to structuralism, because it studies consciousness, imagery and thought; Ulric Neisser named it (1967), and one paper keyed Jean Piaget as its pioneer contributor (not Kohlberg). ASKED TWICE

10 **Mental set** (Einstellung) is the habit of reusing a strategy that worked before, shown by Abraham Luchins's water jars (1942); keyed as the most basic and common obstacle to problem solving. ASKED TWICE

11 **Confirmation bias** is seeking, interpreting and remembering evidence that confirms one's existing belief (Wason's 2-4-6 task, 1960); one paper keyed it as closely related to the representativeness heuristic. ASKED TWICE

12 A **prototype** (Eleanor Rosch) is the idealised best example of a category, abstract and averaged; not a template, a feature or a geon. ASKED TWICE

13 **Telegraphic speech**, two-word utterances without function words ("more milk", "Mummy go"), appears at about 18 to 24 months, after the one-word holophrastic stage at about 12 months. ASKED TWICE

14 The **representativeness heuristic** (Kahneman and Tversky, 1972) misleads when people ignore base rates and judge by similarity alone (Jack the engineer, the taxi-cab problem, Linda's conjunction fallacy). ASKED ONCE

15 **Herbert Simon**: bounded rationality (choice limited by information, mental capacity and time) and satisficing (choosing the first good-enough option); Nobel Prize in Economics, 1978. ASKED ONCE

16 **Hindsight bias**, the "I knew it all along" effect or creeping determinism, belongs to Baruch Fischhoff (Fischhoff and Beyth, 1975, on Nixon's 1972 visits). ASKED ONCE

17 **Allan Paivio's dual-coding theory** (1971): a verbal and an imagery code; visual, concrete information is remembered better because it is coded twice. ASKED ONCE

18 **Jerome Bruner**, with Goodnow and Austin (*A Study of Thinking*, 1956), showed that artificial concepts are formed by developing and testing hypotheses (conservative focusing); Eleanor Rosch is the name for natural concepts. ASKED ONCE

19 The **connectionist** (parallel distributed processing) model of Rumelhart and McClelland (1986) stores knowledge in a network of interconnected units; it is the keyed answer for "a network of interconnected nodes". ASKED ONCE

20 The **linguistic relativity** hypothesis belongs to Benjamin Lee Whorf and his teacher Edward Sapir; the weak version (language influences thought) is supported, the strong version (determinism) is not. ASKED ONCE

21 **Child-directed speech** (motherese) has a shorter mean length of utterance, higher and wider pitch, a slower rate and more pauses than adult-to-adult speech. ASKED ONCE

22 Irreversible language deficits after early deprivation (Genie) support **sensitive (critical) periods**; Eric Lenneberg (1967) placed the window from about age two to puberty. ASKED ONCE

23 **Emotional prosody** (the intonation and emotional tone of speech) depends on right-hemisphere regions homotopic to the left-hemisphere language areas, not on Broca's area. ASKED ONCE

24 Learning to read builds most directly on **spoken language**, above all phonological awareness, not on theory of mind or imagination. ASKED ONCE

25 **Vygotsky** linked adolescent conceptual thinking to the tasks society sets adolescents, to social institutions and to self-reflection; gender differences play no part. ASKED ONCE

26 **Heuristics** were keyed as the option that does NOT reflect inductive reasoning; probabilistic conclusions, decision making and concept formation all do. ASKED ONCE

27 The **availability heuristic** (Tversky and Kahneman, 1973) judges frequency by ease of recall: people wrongly think more words begin with K than have K third. ASKED ONCE

28 **Hill climbing** always takes the move that brings you closest to the goal state, so it fails when a backward move is needed (the hobbits and orcs problem). ASKED ONCE

29 **Dan Slobin's language-making capacity** (1985) is a set of cognitive and perceptual abilities highly specialised for language learning (operating principles), not innate grammar. ASKED ONCE

CHAPTER 9

# Intelligence: Theories, Measurement, Heredity and Environment

IN THIS CHAPTER

9.1 ● What is intelligence?: compare the major definitions of intelligence and distinguish intelligence from aptitude and achievement.

9.2 ●● From Galton to Binet: the birth of mental testing: trace the history of testing from Galton's anthropometric laboratory to the Stanford-Binet and the Army Alpha and Beta.

9.3 ●●● The IQ and the modern tests: calculate ratio and deviation IQs, describe the Wechsler, Stanford-Binet, culture-fair and Indian tests, and judge what IQ scores predict.

9.4 ●● The structure of intelligence: factor theories: explain how factor analysis produced Spearman's g, Thurstone's primary abilities, Guilford's cube, fluid and crystallised intelligence and the CHC model.

9.5 ●● Broader views: multiple, triarchic, PASS and emotional intelligence: describe the process and multiple-intelligence theories and weigh the evidence for each.

9.6 ● Intelligence, speed and the brain: summarise the information-processing and biological correlates of measured intelligence.

9.7 ● Heredity and environment: define heritability correctly and interpret twin, adoption and intervention studies, the Flynn effect and the debate over group differences.

9.8 ●● The extremes: intellectual disability and giftedness: classify intellectual disability, explain the role of adaptive behaviour, and describe giftedness and savant syndrome.

Weightage. 63 of the 819 questions mapped in this book (7.7%) test this chapter, 22 of them from real past papers. 1 of its 8 sections is High yield.

In 1904 the French Ministry of Public Instruction had a problem that any Indian school inspector would recognise. The education laws of the early 1880s had made primary schooling free and compulsory, so every child in Paris now sat in a classroom, and some of them could not keep up. Which of these children were slow because they lacked the capacity to learn at the ordinary pace, and which were simply badly taught, mischievous, often absent or disliked by their teachers? Teachers' opinions were no help: the same child could be called "backward" by one teacher and "lazy" by the next. The ministry set up a commission to study the education of what it called "abnormal" children, and one of its members was a 47-year-old psychologist named Alfred Binet.

Binet had spent years watching children think, including his own two young daughters, and he had become convinced that the measures then in fashion (reaction times, the sharpness of hearing, the size of the head) told you almost nothing about a child's mind. Intelligence, he argued, shows itself in judgement: in understanding a question, keeping a goal in mind and noticing your own mistakes. So, with a young physician named Théodore Simon, he did something disarmingly simple. They collected dozens of small tasks, tried them on children in Paris schools and institutions, and kept the ones that older children passed more often than younger children. The result, published in 1905, was a ladder of 30 tasks. At the bottom, a child had only to follow a lighted match with the eyes; near the top, the child had to explain how two abstract words differed or build a sentence around three given words. How far up the ladder a child climbed, compared with other children of the same age, was the measure.

Binet was proud of the scale and wary of it. It was, he insisted, a practical tool for picking out children who needed special help, not a ruler that measured some fixed quantity inside the head. He prescribed "mental orthopaedics", classroom exercises in attention and self-control, to help slow children improve, and in 1909 he protested against "the brutal pessimism" of those who claimed that a person's intelligence is a fixed quantity that cannot be increased.

Binet died in 1911. Within a decade his ladder had crossed the Atlantic and been turned into something he had warned against. In the United States it was used to label newly arrived immigrants as "feeble-minded", to rank well over a million army recruits, and to argue that intelligence is simply inherited and that whole nations were born less able. Remember Binet's two claims: that intelligence is complex and a test can only sample it, and that it can be raised. By the end of this chapter you will be able to judge, with a century of evidence in hand, how right he was.

## 9.1 What Is Intelligence? LOWER YIELD · 3 QS

Everyone uses the word, and everyone believes they recognise intelligence when they see it: the classmate who solves a puzzle in seconds, the grandmother who keeps the household accounts in her head, the cricket captain who reads a match two overs ahead of everyone else. Yet when psychologists have tried to pin the word down, they have disagreed for more than a

century. It is worth starting with the disagreements, because they shape every theory and every test in this chapter, and because examiners love to ask who said what.

### Definitions from Binet to the Experts

**Alfred Binet** (1857 to 1911) put judgement at the centre: "to judge well, to comprehend well, to reason well, these are the essential activities of intelligence." He also called it good sense, practical sense, initiative and the faculty of adapting oneself to circumstances, and in 1909 he compressed it into four words: comprehension, invention, direction (holding a goal in mind) and censorship (criticising one's own attempts). Binet's view is sometimes classified as a **unifactor theory**, the idea that intelligence is one general, undivided capacity, because his scale produced a single overall level; the label is also attached to Lewis Terman and William Stern, whom you will meet in Section 9.2.

In 1921 the *Journal of Educational Psychology* asked more than a dozen leading American psychologists to define intelligence. The replies varied revealingly. **Lewis Madison Terman** said it was the ability to carry on abstract thinking. **Edward Lee Thorndike** called it "the power of good responses from the point of view of truth or fact." Others stressed the capacity to learn or to adapt to relatively new situations. Two years later, exasperated, the Harvard historian of psychology **Edwin Garrigues Boring** (1886 to 1968) wrote that intelligence is "what the tests test." He was making a serious point about method. Until psychologists agree on a theory, the only definition anyone can check is an **operational definition**, one that defines a concept by the procedure used to measure it.

The definition most often quoted in Indian examinations is that of **David Wechsler** (1896 to 1981): intelligence is "the aggregate or global capacity of the individual to act purposefully, to think rationally and to deal effectively with his environment." It is usually dated 1944, from the third edition of his book *The Measurement of Adult Intelligence*; almost the same words appear in the 1939 first edition, so you may see either year. Notice the two key words. *Aggregate* means that intelligence is made of many distinct abilities; *global* means that they work together as a whole. That double view, many parts but one whole, is exactly how his tests are built: separate subtests whose scores are summed into one full-scale score.

Sixty-five years after the 1921 symposium, **Robert Jeffrey Sternberg** and Douglas Detterman repeated the exercise, putting the same question to about two dozen experts and publishing the replies as *What Is Intelligence?* (1986). The overlap with 1921 was considerable: higher-level abstract reasoning, problem solving and the capacity to learn appeared in both. But the 1986 experts gave more weight to **metacognition** (knowing about, and controlling, one's own thinking) and to the culture that decides which abilities count as intelligent. Sternberg's own definition makes the cultural point explicit: intelligence is mental activity directed toward purposive adaptation to, and selection and shaping of, real-world environments relevant to one's life. Hold on to the three verbs adapt, select and shape; they return as the "contextual" part of his triarchic theory in Section 9.5.

Finally, after the storm over the book *The Bell Curve* (Section 9.7), **Linda Gottfredson** drafted a statement signed by 52 researchers and published in December 1994. Its definition is the working definition most researchers use today: "a very general mental capability that, among other things, involves the ability to reason, plan, solve problems, think abstractly, comprehend complex ideas, learn quickly and learn from experience." It is, the statement added, "not merely book learning, a narrow academic skill, or test-taking smarts."

What do these definitions share? Three themes recur: the capacity to learn from experience, the capacity to reason abstractly, and the capacity to adapt to the environment. Where they part company is on two questions. Is intelligence one thing or many? And how much of it can a test capture? Those two disagreements organise the rest of this chapter. Table 9.1 collects the definitions you are most likely to be asked about.

Table 9.1 Classic definitions of intelligence.

| Who | When | Intelligence is ... |
|---|---|---|
| Alfred Binet | 1905 to 1911 | Judgement, good sense, adapting to circumstances; comprehension, invention, direction, censorship |
| Lewis Terman | 1921 | The ability to carry on abstract thinking |
| Edward Thorndike | 1921 | The power of good responses from the point of view of truth or fact |
| Edwin Boring | 1923 | What the tests test (an operational definition) |
| David Wechsler | 1944 (1939) | The aggregate or global capacity to act purposefully, think rationally and deal effectively with the environment |
| Robert Sternberg | 1985 | Mental activity directed toward adapting to, selecting and shaping real-world environments |
| Linda Gottfredson and 51 cosignatories | 1994 | A very general mental capability to reason, plan, solve problems, think abstractly, learn quickly and learn from experience |

## Intelligence, Aptitude and Achievement ●●

Three words are often used loosely, and examiners exploit the looseness. **Achievement** is what a person has already learned through a specific course of instruction; an achievement test, such as a Class 10 board examination in mathematics, looks backwards at the effects of known teaching. An **aptitude** is a capacity to acquire, with training, some specific knowledge or skill. **Walter Van Dyke Bingham** (1880 to 1952) defined it in 1937 as a condition or set of characteristics regarded as symptomatic of a person's ability to acquire, with training, some knowledge, skill or set of responses, such as the ability to speak a language or produce music. An aptitude test therefore looks forwards: it predicts how well someone will learn to fly, to type or to study engineering before the training begins. The classic battery is the Differential Aptitude Tests of George Bennett, Harold Seashore and Alexander Wesman (1947). Because they predict how well a student will do in different courses and careers, aptitude tests are central tools of educational and vocational guidance.

Intelligence tests sit between the two. They sample broad abilities that schooling influences but does not directly teach, and they predict learning and problem solving in general rather than in one field. The boundary is fuzzy (a vocabulary subtest is partly an achievement measure), so the difference is best understood as one of breadth and purpose, as Table 9.2 shows.

Table 9.2 Achievement, aptitude and intelligence tests compared.

| Type of test | What it measures | Time orientation | Example |
|---|---|---|---|
| Achievement | Effects of a specific, known course of learning | Past | School board examination; reading attainment test |
| Aptitude | Potential to learn a specific skill with training | Future, specific | Differential Aptitude Tests; clerical or mechanical aptitude tests |

| Type of test | What it measures | Time orientation | Example |
|---|---|---|---|
| Intelligence | Broad reasoning and learning ability | Future, general | Wechsler scales; Stanford-Binet; Raven's matrices |

EXAM TRAP

The Wechsler Adult Intelligence Scale (WAIS) is an *intelligence* test, not an aptitude, achievement or attitude test. An attitude test measures a person's evaluative stance towards an object (the Social book). One paper (RML 2022) keyed the aptitude test as "the heart of the guidance programme". The standard phrase in guidance textbooks is that *counselling* is the heart of the guidance programme, so if counselling is offered, choose it; when counselling is not among the options, as in that paper, the aptitude test is the expected answer, because guidance decisions about courses and careers rest on it. The other usual options are guidance tools of a different kind: the case study is an intensive study of one person's history and circumstances, the interview is a face-to-face conversation to gather information and build rapport, and the cumulative record is a running file of a student's marks, test results, health and activities kept across the school years.

## Implicit Theories: What Ordinary People and Cultures Believe

The definitions above are **explicit theories**, formal theories built by psychologists and tested against data. Ordinary people carry **implicit theories of intelligence**, beliefs about what intelligence is that guide how they judge themselves and others. In 1981 Sternberg and his colleagues simply asked people in a railway station, a supermarket and a college library to list the behaviours of an intelligent person. Three clusters emerged: practical problem-solving ability (reasons logically, sees all sides of a problem), verbal ability (speaks clearly, reads widely) and social competence (accepts others for what they are, admits mistakes). Laypeople, in other words, already included the social and practical abilities that the psychometric tests left out.

Implicit theories also differ in a way that changes behaviour. **Carol Susan Dweck** (born 1946) distinguished an **entity theory** of intelligence, the belief that intelligence is a fixed quantity you either have or lack, from an **incremental theory**, the belief that it can be developed through effort and good strategies. In her popular book *Mindset* (2006) these became the **fixed mindset** and the **growth mindset**. Children who hold an entity theory tend to avoid challenges and give up after failure, because failure seems to reveal a permanent lack; children with an incremental theory treat failure as information and persist. Praise can push a child towards one theory or the other. In a well-known series of experiments with American fifth-graders, Claudia Mueller and Dweck (1998) praised some children for being clever ("you must be smart at this") and others for working hard; after a later failure, the children praised for ability chose easier tasks, enjoyed the work less and performed worse than those praised for effort. The mindset research therefore recommends praising effort and strategy, not innate ability. Large school studies of brief mindset interventions have found real but small average effects on grades, with larger effects for low-achieving students, so the idea is useful but not magic.

INDIAN CONTEXT

The Indian-born psychologist J. P. Das (Section 9.5) has pointed out that the closest Indian word, *buddhi*, is broader than the Western test-based idea. It includes discrimination between right and wrong, self-knowledge, determination and feeling, not only speed and accuracy of reasoning. Studies of everyday Indian conceptions likewise give a large place to social competence, responsibility and wisdom. This is one reason a test built on one culture's implicit theory can misjudge people from another.

EXAM TRAP

"Incremental" belongs to Dweck's entity versus incremental theories, not to Sternberg's triarchic theory, whose three parts are componential, *experiential* and contextual. Options such as "incremental" and "experimental" in a Sternberg question are bait. A "growth mindset" is the belief that intelligence is malleable and can be developed through effort.

## 9.2 From Galton to Binet: The Birth of Mental Testing

MEDIUM YIELD · 5 QS

Binet's 1905 scale is usually called the first intelligence test, and it was. But it was a reaction against an earlier attempt that had failed, and you cannot understand why Binet succeeded without knowing why his predecessors did not.

### Francis Galton and the Anthropometric Laboratory

**Francis Galton** (1822 to 1911), a half-cousin of Charles Darwin, was fascinated by the fact that eminent men seemed to come from a small number of families. In *Hereditary Genius* (1869) he traced the family trees of judges, statesmen, scientists and poets and concluded that "natural ability" was inherited. (The obvious objection, that eminent families also pass on wealth, schooling and connections, did not trouble him much.) In 1874 he popularised the phrase "nature and nurture", in 1875 he proposed comparing twins as a way of separating the two, and in 1883 he coined the word *eugenics* for the programme of improving the human stock by selective breeding, an idea that later did great harm.

How was natural ability to be measured? Galton followed the empiricist logic that all knowledge enters through the senses, so the people with the keenest senses and quickest reactions should be the most intelligent. At the International Health Exhibition in London in 1884 he opened an **anthropometric laboratory** (anthropometry is the measurement of the human body) where, for a small fee, about 9,000 visitors had their head size, grip strength, reaction time, visual acuity and highest audible pitch measured. To analyse such data he developed the ideas of regression and correlation, later put into mathematical form by his protégé Karl Pearson. For this work Galton is often called the father of **differential psychology**, the study of individual differences, and of psychometrics. He had, however, measured the wrong things.

### James McKeen Cattell and the "Mental Test"

The American **James McKeen Cattell** (1860 to 1944) took his doctorate under Wilhelm Wundt in Leipzig and then worked with Galton in England. In an 1890 paper in the journal *Mind*, "Mental Tests and Measurements", he coined the term **mental test** and proposed a battery of

ten: grip pressure, rate of hand movement, two-point touch threshold, the pressure needed to cause pain, the smallest noticeable difference in weight, reaction time to sound, time to name colours, bisection of a 50-centimetre line, judgement of a ten-second interval, and the number of letters remembered after one hearing. At Columbia University he tested entering students every year.

The test of these tests came in 1901. Cattell's own student Clark Wissler correlated the scores of Columbia students with their college grades and with one another. The correlations with grades were close to zero, and the tests barely correlated with each other. If reaction time and pitch discrimination were measures of one underlying intelligence, they should have hung together and predicted academic success; they did neither. The sensory approach collapsed, leaving the field open for someone who would measure higher mental processes directly.

### Binet and Simon: Measuring Judgement ●●

That someone was Binet, working with the physician **Théodore Simon** (1873 to 1961; a few sources give 1872). Their approach differed from Galton's in two decisive ways. First, they measured complex functions (comprehension, reasoning, memory, judgement) rather than simple sensations. Second, they chose items empirically: a task stayed in the scale only if older children passed it more often than younger children, so success on the scale tracked mental development.

#### KEY STUDY: THE BINET-SIMON SCALES (1905, 1908, 1911)

**Question.** Could an objective test pick out schoolchildren who needed special education, without relying on teachers' impressions? **Method.** In 1905 Binet and Simon published 30 items arranged in order of increasing difficulty, from following a lighted match with the eyes to defining abstract words. In the 1908 revision they grouped the items by the age at which most normal children passed them (ages 3 to 13). A child's **mental age** was the age level of the items the child could pass (Binet's own term was "mental level"). A 7-year-old who passed the items typical of 9-year-olds had a mental age of 9. The 1911 revision, published the year Binet died, extended the scale into the adult range. **Result.** For the first time a single, standardised procedure could rank children's intellectual development. **Conclusion and criticism.** Binet regarded the score as a practical sample of performance, not a measure of fixed inborn capacity, and he warned against using it to label children for life. Those warnings were largely ignored once the test reached America.

There is a footnote to this story that examiners enjoy. Around 1920 a young Swiss biologist named **Jean Piaget** worked in Simon's laboratory in Paris, standardising reasoning tests on Parisian children. He became less interested in whether the children got the answers right than in the systematic ways they got them wrong, and that curiosity grew into his theory of cognitive development (the Developmental book). So Piaget came out of the Binet tradition, but he did not create the first intelligence test.

### Across the Atlantic: Goddard, Stern and Terman ●●●

**Henry Herbert Goddard** (1866 to 1957), research director of the Vineland Training School for "feeble-minded" children in New Jersey, translated the Binet-Simon scale into English in 1908. He grouped people with intellectual disability into idiots (the most severe), imbeciles, and a new category he named in 1910: the *moron*, from a Greek word for foolish, for people

with mental ages of roughly 8 to 12. His book *The Kallikak Family* (1912) claimed to trace "feeble-mindedness" down two branches of one family, and from 1913 his team tested immigrants arriving at Ellis Island in New York, classifying many as feeble-minded on the strength of a translated test given through interpreters to frightened, exhausted travellers. These terms are now purely historical, but they appear as options in exam questions.

Meanwhile, in Germany, **William Stern** (1871 to 1938) saw a problem with mental age. A child who is two years behind at age 4 is much more retarded in development than one who is two years behind at age 14. In 1912 he proposed dividing mental age by chronological age to give a "mental quotient", in German the *Intelligenzquotient*. Stern therefore coined the term and the ratio of the **intelligence quotient (IQ)**.

It was Lewis Madison Terman (1877 to 1956) of Stanford University who made the IQ famous. His 1916 American revision of the Binet-Simon scale, the **Stanford-Binet Intelligence Scale**, multiplied Stern's quotient by 100 to remove the decimal and reported the result as the IQ:

IQ = (mental age ÷ chronological age) × 100

This is the **ratio IQ**. Work two cases through. A 10-year-old with a mental age of 8 has an IQ of (8 ÷ 10) × 100 = 80. A 10-year-old with a mental age of 12 has an IQ of 120. A child performing exactly at age level always scores 100, whatever the age, which is what made the number so appealing. Terman's Stanford-Binet became the standard American test for decades, and Terman, unlike Binet, believed it measured an inherited capacity.

### Testing Goes to War: Army Alpha and Army Beta

When the United States entered the First World War in 1917, **Robert Mearns Yerkes** (1876 to 1956), then president of the American Psychological Association (and the same Yerkes as the Yerkes-Dodson law of Chapter 10), led a committee, with Terman and Goddard among its members, that built the first large-scale group intelligence tests. The **Army Alpha** was a written, verbal test for recruits who could read English. The **Army Beta** was a non-verbal, pictorial test, given by demonstration and mime, for recruits who were illiterate or did not speak English. About 1.75 million men were tested. Group testing had arrived: one examiner could now test hundreds of people at once.

The aftermath is a lesson in the misuse of data. In *A Study of American Intelligence* (1923) the psychologist Carl Brigham used the army results to rank immigrant groups, with northern Europeans on top, ignoring the obvious fact that recent immigrants scored lower the fewer years they had lived in the United States, that is, the less English and American schooling they had. Such arguments were cited in the debates that led to the restrictive Immigration Act of 1924. Brigham publicly retracted his conclusions in 1930. Table 9.3 gathers the milestones of this history, together with the later tests you will meet in Section 9.3.

> **CLINICAL LINK**
>
> The Ellis Island and army episodes are the origin of three rules every clinical psychologist follows: test a person in a language and format they understand, interpret the score against norms from a comparable population, and never treat a single score as a verdict on a person's worth or future. In India, with its many languages and very unequal schooling, these rules decide which test you choose (Section 9.3).

Table 9.3 Milestones in the measurement of intelligence.

| Year | Who | Milestone |
|---|---|---|
| 1869 | Francis Galton | *Hereditary Genius*: ability runs in families |
| 1884 | Francis Galton | Anthropometric laboratory, London: sensory and motor measures |
| 1890 | James McKeen Cattell | Coins "mental test" |
| 1901 | Clark Wissler | Cattell's tests fail to predict college grades |
| 1904 | Charles Spearman | General factor g (Section 9.4) |
| 1905 | Alfred Binet and Théodore Simon | First intelligence scale: 30 items; revised 1908 (mental age) and 1911 |
| 1908 | Henry Goddard | English translation of the Binet-Simon scale |
| 1912 | William Stern | Intelligence quotient: mental age divided by chronological age |
| 1916 | Lewis Terman | Stanford-Binet; IQ multiplied by 100 |
| 1917 | Robert Yerkes | Army Alpha (verbal) and Army Beta (non-verbal) group tests |
| 1938 | John Raven | Standard Progressive Matrices |
| 1939 | David Wechsler | Wechsler-Bellevue scale; deviation IQ |
| 1955 | David Wechsler | Wechsler Adult Intelligence Scale (WAIS) |
| 2003 | Gale Roid | Stanford-Binet, fifth edition (SB5) |
| 2008 | Pearson | WAIS-IV, with four index scores |

EXAM TRAP

**Stern versus Terman.** William Stern coined "intelligence quotient" (1912); Terman multiplied it by 100 and popularised it in the Stanford-Binet (1916). "Who invented the term IQ?" is Stern. **Two Cattells.** James McKeen Cattell coined "mental test" (1890); Raymond Bernard Cattell, unrelated and a generation later, gave us fluid and crystallised intelligence, the Culture Fair Intelligence Test and the 16PF. **Galton versus Binet.** Galton founded the study of individual differences and ran the first mental-measurement laboratory; Binet (with Simon) built the first practical intelligence test. **Alpha versus Beta.** Alpha is for the literate (verbal); Beta is for the illiterate (non-verbal).

## 9.3 The IQ and the Modern Tests HIGH YIELD · 32 QS

The ratio IQ had a fatal flaw, and the way it was fixed gave us the IQ we use today. To see the flaw, push an adult through the formula.

### From Ratio IQ to Deviation IQ ●●●

Mental age grows quickly in childhood, slows in adolescence and stops growing in the mid to late teens. Chronological age keeps climbing. Take a perfectly ordinary 40-year-old whose mental age, like that of most adults, levelled off at about 16: the ratio formula gives (16 ÷ 40) × 100 = 40, a score that would label an average adult as severely impaired. Terman patched the problem by treating every adult as if he or she were about 16 years old, but the patch hid a second flaw. Even in childhood, the spread of ratio IQs differs from age to age, so an IQ of 115

might put a 6-year-old in a different position relative to his age-mates than a 12-year-old with the same number. A score that means different things at different ages is a poor score.

David Wechsler's solution, introduced with the Wechsler-Bellevue Intelligence Scale in 1939, was to abandon mental age altogether. A person's raw score is compared with the scores of a large, representative sample of people of the same age (the **norms**), and the IQ simply reports how far above or below the average of that age group the person falls. This is the **deviation IQ**; because it is a standard score, some books call it the "standard IQ". The mean of every age group is set at 100 and the **standard deviation (SD)**, the typical distance of scores from the mean, is set at 15. In formula form, if z is the number of standard deviations a person scores above or below the age-group mean (a **z score**):

deviation IQ = 100 + 15z

A 40-year-old who scores exactly at the mean of 40-year-olds gets 100; one who scores one standard deviation above them gets 115. The meaning of 115 is now the same at every age: better than about 84 per cent of one's age-mates. Study guides sometimes describe Wechsler's method as "the actual score divided by the expected mean score for the person's age"; that is a loose way of saying the same thing, that the person is compared with the expected performance of his or her own age group, and if you meet that wording as an option, it is the one intended. Other standard scores work the same way with different numbers: the T score used by many personality and neuropsychological tests has a mean of 50 and an SD of 10, and the Wechsler subtest scaled scores have a mean of 10 and an SD of 3. A percentile rank is different again; it is not a standard score but simply the percentage of the norm group scoring below a person.

Because IQ scores are approximately normally distributed, the familiar properties of the normal curve let you translate any IQ into a percentile, the percentage of people scoring below it. As Figure 9.1 shows, about 68 per cent of people score between 85 and 115 (within one SD of the mean) and about 95 per cent between 70 and 130 (within two SDs).

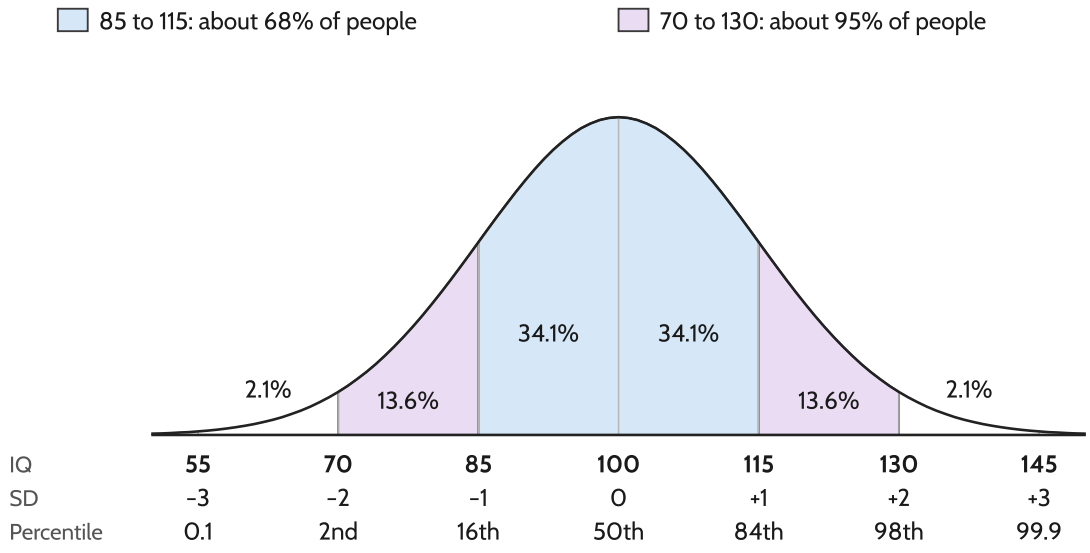


Figure 9.1 The distribution of deviation IQ (mean 100, SD 15). Read across the bottom rows to convert an IQ into standard deviations and percentiles: 85 is the 16th percentile, 115 the 84th, 130 about the 98th, and 70 about the 2nd.

Work two examples. On a Wechsler scale an IQ of 85 is one SD below the mean, so 16 per cent of people score below it (the 16th percentile); an IQ of 100, the mean, is by definition the 50th percentile. Now an older Stanford-Binet style question: on a test with a mean of 100 and an SD of 16, a person scores 136. Then $z = (136 - 100) \div 16 = 2.25$, and the normal-curve table gives about 0.987 of the area below $z = 2.25$, so about 98.7 per cent of people score lower. The SD matters. The Stanford-Binet adopted the deviation IQ in its 1960 revision (Form L-M) with an SD of 16, kept 16 in the fourth edition (1986), and moved to 15, in line with the Wechsler scales, only in the fifth edition (2003). Always check which SD a question gives you.

> **EXAM TRAP**
>
> Ratio IQ = (MA ÷ CA) × 100, from Stern (1912) and Terman (1916); the original Stanford-Binet used it. Deviation IQ is a standard score with mean 100 and SD 15, comparing a person with same-age peers; Wechsler introduced it in 1939, and modern tests (including the SB5) use it. The deviation IQ is not "only for adults", it does not use raw scores without standardisation, and the ratio IQ is not the more reliable of the two (its meaning shifts from age to age, which is why it was abandoned). The old Stanford-Binet SD was 16, not 15.

## The Wechsler Scales ●●●

Wechsler, a Romanian-born American who had tested army recruits in 1918, was chief psychologist at Bellevue Psychiatric Hospital in New York when he built his first scale. He set out to correct several faults he saw in the Stanford-Binet: it was designed for children, so its items had little appeal for adults; it leaned heavily on verbal, word-manipulation tasks; it placed too much weight on speed; and it rested on the concept of mental age, which made no sense for adults. His answer had four features that still define the Wechsler tests. They are **point scales** rather than **age scales**: items are grouped by type into subtests, and points are earned within each subtest, rather than items being grouped by the age at which they are passed, as in Binet's ladder. They combine verbal subtests with non-verbal **performance tests** (block designs, picture puzzles) that require little or no language. They yield subtest and part scores as well as a total, so a clinician can see a profile of strengths and weaknesses. And they use the deviation IQ.

The family has grown with every generation (Table 9.4). Until the 2000s each test produced a **Verbal IQ**, a **Performance IQ** and a **Full Scale IQ (FSIQ)**. The newer editions replaced the verbal-performance split with **index scores** built on factor-analytic research (Section 9.4).

Table 9.4 The Wechsler family of intelligence scales.

| Scale | First edition | Current edition | Age range of current edition |
|---|---|---|---|
| Wechsler-Bellevue | 1939 | (replaced by the WAIS) | Adolescents and adults |
| Wechsler Adult Intelligence Scale (WAIS) | 1955 | WAIS-IV (2008); WAIS-5 (2024) | 16 years to 90 years 11 months |
| Wechsler Intelligence Scale for Children (WISC) | 1949 | WISC-V (2014) | 6 years to 16 years 11 months |
| Wechsler Preschool and Primary Scale of Intelligence (WPPSI) | 1967 | WPPSI-IV (2012) | 2 years 6 months to 7 years 7 months |

The WAIS-IV, still the most widely used adult test in India, has 15 subtests: 10 core subtests that produce the scores and 5 supplemental subtests that can substitute for them. They are organised into four indices (Table 9.5). The **Verbal Comprehension Index (VCI)** measures verbal reasoning and acquired knowledge. The **Perceptual Reasoning Index (PRI)** measures non-verbal, fluid and visual-spatial reasoning. The **Working Memory Index (WMI)** measures the ability to hold information in mind and manipulate it, that is, to attend, concentrate and exert mental control. The **Processing Speed Index (PSI)** measures the speed and accuracy of simple visual scanning and copying. The four indices combine into the FSIQ. An optional **General Ability Index (GAI)**, computed from the six verbal and perceptual core subtests alone, gives an estimate of reasoning ability less affected by working memory and speed, which is useful when, say, a patient's attention has been disrupted by a head injury or depression.

Table 9.5 The structure of the WAIS-IV (2008).

| Index | Core subtests | Supplemental subtests |
|---|---|---|
| Verbal Comprehension (VCI) | Similarities, Vocabulary, Information (3) | Comprehension |
| Perceptual Reasoning (PRI) | Block Design, Matrix Reasoning, Visual Puzzles (3) | Picture Completion, Figure Weights |
| Working Memory (WMI) | Digit Span, Arithmetic (2) | Letter-Number Sequencing |
| Processing Speed (PSI) | Symbol Search, Coding (2) | Cancellation |

The WISC-V for children went further, splitting perceptual reasoning in two. It has five primary indices: Verbal Comprehension, Visual Spatial, Fluid Reasoning, Working Memory and Processing Speed. The WAIS-5 (2024) adopted the same five-index structure for adults. For examination purposes, though, "the four WAIS-IV indices" (VCI, PRI, WMI, PSI) remain the most frequently tested fact in this whole area. Do not confuse any of these with the Wechsler Memory Scale (WMS), first published by Wechsler in 1945, which is a memory test rather than an intelligence test. Its early editions summarised performance as a single memory quotient (MQ) on the IQ metric; the WMS-IV (2009) gives five index scores instead: Auditory Memory, Visual Memory, Visual Working Memory, Immediate Memory and Delayed Memory. An option listing "Verbal IQ, Performance IQ, Full Scale IQ and Memory Quotient" therefore mixes the old WAIS with the old WMS and is not the WAIS-IV.

> **EXAM TRAP**
>
> The WAIS-IV has 3 core verbal (VCI) and 3 core perceptual (PRI) subtests, plus 2 core working-memory and 2 core processing-speed subtests: 10 core in all. The Working Memory Index, built from Digit Span and Arithmetic, is the one that reflects attention, concentration and mental control. One paper asked which subtest is "core in the WAIS-IV but not found in the WISC-V" and keyed Letter-Number Sequencing. Strictly, Letter-Number Sequencing is a *supplemental* subtest in the WAIS-IV (it was part of the Working Memory Index in the WAIS-III) and a secondary subtest in the WISC-V; but it is still the best of the options offered, because Digit Span, Symbol Search and Matrix Reasoning are core in both tests. The WISC-V covers 6 years to 16 years 11 months.

## The Stanford-Binet Today, and Other Individual Tests ●●

The Stanford-Binet was revised in 1937 (by Terman and Maud Merrill), in 1960, in 1986 and most recently as the **Stanford-Binet Fifth Edition (SB5)** (in full, the Stanford-Binet Intelligence Scales, Fifth Edition), developed by Gale Roid and published in 2003. The SB5 covers ages 2 to 85 and over and is organised around five factors drawn from the hierarchical Cattell-Horn-Carroll model of Section 9.4, with the working-memory factor also drawing on Alan Baddeley's working-memory model (Chapter 7): Fluid Reasoning, Knowledge, Quantitative Reasoning, Visual-Spatial Processing and Working Memory. Each factor is measured once in a verbal and once in a non-verbal form, giving 10 subtests and a Full Scale IQ, a Verbal IQ and a Nonverbal IQ, all with mean 100 and SD 15. Like the earlier Stanford-Binets, it uses a **basal level** (the level at which the person passes all items, so easier items need not be given) and a **ceiling level** (the level at which the person fails all or most items, so testing stops).

Several other individual tests appear in questions. Alan and Nadeen Kaufman's Kaufman Assessment Battery for Children (K-ABC, 1983) was one of the first major tests built on Luria's neuropsychology (Section 9.5); their Kaufman Brief Intelligence Test, second edition (KBIT-2, 2004), is a 20-minute screening test with just two scales, a Verbal or crystallised scale and a Nonverbal or fluid scale. For infants, intelligence in the adult sense cannot yet be measured, so developmental scales are used instead: the Bayley Scales of Infant and Toddler Development (Nancy Bayley; the current edition covers roughly the first 42 months) assess cognitive, language and motor development, and the McCarthy Scales of Children's Abilities (Dorothea McCarthy, 1972) cover about 2½ to 8½ years. Developmental scales express results as a **developmental quotient (DQ)**, developmental age divided by chronological age, times 100, a ratio exactly like Stern's. Detailed administration and scoring belong to the Assessment book.

## Classifying IQ Scores ●●

Test publishers attach descriptive labels to ranges of IQ (Table 9.6). The labels have softened over the decades: the WAIS-IV's "borderline" has become "very low" in the WISC-V and WAIS-5, and the old psychiatric category of **borderline intellectual functioning**, which DSM-IV-TR placed at IQ 71 to 84, overlaps the "low average" and "borderline" bands. An IQ in the 70 to 84 region therefore falls into "low average or borderline, depending on the edition."

Table 9.6 IQ classification bands (deviation IQ, mean 100, SD 15).

| IQ range | WAIS-IV label | WISC-V and WAIS-5 label | Theoretical percentage |
|---|---|---|---|
| 130 and above | Very superior | Extremely high | 2.2 |
| 120 to 129 | Superior | Very high | 6.7 |
| 110 to 119 | High average | High average | 16.1 |
| 90 to 109 | Average | Average | 50.0 |
| 80 to 89 | Low average | Low average | 16.1 |
| 70 to 79 | Borderline | Very low | 6.7 |
| 69 and below | Extremely low | Extremely low | 2.2 |

EXAM TRAP

An FSIQ of 70 on its own is *not* a diagnosis of intellectual disability. DSM-5 requires deficits in adaptive functioning as well, and sets severity by adaptive functioning, not by IQ (Section 9.8). On the WAIS-IV a score of exactly 70 falls in the "borderline" band (70 to 79), one point above "extremely low"; so a question asking how to classify an FSIQ of 70 wants "borderline, and intellectual disability only if adaptive functioning is also impaired". With measurement error, scores of about 65 to 75 are compatible with intellectual disability only if adaptive deficits are present.

## Culture-Fair Tests ●●●

A vocabulary item asks, in effect, "Have you met this word?", and whether you have met it depends on your home, school and language. Psychologists have therefore tried to build tests whose content is equally familiar, or equally unfamiliar, to everyone. The early hope was a **culture-free test**, one entirely free of cultural influence; that turned out to be impossible, because even the habit of sitting quietly, working against a clock and treating abstract shapes as a meaningful problem is learned. The realistic goal is a **culture-fair test** (also called culture-reduced), one that minimises language, reading, specific knowledge and speed so that people from different backgrounds are on a more equal footing.

The most famous is **Raven's Progressive Matrices**, devised by the British psychologist **John Carlyle Raven** (1902 to 1970), a student of Spearman. Each item shows a pattern of figures arranged in a matrix with one piece missing, and the examinee chooses the piece that completes it. Solving an item requires working out the rule that governs the rows and columns, which is what Spearman called the eduction of relations (Section 9.4). The Standard Progressive Matrices (SPM), published in 1938 (Raven had drafted it by 1936, so some books give that year), has 60 items in five sets of 12 that grow progressively harder, hence the name. The Coloured Progressive Matrices (36 items) are for children and older or impaired adults, and the Advanced Progressive Matrices (48 items) for high-ability adults. Raven's matrices are the classic measure of fluid intelligence, the ability to reason with novel material (Section 9.4), and they are also the best single marker of Spearman's general factor.

Raymond Bernard Cattell's **Culture Fair Intelligence Test** (CFIT), first issued around 1940 as a "culture-free" test and soon renamed, is a group test in three scales for different ages and abilities. Its subtests (series completion, classification, matrices and "conditions", a topological task) use only figures. Other non-verbal and performance tests that reduce the role of language include Florence Goodenough's Draw-a-Man test (1926), the Leiter International Performance Scale, the Porteus Maze Test, Kohs block designs and Alexander's Pass-Along test; the Army Beta was an early example of the same idea.

EXAM TRAP

Raven's Progressive Matrices measure *fluid* intelligence; choose "fluid" over "crystallised", "verbal" or "social". If "general intelligence (g)" is offered alongside "fluid", the expected answer is still fluid, although Raven's is also the best single measure of g, because Gf and g are very closely related. Raven's is called culture-fair because it is non-verbal, with minimal language demands; it is not culture-free, and schooling still affects scores.

## Intelligence Tests in India ●●●

Most Western tests cannot simply be translated for Indian use: the vocabulary, the general-information items ("Who wrote *Hamlet*?") and even the pictures assume a particular culture, and many children who need assessment have had little or no schooling. Indian psychologists responded in two ways: by adapting the Binet and Wechsler tests with Indian content and norms, and by building performance batteries that need no reading at all. Table 9.7 lists the tests that appear in examinations; full detail belongs to the Assessment book.

**Table 9.7** Intelligence and related tests developed or adapted in India.

| Test | Author | Basis and features |
|---|---|---|
| Binet-Kamat Test | V. V. Kamat (1930s) | Adaptation of the 1916 Stanford-Binet for Marathi- and Kannada-speaking children; ratio IQ; ages about 3 to 22; heavily verbal |
| Hindi adaptation of the Stanford-Binet | S. K. Kulshrestha | Hindi version of the Stanford-Binet; ratio IQ |
| Bhatia's Battery of Performance Tests of Intelligence | C. M. Bhatia (1955) | Five performance subtests: Kohs block design, Alexander's pass-along, pattern drawing, immediate memory (digits), picture construction; normed on about 11 to 16 years; usable with illiterate people; separate rural and urban norms |
| Malin's Intelligence Scale for Indian Children (MISIC) | Arthur J. Malin, Nagpur (1969) | Indian adaptation of the WISC; 6 verbal and 5 performance subtests; ages about 6 to 15 |
| Wechsler Adult Performance Intelligence Scale (WAPIS) | Prabha Ramalingaswamy (1974) | Indian adaptation of the performance (non-verbal) subtests of the WAIS |
| Raven's Progressive Matrices | John C. Raven (Indian norms by several researchers) | Non-verbal; widely used with Indian norms |
| Vineland Social Maturity Scale (Indian adaptation) | Edgar Doll; adapted by A. J. Malin | Adaptive behaviour: social age and social quotient (Section 9.8) |

Now apply the principle. A 10-year-old from a village, who has never been to school and cannot read, is brought for assessment because he struggles to learn. Which test? The WISC-IV and the SB5 depend heavily on verbal knowledge built by schooling and were normed on other populations. Malin's MISIC is an Indian test but half of it is verbal, built on the WISC's information, comprehension, arithmetic and vocabulary items. Bhatia's Battery, a set of performance tasks that need no reading, with rural norms, is the most appropriate choice to minimise cultural and educational bias.

EXAM TRAP

Bhatia's Battery was developed by C. M. Bhatia; the other names often offered, Girishwar Misra, Durganand Sinha and J. B. P. Sinha, are distinguished Indian psychologists known for cultural and indigenous psychology, social psychology and organisational leadership, not for this test. Some notes date the Battery to the 1940s, when Bhatia did the work; 1955 is the commonly cited publication year. Do not confuse Malin's two contributions: MISIC (an intelligence test) and the Indian adaptation of the Vineland Social Maturity Scale (an adaptive-behaviour scale).

## Are IQ Tests Reliable and Valid, and What Do They Predict? ●●

A test is useful only if it is **reliable** (it gives consistent scores) and **valid** (it measures what it claims to measure). The major individual IQ tests are among the most reliable instruments in psychology: the internal-consistency reliability of the WAIS-IV Full Scale IQ is about .98, and retesting after a few weeks typically gives correlations around .9. Individual subtests are much less reliable than the total, which is why clinicians interpret composite scores first.

Validity comes in several kinds (the Assessment and Research Methodology books treat them fully). **Content validity** asks whether the items sample the domain adequately. **Criterion-related validity** asks whether scores relate to an outside criterion; it is called **concurrent validity** when test and criterion are measured at about the same time, and **predictive validity** when the criterion comes later. **Construct validity** asks whether the test behaves as the theory of the construct predicts, for example correlating with other measures of reasoning but not with unrelated traits. So if a new intelligence test is given alongside a well-established one, and the two sets of scores are correlated, the researcher is checking concurrent validity.

What does IQ predict? A great deal, though far from everything. IQ correlates about .5 with school grades and about .55 with the number of years of education completed. In a famous 1998 review of 85 years of research, Frank Schmidt and John Hunter found that general mental ability predicts job performance with a validity of about .5, higher for complex jobs than for simple ones, which makes it the single best predictor of job performance across occupations. Childhood IQ even predicts health and lifespan: in Scotland, people who had been tested at age 11 in the Scottish Mental Survey of 1932 were more likely to be alive decades later if they had scored higher. But a correlation of .5 means IQ accounts for only about a quarter of the variation in grades (.5 squared is .25); motivation, conscientiousness, teaching, opportunity and luck account for the rest.

CLINICAL LINK

Clinical psychologists use intelligence tests to diagnose intellectual disability, to identify specific learning disabilities (where achievement falls well below what ability predicts), to document cognitive decline after brain injury or in dementia, to plan rehabilitation and education, and to establish eligibility for benefits such as a disability allowance or disability certificate. For adults the WAIS is used; for children with special educational needs, the WISC or an Indian equivalent. Every such use rests on current, local norms: a test normed decades earlier inflates scores (the Flynn effect, Section 9.7), which can wrongly deny someone a diagnosis or support.

SCAN YOUR BRAIN

Fill in each blank with the most appropriate term. The answers are below; cover them first.

1. Edwin Boring's statement that intelligence is "what the tests test" is an example of an _____ definition.
2. Wechsler defined intelligence as the aggregate or _____ capacity to act purposefully, think rationally and deal effectively with the environment.
3. The term "mental test" was coined in 1890 by _____.
4. Binet and Simon's first scale (1905) contained _____ items.
5. The term intelligence quotient was coined in 1912 by _____.
6. A 10-year-old with a mental age of 8 has a ratio IQ of _____.
7. The Army _____ was the non-verbal test for illiterate recruits.
8. On a Wechsler scale, an IQ of 85 corresponds to the _____ percentile.
9. The WAIS-IV index built from Digit Span and Arithmetic is the _____ Index.
10. For an illiterate rural Indian child, the most culture-fair of the Indian tests is _____ Battery of Performance Tests.

**Answers:** (1) operational, (2) global, (3) James McKeen Cattell, (4) 30, (5) William Stern, (6) 80, (7) Beta, (8) 16th, (9) Working Memory, (10) Bhatia's.

## 9.4 The Structure of Intelligence: Factor Theories

MEDIUM YIELD · 9 QS

Once there were tests, a new question could be asked with data rather than opinion. If intelligence is one thing, people who do well on one test should do well on all of them. If it is many things, a person could be strong in one and weak in another. The tool for deciding between these possibilities was invented for the purpose, and it shaped the next hundred years of theory.

### Factor Analysis in Plain Words

Suppose 200 Class 9 students take tests in English, Hindi, mathematics, physics and drawing. You correlate every test with every other and lay out the correlations in a table, a correlation matrix. You notice that English and Hindi correlate highly with each other, mathematics and physics correlate highly with each other, and every test correlates at least a little with every other. **Factor analysis** is a family of statistical methods that asks how few underlying variables, called **factors**, are needed to explain such a pattern of correlations. Each test's correlation with a factor is its **factor loading**. Here you might find one factor on which all five tests load (a general ability), plus a "language" factor and a "mathematics-science" factor. The method does not tell you what the factors mean; the researcher names them by looking at which tests load on them, and different methods of extracting and rotating factors can give different-looking answers from the same data. That is why factor analysts have argued for a century.

### Spearman's Two-Factor Theory ●●

The British psychologist **Charles Edward Spearman** (1863 to 1945) invented the first form of factor analysis. In 1904, in a paper titled "General Intelligence, Objectively Determined and

Measured", he analysed the school marks and sensory-discrimination scores of English schoolchildren (in classics, French, English, mathematics, pitch discrimination and music). He found that all the measures correlated positively with one another, a pattern now called the **positive manifold**. Moreover, the correlations fell into a neat hierarchy, a regularity Spearman later expressed as "tetrad differences" close to zero, which is exactly what you would expect if a single common factor ran through every measure.

He called this common factor **general intelligence (g)**. Every test, he proposed, measures g plus a **specific factor (s)** peculiar to that test: the ability to do arithmetic depends partly on g and partly on a specific arithmetical skill. Because it has exactly two kinds of factor, general and specific, this is called the **two-factor theory** (Figure 9.2). Tests differ in how heavily they are "saturated" with g; Spearman found that tasks requiring abstract reasoning were the most saturated, and simple sensory tasks the least. He interpreted g as a kind of general mental energy, and in 1923 he described three "noegenetic" (knowledge-creating) principles that he thought defined it: the apprehension of experience, the **eduction of relations** (seeing the relationship between two things) and the eduction of correlates (applying a relationship to generate a new item). Raven's matrices, built by his student, are a direct test of eduction. In *The Abilities of Man* (1927) Spearman admitted that some tests also share narrower group factors, such as verbal or mechanical ability, but g remained the centre of his theory.

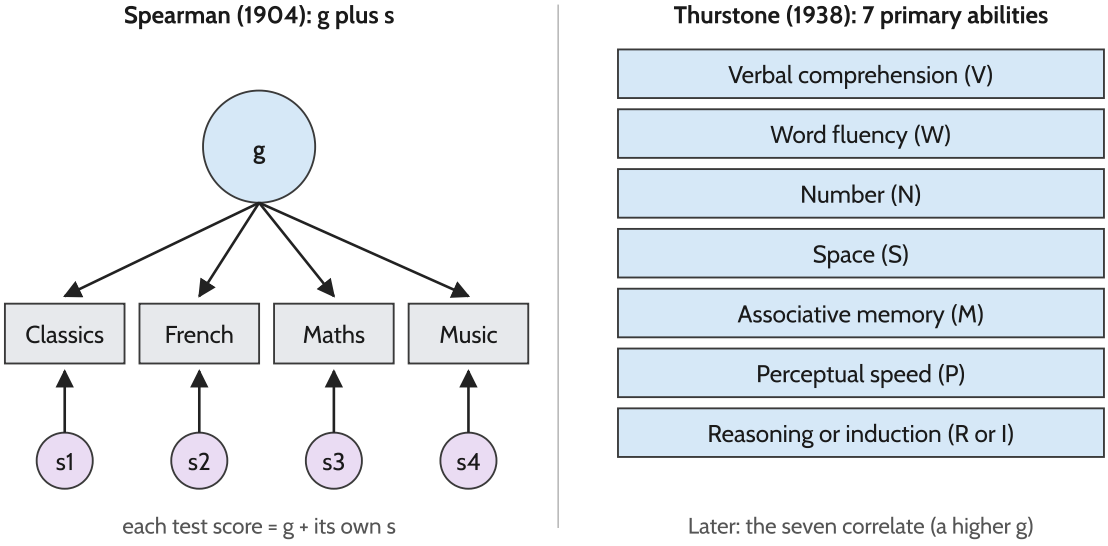


Figure 9.2 Two early answers to "one or many?". In Spearman's two-factor model every test draws on the same general factor g plus its own specific factor s. In Thurstone's model there are seven separate primary abilities and no g, although Thurstone later found that the seven themselves correlate, which brings a general factor back at a higher level.

**EXAM TRAP**

Spearman's g is a *general* factor underlying performance on all cognitive tasks; s is the *specific* factor of each task. Neither is "group factors" (that is Thurstone, Burt and Vernon) nor "genetic predisposition". The two-factor theory is not the triarchic model (Sternberg), multiple intelligences (Gardner), seven primary abilities (Thurstone) or fluid and crystallised intelligence (Cattell). The same Spearman gave us the rank-order correlation coefficient (rho) and, with William Brown, the Spearman-Brown formula used in reliability.

### Thorndike's Multifactor Theory

Edward Lee Thorndike (1874 to 1949), the learning theorist of Chapter 6, rejected g. For him intelligence was the product of a vast number of separate stimulus-response connections, and a person was more intelligent simply by having more of them. This **multifactor theory** is sometimes called the "anarchic" theory, because it allows no governing general factor. Thorndike distinguished three broad kinds of intelligence: **abstract intelligence** (dealing with ideas and symbols), **mechanical intelligence** (also called concrete intelligence: dealing with things and tools) and **social intelligence** (dealing with people). In 1920 he defined social intelligence as the ability to understand and manage men and women, boys and girls, and to act wisely in human relations. His later test, the CAVD (1927), combined four kinds of item, sentence Completion, Arithmetic, Vocabulary and following Directions, and he described intelligence along four dimensions: level (the difficulty of tasks one can solve), range (the number of tasks one can solve at a given level), area (the total of the ranges across levels) and speed.

A related idea came from the English-born psychologist **Godfrey Hilton Thomson** (1881 to 1955), who spent most of his career at Newcastle and then Edinburgh. His **sampling theory** (first proposed in 1916 and developed in *The Factorial Analysis of Human Ability*, 1939) showed that the positive manifold does not prove that a single g exists. Suppose the mind contains a huge number of small, independent "bonds" or elementary abilities, and every test calls on a random sample of them. Any two tests will then share some bonds by chance, and so will correlate, even though nothing general exists. Correlation among tests, Thomson pointed out, could come from overlapping samples rather than from one common cause. Versions of this argument are still debated today.

### Thurstone's Primary Mental Abilities

**Louis Leon Thurstone** (1887 to 1955) of the University of Chicago developed a more powerful method, multiple factor analysis, designed to find several factors at once. In *Primary Mental Abilities* (1938) he gave 56 tests to 240 university students and extracted seven factors that he called the **primary mental abilities**: verbal comprehension (V), word fluency (W), number (N), space (S), associative memory (M), perceptual speed (P) and reasoning, also called induction (R or I). Thurstone argued that these seven were distinct abilities, not reflections of g, and he built the Primary Mental Abilities tests (1941, with his wife Thelma Gwinn Thurstone) to give each person a profile across them. Because it replaces one ruler with a small number of group factors, his view is called a **group-factor theory** (sometimes "oligarchic", rule by a few).

The story has a twist. When the Thurstones tested schoolchildren, whose abilities are less differentiated than those of university students, the seven factors correlated substantially with one another. And a set of correlated factors can itself be factor-analysed, yielding a "second-order" general factor. Spearman and Thurstone were, in the end, both partly right: there are several distinct broad abilities, and there is something general that they share. The hierarchical models below formalise that compromise.

EXAM TRAP

Count carefully. Thurstone: **seven** primary mental abilities (1938). Gardner: **seven** intelligences in 1983, **eight** after the naturalist was added in 1999. Do not mix the lists: "spatial" appears in both, but "word fluency", "perceptual speed" and "associative memory" are Thurstone's, while "musical", "bodily-kinaesthetic" and "intrapersonal" are Gardner's. Thurstone is also the Thurstone of attitude scaling (the method of equal-appearing intervals) in the Social book. In the Indian textbook classification, the unifactor or "monarchic" theory is Binet's (one general ability), the "oligarchic" theory is Thurstone's (a few group factors) and the "anarchic" theory is Thorndike's (many independent abilities).

## Guilford's Structure of Intellect ●●

**Joy Paul Guilford** (1897 to 1987), who had worked on the selection of aircrew during the Second World War, went furthest in the direction of many abilities. His **structure of intellect (SOI) model**, set out fully in *The Nature of Human Intelligence* (1967), classifies every intellectual task along three dimensions, pictured as a cube (Figure 9.3):

- **Operations**, the kind of mental process involved (five): cognition, memory, **divergent production** (generating many possible answers to an open problem, the basis of creativity), **convergent production** (arriving at the one correct answer) and evaluation.
- **Contents**, the kind of material the process works on (four): figural, symbolic, semantic and behavioural.
- **Products**, the form the information takes (six): units, classes, relations, systems, transformations and implications.

Every combination of one operation, one content and one product defines a separate ability, so the model contains 5 × 4 × 6 = 120 abilities. For example, knowing what words mean is the cognition of semantic units; thinking up as many uses as possible for a brick is the divergent production of semantic units. Guilford later split the figural content into visual and auditory, giving 5 × 5 × 6 = 150 abilities (sources date this to 1977 or 1982), and in 1988 he split memory into memory recording and memory retention, giving 6 × 5 × 6 = 180.

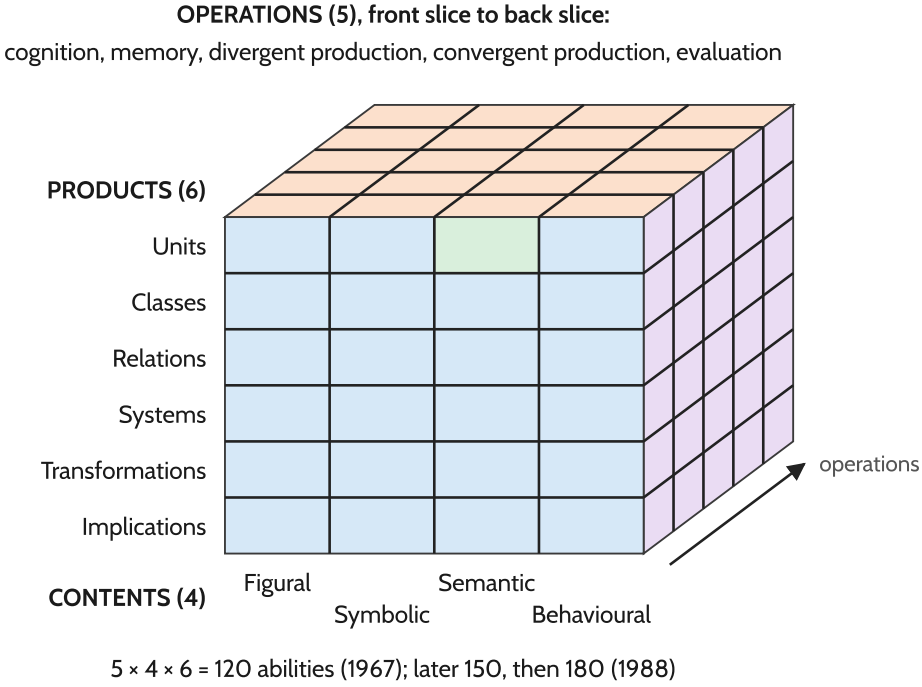


**Figure 9.3** Guilford's structure of intellect. Each small cell is one ability defined by an operation, a content and a product. The front slice shown is cognition; the shaded cell is the cognition of semantic units (CMU), roughly vocabulary knowledge.

Guilford's lasting legacy is the distinction between convergent and divergent thinking, which launched the psychometric study of creativity (Chapter 8). The model itself fared less well. Critics showed that his factor-analytic method could produce "evidence" for almost any structure the researcher had in mind, and that many of the supposedly independent abilities in fact correlate positively, as Spearman would have predicted.

> **EXAM TRAP**
>
> Guilford's model has three *dimensions* (operations, contents, products) but 120 *abilities* (1967), later 150 and then 180. Strictly, then, the model has three dimensions and 120 abilities. One paper (RML 2021) asked "how many dimensions of intelligence" with both "three" and "120" among the options and keyed 120, using "dimensions" loosely to mean abilities. If that exact question returns, 120 is the answer its setters expect; but if a question asks you to name the dimensions, faces or parameters of the cube, the answer is three (operations, contents, products), and if it asks for abilities or factors, it is 120 (or 150 or 180 if the question names the later version). The number of products is six, of operations five, of contents four (later five).

## Hierarchical Models: Burt and Vernon

British psychologists reconciled g with group factors by arranging abilities in a hierarchy, like the branches of a tree. **Cyril Lodowic Burt** (1883 to 1971) proposed one such model in 1949, and **Philip Ewart Vernon** (1905 to 1987) produced the best-known version in *The Structure of Human Abilities* (1950). In Vernon's **hierarchical model**, g sits at the top. Below it are two **major group factors**: **v:ed** (verbal-educational ability) and **k:m** (spatial, mechanical and practical ability). Below these come **minor group factors**: verbal fluency and numerical

ability under v:ed; spatial, mechanical-information and psychomotor abilities under k:m. At the bottom are the specific factors of individual tests. Vernon's model is essentially Spearman's g and s with group factors inserted between them.

## Fluid and Crystallised Intelligence ●●●

**Raymond Bernard Cattell** (1905 to 1998), a British-born student of Spearman who spent his career in the United States, divided g itself into two. He first sketched the idea in the early 1940s and set it out fully in 1963; his student **John Leonard Horn** (1928 to 2006) developed it with him from 1966. **Fluid intelligence (Gf)** is the capacity to reason, see relationships and solve novel problems independently of previously acquired knowledge; it is what you use the first time you meet a puzzle. **Crystallised intelligence (Gc)** is the repertoire of information, cognitive skills and strategies acquired through education and experience: vocabulary, general knowledge, learned methods. Cattell's **investment theory** (1971) linked the two: Gf is "invested" in learning experiences, and the knowledge that results "crystallises" as Gc, rather as water freezes into ice.

The two can be told apart by how they are measured and by how they change across life (Figure 9.4). Gf is measured by Raven's matrices, matrix reasoning, series completion and other novel-problem tests; Gc by vocabulary, information and comprehension subtests. Gf rises through childhood, peaks in the early twenties and then declines gradually. Gc goes on rising, or at least holds steady, well into the sixties, because people continue to learn. (Note that block design and coding, sometimes offered as options, are measures of visual-spatial ability and processing speed, not of Gc.)

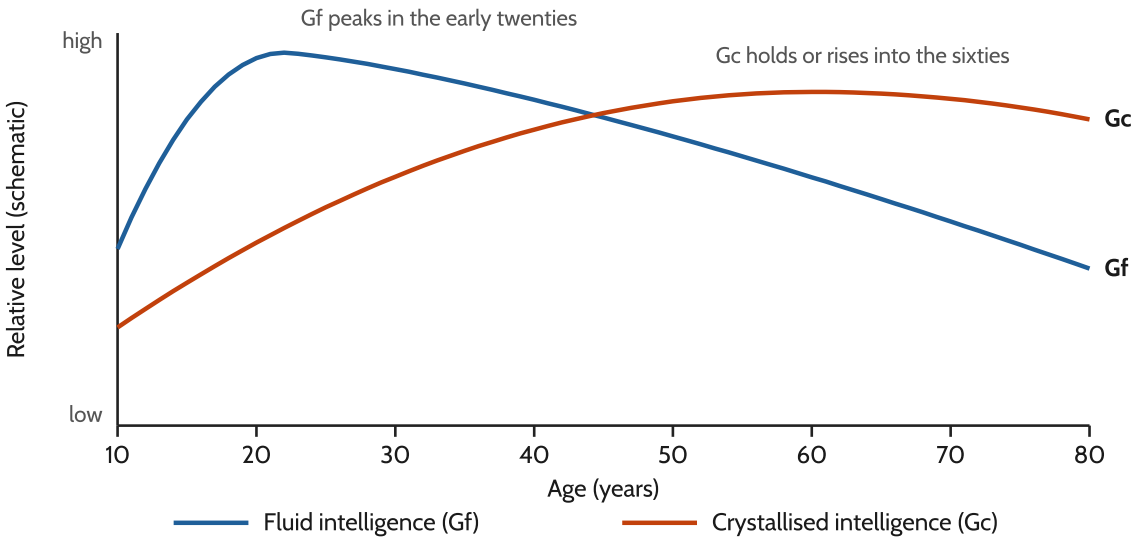


Figure 9.4 A schematic of the typical lifespan pattern for fluid and crystallised intelligence, after Horn and Cattell. Fluid reasoning peaks early and declines; accumulated knowledge holds up or grows into later life. The curves show the general shape only, not actual test scores.

Horn extended the model to nine or ten broad abilities, adding to Gf and Gc visual processing (Gv), auditory processing (Ga), short-term memory (Gsm), long-term storage and retrieval (Glr), processing speed (Gs), decision or reaction speed, quantitative knowledge (Gq) and reading-writing (Grw). Unlike Spearman, Horn did not believe in a single g above them.

EXAM TRAP

Crystallised = a repertoire of information, cognitive skills and strategies; accumulated knowledge and verbal skill; measured best by vocabulary and information. Fluid = the ability to deal with new problems; measured by matrices. "Practical" and "successful" intelligence are Sternberg's terms; "speed of neural processing" and "working memory capacity" are related to, but not the definition of, either. The Gf and Gc theory belongs to *Raymond* Cattell (with Horn), not James McKeen Cattell.

## Carroll's Three Strata and the CHC Model

The most thorough factor-analytic study ever made was the work of **John Bissell Carroll** (1916 to 2003). In *Human Cognitive Abilities* (1993) he re-analysed more than 460 data sets collected over most of the twentieth century. He concluded that abilities form three layers, or strata (Figure 9.5). **Stratum III** is a single general factor, g. **Stratum II** contains eight broad abilities: fluid intelligence, crystallised intelligence, general memory and learning, broad visual perception, broad auditory perception, broad retrieval ability, broad cognitive speediness and processing (decision) speed. **Stratum I** contains about 70 narrow abilities, such as induction, lexical knowledge, memory span and visualisation. This is the **three-stratum theory**.

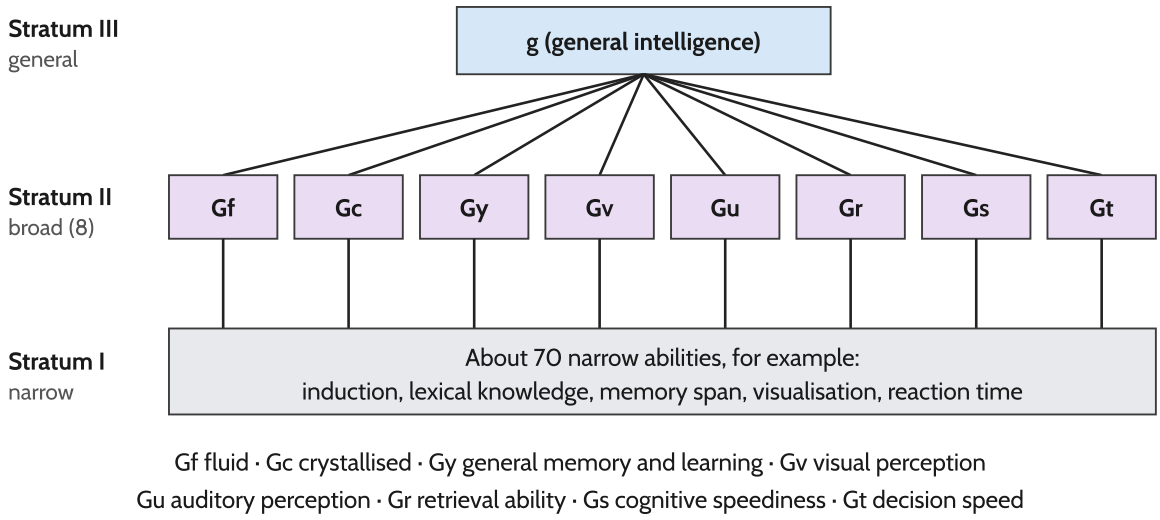


Figure 9.5 Carroll's three-stratum theory (1993). A general factor sits above eight broad abilities, which in turn sit above about 70 narrow ones. Vernon's earlier hierarchy has the same shape, with two major group factors (v:ed and k:m) in place of Carroll's eight.

Because Carroll's broad abilities matched Cattell and Horn's so closely, the psychologist Kevin McGrew merged the two models in the late 1990s into the **Cattell-Horn-Carroll (CHC) theory**. CHC keeps Carroll's g at the top (while noting Horn's doubts about it), recognises roughly ten broad abilities in the middle, and many narrow abilities at the bottom. It is now the dominant framework for test construction: the Woodcock-Johnson batteries, the SB5, the Kaufman tests and the newer Wechsler indices are all organised, more or less explicitly, around CHC abilities.

### Jensen's Level I and Level II Abilities

The American **Arthur Robert Jensen** (1923 to 2012), a champion of g, proposed a simpler two-level scheme. **Level I abilities** are associative: rote learning, memory span (repeating digits) and simple association, with little transformation of the input. **Level II abilities** are conceptual: abstract reasoning, problem solving and the manipulation of concepts, essentially g. Jensen reported that Level I abilities were fairly evenly distributed across social classes and ethnic groups, while groups differed more on Level II tasks.

The scheme is famous because of where he published it. His 1969 article in the *Harvard Educational Review*, "How Much Can We Boost IQ and Scholastic Achievement?", opened with the sentence "Compensatory education has been tried and it apparently has failed." He went on to argue that the heritability of IQ was high and that genetic factors might be partly responsible for the average difference in IQ scores between Black and White Americans. The article provoked a storm of protest (the word "Jensenism" was coined for his position) and one of the longest-running debates in psychology. You will see in Section 9.7 why the step from "heritable within a group" to "genetic differences between groups" is a logical error, and why later research on compensatory education is more encouraging than Jensen claimed. Table 9.8 sets all the factor and structural theories of this section side by side.

Table 9.8 Factor and structural theories of intelligence at a glance.

| Theorist | Year | Theory | Key structure |
|---|---|---|---|
| Charles Spearman | 1904 | Two-factor theory | g (general) plus s (specific) |
| Edward Thorndike | 1920s | Multifactor ("anarchic") theory | Abstract, mechanical and social intelligence; CAVD; no g |
| Godfrey Thomson | 1916, 1939 | Sampling theory | Tests sample overlapping sets of many small bonds |
| L. L. Thurstone | 1938 | Primary mental abilities (group factors) | V, W, N, S, M, P, R: seven abilities |
| J. P. Guilford | 1967 | Structure of intellect | 5 operations × 4 contents × 6 products = 120 (later 150, 180) |
| Burt; P. E. Vernon | 1949; 1950 | Hierarchical model | g; v:ed and k:m; minor group; specific |
| R. B. Cattell and J. L. Horn | 1963; 1966 | Fluid and crystallised (Gf-Gc) | Gf novel reasoning; Gc acquired knowledge; investment theory |
| Arthur Jensen | 1969 | Level I and Level II | Associative versus conceptual abilities |
| J. B. Carroll | 1993 | Three-stratum theory | g; 8 broad; about 70 narrow |
| McGrew (merging the above) | late 1990s | CHC theory | g; about 10 broad; many narrow |

## 9.5 Broader Views: Multiple, Triarchic, PASS and Emotional Intelligence

MEDIUM YIELD · 5 QS

All the theories of Section 9.4 were built the same way: give people a pile of tests, correlate the scores and factor-analyse the correlations. That method has a built-in limit. It can only find structure among the abilities that someone thought to test, and almost everyone tested

the abilities that school rewards. From the 1980s onwards a group of theorists asked what a theory of intelligence would look like if it started somewhere else: from the brain, from the study of experts and prodigies, from how people cope with real life, or from the emotions.

## Gardner's Multiple Intelligences

**Howard Gardner** (born 1943), a Harvard developmental psychologist, spent years in two very different settings: studying artistic development in children with Harvard's Project Zero, and working with adults who had suffered strokes at a Boston aphasia research unit. He was struck by how selectively the brain fails. A stroke could destroy a man's language and leave his musical or spatial abilities intact; a child could be a prodigy at music and ordinary at everything else. In *Frames of Mind* (1983) he proposed that there is no single intelligence but several relatively independent ones. This is the theory of **multiple intelligences**.

Gardner did not use factor analysis. Instead he set eight criteria that a candidate intelligence should meet: it can be isolated by brain damage; it is shown in exceptional form by savants, prodigies or other exceptional individuals; it has an identifiable core operation; it has a distinctive developmental history ending in expert performances; it has a plausible evolutionary history; it is supported by experimental tasks; it is supported by psychometric findings; and it can be encoded in a symbol system (language, number, musical notation). Seven abilities met his criteria in 1983, and an eighth was added in *Intelligence Reframed* (1999), as Table 9.9 shows.

Table 9.9 Gardner's multiple intelligences.

| Intelligence | Core ability | An exemplar |
|---|---|---|
| Linguistic | Sensitivity to the sounds, meanings and order of words | A poet such as Rabindranath Tagore |
| Logical-mathematical | Handling chains of reasoning, numbers and abstract patterns | A mathematician such as Srinivasa Ramanujan |
| Spatial | Perceiving the visual world accurately and transforming it mentally | An architect, navigator or chess player |
| Musical | Sensitivity to pitch, rhythm and timbre; producing music | A musician such as Ravi Shankar |
| Bodily-kinaesthetic | Skilled control of body movement and handling of objects | A dancer, surgeon or batsman such as Sachin Tendulkar |
| Interpersonal | Understanding other people's moods, motives and intentions | A leader, teacher or therapist; Gardner's example was Mahatma Gandhi |
| Intrapersonal | Access to one's own feelings and an accurate model of oneself | A reflective, self-aware person; Gardner's example was Sigmund Freud |
| Naturalist (added 1999) | Recognising and classifying plants, animals and natural patterns | An ornithologist such as Salim Ali |
| Existential (candidate only) | Concern with ultimate questions of life and death | Gardner counts it as "half" an intelligence at most |

Gardner considered and rejected a "spiritual" intelligence; existential intelligence is the nearest he came to admitting one, and he has never fully accepted it. The theory was enormously popular with teachers, because it promised that every child is intelligent in some way, and schools around the world, including many in India, adopted "multiple intelligence" lesson

plans. Researchers were cooler. The criticisms are worth knowing. First, Gardner offered little direct empirical test of the theory; the intelligences were argued for, not measured. Second, when researchers did build tests for them (for example Beth Visser, Michael Ashton and Philip A. Vernon in 2006; this is a different Vernon from the hierarchical theorist of Section 9.4), most of the "cognitive" intelligences correlated substantially with one another and with g, which is the opposite of independence; only the bodily-kinaesthetic and, partly, musical abilities stood apart. Third, critics argued that musical and bodily abilities are better called talents; calling them intelligences changes the word, not the facts.

> **EXAM TRAP**
>
> Gardner's book is *Frames of Mind* (1983): seven intelligences; naturalist added in 1999 to make eight; existential considered, not fully accepted; spiritual rejected. "Linguistic, logical-mathematical and spatial" is a Gardner list, never a Sternberg or Cattell one. Gardner used no factor analysis, so a statement that he "derived the intelligences by factor analysis" is false.

### Sternberg's Triarchic Theory and Successful Intelligence ●●●

Robert Jeffrey Sternberg (born 1949) has a personal reason to distrust IQ tests: as a primary-school pupil he suffered from such severe test anxiety that he scored poorly on group IQ tests, until a teacher who expected more of him changed how he performed. His **triarchic theory of intelligence**, set out in *Beyond IQ* (1985), has three parts, or subtheories (Table 9.10).

The **componential subtheory** describes the mental mechanisms of intelligent thought, the components that analytical intelligence uses. There are three kinds. **Metacomponents** are executive processes that plan what to do, monitor it while it is being done and evaluate it afterwards. **Performance components** carry out the plan: encoding the problem, inferring relations, applying them. **Knowledge-acquisition components** learn new information: selective encoding (picking out the relevant facts), selective combination (putting them together) and selective comparison (relating them to what you already know). This is the kind of intelligence school tests measure best, and Sternberg calls the resulting ability **analytical**.

The **experiential subtheory** concerns how intelligence relates to experience. Intelligence is shown most clearly at two points: when a task is new, where it shows as the ability to cope with novelty and produce insight, and when a task is becoming familiar, where it shows as the ability to automatise performance so that attention is freed for other things. A person strong here has **creative** intelligence.

The **contextual subtheory** relates intelligence to the external world. Intelligent behaviour is purposive **adaptation** to one's environment, **shaping** the environment to suit oneself and, when neither works, **selection** of a better environment. A clerk who learns the unwritten rules of her office adapts; one who persuades her manager to change a pointless procedure shapes; one who leaves for a better job selects. The resulting ability is **practical** intelligence, "street smarts".

Table 9.10 Sternberg's triarchic theory (1985).

| Subtheory | Ability (later name) | What it involves | Everyday example |
|---|---|---|---|

| Subtheory | Ability (later name) | What it involves | Everyday example |
|---|---|---|---|
| Componential | Analytical | Metacomponents, performance components, knowledge-acquisition components | Solving a textbook problem; analysing an argument |
| Experiential | Creative | Coping with novelty; automatising the familiar | Inventing a new solution; insight |
| Contextual | Practical | Adapting to, shaping and selecting environments | Managing a shop, an office or a family budget |

In the late 1990s Sternberg extended the triarchic theory into the theory of **successful intelligence**: the ability to achieve success in life, by one's own standards and within one's culture, by capitalising on strengths and correcting or compensating for weaknesses through a balance of analytical, creative and practical abilities. Central to practical intelligence is **tacit knowledge**, the practical, usually unspoken know-how that is rarely taught directly but is needed to succeed in a particular setting; Richard Wagner and Sternberg (1985) measured it in business managers and found it predicted their success independently of IQ. A striking illustration came from Brazil, where Terezinha Carraher, David Carraher and Analúcia Schliemann (1985) studied children who sold coconuts and snacks in street markets. The children did complex sums in their heads almost flawlessly while selling, but often failed the same sums presented as school arithmetic. A chaiwala's son who never misses a sale can likewise look "weak in maths" on a school test. Critics, notably Linda Gottfredson (2003), argue that tests of practical intelligence correlate with g more than Sternberg admits and that the evidence for a separate practical ability is thin.

> EXAM TRAP
>
> Componential = analytical; experiential (not "experimental") = creative; contextual = practical. The ability to adapt to the present environment, select a better one or modify (shape) it is *contextual*. Sternberg's three are analytical, creative and practical, never "verbal, performance and full scale" (Wechsler), "fluid, crystallised and general" (Cattell and Spearman) or "linguistic, logical-mathematical and spatial" (Gardner).

## The PASS Model

A third alternative started from neuropsychology. The Soviet neuropsychologist **Alexander Romanovich Luria** (1902 to 1977), studying soldiers with brain wounds, described the brain as three "functional units". The first, based in the brainstem and its reticular activating system, regulates arousal and attention. The second, in the occipital, parietal and temporal lobes, receives, processes and stores information. The third, in the frontal lobes, programmes, regulates and verifies behaviour.

Building on Luria, **Jagannath Prasad Das** (born 1931), an Odisha-born psychologist who spent most of his career at the University of Alberta in Canada, together with **Jack A. Naglieri** and **John R. Kirby**, proposed the **PASS model** of intelligence (set out in their 1994 book). PASS stands for four cognitive processes:

- **Planning**: deciding on, carrying out and checking strategies for solving problems (Luria's third unit, the frontal lobes).
- **Attention**: focused, selective and sustained attention while resisting distraction, with the necessary arousal (Luria's first unit).

- **Simultaneous processing**: integrating separate pieces of information into a whole that can be seen at once, as in understanding a map, a matrix or a sentence like "the boy is taller than the girl" (Luria's second unit, chiefly occipital-parietal).
- **Successive processing**: dealing with information in a strict serial order, as in repeating a phone number, following a recipe or sounding out a word (also Luria's second unit, chiefly fronto-temporal).

Naglieri and Das turned the model into a test, the **Cognitive Assessment System (CAS)** (1997; second edition 2014), for school-age children. Because it measures processes rather than acquired knowledge, it shows smaller differences between ethnic and social groups than traditional IQ tests and is designed to link assessment to remediation; Das's PASS Reading Enhancement Programme (PREP) trains successive and simultaneous processing in children with reading difficulties. The Kaufman Assessment Battery for Children (1983) was also built on Luria's distinction between simultaneous and sequential processing.

> INDIAN CONTEXT
>
> The PASS model is one of the few internationally influential theories of intelligence with an Indian-born principal author. J. P. Das's work in the 1970s on simultaneous and successive syntheses led directly to the model, and he also wrote on Indian conceptions of intelligence such as *buddhi* (Section 9.1). Questions sometimes ask you to expand PASS; it is Planning, Attention, Simultaneous, Successive.

### Ceci's Bioecological View

The American developmental psychologist **Stephen J. Ceci** argued in 1990 that intelligence is not one general ability but a set of multiple cognitive potentials, each of which develops only when the context (the knowledge domain, the setting, the culture) and the person's motivation call it out. His **bioecological model** takes its name from Urie Bronfenbrenner's ecological systems theory (the Developmental book). His favourite evidence came from a 1986 study with Jeffrey Liker of expert racetrack handicappers, men who predicted the odds on horse races. The best of them reasoned using a complex model that weighed many interacting variables at once, a feat of cognitive complexity, yet their IQ scores bore no relation to their expertise. Complex thinking, Ceci concluded, depends on knowledge and context, not on IQ alone.

### Emotional Intelligence ●

Thorndike's social intelligence of 1920 and Gardner's two personal intelligences were the ancestors of one of the most popular ideas in modern psychology. In 1990 **Peter Salovey** and **John D. Mayer** introduced the term **emotional intelligence (EI)** in its modern sense, defining it as the ability to monitor one's own and others' feelings and emotions, to discriminate among them and to use this information to guide one's thinking and actions. In 1997 they refined it into the **four-branch model**, which treats EI as a genuine ability, like other intelligences:

1. **Perceiving emotions** accurately in oneself, in others' faces and voices, and in art.
2. **Using emotions to facilitate thought**, for example harnessing a good mood for creative work.

3. **Understanding emotions**: how they blend, change and follow from events.
4. **Managing emotions** in oneself and others.

Their test, the Mayer-Salovey-Caruso Emotional Intelligence Test (MSCEIT, 2002), scores answers against expert or consensus judgements, just as an ability test should.

The idea reached the public through the science journalist **Daniel Goleman** (born 1946), whose bestseller *Emotional Intelligence: Why It Can Matter More Than IQ* (1995) described five domains: knowing one's own emotions, managing one's emotions, motivating oneself, recognising emotions in others, and handling relationships. In *Working with Emotional Intelligence* (1998) he relabelled them self-awareness, self-regulation, motivation, empathy and social skills. Goleman's version is a **mixed model**, because it mixes emotional abilities with personality traits and motivation such as optimism and persistence. Another mixed model came from **Reuven Bar-On**, whose self-report Emotional Quotient Inventory (EQ-i, 1997) measures intrapersonal, interpersonal, adaptability, stress-management and general-mood scales. Finally, Konstantinos Petrides and Adrian Furnham (around 2000) argued that self-report questionnaires cannot measure an ability at all, only how people see their own emotional dispositions; they renamed this **trait emotional intelligence**, a personality trait measured by questionnaires such as the TEIQue. In India, the Emotional Intelligence Scale of Anukool Hyde, Sanjyot Pethe and Upinder Dhar (2002) is widely used in research. Table 9.11 compares the four approaches.

Table 9.11 Models of emotional intelligence.

| Model | Authors and year | Components | Measure |
|---|---|---|---|
| Ability model | Salovey and Mayer, 1990; four branches 1997 | Perceiving, using, understanding, managing emotions | MSCEIT (performance test) |
| Mixed model | Goleman, 1995 (relabelled 1998) | Self-awareness, self-regulation, motivation, empathy, social skills | Competency inventories |
| Mixed model | Bar-On, 1997 | Intrapersonal, interpersonal, adaptability, stress management, general mood | EQ-i (self-report) |
| Trait model | Petrides and Furnham, about 2000 | Emotion-related self-perceptions (a personality trait) | TEIQue (self-report) |

How good is the evidence? Ability EI, measured by the MSCEIT, is only modestly correlated with IQ and the Big Five, so it adds something new, and it predicts the quality of social relationships and some aspects of work performance, though modestly. Mixed and trait measures overlap heavily with personality (especially neuroticism, reversed, and extraversion), so much of what they predict is personality under a new name. And Goleman's claim that EI "can matter more than IQ" has not been supported: across most outcomes, including job performance, general mental ability remains the stronger predictor.

EXAM TRAP

Salovey and Mayer coined the modern concept (1990); Goleman popularised it (1995). Goleman's five parts in the 1995 wording are knowing one's emotions, managing emotions, motivating oneself, recognising emotions in others and handling relationships; the 1998 labels (self-awareness, self-regulation, motivation, empathy, social skills) describe the same five, so if both appear as options, match the wording the question uses. "IQ, EQ, SQ, CQ" is not anyone's model of EI.

## Social, Practical, Spiritual and Other Intelligences

Once intelligence was allowed to be plural, the list grew. Social intelligence, Thorndike's ability to understand and manage people, was measured as early as 1927 by the George Washington Social Intelligence Test, and appears as Guilford's behavioural content and Gardner's interpersonal intelligence. Practical intelligence and tacit knowledge belong to Sternberg. **Spiritual intelligence** was proposed by the physicist-philosopher **Danah Zohar** and the psychiatrist Ian Marshall in *SQ: Spiritual Intelligence, the Ultimate Intelligence* (2000), as the capacity to address problems of meaning and value; the psychologist Robert Emmons made a similar case the same year. Cultural intelligence (CQ), the ability to function effectively across cultures, was proposed by P. Christopher Earley and Soon Ang (2003). These ideas have practical uses, but none has the measurement base of g.

EXAM TRAP

Beware the "quotients". IQ is the intelligence quotient; DQ the developmental quotient of infant scales; EQ the emotional quotient (Bar-On's term); CQ cultural intelligence. "SQ" is ambiguous: in the Vineland Social Maturity Scale it is the *social* quotient (Section 9.8), but in Zohar and Marshall it is *spiritual* intelligence. Read the context.

SCAN YOUR BRAIN

Fill in each blank with the most appropriate term. The answers are below; cover them first.

1. The pattern in which all mental tests correlate positively with one another is called the _____.
2. Thurstone's seven group factors are called the _____.
3. Guilford's original structure-of-intellect model contained _____ abilities.
4. In Vernon's hierarchy the two major group factors are v:ed and _____.
5. Cattell described crystallised intelligence as a repertoire of information, cognitive skills and _____.
6. Carroll's three-stratum theory (1993) places about _____ narrow abilities at Stratum I.
7. Jensen's Level _____ abilities are associative, such as rote memory and digit span.
8. The intelligence Gardner added in 1999 is the _____ intelligence.
9. In Sternberg's theory, adapting to, shaping and selecting environments make up the _____ subtheory.
10. In the PASS model, the two S's stand for simultaneous and _____ processing.

**Answers:** (1) positive manifold, (2) primary mental abilities, (3) 120, (4) k:m, (5) strategies, (6) 70, (7) I, (8) naturalist, (9) contextual, (10) successive.

## 9.6 Intelligence, Speed and the Brain LOWER YIELD · NOT ASKED YET

Factor theories describe the structure of abilities; they say nothing about what goes on in the head when someone solves a problem. From the 1970s, psychologists began asking a different question: which basic information-processing operations distinguish people who score high on IQ tests from those who score low? Galton, you may recall, had bet on simple speed and lost. A century later his bet was partly vindicated.

### Speed of Processing: Reaction Time and Inspection Time

In the **cognitive-correlates approach**, researchers measure performance on simple laboratory tasks and correlate it with IQ. Earl Hunt and his colleagues (1975) showed, for example, that high-verbal university students retrieved the names of letters from memory faster than low-verbal students. Arthur Jensen pursued **reaction time** (RT). Using a panel of lights and buttons, he measured how quickly people lifted a finger from a home button and pressed the button beside whichever light came on, with one to eight possible lights. Choice reaction time rises as the number of alternatives grows (Hick's law), and Jensen found that higher-IQ people were somewhat faster and, even more clearly, less variable from trial to trial. The correlations are modest, typically about −.2 to −.4 (negative because faster times go with higher IQ), but they are remarkable given that the task involves no knowledge at all.

An even purer measure is **inspection time**, introduced by Ted Nettelbeck and M. Lally in 1976. Two vertical lines of clearly different length are flashed for a few hundredths of a second and then covered by a mask; the person simply says which line was longer. Inspection time is the briefest exposure at which the person can still judge accurately. Shorter inspec-

tion times go with higher IQ: across many studies the correlation is about −.3 before correction and about −.5 after statistical correction for unreliability and restricted range. Speed of intake of very simple information, in other words, is part of what IQ tests capture.

### Working Memory

The strongest cognitive correlate of intelligence is **working memory capacity**, the ability to hold information in an active state while processing it (Chapter 7). In an influential 1990 paper, Patrick Kyllonen and Raymond Christal asked whether reasoning ability is "little more than working-memory capacity" after finding very high correlations between the two at the level of latent factors. Later studies put the relationship lower, but working memory remains closely tied to fluid intelligence and much less to crystallised intelligence, which depends on stored knowledge. One reason is that solving a novel problem, such as a Raven's matrix, requires you to hold several partial rules in mind at once while testing them; lose one and the solution collapses. This is why working memory now has its own index on the Wechsler scales and its own factor on the SB5.

### Brain Size, Neural Efficiency and Genes

Are bigger brains smarter? Nineteenth-century craniologists thought so and were famously biased in how they measured skulls. Modern magnetic resonance imaging (MRI) studies of living people find a real but modest positive correlation between total brain volume and IQ, about .2 to .4 depending on the meta-analysis. Brain size explains only a small part of the variation, and it says nothing about differences between groups, whose average head sizes differ for many reasons including body size and nutrition.

How a brain works may matter more than how big it is. In 1988 Richard Haier and his colleagues used positron emission tomography (PET) to measure how much glucose people's brains consumed while they solved Raven's Advanced Progressive Matrices. Surprisingly, the people who scored highest used the *least* glucose: their brains seemed to work less hard to solve the same problems. In a follow-up, people who practised the computer game Tetris for several weeks became better at it while their brains' glucose use fell. This is the **neural efficiency hypothesis**: more intelligent brains accomplish a task with less neural effort. The picture is not simple (on very difficult tasks, high scorers sometimes show more activation, because they keep working when others give up), and in 2007 Rex Jung and Haier summarised imaging studies in the parieto-frontal integration theory, which locates the networks most consistently linked to intelligence in the frontal and parietal lobes and the white-matter tracts between them.

Finally, the genes. Modern genome-wide studies of hundreds of thousands of people show that intelligence is massively **polygenic**: it is influenced by thousands of genetic variants, each with a tiny effect. There is no "gene for intelligence". Earlier searches used *linkage studies*, which track whether a trait travels through families together with known stretches of DNA; that method works well for single-gene disorders such as Huntington's disease but found nothing reliable for intelligence, precisely because no single gene has a large effect. How such genes combine with environments is the subject of the next section.

> **REMEMBER**
>
> Correlates are not causes. Faster reaction time, shorter inspection time, larger working memory, a slightly larger brain and more efficient glucose use all go with higher IQ, but each could be a cause, an effect (years of schooling and reading change the brain) or a shared result of something else, such as better health and nutrition.

## 9.7 Heredity and Environment

LOWER YIELD · 2 QS

No question in psychology has generated more heat than whether differences in intelligence are inherited. The heat comes from politics; the light comes from a small number of research designs that separate the effects of genes from the effects of environment. Learn the designs first, then the one statistic, heritability, that is most often misunderstood.

### Family and Twin Studies

The simplest evidence comes from **family studies**: the more closely two people are related, the more similar their IQs. But relatives share homes as well as genes, so family resemblance alone cannot separate the two. **Twin studies**, first suggested by Galton in 1875, do better. **Monozygotic (MZ) twins** develop from a single fertilised egg and share virtually all their genes; **dizygotic (DZ) twins** develop from two eggs and share on average half their genes, like ordinary siblings. If both kinds of twins grow up together, their environments are about equally similar, so if MZ twins are more alike than DZ twins, the extra similarity is attributed to their extra genetic similarity. The most powerful version studies twins who were separated early in life and reared apart, because any similarity between them cannot come from a shared home.

In 1981 Thomas Bouchard and Matt McGue pooled 111 studies of family resemblance in IQ. Their weighted averages, shown in Figure 9.6, form a clear gradient: the more genes two people share, the more similar their IQs, and for the same degree of relationship, people reared together are more alike than people reared apart. Both genes and shared environment are visible in the same chart.

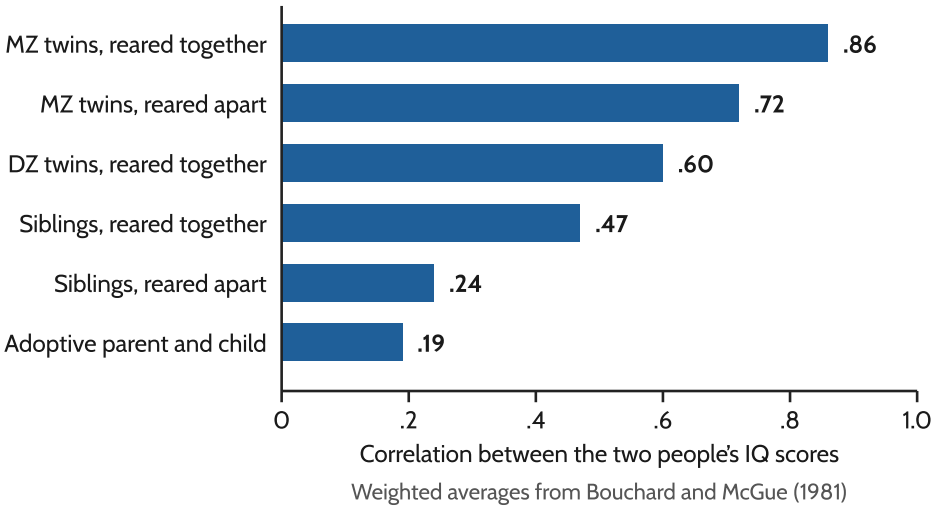


**Figure 9.6** Correlations between the IQ scores of relatives. Identical twins reared apart are more alike than fraternal twins reared together, which points to genes; siblings reared together are more alike than siblings reared apart, and adoptive parents and children resemble each other a little, which points to shared environment.

### KEY STUDY: THE MINNESOTA STUDY OF TWINS REARED APART (FROM 1979)

**Question.** Would identical twins raised in different families turn out alike or different? **Method. Thomas J. Bouchard Jr.** and his colleagues at the University of Minnesota located twins who had been separated in infancy and brought each pair to Minneapolis for about 50 hours of testing on intelligence, personality, interests and health. Pinel and Barnes report 59 pairs of MZ twins and 47 pairs of DZ twins reared apart, aged 19 to 68, plus many twins reared together. **Result.** The MZ twins reared apart correlated about .70 on IQ, nearly as high as MZ twins reared together, and in their 1990 report the team concluded that about 70 per cent of the variance in IQ in their sample was associated with genetic variation. The project became famous for anecdotes such as the "Jim twins", separated at a few weeks of age and reunited at 39, who had both married women named Linda and then women named Betty and had both named a son James Alan or James Allan. **Criticism.** Anecdotes are not data; coincidences are guaranteed when you compare two lives in enough detail. More seriously, many "separated" twins were placed with relatives or similar families, some had met before testing, and all were reared in industrialised countries by families screened for adoption, so the range of environments was narrow. As you will see, a narrow range of environments inflates heritability.

Twin research also carries a warning from its history. The British psychologist Cyril Burt, whose hierarchical model you met in Section 9.4, reported very high IQ correlations for MZ twins reared apart from the 1940s to the 1960s. After his death in 1971, Leon Kamin (1974) noticed that Burt's correlations stayed identical to three decimal places (.771) as his sample supposedly grew, a statistical near-impossibility, and that his collaborators could not be traced. Most historians concluded that the later data were fraudulent, although a few writers have defended him. Modern estimates do not depend on Burt: the independent studies agree with one another.

### Adoption Studies

**Adoption studies** compare adopted children with their biological parents, who gave them their genes but not their upbringing, and with their adoptive parents, who gave them their upbringing but not their genes. Large projects such as the Texas Adoption Project and the Colorado Adoption Project (begun in 1975 by Robert Plomin and John DeFries) produced a consistent and at first surprising result. As adopted children grow older, their IQs come to resemble those of their biological parents more and more, while the resemblance to their adoptive parents fades. Sandra Scarr and Richard Weinberg (1978) found that the IQ correlation between unrelated adolescents reared in the same adoptive home was close to zero.

But adoption studies also show something just as important: adoption changes the *level* of IQ. In a French cross-fostering study, Christiane Capron and Michel Duyme (1989) found that both the social class of the biological parents and that of the adoptive parents made a substantial difference to children's IQ. In another French study, Duyme and his colleagues (1999) followed children with low IQs who were adopted between the ages of 4 and 6; by adolescence, those adopted into high-status homes had gained about 20 points and those adopted into low-status homes about 8. The two findings are not in conflict. Correlations describe who ranks above whom; means describe how high everyone is. Genes can strongly influence the rank order of children within a home while the home raises or lowers the level of all of them.

### Heritability: What It Is and What It Is Not ●●

**Heritability** (written $h^2$) is the proportion of the variation in a trait, within a particular population at a particular time, that is associated with genetic variation among the members of that population. In symbols, $h^2$ = genetic variance ÷ total phenotypic variance. A heritability of .6 for IQ means that 60 per cent of the *differences* among people in that population are associated with their genetic differences. It does not mean that 60 per cent of your intelligence comes from your genes and 40 per cent from your upbringing; for an individual that sentence has no meaning, because every bit of your intelligence comes from genes and environment acting together, as the music of a sitar comes from both the instrument and the player.

You can estimate heritability from twin correlations with a simple formula proposed by Douglas Falconer: $h^2 = 2 \times (r_{MZ} - r_{DZ})$. Push Bouchard and McGue's figures through it: 2 × (.86 − .60) = 2 × .26 = .52. Because MZ twins reared together share all their genes and their home, the rest of their resemblance (.86 − .52 = .34) is attributed to the **shared environment**, the aspects of the home that make siblings alike, and whatever makes even identical twins differ (1 − .86 = .14) to the **non-shared environment** and measurement error. Three findings about heritability are essential:

- **It changes with age.** The heritability of IQ rises from about .2 in infancy to about .4 to .5 in childhood and .6 or more in adulthood, while the effect of the shared family environment shrinks almost to zero by adulthood. Bouchard named this the Wilson effect, after Ronald Wilson's Louisville twin study. One explanation is that, as people grow up, they increasingly choose environments that suit their genetic tendencies.
- **It depends on the environment.** In 2003 **Eric Turkheimer** and his colleagues studied 7-year-old American twins. In middle- and high-income families the heritability of IQ was about .70; in the poorest families it was close to .10, and the shared environment dominated. Where environments are adequate, genetic differences show; where environments are

poor, poverty swamps them. Later meta-analyses found this pattern in the United States but weaker or absent in countries with stronger welfare systems, which is itself evidence that environments matter.

- **Within-group heritability says nothing about between-group differences.** The geneticist **Richard Lewontin** (1929 to 2021) made the point in 1970 with seeds. Take a sack of genetically varied seed corn and plant one handful in rich soil and another in poor soil. Within each tray, differences in height are entirely genetic (heritability 100 per cent), because every plant in the tray received the same soil. Yet the difference between the two trays is entirely environmental. High heritability within groups is therefore no evidence at all that a difference between groups is genetic.

Finally, heritable does not mean unchangeable. **Phenylketonuria (PKU)**, an inherited inability to break down the amino acid phenylalanine, has a heritability close to 100 per cent and once caused severe intellectual disability; a low-phenylalanine diet begun in infancy prevents it. Height is highly heritable too, yet average heights have risen by several centimetres in a few generations as nutrition improved.

> EXAM TRAP
>
> Heritability is a population statistic, not an individual one; it describes differences, not amounts; it is specific to a population and its range of environments; it rises with age; it does not explain differences between groups; and a highly heritable trait can still be changed by the environment. "Individual differences in intelligence are due to genes only" and "environment only" are both wrong: the answer is many factors, genetic and environmental, acting together.

### Reaction Range and the Interplay of Genes and Environment

Behaviour geneticists now speak less of genes *versus* environment and more of how they work together. In 1963 **Irving Gottesman** (1930 to 2016) proposed the concept of **reaction range**: a genotype sets upper and lower limits on how a trait can develop, and the environment determines where within that range a person ends up. A child with a favourable genotype in a deprived home may score lower than a child with a less favourable genotype in a stimulating one. (A related idea is canalisation, the tendency of some traits to develop along the same path across almost any environment.)

Genes and environments are also correlated. Sandra Scarr and Kathleen McCartney (1983) described three kinds of **gene-environment correlation**. It is *passive* when parents provide both genes and a matching environment (bookish parents pass on genes and fill the house with books). It is *evocative* when a child's genetically influenced traits draw responses from others (a curious child gets more answers). It is *active* when people seek out environments that suit them, which Scarr called niche-picking (the curious child grows up and joins the library). Active correlation becomes stronger with age, which helps explain why heritability rises. Turkheimer's finding is an example of **gene-environment interaction**: the effect of genes depends on the environment.

### How the Environment Matters

Which environmental factors make a difference? The list begins before birth: maternal malnutrition, alcohol (the fetal alcohol syndrome of Section 9.8), infections and toxins can all

lower IQ. After birth, nutrition matters. Iodine deficiency is the world's most preventable cause of intellectual impairment; a 1994 meta-analysis found that children in iodine-deficient areas scored about 13 points lower on average, which is why India adopted universal salt iodisation. Exposure to lead, from old paints, fuel and some cookware, lowers IQ even at low levels.

Schooling raises IQ. Children's scores drop slightly over long summer holidays, children who start school later fall behind age-mates, and a 2018 meta-analysis of natural experiments by Stuart Ritchie and Elliot Tucker-Drob estimated that each additional year of education raises IQ by about 1 to 5 points. Family environment matters too: the responsiveness of parents, the availability of books and play materials and the amount of conversation a child hears all correlate with later IQ. Robert Zajonc and Gregory Markus's confluence model (1975) even tried to explain small birth-order effects by the "intellectual climate" of the family, which is diluted as more young children are added. Teachers' expectations can matter as well (the Pygmalion effect, covered in the Social book).

The most dramatic evidence comes from deprivation and its reversal. In the 1930s Harold Skeels moved a group of apparently retarded toddlers from a bleak Iowa orphanage to an institution where women with intellectual disability doted on them. The transferred children gained, on average, close to 30 IQ points; the children left behind lost ground. Skeels followed them up as adults in 1966: the transferred group were self-supporting, while many of the others were still in institutions. More recently, Michael Rutter's study of children adopted into British homes from grossly depriving Romanian orphanages found that those adopted before about 6 months of age largely caught up, while those adopted later often showed lasting cognitive deficits. Early, severe deprivation does lasting harm; early rescue can undo much of it.

## Can We Raise IQ? The Intervention Programmes

If environments matter, can deliberate programmes raise the intelligence of children from poor homes? This was the question Jensen answered so gloomily in 1969. Table 9.12 summarises the best-known attempts.

Table 9.12 Early intervention programmes.

| Programme | Began | Who and what | Main findings |
|---|---|---|---|
| Perry Preschool Project, Michigan | 1962 | David Weikart; two years of high-quality preschool for poor 3- and 4-year-olds | IQ gains faded within a few years, but by age 40 the group had more schooling, higher earnings and fewer arrests |
| Head Start, United States | 1965 | Federal programme of preschool education, health and nutrition for poor children | Early IQ gains fade out by the early grades; lasting benefits in school completion and less grade repetition and special education |
| Milwaukee Project | late 1960s | Rick Heber; intensive day care from infancy for children of mothers with IQs below 75 | Large early IQ gains that shrank later; never fully published in peer-reviewed journals and clouded by Heber's 1981 conviction for misusing grant funds, so treated with caution |
| Abecedarian Project, North Carolina | 1972 | Craig Ramey and Frances Campbell; full-day, year-round enriched care from early infancy to age 5 | IQ and achievement advantages still present in early adulthood; more college attendance |

| Programme | Began | Who and what | Main findings |
|---|---|---|---|
| Integrated Child Development Services (ICDS), India | 1975 | Government anganwadi centres: nutrition, health check-ups and preschool education | One of the world's largest early-childhood programmes; evaluations show benefits for nutrition and school readiness |

Two lessons stand out. First, the earlier and more intensive the programme, the more durable the gains in IQ itself; the Abecedarian children, who started in infancy, kept a measurable advantage. Second, even when IQ gains fade (the **fade-out effect**), good programmes leave lasting benefits in schooling, employment and behaviour. Jensen's "compensatory education has been tried and it apparently has failed" was premature.

## The Flynn Effect

The most startling evidence for environmental influence came from an unexpected source. In 1984 the political philosopher **James Robert Flynn** (1934 to 2020), working in New Zealand, showed that Americans had been gaining about 3 IQ points per decade throughout the twentieth century; in 1987 he reported massive gains in 14 nations. Every time a test was renormed, the new sample had to do better than the old one to earn the same IQ of 100. The gains were largest on tests of fluid reasoning such as Raven's matrices (Dutch military conscripts gained about 20 points between 1952 and 1982) and smallest on vocabulary and general information. The rise in average scores across generations is called the **Flynn effect**, a name coined by Herrnstein and Murray in *The Bell Curve* (1994); Richard Lynn had noticed the same trend in Japan a little earlier.

Genes cannot change that fast, so the cause must be environmental. Candidate explanations include better nutrition and health, more and better schooling, smaller families, a more visually and technologically complex environment and greater familiarity with tests. Flynn's own explanation was that modern life has taught people to put on "scientific spectacles": to classify things abstractly and to take hypothetical problems seriously, exactly what fluid-reasoning tests demand. Since the mid-1990s the effect has stalled or reversed in some countries such as Norway and Denmark; because the reversal appears even within families (younger brothers scoring lower than older ones), it too must be environmental.

CLINICAL LINK

The Flynn effect means test norms go stale. A test normed 15 years ago gives scores about 4 or 5 points too high today. An adult who scores 72 on outdated norms might score 67 or 68 on current norms, and that difference can decide whether a person receives a diagnosis of intellectual disability, a disability certificate or special educational support. In the United States, where the Supreme Court barred the execution of people with intellectual disability in 2002, courts have had to rule on such corrections. Always use the most recent norms available, and say which norms you used in your report.

## Group Differences and Stereotype Threat

In *The Bell Curve* (1994) the psychologist **Richard J. Herrnstein** (1930 to 1994) and the political scientist **Charles Murray** (born 1943) argued that IQ strongly shapes social outcomes such as income, poverty and crime, that American society was stratifying into a "cognitive elite" and a struggling underclass, and that the average difference of about 15 points between Black and

White Americans on IQ tests was probably partly genetic. The book sold hundreds of thousands of copies and provoked fierce criticism.

In response, the American Psychological Association appointed a task force chaired by **Ulric Neisser**, whose report "Intelligence: Knowns and Unknowns" (1996) remains the most balanced summary. Its main conclusions: IQ tests measure something real and predict school and job performance; the tests predict outcomes about equally well for different groups, so they are not "biased" in that narrow statistical sense; the heritability of IQ within groups is substantial; but there was no direct evidence that the average difference between Black and White Americans is genetic, and Lewontin's argument shows that within-group heritability cannot settle the question. Groups differ in history, income, schooling, health and discrimination, and the Flynn effect shows how large purely environmental changes in IQ can be. Since then the gap has narrowed: William Dickens and Flynn (2006) estimated that Black Americans gained 4 to 7 points on White Americans between 1972 and 2002.

One environmental mechanism can act during the test itself. **Claude Steele** and **Joshua Aronson** (1995) gave Black and White Stanford University students a set of difficult verbal items from the Graduate Record Examination, with differences in their entrance-test (SAT) scores statistically controlled. When the test was described as a diagnostic measure of intellectual ability, Black students did markedly worse than White students; when the same test was described as a laboratory problem-solving task unrelated to ability, the difference disappeared. Steele called this **stereotype threat**: the fear of confirming a negative stereotype about one's group, which creates anxiety and uses up the working memory needed for the task. Similar effects were reported for women taking difficult mathematics tests. Later, larger studies have found the effects smaller and less consistent than the early experiments suggested, so stereotype threat is best seen as one contributing factor rather than the whole explanation (the Social book covers it further).

What about men and women? On overall measures of general intelligence the average difference is negligible; the tests were deliberately balanced to avoid one, and independent measures agree. There are small to moderate differences in specific abilities, most reliably an advantage for males in mental rotation and for females in verbal fluency and processing speed, and male scores tend to be more variable, with more men at both extremes.

> EXAM TRAP
>
> The Flynn effect is a rise in *average* scores across generations, not a rise within one person's lifetime, and it is largest on fluid (Raven's-type) tests, not vocabulary. Herrnstein and Murray named it; Flynn documented it. Stereotype threat is Steele and Aronson (1995), not Rosenthal (teacher expectations) or Jensen (Level I and II). *The Bell Curve* (1994) is Herrnstein and Murray; "Intelligence: Knowns and Unknowns" (1996) is Neisser's APA task force.

SCAN YOUR BRAIN

Fill in each blank with the most appropriate term. The answers are below; cover them first.

1. The briefest exposure at which a person can judge which of two lines is longer is the _____ time.
2. Haier's finding that high scorers used less glucose while solving Raven's matrices supports the neural _____ hypothesis.
3. Twins who develop from a single fertilised egg are _____ twins.
4. In Bouchard and McGue's review, identical twins reared apart correlated about _____ on IQ.
5. Heritability is the proportion of _____ in a trait, within a population, associated with genetic variation.
6. Turkheimer found that in the poorest families the heritability of IQ was close to _____.
7. Gottesman's idea that genes set limits within which the environment acts is the _____ range.
8. The rise of about 3 IQ points per decade in average scores is the _____ effect.
9. The early intervention programme that began in infancy in North Carolina in 1972 was the _____ Project.
10. Steele and Aronson's demonstration that a diagnostic label lowers the scores of a stereotyped group is called stereotype _____.

**Answers:** (1) inspection, (2) efficiency, (3) monozygotic (identical), (4) .72, (5) variation (variance), (6) .10 (zero), (7) reaction, (8) Flynn, (9) Abecedarian, (10) threat.

## 9.8 The Extremes: Intellectual Disability and Giftedness MEDIUM YIELD · 7 QS

Binet built his scale to find the children at one end of the distribution. The two tails of the curve in Figure 9.1, the 2 per cent below 70 and the 2 per cent above 130, remain the places where intelligence testing matters most to individual lives.

### Defining Intellectual Disability ●●

The name has changed repeatedly, each time because the old name had become an insult. "Idiot", "imbecile" and "moron" gave way to "mental deficiency", then to **mental retardation**, the term of DSM-IV-TR and of India's Persons with Disabilities Act of 1995, and finally to **intellectual disability**. DSM-5 (2013) uses intellectual disability (intellectual developmental disorder), DSM-5-TR (2022) puts the second name first, and ICD-11 speaks of disorders of intellectual development. Across all modern systems, the definition has three parts:

1. **Deficits in intellectual functions**, such as reasoning, problem solving, planning, abstract thinking, judgement and learning, confirmed by clinical assessment and individually administered standardised intelligence testing. This means an IQ about two standard deviations or more below the mean, that is, about 70 or below; allowing for measurement error, DSM-5 speaks of scores of about 65 to 75.
2. **Deficits in adaptive functioning.** **Adaptive functioning** means the conceptual, social and practical skills that people need to meet the standards of personal independence and

social responsibility expected for their age and culture, such as handling money, communicating, keeping safe and looking after oneself.

3. **Onset during the developmental period.** DSM-IV-TR required onset before 18; the American Association on Intellectual and Developmental Disabilities (AAIDD) extended its own definition to before age 22 in 2021.

The biggest change in DSM-5 concerns severity. Under DSM-IV-TR and ICD-10, the four levels (mild, moderate, severe and profound) were defined by IQ ranges (Table 9.13). DSM-5 keeps the four levels but defines them by adaptive functioning in the conceptual, social and practical domains, because it is adaptive functioning that determines how much support a person needs, and because IQ scores become unreliable at the very low end. About 1 per cent of the population has intellectual disability, and about 85 per cent of them are in the mild range.

Table 9.13 Levels of intellectual disability.

| Level | DSM-IV-TR IQ range | ICD-10 IQ range | Share of cases | Historical label |
|---|---|---|---|---|
| Mild | 50 to 55 to about 70 | 50 to 69 | About 85% | Moron; "educable" |
| Moderate | 35 to 40 to 50 to 55 | 35 to 49 | About 10% | Imbecile; "trainable" |
| Severe | 20 to 25 to 35 to 40 | 20 to 34 | About 3 to 4% | Imbecile or idiot |
| Profound | Below 20 or 25 | Below 20 | About 1 to 2% | Idiot |

Two further DSM-5 categories are often examined. **Global developmental delay** is a provisional diagnosis for children under 5 who fail to meet developmental milestones in several areas but are too young for their severity to be assessed reliably; they must be reassessed later. **Unspecified intellectual disability** is for people over 5 whose level cannot be assessed because of blindness, prelingual deafness, severe motor disability or severe problem behaviour. So a 7-year-old with a full-scale IQ of 65 and severe adaptive impairment does not receive either label: he is old enough and assessable, and he meets the criteria for intellectual disability itself.

> EXAM TRAP
>
> The DSM-IV-TR criterion for mental retardation was an IQ about *two* standard deviations below the mean (about 70). DSM-5 defines severity by *adaptive functioning*, not IQ; IQ cut-offs alone are an old-system answer. Borderline intellectual functioning (IQ about 71 to 84 in DSM-IV-TR) is not intellectual disability. Global developmental delay is for children under 5 only.

## Measuring Adaptive Behaviour ●●

Because the diagnosis depends on everyday functioning, clinicians need a way to measure it. The classic tool was built at the same Vineland Training School where Goddard had worked. **Edgar Arnold Doll** (1889 to 1968) published the **Vineland Social Maturity Scale (VSMS)** in 1935. It is completed by interviewing someone who knows the person well, and it covers self-help (general, eating and dressing), self-direction, occupation, communication, locomotion and socialisation. The scale yields a **social age (SA)**, the age level of the person's social competence, and a **social quotient (SQ)** = (SA ÷ CA) × 100, a ratio exactly like Stern's IQ. A. J. Malin's Indian adaptation of the VSMS is the version most used in Indian clinics and disabili-

ty certification. Its modern American successor, the Vineland Adaptive Behavior Scales (Sara Sparrow, David Balla and Domenic Cicchetti, 1984; third edition 2016), assesses communication, daily living skills, socialisation and motor skills.

Keep the quotients apart: intelligence tests give mental age and IQ; the VSMS gives social age and social quotient; infant scales such as Gesell's and Bayley's give developmental age and developmental quotient.

### Causes, in Brief ●●

Causes are identifiable in perhaps half of all cases, more often in severe than in mild disability. The most common chromosomal cause is **Down syndrome**, described by John Langdon Down in 1866 and shown by Jérôme Lejeune in 1959 to result from an extra copy of chromosome 21 (trisomy 21). The most common inherited cause is **fragile X syndrome**, caused by an expanded repeat in the FMR1 gene on the X chromosome. Metabolic causes include PKU (first described by Asbjørn Følling in 1934, now prevented by newborn screening and diet) and congenital hypothyroidism (cretinism), linked to iodine deficiency. Environmental causes include fetal alcohol syndrome, prenatal infections such as rubella, birth asphyxia, head injury, severe malnutrition and lead poisoning. Edward Zigler (1967) distinguished an "organic" group with an identifiable cause, usually more severe, from a "familial" or "cultural-familial" group, usually mild, representing the lower end of the normal distribution combined with poverty and deprivation. The disorders themselves are covered in the Psychiatric Disorders book.

> INDIAN CONTEXT
>
> The **National Trust for the Welfare of Persons with Autism, Cerebral Palsy, Mental Retardation and Multiple Disabilities Act, 1999** covers exactly the four groups in its title, so autism and intellectual disability are both included. The **Rights of Persons with Disabilities Act, 2016** recognises 21 disabilities and lists specific learning disabilities and autism spectrum disorder under its intellectual disability heading. Disability certification for intellectual disability rests on a standardised IQ test together with an adaptive-behaviour measure such as the VSMS. The Rehabilitation Council of India (RCI Act, 1992) regulates the training of the professionals, including clinical psychologists, who carry out such assessments, and the National Institute for the Empowerment of Persons with Intellectual Disabilities in Secunderabad (founded in 1984) leads national work in the field.

### Giftedness ●●

At the other tail, an IQ of 130 or above (the top 2 per cent) is the most common working cut-off for **giftedness**, but definitions have broadened. An influential 1972 United States government report listed six areas of giftedness: general intellectual ability, specific academic aptitude, creative or productive thinking, leadership, the visual and performing arts, and psychomotor ability.

KEY STUDY: TERMAN'S GENETIC STUDIES OF GENIUS (FROM 1921)

**Question.** What becomes of exceptionally bright children? Popular opinion held that gifted children were frail, odd and doomed to "early ripe, early rot". **Method.** In 1921 Lewis Terman began screening California schoolchildren and selected about 1,500 (1,528) with Stanford-Binet IQs of about 135 or more, most above 140; their average IQ was about 151. The children, nicknamed the "Termites", were then followed for the rest of their lives, the study being carried on after Terman's death by Melita Oden, Robert Sears and others. **Result.** As a group the gifted children were healthier, taller, better adjusted and far more successful in education and careers than average; the stereotype of the sickly genius was wrong. Within the group, Terman and Oden found that the most successful men differed from the least successful not in IQ but in persistence, self-confidence, goal-directedness and family background. Textbook summaries add that the social and behavioural problems that did occur were concentrated among those with extremely high childhood IQs, of about 180 or more. **Criticism.** The sample was selected partly through teacher nomination and was disproportionately white, urban and middle class; many of the women, constrained by their era, never had careers to match their ability. And the screening missed two boys, William Shockley and Luis Alvarez, who were tested but not selected and later won Nobel Prizes in physics, a reminder that an IQ cut-off is not a measure of creative achievement.

Terman's contemporary **Leta Stetter Hollingworth** (1886 to 1939) studied the very top of the range. Her book *Children Above 180 IQ*, published posthumously in 1942, found that such children often struggled socially, because they had few true peers, and she argued that an IQ of about 125 to 155 was "socially optimal". This is consistent with the Terman finding that adjustment problems clustered above 180.

Many researchers now reject the idea that giftedness is simply a high IQ. **Joseph Renzulli** (born 1936) proposed the **three-ring conception of giftedness** (1978): gifted behaviour arises from the interaction of above-average (not necessarily exceptional) ability, task commitment and creativity. Françoys Gagné later distinguished *gifts* (natural abilities) from *talents* (systematically developed skills), with motivation, environment and chance turning one into the other. The relationship between creativity and intelligence, including the threshold hypothesis that IQ matters for creativity only up to about 120, is taken up in Chapter 8.

## Savant Syndrome and Prodigies

**Savant syndrome** is the coexistence of a serious developmental disability, often autism or intellectual disability, with an "island" of exceptional ability. Langdon Down coined the term "idiot savant" in 1887; the psychiatrist Darold Treffert, who studied savants for decades, distinguished splinter skills, talented savants and the very rare prodigious savants. Roughly half of savants are autistic, and perhaps one autistic person in ten shows some savant skill. Typical domains are calendar calculation (naming the weekday of any date), rapid arithmetic, music, art and memory. Kim Peek (1951 to 2009), the inspiration for the film *Rain Man* (1988), memorised thousands of books; the British artist Stephen Wiltshire (born 1974), who is autistic, has drawn detailed panoramas of cities from memory after a single helicopter ride. Gardner used savants as evidence that abilities can be isolated, one of his criteria for separate intelligences.

A savant must be distinguished from a **prodigy**, a child of otherwise typical or high ability who performs at an adult expert level in some domain. India's Shakuntala Devi (1929 to 2013),

the "human computer" who in 1980 multiplied two 13-digit numbers in her head at Imperial College London, was a calculating prodigy, not a savant; Arthur Jensen tested her in 1988 and found extraordinary speed on numerical tasks.

CLINICAL LINK

For a clinical psychologist, the lesson of both tails is the same: the score starts the assessment, it does not end it. A child at IQ 68 with good adaptive skills may need classroom support rather than a diagnosis; a child at IQ 145 may need help with anxiety or social isolation more than an accelerated syllabus; and an autistic adult with a savant skill may need adaptive-skills training to live independently. Always interpret IQ alongside adaptive functioning, history, culture and the person's own goals.

SCAN YOUR BRAIN

Fill in each blank with the most appropriate term. The answers are below; cover them first.

1. DSM-5 sets the severity of intellectual disability by _____ functioning rather than IQ.
2. Under ICD-10, moderate intellectual disability corresponds to IQ 35 to _____.
3. Global developmental delay is diagnosed only in children under the age of _____.
4. The Vineland Social Maturity Scale was published by Edgar Doll in _____.
5. The VSMS yields a social age and a social _____.
6. The most common inherited cause of intellectual disability is _____ syndrome.
7. Terman's gifted children were nicknamed the _____.
8. Renzulli's three rings are above-average ability, task commitment and _____.
9. The term "idiot savant" was coined in 1887 by _____.

**Answers:** (1) adaptive, (2) 49, (3) 5, (4) 1935, (5) quotient, (6) fragile X, (7) Termites, (8) creativity, (9) John Langdon Down.

## The case of Binet's scale, revisited

At the start of this chapter Binet made two claims: that intelligence is complex and a test can only sample it, and that it can be raised. A century of research has been kind to both. The factor theories found that intelligence is neither one thing nor a loose bundle of unrelated things, but a hierarchy: a general factor, broad abilities such as fluid and crystallised intelligence, and many narrow skills, with emotional, practical and creative abilities that standard tests barely touch. And the evidence that it can be raised is overwhelming: the Flynn effect, the gains of adopted children, the effects of schooling, iodine and early intervention, and the recovery of Skeels's toddlers.

The hereditarians who borrowed Binet's scale were not wholly wrong either; within a given population, a large share of the differences in adult IQ is associated with genes. What they missed is that heritability is not destiny: it describes differences within one set of environments, it says nothing about differences between groups, and it coexists with large environmental effects on everyone's level. Binet's original purpose, to find the children who need

help and then help them, survives in the DSM-5's insistence on adaptive functioning and in India's disability laws, which tie an IQ score to a right to support rather than to a label.

## Chapter summary

**9.1 What is intelligence?** Definitions range from Binet's judgement and adaptation, through Terman's abstract thinking, Thorndike's good responses and Boring's operational "what the tests test", to Wechsler's aggregate or global capacity to act purposefully, think rationally and deal effectively with the environment (1944), Sternberg's adapting, shaping and selecting, and Gottfredson's 1994 consensus. Achievement tests measure past learning, aptitude tests predict specific future learning, and intelligence tests predict learning in general. Implicit theories matter: Dweck's entity versus incremental theories (fixed versus growth mindset) shape persistence.

**9.2 From Galton to Binet.** Galton (*Hereditary Genius*, 1869; anthropometric laboratory, 1884) measured sensory acuity and reaction time; James McKeen Cattell coined "mental test" (1890), but Wissler (1901) showed such tests did not predict grades. Binet and Simon's 1905 scale of 30 items, revised in 1908 (mental age) and 1911, measured judgement directly. Goddard translated it, Stern coined the intelligence quotient (1912), and Terman's Stanford-Binet (1916) made IQ = MA ÷ CA × 100 famous. Yerkes's Army Alpha (verbal) and Beta (non-verbal) of 1917 began group testing, and their misuse is a lasting warning.

**9.3 The IQ and the modern tests.** The ratio IQ fails for adults, so Wechsler (1939) introduced the deviation IQ (mean 100, SD 15; IQ = 100 + 15z), which converts directly to percentiles. The WAIS-IV has four indices (VCI, PRI, WMI, PSI) with 10 core subtests; the WISC-V (6 to 16 years 11 months) has five; the SB5 (2003) has five CHC factors. Raven's matrices (1938) and Cattell's Culture Fair test measure fluid reasoning with minimal language; Indian tests include the Binet-Kamat, MISIC and Bhatia's performance battery. IQ tests are highly reliable and predict school and job performance with correlations around .5.

**9.4 Factor theories.** Spearman (1904) found a positive manifold and proposed g plus specific s. Thorndike's multifactor theory named abstract, mechanical and social intelligence; Thomson explained correlations by overlapping samples of bonds; Thurstone (1938) found seven primary mental abilities; Guilford's cube had 120, then 150 and 180 abilities. Burt and Vernon arranged g, v:ed, k:m and specific factors in a hierarchy. Cattell and Horn split g into fluid and crystallised intelligence; Carroll's three strata (1993) and the CHC model now organise most tests. Jensen distinguished associative Level I from conceptual Level II abilities.

**9.5 Broader views.** Gardner (1983) proposed seven, later eight, relatively independent intelligences, attractive to teachers but weakly supported by data. Sternberg's triarchic theory (1985) has componential (analytical), experiential (creative) and contextual (practical) parts and grew into successful intelligence and tacit knowledge. The PASS model of Das, Naglieri and Kirby, based on Luria, measures planning, attention, simultaneous and successive processing with the CAS. Ceci stressed context and knowledge. Emotional intelligence comes in ability (Salovey and Mayer), mixed (Goleman, Bar-On) and trait (Petrides) versions; social, practical and spiritual intelligences extend the list.

**9.6 Intelligence, speed and the brain.** Higher IQ goes with faster and less variable reaction times, shorter inspection times, larger working memory capacity (especially for fluid intelligence),

slightly larger brains and, in Haier's studies, more efficient brain metabolism. Intelligence is highly polygenic. These are correlates, not established causes.

**9.7 Heredity and environment.** Family, twin and adoption studies (Bouchard and McGue's review; the Minnesota study of twins reared apart) show substantial genetic influence. Heritability is a population statistic that rises with age, falls in poverty (Turkheimer) and cannot explain group differences (Lewontin). Gottesman's reaction range and gene-environment correlation and interaction describe how genes and environments combine. Nutrition, schooling, adoption and early intervention (Abecedarian, Perry, Head Start, ICDS) raise IQ or life outcomes, and the Flynn effect shows large environmental gains across generations. The Bell Curve's genetic claims for group differences are not supported; stereotype threat is one environmental factor in test performance.

**9.8 The extremes.** Intellectual disability requires deficits in intellectual and adaptive functioning beginning in the developmental period; DSM-IV-TR graded it by IQ bands, DSM-5 by adaptive functioning. Adaptive behaviour is measured by the VSMS (social age and social quotient) and the Vineland scales; global developmental delay applies under age 5. Causes include Down syndrome, fragile X and PKU. Giftedness is studied through Terman's longitudinal study, Hollingworth's children above 180 and Renzulli's three rings; savant syndrome pairs disability with islands of exceptional skill.

## Who's who

**Table 9.14** The people in this chapter.

| Name | When | Examined for |
|---|---|---|
| Joshua Aronson | 1995 | With Claude Steele, the stereotype threat experiments at Stanford (1995) |
| Alan Baddeley | 1974 | Working-memory model (Chapter 7); with Carroll's theory, the basis of the SB5's five factors |
| Reuven Bar-On | 1997 | Mixed model of emotional intelligence; the self-report Emotional Quotient Inventory (EQ-i); the term EQ |
| Nancy Bayley | 1899 to 1994 | Bayley Scales of Infant (and Toddler) Development: cognitive, language and motor development up to about 42 months |
| C. M. Bhatia | 1955 | Bhatia's Battery of Performance Tests of Intelligence: five performance subtests, rural and urban Indian norms |
| Alfred Binet | 1857 to 1911 | With Simon, the first intelligence test (1905, 30 items); mental age (1908 revision); intelligence as judgement; opposed the idea of fixed intelligence |
| Walter Van Dyke Bingham | 1880 to 1952 | Classic definition of aptitude (1937) |
| Edwin Garrigues Boring | 1886 to 1968 | "Intelligence is what the tests test" (1923), an operational definition |
| Thomas J. Bouchard Jr. | 1981; from 1979 | Review of IQ correlations among relatives with McGue (1981); the Minnesota Study of Twins Reared Apart |
| Carl Campbell Brigham | 1923; 1930 | *A Study of American Intelligence* (1923) ranked immigrant groups from the army data; retracted in 1930 |
| Cyril Lodowic Burt | 1883 to 1971 | Hierarchical model of abilities (1949); twin data exposed as suspect by Kamin (1974) |

| Name | When | Examined for |
|---|---|---|
| John Bissell Carroll | 1916 to 2003 | Three-stratum theory (1993): g, eight broad and about 70 narrow abilities; basis of the CHC model |
| James McKeen Cattell | 1860 to 1944 | Coined "mental test" (1890); sensory and motor tests at Columbia, which Wissler showed did not predict grades (1901) |
| Raymond Bernard Cattell | 1905 to 1998 | Fluid (Gf) versus crystallised (Gc) intelligence (1963); investment theory; Culture Fair Intelligence Test; the 16PF (Personality book) |
| Stephen J. Ceci | 1990 | Bioecological model of intelligence; the racetrack handicappers study with Liker (1986) |
| Jagannath Prasad Das | Born 1931 | PASS model with Naglieri and Kirby (1994); Cognitive Assessment System; PREP; writings on *buddhi* |
| Douglas Detterman | 1986 | With Sternberg, the 1986 expert symposium *What Is Intelligence?* |
| Shakuntala Devi | 1929 to 2013 | Indian calculating prodigy, not a savant; multiplied two 13-digit numbers mentally in 1980 |
| Edgar Arnold Doll | 1889 to 1968 | Vineland Social Maturity Scale (1935): social age and social quotient |
| John Langdon Down | 1828 to 1896 | Described Down syndrome (1866); coined "idiot savant" (1887) |
| Carol Susan Dweck | Born 1946 | Entity versus incremental theories of intelligence; fixed and growth mindsets (*Mindset*, 2006) |
| James Robert Flynn | 1934 to 2020 | The Flynn effect: gains of about 3 IQ points per decade (1984; 14 nations, 1987) |
| Francis Galton | 1822 to 1911 | *Hereditary Genius* (1869); anthropometric laboratory (1884); twin method; "nature and nurture"; eugenics; father of differential psychology |
| Howard Gardner | Born 1943 | Multiple intelligences, *Frames of Mind* (1983): seven intelligences, naturalist added 1999, existential considered |
| Henry Herbert Goddard | 1866 to 1957 | Translated the Binet-Simon scale (1908); coined "moron" (1910); *The Kallikak Family* (1912); testing at Ellis Island |
| Daniel Goleman | Born 1946 | Popularised emotional intelligence (1995); five-part mixed model |
| Irving Gottesman | 1930 to 2016 | Reaction range (1963): genes set limits, environment decides where within them a person ends up |
| Linda Gottfredson | 1994 | Drafted the 52-signatory consensus definition of intelligence (1994); critic of practical intelligence |
| Joy Paul Guilford | 1897 to 1987 | Structure of intellect (1967): 5 operations × 4 contents × 6 products = 120 abilities, later 150 and 180; convergent and divergent production |
| Richard Haier | 1988 | PET study of Raven's solvers: the neural efficiency hypothesis; parieto-frontal integration theory with Rex Jung (2007) |
| Rick Heber | late 1960s | Milwaukee Project of intensive infant day care; results treated with caution |
| Richard J. Herrnstein | 1930 to 1994 | *The Bell Curve* (1994) with Murray; named the Flynn effect |
| Leta Stetter Hollingworth | 1886 to 1939 | *Children Above 180 IQ* (1942); a "socially optimal" IQ of about 125 to 155 |
| John Leonard Horn | 1928 to 2006 | Developed Gf-Gc theory with Cattell (from 1966); about nine broad abilities; doubted a single g |
| Earl Hunt | 1975 | Cognitive-correlates approach: high-verbal students retrieve letter names faster |

| Name | When | Examined for |
|---|---|---|
| Arthur Robert Jensen | 1923 to 2012 | Level I (associative) and Level II (conceptual) abilities; the 1969 *Harvard Educational Review* article; reaction-time studies |
| V. V. Kamat | 1930s | Binet-Kamat Test: Marathi and Kannada adaptation of the 1916 Stanford-Binet |
| Leon Kamin | 1974 | Showed that Burt's twin correlations (.771) were statistically implausible |
| Alan and Nadeen Kaufman | 1983; 2004 | K-ABC (Luria-based); KBIT-2 with a verbal (crystallised) and a nonverbal (fluid) scale |
| John R. Kirby | 1994 | Co-author of the PASS model |
| Richard Lewontin | 1929 to 2021 | Seed-corn argument (1970): heritability within groups says nothing about differences between groups |
| Alexander Romanovich Luria | 1902 to 1977 | Three functional units of the brain, the basis of the PASS model and the K-ABC |
| Arthur J. Malin | 1969 | Malin's Intelligence Scale for Indian Children (MISIC), adapted from the WISC; Indian adaptation of the VSMS |
| John D. Mayer | 1990 | With Salovey, the modern concept of emotional intelligence (1990); four-branch ability model (1997); MSCEIT |
| Kevin McGrew | late 1990s | Merged the Cattell-Horn and Carroll models into the CHC theory |
| Charles Murray | Born 1943 | *The Bell Curve* (1994) with Herrnstein |
| Jack A. Naglieri | 1994; 1997 | Co-author of the PASS model; Cognitive Assessment System with Das (1997) |
| Ulric Neisser | 1928 to 2012 | Chaired the APA task force report "Intelligence: Knowns and Unknowns" (1996) |
| Ted Nettelbeck | 1976 | Inspection time (with M. Lally) |
| Konstantinos Petrides | about 2000 | Trait emotional intelligence (with Adrian Furnham); the TEIQue |
| Jean Piaget | 1896 to 1980 | Standardised reasoning tests in Simon's Paris laboratory around 1920; did not create the first intelligence test (Developmental book) |
| Craig Ramey | 1972 | Abecedarian Project (with Frances Campbell): enriched care from infancy |
| John Carlyle Raven | 1902 to 1970 | Standard Progressive Matrices (1938): non-verbal, culture-fair test of fluid intelligence and g |
| Joseph Renzulli | Born 1936 | Three-ring conception of giftedness (1978): above-average ability, task commitment, creativity |
| Gale Roid | 2003 | Stanford-Binet Fifth Edition (SB5): five CHC factors, verbal and nonverbal, mean 100 and SD 15 |
| Michael Rutter | 1933 to 2021 | Romanian orphans adopted into British homes: near-full recovery if adopted before about 6 months |
| Peter Salovey | 1990 | With Mayer, introduced emotional intelligence in its modern sense (1990); four-branch ability model |
| Sandra Scarr | 1936 to 2021 | Adoption studies with Weinberg (1978); passive, evocative and active gene-environment correlation with McCartney (1983); niche-picking |
| Frank Schmidt and John Hunter | 1998 | General mental ability is the best single predictor of job performance (validity about .5) |
| Théodore Simon | 1873 to 1961 | Binet's collaborator on the 1905, 1908 and 1911 scales |

| Name | When | Examined for |
|---|---|---|
| Harold Skeels | 1930s; 1966 | Orphanage transfer study: toddlers moved to a caring institution gained close to 30 IQ points |
| Charles Edward Spearman | 1863 to 1945 | Two-factor theory (1904): g and s; positive manifold; first factor analysis; eduction of relations |
| Claude Steele | 1995 | Stereotype threat, with Aronson (1995) |
| William Stern | 1871 to 1938 | Coined the intelligence quotient (1912): mental age divided by chronological age |
| Robert Jeffrey Sternberg | Born 1949 | Triarchic theory (*Beyond IQ*, 1985): componential, experiential, contextual; successful intelligence; tacit knowledge; the 1986 expert symposium |
| Lewis Madison Terman | 1877 to 1956 | Stanford-Binet (1916) and IQ = MA ÷ CA × 100; "abstract thinking" definition (1921); gifted "Termites" study (from 1921) |
| Godfrey Hilton Thomson | 1881 to 1955 | Sampling theory (1916; 1939): test correlations arise from overlapping samples of many small bonds |
| Edward Lee Thorndike | 1874 to 1949 | Multifactor ("anarchic") theory: abstract, mechanical and social intelligence; social intelligence (1920); CAVD (1927) |
| Louis Leon Thurstone | 1887 to 1955 | Seven primary mental abilities (1938); multiple factor analysis; group-factor theory |
| Darold Treffert | 1933 to 2020 | Savant syndrome: splinter skills, talented savants and prodigious savants |
| Eric Turkheimer | 2003 | Heritability of IQ about .70 in affluent families but near .10 in the poorest (2003) |
| Philip Ewart Vernon | 1905 to 1987 | Hierarchical model (1950): g, then v:ed and k:m, then minor group and specific factors |
| David Wechsler | 1896 to 1981 | Intelligence as the aggregate or global capacity (1944); Wechsler-Bellevue (1939); deviation IQ; WAIS, WISC and WPPSI |
| David Weikart | 1962 | Perry Preschool Project, Michigan |
| Clark Wissler | 1901 | Showed that Cattell's mental tests neither predicted college grades nor correlated with one another |
| Robert Mearns Yerkes | 1876 to 1956 | Army Alpha (verbal) and Army Beta (non-verbal) group tests (1917); the Yerkes-Dodson law (Chapter 10) |
| Edward Zigler | 1967 | Organic versus familial (cultural-familial) intellectual disability |
| Danah Zohar | 2000 | Spiritual intelligence (SQ), with Ian Marshall (2000) |

## Key terms

unifactor theory, operational definition, metacognition, achievement, aptitude, explicit theories, implicit theories of intelligence, entity theory, incremental theory, fixed mindset, growth mindset, anthropometric laboratory, differential psychology, mental test, mental age, intelligence quotient (IQ), Stanford-Binet Intelligence Scale, ratio IQ, Army Alpha, Army Beta, norms, deviation IQ, standard deviation (SD), z score, point scales, age scales, performance tests, Verbal IQ, Performance IQ, Full Scale IQ (FSIQ), index scores, Verbal Comprehension Index (VCI), Perceptual Reasoning Index (PRI), Working Memory Index (WMI), Processing Speed Index (PSI), General Ability Index (GAI), Stanford-Binet Fifth Edition (SB5), basal level, ceiling level, developmental quotient (DQ), borderline intellectual functioning, culture-free test, culture-fair test, Raven's

Progressive Matrices, Culture Fair Intelligence Test, reliable, valid, content validity, criterion-related validity, concurrent validity, predictive validity, construct validity, factor analysis, factors, factor loading, positive manifold, general intelligence (g), specific factor (s), two-factor theory, eduction of relations, multifactor theory, abstract intelligence, mechanical intelligence, social intelligence, sampling theory, primary mental abilities, group-factor theory, structure of intellect (SOI) model, divergent production, convergent production, hierarchical model, fluid intelligence (Gf), crystallised intelligence (Gc), investment theory, Stratum III, Stratum II, Stratum I, three-stratum theory, Cattell-Horn-Carroll (CHC) theory, Level I abilities, Level II abilities, multiple intelligences, triarchic theory of intelligence, componential subtheory, metacomponents, performance components, knowledge-acquisition components, experiential subtheory, contextual subtheory, adaptation, shaping, selection, successful intelligence, tacit knowledge, PASS model, Cognitive Assessment System (CAS), bioecological model, emotional intelligence (EI), four-branch model, mixed model, trait emotional intelligence, spiritual intelligence, cognitive-correlates approach, reaction time, inspection time, working memory capacity, neural efficiency hypothesis, polygenic, family studies, twin studies, monozygotic (MZ) twins, dizygotic (DZ) twins, adoption studies, heritability, shared environment, non-shared environment, phenylketonuria (PKU), reaction range, gene-environment correlation, gene-environment interaction, fade-out effect, Flynn effect, stereotype threat, mental retardation, intellectual disability, adaptive functioning, global developmental delay, Vineland Social Maturity Scale (VSMS), social age (SA), social quotient (SQ), Down syndrome, fragile X syndrome, giftedness, three-ring conception of giftedness, savant syndrome, prodigy

## From the past papers

1. What type of psychological test is the Wechsler Adult Intelligence Scale (WAIS)? (RML 2025)
   (a) Aptitude test (b) Intelligence test (c) Attitude test (d) Achievement test

   **Answer: (b) Intelligence test.** The WAIS samples broad reasoning, knowledge, working memory and speed and combines them into a Full Scale IQ, which is the job of an intelligence test (Section 9.1). An aptitude test predicts how well someone will learn one specific skill with training (the Differential Aptitude Tests), an achievement test measures what a known course of teaching has already produced (a board examination), and an attitude test measures an evaluative stance towards an object, which is a Social book topic.

2. Which of the following technique/process involved in guidance is called "the heart of the guidance programme"? (RML 2022)
   (a) Case study (b) Interview (c) Cumulative record (d) Aptitude test

   **Answer: (d) Aptitude test (as keyed).** Aptitude tests forecast how well a person will learn a course or an occupation, so the guidance decisions about which stream or career to choose rest on them. The case study, the interview and the cumulative record are also guidance tools, but they describe the person's past and present (an intensive life history, a face-to-face conversation, a running school file) rather than predict future learning. Be aware that the standard textbook phrase is "counselling is the heart of the guidance programme"; counselling is not offered here, but if it appears as an option, choose it.

3. Which of the following psychologists is associated with the development of the first intelligence test? (RML 2025)

   (a) Freud (b) Piaget (c) Binet (d) Skinner

   **Answer: (c) Binet.** Alfred Binet and Théodore Simon published the first practical intelligence scale, 30 items of increasing difficulty, in 1905 (Section 9.2). Piaget is the bait: he worked in Simon's Paris laboratory around 1920 standardising reasoning tests, but his interest in children's errors led to a theory of cognitive development, not to the first test. Freud (psychoanalysis) and Skinner (operant conditioning) had nothing to do with intelligence testing.

4. The term IQ was invented by (JEMAS 2025)

   (a) Lewis Termon (sic) (b) Gordon Allport (c) William Stern (d) John Watson

   **Answer: (c) William Stern.** Stern proposed the *Intelligenzquotient*, mental age divided by chronological age, in 1912. Terman is the tempting wrong answer: he multiplied Stern's quotient by 100 and made the IQ famous through the 1916 Stanford-Binet, but he did not coin it. Allport is a trait theorist of personality and Watson the founder of behaviourism; neither worked on the IQ.

5. IQ if any body has a mental age of 8 and a chronological age of 10 is: (RML 2020)

   (a) 108 (b) 125 (c) 80 (d) 92

   **Answer: (c) 80.** Ratio IQ = (MA ÷ CA) × 100 = (8 ÷ 10) × 100 = 80. The figure 125 comes from turning the ratio upside down (10 ÷ 8 × 100), the commonest slip; 108 and 92 are not produced by any calculation with these numbers. A quick check: whenever mental age is below chronological age, the IQ must be below 100.

6. To determine whether a newly developed intelligence test truly accurately measures intelligence or not, it is necessary to administer the test with one that has been in the field for some time. Which of the following needs to be looked for? (JEMAS 2025)

   (a) Construct validity (b) Concurrent validity (c) Content validity (d) Predictive validity

   **Answer: (b) Concurrent validity.** Giving the new test together with an established test and correlating the two sets of scores is criterion-related validity with the criterion measured at the same time, that is, concurrent validity (Section 9.3). Predictive validity would need a criterion gathered later, such as next year's grades; content validity is judged by examining whether the items sample the domain, not by correlating with another test; and construct validity is the broader question of whether the test behaves as the theory of intelligence predicts across many kinds of evidence.

7. In the WAIS-IV (Wechsler Adult Intelligence Scale, Fourth Edition), how many core subtests are there in the Verbal Comprehension Index and the Perceptual Reasoning Index respectively? (Real PYQ 2025)

   (a) 3 verbal, 4 performance (b) 3 verbal, 3 performance (c) 4 verbal, 3 performance (d) 2 verbal, 4 performance

   **Answer: (b) 3 verbal, 3 performance.** The VCI core is Similarities, Vocabulary and Information; the PRI core is Block Design, Matrix Reasoning and Visual Puzzles (Table 9.5). The options with a 4 are tempting because each index also has supplemental subtests (Comprehension for VCI; Picture Completion and Figure Weights for PRI), but supplemental subtests only substitute for core ones and do not count towards the core total of 10.

8. Which of the following subtests is part of the core battery for the Wechsler Adult Intelligence Scale (WAIS-IV) but NOT typically found in the Wechsler Intelligence Scale for Children (WISC-V)? (Real PYQ 2024)

   (a) Digit Span (b) Symbol Search (c) Matrix Reasoning (d) Letter-Number Sequencing

   **Answer: (d) Letter-Number Sequencing (as keyed).** Digit Span, Symbol Search and Matrix Reasoning are primary subtests in both the WAIS-IV and the WISC-V, so none of them can be the answer. Letter-Number Sequencing is the only option that is not a primary WISC-V subtest (it is a secondary one there). Strictly, it is a supplemental, not core, subtest of the WAIS-IV (it was core in the WAIS-III), so the stem is loose, but (d) is clearly the best of the four.

9. What is the age range valid for Wechsler Intelligence Scale for Children, 5th edition (WISC-V)? (RML 2022)

   (a) 5 years to 12 years (b) 6 years to 16 years 11 months (c) 7 years to 16 years 11 months (d) 6 years to 15 years 11 months

   **Answer: (b) 6 years to 16 years 11 months.** The WISC-V fills the gap between the WPPSI-IV, which ends at 7 years 7 months, and the WAIS-IV, which begins at 16 years (Table 9.4). Options (c) and (d) each move one end of the range by a year, and (a) matches no Wechsler scale.

10. Raven's Progressive Matrices (RPM) is primarily designed to measure which type of intelligence? (Real PYQ 2025)

    (a) Crystallized intelligence (b) Fluid intelligence (c) Verbal intelligence (d) Social intelligence

    **Answer: (b) Fluid intelligence.** Each item requires working out the rule behind a novel pattern of figures, with almost no language or learned knowledge, which is fluid reasoning (Sections 9.3 and 9.4). Crystallised intelligence is acquired knowledge and is measured by vocabulary and information subtests; verbal intelligence is exactly what a non-verbal test avoids; and social intelligence, Thorndike's ability to understand and manage people, is not sampled at all.

11. Raven's Standard Progressive Matrices (SPM) measure which kind of intelligence? (RML 2022)

    (a) Crystallised intelligence (b) Fluid intelligence (c) Both a and b (d) General Intelligence

    **Answer: (b) Fluid intelligence.** The SPM (1938) is the classic measure of Gf. Option (d) is the tempting one, because Raven's is also the best single marker of Spearman's g and Gf and g are closely related; but when fluid intelligence is offered, it is the more precise and expected answer. "Both" is wrong because crystallised knowledge is precisely what the matrices were designed to keep out.

12. A 10-year-old child from a rural area, with no formal schooling, is brought for an intellectual assessment due to academic struggles. The child is illiterate and has had minimal exposure to formal testing materials. Which of the following tests would be the MOST appropriate to use to minimize cultural and educational bias? (Real PYQ 2024)

    (a) Bhatia's Battery of Performance Tests of Intelligence (b) Wechsler Intelligence Scale for Children (WISC-IV) (c) Stanford-Binet Intelligence Scale (5th ed.) (d) Malin's Intelligence Scale for Indian Children (MISIC)

    **Answer: (a) Bhatia's Battery of Performance Tests of Intelligence.** Its five performance subtests (Kohs blocks, pass-along, pattern drawing, immediate memory, picture construction) need no reading, and it has Indian rural norms (Section 9.3). The WISC-IV and SB5 lean heavily on school-built verbal knowledge and were normed on other populations. MISIC is the tempting Indian option, but half of it is verbal, adapted from the WISC's information, comprehension, arithmetic and vocabulary items.

13. The Wechsler Adult Intelligence Scale can be used to assess eligibility for: (Amity 2023)

    (a) Housing benefit (b) Disability allowance (c) Suicide risk (d) Special educational needs

    **Answer: (b) Disability allowance.** A WAIS Full Scale IQ, together with a measure of adaptive behaviour, documents intellectual disability or acquired cognitive impairment for a disability certificate or allowance (Clinical link, Section 9.3). Special educational needs is the tempting option, but it concerns children, who are assessed with the WISC or an Indian equivalent rather than the adult scale. Housing benefit depends on income and need, not on cognitive ability, and suicide risk is assessed by clinical interview and specific risk scales, not by an intelligence test.

14. Cattell considered crystallised intelligence to be a (an) (RML 2021)

    (a) Development Process (b) Ability to deal with new problems (c) Repertoire of information, cognitive skills and strategies (d) Function of brain damage

    **Answer: (c) Repertoire of information, cognitive skills and strategies.** Gc is what fluid ability has "invested" in and crystallised through education and experience: vocabulary, general knowledge and learned methods (Section 9.4). Option (b), the ability to deal with new problems, is the definition of fluid intelligence, the classic swap. Gc does develop, but "a development process" describes how it grows, not what it is; and far from being a function of brain damage, crystallised abilities such as vocabulary are the ones that hold up best with ageing.

15. According to the Guilford model of intelligence, how many dimensions of intelligence are there? (RML 2021)

    (a) Three dimensions (b) 120 dimensions (c) 10 dimensions (d) A variable number depending on I.Q.

    **Answer: (b) 120 dimensions (as keyed).** The key uses "dimensions" loosely to mean the separate abilities, the cells of Guilford's cube: 5 operations × 4 contents × 6 products = 120 (1967), later 150 and then 180 (Section 9.4). Strictly, the cube itself has three dimensions (operations, contents and products), so on a literal reading (a) is the correct fact; the paper's key nevertheless went to 120, so for this exact question mark (b), and whenever a question asks how many abilities or factors, choose 120. Ten and "a variable number depending on IQ" have no basis in the model.

16. Which type of intelligence according to Sternberg's Triarchic theory of intelligence includes the ability to easily adapt to the present environment or select a more favourable environment than the existing one or modify the environment to fit one's need? (JEMAS 2025)

    (a) Componential (b) Contextual (c) Incremental (d) Experimental

    **Answer: (b) Contextual.** Adapting to, selecting and shaping (modifying) the environment are the three functions of the contextual subtheory, whose ability Sternberg later called practical intelligence (Section 9.5). Componential intelligence is the analytical machinery of metacomponents, performance and knowledge-acquisition components. "Incremental" belongs to Dweck's incremental (growth) theory of intelligence, and "experimental" is a corruption of experiential, the subtheory of coping with novelty and automatising, that is, creative intelligence.

17. Individual differences in intelligence are due to: (Amity 2023)

    (a) Only genetic factors (b) Only environmental factors (c) Genetic and economic factors (d) Many factors including genetics and environment

    **Answer: (d) Many factors including genetics and environment.** Twin and adoption studies show substantial heritability, but it is well below 1 and varies with age and circumstances; nutrition, schooling, adoption, early intervention and the Flynn effect show large environmental effects, and the two interact (Section 9.7). Options (a) and (b) each ignore half the evidence, and (c) names only one environmental factor, socio-economic status, leaving out schooling, nutrition, family environment and the rest.

18. The cardinal DSM-IV-TR diagnostic criterion for Mental Retardation is based primarily on an IQ score: (RML 2021)

    (a) Seven standard deviations below the mean (b) One standard deviation below the mean (c) Two standard deviations below the mean (d) Four standard deviations below the mean

    **Answer: (c) Two standard deviations below the mean.** With a mean of 100 and an SD of 15, two SDs below is an IQ of about 70, the DSM-IV-TR cut-off for significantly subaverage intellectual functioning (Section 9.8). One SD below (85) is merely low average; four SDs below (40) already lies at the moderate to severe boundary; and seven SDs below would be an IQ of about −5, which does not exist on the scale.

19. In which of the following situations would a diagnosis of "global developmental delay" be inappropriate? (RML 2022)

    (a) The patient is a child who is too young to fully manifest specific symptoms or to complete requisite assessments. (b) The patient, a 7-year-old boy, has a full-scale IQ of 65 and severe impairment in adaptive functioning. (c) The patient's scores on psychometric tests suggest intellectual disability, but there is insufficient information about the patient's adaptive functional skills. (d) The patient's impaired adaptive functioning suggests intellectual developmental disorder, but there is insufficient information about the level of cognitive impairment measured by standardized instruments.

    **Answer: (b).** Global developmental delay is a provisional DSM-5 diagnosis for children under 5 whose severity cannot yet be assessed reliably (Section 9.8). The 7-year-old is over 5 and has been fully assessed, with both intellectual and adaptive deficits documented, so he receives a diagnosis of intellectual disability itself. Options (a), (c) and (d) all describe an assessment that is incomplete or not yet possible, which is exactly the situation the provisional label exists for.

20. Vineland Social Maturity Scale (VSMS) is a measure of: (RML 2022)

    (a) Mental age and Intelligence quotient (b) Social age and social quotient (c) Developmental age and developmental quotient (d) Emotional age and emotional quotient

    **Answer: (b) Social age and social quotient.** Doll's VSMS (1935) converts social competence into a social age, and SQ = (SA ÷ CA) × 100 (Section 9.8). Mental age and IQ come from intelligence tests; developmental age and DQ come from infant scales such as Gesell's and Bayley's; and "emotional age" is not a standard measure, while EQ is Bar-On's emotional quotient, which has nothing to do with the VSMS.

21. In Terman's study of gifted children, social and behavioural problems were found only in those with: (RML 2022)

    (a) IQs of 150 or higher (b) IQs of 180 or higher (c) IQs of 180 or higher in adulthood (d) IQs of 180 or higher in childhood

    **Answer: (d) IQs of 180 or higher in childhood.** As a group, Terman's "Termites" were healthier and better adjusted than average; the problems that did occur were concentrated among those whose childhood IQs were about 180 or more, in line with Hollingworth's *Children Above 180 IQ* (Section 9.8). Option (a) cannot be right because 150 is about the average of the whole gifted group; (b) omits the key point that the IQs were childhood scores, and (c) wrongly places the testing in adulthood.

22. Which of the following disabilities is explicitly covered under the scope of The National Trust Act (1999) in India for the welfare of persons with the disability? (Real PYQ 2024)

    (a) Autism Spectrum Disorder (b) Specific Learning Disability (c) Intellectual Disability (Mental Retardation) (d) Both (a) and (c)

    **Answer: (d) Both (a) and (c).** The Act's full title names its four groups: autism, cerebral palsy, mental retardation and multiple disabilities (Indian context, Section 9.8). Choosing (a) or (c) alone misses the other named group. Specific learning disability is not among the four; it is recognised under the broader Rights of Persons with Disabilities Act, 2016.

## High-yield sheet

*The facts this chapter's questions ask most, most-asked first. Last read before the exam.*

1 **William Stern** coined the intelligence quotient in 1912 (mental age ÷ chronological age); **Lewis Terman**'s Stanford-Binet (1916) multiplied it by 100, giving the ratio IQ = (MA ÷ CA) × 100, so a mental age of 8 at age 10 gives an IQ of 80. ASKED 4 TIMES

2 **Bhatia's Battery of Performance Tests of Intelligence** (C. M. Bhatia, 1955) has five performance subtests (Kohs blocks, pass-along, pattern drawing, immediate memory, picture construction), needs no reading and has rural norms: the best choice for an illiterate rural Indian child, ahead of the half-verbal MISIC. ASKED 4 TIMES

3 **Fluid intelligence (Gf)** is reasoning with novel problems independent of acquired knowledge; **crystallised intelligence (Gc)** is the repertoire of information, cognitive skills and strategies, measured by vocabulary and information subtests (Raymond Cattell, 1963, with John Horn). ASKED 4 TIMES

4 **Robert Sternberg**'s triarchic theory (*Beyond IQ*, 1985): componential = analytical, experiential = creative, contextual = practical; adapting to, shaping and selecting environments is contextual intelligence. ASKED 4 TIMES

5 **Raven's Progressive Matrices** (John C. Raven; Standard form 1938) is a non-verbal, culture-fair (not culture-free) test of fluid intelligence, and also the best single marker of g. ASKED 3 TIMES

6 The **Wechsler Adult Intelligence Scale** is an individually administered intelligence test (not an aptitude, attitude or achievement test) for adults (WAIS-IV: 16 to 90 years), used among other things to document eligibility for a disability allowance. ASKED 3 TIMES

7 The **WAIS-IV** (2008) yields four index scores, Verbal Comprehension, Perceptual Reasoning, Working Memory and Processing Speed, which combine into the Full Scale IQ; the Working Memory Index (Digit Span, Arithmetic) reflects attention, concentration and mental control. ASKED 4 TIMES

8 **Charles Spearman**'s two-factor theory (1904): a general factor g that underlies all cognitive tasks plus a specific factor s for each task, found by factor analysis of the positive manifold. ASKED 4 TIMES

9 **Edgar Doll**'s Vineland Social Maturity Scale (1935) gives a social age and a social quotient (SQ = SA ÷ CA × 100); the Vineland Adaptive Behavior Scales measure adaptive functioning (communication, daily living skills, socialisation). ASKED 3 TIMES

10 WAIS-IV core subtests: 3 verbal comprehension (Similarities, Vocabulary, Information), 3 perceptual reasoning (Block Design, Matrix Reasoning, Visual Puzzles), 2 working memory and 2 processing speed, 10 in all; Letter-Number Sequencing is supplemental in the WAIS-IV and secondary in the WISC-V (keyed as "WAIS, not WISC"). ASKED TWICE

11 **David Wechsler** replaced mental age with the deviation ("standard") IQ in the Wechsler-Bellevue (1939): the person's score is compared with the expected score of his or her own age group, mean 100, SD 15. ASKED 3 TIMES

12 On a deviation IQ (mean 100, SD 15), 100 is the 50th percentile, 85 the 16th, 115 the 84th, 130 about the 98th and 70 about the 2nd; with an SD of 16, an IQ of 136 (z = 2.25) is about the 98.7th percentile. ASKED 3 TIMES

13 **Alfred Binet** and Théodore Simon published the first intelligence test in 1905 (30 items) for the French ministry; Terman revised it at Stanford as the Stanford-Binet (1916). Piaget did not build the first test. ASKED TWICE

14 DSM-IV-TR mental retardation required an IQ about two standard deviations below the mean (about 70); DSM-5 also requires deficits in adaptive functioning and sets severity by adaptive functioning, so an FSIQ of 70 alone is not intellectual disability. ASKED TWICE

15 Individual differences in intelligence come from many genetic and environmental factors acting together; twin studies (MZ versus DZ twins, especially twins reared apart) estimate heritability, which is a population statistic. ASKED TWICE

16 WAIS-IV bands: 90 to 109 average, 80 to 89 low average, 70 to 79 borderline, 69 and below extremely low; an IQ of 70 to 84 is "low average or borderline, depending on the edition". ASKED TWICE

17 The **SB5** (Gale Roid, 2003) measures five CHC factors (Fluid Reasoning, Knowledge, Quantitative Reasoning, Visual-Spatial Processing, Working Memory), each in verbal and nonverbal form, drawing on Carroll and Baddeley. ASKED TWICE

18 **J. P. Guilford**'s structure of intellect (1967) has three dimensions (5 operations × 4 contents × 6 products) and so 120 abilities, later 150 and 180; one paper keyed "120 dimensions". ASKED ONCE

19 The **WISC-V** (2014) covers 6 years to 16 years 11 months; the WPPSI-IV covers 2 years 6 months to 7 years 7 months and the WAIS-IV 16 to 90 years. ASKED ONCE

20 DSM-5 **global developmental delay** is a provisional diagnosis only for children under 5 whose severity cannot yet be assessed reliably. ASKED ONCE

21 In **Terman**'s gifted study (from 1921; the "Termites", average IQ about 151) the children were healthier and better adjusted than average; social and behavioural problems clustered among those with childhood IQs of 180 or more. ASKED ONCE

22 India's **National Trust Act, 1999** covers autism, cerebral palsy, mental retardation (intellectual disability) and multiple disabilities; specific learning disability is not among them. ASKED ONCE

23 Correlating a new intelligence test with an established one given at the same time tests **concurrent validity**; a later criterion would be predictive validity. ASKED ONCE

24 Keyed answer: the **aptitude test** is "the heart of the guidance programme" (RML 2022); the textbook phrase belongs to counselling, so choose counselling if it is offered. ASKED ONCE

25 **James McKeen Cattell** coined the term "mental test" in 1890. ASKED ONCE

26 Wechsler built the Wechsler-Bellevue to fix the Stanford-Binet's faults: little appeal for adults, too much word manipulation, too much weight on speed and reliance on mental age. ASKED ONCE

27 The **KBIT-2** (Kaufman Brief Intelligence Test) has two scales: Verbal (crystallised) and Nonverbal (fluid). ASKED ONCE

28 The **Bayley Scales** of Infant and Toddler Development assess cognitive, language and motor development up to about 42 months. ASKED ONCE

29 **Daniel Goleman**'s five domains of emotional intelligence (1995): knowing one's emotions, managing them, motivating oneself, recognising emotions in others and handling relationships. ASKED ONCE

30 **Carol Dweck**'s growth mindset is the belief that intelligence is malleable and can be developed through effort; the fixed mindset (entity theory) sees it as unchangeable. ASKED ONCE

CHAPTER 10

# Motivation: Why We Do What We Do

IN THIS CHAPTER

10.1 ●● What Motivation Is: define need, drive, motive, incentive and goal, trace the motivational cycle, and classify motives as primary or secondary and as intrinsic or extrinsic.

10.2 ●● Instinct Theories: explain the instinct theories of James, McDougall and Freud, the ethologists' fixed action patterns, and why instinct theory declined.

10.3 ●● Drives, Homeostasis and Arousal: describe homeostasis, Hull's drive-reduction theory and its limits, optimal arousal, the Yerkes-Dodson law and sensation seeking.

10.4 ● The Pull of the Goal: Incentives, Expectancies and Opponent Processes: contrast push and pull theories, explain expectancy-value thinking, and describe the opponent-process theory of acquired motivation.

10.5 ●●● Hierarchies of Need: Maslow, Alderfer and Other Lists: state Maslow's hierarchy precisely, distinguish deficiency from growth needs, evaluate the theory, and compare it with Alderfer's ERG theory and other lists of needs.

10.6 ●● Psychosocial Motives and Achievement: explain Murray's needs and press, McClelland's three needs, Atkinson's theory of achievement motivation, fear of success and Weiner's attribution theory.

10.7 ●●● Intrinsic Motivation, Competence and the Self: explain curiosity, competence, self-efficacy, the overjustification effect, self-determination theory, flow and the consistency motive.

10.8 ●● Conflict and Frustration: classify motivational conflicts, use Miller's gradients to predict behaviour, and describe the sources of frustration and the ways people react to it.

10.9 ●● Biological Motives: Hunger, Thirst, Sex and Others: explain what starts and stops eating and drinking, how the hypothalamic "centre" theory was revised, and the classic findings on sexual and maternal motivation.

10.10 ●●● Motivation in Practice: Work, School and Clinic: apply Herzberg, Vroom, Adams and Locke to work, apply motivation research to the classroom, and describe motivational interviewing.

Weightage. 76 of the 819 questions mapped in this book (9.3%) test this chapter, 36 of them from real past papers. 3 of its 10 sections are High yield.

In a primate laboratory at the University of Wisconsin in Madison, at the end of the 1940s, a young rhesus monkey sat in its home cage fiddling with a small wooden board. Screwed to the board was a puzzle made of three pieces of hardware: a metal pin, a hook and a hasp. To undo the hasp, the monkey had to pull the pin first, then lift the hook, and only then swing the hasp open. There was nothing behind the hasp. No raisin, no peanut, no sip of juice. The monkey had been fed and watered as usual. Yet it worked at the puzzle, again and again, and when the experimenters put the pieces back together it took them apart again.

The experimenters were Harry Harlow, Margaret Kuenne Harlow and Donald Meyer, and in 1950 they reported what they had seen. Monkeys left with assembled puzzles for about two weeks became fast, nearly error-free solvers. Monkeys given the same puzzles already taken apart, so that there was nothing to solve, did not. Harlow even reported that bringing food into the test tended to disrupt the monkeys' smooth performance rather than improve it. The monkeys, the authors concluded, were driven by the activity itself. They called it a manipulation drive.

Why should this small observation have caused such a stir? Because in 1950 the most powerful theory in American psychology said it should not happen. According to that theory, an animal learns a response only when the response reduces a biological need: hunger, thirst, pain. Harlow's monkeys were not hungry, not thirsty and not in pain. What need was being reduced? If you have ever done the newspaper Sudoku on a bus to Mysuru, or stayed up past midnight to finish a crossword that nobody will ever check, you already know the answer the theory could not give.

This chapter is about that puzzle: why animals and people do what they do. You will meet instincts, drives, arousal, incentives, needs, expectancies and the self, roughly in the order psychology met them. Keep Harlow's monkeys in mind as you read; we will come back to them at the end of the chapter.

## 10.1 What Motivation Is

MEDIUM YIELD · 7 QS

The word **motivation** comes from the Latin *movere*, "to move". In psychology it refers to the processes that start behaviour, give it direction, set its intensity and keep it going until it ends. Notice the four parts of that definition, because questions are built on them. **Activation** is the starting of behaviour (you get up from your desk). **Direction** is the choice of one goal over others (you walk to the kitchen, not the television). **Intensity** is how much effort goes in (you stride, you do not stroll). **Persistence** is how long the behaviour continues in the face of obstacles (you keep searching after finding the biscuit tin empty).

No one has ever seen a motive. You see a student studying until 2 a.m., and you infer that something inside is pushing or pulling him. Motivation is therefore a hypothetical construct, an **intervening variable** placed between what we can observe (hours without food, a promised reward) and what we can measure (speed, effort, persistence). This is why the history of motivation is a history of competing guesses about what that invisible something is.

### Need, drive, motive, incentive and goal

Five words recur in every theory in this chapter, and the exam treats them as distinct. Take one example, thirst, and follow it through.

A **need** is a state of deficiency or lack within the organism. After a long afternoon in a Mysuru summer, your body is short of water. That shortage is a physiological need, and it exists whether or not you feel it. A **drive** is the aroused, tense psychological state that a need produces and that energises behaviour. You feel thirsty, restless and irritable. Need is the biological condition; drive is its psychological consequence. The two are not perfectly correlated: a person can have a need without a strong drive (old people can become dangerously dehydrated without feeling very thirsty), and a drive can outlast the need.

A **motive** is the broader term for any internal state that arouses and directs behaviour towards a goal. In practice "drive" is reserved for biologically based states and "motive" for learned, psychological and social ones, such as the motive to achieve or to be liked. John Atkinson used "motive" in a narrower sense still: a stable personality disposition that a person carries from situation to situation.

An **incentive** is an external stimulus or object that attracts (positive incentive) or repels (negative incentive) the organism. A cold bottle of water in a shop window is an incentive; so is the thought of the chai stall after the exam. A **goal** is the end state towards which motivated behaviour is directed, the incentive as it is represented and pursued: reaching the water and drinking it. The commonly taught shorthand is that drives **push** from inside and incentives **pull** from outside.

### The motivational cycle

Put the five terms in order and you get the **motivational cycle**: a need gives rise to a drive; the drive energises **instrumental behaviour**, that is, goal-directed activity that is a means to an end; the behaviour reaches the goal or incentive; reaching the goal reduces the drive and brings relief; and after a while the need builds up again and the cycle restarts. Figure 10.1 draws the cycle for thirst. The cycle is a useful teaching device, but notice that it is drawn from one theory, drive reduction, which you will meet in Section 10.3. Much of the rest of the chapter is about behaviour that does not fit it.

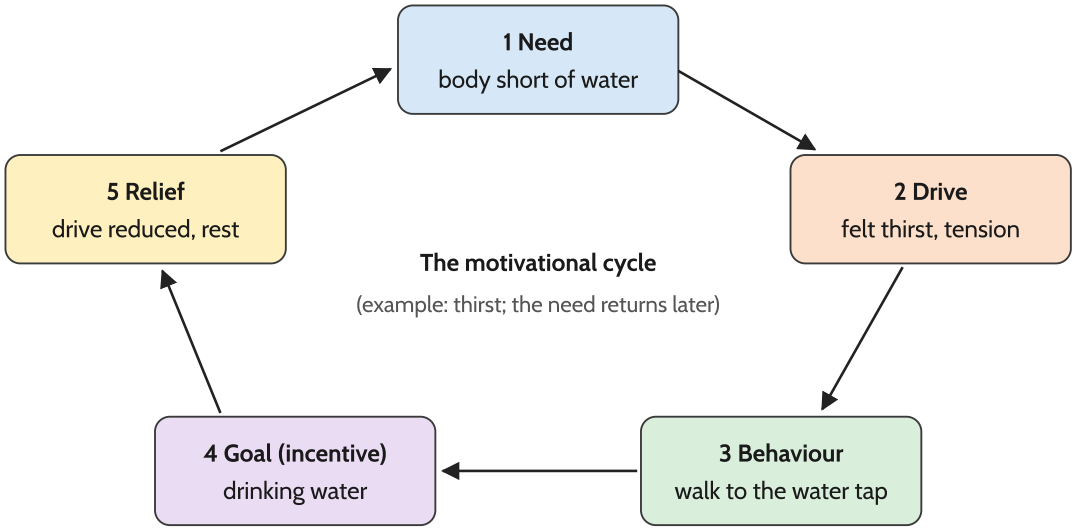


Figure 10.1 The motivational cycle. A need produces a drive, the drive energises instrumental behaviour, the behaviour reaches the goal, and the drive is reduced until the need builds up again. Notice that this picture is drive-reduction theory in miniature.

## Classifying motives ●●

The most common classification divides motives by their origin. The **primary motives**, also called biological, physiological or unlearned motives, are rooted in the body's survival needs and are present without learning: hunger, thirst, the need for oxygen, sleep, avoidance of pain, temperature regulation, elimination, sex and, in many textbooks, the maternal drive. Most of them are **homeostatic** (they correct a bodily imbalance); sex and the maternal drive are the usual exceptions, because no tissue is depleted when they are unsatisfied.

The **secondary motives**, also called psychosocial, social, acquired or learned motives, develop through experience and culture: achievement, affiliation, power, approval, aggression, curiosity about a particular subject, the wish for money. They vary enormously between people and between cultures. Between the two sits a group that some textbooks (following Clifford Morgan and Richard King) call **general** or unlearned non-homeostatic motives: curiosity, activity, manipulation and the need for contact comfort. They seem to be inborn, like primary motives, yet they repair no tissue deficit. Harlow's monkeys belong here. Table 10.1 sets out the three classes side by side.

The personality theorist Henry Murray (you will meet him properly in Section 10.6) drew the same line with his own words. His **viscerogenic needs** are the primary needs tied to bodily events (air, water, food, sex, urination, avoidance of heat, cold and harm). His **psychogenic needs** are the secondary needs that arise from mental life and are only loosely connected to the body (achievement, affiliation, dominance, autonomy, order).

Motives can also be classified by awareness: **conscious** motives are ones the person can report, and **unconscious** motives (the core of Freud's theory) drive behaviour without the person knowing why. Finally, the American personality psychologist **Gordon Willard Allport** (1897 to 1967) pointed out in 1937 that a learned motive can cut loose from its origins. A man who went to sea to earn a living may, when rich, still long for the sea. Allport called this

**functional autonomy**: a means can become an end in itself, so today's motive need not be explained by the need that started it.

Table 10.1 A working classification of motives.

| Class | Other names | Examples | Key feature |
|---|---|---|---|
| Primary | Biological, physiological, unlearned, viscerogenic | Hunger, thirst, sleep, pain avoidance, sex, maternal drive | Rooted in body; mostly homeostatic |
| General | Unlearned non-homeostatic, stimulus motives | Curiosity, exploration, manipulation, activity, contact comfort | Inborn but repair no tissue deficit |
| Secondary | Psychosocial, social, learned, acquired, psychogenic | Achievement, affiliation, power, approval, gregariousness | Learned; vary by person and culture |

EXAM TRAP

**Gregariousness**, the tendency to seek the company of others, is classed as a **social motive** (RML 2022 keyed "social motive" over biological, psychological and personal). The bait is that William McDougall listed gregariousness as one of his instincts (Section 10.2). In the textbook classification of motives, the answer is still "social". Similarly, "psychological motive" and "personal motive" are not standard class names; do not be drawn to them.

## Intrinsic and extrinsic motivation ●●

A second, cross-cutting distinction concerns where the reward lies. The desire to do something for its own sake, because the activity itself is interesting, enjoyable or satisfying, is called **intrinsic motivation**. The desire to do something for a separable outcome (money, marks, praise, or the avoidance of punishment) is called **extrinsic motivation**. A child who reads comics under the blanket is intrinsically motivated; the same child reading the prescribed textbook to escape a scolding is extrinsically motivated.

A classic past-paper example: a toddler goes to great lengths to open a box that she knows is empty. Nothing is inside, so there is no external payoff; the opening is the reward. That is intrinsic motivation. It is not **homeostasis** (no bodily balance is being restored), not extrinsic motivation (there is no separable reward) and not secondary motivation (nothing learned or social is being served). The two kinds of motivation can interact in surprising ways, and Section 10.7 shows how an external reward can sometimes kill an internal interest.

CLINICAL LINK

The intrinsic-extrinsic distinction is used diagnostically. In **malingering** a person deliberately produces or exaggerates symptoms for an obvious external incentive (money, avoiding duty, obtaining drugs). In **factitious disorder** the person also deliberately produces symptoms, but in the absence of obvious external rewards; the motive is internal, typically to occupy the sick role. Same behaviour, different motive, different diagnosis. (The disorders themselves are covered in the Psychiatric Disorders book.)

## Five ways of explaining motivation ●●●

Before we take the theories one by one, it helps to see the whole map. Each major school of psychology produced its own answer to "why do we do what we do?", and examiners love

questions that ask which school said what. Table 10.2 sets out the families in the order this chapter follows them.

Table 10.2 The main families of motivation theory.

| Family | Core claim | Key names | Section |
|---|---|---|---|
| Instinct and psychodynamic | Inborn forces, often unconscious and beyond our control, drive behaviour | James, McDougall, Freud, Lorenz | 10.2 |
| Drive and arousal | We act to reduce tension or to keep arousal at an optimal level | Cannon, Hull, Hebb, Yerkes and Dodson | 10.3 |
| Incentive and expectancy | Goals pull us; we act on what we expect and value | Tolman, Rotter, Atkinson, Vroom | 10.4, 10.6 |
| Humanistic | Free will and an innate tendency to grow towards one's potential | Maslow, Rogers, Alderfer | 10.5 |
| Cognitive and self | Beliefs, attributions, goals and self-efficacy direct behaviour | Weiner, Bandura, Deci and Ryan, Locke | 10.6, 10.7 |

EXAM TRAP

"Which model holds that inner forces largely beyond individual control drive behaviour?" The answer is the **psychodynamic** model (Freud's instincts working through the unconscious). "Which perspective emphasises free will and personal growth" or "an innate tendency to grow and reach one's maximum potential"? The **humanistic** perspective. The cognitive model emphasises thinking and beliefs; the behavioural model emphasises conditioning and reinforcement; "statistical norms" belong to the statistical definition of abnormality (Psychiatric Disorders book), not to any school of motivation. Do not confuse "humanistic" with "existential": Maslow and Rogers are the humanistic pair; Frankl, May and Yalom are the existential names.

## 10.2 Instinct Theories

MEDIUM YIELD · 5 QS

The first scientific answer to the motivation question was the oldest folk answer dressed in new clothes: we do things because we are born to do them. A spider spins a perfect web the first time it tries; a newly hatched turtle heads for the sea. Darwin's theory of evolution (1859) made such inborn behaviour respectable, because a behaviour that helped ancestors survive could be inherited just as a beak or a wing could. Around 1900 it seemed natural to explain human motivation the same way.

### What counts as an instinct

An **instinct** is a complex, unlearned pattern of behaviour that is characteristic of every member of a species and that appears in full form the first time the triggering situation arises. Four criteria follow from the definition: the behaviour is **unlearned** (it appears in animals reared in isolation), **species-specific** or universal within the species, **stereotyped** (performed the same way each time) and **complex** (more than a simple reflex such as the knee jerk). Keep these criteria in mind; you will see that most of what was called "instinct" in human beings fails them.

**William James** (1842 to 1910), in his *Principles of Psychology* of 1890, defined instinct as "the faculty of acting in such a way as to produce certain ends, without foresight of the ends, and without previous education in the performance." James then made a claim that surprised his readers: human beings do not have fewer instincts than animals, they have **more**. His list included sucking, crying, imitation, rivalry, pugnacity, sympathy, fear, acquisitiveness, constructiveness, play, curiosity, sociability, shyness, modesty, jealousy and parental love. Humans look less instinctive, James argued, because our many instincts conflict with one another and because habit quickly modifies them; an instinct, once acted on, gets bound to particular objects and loses its blind quality.

## McDougall's hormic psychology

The theorist most closely tied to the instinct theory of motivation is the British-born psychologist **William McDougall** (1871 to 1938), who moved to Harvard in 1920 and later to Duke University. In *An Introduction to Social Psychology* (1908) McDougall argued that instincts are the prime movers of all human activity: without them, he wrote, the organism would be like a clock without a mainspring.

For McDougall an instinct was an inherited disposition with three aspects, corresponding to the old three-part division of mind. It has a **cognitive** aspect (the animal perceives and pays attention to certain objects), an **affective** aspect (perceiving the object arouses a particular emotion) and a **conative** aspect (the animal feels an impulse to act). Each major instinct was paired with a primary emotion, and Table 10.3 shows the famous pairs. Over time, emotions organise around particular objects into **sentiments** (love of one's country, hatred of a rival), and the most important of these, the self-regarding sentiment, gives character its unity.

Table 10.3 McDougall's principal instincts and their primary emotions (1908).

| Instinct | Primary emotion | Instinct | Primary emotion |
|---|---|---|---|
| Flight | Fear | Self-abasement (subjection) | Negative self-feeling |
| Repulsion | Disgust | Self-assertion (self-display) | Positive self-feeling (elation) |
| Curiosity | Wonder | Parental | Tender emotion |
| Pugnacity | Anger | Also listed | Gregariousness, acquisition, construction, reproduction |

McDougall's list grew across the many editions of his book; by the time of *The Energies of Men* (1932) he preferred the word "propensities" and listed eighteen of them, which is the figure (18 instincts) you will find in most Indian textbooks. His system is called **hormic psychology**, from the Greek *horme*, meaning an urge or impulse, because it put purposive striving at the centre of psychology. McDougall insisted that behaviour is **purposive**: it is directed towards goals, it persists, it varies its means when blocked, and it stops when the goal is reached. This put him in open conflict with John B. Watson's behaviourism, and the two men debated publicly in 1924.

EXAM TRAP

"The instinct theory of motivation was developed by" has the keyed answer **William McDougall**, even though William James also wrote about instincts and Freud built his theory on them. James described instincts; McDougall made them the systematic foundation of motivation and social behaviour. Maslow is the bait from the wrong school (humanistic). A second trap: McDougall's **purposive** (hormic) psychology is not Edward Tolman's **purposive behaviourism**. Both stress goals; McDougall explained goals by instincts, Tolman by learned expectancies (Section 10.4).

### Freud's instincts: life and death ●●●

**Sigmund Freud** (1856 to 1939) also put inborn forces at the root of motivation, though his are better described as drives. His German word was *Trieb*, a driving force or impulse, which his English translators rendered as "instinct". Freud said every instinct has four features: a **source** (a bodily need or excitation), an **aim** (to remove the excitation, which is felt as pleasure), an **impetus** (the amount of force or pressure it exerts) and an **object** (the thing or person through which the aim is achieved). Of the four, only the object is learned and changeable, which is why the same hunger can be satisfied by idli in Mysuru and bread in Vienna.

In his final theory, set out in *Beyond the Pleasure Principle* (1920), Freud grouped all instincts into two classes. The **life instincts** (Eros) serve the survival of the individual and the species: hunger, thirst and above all sex. Their energy is the **libido**. The **death instinct**, which his followers named Thanatos, is an unconscious wish to return to the inorganic state; turned outwards it appears as aggression and destructiveness. Freud developed the idea after the slaughter of the First World War, and he read war and violence as evidence of it. The investment of psychic energy in an object, person or idea he called **cathexis**. The reservoir of all these unorganised instinctual drives is the **id**, which operates on the pleasure principle. (The id, ego and superego, the defence mechanisms and the psychosexual stages are taught in the Personality book; here we need only Freud's view of motivation.)

EXAM TRAP

"According to Freud, human beings are motivated by powerful innate forces known as" has the answer **instincts**, not needs (Maslow, Murray), not drives (Hull's term, though it is a fair translation of *Trieb*) and not the id (the id is the **reservoir** of the instincts, not the forces themselves). "Wars and violence are evidence of" the **death instinct**; libido is the energy of the life instincts. Freud himself did not use the word Thanatos in his published writing; it was popularised by his followers.

### Ethology: fixed action patterns and sign stimuli

While instinct theory was fading in human psychology, it was reborn in zoology. The new discipline was **ethology**, the study of animal behaviour in the natural environment, with an emphasis on its evolution and function. Its founders, the Austrian **Konrad Lorenz** (1903 to 1989), the Dutch-born **Nikolaas Tinbergen** (1907 to 1988) and the Austrian **Karl von Frisch** (1886 to 1982), shared the Nobel Prize in Physiology or Medicine in 1973.

The ethologists replaced the vague word "instinct" with precise terms. A **fixed action pattern** is an unlearned, stereotyped sequence of movements that, once started, runs to comple-

tion. A **sign stimulus** (or releaser) is the simple, specific feature of the environment that triggers it, and the hypothesised neural detector that responds to the sign stimulus was called the **innate releasing mechanism**. Tinbergen found that a male three-spined stickleback in breeding season attacks crude models of fish as long as their underside is red, while ignoring a realistic model without the red belly. Lorenz and Tinbergen showed that a greylag goose that sees an egg outside its nest stretches its neck and rolls the egg back with the underside of its bill; if the egg is removed mid-roll, the goose finishes the rolling movement over empty ground. A herring gull chick pecks at the red spot on its parent's bill, and the pecking brings up food. Lorenz also described **imprinting**, the rapid attachment of newly hatched geese to the first moving object they see, during a brief sensitive period (taught in the Learning and Developmental chapters).

### Why instinct theory declined

As an explanation of human motivation, instinct theory had collapsed by the late 1920s, for three reasons.

First, **naming is not explaining**. Why do people form groups? Because of a gregarious instinct. How do we know there is a gregarious instinct? Because people form groups. The argument is circular: the "instinct" is simply the behaviour renamed and then offered as its own cause. Second, the lists grew out of control. Every new behaviour seemed to earn its own instinct, and in 1924 the sociologist **Luther Lee Bernard** surveyed the literature and counted nearly six thousand proposed human instincts, a reduction to absurdity. Third, the evidence pointed to learning. Anthropologists showed that supposedly universal "instincts" such as pugnacity or acquisitiveness varied enormously between cultures, and the Chinese psychologist **Zing-Yang Kuo** (1898 to 1970) showed in 1930 that even a cat's "instinctive" killing of rats depended on rearing: most kittens that watched their mothers kill rats became rat killers, while few kittens reared together with rats ever killed one.

The idea did not die completely. Ethology showed that fixed action patterns are real in many animals, and modern **evolutionary psychology** asks how natural selection shaped human motives such as mate preferences and fear of snakes. But psychologists stopped treating a list of instincts as an explanation, and looked for a mechanism. The mechanism they found was the drive.

> **REMEMBER**
>
> Instinct theory explained behaviour by naming it. Its replacement, drive theory, tried to say how a bodily need turns into action. Whenever a question asks why instinct theory fell, the expected answer is circularity (naming is not explaining), followed by cultural variability and the role of learning.

## 10.3 Drives, Homeostasis and Arousal MEDIUM YIELD · 7 QS

Drive theory begins not in psychology but in physiology, with a simple observation: the inside of a healthy body stays remarkably constant while the world outside changes. The air around you may be at 45 degrees on a Rajasthan afternoon or at 5 degrees on a Shimla morning, yet your core temperature stays near 37 degrees Celsius. How?

## Homeostasis

The French physiologist **Claude Bernard** (1813 to 1878) argued in the second half of the nineteenth century that the cells of the body live in an internal sea of fluids, which he called the **milieu intérieur** (internal environment). The constancy of this internal environment, he wrote, is the condition for a free and independent life. The American physiologist **Walter Bradford Cannon** (1871 to 1945) of Harvard gave the idea its modern name: he coined the term **homeostasis** in 1926 and popularised it in his book *The Wisdom of the Body* (1932). (Some books date the term to 1932, the year of the book; the word itself first appeared in print in 1926.) Homeostasis is the tendency of the body to maintain a stable internal state by detecting deviations and correcting them.

The usual model is a thermostat. A **set point** is the value the system tries to maintain (the temperature you dial in). When a sensor detects a deviation from it, a correction is switched on (the heater), and when the set point is restored the correction is switched off. This is **negative feedback**: the output of the system feeds back to reduce the deviation that produced it. Some corrections are automatic and need no behaviour at all (sweating, shivering). Others require the whole organism to act: you put on a sweater, walk into the shade or drink water. Those behavioural corrections are what a motivation theory must explain, and that is where Clark Hull came in.

## Hull's drive-reduction theory ●●●

The word "drive" entered psychology in 1918, when **Robert Sessions Woodworth** (1869 to 1962) of Columbia University used it in *Dynamic Psychology* for the energy that powers behaviour, as distinct from the mechanisms that steer it. The theory built on the word belongs to **Clark Leonard Hull** (1884 to 1952) of Yale, whose *Principles of Behavior* (1943) was for a decade the most influential book in American psychology.

Hull's **drive-reduction theory** runs as follows. A physiological need (a departure from homeostasis) creates a drive, an aroused state of tension that energises behaviour. The organism becomes active, and any response that happens to reduce the drive is **reinforced**, that is, strengthened, so that it is more likely to occur the next time the drive appears. Reinforcement, for Hull, simply **is** drive reduction. The goal of motivated behaviour is a return to a state of low tension.

Two features of the theory are often examined. First, drive **energises** but does not **direct**. What decides which response the hungry rat makes is **habit strength**, the learned stimulus-response connection built up by past reinforcement; drive only multiplies it. Second, drive is general: Hull believed that the energy from any drive pooled together and energised whatever habit was strongest at the moment. Hull expressed all this in a formula. In its later form (by 1952) it reads, in simplified notation:

**Reaction potential** ${}_{S}E_{R} = {}_{S}H_{R} \times D \times K \times V$ (minus inhibition)

Because the terms multiply, if any one of them is zero the response does not occur. A rat that has learned a maze perfectly (high habit strength) but has just eaten (drive near zero) will sit in the start box. Hull added that reaction potential must exceed a minimum value, the **reaction threshold** (${}_{S}L_{R}$), before any response appears at all, and that above the threshold its strength shows in four measurable ways: a shorter response latency, a greater **response (re-**

**action) amplitude**, meaning the vigour or size of the response, a higher probability of responding and greater resistance to extinction. Table 10.4 lists the symbols.

Table 10.4 The terms in Hull's equation.

| Symbol | Name | What it stands for |
|---|---|---|
| ${}_{S}E_{R}$ | Reaction (excitatory) potential | The tendency to make the response |
| ${}_{S}H_{R}$ | Habit strength | Learned S-R bond; grows with reinforced trials |
| D | Drive | Energy from need, for example hours without food |
| K | Incentive motivation | Size and quality of the reward |
| V | Stimulus-intensity dynamism | Stronger stimuli evoke the response more readily |
| J | Delay of reinforcement | Longer delays weaken the response |
| $I_{R}$ and ${}_{S}I_{R}$ | Reactive and conditioned inhibition | Fatigue from responding, and learned not-responding |

EXAM TRAP

"The greater the intensity of a stimulus, the greater the probability that a learned response will be elicited" is Hull's **stimulus-intensity dynamism** (V). **Incentive motivation** (K) is about the size of the reward, not the strength of the stimulus. Some coaching notes wrongly gloss V as "need"; need belongs to D. **Response amplitude** and **reaction threshold** are genuine Hullian terms too, but they name a measure of the response and the minimum reaction potential needed to produce it, not the effect of stimulus intensity. Also note that the word "drive" was introduced by **Woodworth (1918)**, but drive-reduction theory is **Hull's (1943)**, and homeostasis is **Cannon's** term (Bernard's concept was the milieu intérieur).

## Primary and secondary drives

Hull distinguished **primary drives**, which arise from tissue needs (hunger, thirst, pain, temperature, sex), from **secondary drives**, which are acquired through learning when a neutral stimulus is repeatedly associated with a primary drive. The clearest demonstration came from **Neal Elgar Miller** (1909 to 2002), Hull's colleague at Yale, in 1948. Rats were shocked in a white compartment and escaped through a door into a black one. Later, with no shock at all, the rats still hurried out of the white compartment, and they learned a completely new response (turning a wheel to open the door) to get out. Fear had become a learned drive, and reducing it was reinforcing enough to teach new behaviour.

The same logic underlies **Orval Hobart Mowrer**'s (1907 to 1982) **two-factor theory** of avoidance learning (1947). Factor one is classical conditioning: a warning signal paired with shock comes to evoke fear. Factor two is operant conditioning: the avoidance response is reinforced because it reduces that fear. Mowrer's theory explains why phobic avoidance is so persistent: each escape from the feared situation reduces anxiety, and so strengthens the avoidance (the Learning chapter develops this).

## What drive reduction cannot explain

Drive-reduction theory works well for thirst after a cricket match. It works badly for much else, and the failures pushed psychology towards new ideas.

**Animals and people seek stimulation.** Harlow's monkeys took puzzles apart for nothing (the opening story). Robert Butler showed in 1953 that rhesus monkeys in a closed box would learn a discrimination task for the sole reward of opening a small window and looking out at the laboratory, and would keep doing so for hours. **Rewards need not reduce any need.** Fred Sheffield and Thornton Roby reported in 1950 that rats learned a maze to obtain saccharin, a sweetener with no nutritional value whatever. **Reducing stimulation can be unbearable.** At McGill University in 1954, in Donald Hebb's laboratory, Woodburn Heron and his colleagues (Bexton, Heron and Scott) paid students a generous daily wage to lie in a quiet cubicle wearing translucent goggles, gloves and cardboard cuffs, with almost nothing to see, hear or touch. By drive-reduction logic this should have been bliss. Most students quit within two or three days, and many reported difficulty thinking and vivid hallucinations. **We eat when we are not hungry**, as anyone who has accepted a second sweet at a wedding knows. And **people deliberately raise their tension**: they climb mountains, ride roller coasters and watch horror films.

These findings did not abolish the concept of drive; they showed that tension reduction is only half the story. Sometimes organisms want less stimulation, and sometimes more. The next idea took that seriously.

## Optimal arousal theory ●●●

Psychologists use **arousal** to mean the general level of physiological and psychological activation of the organism, running from deep sleep at one end to frantic excitement at the other. It is regulated by the brainstem's reticular activating system (the Attention and Consciousness chapter, Chapter 5, covers its anatomy). The **optimal arousal theory** holds that each organism tries to maintain a moderate, optimal level of arousal: when arousal is too low (boredom) it seeks stimulation, and when arousal is too high (anxiety, overload) it seeks to reduce it. The theory was set out most influentially by the Canadian psychologist **Donald Olding Hebb** (1904 to 1985) in a 1955 paper, "Drives and the C.N.S." (the initials punned on "central nervous system" and "conceptual nervous system"). Hebb argued that arousal acts as a general drive and that performance is best at an intermediate level of it. The theory explains sensory-deprivation distress, curiosity and thrill seeking in one stroke, which drive reduction could not.

Optimal arousal theory found its classic evidence in a study that was forty years old by the time Hebb wrote.

### KEY STUDY: YERKES AND DODSON'S DANCING MICE (1908)

**Question.** Does stronger motivation always produce faster learning? **Method. Robert Mearns Yerkes** (1876 to 1956) and **John Dillingham Dodson** (1879 to 1955) at Harvard trained "dancing mice" to choose the white compartment of a discrimination box and avoid the black one; a wrong choice brought an electric shock of weak, medium or strong intensity. They also varied how hard the brightness discrimination was. **Result.** Learning was fastest at an intermediate shock, not the strongest one, and the harder the discrimination, the weaker the shock that produced the fastest learning. **Conclusion.** The relation between motivation and performance is not linear but rises and then falls, and its peak depends on task difficulty. **Criticism.** The original study was about shock strength in mice, not "arousal" or "anxiety" in people; the name "Yerkes-Dodson law" and the arousal interpretation came decades later, and the human evidence for a neat inverted U is weaker than textbooks suggest.

The **Yerkes-Dodson law**, as it is now taught, has two parts. First, the relation between arousal (or anxiety) and performance is an **inverted U**: performance improves as arousal rises from low to moderate, peaks, and then declines as arousal rises further. Second, the optimal level depends on the task: for **simple or well-learned** tasks the optimum is relatively high, while for **complex, difficult or new** tasks the optimum is lower. Figure 10.2 shows both curves. In everyday terms, a little exam anxiety sharpens you, and a lot wrecks you, and it wrecks you sooner on a hard numerical problem than on writing your roll number.

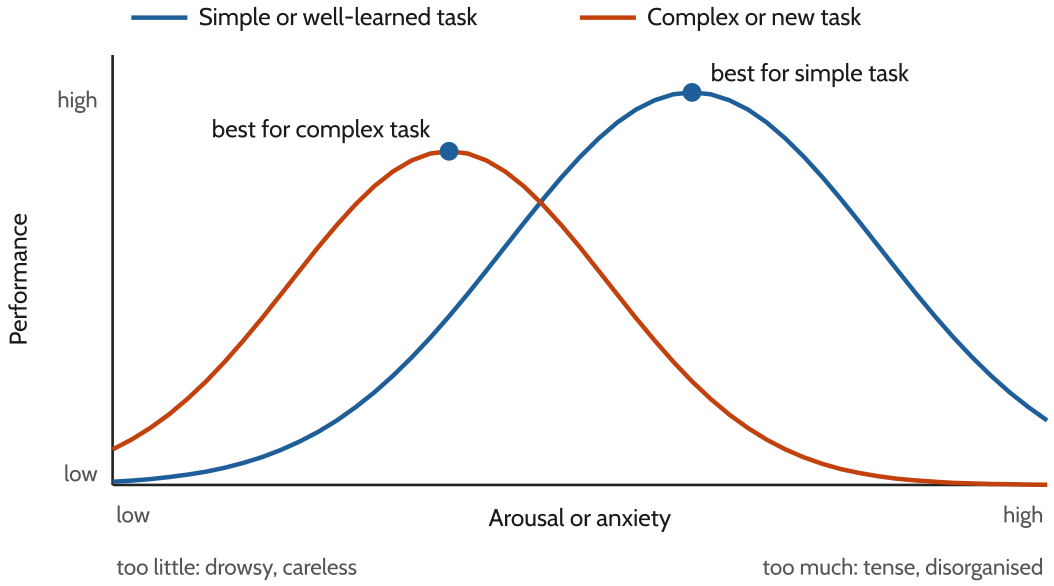


Figure 10.2 The Yerkes-Dodson law. Performance is an inverted-U function of arousal, and the peak for a complex or new task (dashed) lies at a lower arousal level than the peak for a simple or well-learned task (solid).

Why does too much arousal hurt? One answer came from J. A. Easterbrook in 1959: as arousal rises, attention narrows to fewer cues (his **cue-utilisation hypothesis**). At moderate arousal, irrelevant cues are dropped and performance improves; at high arousal, relevant cues are dropped as well and performance falls, and complex tasks, which need more cues, suffer first.

The same arousal logic explains a social phenomenon. Norman Triplett noticed in 1898 that cyclists rode faster when racing others than when racing the clock. **Robert Zajonc** (1923 to 2008) explained such **social facilitation** in 1965 with a drive theory: the mere presence of others raises arousal, and arousal strengthens the **dominant response**, the response most likely in that situation. On an easy or well-learned task the dominant response is the correct one, so an audience improves performance; on a new or difficult task the dominant response is often an error, so an audience impairs it. (Social facilitation and social loafing are developed in the Social Psychology book.)

> EXAM TRAP
>
> "The Yerkes-Dodson law states that between performance and anxiety there is" an **inverted-U shaped relationship**. "Curvilinear" is not false (an inverted U is a kind of curve), but it is less specific, and when both appear the precise answer wins. "Linear" and "M-shaped" are simply wrong. The second half of the law is examined too: on a **complex** task, performance is best at a **moderate arousal level, with the optimum lower** than on a simple task. Never choose "very high arousal" or "independent of task difficulty". An option such as "punishment severity and learning rate" describes what the 1908 mouse experiment actually varied, but when "arousal and performance as an inverted U" is offered, that is the law as it is taught.

## Sensation seeking

If each of us has an optimal arousal level, it need not be the same level for everyone. **Marvin Zuckerman** (1928 to 2018) built a research programme on that idea. In his definition, **sensation seeking** is the tendency to seek varied, novel, complex and intense sensations and experiences, and to take physical, social, legal and financial risks for the sake of them. Zuckerman's Sensation Seeking Scale (first version 1964) measures four components: **thrill and adventure seeking** (desire for risky physical activities such as skydiving), **experience seeking** (novel experiences through the mind, travel, music and a non-conforming lifestyle), **disinhibition** (seeking release through parties, drinking and varied sexual partners) and **boredom susceptibility** (intolerance of repetition and routine). High sensation seekers are over-represented among extreme-sport enthusiasts, early drug users and people who drive fast; biologically they tend to have low levels of the enzyme monoamine oxidase (MAO). Zuckerman's work turned optimal arousal from a principle about all organisms into a dimension on which individuals differ.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. A state of physiological deficiency is a _____; the aroused psychological state it produces is a _____.
2. External objects that pull behaviour towards them are called _____.
3. William McDougall's instinct-based system is called _____ psychology.
4. Freud's energy of the life instincts is the _____.
5. The simple stimulus that triggers a fixed action pattern is a sign stimulus or _____.
6. The term homeostasis was coined by Walter _____.
7. In Hull's equation, K stands for _____ motivation.
8. Rats learned a maze for saccharin, which reduces no need, in a 1950 study by Sheffield and _____.
9. For a complex task, the optimal level of arousal is _____ than for a simple task.
10. Thrill and adventure seeking, experience seeking, disinhibition and boredom susceptibility are the four components of Zuckerman's _____.

**Answers:** (1) need, drive, (2) incentives, (3) hormic, (4) libido, (5) releaser, (6) Cannon, (7) incentive, (8) Roby, (9) lower, (10) sensation seeking.

## 10.4 The Pull of the Goal: Incentives, Expectancies and Opponent Processes

LOWER YIELD · 2 QS

Drive and arousal theories look inside the organism for the source of motivation. Yet a well-fed guest still eats the gulab jamun because it is there. The theories in this section shift the spotlight outwards, to the goal, and then inwards again to what the person expects about the goal.

### Incentive theory: push versus pull ●●

The **incentive theory** of motivation holds that behaviour is motivated primarily by the pull of external rewards and the push away from external punishments, rather than by internal tension. Drives push; incentives pull. The theory explains why behaviour depends on the quality of what is on offer, and not only on how deprived the organism is.

Two classic observations showed incentives at work. In 1928 **Otto Tinklepaugh** let monkeys watch while he hid a piece of banana under a cup; then, out of sight, he swapped it for lettuce, which monkeys normally eat readily. When a monkey lifted the cup and found lettuce, it searched about, shrieked and refused the lettuce. The monkey had been working for a specific expected incentive, not for "food". In 1942 **Leo Crespi** trained rats to run for different amounts of food and then switched them. Rats shifted from a large to a small reward ran more slowly than rats that had always received the small reward, and rats shifted upwards ran faster than rats always given the large one. These **contrast effects** forced Hull to add K, incentive motivation, to his equation.

Pinel and Barnes (Biopsychology) apply the same idea to eating with their **positive-incentive perspective**: we are drawn to eat mainly by the anticipated pleasure of eating, which de-

pends on the food's taste, our past experience with it, the time since we last ate, and the presence of other people, rather than by energy deficits. Section 10.9 returns to it. Modern neuroscience has refined the idea further. **Kent Berridge** and Terry Robinson distinguish the "wanting" of an incentive (its motivational pull, which depends on dopamine) from its "liking" (the pleasure actually felt). Their incentive-sensitisation theory of addiction (1993) proposes that drugs progressively increase wanting without increasing liking, which is why people with addiction can crave a drug they no longer enjoy.

> EXAM TRAP
>
> "In incentive theory, motivation is primarily driven by" **external rewards**. The baits each belong to another theory: intrinsic satisfaction (self-determination theory), biological drives (Hull), self-efficacy beliefs (Bandura). Likewise "drive theory primarily explains motivation as a process to" **reduce physiological needs**; seeking external rewards is incentive theory, setting long-term goals is goal-setting theory, and increasing social connections is affiliation.

## Expectancy-value theories

Incentives do not pull equally hard on everyone. A promotion pulls hard on someone who thinks he can get it and who wants it, and hardly at all on someone who has given up or who does not care. The label **expectancy-value theory** covers the family of theories holding that the strength of motivation depends on two things multiplied together: the person's **expectancy** that an action will lead to an outcome, and the **value** the person places on that outcome. Because the terms multiply, motivation is zero if either is zero: a highly valued goal that seems impossible does not motivate, and neither does a certain outcome that nobody wants.

The idea has several authors. **Edward Chace Tolman** (1886 to 1959) argued against Hull that animals learn expectancies about what leads to what (his "purposive behaviourism"), and that incentives decide whether this knowledge is shown in performance (latent learning is covered in the Learning chapter). **Kurt Lewin** (1890 to 1947) described goals as having positive or negative **valence** in a person's psychological field. **Julian Bernard Rotter** (1916 to 2014), in *Social Learning and Clinical Psychology* (1954), wrote the most quotable version: the potential for a behaviour to occur in a given situation is a function of the person's **expectancy** that the behaviour will be reinforced and the **reinforcement value** of the outcome to that person. Rotter's full theory has four variables: **behaviour potential**, expectancy, reinforcement value and the **psychological situation** (the situation as the person perceives it). The same multiplicative logic reappears in John Atkinson's theory of achievement (Section 10.6), in Victor Vroom's expectancy theory of work (Section 10.10) and in Jacquelynne Eccles's model of students' choices, in which value is split into interest, importance, usefulness and cost.

EXAM TRAP

A bank item reads: "The _____ framework is based on the expectancy, demand and incentive concepts", with the options behavioural, cognitive, social learning and supportive. The wording comes from organisational-behaviour textbooks, which describe three frameworks for explaining behaviour at work: the **cognitive** framework (built on Tolman's expectancies), the **behaviouristic** framework (built on reinforcement) and the **social learning** or social cognitive framework (Rotter, Bandura), which combines the two. The bank keys **social learning**; because expectancy is also the defining idea of Tolman's cognitive approach, "cognitive" has its defenders, so read the options carefully. "Supportive" is not a framework at all but one of Keith Davis's models of organisational behaviour (autocratic, custodial, supportive, collegial). Whatever the key, expectancy and incentive are the shared ideas of Tolman, Rotter and every expectancy-value theory in this section.

### The opponent-process theory of acquired motivation

Why do people keep doing things that hurt at first, like parachuting or running marathons, and keep taking drugs that have stopped giving them pleasure? In 1974 **Richard Lester Solomon** (1918 to 1995) and John D. Corbit proposed an answer in their **opponent-process theory**, which Solomon later (1980) called the opponent-process theory of acquired motivation.

The theory assumes that the brain is built to resist strong emotional departures from neutrality. Any stimulus that produces a strong feeling arouses a **primary process** (the a-process), which produces the initial state, called state A (pleasant or unpleasant). The a-process automatically triggers an **opponent process** (the b-process), which produces the opposite feeling, state B. The two have different time courses. The a-process is quick, closely tied to the stimulus and does not change with repetition. The b-process is sluggish: it starts late, builds slowly and decays slowly after the stimulus stops. What we feel is the difference between them. So during the stimulus, state A peaks and then settles to a lower level (as the b-process grows), and when the stimulus ends, the a-process stops at once and the lingering b-process is felt on its own as an after-reaction in the opposite direction.

The crucial assumption is that **the b-process, unlike the a-process, grows stronger with use**: with repeated exposure it starts sooner, becomes larger and lasts longer. Figure 10.3 shows the consequence. After many exposures, state A becomes weak (this is **tolerance**), and state B becomes strong and long-lasting (this is **withdrawal**, or in the case of frightening activities, exhilaration). Motivation shifts from seeking A to escaping B.

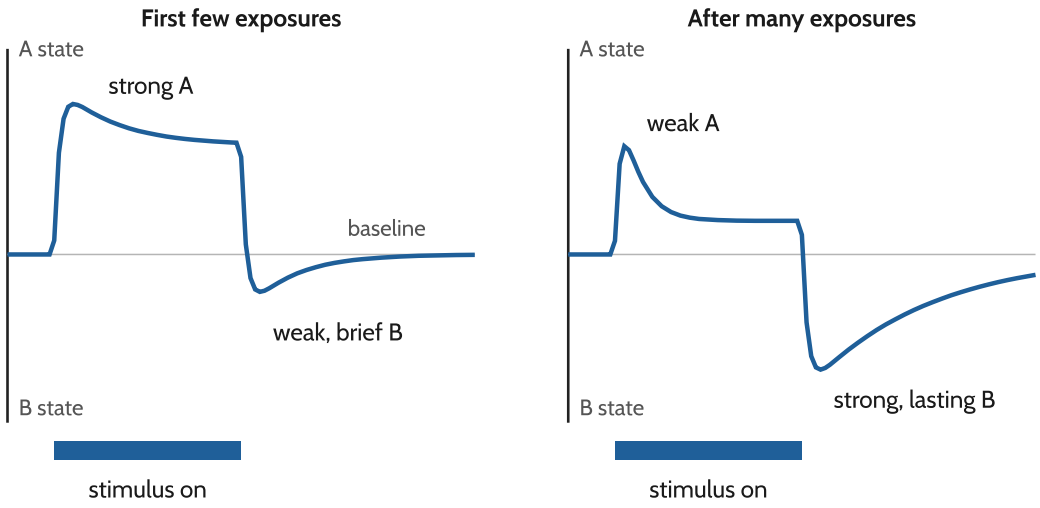


Figure 10.3 The opponent-process theory (after Solomon and Corbit, 1974). The line shows the feeling actually experienced (state A above the baseline, state B below it) while the stimulus is on (black bar) and after it ends. With repetition the opponent b-process grows, so A shrinks (tolerance) and the after-reaction B becomes larger and longer (withdrawal).

Solomon and Corbit's favourite examples make the theory concrete. A novice parachutist is terrified during the first jump (state A) and dazed and stony-faced on landing; a veteran feels only tense eagerness during the jump and, afterwards, an exhilaration that lasts for hours (a strengthened state B). A first dose of an opiate brings intense euphoria (A) followed by mild craving (B); after hundreds of doses the euphoria is faint and the withdrawal is agonising, so the person takes the drug mainly to escape B. The same pattern, the theory claims, describes the comfort of a loved one's presence and the grief of separation, and the relief after a sauna.

CLINICAL LINK

Opponent-process theory gives the clinician a vocabulary for addiction: tolerance is a growing b-process, withdrawal is the b-process exposed when the drug wears off, and craving is the motive to escape B. It also explains why the risk of relapse peaks during early abstinence, when B is strongest. Siegel's conditioned-tolerance research (in the Learning chapter) adds that the b-process can become conditioned to the cues that usually accompany the drug.

EXAM TRAP

There are two "opponent-process theories" in psychology. Ewald Hering's opponent-process theory (1878) is a theory of **colour vision** (red-green, blue-yellow, black-white; see the Sensation chapter). Solomon and Corbit's (1974) is a theory of **acquired motivation and emotion**. Match the author to the domain. In Solomon's theory it is the **b-process**, not the a-process, that strengthens with repetition.

## 10.5 Hierarchies of Need: Maslow, Alderfer and Other Lists HIGH YIELD · 17 QS

By the 1950s American psychology had two grand theories of human nature. Psychoanalysis, the "first force", saw people as driven by unconscious sexual and aggressive instincts; behaviourism, the "second force", saw them as products of conditioning. Both, critics complained, had built their pictures of humanity from neurotic patients and hungry rats. The self-styled "third force", **humanistic psychology**, set out to study healthy, growing people, and to put free will, personal growth and the innate tendency to fulfil one's potential at the centre of psychology. Its two founders were **Abraham Harold Maslow** (1908 to 1970) and Carl Rogers (whose self theory is taught in the Personality book). Of the two, Maslow is the motivation theorist.

There is a neat link to the opening of this chapter. Maslow did his doctoral research at the University of Wisconsin under Harry Harlow, studying dominance in monkeys. He left convinced that motivation had to be studied in its highest as well as its lowest forms.

### The hierarchy of needs ●●●

Maslow first set out his theory in the paper "A Theory of Human Motivation" in *Psychological Review* (1943) and developed it in the book *Motivation and Personality* (1954). His **hierarchy of needs** arranges human needs in five levels, from the most basic to the most distinctively human:

1. **Physiological needs**: food, water, oxygen, sleep, warmth, sex. These are the most powerful of all when unmet.
2. **Safety needs**: security, stability, protection, order, freedom from fear and chaos. Maslow pointed to children's preference for routine and adults' preference for a steady job and insurance.
3. **Belongingness and love needs**: friends, a partner, children, membership in a group, acceptance. (Some exam questions call this level "social needs".)
4. **Esteem needs**: self-respect (competence, mastery, independence) and respect from others (status, recognition, appreciation).
5. **Self-actualisation**: the need to become everything one is capable of becoming. "What a man can be, he must be," Maslow wrote.

The organising principle is **prepotency**: lower needs are stronger than higher needs whenever they are unmet, so they must be at least partly satisfied before the higher needs take over the direction of behaviour. A starving man dreams of food, not of fame. Once food is reliable, safety becomes the dominant concern; once safety is secure, loneliness begins to hurt, and so on upwards. The lower a need stands in the hierarchy, the greater its strength and priority, so physiological and safety needs, at the base, are the **lower needs** (management writers call them the lower-order needs, and belongingness, esteem and self-actualisation the higher-order needs); the four levels below self-actualisation together form the deficiency needs described below. Maslow called all five levels **basic needs**, and he described them as **instinctoid**: inborn in a weak, instinct-like way, although the behaviours used to satisfy them are learned.

Two points of precision are often lost in retelling. First, the hierarchy is not all-or-none. In the 1943 paper Maslow suggested that an average person might be about 85 per cent satisfied

in physiological needs, 70 per cent in safety, 50 per cent in love, 40 per cent in esteem and 10 per cent in self-actualisation. A higher need emerges gradually as lower ones are partly met. Second, Maslow himself acknowledged exceptions, such as the martyr who gives up life for an ideal, or the creative person who pursues work through poverty. The familiar pyramid diagram, incidentally, is not Maslow's own drawing; it was added later by management writers. Figure 10.4 uses it because it is how the theory is examined.

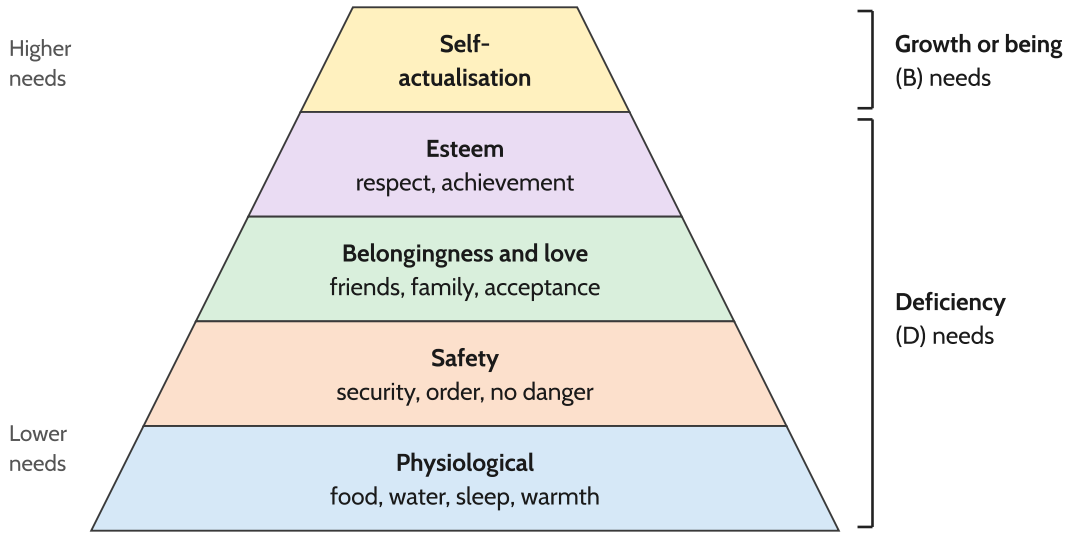


Figure 10.4 Maslow's hierarchy of needs (1943, 1954). The four lower levels are deficiency (D) needs; self-actualisation is a growth or being (B) need. Lower needs are prepotent: when unmet they dominate behaviour, and they must be largely met before higher needs take charge.

EXAM TRAP

Memorise the order bottom to top: **physiological, safety, belongingness and love (social), esteem, self-actualisation**. Distractors swap safety and physiological, or put esteem above self-actualisation. "Which is NOT a level of Maslow's hierarchy?" has the answer **ego**, which belongs to Freud's structural model. "Maslow considers physiological and safety needs as" **lower needs**. The bait "greatest potency needs" is cleverly half-true: Maslow did call the lower needs **prepotent** (strongest when unmet), but the category name for physiological and safety needs is **lower** (lower-order) needs. "Transformational" is a leadership term (Burns, Bass), not Maslow's.

EXAM TRAP

RML 2025 asked which need "must be mostly fulfilled before an individual seeks esteem needs". The level immediately below esteem is **belongingness and love (social belonging)**, so that is the correct answer. The key printed for this question gives **physiological needs**. Physiological needs must indeed be met first, but so must safety and belonging; the question asks which need precedes esteem, and on Maslow's ladder that is belonging. If the same wording appears, choose social belonging; if only physiological appears among lower levels, choose it.

### Deficiency needs, growth needs and metamotivation

In *Toward a Psychology of Being* (1962) Maslow drew his most important distinction. The **deficiency needs** (D-needs) are the four lower levels. They arise from a lack, their satisfaction removes tension, and once satisfied they stop motivating; a fed person is not motivated by food. Behaviour driven by them is **D-motivation**, and it works much like Hull's drive reduction. The **growth needs** or being needs (B-needs) are different. They arise not from a lack but from the wish to grow; their satisfaction does not reduce tension but often increases it, and the more they are satisfied, the stronger they become.

Maslow called the motivation of self-actualising people **metamotivation** (also B-motivation): they are not striving to fill a gap but are drawn towards states of being he called **metaneeds** or B-values, such as truth, goodness, beauty, wholeness, aliveness, uniqueness, justice, order, simplicity, playfulness and meaningfulness. When metaneeds are frustrated, the result is not a neurosis in the ordinary sense but a **metapathology**: cynicism and mistrust when truth is denied, apathy and loss of zest, a feeling of meaninglessness and despair. Maslow also described the **Jonah complex**, the fear of one's own greatness that keeps people from attempting to reach their potential.

Later Maslow added to his list. In *Motivation and Personality* he discussed **cognitive needs** (to know and to understand) and **aesthetic needs** (for order, symmetry and beauty), which some textbooks insert between esteem and self-actualisation to make a seven-level hierarchy. And in his late writing (around 1969, and in the posthumous *The Farther Reaches of Human Nature*, 1971) he wrote of **self-transcendence**: a level beyond self-actualisation at which a person seeks to serve something beyond the self, such as others, a cause, nature or the sacred. An eight-level version (adding cognitive, aesthetic and transcendence needs) circulates widely.

### Self-actualisers and peak experiences

The term **self-actualisation** was coined by the German neurologist **Kurt Goldstein** (1878 to 1965) in *The Organism* (1939), for the single master motive of any organism to realise its capacities. Maslow made it famous. He studied people he judged to be self-actualising, including historical figures such as Abraham Lincoln, Thomas Jefferson, Albert Einstein, Eleanor Roosevelt, William James, Jane Addams and Albert Schweitzer, and concluded that they made up 1 per cent or less of the population. They shared characteristics such as an efficient perception of reality, acceptance of self and others, spontaneity, a focus on problems outside themselves, a need for privacy, autonomy, a continued freshness of appreciation, deep relationships with a few people, a democratic character, a philosophical sense of humour and creativity. They also reported **peak experiences**: moments of intense ecstasy, wonder and awe, resembling mystical experiences, in which the sense of self is transcended. (A questionnaire measure of self-actualisation, Shostrom's Personal Orientation Inventory, is described in the Assessment book.)

EXAM TRAP

"The striving for fulfilment of one's potential is called" **self-actualisation**. Its look-alikes: **self-concept** (one's organised picture of oneself, Rogers), **ideal self** (who one thinks one should be, Rogers), **self-efficacy** (belief in one's ability to perform a task, Bandura). The motivation proposed for self-actualisers is **metamotivation**, not D-motivation (and "E-motivation" and "F-motivation" are invented options). Goldstein coined "self-actualisation"; Maslow popularised it; Rogers spoke of an "actualising tendency" in all organisms.

### Evaluating Maslow

Maslow's hierarchy is probably the best-known motivation theory in the world, and one of the least well supported. Four criticisms recur. **Method**: his self-actualisers were chosen by Maslow himself, according to his own values, so the description of self-actualisers partly describes Maslow's heroes. **Evidence**: in 1976 Mahmoud Wahba and Lawrence Bridwell reviewed the research and found little support for the ranking of the needs or for the idea that satisfying one level activates the next. **Counter-examples**: starving artists, hunger strikers and soldiers who die for comrades put higher needs before lower ones. **Culture**: the ladder reflects an individualistic, Western ideal in which personal fulfilment is the summit; in collectivist cultures, belonging and duty to the group may rank higher, and self-actualisation may be understood in relational terms. A survey of 123 countries by Louis Tay and Ed Diener (2011) found that the needs Maslow named are indeed universal and do predict well-being, but that people gain well-being from social and respect needs even when their basic needs are not fully met: the needs operate largely independently rather than as a strict ladder.

INDIAN CONTEXT

Indian psychologists have often compared Maslow's hierarchy with the four *purusharthas*, the aims of life in Hindu thought: *artha* (material welfare), *kama* (pleasure and desire), *dharma* (duty and right conduct) and *moksha* (liberation). The tradition of the renunciant who sets aside bodily comfort in pursuit of liberation is a standard counter-example to strict prepotency, and Maslow's late idea of self-transcendence is the part of his theory closest to Indian ideas of self-realisation. (Indian psychology is covered in Chapter 2.)

### Alderfer's ERG theory

The American organisational psychologist **Clayton Paul Alderfer** (1940 to 2015) tested Maslow's ideas with workers and in 1969 proposed a simpler, more flexible alternative, **ERG theory**. He collapsed the five levels into three. **Existence** needs cover Maslow's physiological and material safety needs (pay, food, working conditions). **Relatedness** needs cover interpersonal safety, belongingness and the external part of esteem (being respected by others). **Growth** needs cover the internal part of esteem and self-actualisation.

Alderfer then changed the rules of movement. Like Maslow, he accepted a **satisfaction-progression** principle: satisfying a lower need tends to make the next need more important. Unlike Maslow, he allowed **more than one need to operate at once**, and he did not insist that lower needs be satisfied before higher ones can matter. And he added the **frustration-regression** principle: when a higher need is persistently frustrated, the person regresses and puts more effort into the lower need that is easier to satisfy. A clerk with no prospect of growth in his job may become intensely concerned with pay and with office friendships. ERG

theory has found more empirical support than Maslow's strict hierarchy, particularly in organisations.

### Other lists of needs ●●●

Maslow was not the only theorist to propose a list of basic human needs, and exam setters like to mix the lists. Viktor Frankl (1905 to 1997), the Viennese psychiatrist who survived Auschwitz, argued in *Man's Search for Meaning* (1946) that the primary motive in human beings is the **will to meaning**, in contrast to Freud's will to pleasure and Adler's will to power. His therapy is **logotherapy**, the "third Viennese school" of psychotherapy, and the frustration of the will to meaning produces an "existential vacuum". Table 10.5 collects the other lists.

Table 10.5 Who proposed which list of needs.

| Theorist | Year | The needs |
|---|---|---|
| Henry Murray | 1938 | About 20 psychogenic needs, such as achievement, affiliation, dominance, autonomy, order |
| Abraham Maslow | 1943, 1954 | Physiological, safety, belongingness and love, esteem, self-actualisation |
| Erich Fromm | 1955 | Relatedness, transcendence, rootedness, sense of identity, frame of orientation |
| David McClelland | 1953, 1961 | Achievement, affiliation, power (learned needs) |
| Clayton Alderfer | 1969 | Existence, relatedness, growth |
| Deci and Ryan | 1985 | Autonomy, competence, relatedness (innate psychological needs) |
| William Glasser | 1965, 1998 | Survival, love and belonging, power (achievement), freedom, fun |
| Viktor Frankl | 1946 | Will to meaning (the primary motive) |

EXAM TRAP

In William Glasser's reality therapy and choice theory, one of the five basic genetic needs is **power or achievement**. The baits are concepts from other theorists: self-actualisation (Maslow), collective unconscious (Jung), unconditional positive regard (Rogers). "Relatedness, transcendence, rootedness, identity and orientation" are **Fromm's** needs. "Existence, relatedness, growth" is **Alderfer**. "Autonomy, competence, relatedness" is **self-determination theory**. "Achievement, affiliation, power" is **McClelland**. The "will to meaning" belongs to **Frankl's logotherapy**, not to REBT (Ellis), existential-humanistic therapy (May, Bugental) or ACT (Hayes). (The therapies themselves are taught in the Clinical book.)

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Tinklepaugh's monkeys refused lettuce because they _____ banana.
2. Crespi's finding that rats shifted from large to small rewards ran slower than rats always given small rewards is called a _____ effect.
3. Rotter held that behaviour potential depends on expectancy and reinforcement _____.
4. In opponent-process theory, the process that grows stronger with repetition is the _____-process.
5. Maslow's term for the principle that unmet lower needs dominate higher ones is _____.
6. The level immediately below esteem in Maslow's hierarchy is _____ and love.
7. The motivation of self-actualisers, directed at B-values, is called _____.
8. The term self-actualisation was coined by Kurt _____.
9. In ERG theory, falling back to a lower need when a higher one is blocked is called _____-regression.
10. Frankl's primary human motive is the will to _____.

**Answers:** (1) expected, (2) contrast, (3) value, (4) b, (5) prepotency, (6) belongingness, (7) metamotivation, (8) Goldstein, (9) frustration, (10) meaning.

## 10.6 Psychosocial Motives and Achievement MEDIUM YIELD · 8 QS

Maslow's theory says that once you are fed, safe and loved, you strive for esteem and growth. It says much less about why some people strive far harder than others. That question, about individual differences in learned social motives, was taken up at Harvard, beginning with one remarkable man.

### Murray: needs and press ●●●

**Henry Alexander Murray** (1893 to 1988), trained in medicine and biochemistry before turning to psychology, directed the Harvard Psychological Clinic and in *Explorations in Personality* (1938) set out his **personology**, a theory of personality built around motivation. A **need**, for Murray, is a force in the brain that organises perception, thinking and action so as to change an unsatisfying situation; it can be aroused from within or by the environment, it is accompanied by a particular feeling, and it gives behaviour a direction. He distinguished the viscerogenic (bodily) needs from the psychogenic (psychological) needs you met in Section 10.1, and **manifest** needs, expressed openly in behaviour, from **latent** needs, expressed only in fantasy and dreams. His list of psychogenic needs, about twenty in the version most textbooks reproduce, is the ancestor of every later list. Table 10.6 gives the most examined ones; note Murray's "n" prefix for need.

Table 10.6 A selection of Murray's psychogenic needs.

| Need | What the person strives for |
|---|---|
| n Achievement | To accomplish something difficult, overcome obstacles, excel |

| Need | What the person strives for |
|---|---|
| n Affiliation | Friendly social contact, companionship, belonging |
| n Dominance | To control and influence others |
| n Autonomy | Independence, freedom from constraint |
| n Aggression | To overcome opposition forcefully, to attack |
| n Nurturance | To help, feed and protect the helpless |
| n Succorance | To be helped, protected and cared for |
| n Abasement | To submit, accept blame, surrender |
| n Exhibition | To make an impression, be seen and heard |
| n Order | To arrange, organise and keep things tidy |

Murray's second great idea was that needs cannot be understood apart from the environment. A **press** is an environmental force or situation that helps or hinders the satisfaction of a need: a strict father, a rival, a supportive teacher. He distinguished **alpha press**, the environment as it objectively is, from **beta press**, the environment as the person perceives it, and it is beta press that drives behaviour. The combination of a need and a press, a recurring theme in a person's life, he called a **thema**.

To measure needs that people may not be able or willing to report, Christiana Morgan and Murray published the **Thematic Apperception Test** (TAT) in 1935. The person tells stories about ambiguous pictures of people, and the needs and presses of the story's hero are scored as projections of the storyteller's own. The TAT, its logic and its rivals (the Rorschach inkblots, Jung's word association test, sentence completion tests) are taught in the Assessment book. What matters here is that the TAT became the standard measure of achievement motivation, and that Murray's needs were later measured by questionnaires such as Allen Edwards's Personal Preference Schedule.

> EXAM TRAP
>
> Murray's **need-press theory** is the conceptual basis of the **TAT** (not the Rorschach, not the word association test, which is Jung's, not sentence completion). In Murray's system the need for affiliation (n Aff) is the need for social interaction and companionship; power appears in Murray's list as n Dominance (the TAT-scored need for power, n Pow, of McClelland's school descends from it); recognition is n Recognition; achievement is n Ach. "Press" is environmental, "need" is personal, "thema" is their combination.

## McClelland: achievement, power and affiliation ●●

**David Clarence McClelland** (1917 to 1998) took three of Murray's needs and made them the focus of a forty-year research programme. In *The Achievement Motive* (1953), written with John Atkinson, Russell Clark and Edgar Lowell, he developed a way of scoring TAT stories for achievement imagery: concern with a standard of excellence, unique accomplishment or long-term involvement. The resulting **need for achievement** (nAch) is a learned, relatively stable desire to accomplish difficult tasks and to meet standards of excellence through one's own effort.

People high in nAch show a characteristic profile. They prefer tasks of **moderate difficulty** and moderate risk, where success is possible but not certain; they like **personal responsibility** for outcomes; they want **concrete, rapid feedback** on how well they are doing; and they are innovative and future-oriented. They are not the gamblers who bet everything on long odds, nor the timid who pick sure things. McClelland argued that such people make good entrepreneurs, and that nAch is learned in childhood: in Marian Winterbottom's studies of the 1950s, mothers who expected early independence and mastery from their sons tended to have sons high in nAch.

In *The Achieving Society* (1961) McClelland went further, claiming that the level of achievement imagery in a nation's children's stories predicted its later economic growth. If nAch is learned and drives economic development, then perhaps it can be taught.

> **INDIAN CONTEXT: THE KAKINADA EXPERIMENT**
>
> McClelland put this idea to its most famous test in India. In the mid-1960s, working with the Small Industry Extension Training (SIET) Institute in Hyderabad, his team ran achievement motivation training courses for businessmen in Kakinada, a port town in Andhra Pradesh. Participants learned to think, talk and set goals like high achievers. In follow-up comparisons, the trained men started more new businesses, invested more and created more jobs than comparable untrained men. The work was reported in McClelland and David Winter's *Motivating Economic Achievement* (1969) and remains the best-known Indian study in motivation psychology.

McClelland's other two needs complete his **theory of needs** (also called acquired-needs or learned-needs theory). The **need for power** (nPow) is the desire to influence, control and have an impact on others; McClelland and David Winter distinguished **personalised power** (power used for self-aggrandisement) from **socialised power** (power used for the benefit of the group). The **need for affiliation** (nAff) is the desire to establish and maintain warm, friendly relationships. McClelland found that the most effective managers in large organisations show a **leadership motive pattern**: high nPow, low nAff and high self-control. A manager with very high nAff, by contrast, finds it hard to make unpopular decisions. Dan McAdams later added an **intimacy motive**, the desire for warm, close, communicative relationships, as a fourth social motive often scored from the same stories.

> **EXAM TRAP**
>
> McClelland's three needs are **achievement, affiliation and power**, and McClelland insisted they are **learned** (acquired), not innate. The baits come from neighbouring theories: "safety, esteem and self-actualisation" (Maslow), "existence, relatedness and growth" (Alderfer), "autonomy, competence and relatedness" (Deci and Ryan). The author of the "hierarchical theory of motivation" is **A. H. Maslow**, not McClelland, Atkinson or O. H. Mowrer (Mowrer is the two-factor avoidance theorist).

### Atkinson: the motive to succeed and the motive to avoid failure

Why should high achievers prefer tasks of moderate difficulty? McClelland's colleague **John William Atkinson** (1923 to 2003) answered with an expectancy-value theory, published in

1957, that is one of the most elegant in psychology. Atkinson proposed that every achievement situation arouses two opposing tendencies.

The **tendency to approach success** (Ts) is the product of three terms: the **motive to succeed** (Ms, a stable personality disposition, the capacity to feel pride in accomplishment), the **probability of success** (Ps, the person's estimate of the chance of succeeding at this task) and the **incentive value of success** (Is, the pride that success would bring). Atkinson's key insight was that Is is not independent of Ps: success at an easy task brings little pride, and success at a hard one brings a lot. So Is = 1 minus Ps, and

Ts = Ms × Ps × (1 − Ps).

The tendency to avoid failure (Taf) is built the same way from the **motive to avoid failure** (Maf, the capacity to feel shame at failure, usually measured by a test-anxiety questionnaire), the probability of failure and the negative incentive of failure (failing an easy task is the most shameful). The resultant tendency is the difference between the two, and it works out to (Ms − Maf) × Ps × (1 − Ps). Now push some numbers through, as in Table 10.7.

Table 10.7 Worked example of Atkinson's theory: the term Ps × (1 − Ps) and what it predicts.

| Chance of success (Ps) | Incentive (1 − Ps) | Ps × (1 − Ps) | Success-oriented (Ms > Maf) | Failure-avoidant (Maf > Ms) |
|---|---|---|---|---|
| 0.1 (very hard) | 0.9 | 0.09 | Weak approach | Mild avoidance (tolerable) |
| 0.3 | 0.7 | 0.21 | Stronger approach | Stronger avoidance |
| 0.5 (moderate) | 0.5 | 0.25 | Strongest approach: preferred | Strongest avoidance: shunned |
| 0.7 | 0.3 | 0.21 | Stronger approach | Stronger avoidance |
| 0.9 (very easy) | 0.1 | 0.09 | Weak approach | Mild avoidance (tolerable) |

The product Ps × (1 − Ps) peaks at 0.5. So a person whose motive to succeed exceeds his motive to avoid failure is most attracted to tasks of intermediate difficulty, exactly McClelland's observation. A person dominated by fear of failure is most repelled by intermediate tasks, and so, when forced to choose, picks either a very easy task (where failure is unlikely) or a very hard one (where failure is no disgrace, because nobody could succeed). In 1960 Atkinson and George Litwin tested this with a ring-toss game in which students could stand as near to or as far from the peg as they liked. Students high in nAch and low in test anxiety chose intermediate distances; those low in nAch and high in anxiety more often stood very close or very far away. The same logic explains Kurt Lewin's earlier work on **level of aspiration**, the standard a person sets for his own next performance: success-oriented people raise their aspiration modestly after success, while failure-avoidant people set it unrealistically low or unrealistically high.

> **REMEMBER**
>
> High achievers: intermediate risk, personal responsibility, feedback, attribute success to ability and effort and failure to lack of effort, persist after failure. Low achievers (high fear of failure): avoid evaluation, choose very easy or very hard tasks, attribute failure to lack of ability, give up. The link between the two halves of this box is Weiner's attribution theory, below.

### Fear of success

Atkinson's theory was built almost entirely on men; women's achievement data were messy and were often set aside. In her 1968 doctoral research at the University of Michigan, **Matina Souretis Horner** (born 1939) asked why. She gave students the opening line of a story to complete. Women were given: "After first-term finals, Anne finds herself at the top of her medical school class." Men were given the same line about John. About two-thirds of the women wrote stories full of conflict and loss: Anne was lonely, unpopular, unfeminine, or deliberately dropped her grades; a few stories even had her physically disfigured. Fewer than one in ten of the men wrote such stories about John.

Horner proposed a **fear of success** (the motive to avoid success): a learned expectation, stronger in women in her culture, that success, especially in competitive, traditionally male fields, would bring social rejection and doubts about one's femininity. Her paper in the *Journal of Social Issues* (1972) made the idea famous. Later research was less kind. When men wrote about a man succeeding in a field seen as feminine (nursing, for instance), they produced just as much negative imagery, which suggested that the stories reflected cultural stereotypes about who belongs where, not a stable motive peculiar to women. The concept survives as an example of how social norms shape achievement behaviour. Horner herself went on to become, at 32, the president of Radcliffe College.

### Weiner: attributions and achievement

The newest strand of achievement research asks not "how strong is your motive?" but "how do you explain your successes and failures?" Its roots lie in the work of the Austrian-American social psychologist **Fritz Heider** (1896 to 1988), who argued in *The Psychology of Interpersonal Relations* (1958) that people act like naive scientists, explaining behaviour by internal (personal) or external (situational) causes. Such explanations are called **attributions**.

A related idea came from Julian Rotter. His **locus of control** (1966) is a generalised expectancy about what controls the outcomes in one's life. People with an **internal** locus of control believe that outcomes depend on their own effort and ability; people with an **external** locus of control believe that outcomes depend on luck, fate or powerful others. Rotter measured it with his Internal-External (I-E) Scale. A student who believes his results depend on his own preparation rather than on luck or an easy examiner has an internal locus of control. (Low self-efficacy is a different idea, a belief that one cannot perform a task, below; the inferiority complex is Alfred Adler's concept, in the Personality book.)

**Bernard Weiner** (born 1935) of the University of California, Los Angeles, built these strands into an **attribution theory of achievement motivation**, developed through the 1970s and set out in full in 1985. Weiner proposed that the causes people give for success and failure differ along three dimensions. **Locus** (borrowed from Rotter) asks whether the cause is internal or external to the person. **Stability** asks whether the cause is stable over time or variable. **Controllability**, added later, asks whether the person can control the cause. Table 10.8 places the four classic causes.

Table 10.8 Weiner's classification of the causes of success and failure.

| Cause | Locus | Stability | Controllability |
|---|---|---|---|
| Ability | Internal | Stable | Uncontrollable |

| Cause | Locus | Stability | Controllability |
|---|---|---|---|
| Effort | Internal | Unstable | Controllable |
| Task difficulty | External | Stable | Uncontrollable |
| Luck | External | Unstable | Uncontrollable |

Each dimension has its own consequences. Locus governs **self-esteem emotions**: success attributed to internal causes produces pride, while success attributed to luck produces only gratitude or surprise. Stability governs **expectancy**: if you fail because of a stable cause (low ability, a hard examiner), you expect to fail again and lose hope; if you fail because of an unstable cause (insufficient effort, bad luck), you can expect to do better next time. Controllability governs **social emotions**: we feel guilt about our own failures due to controllable causes (I did not study) and shame about uncontrollable ones (I am not clever enough); and we feel anger towards others whose failure was controllable and pity or sympathy towards those whose failure was not.

Work it through with a student who has failed a statistics test. If he says, "I am useless at maths" (internal, stable, uncontrollable), he feels shame, expects to fail again and is likely to give up. If he says, "I did not practise enough problems" (internal, unstable, controllable), he feels guilty but hopeful and is likely to work harder. Same failure, different attribution, opposite motivation. Seligman's learned helplessness research, reformulated in 1978 by Abramson, Seligman and Teasdale, applies the same logic to depression: people who attribute bad events to internal, stable and global causes are at risk (the Learning chapter and the Disorders book develop this).

> EXAM TRAP
>
> Weiner's three dimensions are **locus, stability and controllability**. Do not add "globality" (that is the third dimension of the learned-helplessness reformulation, not Weiner's). **Locus of control** is **Rotter's** generalised expectancy (1966); **locus of causality** is Weiner's dimension (borrowed from Rotter). **Effort** is the only one of the four classic causes that is both internal and controllable, which is why attribution retraining teaches students to attribute failure to effort.

## 10.7 Intrinsic Motivation, Competence and the Self HIGH YIELD · 9 QS

We now return to the puzzle in Harlow's monkey cages. If no drive was being reduced, what was going on? Over the 1950s a group of psychologists began to argue that organisms have built-in motives to explore, to understand and to master their surroundings, and that these motives are as basic as hunger. From that argument grew the modern study of intrinsic motivation.

### Curiosity and exploration

**Harry Frederick Harlow** (1905 to 1981) is better known for his studies of infant monkeys and their cloth and wire "mothers" (in the Developmental book), but his puzzle studies came first. He concluded that a **manipulation drive** and an exploratory drive are primary, unlearned motives, and in a 1953 paper, "Mice, Monkeys, Men, and Motives", he mocked the attempt to

derive all human motives from hunger and thirst. Robert Butler's window-opening monkeys (Section 10.3) showed a matching **visual exploration** drive.

The most systematic account of curiosity came from the British-Canadian psychologist **Daniel Ellis Berlyne** (1924 to 1976), in *Conflict, Arousal and Curiosity* (1960). For Berlyne, **curiosity** is a motive aroused by stimuli with certain **collative variables**, so called because the organism must collate (compare) the stimulus with what it already knows: **novelty**, **complexity**, **surprisingness**, **incongruity** and uncertainty. Moderately novel and complex stimuli are the most attractive, in line with optimal arousal theory. Berlyne distinguished **perceptual curiosity** (the urge to look at or touch something new) from **epistemic curiosity** (the desire for knowledge), and two kinds of exploration: **specific exploration**, aimed at resolving uncertainty about a particular thing (what is making that noise?), and **diversive exploration**, aimed simply at relieving boredom (flicking through television channels). Maslow's cognitive needs to know and understand describe the same territory from the humanistic side.

### Competence or effectance motivation

In 1959 **Robert White** (1904 to 2001) of Harvard published a long review in *Psychological Review* called "Motivation Reconsidered: The Concept of Competence". White pulled together the evidence on exploration, play, manipulation and mastery and argued that all of it reflected a single motive: the urge to interact effectively with the environment. He called this **effectance motivation**, and the feeling it produces, the satisfaction of having an effect, the feeling of efficacy. Watch an infant drop a spoon from a high chair again and again, delighted each time it clatters: the infant is not hungry, it is learning that it can make things happen. White's idea of **competence motivation** is the direct ancestor of two modern theories: Bandura's self-efficacy and Deci and Ryan's need for competence.

### Self-efficacy ●●●

**Albert Bandura** (1925 to 2021), a Canadian-born psychologist at Stanford, introduced **self-efficacy** in a 1977 *Psychological Review* paper. Self-efficacy is a person's belief in his or her capability to organise and carry out the actions required to achieve a particular goal. It is specific to a task (you may have high self-efficacy for cooking biryani and low self-efficacy for public speaking), and it affects which tasks people choose, how much effort they put in, how long they persist in the face of difficulty and how anxious they feel. Bandura distinguished it from **outcome expectancy**, the belief that a given behaviour will produce a given outcome. A student may be sure that solving past papers would raise his marks (high outcome expectancy) and still doubt that he can make himself do it (low self-efficacy).

Bandura named four sources of self-efficacy, in order of power. **Mastery experiences** (performance accomplishments) are the strongest: succeeding at a task builds belief, especially after overcoming difficulty. **Vicarious experiences** come from watching similar others succeed ("if she can do it, so can I"). **Verbal persuasion** is encouragement from others, weaker on its own. **Physiological and emotional states** are interpreted as signs of capability: a pounding heart before a viva may be read as incapacity unless one learns to read it as readiness.

Self-efficacy is the motivational heart of Bandura's **social cognitive theory** (set out in *Social Foundations of Thought and Action*, 1986), the expansion of his earlier social learning theory.

Its organising model is **triadic reciprocal causation**, also called reciprocal determinism: personal factors (beliefs, expectations, biology), behaviour and the environment all influence one another, so none is simply cause and none simply effect. Bandura's work on observational learning, with its attention, retention, reproduction and motivation processes, belongs to the Learning chapter.

EXAM TRAP

"The reciprocal causation model has been described under" **social cognitive theory** (Bandura), not drive-reduction theory, not constructivist theory, and not "two-factor theory". Be ready for the many two-factor theories: **Herzberg** (work: motivators and hygiene, 1959), **Mowrer** (avoidance learning: classical plus operant, 1947), **Schachter and Singer** (emotion: arousal plus cognitive label, 1962) and **Spearman** (intelligence: g plus s, 1904). "Social cognitive theory highlights which factor in motivation?" **Self-efficacy**. Self-efficacy is not self-esteem (a global evaluation of one's worth) and not locus of control (a general belief about what controls outcomes).

## When rewards backfire: the overjustification effect

If intrinsic motivation is real, what happens when you pay someone to do what they already enjoy? Common sense says you get more of it. **Edward L. Deci** (born 1942) tested this in a study published in 1971. College students worked on Soma puzzles, interesting three-dimensional block puzzles, over three sessions. In the second session, half of them were paid a dollar for each puzzle solved; in the third session nobody was paid. The key measure came when the experimenter left the room for a few minutes, leaving the puzzles and some magazines: how long did students spend on the puzzles in this free-choice period? Students who had been paid spent less free time on the puzzles after the payment stopped than students who had never been paid.

KEY STUDY: LEPPER, GREENE AND NISBETT'S MAGIC MARKERS (1973)

**Question.** Can an expected reward undermine a child's interest in an activity the child already enjoys? **Method. Mark Lepper**, David Greene and **Richard Nisbett** observed preschool children at a Stanford nursery school and selected those who, during free play, spent time drawing with magic markers. Each child was then invited to draw in one of three conditions: **expected award** (told in advance that drawing would earn a "Good Player" certificate with a gold seal and ribbon), **unexpected award** (given the same certificate as a surprise afterwards) or **no award**. **Result.** One to two weeks later, the markers were put out again in the classroom. Children in the expected-award group spent about half as much free time drawing as children in the other two groups, and the drawings they had made for the award were judged lower in quality. Unexpected awards did no harm. **Conclusion.** An expected, tangible reward for an already enjoyable activity can undermine intrinsic interest. **Criticism.** The effect depends heavily on conditions; rewards that are unexpected, verbal, or informative about competence do not undermine, and some reviewers (Cameron and Pierce, 1994) argued that the effect is narrower than often claimed.

The authors called the result the **overjustification effect**: when a person is given more external justification for an activity than is needed (a reward for something already enjoyed), the person attributes the behaviour to the reward ("I draw for certificates") rather than to inter-

est ("I draw because I like it"), and when the reward stops, so does the behaviour. A meta-analysis of 128 experiments by Deci, Richard Koestner and Richard Ryan (1999) confirmed that expected, tangible rewards contingent on doing a task reliably reduce intrinsic motivation, while positive verbal feedback tends to increase it. For parents and teachers in a culture of rewards for marks, the message is not "never reward", but "reward in a way that informs rather than controls, and do not pay children for what they already love".

### Self-determination theory and cognitive evaluation theory ●●●

Deci and his Rochester colleague **Richard M. Ryan** (born 1953) built these findings into **self-determination theory** (SDT), presented in *Intrinsic Motivation and Self-Determination in Human Behavior* (1985) and now one of the most widely researched motivation theories. SDT proposes three innate, universal psychological needs whose satisfaction is essential for growth, intrinsic motivation and well-being. **Autonomy** is the need to feel that one's behaviour is self-chosen and volitional, that one is its origin rather than a pawn. **Competence** is the need to feel effective in dealing with the environment (White's idea). **Relatedness** is the need to feel close to and cared for by others. Social conditions that support these needs foster intrinsic motivation; conditions that thwart them undermine it.

The first part of SDT to be developed was **cognitive evaluation theory** (CET), which explains when external events help or harm intrinsic motivation. Any reward, deadline, praise or surveillance has two aspects. Its **controlling** aspect pressures the person to behave in a particular way and shifts the perceived locus of causality from internal to external, lowering autonomy and intrinsic motivation. Its **informational** aspect tells the person about his competence, and if the information is positive it raises intrinsic motivation. Expected tangible rewards are usually experienced as controlling; unexpected praise for good work is usually experienced as informational. That is the theory behind the magic-marker results.

SDT does not treat extrinsic motivation as a single thing. Its **organismic integration theory** arranges motivation along a continuum of increasing self-determination, shown in Table 10.9 for a student preparing for an entrance examination. The more a person internalises the value of an externally prompted activity, the more autonomous, persistent and satisfying the motivation becomes.

Table 10.9 The self-determination continuum (Deci and Ryan).

| Type | Regulation | Example: why I study for the entrance |
|---|---|---|
| Amotivation | None; no intention to act | "I don't see the point; I just sit there." |
| Extrinsic | External regulation | "My parents will be angry if I don't." |
| Extrinsic | Introjected regulation | "I would feel guilty and ashamed if I failed." |
| Extrinsic | Identified regulation | "I value becoming a clinical psychologist." |
| Extrinsic | Integrated regulation | "Helping people is part of who I am." |
| Intrinsic | Intrinsic regulation | "I find psychology fascinating." |

SDT also describes stable individual differences, called **causality orientations**: an autonomy orientation (acting from interest and values), a controlled orientation (acting from rewards, deadlines and others' demands) and an impersonal orientation (feeling that outcomes are beyond one's control, associated with amotivation).

EXAM TRAP

SDT's three needs are **autonomy, competence and relatedness**. The usual wrong options swap one of them for **affiliation** (McClelland), **achievement** (McClelland), **independence**, **superiority** (Adler) or **excellence**. "Which type of motivation in SDT is driven by internal rewards and personal growth?" **Intrinsic motivation**; amotivation is the absence of motivation, and external regulation is the least autonomous form of extrinsic motivation. The overjustification study with magic markers is **Lepper, Greene and Nisbett (1973)**; the Soma puzzle study is **Deci (1971)**.

## Flow ●●

Intrinsic motivation at its most intense has a name. **Mihaly Csikszentmihalyi** (1934 to 2021), a Hungarian-born psychologist at the University of Chicago, began by asking painters why they worked for hours on canvases they then lost interest in once finished. He went on to interview rock climbers, chess players, dancers and surgeons, and in *Beyond Boredom and Anxiety* (1975) and *Flow: The Psychology of Optimal Experience* (1990) he described **flow**: a state of complete absorption in an activity, in which action and awareness merge, attention is fully focused, self-consciousness disappears, the sense of time is distorted (hours pass like minutes) and the activity becomes **autotelic**, rewarding in itself. To study flow in everyday life he and Reed Larson developed the **experience sampling method**, in which people carry a pager (now a phone) and record what they are doing and feeling whenever it signals at random moments.

The central condition for flow is a **balance between challenge and skill**, at a level that stretches the person, together with clear goals and immediate feedback. When the challenge exceeds skill, the person feels anxiety; when skill exceeds the challenge, boredom. Figure 10.5 shows the "flow channel" and how a learner moves through it.

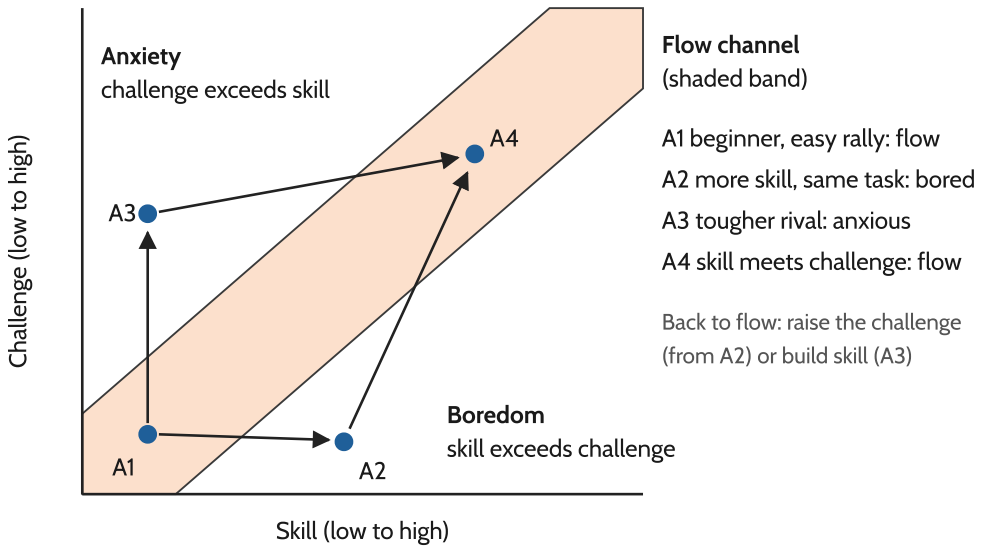


Figure 10.5 The flow channel (after Csikszentmihalyi, 1990). Flow occurs when challenge and skill are in balance. A beginning tennis player is in flow at an easy level (A1); as skill grows he becomes bored (A2), and against a much stronger rival he becomes anxious (A3). He regains flow at a higher level (A4) by seeking harder challenges or by building skill.

EXAM TRAP

"Who introduced the concept of flow?" **Mihaly Csikszentmihalyi**. The baits are near neighbours: Maslow described **peak experiences** (brief moments of ecstasy in self-actualisers, not the same as flow, which can occur in ordinary work), Rogers described the **fully functioning person**, and Frankl described the **will to meaning**.

## The consistency motive: cognitive dissonance ●●

One more cognitive motive deserves a place here, because it appears in motivation questions although its home is social psychology. **Leon Festinger** (1919 to 1989) proposed in *A Theory of Cognitive Dissonance* (1957) that people are motivated to keep their thoughts, beliefs and actions consistent. In his account, **cognitive dissonance** is the unpleasant state of tension that arises when a person holds two inconsistent cognitions, or acts against his own beliefs; like a drive, it motivates the person to reduce it, usually by changing an attitude, changing the behaviour, or adding a justifying thought. Arvind, who believes that smoking is disgusting because it killed his grandfather, smokes at a party with his friends: he will experience dissonance, and may later tell himself that "one cigarette at a party doesn't count". In Festinger and James Carlsmith's 1959 study, students paid only one dollar to tell a waiting student that a boring task was interesting later rated the task as more enjoyable than students paid twenty dollars; with insufficient external justification, they changed their attitude instead. Notice the family resemblance to the overjustification effect: too much external reward weakens internal interest, too little strengthens internal justification. **Effort justification**, the tendency to value what we have suffered for (Elliot Aronson and Judson Mills, 1959), is a special case. The theory and its research are developed in the Social Psychology book.

EXAM TRAP

In the Arvind smoking question, the answer is **cognitive dissonance**. "Justification of effort" is a specific form of dissonance reduction after a costly choice, not the state itself; "drive reduction" is Hull; "increased drive to smoke" is invented. Dissonance motivates the person to **reduce** the inconsistency by changing an attitude or behaviour, never to increase it.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Murray's term for an environmental force that helps or hinders a need is _____.
2. People high in nAch prefer tasks of _____ difficulty.
3. In Atkinson's theory, the incentive value of success equals 1 minus the _____ of success.
4. Matina Horner's "Anne at the top of her medical school class" study proposed a motive to avoid _____.
5. Weiner's three causal dimensions are locus, stability and _____.
6. Robert White called the motive to interact effectively with the environment _____ motivation.
7. Bandura's strongest source of self-efficacy is _____ experience.
8. The undermining of intrinsic interest by expected rewards is the _____ effect.
9. Self-determination theory's three needs are autonomy, competence and _____.
10. In flow, challenge and _____ are balanced.

**Answers:** (1) press, (2) moderate (intermediate), (3) probability, (4) success, (5) controllability, (6) effectance (competence), (7) mastery, (8) overjustification, (9) relatedness, (10) skill.

## 10.8 Conflict and Frustration

MEDIUM YIELD · 5 QS

So far we have treated motives one at a time. Real life is not so tidy. You want the job in Bengaluru and you want to stay near your parents in Mysuru; you want to pass the entrance and you dread another night of revision. Motives collide with each other and with the world, and the study of those collisions is the study of conflict and frustration. It also connects motivation to stress (Chapter 11) and to clinical psychology.

### Frustration and its sources ●●

In psychology, **frustration** is the blocking or thwarting of goal-directed behaviour, and also the unpleasant emotional state that results. Its sources are usually grouped into three classes. **Environmental** (external) sources include physical obstacles (a traffic jam, a closed office), natural events (floods, illness in the family), and social and cultural barriers (rules, prejudice, family disapproval, poverty). **Personal** (internal) sources include physical or mental limitations and, very commonly, a gap between aspiration and ability: the student who sets his heart on a rank his preparation cannot reach. The third source is **conflict** between one's own motives, taken up below. Delays, lack of resources, losses, failure and a sense of meaninglessness are the everyday forms frustration takes.

Frustration is one of several sources of stress that are usually listed together, and questions ask you to tell them apart. **Conflict** is the presence of two or more incompatible motives. **Change** is the need to adjust to new circumstances (the life-event research of Holmes and Rahe is in Chapter 11). Finally, **pressure** is the expectation or demand that one behave in a certain way: pressure to perform (work hard, fast and well) and pressure to conform (meet others' expectations). Ramesh, who must handle a large number of calls at high speed in a

call centre, is under **pressure**; he is not blocked from a goal (frustration), torn between motives (conflict) or facing an unknowable future (unpredictability).

### Lewin's types of conflict

Kurt Lewin, the German-American founder of field theory, described behaviour as a function of the person and the environment, B = f(P, E), played out in the person's **life space**. Objects in the life space have positive or negative **valences** that attract or repel the person, like forces in a physical field. When forces pull in incompatible directions, the result is a **conflict**. In *A Dynamic Theory of Personality* (1935) Lewin distinguished three basic types; a fourth was added by later writers (it is often credited to Carl Hovland and Robert Sears in 1938). Table 10.10 summarises them.

Table 10.10 Types of motivational conflict.

| Type | Structure | Example | Typical behaviour |
|---|---|---|---|
| Approach-approach | Two attractive goals, only one possible | Admission offers from two good colleges | Easiest to resolve; brief hesitation |
| Avoidance-avoidance | Two unpleasant options, one must be chosen | Revise a hated subject or fail it | Vacillation; tries to leave the field |
| Approach-avoidance | One goal with both attractive and unattractive features | A well-paid job far from family | Ambivalence; approach then retreat |
| Double (multiple) approach-avoidance | Two or more goals, each with good and bad features | Job A pays well but is far; job B is near but pays less | Prolonged vacillation; the most common real-life type |

In an **approach-approach conflict** the person is attracted to two goals that cannot both be reached. Such conflicts are the easiest to resolve: once you move even slightly towards one goal, its pull grows and the other's shrinks, so the conflict is unstable and resolves itself. (The medieval puzzle of Buridan's ass, which starved between two equally attractive bales of hay, is a philosopher's joke, not a psychological finding.) In an **avoidance-avoidance conflict** both options are unpleasant. As the person moves towards one, its repulsion grows and pushes him back towards the other, so he vacillates; if he can, he tries to **leave the field** altogether, by daydreaming, procrastinating or physically escaping. In an **approach-avoidance conflict** a single goal both attracts and repels: the cricket fan who wants to watch the final but fears what his boss will say about the day off. The person approaches until the negative features loom large, then retreats, then approaches again. In a **double approach-avoidance conflict** there are two or more such goals, and every option has its own pros and cons. This is the shape of most important real-life decisions: choosing a course, a job, a partner, a city.

### Miller's gradients of approach and avoidance

Why does the approach-avoidance conflict produce such characteristic vacillation? Neal Miller answered with a model that turned Lewin's descriptions into testable predictions. In a 1944 chapter, "Experimental Studies of Conflict", and later in a fuller statement (1959), Miller proposed a set of postulates about **approach and avoidance gradients**, the way the strength of each tendency changes with distance from the goal.

1. The tendency to approach a goal grows stronger the nearer the organism is to it (the approach gradient).

2. The tendency to avoid a feared stimulus also grows stronger the nearer the organism is to it (the avoidance gradient).
3. The avoidance gradient is **steeper** than the approach gradient: fear rises faster than desire as the goal comes closer.
4. The strength of each tendency (the height of each gradient) depends on the strength of the drive on which it is based.
5. When two incompatible tendencies conflict, the stronger one wins.

The evidence came from rats wearing small harnesses that measured how hard they pulled. In a 1948 study, Judson Brown found that hungry rats trained to run to food pulled only a little less hard when stopped far from the goal than when stopped near it, while rats shocked at the goal pulled away very hard near it and hardly at all far from it. The gradients really do differ in slope.

Now look at Figure 10.6. Far from the goal, approach is stronger than avoidance, so the organism moves forward. Close to the goal, avoidance is stronger, so it retreats. At the point where the gradients cross, the two are equal, and the organism hovers, moving forward and back. This is the vacillation of the bride who longs to marry and panics as the wedding day approaches, and of the student who sits down to start a feared assignment and finds himself, ten minutes later, making tea.

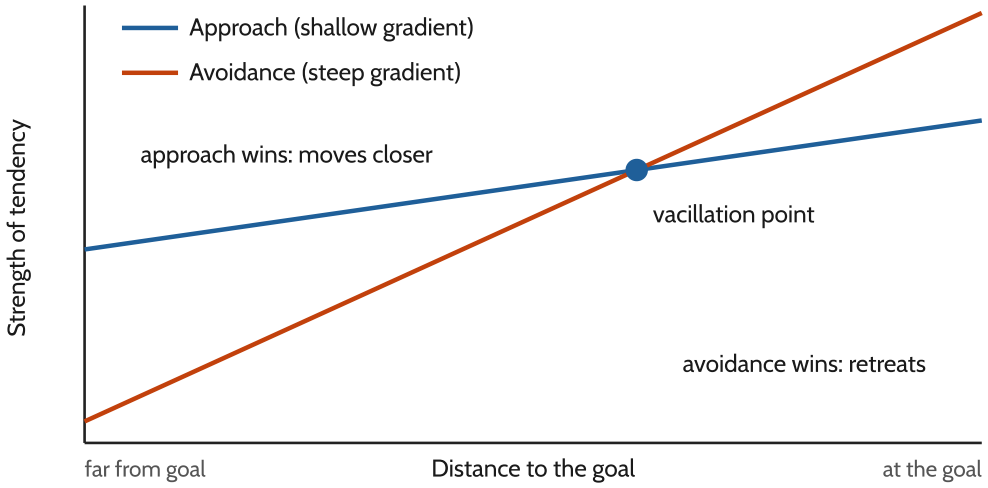


**Figure 10.6** Miller's gradients in an approach-avoidance conflict. Both tendencies grow stronger nearer the goal, but avoidance (dashed) grows faster. Where the lines cross, the two are equal and the person vacillates; lowering the avoidance line moves the crossing point closer to the goal.

CLINICAL LINK

John Dollard and Neal Miller translated these gradients into therapy in *Personality and Psychotherapy* (1950). Suppose a client wants a relationship but fears intimacy. Increasing the approach tendency (urging him on, raising the rewards) raises the whole approach line, so the crossing point moves nearer the goal, where avoidance is steepest; he gets closer and suffers more fear and conflict. Reducing the avoidance tendency (lowering fear through reassurance, exposure and relaxation) lets him approach with less distress. Their rule: in an approach-avoidance conflict, work first on reducing fear, not on increasing desire. Modern exposure therapy follows the same logic.

## Reactions to frustration ●●

How do people respond when a goal is blocked? Some responses are constructive: trying harder, changing the approach, finding a substitute goal, or lowering the goal to a realistic level. Others are less adaptive, and four of them are classic in textbooks.

**Aggression** is the most studied (see the frustration-aggression hypothesis, below). It may be **direct**, aimed at the source of frustration, or redirected onto a safer target when the true source is too powerful or too vague. Such **displaced aggression** is the clerk who, scolded by his boss, goes home and shouts at his children. When a whole group is blamed for a society's frustrations, displacement becomes **scapegoating**. In 1940 Carl Hovland and Robert Sears reported that lynchings in the American South rose in years when the price of cotton fell; later reanalyses found the relationship weaker than first claimed, but the study remains the textbook example of displaced aggression.

**Regression** is a return to behaviour typical of an earlier stage of development. In a 1941 study at the University of Iowa, Roger Barker, Tamara Dembo and Kurt Lewin let young children play with ordinary toys, then showed them a room of far more attractive toys and, after a short time, put those toys behind a wire screen where the children could see but not reach them. When the children went back to the ordinary toys, their play became markedly less constructive, more like the play of much younger children.

**Fixation** is the rigid, stereotyped repetition of a response that no longer works. **Norman Maier** (1900 to 1977) gave rats insoluble discrimination problems on a jumping stand, punishing them unpredictably whatever they chose. Many rats developed a fixed habit, such as always jumping to the left, and kept it even after the problem was made soluble and the correct window was left open in front of them. In *Frustration: The Study of Behavior Without a Goal* (1949) Maier argued that frustration-induced behaviour is qualitatively different from goal-directed behaviour.

**Withdrawal and apathy** are giving up: the person stops trying, loses interest and may become passive and depressed, as in learned helplessness. Finally, people protect themselves from the anxiety of frustration and conflict with the unconscious strategies Freud called **defence mechanisms**. Table 10.11 lists the ones most often examined as reactions to frustration (they are taught in full in the Personality book). The psychologist Saul Rosenzweig, who coined the term **frustration tolerance** for the ability to withstand frustration without maladaptive responses, also classified reactions as **extrapunitive** (blaming others), **intropunitive** (blaming oneself) or **impunitive** (blaming no one), and measured them with his Picture-Frustration Study (in the Assessment book).

Table 10.11 Defence mechanisms often listed as reactions to frustration.

| Mechanism | What happens | Example |
|---|---|---|
| Rationalisation | A socially acceptable reason is constructed, after the act, for behaviour driven by other motives | "I didn't want that seat anyway; the college is overrated." |
| Displacement | An impulse is redirected from a threatening target to a safer one | Shouting at a younger brother after a bad viva |
| Projection | One's own unacceptable impulse is attributed to others | A jealous student accuses classmates of jealousy |
| Sublimation | Sexual or aggressive energy is re-channelled into socially valued activity | Aggression channelled into boxing or surgery |
| Reaction formation | An impulse is replaced by its exaggerated opposite | Excessive sweetness towards a disliked rival |
| Repression | A threatening thought is pushed out of awareness | No memory of a humiliating failure |
| Denial | An unpleasant reality is refused acknowledgement | Insisting the result list must be wrong |
| Regression | Return to an earlier developmental level | An adult sulking and refusing to eat |
| Compensation | Weakness in one area is offset by excelling in another; pushed to excess it is Adler's overcompensation | A poor athlete becoming the class topper |

EXAM TRAP

Three bank items describe the same mechanism in different words: "an intellectual reason to justify an action, constructed in the mind after the behaviour is completed", "logical and socially acceptable explanations, after the event, for behaviour actually driven by unconscious motives" and "a socially acceptable reason for actions really based on unacceptable motives". All three are **rationalisation**. The giveaway is the **after-the-fact reason**. Re-channelling sexual or aggressive energy into a socially valued outlet is **sublimation**; shifting it onto a safer target is **displacement**; attributing it to others is **projection**; turning it into its exaggerated opposite is **reaction formation**; refusing to acknowledge reality is **denial**; pushing a thought out of awareness is **repression**.

## The frustration-aggression hypothesis

In 1939 a group of Yale scholars, John Dollard, Leonard Doob, Neal Miller, Orval Hobart Mowrer and Robert Sears, published *Frustration and Aggression*. Their **frustration-aggression hypothesis** made two sweeping claims: aggression is always a consequence of frustration, and frustration always leads to some form of aggression. Both claims proved too strong. Frustration can lead to regression, fixation, withdrawal or constructive problem solving, and aggression can occur without frustration (a hired fighter, a soldier obeying orders). In 1941 Miller himself revised the hypothesis: frustration produces instigations to several kinds of response, one of which is aggression. In 1989 **Leonard Berkowitz** (1926 to 2016) offered the influential reformulation, which he developed into his cognitive neo-associationist model: frustration produces aggression only to the extent that it produces **negative affect**, and aversive events of any kind (pain, heat, foul smells, insults) can do the same; cues associated with aggression, such as the weapons in Berkowitz and LePage's 1967 study, make an aggressive response more likely. Aggression as a topic belongs to the Social Psychology book.

EXAM TRAP

The frustration-aggression hypothesis is **Dollard, Doob, Miller, Mowrer and Sears (1939)**, not Lewin and not Freud; its revisions are Miller (1941) and Berkowitz (1989). The regression study with toys behind a wire screen is **Barker, Dembo and Lewin (1941)**; the fixation study with rats on a jumping stand is **Maier (1949)**. Lewin described the three basic conflict types; Miller supplied the gradients; the avoidance gradient is **steeper**, not shallower. "Leaving the field" is the hallmark of **avoidance-avoidance** conflict; "approach then retreat" is the hallmark of **approach-avoidance**.

## 10.9 Biological Motives: Hunger, Thirst, Sex and Others

MEDIUM YIELD · 9 QS

We end the survey of theories where drive theory began, in the body. The biological motives are where psychology and physiology meet, and they are where several of the most famous "facts" in introductory psychology turned out to be only half true. The detailed neuroscience of eating and drinking belongs to the Physiological Psychology book; here the aim is to understand what starts and stops these behaviours, and to know the classic findings and the traps around them.

### Hunger: from the stomach to the brain

The most obvious theory of hunger is that we feel hungry because our stomachs are empty. It was tested in one of the most charming experiments in the history of physiology.

KEY STUDY: CANNON AND WASHBURN'S SWALLOWED BALLOON (1912)

**Question.** Is hunger the feeling of an empty stomach contracting? **Method.** Walter Cannon had the idea; his student **A. L. Washburn** had the ability to swallow a balloon. Washburn swallowed an empty balloon tied to the end of a thin tube; Cannon inflated it and connected the tube to a water-filled U-tube, so that each stomach contraction pushed up the water level, which was recorded. Washburn pressed a key whenever he felt a hunger pang. **Result.** Large stomach contractions coincided with Washburn's reports of hunger pangs. **Conclusion.** Cannon and Washburn proposed that hunger is the feeling of stomach contractions and that satiety is the feeling of stomach distension. **Criticism.** Patients whose stomachs had been surgically removed, with the oesophagus joined directly to the small intestine, still reported hunger and satiety and kept a normal body weight by eating smaller, more frequent meals. The stomach contributes to hunger, but it is not its source.

If not the stomach, then what? In the 1940s and 1950s researchers looked for signals in the blood, assuming that eating, like temperature, is regulated around a set point. The **glucostatic theory** (associated with Jean Mayer, 1953) held that we become hungry when blood glucose falls below its set point and stop eating when it is restored; glucose is, after all, the brain's main fuel. The **lipostatic theory** (associated with Gordon Kennedy, 1953) held that each person has a set point for body fat, and that deviations from it produce compensating changes in eating. The two were seen as complementary: a short-term glucostatic system for starting and stopping meals and a long-term lipostatic system for body weight.

Pinel and Barnes, whose Biopsychology textbook is the standard source on this topic, argue that these set-point theories have several serious problems. They cannot explain the world-

wide epidemic of overeating and obesity, since a set-point system should resist it. They fit poorly with the evolutionary pressures on eating: our ancestors faced an unpredictable food supply, so they needed a system that made them eat good food whenever it was available and bank the calories as fat, preventing energy deficits rather than merely responding to them. Their key predictions have not been confirmed: reductions in blood glucose or body fat large enough to trigger eating rarely occur in ordinary life. And they ignore the enormous influence of taste, learning and social factors on eating. Their alternative is the **positive-incentive perspective** introduced in Section 10.4: we are motivated to eat mainly by the anticipated pleasure of eating, and body weight is regulated loosely around a **settling point**, the weight at which the many factors that influence intake and expenditure come into balance. Their "leaky barrel" analogy makes the point: the water level in a leaky barrel stays fairly constant not because anything defends a set level, but because a fuller barrel leaks faster. Change the inflow (a more tempting food environment) and the level drifts to a new settling point. This is why dieters' weight drifts back up, and why the same person can stabilise at different weights in different environments.

### The hypothalamus: two centres and their revision ●●●

Every entrance paper seems to ask about the "satiety centre". Here is the story behind the term. In 1940 Albert Hetherington and Stephen Ranson reported that large bilateral lesions of the **ventromedial hypothalamus** (VMH) made rats eat enormously (**hyperphagia**) and become extremely obese. The **VMH syndrome** has two phases: a **dynamic phase** of several weeks of gross overeating and rapid weight gain, and a **static phase** in which the rat eats just enough to maintain its new, high weight, and defends it against both starvation and forced feeding. In 1951 **Bal Krishan Anand** and John Brobeck at Yale reported that bilateral lesions of the **lateral hypothalamus** (LH) produced **aphagia**, a complete refusal to eat that led to death unless the animal was tube-fed, and later work showed that LH lesions also cause **adipsia**, the refusal to drink. Stimulating the LH made sated animals eat. In 1954 Eliot Stellar put the findings together in the **dual-centre theory**: the LH is a **feeding (hunger) centre** and the VMH a **satiety centre**, working like an accelerator and a brake.

> **INDIAN CONTEXT**
>
> Bal Krishan Anand, the "Anand" of Anand and Brobeck, was an Indian physiologist. After his work at Yale he returned to India and built one of the country's leading neurophysiology departments at the All India Institute of Medical Sciences (AIIMS), New Delhi. With G. S. Chhina and Baldev Singh he also published, in 1961, one of the earliest electroencephalographic studies of yogis in meditation. The "feeding centre" in every psychology textbook carries an Indian name.

The dual-centre theory is still the answer examiners want, but you should know that it has been substantially revised, as Pinel describes. VMH-lesioned rats turned out to overeat **because** they gain weight, rather than gaining weight because they overeat: VMH lesions raise blood insulin, which speeds the conversion of nutrients into fat (lipogenesis) and slows its release (lipolysis), so the rats must keep eating to have enough usable energy. Moreover, many effects of large VMH lesions come from damage to fibres from the nearby **paraventricular nucleus** (PVN), and lesions of the PVN alone produce overeating. LH lesions, for their part, produce a wide range of motor and sensory deficits, not a specific loss of hunger. Modern re-

search focuses on particular cell groups, especially in the **arcuate nucleus**, which receives the body's chemical signals about energy. Table 10.12 lists the hypothalamic regions most often examined in connection with motivation.

Table 10.12 Hypothalamic regions and motivated behaviour.

| Region | Role in motivation | Classic finding or note |
|---|---|---|
| Lateral hypothalamus (LH) | Classic "feeding centre" | Lesions: aphagia and adipsia (Anand and Brobeck 1951); now seen as broader |
| Ventromedial hypothalamus (VMH) | Classic "satiety centre" | Lesions: hyperphagia and obesity, dynamic then static phase (Hetherington and Ranson 1940) |
| Paraventricular nucleus (PVN) | Satiety signals; releases oxytocin and vasopressin | PVN lesions also cause overeating |
| Arcuate nucleus | Reads leptin, insulin and ghrelin | Contains hunger-promoting and satiety-promoting neurons |
| Suprachiasmatic nucleus (SCN) | Master circadian clock | Lesions abolish the daily sleep-wake rhythm (Chapter 5) |
| Preoptic area | Temperature regulation; male sexual behaviour | Contains a sexually dimorphic nucleus |
| Supraoptic nucleus | Water balance | Makes vasopressin (antidiuretic hormone) |

EXAM TRAP

"Satiety centre" is the **ventromedial hypothalamus**; damage gives **hyperphagia and obesity**. "Feeding centre" is the **lateral hypothalamus**; damage gives **aphagia and weight loss** (with adipsia). Mnemonic: **L**ateral lesion, **L**ose weight; **V**entromedial lesion, **V**ery large. Watch the invented baits: "ventromedial **hippocampus**" and "lateral **thalamus**" do not exist as hunger centres. The cerebrum and basal ganglia are not hunger centres either. The hypothalamus as a whole is the body's master homeostatic regulator (hunger, thirst, temperature, circadian rhythm via the SCN, and control of the pituitary); the cerebellum coordinates movement and the pineal gland secretes melatonin. One paper (Amity 2023) keyed "regulation of emotions" as the function of the limbic system rather than "both emotions and hunger and thirst"; classical textbooks treat the hypothalamus (hunger, thirst) as closely linked to the limbic system, so read the options carefully, but expect "emotion" as the limbic answer.

## Hunger and satiety signals

The body tells the brain about its energy state through chemical messengers. In 1950 a spontaneous mutation in a mouse colony at the Jackson Laboratory in Maine produced **ob/ob mice**, which eat excessively and weigh up to three times as much as normal mice. In 1994 Jeffrey Friedman and his colleagues identified the missing product of the *ob* gene: a hormone released by fat cells, which they named **leptin** (from the Greek *leptos*, thin). Leptin signals the size of the body's fat stores to the hypothalamus and, in normal animals, reduces eating; injecting it into ob/ob mice made them eat less and lose weight. Hopes that leptin would cure human obesity faded, however, because most obese people already have high leptin levels and seem to be resistant to it. The pancreatic hormone **insulin** plays a similar long-term signalling role. In the short term, **ghrelin**, a hormone released mainly by the empty stomach (identified in 1999), rises before meals and promotes hunger, while gut peptides such as

**cholecystokinin** (CCK), released from the intestine as food arrives, promote satiety (Gibbs, Young and Smith showed in 1973 that injecting CCK reduced meal size in rats).

Satiety also depends on variety. In **sensory-specific satiety**, the pleasure of a particular food falls as you eat it, while the pleasure of other foods is barely affected; this is why there is always room for dessert, and why rats offered a varied "cafeteria diet" of tasty foods eat far more and grow obese.

### External cues, dieting and eating disorders

If eating were controlled only by internal signals, the clock on the wall would not matter. **Stanley Schachter** (1922 to 1997) of Columbia University suspected that it does matter, and that it matters more for some people. His **externality hypothesis** (1968 to 1971) proposed that obese people are unusually responsive to external food cues (the sight, smell and taste of food, the time of day) and relatively insensitive to internal cues of hunger and fullness. In one study by Schachter and Larry Gross (1968), participants worked in a room with a rigged clock; obese participants ate more crackers when the clock showed that it was past dinner time than when it showed an earlier hour, while normal-weight participants were little affected. Obese participants also ate more when more food was visible and ate less when the food took effort to get. Later research, summarised by Judith Rodin in 1981, showed that responsiveness to external cues is found at all body weights and does not on its own explain obesity, but the idea that environmental cues drive eating survives in the positive-incentive view.

Peter Herman and Janet Polivy added the concept of **restrained eating**: chronic dieters who control their intake by willpower rather than by internal cues. In a classic 1975 study by Herman and Deborah Mack, restrained eaters who were first required to drink a milkshake then ate **more** ice cream in a "taste test" than restrained eaters who had not, as if, once the diet was "broken", there was no reason to hold back. Obesity (a body mass index of 30 or above), **anorexia nervosa** (restriction of intake leading to significantly low weight, with an intense fear of gaining weight) and **bulimia nervosa** (recurrent binge eating with compensatory behaviours such as vomiting) are disorders of this motivational system; their diagnosis and treatment are covered in the Psychiatric Disorders book.

### Thirst

Water makes up more than half of body weight, and two different kinds of loss produce thirst. The first, **osmotic thirst**, arises when the fluid outside the cells becomes too salty (after a salty meal, or when water is lost through sweating). Water then moves out of the cells by osmosis, and the cells shrink. Specialised neurons called **osmoreceptors**, in a region at the front of the hypothalamus (the organum vasculosum of the lamina terminalis), detect the change and trigger both thirst and the release of **vasopressin** (antidiuretic hormone) from the posterior pituitary, which makes the kidneys conserve water. The second, **hypovolaemic thirst**, arises when the volume of blood and other extracellular fluid falls, as after bleeding, diarrhoea or vomiting, even if its saltiness is unchanged. Receptors in the heart and large blood vessels detect the drop in pressure, and the kidneys release renin, which leads to the formation of **angiotensin II**; angiotensin acts on the brain (the subfornical organ) to produce thirst and also an appetite for salt, which is needed to restore volume. Drinking usually stops long before the water has been absorbed, because signals from the mouth, stomach and intestines meter the intake. Cannon had proposed that thirst is simply a

dry mouth, but wetting the mouth does not abolish thirst for long, and the theory failed for much the same reason as his stomach theory of hunger.

## Sexual motivation ●●

Sex is classed as a primary motive, yet it differs from hunger and thirst in a fundamental way: no tissue is depleted when it is unsatisfied, and no individual dies for lack of it. It serves the survival of the species rather than of the individual, and in human beings it is shaped heavily by learning, culture and relationships.

Scientific study of human sexual behaviour began with surveys. **Alfred Charles Kinsey** (1894 to 1956), a zoologist at Indiana University who had spent years classifying gall wasps, turned the same taxonomic zeal on human sexuality. Kinsey and his colleagues interviewed thousands of Americans and published *Sexual Behavior in the Human Male* (1948) and *Sexual Behavior in the Human Female* (1953), together known as the **Kinsey reports**. They shocked the public by showing that practices officially condemned, such as premarital and extramarital sex and homosexual experience, were far commoner than anyone had admitted. Kinsey described sexual orientation not as two boxes but as a continuum, the Kinsey scale, from 0 (exclusively heterosexual) to 6 (exclusively homosexual). His samples were volunteers rather than random samples, a weakness critics stressed.

Where Kinsey asked people what they did, **William Howell Masters** (1915 to 2001), a gynaecologist, and **Virginia Eshelman Johnson** (1925 to 2013) observed it directly. In their laboratory in St. Louis they recorded the physiological responses of several hundred volunteers (382 women and 312 men) during thousands of sexual response cycles and reported the results in *Human Sexual Response* (1966). They described a four-phase **sexual response cycle**: **excitement** (rising arousal, with engorgement of the genitals with blood, erection and vaginal lubrication), **plateau** (arousal levels off at a high level), **orgasm** (rhythmic muscular contractions and intense pleasure) and **resolution** (the body gradually returns to its unaroused state). Men enter a **refractory period** after orgasm during which another orgasm is not possible, while many women can have multiple orgasms if stimulation continues. Their second book, *Human Sexual Inadequacy* (1970), launched modern sex therapy.

The New York sex therapist **Helen Singer Kaplan** (1929 to 1995) noticed that many of her patients' problems began before excitement, with a lack of interest in sex. In *Disorders of Sexual Desire* (1979) she proposed a three-phase (triphasic) model: **desire, excitement, orgasm**. Adding desire as the first phase is the key difference between Kaplan and Masters and Johnson, and diagnostic systems later organised sexual dysfunctions by phase. Rosemary Basson later (2000) proposed a circular model of female sexual response in which desire often follows, rather than precedes, arousal and intimacy.

Hormones matter, but less in humans than in other mammals. In many female mammals sexual receptivity is confined to the **oestrous** period around ovulation, when oestrogen is high. Testosterone supports sexual desire in both men and women, and castration in adulthood reduces but does not always abolish it. Hormones have **organisational** effects before birth, shaping the developing brain and body, and **activational** effects from puberty onwards, switching behaviour on (the Physiological book develops this).

EXAM TRAP

Masters and Johnson: **Excitement, Plateau, Orgasm, Resolution** (EPOR, four phases, 1966). Kaplan: **Desire, Excitement, Orgasm** (three phases, 1979). Options that put desire into a Masters and Johnson sequence, or put plateau before excitement, or end with "refractory" instead of resolution, are scrambles. The reports on the sexual behaviour of men (1948) and women (1953) are **Kinsey's**, not Freud's, Anna Freud's or Karen Horney's.

### Maternal drive, pain avoidance and other biological motives

How strong are the biological drives, compared with each other? In the 1920s and 1930s Carl Warden and his colleagues at Columbia University built the **Columbia obstruction box**: a rat was placed in a start compartment separated from a goal (food, water, a mate, its pups, or a novel area) by an electrified grid, and the experimenters counted how many times in a fixed period the rat would cross the grid to reach the goal. In Warden's results (*Animal Motivation*, 1931) the **maternal drive** produced the most crossings, followed by thirst, hunger, sex and exploration: a mother rat would cross the shock grid to reach her pups more often than a hungry rat would to reach food. Maternal behaviour in rats is primed by the hormones of pregnancy and birth, such as oestrogen, prolactin and oxytocin, but it is not purely hormonal: virgin female rats left with pups for several days begin to build nests and retrieve the pups. In primates, Harlow's later studies of contact comfort showed how much the infant's own motive for attachment matters (the Developmental book covers attachment).

**Pain avoidance** is a powerful primary motive that, unlike hunger, has no cycle: it is aroused by tissue damage and lasts as long as the damage does. Its importance is clearest in its absence: people born with congenital insensitivity to pain suffer repeated injuries, burns and joint damage, and often die young. The need to keep body temperature, oxygen supply and elimination in balance are further primary motives, usually satisfied so automatically that they become noticeable only when blocked. **Sleep** behaves like a drive too, building up with time awake and discharged by sleeping; its stages, rhythms and theories are covered in Chapter 5.

## 10.10 Motivation in Practice: Work, School and Clinic HIGH YIELD · 7 QS

Motivation theories are tested hardest outside the laboratory: in offices and factories, in classrooms and in consulting rooms. This section applies the chapter's ideas to those three settings, and introduces the work-motivation theories that entrance papers ask about every year.

### Motivation at work ●●●

In the late 1950s **Frederick Herzberg** (1923 to 2000) and his colleagues Bernard Mausner and Barbara Snyderman interviewed about two hundred engineers and accountants in the Pittsburgh area using the **critical incident technique**: each person described a time he had felt exceptionally good about his job and a time he had felt exceptionally bad, and what had caused it. The results, published in *The Motivation to Work* (1959), surprised them. The things that made people feel good were not simply the opposite of the things that made them feel bad. Herzberg's **two-factor theory** (also called motivator-hygiene theory) proposed two separate sets of factors, shown in Table 10.13.

Table 10.13 Herzberg's two factors.

| Motivators (satisfiers): job content | Hygiene factors (dissatisfiers): job context |
|---|---|
| Achievement | Company policy and administration |
| Recognition | Supervision |
| The work itself | Relationships with supervisor and peers |
| Responsibility | Working conditions |
| Advancement | Salary |
| Growth | Status and job security |

Herzberg's **motivators** are intrinsic to the work itself, and their presence produces satisfaction and genuine motivation. His **hygiene factors** are features of the job environment, and their absence or poor quality produces dissatisfaction; but improving them merely removes dissatisfaction, without producing positive motivation. The word "hygiene" was chosen deliberately: like clean water, good hygiene prevents disease but does not make you healthy. The opposite of satisfaction, in this theory, is not dissatisfaction but "no satisfaction", and the opposite of dissatisfaction is "no dissatisfaction". The practical recommendation is **job enrichment**: building responsibility, achievement and growth into jobs, rather than relying on pay and perks. Critics point out that the critical-incident method invites self-serving answers (people credit good events to their own achievement and blame bad ones on the company), and that pay can motivate as well as satisfy. Still, Herzberg's lesson that job satisfaction depends on both the content of the work and its environment is the answer examiners expect.

> EXAM TRAP
>
> Herzberg's **motivators** are recognition, achievement, advancement, responsibility, growth and the work itself. **Salary**, benefits, job security, working conditions, company policies and interpersonal relationships are **hygiene factors**, which **prevent dissatisfaction but do not by themselves motivate**. "Job satisfaction depends on" **work motivation and job environment** (both, not either, and not disposition alone). Hygiene factors are not Maslow's self-actualisation needs; if anything they correspond to Maslow's lower needs, and the motivators to his higher needs.

Herzberg asked what motivates; the next theories ask how motivation works. **Victor Harold Vroom** (born 1932), in *Work and Motivation* (1964), applied expectancy-value logic to the workplace in his **expectancy theory**, often called VIE theory. Motivational force depends on three beliefs. **Expectancy** (E) is the belief that effort will lead to successful performance ("if I work harder, I will meet my target"). **Instrumentality** (I) is the belief that performance will lead to outcomes ("if I meet my target, I will get the bonus"). **Valence** (V) is the value of those outcomes to the person ("I want the bonus"). The three multiply: Motivational force = E × I × V. Work it through for Priya, a sales executive in Bengaluru. If she is fairly sure that extra calls will meet her target (E = 0.8), confident that the target brings a bonus (I = 0.9) and keen on the bonus (V = 0.7), her motivation is 0.8 × 0.9 × 0.7, about 0.50. If her company has a history of not paying promised bonuses, instrumentality drops to 0.2 and her motivation falls to about 0.11, even though nothing about her effort or ability has changed. Motivation is highest

when effort leads to performance, performance leads to rewards, and the rewards are valued.

**John Stacy Adams**'s **equity theory** (1963, 1965) adds a social comparison. People compare the ratio of their outcomes (pay, recognition, promotion) to their inputs (effort, skill, experience, time) with the same ratio for a comparison person. When the ratios are equal, they feel equity. When their own ratio is lower (underpayment inequity) they feel angry and resentful; when it is higher (overpayment inequity) they feel guilty, although people tolerate this more easily. Inequity motivates its own reduction: a person may change his inputs (work less), seek higher outcomes (ask for a raise), distort his perceptions ("he deserves more; he has a harder job"), change the comparison person, or leave. If Ravi learns that Suresh, who joined at the same time and does the same work, earns more, equity theory predicts that Ravi's effort will fall, whatever his absolute salary.

The most strongly supported work-motivation theory is **goal-setting theory**, developed over several decades by **Edwin Locke** (born 1938) and **Gary Latham** (born 1945), beginning with Locke's 1968 paper and summarised in their book *A Theory of Goal Setting and Task Performance* (1990). Its central finding is that **specific, difficult (but attainable) goals** lead to higher performance than easy goals, vague goals, or the exhortation to "do your best". Goals work through four mechanisms: they direct attention, mobilise effort, increase persistence and encourage the search for strategies. They work best under four conditions: when the person is committed to the goal, when there is **feedback** on progress, when the person has the ability, and when the task is not so complex that a demanding outcome goal causes panic (for complex new tasks, learning goals work better). Organisational writers later popularised "SMART" goals; the acronym is a management mnemonic, not Locke and Latham's theory.

Two further ideas complete the picture. In *The Human Side of Enterprise* (1960), Douglas McGregor contrasted **Theory X** managers, who assume that workers dislike work and must be controlled, with **Theory Y** managers, who assume that workers seek responsibility and are self-motivated; the assumptions tend to be self-fulfilling. And J. Richard Hackman and Greg Oldham's **job characteristics model** (1976) identified five core features of motivating jobs: skill variety, task identity, task significance, autonomy and feedback.

> EXAM TRAP
>
> Expectancy theory is **Vroom (1964)**; motivation is highest when effort leads to performance, performance leads to outcomes, and the outcomes are **valuable** (not uncertain, difficult or negative). If rewards are guaranteed regardless of performance, instrumentality collapses and so does motivation. Goal-setting theory is **Locke and Latham**: goals should be **specific and challenging**, with feedback; vague, easy or feedback-free goals are the wrong options. Equity theory is **Adams**; the two-factor theory of work is **Herzberg**; ERG is Alderfer.

## Motivation in education

Almost every idea in this chapter has a classroom application. The overjustification research warns against tangible rewards for activities students already enjoy, while cognitive evaluation theory suggests that feedback which informs ("your answer shows clear reasoning") supports intrinsic motivation and feedback which controls ("you must score above 80") undermines it. Attribution theory explains why students who attribute failure to low ability give up

while those who attribute it to effort persist; in 1975 Carol Dweck showed that "helpless" children could be retrained to attribute failure to insufficient effort, after which they persisted longer. Dweck's later work distinguished students who hold an **entity** (fixed) theory of intelligence from those who hold an **incremental** (growth) theory, and showed that praising effort and strategy rather than ability encourages the latter. Achievement goal research contrasts **mastery goals** (to learn and improve) with **performance goals** (to look competent compared with others); mastery goals tend to produce deeper learning and more persistence after failure. Self-efficacy, built through mastery experiences such as graded practice papers, predicts both effort and results, and expectancy-value theory explains why a student works hardest at subjects he both expects to master and values. Finally, the Yerkes-Dodson law reminds teachers and students that a moderate level of exam arousal helps, while high anxiety harms performance, especially on difficult questions.

### Motivation in the clinic

Disturbances of motivation run through psychopathology. **Anhedonia**, the loss of interest or pleasure, is a core symptom of depression, and **avolition**, a reduction in self-initiated purposeful activity, is a negative symptom of schizophrenia (both are covered in the Psychiatric Disorders book). Addiction can be understood through opponent processes and incentive sensitisation (Section 10.4), and treatments such as contingency management deliberately use extrinsic reinforcers (vouchers for drug-free urine samples) to compete with the drug. Behavioural activation treats depression by scheduling rewarding activities, on the reasoning that action can come before motivation and rebuild it.

The best-known clinical method built directly on motivation is **motivational interviewing** (MI), developed by the American psychologist **William Richard Miller** (born 1947) in the early 1980s and set out with the British psychologist Stephen Rollnick in *Motivational Interviewing* (1991). MI is a collaborative, client-centred style of conversation for strengthening a person's own motivation to change. It assumes that people with problems such as harmful drinking are usually not "in denial" but **ambivalent**: they are caught in an approach-avoidance conflict about change, exactly as in Section 10.8. Confrontation tends to push them to defend the status quo. Instead, the counsellor expresses empathy, helps the client see the **discrepancy** between current behaviour and deeper goals and values, avoids arguing (originally phrased as "rolling with resistance"), supports self-efficacy, and listens for and reinforces the client's own **change talk**. Its core skills are summarised as OARS: open questions, affirmations, reflective listening and summaries. MI is often combined with James Prochaska and Carlo DiClemente's **transtheoretical model** of the stages of change: precontemplation, contemplation, preparation, action and maintenance. The detailed technique belongs to the Clinical book; the motivational insight belongs here: people change most readily when change serves their own values, when they believe they can do it, and when they feel autonomous rather than controlled. That sentence summarises self-determination theory, self-efficacy theory and expectancy-value theory at once.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. When both options in a conflict are unpleasant and the person tries to escape the situation, the conflict is _____-avoidance.
2. In Miller's model, the _____ gradient is steeper than the approach gradient.
3. Rats that kept jumping to one side after a problem became soluble showed _____ (Maier, 1949).
4. Bilateral lesions of the lateral hypothalamus produce aphagia and _____.
5. The hormone released by fat cells and missing in ob/ob mice is _____.
6. Thirst caused by a fall in blood volume, as after bleeding, is _____ thirst.
7. Masters and Johnson's four phases are excitement, plateau, orgasm and _____.
8. Kaplan's model begins with the phase of _____.
9. In Herzberg's theory, salary is a _____ factor.
10. Vroom's motivational force is the product of expectancy, instrumentality and _____.

**Answers:** (1) avoidance, (2) avoidance, (3) fixation, (4) adipsia, (5) leptin, (6) hypovolaemic, (7) resolution, (8) desire, (9) hygiene, (10) valence.

## The case of Harlow's puzzle-solving monkeys, revisited

Go back to the rhesus monkey in Madison, pulling the pin, lifting the hook and swinging open a hasp with nothing behind it. In 1950 the most powerful theory in American psychology could not say what need was being reduced. You can now give several answers, and each comes from a different part of this chapter. The monkey was showing one of the general motives of Section 10.1, a manipulation drive that is inborn yet repairs no tissue deficit. A caged monkey with nothing to do is under-aroused, and optimal arousal theory predicts that it will seek stimulation, just as the students in Hebb's sensory-deprivation cubicles at McGill could not bear to go without it. Berlyne would point to the collative properties of the puzzle (its novelty, its complexity, the uncertainty about how it comes apart); Robert White would point to the satisfaction of having an effect on the world; Deci and Ryan would call it the need for competence. In the simplest language of all, the monkey was intrinsically motivated: the reward was the activity itself, exactly as it is for the toddler who opens an empty box or for you, finishing a crossword that nobody will ever check.

Harlow's odd side observation, that bringing food into the test tended to disrupt the monkeys' smooth performance, has a modern echo too. Two decades later Edward Deci's paid puzzle solvers and Lepper, Greene and Nisbett's rewarded preschool artists showed that an expected external reward can crowd out an internal interest, and cognitive evaluation theory explained why: a reward that feels controlling shifts the perceived cause of the behaviour from inside the person to outside. One more thread runs from that laboratory. Harlow's doctoral student Abraham Maslow left Wisconsin convinced that motivation had to be studied at its highest as well as its lowest levels, and he went on to build the hierarchy of needs. So the small wooden board in Madison stands at a crossroads of the whole chapter. It marked the limits of instinct and drive-reduction theory, and it pointed the way to curiosity, competence, growth and the self.

## Chapter summary

**10.1 What Motivation Is.** Motivation (from the Latin *movere*, to move) is the set of processes that activate behaviour, direct it, set its intensity and keep it going; it is never seen directly but inferred as an intervening variable. A need is a physiological deficiency, a drive is the aroused psychological state the need produces, a motive is the broader and often learned state that directs behaviour towards a goal, an incentive is an external object that attracts or repels, and a goal is the end state pursued. Put in order, they form the motivational cycle: need, drive, instrumental behaviour, goal, relief. Primary motives (biological, unlearned and mostly homeostatic, such as hunger and thirst) are contrasted with secondary motives (psychosocial and learned, such as achievement, affiliation and gregariousness), with general motives such as curiosity and manipulation in between; Murray's viscerogenic and psychogenic needs draw the same line, and Allport's functional autonomy (1937) explains how a learned motive can outlive its origin. Intrinsic motivation is doing something for its own sake; extrinsic motivation is doing it for a separable outcome. The main families of theory are instinct and psychodynamic, drive and arousal, incentive and expectancy, humanistic, and cognitive.

**10.2 Instinct Theories.** An instinct is a complex, unlearned, species-specific and stereotyped pattern of behaviour. William James (1890) argued that human beings have more instincts than animals, not fewer. William McDougall (1908) made instincts, each with a cognitive, affective and conative aspect and each paired with a primary emotion, the prime movers of all behaviour in his purposive, hormic psychology, and later listed eighteen propensities. Freud's instincts (*Trieb*) have a source, aim, impetus and object; in 1920 he divided them into the life instincts (Eros, whose energy is the libido) and the death instinct (Thanatos, seen in war and aggression), with the id as their reservoir and cathexis as the investment of their energy. The ethologists Lorenz, Tinbergen and von Frisch (Nobel Prize 1973) described fixed action patterns released by sign stimuli. Instinct theory declined because naming a behaviour does not explain it, because the lists grew to absurd lengths (L. L. Bernard, 1924), and because culture and rearing (Kuo, 1930) shaped behaviour that was supposed to be inborn.

**10.3 Drives, Homeostasis and Arousal.** Claude Bernard described the constant milieu intérieur, and Walter Cannon named the body's self-correcting balance homeostasis (1926; *The Wisdom of the Body*, 1932), a negative-feedback system defending a set point. Woodworth introduced the word drive (1918), and Clark Hull's drive-reduction theory (1943) held that a need creates a drive and that any response which reduces the drive is reinforced; in his equation ${}_{S}E_{R} = {}_{S}H_{R} \times D \times K \times V$, drive energises but habit directs. Secondary drives are learned, as Neal Miller's fearful rats (1948) and Mowrer's two-factor theory of avoidance (1947) showed. Drive reduction could not explain Harlow's puzzle solvers, Butler's window-watching monkeys, Sheffield and Roby's saccharin-running rats or the misery of sensory deprivation, which led to Hebb's optimal arousal theory (1955). The Yerkes-Dodson law (1908) describes an inverted-U relation between arousal and performance, with a lower optimum for complex tasks; Zajonc's drive theory of social facilitation (1965) applies the same logic to audiences; and Zuckerman's sensation seeking shows that people differ in the level of arousal they seek.

**10.4 The Pull of the Goal: Incentives, Expectancies and Opponent Processes.** Incentive theory holds that behaviour is pulled by external rewards rather than pushed by internal tension; Tinklepaugh's monkeys (1928) refused lettuce when they expected banana, and Crespi's contrast effects (1942) forced Hull to add incentive motivation (K) to his equation. Berridge and Robinson separated wanting from liking in their incentive-sensitisation theory of addiction

(1993). Expectancy-value theories, from Tolman, Lewin's valences and Rotter's formula (1954) to Atkinson, Vroom and Eccles, make motivation the product of the expectancy that an action will succeed and the value of its outcome, so that motivation vanishes if either is zero. Solomon and Corbit's opponent-process theory of acquired motivation (1974) proposes that every strong emotional state (the a-process) triggers a sluggish opposing b-process that grows stronger with repetition, which explains tolerance, withdrawal and the exhilaration of the veteran parachutist; it must not be confused with Hering's opponent-process theory of colour vision.

**10.5 Hierarchies of Need: Maslow, Alderfer and Other Lists.** Humanistic psychology, the third force, emphasises free will and an innate tendency to grow. Abraham Maslow (1943, 1954) ordered human needs from physiological through safety, belongingness and love, and esteem to self-actualisation, with the principle of prepotency: unmet lower needs dominate and must be largely satisfied before higher needs take charge. The four lower levels are deficiency (D) needs; self-actualisation is a growth or being (B) need. Self-actualisers show metamotivation towards meta-needs (B-values) whose frustration produces metapathologies, they report peak experiences, and Maslow later added cognitive and aesthetic needs and self-transcendence. Goldstein coined the term self-actualisation. The theory has been criticised for its hand-picked sample, weak evidence for the ordering (Wahba and Bridwell, 1976), counter-examples and cultural bias. Alderfer's ERG theory (1969) reduces the levels to existence, relatedness and growth, lets several needs operate at once and adds frustration-regression. Other lists include those of Murray, Fromm, McClelland, Deci and Ryan, Glasser (whose five needs include power or achievement) and Frankl, whose will to meaning is the basis of logotherapy.

**10.6 Psychosocial Motives and Achievement.** Henry Murray's *Explorations in Personality* (1938) described about twenty psychogenic needs, manifest and latent, and the press of the environment (alpha press as it is, beta press as perceived) that helps or hinders them; the Thematic Apperception Test of Morgan and Murray (1935) was built to measure them. David McClelland scored TAT stories for the need for achievement (nAch), whose holders prefer moderate risk, personal responsibility and quick feedback, and added the needs for power and affiliation, all three learned; his Kakinada training studies tested whether nAch can be taught. John Atkinson's theory (1957) makes the tendency to approach success equal to $M_s \times P_s \times (1 - P_s)$, which peaks at intermediate difficulty, so success-oriented people prefer moderate tasks while failure-avoidant people choose very easy or very hard ones. Matina Horner proposed a fear of success in women (1968, 1972), later reinterpreted as a reflection of cultural stereotypes. Building on Heider's attributions and Rotter's locus of control (1966), Bernard Weiner classified the causes of success and failure by locus, stability and controllability, and showed how each dimension shapes emotion and expectancy.

**10.7 Intrinsic Motivation, Competence and the Self.** Harlow's manipulation drive, Butler's visual exploration and Berlyne's curiosity (1960), aroused by the collative variables of novelty, complexity, surprise and incongruity, showed that organisms seek stimulation and knowledge for their own sake. Robert White (1959) unified these findings as effectance or competence motivation. Albert Bandura's self-efficacy (1977), a task-specific belief in one's capability built mainly through mastery experiences, is the motivational core of his social cognitive theory and its triadic reciprocal causation. Deci's Soma puzzles (1971) and Lepper, Greene and Nisbett's magic markers (1973) demonstrated the overjustification effect, in which expected tangible rewards undermine intrinsic interest. Deci and Ryan's self-determination theory (1985) proposes three innate needs, autonomy, competence and relatedness; its cognitive evaluation theory separates the controlling from the informational aspect of rewards, and its continuum runs from amoti-

vation through external, introjected, identified and integrated regulation to intrinsic motivation. Csikszentmihalyi's flow is total absorption when challenge and skill are balanced, and Festinger's cognitive dissonance (1957) is a drive-like tension that motivates people to make their beliefs and actions consistent.

**10.8 Conflict and Frustration.** Frustration is the blocking of goal-directed behaviour, arising from environmental obstacles, personal limitations or conflict between motives; it differs from pressure, the demand to perform or conform, and from change. Kurt Lewin (1935) described approach-approach conflicts (the easiest to resolve), avoidance-avoidance conflicts (vacillation and attempts to leave the field) and approach-avoidance conflicts (ambivalence), and later writers added the double approach-avoidance conflict typical of real-life decisions. Neal Miller's gradients (1944, 1959) showed that both approach and avoidance strengthen near the goal but that avoidance is steeper, so the organism vacillates where the lines cross; Dollard and Miller's clinical rule is to reduce fear before increasing desire. Classic reactions to frustration are aggression (direct or displaced, as in scapegoating), regression (Barker, Dembo and Lewin, 1941), fixation (Maier, 1949), withdrawal and apathy, and defence mechanisms. The frustration-aggression hypothesis of Dollard, Doob, Miller, Mowrer and Sears (1939) was softened by Miller (1941) and reformulated by Berkowitz (1989) in terms of negative affect and aggressive cues.

**10.9 Biological Motives: Hunger, Thirst, Sex and Others.** Cannon and Washburn's swallowed balloon (1912) linked hunger pangs to stomach contractions, but people without stomachs still feel hunger. Glucostatic and lipostatic set-point theories gave way, in Pinel's account, to a positive-incentive view with a settling point for body weight. In the classic dual-centre theory (Stellar, 1954), lesions of the ventromedial hypothalamus, the satiety centre, cause hyperphagia and obesity (Hetherington and Ranson, 1940), and lesions of the lateral hypothalamus, the feeding centre, cause aphagia and adipsia (Anand and Brobeck, 1951); modern work has revised both claims and emphasises insulin, the paraventricular and arcuate nuclei, and signals such as leptin, ghrelin and cholecystokinin. Schachter's externality hypothesis and Herman's restrained eating show the power of external cues. Thirst is osmotic (cell dehydration detected by osmoreceptors) or hypovolaemic (loss of blood volume, signalled through angiotensin II). Kinsey's reports (1948, 1953) surveyed sexual behaviour, Masters and Johnson (1966) described excitement, plateau, orgasm and resolution, and Kaplan (1979) added desire. In Warden's obstruction box the maternal drive was the strongest; pain avoidance, temperature regulation and sleep complete the list of primary motives.

**10.10 Motivation in Practice: Work, School and Clinic.** Herzberg's two-factor theory (1959) separates motivators intrinsic to the work (achievement, recognition, the work itself, responsibility, advancement, growth), which create satisfaction, from hygiene factors in the job environment (policy, supervision, relationships, conditions, salary, security), which only prevent dissatisfaction; job satisfaction depends on both. Vroom's expectancy theory (1964) makes motivational force the product of expectancy, instrumentality and valence; Adams' equity theory explains reactions to unfair outcome-to-input ratios; and Locke and Latham's goal-setting theory shows that specific, challenging goals with commitment and feedback raise performance. McGregor's Theory X and Theory Y and Hackman and Oldham's job characteristics model complete the work picture. In schools, informational feedback, attribution retraining (Dweck, 1975), incremental beliefs about ability, mastery goals, self-efficacy and moderate arousal all support learning. In the clinic, anhedonia, avolition and addiction are disturbances of motivation, and William R. Miller and Stephen Rollnick's motivational interviewing (1991) resolves ambiva-

lence by evoking the client's own reasons for change, often alongside Prochaska and DiClemente's stages of change.

## Who's who

Table 10.14 The people in this chapter.

| Name | When | Examined for |
|---|---|---|
| John Stacy Adams | 1963, 1965 | Equity theory: comparing one's ratio of outcomes to inputs with another person's |
| Clayton Paul Alderfer | 1940 to 2015 | ERG theory (1969): existence, relatedness, growth; frustration-regression |
| Gordon Willard Allport | 1897 to 1967 | Functional autonomy of motives (1937) |
| Bal Krishan Anand | 1951 (with Brobeck) | Lateral hypothalamic lesions cause aphagia: the "feeding centre"; later built neurophysiology at AIIMS |
| Elliot Aronson | born 1932 | Effort justification: valuing what we suffered for (with Mills, 1959) |
| John William Atkinson | 1923 to 2003 | Achievement motivation theory (1957): motive to succeed versus motive to avoid failure; preference for intermediate risk |
| Albert Bandura | 1925 to 2021 | Self-efficacy (1977) and its four sources; social cognitive theory and triadic reciprocal causation (1986) |
| Roger Barker | 1903 to 1990 | Regression after frustration: toys behind a wire screen (with Dembo and Lewin, 1941) |
| Leonard Berkowitz | 1926 to 2016 | Reformulated frustration-aggression hypothesis (1989): negative affect and aggressive cues |
| Daniel Ellis Berlyne | 1924 to 1976 | Curiosity and collative variables (1960); perceptual versus epistemic curiosity; specific versus diversive exploration |
| Claude Bernard | 1813 to 1878 | The milieu intérieur, the constant internal environment |
| Luther Lee Bernard | 1924 | Counted nearly six thousand proposed instincts, helping to discredit instinct theory |
| Kent Berridge | 1993 (with Robinson) | "Wanting" versus "liking"; incentive-sensitisation theory of addiction |
| John Brobeck | 1951 (with Anand) | Lateral hypothalamic lesions cause aphagia |
| Judson Brown | 1948 | Harness studies showing that the avoidance gradient is steeper than the approach gradient |
| Robert Butler | 1953 | Monkeys learned a task for the chance to look out of a window: visual exploration drive |
| Walter Bradford Cannon | 1871 to 1945 | Coined homeostasis (1926; *The Wisdom of the Body*, 1932); stomach-contraction theory of hunger (1912) |
| Leo Crespi | 1942 | Contrast effects after shifts in reward size |
| Mihaly Csikszentmihalyi | 1934 to 2021 | Flow (1975, 1990): challenge-skill balance; experience sampling method |
| Edward L. Deci | born 1942 | Soma puzzle study (1971); self-determination and cognitive evaluation theories with Ryan (1985) |
| John Dillingham Dodson | 1879 to 1955 | Yerkes-Dodson law (1908) |

| Name | When | Examined for |
|---|---|---|
| John Dollard | 1900 to 1980 | Frustration-aggression hypothesis (1939); *Personality and Psychotherapy* with Miller (1950) |
| Leonard Doob | 1909 to 2000 | Co-author of *Frustration and Aggression* (1939) |
| Carol Dweck | born 1946 | Attribution retraining of helpless children (1975); entity versus incremental theories of ability |
| J. A. Easterbrook | 1959 | Cue-utilisation hypothesis: high arousal narrows attention |
| Leon Festinger | 1919 to 1989 | Cognitive dissonance (1957); one-dollar versus twenty-dollar study with Carlsmith (1959) |
| Viktor Frankl | 1905 to 1997 | Will to meaning; logotherapy; *Man's Search for Meaning* (1946) |
| Sigmund Freud | 1856 to 1939 | Instincts: life instincts (Eros, libido) and the death instinct (1920); cathexis; the id as reservoir of instincts |
| Jeffrey Friedman | born 1954 | Identified leptin, the hormone missing in ob/ob mice (1994) |
| Karl von Frisch | 1886 to 1982 | Co-founder of ethology; Nobel Prize 1973 |
| Erich Fromm | 1900 to 1980 | Five needs: relatedness, transcendence, rootedness, identity, frame of orientation (1955) |
| William Glasser | 1925 to 2013 | Reality therapy and choice theory: five basic needs, including power or achievement |
| Kurt Goldstein | 1878 to 1965 | Coined self-actualisation (*The Organism*) |
| J. Richard Hackman | 1976 (with Oldham) | Job characteristics model: skill variety, task identity, task significance, autonomy, feedback |
| Harry Frederick Harlow | 1905 to 1981 | Manipulation drive in puzzle-solving monkeys (1950); Maslow's doctoral adviser |
| Donald Olding Hebb | 1904 to 1985 | Optimal arousal theory ("Drives and the C.N.S.", 1955); sensory-deprivation research at McGill |
| Fritz Heider | 1896 to 1988 | Naive psychology: internal versus external attributions (1958) |
| Ewald Hering | 1834 to 1918 | Opponent-process theory of colour vision (not of motivation) |
| C. Peter Herman | 1975 (with Mack) | Restrained eating: dieters eat more after a milkshake preload |
| Woodburn Heron | 1954 (with Bexton and Scott) | Sensory-deprivation studies at McGill |
| Frederick Herzberg | 1923 to 2000 | Two-factor (motivator-hygiene) theory (1959) |
| Albert Hetherington | 1940 (with Ranson) | VMH lesions cause hyperphagia and obesity |
| Matina Souretis Horner | born 1939 | Fear of success (1968, 1972): "Anne at the top of her medical school class" |
| Carl Hovland | 1912 to 1961 | Double approach-avoidance conflict (with Sears, 1938); lynchings and cotton prices as displaced aggression (1940) |
| Clark Leonard Hull | 1884 to 1952 | Drive-reduction theory (*Principles of Behavior*, 1943); habit strength, K and V |
| William James | 1842 to 1910 | Defined instinct (1890); humans have more instincts than animals |
| Virginia Eshelman Johnson | 1925 to 2013 | Four-phase sexual response cycle with Masters (1966) |
| Helen Singer Kaplan | 1929 to 1995 | Triphasic model: desire, excitement, orgasm (1979) |

| Name | When | Examined for |
|---|---|---|
| Gordon Kennedy | 1953 | Lipostatic (body-fat set-point) theory of hunger |
| Alfred Charles Kinsey | 1894 to 1956 | Reports on male (1948) and female (1953) sexual behaviour; the Kinsey scale |
| Zing-Yang Kuo | 1898 to 1970 | Kittens reared with rats rarely killed them (1930): rearing, not instinct |
| Gary Latham | born 1945 | Goal-setting theory with Locke (1990) |
| Mark Lepper | 1973 | Overjustification effect: the magic-marker study with Greene and Nisbett |
| Kurt Lewin | 1890 to 1947 | Field theory and valence; types of conflict (1935); regression study (1941); level of aspiration |
| Edwin Locke | born 1938 | Goal-setting theory (from 1968): specific, difficult goals with feedback |
| Konrad Lorenz | 1903 to 1989 | Ethology: fixed action patterns, egg-rolling, imprinting; Nobel Prize 1973 |
| Norman Maier | 1900 to 1977 | Frustration-induced fixation in rats (1949) |
| Abraham Harold Maslow | 1908 to 1970 | Hierarchy of needs (1943, 1954); D- and B-needs; metamotivation; peak experiences; self-transcendence |
| William Howell Masters | 1915 to 2001 | Sexual response cycle: excitement, plateau, orgasm, resolution (1966) |
| Jean Mayer | 1920 to 1993 | Glucostatic theory of hunger (1953) |
| David Clarence McClelland | 1917 to 1998 | Needs for achievement, power and affiliation (1953, 1961); the Kakinada training studies |
| William McDougall | 1871 to 1938 | Instinct theory of motivation (1908); hormic (purposive) psychology; instincts paired with emotions |
| Douglas McGregor | 1906 to 1964 | Theory X and Theory Y (*The Human Side of Enterprise*, 1960) |
| Neal Elgar Miller | 1909 to 2002 | Learned fear as a drive (1948); approach and avoidance gradients (1944, 1959); 1941 revision of frustration-aggression |
| William Richard Miller | born 1947 | Motivational interviewing (with Rollnick, 1991) |
| Christiana Morgan | 1897 to 1967 | Thematic Apperception Test with Murray (1935) |
| Orval Hobart Mowrer | 1907 to 1982 | Two-factor theory of avoidance learning (1947); co-author of *Frustration and Aggression* |
| Henry Alexander Murray | 1893 to 1988 | Needs and press (1938); viscerogenic and psychogenic needs; TAT with Morgan (1935) |
| Richard Nisbett | born 1941 | Overjustification study with Lepper and Greene (1973) |
| James Prochaska | early 1980s (with DiClemente) | Transtheoretical model: precontemplation, contemplation, preparation, action, maintenance |
| Stephen Walter Ranson | 1880 to 1942 | VMH lesions cause hyperphagia and obesity (with Hetherington, 1940) |
| Judith Rodin | 1981 | Review showing that external-cue responsiveness does not explain obesity |
| Carl Rogers | 1902 to 1987 | Co-founder of humanistic psychology with Maslow; actualising tendency (Personality book) |
| Stephen Rollnick | 1991 (with Miller) | Motivational interviewing |
| Saul Rosenzweig | 1907 to 2004 | Frustration tolerance; extrapunitive, intropunitive and impunitive reactions; Picture-Frustration Study |

| Name | When | Examined for |
|---|---|---|
| Julian Bernard Rotter | 1916 to 2014 | Expectancy and reinforcement value (1954); locus of control and the I-E Scale (1966) |
| Richard M. Ryan | born 1953 | Self-determination theory with Deci (1985): autonomy, competence, relatedness |
| Stanley Schachter | 1922 to 1997 | Externality hypothesis of obesity; the rigged-clock study (1968) |
| Robert Sears | 1908 to 1989 | Co-author of *Frustration and Aggression* (1939); with Hovland, double approach-avoidance (1938) and lynchings and cotton prices (1940) |
| Martin Seligman | born 1942 | Learned helplessness; attributional reformulation with Abramson and Teasdale (1978) |
| Fred Sheffield | 1950 (with Roby) | Rats learned a maze for saccharin, which reduces no need |
| Richard Lester Solomon | 1918 to 1995 | Opponent-process theory of acquired motivation (with Corbit, 1974) |
| Eliot Stellar | 1919 to 1993 | Dual-centre theory of hunger (1954) |
| Nikolaas Tinbergen | 1907 to 1988 | Sign stimuli: the stickleback's red belly; Nobel Prize 1973 |
| Otto Tinklepaugh | 1928 | Monkeys expecting banana rejected lettuce: specific incentive expectancy |
| Edward Chace Tolman | 1886 to 1959 | Purposive behaviourism; learned expectancies |
| Norman Triplett | 1861 to 1931 | Cyclists raced faster against others (1898): the first social facilitation study |
| Victor Harold Vroom | born 1932 | Expectancy (VIE) theory of work motivation (1964) |
| Mahmoud Wahba | 1976 (with Bridwell) | Review finding little support for Maslow's ordering of needs |
| Carl Warden | 1931 | Columbia obstruction box: maternal drive strongest |
| A. L. Washburn | 1912 | Swallowed the balloon in Cannon's hunger study |
| Bernard Weiner | born 1935 | Attribution theory of achievement motivation: locus, stability, controllability |
| Robert White | 1904 to 2001 | Effectance or competence motivation (1959) |
| Robert Sessions Woodworth | 1869 to 1962 | Introduced the term "drive" (*Dynamic Psychology*, 1918) |
| Robert Mearns Yerkes | 1876 to 1956 | Yerkes-Dodson law (1908): the dancing-mice study |
| Robert Zajonc | 1923 to 2008 | Drive theory of social facilitation (1965): arousal strengthens the dominant response |
| Marvin Zuckerman | 1928 to 2018 | Sensation seeking and its four components; Sensation Seeking Scale (1964) |

## Key terms

approach and avoidance gradients, approach-approach conflict, approach-avoidance conflict, arousal, attribution theory of achievement motivation, avoidance-avoidance conflict, cognitive dissonance, cognitive evaluation theory, collative variables, curiosity, death instinct, deficiency needs, displaced aggression, double approach-avoidance conflict, drive, drive-reduction theory, dual-centre theory, effectance motivation, equity theory, ERG theory, ethology, expectancy theory, expectancy-value theory, externality hypothesis, extrinsic motivation, fear of success, fixed action pattern, flow, frustration, frustration-aggression hypothesis, frustration-regression, functional

autonomy, glucostatic theory, goal, goal-setting theory, growth needs, hierarchy of needs, homeostasis, hormic psychology, humanistic psychology, hygiene factors, hypovolaemic thirst, incentive, incentive theory, instinct, intrinsic motivation, lateral hypothalamus, leptin, lipostatic theory, locus of control, maternal drive, metamotivation, metaneeds, metapathology, milieu intérieur, motivation, motivational cycle, motivational interviewing, motivators, motive, need, need for achievement, opponent-process theory, optimal arousal theory, osmotic thirst, overjustification effect, peak experiences, prepotency, press, pressure, primary motives, psychogenic needs, secondary motives, self-actualisation, self-determination theory, self-efficacy, self-transcendence, sensation seeking, set point, settling point, sexual response cycle, sign stimulus, social cognitive theory, triadic reciprocal causation, two-factor theory, ventromedial hypothalamus, viscerogenic needs, will to meaning, Yerkes-Dodson law

## From the past papers

1. A toddler goes to great lengths to open a box. She knows there is nothing in the box. This is an example of: (RML 2024)

   (a) Intrinsic motivation (b) Homeostasis (c) Extrinsic motivation (d) Secondary motivation

   **Answer: (a) Intrinsic motivation.** The box is empty, so the only reward is the activity of opening it, which is what doing something for its own sake means (compare Harlow's puzzle-solving monkeys). Homeostasis would need a bodily imbalance to correct, extrinsic motivation needs a separable payoff such as a sweet or praise, and secondary motivation refers to learned social motives such as achievement or approval, none of which is being served here.

2. The phenomenon of "gregariousness" has a: (RML 2022)

   (a) Social motive (b) Biological motive (c) Psychological motive (d) Personal motive

   **Answer: (a) Social motive.** Gregariousness, the tendency to seek the company of others, is classed with affiliation among the social (secondary or psychosocial) motives. "Biological" tempts only because McDougall listed gregariousness among his instincts; "psychological motive" and "personal motive" are not standard class names in the classification of motives.

3. Which model asserts that inner forces largely beyond individual control drive human behavior? (RML 2025)

   (a) Cognitive model (b) Behavioral model (c) Humanistic model (d) Psychodynamic model

   **Answer: (d) Psychodynamic model.** Freud's model places the causes of behaviour in unconscious instincts and conflicts that the person neither sees nor controls. The cognitive model locates the causes in conscious thoughts and beliefs, the behavioural model in conditioning and reinforcement from the environment, and the humanistic model in free will and personal growth, which is the opposite of control by hidden forces.

4. According to Freud, human beings are motivated by powerful 'innate forces' which are known as: (Amity 2023)

   (a) Needs (b) Drives (c) Instincts (d) Id

   **Answer: (c) Instincts.** Freud's word *Trieb* is conventionally translated "instinct", and he grouped the instincts into the life instincts (Eros) and the death instinct. "Needs" is the vocabulary of Maslow and Murray; "drives" is Hull's term (and a fair literal translation of *Trieb*, but not the classical answer); the id is the reservoir that holds the instincts, not the forces themselves.

5. Deviations from homeostasis can create an internal state of biological and psychological tension. What is the best term for this state? (RML 2021)

   (a) Regulation (b) Drive (c) Dissonance (d) Equilibrium

   **Answer: (b) Drive.** In Hull's theory a departure from homeostasis is a need, and the aroused, tense state it produces is a drive, which energises behaviour until the need is reduced. Equilibrium is the balanced state that homeostasis restores, the opposite of tension; regulation is the corrective process (negative feedback), not the tension itself; dissonance is Festinger's tension between inconsistent cognitions, not a bodily deviation.

6. The Yerkes-Dodson law states that between performance and anxiety, there is: (RML 2024)

   (a) Inverted-U shaped relationship (b) Curvilinear relationship (c) Linear relationship (d) M-shaped relationship

   **Answer: (a) Inverted-U shaped relationship.** Performance rises with arousal up to a moderate level and then falls, and the peak comes at a lower level of arousal for complex tasks (Figure 10.2). "Curvilinear" is true but less specific, so the precise inverted U wins; a linear relationship would mean that more anxiety always helps or always hurts; no two-peaked (M-shaped) version of the law exists.

7. According to Zajonc's drive theory of social facilitation, the presence of others raises arousal, which strengthens the dominant response. This means an audience will MOST improve performance on a task that is: (Amity recall)

   (a) Novel and difficult (b) Easy and well-learned (c) Creative and open-ended (d) Performed for the first time

   **Answer: (b) Easy and well-learned.** On an easy or well-practised task the dominant (most probable) response is the correct one, so the extra arousal from an audience strengthens it and performance improves. On novel, difficult or first-time tasks the dominant response is more often an error, and on creative, open-ended tasks it is the obvious, stereotyped idea, so an audience tends to impair performance there.

8. In Incentive Theory, motivation is primarily driven by: (RML 2025)

   (a) Intrinsic satisfaction (b) External rewards (c) Biological drives (d) Self-efficacy beliefs

   **Answer: (b) External rewards.** Incentive theory says behaviour is pulled by external goals, as Tinklepaugh's lettuce-refusing monkeys and Crespi's contrast effects showed. Intrinsic satisfaction is the core of self-determination theory, biological drives push from within in Hull's theory, and self-efficacy beliefs belong to Bandura's social cognitive theory. The same paper asked what drive theory explains motivation as a process to do: **reduce physiological needs**, not seek external rewards (incentive theory), set long-term goals (goal-setting theory) or increase social connections (affiliation).

9. Which stage is characterised by realising your full potential according to Maslow's Hierarchy of Needs? (RML 2021)

   (a) Safety Needs (b) Social Needs (c) Esteem Needs (d) Self-Actualisation Needs

   **Answer: (d) Self-Actualisation Needs.** Self-actualisation, the top of the hierarchy, is the need to become everything one is capable of becoming. Safety needs concern security and order, social needs concern belonging and love, and esteem needs concern respect and achievement; all three are deficiency needs that must be largely met before self-actualisation takes charge.

10. According to Maslow's Hierarchy of Needs, which need must be mostly fulfilled before an individual seeks esteem needs? (RML 2025)

    (a) Self-actualization needs (b) Physiological needs (c) Social belonging needs (d) Safety needs

    **Answer: (c) Social belonging needs.** The printed key gives (b). The level immediately below esteem is belongingness and love, so it is the need that must be mostly fulfilled just before esteem needs take over. Physiological and safety needs must be met as well, but they come earlier in the sequence and so are not the best answer to "before esteem"; self-actualisation lies above esteem and cannot be its precondition (see the Exam trap in Section 10.5).

11. In Reality Therapy, developed by William Glasser, which of the following is considered one of the five basic genetic needs? (Real PYQ 2025)

    (a) Self-actualization (b) Power or Achievement (c) Collective Unconscious (d) Unconditional positive regard

    **Answer: (b) Power or Achievement.** Glasser's five basic needs are survival, love and belonging, power (achievement and recognition), freedom and fun. Self-actualisation is the top of Maslow's hierarchy, the collective unconscious is Jung's inherited layer of the psyche, and unconditional positive regard is one of Rogers's therapeutic conditions; none of them is a Glasser need.

12. Viktor Frankl's concept of the "will to meaning," the fundamental motivational force in humans to find purpose in life, is central to which therapeutic school? (Real PYQ 2024)

    (a) Rational Emotive Behavior Therapy (REBT) (b) Logotherapy (c) Existential-Humanistic Therapy (d) Acceptance and Commitment Therapy (ACT)

    **Answer: (b) Logotherapy.** Frankl held that the primary human motive is the will to meaning, in contrast to Freud's will to pleasure and Adler's will to power, and built logotherapy on it. Existential-humanistic therapy (Rollo May, James Bugental) shares the concern with meaning but is not Frankl's school; REBT (Albert Ellis) disputes irrational beliefs, and ACT (Steven Hayes) trains acceptance and values-based action.

13. McClelland's theory of needs proposes three acquired needs that drive behaviour. They are the needs for: (Amity recall)

    (a) Achievement, affiliation, and power (b) Safety, esteem, and self-actualization (c) Existence, relatedness, and growth (d) Autonomy, competence, and relatedness

    **Answer: (a) Achievement, affiliation, and power.** McClelland scored achievement, affiliation and power imagery in TAT stories and insisted that these needs are learned. Option (b) lists three of Maslow's levels, (c) is Alderfer's ERG theory, and (d) gives the three innate needs of Deci and Ryan's self-determination theory.

14. Which of the following is NOT one of the three fundamental psychological needs according to Self-Determination Theory? (RML 2025)

    (a) Autonomy (b) Competence (c) Affiliation (d) Relatedness

    **Answer: (c) Affiliation.** The three innate needs of self-determination theory are autonomy, competence and relatedness. Affiliation looks like relatedness, which is why it is the bait, but it is McClelland's learned need (nAff), not an SDT need. The same paper asked which SDT motivation is driven by internal rewards and personal growth: **intrinsic motivation**, not extrinsic motivation, not amotivation (the absence of motivation) and not external regulation (the least autonomous form of extrinsic motivation).

15. Reciprocal causation model has been described under: (IHBAS 2020)

    (a) Social cognitive theory (b) Two-factor theory (c) Drive reduction theory (d) Constructivist theory

    **Answer: (a) Social cognitive theory.** Bandura's social cognitive theory (1986) rests on triadic reciprocal causation, in which personal factors, behaviour and environment influence one another. "Two-factor theory" names several other theories (Herzberg, Mowrer, Schachter and Singer, Spearman); drive-reduction theory is Hull's one-way chain from need to drive to behaviour; constructivist theory (Piaget, Vygotsky, Bruner) is a theory of learning in which learners actively build their own knowledge, not a causal model of motivation.

16. Social Cognitive Theory highlights the importance of which factor in motivation? (RML 2025)

    (a) External rewards only (b) Biological drives (c) Self-efficacy (d) Goal difficulty

    **Answer: (c) Self-efficacy.** In Bandura's theory, belief in one's capability to carry out a task decides which tasks people attempt, how hard they try and how long they persist. External rewards do matter in the theory, but not "only"; biological drives belong to Hull; goal difficulty is the central variable of Locke and Latham's goal-setting theory.

17. Who introduced the concept of "flow" in psychology? (RML 2023)

    (a) Carl Rogers (b) Abraham Maslow (c) Mihaly Csikszentmihalyi (d) Viktor Frankl

    **Answer: (c) Mihaly Csikszentmihalyi.** Csikszentmihalyi described flow (1975, 1990), the absorbed, autotelic state that arises when challenge and skill are in balance. Maslow's peak experiences are brief, ecstatic moments in self-actualisers, Rogers described the fully functioning person, and Frankl the will to meaning.

18. Arvind is a college student who has strong negative feelings about smoking cigarettes. His grandfather died of lung cancer and he believes smoking is disgusting. One night at a party, Arvind's friends all go outside for a cigarette and he decides to smoke as well. What will he most likely experience as a result? (RML 2021)

    (a) Justification of efforts (b) Increased drive to smoke (c) Cognitive Dissonance (d) Drive Reduction

    **Answer: (c) Cognitive Dissonance.** Acting against a strongly held belief creates the tension Festinger (1957) called cognitive dissonance, which Arvind will be motivated to reduce, perhaps by telling himself that one cigarette at a party does not count. Justification of effort is a particular kind of dissonance reduction after suffering for a goal, drive reduction is Hull's account of satisfying a bodily need, and nothing in the story suggests an increased drive to smoke.

19. If Ramesh works in a call centre, and has to work very hard, do a lot of work, and that too at a fast rate, he experiences: (RML 2020)

    (a) Conflict (b) Frustration (c) Unpredictability (d) Pressure

    **Answer: (d) Pressure.** Pressure is the demand to perform at a high level, quickly and well (or to conform to others' expectations). Frustration would mean that a goal is blocked, conflict that incompatible motives pull him in different directions, and unpredictability that he cannot foresee what will happen; Ramesh's problem is simply the demand itself.

20. The 'satiety centre' of the brain is located in the: (RML 2024)

    (a) Lateral hypothalamus (b) Ventromedial hippocampus (c) Lateral thalamus (d) Ventromedial hypothalamus

    **Answer: (d) Ventromedial hypothalamus.** Bilateral VMH lesions produce hyperphagia and obesity (Hetherington and Ranson, 1940), so Stellar's dual-centre theory (1954) called the VMH the satiety centre. The lateral hypothalamus is the opposite, the feeding centre, whose destruction causes aphagia (Anand and Brobeck, 1951); "ventromedial hippocampus" and "lateral thalamus" are invented by changing one word. JEMAS 2025 asked the same question with the cerebrum and the basal ganglia as baits; neither is a hunger centre.

21. Masters and Johnson's sexual response cycle is distinct from Kaplan's model. Which of the following correctly reflects Masters and Johnson's sequence? (IHBAS 2025)

    (a) Excitement → Plateau → Orgasm → Resolution (b) Plateau → Excitement → Orgasm → Resolution (c) Excitement → Desire → Plateau → Orgasm → Refractory (d) Desire → Arousal → Plateau → Orgasm → Resolution

    **Answer: (a) Excitement → Plateau → Orgasm → Resolution.** Masters and Johnson (1966) observed four phases in that order. Option (b) reverses excitement and plateau; options (c) and (d) insert desire, which is Helen Singer Kaplan's addition (desire, excitement, orgasm, 1979), and (c) also ends with "refractory", which is the male recovery period after orgasm, not a phase of the cycle.

22. According to Herzberg's Two-Factor Theory, which of the following factors are considered motivators that lead to job satisfaction? (RML 2023)

    (a) Salary and benefits (b) Job security and work conditions (c) Recognition and opportunities for advancement (d) Company policies and interpersonal relationships

    **Answer: (c) Recognition and opportunities for advancement.** Herzberg's motivators belong to the content of the work: achievement, recognition, the work itself, responsibility, advancement and growth. Salary, benefits, job security, working conditions, company policies and interpersonal relationships are all hygiene factors: when they are poor they cause dissatisfaction, but improving them only removes the dissatisfaction without creating motivation.

23. Job satisfaction depends on (JEMAS 2025)

    (a) work motivation or job environment. (b) only work motivation. (c) personal disposition only. (d) work motivation and job environment.

    **Answer: (d) work motivation and job environment.** Herzberg's research showed that satisfaction comes from the content of the work (motivators) while dissatisfaction comes from its context (hygiene factors), so the two together determine how a person feels about the job. "Or" and "only" make a single source sufficient, and personal disposition plays a part but is not the only determinant.

24. In Expectancy Theory, motivation is highest when an individual believes: (1) effort leads to performance, (2) performance leads to outcomes, and (3) outcomes are: (RML 2025)

    (a) Uncertain (b) Valuable (c) Difficult (d) Negative

    **Answer: (b) Valuable.** Vroom's motivational force is expectancy × instrumentality × valence, and the third term, valence, is the value of the outcome to the person. Uncertain outcomes lower instrumentality, "difficult" describes the task rather than the outcome, and a negative valence repels. A second item in the same paper turned the idea round: motivation is highest when effort leads to performance and performance to desired rewards, not when rewards are guaranteed regardless of performance (instrumentality collapses), when effort is unrelated to performance (expectancy is zero) or when rewards are unimportant (valence is zero).

25. In Goal-Setting Theory, what characteristic makes a goal most effective at motivating behavior? (RML 2025)

    (a) Being vague and flexible (b) Being specific and challenging (c) Being easy to accomplish (d) Avoiding feedback

    **Answer: (b) Being specific and challenging.** Locke and Latham found that specific, difficult but attainable goals, accepted by the person and accompanied by feedback, produce higher performance than vague "do your best" goals or easy goals. Avoiding feedback is wrong because feedback is one of the conditions that makes goals work. A parallel item in the same paper offered "unrelated to personal values" as a bait; a goal the person does not value wins little commitment, and commitment is another of the theory's conditions.

## High-yield sheet

*The facts this chapter's questions ask most, most-asked first. Last read before the exam.*

1 **Abraham Maslow** is the author of the hierarchical theory of motivation ("A Theory of Human Motivation", 1943; *Motivation and Personality*, 1954). Bottom to top: physiological, safety, belongingness and love (social), esteem, self-actualisation. "Ego" is not a level; it is Freud's. ASKED 5 TIMES

2 **Henry Murray**'s need-press theory (*Explorations in Personality*, 1938) underlies the **Thematic Apperception Test** (Christiana Morgan and Murray, 1935): a projective test in which stories about ambiguous pictures reveal the teller's needs, conflicts and motives. ASKED 5 TIMES

3 The **ventromedial hypothalamus** is the classic satiety centre: lesions cause hyperphagia and obesity (Hetherington and Ranson, 1940). The **lateral hypothalamus** is the feeding centre: lesions cause aphagia, adipsia and weight loss (Anand and Brobeck, 1951). ASKED 4 TIMES

4 **Sigmund Freud**: the innate forces that motivate us are **instincts**; war and violence express the **death instinct**, libido is the energy of the life instincts, the **id** is the reservoir of unorganised instinctual drives, and **cathexis** is the investment of psychic energy in an object. ASKED 4 TIMES

5 **Frederick Herzberg**'s two-factor theory (1959): motivators (achievement, recognition, the work itself, responsibility, advancement, growth) create satisfaction; hygiene factors (salary, security, conditions, policy, supervision, relationships) only prevent dissatisfaction. Job satisfaction depends on work motivation **and** job environment. ASKED 3 TIMES

6 **Self-actualisation**, the striving to realise one's full potential, is the top of Maslow's hierarchy; the term was coined by Kurt Goldstein (1939) and made famous by Maslow. ASKED 4 TIMES

7 **Humanistic psychology**, the third force of Abraham Maslow and Carl Rogers, emphasises free will, personal growth and an innate tendency to grow and reach one's maximum potential. ASKED 4 TIMES

8 **Self-determination theory** (Edward Deci and Richard Ryan, 1985) names three innate needs: autonomy, competence and relatedness (not affiliation). Intrinsic motivation, driven by internal reward and growth, is its most self-determined form. ASKED 3 TIMES

9 The **Yerkes-Dodson law** (Robert Yerkes and John Dodson, 1908): an inverted-U relation between arousal or anxiety and performance, with a lower optimum for complex or difficult tasks. ASKED 3 TIMES

10 **Victor Vroom**'s expectancy (VIE) theory (1964): motivational force = expectancy × instrumentality × valence, highest when effort leads to performance, performance leads to rewards, and the rewards are valued. ASKED 2 TIMES

11 **Edwin Locke and Gary Latham**'s goal-setting theory: specific, challenging (but attainable) goals, with commitment and feedback, raise performance more than vague or easy goals. ASKED 2 TIMES

12 A deviation from homeostasis creates a **drive**, a state of tension; **Clark Hull**'s drive-reduction theory (1943) explains motivation as reducing physiological needs. ASKED 2 TIMES

13 **Albert Bandura**'s social cognitive theory (1986) makes **self-efficacy** (1977) the key motivational factor and rests on triadic reciprocal causation of person, behaviour and environment. ASKED 2 TIMES

14 **Rationalisation** is an after-the-fact, socially acceptable reason for behaviour that was really driven by other (unacceptable or unconscious) motives. ASKED 3 TIMES

15 **Leon Festinger**'s cognitive dissonance (1957): acting against one's beliefs, like Arvind smoking at the party, creates tension that motivates a change of attitude or behaviour to reduce it. ASKED 2 TIMES

16 **Viktor Frankl**'s will to meaning, the primary human motive, is the basis of logotherapy. ASKED 2 TIMES

17 **Masters and Johnson** (1966): excitement, plateau, orgasm, resolution; Helen Singer Kaplan (1979) put desire first. The reports on male (1948) and female (1953) sexual behaviour are **Alfred Kinsey**'s. ASKED 2 TIMES

18 The **hypothalamus** is the master homeostatic regulator (hunger, thirst, temperature, control of the pituitary); its suprachiasmatic nucleus is the circadian clock. ASKED 2 TIMES

19 In Maslow's hierarchy the need immediately below esteem is **belongingness and love** (social belonging); RML 2025 printed "physiological" as the key. ASKED ONCE

20 **Gregariousness**, seeking the company of others, is a social (secondary, psychosocial) motive. ASKED ONCE

21 A toddler who works to open a box she knows is empty shows **intrinsic motivation**: the activity is its own reward. ASKED ONCE

22 **Incentive theory**: behaviour is pulled by external rewards, not pushed by internal drives. ASKED ONCE

23 **Flow**, total absorption when challenge and skill are balanced, was introduced by **Mihaly Csikszentmihalyi**. ASKED ONCE

24 **Pressure** is the demand to perform (work hard, fast and a lot, like Ramesh in the call centre) or to conform; it differs from frustration, conflict and change. ASKED ONCE

25 **David McClelland**'s three learned needs: achievement, affiliation and power. ASKED ONCE

26 **William Glasser**'s five basic needs: survival, love and belonging, power (achievement), freedom and fun. ASKED ONCE

27 **Robert Zajonc**'s drive theory of social facilitation (1965): an audience raises arousal and strengthens the dominant response, so it most improves easy, well-learned tasks. ASKED ONCE

28 The **psychodynamic** model holds that inner forces largely beyond individual control drive behaviour. ASKED ONCE

29 **William McDougall** (*An Introduction to Social Psychology*, 1908) developed the instinct theory of motivation (hormic psychology). ASKED ONCE

30 In Hull's equation, **stimulus-intensity dynamism** (V) means a stronger stimulus makes the learned response more likely; incentive motivation (K) is reward size. ASKED ONCE

CHAPTER 11

# Emotion and Stress

IN THIS CHAPTER

11.1 ●● What is an emotion?: Define emotion and separate it from feeling, mood, affect and temperament, and compare the basic-emotion lists of Ekman, Izard, Tomkins and Plutchik with Russell's circumplex.

11.2 ●● The classic theories: from the body to the label: Explain the James-Lange, Cannon-Bard and Schachter-Singer theories, Cannon's five criticisms, and the misattribution of arousal.

11.3 ● Appraisal, feedback, evolution and opposing processes: Describe Lazarus's appraisal theory, the Lazarus-Zajonc debate, the facial feedback hypothesis, Darwin's evolutionary view and Solomon's opponent-process theory.

11.4 ●●● The emotional brain and body: Trace emotion through the autonomic nervous system, hypothalamus, limbic system, amygdala and prefrontal cortex, and judge the polygraph.

11.5 ● Expressing and reading emotion: Evaluate the evidence for universal facial expressions, and explain display rules, FACS, the Duchenne smile and cultural and gender differences.

11.6 ● Four emotions up close: Distinguish fear from anxiety, explain why catharsis fails, and describe the science of happiness and of disgust.

11.7 ●● Emotional development and regulation: Chart how emotions emerge in infancy and apply Gross's process model of emotion regulation; define alexithymia.

11.8 ●● Stress and the body's emergency systems: Define stress three ways and describe fight or flight, Selye's general adaptation syndrome, the HPA axis and tend-and-befriend.

11.9 ●●● Sources of stress and the transactional model: Compare life events, daily hassles, chronic stressors and work stress, and apply Lazarus and Folkman's transactional model.

11.10 ●●● Coping, personality and health: Classify coping, describe hardiness, optimism, resilience and sense of coherence, link Types A to D to heart disease, and explain psychoneuroimmunology and stress management.

Weightage. 73 of the 819 questions mapped in this book (8.9%) test this chapter, 48 of them from real past papers. 3 of its 10 sections are High yield.

North of Vancouver, in Canada, a footbridge of wooden planks and wire cable hangs across the Capilano Canyon. It is about 140 metres long and barely a metre and a half wide. It tilts and sways under every step, its cable handrails are low, and the rocks and rapids of the Capilano River lie some 70 metres below. Most people who cross it grip the cables and walk faster than they meant to. In 1974 two psychologists, Donald Dutton and Arthur Aron, decided that this frightening bridge would make an excellent laboratory.

Their procedure was simple. An attractive young woman stood on the bridge and stopped men aged roughly 18 to 35 who were crossing alone. She explained that she was doing a class project on the effects of scenic places on creativity, and asked each man to fill in a short questionnaire, which included writing a brief story about a picture of a woman. When he had finished, she tore a corner off a sheet of paper, wrote down her name and telephone number, and said that he was welcome to call her if he wanted to talk about the study. Exactly the same thing was done a little further up the river, on a second bridge: a solid cedar structure only about 3 metres above a shallow stream, with high handrails, that did not sway at all.

The men on the swaying bridge wrote stories with noticeably more sexual imagery than the men on the solid bridge. And more of them picked up the telephone: nine of the men interviewed on the suspension bridge called the woman, against only two from the solid bridge. When the interviewer was a man, almost nobody called from either bridge.

Why should a frightening bridge make a stranger more attractive? The men's hearts were pounding because of the bridge, yet at least some of them seem to have read that pounding as attraction to the woman. Think about what this implies. If the same racing heart can be "fear" on one reading and "attraction" on another, where exactly is an emotion? Is it in the body, in the brain, or in the story we tell ourselves about our body? That question has driven more than a century of research, and it leads directly to the second subject of this chapter: stress, which is what happens when the body's emergency system is switched on and then left on. Keep the men on the Capilano bridge in mind; you will meet them again at the end of the chapter.

## 11.1 What is an emotion? MEDIUM YIELD · 4 QS

Everyone knows what an emotion is until asked to define one. You know what it is like to be furious when a scooter cuts across you at a Mysuru signal, or to feel your stomach drop when the question paper is handed out. But what, exactly, has happened in each case? Psychologists answer by taking an emotion apart.

### The three components of an emotion

An **emotion** is a relatively brief, whole-person response to an event that matters to you, made up of three components that normally occur together. The first is **physiological arousal**: the bodily changes, most of them controlled by the autonomic nervous system, such as a pounding heart, fast breathing, sweating palms and a dry mouth. The second is **expres-**

**sive behaviour**: what can be seen and heard from outside, such as the face, the voice, posture and actions like running or striking. The third is **conscious experience**: the private, subjective awareness of the state ("I am afraid"), together with the thoughts that go with it. Many modern theorists add two further ingredients that are really part of the same package: an *appraisal* (an evaluation of what the event means for you) and an *action tendency* (an urge to do something, such as to flee, attack or approach).

Notice how well the three components map onto the theories you are about to meet. The great theoretical quarrels about emotion are mostly quarrels about the *order* and the *causal links* among these components. Does the body react first and the feeling follow? Do they occur together? Does a thought have to come first? Keep the three components in view and the theories will sort themselves out.

## Emotion, feeling, mood, affect and temperament ●●

Five words are used loosely in everyday speech but carefully in psychology, and examiners like to test the difference. A **feeling** is the subjective, consciously experienced part of an emotion: the "what it is like" component. An emotion is short (seconds to minutes) and is usually about something: you are angry *at* a person, afraid *of* a dog. A **mood** is longer, lower in intensity and more diffuse; it often has no clear object or cause, and it colours everything else. You can wake up irritable after a bad night and stay that way all day without being angry at anyone in particular.

The word **affect** has two uses. In research psychology it is the umbrella term for all states that feel good or bad, so emotions, moods and feelings are all forms of affect, and "positive affect" and "negative affect" describe your current feeling state as a whole. In clinical psychology and psychiatry, affect has a narrower meaning: the observable, moment-to-moment expression of emotion, as a clinician sees it in the face, voice and gestures during an interview. Finally, **temperament** is the biologically based, early-appearing and fairly stable individual difference in how easily a person becomes emotionally aroused and how well he or she calms down. A baby who startles at every noise and is hard to soothe differs in temperament from one who takes the world in her stride. (The famous temperament classification of Alexander Thomas and Stella Chess, easy, difficult and slow-to-warm-up, is taught in the Developmental book; the older "four temperaments" of Hippocrates and Galen, sanguine, choleric, melancholic and phlegmatic, belong to the history of personality theory.) Table 11.1 sets the five terms side by side.

Table 11.1 Five terms that are easy to confuse.

| Term | What it is | Typical time span | Example |
|---|---|---|---|
| Emotion | Brief, multi-component response to a specific event | Seconds to minutes | Anger when someone pushes ahead in the queue |
| Feeling | The subjective, conscious component of an emotion | As long as the emotion | "I feel humiliated" |
| Mood | Diffuse, low-intensity state, often without a clear object | Hours to days | Gloomy all day for no obvious reason |
| Affect | Umbrella term for good or bad feeling states; clinically, the observable expression of emotion | Current, moment to moment | "Affect restricted, mood low" |

| Term | What it is | Typical time span | Example |
|---|---|---|---|
| Temperament | Inborn style of emotional reactivity and self-regulation | Years to a lifetime | A child who is slow to warm up to strangers |

CLINICAL LINK

In the mental status examination, clinicians record *mood* and *affect* separately. Mood is the patient's sustained, pervasive emotional state, usually in the patient's own words ("I've been low for weeks"); its key feature is its persistence. Affect is what the clinician observes now, and it is described by its *quality* (anxious, sad, cheerful), its *range* (full, restricted, blunted or flat), its *reactivity* or mobility (does it change appropriately as the topic changes, or is it fixed or labile?), its intensity, and whether it is congruent with the mood and appropriate to what is being said. A common teaching image: mood is the climate, affect is the weather.

## Basic emotions: is there a short list? ●●

If emotions are products of evolution, there might be a small set of **basic emotions** (also called primary emotions): emotions that are innate, appear early in life, are found in all cultures, have distinctive facial expressions, and serve a clear survival function. More complex emotions such as jealousy, pride or nostalgia would then be blends of these or would require more sophisticated thought. Several theorists have proposed such lists, and the lists are a favourite source of exam questions.

**Silvan Solomon Tomkins** (1911 to 1991), in his *Affect Imagery Consciousness* (first two volumes, 1962 and 1963), proposed an **affect theory** in which nine innate affects, each with its own facial pattern, are the primary motivators of behaviour: interest-excitement, enjoyment-joy, surprise-startle, distress-anguish, fear-terror, shame-humiliation, anger-rage, disgust and "dissmell" (the reaction to a bad smell). Tomkins argued that the face is the main site of emotion, an idea that led directly to the facial feedback hypothesis (Section 11.3). Two younger researchers whom he mentored and inspired, Ekman and Izard, carried the programme forward.

**Paul Ekman** (born 1934) and his colleague **Wallace V. Friesen** concluded from cross-cultural research in the late 1960s and early 1970s (Section 11.5) that six emotions have universally recognised facial expressions: *happiness*, *sadness*, *fear*, *anger*, *surprise* and *disgust*. In 1986 they added a seventh, *contempt*, whose expression is a one-sided tightening and raising of the lip corner. **Carroll Ellis Izard** (1923 to 2017) developed **differential emotions theory** (1971, 1977), which names ten fundamental emotions, each with its own feeling, expression and motivational function: interest, joy, surprise, sadness (distress), anger, disgust, contempt, fear, shame (shyness) and guilt. Izard also built coding systems for reading emotion in infants' faces.

Table 11.2 The best-known lists of basic emotions.

| Theorist | Year | Number | The list |
|---|---|---|---|
| John B. Watson | 1919 | 3 | Fear, rage, love |
| Silvan Tomkins | 1962 | 9 | Interest, enjoyment, surprise, distress, fear, shame, anger, disgust, dissmell |

| Theorist | Year | Number | The list |
|---|---|---|---|
| Carroll Izard | 1971 to 1977 | 10 | Interest, joy, surprise, sadness, anger, disgust, contempt, fear, shame, guilt |
| Paul Ekman | 1971 to 1972; contempt 1986 | 6, then 7 | Happiness, sadness, fear, anger, surprise, disgust (then contempt) |
| Robert Plutchik | 1980 | 8 | Joy, trust, fear, surprise, sadness, disgust, anger, anticipation |
| Jaak Panksepp | 1998 | 7 brain systems | Seeking, rage, fear, lust, care, panic (grief), play |

Two more names complete the table. **John Broadus Watson** (1878 to 1958), the founder of behaviourism, held that newborns show only three unlearned emotional reactions, fear, rage and love, and that every other emotion is built from them by conditioning (Chapter 6). **Jaak Panksepp** (1943 to 2017), working from brain stimulation in animals, identified seven subcortical emotional systems, which he wrote in capitals to show they were brain circuits rather than everyday words: SEEKING, RAGE, FEAR, LUST, CARE, PANIC (later PANIC/GRIEF) and PLAY.

INDIAN CONTEXT

The idea of a short list of basic emotions is very old in India. The *Natyashastra*, the Sanskrit treatise on drama attributed to Bharata Muni (usually dated somewhere between about 200 BCE and 200 CE), describes eight *sthayibhavas* (enduring emotional states), each of which an audience savours as a *rasa*: *rati* (love, giving the erotic rasa *shringara*), *hasa* (mirth), *shoka* (sorrow), *krodha* (anger), *utsaha* (energy or heroism), *bhaya* (fear), *jugupsa* (disgust) and *vismaya* (wonder or astonishment). The philosopher Abhinavagupta (about the tenth to eleventh century) argued for a ninth, *shanta* (peace). Notice how closely the list overlaps with Ekman's: anger, fear, disgust, sorrow, joy and surprise are all there, nearly two thousand years earlier.

## Plutchik's wheel of emotions

**Robert Plutchik** (1927 to 2006) offered the most elaborate basic-emotion model in his **psychoevolutionary theory** of emotion (1980). He proposed eight primary emotions, each tied to an adaptive behaviour that helped our ancestors survive, arranged in a circle so that each emotion sits opposite its psychological opposite: joy opposite sadness, trust opposite disgust, fear opposite anger, and surprise opposite anticipation (Figure 11.1, right). Plutchik's famous three-dimensional "cone" adds intensity: each primary emotion has a milder and a stronger form, so that annoyance intensifies into anger and then rage (Table 11.3). Emotions next to each other mix to form **primary dyads**: joy and trust give *love*; trust and fear give *submission*; fear and surprise give *awe*; surprise and sadness give *disapproval*; sadness and disgust give *remorse*; disgust and anger give *contempt*; anger and anticipation give *aggressiveness*; and anticipation and joy give *optimism*. Plutchik compared emotions to colours: a few primaries, mixed in different proportions and intensities, produce the whole palette.

Table 11.3 Plutchik's eight primary emotions, their intensities, opposites and functions.

| Primary | Milder form | Stronger form | Opposite | Adaptive function |
|---|---|---|---|---|

| Primary | Milder form | Stronger form | Opposite | Adaptive function |
|---|---|---|---|---|
| Joy | Serenity | Ecstasy | Sadness | Reproduction |
| Trust | Acceptance | Admiration | Disgust | Incorporation |
| Fear | Apprehension | Terror | Anger | Protection |
| Surprise | Distraction | Amazement | Anticipation | Orientation |
| Sadness | Pensiveness | Grief | Joy | Reintegration |
| Disgust | Boredom | Loathing | Trust | Rejection |
| Anger | Annoyance | Rage | Fear | Destruction |
| Anticipation | Interest | Vigilance | Surprise | Exploration |

### Dimensions instead of categories: Russell's circumplex

A rival tradition says that emotions are not separate boxes at all but points in a continuous space. The idea goes back to **Wilhelm Wundt** (1832 to 1920), whose tridimensional theory of feeling described every feeling by its position on three dimensions: pleasure-displeasure, tension-relaxation and excitement-depression (Chapter 1). The modern version is the **circumplex model** of **James A. Russell** (1980). Russell asked people to judge how similar emotion words were and found that the words fell in a circle defined by just two dimensions: **valence** (how pleasant or unpleasant the state is) and **arousal** (how activated or calm it is). "Excited" is pleasant and high in arousal; "calm" is pleasant and low; "afraid" is unpleasant and high; "bored" and "sad" are unpleasant and low (Figure 11.1, left). Russell later called the blend of valence and arousal you are in at any moment **core affect**. A closely related scheme, from David Watson and Auke Tellegen (1985), rotates the same circle by 45 degrees and describes it with two independent axes, positive affect and negative affect, the basis of the widely used PANAS scale (Psychological Assessment book).

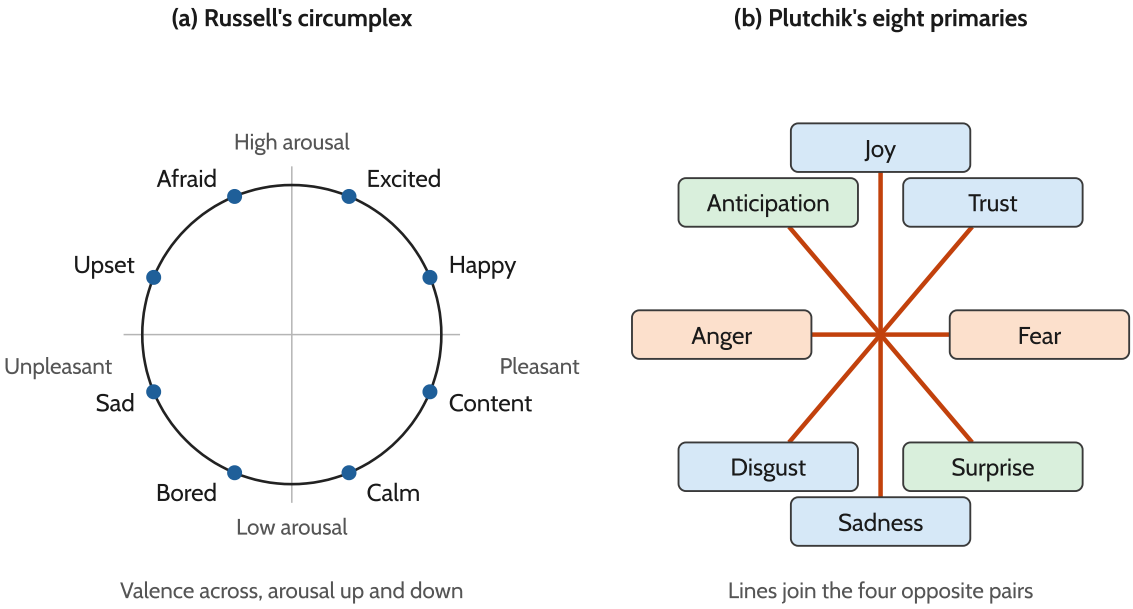


Figure 11.1 Two ways of mapping emotion. (a) Russell's circumplex (1980) places every emotion on a circle defined by valence and arousal. (b) Plutchik's model (1980) has eight primary emotions; each sits across from its opposite (joy and sadness, trust and disgust, fear and anger, anticipation and surprise), and neighbours blend into dyads such as love and awe.

Which view is right, categories or dimensions? Probably both capture something. Basic-emotion theorists point to universal expressions and to brain circuits such as the fear system of the amygdala (Section 11.4). Dimensional theorists point out that in everyday life emotions blend and shade into one another, and that brain imaging has not found one neat "spot" for each emotion word. The most radical dimensional view is the **theory of constructed emotion** of **Lisa Feldman Barrett** (set out in her 2017 book *How Emotions Are Made*). Barrett argues that the brain continually produces core affect, and that we *construct* specific emotions such as "anger" by categorising that core affect with concepts learned from our language and culture. On this view "fear" is less like a reflex and more like a word we have learned to apply, which is why the same pounding heart could be called fear on one bridge and attraction on another.

EXAM TRAP

**Jealousy, shame, guilt, pride, love and envy are not on Ekman's list.** Ekman's six are happiness, sadness, fear, anger, surprise and disgust; contempt was added as a seventh in 1986. Jealousy is a complex, social emotion (a blend of fear, anger and sadness about losing a relationship to a rival). Shame and guilt are on Izard's list of ten, not Ekman's. Trust and anticipation appear only in Plutchik's eight. "Dissmell" is Tomkins's.

**Do not mix up the two-dimension schemes.** Russell's circumplex uses valence by arousal; Watson and Tellegen use positive affect by negative affect; Wundt's older theory has three dimensions (pleasure-displeasure, tension-relaxation, excitement-depression).

## 11.2 The classic theories: from the body to the label MEDIUM YIELD · 6 QS

Now return to the three components: arousal, expression and experience. Common sense has a firm view of how they are related. You see a snake on a village path, you feel afraid, and *because* you feel afraid your heart races and you jump back. Feeling comes first and the body follows. The first scientific theory of emotion turned this common sense on its head.

### The James-Lange theory: we feel afraid because we tremble ●●●

In 1884 the American psychologist and philosopher **William James** (1842 to 1910) published an article in the journal *Mind* with the deceptively simple title "What is an emotion?". A year later, in 1885, the Danish physician **Carl Georg Lange** (1834 to 1900) independently published a very similar account, so the theory carries both names. James put it memorably in his *Principles of Psychology* (1890): common sense says that we lose our fortune, are sorry and weep; we meet a bear, are frightened and run. James argued that this order is wrong: "the bodily changes follow directly the perception of the exciting fact, and … our feeling of the same changes as they occur IS the emotion." In his most quoted line, "we feel sorry because we cry, angry because we strike, afraid because we tremble."

The **James-Lange theory**, then, says that an emotion-provoking stimulus first triggers bodily responses, autonomic changes in the viscera and actions of the skeletal muscles, and that the emotion we feel is simply our perception of those bodily changes. Take away the racing heart, the trembling and the running, James said, and nothing would be left of fear but a "cold and neutral state of intellectual perception." Lange put more weight on changes in the blood vessels; James included all bodily changes. Notice what the theory implies: if each emotion is the feeling of a particular bodily state, then different emotions must have different bodily patterns. Fear must feel like fear because its pattern of arousal differs from that of anger.

Textbooks often draw the James-Lange sequence as *stimulus, then bodily arousal, then awareness or "interpretation" of the arousal, then emotion* (Figure 11.2, top row). The "interpretation" in this diagram is not a reasoned judgement about the situation. It is just the brain noticing what the body is doing: "my heart is pounding and my knees are shaking." The feeling of those changes is the fear.

Is there evidence for James? Some. In 1966 the psychologist **George W. Hohmann** interviewed men with spinal cord injuries and reported that those with higher injuries, who received less feedback from the body, described their anger and fear as less intense than before. Later and better-controlled studies, however, found that people with severe spinal injuries still experience a full range of emotions, with at most some dampening of fear and anger. In 1983 Paul Ekman, **Robert W. Levenson** and Friesen found modest differences in autonomic patterns between emotions: anger and fear both raised heart rate, but anger raised finger temperature while fear lowered it. The modern verdict is that bodily feedback is not necessary for emotion but can colour and intensify it, a point you will meet again in the facial feedback hypothesis.

### Cannon's critique and the Cannon-Bard theory

The most powerful attack on James came from the Harvard physiologist **Walter Bradford Cannon** (1871 to 1945), the man who gave us the phrase "fight or flight" (Section 11.8). In a fa-

mous paper of 1927, Cannon listed five objections to the James-Lange theory:

1. **Cutting the body off from the brain does not abolish emotion.** Animals whose viscera had been surgically separated from the central nervous system (Cannon's own sympathectomised cats, and dogs studied earlier by the physiologist Charles Sherrington) still showed rage, fear and pleasure in appropriate situations.
2. **The same visceral changes occur in very different emotions and in non-emotional states.** A racing heart, faster breathing and sweating accompany fear, anger and excitement alike, and also fever, cold and exercise. Such a general pattern cannot tell fear from anger.
3. **The viscera are relatively insensitive.** The internal organs have few sensory nerve fibres, so the brain receives only vague information about them, too vague to support fine distinctions among emotions.
4. **Visceral changes are too slow.** Autonomic responses take a second or more to develop, but we often feel an emotion almost instantly.
5. **Producing the bodily changes artificially does not produce the emotion.** When the Spanish physician **Gregorio Marañón** injected 210 people with adrenaline in 1924, about 71 percent reported only bodily symptoms and no emotion at all; the rest described a cold, "as if" state ("I feel as if I were afraid"), but not genuine fear.

In place of James's theory Cannon proposed that an emotional stimulus is processed in the thalamus, which then sends signals *simultaneously* in two directions: up to the cerebral cortex, producing the conscious experience of emotion, and down to the body through the hypothalamus and autonomic nervous system, producing the physiological arousal. His student and collaborator **Philip Bard** (1898 to 1977) extended the theory and showed that the hypothalamus is critical for the bodily expression of rage (Section 11.4). The **Cannon-Bard theory** (also called the thalamic theory) therefore holds that the feeling of emotion and the bodily reaction occur at the same time and independently; neither causes the other (Figure 11.2, second row). You see the snake, and your fear and your pounding heart arise together.

The Cannon-Bard theory, too, turned out to be only partly right. Its anatomy was wrong in detail (the thalamus is a relay, not the seat of emotion), and the facial feedback and misattribution findings you are about to meet show that bodily states *do* influence feelings, which the theory denies. But Cannon's criticisms were valuable. They forced the next generation to ask how a single, rather general bodily arousal could give rise to many different emotions. The answer that came next was: through thinking.

### Schachter and Singer's two-factor theory ●●

In 1962 **Stanley Schachter** (1922 to 1997) and **Jerome E. Singer** (1934 to 2010) proposed a theory that took something from both sides. Like Cannon, they accepted that physiological arousal is much the same from one emotion to another. Like James, they accepted that arousal matters. Their **two-factor theory** of emotion (also called the cognitive labelling or cognitive arousal theory) says that an emotion requires two ingredients: *physiological arousal*, which determines how intense the emotion is, and a *cognitive label*, an explanation of the arousal drawn from the situation, which determines which emotion it is. When you notice that you are aroused, you look around for a reason; if the reason you find is a growling dog, the arousal becomes fear; if it is a person you like, the same arousal becomes attraction.

Students sometimes remember it as the "jukebox" theory: arousal is the coin that makes the machine play, and the situation selects the song.

### KEY STUDY: SCHACHTER AND SINGER'S ADRENALINE EXPERIMENT (1962)

**Question.** If people are physiologically aroused but have no explanation for it, will they label the arousal according to the emotional cues around them? **Method.** Male university students were told the study tested the effects of a vitamin compound, "Suproxin", on vision. Most actually received an injection of adrenaline (epinephrine), which produces a racing heart, trembling hands and a flushed face; a placebo group received saline. The adrenaline groups were told different things: the *informed* group was correctly told what symptoms to expect; the *ignorant* group was told nothing; the *misinformed* group was told to expect irrelevant symptoms such as numb feet and itching. Each participant then waited with a stooge who was supposedly another participant. In the *euphoria* condition the stooge played paper basketball, flew paper aeroplanes and twirled a hula hoop; in the *anger* condition the stooge complained bitterly about an insulting questionnaire that both had to fill in (one item asked how many men, other than his father, the participant's mother had had extramarital affairs with) and finally tore it up and stormed out.

**Results.** Participants who were aroused but had no correct explanation for their arousal (the ignorant and misinformed groups) caught the stooge's mood more: they behaved more euphorically and reported feeling happier in the euphoria condition, and acted angrier in the anger condition, than the informed participants, who could put their symptoms down to the injection. **Conclusion.** Unexplained arousal is labelled using situational cues. **Criticisms.** The effects were small and some were not statistically reliable; the placebo group did not differ as much from the adrenaline groups as the theory predicted; and later attempts to repeat the study, by Gary Marshall and Philip Zimbardo (1979) and by Christina Maslach (1979), found that unexplained arousal tended to be felt as unpleasant and anxious whatever the stooge did. A review by Rainer Reisenzein (1983) concluded that arousal can intensify emotions but that there is little evidence that it is necessary for them. Table 11.4 summarises the logic of the design.

Table 11.4 The logic of Schachter and Singer's design.

| Group | Injection | Told about symptoms? | Predicted to catch the stooge's emotion? |
|---|---|---|---|
| Informed | Adrenaline | Yes, correctly | Least: arousal already explained |
| Ignorant | Adrenaline | No | Most: arousal needs a label |
| Misinformed | Adrenaline | Yes, wrongly | Most (euphoria condition only) |
| Placebo | Saline | No | Little: no arousal to label |

## Misattribution of arousal and excitation transfer ●●

If arousal is labelled from the situation, it can be labelled *wrongly*. **Misattribution of arousal** occurs when people mistake the true source of their physiological arousal and attribute it to something else that happens to be present. The Capilano bridge study of **Donald G. Dutton** and **Arthur P. Aron** (1974), described at the start of this chapter, is the classic demonstration: arousal produced by fear of the bridge was misattributed to attraction to the interviewer. A related experiment by **Stuart Valins** (1966) showed that even the *belief* that one is aroused can be enough. Men looked at photographs of women while hearing what they were told was

their own heartbeat; when the false heartbeat sped up for certain photographs, the men later rated those women as more attractive.

**Dolf Zillmann**, from the early 1970s onward, developed **excitation transfer theory**, which explains why arousal from one situation can spill over into the next. Physiological arousal dies away slowly, more slowly than our awareness of its cause. So for a while after one arousing event, residual arousal remains, and if a new emotional event occurs in that window, the leftover arousal is added to the new emotion and intensifies it. In Zillmann's experiments, people who had just pedalled hard on an exercise bicycle and were then provoked retaliated more aggressively than rested people, especially once enough time had passed that they no longer connected their arousal with the exercise. Everyday examples are easy to find: the near miss in traffic that makes you snap at the shopkeeper a few minutes later, or the post-match crowd that turns a minor argument into a fight. Excitation transfer is also the bridge between this chapter and the study of aggression in the Social book.

EXAM TRAP

**Match the theory to its sequence.** Commonsense: stimulus, emotion, then arousal. James-Lange: stimulus, arousal, then emotion (the emotion *is* the perceived bodily change). Cannon-Bard: stimulus, then arousal and emotion *simultaneously and independently*, via the thalamus. Schachter-Singer: arousal *plus* a cognitive label. Lazarus: appraisal first (Section 11.3). An option such as "emotions and visceral reactions are simultaneous" describes Cannon-Bard, not James-Lange; "you feel the emotion and then the body responds" is the commonsense view that James rejected.

**Misattribution of arousal is explained by the two-factor theory**, not by James-Lange, Cannon-Bard or facial feedback. What makes Schachter and Singer different from both earlier theories is that their theory includes *cognition and arousal*.

**"Two-factor theory" is a name with four owners.** Schachter and Singer's two-factor theory of emotion (1962); Spearman's two-factor theory of intelligence (g and s, 1904; Chapter 9); Mowrer's two-factor theory of avoidance learning (classical plus operant conditioning; Chapter 6); and Herzberg's two-factor (motivation-hygiene) theory of job satisfaction (Chapter 10). Read the stem for the subject.

**Cannon-Bard named the thalamus** as the relay; the *hypothalamus* came in through Bard's later work on sham rage. If an option says "the somatic nervous system is the seat of emotion", reject it: no classic theory says so.

## 11.3 Appraisal, feedback, evolution and opposing processes

LOWER YIELD · NOT ASKED YET

Schachter and Singer brought thinking into the theory of emotion, but in their account thought arrives *after* arousal, to explain it. The next step was to ask whether thought comes first: whether we become aroused at all only because we have judged that something matters.

### Cognitive appraisal: Arnold and Lazarus

The word *appraisal* was introduced into emotion theory by **Magda B. Arnold** (1903 to 2002) in her book *Emotion and Personality* (1960). Arnold defined appraisal as an immediate, intuitive evaluation of whether something is good or bad for me, and argued that this evaluation is

what turns a perception into an emotion: the snake frightens you only once you have appraised it as dangerous. The idea was developed into the most influential cognitive theory of emotion and stress by the American psychologist **Richard S. Lazarus** (1922 to 2002), first in *Psychological Stress and the Coping Process* (1966) and later in *Emotion and Adaptation* (1991).

According to Lazarus's **cognitive appraisal theory**, emotions are produced by the way a person evaluates a situation in relation to his or her own goals and well-being. Lazarus distinguished two kinds of appraisal. In **primary appraisal** you ask, "Does this matter to me, and how?" The answer may be that the event is irrelevant, that it is benign or positive, or that it is stressful; and a stressful event may be appraised as *harm or loss* (damage already done), *threat* (harm expected in the future) or *challenge* (an opportunity to gain or grow). In **secondary appraisal** you ask, "What can I do about it?": you evaluate your resources and options for coping. Because situations and resources change, appraisal is continually revised; Lazarus called this **reappraisal**. The specific emotion depends on the pattern of appraisal. Lazarus summed up each emotion as a "core relational theme": anger is the appraisal of a demeaning offence against me and mine; anxiety is facing an uncertain, existential threat; guilt is having broken a moral rule; sadness is having suffered an irrevocable loss.

Because the appraisal stands between the stimulus and the response, Lazarus's account is often called the **cognitive-mediational theory** of emotion: stimulus, then appraisal, then emotion together with its bodily arousal (Figure 11.2, bottom row). The same theory reappears, with only small changes, as Lazarus and Folkman's transactional model of stress in Section 11.9, which is why the terms primary and secondary appraisal are examined under both emotion and stress.

KEY STUDY: THE SUBINCISION FILM (SPEISMAN, LAZARUS AND COLLEAGUES, 1964)

**Question.** Can changing how people appraise an event change their emotional and bodily response to it? **Method.** Joseph Speisman, Lazarus and their colleagues showed participants a distressing anthropological film of a painful initiation rite (subincision of the penis) performed on adolescent boys in an Aboriginal Australian community, while recording skin conductance and heart rate. The film was shown with one of four soundtracks: a *trauma* track that emphasised the pain and mutilation; a *denial* track that described the boys as willing and the occasion as happy; an *intellectualisation* track that took a detached, anthropological view; or no soundtrack. **Results.** Physiological arousal was highest with the trauma track and lowest with the denial and intellectualisation tracks. **Conclusion.** The same images produced different emotional responses depending on how viewers were led to appraise them. This is the experimental heart of appraisal theory, and it is also the ancestor of the "reappraisal" strategy of emotion regulation (Section 11.7).

### Does feeling need thinking? The Lazarus-Zajonc debate

Not everyone accepted that appraisal must come first. In 1980 the social psychologist **Robert B. Zajonc** (1923 to 2008) published a paper with the provocative title "Feeling and thinking: Preferences need no inferences." Zajonc argued for **affective primacy**: affective reactions can occur before, and independently of, any cognitive processing, and are sometimes faster than it. His main evidence came from the **mere exposure effect**, which he had described in 1968: simply being exposed to a stimulus repeatedly makes people like it more, whether it is a nonsense word, a Chinese character or a face. In a striking version, William Kunst-Wilson and Zajonc (1980) flashed irregular octagons so briefly that participants later could not recog-

nise them better than chance; yet when asked which of two shapes they *liked*, they chose the previously seen one about 60 percent of the time. Liking had formed without recognition.

Lazarus replied in 1982 and again in 1984 ("On the primacy of cognition"), and Zajonc answered in the same year ("On the primacy of affect"), both in the journal *American Psychologist*. Much of the disagreement turned out to be about definitions. Lazarus insisted that appraisal need not be conscious or deliberate; a fast, automatic evaluation counts as cognition. Zajonc meant by cognition something closer to conscious, deliberate processing. Joseph LeDoux's discovery of a fast subcortical route to the amygdala (Section 11.4) later gave both sides something: there is indeed a rapid emotional evaluation that bypasses the cortex, but it is itself a form of information processing. The modern consensus is that some emotional reactions are quick and automatic while others depend on slower, more elaborate thought.

### The facial feedback hypothesis

Is there anything in the old advice to "put on a happy face"? Charles Darwin thought so, writing in 1872 that "the free expression by outward signs of an emotion intensifies it," while suppressing the outward signs softens it. Tomkins later made the face the centre of his theory of affect, and Izard developed the idea further. The **facial feedback hypothesis** states that feedback from our own facial expressions influences our emotional experience. In its strong form, the face is necessary to produce emotion; in its weak form, which is the version with evidence behind it, facial expression modulates (strengthens or weakens) emotions that are already under way. You can see that this is a modern descendant of the James-Lange theory, with the face taking the place of the viscera.

The best-known test came from **Fritz Strack**, Leonard Martin and Sabine Stepper (1988). To prevent participants from realising that their faces were being manipulated, they were told the study concerned how people with disabilities might write by holding a pen in the mouth. Some held the pen between their teeth, which forces the muscles of a smile; others held it with their lips, which makes smiling impossible and produces something closer to a pout. Those holding the pen in their teeth rated cartoons as funnier. Other studies found similar effects: people who are made to frown while viewing slides report more anger, and people who hold a smile report more happiness.

The story has a modern twist that examiners at postgraduate level increasingly expect you to know. In 2016 a large, pre-registered replication of the pen study, run in 17 laboratories, found no effect at all. Later work suggested that the effect may depend on details such as whether participants know they are being video-recorded, and a 2019 meta-analysis of many facial feedback studies concluded that the effect is real but small. The fair summary is that facial feedback can nudge emotions but is not a powerful cause of them.

> **CLINICAL LINK**
>
> If frowning feeds sadness, could stopping people from frowning help depression? Several small randomised trials have injected botulinum toxin into the frown muscles between the eyebrows of patients with depression and found improvements in mood compared with placebo injections. The results are intriguing but preliminary. The same logic underlies simpler techniques: a therapist may coach a socially anxious client to adopt an open posture and a relaxed face, on the reasoning that expression and feeling influence each other in both directions.

### The evolutionary view: Darwin's legacy

The first major work on the science of emotion was not by a psychologist at all. In 1872 **Charles Darwin** (1809 to 1882) published *The Expression of the Emotions in Man and Animals*, in which he argued that emotional expressions are products of evolution, shared across human cultures and continuous with those of other animals. A snarling dog and a sneering man both bare the teeth. To explain expressions, Darwin proposed three principles. The *principle of serviceable associated habits* holds that expressions began as useful actions: the raised upper lip of disgust and the narrowed nostrils once helped reject a bad smell or taste. The **principle of antithesis** holds that opposite emotional states are signalled by opposite postures: a dog threatening attack stands tall with ears forward, hair and tail raised, whereas a submissive dog crouches with ears back and tail down, so that the signals cannot be confused. The *principle of the direct action of the nervous system* holds that some expressions, such as trembling with fear, are simply overflows of nervous excitation.

The **evolutionary view of emotion** that grew from Darwin's book treats emotions as adaptations: packages of physiology, expression and action that solved recurring survival problems. Fear prepares escape from predators, disgust protects against contamination and disease, anger mobilises the defence of resources, and expressions communicate our state to others so that they can respond. Plutchik's psychoevolutionary theory (Table 11.3) and Ekman's work on universal expressions (Section 11.5) are both direct descendants of Darwin's approach.

### Opponent-process theory: why pleasures fade and pains turn sweet

Why does a first-time parachutist feel terror before the jump and an elated high afterwards, while a veteran parachutist feels only mild tension before and a long glow of exhilaration after? **Richard L. Solomon** (1918 to 1995) and **John D. Corbit** answered with the **opponent-process theory** of emotion and acquired motivation (1974). They proposed that any strong emotional reaction (the *a-process*) automatically triggers an opposing reaction (the *b-process*) that restores emotional balance. The a-process is quick and closely tied to the stimulus. The b-process is sluggish: it starts late, builds slowly and outlasts the stimulus, so when the stimulus ends we are left with an after-reaction opposite to the original emotion. Crucially, the b-process grows stronger with repeated exposure, while the a-process stays the same.

This simple machinery explains a great deal. With repetition the strengthened b-process cancels more of the a-process, so the initial reaction weakens (the drug user's high fades, a sign of tolerance), while the after-reaction grows (the drug user's craving and misery when the drug wears off, a sign of withdrawal). For painful experiences the pattern is reversed: the fear of the parachute jump weakens with practice, and the pleasant after-reaction, relief and exhilaration, grows until it becomes the reason for jumping. Solomon used the same idea to explain why separation from a loved one becomes more painful as a relationship matures. Note that this is a different theory from Ewald Hering's opponent-process theory of colour vision (Chapter 3); only the name is shared. The motivational side of Solomon's theory, including addiction, is taken up in Chapter 10.

### Putting the theories side by side

Figure 11.2 lines up the four classic theories. Read each row from left to right, and notice where the arrows split or where a box is added. James-Lange puts the body first. Cannon-Bard splits the response in two at the thalamus. Schachter and Singer add a cognitive label af-

ter the arousal. Lazarus moves cognition to the front, as an appraisal that precedes and shapes both the feeling and the arousal.

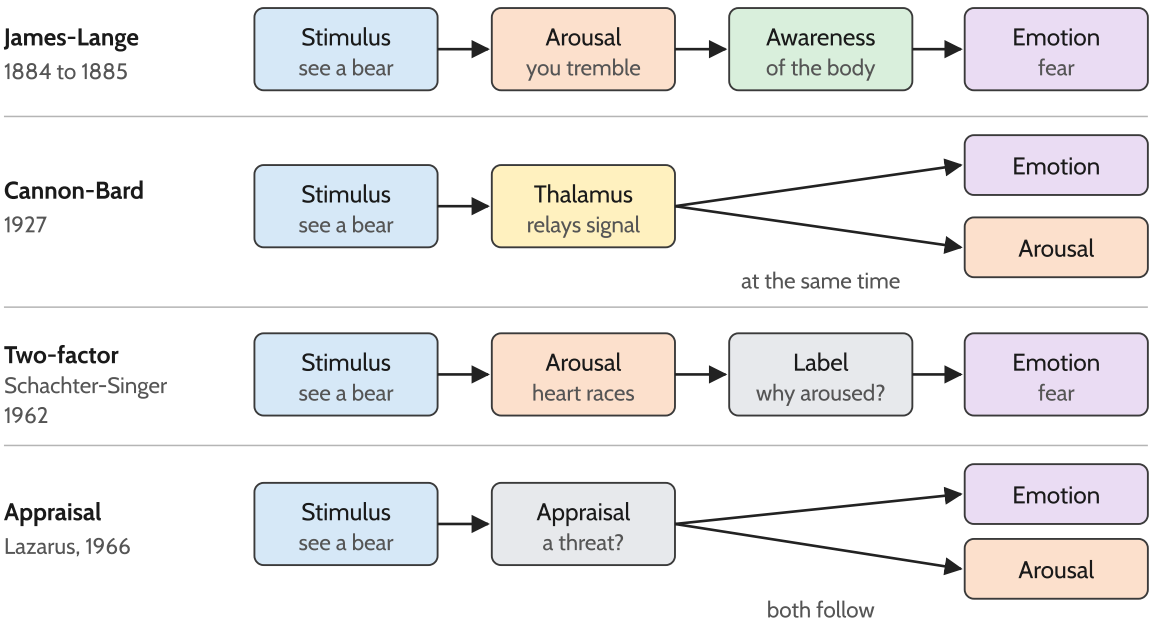


Figure 11.2 Four classic theories of emotion compared. Follow each row from the stimulus to the emotion. James-Lange: the body reacts first and the felt emotion is awareness of that reaction. Cannon-Bard: the thalamus sends signals that produce feeling and arousal at the same time. Two-factor: arousal plus a cognitive label. Appraisal (cognitive-mediational): an evaluation of the stimulus comes first, and both emotion and arousal follow from it.

Where does the evidence leave us? Each theory captured a piece of the truth. Bodily feedback can colour emotion (James), feeling and bodily response are partly separable (Cannon), the meaning we assign to arousal matters (Schachter and Singer), and appraisals shape which emotion we feel (Lazarus), although some evaluations are fast and automatic (Zajonc). Pinel and Barnes call the result the modern biopsychological view: the perception of the stimulus, the autonomic and somatic responses, and the experience of emotion can each influence the other two. Emotion is a loop, not a line.

EXAM TRAP

**Two Lazaruses.** Richard S. Lazarus (1922 to 2002) gave us cognitive appraisal, the transactional model of stress and the daily hassles scale. Arnold A. Lazarus (1932 to 2013) was a clinical psychologist who founded multimodal therapy (the BASIC ID). If the question is about appraisal, coping or hassles, the answer is Richard.

**Who said affect comes first?** Zajonc (affective primacy, mere exposure, 1980 and 1984). Lazarus said cognition (appraisal) comes first. Arnold introduced the word appraisal in 1960.

**Facial feedback is credited to Tomkins and Izard as theory, and to Strack, Martin and Stepper (1988) for the pen study**; Darwin anticipated it in 1872. Do not confuse the facial feedback hypothesis (my face affects my feeling) with the universality hypothesis (faces express the same emotions everywhere).

**Opponent-process theory** of emotion is Solomon and Corbit (1974); the opponent-process theory of colour vision is Hering (1878).

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. The three components of an emotion are physiological arousal, expressive behaviour and _____ experience.
2. A diffuse, low-intensity state that lasts hours or days and often has no clear object is a _____.
3. Ekman's six basic emotions are happiness, sadness, fear, anger, disgust and _____; in 1986 he added contempt.
4. In Plutchik's wheel, the opposite of trust is _____.
5. Russell's circumplex arranges emotions on two dimensions, valence and _____.
6. "We feel afraid because we tremble" summarises the _____ theory.
7. Cannon argued that the _____ sends signals to the cortex and to the body at the same time.
8. In Schachter and Singer's experiment, participants injected with _____ who had no explanation for their symptoms caught the stooge's mood most.
9. In Lazarus's theory, asking "What can I do about it?" is _____ appraisal.
10. Zajonc's evidence for affective primacy came mainly from the _____ effect.

**Answers:** (1) conscious (subjective), (2) mood, (3) surprise, (4) disgust, (5) arousal, (6) James-Lange, (7) thalamus, (8) adrenaline (epinephrine), (9) secondary, (10) mere exposure.

## 11.4 The emotional brain and body HIGH YIELD · 25 QS

Theories of emotion make claims about the body and the brain, so it is time to look at them directly. This section gives you the essentials; the full anatomy is in the Physiological book.

### The autonomic nervous system ●●

The bodily side of emotion is run mainly by the **autonomic nervous system** (ANS), the part of the peripheral nervous system that controls the heart, blood vessels, glands and internal

organs without our deliberate control. It has two divisions that usually work in opposition. The **sympathetic nervous system** prepares the body for action: it is the "fight or flight" division, most active in fear, anger and excitement. The **parasympathetic nervous system** conserves and restores energy: it is the "rest and digest" division, dominant when you are calm and after a meal. Table 11.5 lists the main effects.

Table 11.5 What the two divisions of the autonomic nervous system do.

| Organ or function | Sympathetic (arousing) | Parasympathetic (calming) |
|---|---|---|
| Pupils | Dilate | Constrict |
| Heart rate and blood pressure | Increase | Decrease |
| Breathing | Faster; airways widen | Slower; airways narrow |
| Salivation | Reduced (dry mouth) | Increased |
| Digestion | Inhibited | Stimulated |
| Sweat glands | Activated (skin resistance falls) | Not innervated |
| Liver | Releases glucose | Stores glucose as glycogen |
| Adrenal medulla | Releases adrenaline and noradrenaline | No effect |
| Blood flow | To skeletal muscles | To digestive organs |

One item in the table deserves special attention because it is the basis of lie detection. When the sympathetic system activates the sweat glands, even slightly, the skin conducts electricity better. This change is called the **galvanic skin response** (GSR), now usually called electrodermal activity or skin conductance response: skin *resistance* falls and skin *conductance* rises. It is one of the most sensitive measures of emotional arousal available.

Where are the bodily responses actually controlled? The vital, involuntary functions of heartbeat, blood pressure and breathing are paced by centres in the **medulla oblongata**, the lowest part of the brainstem, which belongs to the hindbrain together with the pons and cerebellum. The hypothalamus, higher up, acts as the head ganglion of the ANS: during emotion it sends commands down to these brainstem centres and to the spinal cord. So the hindbrain carries out the adjustments and the hypothalamus orchestrates them.

### The hypothalamus and sham rage

In the late 1920s Philip Bard, Cannon's student, removed the cerebral cortex of cats. The decorticate cats reacted to the slightest touch with a full display of rage: arched back, erect hair, hissing, bared teeth, lashing tail and dilated pupils. But the rage was oddly empty. It was out of all proportion to the provocation, it was not directed at the source of the touch, and it stopped the moment the stimulus stopped. Bard called it **sham rage**. Sham rage could still be produced when the brain was removed down to, but not including, the hypothalamus; once the hypothalamus was also removed, it disappeared. Bard concluded that the hypothalamus organises the expression of rage, and that a normal function of the cortex is to inhibit and direct it.

At about the same time, in Zurich, the Swiss physiologist **Walter Rudolf Hess** (1881 to 1973) implanted electrodes in the hypothalamus of freely moving cats and found that stimulating certain points produced an attack-like defensive reaction, with hissing, raised hair and dilat-

ed pupils, while other points produced calm and sleepiness. Hess shared the 1949 Nobel Prize in Physiology or Medicine for this mapping of the brain's control of the internal organs.

### The Papez circuit and the limbic system ●●●

Where, then, does the experience of emotion arise? In 1937 the American neuroanatomist **James Papez** (1883 to 1958; pronounced "payps") proposed that emotion depends on a loop of interconnected structures on the inner surface of the brain. The **Papez circuit** runs from the hippocampus through the fornix to the mammillary bodies of the hypothalamus, then by the mammillothalamic tract to the anterior nuclei of the thalamus, then to the cingulate gyrus (cingulate cortex), and back through the parahippocampal region (entorhinal cortex) to the hippocampus. Papez suggested that emotional *expression* depends on this circuit's action on the hypothalamus, and emotional *experience* on its action on the cortex.

The French surgeon **Paul Broca** (1824 to 1880) had already, in 1878, named the ring of cortex bordering the brainstem the "great limbic lobe" (from the Latin *limbus*, border). In 1949 the American physician **Paul D. MacLean** (1913 to 2007) called these structures the "visceral brain", and in 1952 he renamed them the **limbic system**, enlarging Papez's circuit by adding the amygdala, septum, hypothalamus and parts of the prefrontal cortex. MacLean's limbic system theory, which made the limbic system "the seat of emotion", became the textbook view for half a century. MacLean later placed it within his **triune brain** theory, in which the human brain contains three layers laid down in evolution: a reptilian complex (basal ganglia and brainstem) for instinctive routines, a paleomammalian brain (the limbic system) for emotion, and a neomammalian brain (the neocortex) for reason.

Two corrections are needed. First, the Papez circuit is now known to matter more for memory than for emotion: damage to the hippocampus, fornix or mammillary bodies (as in Korsakoff's syndrome) produces amnesia rather than emotional change. Second, the triune brain is an outdated evolutionary story; brains did not evolve by adding new layers on top of old ones, and the limbic system is not a sharply bounded unit. But the core point survives: structures such as the amygdala, hypothalamus and cingulate cortex are central to emotion, motivation and emotional memory.

### The amygdala: Klüver-Bucy syndrome and emotional memory ●●●

In 1937, and in full in 1939, the psychologist **Heinrich Klüver** (1897 to 1979) and the neurosurgeon **Paul Bucy** (1904 to 1992) described a remarkable change in rhesus monkeys after they removed both anterior temporal lobes. (Klüver had been studying the hallucinogen mescaline, and the surgery began as a test of whether the temporal lobes were involved in its effects.) Monkeys that had been wild and aggressive became tame and placid. They showed no fear of humans or of snakes, which normal monkeys find terrifying. They examined every object by putting it in their mouths, ate things they would normally reject, including meat and inedible objects, attended compulsively to every object in view, and directed sexual behaviour at inappropriate partners and objects. Although they could see, they seemed not to recognise what objects were or meant.

This pattern is now called **Klüver-Bucy syndrome**. Its features are best remembered as a list: *psychic blindness* or visual agnosia (seeing without recognising); *hyperorality* (examining things by mouth); *hypermetamorphosis* (a compulsion to attend to and touch every visual stimulus); *dietary change and hyperphagia* (overeating, eating inedible things); *hypersexuality*;

and *placidity*, that is, loss of fear and anger. Later lesion studies traced the emotional part of the syndrome, the taming and loss of fear, to bilateral damage to the **amygdala**, an almond-shaped cluster of nuclei deep in each temporal lobe. In humans the syndrome is rare; it follows bilateral temporal damage from herpes simplex encephalitis, head injury or some dementias, and because the surrounding temporal cortex is usually damaged too, human patients often also show amnesia and aphasia (difficulty understanding or producing language).

Human cases make the amygdala's role in fear vivid. A woman known as S.M., whose amygdalae were destroyed on both sides by a rare genetic disorder (Urbach-Wiethe disease), cannot recognise fear in other people's faces, and when researchers at the University of Iowa took her to an exotic pet shop and a "haunted house" she showed curiosity but no fear at all. The amygdala is also the brain's tool for making emotional events memorable. When something arousing happens, stress hormones (adrenaline and cortisol) and the amygdala's own activity strengthen the consolidation of the memory by the hippocampus. In one well-known experiment by **Larry Cahill**, **James L. McGaugh** and colleagues (1994), people heard either an emotional or a neutral version of a story; the emotional version was remembered better, unless participants had taken propranolol, a drug that blocks noradrenaline receptors, in which case the memory advantage disappeared. This emotional modulation of memory explains why you remember the day of a frightening accident in vivid detail (and why "flashbulb memories", Chapter 7, feel so clear).

**KEY STUDY: A DOUBLE DISSOCIATION OF EMOTIONAL AND FACTUAL MEMORY (BECHARA AND COLLEAGUES, 1995)**

Antoine Bechara, Antonio Damasio and their colleagues gave a fear-conditioning task to three patients: S.M., with bilateral amygdala damage; a patient with bilateral hippocampal damage; and a patient with damage to both. A coloured slide (the conditioned stimulus) was paired with a loud, startling boat horn, and the skin conductance response was recorded. S.M. never developed a conditioned skin conductance response to the slide, yet she could tell the researchers exactly which colour had been followed by the horn. The hippocampal patient showed the opposite: a normal conditioned skin conductance response, but no conscious memory of which colour went with the horn. The patient with both lesions showed neither. The amygdala is needed for emotional (conditioned) memory, the hippocampus for declarative memory of the facts.

### LeDoux's two roads to fear

How does the amygdala get its information so quickly? **Joseph LeDoux** (born 1949), working with rats in auditory fear conditioning (a tone paired with a mild shock), traced the pathways in detail and described them for a general audience in *The Emotional Brain* (1996). Sounds reach the auditory part of the thalamus (the medial geniculate nucleus). LeDoux found that lesions of the auditory thalamus or of the amygdala blocked fear conditioning to a tone, but lesions of the auditory cortex did not. The reason is that information travels from the thalamus to the amygdala by two routes. The **low road** runs directly from the sensory thalamus to the amygdala. It is fast (in rats, a matter of milliseconds) but crude, "quick and dirty" in LeDoux's phrase: it can tell that something is sudden or snake-like, but not what it is. The **high road** runs from the thalamus to the sensory cortex and only then to the amygdala. It is slower but detailed, able to identify the object and its context (Figure 11.3). The amygdala then

triggers the fear response through its outputs: to the periaqueductal grey of the midbrain for freezing, to the lateral hypothalamus for sympathetic arousal, and to the paraventricular hypothalamus for the stress hormone response.

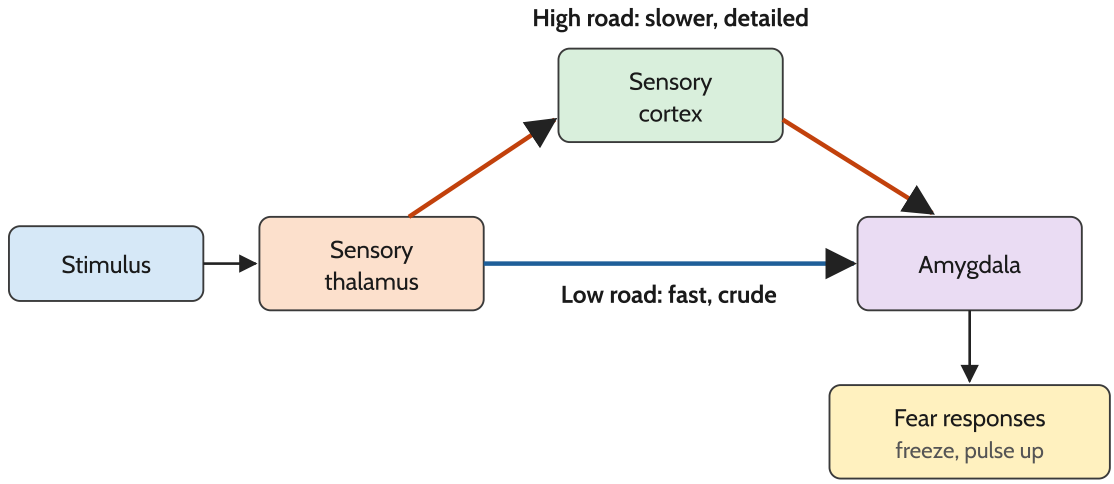


**Figure 11.3** LeDoux's two routes to fear. Signals from the sensory thalamus reach the amygdala directly by the fast, crude low road (solid line) and indirectly through the sensory cortex by the slower, detailed high road (broken lines). On a forest path, the low road makes you freeze at a curved shape; a fraction of a second later the high road tells you whether it is a snake or a stick.

The two-road model explains a familiar experience. Walking through a forest you freeze at a long, curved shape on the path, and only a moment later realise it is a stick. The freezing came from the low road; the correction came from the high road. It also explains why phobias and traumatic fears can be so resistant to reasoning: the subcortical route can trigger fear without consulting the cortex, which "knows" that the spider is harmless. A striking related phenomenon is **affective blindsight**. Beatrice de Gelder and colleagues (1999) studied a patient, G.Y., whose primary visual cortex was damaged on one side, leaving him consciously blind in half of his visual field. When happy, sad, angry or fearful faces were presented in the blind half, he could still guess their emotional expression well above chance, presumably through subcortical routes to the amygdala, although he reported seeing nothing.

## The prefrontal cortex and the two hemispheres ●●

The amygdala raises the alarm; the prefrontal cortex helps decide what to do about it. The classic case is that of **Phineas Gage** (1823 to 1860), a responsible and well-liked railway construction foreman in Vermont who, in 1848, was tamping explosive into a rock with an iron rod about a metre long when the charge exploded and drove the rod up through his left cheek and out of the top of his skull. Gage survived and kept his intelligence, but he became impulsive, profane and unreliable, so much so that his friends said he was "no longer Gage," as his physician **John Martyn Harlow** reported in 1868. A computer reconstruction of his skull in 1994 by Hanna Damasio and colleagues placed the damage in the medial (ventromedial) prefrontal cortex of both hemispheres.

**Antonio Damasio** (born 1944) studied modern patients with similar damage, such as a man he called Elliot, whose reasoning and IQ were intact but whose personal and financial deci-

sions became disastrous. In *Descartes' Error* (1994), Damasio proposed the **somatic marker hypothesis**: through experience, the ventromedial prefrontal cortex attaches bodily feelings ("somatic markers", a gut feeling of unease) to the likely outcomes of choices, and these markers steer decisions quickly, often before we can say why. In the Iowa Gambling Task, normal participants begin to show skin conductance responses before choosing from risky card decks, and soon avoid them; patients with ventromedial damage show no such anticipatory responses and keep choosing the risky decks. Emotion, on this view, is not the enemy of good decisions but a necessary part of them.

The two cerebral hemispheres also differ in emotion. The older *right-hemisphere hypothesis* held that the right hemisphere is specialised for all emotion; the *valence hypothesis* holds that the right hemisphere is specialised for negative emotions and the left for positive ones. Richard Davidson's EEG studies support a related *approach-withdrawal* model: greater left frontal activity accompanies approach-related, mostly positive emotion, and greater right frontal activity accompanies withdrawal-related, mostly negative emotion, and people prone to depression tend to show relatively low left frontal activity. Facial expressions are usually stronger on the left side of the face, which is controlled mainly by the right hemisphere.

CLINICAL LINK

Emotional expression and emotional experience can come apart after brain damage. In **pseudobulbar affect** (pathological laughing and crying), patients with stroke, multiple sclerosis, motor neurone disease (ALS) or traumatic brain injury have sudden, brief, uncontrollable outbursts of crying or laughing that do not match how they actually feel. Damage to the pathways by which the cortex normally controls the brainstem's emotional-motor centres "releases" the expression from the feeling, much as in Bard's decorticate cats. It is not depression, and it should not be mistaken for it.

Table 11.6 gathers the brain structures of this section, with the evidence to remember for each.

Table 11.6 Brain structures in emotion: what to remember for each.

| Structure | Role in emotion | Classic evidence |
|---|---|---|
| Medulla (hindbrain) | Vital centres for heartbeat and breathing | Brainstem physiology |
| Hypothalamus | Head ganglion of the ANS; organises rage and the stress hormone response | Bard's sham rage (late 1920s); Hess's stimulation |
| Amygdala | Threat detection, fear conditioning, emotional memory | Klüver-Bucy (1937, 1939); LeDoux; patient S.M. |
| Hippocampus | Declarative memory and context of fear | Bechara and colleagues (1995) |
| Cingulate cortex | Part of Papez's circuit; emotional attention and pain distress | Papez (1937) |
| Prefrontal cortex | Regulation of emotion, decisions, social conduct | Phineas Gage (1848); Damasio (1994) |

## Lie detection: the polygraph and the guilty knowledge test

If lying makes people anxious, and anxiety shows in the body, then measuring the body should reveal lies. That is the logic of the **polygraph** (literally "many writings"), commonly called the lie detector, which records several measures of autonomic arousal at once: heart

rate, blood pressure, breathing and skin conductance. The usual procedure is the **control question technique**. The examiner compares the suspect's responses to relevant questions ("Did you take the money from the office?") with responses to control questions designed to make almost anyone slightly uncomfortable ("Before the age of 25, did you ever take something that did not belong to you?"). A bigger response to the relevant questions is taken as a sign of deception.

The problem is that the polygraph detects arousal, not lies. An innocent suspect accused of a crime is likely to react strongly to the relevant question simply out of fear of being disbelieved, so innocent people are often wrongly judged deceptive. Guilty people can learn countermeasures, such as biting the tongue or pressing the toes to the floor during control questions, which raise their arousal and mask the difference. Reviews of real criminal cases estimate the accuracy of the control question technique at not much better than chance, and a major review by the US National Research Council in 2003 concluded that polygraph testing is too inaccurate to be relied on for security screening.

A better method changes the question. The **guilty knowledge test** (also called the concealed information test) does not try to catch a lie. Instead it presents details of the crime that only the guilty person would know, mixed with equally plausible false details, and looks for a consistent response to the true ones. In the classic study by **David T. Lykken** (1959), volunteers carried out a mock theft of a purse from an office. They were then asked, for example, "Where did we find the purse? In the washroom? In a locker? Hanging on a coat rack?" Using skin conductance alone, 88 percent of the mock criminals were correctly identified, and none of the innocent participants was judged guilty. Innocent people react to all the alternatives alike, because to them all are equally meaningless.

INDIAN CONTEXT

Indian investigators have used polygraph tests, narco-analysis (questioning under a sedative drug) and brain-based tests such as the brain electrical activation profile. In *Selvi v. State of Karnataka* (2010), the Supreme Court of India held that administering these tests to a person without consent violates the right against self-incrimination under Article 20(3) of the Constitution and personal liberty under Article 21. They may be given only with the person's consent, and even then the results themselves are not admissible as evidence, although facts later discovered with their help may be. The judgement drew directly on the scientific doubts about such tests summarised above.

EXAM TRAP

**The limbic system is the classic "seat of emotion"**, so when the options are "limbic system" versus one of its parts (hippocampus, thalamus, hypothalamus), choose the limbic system for a general question about emotional behaviour. Reject the reticular formation (arousal, wakefulness and attention), the pituitary (the "master" endocrine gland), the cerebellum (coordination and balance) and the corpus callosum (connects the hemispheres). Technically the limbic system is part of the forebrain, but "forebrain" is too broad to be the keyed answer.

**Amygdala, not hippocampus**, for fear, aggression, threat detection, fear conditioning and the *emotional* enhancement of memory. The hippocampus stores the facts; the amygdala adds the emotional boost. The suprachiasmatic nucleus is the circadian clock, not an emotion structure.

**Lesion effects that are easy to swap.** Bilateral amygdala (temporal lobe) removal produces Klüver-Bucy tameness, hyperorality and hypersexuality, not fear. Intense rage, attack posture and raised hair come from removing the cortex (sham rage) or stimulating the hypothalamus. Overeating and obesity come from damage to the ventromedial hypothalamus (Chapter 10). Severe anterograde amnesia "as in H.M." comes from the hippocampus and medial temporal lobe (Chapter 7). Cerebellar damage disturbs balance and coordination, and caudate nucleus (basal ganglia) damage disturbs movement and habit learning; neither produces Klüver-Bucy syndrome. A Klüver-Bucy option that includes "hyperalertness" or "avolition and alogia" is wrong: placidity, not alertness, and the "A" symptoms are schizophrenia's.

**Sympathetic arousal dilates the pupils.** In an "odd one out" list of sympathetic effects (raised heart rate, lowered skin resistance, faster breathing, pupil constriction), the odd one is pupil constriction, which is parasympathetic.

**Heartbeat and breathing are paced by the hindbrain (medulla)**; the hypothalamus regulates them indirectly. If the question asks which division "modulates" involuntary heartbeat and respiration and offers hindbrain, that is the keyed answer.

**Papez circuit order:** hippocampus, fornix, mammillary bodies, (mammillothalamic tract), anterior thalamus, cingulate gyrus, entorhinal cortex, back to hippocampus. The amygdala is *not* in Papez's original loop (some textbooks, Pinel and Barnes among them, draw it with the neighbouring structures); MacLean added it to the limbic system.

**Limbic system versus hypothalamus.** When a question asks for *the* function of the limbic system, the keyed answer is the regulation of emotion (with motivation and memory as runners-up). Hunger, thirst and temperature are the classic functions of the hypothalamus, so an option such as "both emotion and hunger and thirst" was not keyed (Amity 2023), even though MacLean counted the hypothalamus as part of the limbic system.

## 11.5 Expressing and reading emotion

LOWER YIELD · NOT ASKED YET

Darwin claimed in 1872 that emotional expressions are inherited and universal. For most of the twentieth century, however, the dominant view in the social sciences was the opposite. Anthropologists such as Margaret Mead and Ray Birdwhistell, the founder of *kinesics* (the study of body movement as communication), argued that expressions are learned like a lan-

guage and vary from culture to culture. A smile in one society, they suggested, might signal something quite different in another. Who was right?

## Are facial expressions universal?

The **universality hypothesis** states that certain basic emotions are expressed by the same facial movements, and recognised from them, in all human cultures. Ekman's first test, published with E. Richard Sorenson and Friesen in 1969, showed photographs of posed expressions to people in the United States, Brazil, Japan, New Guinea and Borneo, and found strong agreement about which emotion each face showed, especially among the literate groups. But literate people might simply have learned Western expressions from films and magazines. The decisive test needed people who had not.

### KEY STUDY: EKMAN AND FRIESEN AMONG THE FORE OF NEW GUINEA (1971)

**Question.** Would people with almost no contact with outsiders recognise the same facial expressions as Westerners? **Method.** In the late 1960s Ekman and Friesen worked with the South Fore, a preliterate people in the highlands of Papua New Guinea who had seen no films or magazines and had little contact with Westerners. Through a translator, a participant heard a short story ("her friends have come, and she is happy"; "her child has died, and she feels very sad"; "she is angry and about to fight") and then chose which of three photographed faces fitted the story. In a second part of the study, Fore people were asked to show the face they would make in each story; their expressions were video-recorded and later shown to American students. **Results.** The Fore chose the expected face far above chance for happiness, sadness, anger, disgust and surprise. The one consistent confusion was between fear and surprise, which the Fore often mixed up. American students also recognised the Fore's posed expressions. **Conclusion.** At least six expressions are universal, as Darwin had argued. **Criticisms.** The forced-choice format can exaggerate agreement, the photographs were posed and exaggerated, and translation may have helped. More recent fieldwork in other isolated societies (for example, Carlos Crivelli and colleagues' 2016 study in the Trobriand Islands, where a wide-eyed "gasping" face was read as a threat rather than as fear) suggests that agreement is weaker than once believed.

Further support for an inborn basis comes from people who could never have learned expressions by watching. Children born blind smile, cry and frown much like sighted children. In a neat natural experiment, David Matsumoto and Bob Willingham (2009) compared the faces of judo athletes at the moment of winning or losing a medal: congenitally blind athletes at the 2004 Paralympic Games produced the same expressions as sighted athletes at the 2004 Olympic Games.

## Display rules and cultural differences

If expressions are universal, why do people from different cultures so often seem to express emotion differently? Ekman and Friesen's answer was **display rules**: culturally learned norms about which emotions may be shown, by whom, to whom and when. Display rules can lead people to *intensify* an expression (looking delighted with a gift you do not like), *de-intensify* it (looking only mildly pleased when you have beaten a friend), *neutralise* it (keeping a poker face) or *mask* it with a different expression (smiling to hide anger). At an Indian wedding, the bride's tears at her *bidaai* are expected and appropriate; at a funeral, laughter must be hidden; and many young people learn to mask anger in front of elders.

The classic demonstration was Friesen's study of 1972. Japanese and American students watched unpleasant film clips of surgery and accidents while their faces were secretly recorded. When they watched alone, both groups showed very similar expressions of disgust and distress. When a higher-status experimenter sat with them, the Japanese students more often masked their negative expressions with polite smiles. The underlying expression was universal; its public display was governed by culture. The sociologist Arlie Hochschild extended the idea to work in *The Managed Heart* (1983): flight attendants, receptionists and call-centre staff must follow workplace display rules, smiling whatever they feel, and this "emotional labour" is a known contributor to burnout (Section 11.9).

Culture also shapes how expressions are read. A meta-analysis by Hillary Elfenbein and Nalini Ambady (2002) found that emotions are recognised across cultures well above chance, but more accurately within one's own culture, an *in-group advantage* rather like understanding your own regional accent best. Rachael Jack and colleagues (2012) showed that East Asian observers rely more on the eyes, and Western observers more on the mouth, when reading faces; tellingly, a Japanese-style emoticon puts the emotion in the eyes (^_^) whereas the Western one puts it in the mouth :). Cultures also differ in which emotions they *value*. Jeanne Tsai has shown that North Americans tend to prize high-arousal positive states such as excitement, whereas East Asians tend to prize low-arousal positive states such as calm, a difference visible even in the smiles shown in children's storybooks.

### Measuring the face: FACS, the Duchenne smile and microexpressions

To study expressions precisely, Ekman and Friesen published the **Facial Action Coding System** (FACS) in 1978. FACS describes any facial movement as a combination of numbered *action units*, each corresponding to the action of one or more facial muscles; for example, action unit 12 is the lip corner puller (zygomaticus major) and action unit 6 is the cheek raiser (the outer part of orbicularis oculi). Trained coders can describe any expression, real or faked, in these anatomical terms, without guessing at the emotion behind it. FACS is used in research on depression, pain, deception, infant development and animation.

FACS made it possible to settle an old question: how can you tell a real smile from a polite one? In 1862 the French neurologist **Guillaume-Benjamin-Amand Duchenne de Boulogne** (1806 to 1875) stimulated the facial muscles of volunteers electrically and photographed the results. He noticed that a smile of genuine enjoyment involves two muscles: the zygomaticus major, which pulls the lip corners up, and the orbicularis oculi, which encircles the eye, raising the cheeks and producing "crow's feet" wrinkles at the outer corners. The zygomaticus is easy to control voluntarily; the outer part of the orbicularis oculi is not. A genuine smile, which Ekman named the **Duchenne smile**, therefore involves the eyes; a polite or forced smile (sometimes called a social smile) involves only the mouth. Look at the eyes of anyone posing for a photograph.

When people try to hide an emotion, the true feeling may leak out in a **microexpression**, a full facial expression that flashes across the face for only a fraction of a second (roughly between a twenty-fifth and a fifth of a second) before it is suppressed. Microexpressions were first described by Ernest Haggard and Kenneth Isaacs in 1966 and popularised by Ekman's work on deception. A caution is in order, however. When people are tested on whether they can tell liars from truth-tellers, accuracy is disappointing: a large review by Charles Bond

and Bella DePaulo (2006) found an average of about 54 percent, barely above the 50 percent expected by guessing, and professionals such as police officers do little better.

### Beyond the face: body, voice and gesture

Faces are not the whole story. Posture, gesture, gaze, touch and the tone of the voice (prosody) all carry emotion. Pride, for example, has a recognisable display involving the whole body: a small smile, the head tilted slightly back, an expanded chest and the arms raised or the hands on the hips; Jessica Tracy and Matsumoto found the same display in congenitally blind athletes after victories. Indeed, when emotions are very intense, the body can be more informative than the face. Hillel Aviezer and colleagues (2012) showed people photographs of professional tennis players at the moment of winning or losing a crucial point. From the faces alone, viewers could not reliably tell winners from losers; from the bodies, they could. Some body language is culture-specific: a sideways head-wobble that signals agreement in much of South India can confuse a visitor who reads it as "no".

EXAM TRAP

**The "7-38-55 rule" is a misquotation.** Albert Mehrabian's experiments (1967, 1971) found that when a speaker's words, tone of voice and facial expression *conflict* about feelings or attitudes (saying "I like you" in a hostile tone, for example), listeners rely more on the face (55 percent) and tone (38 percent) than on the words (7 percent). Mehrabian himself warned that this applies only to such inconsistent messages about feelings, not to communication in general.

### Gender differences in emotion

Do women and men differ in emotion? The answer depends on which component you measure. On *expression*, the differences are reliable: women are, on average, more facially expressive and smile more than men. On *recognition*, women are on average more accurate at decoding nonverbal cues, a difference documented in Judith Hall's review of 1978 and many later studies. On *experience* and *physiology*, the differences are much smaller and less consistent; Ann Kring and Albert Gordon (1998), for instance, found that women were more expressive while watching emotional films but did not report experiencing more emotion than men. Most researchers attribute the expressive differences largely to socialisation and display rules ("boys don't cry"; girls should not show anger) layered on top of a shared emotional system, and they note that the stereotype of the "emotional woman" exaggerates differences that are modest and vary by emotion; anger, for example, is not more often expressed by women.

EXAM TRAP

**Ekman's universality research used the Fore of Papua New Guinea**, not the Trobriand Islanders (Crivelli's later challenge) and not Mead's Samoans. The usual confusion in the Fore data was fear with surprise. **Display rules** govern *showing* emotions; they do not change the emotion felt. The Duchenne smile is identified by the *eye* muscle (orbicularis oculi); the zygomaticus alone gives a social smile. FACS codes muscle *action units*, not emotions.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. The division of the autonomic nervous system that dilates the pupils and dries the mouth is the _____ division.
2. The violent but undirected rage of a cat whose cortex has been removed is called _____ rage.
3. The loop of hippocampus, fornix, mammillary bodies, anterior thalamus and cingulate gyrus is the _____ circuit.
4. Paul MacLean coined the term _____ system in 1952.
5. Tameness, hyperorality, hypersexuality and psychic blindness after bilateral temporal lobe removal make up _____ syndrome.
6. LeDoux's direct pathway from the sensory thalamus to the amygdala is called the _____ road.
7. Damasio's idea that bodily "gut feelings" guide decisions is the _____ marker hypothesis.
8. The lie-detection method that tests for knowledge only the culprit would have is the _____ knowledge test.
9. Culturally learned norms about when emotions may be shown are called _____ rules.
10. A genuine smile that involves the orbicularis oculi muscle around the eyes is a _____ smile.

**Answers:** (1) sympathetic, (2) sham, (3) Papez, (4) limbic, (5) Klüver-Bucy, (6) low, (7) somatic, (8) guilty, (9) display, (10) Duchenne.

## 11.6 Four emotions up close

LOWER YIELD · 3 QS

General theories are only half of the story of emotion. Each emotion also has its own causes, functions and disorders. Four have attracted the most research and the most exam questions: fear, anger, happiness and disgust.

### Fear and anxiety

**Fear** is the emotional response to a present, identifiable danger: the dog that is snarling at you now. It is intense, rises quickly, prepares the body for fight or flight, and fades when the danger passes. **Anxiety** is a more diffuse, future-oriented state of apprehension about possible threats: the worry, muscle tension and vigilance of the night before an exam. When anxiety has no particular object at all, it is described as *free-floating*. A useful distinction for the clinic, made by Charles Spielberger in his State-Trait Anxiety Inventory (1970), separates *state anxiety* (how anxious you are at this moment) from *trait anxiety* (a stable tendency to respond to many situations with anxiety).

Both fear and anxiety are adaptive in moderate doses: they keep us away from cliffs and make us prepare for examinations. They become clinical problems when they are out of proportion to the real danger, persistent and disabling. A *phobia* is a persistent, irrational fear of a specific object or situation that leads to avoidance; generalised anxiety disorder involves continued, uncontrollable apprehension about many unspecified things; and panic disorder in-

volves sudden surges of intense fear and a "fear of fear" (these disorders are covered in the Disorders book). Why are phobias of snakes, heights and spiders so much more common than phobias of electric sockets or cars, which are far more dangerous today? Martin Seligman's idea of *preparedness* (Chapter 6) proposes that evolution has prepared us to learn fears of the dangers our ancestors faced, and Arne Öhman's research shows that such fear-relevant stimuli are detected and conditioned especially quickly, partly through LeDoux's low road. The link between arousal and performance, the inverted U of the Yerkes-Dodson law, is discussed in Chapter 10.

## Anger, aggression and the myth of catharsis

Three terms need separating. *Anger* is an emotion, a feeling of displeasure and antagonism, usually in response to a perceived wrong. *Aggression* is behaviour intended to harm someone. *Hostility* is an attitude, a lasting, cynical mistrust of and ill will towards others. You can be angry without being aggressive, and aggressive without being angry (a hired thug). The distinction matters later in this chapter, because it is hostility, rather than anger or the broader Type A pattern, that has turned out to be most dangerous for the heart.

What should an angry person do with the anger? A long tradition says: let it out. The idea of catharsis goes back to Aristotle, who wrote that tragedy purges the audience of pity and fear; Sigmund Freud and Josef Breuer used a "cathartic method" in which patients relived and discharged buried feelings; and the ethologist Konrad Lorenz compared aggressive energy to water building up behind a dam. The **catharsis hypothesis** holds that expressing anger or aggression in a safe way, by hitting a pillow, shouting, playing violent games or watching violence, drains it and so reduces later aggression. It is one of the most popular ideas in pop psychology, and one of the most thoroughly disconfirmed.

### KEY STUDY: DOES VENTING FEED THE FLAME? (BUSHMAN, 2002)

**Method.** Brad Bushman first made participants angry: each wrote an essay and received a harsh evaluation from a supposed fellow student ("This is one of the worst essays I have read!"). Participants were then assigned to one of three groups. The *rumination* group hit a punching bag while thinking about the person who had insulted them (a photograph of the "evaluator" was shown). The *distraction* group hit the punching bag while thinking about becoming physically fit (a photograph of an athlete was shown). The *control* group simply sat quietly for two minutes. All then had the chance to blast the evaluator with unpleasant noise. **Results.** The rumination group was the angriest and the most aggressive; the control group, who did nothing, was the least angry and least aggressive; the distraction group fell in between. **Conclusion.** Venting while thinking about the provocation fed anger rather than draining it. Doing nothing at all worked better than "letting it out".

Why does catharsis fail? Venting keeps the provocation in mind (rumination), rehearses aggressive responses so that they become more available, and keeps arousal high, which, as excitation transfer predicts, can be added to the next provocation. What does work is reducing arousal (relaxation, slow breathing, a time-out), reappraising the provocation (perhaps the other person was under pressure), distraction, humour, and assertive rather than aggressive communication. The theories of aggression themselves, including the frustration-aggression hypothesis, are covered in Chapter 10 and in the Social book.

## Happiness and subjective well-being ●●●

For most of its history psychology studied misery far more than happiness. That changed in the 1980s, largely through the work of **Ed Diener** (1946 to 2021), whose landmark review of 1984 defined **subjective well-being** (SWB), the scientific term for what people call happiness. Diener's **tripartite model** says that subjective well-being has three components: a *cognitive* one, **life satisfaction**, a person's overall judgement of how his or her life is going; and two *affective* ones, the frequent presence of *positive affect* and the relative absence of *negative affect*. A happy person, in this sense, judges life to be satisfying and experiences frequent pleasant and infrequent unpleasant emotions. Diener and his colleagues measured the cognitive component with the five-item Satisfaction With Life Scale (1985).

What makes people happy? Research has produced some surprises. Close relationships are among the strongest correlates of happiness; income matters a good deal when it lifts people out of poverty but much less once basic needs are met. One reason for the weak effect of good fortune is adaptation. According to the **adaptation-level theory** of **Harry Helson** (1898 to 1977), set out in 1964, we judge every new experience against a neutral point set by our recent experience, and that neutral point moves as experience changes. Philip Brickman and Donald Campbell (1971) applied the idea to happiness and called the result the **hedonic treadmill**: as our circumstances improve, our expectations rise to match, so that we keep running without getting any happier. In a famous study, Brickman, Dan Coates and Ronnie Janoff-Bulman (1978) found that lottery winners were not significantly happier than a comparison group a year or so later and took less pleasure in everyday events, while people paralysed in accidents were less happy than the comparison group but not nearly as unhappy as most people would predict. Later research has softened the treadmill: some events, such as long unemployment, widowhood or severe disability, can lower well-being for years, and people differ in how fully they adapt.

A second reason is comparison. **Relative deprivation** is the feeling that one is worse off than the people one compares oneself with, whatever one's absolute position. The term comes from the wartime research of Samuel Stouffer and colleagues, published as *The American Soldier* (1949): military policemen, who had poor chances of promotion, were more satisfied with their promotion prospects than men in the Army Air Corps, who were promoted far more often, because the airmen compared themselves with the many colleagues who had been promoted faster. If you receive a ten percent salary rise at your Mysuru office, you are pleased, until you discover that your colleague received fifteen. Comparison even works backwards in time and possibility: Victoria Medvec, Scott Madey and Thomas Gilovich (1995) found that Olympic bronze medallists looked happier than silver medallists, because silver medallists compare themselves with the gold they just missed, whereas bronze medallists compare themselves with finishing without any medal.

What are positive emotions *for*? Negative emotions have obvious survival value: fear narrows attention and prepares escape. **Barbara Fredrickson** proposed in 1998, and developed in 2001, the **broaden-and-build theory** of positive emotions. Its first claim (the broaden part) is that positive emotions such as joy, interest, contentment and love widen our momentary "thought-action repertoire": they make attention and thinking broader, more flexible and more creative, and make us more open to people and ideas. Its second claim (the build part) is that, over time, these broadened moments build lasting personal resources, physical (health), intellectual (knowledge), social (friendships) and psychological (resilience, opti-

mism), which help us cope with later adversity. Fredrickson and Robert Levenson (1998) also found an *undo effect*: after a fear-inducing film, participants who then watched a film evoking contentment or amusement recovered their resting heart rate and blood pressure faster than those who watched a neutral or sad film. (A later claim that flourishing requires a precise ratio of about three positive emotions to every negative one turned out to rest on faulty mathematics, and that part of the work was formally withdrawn.)

Positive psychology, the movement launched by Martin Seligman in 1998, also studies how people make the most of good experiences. **Savoring**, a concept developed by Fred Bryant and Joseph Veroff (their book appeared in 2007), is the capacity to attend to, appreciate and enhance positive experiences. It is the mirror image of coping: coping manages bad experiences, savoring amplifies good ones. We can savour the past (reminiscing), the present (savouring the moment) and the future (anticipation). Bryant and Veroff described savouring processes such as basking (in praise or pride), thanksgiving (gratitude), marvelling (awe) and luxuriating (physical pleasure), and strategies such as sharing the experience with others, building memories, sharpening the senses and absorbing oneself fully. Lingering over the first sip of filter coffee on a cold morning, rather than gulping it while scrolling your phone, is savouring.

## Disgust

**Disgust** is the emotion of rejection: in the words of **Paul Rozin** and April Fallon (1987), revulsion at the prospect of taking an offensive object into the body. Its facial expression, a wrinkled nose, raised upper lip and sometimes a protruding tongue, is literally the face of spitting something out, exactly as Darwin's principle of serviceable habits predicts, and its function is to protect us from contamination and disease. Disgust obeys what Rozin calls the laws of "sympathetic magic". By the *law of contagion*, anything that has touched a disgusting thing becomes disgusting itself: most people refuse a glass of juice into which a completely sterilised cockroach has been dipped. By the *law of similarity*, things that look like disgusting things are treated as disgusting: people avoid chocolate fudge shaped like dog faeces even though they know exactly what it is.

Disgust develops late. Toddlers will put almost anything in their mouths, and a full adult sense of disgust appears only during the school years, learned largely from the reactions of others. Over development, disgust spreads outward from *core* disgust (spoiled food, body products, vermin) to *animal-reminder* disgust (death, wounds and anything that reminds us we are animals), *interpersonal* disgust (contact with strangers or the "unclean") and finally *moral* disgust at cheating or cruelty. Brain imaging shows that both feeling disgust and seeing disgust on another person's face activate the anterior insula. Clinically, heightened disgust sensitivity plays a part in contamination-type obsessive-compulsive disorder and in blood-injection-injury phobia.

EXAM TRAP

**Diener's tripartite model of subjective well-being** is life satisfaction (cognitive) plus high positive affect plus low negative affect. Options that replace life satisfaction with "physical wellness", "spiritual wellness" or "social wellness" are distractors. Do not confuse it with Seligman's PERMA model (positive emotion, engagement, relationships, meaning, accomplishment) or Carol Ryff's six-part model of psychological well-being.

**Barbara Fredrickson = broaden-and-build** (and the undo effect). Flow theory is Mihaly Csikszentmihalyi's; cognitive distortions are Aaron Beck's (and David Burns's popular list); psychodynamic theory is Freud's.

**Savoring** (Bryant and Veroff) means enhancing and prolonging the enjoyment of positive experiences. It is not "avoiding negative experiences", "minimising stress through relaxation" or "reflecting on past traumas".

**Hedonic treadmill** is Brickman and Campbell (1971), building on Helson's adaptation-level theory (1964). **Relative deprivation** began with Stouffer's *The American Soldier* (1949). **The catharsis hypothesis** predicts that venting reduces aggression; the evidence shows it usually increases it.

## 11.7 Emotional development and regulation

MEDIUM YIELD · 6 QS

Emotions are not fully formed at birth, and they do not run their course untouched once they begin. This section looks at how emotions emerge in infancy, and at how people of every age manage the emotions they have.

### How emotions develop ●●●

John B. Watson, as you saw in Table 11.2, believed that newborns have only three innate emotional reactions: fear (to loud noises and loss of support), rage (to restriction of movement) and love (to stroking and cuddling). A more careful observational account came from the psychologist **Katharine M. Banham Bridges**, who, from observations of infants in Montreal, in 1932 described emotional development as a process of *differentiation*, like a branching tree. At birth, she reported, the infant shows only a diffuse *general excitement*. Within the first three months this splits into two broad states, *distress* and *delight*. Distress then branches into anger, disgust and fear in the following months, and delight into elation and affection; by about two years the child shows a wide range of emotions, including jealousy and joy.

Modern research, using methods such as Izard's coding of infant faces, finds that differentiation happens faster than Bridges thought. By about six months infants show most of the *primary emotions*: joy, interest, surprise, sadness, anger, disgust and fear. The *social smile*, a smile directed at a human face, appears at about six to eight weeks, and wariness of strangers (stranger anxiety) at around six to eight months. A second group, the **self-conscious emotions** (embarrassment, pride, shame, guilt and envy), requires something more: a sense of oneself as a separate person who can be evaluated. Michael Lewis and Jeanne Brooks-Gunn (1979) showed that infants first recognise themselves in a mirror (touching a spot of rouge secretly placed on their own nose) between about 15 and 24 months, and the self-conscious emotions appear only after this, from about 18 months onwards; pride, shame and guilt, which require knowing the rules by which one is judged, become clear between about two and three years. Self-conscious emotions therefore appear well after 12 months.

Infants also learn to use other people's emotions as information. In **social referencing**, a child faced with an uncertain situation looks to a caregiver's face to decide how to feel and act. In a classic study by James Sorce, Joseph Campos and colleagues (1985), one-year-olds crawled towards a "visual cliff" (a glass-covered drop) of uncertain depth while their mothers, on the far side, posed either a happy or a fearful face. About three-quarters of the infants crossed when the mother looked happy; none crossed when she looked afraid. Infants also expect emotional responsiveness from caregivers. In Edward Tronick's "still-face" procedure (first described in 1978), a mother interacts normally with her infant and then suddenly holds her face still and unresponsive; within moments the infant tries harder and harder to re-engage her, then becomes distressed and turns away. Even very young infants actively expect emotional give-and-take. (Temperament, attachment and the rest of emotional development are covered in the Developmental book.)

### Emotion regulation: Gross's process model ●●

From the first months, caregivers help infants manage their emotions by soothing, distracting and feeding them; gradually children take over the job themselves. **Emotion regulation** refers to the processes by which we influence which emotions we have, when we have them, and how we experience and express them. The most influential framework is the **process model of emotion regulation** of **James J. Gross** (1998). Gross pointed out that an emotion unfolds over time: we enter a situation, attend to some aspect of it, appraise its meaning, and then respond. We can intervene at each of these points, which gives five families of strategies (Figure 11.4). Consider a student on the night before an entrance examination.

1. **Situation selection**: choosing which situations to enter. The student decides not to join the anxious group chat where everyone is comparing marks on mock tests.
2. **Situation modification**: changing the situation one is in. When a friend keeps quizzing her on topics she has not revised, she asks him to stop.
3. **Attentional deployment**: directing attention within the situation. She distracts herself with a walk, or concentrates on the topics she knows well. Rumination, dwelling on the feeling and its causes, is also a form of attentional deployment, an unhelpful one.
4. **Cognitive change**: changing how one appraises the situation. The central strategy here is **cognitive reappraisal**: "This exam is a chance to show what I have learned, and there are other colleges if it goes badly"; or "My racing heart is my body getting ready to perform."
5. **Response modulation**: acting on the emotional response once it has arisen. Examples are slow breathing, exercise, drinking, and **expressive suppression**, keeping a calm face while feeling panic inside.

The first four families act before the emotional response is fully formed, so Gross called them *antecedent-focused*; the fifth acts on the response after it has formed, so it is *response-focused*.

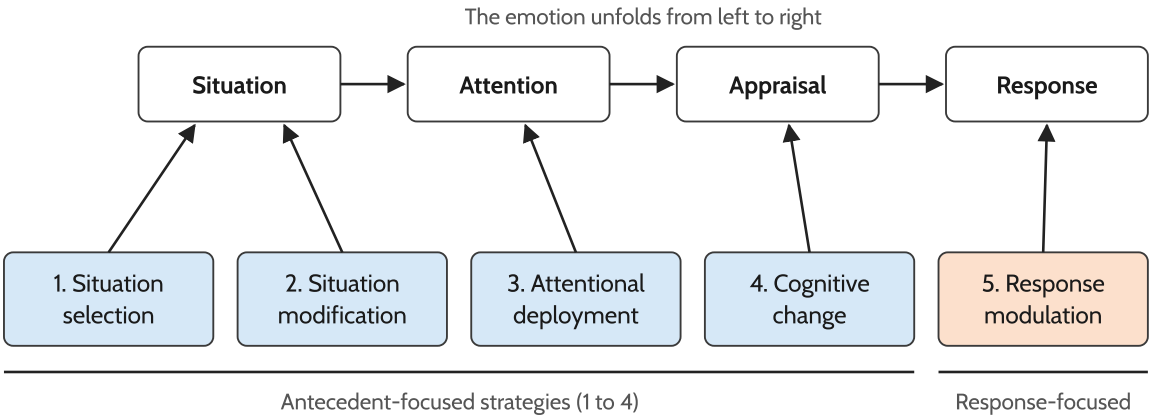


**Figure 11.4** Gross's process model of emotion regulation (1998). An emotion unfolds from situation to attention to appraisal to response, and each of the five families of strategies acts at a different point. Reappraisal belongs to cognitive change (4); suppression belongs to response modulation (5).

Are all strategies equally good? Gross's research compared the two most studied ones, reappraisal and suppression. In experiments by Gross and Robert Levenson in the 1990s, participants watched a disgusting film; those told to suppress their expressions showed less disgust on their faces but felt just as disgusted, and their sympathetic arousal actually *increased*. Suppression is effortful: later studies found that it also impairs memory for what happened during the suppression and makes conversation partners feel less connected. Reappraisal, by contrast, reduces both the feeling and the expression at little cost. Using the Emotion Regulation Questionnaire, Gross and Oliver John (2003) found that habitual reappraisers report more positive emotion, closer relationships and greater well-being, and habitual suppressors the reverse. Reappraisal can even improve performance under pressure: Jeremy Jamieson and colleagues (2010) told some students preparing for a graduate admission test that feeling anxious and aroused could help them perform; those students scored higher on the mathematics section than students given no such message.

CLINICAL LINK

Difficulties in emotion regulation run through many disorders: depression (rumination), anxiety disorders (avoidance, a form of situation selection taken too far), substance use (response modulation with a drug) and borderline personality disorder. Dialectical behaviour therapy, developed by Marsha Linehan, teaches four skills modules, mindfulness, distress tolerance, emotion regulation and interpersonal effectiveness, and cognitive behaviour therapy is in large part systematic training in reappraisal (Clinical book).

## Alexithymia: no words for feelings

Some people find it hard to know what they are feeling at all. The psychiatrist **Peter E. Sifneos** (1920 to 2008) coined the term **alexithymia** in 1973, from the Greek *a* (lack), *lexis* (word) and *thymos* (emotion): literally, "no words for emotions". He had noticed that many of his patients with psychosomatic illnesses could describe their physical symptoms in detail but could not describe their feelings. Alexithymia is a personality trait rather than a diagnosis, and it has four features: difficulty identifying feelings and telling them apart from bodily sensations ("I feel sick" rather than "I feel anxious"); difficulty describing feelings to others;

an externally oriented, concrete style of thinking focused on practical details rather than inner experience; and a limited fantasy life. It is usually measured with the Toronto Alexithymia Scale (TAS-20), and it is more common in people with psychosomatic complaints, depression, eating disorders, substance use, post-traumatic stress and autism spectrum conditions. For the therapist, it means that "How did that make you feel?" may draw a blank, and that helping the client label emotions is often the first task.

## Emotional intelligence ●●

If people differ in recognising and managing emotions, perhaps this is a kind of intelligence. Peter Salovey and John Mayer introduced the scientific concept of emotional intelligence in 1990, and Daniel Goleman's best-selling *Emotional Intelligence* (1995) made it famous. Goleman's popular model has five domains: knowing one's own emotions (self-awareness), managing one's emotions, motivating oneself, recognising emotions in others (empathy), and handling relationships; later he relabelled them self-awareness, self-regulation, motivation, empathy and social skills.

Goleman's empathy domain deserves a closer look, because empathy is itself a part of emotion. Social psychologists such as Robert Baron and Nyla Branscombe describe it as having three distinct components. *Emotional empathy* is sharing the feelings of another person, feeling distressed when a friend is distressed. The *cognitive component*, sometimes called empathic accuracy or perspective taking, is perceiving accurately what another person is thinking and feeling. *Empathic concern* is feeling concern and compassion for the other person's welfare, the component most closely tied to helping (Social book). A clinician needs all three: accuracy to understand the client, concern to want to help, and enough control over emotional empathy not to be swamped by the client's distress. The ability model of Mayer and Salovey, the trait models, their measurement and the debate over whether emotional intelligence is really an intelligence are covered in Chapter 9.

EXAM TRAP

**Gross's five families, in order:** situation selection, situation modification, attentional deployment, cognitive change, response modulation. **Reappraisal is cognitive change**, not response modulation; **suppression is response modulation**. Distraction and rumination are attentional deployment.

**Expressive suppression (Gross) is not Freud's suppression or repression.** In Gross's sense it is hiding the outward expression of a feeling. In defence-mechanism language, repression is the unconscious, involuntary banishing of a thought or feeling from awareness, while suppression is the conscious, deliberate pushing of it out of mind.

**First emotions in infancy.** In Bridges's (1932) scheme, the first differentiation of general excitement is into *distress and delight*. Watson's three innate emotions were fear, rage and love. Self-conscious emotions (embarrassment, pride, shame, guilt) appear after 12 months, from about 18 months, because they need self-recognition.

**Alexithymia** (Sifneos, 1973) is difficulty identifying and describing one's own feelings; it is not the absence of emotion, and it is not anhedonia (the inability to feel pleasure, a symptom of depression).

**Goleman's five domains come in two wordings.** The 1995 book lists knowing one's emotions, managing emotions, motivating oneself, recognising emotions in others and handling relationships; the later relabelling is self-awareness, self-regulation, motivation, empathy and social skills. A practice mock keyed the 1995 wording, so prefer it when both appear. **Empathy's three components** are emotional empathy, a cognitive component (empathic accuracy) and empathic concern; "behavioural aspect" and "positivity" are not among them.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Fear responds to a present danger, whereas _____ is a diffuse, future-oriented apprehension.
2. The idea that venting anger reduces later aggression is the _____ hypothesis.
3. Diener's three components of subjective well-being are life satisfaction, positive affect and the absence of _____ affect.
4. Brickman and Campbell's term for adapting to good fortune so that happiness returns to its old level is the hedonic _____.
5. Feeling worse off than the people you compare yourself with is called relative _____.
6. Fredrickson's theory that positive emotions widen thinking and build resources is the _____ theory.
7. Looking to a caregiver's face to decide how to react to an uncertain situation is social _____.
8. In Gross's process model, reappraisal belongs to the family called _____ change.
9. Difficulty identifying and describing one's own feelings is called _____.

**Answers:** (1) anxiety, (2) catharsis, (3) negative, (4) treadmill, (5) deprivation, (6) broaden-and-build, (7) referencing, (8) cognitive, (9) alexithymia.

## 11.8 Stress and the body's emergency systems MEDIUM YIELD · 6 QS

Think back to the men on the Capilano bridge. For the minute or two it took to cross, their hearts pounded, their palms sweated and their blood carried a surge of adrenaline. Then they reached the far side and it was over. Now imagine a bridge that never ends: a threat or a demand that goes on for weeks, months or years. That is the territory of stress. Everything you have learned about the bodily side of emotion, the sympathetic nervous system, the hypothalamus and the amygdala, is also the machinery of stress. What changes is time. A system built for emergencies lasting minutes is asked to run for months, and the price is paid in health.

### What is stress? Three ways to define it

The word is used so loosely ("I'm so stressed about Monday") that psychologists have had to decide what it should mean. Three answers have been given, and each one is tied to a research tradition you will meet in this chapter.

The *stimulus definition* treats stress as something out in the environment: a demand or event that puts pressure on a person, as a load puts pressure on a bridge. A death in the family, an examination or a flood *is* the stress, and the research question is how much of it a person has faced. To avoid confusion, the event itself is now called a **stressor**: any event or circumstance that threatens, challenges or harms a person's well-being and so calls for adjustment. Holmes and Rahe's counting of life events (Section 11.9) is the classic stimulus approach.

The *response definition* treats stress as what happens inside the person: the physiological and psychological reaction to a stressor. Hans Selye, whom you are about to meet, defined stress as "the nonspecific response of the body to any demand made upon it." Cannon's and Selye's work on the body's emergency reactions is the classic response approach, and some writers borrow the engineer's word *strain* for this reaction.

The *transactional definition*, associated with Richard Lazarus, says that stress lies neither in the event alone nor in the response alone but in the relationship between person and environment. A situation becomes stressful when a person appraises its demands as taxing or exceeding his or her resources. The same viva voce examination is a crushing threat to one candidate and an exciting challenge to another. This is the definition most psychologists use today, and Section 11.9 develops it. Putting the three together, **stress** can be defined as the process by which we perceive and respond to events, called stressors, that we appraise as threatening or challenging. Stressors can be physical (cold, injury, infection, noise) or psychological (an insult, a deadline, a loss). One of the boldest claims of early stress research was that the body responds to both kinds in much the same way.

### Cannon: fight or flight and homeostasis

Walter Cannon, whose attack on the James-Lange theory you met in Section 11.2, began his career at Harvard watching digestion with the newly discovered X-rays. He noticed that the rhythmic movements of a cat's stomach stopped dead whenever the animal was frightened or enraged. Over the following years he traced the reason. In *Bodily Changes in Pain, Hunger, Fear and Rage* (1915) he described how pain, fear and rage activate the sympathetic nervous system and the adrenal glands, which together prepare the animal for violent action. He called this the **fight-or-flight response**: an immediate, coordinated mobilisation of the body

either to attack a threat or to escape from it. The heart beats faster and harder, blood pressure rises, breathing quickens and the airways widen, blood is diverted from the skin and gut to the skeletal muscles and brain, the liver releases glucose for fuel, the pupils dilate, digestion stops, and the blood clots more readily, which limits bleeding from a wound. Every item is a preparation for running or fighting (compare Table 11.5).

The machinery behind this response is now called the **sympathetic-adrenal-medullary axis** (SAM axis). The hypothalamus activates the sympathetic nervous system. Sympathetic nerves act directly on the organs within a second or two, and they also stimulate the *adrenal medulla*, the inner core of the adrenal glands that sit on top of each kidney, to release **adrenaline** (epinephrine) and noradrenaline (norepinephrine) into the bloodstream. Because these hormones circulate, they spread and prolong the sympathetic effects throughout the body. The SAM axis is fast: it acts in seconds.

Cannon also gave physiology one of its central ideas. He coined the term **homeostasis** in 1926 and popularised it in *The Wisdom of the Body* (1932): the tendency of the body to keep its internal environment, such as temperature, blood sugar and the balance of salt and water, stable within narrow limits (the motivational side of homeostasis, drive reduction, is in Chapter 10). Fight or flight is a temporary, deliberate departure from resting balance in the service of survival; when the emergency passes, the parasympathetic division restores it. Cannon even argued, in a famous essay on "voodoo death" (1942), that a person convinced of having been fatally cursed could die from a prolonged, unrelieved emergency reaction. Modern researchers add a third option that often comes before fighting or fleeing: *freezing*, the motionless watchfulness of a deer caught in headlights, organised by the periaqueductal grey of the midbrain (Section 11.4). Hence the modern phrase "fight, flight or freeze".

## Selye's general adaptation syndrome ●●

Cannon studied the emergency. The scientist who asked what happens when the emergency does not end was an endocrinologist working in Montreal, and he came to the question by accident.

### KEY STUDY: SELYE'S RATS AND "A SYNDROME PRODUCED BY DIVERSE NOCUOUS AGENTS" (1936)

**Question.** In the mid-1930s **Hans Selye** (1907 to 1982), a young Vienna-born researcher at McGill University, was hunting for a new sex hormone. **Method.** He injected rats with extracts of ovary and placenta and examined their organs. **Result.** The rats showed a striking triad of changes: the adrenal cortex was enlarged; the thymus, spleen and lymph nodes had shrunk; and the lining of the stomach and intestine was covered in bleeding ulcers. Selye was thrilled, until he found that extracts of other organs produced the same triad, and so did a plain toxic chemical, formalin. His hormone had vanished. Then came the insight: perhaps he had found not a hormone but the body's *general* response to any damaging agent. He exposed rats to cold, heat, surgical injury, forced exercise and restraint, and the same triad appeared every time. **Conclusion.** In a short letter to *Nature* in 1936, "A syndrome produced by diverse nocuous agents," Selye proposed that the body reacts to all such agents with one nonspecific pattern. He later named it the general adaptation syndrome and brought it to a wide audience in *The Stress of Life* (1956). **Criticism.** The stressors were physical and often extreme, and the subjects were rats; whether psychological stressors in people work the same way had still to be shown.

The **general adaptation syndrome** (GAS) is Selye's description of how the body responds to prolonged stress, in three stages that always occur in the same order: alarm, resistance, exhaustion (Figure 11.5).

1. **Stage 1.** In the **alarm reaction** the body first recognises the stressor and mobilises against it. Selye described a brief *shock* phase in which resistance actually dips below normal (body temperature and blood pressure may fall), followed by a *countershock* phase in which the sympathetic system and the adrenal glands spring into action. The alarm reaction is, in effect, Cannon's fight-or-flight response. An overwhelming stressor can kill during this stage.
2. **Stage 2.** In the **stage of resistance**, if the stressor continues, the body settles into sustained adaptation. Stress hormones, especially cortisol, stay high; resistance to *this* stressor rises above normal, and outwardly the person seems to be coping. But resources are being spent all the time, and resistance to *other*, new stressors falls. Selye's rats that had adapted to cold, for example, became unusually vulnerable to other stressors.
3. **Stage 3.** In the **stage of exhaustion**, if the stress goes on long enough, the body's capacity to adapt, which Selye pictured as a limited store of "adaptation energy", is used up. Resistance collapses, signs of the alarm reaction return, and the organism becomes vulnerable to what Selye called **diseases of adaptation**: high blood pressure, heart disease, ulcers, kidney disease, arthritis and lowered resistance to infection. In the extreme, the outcome is death.

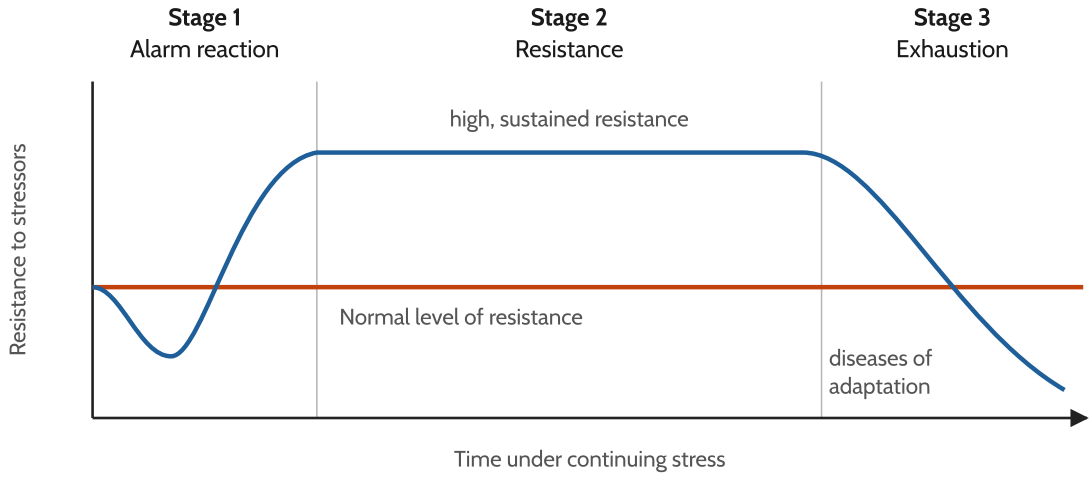


Figure 11.5 Selye's general adaptation syndrome. In the alarm reaction, resistance first dips below normal (shock) and then rises (countershock). During resistance it stays above normal while the body adapts. If the stressor continues, resistance collapses in exhaustion and diseases of adaptation appear. The thick curve shows resistance; the labelled horizontal line marks the normal resting level.

Selye also insisted that stress is not always bad. In *Stress Without Distress* (1974) he distinguished **eustress** (from the Greek *eu*, good), stress that is pleasant, motivating or beneficial, from **distress**, stress that is unpleasant and harmful. A wedding, a promotion, a hard-fought cricket match or a well-prepared examination can all be eustress; a bereavement or a long illness is distress. Whether a stressor is experienced as eustress depends on its intensity, on how long it lasts and, above all, on how it is appraised. Eustress is not the absence of stress; it is the optimal amount, the moderate arousal that the Yerkes-Dodson principle (Chapter 10) says

helps performance. Pinel and Barnes note a biological echo: brief stressors actually strengthen some parts of the immune system (Section 11.10).

How well has the theory lasted? Its central claims, that psychological and physical stressors can trigger the same bodily response and that long-lasting stress can make us ill, have stood. Three of its details have not. First, the response is less nonspecific than Selye thought. The American physiologist John W. Mason showed in the early 1970s that different stressors produce different patterns of hormones, and that when the psychological upset is removed (monkeys deprived of food but given flavoured, non-nutritive pellets, so that they were not upset by the deprivation), the usual hormonal stress response largely fails to appear. Emotional appraisal, in other words, is a key ingredient, which is exactly Lazarus's point. Second, Selye largely ignored the sympathetic nervous system; modern accounts give the SAM axis and the HPA axis equal billing. Third, exhaustion is now thought to come less from running out of hormones than from the damage done by prolonged exposure to them, as the next subsection explains.

### The HPA axis: the slower stress system ●●●

The body's second stress system, the one whose effects Selye saw in his rats' enlarged adrenal glands, is the **hypothalamic-pituitary-adrenal axis** (HPA axis). It works in three hormonal steps. First, the paraventricular nucleus of the hypothalamus releases *corticotropin-releasing hormone* (CRH, also called corticotropin-releasing factor). Second, CRH travels a short distance through a private blood supply to the *anterior* pituitary gland, which responds by releasing *adrenocorticotropic hormone* (ACTH) into the general circulation. Third, ACTH reaches the *adrenal cortex*, the outer layer of the adrenal glands, and stimulates it to secrete **glucocorticoids**, the steroid hormones that help regulate glucose metabolism. The main glucocorticoid in humans is **cortisol** (in rats it is corticosterone). The level of circulating glucocorticoids is the most commonly used physiological measure of stress.

The HPA axis is slower than the SAM axis: cortisol levels peak roughly 20 to 40 minutes after a stressor begins, and its effects last for hours. Cortisol raises blood sugar by making glucose from protein and fat, which fuels sustained effort; it damps down inflammation and immune activity; and it slows digestion, growth and reproduction, which can wait until the crisis is over. It also feeds back on the brain. Cortisol acts on receptors in the hypothalamus and the anterior pituitary (and the hippocampus) to switch off further release of CRH and ACTH, a *negative feedback* loop that works like a thermostat. In many patients with severe (melancholic) depression this feedback is weakened: given dexamethasone, a synthetic glucocorticoid that normally suppresses the body's own cortisol, their cortisol stays high (the dexamethasone suppression test is discussed in the Disorders book).

The trouble is that the HPA axis was built for the zebra's problem, not ours. The neuroscientist Robert Sapolsky made the point in the title of his book *Why Zebras Don't Get Ulcers* (1994): a zebra's stress is acute, a desperate sprint from a lion that is over in minutes, one way or the other. Human beings switch on the same response for loans, traffic, deadlines and family quarrels, and leave it switched on. Prolonged high cortisol suppresses immune function, raises blood pressure and blood sugar, encourages fat storage around the abdomen and damages the hippocampus, which is packed with glucocorticoid receptors: chronic stress reduces dendritic branching and the birth of new neurons there and impairs hippocampus-dependent memory. Peter Sterling and Joseph Eyer (1988) called the body's active adjustment to de-

mands *allostasis* ("stability through change"), and **Bruce S. McEwen** (1938 to 2020) named its cost **allostatic load**: the cumulative wear and tear on the body produced by repeated or chronic activation of the stress systems. Allostatic load is the modern replacement for Selye's stage of exhaustion. Table 11.7 sets the two stress systems side by side.

Table 11.7 The body's two stress systems.

| Feature | SAM axis (fast) | HPA axis (slow) |
|---|---|---|
| Full name | Sympathetic-adrenal-medullary axis | Hypothalamic-pituitary-adrenal axis |
| Route | Hypothalamus to sympathetic nerves to adrenal medulla | Hypothalamus (CRH) to anterior pituitary (ACTH) to adrenal cortex |
| Main messengers | Adrenaline and noradrenaline | Glucocorticoids, chiefly cortisol |
| Speed and duration | Seconds; short-lived | Minutes; lasts hours |
| Main effects | Heart rate, blood pressure, breathing and glucose up; fight or flight | Energy mobilised; inflammation and immunity damped; negative feedback on the brain |
| Classic researcher | Walter Cannon (1915) | Hans Selye (1936) |

## Tend and befriend

Fight or flight was mapped almost entirely on males. **Shelley E. Taylor** (born 1946) and her colleagues at the University of California, Los Angeles, pointed out in 2000 that until the mid-1990s women had made up only about 17 percent, fewer than one in five, of the participants in laboratory studies of the physiological stress response. Looking at the evidence from animals and humans with both sexes in mind, they proposed that females, in particular, often respond to stress with a pattern they called **tend-and-befriend**. *Tending* means protecting and calming offspring; *befriending* means seeking out and maintaining social networks that offer protection and support. The evolutionary logic is that a pregnant or nursing female who fights or flees puts her young at risk, whereas one who shields them and gathers allies protects both. The proposed biology centres on the hormone oxytocin, whose effects are enhanced by oestrogen, together with the brain's own opioids, which promote affiliation and calm. Taylor did not claim that women never fight or flee, or that men never seek company; tend-and-befriend is a difference of emphasis that sits on top of the shared fight-or-flight machinery. It also explains part of the protective power of social support, discussed in Section 11.10.

EXAM TRAP

**The order of the GAS is alarm, resistance, exhaustion.** "Illness" is not a stage (illness is what exhaustion leads to), and resistance never comes first. Alarm, resistance and exhaustion belong to **Selye's general adaptation syndrome**, not to life events theory, the health belief model or protection motivation theory (Section 11.10).

**Match the person to the idea.** Cannon: fight or flight (1915) and homeostasis (1926). Selye: general adaptation syndrome (1936; *The Stress of Life*, 1956) and eustress versus distress (1974). Taylor: tend-and-befriend (2000). McEwen: allostatic load.

**Cortex versus medulla.** Cortisol comes from the adrenal *cortex* (HPA axis); adrenaline and noradrenaline come from the adrenal *medulla* (SAM axis). Asked which gland is primarily responsible for the stress response, answer the **adrenal glands**. The pituitary is the "master gland" and releases ACTH, but it is not the stress gland. Distractor hormones: insulin (pancreas, lowers blood sugar), thyroxine (thyroid, sets metabolic rate), melatonin (pineal, signals darkness and sleep).

**The HPA sequence** is hypothalamus (CRH), then *anterior* pituitary (ACTH), then adrenal cortex (cortisol); cortisol's negative feedback acts mainly on the hypothalamus and anterior pituitary. The posterior pituitary releases oxytocin and vasopressin and is not part of the HPA axis.

**Eustress is good stress, not zero stress.**

## 11.9 Sources of stress and the transactional model HIGH YIELD · 8 QS

What kinds of events cause stress? Researchers have tried three ways of answering: counting the big events in a person's life, counting the small ones, and sorting stressors by how long they last and how many people they strike. Each approach found something real, and each ran into the same awkward fact: the same event affects different people very differently. That fact leads to the transactional model at the end of this section.

### Life events: Holmes and Rahe's Social Readjustment Rating Scale ●●

In the 1950s and 1960s the psychiatrists **Thomas H. Holmes** and **Richard H. Rahe**, at the University of Washington School of Medicine in Seattle, noticed in their patients' case histories that illnesses seemed to follow clusters of major changes in their lives. (The idea of charting a patient's life events alongside the course of illness went back to the psychiatrist Adolf Meyer's "life chart".) In 1967 they published a way of measuring such change. They gave 394 people a list of 43 life events and asked them to rate how much readjustment each one required, compared with marriage, which was arbitrarily given a value of 500. The average ratings, divided by 10, became **life change units** (LCUs), ranging from 100 for the death of a spouse to 11 for minor violations of the law. The resulting instrument is the **Social Readjustment Rating Scale** (SRRS). A respondent ticks the events experienced in the past year and adds up their life change units (Table 11.8).

Table 11.8 Selected items from Holmes and Rahe's Social Readjustment Rating Scale (1967).

| Rank | Life event | Life change units |
|---|---|---|
| 1 | Death of a spouse | 100 |

| Rank | Life event | Life change units |
|---|---|---|
| 2 | Divorce | 73 |
| 3 | Marital separation | 65 |
| 4 | Jail term | 63 |
| 5 | Death of a close family member | 63 |
| 6 | Personal injury or illness | 53 |
| 7 | Marriage (the anchor event) | 50 |
| 8 | Fired at work | 47 |
| 10 | Retirement | 45 |
| 12 | Pregnancy | 40 |
| 17 | Death of a close friend | 37 |
| 25 | Outstanding personal achievement | 28 |
| 27 | Beginning or ending school | 26 |
| 32 | Change in residence | 20 |
| 41 | Vacation | 13 |
| 42 | Christmas | 12 |
| 43 | Minor violations of the law | 11 |

Look down the list and something stands out. Marriage, pregnancy, outstanding personal achievement, a vacation and Christmas are all there, alongside death and divorce. For Holmes and Rahe the stressor was *change itself*, whether welcome or not, because any change requires readjustment. This is the stimulus definition of stress in its purest form. In their research, people whose scores totalled more than about 300 in a year were classed as facing a major life crisis and were considerably more likely to fall ill in the following year; scores of 150 to 299 carried a moderate risk. In prospective studies, Rahe gave the scale to US Navy sailors before long tours of duty and found that those with higher life-change scores reported somewhat more illness while at sea.

The SRRS was a landmark because it made stress countable, and thousands of studies used it. But it has serious weaknesses, and examiners like them:

1. **It lumps good and bad events together.** Later research found that it is undesirable, uncontrollable change that predicts illness. The Life Experiences Survey of Irwin Sarason, James Johnson and Judith Siegel (1978) let respondents rate each event as positive or negative and how much it affected them; negative change predicted problems, positive change did not.
2. **It ignores meaning and appraisal.** A divorce is a catastrophe for one person and a release for another, yet both score 73.
3. **It ignores resources and coping**, such as social support, which change the impact of the same event.
4. **It relies on memory.** Recall of events over a year is unreliable, and people who are ill may search their memories harder for causes.

5. **Some items are confounded with illness.** "Personal injury or illness", "change in sleeping habits" and "change in eating habits" may be early signs or results of illness rather than causes of it.
6. **It is culture-bound.** Christmas and mortgages of over 10,000 dollars reflect American life in the 1960s.
7. **The correlations are small.** Most people with high scores do not become ill, so the scale is a weak predictor for any individual.

> **INDIAN CONTEXT**
>
> An American list of life events fits Indian lives imperfectly: few Indian respondents would rate Christmas or a large mortgage as typical stressors, while events such as arranging a daughter's marriage, a family dispute over property or a failed examination may weigh more heavily. In 1984 Gurmeet Singh, Dalbir Kaur and Harsharan Kaur published the *Presumptive Stressful Life Events Scale* (PSLES), a 51-item life-events scale developed with Indian respondents, in the *Indian Journal of Psychiatry*. It remains one of the most widely used life-events measures in Indian clinical research.

## Daily hassles and uplifts

Richard Lazarus thought that major life events were the wrong unit of stress. On most days nobody dies, marries or is sacked. But the bus is late, the phone battery dies in the middle of a call, the neighbour's dog barks all night and there are three assignments due on Friday. In 1981 **Allen D. Kanner**, James C. Coyne, Catherine Schaefer and Lazarus defined **daily hassles** as the irritating, frustrating and distressing demands that to some degree characterise everyday transactions with the environment, and **uplifts** as the positive experiences that can offset them, such as relating well with friends, completing a task, getting enough sleep or receiving good news. They built a 117-item Hassles Scale and a 135-item Uplifts Scale and had a group of 100 middle-aged adults complete them every month for most of a year. The most frequent hassles were concerns about weight, the health of a family member, rising prices of common goods, home maintenance, having too many things to do, and misplacing or losing things.

The key finding was that hassles predicted psychological symptoms, such as anxiety and depression, *better* than life-event scores did, and they went on predicting symptoms even when life events were statistically controlled. Later work by Anita DeLongis, Coyne, Lazarus and their colleagues (1982) found that hassles also predicted physical health better than major events. Part of the reason is that big events act partly through small ones: a divorce brings a hundred daily hassles about money, housing and children. Uplifts had weaker and less consistent effects than hassles; the bad, it seems, is stronger than the good.

## Acute, chronic and cataclysmic stressors ●●

Stressors can also be sorted by how long they last. An **acute stressor** is short-lived, with a clear beginning and end: a surprise test (a "pop quiz"), a job interview, a near miss on the highway. A **chronic stressor** persists for months or years, often with no foreseeable end: living with HIV disease or cancer, a long-running marital conflict, caring for a parent with dementia, poverty, unemployment or discrimination. The body handles acute stressors well; as

Section 11.8 showed, it is chronic stress that produces allostatic load and Selye's exhaustion, and Section 11.10 shows that the two kinds even have opposite effects on immunity.

Stressors can be sorted, too, by how many people they strike. Lazarus and Cohen (1977) distinguished three categories. **Cataclysmic events** are sudden, powerful catastrophes that affect many people at once: earthquakes, floods, wars, terrorist attacks and pandemics, such as the 2004 Indian Ocean tsunami on the Tamil Nadu coast or the Kedarnath floods of 2013. Because the suffering is shared, survivors can support one another, and for many people the stress subsides over time, although some develop lasting problems. *Personal stressors* are major life events, such as bereavement, divorce or job loss, that strike one person or family. *Background stressors* are the daily hassles of the previous subsection.

When a stressor is extreme, involving actual or threatened death, serious injury or sexual violence, some of those exposed develop post-traumatic stress disorder (PTSD). ICD-11 places PTSD in a group of *disorders specifically associated with stress*, together with complex PTSD, prolonged grief disorder and adjustment disorder, whose diagnosis requires an identifiable stressor; generalised anxiety disorder, panic disorder and obsessive-compulsive disorder do not. (The disorders themselves are covered in the Disorders book.)

### Frustration, conflict, pressure and change

Many textbooks, following Wayne Weiten, group the psychological sources of stress into four types: frustration, conflict, change and pressure. You studied the first and second in Chapter 10 (Section 10.8), so a short reminder is enough here. *Frustration* is the blocking of goal-directed behaviour. *Conflict* is the simultaneous presence of incompatible motives. **Kurt Lewin** (1890 to 1947) described three basic types in 1935: *approach-approach* (two attractive goals, such as two good colleges; the mildest and quickest to resolve), *avoidance-avoidance* (two unattractive options, such as a painful dental visit or a worsening toothache; the person vacillates and tries to "leave the field") and *approach-avoidance* (one goal with both attractive and repellent features, such as a well-paid job far from home). Later writers added the *double approach-avoidance* conflict, two goals each with good and bad features, which is the shape of most real-life decisions. **Neal E. Miller** (1909 to 2002) showed that the avoidance gradient is steeper than the approach gradient, which is why people in approach-avoidance conflicts move towards the goal and then retreat as they near it (Figure 10.6). Approach-avoidance and avoidance-avoidance conflicts are the most stressful; approach-approach conflicts the least. *Change* is the life-event stress measured by Holmes and Rahe. *Pressure* is the expectation or demand that one behave in a certain way: pressure to perform (a call-centre worker who must handle a quota of calls at high speed) and pressure to conform (a young adult expected to choose the career or marriage the family has in mind).

### Stress at work and burnout ●●

For most adults the workplace is a major source of stress, and organisational psychologists have catalogued its sources. A widely used scheme from Stephen Robbins's textbook of organisational behaviour sorts potential stressors into three groups. *Environmental factors* lie outside the organisation: economic uncertainty, political uncertainty and technological change. *Organisational factors* lie within the job and the organisation: *task demands* (workload, pace, working conditions, the design of the job), *role demands*, *interpersonal demands* (poor relationships with colleagues, lack of support), the structure of the organisation and its

style of leadership. *Individual factors* are those the employee brings from personal life: family problems, economic problems such as debt, and personality.

Role demands deserve a closer look. In *Organizational Stress* (1964), Robert Kahn and his colleagues described **role conflict**, the presence of incompatible expectations (a supervisor told to cut costs and also to improve service, or a working mother pulled between office and home), and **role ambiguity**, uncertainty about what one is expected to do, how one's work will be judged or where one's authority ends. To these is added *role overload*, simply having too much to do. In 1979 **Robert A. Karasek** proposed the **job demand-control model**: strain is greatest not in jobs with the heaviest demands but in jobs that combine *high demands* with *low control* (little decision latitude). The assembly-line worker, the call-centre agent with a scripted screen and the waiter at a busy restaurant are in "high-strain" jobs; the busy senior executive, who can decide how and when to work, often is not. Jeffrey Johnson and Ellen Hall (1988) later added social support as a third dimension: isolated high-strain jobs are the worst of all.

When work stress becomes chronic, the result may be **burnout**. The term was introduced in 1974 by the psychologist **Herbert J. Freudenberger** (1926 to 1999), who saw it in himself and in fellow volunteers at a free clinic in New York. **Christina Maslach** (born 1946), whose research on human-service workers made the concept scientific, and Susan Jackson published the Maslach Burnout Inventory in 1981. It measures three dimensions: *emotional exhaustion* (feeling drained and used up), *depersonalisation* (a cynical, detached, even callous attitude towards the people one serves) and *reduced personal accomplishment* (feeling ineffective). Burnout is most common in the helping professions: nursing, medicine, teaching, social work, policing and counselling, where the emotional labour of Section 11.5 is part of the job. In ICD-11 (2019) the World Health Organisation describes burnout as an *occupational phenomenon*, not a medical condition, resulting from chronic workplace stress that has not been successfully managed, and characterised by exhaustion, mental distance or cynicism about one's job, and reduced professional efficacy.

CLINICAL LINK

Clinical psychologists are themselves at high risk of burnout. They spend their working days absorbing other people's distress, often with heavy caseloads and little control over the system they work in, which is Karasek's high-strain recipe. The protections are those this chapter describes: regular supervision and peer support (social support), realistic caseloads and some control over one's schedule (control), and deliberate attention to sleep, exercise and life outside work. Training programmes increasingly treat self-care as a professional skill rather than a luxury.

## Control and predictability

Why do some stressors do so much more damage than others of the same physical size? A series of animal experiments gave an answer that has run through stress research ever since.

**KEY STUDY: EXECUTIVE MONKEYS AND HELPLESS RATS (BRADY, 1958; WEISS, 1971)**

In 1958 **Joseph V. Brady** reported a famous experiment on "executive" monkeys. Pairs of monkeys received electric shocks; the "executive" in each pair could prevent shocks for both of them by pressing a lever at regular intervals, while its yoked partner could do nothing. The executives developed stomach ulcers, and some died; the yoked monkeys did not. The study was taken as proof that the burden of responsibility causes ulcers. But the executives had not been chosen at random: they were the monkeys that had learned the avoidance task fastest in a pretest, and these were probably the most emotionally reactive animals.

In 1971 **Jay M. Weiss** repeated the logic with proper controls, using rats in triads. One rat could avoid or turn off a shock by turning a wheel; a yoked rat received exactly the same shocks but had no control; a third rat received no shocks. The results reversed Brady's: the rats with control developed far *fewer* stomach ulcers than the helpless yoked rats. Weiss also found that rats that heard a warning tone before each shock developed fewer ulcers than rats shocked without warning. **Conclusion.** The same physical stressor does less damage when it is *controllable* and *predictable*. This is the same lesson as Seligman's learned helplessness experiments (Chapter 6).

Human research tells the same story. In the experiments David Glass and Jerome Singer (the same Singer as the two-factor theory) reported in *Urban Stress* (1972), people exposed to loud, *unpredictable* bursts of noise afterwards gave up sooner on insoluble puzzles and made more proofreading errors than people exposed to the same noise at predictable intervals. When participants were told they could stop the noise by pressing a button, the after-effects shrank, even though hardly anyone pressed it. Simply believing that one has control helped. In a field study, Ellen Langer and Judith Rodin (1976) gave residents on one floor of a nursing home small choices about their daily lives and a plant to look after; compared with residents on another floor whose lives were managed for them, they became more active and happier, and 18 months later fewer of them had died. Perceived control is a thread you will see again in secondary appraisal, hardiness and self-efficacy.

## Lazarus and Folkman's transactional model ●●

Every finding in this section points the same way. Life events matter less than how they are taken; hassles matter because of how they accumulate in a particular life; control and predictability change the impact of identical shocks. Richard Lazarus drew the conclusion in his appraisal theory of emotion (Section 11.3) and, with **Susan Folkman**, built it into a full theory of stress in *Stress, Appraisal, and Coping* (1984). Their **transactional model of stress** defines psychological stress as "a particular relationship between the person and the environment that is appraised by the person as taxing or exceeding his or her resources and endangering his or her well-being." Stress is a *transaction*: it exists only in the meeting of a particular person with a particular situation.

The model runs through a sequence of appraisals (Figure 11.6). In *primary appraisal* the person asks, "Is this relevant to my well-being, and how?" The event may be judged *irrelevant*, *benign-positive* or *stressful*, and a stressful event may be appraised as *harm or loss* (damage already done), *threat* (harm anticipated) or *challenge* (an opportunity for gain or growth, which brings eagerness rather than dread). In *secondary appraisal* the person asks, "What can I do about it?", evaluating his or her coping options and resources: physical (health and energy), social (people who can help), psychological (skills, beliefs, self-efficacy) and material (money,

time). Stress is greatest when the demands appraised in primary appraisal seem to outweigh the resources appraised in secondary appraisal. The person then copes (Section 11.10), and the results of coping, together with any new information, feed back into *reappraisal*, so that the whole transaction keeps changing.

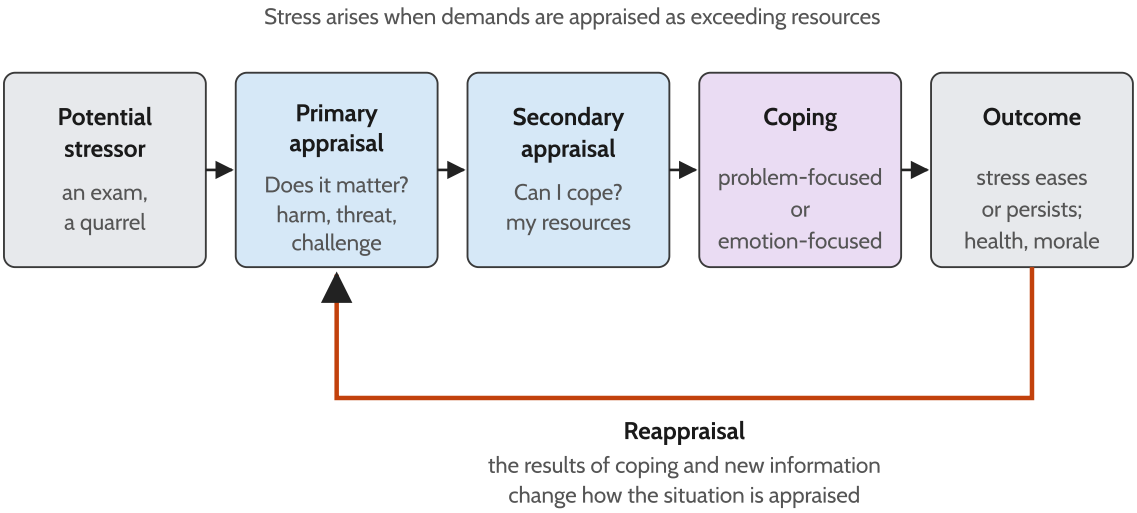


**Figure 11.6** Lazarus and Folkman's transactional model of stress (1984). A potential stressor becomes stressful only through primary appraisal (is it harm, threat or challenge?) and secondary appraisal (can I cope?). Coping follows, and its outcome feeds back through reappraisal (the line returning along the bottom). "Primary" and "secondary" name the questions asked, not a fixed order in time.

Push two people through the model and its power becomes clear. On the morning of the entrance examination, Ananya and Kiran sit in the same Mysuru hall with the same question paper. Ananya's primary appraisal is *threat*: "If I fail, my parents will be humiliated and my year is wasted." Her secondary appraisal is bleak: "I never finished revising developmental psychology, and nobody can help me now." Demands far exceed resources, and she is shaking before the first question. Kiran's primary appraisal is *challenge*: "This matters, but it is my chance to show what I know." His secondary appraisal is reassuring: "I have practised twenty past papers, I know my weak areas, and there is another attempt next year if I need it." He feels keyed up but focused, which is Selye's eustress. Then Ananya reads the first five questions and finds she knows every answer. Her reappraisal ("Maybe I am better prepared than I feared") lowers her stress, and she settles. Same stressor, different transactions, different stress. Later research by Jim Blascovich and his colleagues has shown that challenge and threat appraisals even produce different cardiovascular patterns: in challenge the heart pumps more blood and the blood vessels relax, in threat the vessels constrict.

Two features of the model are often misunderstood. First, "primary" and "secondary" do not mean first and second in time, or more and less important; the two appraisals interact continuously, and a good secondary appraisal ("I can handle this") can turn a threat into a challenge. Second, appraisals need not be slow or conscious. As Lazarus argued in his debate with Zajonc (Section 11.3), an appraisal can be an instant, automatic evaluation.

### Stress meets vulnerability: the diathesis-stress model ●●●

The transactional model explains why the same event is stressful for one person and not another. A related idea explains why the same amount of stress makes one person ill and leaves another well. The **diathesis-stress model** holds that psychological disorders develop when a *diathesis* (from the Greek for "disposition"), a pre-existing vulnerability, is triggered by *stress*. The diathesis may be genetic or biological (a family history of depression, an overreactive HPA axis), psychological (a negative way of thinking about oneself) or the result of early experience (childhood maltreatment). Neither ingredient is usually enough on its own: a person with a strong vulnerability may stay well in a protected life, and a person with little vulnerability may weather severe stress. The two trade off: the larger the diathesis, the less stress is needed to trigger a disorder. The idea was set out for schizophrenia by **Paul E. Meehl** (1920 to 2003) in 1962 and developed by Joseph Zubin and Bonnie Spring as the vulnerability-stress model in 1977; it is now applied to depression, anxiety disorders, PTSD and much else.

The diathesis-stress model is often set against single-cause models of abnormality, which examiners use as distractors. The *biological* model locates disorders in genes and brain chemistry alone; the *psychodynamic* model in unconscious conflicts; the *behavioural* model in learned experiences; the *cognitive* model in maladaptive thinking (Aaron Beck, Albert Ellis); the *humanistic-existential* model in blocked growth, lack of self-acceptance or failure to find meaning (Carl Rogers, Abraham Maslow, Rollo May); and the *sociocultural* model in social and cultural forces such as poverty, family structure and discrimination. The oldest view of all, the *supernatural* (demonological) model, blamed possession by spirits or divine punishment, and no modern model accepts it. The diathesis-stress model is the one that explicitly says a biological or psychological *predisposition* is set off by a *stressful situation*. Its broader cousin, the biopsychosocial model, appears in Section 11.10. (The models of abnormality are covered fully in the Disorders book.)

EXAM TRAP

**The SRRS measures life events.** Holmes and Rahe (1967) counted major life changes in life change units; death of a spouse scores 100 and marriage, the anchor, 50. It does not measure daily hassles (Kanner, Coyne, Schaefer and Lazarus, 1981), physiological reactions (Selye's tradition) or cognitive appraisal (Lazarus). Positive events such as marriage, vacations and Christmas count, because the scale measures change. Some revision notes credit Holmes and Rahe with "catastrophes and daily hassles" as types of stressor; that threefold scheme (cataclysmic, personal, background) is Lazarus and Cohen's (1977).

**Hassles beat life events** as predictors of symptoms and health. **Acute or chronic?** A pop quiz, an interview or a near miss is acute; HIV disease, cancer, marital discord, caregiving and poverty are chronic.

**Secondary appraisal** is the assessment of whether one has the resources to cope. Primary appraisal judges relevance and whether the event is harm-loss, threat or challenge. Phrases such as "initial threat evaluation", "coping resource assessment" or "situational capacity evaluation" sound plausible but are not Lazarus and Folkman's terms.

**Organisational stressors** are task demands, role demands, interpersonal demands, structure and leadership. Family and economic problems and the number of dependants are *individual* factors; political and economic uncertainty are *environmental* factors. Role conflict means incompatible expectations; role ambiguity means unclear ones.

**Control reduces damage.** Brady's "executive monkeys" (1958) seemed to show the opposite, but the executives were not randomly assigned; Weiss's properly controlled rats (1971) with control got fewer ulcers. Karasek: high demands plus *low* control produce the most strain.

**Diathesis-stress** means vulnerability *plus* stress. Options saying "genetic factors only", "environmental factors only", "learned experiences alone", "unconscious conflicts only" or "supernatural influences" are the distractors.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Walter Cannon named the body's emergency mobilisation the _____ response and coined the term homeostasis.
2. The three stages of Selye's general adaptation syndrome, in order, are alarm, _____ and exhaustion.
3. Selye called pleasant, beneficial stress _____.
4. In the HPA axis, the anterior pituitary releases _____, which makes the adrenal cortex secrete cortisol.
5. McEwen's term for the cumulative wear and tear of chronic stress is allostatic _____.
6. Taylor's term for the affiliative stress response more typical of females is _____.
7. On Holmes and Rahe's scale, the death of a spouse is worth 100 life change _____.
8. Kanner, Coyne, Schaefer and Lazarus (1981) found that daily _____ predicted symptoms better than major life events.
9. Karasek's model says strain is greatest in jobs with high demands and low _____.
10. In the transactional model, judging whether you have the resources to cope is _____ appraisal.

**Answers:** (1) fight-or-flight, (2) resistance, (3) eustress, (4) ACTH (adrenocorticotropic hormone), (5) load, (6) tend-and-befriend, (7) units, (8) hassles, (9) control (decision latitude), (10) secondary.

## 11.10 Coping, personality and health HIGH YIELD · 15 QS

Two people meet the same stressor and appraise it in much the same way, yet one stays well and the other falls ill. Part of the difference lies in what they do about it (coping), part in the people around them (social support), part in stable features of their personalities, and part in what stress does to the heart and the immune system. This final section follows that path, from the first attempt to cope to the cells of the immune system, and ends with what can be done to manage stress.

### What coping is: problem-focused and emotion-focused ●●●

Lazarus and Folkman (1984) defined **coping** as "constantly changing cognitive and behavioral efforts to manage specific external and/or internal demands that are appraised as taxing or exceeding the resources of the person." Notice three things about the definition. Coping is a *process*: it is what a person actually thinks and does in a particular encounter, and it changes as the encounter unfolds, rather than being a fixed trait. Coping is defined by *effort*, not by success: a strategy that fails is still coping. And coping may be aimed at the outside demand or at the inner turmoil it causes, which gives the most important distinction in the field.

**Problem-focused coping** is directed at the stressor itself: defining the problem, gathering information, generating and weighing solutions, making a plan, managing time, confronting the person responsible, or learning a new skill. **Emotion-focused coping** is directed at the emotional response to the stressor: reappraising the situation in a more positive

light, seeking sympathy and emotional support, accepting what cannot be changed, distancing oneself, humour, prayer, relaxation and distraction, but also venting, wishful thinking, denial and drinking. In a study of 100 middle-aged adults who described more than 1,300 stressful episodes, Folkman and Lazarus (1980) found that people used *both* kinds of coping in 98 percent of them. Problem-focused coping rose when people saw something that could be done, such as a difficulty at work, and emotion-focused coping rose when they saw the situation as one that had to be accepted, such as a health problem.

Which is better? Neither, in general. The best predictor of good outcomes is the *goodness of fit* between the strategy and the controllability of the stressor: change what can be changed, and manage your feelings about what cannot. A candidate who learns three weeks before the entrance examination that she is weak in statistics should use problem-focused coping (a revision plan, extra practice papers). The same candidate, the evening after the results are declared, can no longer change the outcome; emotion-focused coping (talking it over with a friend, reappraising the result as one attempt among several) is now what helps. Studying caregivers of partners dying of AIDS in the 1990s, Folkman found that even in the midst of chronic grief they reported positive emotions, which they drew from finding meaning in caregiving, revising their goals and noticing small good moments. She later called this **meaning-focused coping**: drawing on beliefs, values and goals to find benefit and purpose in a stressful situation that cannot be changed.

A second distinction cuts across the first. **Approach coping** turns towards the stressor and one's reactions to it (seeking information, planning, expressing feelings, reappraisal); **avoidance coping** turns away (denial, distraction, wishful thinking, withdrawal, substance use). Susan Roth and Lawrence Cohen (1986) reviewed the evidence, and a meta-analysis by Jerry Suls and Barbara Fletcher (1985) gave the most useful summary: avoidance can help in the short term, for example in the hours before an operation, when worrying about details only raises anxiety, but approach coping does better in the long run, and chronic avoidance is linked to poorer mental and physical health. Researchers measure these strategies with questionnaires such as Folkman and Lazarus's Ways of Coping Questionnaire and the COPE Inventory of Charles Carver, Michael Scheier and Jagdish Weintraub (1989). Table 11.9 classifies some common strategies.

Table 11.9 Classifying coping strategies, with an exam candidate as the example.

| Type | Aimed at | Examples | Most useful when |
|---|---|---|---|
| Problem-focused (approach) | The stressor | Making a revision timetable; asking the teacher about weak topics; taking a mock test | Something can still be done |
| Emotion-focused, approach | One's feelings, facing them | Reappraising the exam as a challenge; talking to a friend; acceptance; relaxation | The stressor cannot be changed, or while waiting |
| Emotion-focused, avoidant | One's feelings, escaping them | Denial ("it will be fine"); endless scrolling; drinking; sleeping it off | Briefly, at most; harmful if chronic |
| Meaning-focused | Goals and values | "Whatever the result, I have learned how to study"; revising career goals | Lasting, uncontrollable stress or loss |

Where do defence mechanisms fit? Sigmund Freud introduced the idea, and his daughter Anna Freud systematised it in her book *The Ego and the Mechanisms of Defence* (1936), describing defence mechanisms as *unconscious*, automatic strategies that protect the ego from anxiety by distorting or denying reality. Coping, in Lazarus's sense, is *conscious*, deliberate and

flexible. The two overlap at the edges, and George Vaillant, in *Adaptation to Life* (1977), arranged defences in a hierarchy from immature (such as projection and acting out) to mature (sublimation, humour, altruism, anticipation and suppression), and found that men who relied on mature defences enjoyed better physical and mental health over decades. The defences examined most often are easy to tell apart by their key words: *displacement* redirects a feeling from its true target to a safer one (a woman angry with her husband shouts at her child); *sublimation* channels an unacceptable impulse into a socially valued activity (an angry young man takes up kickboxing); *repression* is the involuntary banishing of a painful memory or feeling from awareness, whereas *suppression* is deliberate; *projection* attributes one's own unacceptable feelings to someone else; *isolation of affect* keeps the idea but strips away its feeling (a patient calmly describes a gruesome fantasy); *rationalisation* offers a plausible excuse; *denial* refuses to acknowledge a painful reality at all (insisting that a frightening diagnosis must be a mistake); and *reaction formation* turns a feeling into its opposite. Notice the difference between denial and isolation: the patient who calmly describes a gruesome fantasy is not denying the idea, which he reports in full, but has cut it off from its feeling. The full list is in Table 10.11 and the Personality book.

## Social support ●●

Of all the resources that people bring to stress, the most consistently protective is other people. **Social support** is the perception or experience that one is loved and cared for, valued and esteemed, and part of a network of mutual help. James House (1981) distinguished four kinds: *emotional* support (empathy, love, trust and caring), *instrumental* or tangible support (money, a lift to the hospital, help with the cooking), *informational* support (advice, suggestions and information) and *appraisal* support (feedback that helps a person evaluate himself or herself and the situation). An important finding is that *perceived* support, the belief that help would be there if needed, predicts well-being more consistently than *received* support, the help actually given. Highly visible help can even backfire, by making the receiver feel indebted or incompetent.

How does support protect? Sheldon Cohen and Thomas Wills (1985) reviewed the evidence for two answers. The *main-effect* (or direct-effect) hypothesis says that support is good for people whether or not they are under stress, because belonging, a sense of being valued and encouragement towards healthy habits are beneficial in themselves. The **stress-buffering hypothesis** says that support protects mainly when stress is high, by improving secondary appraisal ("I am not alone with this") and by supplying coping resources. Both turned out to be true: measures of how integrated a person is in a social network tend to show main effects, and measures of perceived support that matches the person's needs tend to show buffering.

The health effects are large. In the Alameda County study in California, Lisa Berkman and Leonard Syme (1979) followed nearly 7,000 adults for nine years. People with the fewest social ties (marriage, friends and relatives, membership of religious and other groups) had roughly two to three times the death rate of those with the most, even after allowance was made for their initial health and health habits. A meta-analysis by Julianne Holt-Lunstad and her colleagues (2010), combining 148 studies of more than 300,000 people, found that people with stronger social relationships had about a 50 percent greater likelihood of survival over the periods studied, an effect comparable in size to that of giving up smoking. In Sheldon Cohen's laboratory, people with more diverse social networks were less likely to develop colds after being deliberately exposed to a cold virus (1997). Support has a darker twin, too:

relationships full of conflict are themselves a major source of stress, a point made clear in the Indian joint family, which can provide abundant support and abundant daily hassles at the same time.

The absence of support is loneliness, and the sociologist **Robert S. Weiss**, in *Loneliness: The Experience of Emotional and Social Isolation* (1973), showed that it comes in two forms. **Emotional isolation** (emotional loneliness) is the absence of a close, intimate attachment to one specific person, such as a partner or a confidant; it produces a sense of utter aloneness that no number of acquaintances can cure. **Social isolation** (social loneliness) is the absence of a wider network of friends, colleagues and community, a place to belong; it produces boredom and a feeling of being on the margins, and a loving partner alone does not cure it. A newly married woman who moves to a new city with her husband may have no emotional isolation but severe social isolation; a widower surrounded by kind relatives may have the reverse. Neither is the same as feeling incompetent or having low self-esteem, which are problems of self-evaluation rather than of attachment.

### Hardiness, optimism, resilience and sense of coherence ●●

Some people seem to stay healthy under stress that flattens others. Four related ideas try to say what they have.

The first is **hardiness**, introduced by **Suzanne C. Kobasa** in 1979. Kobasa studied middle- and upper-level executives at a large public utility in Chicago (Illinois Bell Telephone) and picked out two groups who had all experienced many stressful life events in the previous three years: those who had fallen ill and those who had stayed well. The healthy executives scored higher on three attitudes, often remembered as the three Cs. *Commitment* is a sense of involvement and purpose in work, family and life, rather than alienation. *Control* is the belief that one can influence events, rather than being powerless. *Challenge* is the tendency to see change as a normal part of life and an opportunity for growth, rather than a threat. A later prospective study by Kobasa, Salvatore Maddi and Stephen Kahn (1982) found that hardiness predicted who would stay healthy in the years that followed. In transactional terms, hardy people appraise stressors as challenges (primary appraisal) and believe they can act (secondary appraisal), and so they tend to cope actively. The main criticism is that hardiness scales overlap heavily with low neuroticism, so part of what they measure may simply be a lack of negative emotionality.

The second idea is optimism, studied in two traditions. **Michael F. Scheier** and **Charles S. Carver** (1985) defined **dispositional optimism** as a generalised expectancy that good things rather than bad will happen, and measured it with the Life Orientation Test. Optimists use more problem-focused coping and positive reappraisal and less denial, and in one study by Scheier and colleagues (1989), optimistic men recovered faster after coronary bypass surgery and returned sooner to normal life. The second tradition began with helplessness. **Martin E. P. Seligman** (born 1942) and **Steven F. Maier** showed in 1967 that dogs exposed to inescapable shock later failed to escape shock they could have avoided, the phenomenon of learned helplessness (Chapter 6). When the theory was reformulated in 1978 in terms of how people explain bad events, it led to the idea of **explanatory style**, a person's habitual way of explaining the causes of events. A *pessimistic* explanatory style explains bad events as internal, stable and global ("It's me, it will always be like this, and it ruins everything"); an *optimistic* style explains them as external, unstable (temporary) and specific ("That paper was unusually hard,

it was one bad day, and it is only one subject"). Christopher Peterson, Seligman and George Vaillant (1988) found that the explanatory style of Harvard graduates at about age 25 predicted their physical health between 45 and 60. In *Learned Optimism* (1990) Seligman argued that pessimism can be unlearned, by catching and disputing pessimistic explanations, a technique he adapted from Albert Ellis; he called the result **learned optimism**. One caution: *unrealistic* optimism, the belief that bad things happen to other people, can lead people to ignore genuine health risks.

The third idea is **resilience**: the process of adapting well, or bouncing back, in the face of serious adversity, trauma or risk. Its classic evidence comes from the Kauai Longitudinal Study of **Emmy E. Werner** (1929 to 2017) and Ruth Smith, who followed all 698 children born in 1955 on the Hawaiian island of Kauai from before birth into middle age. About a third of the children were at high risk, born into poverty, family discord, parental alcoholism or mental illness, or after a difficult birth. Yet about one in three of these high-risk children grew into competent, confident and caring adults. What protected them was ordinary: an easy-going temperament, at least one close bond with a caring adult (often a grandparent, aunt or teacher rather than a parent), competence in some valued activity, faith, and a supportive community. Ann Masten (2001) summed up such findings by calling resilience "ordinary magic": it comes from ordinary human adaptive systems, not from rare gifts. George Bonanno (2004) showed that after bereavement or trauma, a resilient course, with a short period of disturbance and a steady return to normal functioning, is the most common outcome, not the exception. Resilience is not invulnerability, and it can be strengthened.

The fourth idea came from an unusual question. In a 1970 study of the menopause among women in Israel, the medical sociologist **Aaron Antonovsky** (1923 to 1994) noticed that some of the women had survived Nazi concentration camps, and that, astonishingly, a substantial minority of these survivors (about 29 percent, against about 51 percent of a comparison group who had not been in the camps) were in good emotional health. Medicine, he argued, had always asked what makes people ill (*pathogenesis*); he proposed instead to ask what keeps people healthy, which he called **salutogenesis** (from the Latin *salus*, health). His answer, in *Health, Stress and Coping* (1979) and *Unraveling the Mystery of Health* (1987), was the **sense of coherence** (SOC): a pervasive, enduring yet dynamic feeling of confidence that life is *comprehensible* (events are structured, predictable and explicable), *manageable* (resources are available to meet its demands) and *meaningful* (its demands are challenges worth investing in). Meaningfulness, Antonovsky thought, is the motivational heart of the three. People draw on *generalised resistance resources*, such as money, knowledge, social support and a stable culture, and a strong sense of coherence lets them use these resources well. Two further personal resources, self-efficacy (Albert Bandura, 1977) and an internal locus of control (Julian Rotter, 1966), work in much the same way and are covered in Chapter 10.

EXAM TRAP

**Three-part lists that are easy to swap.** Hardiness (Kobasa, 1979): commitment, control, challenge. Sense of coherence (Antonovsky, 1979 and 1987): comprehensibility, manageability, meaningfulness. Diener's subjective well-being: life satisfaction, positive affect, negative affect. Do not give "confidence" or "coherence" as one of Kobasa's Cs.

**Learned helplessness is Seligman and Maier** (1967), later reformulated by Abramson, Seligman and Teasdale (1978). It is not Aaron Beck (cognitive therapy and the negative cognitive triad), Harry Stack Sullivan (the interpersonal theory of psychiatry) or Daniel Kahneman (judgement heuristics and, with Amos Tversky, prospect theory). Dispositional optimism and the Life Orientation Test are Scheier and Carver (1985); learned optimism and explanatory style are Seligman.

**Emotional isolation** (Weiss, 1973) is the lack of a deep attachment to one specific person; the lack of a circle of friends is **social isolation**.

## Personality and heart disease: Types A, B, C and D ●●●

In the mid-1950s two San Francisco cardiologists, **Meyer Friedman** (1910 to 2001) and **Ray H. Rosenman** (1920 to 2013), had the chairs in their waiting room reupholstered. The upholsterer remarked that only the front edges of the seats were worn out, as if their patients spent their waiting time perched on the edge, ready to spring up. Friedman later said that he did not grasp the significance of the remark until years afterwards, but he and Rosenman did come to notice that many of their coronary patients shared a distinctive style: hurried, impatient, intensely competitive and quick to anger. In 1959, in the *Journal of the American Medical Association*, they reported that men showing this pattern had higher blood cholesterol, faster blood clotting and more coronary heart disease than men who did not.

They called it the **Type A behaviour pattern**, and they described three core components. The first is *competitive achievement striving*: a relentless drive to accomplish more and more, and to win. The second is an exaggerated *sense of time urgency* ("hurry sickness"): doing two things at once, finishing other people's sentences, fuming in queues and traffic. The third is *aggressiveness and hostility*, easily aroused. The opposite pattern, **Type B**, is relaxed, patient and easy-going, unhurried and less driven by competition, although not necessarily less ambitious or less successful. Friedman and Rosenman assessed the pattern mainly with the Structured Interview, in which the interviewer deliberately provokes impatience (by speaking slowly, pausing or interrupting) and codes the person's speech and behaviour as much as the content of the answers; a questionnaire, the Jenkins Activity Survey, was developed later. Their popular book was *Type A Behavior and Your Heart* (1974).

### KEY STUDY: THE WESTERN COLLABORATIVE GROUP STUDY (1960 TO 1969)

**Question.** Does Type A behaviour predict heart disease in men who are healthy at the start? **Method.** In 1960 and 1961 Rosenman, Friedman and their colleagues classified 3,154 healthy employed men, aged 39 to 59 and working in ten Californian companies, as Type A or Type B using the Structured Interview, and followed them for eight and a half years. **Results.** Type A men developed coronary heart disease about twice as often as Type B men, and the difference remained after allowing for the standard risk factors of cholesterol, blood pressure and smoking. In 1981 a review panel of the US National Heart, Lung and Blood Institute accepted Type A behaviour as an independent risk factor for coronary heart disease. **The twist.** Later large studies, notably the Multiple Risk Factor Intervention Trial (1985), found no link between global Type A behaviour and heart disease, and meta-analyses showed that the association had weakened or vanished in later research. The question became: is one ingredient of Type A doing the damage?

The answer that emerged in the 1980s was **hostility**, in particular *cynical hostility*: a mistrustful, cynical view of other people combined with a readiness to become angry. When the Western Collaborative Group interviews were reanalysed component by component, first by Karen Matthews and her colleagues (1977) and later by Michael Hecker and his colleagues (1988), "potential for hostility" stood out as the strongest predictor of heart disease; in the 1988 analysis it still predicted heart disease after the standard risk factors and the other Type A components were taken into account, and it accounted for much of the link between global Type A and heart disease. John Barefoot, Grant Dahlstrom and **Redford B. Williams** (1983) traced 255 doctors who had taken the MMPI as medical students 25 years earlier; those with high scores on its Cook-Medley Hostility scale had developed coronary heart disease several times more often, and had a higher death rate from all causes. Williams popularised the finding in *Anger Kills* (1993), and a meta-analysis by Yoichi Chida and Andrew Steptoe (2009) confirmed that anger and hostility predict coronary heart disease in both healthy and already ill people. How might hostility harm the heart? Hostile people show larger and more frequent surges of blood pressure and stress hormones in everyday encounters; they tend to have poorer health habits; and they drive away the social support that would buffer their stress. The good news is that the pattern can change: in Friedman's Recurrent Coronary Prevention Project (1986), heart attack survivors who received counselling to reduce their Type A behaviour had significantly fewer second heart attacks over the next four and a half years than patients who received only standard cardiological advice.

Two later "types" extended the idea to other illnesses. A **Type C** (cancer-prone) pattern was described in the 1980s, most influentially by the psychologist **Lydia Temoshok** from her work with patients with melanoma: cooperative, appeasing, unassertive and patient, compliant with authority and, above all, inclined to suppress negative emotions, especially anger. (Some revision notes describe Type C people as introverted, detail-minded perfectionists; the research construct is the emotionally suppressed, unassertive style.) The evidence that personality causes cancer is weak: large prospective studies have not found a reliable link, and the most striking supporting data, from Hans Eysenck and Ronald Grossarth-Maticek, were judged unsafe by an enquiry at King's College London in 2019, after which a number of the papers were retracted. Coping style may, however, affect quality of life after a diagnosis.

The most important recent addition is the **Type D personality** ("D" for distressed), introduced in the mid-1990s by the Belgian psychologist **Johan Denollet**. Type D is the *joint* pres-

ence of two traits. **Negative affectivity** is the tendency to experience negative emotions, such as worry, gloom, irritability and tension, across time and situations. **Social inhibition** is the tendency to inhibit the expression of those emotions and behaviours in social interaction, for fear of disapproval. A Type D person, in other words, feels bad a great deal and keeps it to himself or herself. In Denollet's best-known study, published in *The Lancet* in 1996, 303 patients with coronary heart disease were followed for six to ten years: 27 percent of the Type D patients died, against 7 percent of the others, and Type D predicted death independently of the severity of the heart disease. Type D is measured with the 14-item DS14 scale. Later and larger studies have found weaker effects than the early ones, and the debate about how much Type D adds to depression and anxiety continues, but it remains the personality type most often examined in cardiac psychology today. Table 11.10 compares the four types.

Table 11.10 Personality "types" and health.

| Type | Proposed by | Core features | Health link and status |
|---|---|---|---|
| Type A | Friedman and Rosenman (1959) | Competitive achievement striving, time urgency, impatience, hostility | Coronary heart disease; the global pattern has weak support, hostility is the toxic component |
| Type B | Friedman and Rosenman (1959) | Relaxed, patient, easy-going, less hurried | Lower coronary risk in early studies |
| Type C | Temoshok and others (1980s) | Cooperative, unassertive, compliant, suppresses negative emotion | Cancer proneness claimed; evidence weak |
| Type D | Denollet (mid-1990s) | Negative affectivity plus social inhibition | Poorer prognosis in heart disease; effect smaller in later studies |

EXAM TRAP

**Type A versus hostility.** RML 2024 asked which factor is "most directly linked to a higher risk for coronary heart disease", offering Type A personality, Type B personality, hostility and self-efficacy, and keyed **Type A**. That is the classic textbook answer and the one to give when the question is about personality types. Research since the 1980s, however, shows that it is **hostility**, one component of Type A, that carries most of the risk. If a stem asks for the "toxic", "critical" or "most predictive component" of Type A, the answer is hostility (cynical hostility, anger).

**Type A characteristics** are competitiveness, impatience (time urgency) and hostility; any option containing "non-competitiveness" or "lack of aggression" is wrong. Introversion and extraversion are not coronary risk factors, and Type B is the low-risk pattern.

**Type D** is negative affectivity *plus* social inhibition (Denollet). "Hostility and competitiveness" is Type A; "an optimistic and resilient outlook" describes hardiness or dispositional optimism, which protect; "perfectionism and rigidity" belongs to obsessive-compulsive personality descriptions. Do not confuse Denollet's Type D personality with Main and Solomon's Type D (disorganised) attachment in infants (Developmental book), or the "D" with depression.

## Psychoneuroimmunology: how stress gets under the skin ●●

For most of the twentieth century medicine worked with a *biomedical model*: disease is a biological malfunction, to be explained by germs, genes and chemistry. In 1977, in the journal *Science*, the American psychiatrist **George L. Engel** (1913 to 1999) argued that this model was too narrow and proposed the **biopsychosocial model**: health and illness arise from the in-

teraction of *biological* factors (genes, physiology, infection), *psychological* factors (emotions, beliefs, coping, behaviour) and *social* factors (family, relationships, social class, culture, access to care). The model is now the framework of health psychology and of clinical psychology alike. Within it, disorders in which psychological factors play a causal role are often called *psychosomatic disorders*. Pinel and Barnes point out that so many illnesses, from heart disease and asthma to skin disorders, have turned out to be affected by stress that it is more sensible to think of most medical disorders as psychosomatic to some degree. Gastric ulcers, the classic example, show the point: most are associated with infection by the bacterium *Helicobacter pylori*, but many infected people never develop ulcers, and stress appears to increase the stomach's vulnerability.

The most surprising link between mind and body runs through the immune system. **Psychoneuroimmunology** (PNI) is the study of interactions among psychological factors, the nervous and endocrine systems, and the immune system. A brief reminder of the immune system helps. The *innate* immune system reacts quickly and in the same way to all invaders, through inflammation and cells such as phagocytes and natural killer cells. The *adaptive* immune system is slower but specific and has a memory: its lymphocytes, the T cells and B cells, target particular antigens, and B cells produce antibodies. Until the 1970s the immune system was thought to work on its own, untouched by the brain. The experiment that changed this began as a study of taste aversion.

### KEY STUDY: CONDITIONED IMMUNOSUPPRESSION (ADER AND COHEN, 1975)

**Background.** The psychologist **Robert Ader** (1932 to 2011), at the University of Rochester, was studying conditioned taste aversion (Chapter 6). Rats drank saccharin-flavoured water and were then injected with cyclophosphamide, a drug that causes nausea, so that they learned to avoid the sweet taste. When he went on giving the rats saccharin water without the drug, to see how long the aversion would last, some of them unexpectedly sickened and died, and the more saccharin they had drunk during conditioning, the more likely they were to die. Ader knew that cyclophosphamide also suppresses the immune system. Could the taste alone have become a conditioned stimulus for immune suppression? **Method.** With the immunologist **Nicholas Cohen**, he paired saccharin with cyclophosphamide, then injected the rats with an antigen (sheep red blood cells) and gave some of them saccharin alone. **Results.** Rats re-exposed to the saccharin produced fewer antibodies than control rats. **Conclusion.** The immune system can be classically conditioned, so the brain must be able to influence it. Ader and Cohen's paper of 1975 is usually taken as the founding study of psychoneuroimmunology, a name Ader helped establish with his edited book *Psychoneuroimmunology* (1981).

Evidence in people soon followed. **Janice K. Kiecolt-Glaser** and her husband, the immunologist Ronald Glaser, at Ohio State University, found that the activity of medical students' natural killer cells fell during examinations (1984). In a study published in 1995, they made a small, standard wound on the arm of women who were caring for a relative with dementia and of matched women who were not; the caregivers' wounds took about nine days, or 24 percent, longer to heal. Later they found that a hostile discussion between married partners was enough to slow the healing of such wounds. **Sheldon Cohen**, David Tyrrell and Andrew Smith (1991) measured psychological stress in 394 healthy volunteers, then gave them nasal drops containing one of five common cold viruses (a further 26 volunteers received plain saline drops) and kept them in quarantine. The higher the stress score, the more likely a per-

son was to develop a cold, in a smooth dose-response pattern. Elissa Epel and her colleagues (2004) found that mothers with the highest levels of perceived stress, many of them caring for a chronically ill child, had telomeres (the protective caps on chromosomes that shorten with age) as short as those of women about a decade older.

Does stress, then, simply weaken immunity? Not quite. A meta-analysis of about 300 studies by Suzanne Segerstrom and Gregory Miller (2004) found that the answer depends on the kind of stress. *Acute* stressors lasting minutes, such as public speaking or a mental arithmetic task, actually boosted parts of the innate immune system, as if preparing the body for injury; this is the biological face of Selye's eustress. *Chronic* stressors, such as caregiving, unemployment, disability or trauma, suppressed immune function, above all the adaptive (cellular and antibody) system, and the longest-lasting stressors suppressed almost every measure of immunity. Stress acts through two kinds of pathway. The *direct* pathway is physiological: immune cells carry receptors for cortisol, adrenaline and noradrenaline, so the HPA and SAM axes speak to them directly, and chronic stress also promotes a long-lasting, low-grade inflammation, driven by chemical messengers called cytokines, that is now linked to heart disease, diabetes and depression. The *indirect* pathway is behavioural: stressed people sleep less, smoke and drink more, exercise less, eat worse and delay seeing a doctor. A caution runs through this literature. Most human studies are correlational, and people under stress may simply notice and report more symptoms, which is why experiments such as Cohen's cold-virus study carry so much weight.

## Health beliefs and health behaviour ●●

Because so much of stress's effect on health runs through behaviour, health psychologists have built models to predict whether people will act to protect their health: get screened, take medicine, stop smoking, exercise. Three are examined most often, and they are frequently offered as wrong options in questions about Selye or life events (Table 11.11).

The **health belief model** was developed in the 1950s by social psychologists in the United States Public Health Service, among them Godfrey Hochbaum and **Irwin M. Rosenstock**, to explain why so few people took up free screening for tuberculosis. It says that people act to protect their health when they believe they are susceptible to a condition (*perceived susceptibility*), that it would be serious (*perceived severity*), that the recommended action would help (*perceived benefits*) and that its costs are acceptable (*perceived barriers*), and when something prompts them (*cues to action*, such as a symptom or a reminder); self-efficacy was added in 1988. **Protection motivation theory**, proposed by Ronald W. Rogers in 1975 to explain how fear appeals work and revised in 1983, says that motivation to protect oneself depends on a *threat appraisal* (how severe the danger is and how vulnerable one is) and a *coping appraisal* (whether the recommended action works, which is response efficacy, and whether one can carry it out, which is self-efficacy). The **theory of planned behaviour** of **Icek Ajzen** (1985, 1991) says that behaviour follows from intention, and intention from three factors: the person's *attitude* towards the behaviour, the *subjective norm* (perceived social pressure from important others) and *perceived behavioural control*, a concept very close to self-efficacy (the theory is taught with attitudes in the Social book). The stages-of-change, or transtheoretical, model of Prochaska and DiClemente is in Chapter 10.

Table 11.11 Models of health behaviour.

| Model | Proposed by | Key components |
|---|---|---|
| Health belief model | Hochbaum, Rosenstock and colleagues (1950s) | Perceived susceptibility, severity, benefits and barriers; cues to action; later self-efficacy |
| Protection motivation theory | Rogers (1975; revised 1983) | Threat appraisal (severity, vulnerability) and coping appraisal (response efficacy, self-efficacy) |
| Theory of planned behaviour | Ajzen (1985, 1991) | Attitude, subjective norm and perceived behavioural control, leading to intention |
| Transtheoretical model | Prochaska and DiClemente (1980s) | Precontemplation, contemplation, preparation, action, maintenance |

## Managing stress

If stress is a transaction between demands, appraisals and the body, it can be managed at each point: by changing the stressor (problem solving, time management, assertiveness), by changing the appraisal (cognitive restructuring) and by calming the body (relaxation, meditation, exercise). The techniques below are the ones most often named in examinations; their use in therapy is covered in the Clinical book.

*Relaxation training*. The American physician **Edmund Jacobson** (1888 to 1983) noticed in the 1920s that anxious patients carried chronic tension in their muscles, and he reasoned that a relaxed body cannot also be an anxious one. His **progressive muscle relaxation** (PMR), set out in *Progressive Relaxation* (1929), teaches people to tense and then release one muscle group after another (hands, arms, face, neck, shoulders, chest, stomach, legs) while attending to the contrast between tension and relaxation, until they can relax deeply at will. Joseph Wolpe (1958) used a shortened version as the response incompatible with anxiety in systematic desensitisation, in which clients rate their anxiety on a 0 to 100 Subjective Units of Distress (SUD) scale as they work up a graded fear hierarchy. The cardiologist **Herbert Benson** (1935 to 2022), after studying practitioners of Transcendental Meditation, described the **relaxation response** in a 1975 book of that name: an integrated physiological state that is the opposite of fight or flight, with lower heart rate, breathing rate, oxygen consumption and blood pressure. It can be produced, Benson said, by four simple elements: a quiet environment, a mental device (a word or sound repeated silently), a passive attitude and a comfortable position.

*Biofeedback*. In **biofeedback** a person is given continuous information, as a tone, a light or a moving line on a screen, about a bodily process that is normally outside awareness, such as muscle tension (recorded by electromyography), skin temperature, skin conductance, heart rate, blood pressure or brain waves (neurofeedback), so that he or she can learn to control it. The idea drew on Neal Miller's reports in the 1960s that autonomic responses such as heart rate could be operantly conditioned in rats, results that later proved hard to repeat. Biofeedback helps with tension headache, migraine (by learning to warm the hands) and some other stress-related conditions, although it is often no more effective than relaxation training alone.

*Meditation and mindfulness*. Concentrative meditation focuses attention on a single object, such as a mantra or the breath; Transcendental Meditation, brought to the West by Maharishi Mahesh Yogi, is the best-known example. **Mindfulness** is the practice of paying attention, on purpose and without judgement, to experience in the present moment, an idea drawn from Buddhist *vipassana* meditation. In 1979 **Jon Kabat-Zinn** (born 1944) founded an eight-week Mindfulness-Based Stress Reduction programme at the University of

Massachusetts Medical School, and mindfulness-based cognitive therapy (Zindel Segal, Mark Williams and John Teasdale, 2002) later adapted it to prevent relapse in depression. Meta-analyses find moderate benefits for anxiety, depression and pain. A caution about evidence applies here, too: a survey showing that regular meditators report less stress than non-meditators cannot show that meditation *reduces* stress, because calmer people may be the ones who choose to meditate; only experiments with random assignment can establish cause (Research Methods book).

> **INDIAN CONTEXT**
>
> Many of the techniques in this subsection have Indian roots. Patanjali's *Yoga Sutras* describe an eightfold path whose limbs include *asana* (posture), *pranayama* (regulation of the breath), *dharana* (concentration) and *dhyana* (meditation); slow, controlled breathing is one of the quickest ways to shift the autonomic balance from sympathetic towards parasympathetic activity. Transcendental Meditation came from an Indian teacher, and the *vipassana* tradition that inspired mindfulness was revived and taught worldwide from India by S. N. Goenka. Indian research centres, including NIMHANS in Bengaluru, have run clinical trials of yoga-based breathing for depression and anxiety, and yoga is now part of many hospital and workplace stress-management programmes.

*Exercise*. Regular aerobic exercise lowers resting heart rate and blood pressure, reduces the physiological reaction to later stressors, improves sleep and mood, and is an effective treatment for mild to moderate depression. It also offers distraction and a sense of mastery.

*Cognitive restructuring and stress inoculation*. Because stress depends on appraisal, changing appraisals is one of the most powerful tools. **Cognitive restructuring** means identifying distorted, stress-producing thoughts and replacing them with more realistic ones; it is the core of Albert Ellis's rational emotive behaviour therapy (with its ABC model: an *activating event* leads to emotional *consequences* only through the *beliefs* held about it) and of Aaron Beck's cognitive therapy, and it is reappraisal (Section 11.7) made systematic. **Donald Meichenbaum** (born 1940) combined it with skills training in **stress inoculation training**, developed in the 1970s and set out in his book *Stress Inoculation Training* (1985). The name comes from the logic of vaccination: people are exposed to manageable doses of stress while they build their defences. It has three phases. In the *conceptualisation* (education) phase, the client learns how stress works and how his or her own appraisals and self-statements feed it. In the *skill acquisition and rehearsal* phase, the client learns and practises coping skills: relaxation, problem solving and coping self-statements such as "One step at a time; I can handle this." In the *application and follow-through* phase, the client uses the skills in imagined, role-played and then real, gradually harder situations, with planning to prevent relapse.

CLINICAL LINK

A clinical psychologist assessing a stressed client works through the same transactional map. What are the stressors, acute and chronic? How does the client appraise them, and with what explanatory style? What resources are available: support, money, skills, self-efficacy? Which coping strategies are being used, and do they fit the controllability of the problem? Is there a vulnerability, such as a history of depression, that makes this stress more dangerous? Intervention then targets the weakest link: problem-solving training for the controllable stressors, cognitive restructuring for the threatening appraisals, relaxation or mindfulness for the over-aroused body, and help in mobilising social support. Asking "What do you do when you are under stress?" is one of the most informative questions in any intake interview.

EXAM TRAP

**Who gave which technique?** Progressive muscle relaxation: Edmund Jacobson (1920s; *Progressive Relaxation*, 1929). Systematic desensitisation and the SUD scale: Joseph Wolpe, who borrowed Jacobson's relaxation. Relaxation response: Herbert Benson (1975). Mindfulness-Based Stress Reduction: Jon Kabat-Zinn (1979). Stress inoculation training: Donald Meichenbaum, with three phases (conceptualisation, skill acquisition and rehearsal, application and follow-through). Biofeedback gives information *about one's own body*; it is not a drug or a form of hypnosis.

**The biopsychosocial model** (George Engel, 1977) integrates biological, psychological and social factors. "Biological, psychological and cultural" is the trap; culture is part of the social.

**Psychoneuroimmunology began with Ader and Cohen's conditioned immunosuppression** in rats (1975), using saccharin and cyclophosphamide. Acute stress tends to boost innate immunity; chronic stress suppresses immunity (Segerstrom and Miller, 2004).

**Health behaviour models.** The theory of planned behaviour (Ajzen) has attitude, subjective norm and perceived behavioural control (an option listing "attitude, subjective norms and self-efficacy" is the keyed answer, because perceived control and self-efficacy are close). The health belief model is about perceived susceptibility, severity, benefits and barriers; protection motivation theory about threat and coping appraisal. None of them describes stages of alarm, resistance and exhaustion.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Making a study timetable to deal with an approaching examination is _____-focused coping.
2. Cohen and Wills's idea that social support protects mainly when stress is high is the stress-_____ hypothesis.
3. Weiss called the lack of a close attachment to one specific person _____ isolation.
4. Kobasa's hardiness consists of commitment, control and _____.
5. Antonovsky's sense of coherence has three components: comprehensibility, manageability and _____.
6. Friedman and Rosenman's coronary-prone pattern of competitiveness, time urgency and hostility is the Type _____ behaviour pattern.
7. Denollet's Type D personality combines negative affectivity with social _____.
8. Ader and Cohen showed in 1975 that immune suppression can be _____ conditioned.
9. Engel's model of health and illness, proposed in 1977, is the _____ model.
10. Tensing and releasing one muscle group after another is Jacobson's progressive muscle _____.

**Answers:** (1) problem, (2) buffering, (3) emotional, (4) challenge, (5) meaningfulness, (6) A, (7) inhibition, (8) classically, (9) biopsychosocial, (10) relaxation.

## The case of the Capilano bridge, revisited

Go back to the men swaying above the Capilano River in 1974, and you can now say a great deal about what was happening to them. As each man stepped onto the bridge, his low road from thalamus to amygdala flagged the drop as dangerous before he had thought about it, and his sympathetic nervous system and adrenal medulla, Cannon's fight-or-flight machinery, set his heart pounding and his palms sweating. James would have said that his fear simply *was* the feeling of those changes; Cannon would have said that feeling and arousal rose together. Then a smiling woman appeared with a questionnaire. Here Schachter and Singer's two-factor theory does the work: a man aroused for one reason found a second, attractive explanation standing in front of him, and some of the arousal was labelled as attraction. Dutton and Aron did not rest on one field study, which had an obvious weakness (perhaps the men who choose a frightening bridge are simply bolder). In a laboratory experiment reported in the same paper, men who expected to receive strong electric shocks found a female confederate more attractive than men who expected only mild ones. The misattribution of arousal, and Zillmann's excitation transfer, have been part of emotion theory ever since.

The bridge also illustrates the second half of the chapter. The crossing was an acute stressor, a few minutes of alarm reaction followed by recovery, and for many of the men it was probably eustress, the kind of thrill that tourists pay to experience. Whether a man felt threat or challenge depended on his appraisal and on his secondary appraisal of whether he could cope with the sway, exactly as Lazarus and Folkman would predict. Solomon's opponent-process theory would add that the relief on reaching solid ground was the b-process, and that a man

who crossed every day would come to feel less fear and more exhilaration. And if the bridge had been a permanent condition of life, with no far side to reach, the HPA axis, allostatic load and the stage of exhaustion would have taken over from the brief, harmless surge that made a stranger's telephone number so appealing.

## Chapter summary

**11.1 What is an emotion?** An emotion is a brief, whole-person response to a significant event, with three components: physiological arousal, expressive behaviour and conscious experience (modern theorists add appraisal and action tendency). A feeling is the subjective component; a mood is longer, weaker and often objectless; affect is the umbrella term for good or bad states and, clinically, the observable expression of emotion (rated for quality, range and reactivity in the mental status examination); temperament is the inborn style of emotional reactivity. Basic-emotion lists include Watson's three (fear, rage, love), Tomkins's nine affects (1962), Izard's ten (differential emotions theory), Ekman's six (happiness, sadness, fear, anger, surprise, disgust; contempt added in 1986), Plutchik's eight primaries arranged as opposites on a wheel with dyads such as love and awe (1980) and Panksepp's seven brain systems. Dimensional models include Wundt's tridimensional theory, Russell's circumplex of valence by arousal (1980) and Barrett's theory of constructed emotion.

**11.2 The classic theories: from the body to the label.** The James-Lange theory (1884, 1885) says that bodily changes come first and that the feeling of them is the emotion ("we feel afraid because we tremble"). Cannon (1927) objected that emotion survives separation of viscera from brain, that visceral changes are similar across emotions, insensitive and slow, and that artificial arousal (Marañón's adrenaline injections) produces bodily symptoms and at most an "as if" emotion; the Cannon-Bard theory holds that the thalamus triggers feeling and arousal simultaneously and independently. Schachter and Singer's two-factor theory (1962) says that emotion requires arousal plus a cognitive label; in their adrenaline experiment, aroused participants without an explanation caught the euphoria or anger of a stooge. Misattribution of arousal (Dutton and Aron's bridge, 1974; Valins's false heartbeats) and Zillmann's excitation transfer follow from the theory.

**11.3 Appraisal, feedback, evolution and opposing processes.** Arnold (1960) introduced appraisal, and Lazarus built the cognitive appraisal (cognitive-mediational) theory: primary appraisal asks whether an event matters and whether it is harm, threat or challenge, secondary appraisal asks what can be done, and reappraisal revises both; the subincision film study showed that changing appraisals changes arousal. Zajonc argued for affective primacy from the mere exposure effect (1980, 1984), and Lazarus replied that appraisal can be automatic. The facial feedback hypothesis (Tomkins, Izard; Strack, Martin and Stepper's pen study, 1988) holds that expressions modulate feelings, although the effect is small. Darwin (1872) explained expressions by serviceable habits, antithesis and direct nervous action and founded the evolutionary view. Solomon and Corbit's opponent-process theory (1974) explains tolerance, withdrawal and acquired pleasures through a slow, strengthening b-process.

**11.4 The emotional brain and body.** The sympathetic division of the autonomic nervous system arouses (dilating the pupils, lowering skin resistance) and the parasympathetic division calms; the medulla of the hindbrain paces heartbeat and breathing, and the hypothalamus orchestrates. Bard's decorticate cats showed sham rage, organised by the hypothalamus; Hess elicited

attack by hypothalamic stimulation. Papez (1937) proposed a circuit from hippocampus through fornix, mammillary bodies, anterior thalamus and cingulate gyrus; MacLean named the limbic system (1952), the classic "seat of emotion", and the triune brain. Bilateral temporal lobe removal produces Klüver-Bucy syndrome (tameness, psychic blindness, hyperorality, hypersexuality); the amygdala detects threat, supports fear conditioning and enhances emotional memory. LeDoux described a fast low road and a slower high road to the amygdala; affective blindsight shows emotional processing without awareness. Damasio's somatic marker hypothesis gives the ventromedial prefrontal cortex a role in decisions. The polygraph detects arousal, not lies; the guilty knowledge test (Lykken, 1959) is more accurate.

**11.5 Expressing and reading emotion.** Ekman and Friesen's studies of the Fore of New Guinea supported the universality of six facial expressions, with fear and surprise most often confused; blind athletes produce the same expressions as sighted ones. Display rules, cultural norms for showing emotion, explain cultural differences (Friesen's Japanese and American students, 1972), and emotional labour at work contributes to burnout. Recognition shows an in-group advantage. FACS (1978) codes expressions as muscle action units; the Duchenne smile involves the orbicularis oculi around the eyes; microexpressions may betray concealed feelings, but people detect lies at little better than chance. Body posture can be more informative than the face at high intensity. Women are on average more facially expressive and better at decoding nonverbal cues, differences attributed mainly to socialisation.

**11.6 Four emotions up close.** Fear is a response to present danger; anxiety is diffuse and future-oriented, and Spielberger separated state from trait anxiety. Anger is an emotion, aggression a behaviour and hostility an attitude; the catharsis hypothesis fails, because venting while ruminating increases aggression (Bushman, 2002). Diener's subjective well-being has three components: life satisfaction, positive affect and low negative affect. Helson's adaptation-level theory and Brickman and Campbell's hedonic treadmill explain why good fortune fades (lottery winners), and relative deprivation (Stouffer, 1949) explains why comparisons matter. Fredrickson's broaden-and-build theory says positive emotions widen thought and build resources, and savoring (Bryant and Veroff) prolongs positive experience. Disgust, studied by Rozin, protects against contamination and follows the laws of contagion and similarity.

**11.7 Emotional development and regulation.** Bridges (1932) described differentiation from general excitement into distress and delight and then into specific emotions; primary emotions appear in the first six months, while self-conscious emotions such as embarrassment, pride, shame and guilt appear after 12 months, once self-recognition develops. Infants use social referencing (the visual cliff) and expect emotional reciprocity (Tronick's still face). Gross's process model (1998) lists five families of regulation strategies: situation selection, situation modification, attentional deployment, cognitive change (including reappraisal) and response modulation (including suppression); reappraisal is generally healthier than suppression. Alexithymia (Sifneos, 1973) is difficulty identifying and describing one's feelings. Goleman's five domains of emotional intelligence are pointed to Chapter 9.

**11.8 Stress and the body's emergency systems.** Stress has been defined as a stimulus (the stressor), as a response (Selye: the nonspecific response of the body to any demand) and, most usefully, as a transaction in which demands are appraised as exceeding resources. Cannon described the fight-or-flight response (1915), run by the fast sympathetic-adrenal-medullary axis with adrenaline and noradrenaline, and coined homeostasis. Selye's general adaptation syndrome (1936; *The Stress of Life*, 1956) has three stages: alarm, resistance and exhaustion, the last bringing diseases of adaptation; he also distinguished eustress from distress (1974). The slower HPA axis

runs from hypothalamic CRH to anterior pituitary ACTH to cortisol from the adrenal cortex, with negative feedback; chronic activation produces allostatic load (McEwen). Taylor's tend-and-befriend (2000) describes an affiliative stress response, especially in females, involving oxytocin.

**11.9 Sources of stress and the transactional model.** Holmes and Rahe's Social Readjustment Rating Scale (1967) scores 43 life events in life change units (death of a spouse 100, marriage 50), counting positive as well as negative change, but it ignores appraisal and predicts illness only weakly; India has its own Presumptive Stressful Life Events Scale. Daily hassles (Kanner, Coyne, Schaefer and Lazarus, 1981) predict symptoms better than major events; uplifts matter less. Stressors may be acute or chronic, and cataclysmic, personal or background (Lazarus and Cohen, 1977). Frustration, conflict (Lewin's types and Miller's gradients), pressure and change are further sources. Work stress arises from environmental, organisational (task and role demands) and individual factors; role conflict, role ambiguity and Karasek's high-demand, low-control jobs are especially harmful, and chronic work stress can lead to burnout (Freudenberger, 1974; Maslach's three dimensions). Control and predictability reduce the damage of stressors (Weiss's rats). Lazarus and Folkman's transactional model (1984) explains stress through primary appraisal, secondary appraisal, coping and reappraisal, and the diathesis-stress model explains disorder as vulnerability triggered by stress.

**11.10 Coping, personality and health.** Coping is the changing effort to manage demands appraised as taxing. Problem-focused coping acts on the stressor and emotion-focused coping on the feelings; people use both, and the best strategy fits the controllability of the situation. Approach coping wins in the long run over avoidance coping; defence mechanisms are unconscious, coping is conscious. Social support (emotional, instrumental, informational and appraisal) protects health directly and by buffering stress; Weiss separated emotional from social isolation. Hardiness (Kobasa, 1979: commitment, control, challenge), dispositional optimism (Scheier and Carver), optimistic explanatory style and learned optimism (Seligman, building on learned helplessness), resilience (Werner's Kauai study) and Antonovsky's sense of coherence (comprehensibility, manageability, meaningfulness) protect against stress. Friedman and Rosenman's Type A pattern (competitiveness, time urgency, hostility) predicted heart disease in the Western Collaborative Group Study, but hostility is the toxic component; Type B is relaxed, Type C (cancer-prone) has weak support, and Denollet's Type D combines negative affectivity with social inhibition. Engel's biopsychosocial model (1977) frames health psychology; Ader and Cohen's conditioned immunosuppression (1975) founded psychoneuroimmunology, and chronic stress suppresses immunity and slows healing, while acute stress can boost innate immunity. Health behaviour is predicted by the health belief model, protection motivation theory and the theory of planned behaviour. Stress management uses progressive muscle relaxation (Jacobson), the relaxation response (Benson), biofeedback, meditation and mindfulness (Kabat-Zinn), exercise, cognitive restructuring and stress inoculation training (Meichenbaum).

## Who's who

Table 11.12 The people in this chapter.

| Name | When | Examined for |
|---|---|---|

| Name | When | Examined for |
|---|---|---|
| Robert Ader | 1932 to 2011 | Conditioned immunosuppression with Cohen (1975); founder of psychoneuroimmunology |
| Icek Ajzen | 1985, 1991 | Theory of planned behaviour |
| Aaron Antonovsky | 1923 to 1994 | Salutogenesis and the sense of coherence (1979, 1987) |
| Magda B. Arnold | 1903 to 2002 | Introduced appraisal into emotion theory (1960) |
| Arthur P. Aron | 1974 | Capilano bridge study of misattribution, with Dutton |
| Philip Bard | 1898 to 1977 | Cannon-Bard theory; sham rage and the hypothalamus |
| Lisa Feldman Barrett | 2017 | Theory of constructed emotion |
| Herbert Benson | 1935 to 2022 | The relaxation response (1975) |
| Joseph V. Brady | 1958 | "Executive monkey" ulcer study, flawed by non-random assignment |
| Katharine M. Banham Bridges | 1932 | Infant emotions differentiate from general excitement into distress and delight |
| Paul Broca | 1824 to 1880 | Named the "great limbic lobe" (1878) |
| Paul Bucy | 1904 to 1992 | Klüver-Bucy syndrome (1937, 1939) |
| Larry Cahill | 1994 | Propranolol abolishes the memory advantage of emotional stories (with McGaugh) |
| Walter Bradford Cannon | 1871 to 1945 | Fight or flight (1915); homeostasis; five criticisms of James-Lange and the Cannon-Bard theory (1927) |
| Charles S. Carver | 1985 | Dispositional optimism and the Life Orientation Test (with Scheier); COPE |
| Nicholas Cohen | 1975 | Conditioned immunosuppression, with Ader |
| Sheldon Cohen | 1985, 1991 | Stress-buffering hypothesis (with Wills); stress and susceptibility to colds |
| John D. Corbit | 1974 | Opponent-process theory, with Solomon |
| Antonio Damasio | born 1944 | Somatic marker hypothesis; *Descartes' Error* (1994) |
| Charles Darwin | 1809 to 1882 | *The Expression of the Emotions in Man and Animals* (1872); principle of antithesis |
| Johan Denollet | 1996 | Type D (distressed) personality: negative affectivity plus social inhibition |
| Ed Diener | 1946 to 2021 | Subjective well-being and its tripartite model (1984) |
| Guillaume-Benjamin-Amand Duchenne de Boulogne | 1806 to 1875 | Genuine smile involves the muscle around the eye (1862) |
| Donald G. Dutton | 1974 | Capilano bridge study of misattribution, with Aron |
| Paul Ekman | born 1934 | Six (later seven) basic emotions; Fore study; display rules; FACS (1978) |

| Name | When | Examined for |
|---|---|---|
| George L. Engel | 1913 to 1999 | Biopsychosocial model (1977) |
| Susan Folkman | 1984 | Transactional model and ways of coping, with Lazarus; meaning-focused coping |
| Barbara Fredrickson | 1998, 2001 | Broaden-and-build theory of positive emotions; undo effect |
| Herbert J. Freudenberger | 1926 to 1999 | Coined the term burnout (1974) |
| Meyer Friedman | 1910 to 2001 | Type A behaviour pattern, with Rosenman (1959) |
| Wallace V. Friesen | 1969 to 1978 | Ekman's collaborator: Fore study, display rules (1972), FACS |
| Phineas Gage | 1823 to 1860 | Frontal lobe injury (1848) and change of personality |
| James J. Gross | 1998 | Process model of emotion regulation; reappraisal versus suppression |
| John Martyn Harlow | 1868 | Physician who reported Gage's case |
| Harry Helson | 1898 to 1977 | Adaptation-level theory (1964) |
| Walter Rudolf Hess | 1881 to 1973 | Hypothalamic stimulation produces attack; Nobel Prize (1949) |
| George W. Hohmann | 1966 | Spinal cord injury and reduced intensity of emotion |
| Thomas H. Holmes | 1967 | Social Readjustment Rating Scale, with Rahe |
| Carroll Ellis Izard | 1923 to 2017 | Differential emotions theory (ten fundamental emotions) |
| Edmund Jacobson | 1888 to 1983 | Progressive muscle relaxation (1929) |
| William James | 1842 to 1910 | James-Lange theory (1884): "we feel afraid because we tremble" |
| Jon Kabat-Zinn | born 1944 | Mindfulness-Based Stress Reduction (1979) |
| Allen D. Kanner | 1981 | Hassles and Uplifts Scales, with Coyne, Schaefer and Lazarus |
| Robert A. Karasek | 1979 | Job demand-control model |
| Janice K. Kiecolt-Glaser | 1984, 1995 | Stress, immunity and slower wound healing |
| Heinrich Klüver | 1897 to 1979 | Klüver-Bucy syndrome (1937, 1939) |
| Suzanne C. Kobasa | 1979 | Hardiness: commitment, control, challenge |
| Carl Georg Lange | 1834 to 1900 | James-Lange theory (1885) |
| Richard S. Lazarus | 1922 to 2002 | Cognitive appraisal; transactional model (1984); daily hassles; debate with Zajonc |
| Joseph LeDoux | born 1949 | Low road and high road to the amygdala; *The Emotional Brain* (1996) |

| Name | When | Examined for |
|---|---|---|
| Robert W. Levenson | 1983 | Different autonomic patterns for different emotions (with Ekman and Friesen) |
| Kurt Lewin | 1890 to 1947 | Approach-approach, avoidance-avoidance and approach-avoidance conflicts (1935) |
| David T. Lykken | 1959 | Guilty knowledge test |
| Paul D. MacLean | 1913 to 2007 | Named the limbic system (1952); triune brain |
| Steven F. Maier | 1967 | Learned helplessness, with Seligman |
| Gregorio Marañón | 1924 | Adrenaline injections produce bodily symptoms and at most "as if" emotion |
| Christina Maslach | born 1946 | Maslach Burnout Inventory (1981): exhaustion, depersonalisation, reduced accomplishment |
| Bruce S. McEwen | 1938 to 2020 | Allostatic load |
| James L. McGaugh | 1994 | Stress hormones, the amygdala and emotional memory |
| Paul E. Meehl | 1920 to 2003 | Diathesis-stress account of schizophrenia (1962) |
| Donald Meichenbaum | born 1940 | Stress inoculation training (book, 1985) |
| Neal E. Miller | 1909 to 2002 | Approach and avoidance gradients; early work behind biofeedback |
| Jaak Panksepp | 1943 to 2017 | Seven subcortical emotional systems |
| James Papez | 1883 to 1958 | Papez circuit (1937) |
| Robert Plutchik | 1927 to 2006 | Wheel of eight primary emotions and dyads (1980) |
| Richard H. Rahe | 1967 | Social Readjustment Rating Scale, with Holmes |
| Ray H. Rosenman | 1920 to 2013 | Type A behaviour pattern; Western Collaborative Group Study |
| Irwin M. Rosenstock | 1950s | Health belief model |
| Paul Rozin | 1987 | Disgust; laws of contagion and similarity |
| James A. Russell | 1980 | Circumplex model of valence and arousal; core affect |
| Stanley Schachter | 1922 to 1997 | Two-factor theory of emotion (1962) |
| Michael F. Scheier | 1985 | Dispositional optimism and the Life Orientation Test (with Carver) |
| Martin E. P. Seligman | born 1942 | Learned helplessness (1967); explanatory style; *Learned Optimism* (1990) |
| Hans Selye | 1907 to 1982 | General adaptation syndrome (1936): alarm, resistance, exhaustion; eustress and distress |
| Peter E. Sifneos | 1920 to 2008 | Coined alexithymia (1973) |
| Jerome E. Singer | 1934 to 2010 | Two-factor theory, with Schachter (1962); *Urban Stress*, with Glass (1972) |

| Name | When | Examined for |
|---|---|---|
| Richard L. Solomon | 1918 to 1995 | Opponent-process theory of emotion (1974) |
| Fritz Strack | 1988 | Pen-in-the-mouth test of facial feedback |
| Shelley E. Taylor | born 1946 | Tend-and-befriend (2000) |
| Lydia Temoshok | 1980s | Type C (cancer-prone) pattern |
| Silvan Solomon Tomkins | 1911 to 1991 | Affect theory; nine innate affects (1962) |
| Stuart Valins | 1966 | False heartbeat feedback increases attraction |
| John Broadus Watson | 1878 to 1958 | Three innate emotions: fear, rage, love |
| Jay M. Weiss | 1971 | Control and predictability reduce stress ulcers in rats |
| Robert S. Weiss | 1973 | Emotional isolation versus social isolation |
| Emmy E. Werner | 1929 to 2017 | Kauai longitudinal study of resilience |
| Redford B. Williams | 1983, 1993 | Hostility as the toxic component of Type A; *Anger Kills* |
| Wilhelm Wundt | 1832 to 1920 | Tridimensional theory of feeling |
| Robert B. Zajonc | 1923 to 2008 | Affective primacy and the mere exposure effect (1968, 1980) |
| Dolf Zillmann | 1970s | Excitation transfer theory |

## Key terms

acute stressor, adaptation-level theory, affect, affect theory, affective blindsight, affective primacy, alarm reaction, alexithymia, allostatic load, amygdala, anxiety, approach coping, arousal, attentional deployment, autonomic nervous system, avoidance coping, basic emotions, biofeedback, biopsychosocial model, broaden-and-build theory, burnout, Cannon-Bard theory, cataclysmic events, catharsis hypothesis, chronic stressor, circumplex model, cognitive appraisal theory, cognitive change, cognitive reappraisal, cognitive restructuring, cognitive-mediational theory, conscious experience, control question technique, coping, core affect, cortisol, daily hassles, diathesis-stress model, differential emotions theory, diseases of adaptation, disgust, display rules, dispositional optimism, distress, Duchenne smile, emotion, emotion regulation, emotion-focused coping, emotional isolation, eustress, evolutionary view of emotion, excitation transfer theory, explanatory style, expressive behaviour, expressive suppression, Facial Action Coding System, facial feedback hypothesis, fear, feeling, fight-or-flight response, galvanic skin response, general adaptation syndrome, guilty knowledge test, hardiness, health belief model, hedonic treadmill, high road, homeostasis, hostility, hypothalamic-pituitary-adrenal axis, James-Lange theory, job demand-control model, Klüver-Bucy syndrome, learned optimism, life change units, life satisfaction, limbic system, low road, medulla oblongata, mere exposure effect, microexpression, mindfulness, misattribution of arousal, mood, negative affectivity, opponent-process theory, Papez circuit, parasympathetic nervous system, physiological arousal, polygraph, primary appraisal, primary dyads, principle of antithesis, problem-focused coping, process model of emotion regulation, progressive muscle relaxation, protection motivation theory, pseudobulbar affect, psychoevolutionary theory, psychoneuroimmunology, reappraisal, relative deprivation, relaxation response, resilience, re-

sponse modulation, role ambiguity, role conflict, salutogenesis, savoring, secondary appraisal, self-conscious emotions, sense of coherence, sham rage, situation modification, situation selection, social inhibition, social isolation, Social Readjustment Rating Scale, social referencing, social support, somatic marker hypothesis, stage of exhaustion, stage of resistance, stress, stress inoculation training, stress-buffering hypothesis, stressor, subjective well-being, sympathetic nervous system, sympathetic-adrenal-medullary axis, temperament, tend-and-befriend, theory of constructed emotion, theory of planned behaviour, transactional model of stress, tripartite model, triune brain, two-factor theory, Type A behaviour pattern, Type B, Type C, Type D personality, universality hypothesis, uplifts, valence

## From the past papers

1. Which of the following is NOT one of the six universally recognized basic emotions described by Ekman? (RML 2025)

   (a) Anger (b) Fear (c) Jealousy (d) Disgust

   **Answer: (c) Jealousy.** Ekman's six are happiness, sadness, fear, anger, surprise and disgust (contempt was added in 1986). Jealousy is a complex social emotion, a blend of fear, anger and sadness about losing a relationship to a rival, and has no universal facial expression.

2. Our current mood and feelings are best described by the term: (RML 2024)

   (a) Emotion (b) Mood (c) Affect (d) Feeling

   **Answer: (c) Affect.** Affect is the umbrella term for one's present feeling state and, clinically, its moment-to-moment expression. A mood is a sustained, pervasive state lasting hours or days; an emotion is a brief response to a specific event; a feeling is only the subjective part of an emotion.

3. According to the James-Lange theory of emotion, what is the correct order of events? (RML 2025)

   (a) Event → Emotion → Interpretation → Arousal (b) Event → Arousal → Emotion (c) Event → Arousal → Interpretation → Emotion (d) Interpretation → Arousal → Emotion

   **Answer: (c) Event, arousal, interpretation, emotion.** In James-Lange the body reacts first and the felt emotion is the awareness (the "interpretation") of the bodily change. Option (b) is not false but leaves out that step, so the fuller textbook sequence (c) is keyed. Options (a) and (d) put emotion or interpretation before arousal, which reverses the theory.

4. Schachter and Singer's theory of emotion differs from the theories of James-Lange and Cannon-Bard. How does it differ? (RML 2020)

   (a) It includes development and arousal (b) It includes the limbic system (c) It includes cognition and arousal (d) It has never been criticised

   **Answer: (c) It includes cognition and arousal.** The two-factor theory requires both physiological arousal and a cognitive label taken from the situation. It says nothing about development or the limbic system, and it has been criticised heavily, notably by the failed replications of Marshall and Zimbardo and of Maslach (1979).

5. Which theory of emotion explains the phenomenon of misattribution of arousal (for example, feeling romantic attraction when actually scared)? (RML 2025)

   (a) James-Lange theory (b) Two-factor theory (c) Cannon-Bard theory (d) Facial feedback hypothesis

   **Answer: (b) Two-factor theory.** If emotion is arousal plus a label, the label can be wrong, as on the Capilano bridge. James-Lange and Cannon-Bard have no labelling step, and the facial feedback hypothesis concerns the effect of one's own expressions on feelings.

6. According to experts on socioemotional development like Campos (2009) and Kagan (2010), self-conscious emotions appear at which age? (RML 2022)

   (a) 6 to 9 months (b) Before 9 months (c) After 12 months (d) 3 months

   **Answer: (c) After 12 months.** Embarrassment, pride, shame and guilt require a sense of self, which appears with mirror self-recognition at about 15 to 24 months; these emotions emerge from about 18 months. The earlier ages fit the primary emotions, which are present by about six months.

7. The first emotions expressed by infants are: (IHBAS 2020)

   (a) Pain and anger (b) Rage and fear (c) Fear and euphoria (d) Delight and distress

   **Answer: (d) Delight and distress.** In Bridges's differentiation scheme (1932) the newborn's general excitement splits first into distress and delight, and only later into anger, fear, elation and the rest. Rage and fear (with love) are Watson's three innate emotions, a different and older claim.

8. In James Gross's process model of emotion regulation, cognitive reappraisal belongs to which family of strategies? (Recall set)

   (a) Response modulation (b) Cognitive change (c) Situation selection (d) Attentional deployment

   **Answer: (b) Cognitive change.** Reappraisal changes the meaning of the situation before the emotional response is fully formed. Response modulation includes expressive suppression; situation selection is choosing which situations to enter; attentional deployment includes distraction and rumination.

9. Savoring is a concept in Positive Psychology that involves: (IHBAS 2024)

   (a) Avoiding negative experiences (b) Enhancing the enjoyment of positive experiences (c) Minimizing stress through relaxation (d) Reflecting on past traumas

   **Answer: (b) Enhancing the enjoyment of positive experiences.** Bryant and Veroff defined savoring as attending to, appreciating and prolonging good experiences, past, present or future. Avoiding bad experiences and relaxing away stress are forms of coping, not savoring, and dwelling on trauma is rumination.

10. Barbara Fredrickson is known for her work on: (IHBAS 2024)

    (a) Cognitive distortions (b) Flow theory (c) Broaden-and-build theory (d) Psychodynamic theory

    **Answer: (c) Broaden-and-build theory.** Fredrickson proposed that positive emotions broaden thought-action repertoires and build lasting resources. Flow is Csikszentmihalyi's, cognitive distortions are Beck's (and Burns's), and psychodynamic theory is Freud's.

11. Choose the odd one out regarding the pattern of sympathetic activation during emotion: (Amity 2023)

    (a) Increased heart rate (b) Pupil constriction (c) Lowered skin resistance (d) Increased breathing rate

    **Answer: (b) Pupil constriction.** Sympathetic arousal dilates the pupils; constriction is parasympathetic. Faster heart rate and breathing, and lowered skin resistance from sweating (the galvanic skin response), are all sympathetic effects.

12. Non-voluntary coping responses like heart-beat, respiration are modulated by: (JEMAS 2025)

    (a) Hypothalamus (b) Cerebellum (c) Hind brain (d) Amygdala

    **Answer: (c) Hind brain.** The cardiovascular and respiratory centres lie in the medulla oblongata, part of the hindbrain. The hypothalamus orchestrates them only indirectly, the amygdala triggers fear responses through the hypothalamus and brainstem, and the cerebellum coordinates movement.

13. The seat of emotion is to be found in: (RML 2024)

    (a) Reticular formation (b) Hind brain (c) Limbic system (d) Forebrain

    **Answer: (c) Limbic system.** MacLean's limbic system (1952), including the amygdala, hypothalamus, hippocampus and cingulate gyrus, is the classic seat of emotion. The reticular formation governs arousal and wakefulness, the hindbrain vital functions, and "forebrain" is too broad an answer, although the limbic system lies within it.

14. Which cluster of symptoms is seen in Klüver-Bucy syndrome? (JEMAS 2025)

    (a) Agnosia, aphasia, hyperorality, hyperphagia, hypersexuality (b) Agnosia, avolition, alogia, asexuality (c) Hypersexuality, altered emotional response, hyperorality, hyperalertness (d) Hypersexuality, hypo-sensitivity, delayed reaction

    **Answer: (a).** Bilateral temporal lobe (amygdala) damage produces psychic blindness (agnosia), hyperorality, hyperphagia and hypersexuality, and in humans often aphasia because the nearby temporal cortex is damaged. Option (c) is tempting, but the monkeys became placid, not hyperalert; avolition and alogia are negative symptoms of schizophrenia, and "asexuality" is the opposite of what happens.

15. 'Psychological disorders develop when a biological predisposition is set off by a stressful situation', is proposed by: (JEMAS 2025)

    (a) Humanistic existential models (b) Socio-cultural model (c) Diathesis-stress model (d) Cognitive model

    **Answer: (c) Diathesis-stress model.** A diathesis (vulnerability) plus a stressful trigger is the definition of the model (Meehl, 1962; Zubin and Spring, 1977). Humanistic-existential models stress blocked growth and meaning, sociocultural models social forces, and cognitive models maladaptive thinking; none of them is defined by predisposition plus stress.

16. In order, starting from the first stage, what are the stages of the general adaptation syndrome? (RML 2024)

    (a) Resistance, alarm, exhaustion (b) Alarm, resistance, exhaustion (c) Resistance, illness, alarm (d) Alarm, exhaustion, illness

    **Answer: (b) Alarm, resistance, exhaustion.** Selye's body first mobilises (alarm), then adapts (resistance), then, if stress continues, runs down (exhaustion). Illness is a consequence of exhaustion, not a stage, and resistance cannot come before the alarm that triggers it.

17. Which endocrine gland is primarily responsible for controlling the body's stress response and stress-related metabolic changes? (Real PYQ 2025)

    (a) Thyroid gland (b) Adrenal glands (c) Pituitary gland (d) Pancreas

    **Answer: (b) Adrenal glands.** The adrenal medulla releases adrenaline and noradrenaline (SAM axis) and the adrenal cortex releases cortisol (HPA axis). The pituitary releases ACTH but is not itself the stress gland; the thyroid sets metabolic rate and the pancreas controls blood sugar through insulin and glucagon.

18. Which of the following would be considered an acute stressor? (RML 2024)

    (a) A pop quiz (b) HIV disease (c) Marital discord (d) Colon cancer

    **Answer: (a) A pop quiz.** An acute stressor is brief, with a clear beginning and end. HIV disease, cancer and a troubled marriage last for months or years with no clear end, so they are chronic stressors.

19. Social Readjustment Rating Scale of Holmes and Rahe focuses upon: (IHBAS 2025)

    (a) Life events (b) Physiological reactions (c) Cognitive changes (d) Daily hassles

    **Answer: (a) Life events.** The SRRS (1967) scores 43 major life changes in life change units, from death of a spouse (100) to minor violations of the law (11). Daily hassles were measured by Kanner and colleagues (1981); physiological reactions belong to Selye's tradition and cognitive appraisal to Lazarus's.

20. According to the Transactional Model of Stress, which term refers to the individual's assessment of whether they have the resources to cope with a stressor? (IHBAS 2024)

    (a) Initial threat evaluation (b) Secondary appraisal (c) Coping resource assessment (d) Situational capacity evaluation

    **Answer: (b) Secondary appraisal.** In Lazarus and Folkman's model (1984), primary appraisal judges whether an event is relevant and whether it is harm, threat or challenge; secondary appraisal judges the options and resources for coping. The other three phrases sound plausible but are not terms of the model.

21. Which factor is an organizational stressor leading to job related stress? (IHBAS 2024)

    (a) Family and economic problems (b) Political uncertainty (c) Task and role demands (d) Number of dependents in family

    **Answer: (c) Task and role demands.** Organisational stressors arise within the job: task demands, role demands (role conflict, ambiguity, overload), interpersonal demands, structure and leadership. Family and economic problems and the number of dependants are individual factors; political uncertainty is an environmental factor.

22. Type A behavior pattern is characterized by: (IHBAS 2020)

    (a) Non-competitiveness, impatience and hostility (b) Impatience, non-competitiveness, and aggression (c) Competitiveness, impatience and hostility (d) Hostility, lack of aggression and competitiveness

    **Answer: (c) Competitiveness, impatience and hostility.** Friedman and Rosenman's Type A combines competitive achievement striving, time urgency (impatience) and hostility. Every other option contains a "non-" or "lack of" that contradicts the pattern.

23. Which of the following factor is most directly linked to a higher risk for coronary heart disease? (RML 2024)

    (a) Type A personality (b) Type B personality (c) Hostility (d) Self-efficacy

    **Answer (as keyed): (a) Type A personality.** Type A was the classic coronary-prone pattern of the Western Collaborative Group Study, and it is the keyed answer. Be aware, though, that later research identifies hostility, one component of Type A, as the part that carries most of the risk, so a stem asking for the "toxic component" of Type A should be answered with hostility. Type B is the low-risk pattern, and self-efficacy is a protective belief.

24. The "D-type" personality, identified as a risk factor for poor prognosis in cardiovascular disease, is characterized by: (Real PYQ 2024)

    (a) Hostility and competitiveness (b) The joint tendency to experience negative emotions and to inhibit self-expression in social situations (social inhibition) (c) An optimistic and resilient outlook on life (d) Perfectionism and rigidity

    **Answer: (b).** Denollet's Type D (distressed) personality is negative affectivity combined with social inhibition. Hostility and competitiveness describe Type A; optimism and resilience are protective factors; perfectionism and rigidity belong to obsessive-compulsive personality descriptions.

25. Learned helplessness phenomenon is offered by: (JEMAS 2025)

    (a) Aaron T. Beck (b) Martin Seligman and Steven F. Maier (c) Harry Stack Sullivan (d) Daniel Kahneman

    **Answer: (b) Martin Seligman and Steven F. Maier.** Their dogs (1967), given inescapable shocks, later failed to escape avoidable ones; the idea became a model of depression and, through explanatory style, of learned optimism. Beck is known for cognitive therapy, Sullivan for the interpersonal theory of psychiatry and Kahneman for judgement heuristics and prospect theory.

## High-yield sheet

*The facts this chapter's questions ask most, most-asked first. Last read before the exam.*

1 **The limbic system** is the classic "seat of emotion": Paul MacLean named it in 1952, enlarging James Papez's 1937 circuit, and it regulates emotion, motivation and memory. Hunger and thirst are the hypothalamus; arousal is the reticular formation; coordination is the cerebellum. ASKED 7 TIMES

2 **The amygdala** is the structure for fear and aggression: threat detection, fear conditioning (LeDoux's low and high roads) and the emotional enhancement of memory (Cahill and McGaugh, 1994). The hippocampus stores the facts; the amygdala adds the emotional boost. ASKED 7 TIMES

3 **The diathesis-stress model**: a disorder appears when a pre-existing vulnerability (diathesis, often biological) is set off by a stressful situation (Meehl, 1962; Zubin and Spring, 1977). Not genes alone, environment alone or unconscious conflict alone. ASKED 4 TIMES

4 **Klüver-Bucy syndrome** (Heinrich Klüver and Paul Bucy, 1937 and 1939): bilateral removal of the anterior temporal lobes (amygdala) in monkeys produces tameness and loss of fear, hyperorality, hypersexuality, psychic blindness (visual agnosia) and hyperphagia. ASKED 4 TIMES

5 **James-Lange theory** (William James, 1884; Carl Lange, 1885): event, bodily arousal, awareness ("interpretation") of the arousal, emotion; "we feel afraid because we tremble". Feeling and arousal at the same time, via the thalamus, is Cannon-Bard (1927). ASKED 4 TIMES

6 **Type A behaviour pattern** (Meyer Friedman and Ray Rosenman, 1959): competitiveness, time urgency (impatience) and hostility; it predicted coronary heart disease (about twice the risk) in the Western Collaborative Group Study. Hostility is the toxic component; Type B is the low-risk pattern. ASKED 3 TIMES

7 **Repression** is the involuntary blocking of painful feelings from awareness (suppression is deliberate); **isolation of affect** keeps the idea but strips its feeling (a gruesome fantasy told calmly); **projection** attributes one's own unacceptable feelings to someone else. ASKED 3 TIMES

8 **The adrenal glands** are the stress glands. HPA axis: hypothalamus (CRH), anterior pituitary (ACTH), adrenal cortex (cortisol), with cortisol's negative feedback on the hypothalamus and anterior pituitary; the adrenal medulla releases adrenaline and noradrenaline (SAM axis). ASKED 4 TIMES

9 **Displacement** redirects a feeling onto a safer target (angry with her husband, a woman shouts at her child); **sublimation** channels it into a socially valued activity (anger into a kickboxing class). ASKED 3 TIMES

10 **Schachter and Singer's two-factor theory** (1962): emotion is physiological arousal plus a cognitive label, so it adds cognition to arousal, and it explains misattribution of arousal (Dutton and Aron's Capilano bridge, 1974). ASKED TWICE

11 **Selye's general adaptation syndrome** (1936; *The Stress of Life*, 1956): alarm, resistance, exhaustion, always in that order. Illness is not a stage, and the GAS is not life events theory or the health belief model. ASKED TWICE

12 **Sympathetic arousal** dilates the pupils, speeds heart rate and breathing and lowers skin resistance; pupil constriction is parasympathetic. Heartbeat and breathing are paced by the medulla of the **hindbrain**. ASKED TWICE

13 **Infant emotions**: in Bridges's scheme (1932) general excitement differentiates first into **distress and delight**; **self-conscious emotions** (embarrassment, pride, shame, guilt) appear after 12 months, from about 18 months, once mirror self-recognition develops. ASKED TWICE

14 **Type D personality** (Johan Denollet, mid-1990s; *Lancet* 1996): negative affectivity plus social inhibition, a predictor of poor prognosis in coronary heart disease. ASKED TWICE

15 **Affect** is the current feeling state and its observable expression, rated for quality, range and reactivity; **mood** is the sustained, pervasive state, defined by its persistence. ASKED TWICE

16 **Papez circuit** (1937): hippocampus, fornix, mammillary bodies, anterior thalamus, cingulate gyrus, entorhinal cortex, back to the hippocampus. ASKED TWICE

17 **Pseudobulbar affect**: sudden, brief, involuntary outbursts of laughing or crying out of proportion to the actual mood, after stroke, multiple sclerosis, ALS or brain injury; it is not depression. ASKED TWICE

18 **Ekman's six** universal emotions: happiness, sadness, fear, anger, surprise and disgust (contempt added in 1986). Jealousy, shame and guilt are not on the list. ASKED ONCE

19 **Gross's process model** (1998): situation selection, situation modification, attentional deployment, cognitive change (reappraisal) and response modulation (suppression). ASKED ONCE

20 **Lazarus and Folkman's transactional model** (1984): primary appraisal asks whether an event matters (harm, threat or challenge); **secondary appraisal** asks whether I have the resources to cope. ASKED ONCE

21 **Holmes and Rahe's Social Readjustment Rating Scale** (1967) measures life events: 43 changes in life change units, death of a spouse 100, marriage 50. Daily hassles are Kanner, Coyne, Schaefer and Lazarus (1981). ASKED ONCE

22 **Acute versus chronic stressors**: a pop quiz is acute; HIV disease, cancer and marital discord are chronic. ASKED ONCE

23 **Organisational stressors** are task and role demands (and interpersonal demands, structure, leadership); family and economic problems are individual factors, political uncertainty an environmental one. ASKED ONCE

24 **Biopsychosocial model** (George Engel, 1977): biological, psychological and social factors, not "cultural" in place of social. ASKED ONCE

25 **Barbara Fredrickson**: the broaden-and-build theory of positive emotions (1998, 2001) and the undo effect. Flow is Csikszentmihalyi's. ASKED ONCE

26 **Savoring** (Fred Bryant and Joseph Veroff): attending to, enhancing and prolonging the enjoyment of positive experiences. ASKED ONCE

27 **Emotional isolation** (Robert Weiss, 1973) is the lack of a deep attachment to one specific person; lacking a circle of friends is social isolation. ASKED ONCE

28 **Learned helplessness**: Martin Seligman and Steven Maier (1967), dogs given inescapable shock; not Beck, Sullivan or Kahneman. ASKED ONCE

29 **Theory of planned behaviour** (Icek Ajzen): attitude, subjective norm and perceived behavioural control (keyed as "self-efficacy") shape intention. ASKED ONCE

30 **Diener's tripartite model** of subjective well-being: life satisfaction (cognitive), positive affect and low negative affect. ASKED ONCE

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Page numbers point to where a term is defined (in bold in the text) or where a person is introduced. The two editions share these page numbers.

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